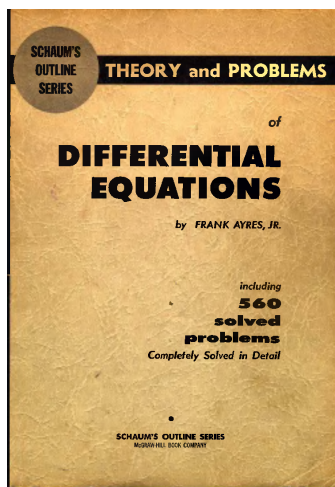


A Solution Manual For

Schaums Outline. Theory and problems of
Differential Equations, 1st edition. Frank Ayres.
McGraw Hill 1952



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December 3, 2025

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Contents

1	Lookup tables for all problems in current book	5
2	Book Solved Problems	31

Lookup tables for all problems in current book

Local contents

1.1	Chapter 2. Solutions of differential equations. Supplementary problems. Page 11	7
1.2	Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22	7
1.3	Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33	10
1.4	Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39	12
1.5	Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65	15
1.6	Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74	16
1.7	Chapter 12. Linear equations of order n. Supplementary problems. Page 81	17
1.8	Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86	18
1.9	Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92	19
1.10	Chapter 15. Linear equations with constant coefficients (Variation of parameters). Supplementary problems. Page 98	20
1.11	Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107	21
1.12	Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre). Supplementary problems. Page 110	22

1.13	Chapter 18. Linear equations with variable coefficients (Equations of second order). Supplementary problems. Page 120	23
1.14	Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132	25
1.15	Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163	26
1.16	Chapter 25. Integration in series. Supplementary problems. Page 205	27
1.17	Chapter 26. Integration in series (singular points). Supplementary problems. Page 218	28
1.18	Chapter 27. The Legendre, Bessel and Gauss Equations. Supplementary problems. Page 230	29

1.1 Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Table 1.1: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7845	13	$y'x = 2y$	✓	✓	✓	✓
7846	14	$y'y + x = 0$	✓	✓	✓	✓
7847	15	$y = y'x + y'^4$	✓	✓	✓	✗
7848	16	$2x^3y' = y(3x^2 + y^2)$	✓	✓	✓	✓
7849	17	$y - 2y' + y'' = 0$	✓	✓	✓	✓
7850	18	$(-1 + x)y'' - y'x + y = 0$	✓	✓	✓	✗
7851	19	$y'' - y = 0$	✓	✓	✓	✓
7852	20	$y'' - y = 4 - x$	✓	✓	✓	✓
7853	21	$2y - 3y' + y'' = 0$	✓	✓	✓	✓
7854	22	$2y - 3y' + y'' = 2e^x(1 - x)$	✓	✓	✓	✓

1.2 Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Table 1.2 Lookup table

Continued from previous page

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7869	40	$xyy' = (1 + y)(1 - x)$	✓	✓	✓	✓
7870	41	$y^2 - x^2 + xyy' = 0$	✓	✓	✓	✓
7871	42	$y(1 + 2yx) + x(-yx + 1)y' = 0$	✓	✓	✓	✓
7872	43	$1 + (-x^2 + 1) \cot(y) y' = 0$	✓	✓	✓	✓
7873	44	$x^3 + y^3 + 3xy^2y' = 0$	✓	✓	✓	✓
7874	45	$3x + 2y + 1 - (3x + 2y - 1)y' = 0$	✓	✓	✓	✓
7875	46	$y'x + 2y = 0$ $y(2) = 1$	✓	✓	✓	✓
7876	47	$xyy' + x^2 + y^2 = 0$ $y(1) = -1$	✓	✓	✓	✓
7877	48	$\cos(y) + (1 + e^{-x}) \sin(y) y' = 0$ $y(0) = \frac{\pi}{4}$	✓	✓	✓	✓
7878	49	$y^2 + yx - y'x = 0$ $y(1) = 1$	✓	✓	✓	✓
7879	51	$y' = -2(2x + 3y)^2$	✓	✓	✓	✓
7880	52	$x - 2 \sin(y) + 3 + (2x - 4 \sin(y) - 3) \cos(y) y' = 0$	✓	✓	✓	✓

1.3 Chapter 5. Equations of first order and first degree (Exact equations). Supplemetary problems.

Page 33

Table 1.3: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7881	23 (a)	$x^2 - y - y'x = 0$	✓	✓	✓	✓
7882	23 (d)	$x^2 + y^2 + 2xyy' = 0$	✓	✓	✓	✓
7883	23 (e)	$x + y \cos(x) + \sin(x) y' = 0$	✓	✓	✓	✓
7884	23 (h)	$2x + 3y + 4 + (3x + 4y + 5) y' = 0$	✓	✓	✓	✓
7885	23 (i)	$4x^3y^3 + \frac{1}{x} + \left(3x^4y^2 - \frac{1}{y}\right) y' = 0$	✓	✓	✓	✓
7886	23 (j)	$2u^2 + 2uv + (u^2 + v^2) v' = 0$	✓	✓	✓	✗
7887	23 (k)	$x\sqrt{x^2 + y^2} - y + \left(y\sqrt{x^2 + y^2} - x\right) y' = 0$	✓	✓	✓	✗
7888	23 (m)	$x + y + 1 - (3 - x + y) y' = 0$	✓	✓	✓	✓
7889	23 (o)	$y^2 - \frac{y}{x(x+y)} + 2 + \left(\frac{1}{x+y} + 2(1+x)y\right) y' = 0$	✓	✓	✓	✗
7890	23 (p)	$2xy e^{x^2y} + y^2 e^{xy^2} + 1 + \left(x^2 e^{x^2y} + 2xy e^{xy^2} - 2y\right) y' = 0$	✓	✓	✓	✗
7891	24 (p)	$y(x - 2y) - x^2y' = 0$	✓	✓	✓	✓

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
7892	24 (c)	$xyy' + x^2 + y^2 = 0$	✓	✓	✓	✓
7893	24 (d)	$x^2 + y^2 + 2xyy' = 0$	✓	✓	✓	✓
7894	24 (g)	$1 - \sqrt{a^2 - x^2} y' = 0$	✓	✓	✓	✓
7895	24 (L)	$x + y + 1 - (-3 + x - y) y' = 0$	✓	✓	✓	✓
7896	25 (a)	$x - x^2 - y^2 + y'y = 0$	✓	✓	✓	✓
7897	25 (b)	$2y - 3x + y'x = 0$	✓	✓	✓	✓
7898	25 (c)	$x - y^2 + 2xyy' = 0$	✓	✓	✓	✓
7899	25 (d)	$-y - 3(x^2 + y^2) x^2 + y'x = 0$	✓	✓	✓	✓
7900	25 (e)	$y - \ln(x) - y'x = 0$	✓	✓	✓	✓
7901	25 (f)	$3x^2 + y^2 - 2xyy' = 0$	✓	✓	✓	✓
7902	25 (g)	$yx - 2y^2 - (x^2 - 3yx) y' = 0$	✓	✓	✓	✓
7903	25 (h)	$x + y - (x - y) y' = 0$	✓	✓	✓	✓
7904	25 (L)	$2y - 3xy^2 - y'x = 0$	✓	✓	✓	✓
7905	25 (j)	$y + x(x^2y - 1) y' = 0$	✓	✓	✓	✗
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Table 1.3 Lookup table

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
7906	25 (k)	$y + x^3y + 2x^2 + (x + 4xy^4 + 8y^3) y' = 0$	✓	✓	✓	✗
7907	26 (a)	$-y - e^x x^2 + y'x = 0$	✓	✓	✓	✓
7908	26 (b)	$1 + y^2 = (x^2 + x) y'$	✓	✓	✓	✓
7909	26 (c)	$2y - x^3 + y'x = 0$	✓	✓	✓	✓
7910	26 (d)	$y + (y^2 - x) y' = 0$	✓	✓	✓	✓
7911	26 (e)	$3y^3 - yx - (x^2 + 6xy^2) y' = 0$	✓	✓	✓	✓
7912	26 (f)	$3x^2y^2 + 4(x^3y - 3) y' = 0$	✓	✓	✓	✗
7913	26 (g)	$y(x + y) - x^2y' = 0$	✓	✓	✓	✓
7914	26 (h)	$2y + 3xy^2 + (x + 2x^2y) y' = 0$	✓	✓	✓	✓
7915	26 (i)	$y(y^2 - 2x^2) + x(2y^2 - x^2) y' = 0$	✓	✓	✓	✓
7916	27	$-y + y'x = 0$	✓	✓	✓	✓

1.4 Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Table 1.4: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7917	19 (a)	$y' + y = 2x + 2$	✓	✓	✓	✓
7918	19 (c)	$y' - y = yx$	✓	✓	✓	✓
7919	19 (d)	$-3y - (x - 2)e^x + y'x = 0$	✓	✓	✓	✓
7920	19 (e)	$i' - 6i = 10 \sin(2t)$	✓	✓	✓	✓
7921	19 (f)	$y' + y = y^2 e^x$	✓	✓	✓	✓
7922	19 (g)	$y + (yx + x - 3y)y' = 0$	✓	✓	✓	✓
7923	19 (h)	$(2s - e^{2t})s' = 2s e^{2t} - 2 \cos(2t)$	✓	✓	✓	✗
7924	19 (i)	$y'x + y - x^3 y^6 = 0$	✓	✓	✓	✓
7925	19 (j)	$r' + 2r \cos(\theta) + \sin(2\theta) = 0$	✓	✓	✓	✓
7926	19 (k)	$y(1 + y^2) = 2(1 - 2xy^2)y'$	✓	✓	✓	✓
7927	19 (L)	$y'y - xy^2 + x = 0$	✓	✓	✓	✓
7928	19 (m)	$(x - x\sqrt{x^2 - y^2})y' - y = 0$	✓	✓	✓	✗
7929	19 (o)	$2x' - \frac{x}{y} + x^3 \cos(y) = 0$	✓	✓	✓	✓
7930	19 (p)	$y'x = y(1 - x \tan(x)) + x^2 \cos(x)$	✓	✓	✓	✓

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Table 1.4 Lookup table

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
7931	19 (q)	$2 + y^2 - (yx + 2y + y^3) y' = 0$	✓	✓	✓	✓
7932	19 (r)	$1 + y^2 = (\arctan(y) - x) y'$	✓	✓	✓	✓
7933	19 (s)	$2y^5x - y + 2y'x = 0$	✓	✓	✓	✓
7934	19 (t)	$1 + \sin(y) = (2y \cos(y) - x(\sec(y) + \tan(y))) y'$	✓	✓	✓	✗
7935	22 (a)	$y'x = 2y + e^x x^3$ $y(1) = 0$	✓	✓	✓	✓
7936	22 (b)	$Li' + Ri = E \sin(2t)$ $i(0) = 0$	✓	✓	✓	✓
7937	23 (a)	$x^2 \cos(y) y' = 2x \sin(y) - 1$	✓	✓	✓	✗
7938	23 (b)	$4x^2yy' = 3x(3y^2 + 2) + 2(3y^2 + 2)^3$	✗	✓	✓	✓
7939	23 (c)	$xy^3 - y^3 - e^x x^2 + 3xy^2y' = 0$	✓	✓	✓	✓
7940	23 (d)	$y' + x(x + y) = x^3(x + y)^3 - 1$	✓	✓	✓	✓
7941	23 (e)	$y + e^y - e^{-x} + (1 + e^y) y' = 0$	✓	✓	✓	✓

Table 1.6 Lookup table

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
7964	17	$y(3 - 4y)^2 y'^2 = 4 - 4y$	✓	✓	✓	✓
7965	18	$y'^3 - 4x^4 y' + 8x^3 y = 0$	✓	✓	✗	✗
7966	19	$(1 + y'^2)(x - y)^2 = (y'y + x)^2$	✓	✓	✓	✗

1.7 Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Table 1.7: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7967	10	$y'' + y' - 6y = 0$	✓	✓	✓	✓
7968	11	$y''' - 6y'' + 12y' - 8y = 0$	✓	✓	✓	✓
7969	12	$2y - 3y' + y'' = e^{5x}$	✓	✓	✓	✓
7970	13	$y'' + 9y = \cos(x)x$	✓	✓	✓	✓
7971	14	$x^2 y'' - 3y'x + 4y = 0$	✓	✓	✓	✓
7972	15	$-y + y'x + x^3 y''' = 3x^4$	✓	✓	✓	✓
7973	16	$xy'' - y' + 4x^3 y = 0$	✓	✓	✓	✓
7974	17	$y'' + y'^2 + 1 = 0$	✓	✓	✓	✓

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Table 1.9 Lookup table

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
7996	20	$y'' - y = \frac{1}{(1 + e^{-x})^2}$	✓	✓	✓	✓
7997	21	$y'' + y = \csc(x)$	✓	✓	✓	✓
7998	22	$2y - 3y' + y'' = \sin(e^{-x})$	✓	✓	✓	✓

1.10 Chapter 15. Linear equations with constant coefficients (Variation of parameters). Supplementary problems. Page 98

Table 1.10: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
7999	10	$y'' + y = \csc(x)$	✓	✓	✓	✓
8000	11	$4y + y'' = 4 \sec(x)^2$	✓	✓	✓	✓
8001	12	$y'' - 4y' + 3y = \frac{1}{1 + e^{-x}}$	✓	✓	✓	✗
8002	13	$y'' - y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$	✓	✓	✓	✓
8003	14	$y'' - y = \frac{1}{(1 + e^{-x})^2}$	✓	✓	✓	✓
8004	15	$y'' + 2y = e^x + 2$	✓	✓	✓	✓

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Table 1.11 Lookup table

Continued from previous page

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8015	30	$y''' + y = \cos(x)$	✓	✓	✓	✓
8016	31	$4y + y'' = \sin(2x)$	✓	✓	✓	✓
8017	32	$y'' + 5y = \cos(\sqrt{5}x)$	✓	✓	✓	✓
8018	33	$y''' + y'' + y' + y = e^x + e^{-x} + \sin(x)$	✓	✓	✓	✓
8019	34	$y'' - y = x^2$	✓	✓	✓	✓
8020	36	$y'' + 2y = x^3 + x^2 + e^{-2x} + \cos(3x)$	✓	✓	✓	✓
8021	37	$y'' - 2y' - y = \cos(x) e^x$	✓	✓	✓	✓
8022	38	$y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$	✓	✓	✓	✓
8023	39	$y'' - y = x e^{3x}$	✓	✓	✓	✓
8024	40	$y'' + 5y' + 6y = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$	✓	✓	✓	✓

1.12 Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre). Supplementary problems. Page 110

Table 1.12: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8025	6	$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$	✓	✓	✓	✓
8026	7	$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$	✓	✓	✓	✓
8027	8	$x^3 y''' + 2x^2 y'' = x + \sin(\ln(x))$	✓	✓	✓	✓
8028	9	$-y + y'x + x^3 y''' = 3x^4$	✓	✓	✓	✓
8029	10	$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$	✓	✓	✓	✗
8030	11	$-12y - 2(2x+1) y' + (2x+1)^2 y'' = 6x$	✓	✓	✓	✗

**1.13 Chapter 18. Linear equations with variable coefficients (Equations of second order).
Supplementary problems. Page 120**

Table 1.13: Lookup table for all problems in current section.

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8031	21	$xy'' - (x + 2)y' + 2y = 0$	✓	✓	✓	✗
8032	22	$2y - 2y'x + (x^2 + 1)y'' = 2$	✓	✓	✓	✗
8033	23	$(x^2 + 4)y'' - 2y'x + 2y = 8$	✓	✓	✓	✗
8034	24	$(1 + x)y'' - (2x + 3)y' + (x + 2)y = (x^2 + 2x + 1)e^{2x}$	✓	✓	✓	✗

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Table 1.13 Lookup table

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ID	problem	ODE	Solved?	Maple	Mma	Sympy
8035	25	$y'' - 2 \tan(x) y' - 10y = 0$	✓	✓	✓	✗
8036	26	$x^2 y'' - x(2x + 3) y' + (x^2 + 3x + 3) y = (-x^2 + 6) e^x$	✓	✓	✓	✗
8037	27	$4x^2 y'' + 4x^3 y' + (x^2 + 1)^2 y = 0$	✓	✓	✓	✗
8038	28	$x^2 y'' + (-4x^2 + x) y' + (4x^2 - 2x + 1) y = (x^2 - x + 1) e^x$	✓	✓	✓	✗
8039	29	$xy'' - y' + 4x^3 y = 0$	✓	✓	✓	✓
8040	31	$x^8 y'' + 4x^7 y' + y = \frac{1}{x^3}$	✓	✓	✓	✗
8041	32	$(x \sin(x) + \cos(x)) y'' - x \cos(x) y' + y \cos(x) = x$	✓	✓	✓	✗
8042	33	$xy'' - 3y' + \frac{3y}{x} = x + 2$	✓	✓	✓	✓
8043	35	$(1 + x) y'' - (3x + 4) y' + 3y = (3x + 2) e^{3x}$	✓	✓	✓	✗
8044	36	$x^2 y'' - 4y' x + (9x^2 + 6) y = 0$	✓	✓	✓	✓
8045	37	$xy'' + 2y' + 4yx = 4$	✓	✓	✓	✗
8046	38	$2y - 2y' x + (x^2 + 1) y'' = \frac{-x^2 + 1}{x}$	✓	✓	✓	✗

Table 1.14 Lookup table

Continued from previous page

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8058	33	$(1 + 2y + 3y^2) y''' + 6y'(y'' + y'^2 + 3yy'') = x$	✗	✓	✗	✗
8059	34	$3x(y^2 y''' + 6yy'y'' + 2y'^3) - 3y(yy'' + 2y'^2) = -\frac{2}{x}$	✗	✓	✗	✗
8060	35	$yy''' + 3y'y'' - 2yy'' - 2y'^2 + y'y = e^{2x}$	✗	✓	✗	✗
8061	36	$2(1 + y)y'' + 2y'^2 + y^2 + 2y = 0$	✓	✓	✓	✗

1.15 Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Table 1.15: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8062	10	$x'(t) - y'(t) + y(t) = -e^t$ $x(t) + y'(t) - y(t) = e^{2t}$	✓	✓	✓	✓
8063	11	$x'(t) + 2x(t) + y'(t) + y(t) = t$ $5x(t) + y'(t) + 3y(t) = t^2$	✓	✓	✓	✓
8064	12	$x'(t) + x(t) + 2y'(t) + 7y(t) = e^t + 2$ $-2x(t) + y'(t) + 3y(t) = e^t - 1$	✓	✓	✓	✓
8065	13	$x'(t) - x(t) + y'(t) + 3y(t) = e^{-t} - 1$ $x'(t) + 2x(t) + y'(t) + 3y(t) = 1 + e^{2t}$	✓	✓	✓	✗
8066	17	$x'(t) - x(t) + y'(t) + 2y(t) = 1 + e^t$ $y'(t) + 2y(t) + z'(t) + z(t) = e^t + 2$ $x'(t) - x(t) + z'(t) + z(t) = 3 + e^t$	✓	✓	✓	✓

Table 1.16 Lookup table

Continued from previous page

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8075	16	$p(1+p)y - 2y'x + (-x^2 + 1)y'' = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8076	17	$y'' + x^2y = x^2 + x + 1$ Series expansion around $x = 0$.	✓	✓	✓	✗

1.17 Chapter 26. Integration in series (singular points). Supplemetary problems. Page 218

Table 1.17: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8077	11	$2(x^3 + x^2)y'' - (-3x^2 + x)y' + y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8078	12	$4xy'' + 2(1 - x)y' - y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8079	13	$2x^2y'' - y'x + (-x^2 + 1)y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8080	14	$xy'' + y' + yx = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8081	15	$x^2y'' - y'x + (x^2 + 1)y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓

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Table 1.17 Lookup table

Continued from previous page

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8082	16	$xy'' - 2y' + y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8083	17	$xy'' + 2y' + yx = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8084	18	$x^2(1+x)y'' + x(1+x)y' - y = 0$ Series expansion around $x = 0$.	✓	✓	✓	✓
8085	19	$2xy'' + y' - y = 1 + x$ Series expansion around $x = 0$.	✓	✓	✓	✗
8086	20	$2x^3y'' + x^2y' + y = 0$ Series expansion around $x = \infty$.	✓	✓	✓	✓
8087	21	$x^3y'' + (x^2 + x)y' - y = 0$ Series expansion around $x = \infty$.	✓	✓	✓	✓

1.18 Chapter 27. The Legendre, Bessel and Gauss Equations. Supplementary problems. Page 230

Table 1.18: Lookup table for all problems in current section

ID	problem	ODE	Solved?	Maple	Mma	Sympy
8088	20	$z'' + tz' + \left(t^2 - \frac{1}{9}\right)z = 0$ Series expansion around $t = 0$.	✓	✓	✓	✓

Book Solved Problems

Local contents

2.1	Chapter 2. Solutions of differential equations. Supplemetary problems. Page 11	33
2.2	Chapter 4. Equations of first order and first degree (Variable separable). Supplemetary problems. Page 22	191
2.3	Chapter 5. Equations of first order and first degree (Exact equations). Supplemetary problems. Page 33	714
2.4	Chapter 6. Equations of first order and first degree (Linear equations). Supplemetary problems. Page 39	1061
2.5	Chapter 9. Equations of first order and higher degree. Supplemetary problems. Page 65	1318
2.6	Chapter 10. Singular solutions, Extraneous loci. Supplemetary problems. Page 74	1416
2.7	Chapter 12. Linear equations of order n. Supplemetary problems. Page 81	1488
2.8	Chapter 13. Homogeneous Linear equations with constant coefficients. Supplemetary problems. Page 86	1582
2.9	Chapter 14. Linear equations with constant coefficients. Supplemetary problems. Page 92	1635
2.10	Chapter 15. Linear equations with constant coefficients (Variation of parameters). Supplemetary problems. Page 98	1761
2.11	Chapter 16. Linear equations with constant coefficients (Short methods). Supplemetary problems. Page 107	1884
2.12	Chapter 17. Linear equations with variable coefficients (Cauchy and Legndre). Supplemetary problems. Page 110	2025
2.13	Chapter 18. Linear equations with variable coefficients (Equations of second order). Supplemetary problems. Page 120	2140

2.14	Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132	2347
2.15	Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163	2463
2.16	Chapter 25. Integration in series. Supplementary problems. Page 205	2500
2.17	Chapter 26. Integration in series (singular points). Supplementary problems. Page 218	2600
2.18	Chapter 27. The Legendre, Bessel and Gauss Equations. Supplementary problems. Page 230	2723

2.1 Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Local contents

2.1.1	Problem 13	34
2.1.2	Problem 14	57
2.1.3	Problem 15	87
2.1.4	Problem 16	91
2.1.5	Problem 17	131
2.1.6	Problem 18	139
2.1.7	Problem 19	153
2.1.8	Problem 20	164
2.1.9	Problem 21	174
2.1.10	Problem 22	181

2.1.1 Problem 13

Local contents

Solved using first_order_ode_linear	34
Solved using first_order_ode_separable	36
Solved using first_order_ode_exact	37
Solved using first_order_ode_homog_A	41
Solved using first_order_ode_homog_type_D2	44
Solved using first_order_ode_homog_type_G	46
Solved using first_order_ode_homog_type_maple_C	47
Solved using first_order_ode_LIE	51
✓ Maple	55
✓ Mathematica	56
✓ Sympy	56

Internal problem ID [7845]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 13

Date solved : Thursday, November 27, 2025 at 06:20:17 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.043 (sec)

Solve

$$y'x = 2y$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{2}{x}$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}\left(\frac{y}{x^2}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^2} &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_1 x^2$$

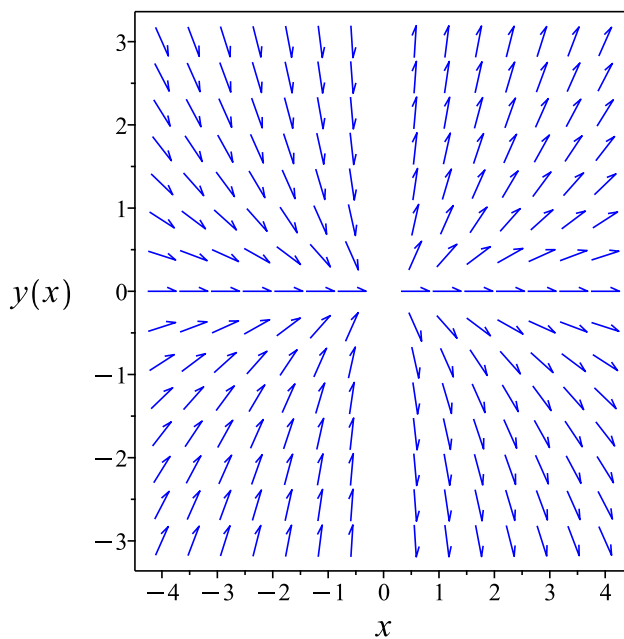


Figure 2.1: Slope field $y'x = 2y$

Summary of solutions found

$$y = c_1 x^2$$

Solved using first_order_ode_separable

Time used: 0.131 (sec)

Solve

$$y'x = 2y$$

The ode

$$y' = \frac{2y}{x} \tag{2.1}$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{2y}{x} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{2}{x} \\ g(y) &= y \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{y} dy &= \int \frac{2}{x} dx \end{aligned}$$

$$\ln(y) = \ln(x^2) + c_1$$

Taking the exponential of both sides the solution becomes

$$y = c_1 x^2$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y = 0$$

for y gives

$$y = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$y = c_1 x^2$$

$$y = 0$$

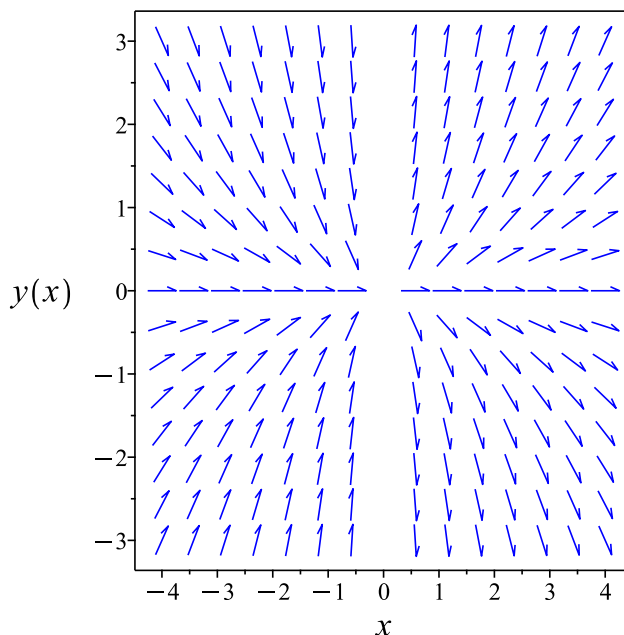


Figure 2.2: Slope field $y'x = 2y$

Summary of solutions found

$$y = 0$$

$$y = c_1 x^2$$

Solved using first_order_ode_exact

Time used: 0.080 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N\end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned}(x) dy &= (2y) dx \\ (-2y) dx + (x) dy &= 0\end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= -2y \\ N(x, y) &= x\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-2y) \\ &= -2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((-2) - (1)) \\ &= -\frac{3}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{3}{x} \, dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-3 \ln(x)} \\ &= \frac{1}{x^3} \end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{1}{x^3} (-2y) \\ &= -\frac{2y}{x^3} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{1}{x^3} (x) \\ &= \frac{1}{x^2} \end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(-\frac{2y}{x^3} \right) + \left(\frac{1}{x^2} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\int \frac{\partial \phi}{\partial y} dy = \int \overline{N} dy$$

$$\int \frac{\partial \phi}{\partial y} dy = \int \frac{1}{x^2} dy$$

$$\phi = \frac{y}{x^2} + f(x) \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{2y}{x^3} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = -\frac{2y}{x^3}$. Therefore equation (4) becomes

$$-\frac{2y}{x^3} = -\frac{2y}{x^3} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

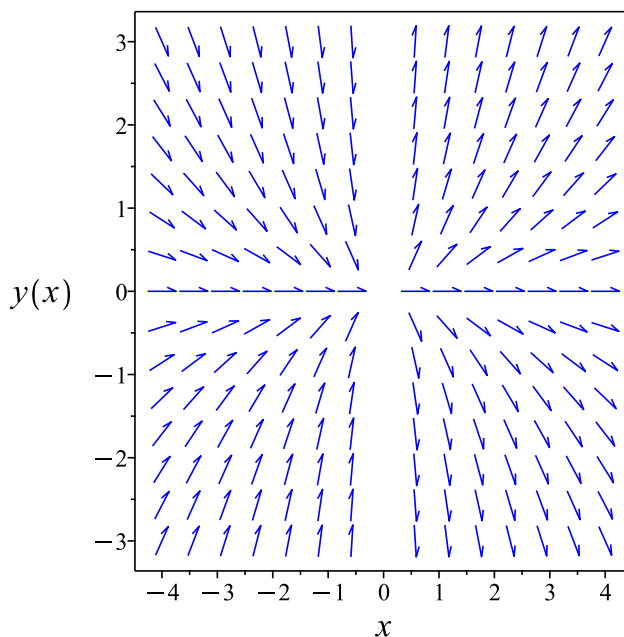
$$\phi = \frac{y}{x^2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y}{x^2}$$

Solving for y gives

$$y = c_1 x^2$$

Figure 2.3: Slope field $y'x = 2y$ Summary of solutions found

$$y = c_1 x^2$$

Solved using first_order_ode_homog_A

Time used: 0.375 (sec)

Solve

$$y'x = 2y$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{2y}{x} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = 2y$ and $N = x$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\frac{du}{dx}x + u = 2u$$

$$\frac{du}{dx} = \frac{u(x)}{x}$$

Or

$$u'(x) - \frac{u(x)}{x} = 0$$

Or

$$u'(x)x - u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = \frac{u(x)}{x} \tag{2.2}$$

is separable as it can be written as

$$u'(x) = \frac{u(x)}{x}$$

$$= f(x)g(u)$$

Where

$$f(x) = \frac{1}{x}$$

$$g(u) = u$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u} du = \int \frac{1}{x} dx$$

$$\ln(u(x)) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = c_1 x$$

$$u(x) = 0$$

Converting $u(x) = c_1 x$ back to y gives

$$y = c_1 x^2$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

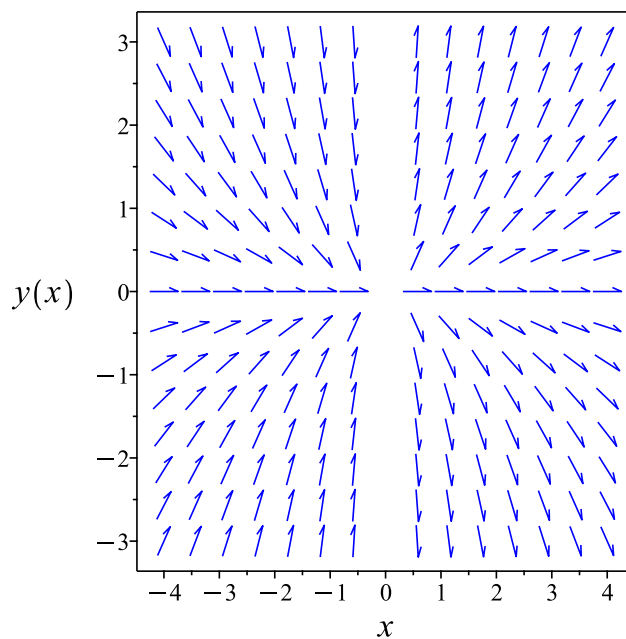


Figure 2.4: Slope field $y'x = 2y$

Summary of solutions found

$$y = 0$$

$$y = c_1 x^2$$

Solved using first_order_ode_homog_type_D2

Time used: 0.116 (sec)

Solve

$$y'x = 2y$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$(u'(x)x + u(x))x = 2u(x)x$$

Which is now solved The ode

$$u'(x) = \frac{u(x)}{x} \tag{2.3}$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u} du &= \int \frac{1}{x} dx \end{aligned}$$

$$\ln(u(x)) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = c_1 x$$

$$u(x) = 0$$

Converting $u(x) = c_1 x$ back to y gives

$$y = c_1 x^2$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

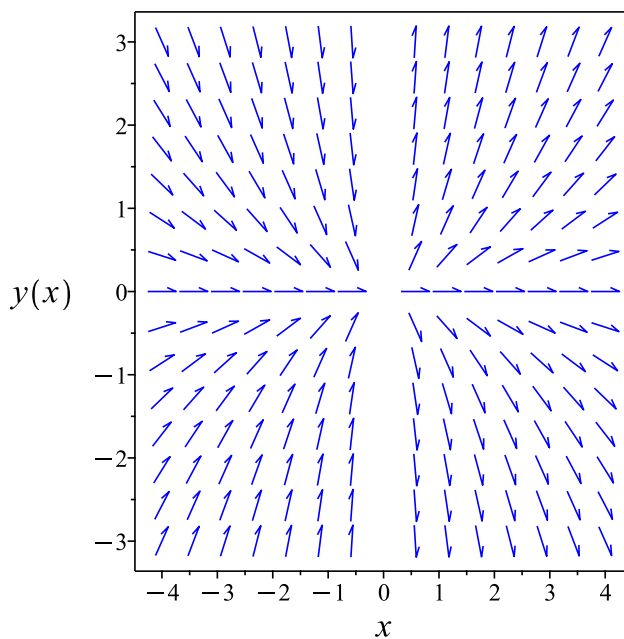


Figure 2.5: Slope field $y'x = 2y$

Summary of solutions found

$$y = 0$$

$$y = c_1 x^2$$

Solved using first_order_ode_homog_type_G

Time used: 0.046 (sec)

Solve

$$y'x = 2y$$

Multiplying the right side of the ode, which is $\frac{2y}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{2y}{x} \\ &= 2 \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ does not have y and does not have x then

$$\alpha = 0$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = 2$$

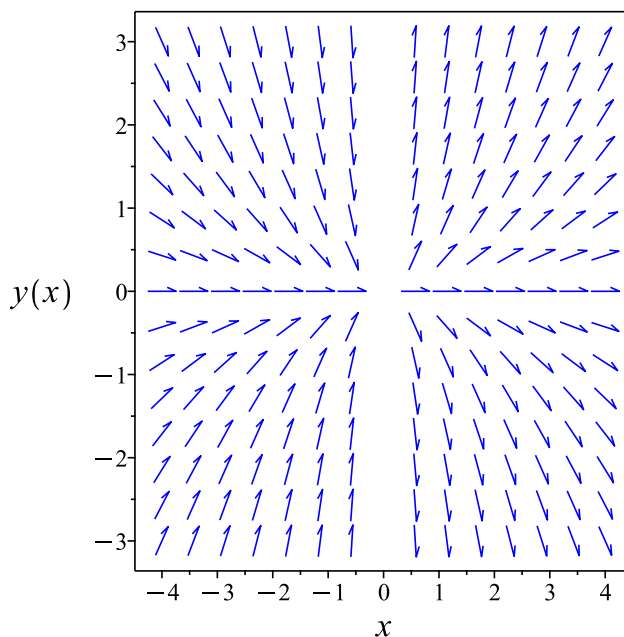
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^y -\frac{1}{2z} dz = 0$$

Figure 2.6: Slope field $y'x = 2y$ Summary of solutions found

$$\ln(x) - c_1 + \int^y -\frac{1}{2z} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.517 (sec)

Solve

$$y'x = 2y$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{2Y(X) + 2y_0}{X + x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{2Y(X)}{X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= \frac{2Y}{X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = 2Y$ and $N = X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= 2u \\ \frac{du}{dX} &= \frac{u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) X - u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = \frac{u(X)}{X} \quad (2.4)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= \frac{u(X)}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= \frac{1}{X} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{1}{u} du = \int \frac{1}{X} dX$$

$$\ln(u(X)) = \ln(X) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(X) = c_1 X$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(X) = c_1 X$$

$$u(X) = 0$$

Converting $u(X) = c_1 X$ back to $Y(X)$ gives

$$Y(X) = X^2 c_1$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$Y(X) = X^2 c_1 \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = c_1 x^2$$

Using the solution for $Y(X)$

$$Y(X) = 0 \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

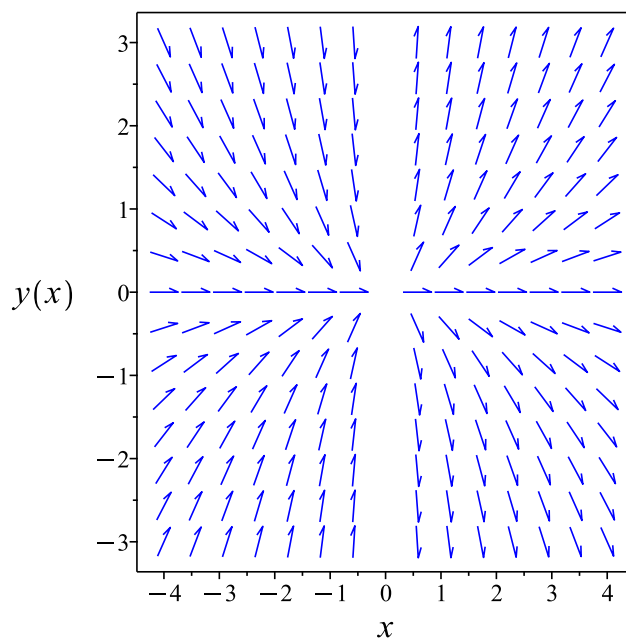


Figure 2.7: Slope field $y'x = 2y$

Summary of solutions found

$$y = 0$$

$$y = c_1 x^2$$

Solved using first_order_ode_LIE

Time used: 0.721 (sec)

Solve

$$y'x = 2y$$

Writing the ode as

$$y' = \frac{2y}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 + \frac{2y(b_3 - a_2)}{x} - \frac{4y^2 a_3}{x^2} + \frac{2y(xa_2 + ya_3 + a_1)}{x^2} - \frac{2(xb_2 + yb_3 + b_1)}{x} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{b_2 x^2 + 2y^2 a_3 + 2xb_1 - 2ya_1}{x^2} = 0$$

Setting the numerator to zero gives

$$-b_2 x^2 - 2y^2 a_3 - 2xb_1 + 2ya_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-2a_3v_2^2 - b_2v_1^2 + 2a_1v_2 - 2b_1v_1 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-2a_3v_2^2 - b_2v_1^2 + 2a_1v_2 - 2b_1v_1 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} 2a_1 &= 0 \\ -2a_3 &= 0 \\ -2b_1 &= 0 \\ -b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= a_2 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 0 \\ \eta &= y \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y} dy \end{aligned}$$

Which results in

$$S = \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{2y}{x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{2}{x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{2}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

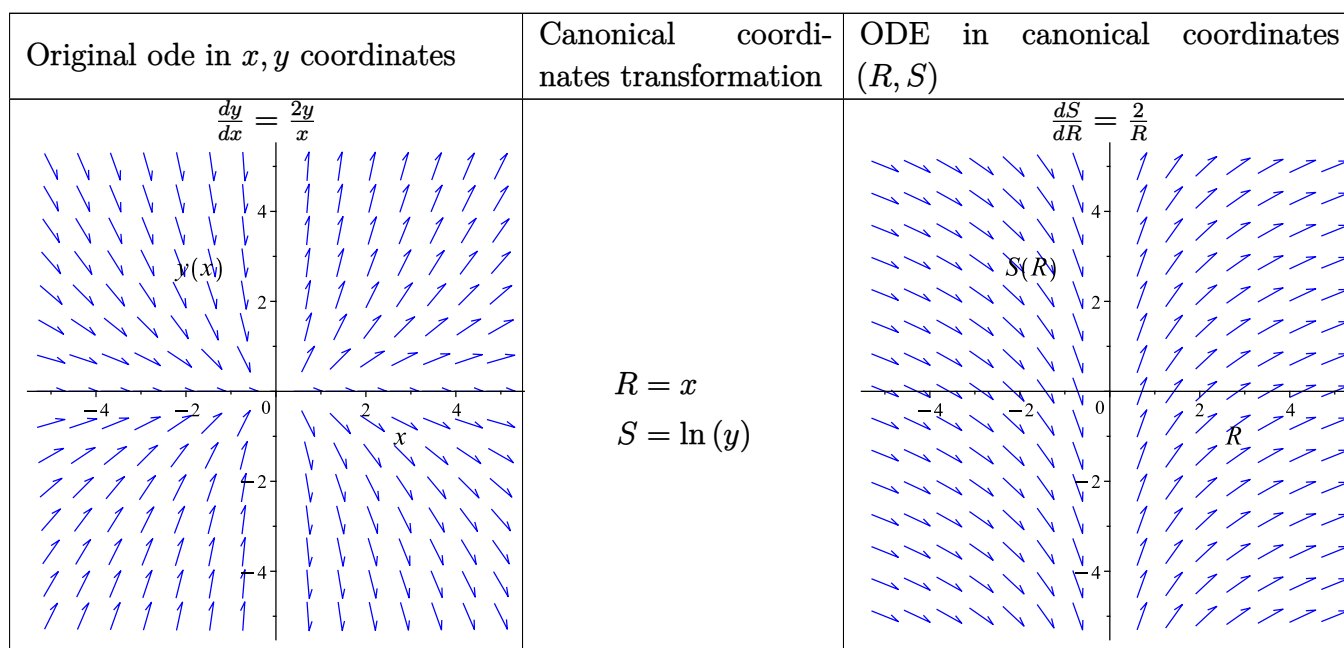
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{2}{R} dR \\ S(R) &= 2 \ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

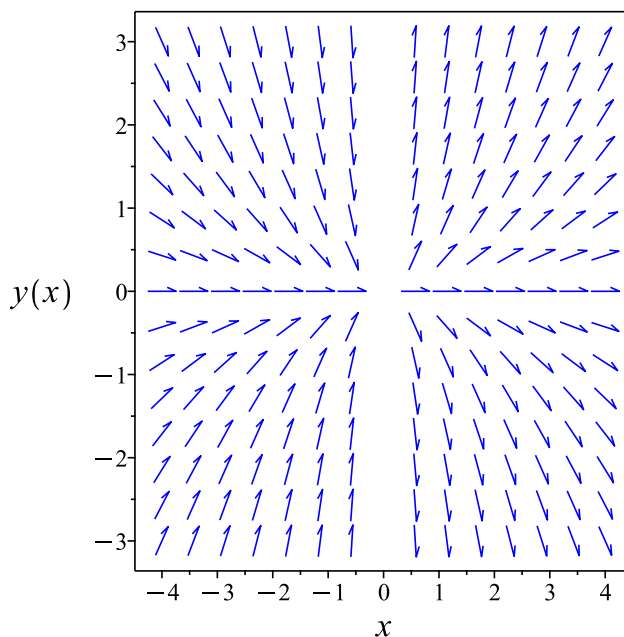
$$\ln(y) = 2 \ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = e^{c_2} x^2$$

Figure 2.8: Slope field $y'x = 2y$ Summary of solutions found

$$y = e^{c_2 x^2}$$

✓ Maple. Time used: 0.001 (sec). Leaf size: 9

```
ode:=diff(y(x),x)*x = 2*y(x);
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x^2$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right) = 2y(x)$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{2y(x)}{x}$$
- Separate variables

$$\frac{\frac{d}{dx}y(x)}{y(x)} = \frac{2}{x}$$
- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int \frac{2}{x} dx + C1$$
- Evaluate integral

$$\ln(y(x)) = 2\ln(x) + C1$$
- Solve for $y(x)$

$$y(x) = e^{C1} x^2$$
- Redefine the integration constant(s)

$$y(x) = C1 x^2$$

✓ **Mathematica.** Time used: 0.014 (sec). Leaf size: 16

```
ode=x*D[y[x],x]==2*y[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 x^2$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.061 (sec). Leaf size: 7

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) - 2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 x^2$$

2.1.2 Problem 14

Local contents

Solved using first_order_ode_separable	57
Solved using first_order_ode_bernoulli	59
Solved using first_order_ode_exact	61
Solved using first_order_ode_isobaric	64
Solved using first_order_ode_homog_A	67
Solved using first_order_ode_homog_type_D2	70
Solved using first_order_ode_homog_type_G	73
Solved using first_order_ode_homog_type_maple_C	75
Solved using first_order_ode_LIE	80
✓ Maple	85
✓ Mathematica	86
✓ Sympy	86

Internal problem ID [7846]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 14

Date solved : Thursday, November 27, 2025 at 06:20:25 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.061 (sec)

Solve

$$x + yy' = 0$$

The ode

$$y' = -\frac{x}{y} \quad (2.5)$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{x}{y} \\ &= f(x)g(y) \end{aligned}$$

Where

$$f(x) = -x$$

$$g(y) = \frac{1}{y}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int y dy = \int -x dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + c_1$$

Solving for y gives

$$y = \sqrt{-x^2 + 2c_1}$$

$$y = -\sqrt{-x^2 + 2c_1}$$

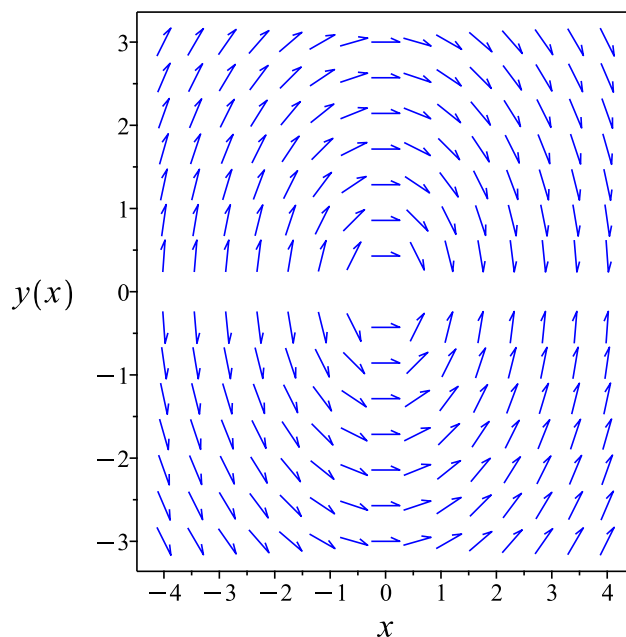


Figure 2.9: Slope field $x + yy' = 0$

Summary of solutions found

$$y = \sqrt{-x^2 + 2c_1}$$

$$y = -\sqrt{-x^2 + 2c_1}$$

Solved using first_order_ode_bernoulli

Time used: 0.132 (sec)

Solve

$$x + yy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x}{y} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = (-x) \frac{1}{y} \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= 0 \\ f_1 &= -x \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_1(x) \tag{3}$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= 0 \\ f_1(x) &= -x \\ n &= -1 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = 0 - x \tag{4}$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= y^2 \end{aligned} \tag{5}$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(x)}{2} &= -x \\ v' &= -2x \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dv &= \int -2x dx \\ v(x) &= -x^2 + c_1 \end{aligned}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^2 = -x^2 + c_1$$

Solving for y gives

$$\begin{aligned} y &= \sqrt{-x^2 + c_1} \\ y &= -\sqrt{-x^2 + c_1} \end{aligned}$$

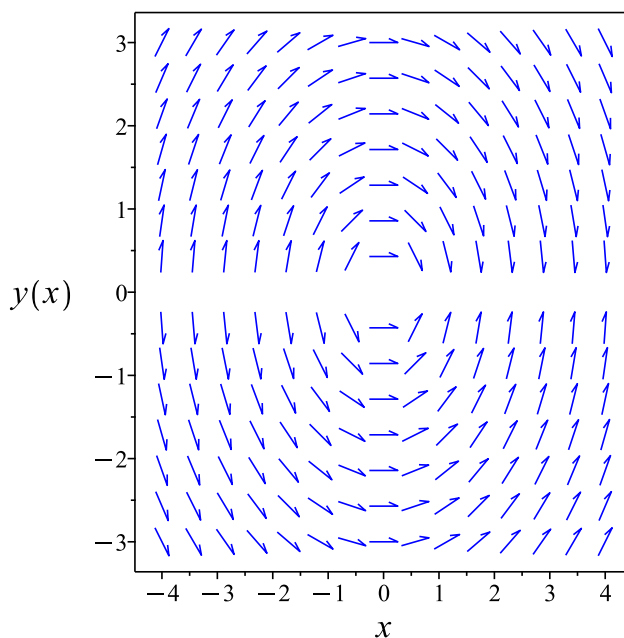


Figure 2.10: Slope field $x + yy' = 0$

Summary of solutions found

$$y = \sqrt{-x^2 + c_1}$$

$$y = -\sqrt{-x^2 + c_1}$$

Solved using first_order_ode_exact

Time used: 0.060 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (y) dy &= (-x) dx \\ (x) dx + (y) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x \\ N(x, y) &= y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x) \\ &= 0 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(y) \\ &= 0 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x dx \\ \phi &= \frac{x^2}{2} + f(y) \end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 0 + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = y$. Therefore equation (4) becomes

$$y = 0 + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = y$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int (y) dy$$

$$f(y) = \frac{y^2}{2} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x^2}{2} + \frac{y^2}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x^2}{2} + \frac{y^2}{2}$$

Solving for y gives

$$y = \sqrt{-x^2 + 2c_1}$$

$$y = -\sqrt{-x^2 + 2c_1}$$

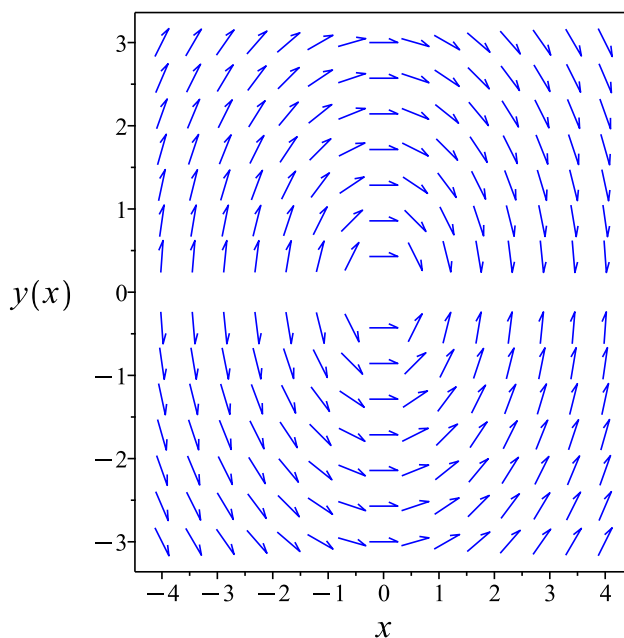


Figure 2.11: Slope field $x + yy' = 0$

Summary of solutions found

$$y = \sqrt{-x^2 + 2c_1}$$

$$y = -\sqrt{-x^2 + 2c_1}$$

Solved using first_order_ode_isobaric

Time used: 0.332 (sec)

Solve

$$x + yy' = 0$$

Solving for y' gives

$$y' = -\frac{x}{y} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{x}{y} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{1}{u(x)}$$

The ode

$$u'(x) = -\frac{u(x)^2 + 1}{u(x)x} \quad (2.6)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(u) = \frac{u^2 + 1}{u}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{u}{u^2 + 1} du = \int -\frac{1}{x} dx$$

$$\frac{\ln(u(x)^2 + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -i$$

$$u(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

$$u(x) = -i$$

$$u(x) = i$$

Converting $\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{y^2}{x^2} + 1} = \frac{c_1}{x}$$

Converting $u(x) = -i$ back to y gives

$$\frac{y}{x} = -i$$

Converting $u(x) = i$ back to y gives

$$\frac{y}{x} = i$$

Simplifying the above gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} = \frac{c_1}{x}$$

$$\frac{y}{x} = -i$$

$$\frac{y}{x} = i$$

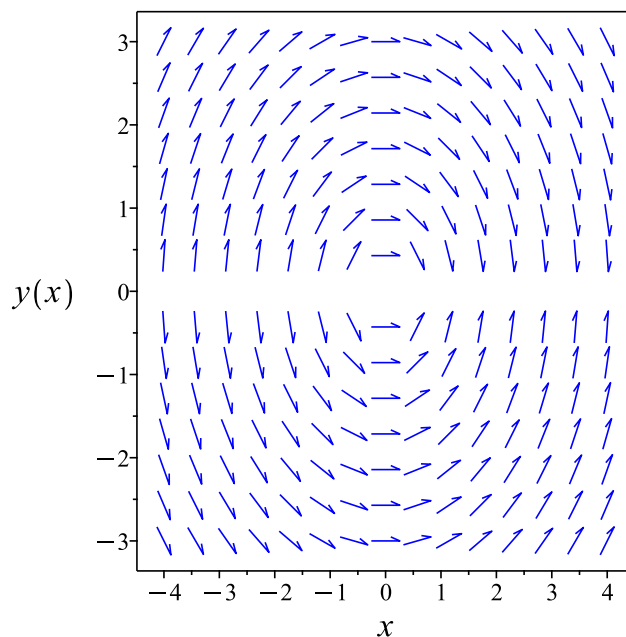
Solving for y gives

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Figure 2.12: Slope field $x + yy' = 0$ Summary of solutions found

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Solved using first_order_ode_homog_A

Time used: 0.587 (sec)

Solve

$$x + yy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x}{y} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x$ and $N = y$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= -\frac{1}{u} \\ \frac{du}{dx} &= \frac{-\frac{1}{u(x)} - u(x)}{x}\end{aligned}$$

Or

$$u'(x) - \frac{-\frac{1}{u(x)} - u(x)}{x} = 0$$

Or

$$u'(x)u(x)x + u(x)^2 + 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{u(x)^2 + 1}{u(x)x} \tag{2.7}$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= -\frac{u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= -\frac{1}{x} \\ g(u) &= \frac{u^2 + 1}{u}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{u^2 + 1} du &= \int -\frac{1}{x} dx\end{aligned}$$

$$\frac{\ln(u(x)^2 + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -i$$

$$u(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

$$u(x) = -i$$

$$u(x) = i$$

Converting $\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} = \frac{c_1}{x}$$

Converting $u(x) = -i$ back to y gives

$$y = -ix$$

Converting $u(x) = i$ back to y gives

$$y = ix$$

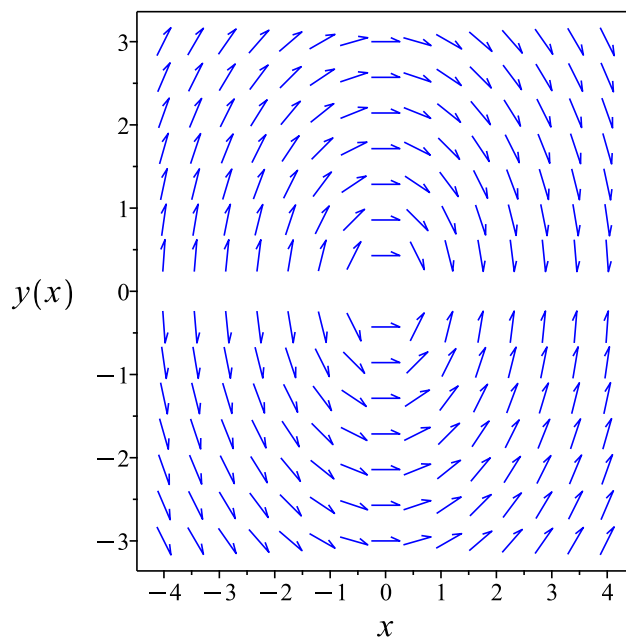
Solving for y gives

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Figure 2.13: Slope field $x + yy' = 0$ Summary of solutions found

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.203 (sec)

Solve

$$x + yy' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$x + u(x)x(u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{u(x)^2 + 1}{u(x)x} \quad (2.8)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{u^2 + 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(u(x)^2 + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$$

$$u(x) = -i$$

$$u(x) = i$$

Converting $\sqrt{u(x)^2 + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{y^2}{x^2} + 1} = \frac{c_1}{x}$$

Converting $u(x) = -i$ back to y gives

$$y = -ix$$

Converting $u(x) = i$ back to y gives

$$y = ix$$

Simplifying the above gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} = \frac{c_1}{x}$$

$$y = -ix$$

$$y = ix$$

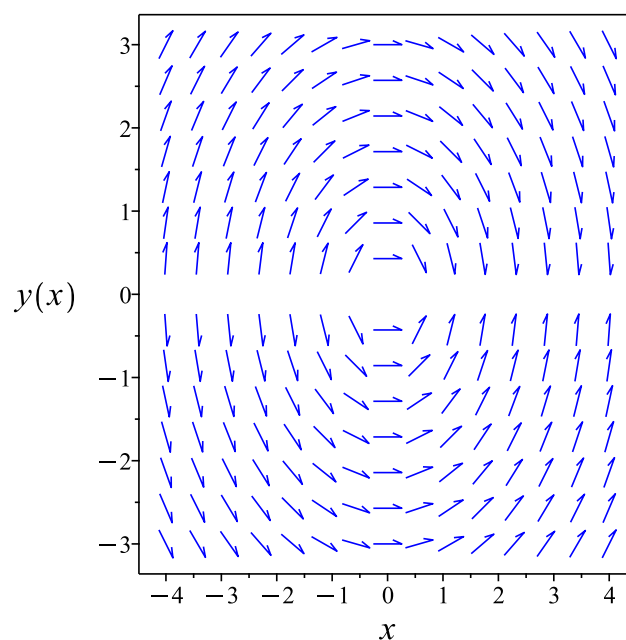
Solving for y gives

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Figure 2.14: Slope field $x + yy' = 0$ Summary of solutions found

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Solved using first_order_ode_homog_type_G

Time used: 0.079 (sec)

Solve

$$x + yy' = 0$$

Multiplying the right side of the ode, which is $-\frac{x}{y}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x}{y} \\ &= -\frac{x^2}{y^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{2x^2}{y^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{2x^2}{y^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{1}{z^2}$$

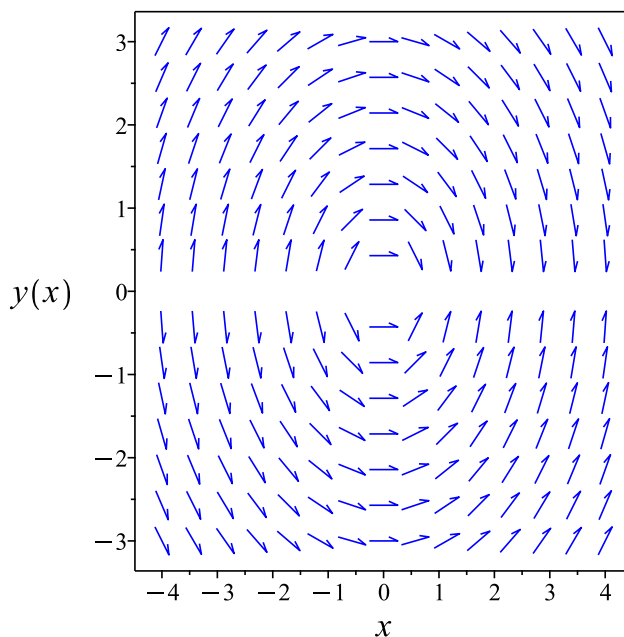
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1 + \frac{1}{z^2})} dz = 0$$

Figure 2.15: Slope field $x + yy' = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1 + \frac{1}{z^2})} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.808 (sec)

Solve

$$x + yy' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{X + x_0}{Y(X) + y_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X}{Y(X)}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X}{Y} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X$ and $N = Y$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -\frac{1}{u} \\ \frac{du}{dX} &= \frac{-\frac{1}{u(X)} - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-\frac{1}{u(X)} - u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) u(X) X + u(X)^2 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{u(X)^2 + 1}{u(X) X} \quad (2.9)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{u(X)^2 + 1}{u(X) X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{1}{X} \\ g(u) &= \frac{u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{u}{u^2 + 1} du = \int -\frac{1}{X} dX$$

$$\frac{\ln(u(X)^2 + 1)}{2} = \ln\left(\frac{1}{X}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{u(X)^2 + 1} = \frac{c_1}{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2 + 1}{u} = 0$$

for $u(X)$ gives

$$u(X) = -i$$

$$u(X) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(X)^2 + 1} = \frac{c_1}{X}$$

$$u(X) = -i$$

$$u(X) = i$$

Converting $\sqrt{u(X)^2 + 1} = \frac{c_1}{X}$ back to $Y(X)$ gives

$$\sqrt{\frac{Y(X)^2 + X^2}{X^2}} = \frac{c_1}{X}$$

Converting $u(X) = -i$ back to $Y(X)$ gives

$$Y(X) = -iX$$

Converting $u(X) = i$ back to $Y(X)$ gives

$$Y(X) = iX$$

Using the solution for $Y(X)$

$$\sqrt{\frac{Y(X)^2 + X^2}{X^2}} = \frac{c_1}{X} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\sqrt{\frac{x^2 + y^2}{x^2}} = \frac{c_1}{x}$$

Using the solution for $Y(X)$

$$Y(X) = -iX \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -ix$$

Using the solution for $Y(X)$

$$Y(X) = iX \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = ix$$

Solving for y gives

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

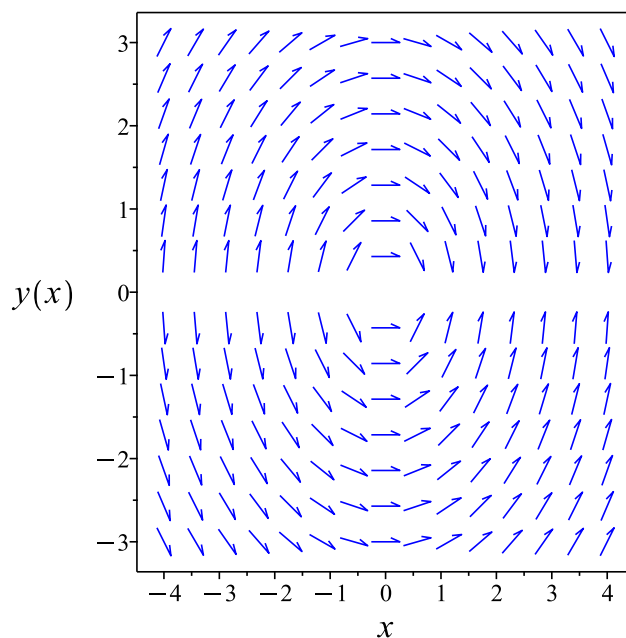


Figure 2.16: Slope field $x + yy' = 0$

Summary of solutions found

$$y = \sqrt{c_1^2 - x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = -\sqrt{c_1^2 - x^2}$$

Solved using first_order_ode_LIE

Time used: 0.780 (sec)

Solve

$$x + yy' = 0$$

Writing the ode as

$$y' = -\frac{x}{y}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{x(b_3 - a_2)}{y} - \frac{x^2 a_3}{y^2} + \frac{xa_2 + ya_3 + a_1}{y} - \frac{x(xb_2 + yb_3 + b_1)}{y^2} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{x^2 a_3 + x^2 b_2 - 2yxa_2 + 2yxb_3 - y^2 a_3 - b_2 y^2 + xb_1 - ya_1}{y^2} = 0$$

Setting the numerator to zero gives

$$-x^2a_3 - x^2b_2 + 2yxa_2 - 2yxb_3 + y^2a_3 + b_2y^2 - xb_1 + ya_1 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$2a_2v_1v_2 - a_3v_1^2 + a_3v_2^2 - b_2v_1^2 + b_2v_2^2 - 2b_3v_1v_2 + a_1v_2 - b_1v_1 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-a_3 - b_2)v_1^2 + (2a_2 - 2b_3)v_1v_2 - b_1v_1 + (a_3 + b_2)v_2^2 + a_1v_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 &= 0 \\ -b_1 &= 0 \\ 2a_2 - 2b_3 &= 0 \\ -a_3 - b_2 &= 0 \\ a_3 + b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= -b_2 \\ b_1 &= 0 \\ b_2 &= b_2 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = y$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x}{y}\right) (x) \\ &= y + \frac{x^2}{y} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x^2}{y}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(x^2 + y^2)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x}{y}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{x}{x^2 + y^2} \\ S_y &= \frac{y}{x^2 + y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \tag{2A}$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

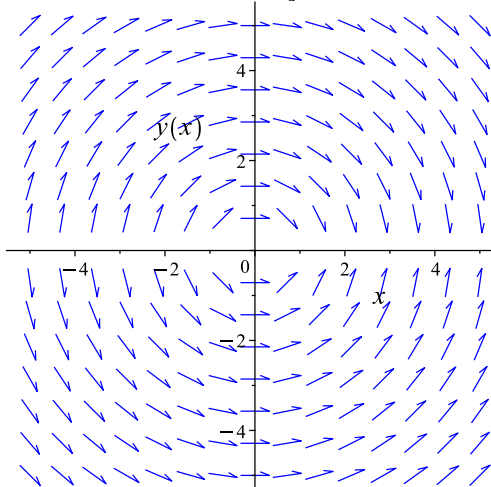
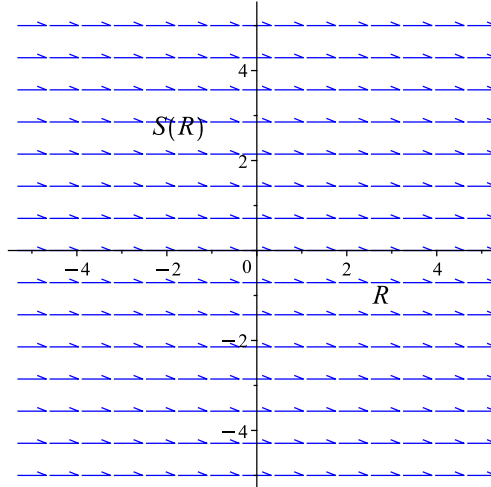
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(x^2 + y^2)}{2} = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x}{y}$ 	$R = x$ $S = \frac{\ln(x^2 + y^2)}{2}$	$\frac{dS}{dR} = 0$ 

Solving for y gives

$$y = \sqrt{-x^2 + e^{2c_2}}$$

$$y = -\sqrt{-x^2 + e^{2c_2}}$$

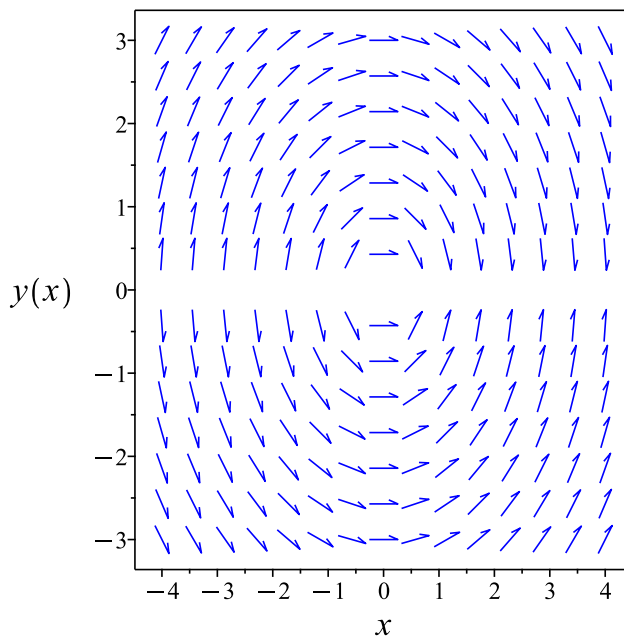


Figure 2.17: Slope field $x + yy' = 0$

Summary of solutions found

$$y = \sqrt{-x^2 + e^{2c_2}}$$

$$y = -\sqrt{-x^2 + e^{2c_2}}$$

✓ Maple. Time used: 0.003 (sec). Leaf size: 27

```
ode:=diff(y(x),x)*y(x)+x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{-x^2 + c_1}$$

$$y = -\sqrt{-x^2 + c_1}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

<- Bernoulli successful

Maple step by step

Let's solve

$$y(x) \left(\frac{d}{dx} y(x) \right) + x = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Separate variables

$$y(x) \left(\frac{d}{dx} y(x) \right) = -x$$

- Integrate both sides with respect to x

$$\int y(x) \left(\frac{d}{dx} y(x) \right) dx = \int -x dx + C1$$

- Evaluate integral

$$\frac{y(x)^2}{2} = -\frac{x^2}{2} + C1$$

- Solve for $y(x)$

$$\{y(x) = \sqrt{-x^2 + 2C1}, y(x) = -\sqrt{-x^2 + 2C1}\}$$

- Redefine the integration constant(s)

$$\{y(x) = \sqrt{-x^2 + C1}, y(x) = -\sqrt{-x^2 + C1}\}$$

✓ **Mathematica.** Time used: 0.054 (sec). Leaf size: 39

```
ode=y[x]*D[y[x],x]+x==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{-x^2 + 2c_1}$$

$$y(x) \rightarrow \sqrt{-x^2 + 2c_1}$$

✓ **Sympy.** Time used: 0.192 (sec). Leaf size: 22

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x + y(x)*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\sqrt{C_1 - x^2}, y(x) = \sqrt{C_1 - x^2} \right]$$

2.1.3 Problem 15

Local contents

Solved using first_order_ode_clairaut	87
✓ Maple	89
✓ Mathematica	90
✗ Sympy	90

Internal problem ID [7847]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 15

Date solved : Thursday, November 27, 2025 at 06:20:35 AM

CAS classification : `[_1st_order, _with_linear_symmetries], _Clairaut]`

Solved using first_order_ode_clairaut

Time used: 0.093 (sec)

Solve

$$y = y'x + y'^4$$

This is Clairaut ODE. It has the form

$$y = y'x + g(y')$$

Where g is function of $y'(x)$. Let $p = y'$ the ode becomes

$$y = p^4 + px$$

Solving for y from the above results in

$$y = p^4 + px \tag{1A}$$

The above ode is a Clairaut ode which is now solved.

Writing the equation (1A) as

$$y = px + g(p)$$

We now write $g \equiv g(p)$ to make notation simpler but we should always remember that g is function of p which in turn is function of x . Hence the above becomes

$$y = px + g \tag{1}$$

Where

$$g = p^4$$

Taking derivative of (1) w.r.t. x gives

$$\begin{aligned} p &= \frac{d}{dx}(xp + g) \\ p &= \left(p + x \frac{dp}{dx}\right) + \left(g' \frac{dp}{dx}\right) \\ p &= p + (x + g') \frac{dp}{dx} \\ 0 &= (x + g') \frac{dp}{dx} \end{aligned}$$

Where g' is derivative of $g(p)$ w.r.t. p .

The general solution is given by

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ p &= c_1 \end{aligned}$$

Substituting this in (1) gives the general solution as

$$y = c_1^4 + c_1 x$$

The singular solution is found from solving for p from

$$x + g'(p) = 0$$

And substituting the result back in (1). Since we found above that $g = p^4$, then the above equation becomes

$$\begin{aligned} x + g'(p) &= 4p^3 + x \\ &= 0 \end{aligned}$$

Solving the above for p results in

$$\begin{aligned} p_1 = y &= -\frac{3 \cdot 2^{1/3} (-x)^{4/3}}{8} \\ p_2 = y &= \frac{3 \cdot 2^{1/3} (-x)^{4/3} (1 + i\sqrt{3})}{16} \\ p_3 = y &= -\frac{3 \cdot 2^{1/3} (-x)^{4/3} (i\sqrt{3} - 1)}{16} \end{aligned}$$

Substituting the above back in (1) results in

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3}}{8}$$

$$y = \frac{3 \cdot 2^{1/3} (-x)^{4/3} (1 + i\sqrt{3})}{16}$$

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3} (i\sqrt{3} - 1)}{16}$$

Summary of solutions found

$$y = c_1(c_1^3 + x)$$

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3}}{8}$$

$$y = \frac{3 \cdot 2^{1/3} (-x)^{4/3} (1 + i\sqrt{3})}{16}$$

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3} (i\sqrt{3} - 1)}{16}$$

✓ **Maple.** Time used: 0.040 (sec). Leaf size: 66

```
ode:=y(x) = diff(y(x),x)*x+diff(y(x),x)^4;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3}}{8}$$

$$y = \frac{3 \cdot 2^{1/3} (-x)^{4/3} (1 + i\sqrt{3})}{16}$$

$$y = -\frac{3 \cdot 2^{1/3} (-x)^{4/3} (i\sqrt{3} - 1)}{16}$$

$$y = c_1(c_1^3 + x)$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

```

trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful

```

Maple step by step

Let's solve

$$y(x) = x \left(\frac{d}{dx} y(x) \right) + \left(\frac{d}{dx} y(x) \right)^4$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \text{RootOf}(-Z^4 + xZ - y(x))$$

✓ **Mathematica.** Time used: 0.005 (sec). Leaf size: 75

```

ode=y[x]==x*D[y[x],x]+(D[y[x],x])^4;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow c_1(x + c_1^3)$$

$$y(x) \rightarrow -\frac{3}{4} \left(-\frac{1}{2} \right)^{2/3} x^{4/3}$$

$$y(x) \rightarrow -\frac{3x^{4/3}}{4 \cdot 2^{2/3}}$$

$$y(x) \rightarrow \frac{3\sqrt[3]{-1}x^{4/3}}{4 \cdot 2^{2/3}}$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) + y(x) - Derivative(y(x), x)**4, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)

```

Timed Out

2.1.4 Problem 16

Local contents

Solved using first_order_ode_bernoulli	91
Solved using first_order_ode_exact	94
Solved using first_order_ode_isobaric	98
Solved using first_order_ode_homog_A	102
Solved using first_order_ode_dAlembert	105
Solved using first_order_ode_homog_type_D2	115
Solved using first_order_ode_homog_type_G	118
Solved using first_order_ode_homog_type_maple_C	119
Solved using first_order_ode_LIE	124
✓ Maple	129
✓ Mathematica	130
✓ Sympy	130

Internal problem ID [7848]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 16

Date solved : Thursday, November 27, 2025 at 06:20:42 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.112 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y(3x^2 + y^2)}{2x^3} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(\frac{3}{2x}\right)y + \left(\frac{1}{2x^3}\right)y^3 \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{3}{2x} \\ f_1 &= \frac{1}{2x^3} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= \frac{3}{2x} \\ f_1(x) &= \frac{1}{2x^3} \\ n &= 3 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^3$ gives

$$y' \frac{1}{y^3} = \frac{3}{2x y^2} + \frac{1}{2x^3} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= \frac{1}{y^2} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{2}{y^3} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -\frac{v'(x)}{2} &= \frac{3v(x)}{2x} + \frac{1}{2x^3} \\ v' &= -\frac{3v}{x} - \frac{1}{x^3} \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{3}{x} \\ p(x) &= -\frac{1}{x^3} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{3}{x} \, dx} \\ &= x^3 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) \left(-\frac{1}{x^3} \right) \\ \frac{d}{dx}(v x^3) &= (x^3) \left(-\frac{1}{x^3} \right) \\ d(v x^3) &= -1 \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} v x^3 &= \int -1 \, dx \\ &= -x + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x^3 gives the final solution

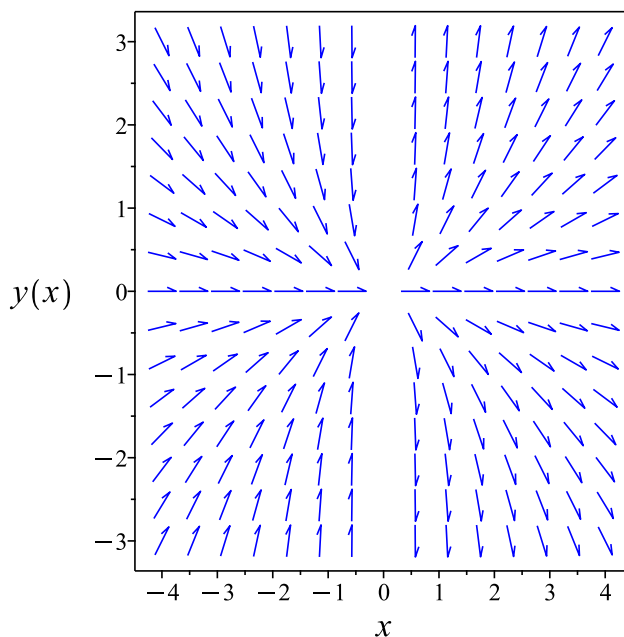
$$v(x) = \frac{c_1 - x}{x^3}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y^2} = \frac{c_1 - x}{x^3}$$

Solving for y gives

$$\begin{aligned} y &= \frac{\sqrt{x(c_1 - x)} \, x}{c_1 - x} \\ y &= -\frac{\sqrt{x(c_1 - x)} \, x}{c_1 - x} \end{aligned}$$

Figure 2.18: Slope field $2x^3 y' = y(y^2 + 3x^2)$ Summary of solutions found

$$y = \frac{\sqrt{x(c_1 - x)} x}{c_1 - x}$$

$$y = -\frac{\sqrt{x(c_1 - x)} x}{c_1 - x}$$

Solved using first_order_ode_exact

Time used: 0.140 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\frac{\partial \phi}{\partial x} = M$$

$$\frac{\partial \phi}{\partial y} = N$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2x^3) dy &= (y(3x^2 + y^2)) dx \\ (-y(3x^2 + y^2)) dx &+ (2x^3) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -y(3x^2 + y^2) \\ N(x, y) &= 2x^3 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-y(3x^2 + y^2)) \\ &= -3x^2 - 3y^2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(2x^3) \\ &= 6x^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{2x^3} ((-3x^2 - 3y^2) - (6x^2)) \\ &= \frac{-9x^2 - 3y^2}{2x^3} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= -\frac{1}{3x^2y + y^3} ((6x^2) - (-3x^2 - 3y^2)) \\ &= -\frac{3}{y} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int -\frac{3}{y} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-3 \ln(y)} \\ &= \frac{1}{y^3} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \bar{M} &= \mu M \\ &= \frac{1}{y^3} (-y(3x^2 + y^2)) \\ &= \frac{-3x^2 - y^2}{y^2} \end{aligned}$$

And

$$\begin{aligned} \bar{N} &= \mu N \\ &= \frac{1}{y^3} (2x^3) \\ &= \frac{2x^3}{y^3} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-3x^2 - y^2}{y^2} \right) + \left(\frac{2x^3}{y^3} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-3x^2 - y^2}{y^2} dx \\ \phi &= -\frac{x(x^2 + y^2)}{y^2} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned} \frac{\partial \phi}{\partial y} &= -\frac{2x}{y} + \frac{2x(x^2 + y^2)}{y^3} + f'(y) \\ &= \frac{2x^3}{y^3} + f'(y) \end{aligned} \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{2x^3}{y^3}$. Therefore equation (4) becomes

$$\frac{2x^3}{y^3} = \frac{2x^3}{y^3} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{x(x^2 + y^2)}{y^2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{x(x^2 + y^2)}{y^2}$$

Solving for y gives

$$y = \frac{\sqrt{-x(x+c_1)}x}{x+c_1}$$

$$y = -\frac{\sqrt{-x(x+c_1)}x}{x+c_1}$$

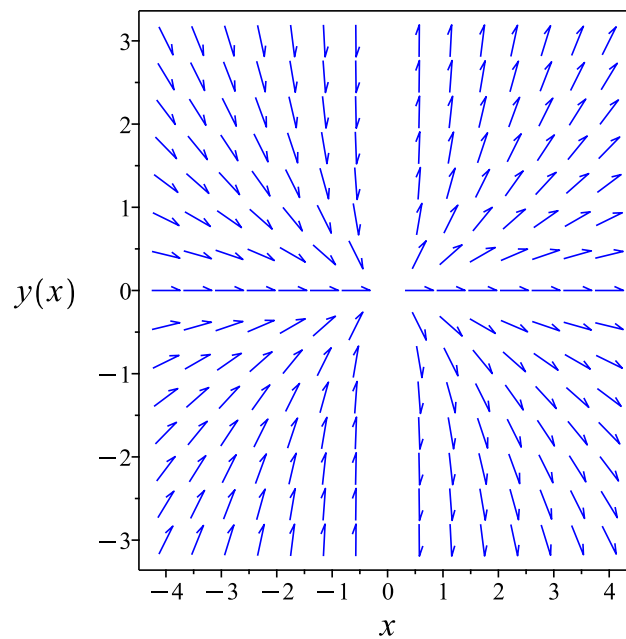


Figure 2.19: Slope field $2x^3y' = y(y^2 + 3x^2)$

Summary of solutions found

$$y = \frac{\sqrt{-x(x+c_1)}x}{x+c_1}$$

$$y = -\frac{\sqrt{-x(x+c_1)}x}{x+c_1}$$

Solved using first_order_ode_isobaric

Time used: 0.348 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

Solving for y' gives

$$y' = \frac{y(y^2 + 3x^2)}{2x^3} \tag{1}$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{y(y^2 + 3x^2)}{2x^3} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = \frac{u(x) (x^2 u(x)^2 + 3x^2)}{2x^2}$$

The ode

$$u'(x) = \frac{u(x) (u(x)^2 + 1)}{2x} \quad (2.10)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x) (u(x)^2 + 1)}{2x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{2x} \\ g(u) &= u(u^2 + 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(u^2 + 1)} du &= \int \frac{1}{2x} dx \end{aligned}$$

$$\ln \left(\frac{u(x)}{\sqrt{u(x)^2 + 1}} \right) = \ln(\sqrt{x}) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u^2 + 1) = 0$$

for $u(x)$ gives

$$\begin{aligned} u(x) &= 0 \\ u(x) &= -i \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \frac{u(x)}{\sqrt{u(x)^2 + 1}} &= \sqrt{x} c_1 \\ u(x) &= 0 \\ u(x) &= -i \end{aligned}$$

Converting $\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$ back to y gives

$$\frac{y}{\sqrt{\frac{y^2}{x^2} + 1} x} = \sqrt{x} c_1$$

Converting $u(x) = 0$ back to y gives

$$\frac{y}{x} = 0$$

Converting $u(x) = -i$ back to y gives

$$\frac{y}{x} = -i$$

Simplifying the above gives

$$\frac{y}{\sqrt{\frac{x^2+y^2}{x^2}} x} = \sqrt{x} c_1$$

$$\frac{y}{x} = 0$$

$$\frac{y}{x} = -i$$

Solving for y gives

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2} c_1 \sqrt{-\frac{1}{c_1^2 x} - 1}$$

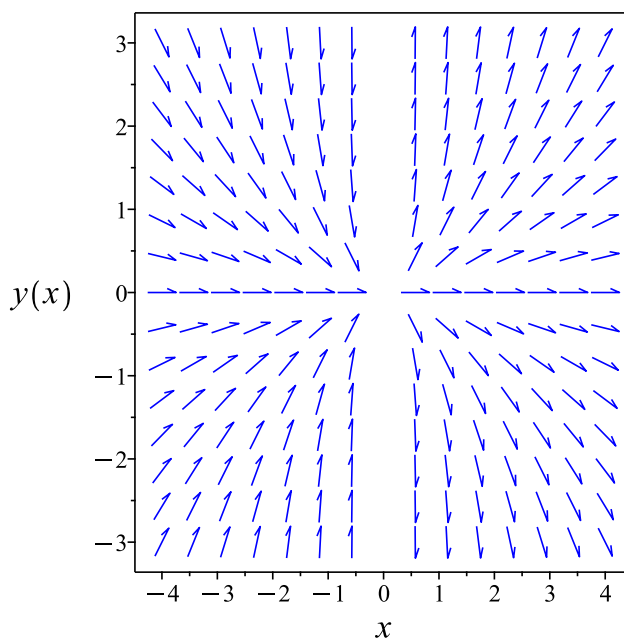


Figure 2.20: Slope field $2x^3 y' = y(y^2 + 3x^2)$

Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2} c_1 \sqrt{-\frac{1}{c_1^2 x} - 1}$$

Solved using first_order_ode_homog_A

Time used: 0.625 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y(3x^2 + y^2)}{2x^3} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = y(3x^2 + y^2)$ and $N = 2x^3$ are both homogeneous and of the same order $n = 3$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= \frac{3}{2}u + \frac{1}{2}u^3 \\ \frac{du}{dx} &= \frac{\frac{u(x)}{2} + \frac{u(x)^3}{2}}{x} \end{aligned}$$

Or

$$u'(x) - \frac{\frac{u(x)}{2} + \frac{u(x)^3}{2}}{x} = 0$$

Or

$$-u(x)^3 + 2u'(x)x - u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = \frac{u(x)(u(x)^2 + 1)}{2x} \quad (2.11)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(u(x)^2 + 1)}{2x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = \frac{1}{2x}$$

$$g(u) = u(u^2 + 1)$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u(u^2 + 1)} du = \int \frac{1}{2x} dx$$

$$\ln \left(\frac{u(x)}{\sqrt{u(x)^2 + 1}} \right) = \ln(\sqrt{x}) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u^2 + 1) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = -i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$$

$$u(x) = 0$$

$$u(x) = -i$$

Converting $\frac{u(x)}{\sqrt{u(x)^2+1}} = \sqrt{x} c_1$ back to y gives

$$\frac{y}{\sqrt{\frac{x^2+y^2}{x^2}} x} = \sqrt{x} c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Converting $u(x) = -i$ back to y gives

$$y = -ix$$

Solving for y gives

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2} c_1 \sqrt{-\frac{1}{c_1^2 x - 1}}$$

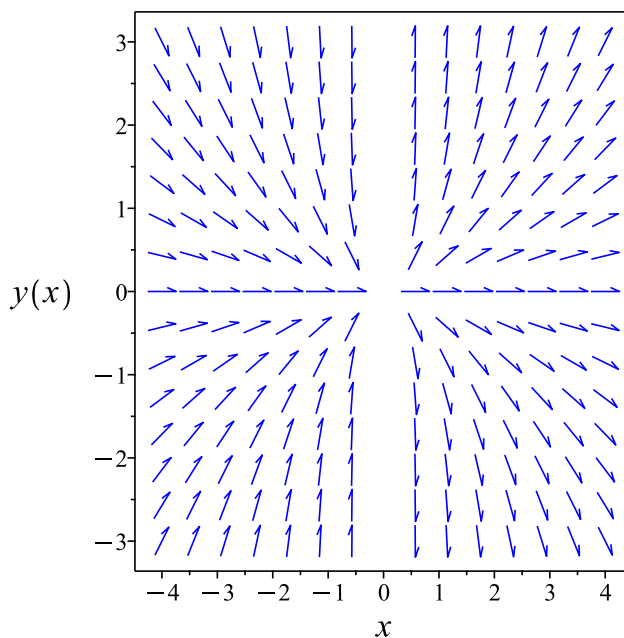


Figure 2.21: Slope field $2x^3 y' = y(y^2 + 3x^2)$

Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2} c_1 \sqrt{-\frac{1}{c_1^2 x - 1}}$$

Solved using first_order_ode_dAlembert

Time used: 2.431 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

Let $p = y'$ the ode becomes

$$2x^3p = y(3x^2 + y^2)$$

Solving for y from the above results in

$$y = \left((p + \sqrt{p^2 + 1})^{1/3} - \frac{1}{(p + \sqrt{p^2 + 1})^{1/3}} \right) x \quad (1)$$

$$y = \left(-\frac{(p + \sqrt{p^2 + 1})^{1/3}}{2} + \frac{1}{2(p + \sqrt{p^2 + 1})^{1/3}} + \frac{i\sqrt{3} \left((p + \sqrt{p^2 + 1})^{1/3} + \frac{1}{(p + \sqrt{p^2 + 1})^{1/3}} \right)}{2} \right) x \quad (2)$$

$$y = \left(-\frac{(p + \sqrt{p^2 + 1})^{1/3}}{2} + \frac{1}{2(p + \sqrt{p^2 + 1})^{1/3}} - \frac{i\sqrt{3} \left((p + \sqrt{p^2 + 1})^{1/3} + \frac{1}{(p + \sqrt{p^2 + 1})^{1/3}} \right)}{2} \right) x \quad (3)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is dAlembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{(p + \sqrt{p^2 + 1})^{2/3} - 1}{(p + \sqrt{p^2 + 1})^{1/3}} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{(p + \sqrt{p^2 + 1})^{2/3} - 1}{(p + \sqrt{p^2 + 1})^{1/3}} = \left(\frac{x}{3(p + \sqrt{p^2 + 1})^{2/3}} + \frac{xp}{3(p + \sqrt{p^2 + 1})^{2/3} \sqrt{p^2 + 1}} + \frac{x}{3(p + \sqrt{p^2 + 1})^{4/3}} + \frac{xp}{3(p + \sqrt{p^2 + 1})^{4/3} \sqrt{p^2 + 1}} \right) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{(p + \sqrt{p^2 + 1})^{2/3} - 1}{(p + \sqrt{p^2 + 1})^{1/3}} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= 0 \\ p_2 &= \frac{(2 + 2i\sqrt{3})^{3/2}}{8} \\ p_3 &= \frac{(-2i\sqrt{3} + 2)^{3/2}}{8} \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= 0 \\ y &= ix \\ y &= -ix \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$\begin{aligned} p'(x) &= \frac{p(x) - \frac{(p(x) + \sqrt{p(x)^2 + 1})^{2/3} - 1}{(p(x) + \sqrt{p(x)^2 + 1})^{1/3}}}{\frac{x}{3(p(x) + \sqrt{p(x)^2 + 1})^{2/3}} + \frac{xp(x)}{3(p(x) + \sqrt{p(x)^2 + 1})^{2/3} \sqrt{p(x)^2 + 1}} + \frac{x}{3(p(x) + \sqrt{p(x)^2 + 1})^{4/3}} + \frac{xp(x)}{3(p(x) + \sqrt{p(x)^2 + 1})^{4/3} \sqrt{p(x)^2 + 1}}} \end{aligned} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{3\sqrt{p(x)^2 + 1} \left(p(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} - \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + 1 \right)}{\left(\left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + 1 \right) x} \quad (2.12)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{3\sqrt{p(x)^2 + 1} \left(p(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} - \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + 1 \right)}{\left(\left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + 1 \right) x} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{3}{x} \\ g(p) &= \frac{\sqrt{p^2 + 1} \left(- (p + \sqrt{p^2 + 1})^{2/3} + p(p + \sqrt{p^2 + 1})^{1/3} + 1 \right)}{(p + \sqrt{p^2 + 1})^{2/3} + 1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{(p + \sqrt{p^2 + 1})^{2/3} + 1}{\sqrt{p^2 + 1} \left(- (p + \sqrt{p^2 + 1})^{2/3} + p(p + \sqrt{p^2 + 1})^{1/3} + 1 \right)} dp &= \int \frac{3}{x} dx \\ \int^{p(x)} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(- (\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau &= \ln(x^3) + c_1 \end{aligned}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\sqrt{p^2 + 1} \left(- (p + \sqrt{p^2 + 1})^{2/3} + p(p + \sqrt{p^2 + 1})^{1/3} + 1 \right)}{(p + \sqrt{p^2 + 1})^{2/3} + 1} = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= 0 \\ p(x) &= -i \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau = \ln(x^3) + c_1$$

$$p(x) = 0$$

$$p(x) = -i$$

Substituting the above solution for p in (2A) gives

y

$$\begin{aligned} & x \left(\left(\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right) + \sqrt{\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right)} \right) \\ &= \frac{\left(\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right) + \sqrt{\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right)} \right)}{\left(\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right) + \sqrt{\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right)} \right)} \end{aligned}$$

$$y = 0$$

$$y = -ix$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{i(p + \sqrt{p^2 + 1})^{2/3} \sqrt{3} + i\sqrt{3} - (p + \sqrt{p^2 + 1})^{2/3} + 1}{2(p + \sqrt{p^2 + 1})^{1/3}}$$

$$g = 0$$

Hence (2) becomes

$$p - \frac{i(p + \sqrt{p^2 + 1})^{2/3} \sqrt{3} + i\sqrt{3} - (p + \sqrt{p^2 + 1})^{2/3} + 1}{2(p + \sqrt{p^2 + 1})^{1/3}} = \left(\frac{ix\sqrt{3}}{6(p + \sqrt{p^2 + 1})^{2/3}} + \frac{ix\sqrt{3}p}{6(p + \sqrt{p^2 + 1})^{2/3} \sqrt{p^2 + 1}} \right) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{i(p + \sqrt{p^2 + 1})^{2/3} \sqrt{3} + i\sqrt{3} - (p + \sqrt{p^2 + 1})^{2/3} + 1}{2(p + \sqrt{p^2 + 1})^{1/3}} = 0$$

Unable to solve for p from the above. No singular solution can be found.

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{i(p(x) + \sqrt{p(x)^2 + 1})^{2/3} \sqrt{3} + i\sqrt{3} - (p(x) + \sqrt{p(x)^2 + 1})^{2/3} + 1}{2(p(x) + \sqrt{p(x)^2 + 1})^{1/3}}}{\frac{ix\sqrt{3}}{6(p(x) + \sqrt{p(x)^2 + 1})^{2/3}} + \frac{ix\sqrt{3}p(x)}{6(p(x) + \sqrt{p(x)^2 + 1})^{2/3} \sqrt{p(x)^2 + 1}} - \frac{x}{6(p(x) + \sqrt{p(x)^2 + 1})^{2/3}} - \frac{xp(x)}{6(p(x) + \sqrt{p(x)^2 + 1})^{2/3} \sqrt{p(x)^2 + 1}} - \frac{1}{6(p(x) + \sqrt{p(x)^2 + 1})^{2/3}}}$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{6\sqrt{p(x)^2 + 1} \left(ip(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} \sqrt{3} - i\sqrt{3} + 2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + p(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} \right)}{\left(-2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} + i - \sqrt{3} \right) \left(2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} + i - \sqrt{3} \right) x} \quad (2.13)$$

is separable as it can be written as

$$p'(x) = \frac{6\sqrt{p(x)^2 + 1} \left(ip(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} \sqrt{3} - i\sqrt{3} + 2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{2/3} + p(x) \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} \right)}{\left(-2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} + i - \sqrt{3} \right) \left(2 \left(p(x) + \sqrt{p(x)^2 + 1} \right)^{1/3} + i - \sqrt{3} \right) x}$$

$$= f(x)g(p)$$

Where

$$f(x) = \frac{6}{x}$$

$$g(p) = \frac{\sqrt{p^2+1} \left(ip(p+\sqrt{p^2+1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(p+\sqrt{p^2+1})^{2/3} + p(p+\sqrt{p^2+1})^{1/3} + 1 \right)}{\left(-2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right) \left(2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right)}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{\left(-2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right) \left(2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right)}{\sqrt{p^2+1} \left(ip(p+\sqrt{p^2+1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(p+\sqrt{p^2+1})^{2/3} + p(p+\sqrt{p^2+1})^{1/3} + 1 \right)} dp = \int \frac{6}{x} dx$$

$$\int^{p(x)} \frac{\left(-2(\tau+\sqrt{\tau^2+1})^{1/3} + i - \sqrt{3} \right) \left(2(\tau+\sqrt{\tau^2+1})^{1/3} + i - \sqrt{3} \right)}{\sqrt{\tau^2+1} \left(i\tau(\tau+\sqrt{\tau^2+1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(\tau+\sqrt{\tau^2+1})^{2/3} + \tau(\tau+\sqrt{\tau^2+1})^{1/3} + 1 \right)} d\tau = \ln(x^6)$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\sqrt{p^2+1} \left(ip(p+\sqrt{p^2+1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(p+\sqrt{p^2+1})^{2/3} + p(p+\sqrt{p^2+1})^{1/3} + 1 \right)}{\left(-2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right) \left(2(p+\sqrt{p^2+1})^{1/3} + i - \sqrt{3} \right)} = 0$$

for $p(x)$ gives

$$p(x) = -i$$

$$p(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{\left(-2(\tau+\sqrt{\tau^2+1})^{1/3} + i - \sqrt{3} \right) \left(2(\tau+\sqrt{\tau^2+1})^{1/3} + i - \sqrt{3} \right)}{\sqrt{\tau^2+1} \left(i\tau(\tau+\sqrt{\tau^2+1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(\tau+\sqrt{\tau^2+1})^{2/3} + \tau(\tau+\sqrt{\tau^2+1})^{1/3} + 1 \right)} d\tau = \ln(x^6)$$

$$p(x) = -i$$

$$p(x) = i$$

Substituting the above solution for p in (2A) gives

$$y = x \left(i \left(\text{RootOf} \left(- \int^{-Z} \frac{\left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} - i + \sqrt{3} \right) \left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} - i + \sqrt{3} \right)}{\sqrt{\tau^2 + 1} \left(i\tau(\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^6) + c_2 \right) + \sqrt{\dots} \right)$$

$$y = 2ix$$

$$y = ix$$

Solving ode 3A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= - \frac{(1 + i\sqrt{3}) (p + \sqrt{p^2 + 1})^{2/3} + i\sqrt{3} - 1}{2 (p + \sqrt{p^2 + 1})^{1/3}} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p + \frac{(1 + i\sqrt{3}) (p + \sqrt{p^2 + 1})^{2/3} + i\sqrt{3} - 1}{2 (p + \sqrt{p^2 + 1})^{1/3}} = \left(- \frac{ix\sqrt{3}}{6 (p + \sqrt{p^2 + 1})^{2/3}} - \frac{ix\sqrt{3}p}{6 (p + \sqrt{p^2 + 1})^{2/3} \sqrt{p^2 + 1}} + \frac{ix\sqrt{3}}{6 (p + \sqrt{p^2 + 1})^{1/3}} \right) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p + \frac{(1 + i\sqrt{3}) (p + \sqrt{p^2 + 1})^{2/3} + i\sqrt{3} - 1}{2 (p + \sqrt{p^2 + 1})^{1/3}} = 0$$

Solving the above for p results in

$$p_1 = - \frac{i\sqrt{3} \left(-\frac{i}{2} + \frac{\sqrt{3}}{2} \right)^3}{2} + \frac{\left(-\frac{i}{2} + \frac{\sqrt{3}}{2} \right)^3}{2} - \frac{i}{2} + \frac{\sqrt{3}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -ix$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$\begin{aligned} p'(x) &= \frac{p(x) + \frac{(1+i\sqrt{3})(p(x)+\sqrt{p(x)^2+1})}{2(p(x)+\sqrt{p(x)^2+1})}}{6(p(x)+\sqrt{p(x)^2+1})^{2/3} - \frac{ix\sqrt{3}}{6(p(x)+\sqrt{p(x)^2+1})^{2/3}} - \frac{ix\sqrt{3}p(x)}{6(p(x)+\sqrt{p(x)^2+1})^{2/3}\sqrt{p(x)^2+1}} + \frac{ix\sqrt{3}}{6(p(x)+\sqrt{p(x)^2+1})^{4/3}} + \frac{ix\sqrt{3}p(x)}{6(p(x)+\sqrt{p(x)^2+1})^{4/3}\sqrt{p(x)^2+1}} - \frac{ix\sqrt{3}}{6(p(x)+\sqrt{p(x)^2+1})^{4/3}}} \end{aligned} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$\begin{aligned} p'(x) = & \frac{6\sqrt{p(x)^2+1} \left(ip(x) \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} \sqrt{3} - i\sqrt{3} - 2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{2/3} - p(x) \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} - 1 \right)}{\left(-2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} + i + \sqrt{3} \right) \left(2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} + i + \sqrt{3} \right)} \end{aligned} \quad (2.14)$$

is separable as it can be written as

$$\begin{aligned} p'(x) = & - \frac{6\sqrt{p(x)^2+1} \left(ip(x) \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} \sqrt{3} - i\sqrt{3} - 2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{2/3} - p(x) \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} - 1 \right)}{\left(-2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} + i + \sqrt{3} \right) \left(2 \left(p(x) + \sqrt{p(x)^2+1} \right)^{1/3} + i + \sqrt{3} \right)} \\ & = f(x)g(p) \end{aligned}$$

Where

$$f(x) = -\frac{6}{x}$$

$$g(p) = \frac{\sqrt{p^2+1} \left(ip(p + \sqrt{p^2+1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(p + \sqrt{p^2+1})^{2/3} - p(p + \sqrt{p^2+1})^{1/3} - 1 \right)}{\left(-2(p + \sqrt{p^2+1})^{1/3} + i + \sqrt{3} \right) \left(2(p + \sqrt{p^2+1})^{1/3} + i + \sqrt{3} \right)}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{\left(-2(p + \sqrt{p^2+1})^{1/3} + i + \sqrt{3} \right) \left(2(p + \sqrt{p^2+1})^{1/3} + i + \sqrt{3} \right)}{\sqrt{p^2+1} \left(ip(p + \sqrt{p^2+1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(p + \sqrt{p^2+1})^{2/3} - p(p + \sqrt{p^2+1})^{1/3} - 1 \right)} dp = \int -\frac{6}{x} dx$$

$$\int^{p(x)} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right)}{\sqrt{\tau^2 + 1} \left(i\tau (\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} - 1\right)} d\tau = \ln \left(\frac{1}{x^6}\right)$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\sqrt{p^2 + 1} \left(ip(p + \sqrt{p^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(p + \sqrt{p^2 + 1})^{2/3} - p(p + \sqrt{p^2 + 1})^{1/3} - 1\right)}{\left(-2(p + \sqrt{p^2 + 1})^{1/3} + i + \sqrt{3}\right) \left(2(p + \sqrt{p^2 + 1})^{1/3} + i + \sqrt{3}\right)} = 0$$

for $p(x)$ gives

$$p(x) = -i$$

$$p(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right)}{\sqrt{\tau^2 + 1} \left(i\tau (\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} - 1\right)} d\tau = \ln \left(\frac{1}{x^6}\right)$$

$$p(x) = -i$$

$$p(x) = i$$

Substituting the above solution for p in (2A) gives

$$y =$$

$$x \left((1 + i\sqrt{3}) \left(\text{RootOf} \left(-\int^{-Z} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right)}{\sqrt{\tau^2 + 1} \left(i\tau (\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} - 1\right)} d\tau + \ln \left(\frac{1}{x^6}\right) \right) \right. \right. \\ \left. \left. - 2 \left(\text{RootOf} \left(-\int^{-Z} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3}\right)}{\sqrt{\tau^2 + 1} \left(i\tau (\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau (\tau + \sqrt{\tau^2 + 1})^{1/3} - 1\right)} d\tau + \ln \left(\frac{1}{x^6}\right) \right) \right) \right)$$

$$y = -ix$$

$$y = -2ix$$

The solution

$$y = -2ix$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = 2ix$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$y =$

$$x \left((1 + i\sqrt{3}) \left(\text{RootOf} \left(-\int^{-Z} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3} \right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3} \right)}{\sqrt{\tau^2 + 1} \left(i\tau(\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} - 1 \right)} d\tau + \ln \left(\frac{1}{x^6} \right) \right) \right. \right. \\ \left. \left. - \frac{2 \left(\text{RootOf} \left(-\int^{-Z} \frac{\left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3} \right) \left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} + i + \sqrt{3} \right)}{\sqrt{\tau^2 + 1} \left(i\tau(\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} - 2(\tau + \sqrt{\tau^2 + 1})^{2/3} - \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} - 1 \right)} d\tau + \ln \left(\frac{1}{x^6} \right) \right) \right)}{2} \right)$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

y

$$x \left(i \left(\text{RootOf} \left(-\int^{-Z} \frac{\left(2(\tau + \sqrt{\tau^2 + 1})^{1/3} - i + \sqrt{3} \right) \left(-2(\tau + \sqrt{\tau^2 + 1})^{1/3} - i + \sqrt{3} \right)}{\sqrt{\tau^2 + 1} \left(i\tau(\tau + \sqrt{\tau^2 + 1})^{1/3} \sqrt{3} - i\sqrt{3} + 2(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^6) + c_2 \right) \right) + \sqrt{\dots} \right)$$

was found not to satisfy the ode or the IC. Hence it is removed.

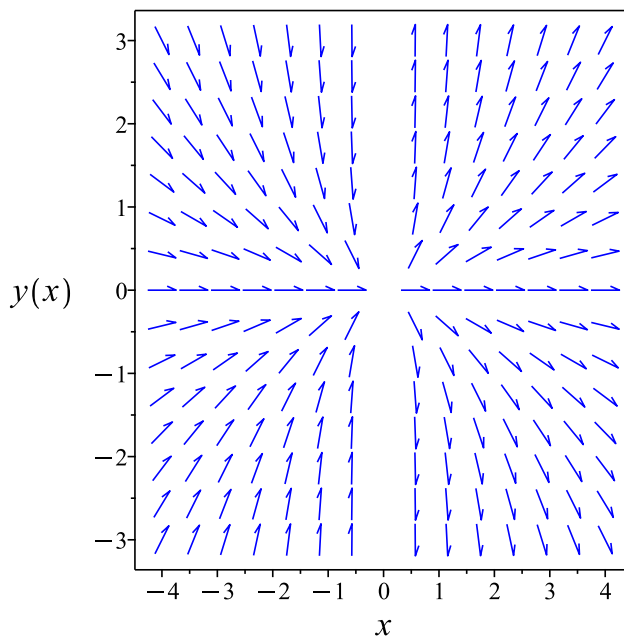


Figure 2.22: Slope field $2x^3y' = y(y^2 + 3x^2)$

Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = ix$$

$$y$$

$$x \left(\left(\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right) + \sqrt{\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right)} + \sqrt{\text{RootOf} \left(-\int^{-Z} \frac{(\tau + \sqrt{\tau^2 + 1})^{2/3} + 1}{\sqrt{\tau^2 + 1} \left(-(\tau + \sqrt{\tau^2 + 1})^{2/3} + \tau(\tau + \sqrt{\tau^2 + 1})^{1/3} + 1 \right)} d\tau + \ln(x^3) + c_1 \right)} \right)$$

Solved using first_order_ode_homog_type_D2

Time used: 0.185 (sec)

Solve

$$2x^3 y' = y(y^2 + 3x^2)$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$2x^3(u'(x)x + u(x)) = u(x)x(u(x)^2 x^2 + 3x^2)$$

Which is now solved The ode

$$u'(x) = \frac{u(x)(u(x)^2 + 1)}{2x} \quad (2.15)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(u(x)^2 + 1)}{2x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{2x} \\ g(u) &= u(u^2 + 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(u^2 + 1)} du &= \int \frac{1}{2x} dx \end{aligned}$$

$$\ln \left(\frac{u(x)}{\sqrt{u(x)^2 + 1}} \right) = \ln(\sqrt{x}) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u^2 + 1) = 0$$

for $u(x)$ gives

$$\begin{aligned} u(x) &= 0 \\ u(x) &= -i \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \frac{u(x)}{\sqrt{u(x)^2 + 1}} &= \sqrt{x} c_1 \\ u(x) &= 0 \\ u(x) &= -i \end{aligned}$$

Converting $\frac{u(x)}{\sqrt{u(x)^2 + 1}} = \sqrt{x} c_1$ back to y gives

$$\frac{y}{\sqrt{\frac{y^2}{x^2} + 1} x} = \sqrt{x} c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Converting $u(x) = -i$ back to y gives

$$y = -ix$$

Simplifying the above gives

$$\frac{y}{\sqrt{\frac{x^2+y^2}{x^2}}x} = \sqrt{x} c_1$$

$$y = 0$$

$$y = -ix$$

Solving for y gives

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2}c_1\sqrt{-\frac{1}{c_1^2x}-1}$$

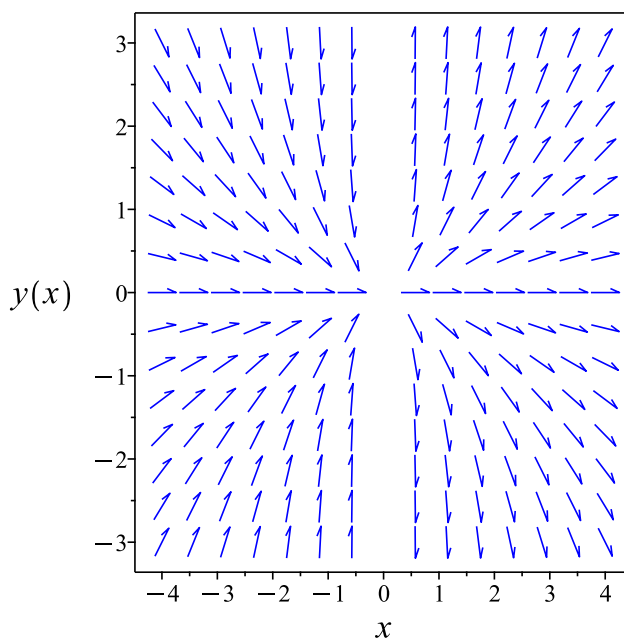


Figure 2.23: Slope field $2x^3y' = y(y^2 + 3x^2)$

Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2}c_1\sqrt{-\frac{1}{c_1^2x}-1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.081 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

Multiplying the right side of the ode, which is $\frac{y(3x^2+y^2)}{2x^3}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y(3x^2+y^2)}{2x^3} \\ &= \frac{3x^2+y^2}{2x^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{y^2}{x^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{y^2}{x^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{z^2}{2} + \frac{3}{2}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z\left(-\frac{1}{2} - \frac{z^2}{2}\right)} dz = 0$$

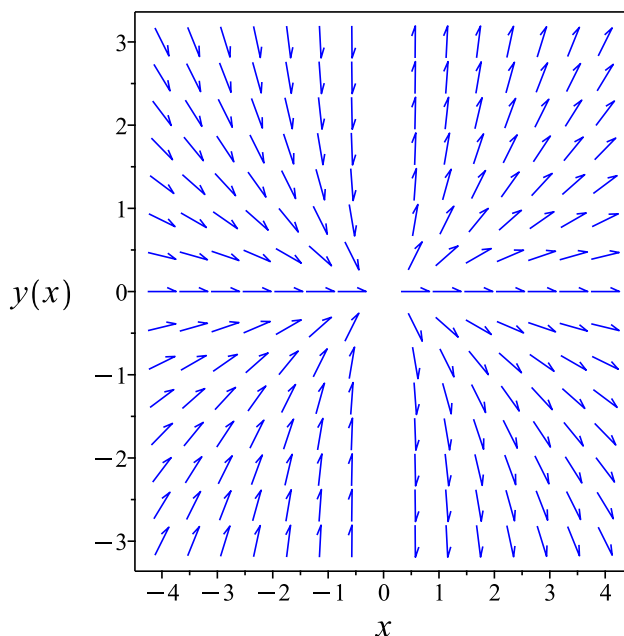


Figure 2.24: Slope field $2x^3y' = y(y^2 + 3x^2)$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z\left(-\frac{1}{2} - \frac{z^2}{2}\right)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.987 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{(Y(X) + y_0)((Y(X) + y_0)^2 + 3(X + x_0)^2)}{2(X + x_0)^3}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$\begin{aligned}x_0 &= 0 \\y_0 &= 0\end{aligned}$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{3Y(X)X^2 + Y(X)^3}{2X^3}$$

In canonical form, the ODE is

$$\begin{aligned}Y' &= F(X, Y) \\&= \frac{Y(3X^2 + Y^2)}{2X^3}\end{aligned}\tag{1}$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = Y(3X^2 + Y^2)$ and $N = 2X^3$ are both homogeneous and of the same order $n = 3$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dX}X + u &= \frac{3}{2}u + \frac{1}{2}u^3 \\ \frac{du}{dX} &= \frac{\frac{u(X)}{2} + \frac{u(X)^3}{2}}{X}\end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{u(X)}{2} + \frac{u(X)^3}{2}}{X} = 0$$

Or

$$-u(X)^3 + 2\left(\frac{d}{dX}u(X)\right)X - u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = \frac{u(X)(u(X)^2 + 1)}{2X}\tag{2.16}$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= \frac{u(X)(u(X)^2 + 1)}{2X} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= \frac{1}{2X} \\ g(u) &= u(u^2 + 1)\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{1}{u(u^2 + 1)} du &= \int \frac{1}{2X} dX\end{aligned}$$

$$\ln \left(\frac{u(X)}{\sqrt{u(X)^2 + 1}} \right) = \ln(\sqrt{X}) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{u(X)}{\sqrt{u(X)^2 + 1}} = c_1 \sqrt{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u^2 + 1) = 0$$

for $u(X)$ gives

$$\begin{aligned}u(X) &= 0 \\ u(X) &= -i\end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{u(X)}{\sqrt{u(X)^2 + 1}} = c_1 \sqrt{X}$$

$$u(X) = 0$$

$$u(X) = -i$$

Converting $\frac{u(X)}{\sqrt{u(X)^2 + 1}} = c_1 \sqrt{X}$ back to $Y(X)$ gives

$$\frac{Y(X)}{\sqrt{\frac{Y(X)^2 + X^2}{X^2}} X} = c_1 \sqrt{X}$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Converting $u(X) = -i$ back to $Y(X)$ gives

$$Y(X) = -iX$$

Using the solution for $Y(X)$

$$\frac{Y(X)}{\sqrt{\frac{Y(X)^2 + X^2}{X^2}} X} = c_1 \sqrt{X} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\frac{y}{\sqrt{\frac{x^2 + y^2}{x^2}} x} = \sqrt{x} c_1$$

Using the solution for $Y(X)$

$$Y(X) = 0 \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

Using the solution for $Y(X)$

$$Y(X) = -iX \tag{A}$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

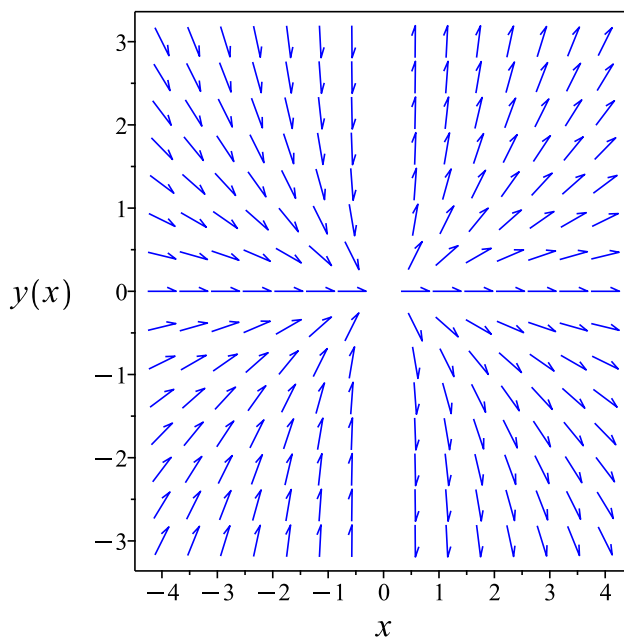
$$y = -ix$$

Solving for y gives

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2} c_1 \sqrt{-\frac{1}{c_1^2 x - 1}}$$

Figure 2.25: Slope field $2x^3y' = y(y^2 + 3x^2)$ Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = x^{3/2}c_1\sqrt{-\frac{1}{c_1^2x} - 1}$$

Solved using first_order_ode_LIE

Time used: 0.308 (sec)

Solve

$$2x^3y' = y(y^2 + 3x^2)$$

Writing the ode as

$$y' = \frac{y(3x^2 + y^2)}{2x^3}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2\xi_y - \omega_x\xi - \omega_y\eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1E)$$

$$\eta = \frac{Ax + By}{Cx} \quad (2E)$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\frac{A}{Cx} - \frac{Ax + By}{Cx^2} + \frac{y(3x^2 + y^2)B}{2x^4C} - \frac{3y}{x^2} + \frac{3y(3x^2 + y^2)}{2x^4} - \frac{\left(\frac{3x^2 + y^2}{2x^3} + \frac{y^2}{x^3}\right)(Ax + By)}{Cx} = 0 \quad (5E)$$

Putting the above in normal form gives

$$-\frac{3Ax^3 + 3Axy^2 + 2Bx^2y + 2By^3 - 3yx^2C - 3Cy^3}{2x^4C} = 0$$

Setting the numerator to zero gives

$$-3Ax^3 - 3Axy^2 - 2Bx^2y - 2By^3 + 3yx^2C + 3Cy^3 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-3Av_1^3 - 3Av_1v_2^2 - 2Bv_1^2v_2 - 2Bv_2^3 + 3Cv_1^2v_2 + 3Cv_2^3 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-3Av_1^3 + (-2B + 3C)v_1^2v_2 - 3Av_1v_2^2 + (-2B + 3C)v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -3A &= 0 \\ -2B + 3C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= 0 \\ B &= \frac{3C}{2} \\ C &= C \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 1 \\ \eta &= \frac{3y}{2x} \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \tag{1}$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{\frac{3y}{2x}}{1} \\ &= \frac{3y}{2x} \end{aligned}$$

This is easily solved to give

$$y = x^{3/2} c_1$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = \frac{y}{x^{3/2}}$$

And S is found from

$$\begin{aligned} dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1} \end{aligned}$$

Integrating gives

$$\begin{aligned} S &= \int \frac{dx}{T} \\ &= x \end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y(3x^2 + y^2)}{2x^3}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= -\frac{3y}{2x^{5/2}} \\ R_y &= \frac{1}{x^{3/2}} \\ S_x &= 1 \\ S_y &= 0 \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{2x^{9/2}}{y^3} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{2}{R^3}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

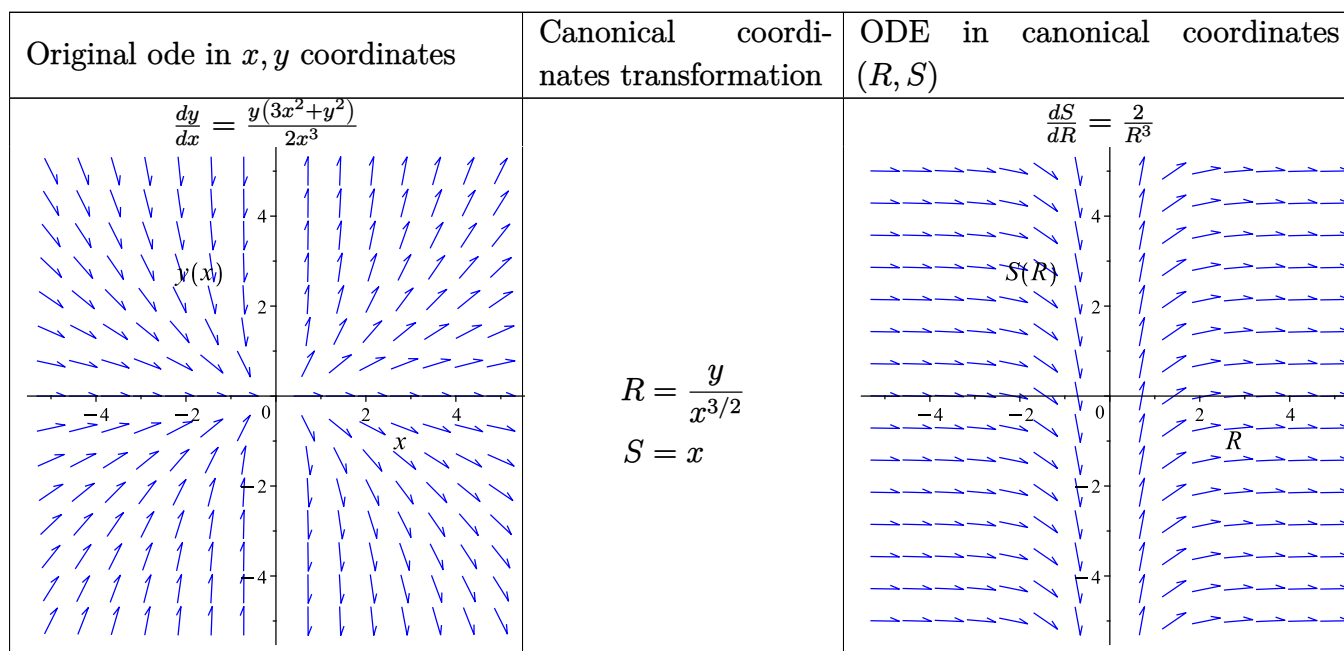
$$\int dS = \int \frac{2}{R^3} dR$$

$$S(R) = -\frac{1}{R^2} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = -\frac{x^3}{y^2} + c_2$$

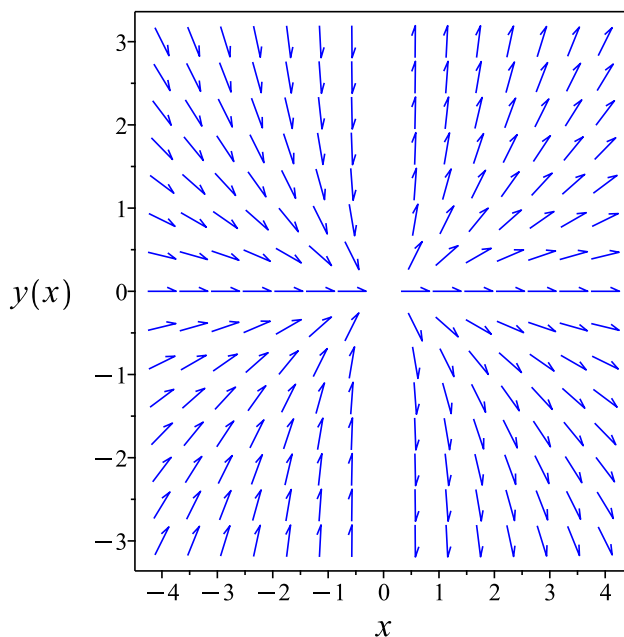
The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = \frac{\sqrt{x(-x+c_2)}x}{-x+c_2}$$

$$y = -\frac{\sqrt{x(-x+c_2)}x}{-x+c_2}$$

Figure 2.26: Slope field $2x^3 y' = y(y^2 + 3x^2)$ Summary of solutions found

$$y = \frac{\sqrt{x(-x + c_2)} x}{-x + c_2}$$

$$y = -\frac{\sqrt{x(-x + c_2)} x}{-x + c_2}$$

✓ Maple. Time used: 0.002 (sec). Leaf size: 43

```
ode:=2*x^3*diff(y(x),x) = y(x)*(3*x^2+y(x)^2);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\sqrt{(-x + c_1) x} x}{-x + c_1}$$

$$y = \frac{\sqrt{(-x + c_1) x} x}{-c_1 + x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
```

<- Bernoulli successful

Maple step by step

Let's solve

$$2x^3 \left(\frac{d}{dx} y(x) \right) = y(x) (y(x)^2 + 3x^2)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{y(x)(y(x)^2 + 3x^2)}{2x^3}$$

✓ **Mathematica.** Time used: 0.105 (sec). Leaf size: 47

```
ode=2*x^3*D[y[x],x]==y[x]*(y[x]^2+3*x^2);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{x^{3/2}}{\sqrt{-x + c_1}}$$

$$y(x) \rightarrow \frac{x^{3/2}}{\sqrt{-x + c_1}}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.443 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x**3*Derivative(y(x), x) - (3*x**2 + y(x)**2)*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\sqrt{\frac{x^3}{C_1 - x}}, \quad y(x) = \sqrt{\frac{x^3}{C_1 - x}} \right]$$

2.1.5 Problem 17

Local contents

Solved as second order linear constant coeff ode	131
Solved as second order solved by an integrating factor	132
Solved as second order ode using Kovacic algorithm	134
✓ Maple	137
✓ Mathematica	138
✓ Sympy	138

Internal problem ID [7849]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:21:20 AM

CAS classification : `[[_2nd_order, _missing_x]]`

Solved as second order linear constant coeff ode

Time used: 0.049 (sec)

Solve

$$y'' - 2y' + y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -2, C = 1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 2\lambda e^{x\lambda} + e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 2\lambda + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -2, C = 1$ into the above gives

$$\begin{aligned}\lambda_{1,2} &= \frac{2}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{(-2)^2 - (4)(1)(1)} \\ &= 1\end{aligned}$$

Hence this is the case of a double root $\lambda_{1,2} = -1$. Therefore the solution is

$$y = c_1 e^x + c_2 x e^x \quad (1)$$

Summary of solutions found

$$y = e^x c_1 + x e^x c_2$$

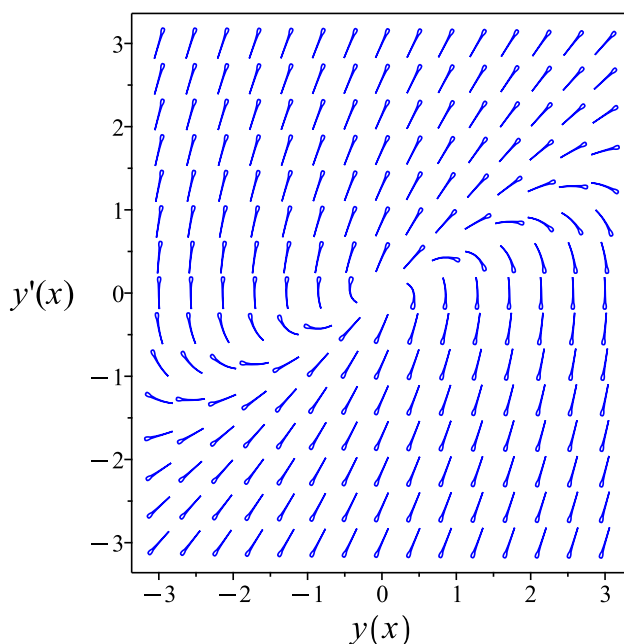


Figure 2.27: Slope field $y'' - 2y' + y = 0$

Solved as second order solved by an integrating factor

Time used: 0.062 (sec)

Solve

$$y'' - 2y' + y = 0$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x)^2 + p'(x))y}{2} = f(x)$$

Where $p(x) = -2$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int -2 \, dx} \\ &= e^{-x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$\begin{aligned} (M(x)y)'' &= 0 \\ (y e^{-x})'' &= 0 \end{aligned}$$

Integrating once gives

$$(y e^{-x})' = c_1$$

Integrating again gives

$$(y e^{-x}) = c_1 x + c_2$$

Hence the solution is

$$y = \frac{c_1 x + c_2}{e^{-x}}$$

Or

$$y = c_1 x e^x + c_2 e^x$$

Summary of solutions found

$$y = c_1 x e^x + c_2 e^x$$

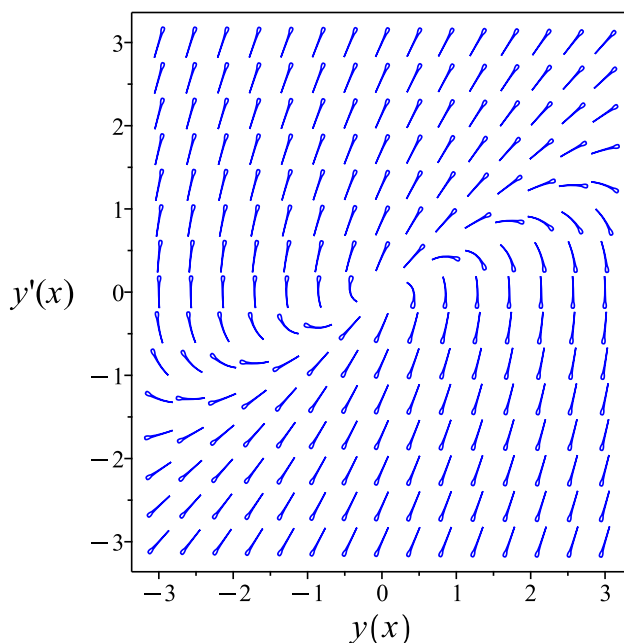


Figure 2.28: Slope field $y'' - 2y' + y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.052 (sec)

Solve

$$y'' - 2y' + y = 0$$

Writing the ode as

$$y'' - 2y' + y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -2 \\ C &= 1 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 0 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.8: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2}{1} dx} \\ &= z_1 e^x \\ &= z_1(e^x) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{2x}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(e^x) + c_2(e^x(x)) \end{aligned}$$

Summary of solutions found

$$y = e^x c_1 + x e^x c_2$$

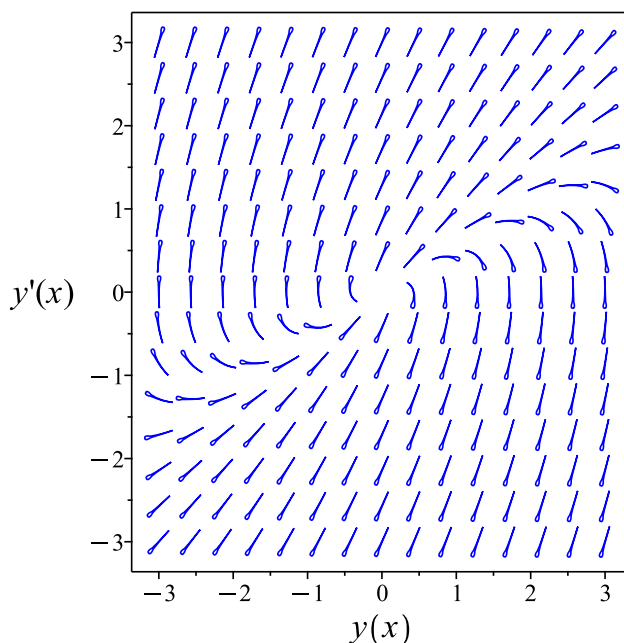


Figure 2.29: Slope field $y'' - 2y' + y = 0$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 12

```
ode:=y(x)-2*diff(y(x),x)+diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x(c_2x + c_1)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 2\frac{d}{dx}y(x) + y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 - 2r + 1 = 0$
- Factor the characteristic polynomial
 $(r - 1)^2 = 0$
- Root of the characteristic polynomial
 $r = 1$
- 1st solution of the ODE
 $y_1(x) = e^x$
- Repeated root, multiply $y_1(x)$ by x to ensure linear independence
 $y_2(x) = x e^x$
- General solution of the ODE
 $y(x) = C1 y_1(x) + C2 y_2(x)$
- Substitute in solutions
 $y(x) = C1 e^x + C2 x e^x$

✓ **Mathematica.** Time used: 0.008 (sec). Leaf size: 16

```
ode=D[y[x],{x,2}]-2*D[y[x],x]+y[x]==0;  
ic={};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x(c_2x + c_1)$$

✓ **Sympy.** Time used: 0.080 (sec). Leaf size: 10

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(y(x) - 2*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + C_2x)e^x$$

2.1.6 Problem 18

Local contents

Solved as second order ode using change of variable on y method 2 . . .	139
Solved as second order ode using non constant coeff transformation on B method	141
Solved as second order ode using Kovacic algorithm	144
✓ Maple	150
✓ Mathematica	152
✗ Sympy	152

Internal problem ID [7850]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:21:25 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using change of variable on y method 2

Time used: 0.320 (sec)

Solve

$$(x-1)y'' - y'x + y = 0$$

In normal form the ode

$$(x-1)y'' - y'x + y = 0 \tag{1}$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = -\frac{x}{x-1}$$

$$q(x) = \frac{1}{x-1}$$

Applying change of variables on the dependnt variable $y = v(x)x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right)v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right)v(x) = 0 \tag{3}$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{n}{x-1} + \frac{1}{x-1} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$\begin{aligned} v''(x) + \left(\frac{2}{x} - \frac{x}{x-1}\right)v'(x) &= 0 \\ v''(x) + \left(\frac{2}{x} - \frac{x}{x-1}\right)v'(x) &= 0 \end{aligned} \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \left(\frac{2}{x} - \frac{x}{x-1}\right)u(x) = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{x^2 - 2x + 2}{(x-1)x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{x^2 - 2x + 2}{(x-1)x} dx} \\ &= \frac{x^2 e^{-x}}{x-1} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu u &= 0 \\ \frac{d}{dx}\left(\frac{u x^2 e^{-x}}{x-1}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{u x^2 e^{-x}}{x-1} &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2 e^{-x}}{x-1}$ gives the final solution

$$u(x) = \frac{(x-1) e^x c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned}v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= \frac{e^x c_1}{x} + c_2\end{aligned}$$

Hence

$$\begin{aligned}y &= v(x) x^n \\ &= \left(\frac{e^x c_1}{x} + c_2\right) x \\ &= e^x c_1 + c_2 x\end{aligned}$$

Summary of solutions found

$$y = \left(\frac{e^x c_1}{x} + c_2\right) x$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.197 (sec)

Solve

$$(x-1)y'' - y'x + y = 0$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \quad (1)$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$A = x - 1$$

$$B = -x$$

$$C = 1$$

$$F = 0$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x - 1)(0) + (-x)(-1) + (1)(-x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-(x - 1)xv'' + (x^2 - 2x + 2)v' = 0$$

Now by applying $v' = u$ the above becomes

$$(-x^2 + x) u'(x) + (x^2 - 2x + 2) u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{x^2 - 2x + 2}{(x-1)x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{x^2 - 2x + 2}{(x-1)x} dx} \\ &= \frac{x^2 e^{-x}}{x-1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2 e^{-x}}{x-1} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2 e^{-x}}{x-1} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2 e^{-x}}{x-1}$ gives the final solution

$$u(x) = \frac{(x-1) e^x c_1}{x^2}$$

The ode for v now becomes

$$v'(x) = \frac{(x-1) e^x c_1}{x^2}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dv &= \int \frac{(x-1) e^x c_1}{x^2} dx \\ v(x) &= \frac{e^x c_1}{x} + c_2 \end{aligned}$$

Replacing $v(x)$ above by $-\frac{y}{x}$, then the solution becomes

$$\begin{aligned} y(x) &= Bv \\ &= -e^x c_1 - c_2 x \end{aligned}$$

Summary of solutions found

$$y = -e^x c_1 - c_2 x$$

Solved as second order ode using Kovacic algorithm

Time used: 0.321 (sec)

Solve

$$(x-1)y'' - y'x + y = 0$$

Writing the ode as

$$(x-1)y'' - y'x + y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x-1 \\ B &= -x \\ C &= 1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{x^2 - 4x + 6}{4(x-1)^2} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned}s &= x^2 - 4x + 6 \\ t &= 4(x - 1)^2\end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{x^2 - 4x + 6}{4(x - 1)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.10: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}\mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 2 \\ &= 0\end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4(x - 1)^2$. There is a pole at $x = 1$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Therefore

$$L = [1, 2]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{1}{4} - \frac{1}{2(x-1)} + \frac{3}{4(x-1)^2}$$

For the pole at $x = 1$ let b be the coefficient of $\frac{1}{(x-1)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $O_r(\infty) = 0$ then

$$v = \frac{-O_r(\infty)}{2} = \frac{0}{2} = 0$$

$[\sqrt{r}]_\infty$ is the sum of terms involving x^i for $0 \leq i \leq v$ in the Laurent series for $\sqrt{r}$ at ∞ . Therefore

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^v a_i x^i \\ &= \sum_{i=0}^0 a_i x^i \end{aligned} \tag{8}$$

Let a be the coefficient of $x^v = x^0$ in the above sum. The Laurent series of $\sqrt{r}$ at ∞ is

$$\sqrt{r} \approx \frac{1}{2} - \frac{1}{2x} + \frac{1}{x^3} + \frac{11}{4x^4} + \frac{21}{4x^5} + \frac{15}{2x^6} + \frac{6}{x^7} - \frac{117}{16x^8} + \dots \tag{9}$$

Comparing Eq. (9) with Eq. (8) shows that

$$a = \frac{1}{2}$$

From Eq. (9) the sum up to $v = 0$ gives

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^0 a_i x^i \\ &= \frac{1}{2} \end{aligned} \tag{10}$$

Now we need to find b , where b be the coefficient of $x^{v-1} = x^{-1} = \frac{1}{x}$ in r minus the coefficient of same term but in $([\sqrt{r}]_\infty)^2$ where $[\sqrt{r}]_\infty$ was found above in Eq (10). Hence

$$([\sqrt{r}]_\infty)^2 = \frac{1}{4}$$

This shows that the coefficient of $\frac{1}{x}$ in the above is 0. Now we need to find the coefficient of $\frac{1}{x}$ in r . How this is done depends on if $v = 0$ or not. Since $v = 0$ then starting from $r = \frac{s}{t}$ and doing long division in the form

$$r = Q + \frac{R}{t}$$

Where Q is the quotient and R is the remainder. Then the coefficient of $\frac{1}{x}$ in r will be the coefficient in R of the term in x of degree of t minus one, divided by the leading coefficient in t . Doing long division gives

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{x^2 - 4x + 6}{4x^2 - 8x + 4} \\ &= Q + \frac{R}{4x^2 - 8x + 4} \\ &= \left(\frac{1}{4}\right) + \left(\frac{5 - 2x}{4x^2 - 8x + 4}\right) \\ &= \frac{1}{4} + \frac{5 - 2x}{4x^2 - 8x + 4} \end{aligned}$$

Since the degree of t is 2, then we see that the coefficient of the term x in the remainder R is -2 . Dividing this by leading coefficient in t which is 4 gives $-\frac{1}{2}$. Now b can be found.

$$\begin{aligned} b &= \left(-\frac{1}{2}\right) - (0) \\ &= -\frac{1}{2} \end{aligned}$$

Hence

$$\begin{aligned} [\sqrt{r}]_{\infty} &= \frac{1}{2} \\ \alpha_{\infty}^{+} &= \frac{1}{2} \left(\frac{b}{a} - v \right) = \frac{1}{2} \left(\frac{-\frac{1}{2}}{\frac{1}{2}} - 0 \right) = -\frac{1}{2} \\ \alpha_{\infty}^{-} &= \frac{1}{2} \left(-\frac{b}{a} - v \right) = \frac{1}{2} \left(-\frac{-\frac{1}{2}}{\frac{1}{2}} - 0 \right) = \frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{x^2 - 4x + 6}{4(x - 1)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^{+}	α_c^{-}
1	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_{\infty}$	α_{∞}^{+}	α_{∞}^{-}
0	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^+ = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^+ - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

Substituting the above values in the above results in

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (+)[\sqrt{r}]_\infty \\ &= -\frac{1}{2(x-1)} + \left(\frac{1}{2}\right) \\ &= -\frac{1}{2(x-1)} + \frac{1}{2} \\ &= \frac{x-2}{2x-2} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2 \left(-\frac{1}{2(x-1)} + \frac{1}{2} \right) (0) + \left(\left(\frac{1}{2(x-1)^2} \right) + \left(-\frac{1}{2(x-1)} + \frac{1}{2} \right)^2 - \left(\frac{x^2 - 4x + 6}{4(x-1)^2} \right) \right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2(x-1)} + \frac{1}{2}\right) dx} \\ &= \frac{e^{\frac{x}{2}}}{\sqrt{x-1}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-x}{x-1} dx} \\ &= z_1 e^{\frac{x}{2} + \frac{\ln(x-1)}{2}} \\ &= z_1 (\sqrt{x-1} e^{\frac{x}{2}}) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-x}{x-1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{x+\ln(x-1)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{x e^{x+\ln(x-1)} e^{-2x}}{x-1} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 \left(e^x \left(-\frac{x e^{x+\ln(x-1)} e^{-2x}}{x-1} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = e^x c_1 - c_2 x$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 12

```
ode:=(-1+x)*diff(diff(y(x),x),x)-diff(y(x),x)*x+y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x + c_2 e^x$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

checking if the LODE is missing y

-> Trying a Liouvillian solution using Kovacic's algorithm

A Liouvillian solution exists

Reducible group (found an exponential solution)

Reducible group (found another exponential solution)

<- Kovacic's algorithm successful

Maple step by step

Let's solve

$$(x-1) \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{y(x)}{x-1} + \frac{x \left(\frac{d}{dx} y(x) \right)}{x-1}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{x \left(\frac{d}{dx} y(x) \right)}{x-1} + \frac{y(x)}{x-1} = 0$$

- Check to see if $x_0 = 1$ is a regular singular point

- o Define functions

$$[P_2(x) = -\frac{x}{x-1}, P_3(x) = \frac{1}{x-1}]$$

- o $(x-1) \cdot P_2(x)$ is analytic at $x = 1$

- $$\left. ((x-1) \cdot P_2(x)) \right|_{x=1} = -1$$
- $(x-1)^2 \cdot P_3(x)$ is analytic at $x = 1$

$$\left. ((x-1)^2 \cdot P_3(x)) \right|_{x=1} = 0$$
- $x = 1$ is a regular singular point
 Check to see if $x_0 = 1$ is a regular singular point
 $x_0 = 1$
- Multiply by denominators

$$(x-1) \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + y(x) = 0$$
- Change variables using $x = u + 1$ so that the regular singular point is at $u = 0$

$$u \left(\frac{d^2}{du^2} y(u) \right) + (-u-1) \left(\frac{d}{du} y(u) \right) + y(u) = 0$$
- Assume series solution for $y(u)$

$$y(u) = \sum_{k=0}^{\infty} a_k u^{k+r}$$
- $\square$ Rewrite ODE with series expansions
 - Convert $u^m \cdot \left(\frac{d}{du} y(u) \right)$ to series expansion for $m = 0..1$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) u^{k+r-1+m}$$
 - Shift index using $k \rightarrow k+1-m$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) u^{k+r}$$
 - Convert $u \cdot \left(\frac{d^2}{du^2} y(u) \right)$ to series expansion

$$u \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) u^{k+r-1}$$
 - Shift index using $k \rightarrow k+1$

$$u \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) (k+r) u^{k+r}$$
- Rewrite ODE with series expansions

$$a_0 r(-2+r) u^{-1+r} + \left(\sum_{k=0}^{\infty} (a_{k+1} (k+1+r) (k+r-1) - a_k (k+r-1)) u^{k+r} \right) = 0$$
- a_0 cannot be 0 by assumption, giving the indicial equation

$$r(-2+r) = 0$$
- Values of r that satisfy the indicial equation

$$r \in \{0, 2\}$$
- Each term in the series must be 0, giving the recursion relation

$$(k+r-1) (a_{k+1} (k+1+r) - a_k) = 0$$
- Recursion relation that defines series solution to ODE

$$a_{k+1} = \frac{a_k}{k+1+r}$$
- Recursion relation for $r = 0$

$$a_{k+1} = \frac{a_k}{k+1}$$
- Solution for $r = 0$

- $$\left[y(u) = \sum_{k=0}^{\infty} a_k u^k, a_{k+1} = \frac{a_k}{k+1} \right]$$
- Revert the change of variables $u = x - 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x-1)^k, a_{k+1} = \frac{a_k}{k+1} \right]$$
- Recursion relation for $r = 2$

$$a_{k+1} = \frac{a_k}{k+3}$$
- Solution for $r = 2$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^{k+2}, a_{k+1} = \frac{a_k}{k+3} \right]$$
- Revert the change of variables $u = x - 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x-1)^{k+2}, a_{k+1} = \frac{a_k}{k+3} \right]$$
- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k (x-1)^k \right) + \left(\sum_{k=0}^{\infty} b_k (x-1)^{k+2} \right), a_{k+1} = \frac{a_k}{k+1}, b_{k+1} = \frac{b_k}{k+3} \right]$$

✓ **Mathematica.** Time used: 0.098 (sec). Leaf size: 90

```
ode=(x-1)*D[y[x],{x,2}]-x*D[y[x],x]+y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \exp \left(\int_1^x \frac{K[1] - 2}{2(K[1] - 1)} dK[1] - \frac{1}{2} \int_1^x -\frac{K[2]}{K[2] - 1} dK[2] \right) \left(c_2 \int_1^x \exp \left(-2 \int_1^{K[3]} \frac{K[1] - 2}{2(K[1] - 1)} dK[1] \right) \right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) + (x - 1)*Derivative(y(x), (x, 2)) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

False

2.1.7 Problem 19

Local contents

Solved as second order linear constant coeff ode	153
Solved as second order can be made integrable	155
Solved as second order ode using Kovacic algorithm	159
✓ Maple	162
✓ Mathematica	163
✓ Sympy	163

Internal problem ID [7851]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:21:31 AM

CAS classification : `[[_2nd_order, _missing_x]]`

Solved as second order linear constant coeff ode

Time used: 0.049 (sec)

Solve

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = e^x c_1 + e^{-x} c_2$$

Summary of solutions found

$$y = e^x c_1 + e^{-x} c_2$$

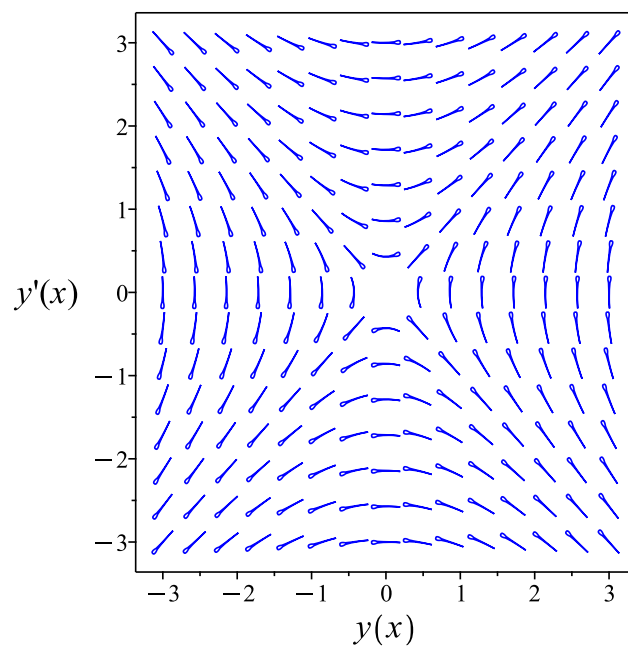


Figure 2.30: Slope field $y'' - y = 0$

Solved as second order can be made integrable

Time used: 1.109 (sec)

Solve

$$y'' - y = 0$$

Multiplying the ode by y' gives

$$y'y'' - yy' = 0$$

Integrating the above w.r.t x gives

$$\int (y'y'' - yy') dx = 0$$

$$\frac{y'^2}{2} - \frac{y^2}{2} = c_1$$

Which is now solved for y . Solve Let $p = y'$ the ode becomes

$$\frac{p^2}{2} - \frac{y^2}{2} = c_1$$

Solving for y from the above results in

$$y = \sqrt{p^2 - 2c_1} \quad (1)$$

$$y = -\sqrt{p^2 - 2c_1} \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.Solving ode 1ATaking derivative of $(*)$ w.r.t. x gives

$$p = f + (xf' + g')\frac{dp}{dx}$$

$$p - f = (xf' + g')\frac{dp}{dx} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = 0$$

$$g = \sqrt{p^2 - 2c_1}$$

Hence (2) becomes

$$p = \frac{pp'(x)}{\sqrt{p^2 - 2c_1}} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \sqrt{2} \sqrt{-c_1}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \sqrt{p(x)^2 - 2c_1} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\int \frac{1}{\sqrt{p^2 - 2c_1}} dp = dx$$

$$\ln \left(p + \sqrt{p^2 - 2c_1} \right) = x + c_2$$

Singular solutions are found by solving

$$\sqrt{p^2 - 2c_1} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = \sqrt{2} \sqrt{c_1}$$

$$p(x) = -\sqrt{2} \sqrt{c_1}$$

Substituting the above solution for p in (2A) gives

$$y = \sqrt{\frac{(e^{2x+2c_2} + 2c_1)^2 e^{-2x-2c_2}}{4} - 2c_1}$$

$$y = 0$$

$$y = 0$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= 0 \\ g &= -\sqrt{p^2 - 2c_1} \end{aligned}$$

Hence (2) becomes

$$p = -\frac{pp'(x)}{\sqrt{p^2 - 2c_1}} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\sqrt{2}\sqrt{-c_1}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = -\sqrt{p(x)^2 - 2c_1} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\begin{aligned} \int -\frac{1}{\sqrt{p^2 - 2c_1}} dp &= dx \\ -\ln\left(p + \sqrt{p^2 - 2c_1}\right) &= x + c_3 \end{aligned}$$

Singular solutions are found by solving

$$-\sqrt{p^2 - 2c_1} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = \sqrt{2} \sqrt{c_1}$$

$$p(x) = -\sqrt{2} \sqrt{c_1}$$

Substituting the above solution for p in (2A) gives

$$y = -\sqrt{\frac{(e^{-2x-2c_3} + 2c_1)^2 e^{2x+2c_3}}{4} - 2c_1}$$

$$y = 0$$

$$y = 0$$

Simplifying the above gives

$$y = \sqrt{2} \sqrt{-c_1}$$

$$y = \frac{\sqrt{e^{-2x-2c_2} (e^{2x+2c_2} - 2c_1)^2}}{2}$$

$$y = 0$$

$$y = 0$$

$$y = -\sqrt{2} \sqrt{-c_1}$$

$$y = -\frac{\sqrt{e^{-2x-2c_3} (2e^{2x+2c_3} c_1 - 1)^2}}{2}$$

$$y = 0$$

$$y = 0$$

The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = \sqrt{2} \sqrt{-c_1}$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = -\sqrt{2} \sqrt{-c_1}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = \frac{\sqrt{e^{-2x-2c_2} (e^{2x+2c_2} - 2c_1)^2}}{2}$$

$$y = -\frac{\sqrt{e^{-2x-2c_3} (2e^{2x+2c_3}c_1 - 1)^2}}{2}$$

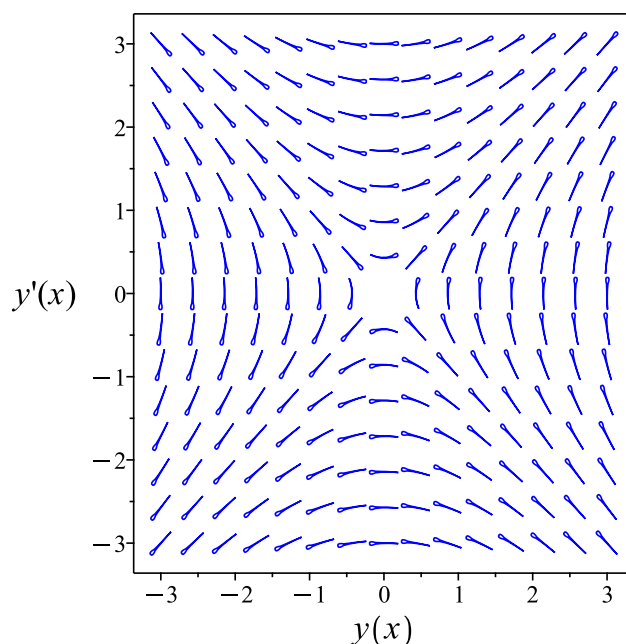


Figure 2.31: Slope field $y'' - y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.053 (sec)

Solve

$$y'' - y = 0$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.12: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(e^{-x}) + c_2\left(e^{-x}\left(\frac{e^{2x}}{2}\right)\right) \end{aligned}$$

Summary of solutions found

$$y = e^{-x}c_1 + \frac{c_2 e^x}{2}$$

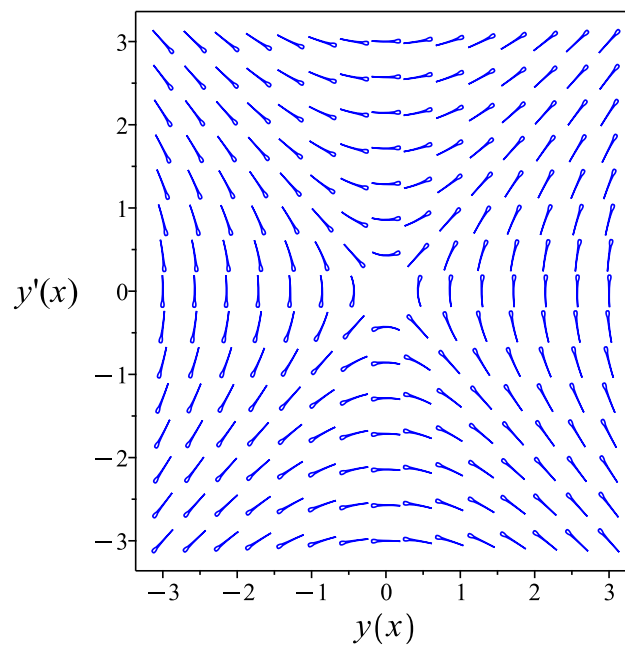


Figure 2.32: Slope field $y'' - y = 0$

✓ Maple. Time used: 0.001 (sec). Leaf size: 15

```
ode:=diff(diff(y(x),x),x)-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 e^{-x} + c_2 e^x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
```

<- constant coefficients successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

- Substitute in solutions

$$y(x) = C_1 e^{-x} + C_2 e^x$$

✓ **Mathematica.** Time used: 0.007 (sec). Leaf size: 20

```
ode=D[y[x],{x,2}]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 e^x + c_2 e^{-x}$$

✓ **Sympy.** Time used: 0.025 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x$$

2.1.8 Problem 20

Local contents

Solved as second order linear constant coeff ode	164
Solved as second order ode using Kovacic algorithm	167
✓ Maple	171
✓ Mathematica	173
✓ Sympy	173

Internal problem ID [7852]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 20

Date solved : Thursday, November 27, 2025 at 06:21:36 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.115 (sec)

Solve

$$y'' - y = -x + 4$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = -x + 4$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = e^x c_1 + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = e^x c_1 + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x + 1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_2x + A_1$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-A_2x - A_1 = -x + 4$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = -4, A_2 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

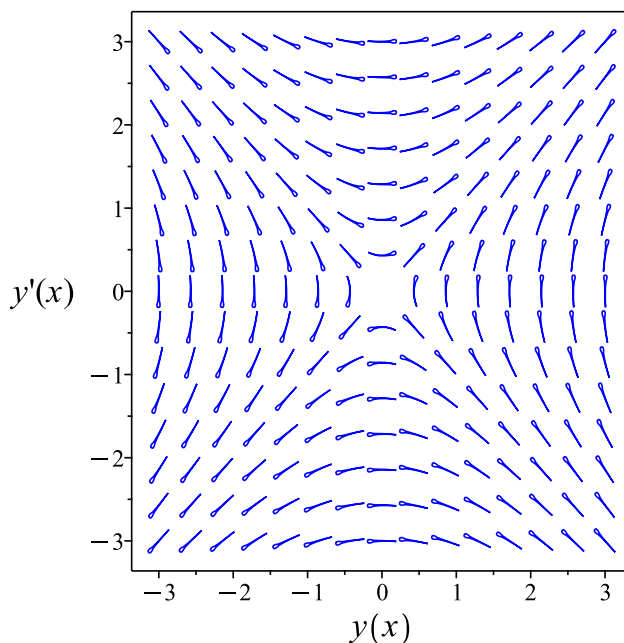
$$y_p = x - 4$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{-x} c_2) + (x - 4) \end{aligned}$$

Summary of solutions found

$$y = x - 4 + e^x c_1 + e^{-x} c_2$$

Figure 2.33: Slope field $y'' - y = -x + 4$

Solved as second order ode using Kovacic algorithm

Time used: 0.101 (sec)

Solve

$$y'' - y = -x + 4$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.14: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{-x}c_1 + \frac{c_2 e^x}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x + 1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{2}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_2x + A_1$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-A_2x - A_1 = -x + 4$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = -4, A_2 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = x - 4$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-x} c_1 + \frac{c_2 e^x}{2} \right) + (x - 4) \end{aligned}$$

Summary of solutions found

$$y = x - 4 + e^{-x} c_1 + \frac{c_2 e^x}{2}$$

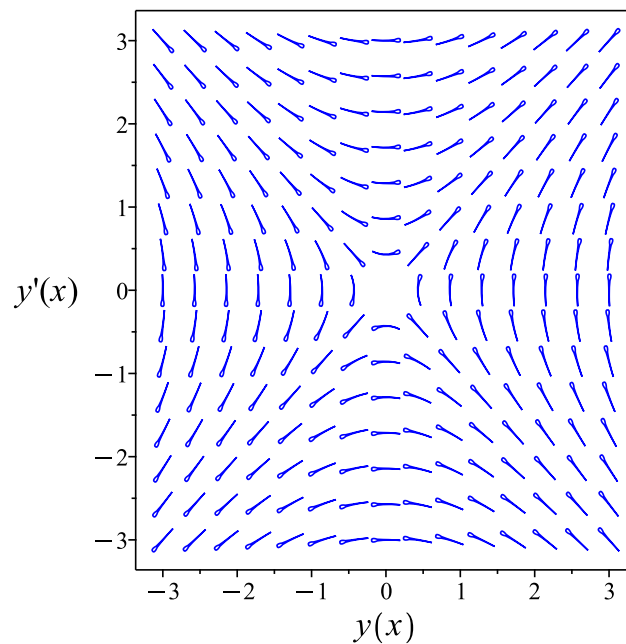


Figure 2.34: Slope field $y'' - y = -x + 4$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 17

```
ode:=diff(diff(y(x),x),x)-y(x) = 4-x;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x c_2 + e^{-x} c_1 - 4 + x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
```

```

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = 4 - x$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = 4 - x \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = \frac{e^{-x} \int e^x (-4+x) dx}{2} - \frac{e^x \int e^{-x} (-4+x) dx}{2}$$

- Compute integrals

$$y_p(x) = -4 + x$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x - 4 + x$$

✓ **Mathematica.** Time used: 0.009 (sec). Leaf size: 22

```
ode=D[y[x],{x,2}]-y[x]==4-x;  
ic={};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x + c_1 e^x + c_2 e^{-x} - 4$$

✓ **Sympy.** Time used: 0.034 (sec). Leaf size: 15

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(x - y(x) + Derivative(y(x), (x, 2)) - 4, 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x + x - 4$$

2.1.9 Problem 21

Local contents

Solved as second order linear constant coeff ode	174
Solved as second order ode using Kovacic algorithm	176
✓ Maple	179
✓ Mathematica	180
✓ Sympy	180

Internal problem ID [7853]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 21

Date solved : Thursday, November 27, 2025 at 06:21:41 AM

CAS classification : `[_2nd_order, _missing_x]`

Solved as second order linear constant coeff ode

Time used: 0.055 (sec)

Solve

$$y'' - 3y' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -3, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 3\lambda e^{x\lambda} + 2e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 3\lambda + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -3, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{3}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-3^2 - (4)(1)(2)} \\ &= \frac{3}{2} \pm \frac{1}{2} \end{aligned}$$

Hence

$$\lambda_1 = \frac{3}{2} + \frac{1}{2}$$

$$\lambda_2 = \frac{3}{2} - \frac{1}{2}$$

Which simplifies to

$$\lambda_1 = 2$$

$$\lambda_2 = 1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(2)x} + c_2 e^{(1)x}$$

Or

$$y = e^{2x} c_1 + c_2 e^x$$

Summary of solutions found

$$y = e^{2x} c_1 + c_2 e^x$$

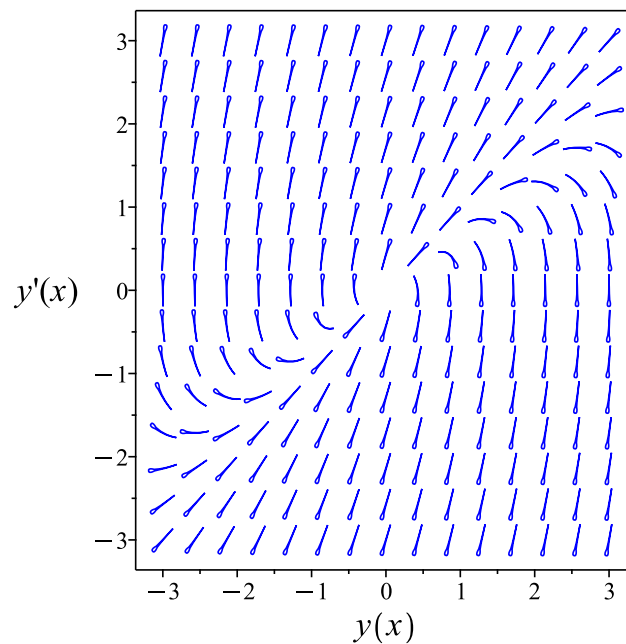


Figure 2.35: Slope field $y'' - 3y' + 2y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.060 (sec)

Solve

$$y'' - 3y' + 2y = 0$$

Writing the ode as

$$y'' - 3y' + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -3 \\ C &= 2 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{4} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 4 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \frac{z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.16: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{1}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3}{1} dx} \\ &= z_1 e^{\frac{3x}{2}} \\ &= z_1 \left(e^{\frac{3x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3x}}{(y_1)^2} dx \\ &= y_1 (e^{3x} e^{-2x}) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 (e^x (e^{3x} e^{-2x})) \end{aligned}$$

Summary of solutions found

$$y = e^x c_1 + e^{2x} c_2$$

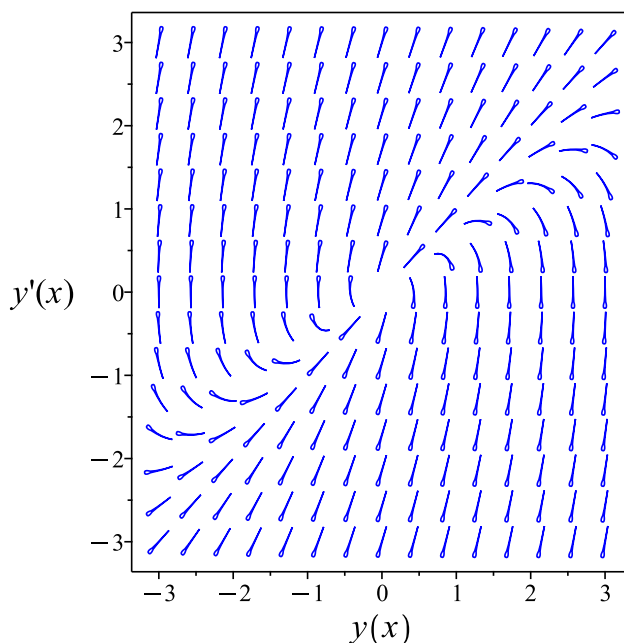


Figure 2.36: Slope field $y'' - 3y' + 2y = 0$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 13

```
ode:=2*y(x)-3*diff(y(x),x)+diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x(c_1 + c_2 e^x)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 3\frac{d}{dx}y(x) + 2y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of ODE

$$r^2 - 3r + 2 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r - 2) = 0$$

- Roots of the characteristic polynomial

$$r = (1, 2)$$

- 1st solution of the ODE

$$y_1(x) = e^x$$

- 2nd solution of the ODE

$$y_2(x) = e^{2x}$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

- Substitute in solutions

$$y(x) = C_1 e^x + C_2 e^{2x}$$

✓ **Mathematica.** Time used: 0.008 (sec). Leaf size: 18

```
ode=D[y[x],{x,2}]-3*D[y[x],x]+2*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x(c_2 e^x + c_1)$$

✓ **Sympy.** Time used: 0.086 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*y(x) - 3*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + C_2 e^x) e^x$$

2.1.10 Problem 22

Local contents

Solved as second order linear constant coeff ode	181
Solved as second order ode using Kovacic algorithm	184
✓ Maple	188
✓ Mathematica	190
✓ Sympy	190

Internal problem ID [7854]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 2. Solutions of differential equations. Supplementary problems. Page 11

Problem number : 22

Date solved : Thursday, November 27, 2025 at 06:21:46 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.142 (sec)

Solve

$$y'' - 3y' + 2y = 2e^x(1 - x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -3, C = 2, f(x) = -2e^x(x - 1)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -3, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 3\lambda e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 3\lambda + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -3, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{3}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-3^2 - (4)(1)(2)} \\ &= \frac{3}{2} \pm \frac{1}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= \frac{3}{2} + \frac{1}{2} \\ \lambda_2 &= \frac{3}{2} - \frac{1}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= 2 \\ \lambda_2 &= 1 \end{aligned}$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(2)x} + c_2 e^{(1)x} \end{aligned}$$

Or

$$y = e^{2x} c_1 + c_2 e^x$$

Therefore the homogeneous solution y_h is

$$y_h = e^{2x} c_1 + c_2 e^x$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$-2e^x(x-1)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x e^x, e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{2x}\}$$

Since e^x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2 e^x, x e^x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x^2 e^x + A_2 x e^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$2A_1 e^x - 2A_1 x e^x - A_2 e^x = 2e^x(1 - x)$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 1, A_2 = 0]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

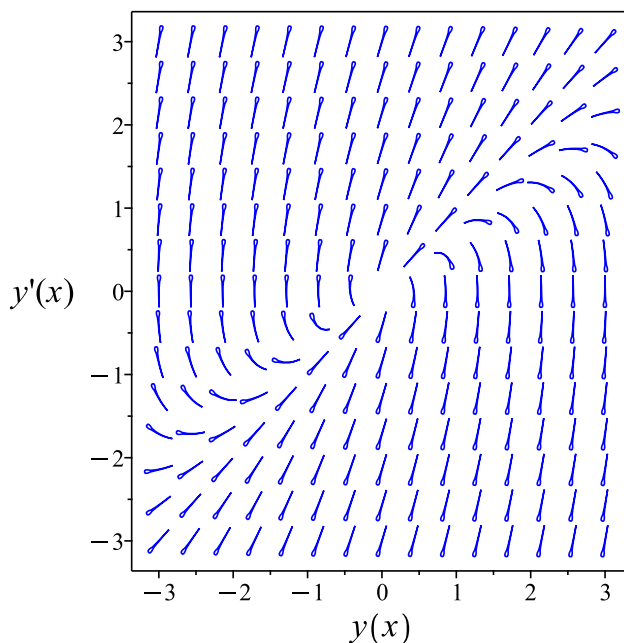
$$y_p = x^2 e^x$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x} c_1 + c_2 e^x) + (x^2 e^x) \end{aligned}$$

Summary of solutions found

$$y = x^2 e^x + e^{2x} c_1 + c_2 e^x$$

Figure 2.37: Slope field $y'' - 3y' + 2y = 2e^x(1-x)$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.124 (sec)

Solve

$$y'' - 3y' + 2y = 2e^x(1-x)$$

Writing the ode as

$$y'' - 3y' + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -3 \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{4} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 4 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \frac{z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.18: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{1}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3}{1} dx} \\ &= z_1 e^{\frac{3x}{2}} \\ &= z_1 \left(e^{\frac{3x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3x}}{(y_1)^2} dx \\ &= y_1 (e^{3x} e^{-2x}) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 (e^x (e^{3x} e^{-2x})) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^x c_1 + e^{2x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$-2e^x(x-1)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{xe^x, e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{2x}\}$$

Since e^x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2e^x, xe^x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1x^2e^x + A_2xe^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$2A_1e^x - 2A_1xe^x - A_2e^x = 2e^x(1-x)$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 1, A_2 = 0]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = x^2 e^x$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{2x} c_2) + (x^2 e^x) \end{aligned}$$

Summary of solutions found

$$y = x^2 e^x + e^x c_1 + e^{2x} c_2$$

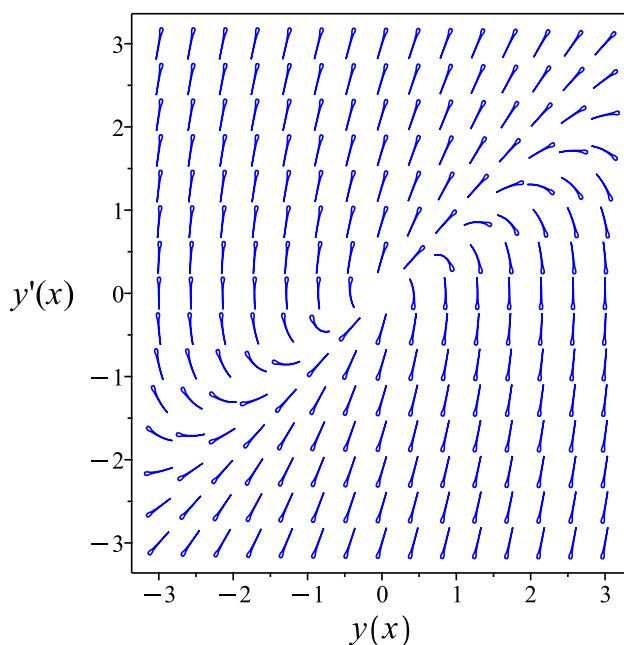


Figure 2.38: Slope field $y'' - 3y' + 2y = 2e^x(1-x)$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 16

```
ode:=2*y(x)-3*diff(y(x),x)+diff(diff(y(x),x),x) = 2*exp(x)*(1-x);
dsolve(ode,y(x), singsol=all);
```

$$y = (e^x c_1 + x^2 + c_2) e^x$$

Maple trace

Methods for second order ODEs:


```

--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
<- double symmetry of the form [xi=0, eta=F(x)] successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 3\frac{d}{dx}y(x) + 2y(x) = 2e^x(1-x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2}y(x) = -2y(x) - 2e^xx + 3\frac{d}{dx}y(x) + 2e^x$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2}y(x) - 3\frac{d}{dx}y(x) + 2y(x) = -2e^x(-1+x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 3r + 2 = 0$$

- Factor the characteristic polynomial

$$(r-1)(r-2) = 0$$

- Roots of the characteristic polynomial

$$r = (1, 2)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^x$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{2x}$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^x + C_2 e^{2x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = -2e^x(-1+x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{3x}$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -2e^x(e^x \int (-1+x)e^{-x} dx - \int (-1+x) dx)$$

- Compute integrals
 $y_p(x) = e^x x^2$
- Substitute particular solution into general solution to ODE
 $y(x) = C_1 e^x + C_2 e^{2x} + e^x x^2$

✓ **Mathematica.** Time used: 0.014 (sec). Leaf size: 21

```
ode=D[y[x],{x,2}]-3*D[y[x],x]+2*y[x]==2*Exp[x]*(1-x);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x (x^2 + c_2 e^x + c_1)$$

✓ **Sympy.** Time used: 0.131 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((2*x - 2)*exp(x) + 2*y(x) - 3*Derivative(y(x), x) + Derivative(y(x), (x,
2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + C_2 e^x + x^2) e^x$$

2.2 Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Local contents

2.2.1	Problem 24	192
2.2.2	Problem 25	215
2.2.3	Problem 26	231
2.2.4	Problem 27	262
2.2.5	Problem 28	279
2.2.6	Problem 29	300
2.2.7	Problem 30	322
2.2.8	Problem 31	326
2.2.9	Problem 32	349
2.2.10	Problem 34	362
2.2.11	Problem 35	381
2.2.12	Problem 37	397
2.2.13	Problem 38	430
2.2.14	Problem 39	451
2.2.15	Problem 40	477
2.2.16	Problem 41	485
2.2.17	Problem 42	523
2.2.18	Problem 43	544
2.2.19	Problem 44	558
2.2.20	Problem 45	600
2.2.21	Problem 46	614
2.2.22	Problem 47	640
2.2.23	Problem 48	673
2.2.24	Problem 49	683
2.2.25	Problem 51	689
2.2.26	Problem 52	708

2.2.1 Problem 24

Local contents

Solved using first_order_ode_linear	192
Solved using first_order_ode_separable	194
Solved using first_order_ode_exact	195
Solved using first_order_ode_homog_A	199
Solved using first_order_ode_homog_type_D2	202
Solved using first_order_ode_homog_type_G	204
Solved using first_order_ode_homog_type_maple_C	205
Solved using first_order_ode_LIE	209
✓ Maple	213
✓ Mathematica	214
✓ Sympy	214

Internal problem ID [7855]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 24

Date solved : Thursday, November 27, 2025 at 06:21:51 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.026 (sec)

Solve

$$y'x + 4y = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = \frac{4}{x}$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int \frac{4}{x} dx} \\ &= x^4\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}(x^4 y) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}x^4 y &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor x^4 gives the final solution

$$y = \frac{c_1}{x^4}$$

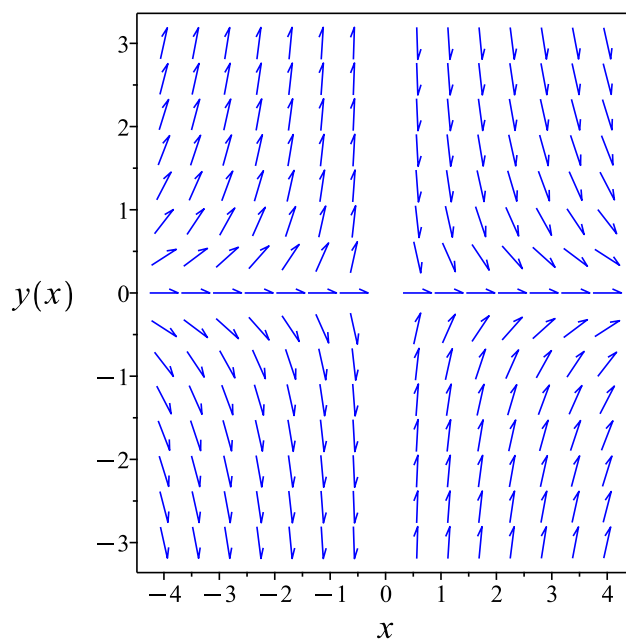


Figure 2.39: Slope field $y'x + 4y = 0$

Summary of solutions found

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_separable

Time used: 0.131 (sec)

Solve

$$y'x + 4y = 0$$

The ode

$$y' = -\frac{4y}{x} \tag{2.17}$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{4y}{x} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{4}{x} \\ g(y) &= y \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{y} dy &= \int -\frac{4}{x} dx \end{aligned}$$

$$\ln(y) = \ln\left(\frac{1}{x^4}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$y = \frac{c_1}{x^4}$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y = 0$$

for y gives

$$y = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$y = \frac{c_1}{x^4}$$

$$y = 0$$

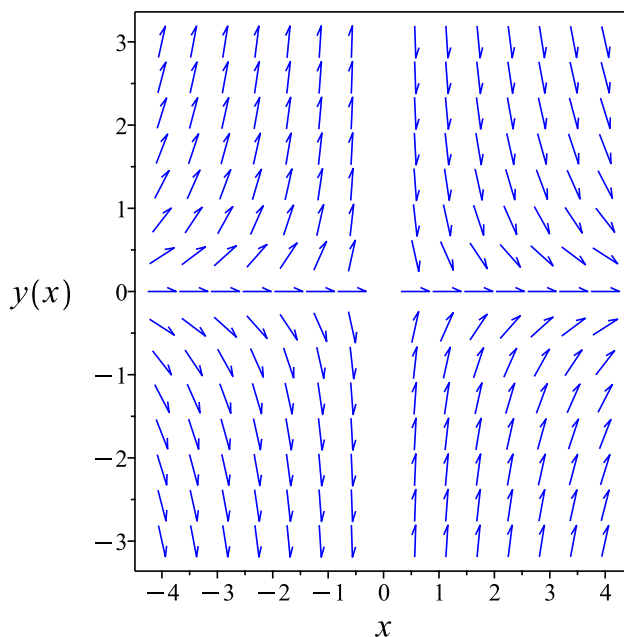


Figure 2.40: Slope field $y'x + 4y = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_exact

Time used: 0.119 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (-4y) dx \\ (4y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 4y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(4y) \\ &= 4 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((4) - (1)) \\ &= \frac{3}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int \frac{3}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{3 \ln(x)} \\ &= x^3 \end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= x^3(4y) \\ &= 4y x^3 \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= x^3(x) \\ &= x^4 \end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (4y x^3) + (x^4) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int x^4 dy \\ \phi &= x^4 y + f(x)\end{aligned}\tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = 4y x^3 + f'(x)\tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = 4y x^3$. Therefore equation (4) becomes

$$4y x^3 = 4y x^3 + f'(x)\tag{5}$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

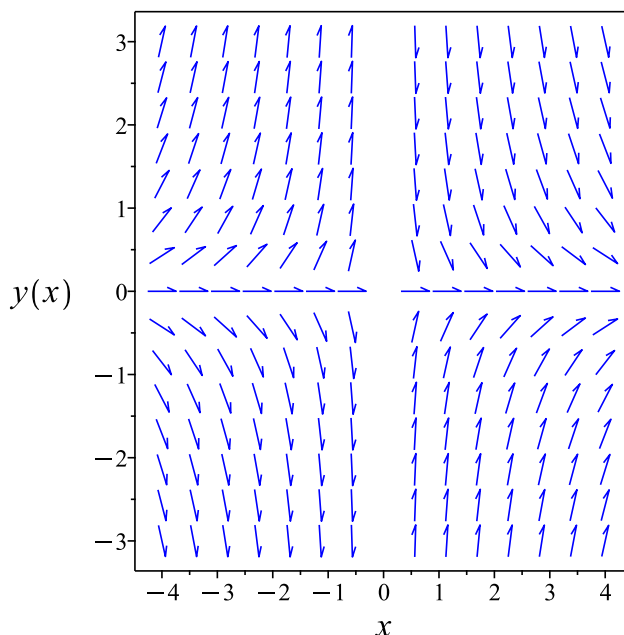
$$\phi = x^4 y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x^4 y$$

Solving for y gives

$$y = \frac{c_1}{x^4}$$

Figure 2.41: Slope field $y'x + 4y = 0$ Summary of solutions found

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_homog_A

Time used: 0.408 (sec)

Solve

$$y'x + 4y = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{4y}{x} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -4y$ and $N = x$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= -4u \\ \frac{du}{dx} &= -\frac{5u(x)}{x}\end{aligned}$$

Or

$$u'(x) + \frac{5u(x)}{x} = 0$$

Or

$$u'(x)x + 5u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{5u(x)}{x} \tag{2.18}$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= -\frac{5u(x)}{x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= -\frac{5}{x} \\ g(u) &= u\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u} du &= \int -\frac{5}{x} dx\end{aligned}$$

$$\ln(u(x)) = \ln\left(\frac{1}{x^5}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = \frac{c_1}{x^5}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = \frac{c_1}{x^5}$$

$$u(x) = 0$$

Converting $u(x) = \frac{c_1}{x^5}$ back to y gives

$$y = \frac{c_1}{x^4}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

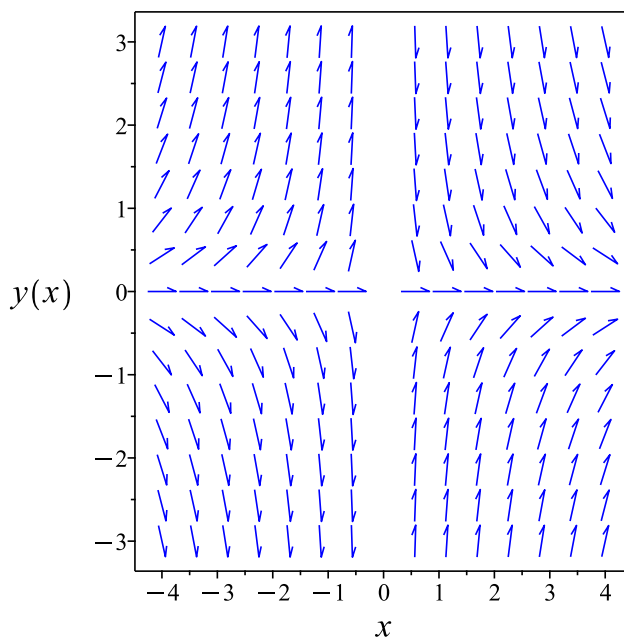


Figure 2.42: Slope field $y'x + 4y = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.149 (sec)

Solve

$$y'x + 4y = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$(u'(x)x + u(x))x + 4u(x)x = 0$$

Which is now solved The ode

$$u'(x) = -\frac{5u(x)}{x} \quad (2.19)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{5u(x)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{5}{x} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u} du &= \int -\frac{5}{x} dx \end{aligned}$$

$$\ln(u(x)) = \ln\left(\frac{1}{x^5}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = \frac{c_1}{x^5}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = \frac{c_1}{x^5}$$

$$u(x) = 0$$

Converting $u(x) = \frac{c_1}{x^5}$ back to y gives

$$y = \frac{c_1}{x^4}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

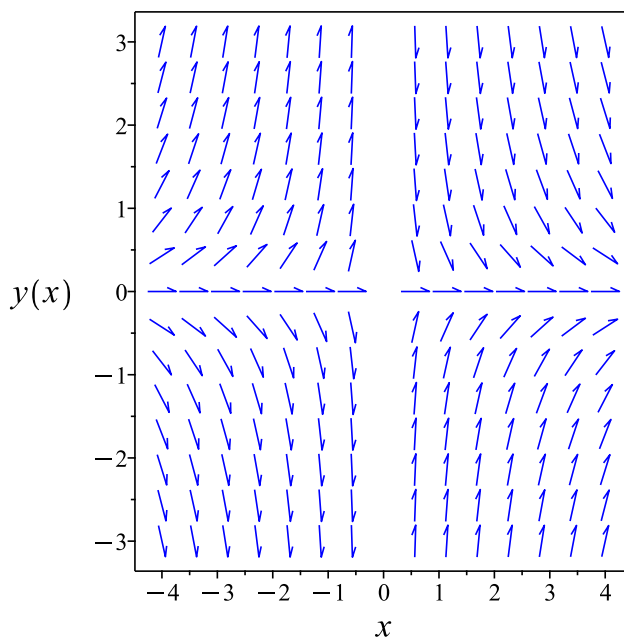


Figure 2.43: Slope field $y'x + 4y = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_homog_type_G

Time used: 0.046 (sec)

Solve

$$y'x + 4y = 0$$

Multiplying the right side of the ode, which is $-\frac{4y}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{4y}{x} \\ &= -4 \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ does not have y and does not have x then

$$\alpha = 0$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -4$$

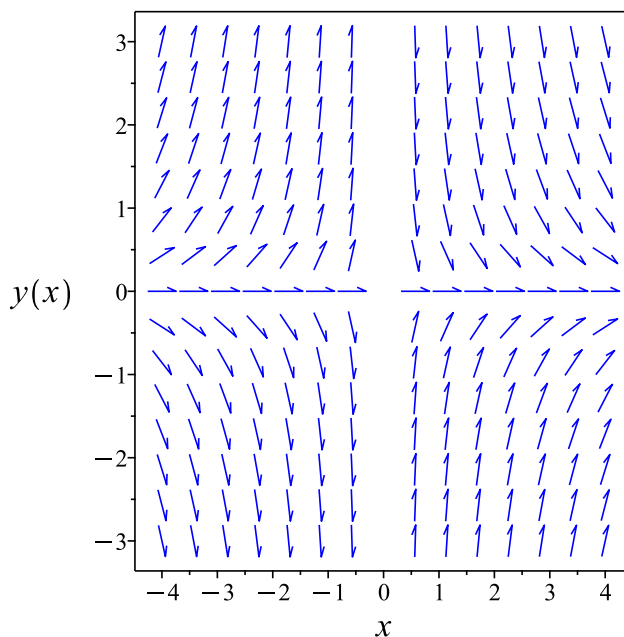
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^y \frac{1}{4z} dz = 0$$

Figure 2.44: Slope field $y'x + 4y = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^y \frac{1}{4z} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.548 (sec)

Solve

$$y'x + 4y = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{4(Y(X) + y_0)}{X + x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{4Y(X)}{X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{4Y}{X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -4Y$ and $N = X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -4u \\ \frac{du}{dX} &= -\frac{5u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) + \frac{5u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) X + 5u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{5u(X)}{X} \quad (2.20)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{5u(X)}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{5}{X} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{1}{u} du = \int -\frac{5}{X} dX$$

$$\ln(u(X)) = \ln\left(\frac{1}{X^5}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(X) = \frac{c_1}{X^5}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(X) = \frac{c_1}{X^5}$$

$$u(X) = 0$$

Converting $u(X) = \frac{c_1}{X^5}$ back to $Y(X)$ gives

$$Y(X) = \frac{c_1}{X^4}$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$Y(X) = \frac{c_1}{X^4} \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = \frac{c_1}{x^4}$$

Using the solution for $Y(X)$

$$Y(X) = 0 \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

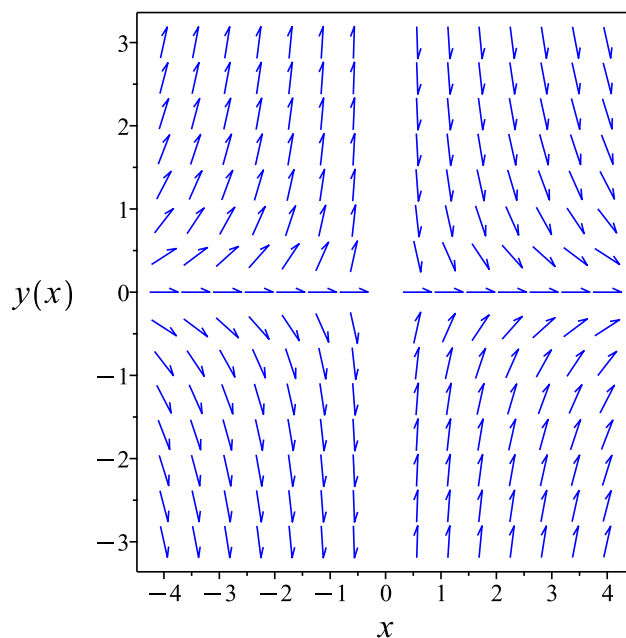


Figure 2.45: Slope field $y'x + 4y = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{x^4}$$

Solved using first_order_ode_LIE

Time used: 0.711 (sec)

Solve

$$y'x + 4y = 0$$

Writing the ode as

$$y' = -\frac{4y}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{4y(b_3 - a_2)}{x} - \frac{16y^2a_3}{x^2} - \frac{4y(xa_2 + ya_3 + a_1)}{x^2} + \frac{4xb_2 + 4yb_3 + 4b_1}{x} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{5b_2x^2 - 20y^2a_3 + 4xb_1 - 4ya_1}{x^2} = 0$$

Setting the numerator to zero gives

$$5b_2x^2 - 20y^2a_3 + 4xb_1 - 4ya_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-20a_3v_2^2 + 5b_2v_1^2 - 4a_1v_2 + 4b_1v_1 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-20a_3v_2^2 + 5b_2v_1^2 - 4a_1v_2 + 4b_1v_1 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -4a_1 &= 0 \\ -20a_3 &= 0 \\ 4b_1 &= 0 \\ 5b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= a_2 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 0 \\ \eta &= y \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y} dy \end{aligned}$$

Which results in

$$S = \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{4y}{x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{4}{x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{4}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

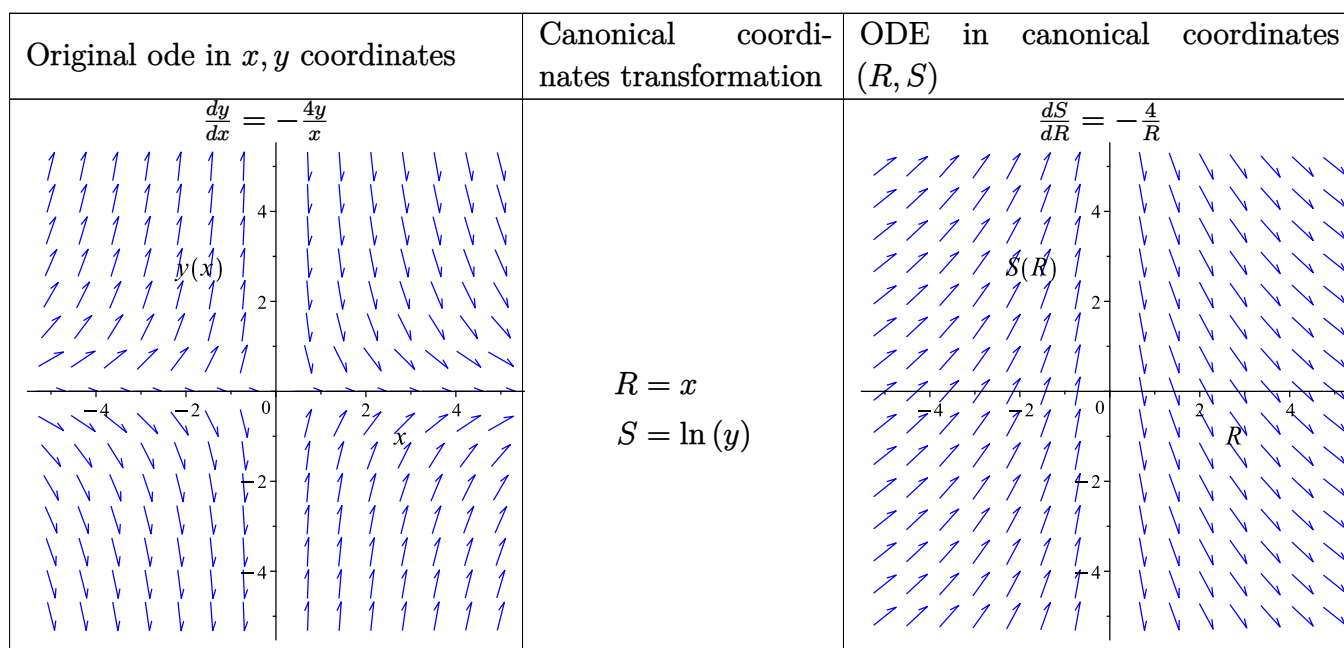
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned}\int dS &= \int -\frac{4}{R} dR \\ S(R) &= -4 \ln(R) + c_2\end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

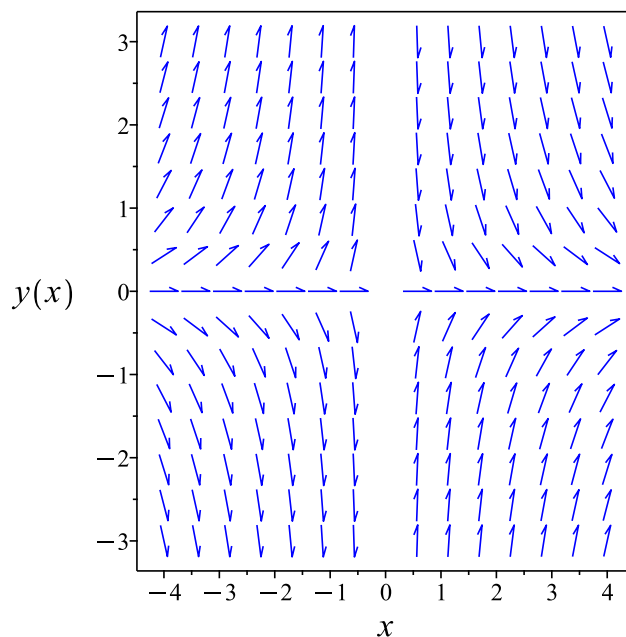
$$\ln(y) = -4 \ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = \frac{e^{c_2}}{x^4}$$

Figure 2.46: Slope field $y'x + 4y = 0$ Summary of solutions found

$$y = \frac{e^{c_2}}{x^4}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 9

```
ode:=4*y(x)+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{c_1}{x^4}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$4y(x) + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Solve for the highest derivative
 $\frac{d}{dx}y(x) = -\frac{4y(x)}{x}$
- Separate variables
 $\frac{\frac{d}{dx}y(x)}{y(x)} = -\frac{4}{x}$
- Integrate both sides with respect to x
 $\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int -\frac{4}{x} dx + C1$
- Evaluate integral
 $\ln(y(x)) = -4 \ln(x) + C1$
- Solve for $y(x)$
 $y(x) = \frac{e^{C1}}{x^4}$
- Redefine the integration constant(s)
 $y(x) = \frac{C1}{x^4}$

✓ **Mathematica.** Time used: 0.014 (sec). Leaf size: 16

```
ode=4*y[x]+x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{C_1}{x^4}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.063 (sec). Leaf size: 7

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) + 4*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{C_1}{x^4}$$

2.2.2 Problem 25

Local contents

Solved using first_order_ode_linear	215
Solved using first_order_ode_separable	217
Solved using first_order_ode_exact	219
Solved using first_order_ode_LIE	223
✓ Maple	228
✓ Mathematica	229
✓ Sympy	230

Internal problem ID [7856]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 25

Date solved : Thursday, November 27, 2025 at 06:21:59 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.059 (sec)

Solve

$$1 + 2y + (-x^2 + 4) y' = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{2}{x^2 - 4}$$

$$p(x) = \frac{1}{x^2 - 4}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{2}{x^2-4} dx} \\ &= \frac{\sqrt{x+2}}{\sqrt{x-2}} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{1}{x^2 - 4} \right) \\ \frac{d}{dx} \left(\frac{y\sqrt{x+2}}{\sqrt{x-2}} \right) &= \left(\frac{\sqrt{x+2}}{\sqrt{x-2}} \right) \left(\frac{1}{x^2 - 4} \right) \\ d \left(\frac{y\sqrt{x+2}}{\sqrt{x-2}} \right) &= \left(\frac{\sqrt{x+2}}{(x^2 - 4)\sqrt{x-2}} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y\sqrt{x+2}}{\sqrt{x-2}} &= \int \frac{\sqrt{x+2}}{(x^2 - 4)\sqrt{x-2}} dx \\ &= -\frac{\sqrt{x+2}}{2\sqrt{x-2}} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{\sqrt{x+2}}{\sqrt{x-2}}$ gives the final solution

$$y = \frac{2\sqrt{x-2}c_1 - \sqrt{x+2}}{2\sqrt{x+2}}$$

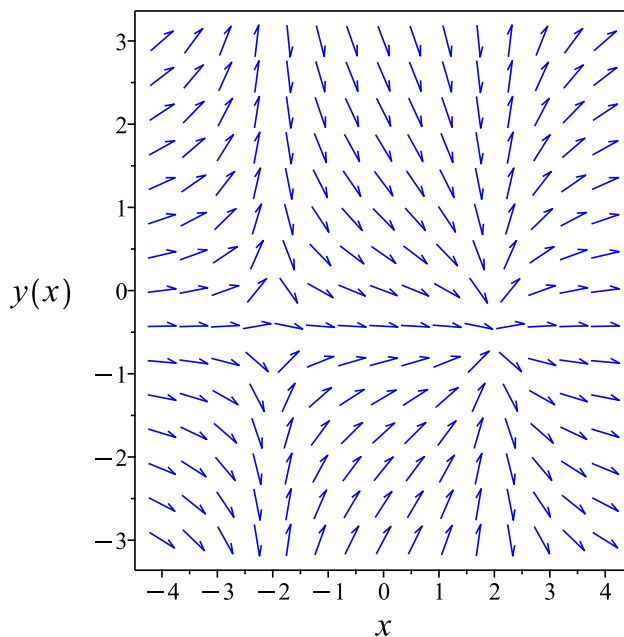


Figure 2.47: Slope field $1 + 2y + (-x^2 + 4)y' = 0$

Summary of solutions found

$$y = \frac{2\sqrt{x-2}c_1 - \sqrt{x+2}}{2\sqrt{x+2}}$$

Solved using first_order_ode_separable

Time used: 0.144 (sec)

Solve

$$1 + 2y + (-x^2 + 4) y' = 0$$

The ode

$$y' = \frac{1 + 2y}{(x - 2)(x + 2)} \quad (2.21)$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{1 + 2y}{(x - 2)(x + 2)} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{(x - 2)(x + 2)} \\ g(y) &= 1 + 2y \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{1 + 2y} dy &= \int \frac{1}{(x - 2)(x + 2)} dx \end{aligned}$$

$$\frac{\ln(1 + 2y)}{2} = \ln \left(\frac{(x - 2)^{1/4}}{(x + 2)^{1/4}} \right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{1 + 2y} = \frac{c_1(x - 2)^{1/4}}{(x + 2)^{1/4}}$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$1 + 2y = 0$$

for y gives

$$y = -\frac{1}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{1 + 2y} = \frac{c_1(x - 2)^{1/4}}{(x + 2)^{1/4}}$$

$$y = -\frac{1}{2}$$

Solving for y gives

$$y = -\frac{1}{2}$$

$$y = \frac{c_1^2 \sqrt{x - 2} - \sqrt{x + 2}}{2\sqrt{x + 2}}$$

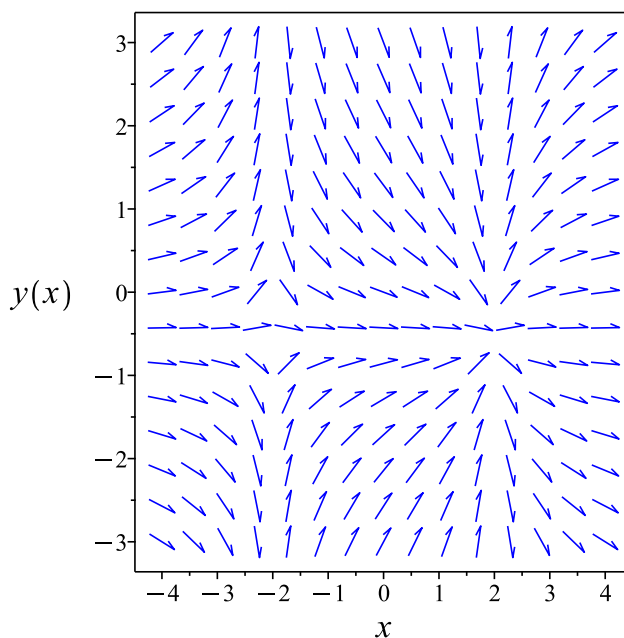


Figure 2.48: Slope field $1 + 2y + (-x^2 + 4)y' = 0$

Summary of solutions found

$$y = -\frac{1}{2}$$

$$y = \frac{c_1^2 \sqrt{x-2} - \sqrt{x+2}}{2\sqrt{x+2}}$$

Solved using first_order_ode_exact

Time used: 0.135 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (-x^2 + 4) dy &= (-2y - 1) dx \\ (1 + 2y) dx + (-x^2 + 4) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= 1 + 2y \\N(x, y) &= -x^2 + 4\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(1 + 2y) \\&= 2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x^2 + 4) \\&= -2x\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\&= -\frac{1}{x^2 - 4} ((2) - (-2x)) \\&= \frac{-2x - 2}{x^2 - 4}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\&= e^{\int \frac{-2x-2}{x^2-4} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-\frac{3 \ln(x-2)}{2} - \frac{\ln(x+2)}{2}} \\&= \frac{1}{(x-2)^{3/2} \sqrt{x+2}}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\&= \frac{1}{(x-2)^{3/2} \sqrt{x+2}} (1 + 2y) \\&= \frac{1 + 2y}{(x-2)^{3/2} \sqrt{x+2}}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{(x-2)^{3/2} \sqrt{x+2}} (-x^2 + 4) \\ &= -\frac{\sqrt{x+2}}{\sqrt{x-2}}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1+2y}{(x-2)^{3/2} \sqrt{x+2}} \right) + \left(-\frac{\sqrt{x+2}}{\sqrt{x-2}} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int -\frac{\sqrt{x+2}}{\sqrt{x-2}} dy \\ \phi &= -\frac{y\sqrt{x+2}}{\sqrt{x-2}} + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= \frac{y\sqrt{x+2}}{2(x-2)^{3/2}} - \frac{y}{2\sqrt{x-2}\sqrt{x+2}} + f'(x) \\ &= \frac{2y}{(x-2)^{3/2}\sqrt{x+2}} + f'(x)\end{aligned} \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{1+2y}{(x-2)^{3/2}\sqrt{x+2}}$. Therefore equation (4) becomes

$$\frac{1+2y}{(x-2)^{3/2}\sqrt{x+2}} = \frac{2y}{(x-2)^{3/2}\sqrt{x+2}} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = \frac{1}{(x-2)^{3/2} \sqrt{x+2}}$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int \left(\frac{1}{(x-2)^{3/2} \sqrt{x+2}} \right) dx$$

$$f(x) = -\frac{\sqrt{x+2}}{2\sqrt{x-2}} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = -\frac{y\sqrt{x+2}}{\sqrt{x-2}} - \frac{\sqrt{x+2}}{2\sqrt{x-2}} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

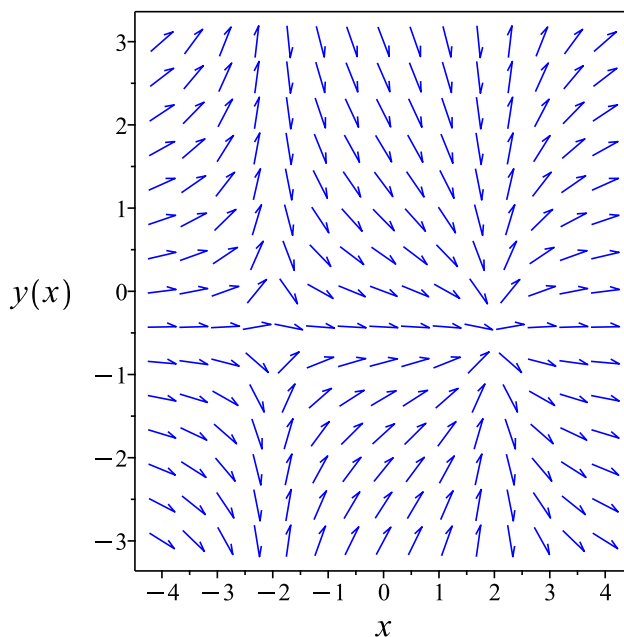
$$c_1 = -\frac{y\sqrt{x+2}}{\sqrt{x-2}} - \frac{\sqrt{x+2}}{2\sqrt{x-2}}$$

Simplifying the above gives

$$-\frac{\sqrt{x+2}(1+2y)}{2\sqrt{x-2}} = c_1$$

Solving for y gives

$$y = -\frac{2\sqrt{x-2}c_1 + \sqrt{x+2}}{2\sqrt{x+2}}$$

Figure 2.49: Slope field $1 + 2y + (-x^2 + 4)y' = 0$ Summary of solutions found

$$y = -\frac{2\sqrt{x-2}c_1 + \sqrt{x+2}}{2\sqrt{x+2}}$$

Solved using first_order_ode_LIE

Time used: 0.848 (sec)

Solve

$$1 + 2y + (-x^2 + 4)y' = 0$$

Writing the ode as

$$y' = \frac{1 + 2y}{x^2 - 4}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned}
 b_2 &= 0 \\
 2a_2 &= 0 \\
 -4a_3 &= 0 \\
 4a_3 &= 0 \\
 -2b_2 &= 0 \\
 2a_1 + 8b_2 &= 0 \\
 4a_1 + 2a_3 &= 0 \\
 8a_2 - 4a_3 &= 0 \\
 a_2 - 2b_1 - 8b_2 + b_3 &= 0 \\
 4a_2 - a_3 + 8b_1 + 16b_2 - 4b_3 &= 0
 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}
 a_1 &= 0 \\
 a_2 &= 0 \\
 a_3 &= 0 \\
 b_1 &= b_1 \\
 b_2 &= 0 \\
 b_3 &= 2b_1
 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}
 \xi &= 0 \\
 \eta &= 1 + 2y
 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \tag{1}$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{1+2y} dy \end{aligned}$$

Which results in

$$S = \frac{\ln(1+2y)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{1+2y}{x^2-4}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{1+2y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{R^2-4} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R^2-4}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

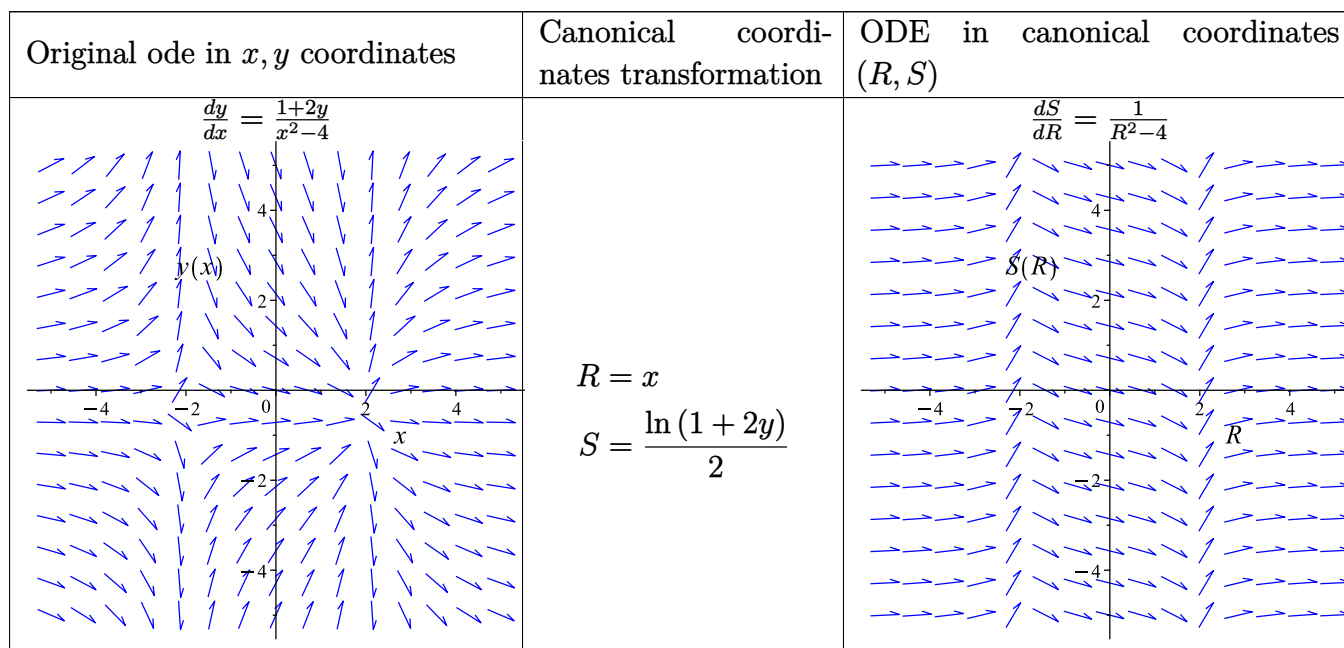
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{1}{R^2-4} dR \\ S(R) &= \frac{\ln(R-2)}{4} - \frac{\ln(R+2)}{4} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(1+2y)}{2} = \frac{\ln(x-2)}{4} - \frac{\ln(x+2)}{4} + c_2$$

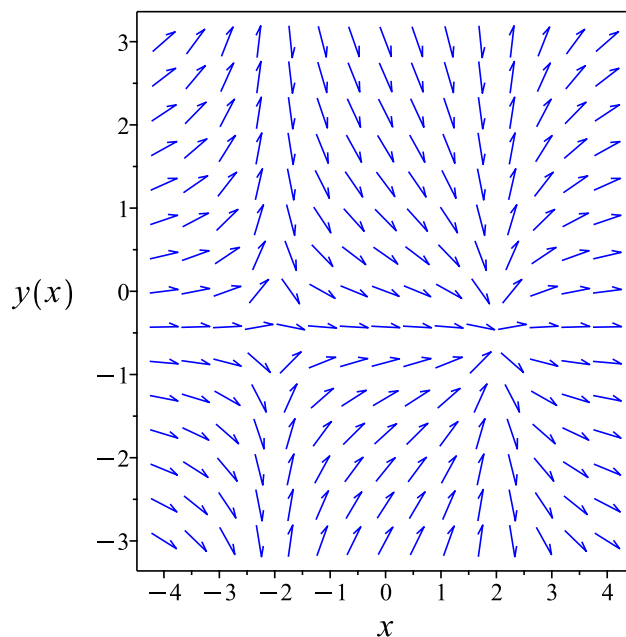
The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{(e^{-4c_2}x + 2e^{-4c_2} - \sqrt{e^{-4c_2}x^2 - 4e^{-4c_2}})e^{4c_2}}{2(x+2)}$$

$$y = -\frac{(e^{-4c_2}x + 2e^{-4c_2} + \sqrt{e^{-4c_2}x^2 - 4e^{-4c_2}})e^{4c_2}}{2(x+2)}$$

Figure 2.50: Slope field $1 + 2y + (-x^2 + 4)y' = 0$ Summary of solutions found

$$y = -\frac{(e^{-4c_2}x + 2e^{-4c_2} - \sqrt{e^{-4c_2}x^2 - 4e^{-4c_2}})e^{4c_2}}{2(x+2)}$$

$$y = -\frac{(e^{-4c_2}x + 2e^{-4c_2} + \sqrt{e^{-4c_2}x^2 - 4e^{-4c_2}})e^{4c_2}}{2(x+2)}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 18

```
ode:=1+2*y(x)+(-x^2+4)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{1}{2} + \frac{\sqrt{x-2}c_1}{\sqrt{x+2}}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```


Maple step by step

Let's solve

$$1 + 2y(x) + (-x^2 + 4) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-1-2y(x)}{-x^2+4}$$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{-1-2y(x)} = \frac{1}{-x^2+4}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{-1-2y(x)} dx = \int \frac{1}{-x^2+4} dx + C1$$

- Evaluate integral

$$-\frac{\ln(-1-2y(x))}{2} = \frac{\ln(x+2)}{4} - \frac{\ln(x-2)}{4} + C1$$

- Solve for $y(x)$

$$\left\{ y(x) = \frac{-e^{4C1}x + \sqrt{e^{4C1}x^2 - 4e^{4C1}} - 2e^{4C1}}{2e^{4C1}(x+2)}, y(x) = \frac{-e^{4C1}x + 2e^{4C1} + \sqrt{e^{4C1}x^2 - 4e^{4C1}}}{2e^{4C1}(x+2)} \right\}$$

- Simplify

$$\left\{ y(x) = \frac{-x-2+e^{-4C1}\sqrt{e^{4C1}(x^2-4)}}{2x+4}, y(x) = \frac{-e^{-4C1}\sqrt{e^{4C1}(x^2-4)}-x-2}{2x+4} \right\}$$

✓ **Mathematica.** Time used: 0.077 (sec). Leaf size: 73

```
ode=(1+2*y[x])+(4-x^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \exp \left(\int_1^x \frac{2}{K[1]^2 - 4} dK[1] \right) \left(\int_1^x \frac{\exp \left(- \int_1^{K[2]} \frac{2}{K[1]^2 - 4} dK[1] \right)}{K[2]^2 - 4} dK[2] + c_1 \right)$$

$$y(x) \rightarrow -\frac{1}{2}$$

✓ Sympy. Time used: 0.193 (sec). Leaf size: 20

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq((4 - x**2)*Derivative(y(x), x) + 2*y(x) + 1,0)  
ics = {}  
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{C_1\sqrt{x-2}}{\sqrt{x+2}} - \frac{1}{2}$$

2.2.3 Problem 26

Local contents

Solved using first_order_ode_separable	231
Solved using first_order_ode_bernoulli	233
Solved using first_order_ode_exact	236
Solved using first_order_ode_isobaric	240
Solved using first_order_ode_homog_A	243
Solved using first_order_ode_homog_type_D2	246
Solved using first_order_ode_homog_type_G	249
Solved using first_order_ode_homog_type_maple_C	250
Solved using first_order_ode_LIE	255
✓ Maple	259
✓ Mathematica	260
✓ Sympy	261

Internal problem ID [7857]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 26

Date solved : Thursday, November 27, 2025 at 06:22:09 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.073 (sec)

Solve

$$y^2 - y'x^2 = 0$$

The ode

$$y' = \frac{y^2}{x^2} \tag{2.22}$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{y^2}{x^2} \\ &= f(x)g(y) \end{aligned}$$

Where

$$f(x) = \frac{1}{x^2}$$
$$g(y) = y^2$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$
$$\int \frac{1}{y^2} dy = \int \frac{1}{x^2} dx$$

$$\frac{1}{y} = \frac{1}{x} - c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y^2 = 0$$

for y gives

$$y = 0$$

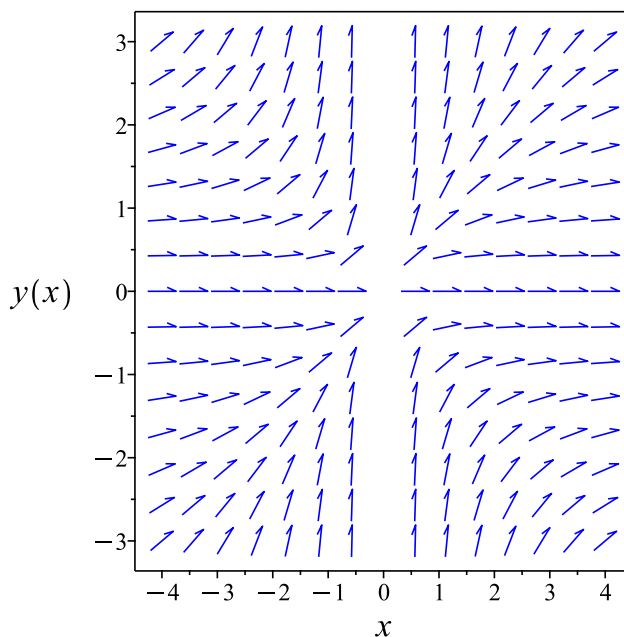
Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{y} = \frac{1}{x} - c_1$$
$$y = 0$$

Solving for y gives

$$y = 0$$
$$y = -\frac{x}{c_1 x - 1}$$

Figure 2.51: Slope field $y^2 - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = -\frac{x}{c_1x - 1}$$

Solved using first_order_ode_bernoulli

Time used: 0.132 (sec)

Solve

$$y^2 - y'x^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y^2}{x^2} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(\frac{1}{x^2}\right) y^2 \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= 0 \\ f_1 &= \frac{1}{x^2} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= 0 \\ f_1(x) &= \frac{1}{x^2} \\ n &= 2 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^2$ gives

$$y' \frac{1}{y^2} = 0 + \frac{1}{x^2} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= \frac{1}{y} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{1}{y^2} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -v'(x) &= \frac{1}{x^2} \\ v' &= -\frac{1}{x^2} \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\int dv = \int -\frac{1}{x^2} dx$$

$$v(x) = \frac{1}{x} + c_1$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y} = c_1 + \frac{1}{x}$$

Solving for y gives

$$y = \frac{x}{c_1 x + 1}$$

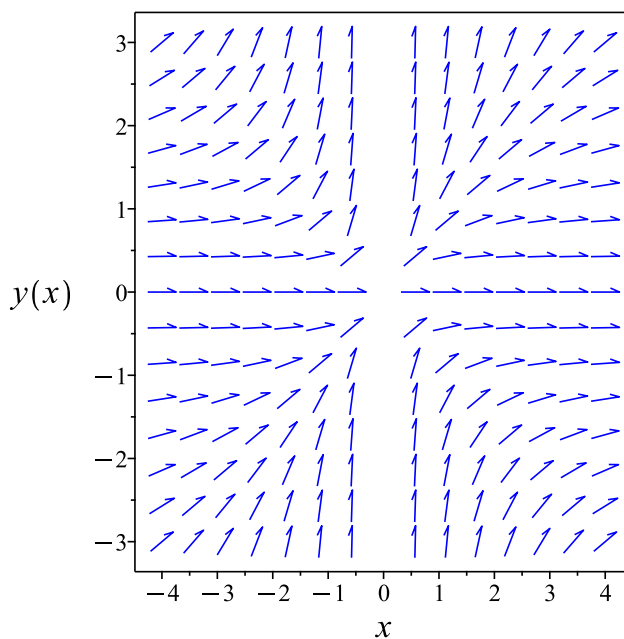


Figure 2.52: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$y = \frac{x}{c_1 x + 1}$$

Solved using first_order_ode_exact

Time used: 0.155 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (-x^2) dy &= (-y^2) dx \\ (y^2) dx + (-x^2) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y^2 \\ N(x, y) &= -x^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y^2) \\ &= 2y\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x^2) \\ &= -2x\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{x^2} ((2y) - (-2x)) \\ &= \frac{-2x - 2y}{x^2}\end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned}B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y^2} ((-2x) - (2y)) \\ &= \frac{-2x - 2y}{y^2}\end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned}R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(-2x) - (2y)}{x(y^2) - y(-x^2)} \\ &= -\frac{2}{yx}\end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = -\frac{2}{t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned}\mu &= e^{\int R dt} \\ &= e^{\int (-\frac{2}{t}) dt}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2 \ln(t)} \\ &= \frac{1}{t^2}\end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{y^2 x^2}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{y^2 x^2} (y^2) \\ &= \frac{1}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{y^2 x^2} (-x^2) \\ &= -\frac{1}{y^2}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1}{x^2} \right) + \left(-\frac{1}{y^2} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{1}{x^2} dx \\ \phi &= -\frac{1}{x} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 0 + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{1}{y^2}$. Therefore equation (4) becomes

$$-\frac{1}{y^2} = 0 + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{1}{y^2}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int \left(-\frac{1}{y^2}\right) dy \\ f(y) &= \frac{1}{y} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{1}{x} + \frac{1}{y} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{1}{x} + \frac{1}{y}$$

Solving for y gives

$$y = \frac{x}{c_1 x + 1}$$

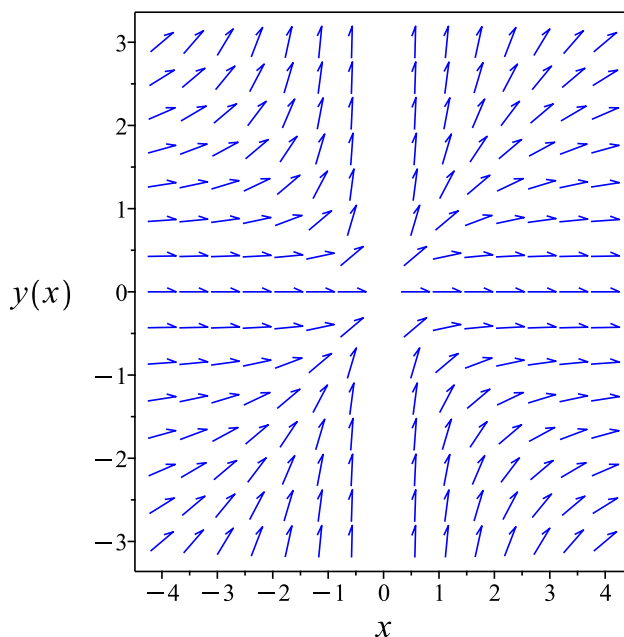


Figure 2.53: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$y = \frac{x}{c_1 x + 1}$$

Solved using first_order_ode_isobaric

Time used: 0.427 (sec)

Solve

$$y^2 - y'x^2 = 0$$

Solving for y' gives

$$y' = \frac{y^2}{x^2} \tag{1}$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{y^2}{x^2} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = u(x)^2$$

The ode

$$u'(x) = \frac{u(x)(u(x) - 1)}{x} \quad (2.23)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(u(x) - 1)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(u) &= u(u - 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(u - 1)} du &= \int \frac{1}{x} dx \end{aligned}$$

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u - 1) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

$$u(x) = 0$$

$$u(x) = 1$$

Converting $\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$ back to y gives

$$\ln\left(\frac{y}{x} - 1\right) - \ln\left(\frac{y}{x}\right) = \ln(x) + c_1$$

Converting $u(x) = 0$ back to y gives

$$\frac{y}{x} = 0$$

Converting $u(x) = 1$ back to y gives

$$\frac{y}{x} = 1$$

Simplifying the above gives

$$\ln\left(\frac{-x + y}{x}\right) - \ln\left(\frac{y}{x}\right) = \ln(x) + c_1$$

$$\frac{y}{x} = 0$$

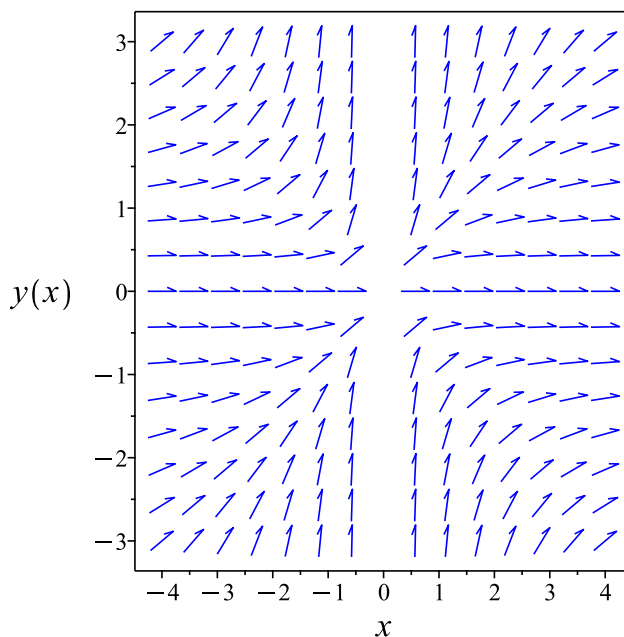
$$\frac{y}{x} = 1$$

Solving for y gives

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Figure 2.54: Slope field $y^2 - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Solved using first_order_ode_homog_A

Time used: 0.675 (sec)

Solve

$$y^2 - y'x^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y^2}{x^2} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x,y)$ and $N(x,y)$ are both homogeneous functions and of the same order. Recall that a function $f(x,y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = y^2$ and $N = x^2$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= u^2 \\ \frac{du}{dx} &= \frac{u(x)^2 - u(x)}{x}\end{aligned}$$

Or

$$u'(x) - \frac{u(x)^2 - u(x)}{x} = 0$$

Or

$$-u(x)^2 + u'(x)x + u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = \frac{u(x)(u(x) - 1)}{x} \tag{2.24}$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= \frac{u(x)(u(x) - 1)}{x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= \frac{1}{x} \\ g(u) &= u(u - 1)\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(u - 1)} du &= \int \frac{1}{x} dx\end{aligned}$$

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u - 1) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

$$u(x) = 0$$

$$u(x) = 1$$

Converting $\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$ back to y gives

$$\ln\left(\frac{-x + y}{x}\right) - \ln\left(\frac{y}{x}\right) = \ln(x) + c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Converting $u(x) = 1$ back to y gives

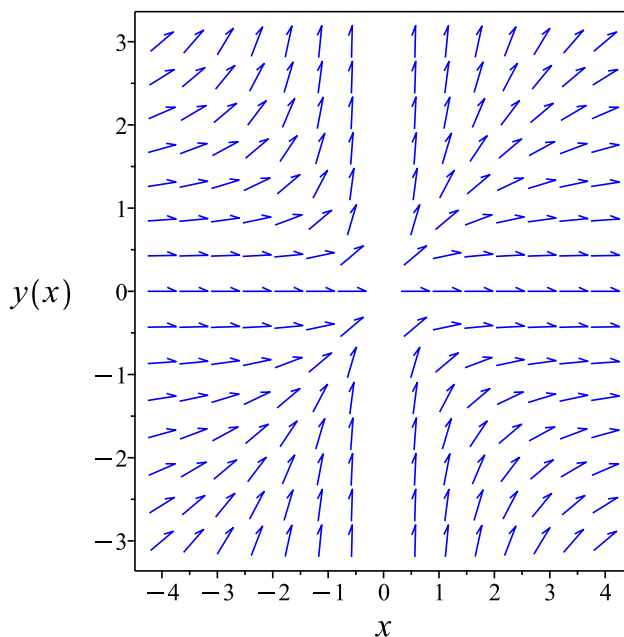
$$y = x$$

Solving for y gives

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Figure 2.55: Slope field $y^2 - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.257 (sec)

Solve

$$y^2 - y'x^2 = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)^2 x^2 - (u'(x)x + u(x))x^2 = 0$$

Which is now solved The ode

$$u'(x) = \frac{u(x)(u(x) - 1)}{x} \quad (2.25)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(u(x) - 1)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = \frac{1}{x}$$

$$g(u) = u(u - 1)$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u(u - 1)} du = \int \frac{1}{x} dx$$

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u - 1) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$$

$$u(x) = 0$$

$$u(x) = 1$$

Converting $\ln(u(x) - 1) - \ln(u(x)) = \ln(x) + c_1$ back to y gives

$$\ln\left(\frac{y}{x} - 1\right) - \ln\left(\frac{y}{x}\right) = \ln(x) + c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Converting $u(x) = 1$ back to y gives

$$y = x$$

Simplifying the above gives

$$\ln\left(\frac{-x+y}{x}\right) - \ln\left(\frac{y}{x}\right) = \ln(x) + c_1$$

$$y = 0$$

$$y = x$$

Solving for y gives

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

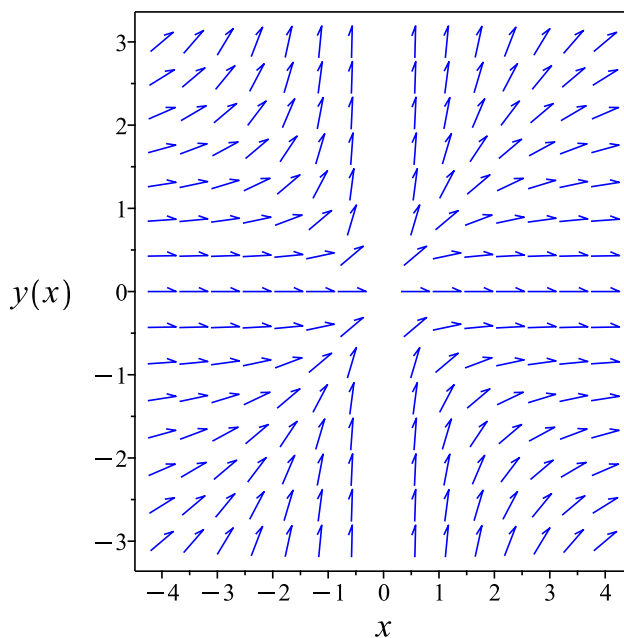


Figure 2.56: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.078 (sec)

Solve

$$y^2 - y'x^2 = 0$$

Multiplying the right side of the ode, which is $\frac{y^2}{x^2}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y^2}{x^2} \\ &= \frac{y}{x} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{y}{x} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{y}{x} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = z$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1-z)} dz = 0$$

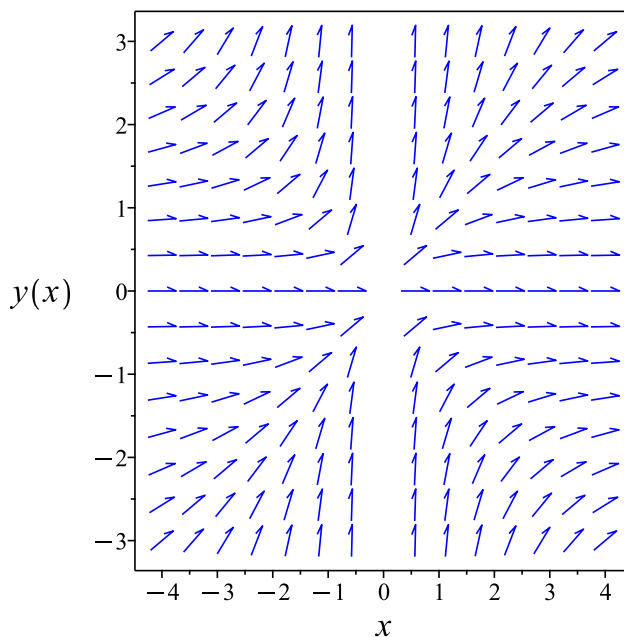


Figure 2.57: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1-z)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.975 (sec)

Solve

$$y^2 - y'x^2 = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{(Y(X) + y_0)^2}{(X + x_0)^2}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{Y(X)^2}{X^2}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= \frac{Y^2}{X^2} \end{aligned} \tag{1}$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = Y^2$ and $N = X^2$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= u^2 \\ \frac{du}{dX} &= \frac{u(X)^2 - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{u(X)^2 - u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) X - u(X)^2 + u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = \frac{u(X)(u(X) - 1)}{X} \tag{2.26}$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= \frac{u(X)(u(X) - 1)}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= \frac{1}{X} \\ g(u) &= u(u-1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{1}{u(u-1)} du &= \int \frac{1}{X} dX \end{aligned}$$

$$\ln(u(X)-1) - \ln(u(X)) = \ln(X) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u-1) = 0$$

for $u(X)$ gives

$$\begin{aligned} u(X) &= 0 \\ u(X) &= 1 \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \ln(u(X)-1) - \ln(u(X)) &= \ln(X) + c_1 \\ u(X) &= 0 \\ u(X) &= 1 \end{aligned}$$

Converting $\ln(u(X)-1) - \ln(u(X)) = \ln(X) + c_1$ back to $Y(X)$ gives

$$\ln\left(\frac{Y(X)-X}{X}\right) - \ln\left(\frac{Y(X)}{X}\right) = \ln(X) + c_1$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Converting $u(X) = 1$ back to $Y(X)$ gives

$$Y(X) = X$$

Using the solution for $Y(X)$

$$\ln \left(\frac{Y(X) - X}{X} \right) - \ln \left(\frac{Y(X)}{X} \right) = \ln(X) + c_1 \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$\ln \left(\frac{-x + y}{x} \right) - \ln \left(\frac{y}{x} \right) = \ln(x) + c_1$$

Using the solution for $Y(X)$

$$Y(X) = 0 \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

Using the solution for $Y(X)$

$$Y(X) = X \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = x$$

Solving for y gives

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

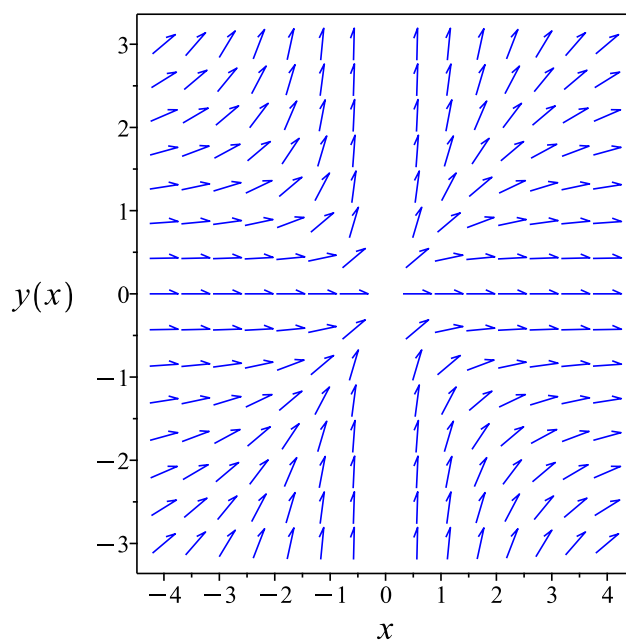


Figure 2.58: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$y = 0$$

$$y = x$$

$$y = -\frac{x}{x e^{c_1} - 1}$$

Solved using first_order_ode_LIE

Time used: 0.281 (sec)

Solve

$$y^2 - y'x^2 = 0$$

Writing the ode as

$$y' = \frac{y^2}{x^2}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1\text{E})$$

$$\eta = \frac{Ax + By}{Cx} \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\frac{A}{Cx} - \frac{Ax + By}{Cx^2} + \frac{y^2 B}{x^3 C} + \frac{2y^2}{x^3} - \frac{2y(Ax + By)}{x^3 C} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{y(2Ax + Bx + By - 2Cy)}{x^3 C} = 0$$

Setting the numerator to zero gives

$$-y(2Ax + Bx + By - 2Cy) = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-v_2(2Av_1 + Bv_1 + Bv_2 - 2Cv_2) = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-2A - B)v_2v_1 + (-B + 2C)v_2^2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -2A - B &= 0 \\ -B + 2C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= -C \\ B &= 2C \\ C &= C \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 1 \\ \eta &= -\frac{x - 2y}{x} \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{-\frac{x-2y}{x}}{1} \\ &= \frac{-x+2y}{x}\end{aligned}$$

This is easily solved to give

$$y = c_1 x^2 + x$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = \frac{-x+y}{x^2}$$

And S is found from

$$\begin{aligned}dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1}\end{aligned}$$

Integrating gives

$$\begin{aligned}S &= \int \frac{dx}{T} \\ &= x\end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y^2}{x^2}$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= \frac{x-2y}{x^3} \\ R_y &= \frac{1}{x^2} \\ S_x &= 1 \\ S_y &= 0\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{x^4}{(x-y)^2} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R^2}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

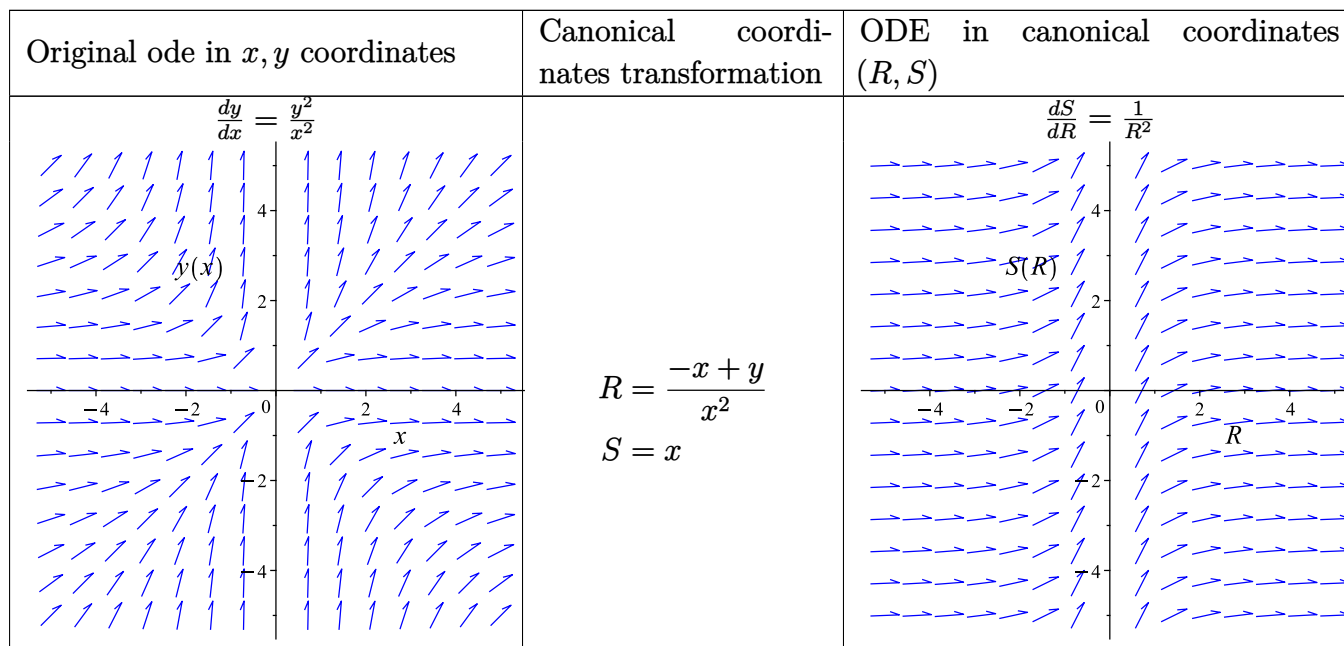
$$\int dS = \int \frac{1}{R^2} dR$$

$$S(R) = -\frac{1}{R} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = -\frac{x^2}{-x+y} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Simplifying the above gives

$$x = \frac{x^2}{x - y} + c_2$$

Solving for y gives

$$y = \frac{c_2 x}{-x + c_2}$$

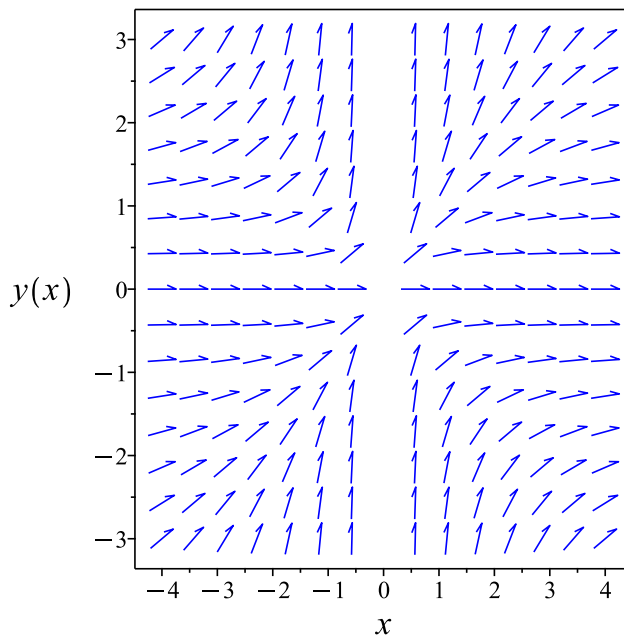


Figure 2.59: Slope field $y^2 - y'x^2 = 0$

Summary of solutions found

$$y = \frac{c_2 x}{-x + c_2}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 13

```
ode:=y(x)^2-x^2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x}{c_1 x + 1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
```

```
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$y(x)^2 - x^2 \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{y(x)^2}{x^2}$$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{y(x)^2} = \frac{1}{x^2}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{y(x)^2} dx = \int \frac{1}{x^2} dx + C1$$

- Evaluate integral

$$-\frac{1}{y(x)} = -\frac{1}{x} + C1$$

- Solve for $y(x)$

$$y(x) = -\frac{x}{C1x-1}$$

✓ **Mathematica.** Time used: 0.071 (sec). Leaf size: 21

```
ode=y[x]^2-x^2*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x}{1 - c_1 x}$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 0.085 (sec). Leaf size: 10

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-x**2*Derivative(y(x), x) + y(x)**2,0)  
ics = {}  
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = -\frac{x}{C_1x - 1}$$

2.2.4 Problem 27

Local contents

Solved using first_order_ode_linear	262
Solved using first_order_ode_separable	264
Solved using first_order_ode_exact	266
Solved using first_order_ode_homog_type_D2	270
Solved using first_order_ode_LIE	273
✓ Maple	277
✓ Mathematica	278
✓ Sympy	278

Internal problem ID [7858]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 27

Date solved : Thursday, November 27, 2025 at 06:22:18 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.039 (sec)

Solve

$$1 + y - (x + 1)y' = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1}{x+1}$$

$$p(x) = \frac{1}{x+1}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{x+1} dx} \\ &= \frac{1}{x+1}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{1}{x+1} \right) \\ \frac{d}{dx} \left(\frac{y}{x+1} \right) &= \left(\frac{1}{x+1} \right) \left(\frac{1}{x+1} \right) \\ d \left(\frac{y}{x+1} \right) &= \frac{1}{(x+1)^2} dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x+1} &= \int \frac{1}{(x+1)^2} dx \\ &= -\frac{1}{x+1} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x+1}$ gives the final solution

$$y = c_1 x + c_1 - 1$$

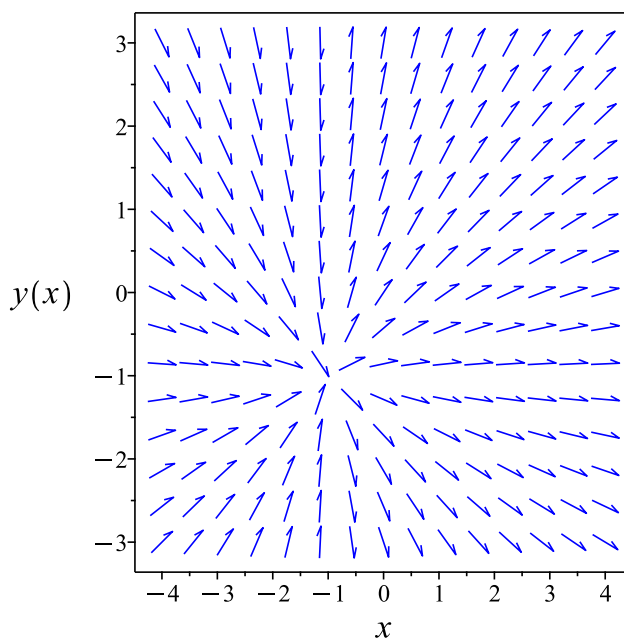


Figure 2.60: Slope field $1 + y - (x + 1)y' = 0$

Summary of solutions found

$$y = c_1 x + c_1 - 1$$

Solved using first_order_ode_separable

Time used: 0.071 (sec)

Solve

$$1 + y - (x + 1) y' = 0$$

The ode

$$y' = \frac{1 + y}{x + 1} \quad (2.27)$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{1 + y}{x + 1} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x + 1} \\ g(y) &= y + 1 \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{y + 1} dy &= \int \frac{1}{x + 1} dx \end{aligned}$$

$$\ln(1 + y) = \ln(x + 1) + c_1$$

Taking the exponential of both sides the solution becomes

$$1 + y = c_1(x + 1)$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y + 1 = 0$$

for y gives

$$y = -1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$1 + y = c_1(x + 1)$$

$$y = -1$$

Solving for y gives

$$y = -1$$

$$y = c_1x + c_1 - 1$$

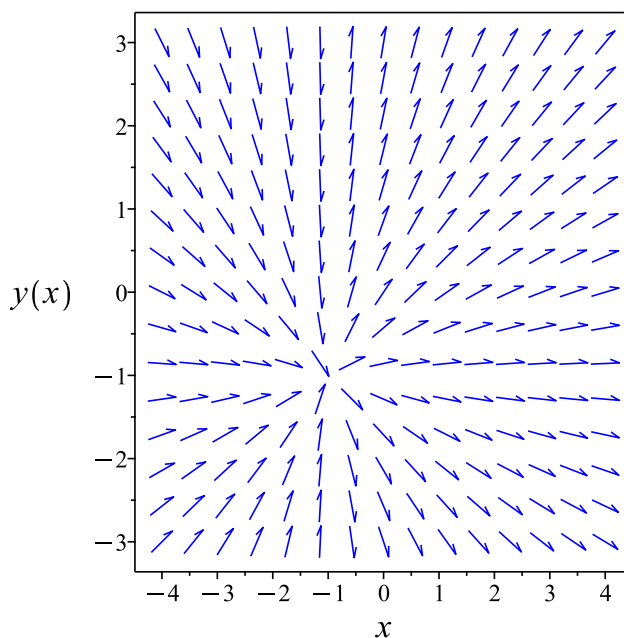


Figure 2.61: Slope field $1 + y - (x + 1)y' = 0$

Summary of solutions found

$$y = -1$$

$$y = c_1x + c_1 - 1$$

Solved using first_order_ode_exact

Time used: 0.112 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (-1 - x) dy &= (-y - 1) dx \\ (y + 1) dx + (-1 - x) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y + 1 \\ N(x, y) &= -1 - x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y+1) \\ &= 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-1-x) \\ &= -1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{-1-x} ((1) - (-1)) \\ &= -\frac{2}{x+1}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{2}{x+1} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2 \ln(x+1)} \\ &= \frac{1}{(x+1)^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= \frac{1}{(x+1)^2}(y+1) \\ &= \frac{y+1}{(x+1)^2}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{(x+1)^2}(-1-x) \\ &= -\frac{1}{x+1}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{y+1}{(x+1)^2} \right) + \left(-\frac{1}{x+1} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int -\frac{1}{x+1} dy \\ \phi &= -\frac{y}{x+1} + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = \frac{y}{(x+1)^2} + f'(x) \tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{y+1}{(x+1)^2}$. Therefore equation (4) becomes

$$\frac{y+1}{(x+1)^2} = \frac{y}{(x+1)^2} + f'(x) \tag{5}$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = \frac{1}{(x+1)^2}$$

Integrating the above w.r.t x gives

$$\int f'(x) \, dx = \int \left(\frac{1}{(x+1)^2} \right) \, dx$$

$$f(x) = -\frac{1}{x+1} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = -\frac{y}{x+1} - \frac{1}{x+1} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

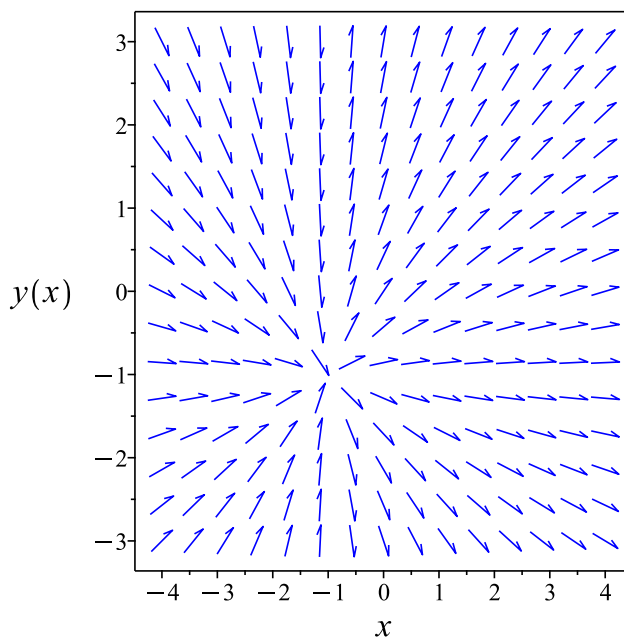
$$c_1 = -\frac{y}{x+1} - \frac{1}{x+1}$$

Simplifying the above gives

$$\frac{-1-y}{x+1} = c_1$$

Solving for y gives

$$y = -c_1x - c_1 - 1$$

Figure 2.62: Slope field $1 + y - (x + 1)y' = 0$ Summary of solutions found

$$y = -c_1x - c_1 - 1$$

Solved using first_order_ode_homog_type_D2

Time used: 0.216 (sec)

Solve

$$1 + y - (x + 1)y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$1 + u(x)x - (x + 1)(u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{u(x) - 1}{x(x + 1)} \quad (2.28)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x) - 1}{x(x + 1)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = \frac{1}{x(x+1)}$$

$$g(u) = -u + 1$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{-u+1} du = \int \frac{1}{x(x+1)} dx$$

$$-\ln(-u(x)+1) = \ln\left(\frac{x}{x+1}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{1}{-u(x)+1} = \frac{c_1 x}{x+1}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$-u + 1 = 0$$

for $u(x)$ gives

$$u(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{-u(x)+1} = \frac{c_1 x}{x+1}$$

$$u(x) = 1$$

Converting $\frac{1}{-u(x)+1} = \frac{c_1 x}{x+1}$ back to y gives

$$\frac{1}{-\frac{y}{x}+1} = \frac{c_1 x}{x+1}$$

Converting $u(x) = 1$ back to y gives

$$y = x$$

Simplifying the above gives

$$\frac{x}{x-y} = \frac{c_1 x}{x+1}$$

$$y = x$$

Solving for y gives

$$y = x$$

$$y = \frac{c_1 x - x - 1}{c_1}$$

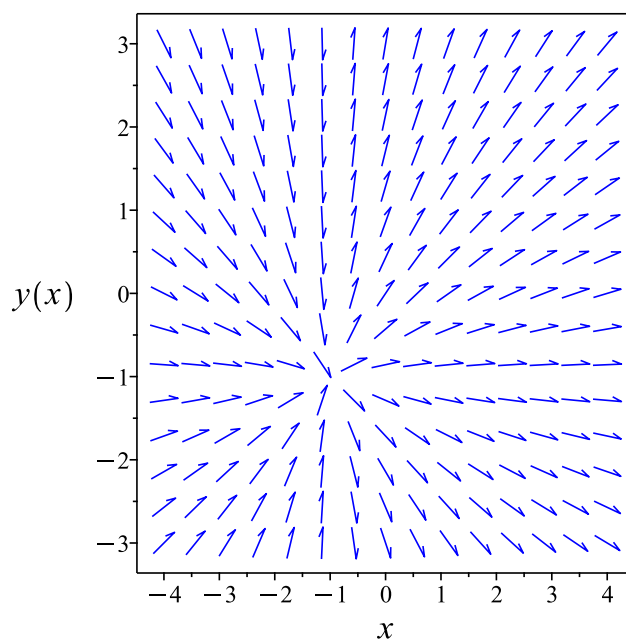


Figure 2.63: Slope field $1 + y - (x + 1)y' = 0$

Summary of solutions found

$$y = x$$

$$y = \frac{c_1 x - x - 1}{c_1}$$

Solved using first_order_ode_LIE

Time used: 0.581 (sec)

Solve

$$1 + y - (x + 1) y' = 0$$

Writing the ode as

$$y' = \frac{y + 1}{x + 1}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 + \frac{(y + 1)(b_3 - a_2)}{x + 1} - \frac{(y + 1)^2 a_3}{(x + 1)^2} + \frac{(y + 1)(xa_2 + ya_3 + a_1)}{(x + 1)^2} - \frac{xb_2 + yb_3 + b_1}{x + 1} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{xb_1 - xb_2 - xb_3 - ya_1 + ya_2 + ya_3 - a_1 + a_2 + a_3 + b_1 - b_2 - b_3}{(x + 1)^2} = 0$$

Setting the numerator to zero gives

$$-xb_1 + xb_2 + xb_3 + ya_1 - ya_2 - ya_3 + a_1 - a_2 - a_3 - b_1 + b_2 + b_3 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$a_1 v_2 - a_2 v_2 - a_3 v_2 - b_1 v_1 + b_2 v_1 + b_3 v_1 + a_1 - a_2 - a_3 - b_1 + b_2 + b_3 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-b_1 + b_2 + b_3) v_1 + (a_1 - a_2 - a_3) v_2 + a_1 - a_2 - a_3 - b_1 + b_2 + b_3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 - a_2 - a_3 &= 0 \\ -b_1 + b_2 + b_3 &= 0 \\ a_1 - a_2 - a_3 - b_1 + b_2 + b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= a_2 + a_3 \\ a_2 &= a_2 \\ a_3 &= a_3 \\ b_1 &= b_2 + b_3 \\ b_2 &= b_2 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 0 \\ \eta &= x + 1 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{x+1} dy \end{aligned}$$

Which results in

$$S = \frac{y}{x+1}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y+1}{x+1}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{y}{(x+1)^2} \\ S_y &= \frac{1}{x+1} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{(x+1)^2} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{(R+1)^2}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

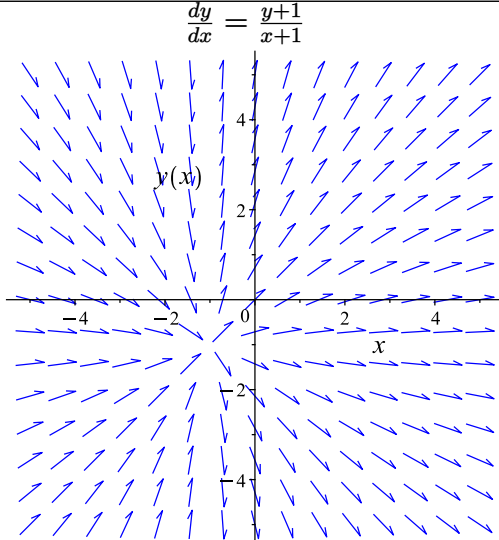
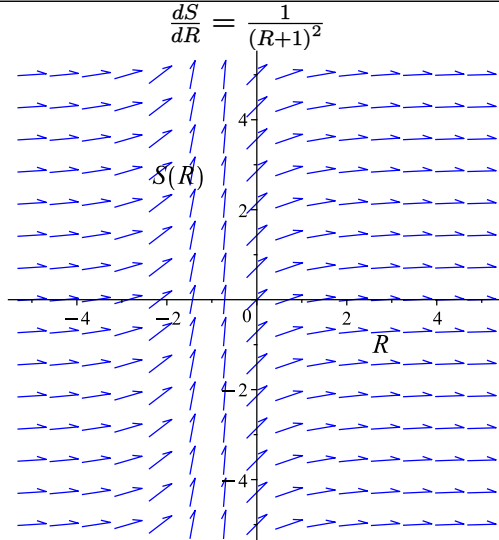
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{1}{(R+1)^2} dR \\ S(R) &= -\frac{1}{R+1} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

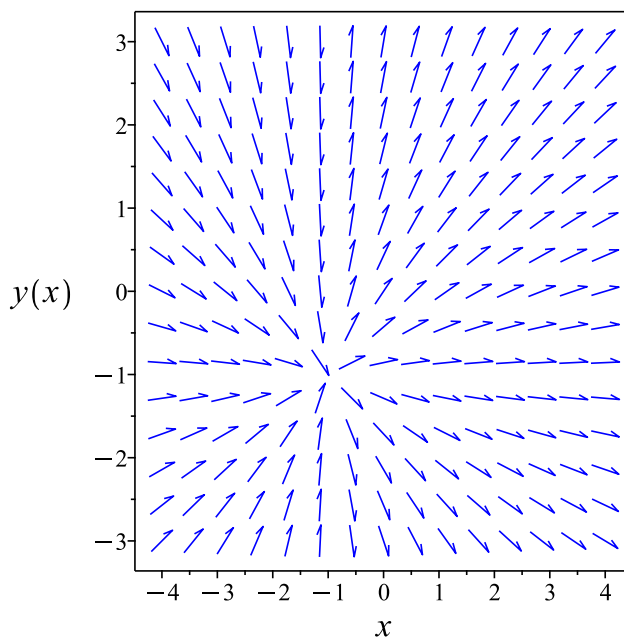
$$\frac{y}{x+1} = -\frac{1}{x+1} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{y+1}{x+1}$ 	$R = x$ $S = \frac{y}{x+1}$	$\frac{dS}{dR} = \frac{1}{(R+1)^2}$ 

Solving for y gives

$$y = c_2 x + c_2 - 1$$

Figure 2.64: Slope field $1 + y - (x + 1)y' = 0$ Summary of solutions found

$$y = c_2x + c_2 - 1$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 10

```
ode:=1+y(x)-(1+x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1x + c_1 - 1$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$1 + y(x) - (1 + x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx} y(x)$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{-1-y(x)}{1+x}$$
- Separate variables

$$\frac{\frac{d}{dx}y(x)}{-1-y(x)} = -\frac{1}{1+x}$$
- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}y(x)}{-1-y(x)} dx = \int -\frac{1}{1+x} dx + C1$$
- Evaluate integral

$$-\ln(-1-y(x)) = -\ln(1+x) + C1$$
- Solve for $y(x)$

$$y(x) = -\frac{e^{C1}+x+1}{e^{C1}}$$
- Simplify

$$y(x) = -1 + (-1-x)e^{-C1}$$
- Redefine the integration constant(s)

$$y(x) = -1 + (-1-x)C1$$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 18

```
ode=(1+y[x])-(1+x)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -1 + c_1(x+1)$$

$$y(x) \rightarrow -1$$

✓ **Sympy.** Time used: 0.138 (sec). Leaf size: 8

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-x - 1)*Derivative(y(x), x) + y(x) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1x + C_1 - 1$$

2.2.5 Problem 28

Local contents

Solved using first_order_ode_exact	279
Solved using first_order_ode_isobaric	284
Solved using first_order_ode_homog_type_G	287
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	289
Solved using first_order_ode_LIE	293
✓ Maple	298
✓ Mathematica	299
✓ Sympy	299

Internal problem ID [7859]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 28

Date solved : Thursday, November 27, 2025 at 06:22:25 AM

CAS classification :

[[_homogeneous, 'class G'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.177 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x^2 y - x) dy &= (-x y^2 - y) dx \\ (x y^2 + y) dx + (x^2 y - x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x y^2 + y \\ N(x, y) &= x^2 y - x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x y^2 + y) \\ &= 2yx + 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x^2 y - x) \\ &= 2yx - 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{(yx - 1)x} ((2yx + 1) - (2yx - 1)) \\ &= \frac{2}{(yx - 1)x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{x y^2 + y} ((2yx - 1) - (2yx + 1)) \\ &= -\frac{2}{y(yx + 1)} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(2yx - 1) - (2yx + 1)}{x(xy^2 + y) - y(x^2y - x)} \\ &= -\frac{1}{yx} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = -\frac{1}{t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int (-\frac{1}{t}) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\ln(t)} \\ &= \frac{1}{t} \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{yx}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{yx} (x y^2 + y) \\ &= \frac{yx + 1}{x}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{yx} (x^2 y - x) \\ &= \frac{yx - 1}{y}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{yx + 1}{x} \right) + \left(\frac{yx - 1}{y} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{yx + 1}{x} dx \\ \phi &= yx + \ln(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{yx-1}{y}$. Therefore equation (4) becomes

$$\frac{yx-1}{y} = x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{1}{y}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) \, dy &= \int \left(-\frac{1}{y}\right) \, dy \\ f(y) &= -\ln(y) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

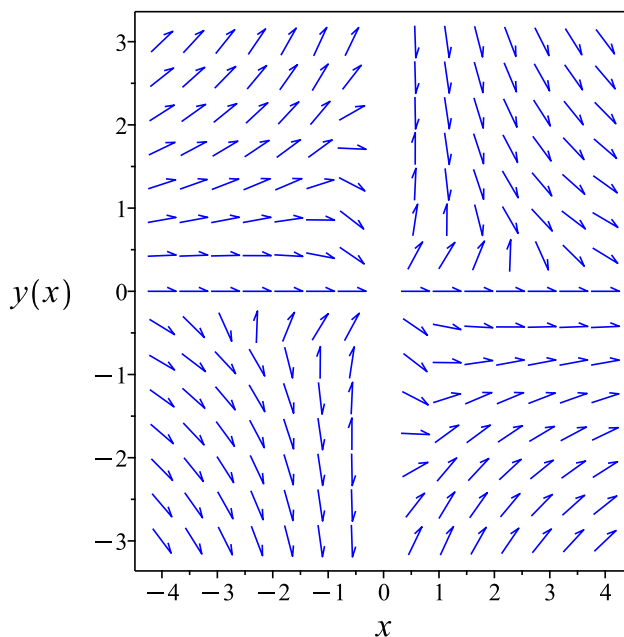
$$\phi = yx + \ln(x) - \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = yx + \ln(x) - \ln(y)$$

Solving for y gives

$$y = -\frac{\text{LambertW}(-x^2 e^{-c_1})}{x}$$

Figure 2.65: Slope field $y^2x + y + (x^2y - x)y' = 0$ Summary of solutions found

$$y = -\frac{\text{LambertW}(-x^2e^{-c_1})}{x}$$

Solved using first_order_ode_isobaric

Time used: 0.511 (sec)

Solve

$$y^2x + y + (x^2y - x)y' = 0$$

Solving for y' gives

$$y' = -\frac{y(yx + 1)}{x(yx - 1)} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{y(yx + 1)}{x(yx - 1)} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = -1$$

Since the ode is isobaric of order $m = -1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= \frac{u}{x} \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$-\frac{u(x)}{x^2} + \frac{u'(x)}{x} = -\frac{u(x)(u(x)+1)}{x^2(u(x)-1)}$$

The ode

$$u'(x) = -\frac{2u(x)}{x(u(x)-1)} \quad (2.29)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)}{x(u(x)-1)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{2}{x} \\ g(u) &= \frac{u}{u-1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u-1}{u} du &= \int -\frac{2}{x} dx \end{aligned}$$

$$u(x) - \ln(u(x)) = \ln\left(\frac{1}{x^2}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u}{u-1} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) - \ln(u(x)) = \ln\left(\frac{1}{x^2}\right) + c_1$$

$$u(x) = 0$$

Converting $u(x) - \ln(u(x)) = \ln\left(\frac{1}{x^2}\right) + c_1$ back to y gives

$$yx - \ln(yx) = \ln\left(\frac{1}{x^2}\right) + c_1$$

Converting $u(x) = 0$ back to y gives

$$yx = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{\text{LambertW}(-x^2 e^{-c_1})}{x}$$

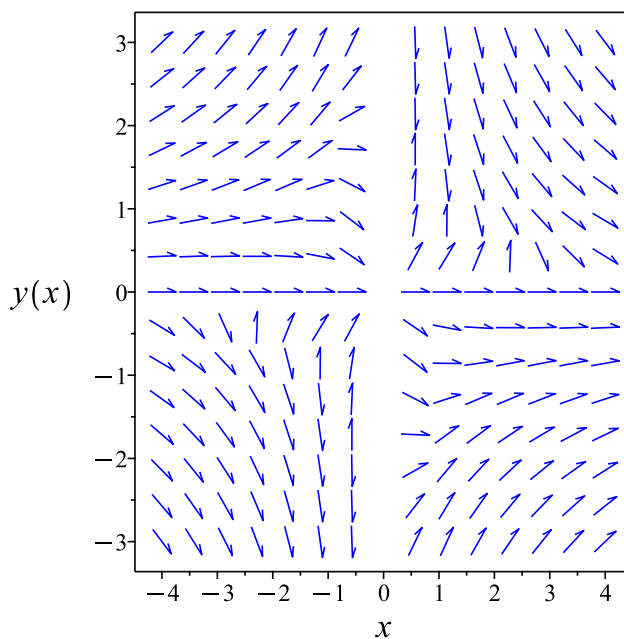


Figure 2.66: Slope field $y^2x + y + (x^2y - x)y' = 0$

Summary of solutions found

$$y = 0$$

$$y = -\frac{\text{LambertW}(-x^2 e^{-c_1})}{x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.083 (sec)

Solve

$$y^2 x + y + (x^2 y - x) y' = 0$$

Multiplying the right side of the ode, which is $-\frac{y(yx+1)}{x(yx-1)}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{y(yx+1)}{x(yx-1)} \\ &= -\frac{yx+1}{yx-1} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{2yx}{(yx-1)^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{2yx}{(yx-1)^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= 1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z+1}{z-1}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{yx} \frac{1}{z\left(-1 + \frac{z+1}{z-1}\right)} dz = 0$$

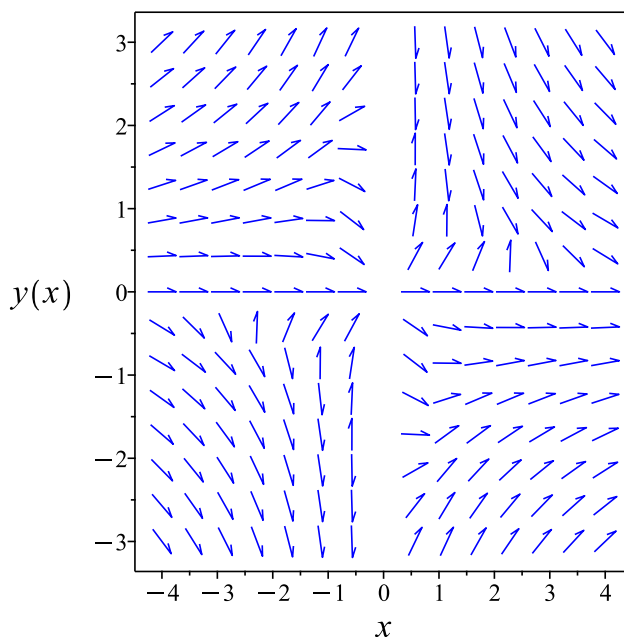


Figure 2.67: Slope field $y^2x + y + (x^2y - x)y' = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{yx} \frac{1}{z\left(-1 + \frac{z+1}{z-1}\right)} dz = 0$$

Solve

$$y^2x + y + (x^2y - x)y' = 0$$

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 1.861 (sec)

This is Abel second kind ODE, it has the form

$$(y + g) y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$y^2x + y + (x^2y - x) y' = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= -\frac{1}{x} \\ f_0 &= 0 \\ f_1 &= -\frac{1}{x^2} \\ f_2 &= -\frac{1}{x} \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = \frac{2u(x)^3}{x^3} + \frac{2u(x)^2}{x^2} + \frac{u(x)}{x}$$

Which is now solved.

Solve Solving for u' gives

$$u' = \frac{u(x) (2u(x)^2 + 2u(x)x + x^2)}{x^3} \quad (1)$$

Each of the above ode's is now solved An ode $u'(x) = f(x, u)$ is isobaric if

$$f(tx, t^m u) = t^{m-1} f(x, u) \quad (1)$$

Where here

$$f(x, u) = \frac{u(x) (2u(x)^2 + 2u(x)x + x^2)}{x^3} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} u &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = \frac{u(x) (2x^2u(x)^2 + 2x^2u(x) + x^2)}{x^2}$$

The ode

$$u'(x) = \frac{2u(x)^2(u(x) + 1)}{x} \quad (2.30)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{2u(x)^2(u(x) + 1)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{2}{x} \\ g(u) &= u^2(u + 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u^2(u + 1)} du &= \int \frac{2}{x} dx \end{aligned}$$

$$\ln(u(x) + 1) - \frac{1}{u(x)} - \ln(u(x)) = \ln(x^2) + c_2$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2(u + 1) = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) + 1) - \frac{1}{u(x)} - \ln(u(x)) = \ln(x^2) + c_2$$

$$u(x) = -1$$

$$u(x) = 0$$

Converting $\ln(u(x) + 1) - \frac{1}{u(x)} - \ln(u(x)) = \ln(x^2) + c_2$ back to $u(x)$ gives

$$\ln\left(\frac{u(x)}{x} + 1\right) - \frac{x}{u(x)} - \ln\left(\frac{u(x)}{x}\right) = \ln(x^2) + c_2$$

Converting $u(x) = -1$ back to $u(x)$ gives

$$\frac{u(x)}{x} = -1$$

Converting $u(x) = 0$ back to $u(x)$ gives

$$\frac{u(x)}{x} = 0$$

Solving for $u(x)$ gives

$$\ln\left(\frac{u(x) + x}{x}\right) - \frac{x}{u(x)} - \ln\left(\frac{u(x)}{x}\right) = \ln(x^2) + c_2$$

$$u(x) = 0$$

$$u(x) = -x$$

Substituting $u(x) = \frac{1}{y - \frac{1}{x}}$ in the above solution gives

$$\ln\left(\frac{\frac{1}{y - \frac{1}{x}} + x}{x}\right) - x\left(y - \frac{1}{x}\right) - \ln\left(\frac{1}{x\left(y - \frac{1}{x}\right)}\right) = \ln(x^2) + c_2$$

Now we transform the solution $u(x) = -x$ to y using $u(x) = \frac{1}{y - \frac{1}{x}}$ which gives

$$y = 0$$

Simplifying the above gives

$$-yx + \ln\left(\frac{xy}{yx-1}\right) - \ln\left(\frac{1}{yx-1}\right) + 1 = \ln(x^2) + c_2$$

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{\text{LambertW}(-x^2 e^{c_2-1})}{x}$$

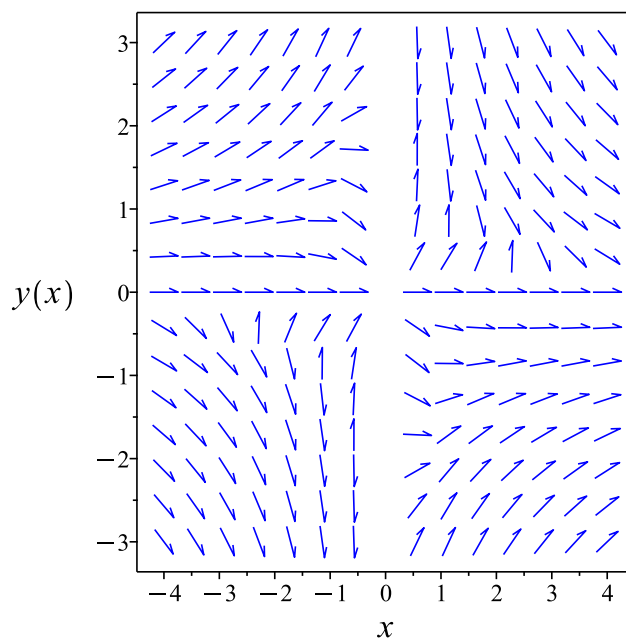
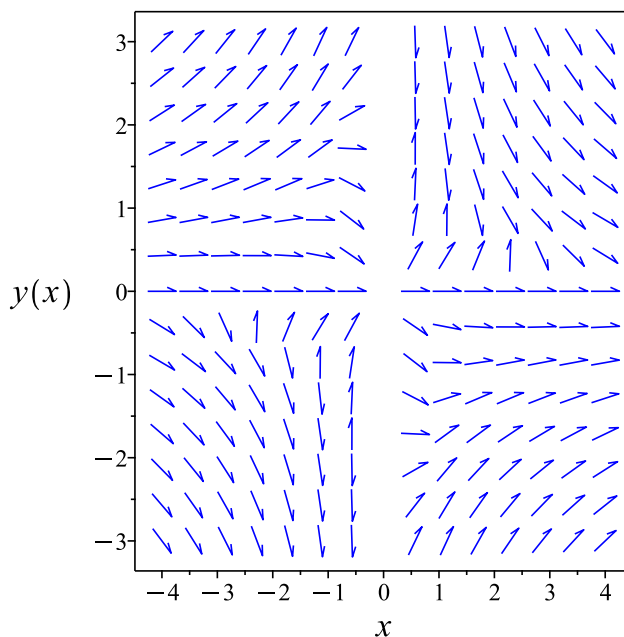


Figure 2.68: Slope field $y^2x + y + (x^2y - x)y' = 0$

Figure 2.69: Slope field $y^2x + y + (x^2y - x)y' = 0$ Summary of solutions found

$$y = 0$$

$$y = -\frac{\text{LambertW}(-x^2e^{x^2-1})}{x}$$

Solved using first_order_ode_LIE

Time used: 1.147 (sec)

Solve

$$y^2x + y + (x^2y - x)y' = 0$$

Writing the ode as

$$y' = -\frac{y(yx + 1)}{x(yx - 1)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2\xi_y - \omega_x\xi - \omega_y\eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{y(yx+1)(b_3-a_2)}{x(yx-1)} - \frac{y^2(yx+1)^2 a_3}{x^2(yx-1)^2} \\ - \left(-\frac{y^2}{x(yx-1)} + \frac{y(yx+1)}{x^2(yx-1)} + \frac{y^2(yx+1)}{x(yx-1)^2} \right) (xa_2 + ya_3 + a_1) \\ - \left(-\frac{yx+1}{x(yx-1)} - \frac{y}{yx-1} + \frac{(yx+1)y}{(yx-1)^2} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{2x^4y^2b_2 - 2x^2y^4a_3 + x^3y^2b_1 - x^2y^3a_1 - 4x^3yb_2 - 2x^2y^2a_2 - 2x^2y^2b_3 - 4xy^3a_3 - 2x^2yb_1 - 2xy^2a_1 - xb_1 + ya_1}{x^2(yx-1)^2} = 0$$

Setting the numerator to zero gives

$$\begin{aligned} 2x^4y^2b_2 - 2x^2y^4a_3 + x^3y^2b_1 - x^2y^3a_1 - 4x^3yb_2 - 2x^2y^2a_2 \\ - 2x^2y^2b_3 - 4xy^3a_3 - 2x^2yb_1 - 2xy^2a_1 - xb_1 + ya_1 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} -2a_3v_1^2v_2^4 + 2b_2v_1^4v_2^2 - a_1v_1^2v_2^3 + b_1v_1^3v_2^2 - 2a_2v_1^2v_2^2 - 4a_3v_1v_2^3 \\ - 4b_2v_1^3v_2 - 2b_3v_1^2v_2^2 - 2a_1v_1v_2^2 - 2b_1v_1^2v_2 + a_1v_2 - b_1v_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} 2b_2v_1^4v_2^2 + b_1v_1^3v_2^2 - 4b_2v_1^3v_2 - 2a_3v_1^2v_2^4 - a_1v_1^2v_2^3 + (-2a_2 - 2b_3)v_1^2v_2^2 \\ - 2b_1v_1^2v_2 - 4a_3v_1v_2^3 - 2a_1v_1v_2^2 - b_1v_1 + a_1v_2 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 &= 0 \\ b_1 &= 0 \\ -2a_1 &= 0 \\ -a_1 &= 0 \\ -4a_3 &= 0 \\ -2a_3 &= 0 \\ -2b_1 &= 0 \\ -b_1 &= 0 \\ -4b_2 &= 0 \\ 2b_2 &= 0 \\ -2a_2 - 2b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= -b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= -x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{y(yx + 1)}{x(yx - 1)} \right) (-x) \\ &= -\frac{2y}{yx - 1} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{-\frac{2y}{yx-1}} dy \end{aligned}$$

Which results in

$$S = -\frac{yx}{2} + \frac{\ln(y)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{y(yx+1)}{x(yx-1)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{y}{2} \\ S_y &= \frac{-yx+1}{2y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{2R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{2R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

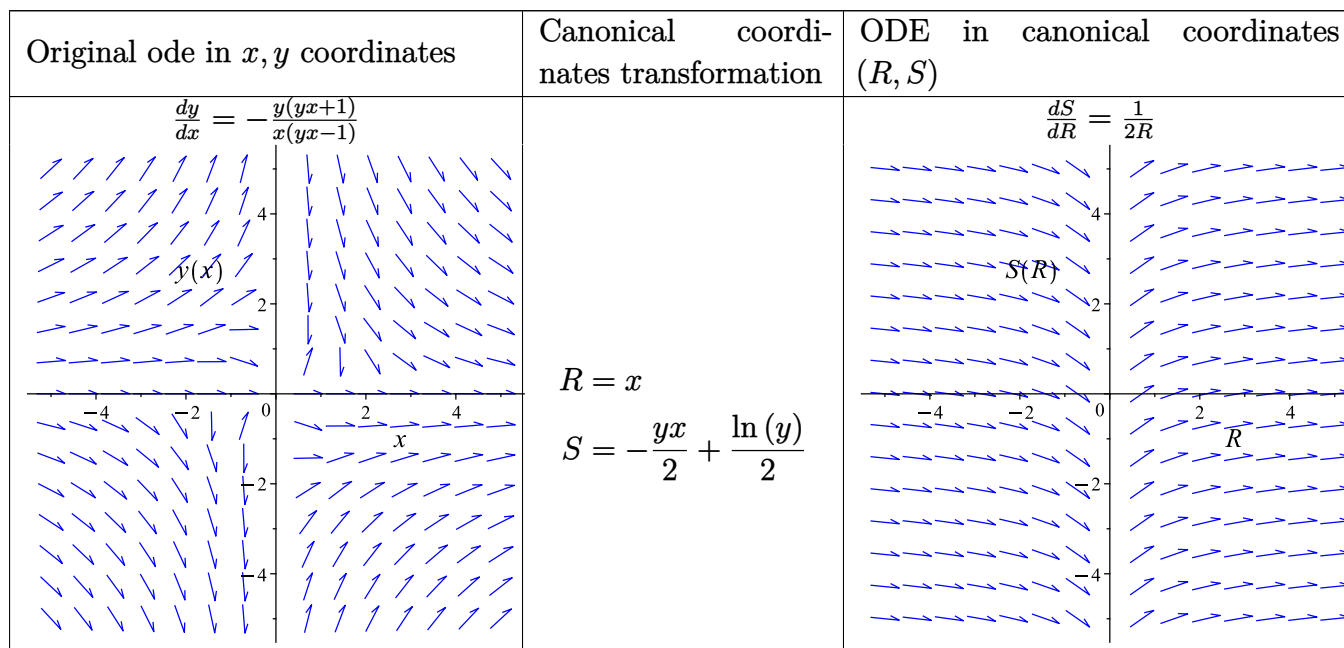
$$\int dS = \int \frac{1}{2R} dR$$

$$S(R) = \frac{\ln(R)}{2} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\frac{yx}{2} + \frac{\ln(y)}{2} = \frac{\ln(x)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{\text{LambertW}(-x^2 e^{2c_2})}{x}$$

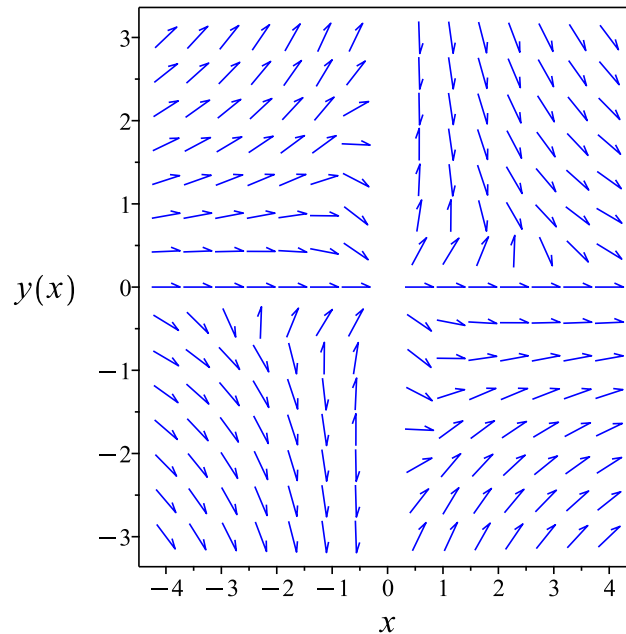


Figure 2.70: Slope field $y^2 x + y + (x^2 y - x) y' = 0$

Summary of solutions found

$$y = -\frac{\text{LambertW}(-x^2 e^{2c_2})}{x}$$

✓ **Maple.** Time used: 0.016 (sec). Leaf size: 19

```
ode:=x*y(x)^2+y(x)+(x^2*y(x)-x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\text{LambertW}(-x^2 e^{-2c_1})}{x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
```

```
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful
```

Maple step by step

Let's solve

$$xy(x)^2 + y(x) + (x^2y(x) - x) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{xy(x)^2 + y(x)}{x^2y(x) - x}$$

✓ **Mathematica.** Time used: 5.957 (sec). Leaf size: 33

```
ode=(x*y[x]^2+y[x])+(x^2*y[x]-x)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{W\left(e^{-1+\frac{9c_1}{2^{2/3}}}x^2\right)}{x}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.332 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x)**2 + (x**2*y(x) - x)*Derivative(y(x), x) + y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = -\frac{W(C_1x^2)}{x}$$

2.2.6 Problem 29

Local contents

Solved using first_order_ode_exact	300
Solved using first_order_ode_isobaric	304
Solved using first_order_ode_homog_A	307
Solved using first_order_ode_homog_type_D2	310
Solved using first_order_ode_homog_type_G	312
Solved using first_order_ode_LIE	314
✓ Maple	320
✓ Mathematica	321
✓ Sympy	321

Internal problem ID [7860]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 29

Date solved : Thursday, November 27, 2025 at 06:22:37 AM

CAS classification : [[_homogeneous, 'class A'], _dAlembert]

Solved using first_order_ode_exact

Time used: 0.203 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(x \cos\left(\frac{y}{x}\right)\right) dy &= \left(y \cos\left(\frac{y}{x}\right) - x \sin\left(\frac{y}{x}\right)\right) dx \\ \left(x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)\right) dx &+ \left(x \cos\left(\frac{y}{x}\right)\right) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) \\ N(x, y) &= x \cos\left(\frac{y}{x}\right) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) \right) \\ &= \frac{y \sin\left(\frac{y}{x}\right)}{x} \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(x \cos\left(\frac{y}{x}\right) \right) \\ &= \cos\left(\frac{y}{x}\right) + \frac{y \sin\left(\frac{y}{x}\right)}{x} \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{\sec\left(\frac{y}{x}\right)}{x} \left(\left(\frac{y \sin\left(\frac{y}{x}\right)}{x} \right) - \left(\cos\left(\frac{y}{x}\right) + \frac{y \sin\left(\frac{y}{x}\right)}{x} \right) \right) \\ &= -\frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{1}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-\ln(x)} \\ &= \frac{1}{x}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x} \left(x \sin \left(\frac{y}{x} \right) - y \cos \left(\frac{y}{x} \right) \right) \\ &= \frac{x \sin \left(\frac{y}{x} \right) - y \cos \left(\frac{y}{x} \right)}{x}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x} \left(x \cos \left(\frac{y}{x} \right) \right) \\ &= \cos \left(\frac{y}{x} \right)\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{x \sin \left(\frac{y}{x} \right) - y \cos \left(\frac{y}{x} \right)}{x} \right) + \left(\cos \left(\frac{y}{x} \right) \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x} dx \\ \phi &= x \sin\left(\frac{y}{x}\right) + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \cos\left(\frac{y}{x}\right) + f'(y)\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \cos\left(\frac{y}{x}\right)$. Therefore equation (4) becomes

$$\cos\left(\frac{y}{x}\right) = \cos\left(\frac{y}{x}\right) + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = x \sin\left(\frac{y}{x}\right) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x \sin\left(\frac{y}{x}\right)$$

Solving for y gives

$$y = \arcsin\left(\frac{c_1}{x}\right) x$$

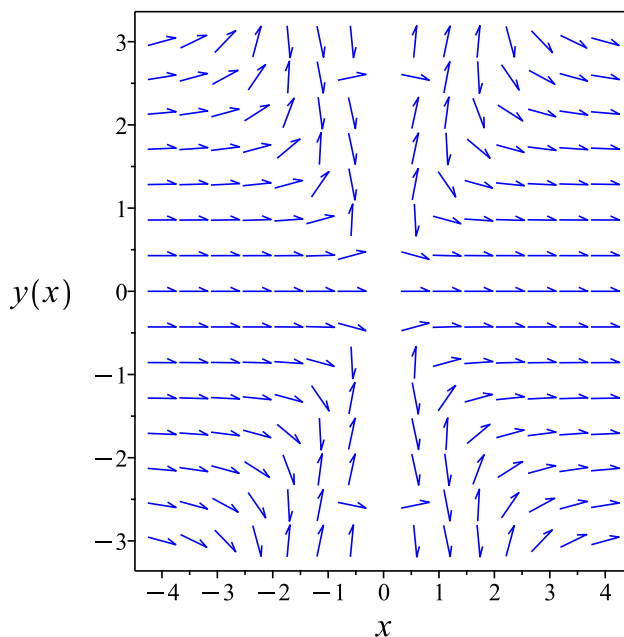


Figure 2.71: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$

Summary of solutions found

$$y = \arcsin\left(\frac{c_1}{x}\right) x$$

Solved using first_order_ode_isobaric

Time used: 0.280 (sec)

Solve

$$x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$$

Solving for y' gives

$$y' = -\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x \sin(u(x)) - xu(x) \cos(u(x))}{x \cos(u(x))}$$

The ode

$$u'(x) = -\frac{\sin(u(x))}{\cos(u(x))x} \quad (2.31)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{\sin(u(x))}{\cos(u(x))x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{\sin(u)}{\cos(u)} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{\cos(u)}{\sin(u)} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\ln(\sin(u(x))) = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sin(u(x)) = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{\sin(u)}{\cos(u)} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sin(u(x)) = \frac{c_1}{x}$$

$$u(x) = 0$$

Converting $\sin(u(x)) = \frac{c_1}{x}$ back to y gives

$$\sin\left(\frac{y}{x}\right) = \frac{c_1}{x}$$

Converting $u(x) = 0$ back to y gives

$$\frac{y}{x} = 0$$

Solving for y gives

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right)x$$

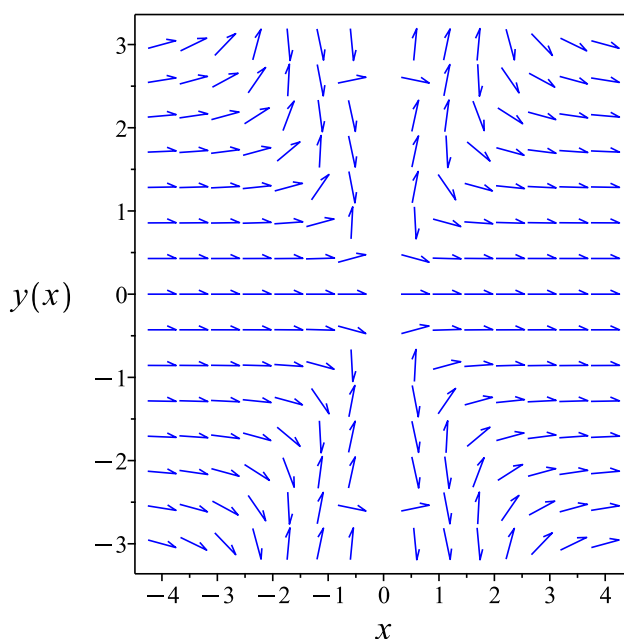


Figure 2.72: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$

Summary of solutions found

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right) x$$

Solved using first_order_ode_homog_A

Time used: 0.526 (sec)

Solve

$$x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = y \cos\left(\frac{y}{x}\right) - x \sin\left(\frac{y}{x}\right)$ and $N = x \cos\left(\frac{y}{x}\right)$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= u - \frac{\sin(u)}{\cos(u)} \\ \frac{du}{dx} &= -\frac{\sin(u(x))}{\cos(u(x))x} \end{aligned}$$

Or

$$u'(x) + \frac{\sin(u(x))}{\cos(u(x))x} = 0$$

Or

$$u'(x) \cos(u(x))x + \sin(u(x)) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{\sin(u(x))}{\cos(u(x))x} \quad (2.32)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{\sin(u(x))}{\cos(u(x))x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{\sin(u)}{\cos(u)} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{\cos(u)}{\sin(u)} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\ln(\sin(u(x))) = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sin(u(x)) = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{\sin(u)}{\cos(u)} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sin(u(x)) = \frac{c_1}{x}$$

$$u(x) = 0$$

Converting $\sin(u(x)) = \frac{c_1}{x}$ back to y gives

$$\sin\left(\frac{y}{x}\right) = \frac{c_1}{x}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right)x$$

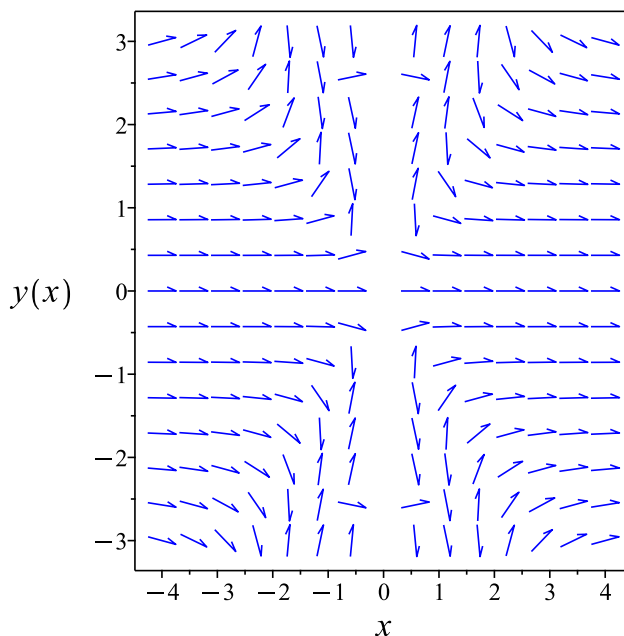


Figure 2.73: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$

Summary of solutions found

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right)x$$

Solved using first_order_ode_homog_type_D2

Time used: 0.207 (sec)

Solve

$$x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$x \sin(u(x)) - u(x)x \cos(u(x)) + x \cos(u(x))(u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{\sin(u(x))}{\cos(u(x))x} \quad (2.33)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{\sin(u(x))}{\cos(u(x))x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{\sin(u)}{\cos(u)} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{\cos(u)}{\sin(u)} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\ln(\sin(u(x))) = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sin(u(x)) = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{\sin(u)}{\cos(u)} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sin(u(x)) = \frac{c_1}{x}$$

$$u(x) = 0$$

Converting $\sin(u(x)) = \frac{c_1}{x}$ back to y gives

$$\sin\left(\frac{y}{x}\right) = \frac{c_1}{x}$$

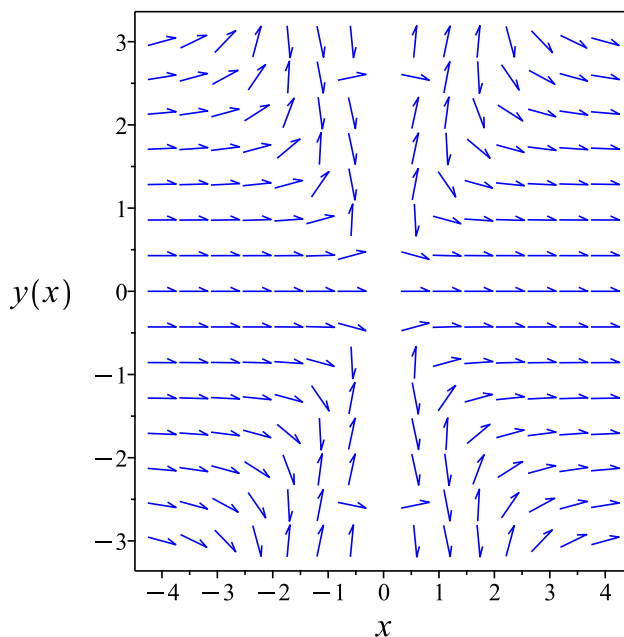
Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right)x$$

Figure 2.74: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$ Summary of solutions found

$$y = 0$$

$$y = \arcsin\left(\frac{c_1}{x}\right) x$$

Solved using first_order_ode_homog_type_G

Time used: 0.093 (sec)

Solve

$$x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$$

Multiplying the right side of the ode, which is $-\frac{x \sin(\frac{y}{x}) - y \cos(\frac{y}{x})}{x \cos(\frac{y}{x})}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)} \\ &= \frac{y \cos\left(\frac{y}{x}\right) - x \sin\left(\frac{y}{x}\right)}{y \cos\left(\frac{y}{x}\right)} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned}
 f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\
 &= \frac{\cos\left(\frac{y}{x}\right)^2 y - \cos\left(\frac{y}{x}\right) \sin\left(\frac{y}{x}\right) x + \sin\left(\frac{y}{x}\right)^2 y}{y \cos\left(\frac{y}{x}\right)^2} \\
 f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\
 &= - \frac{\cos\left(\frac{y}{x}\right)^2 y - \cos\left(\frac{y}{x}\right) \sin\left(\frac{y}{x}\right) x + \sin\left(\frac{y}{x}\right)^2 y}{y \cos\left(\frac{y}{x}\right)^2} \\
 \alpha &= \frac{f_x}{f_y} \\
 &= -1
 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned}
 y &= \frac{z}{x^\alpha} \\
 &= \frac{z}{1} \\
 &= \frac{z}{x}
 \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{\cos(z)z - \sin(z)}{z \cos(z)}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\cos(z)z - \sin(z)}{z \cos(z)} \right)} dz = 0$$

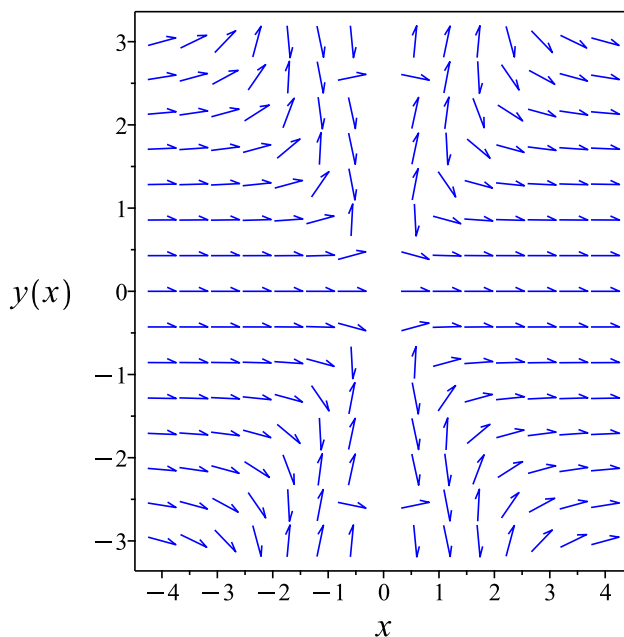


Figure 2.75: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\cos(z)z - \sin(z)}{z \cos(z)}\right)} dz = 0$$

Solved using first_order_ode_LIE

Time used: 1.313 (sec)

Solve

$$x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$$

Writing the ode as

$$y' = -\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{(x \sin(\frac{y}{x}) - y \cos(\frac{y}{x}))(b_3 - a_2)}{x \cos(\frac{y}{x})} - \frac{(x \sin(\frac{y}{x}) - y \cos(\frac{y}{x}))^2 a_3}{x^2 \cos(\frac{y}{x})^2} \\ - \left(-\frac{\sin(\frac{y}{x}) - \frac{y \cos(\frac{y}{x})}{x} - \frac{y^2 \sin(\frac{y}{x})}{x^2}}{x \cos(\frac{y}{x})} + \frac{x \sin(\frac{y}{x}) - y \cos(\frac{y}{x})}{x^2 \cos(\frac{y}{x})} \right. \\ \left. + \frac{(x \sin(\frac{y}{x}) - y \cos(\frac{y}{x})) y \sin(\frac{y}{x})}{x^3 \cos(\frac{y}{x})^2} \right) (xa_2 + ya_3 + a_1) \\ - \left(-\frac{y \sin(\frac{y}{x})}{x^2 \cos(\frac{y}{x})} - \frac{(x \sin(\frac{y}{x}) - y \cos(\frac{y}{x})) \sin(\frac{y}{x})}{x^2 \cos(\frac{y}{x})^2} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\begin{aligned} \frac{b_2 x^2 \cos(\frac{y}{x})^2 - \cos(\frac{y}{x})^2 x y a_2 + \cos(\frac{y}{x})^2 x y b_3 - \cos(\frac{y}{x})^2 y^2 a_3 + \cos(\frac{y}{x}) \sin(\frac{y}{x}) x^2 a_2 - \cos(\frac{y}{x}) \sin(\frac{y}{x}) x^2 b_3 +}{2} \\ = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned} b_2 x^2 \cos(\frac{y}{x})^2 - \cos(\frac{y}{x})^2 x y a_2 + \cos(\frac{y}{x})^2 x y b_3 - \cos(\frac{y}{x})^2 y^2 a_3 \\ + \cos(\frac{y}{x}) \sin(\frac{y}{x}) x^2 a_2 - \cos(\frac{y}{x}) \sin(\frac{y}{x}) x^2 b_3 + 2 \cos(\frac{y}{x}) \sin(\frac{y}{x}) x y a_3 \\ - \sin(\frac{y}{x})^2 x^2 a_3 + \sin(\frac{y}{x})^2 x^2 b_2 - \sin(\frac{y}{x})^2 x y a_2 + \sin(\frac{y}{x})^2 x y b_3 \\ - \sin(\frac{y}{x})^2 y^2 a_3 + \sin(\frac{y}{x})^2 x b_1 - \sin(\frac{y}{x})^2 y a_1 = 0 \end{aligned} \quad (6E)$$

Simplifying the above gives

$$\frac{x(x^2 a_3 \cos(\frac{2y}{x}) + x^2 a_2 \sin(\frac{2y}{x}) - x^2 b_3 \sin(\frac{2y}{x}) + 2 x y a_3 \sin(\frac{2y}{x}) - x b_1 \cos(\frac{2y}{x}) + y a_1 \cos(\frac{2y}{x}) - x^2 a_3 + 2 b_2 x^2}{2} = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\left\{ x, y, \cos\left(\frac{2y}{x}\right), \sin\left(\frac{2y}{x}\right) \right\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\left\{ x = v_1, y = v_2, \cos\left(\frac{2y}{x}\right) = v_3, \sin\left(\frac{2y}{x}\right) = v_4 \right\}$$

The above PDE (6E) now becomes

$$\frac{v_1(v_1^2 a_2 v_4 + v_1^2 a_3 v_3 + 2v_1 v_2 a_3 v_4 - v_1^2 b_3 v_4 + v_2 a_1 v_3 - 2v_1 v_2 a_2 - v_1^2 a_3 - 2v_2^2 a_3 - v_1 b_1 v_3 + 2b_2 v_1^2 + 2v_1 v_2 b_3 - v_2^2)}{2} = 0$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3, v_4\}$$

Equation (7E) now becomes

$$\begin{aligned} & \left(-\frac{a_3}{2} + b_2\right) v_1^3 + \frac{a_3 v_3 v_1^3}{2} + \left(\frac{a_2}{2} - \frac{b_3}{2}\right) v_4 v_1^3 + \frac{b_1 v_1^2}{2} + a_3 v_4 v_2 v_1^2 \\ & + (b_3 - a_2) v_2 v_1^2 - \frac{b_1 v_3 v_1^2}{2} + \frac{a_1 v_2 v_3 v_1}{2} - a_3 v_2^2 v_1 - \frac{a_1 v_2 v_1}{2} = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_3 &= 0 \\ -\frac{a_1}{2} &= 0 \\ \frac{a_1}{2} &= 0 \\ -a_3 &= 0 \\ \frac{a_3}{2} &= 0 \\ -\frac{b_1}{2} &= 0 \\ \frac{b_1}{2} &= 0 \\ \frac{a_2}{2} - \frac{b_3}{2} &= 0 \\ -\frac{a_3}{2} + b_2 &= 0 \\ b_3 - a_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$a_1 = 0$$

$$a_2 = b_3$$

$$a_3 = 0$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = b_3$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = y$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)} \right) (x) \\ &= \tan\left(\frac{y}{x}\right) x \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\tan\left(\frac{y}{x}\right) x} dy\end{aligned}$$

Which results in

$$S = -\frac{\ln\left(1 + \tan\left(\frac{y}{x}\right)^2\right)}{2} + \ln\left(\tan\left(\frac{y}{x}\right)\right)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right)}{x \cos\left(\frac{y}{x}\right)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{y \cot\left(\frac{y}{x}\right)}{x^2} \\ S_y &= \frac{\cot\left(\frac{y}{x}\right)}{x} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

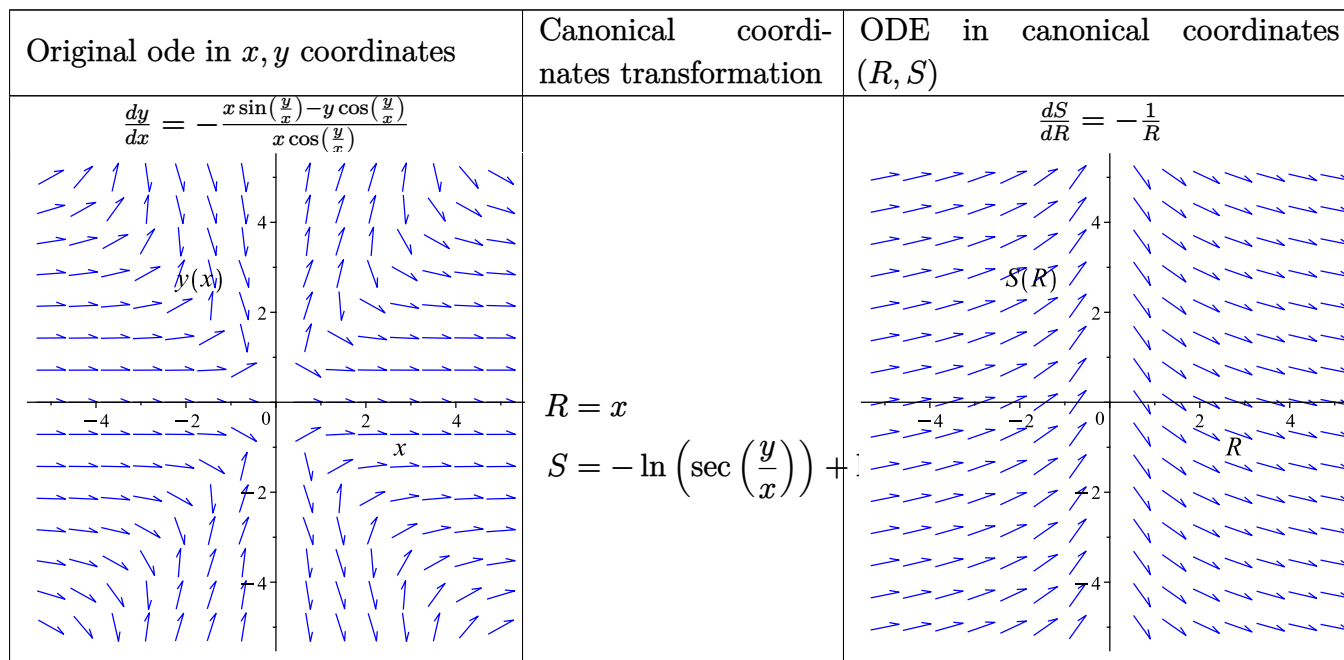
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{R} dR \\ S(R) &= -\ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\ln\left(\sec\left(\frac{y}{x}\right)\right) + \ln\left(\tan\left(\frac{y}{x}\right)\right) = -\ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = \arctan\left(\frac{e^{c_2}}{x}, \frac{\sqrt{x^2 - e^{2c_2}}}{x}\right) x$$

$$y = \arctan\left(\frac{e^{c_2}}{x}, -\frac{\sqrt{x^2 - e^{2c_2}}}{x}\right) x$$

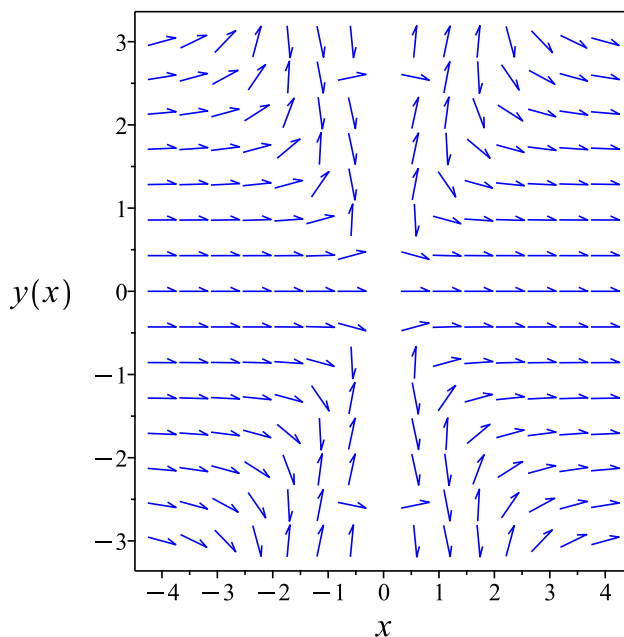


Figure 2.76: Slope field $x \sin\left(\frac{y}{x}\right) - y \cos\left(\frac{y}{x}\right) + x \cos\left(\frac{y}{x}\right) y' = 0$

Summary of solutions found

$$y = \arctan\left(\frac{e^{c_2}}{x}, \frac{\sqrt{x^2 - e^{2c_2}}}{x}\right) x$$

$$y = \arctan\left(\frac{e^{c_2}}{x}, -\frac{\sqrt{x^2 - e^{2c_2}}}{x}\right) x$$

✓ Maple. Time used: 0.007 (sec). Leaf size: 14

```
ode:=x*sin(y(x)/x)-y(x)*cos(y(x)/x)+x*cos(y(x)/x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = x \arcsin\left(\frac{1}{xc_1}\right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
```

```
trying inverse linear
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
```

Maple step by step

Let's solve

$$x \sin\left(\frac{y(x)}{x}\right) - y(x) \cos\left(\frac{y(x)}{x}\right) + x \cos\left(\frac{y(x)}{x}\right) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-x \sin\left(\frac{y(x)}{x}\right) + y(x) \cos\left(\frac{y(x)}{x}\right)}{x \cos\left(\frac{y(x)}{x}\right)}$$

✓ **Mathematica.** Time used: 12.885 (sec). Leaf size: 21

```
ode=(x*Sin[y[x]/x]-y[x]*Cos[y[x]/x])+(x*Cos[y[x]/x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x \arcsin\left(\frac{e^{c_1}}{x}\right)$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.827 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*sin(y(x)/x) + x*cos(y(x)/x)*Derivative(y(x), x) - y(x)*cos(y(x)/x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = x \left(\pi - \arcsin\left(\frac{C_1}{x}\right) \right), y(x) = x \arcsin\left(\frac{C_1}{x}\right) \right]$$

2.2.7 Problem 30

Local contents

Solved using first_order_ode_homog_type_D2	322
✓ Maple	324
✓ Mathematica	325
✗ Sympy	325

Internal problem ID [7861]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 30

Date solved : Thursday, November 27, 2025 at 06:22:46 AM

CAS classification : [[_homogeneous, 'class D'], _rational]

Solved using first_order_ode_homog_type_D2

Time used: 0.177 (sec)

Solve

$$y^2(x^2 + 2) + (y^3 + x^3)(y - y'x) = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)^2 x^2 (x^2 + 2) + (u(x)^3 x^3 + x^3)(u(x)x - (u'(x)x + u(x))x) = 0$$

Which is now solved The ode

$$u'(x) = \frac{u(x)^2 (x^2 + 2)}{(u(x) + 1)(u(x)^2 - u(x) + 1)x^3} \quad (2.34)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)^2 (x^2 + 2)}{(u(x) + 1)(u(x)^2 - u(x) + 1)x^3} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{x^2 + 2}{x^3} \\ g(u) &= \frac{u^2}{(u + 1)(u^2 - u + 1)} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{(u+1)(u^2-u+1)}{u^2} du = \int \frac{x^2+2}{x^3} dx$$

$$\frac{u(x)^3 - 2}{2u(x)} = \ln(x) - \frac{1}{x^2} + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2}{(u+1)(u^2-u+1)} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{u(x)^3 - 2}{2u(x)} = \ln(x) - \frac{1}{x^2} + c_1$$

$$u(x) = 0$$

Converting $\frac{u(x)^3 - 2}{2u(x)} = \ln(x) - \frac{1}{x^2} + c_1$ back to y gives

$$\frac{\left(\frac{y^3}{x^3} - 2\right)x}{2y} = \ln(x) - \frac{1}{x^2} + c_1$$

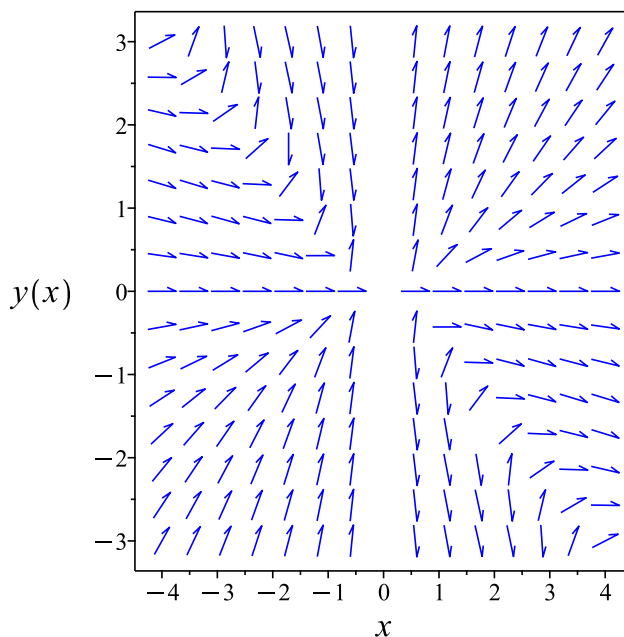
Converting $u(x) = 0$ back to y gives

$$y = 0$$

Simplifying the above gives

$$\frac{y^3 - 2x^3}{2x^2y} = \ln(x) - \frac{1}{x^2} + c_1$$

$$y = 0$$

Figure 2.77: Slope field $y^2(x^2 + 2) + (y^3 + x^3)(y - y'x) = 0$ Summary of solutions found

$$\frac{y^3 - 2x^3}{2x^2y} = \ln(x) - \frac{1}{x^2} + c_1$$

$$y = 0$$

✓ **Maple.** Time used: 0.008 (sec). Leaf size: 659

```
ode:=y(x)^2*(x^2+2)+(x^3+y(x)^3)*(y(x)-diff(y(x),x)*x) = 0;
dsolve(ode,y(x), singsol=all);
```

Solution too large to show

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
```



```
trying homogeneous D
<- homogeneous successful
```

Maple step by step

Let's solve

$$y(x)^2(x^2 + 2) + (x^3 + y(x)^3)(y(x) - x(\frac{d}{dx}y(x))) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x)(y(x)^3 + y(x)x^2 + x^3 + 2y(x))}{x(x^3 + y(x)^3)}$$

✓ **Mathematica.** Time used: 54.804 (sec). Leaf size: 423

```
ode=y[x]^2*(x^2+2)+(x^3+y[x]^3)*(y[x]-x*D[y[x],x])==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\sqrt[3]{2}(2x^2 \log(x) + (-3 + 2c_1)x^2 - 2)}{\sqrt[3]{54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3}}} + \frac{\sqrt[3]{54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3}}}{3\sqrt[3]{2}}$$

$$y(x) \rightarrow \frac{2\sqrt[3]{2}(1 + i\sqrt{3})(-6x^2 \log(x) + (9 - 6c_1)x^2 + 6) + 2^{2/3}(-1 + i\sqrt{3})(54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3})}{12\sqrt[3]{54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3}}}$$

$$y(x) \rightarrow \frac{i(\sqrt{3} + i)(2x^2 \log(x) + (-3 + 2c_1)x^2 - 2)}{2^{2/3}\sqrt[3]{54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3}}} - \frac{(1 + i\sqrt{3})\sqrt[3]{54x^3 + 2\sqrt{729x^6 + (-6x^2 \log(x) + (9 - 6c_1)x^2 + 6)^3}}}{6\sqrt[3]{2}}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x**2 + 2)*y(x)**2 + (x**3 + y(x)**3)*(-x*Derivative(y(x), x) + y(x)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (x**3 + x**2*y(x) + y(x)
)**3 + 2*y(x))*y(x)/(x*(x**3 + y(x)**3)) cannot be solved by the factorable
group method
```

2.2.8 Problem 31

Local contents

Solved using first_order_ode_exact	326
Solved using first_order_ode_isobaric	331
Solved using first_order_ode_homog_A	334
Solved using first_order_ode_homog_type_maple_C	337
Solved using first_order_ode_LIE	341
✓ Maple	347
✓ Mathematica	348
✗ Sympy	348

Internal problem ID [7862]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 31

Date solved : Thursday, November 27, 2025 at 06:22:54 AM

CAS classification : [[_homogeneous, 'class G'], _dAlembert]

Solved using first_order_ode_exact

Time used: 0.476 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} & \left(-x \left(x + \sqrt{x^2 + y^2} \right) \right) dy = \left(-\sqrt{x^2 + y^2} y \right) dx \\ & \left(\sqrt{x^2 + y^2} y \right) dx + \left(-x \left(x + \sqrt{x^2 + y^2} \right) \right) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \sqrt{x^2 + y^2} y \\ N(x, y) &= -x \left(x + \sqrt{x^2 + y^2} \right) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\sqrt{x^2 + y^2} y \right) \\ &= \frac{x^2 + 2y^2}{\sqrt{x^2 + y^2}} \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(-x \left(x + \sqrt{x^2 + y^2} \right) \right) \\ &= - \frac{2 \left(x^2 + x \sqrt{x^2 + y^2} + \frac{y^2}{2} \right)}{\sqrt{x^2 + y^2}} \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{x^2 y}$ is an integrating factor. Therefore by multiplying $M = y\sqrt{x^2 + y^2}$ and $N = -x(x + \sqrt{x^2 + y^2})$ by this integrating

factor the ode becomes exact. The new M, N are

$$M = \frac{\sqrt{x^2 + y^2}}{x^2}$$

$$N = -\frac{x + \sqrt{x^2 + y^2}}{xy}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\frac{\partial \phi}{\partial x} = M$$

$$\frac{\partial \phi}{\partial y} = N$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\left(-\frac{x + \sqrt{x^2 + y^2}}{xy} \right) dy = \left(-\frac{\sqrt{x^2 + y^2}}{x^2} \right) dx$$

$$\left(\frac{\sqrt{x^2 + y^2}}{x^2} \right) dx + \left(-\frac{x + \sqrt{x^2 + y^2}}{xy} \right) dy = 0 \quad (2A)$$

Comparing (1A) and (2A) shows that

$$M(x, y) = \frac{\sqrt{x^2 + y^2}}{x^2}$$

$$N(x, y) = -\frac{x + \sqrt{x^2 + y^2}}{xy}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{\sqrt{x^2 + y^2}}{x^2} \right) \\ &= \frac{y}{x^2 \sqrt{x^2 + y^2}} \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(-\frac{x + \sqrt{x^2 + y^2}}{xy} \right) \\ &= \frac{y}{x^2 \sqrt{x^2 + y^2}} \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{\sqrt{x^2 + y^2}}{x^2} dx \\ \phi &= \frac{\ln(x + \sqrt{x^2 + y^2})}{x} - \frac{\sqrt{x^2 + y^2}}{x} + f(y) \end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y .

Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= \frac{\frac{yx}{\sqrt{x^2+y^2}(x+\sqrt{x^2+y^2})} - \frac{y}{\sqrt{x^2+y^2}}}{x} + f'(y) \\ &= \frac{y}{(-x - \sqrt{x^2+y^2})x} + f'(y)\end{aligned}\quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{x+\sqrt{x^2+y^2}}{xy}$. Therefore equation (4) becomes

$$-\frac{x + \sqrt{x^2+y^2}}{xy} = \frac{y}{(-x - \sqrt{x^2+y^2})x} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{2}{y}$$

Integrating the above w.r.t y gives

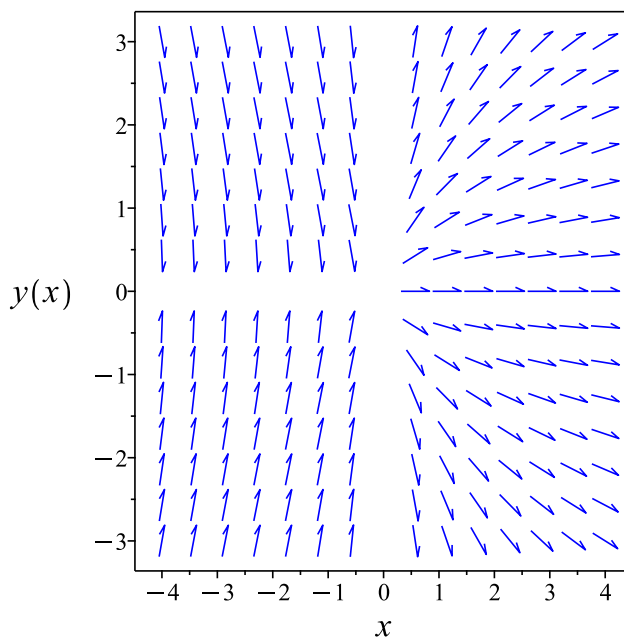
$$\begin{aligned}\int f'(y) dy &= \int \left(-\frac{2}{y}\right) dy \\ f(y) &= -2 \ln(y) + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{\ln(x + \sqrt{x^2+y^2})x - \sqrt{x^2+y^2}}{x} - 2 \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{\ln(x + \sqrt{x^2+y^2})x - \sqrt{x^2+y^2}}{x} - 2 \ln(y)$$

Figure 2.78: Slope field $y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2})y' = 0$ Summary of solutions found

$$\frac{\ln(x + \sqrt{x^2 + y^2})x - 2x \ln(y) - \sqrt{x^2 + y^2}}{x} = c_1$$

Solved using first_order_ode_isobaric

Time used: 0.992 (sec)

Solve

$$y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2})y' = 0$$

Solving for y' gives

$$y' = \frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = \frac{u(x) \sqrt{x^2 + x^2 u(x)^2}}{x + \sqrt{x^2 + x^2 u(x)^2}}$$

The ode

$$u'(x) = -\frac{u(x)}{x \left(\sqrt{u(x)^2 + 1} + 1 \right)} \quad (2.35)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)}{x \left(\sqrt{u(x)^2 + 1} + 1 \right)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{u}{\sqrt{u^2 + 1} + 1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{\sqrt{u^2 + 1} + 1}{u} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln \left(\frac{1}{x} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u}{\sqrt{u^2 + 1} + 1} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln\left(\frac{1}{x}\right) + c_1$$

$$u(x) = 0$$

Converting $\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln\left(\frac{1}{x}\right) + c_1$ back to y gives

$$\sqrt{\frac{y^2}{x^2} + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{y^2}{x^2} + 1}} \right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

Converting $u(x) = 0$ back to y gives

$$\frac{y}{x} = 0$$

Simplifying the above gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}} \right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

$$\frac{y}{x} = 0$$

Solving for y gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}} \right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

$$y = 0$$

The solution

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}} \right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

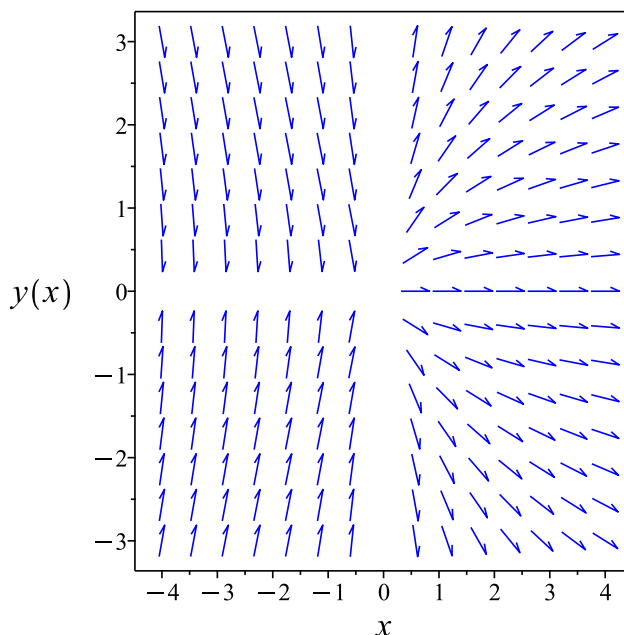


Figure 2.79: Slope field $y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$

Summary of solutions found

$$y = 0$$

Solved using first_order_ode_homog_A

Time used: 1.589 (sec)

Solve

$$y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x, y)}{N(x, y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = \sqrt{x^2 + y^2} y$ and $N = x(x + \sqrt{x^2 + y^2})$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since

this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= \frac{u\sqrt{u^2 + 1}}{\sqrt{u^2 + 1} + 1} \\ \frac{du}{dx} &= \frac{\frac{u(x)\sqrt{u(x)^2 + 1}}{\sqrt{u(x)^2 + 1} + 1} - u(x)}{x} \end{aligned}$$

Or

$$u'(x) - \frac{\frac{u(x)\sqrt{u(x)^2 + 1}}{\sqrt{u(x)^2 + 1} + 1} - u(x)}{x} = 0$$

Or

$$u'(x)x\sqrt{u(x)^2 + 1} + u'(x)x + u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{u(x)}{\left(\sqrt{u(x)^2 + 1} + 1\right)x} \quad (2.36)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)}{\left(\sqrt{u(x)^2 + 1} + 1\right)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{u}{\sqrt{u^2 + 1} + 1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{\sqrt{u^2 + 1} + 1}{u} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln \left(\frac{1}{x} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u}{\sqrt{u^2 + 1} + 1} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln \left(\frac{1}{x} \right) + c_1$$

$$u(x) = 0$$

Converting $\sqrt{u(x)^2 + 1} - \operatorname{arctanh} \left(\frac{1}{\sqrt{u(x)^2 + 1}} \right) + \ln(u(x)) = \ln \left(\frac{1}{x} \right) + c_1$ back to y gives

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}} \right) + \ln \left(\frac{y}{x} \right) = \ln \left(\frac{1}{x} \right) + c_1$$

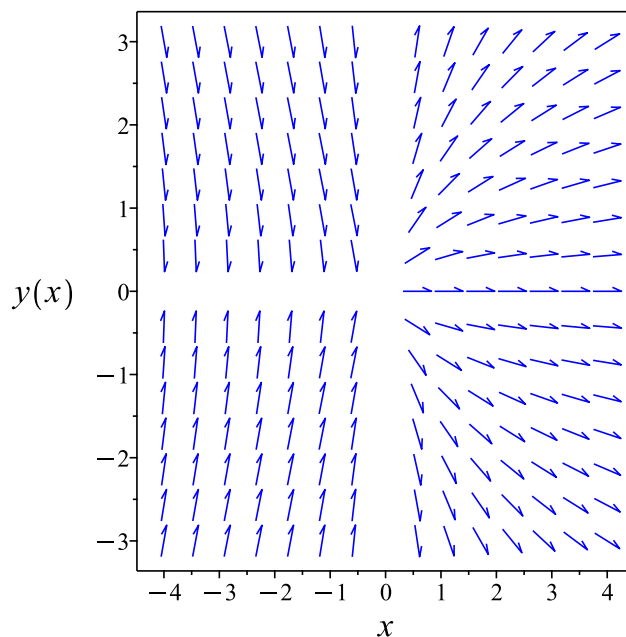
Converting $u(x) = 0$ back to y gives

$$y = 0$$

The solution

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh} \left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}} \right) + \ln \left(\frac{y}{x} \right) = \ln \left(\frac{1}{x} \right) + c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

Figure 2.80: Slope field $y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$ Summary of solutions found

$$y = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.972 (sec)

Solve

$$y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{(Y(X) + y_0) \sqrt{(X + x_0)^2 + (Y(X) + y_0)^2}}{(X + x_0) \left(X + x_0 + \sqrt{(X + x_0)^2 + (Y(X) + y_0)^2} \right)}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{\sqrt{X^2 + Y(X)^2} Y(X)}{X^2 + X\sqrt{X^2 + Y(X)^2}}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= \frac{\sqrt{X^2 + Y^2} Y}{X(\sqrt{X^2 + Y^2} + X)} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = \sqrt{X^2 + Y^2} Y$ and $N = X(\sqrt{X^2 + Y^2} + X)$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= \frac{\sqrt{u^2 + 1} u}{\sqrt{u^2 + 1} + 1} \\ \frac{du}{dX} &= \frac{\frac{\sqrt{u(X)^2 + 1} u(X)}{\sqrt{u(X)^2 + 1} + 1} - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{\sqrt{u(X)^2 + 1} u(X)}{\sqrt{u(X)^2 + 1} + 1} - u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)X\sqrt{u(X)^2 + 1} + \left(\frac{d}{dX}u(X)\right)X + u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{u(X)}{\left(\sqrt{u(X)^2 + 1} + 1\right)X} \quad (2.37)$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= -\frac{u(X)}{\left(\sqrt{u(X)^2+1}+1\right)X} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= -\frac{1}{X} \\ g(u) &= \frac{u}{\sqrt{u^2+1}+1}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{\sqrt{u^2+1}+1}{u} du &= \int -\frac{1}{X} dX\end{aligned}$$

$$\sqrt{u(X)^2+1} - \operatorname{arctanh}\left(\frac{1}{\sqrt{u(X)^2+1}}\right) + \ln(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u}{\sqrt{u^2+1}+1} = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{u(X)^2+1} - \operatorname{arctanh}\left(\frac{1}{\sqrt{u(X)^2+1}}\right) + \ln(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$$

$$u(X) = 0$$

Converting $\sqrt{u(X)^2 + 1} - \operatorname{arctanh}\left(\frac{1}{\sqrt{u(X)^2 + 1}}\right) + \ln(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$ back to $Y(X)$ gives

$$\sqrt{\frac{X^2 + Y(X)^2}{X^2}} - \operatorname{arctanh}\left(\frac{1}{\sqrt{\frac{X^2 + Y(X)^2}{X^2}}}\right) + \ln\left(\frac{Y(X)}{X}\right) = \ln\left(\frac{1}{X}\right) + c_1$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$\sqrt{\frac{X^2 + Y(X)^2}{X^2}} - \operatorname{arctanh}\left(\frac{1}{\sqrt{\frac{X^2 + Y(X)^2}{X^2}}}\right) + \ln\left(\frac{Y(X)}{X}\right) = \ln\left(\frac{1}{X}\right) + c_1 \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh}\left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}}\right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

Using the solution for $Y(X)$

$$Y(X) = 0 \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

The solution

$$\sqrt{\frac{x^2 + y^2}{x^2}} - \operatorname{arctanh}\left(\frac{1}{\sqrt{\frac{x^2 + y^2}{x^2}}}\right) + \ln\left(\frac{y}{x}\right) = \ln\left(\frac{1}{x}\right) + c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

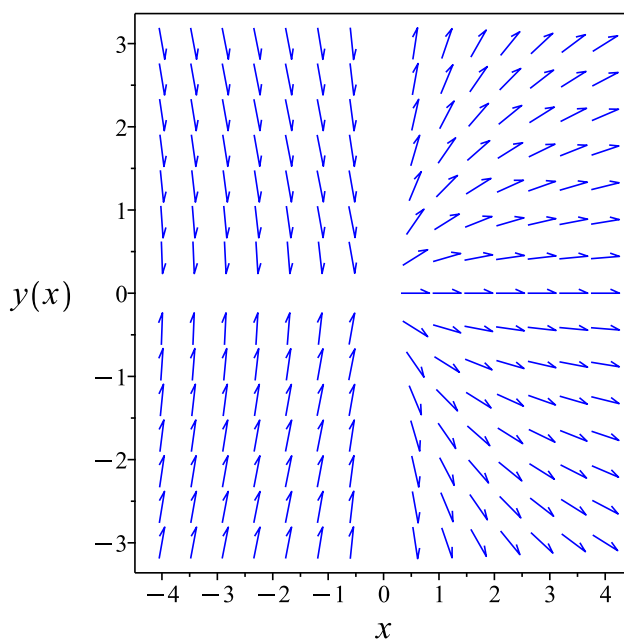


Figure 2.81: Slope field $y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$

Summary of solutions found

$$y = 0$$

Solved using first_order_ode_LIE

Time used: 3.648 (sec)

Solve

$$y\sqrt{x^2 + y^2} - x(x + \sqrt{x^2 + y^2}) y' = 0$$

Writing the ode as

$$y' = \frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 + \frac{y\sqrt{x^2+y^2}(b_3-a_2)}{x(x+\sqrt{x^2+y^2})} - \frac{y^2(x^2+y^2)a_3}{x^2(x+\sqrt{x^2+y^2})^2} \\ - \left(\frac{y}{\sqrt{x^2+y^2}(x+\sqrt{x^2+y^2})} - \frac{y\sqrt{x^2+y^2}}{x^2(x+\sqrt{x^2+y^2})} \right. \\ \left. - \frac{y\sqrt{x^2+y^2}\left(1+\frac{x}{\sqrt{x^2+y^2}}\right)}{x(x+\sqrt{x^2+y^2})^2} \right) (xa_2 + ya_3 + a_1) - \left(\frac{\sqrt{x^2+y^2}}{x(x+\sqrt{x^2+y^2})} \right. \\ \left. + \frac{y^2}{\sqrt{x^2+y^2}x(x+\sqrt{x^2+y^2})} - \frac{y^2}{x(x+\sqrt{x^2+y^2})^2} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{(x^2+y^2)^{3/2}y^2a_3 - \sqrt{x^2+y^2}x^4b_2 + \sqrt{x^2+y^2}x^2y^2a_3 + \sqrt{x^2+y^2}y^4a_3 - x^5b_2 - x^3y^2a_3 - x^2y^3a_2 + x^2y^3b_3}{\sqrt{x^2+y^2}(x+\sqrt{x^2+y^2})}$$

Setting the numerator to zero gives

$$\begin{aligned} (x^2+y^2)^{3/2}y^2a_3 + \sqrt{x^2+y^2}x^4b_2 - \sqrt{x^2+y^2}x^2y^2a_3 \\ - \sqrt{x^2+y^2}y^4a_3 + x^5b_2 + x^3y^2a_3 + x^2y^3a_2 - x^2y^3b_3 + 2xy^4a_3 \\ - (x^2+y^2)^{3/2}xb_1 + (x^2+y^2)^{3/2}ya_1 - x^4b_1 + x^3ya_1 - 2x^2y^2b_1 + 2xy^3a_1 = 0 \end{aligned} \quad (6\text{E})$$

Simplifying the above gives

$$\begin{aligned}
 & (x^2 + y^2)^{3/2} y^2 a_3 + (x^2 + y^2) x^3 b_2 + (x^2 + y^2) x^2 y a_2 + 2(x^2 + y^2) x y^2 a_3 \\
 & + \sqrt{x^2 + y^2} x^4 b_2 - \sqrt{x^2 + y^2} x^2 y^2 a_3 - \sqrt{x^2 + y^2} y^4 a_3 \\
 & - x^4 y a_2 - x^3 y^2 a_3 - x^3 y^2 b_2 - x^2 y^3 b_3 - (x^2 + y^2)^{3/2} x b_1 \\
 & + (x^2 + y^2)^{3/2} y a_1 - (x^2 + y^2) x^2 b_1 + 2(x^2 + y^2) x y a_1 - x^3 y a_1 - x^2 y^2 b_1 = 0
 \end{aligned} \tag{6E}$$

Since the PDE has radicals, simplifying gives

$$\begin{aligned}
 & x^5 b_2 + \sqrt{x^2 + y^2} x^4 b_2 + x^3 y^2 a_3 + x^2 y^3 a_2 - x^2 y^3 b_3 + 2x y^4 a_3 - x^4 b_1 - x^3 \sqrt{x^2 + y^2} b_1 \\
 & + x^3 y a_1 + x^2 \sqrt{x^2 + y^2} y a_1 - 2x^2 y^2 b_1 - x \sqrt{x^2 + y^2} y^2 b_1 + 2x y^3 a_1 + \sqrt{x^2 + y^2} y^3 a_1 = 0
 \end{aligned}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, \sqrt{x^2 + y^2}\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, \sqrt{x^2 + y^2} = v_3\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
 & v_1^2 v_2^3 a_2 + v_1^3 v_2^2 a_3 + 2v_1 v_2^4 a_3 + v_1^5 b_2 + v_3 v_1^4 b_2 - v_1^2 v_2^3 b_3 + v_1^3 v_2 a_1 + v_1^2 v_3 v_2 a_1 \\
 & + 2v_1 v_2^3 a_1 + v_3 v_2^3 a_1 - v_1^4 b_1 - v_1^3 v_3 b_1 - 2v_1^2 v_2^2 b_1 - v_1 v_3 v_2^2 b_1 = 0
 \end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$\begin{aligned}
 & v_1^5 b_2 + v_3 v_1^4 b_2 - v_1^4 b_1 + v_1^3 v_2^2 a_3 + v_1^3 v_2 a_1 - v_1^3 v_3 b_1 + (-b_3 + a_2) v_1^2 v_2^3 \\
 & - 2v_1^2 v_2^2 b_1 + v_1^2 v_3 v_2 a_1 + 2v_1 v_2^4 a_3 + 2v_1 v_2^3 a_1 - v_1 v_3 v_2^2 b_1 + v_3 v_2^3 a_1 = 0
 \end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned}
 a_1 &= 0 \\
 a_3 &= 0 \\
 b_2 &= 0 \\
 2a_1 &= 0 \\
 2a_3 &= 0 \\
 -2b_1 &= 0 \\
 -b_1 &= 0 \\
 -b_3 + a_2 &= 0
 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}
 a_1 &= 0 \\
 a_2 &= b_3 \\
 a_3 &= 0 \\
 b_1 &= 0 \\
 b_2 &= 0 \\
 b_3 &= b_3
 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}
 \xi &= x \\
 \eta &= y
 \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}
 \eta &= \eta - \omega(x, y) \xi \\
 &= y - \left(\frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})} \right) (x) \\
 &= \frac{xy}{x + \sqrt{x^2 + y^2}} \\
 \xi &= 0
 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \tag{1}$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{xy}{x + \sqrt{x^2 + y^2}}} dy \end{aligned}$$

Which results in

$$S = \frac{\sqrt{x^2 + y^2}}{x} - \frac{x \ln \left(\frac{2x^2 + 2\sqrt{x^2} \sqrt{x^2 + y^2}}{y} \right)}{\sqrt{x^2}} + \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{-x - \sqrt{x^2 + y^2}}{x^2} \\ S_y &= \frac{2x^2 + y^2 + 2x\sqrt{x^2 + y^2}}{y(x + \sqrt{x^2 + y^2})x} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{-2x^2 - y^2 - 2x\sqrt{x^2 + y^2}}{x(x + \sqrt{x^2 + y^2})^2} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

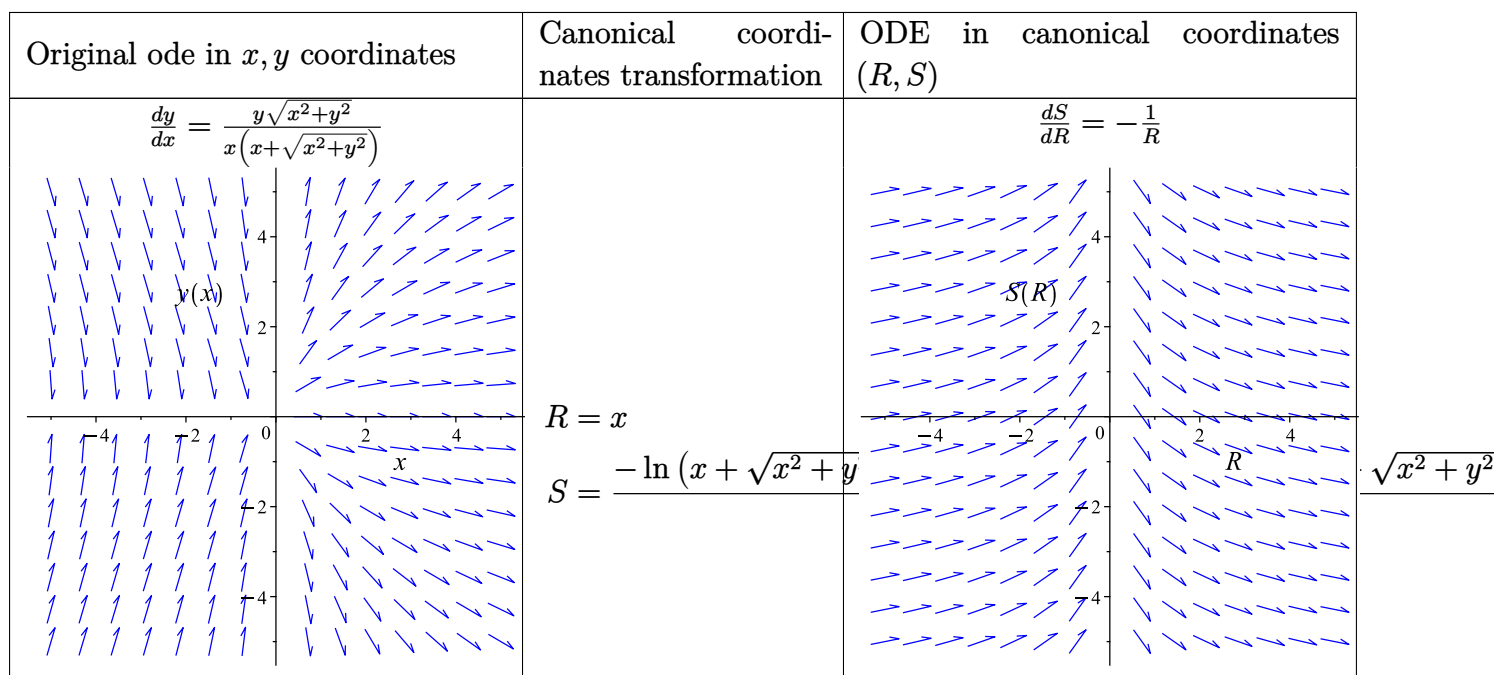
$$\int dS = \int -\frac{1}{R} dR$$

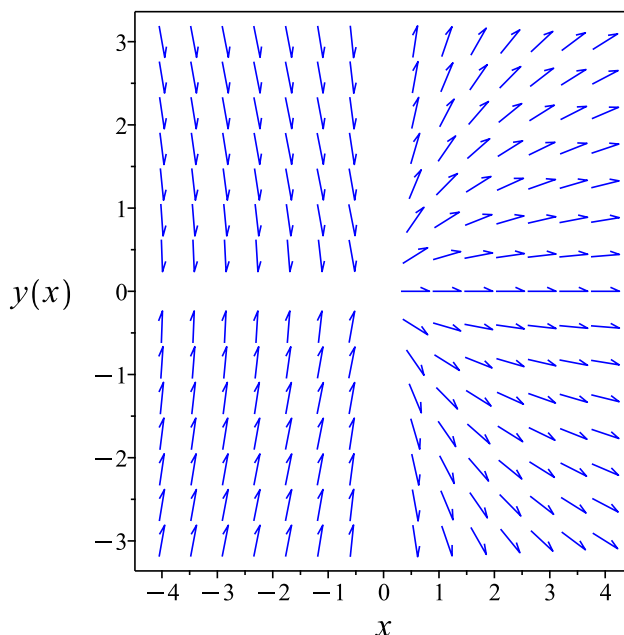
$$S(R) = -\ln(R) + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{-\ln(x + \sqrt{x^2 + y^2})x - x \ln(x) + 2x \ln(y) - \ln(2)x + \sqrt{x^2 + y^2}}{x} = -\ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Figure 2.82: Slope field $y\sqrt{x^2+y^2} - x(x+\sqrt{x^2+y^2})y' = 0$ Summary of solutions found

$$\frac{-\ln(x+\sqrt{x^2+y^2})x - x\ln(x) + 2x\ln(y) - \ln(2)x + \sqrt{x^2+y^2}}{x} = -\ln(x) + c_2$$

✓ **Maple.** Time used: 0.025 (sec). Leaf size: 60

```
ode:=y(x)*(x^2+y(x)^2)^(1/2)-x*(x+(x^2+y(x)^2)^(1/2))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$\frac{x \ln\left(\frac{x(x+\sqrt{x^2+y^2})}{y}\right) - \ln(x)x + x \ln(2) - \ln(y)x - c_1x - \sqrt{x^2+y^2}}{x} = 0$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

trying homogeneous G

1st order, trying the canonical coordinates of the invariance group

<- 1st order, canonical coordinates successful

<- homogeneous successful

Maple step by step

Let's solve

$$y(x) \sqrt{x^2 + y(x)^2} - x \left(x + \sqrt{x^2 + y(x)^2} \right) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{y(x) \sqrt{x^2 + y(x)^2}}{x \left(x + \sqrt{x^2 + y(x)^2} \right)}$$

✓ **Mathematica.** Time used: 0.189 (sec). Leaf size: 43

```
ode=y[x]*Sqrt[x^2+y[x]^2]-x*(x+Sqrt[x^2+y[x]^2])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\sqrt{\frac{y(x)^2}{x^2} + 1} + \log \left(\sqrt{\frac{y(x)^2}{x^2} + 1} - 1 \right) = -\log(x) + c_1, y(x) \right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*(x + sqrt(x**2 + y(x)**2))*Derivative(y(x), x) + sqrt(x**2 + y(x)**2)
        *y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
TypeError : cannot determine truth value of Relational: 2*x**2*asinh(x/_X0) < 2*
C1*x**2
```


2.2.9 Problem 32

Local contents

Solved using first_order_ode_dAlembert	349
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	352
Solved using first_order_ode_LIE	355
✓ Maple	360
✓ Mathematica	361
✓ Sympy	361

Internal problem ID [7863]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 32

Date solved : Thursday, November 27, 2025 at 06:23:13 AM

CAS classification :

[[_homogeneous, 'class C'], _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_dAlembert

Time used: 0.476 (sec)

Solve

$$x + y + 1 + (2x + 2y + 1)y' = 0$$

Let $p = y'$ the ode becomes

$$x + y + 1 + (2x + 2y + 1)p = 0$$

Solving for y from the above results in

$$y = -x - \frac{p+1}{1+2p} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -1 \\ g &= \frac{-p-1}{1+2p} \end{aligned}$$

Hence (2) becomes

$$p+1 = \left(-\frac{1}{1+2p} + \frac{2p}{(1+2p)^2} + \frac{2}{(1+2p)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p+1=0$$

Solving the above for p results in

$$p_1 = -1$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -x$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x)+1}{-\frac{1}{1+2p(x)} + \frac{2p(x)}{(1+2p(x))^2} + \frac{2}{(1+2p(x))^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\begin{aligned} \int \frac{1}{(p+1)(1+2p)^2} dp &= dx \\ \ln(p+1) - \frac{1}{1+2p} - \ln(1+2p) &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$(p+1)(1+2p)^2 = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = -1$$

$$p(x) = -\frac{1}{2}$$

Substituting the above solution for p in (2A) gives

$$y = -x + \left(-\frac{e^{\text{RootOf}\left(-\ln\left(\frac{e^{-Z}}{2} + \frac{1}{2}\right)\right)e^{-Z} + e^{-Z}c_1 + _Z e^{-Z} + x e^{-Z} + 1}}{2} - \frac{1}{2} \right) e^{-\text{RootOf}\left(-\ln\left(\frac{e^{-Z}}{2} + \frac{1}{2}\right)\right)e^{-Z} + e^{-Z}c_1 + _Z e^{-Z} + x e^{-Z} + 1}$$

$$y = -x$$

Simplifying the above gives

$$y = -x$$

$$y = -x - \frac{1}{2} - \frac{e^{-\text{RootOf}(e^{-Z}\ln(2) - e^{-Z}\ln(1 + e^{-Z}) + e^{-Z}c_1 + _Z e^{-Z} + x e^{-Z} + 1)}}{2}$$

$$y = -x$$

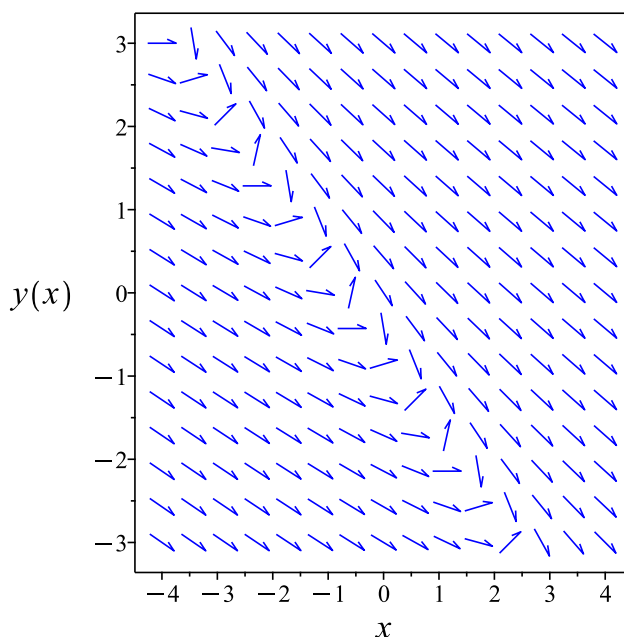


Figure 2.83: Slope field $x + y + 1 + (2x + 2y + 1)y' = 0$

Summary of solutions found

$$y = -x$$

$$y = -x - \frac{1}{2} - \frac{e^{-\text{RootOf}(e^{-Z}\ln(2)-e^{-Z}\ln(1+e^{-Z})+e^{-Z}c_1+Ze^{-Z}+xe^{-Z}+1)}}{2}$$

Solve

$$x + y + 1 + (2x + 2y + 1) y' = 0$$

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 0.844 (sec)

This is Abel second kind ODE, it has the form

$$(y + g) y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$x + y + 1 + (2x + 2y + 1) y' = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= x + \frac{1}{2} \\ f_0 &= -\frac{1}{2} - \frac{x}{2} \\ f_1 &= -\frac{1}{2} \\ f_2 &= 0 \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = \frac{u(x)^3}{4} - \frac{u(x)^2}{2}$$

Which is now solved.

Solve Integrating gives

$$\begin{aligned} \int \frac{1}{\frac{1}{4}u^3 - \frac{1}{2}u^2} du &= dx \\ \ln(u - 2) + \frac{2}{u} - \ln(u) &= x + c_3 \end{aligned}$$

Singular solutions are found by solving

$$\frac{1}{4}u^3 - \frac{1}{2}u^2 = 0$$

for $u(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$u(x) = 0$$

$$u(x) = 2$$

Substituting $u(x) = \frac{1}{x+y+\frac{1}{2}}$ in the above solution gives

$$\ln\left(\frac{1}{x+y+\frac{1}{2}} - 2\right) + 2x + 2y + 1 - \ln\left(\frac{1}{x+y+\frac{1}{2}}\right) = x + c_3$$

Now we transform the solution $u(x) = 2$ to y using $u(x) = \frac{1}{x+y+\frac{1}{2}}$ which gives

$$y = -x$$

Simplifying the above gives

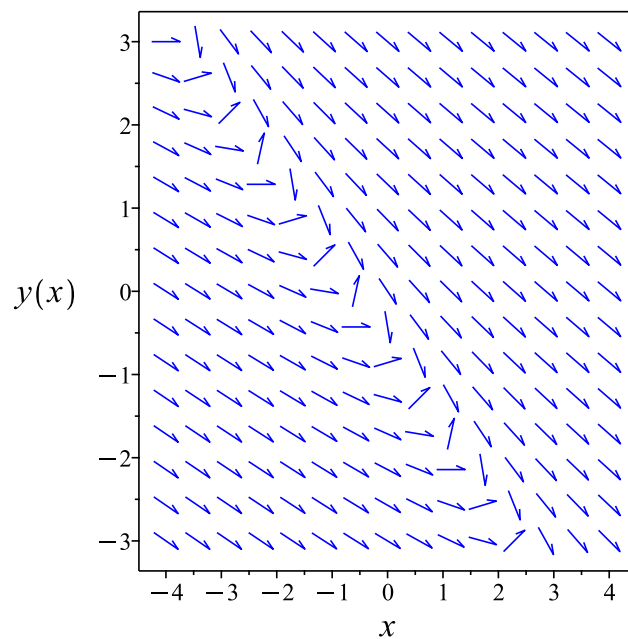
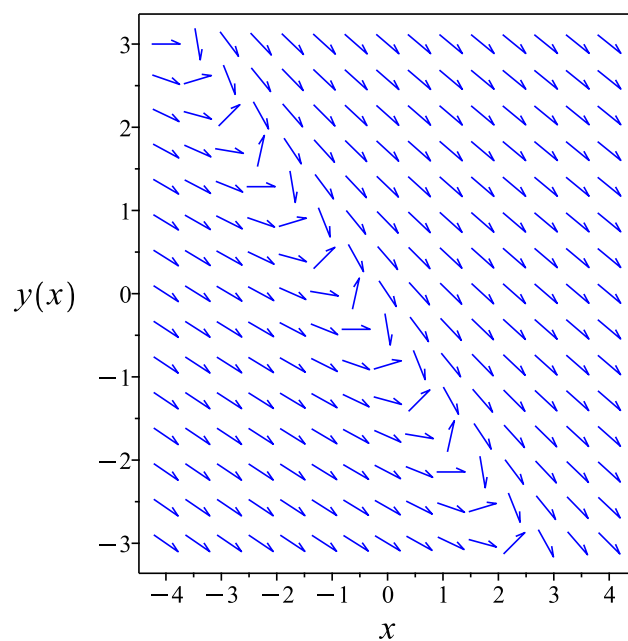
$$\ln(2) + \ln\left(\frac{-x-y}{2x+2y+1}\right) + 2x + 2y + 1 - \ln\left(\frac{1}{2x+2y+1}\right) = x + c_3$$

$$y = -x$$

Solving for y gives

$$y = -x$$

$$y = -x + \frac{\text{LambertW}(-e^{x+c_3-1})}{2}$$

Figure 2.84: Slope field $x + y + 1 + (2x + 2y + 1)y' = 0$ Figure 2.85: Slope field $x + y + 1 + (2x + 2y + 1)y' = 0$

Summary of solutions found

$$y = -x$$

$$y = -x + \frac{\text{LambertW}(-e^{x+c_3-1})}{2}$$

Solved using first_order_ode_LIE

Time used: 0.287 (sec)

Solve

$$x + y + 1 + (2x + 2y + 1) y' = 0$$

Writing the ode as

$$y' = -\frac{x + y + 1}{2x + 2y + 1}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1\text{E})$$

$$\eta = \frac{Ax + By}{Cx} \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & \frac{A}{Cx} - \frac{Ax + By}{Cx^2} - \frac{(x + y + 1)B}{(2x + 2y + 1)Cx} + \frac{1}{2x + 2y + 1} \\ & - \frac{2(x + y + 1)}{(2x + 2y + 1)^2} - \frac{\left(-\frac{1}{2x + 2y + 1} + \frac{2x + 2y + 2}{(2x + 2y + 1)^2}\right)(Ax + By)}{Cx} = 0 \end{aligned} \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{2Bx^3 + 8Bx^2y + 10Bxy^2 + 4By^3 + Ax^2 + 3Bx^2 + 8Bxy + 4By^2 + Cx^2 + Bx + By}{Cx^2(2x + 2y + 1)^2} = 0$$

Setting the numerator to zero gives

$$-2Bx^3 - 8Bx^2y - 10Bxy^2 - 4By^3 - Ax^2 - 3Bx^2 - 8Bxy - 4By^2 - Cx^2 - Bx - By = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-2Bv_1^3 - 8Bv_1^2v_2 - 10Bv_1v_2^2 - 4Bv_2^3 - Av_1^2 - 3Bv_1^2 - 8Bv_1v_2 - 4Bv_2^2 - Cv_1^2 - Bv_1 - Bv_2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-2Bv_1^3 - 8Bv_1^2v_2 + (-A - 3B - C)v_1^2 - 10Bv_1v_2^2 - 8Bv_1v_2 - Bv_1 - 4Bv_2^3 - 4Bv_2^2 - Bv_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -10B &= 0 \\ -8B &= 0 \\ -4B &= 0 \\ -2B &= 0 \\ -B &= 0 \\ -A - 3B - C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= -C \\ B &= 0 \\ C &= C \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 1 \\ \eta &= -1 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{-1}{1} \\ &= -1 \end{aligned}$$

This is easily solved to give

$$y = c_1 - x$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = x + y$$

And S is found from

$$\begin{aligned} dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1} \end{aligned}$$

Integrating gives

$$\begin{aligned} S &= \int \frac{dx}{T} \\ &= x \end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x + y + 1}{2x + 2y + 1}$$

Evaluating all the partial derivatives gives

$$R_x = 1$$

$$R_y = 1$$

$$S_x = 1$$

$$S_y = 0$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{2x + 2y + 1}{x + y} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{2R + 1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

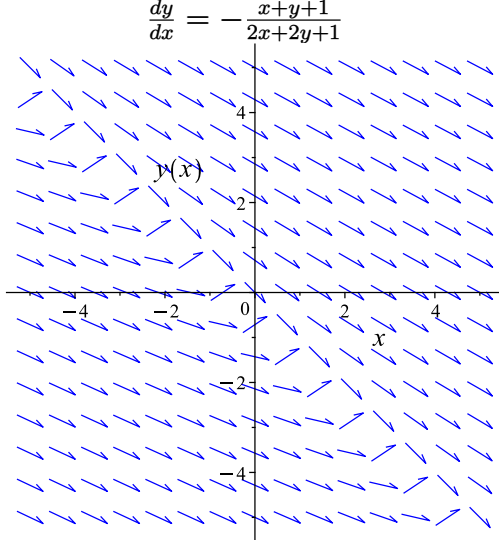
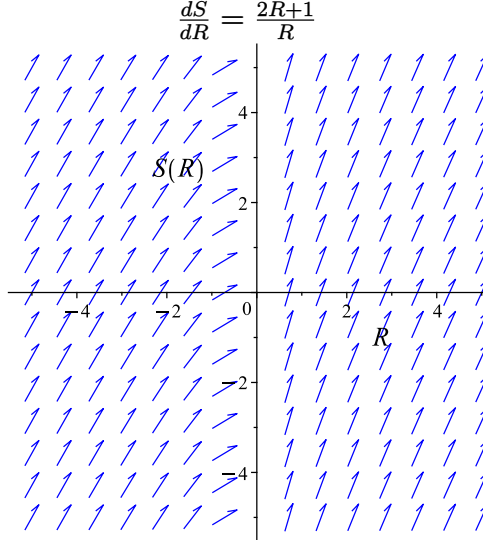
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{2R + 1}{R} dR \\ S(R) &= 2R + \ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = 2x + 2y + \ln(x + y) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x+y+1}{2x+2y+1}$ 	$R = x + y$ $S = x$	$\frac{dS}{dR} = \frac{2R+1}{R}$ 

Solving for y gives

$$y = \frac{\text{LambertW}(2e^{x-c_2})}{2} - x$$

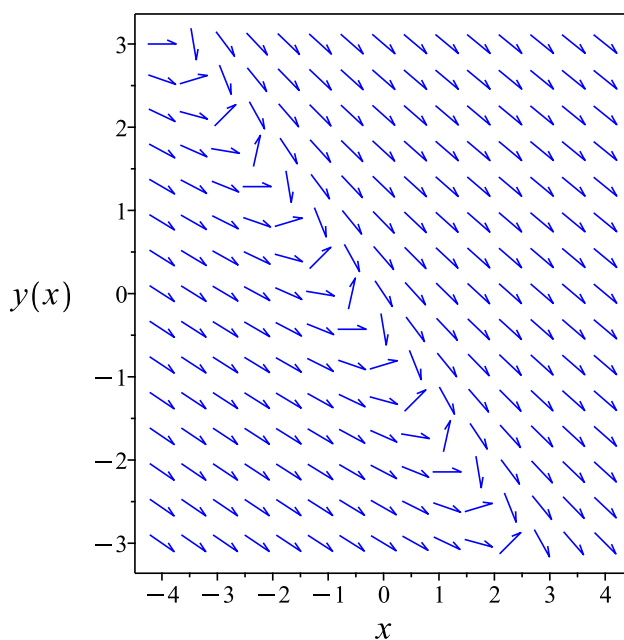


Figure 2.86: Slope field $x + y + 1 + (2x + 2y + 1)y' = 0$

Summary of solutions found

$$y = \frac{\text{LambertW}(2e^{x-c_2})}{2} - x$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 19

```
ode:=x+y(x)+1+(2*x+2*y(x)+1)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\text{LambertW}(2e^{x-c_1})}{2} - x$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous C
1st order, trying the canonical coordinates of the invariance group
  -> Calling odsolve with the ODE, diff(y(x),x) = -1, y(x)
      *** Sublevel 2 ***
      Methods for first order ODEs:
      --- Trying classification methods ---
      trying a quadrature
      trying 1st order linear
      <- 1st order linear successful
<- 1st order, canonical coordinates successful
<- homogeneous successful
```

Maple step by step

Let's solve

$$x + y(x) + 1 + (2x + 2y(x) + 1) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx} y(x)$
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-x-y(x)-1}{2x+2y(x)+1}$$

✓ **Mathematica.** Time used: 1.352 (sec). Leaf size: 30

```
ode=(x+y[x]+1)+(2*x+2*y[x]+1)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}(-2x + W(-e^{x-1+c_1}))$$

$$y(x) \rightarrow -x$$

✓ **Sympy.** Time used: 0.564 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x + (2*x + 2*y(x) + 1)*Derivative(y(x), x) + y(x) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = -x + \frac{W(C_1 e^x)}{2}$$

2.2.10 Problem 34

Local contents

Solved using first_order_ode_linear	362
Solved using first_order_ode_separable	364
Solved using first_order_ode_exact	366
Solved using first_order_ode_homog_type_maple_C	370
Solved using first_order_ode_LIE	374
✓ Maple	378
✓ Mathematica	379
✓ Sympy	380

Internal problem ID [7864]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 34

Date solved : Thursday, November 27, 2025 at 06:23:25 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.070 (sec)

Solve

$$1 + 2y - y'(-x + 4) = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = \frac{2}{x-4}$$

$$p(x) = -\frac{1}{x-4}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x-4} dx} \\ &= (x-4)^2\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(-\frac{1}{x-4} \right) \\ \frac{d}{dx}(y(x-4)^2) &= ((x-4)^2) \left(-\frac{1}{x-4} \right) \\ d(y(x-4)^2) &= (-x+4) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}y(x-4)^2 &= \int -x+4 dx \\ &= -\frac{1}{2}x^2 + 4x + c_1\end{aligned}$$

Dividing throughout by the integrating factor $(x-4)^2$ gives the final solution

$$y = \frac{4x - \frac{1}{2}x^2 + c_1}{(x-4)^2}$$

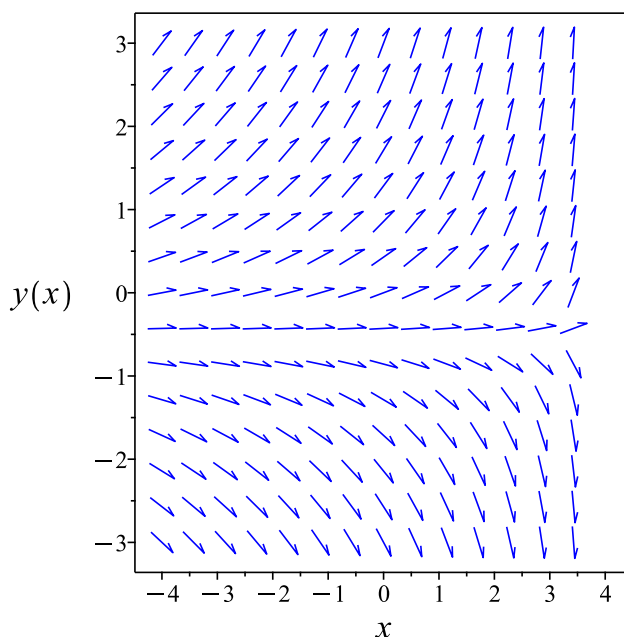


Figure 2.87: Slope field $1 + 2y - y'(-x + 4) = 0$

Summary of solutions found

$$y = \frac{4x - \frac{1}{2}x^2 + c_1}{(x-4)^2}$$

Solved using first_order_ode_separable

Time used: 0.104 (sec)

Solve

$$1 + 2y - y'(-x + 4) = 0$$

The ode

$$y' = -\frac{1 + 2y}{x - 4} \quad (2.38)$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{1 + 2y}{x - 4} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x - 4} \\ g(y) &= -2y - 1 \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{-2y - 1} dy &= \int \frac{1}{x - 4} dx \end{aligned}$$

$$-\frac{\ln(-2y - 1)}{2} = \ln(x - 4) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{1}{\sqrt{-2y - 1}} = c_1(x - 4)$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$-2y - 1 = 0$$

for y gives

$$y = -\frac{1}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{\sqrt{-2y-1}} = c_1(x-4)$$

$$y = -\frac{1}{2}$$

Solving for y gives

$$y = -\frac{1}{2}$$

$$y = -\frac{c_1^2 x^2 - 8c_1^2 x + 16c_1^2 + 1}{2c_1^2 (x-4)^2}$$

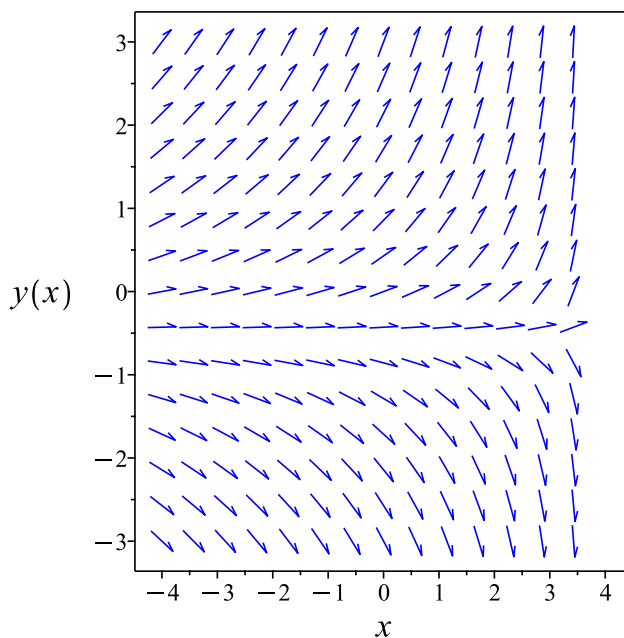


Figure 2.88: Slope field $1 + 2y - y'(-x + 4) = 0$

Summary of solutions found

$$y = -\frac{1}{2}$$

$$y = -\frac{c_1^2 x^2 - 8c_1^2 x + 16c_1^2 + 1}{2c_1^2 (x-4)^2}$$

Solved using first_order_ode_exact

Time used: 0.141 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (x - 4) dy &= (-2y - 1) dx \\ (1 + 2y) dx + (x - 4) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 1 + 2y \\ N(x, y) &= x - 4 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(1 + 2y) \\ &= 2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x - 4) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x-4} ((2) - (1)) \\ &= \frac{1}{x-4}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{1}{x-4} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x-4)} \\ &= x - 4\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= (x - 4)(1 + 2y) \\ &= (1 + 2y)(x - 4)\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= x - 4(x - 4) \\ &= (x - 4)^2\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ ((1 + 2y)(x - 4)) + ((x - 4)^2) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int (x - 4)^2 dy \\ \phi &= y(x - 4)^2 + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = 2y(x - 4) + f'(x) \tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = (1 + 2y)(x - 4)$. Therefore equation (4) becomes

$$(1 + 2y)(x - 4) = 2y(x - 4) + f'(x) \tag{5}$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = x - 4$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int (x - 4) dx$$

$$f(x) = \frac{1}{2}x^2 - 4x + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = y(x - 4)^2 + \frac{x^2}{2} - 4x + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = y(x - 4)^2 + \frac{x^2}{2} - 4x$$

Solving for y gives

$$y = \frac{-x^2 + 2c_1 + 8x}{2(x - 4)^2}$$

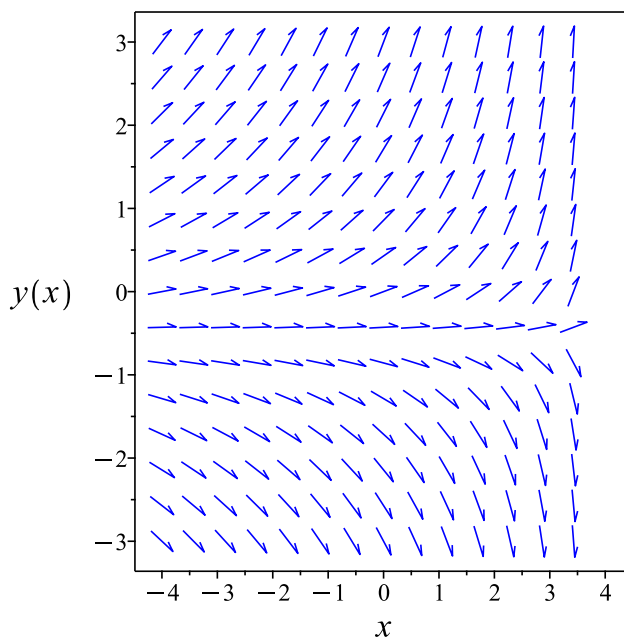


Figure 2.89: Slope field $1 + 2y - y'(-x + 4) = 0$

Summary of solutions found

$$y = \frac{-x^2 + 2c_1 + 8x}{2(x - 4)^2}$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.553 (sec)

Solve

$$1 + 2y - y'(-x + 4) = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{1 + 2Y(X) + 2y_0}{X + x_0 - 4}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$\begin{aligned} x_0 &= 4 \\ y_0 &= -\frac{1}{2} \end{aligned}$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{2Y(X)}{X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{2Y}{X} \end{aligned} \tag{1}$$

An ode of the form $Y' = \frac{M(X,Y)}{N(X,Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -2Y$ and $N = X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -2u \\ \frac{du}{dX} &= -\frac{3u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) + \frac{3u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)X + 3u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{3u(X)}{X} \quad (2.39)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{3u(X)}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{3}{X} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{1}{u} du &= \int -\frac{3}{X} dX \end{aligned}$$

$$\ln(u(X)) = \ln\left(\frac{1}{X^3}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(X) = \frac{c_1}{X^3}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(X) = \frac{c_1}{X^3}$$

$$u(X) = 0$$

Converting $u(X) = \frac{c_1}{X^3}$ back to $Y(X)$ gives

$$Y(X) = \frac{c_1}{X^2}$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$Y(X) = \frac{c_1}{X^2} \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - \frac{1}{2}$$

$$X = 4 + x$$

Then the solution in y becomes using EQ (A)

$$y + \frac{1}{2} = \frac{c_1}{(x - 4)^2}$$

Using the solution for $Y(X)$

$$Y(X) = 0 \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - \frac{1}{2}$$

$$X = 4 + x$$

Then the solution in y becomes using EQ (A)

$$y + \frac{1}{2} = 0$$

Solving for y gives

$$y = -\frac{1}{2}$$

$$y = \frac{-x^2 + 2c_1 + 8x - 16}{2(x-4)^2}$$

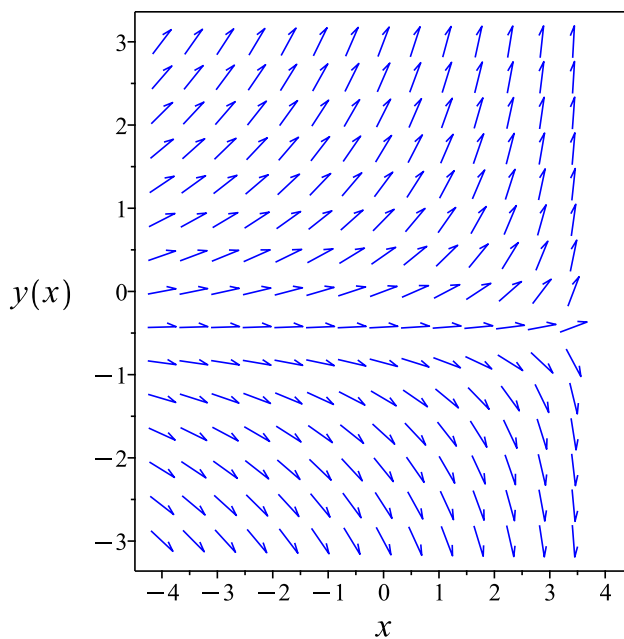


Figure 2.90: Slope field $1 + 2y - y'(-x + 4) = 0$

Summary of solutions found

$$y = -\frac{1}{2}$$

$$y = \frac{-x^2 + 2c_1 + 8x - 16}{2(x-4)^2}$$

Solved using first_order_ode_LIE

Time used: 0.789 (sec)

Solve

$$1 + 2y - y'(-x + 4) = 0$$

Writing the ode as

$$y' = -\frac{1 + 2y}{x - 4}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(1 + 2y)(b_3 - a_2)}{x - 4} - \frac{(1 + 2y)^2 a_3}{(x - 4)^2} - \frac{(1 + 2y)(xa_2 + ya_3 + a_1)}{(x - 4)^2} + \frac{2xb_2 + 2yb_3 + 2b_1}{x - 4} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{3x^2b_2 - 6y^2a_3 + 2xb_1 - 16xb_2 - xb_3 - 2ya_1 - 8ya_2 - 5ya_3 - a_1 - 4a_2 - a_3 - 8b_1 + 16b_2 + 4b_3}{(x - 4)^2} = 0$$

Setting the numerator to zero gives

$$3x^2b_2 - 6y^2a_3 + 2xb_1 - 16xb_2 - xb_3 - 2ya_1 - 8ya_2 - 5ya_3 - a_1 - 4a_2 - a_3 - 8b_1 + 16b_2 + 4b_3 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} -6a_3v_2^2 + 3b_2v_1^2 - 2a_1v_2 - 8a_2v_2 - 5a_3v_2 + 2b_1v_1 \\ - 16b_2v_1 - b_3v_1 - a_1 - 4a_2 - a_3 - 8b_1 + 16b_2 + 4b_3 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} 3b_2v_1^2 + (2b_1 - 16b_2 - b_3)v_1 - 6a_3v_2^2 + (-2a_1 - 8a_2 - 5a_3)v_2 \\ - a_1 - 4a_2 - a_3 - 8b_1 + 16b_2 + 4b_3 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -6a_3 &= 0 \\ 3b_2 &= 0 \\ -2a_1 - 8a_2 - 5a_3 &= 0 \\ 2b_1 - 16b_2 - b_3 &= 0 \\ -a_1 - 4a_2 - a_3 - 8b_1 + 16b_2 + 4b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= -4a_2 \\ a_2 &= a_2 \\ a_3 &= 0 \\ b_1 &= b_1 \\ b_2 &= 0 \\ b_3 &= 2b_1 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}\xi &= 0 \\ \eta &= 1 + 2y\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{1 + 2y} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(1 + 2y)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{1 + 2y}{x - 4}$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{1 + 2y}\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{R-4} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R-4}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

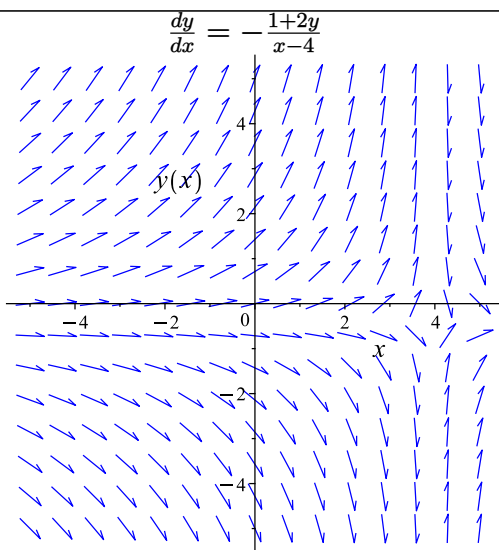
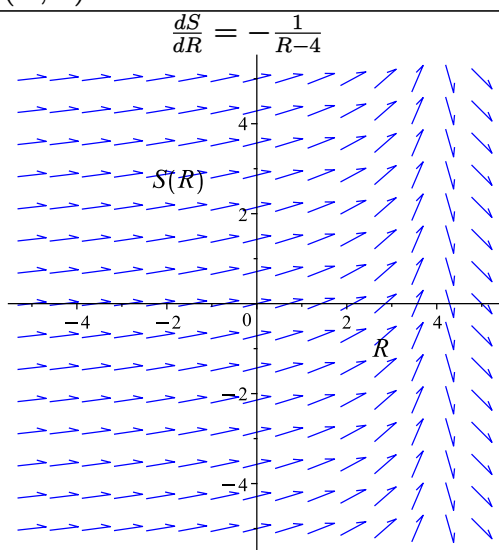
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{R-4} dR \\ S(R) &= -\ln(R-4) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(1+2y)}{2} = -\ln(x-4) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{1+2y}{x-4}$ 	$\begin{aligned} R &= x \\ S &= \frac{\ln(1+2y)}{2} \end{aligned}$	$\frac{dS}{dR} = -\frac{1}{R-4}$ 

Solving for y gives

$$y = \frac{-x^2 + e^{2c_2} + 8x - 16}{2x^2 - 16x + 32}$$

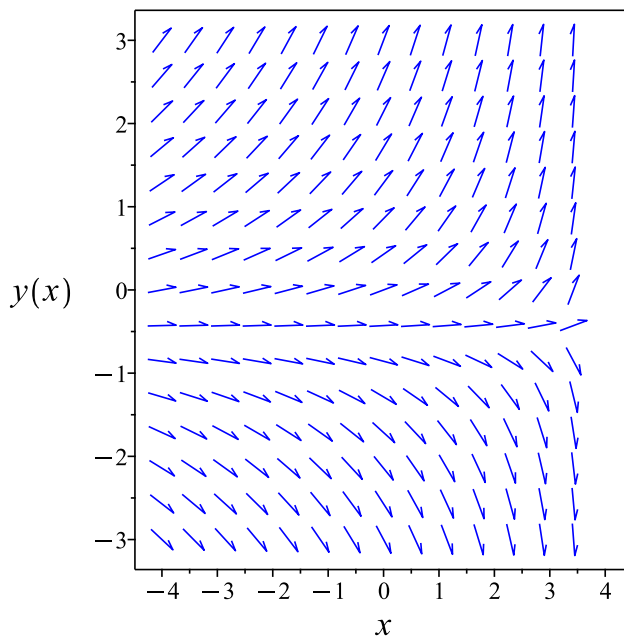


Figure 2.91: Slope field $1 + 2y - y'(-x + 4) = 0$

Summary of solutions found

$$y = \frac{-x^2 + e^{2c_2} + 8x - 16}{2x^2 - 16x + 32}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 20

```
ode:=1+2*y(x)-(4-x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{4x - \frac{1}{2}x^2 + c_1}{(x - 4)^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$1 + 2y(x) - (4 - x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-1-2y(x)}{4-x}$$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{-1-2y(x)} = -\frac{1}{4-x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{-1-2y(x)} dx = \int -\frac{1}{4-x} dx + C1$$

- Evaluate integral

$$-\frac{\ln(-1-2y(x))}{2} = \ln(4-x) + C1$$

- Solve for $y(x)$

$$y(x) = -\frac{x^2 + e^{-2C1} - 8x + 16}{2(x^2 - 8x + 16)}$$

- Simplify

$$y(x) = \frac{-x^2 - e^{-2C1} + 8x - 16}{2(-4+x)^2}$$

- Redefine the integration constant(s)

$$y(x) = \frac{C1 - \frac{1}{2}x^2 + 4x}{(-4+x)^2}$$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 34

```
ode=(1+2*y[x])-(4-x)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{-x^2 + 8x + 2c_1}{2(x-4)^2}$$

$$y(x) \rightarrow -\frac{1}{2}$$

✓ Sympy. Time used: 0.186 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x - 4)*Derivative(y(x), x) + 2*y(x) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1}{2(x^2 - 8x + 16)} - \frac{1}{2}$$

2.2.11 Problem 35

Local contents

Solved using first_order_ode_linear	381
Solved using first_order_ode_separable	383
Solved using first_order_ode_exact	384
Solved using first_order_ode_homog_type_D2	388
Solved using first_order_ode_LIE	390
✓ Maple	395
✓ Mathematica	396
✓ Sympy	396

Internal problem ID [7865]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 35

Date solved : Thursday, November 27, 2025 at 06:23:33 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.045 (sec)

Solve

$$yx + (x^2 + 1)y' = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{x}{x^2 + 1} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{x}{x^2 + 1} \, dx} \\ &= \sqrt{x^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}\left(\sqrt{x^2+1}y\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\sqrt{x^2+1}y &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{x^2+1}$ gives the final solution

$$y = \frac{c_1}{\sqrt{x^2+1}}$$

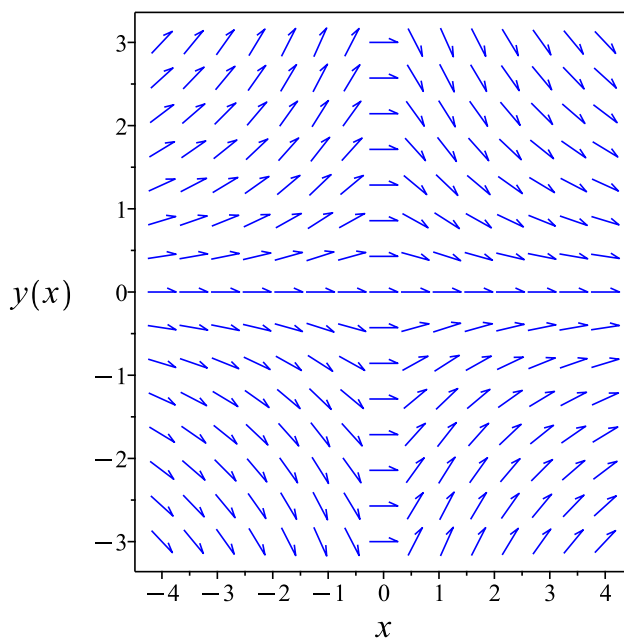


Figure 2.92: Slope field $yx + (x^2 + 1)y' = 0$

Summary of solutions found

$$y = \frac{c_1}{\sqrt{x^2+1}}$$

Solved using first_order_ode_separable

Time used: 0.076 (sec)

Solve

$$yx + (x^2 + 1) y' = 0$$

The ode

$$y' = -\frac{xy}{x^2 + 1} \quad (2.40)$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{xy}{x^2 + 1} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{x}{x^2 + 1} \\ g(y) &= y \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{y} dy &= \int -\frac{x}{x^2 + 1} dx \end{aligned}$$

$$\ln(y) = \ln\left(\frac{1}{\sqrt{x^2 + 1}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y = 0$$

for y gives

$$y = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

$$y = 0$$

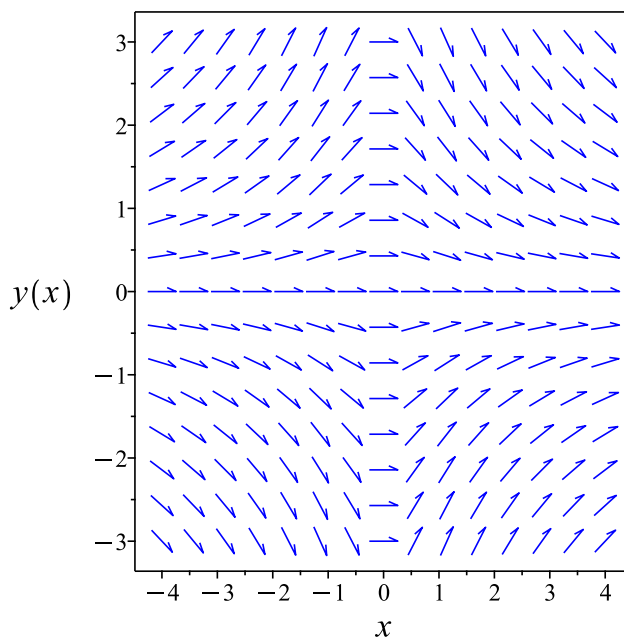


Figure 2.93: Slope field $yx + (x^2 + 1)y' = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Solved using first_order_ode_exact

Time used: 0.092 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \tag{A}$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x^2 + 1) dy &= (-yx) dx \\ (yx) dx + (x^2 + 1) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= yx \\ N(x, y) &= x^2 + 1 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(yx) \\ &= x \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x^2 + 1) \\ &= 2x \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x^2 + 1} ((x) - (2x)) \\ &= -\frac{x}{x^2 + 1} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{x}{x^2+1} \, dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\frac{\ln(x^2+1)}{2}} \\ &= \frac{1}{\sqrt{x^2+1}} \end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned} \bar{M} &= \mu M \\ &= \frac{1}{\sqrt{x^2+1}} (yx) \\ &= \frac{yx}{\sqrt{x^2+1}} \end{aligned}$$

And

$$\begin{aligned} \bar{N} &= \mu N \\ &= \frac{1}{\sqrt{x^2+1}} (x^2+1) \\ &= \sqrt{x^2+1} \end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned} \bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{yx}{\sqrt{x^2+1}} \right) + \left(\sqrt{x^2+1} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \sqrt{x^2 + 1} dy \\ \phi &= \sqrt{x^2 + 1} y + f(x) \end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = \frac{yx}{\sqrt{x^2 + 1}} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{yx}{\sqrt{x^2 + 1}}$. Therefore equation (4) becomes

$$\frac{yx}{\sqrt{x^2 + 1}} = \frac{yx}{\sqrt{x^2 + 1}} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

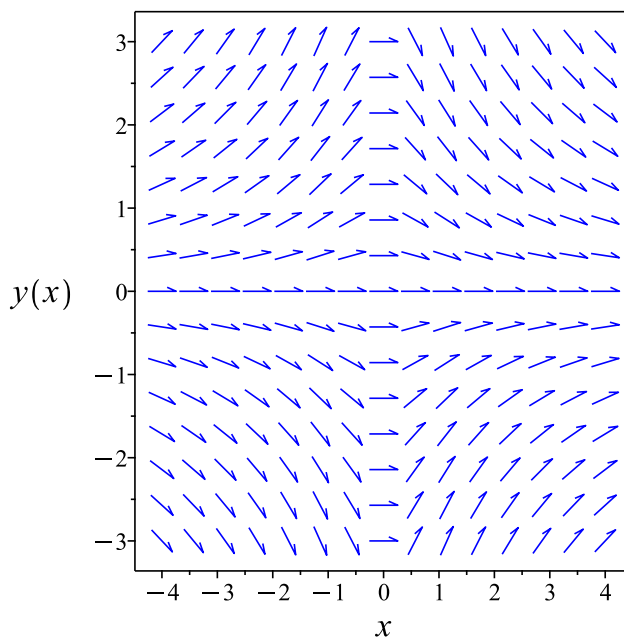
$$\phi = \sqrt{x^2 + 1} y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \sqrt{x^2 + 1} y$$

Solving for y gives

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Figure 2.94: Slope field $yx + (x^2 + 1)y' = 0$ Summary of solutions found

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.170 (sec)

Solve

$$yx + (x^2 + 1)y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)x^2 + (x^2 + 1)(u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{u(x)(2x^2 + 1)}{x(x^2 + 1)} \quad (2.41)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)(2x^2 + 1)}{x(x^2 + 1)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{2x^2 + 1}{x(x^2 + 1)}$$

$$g(u) = u$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u} du = \int -\frac{2x^2 + 1}{x(x^2 + 1)} dx$$

$$\ln(u(x)) = \ln\left(\frac{1}{x\sqrt{x^2 + 1}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = \frac{c_1}{x\sqrt{x^2 + 1}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = \frac{c_1}{x\sqrt{x^2 + 1}}$$

$$u(x) = 0$$

Converting $u(x) = \frac{c_1}{x\sqrt{x^2 + 1}}$ back to y gives

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

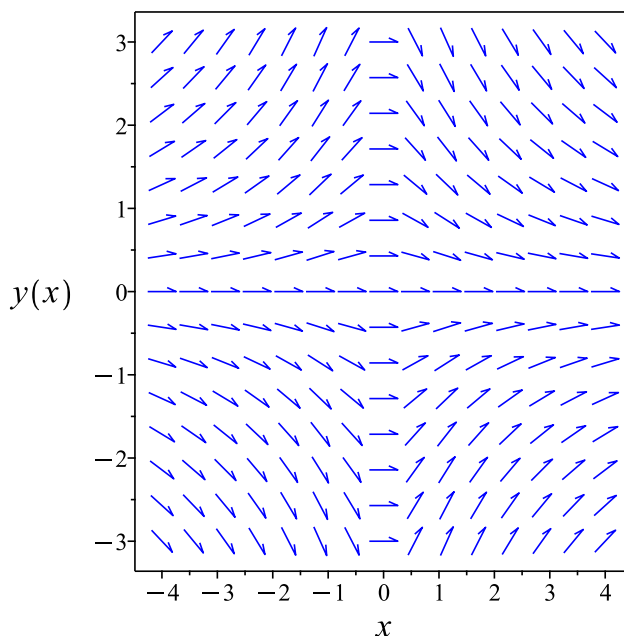


Figure 2.95: Slope field $yx + (x^2 + 1)y' = 0$

Summary of solutions found

$$y = 0$$

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Solved using first_order_ode_LIE

Time used: 0.747 (sec)

Solve

$$yx + (x^2 + 1)y' = 0$$

Writing the ode as

$$y' = -\frac{xy}{x^2 + 1}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1E)$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{xy(b_3 - a_2)}{x^2 + 1} - \frac{x^2y^2a_3}{(x^2 + 1)^2} \\ - \left(-\frac{y}{x^2 + 1} + \frac{2x^2y}{(x^2 + 1)^2} \right) (xa_2 + ya_3 + a_1) + \frac{x(xb_2 + yb_3 + b_1)}{x^2 + 1} = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{2x^4b_2 - 2x^2y^2a_3 + x^3b_1 - x^2ya_1 + 3x^2b_2 + 2xya_2 + y^2a_3 + xb_1 + ya_1 + b_2}{(x^2 + 1)^2} = 0$$

Setting the numerator to zero gives

$$2x^4b_2 - 2x^2y^2a_3 + x^3b_1 - x^2ya_1 + 3x^2b_2 + 2xya_2 + y^2a_3 + xb_1 + ya_1 + b_2 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-2a_3v_1^2v_2^2 + 2b_2v_1^4 - a_1v_1^2v_2 + b_1v_1^3 + 2a_2v_1v_2 + a_3v_2^2 + 3b_2v_1^2 + a_1v_2 + b_1v_1 + b_2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-2a_3v_1^2v_2^2 + 2b_2v_1^4 - a_1v_1^2v_2 + b_1v_1^3 + 2a_2v_1v_2 + a_3v_2^2 + 3b_2v_1^2 + a_1v_2 + b_1v_1 + b_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 &= 0 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ -a_1 &= 0 \\ 2a_2 &= 0 \\ -2a_3 &= 0 \\ 2b_2 &= 0 \\ 3b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= 0 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 0 \\ \eta &= y \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y} dy \end{aligned}$$

Which results in

$$S = \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{xy}{x^2 + 1}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{x}{x^2 + 1} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{R}{R^2 + 1}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

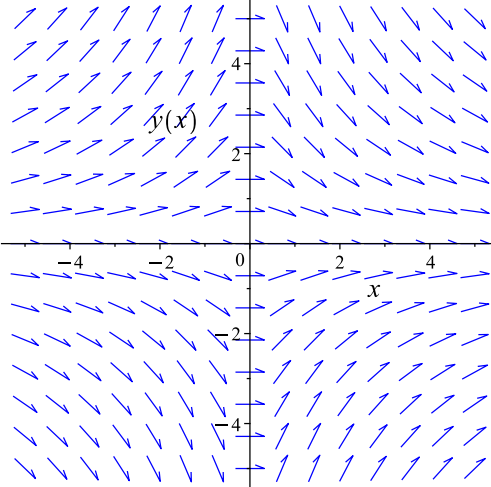
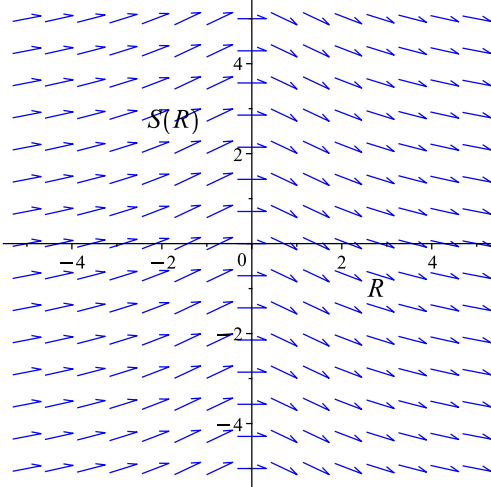
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{R}{R^2 + 1} dR \\ S(R) &= -\frac{\ln(R^2 + 1)}{2} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

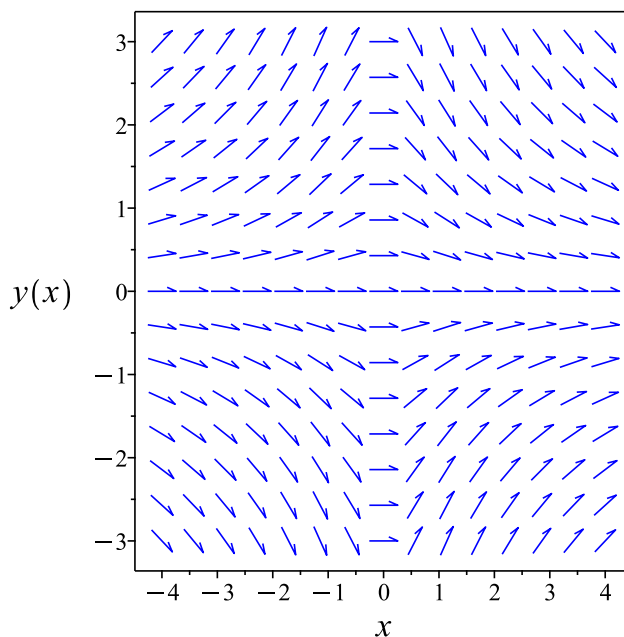
$$\ln(y) = -\frac{\ln(x^2 + 1)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{xy}{x^2+1}$ 	$R = x$ $S = \ln(y)$	$\frac{dS}{dR} = -\frac{R}{R^2+1}$ 

Solving for y gives

$$y = e^{-\frac{\ln(x^2+1)}{2} + c_2}$$

Figure 2.96: Slope field $yx + (x^2 + 1)y' = 0$ Summary of solutions found

$$y = e^{-\frac{\ln(x^2+1)}{2} + c_2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 13

```
ode:=(x^2+1)*diff(y(x),x)+y(x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{c_1}{\sqrt{x^2 + 1}}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve
 $xy(x) + (x^2 + 1) \left(\frac{d}{dx} y(x) \right) = 0$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Solve for the highest derivative
 $\frac{d}{dx}y(x) = -\frac{xy(x)}{x^2+1}$
- Separate variables
 $\frac{\frac{d}{dx}y(x)}{y(x)} = -\frac{x}{x^2+1}$
- Integrate both sides with respect to x
 $\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int -\frac{x}{x^2+1} dx + C1$
- Evaluate integral
 $\ln(y(x)) = -\frac{\ln(x^2+1)}{2} + C1$
- Solve for $y(x)$
 $y(x) = e^{-\frac{\ln(x^2+1)}{2} + C1}$
- Simplify
 $y(x) = \frac{e^{C1}}{\sqrt{x^2+1}}$
- Redefine the integration constant(s)
 $y(x) = \frac{C1}{\sqrt{x^2+1}}$

✓ **Mathematica.** Time used: 0.019 (sec). Leaf size: 22

```
ode=(x*y[x])+(1+x^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{c_1}{\sqrt{x^2+1}}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.141 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) + (x**2 + 1)*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1}{\sqrt{x^2+1}}$$

2.2.12 Problem 37

Local contents

Solved using first_order_ode_exact	397
Solved using first_order_ode_isobaric	401
Solved using first_order_ode_homog_A	404
Solved using first_order_ode_dAlembert	407
Solved using first_order_ode_homog_type_D2	411
Solved using first_order_ode_homog_type_G	414
Solved using first_order_ode_homog_type_maple_C	415
Solved using first_order_ode_abel_second_kind_case_5	420
Solved using first_order_ode_LIE	422
✓ Maple	427
✓ Mathematica	428
✓ Sympy	429

Internal problem ID [7866]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 37

Date solved : Thursday, November 27, 2025 at 06:23:39 AM

CAS classification :

[[_homogeneous, 'class A'], _exact, _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_exact

Time used: 0.078 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N\end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned}(3y + 2x) dy &= (-x - 2y) dx \\ (x + 2y) dx + (3y + 2x) dy &= 0\end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= x + 2y \\ N(x, y) &= 3y + 2x\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x + 2y) \\ &= 2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(3y + 2x) \\ &= 2\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x + 2y dx \\ \phi &= \frac{x(x + 4y)}{2} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 2x + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 3y + 2x$. Therefore equation (4) becomes

$$3y + 2x = 2x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 3y$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (3y) dy \\ f(y) &= \frac{3y^2}{2} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x(x + 4y)}{2} + \frac{3y^2}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x(x+4y)}{2} + \frac{3y^2}{2}$$

Simplifying the above gives

$$\frac{(x+3y)(x+y)}{2} = c_1$$

Solving for y gives

$$y = -\frac{2x}{3} - \frac{\sqrt{x^2 + 6c_1}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{x^2 + 6c_1}}{3}$$

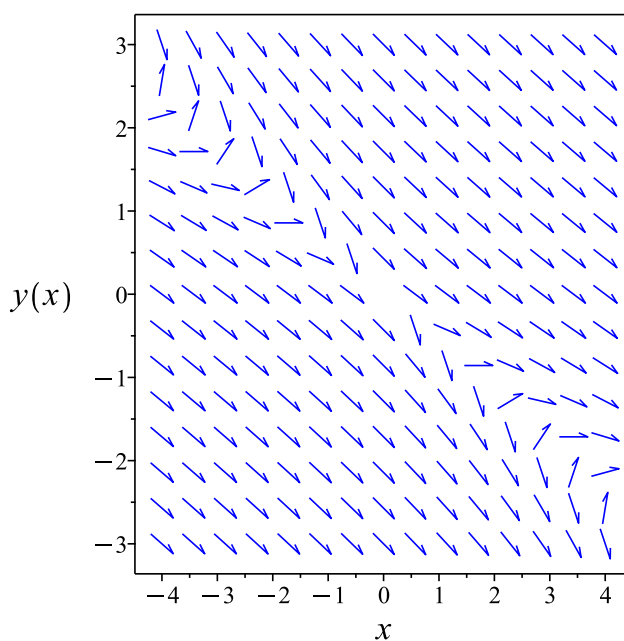


Figure 2.97: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -\frac{2x}{3} - \frac{\sqrt{x^2 + 6c_1}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{x^2 + 6c_1}}{3}$$

Solved using first_order_ode_isobaric

Time used: 0.276 (sec)

Solve

$$x + 2y + (3y + 2x) y' = 0$$

Solving for y' gives

$$y' = -\frac{x + 2y}{3y + 2x} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{x + 2y}{3y + 2x} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x + 2xu(x)}{3xu(x) + 2x}$$

The ode

$$u'(x) = -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \quad (2.42)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{(u + 1)(3u + 1)}{3u + 2} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{3u+2}{(u+1)(3u+1)} du = \int -\frac{1}{x} dx$$

$$\frac{\ln(3u(x)^2 + 4u(x) + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{(u+1)(3u+1)}{3u+2} = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Converting $\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{3y^2}{x^2} + \frac{4y}{x} + 1} = \frac{c_1}{x}$$

Converting $u(x) = -1$ back to y gives

$$\frac{y}{x} = -1$$

Converting $u(x) = -\frac{1}{3}$ back to y gives

$$\frac{y}{x} = -\frac{1}{3}$$

Simplifying the above gives

$$\sqrt{\frac{3y^2 + 4yx + x^2}{x^2}} = \frac{c_1}{x}$$

$$\frac{y}{x} = -1$$

$$\frac{y}{x} = -\frac{1}{3}$$

Solving for y gives

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

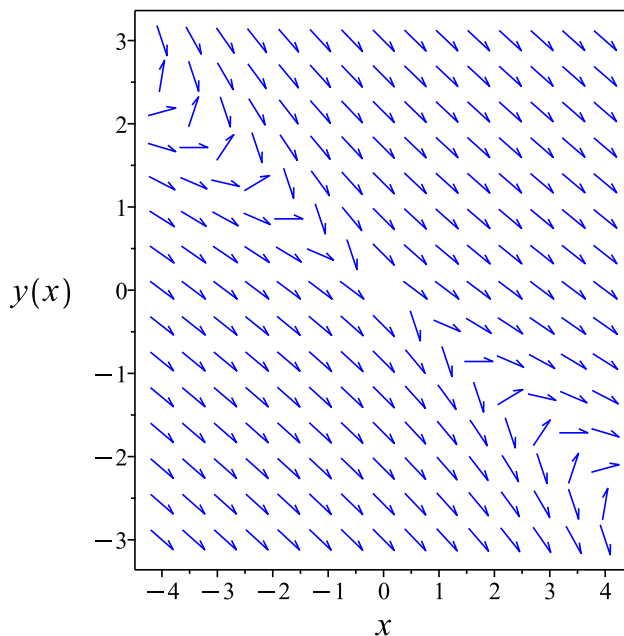


Figure 2.98: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Solved using first_order_ode_homog_A

Time used: 0.584 (sec)

Solve

$$x + 2y + (3y + 2x)y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x + 2y}{3y + 2x} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x - 2y$ and $N = 3y + 2x$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= \frac{-2u - 1}{3u + 2} \\ \frac{du}{dx} &= \frac{\frac{-2u(x)-1}{3u(x)+2} - u(x)}{x} \end{aligned}$$

Or

$$u'(x) - \frac{\frac{-2u(x)-1}{3u(x)+2} - u(x)}{x} = 0$$

Or

$$3u'(x) xu(x) + 2u'(x) x + 3u(x)^2 + 4u(x) + 1 = 0$$

Or

$$1 + x(3u(x) + 2) u'(x) + 3u(x)^2 + 4u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \quad (2.43)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{(u + 1)(3u + 1)}{3u + 2} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{3u + 2}{(u + 1)(3u + 1)} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(3u(x)^2 + 4u(x) + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{(u + 1)(3u + 1)}{3u + 2} = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Converting $\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{(x+y)(x+3y)}{x^2}} = \frac{c_1}{x}$$

Converting $u(x) = -1$ back to y gives

$$y = -x$$

Converting $u(x) = -\frac{1}{3}$ back to y gives

$$y = -\frac{x}{3}$$

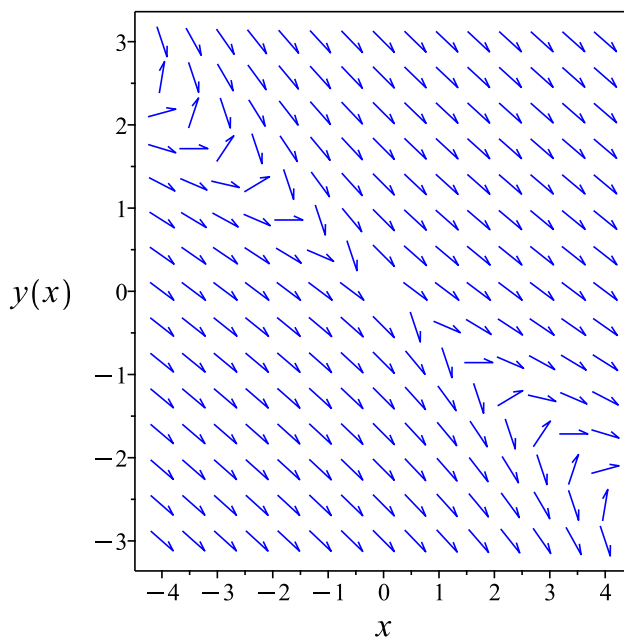
Solving for y gives

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Figure 2.99: Slope field $x + 2y + (3y + 2x)y' = 0$ Summary of solutions found

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Solved using first_order_ode_dAlembert

Time used: 0.976 (sec)

Solve

$$x + 2y + (3y + 2x)y' = 0$$

Let $p = y'$ the ode becomes

$$x + 2y + (3y + 2x)p = 0$$

Solving for y from the above results in

$$y = -\frac{x(2p + 1)}{2 + 3p} \tag{1}$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{-2p - 1}{2 + 3p} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{-2p - 1}{2 + 3p} = \left(-\frac{2x}{2 + 3p} + \frac{6xp}{(2 + 3p)^2} + \frac{3x}{(2 + 3p)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{-2p - 1}{2 + 3p} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= -\frac{1}{3} \\ p_2 &= -1 \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= -\frac{x}{3} \\ y &= -x \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{-2p(x)-1}{2+3p(x)}}{-\frac{2x}{2+3p(x)} + \frac{6xp(x)}{(2+3p(x))^2} + \frac{3x}{(2+3p(x))^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{(2+3p(x))(3p(x)+1)(p(x)+1)}{x} \quad (2.44)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{(2+3p(x))(3p(x)+1)(p(x)+1)}{x} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(p) &= (2+3p)(3p+1)(p+1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{(2+3p)(3p+1)(p+1)} dp &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(p(x)+1)}{2} + \frac{\ln(3p(x)+1)}{2} - \ln(2+3p(x)) = \ln\left(\frac{1}{x}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$(2+3p)(3p+1)(p+1) = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= -1 \\ p(x) &= -\frac{2}{3} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(p(x)+1)}{2} + \frac{\ln(3p(x)+1)}{2} - \ln(2+3p(x)) = \ln\left(\frac{1}{x}\right) + c_1$$

$$p(x) = -1$$

$$p(x) = -\frac{2}{3}$$

Substituting the above solution for p in (2A) gives

$$y = \frac{x \left(-\frac{2(3e^{2c_1} - x^2 - \sqrt{-3x^2e^{2c_1} + x^4})}{3(3e^{2c_1} - x^2)} + 1 \right)}{-1 + \frac{3e^{2c_1} - x^2 - \sqrt{-3x^2e^{2c_1} + x^4}}{3e^{2c_1} - x^2}}$$

$$y = -x$$

Simplifying the above gives

$$y = -\frac{x}{3}$$

$$y = -x$$

$$y = \frac{\left(x^2 - 3e^{2c_1} - 2\sqrt{x^2(-3e^{2c_1} + x^2)} \right) x}{3\sqrt{x^2(-3e^{2c_1} + x^2)}}$$

$$y = -x$$

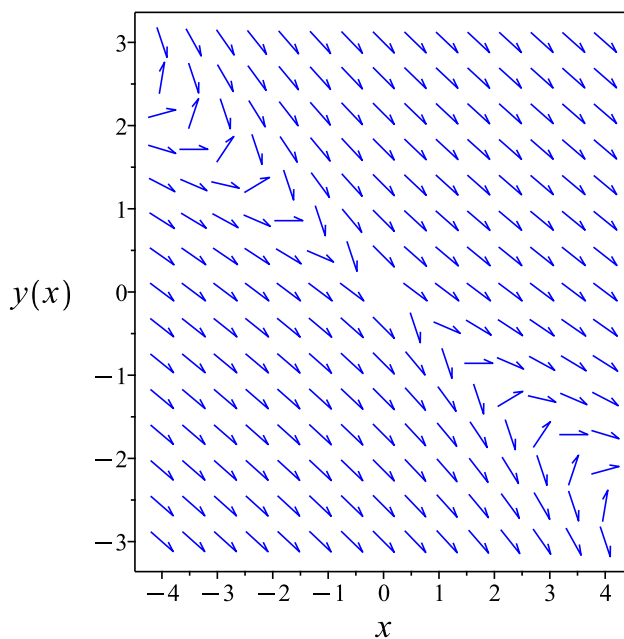


Figure 2.100: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = \frac{\left(x^2 - 3e^{2c_1} - 2\sqrt{x^2(-3e^{2c_1} + x^2)}\right)x}{3\sqrt{x^2(-3e^{2c_1} + x^2)}}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.215 (sec)

Solve

$$x + 2y + (3y + 2x)y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$x + 2u(x)x + (3u(x)x + 2x)(u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \quad (2.45)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{(u(x) + 1)(3u(x) + 1)}{x(3u(x) + 2)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{(u + 1)(3u + 1)}{3u + 2} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{3u + 2}{(u + 1)(3u + 1)} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(3u(x)^2 + 4u(x) + 1)}{2} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{(u+1)(3u+1)}{3u+2} = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$$

$$u(x) = -1$$

$$u(x) = -\frac{1}{3}$$

Converting $\sqrt{3u(x)^2 + 4u(x) + 1} = \frac{c_1}{x}$ back to y gives

$$\sqrt{\frac{3y^2}{x^2} + \frac{4y}{x} + 1} = \frac{c_1}{x}$$

Converting $u(x) = -1$ back to y gives

$$y = -x$$

Converting $u(x) = -\frac{1}{3}$ back to y gives

$$y = -\frac{x}{3}$$

Simplifying the above gives

$$\sqrt{\frac{3y^2 + 4yx + x^2}{x^2}} = \frac{c_1}{x}$$

$$y = -x$$

$$y = -\frac{x}{3}$$

Solving for y gives

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

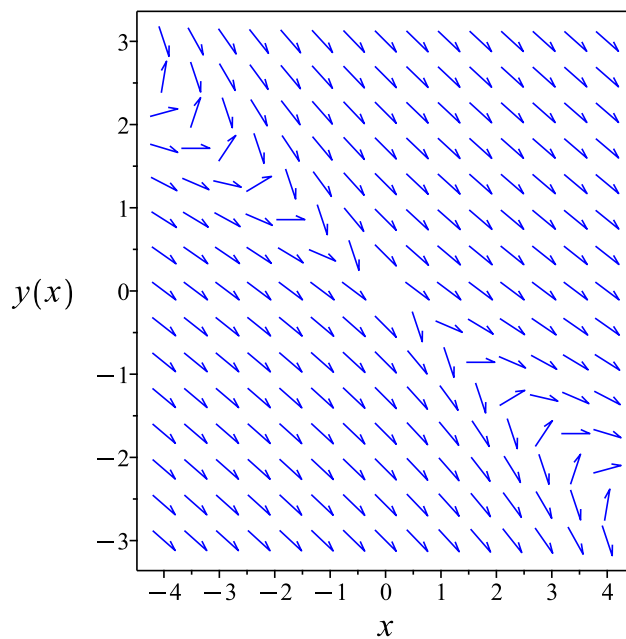


Figure 2.101: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Solved using first_order_ode_homog_type_G

Time used: 0.088 (sec)

Solve

$$x + 2y + (3y + 2x)y' = 0$$

Multiplying the right side of the ode, which is $-\frac{x+2y}{3y+2x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x+2y}{3y+2x} \\ &= -\frac{x(x+2y)}{y(3y+2x)} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{2x(x^2 + 3yx + 3y^2)}{y(3y + 2x)^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{2x(x^2 + 3yx + 3y^2)}{y(3y + 2x)^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{2z + 1}{z(3z + 2)}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{2z+1}{z(3z+2)}\right)} dz = 0$$

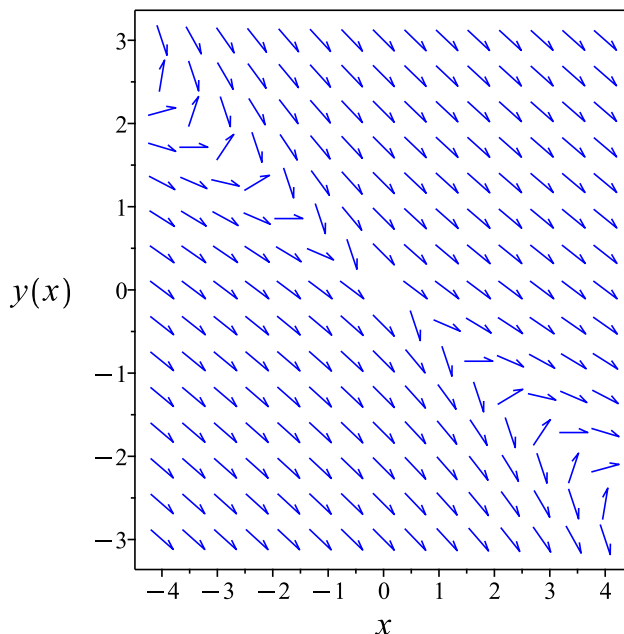


Figure 2.102: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{2z+1}{z(3z+2)}\right)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.849 (sec)

Solve

$$x + 2y + (3y + 2x)y' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{X + x_0 + 2Y(X) + 2y_0}{3Y(X) + 3y_0 + 2X + 2x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X + 2Y(X)}{3Y(X) + 2X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X + 2Y}{3Y + 2X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X - 2Y$ and $N = 3Y + 2X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= \frac{-2u - 1}{3u + 2} \\ \frac{du}{dX} &= \frac{\frac{-2u(X)-1}{3u(X)+2} - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{-2u(X)-1}{3u(X)+2} - u(X)}{X} = 0$$

Or

$$3\left(\frac{d}{dX}u(X)\right)Xu(X) + 2\left(\frac{d}{dX}u(X)\right)X + 3u(X)^2 + 4u(X) + 1 = 0$$

Or

$$1 + X(3u(X) + 2)\left(\frac{d}{dX}u(X)\right) + 3u(X)^2 + 4u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{(u(X) + 1)(3u(X) + 1)}{X(3u(X) + 2)} \quad (2.46)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{(u(X) + 1)(3u(X) + 1)}{X(3u(X) + 2)} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{1}{X} \\ g(u) &= \frac{(u + 1)(3u + 1)}{3u + 2} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{3u + 2}{(u + 1)(3u + 1)} du &= \int -\frac{1}{X} dX \end{aligned}$$

$$\frac{\ln(3u(X)^2 + 4u(X) + 1)}{2} = \ln\left(\frac{1}{X}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{3u(X)^2 + 4u(X) + 1} = \frac{c_1}{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{(u + 1)(3u + 1)}{3u + 2} = 0$$

for $u(X)$ gives

$$\begin{aligned} u(X) &= -1 \\ u(X) &= -\frac{1}{3} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{3u(X)^2 + 4u(X) + 1} = \frac{c_1}{X}$$

$$u(X) = -1$$

$$u(X) = -\frac{1}{3}$$

Converting $\sqrt{3u(X)^2 + 4u(X) + 1} = \frac{c_1}{X}$ back to $Y(X)$ gives

$$\sqrt{\frac{(Y(X) + X)(3Y(X) + X)}{X^2}} = \frac{c_1}{X}$$

Converting $u(X) = -1$ back to $Y(X)$ gives

$$Y(X) = -X$$

Converting $u(X) = -\frac{1}{3}$ back to $Y(X)$ gives

$$Y(X) = -\frac{X}{3}$$

Using the solution for $Y(X)$

$$\sqrt{\frac{(Y(X) + X)(3Y(X) + X)}{X^2}} = \frac{c_1}{X} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\sqrt{\frac{(x+y)(x+3y)}{x^2}} = \frac{c_1}{x}$$

Using the solution for $Y(X)$

$$Y(X) = -X \quad (A)$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -x$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{X}{3} \quad (A)$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{x}{3}$$

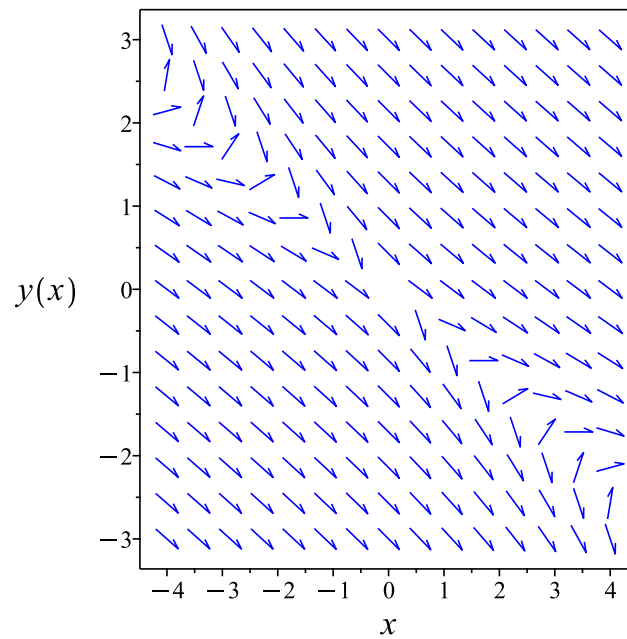
Solving for y gives

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Figure 2.103: Slope field $x + 2y + (3y + 2x)y' = 0$ Summary of solutions found

$$y = -x$$

$$y = -\frac{x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3c_1^2 + x^2}}{3}$$

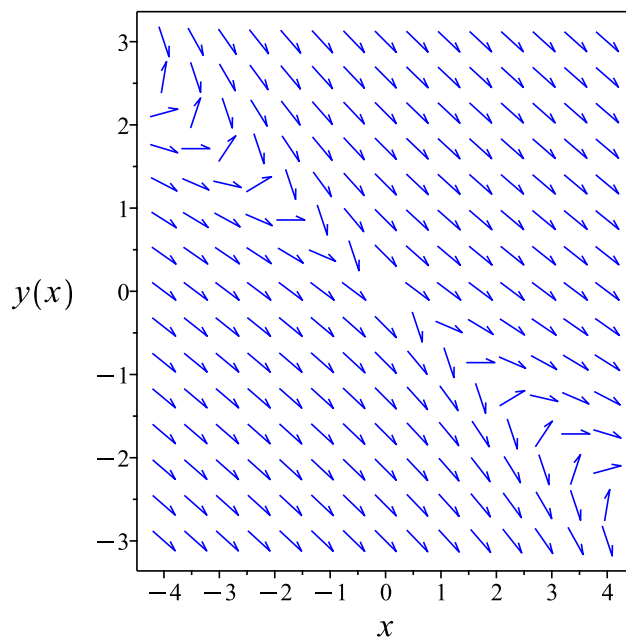
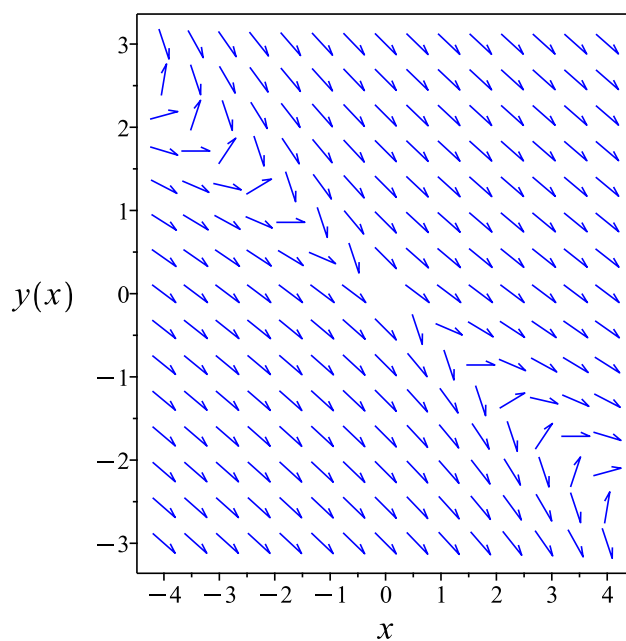
$$y = -\frac{2x}{3} + \frac{\sqrt{3c_1^2 + x^2}}{3}$$

Solve

$$x + 2y + (3y + 2x)y' = 0$$

Solved using first_order_ode_abel_second_kind_case_5

Time used: 0.036 (sec)

Figure 2.104: Slope field $x + 2y + (3y + 2x)y' = 0$ Figure 2.105: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -\frac{2x}{3} - \frac{\sqrt{x^2 + 9c_1}}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{x^2 + 9c_1}}{3}$$

Solved using first_order_ode_LIE

Time used: 1.247 (sec)

Solve

$$x + 2y + (3y + 2x) y' = 0$$

Writing the ode as

$$y' = -\frac{x + 2y}{3y + 2x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(x + 2y)(b_3 - a_2)}{3y + 2x} - \frac{(x + 2y)^2 a_3}{(3y + 2x)^2} - \left(-\frac{1}{3y + 2x} + \frac{2x + 4y}{(3y + 2x)^2} \right) (xa_2 + ya_3 + a_1) \quad (5\text{E})$$

$$- \left(-\frac{2}{3y + 2x} + \frac{3x + 6y}{(3y + 2x)^2} \right) (xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$\frac{2x^2a_2 - x^2a_3 + 5x^2b_2 - 2x^2b_3 + 6xya_2 - 4xya_3 + 12xyb_2 - 6xyb_3 + 6y^2a_2 - 5y^2a_3 + 9y^2b_2 - 6y^2b_3 + xb_1 -}{(3y + 2x)^2} = 0$$

Setting the numerator to zero gives

$$2x^2a_2 - x^2a_3 + 5x^2b_2 - 2x^2b_3 + 6xya_2 - 4xya_3 + 12xyb_2 - 6xyb_3 + 6y^2a_2 - 5y^2a_3 + 9y^2b_2 - 6y^2b_3 + xb_1 - ya_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} 2a_2v_1^2 + 6a_2v_1v_2 + 6a_2v_2^2 - a_3v_1^2 - 4a_3v_1v_2 - 5a_3v_2^2 + 5b_2v_1^2 \\ + 12b_2v_1v_2 + 9b_2v_2^2 - 2b_3v_1^2 - 6b_3v_1v_2 - 6b_3v_2^2 - a_1v_2 + b_1v_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} (2a_2 - a_3 + 5b_2 - 2b_3)v_1^2 + (6a_2 - 4a_3 + 12b_2 - 6b_3)v_1v_2 \\ + b_1v_1 + (6a_2 - 5a_3 + 9b_2 - 6b_3)v_2^2 - a_1v_2 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} b_1 &= 0 \\ -a_1 &= 0 \\ 2a_2 - a_3 + 5b_2 - 2b_3 &= 0 \\ 6a_2 - 5a_3 + 9b_2 - 6b_3 &= 0 \\ 6a_2 - 4a_3 + 12b_2 - 6b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= -4b_2 + b_3 \\ a_3 &= -3b_2 \\ b_1 &= 0 \\ b_2 &= b_2 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = y$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x+2y}{3y+2x} \right) (x) \\ &= y + \frac{x(x+2y)}{3y+2x} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x(x+2y)}{3y+2x}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(x^2 + 4yx + 3y^2)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x + 2y}{3y + 2x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{x + 2y}{(x + 3y)(x + y)} \\ S_y &= \frac{3y + 2x}{(x + 3y)(x + y)} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \tag{2A}$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

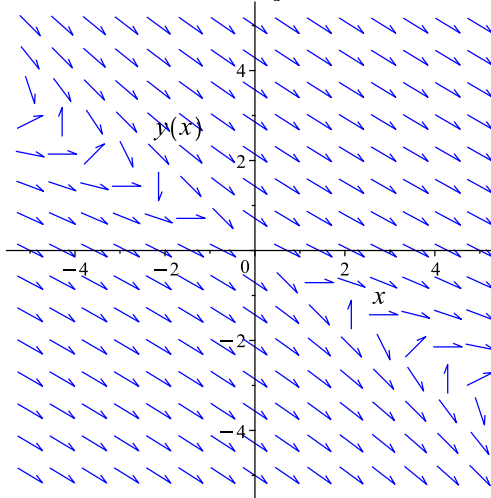
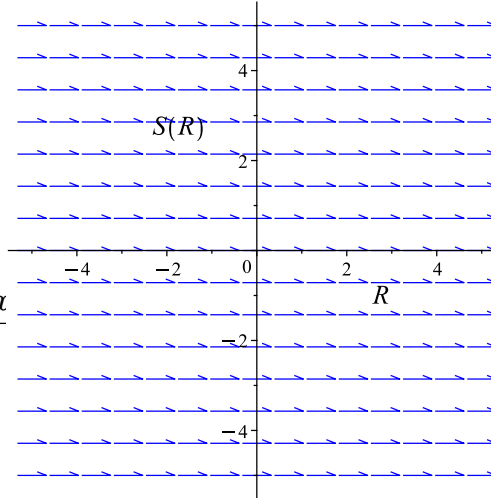
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(x + 3y)}{2} + \frac{\ln(x + y)}{2} = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x+2y}{3y+2x}$ 	$R = x$ $S = \frac{\ln(x+3y)}{2} + \ln(x)$	$\frac{dS}{dR} = 0$ 

Solving for y gives

$$y = -\frac{x(-x - \sqrt{x^2 + 3e^{2c_2}}) - e^{2c_2}}{-x - \sqrt{x^2 + 3e^{2c_2}}}$$

$$y = -\frac{x(-x + \sqrt{x^2 + 3e^{2c_2}}) - e^{2c_2}}{-x + \sqrt{x^2 + 3e^{2c_2}}}$$

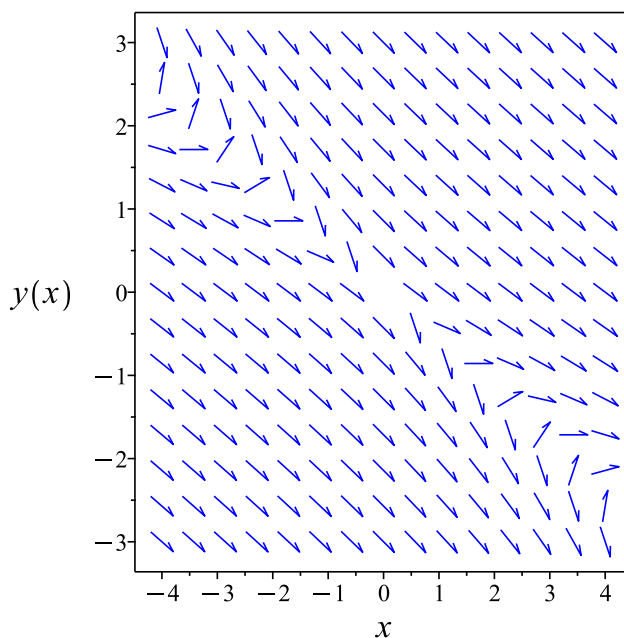


Figure 2.106: Slope field $x + 2y + (3y + 2x)y' = 0$

Summary of solutions found

$$y = -\frac{x(-x - \sqrt{x^2 + 3e^{2c_2}}) - e^{2c_2}}{-x - \sqrt{x^2 + 3e^{2c_2}}}$$

$$y = -\frac{x(-x + \sqrt{x^2 + 3e^{2c_2}}) - e^{2c_2}}{-x + \sqrt{x^2 + 3e^{2c_2}}}$$

✓ Maple. Time used: 0.014 (sec). Leaf size: 51

```
ode:=x+2*y(x)+(2*x+3*y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{-2c_1x - \sqrt{x^2c_1^2 + 3}}{3c_1}$$

$$y = \frac{-2c_1x + \sqrt{x^2c_1^2 + 3}}{3c_1}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

trying homogeneous types:

trying homogeneous D

<- homogeneous successful

Maple step by step

Let's solve

$$x + 2y(x) + (2x + 3y(x)) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
 - $\frac{d}{dx} y(x)$
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{dx} G(x, y(x)) = 0$
 - Compute derivative of lhs

- $$\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y)\right) \left(\frac{d}{dx}y(x)\right) = 0$$
- Evaluate derivatives
 $2 = 2$
 - Condition met, ODE is exact
 - Exact ODE implies solution will be of this form
 $\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y)\right]$
 - Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x
 $G(x, y) = \int (x + 2y) dx + _F1(y)$
 - Evaluate integral
 $G(x, y) = \frac{x^2}{2} + 2yx + _F1(y)$
 - Take derivative of $G(x, y)$ with respect to y
 $N(x, y) = \frac{\partial}{\partial y}G(x, y)$
 - Compute derivative
 $2x + 3y = 2x + \frac{d}{dy}_F1(y)$
 - Isolate for $\frac{d}{dy}_F1(y)$
 $\frac{d}{dy}_F1(y) = 3y$
 - Solve for $_F1(y)$
 $_F1(y) = \frac{3y^2}{2}$
 - Substitute $_F1(y)$ into equation for $G(x, y)$
 $G(x, y) = \frac{1}{2}x^2 + 2yx + \frac{3}{2}y^2$
 - Substitute $G(x, y)$ into the solution of the ODE
 $\frac{1}{2}x^2 + 2yx + \frac{3}{2}y^2 = C1$
 - Solve for $y(x)$
 $\left\{y(x) = -\frac{2x}{3} - \frac{\sqrt{x^2+6C1}}{3}, y(x) = -\frac{2x}{3} + \frac{\sqrt{x^2+6C1}}{3}\right\}$

✓ **Mathematica.** Time used: 0.251 (sec). Leaf size: 99

```
ode=(x+2*y[x])+(2*x+3*y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{3} \left(-2x - \sqrt{x^2 + 3e^{2c_1}} \right)$$

$$y(x) \rightarrow \frac{1}{3} \left(-2x + \sqrt{x^2 + 3e^{2c_1}} \right)$$

$$y(x) \rightarrow \frac{1}{3} \left(-\sqrt{x^2} - 2x \right)$$

$$y(x) \rightarrow \frac{1}{3} \left(\sqrt{x^2} - 2x \right)$$

✓ Sympy. Time used: 0.768 (sec). Leaf size: 36

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x + (2*x + 3*y(x))*Derivative(y(x), x) + 2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\frac{2x}{3} - \frac{\sqrt{C_1 + x^2}}{3}, y(x) = -\frac{2x}{3} + \frac{\sqrt{C_1 + x^2}}{3} \right]$$

2.2.13 Problem 38

Local contents

Solved using first_order_ode_isobaric	430
Solved using first_order_ode_homog_A	433
Solved using first_order_ode_dAlembert	436
Solved using first_order_ode_homog_type_maple_C	439
Solved using first_order_ode_LIE	444
✓ Maple	449
✓ Mathematica	449
✓ Sympy	450

Internal problem ID [7867]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 38

Date solved : Thursday, November 27, 2025 at 06:23:51 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_isobaric

Time used: 0.998 (sec)

Solve

$$2y'x - 2y = \sqrt{x^2 + 4y^2}$$

Solving for y' gives

$$y' = \frac{2y + \sqrt{x^2 + 4y^2}}{2x} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{2y + \sqrt{x^2 + 4y^2}}{2x} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = \frac{2xu(x) + \sqrt{x^2 + 4x^2u(x)^2}}{2x}$$

The ode

$$u'(x) = \frac{\sqrt{4u(x)^2 + 1}}{2x} \quad (2.47)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{\sqrt{4u(x)^2 + 1}}{2x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{2x} \\ g(u) &= \sqrt{4u^2 + 1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{\sqrt{4u^2 + 1}} du &= \int \frac{1}{2x} dx \end{aligned}$$

$$\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\sqrt{4u^2 + 1} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i}{2}$$

$$u(x) = \frac{i}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$$

$$u(x) = -\frac{i}{2}$$

$$u(x) = \frac{i}{2}$$

Converting $\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$ back to y gives

$$\frac{\operatorname{arcsinh}\left(\frac{2y}{x}\right)}{2} = \ln(\sqrt{x}) + c_1$$

Converting $u(x) = -\frac{i}{2}$ back to y gives

$$\frac{y}{x} = -\frac{i}{2}$$

Converting $u(x) = \frac{i}{2}$ back to y gives

$$\frac{y}{x} = \frac{i}{2}$$

Solving for y gives

$$\frac{\operatorname{arcsinh}\left(\frac{2y}{x}\right)}{2} = \frac{\ln(x)}{2} + c_1$$

$$y = -\frac{ix}{2}$$

$$y = \frac{ix}{2}$$

The solution

$$\frac{\operatorname{arcsinh}\left(\frac{2y}{x}\right)}{2} = \frac{\ln(x)}{2} + c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

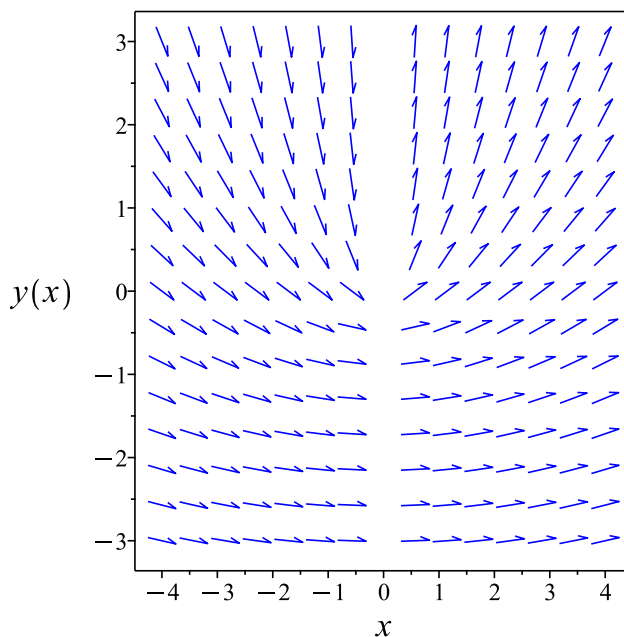


Figure 2.107: Slope field $2y'x - 2y = \sqrt{x^2 + 4y^2}$

Summary of solutions found

$$y = -\frac{ix}{2}$$

$$y = \frac{ix}{2}$$

Solved using first_order_ode_homog_A

Time used: 1.023 (sec)

Solve

$$2y'x - 2y = \sqrt{x^2 + 4y^2}$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{2y + \sqrt{x^2 + 4y^2}}{2x} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = 2y + \sqrt{x^2 + 4y^2}$ and $N = 2x$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= u + \frac{\sqrt{4u^2 + 1}}{2} \\ \frac{du}{dx} &= \frac{\sqrt{4u(x)^2 + 1}}{2x}\end{aligned}$$

Or

$$u'(x) - \frac{\sqrt{4u(x)^2 + 1}}{2x} = 0$$

Or

$$2u'(x)x - \sqrt{4u(x)^2 + 1} = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = \frac{\sqrt{4u(x)^2 + 1}}{2x} \quad (2.48)$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= \frac{\sqrt{4u(x)^2 + 1}}{2x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= \frac{1}{2x} \\ g(u) &= \sqrt{4u^2 + 1}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{\sqrt{4u^2 + 1}} du &= \int \frac{1}{2x} dx\end{aligned}$$

$$\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\sqrt{4u^2 + 1} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i}{2}$$

$$u(x) = \frac{i}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$$

$$u(x) = -\frac{i}{2}$$

$$u(x) = \frac{i}{2}$$

Converting $\frac{\operatorname{arcsinh}(2u(x))}{2} = \ln(\sqrt{x}) + c_1$ back to y gives

$$\operatorname{arcsinh}\left(\frac{2y}{x}\right) = \ln(x) + 2c_1$$

Converting $u(x) = -\frac{i}{2}$ back to y gives

$$y = -\frac{ix}{2}$$

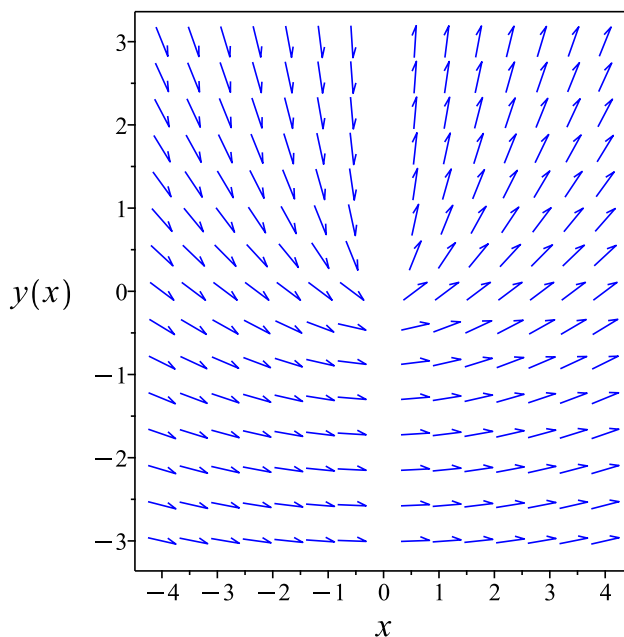
Converting $u(x) = \frac{i}{2}$ back to y gives

$$y = \frac{ix}{2}$$

The solution

$$\operatorname{arcsinh}\left(\frac{2y}{x}\right) = \ln(x) + 2c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

Figure 2.108: Slope field $2y'x - 2y = \sqrt{x^2 + 4y^2}$ Summary of solutions found

$$y = -\frac{ix}{2}$$

$$y = \frac{ix}{2}$$

Solved using first_order_ode_dAlembert

Time used: 0.443 (sec)

Solve

$$2y'x - 2y = \sqrt{x^2 + 4y^2}$$

Let $p = y'$ the ode becomes

$$2px - 2y = \sqrt{x^2 + 4y^2}$$

Solving for y from the above results in

$$y = \frac{x(4p^2 - 1)}{8p} \tag{1}$$

This has the form

$$y = xf(p) + g(p) \tag{*}$$

Where f, g are functions of $p = y'(x)$. The above ode is d'Alembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{4p^2 - 1}{8p} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{4p^2 - 1}{8p} = \left(\frac{x}{2} + \frac{x}{8p^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{4p^2 - 1}{8p} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= \frac{i}{2} \\ p_2 &= -\frac{i}{2} \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= \frac{ix}{2} \\ y &= -\frac{ix}{2} \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{4p(x)^2 - 1}{8p(x)}}{\frac{x}{2} + \frac{x}{8p(x)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{p(x)}{x} \quad (2.49)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{p(x)}{x} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(p) &= p \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{p} dp &= \int \frac{1}{x} dx \end{aligned}$$

$$\ln(p(x)) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$p(x) = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p = 0$$

for $p(x)$ gives

$$p(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

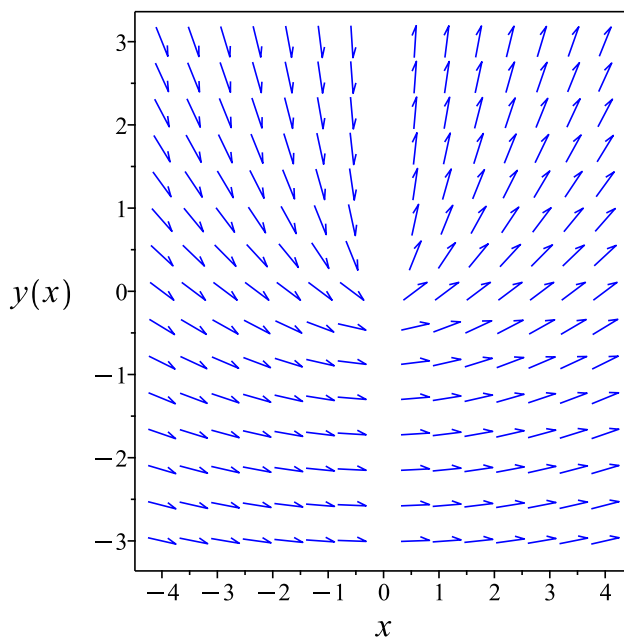
Therefore the solutions found are

$$p(x) = c_1 x$$

$$p(x) = 0$$

Substituting the above solution for p in (2A) gives

$$y = \frac{4c_1^2 x^2 - 1}{8c_1}$$

Figure 2.109: Slope field $2y'x - 2y = \sqrt{x^2 + 4y^2}$ Summary of solutions found

$$y = -\frac{ix}{2}$$

$$y = \frac{ix}{2}$$

$$y = \frac{4c_1^2x^2 - 1}{8c_1}$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.533 (sec)

Solve

$$2y'x - 2y = \sqrt{x^2 + 4y^2}$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{2Y(X) + 2y_0 + \sqrt{(X + x_0)^2 + 4(Y(X) + y_0)^2}}{2X + 2x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{2Y(X) + \sqrt{X^2 + 4Y(X)^2}}{2X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= \frac{2Y + \sqrt{X^2 + 4Y^2}}{2X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = 2Y + \sqrt{X^2 + 4Y^2}$ and $N = 2X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= u + \frac{\sqrt{4u^2 + 1}}{2} \\ \frac{du}{dX} &= \frac{\sqrt{4u(X)^2 + 1}}{2X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\sqrt{4u(X)^2 + 1}}{2X} = 0$$

Or

$$2\left(\frac{d}{dX}u(X)\right)X - \sqrt{4u(X)^2 + 1} = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = \frac{\sqrt{4u(X)^2 + 1}}{2X} \quad (2.50)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= \frac{\sqrt{4u(X)^2 + 1}}{2X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$f(X) = \frac{1}{2X}$$

$$g(u) = \sqrt{4u^2 + 1}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{1}{\sqrt{4u^2 + 1}} du = \int \frac{1}{2X} dX$$

$$\frac{\operatorname{arcsinh}(2u(X))}{2} = \ln(\sqrt{X}) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\sqrt{4u^2 + 1} = 0$$

for $u(X)$ gives

$$u(X) = -\frac{i}{2}$$

$$u(X) = \frac{i}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\operatorname{arcsinh}(2u(X))}{2} = \ln(\sqrt{X}) + c_1$$

$$u(X) = -\frac{i}{2}$$

$$u(X) = \frac{i}{2}$$

Converting $\frac{\operatorname{arcsinh}(2u(X))}{2} = \ln(\sqrt{X}) + c_1$ back to $Y(X)$ gives

$$\operatorname{arcsinh}\left(\frac{2Y(X)}{X}\right) = \ln(X) + 2c_1$$

Converting $u(X) = -\frac{i}{2}$ back to $Y(X)$ gives

$$Y(X) = -\frac{iX}{2}$$

Converting $u(X) = \frac{i}{2}$ back to $Y(X)$ gives

$$Y(X) = \frac{iX}{2}$$

Using the solution for $Y(X)$

$$\operatorname{arcsinh}\left(\frac{2Y(X)}{X}\right) = \ln(X) + 2c_1 \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$\operatorname{arcsinh}\left(\frac{2y}{x}\right) = \ln(x) + 2c_1$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{iX}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{ix}{2}$$

Using the solution for $Y(X)$

$$Y(X) = \frac{iX}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = \frac{ix}{2}$$

The solution

$$\operatorname{arcsinh}\left(\frac{2y}{x}\right) = \ln(x) + 2c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

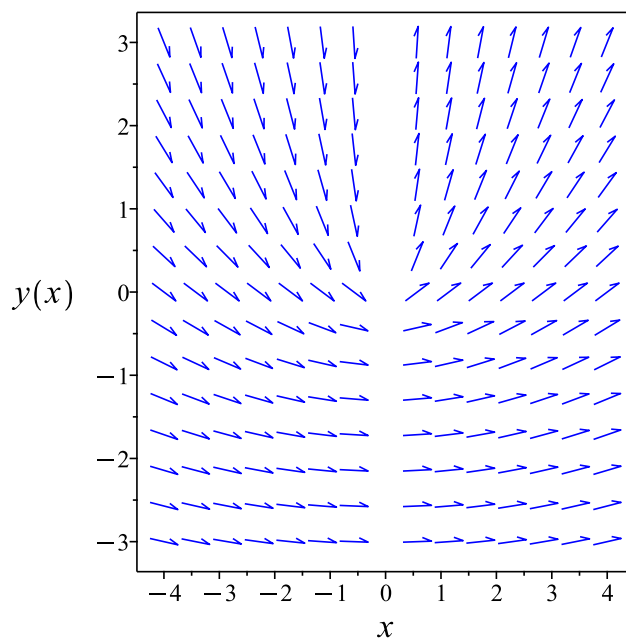


Figure 2.110: Slope field $2y'x - 2y = \sqrt{x^2 + 4y^2}$

Summary of solutions found

$$y = -\frac{ix}{2}$$

$$y = \frac{ix}{2}$$

Solved using first_order_ode_LIE

Time used: 1.437 (sec)

Solve

$$2y'x - 2y = \sqrt{x^2 + 4y^2}$$

Writing the ode as

$$y' = \frac{2y + \sqrt{x^2 + 4y^2}}{2x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = \frac{Ax + By}{Cx} \quad (1\text{E})$$

$$\eta = 1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & \frac{(2y + \sqrt{x^2 + 4y^2}) \left(-\frac{A}{Cx} + \frac{Ax+By}{Cx^2} \right)}{2x} - \frac{(2y + \sqrt{x^2 + 4y^2})^2 B}{4x^3 C} \\ & - \frac{\left(\frac{1}{2\sqrt{x^2+4y^2}} - \frac{2y+\sqrt{x^2+4y^2}}{2x^2} \right) (Ax + By)}{Cx} - \frac{2 + \frac{4y}{\sqrt{x^2+4y^2}}}{2x} = 0 \end{aligned} \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{4A\sqrt{x^2 + 4y^2} xy + 8Ax y^2 - B(x^2 + 4y^2)^{3/2} + 4B\sqrt{x^2 + 4y^2} y^2 - 2B x^2 y - 4\sqrt{x^2 + 4y^2} x^2 C - 8C x^2 y}{4\sqrt{x^2 + 4y^2} x^3 C} = 0$$

Setting the numerator to zero gives

$$4A\sqrt{x^2 + 4y^2}xy + 8Ax y^2 - B(x^2 + 4y^2)^{3/2} + 4B\sqrt{x^2 + 4y^2}y^2 - 2Bx^2y - 4\sqrt{x^2 + 4y^2}x^2C - 8Cx^2y = 0 \quad (6E)$$

Simplifying the above gives

$$2A(x^2 + 4y^2)x + 4A\sqrt{x^2 + 4y^2}xy - 2Ax^3 - B(x^2 + 4y^2)^{3/2} + 4B\sqrt{x^2 + 4y^2}y^2 - 2Bx^2y - 4\sqrt{x^2 + 4y^2}x^2C - 8Cx^2y = 0 \quad (6E)$$

Since the PDE has radicals, simplifying gives

$$x(4A\sqrt{x^2 + 4y^2}y + 8Ay^2 - B\sqrt{x^2 + 4y^2}x - 2Bxy - 4C\sqrt{x^2 + 4y^2}x - 8xCy) = 0$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, \sqrt{x^2 + 4y^2}\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, \sqrt{x^2 + 4y^2} = v_3\}$$

The above PDE (6E) now becomes

$$v_1(8Av_2^2 + 4Av_3v_2 - 2Bv_1v_2 - Bv_3v_1 - 8v_1Cv_2 - 4Cv_3v_1) = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$(-2B - 8C)v_2v_1^2 + (-B - 4C)v_3v_1^2 + 8Av_2^2v_1 + 4Av_2v_3v_1 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$4A = 0$$

$$8A = 0$$

$$-2B - 8C = 0$$

$$-B - 4C = 0$$

Solving the above equations for the unknowns gives

$$A = 0$$

$$B = -4C$$

$$C = C$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = -\frac{4y}{x}$$

$$\eta = 1$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= 1 - \left(\frac{2y + \sqrt{x^2 + 4y^2}}{2x} \right) \left(-\frac{4y}{x} \right) \\ &= \frac{2\sqrt{x^2 + 4y^2}y + x^2 + 4y^2}{x^2} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{2\sqrt{x^2 + 4y^2}y + x^2 + 4y^2}{x^2}} dy\end{aligned}$$

Which results in

$$S = x^2 \left(\frac{y}{x^2} - \frac{\sqrt{x^2 + 4y^2}}{2x^2} \right)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{2y + \sqrt{x^2 + 4y^2}}{2x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{x}{2\sqrt{x^2 + 4y^2}} \\ S_y &= 1 - \frac{2y}{\sqrt{x^2 + 4y^2}} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

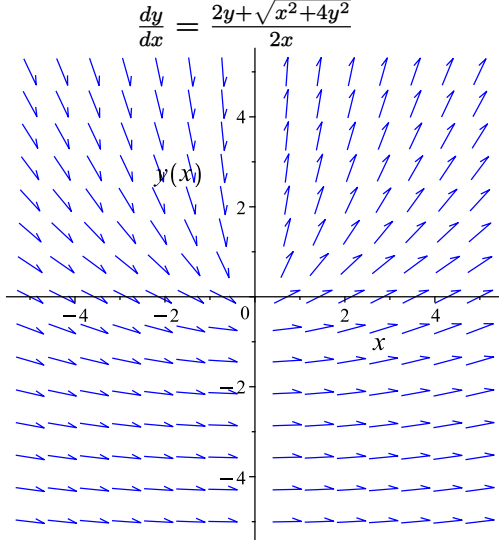
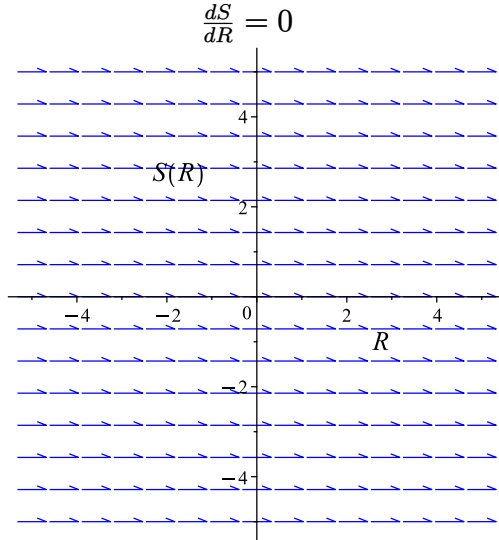
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\frac{\sqrt{x^2 + 4y^2}}{2} + y = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{2y + \sqrt{x^2 + 4y^2}}{2x}$ 	$R = x$ $S = -\frac{\sqrt{x^2 + 4y^2}}{2} + y$	$\frac{dS}{dR} = 0$ 

Solving for y gives

$$y = \frac{4c_2^2 - x^2}{8c_2}$$

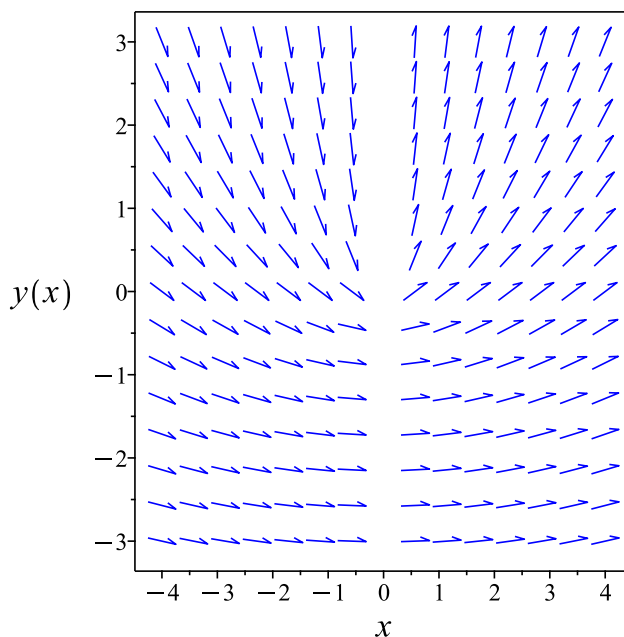


Figure 2.111: Slope field $2y'x - 2y = \sqrt{x^2 + 4y^2}$

Summary of solutions found

$$y = \frac{4c_2^2 - x^2}{8c_2}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 30

```
ode:=2*diff(y(x),x)*x-2*y(x) = (x^2+4*y(x)^2)^(1/2);
dsolve(ode,y(x), singsol=all);
```

$$\frac{-c_1 x^2 + 2y + \sqrt{x^2 + 4y^2}}{x^2} = 0$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

trying homogeneous G

1st order, trying the canonical coordinates of the invariance group

<- 1st order, canonical coordinates successful

<- homogeneous successful

Maple step by step

Let's solve

$$2x\left(\frac{d}{dx}y(x)\right) - 2y(x) = \sqrt{x^2 + 4y(x)^2}$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{2y(x) + \sqrt{x^2 + 4y(x)^2}}{2x}$$

✓ **Mathematica.** Time used: 0.194 (sec). Leaf size: 18

```
ode=2*x*D[y[x],x]-2*y[x]== Sqrt[x^2+4*y[x]^2];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}x \sinh(\log(x) + 2c_1)$$

✓ Sympy. Time used: 0.783 (sec). Leaf size: 14

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*Derivative(y(x), x) - sqrt(x**2 + 4*y(x)**2) - 2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = -\frac{x \sinh(C_1 - \log(x))}{2}$$

2.2.14 Problem 39

Local contents

Solved using first_order_ode_dAlembert	451
Solved using first_order_ode_homog_type_maple_C	455
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	460
Solved using first_order_ode_LIE	470
✓ Maple	475
✓ Mathematica	475
✓ Sympy	476

Internal problem ID [7868]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 39

Date solved : Thursday, November 27, 2025 at 06:24:07 AM

CAS classification :

[[_homogeneous, 'class C'], _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_dAlembert

Time used: 5.525 (sec)

Solve

$$3y - 7x + 7 + (7y - 3x + 3)y' = 0$$

Let $p = y'$ the ode becomes

$$3y - 7x + 7 + (7y - 3x + 3)p = 0$$

Solving for y from the above results in

$$y = \frac{(3p + 7)x}{3 + 7p} + \frac{-3p - 7}{3 + 7p} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{3p + 7}{3 + 7p} \\ g &= \frac{-3p - 7}{3 + 7p} \end{aligned}$$

Hence (2) becomes

$$p - \frac{3p + 7}{3 + 7p} = \left(\frac{3x}{3 + 7p} - \frac{21xp}{(3 + 7p)^2} - \frac{49x}{(3 + 7p)^2} - \frac{3}{3 + 7p} + \frac{21p}{(3 + 7p)^2} + \frac{49}{(3 + 7p)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{3p + 7}{3 + 7p} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= 1 \\ p_2 &= -1 \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= x - 1 \\ y &= 1 - x \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{3p(x)+7}{3+7p(x)}}{\frac{3x}{3+7p(x)} - \frac{21xp(x)}{(3+7p(x))^2} - \frac{49x}{(3+7p(x))^2} - \frac{3}{3+7p(x)} + \frac{21p(x)}{(3+7p(x))^2} + \frac{49}{(3+7p(x))^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{7(p(x) - 1)(3 + 7p(x))(p(x) + 1)}{40(x - 1)} \quad (2.51)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{7(p(x)-1)(3+7p(x))(p(x)+1)}{40(x-1)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{7}{40(x-1)} \\ g(p) &= (p-1)(3+7p)(p+1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{(p-1)(3+7p)(p+1)} dp &= \int -\frac{7}{40(x-1)} dx \end{aligned}$$

$$\frac{\ln(p(x)+1)}{8} + \frac{\ln(p(x)-1)}{20} - \frac{7 \ln(3+7p(x))}{40} = \ln \left(\frac{1}{(x-1)^{7/40}} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$(p-1)(3+7p)(p+1) = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= -1 \\ p(x) &= 1 \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \frac{\ln(p(x)+1)}{8} + \frac{\ln(p(x)-1)}{20} - \frac{7 \ln(3+7p(x))}{40} &= \ln \left(\frac{1}{(x-1)^{7/40}} \right) + c_1 \\ p(x) &= -1 \\ p(x) &= 1 \end{aligned}$$

Substituting the above solution for p in (2A) gives

Expression too large to display

$$y = 1 - x$$

$$y = x - 1$$

Simplifying the above gives

$$y = x - 1$$

$$y = 1 - x$$

Expression too large to display

$$y = 1 - x$$

$$y = x - 1$$

The solution

Expression too large to display

was found not to satisfy the ode or the IC. Hence it is removed.

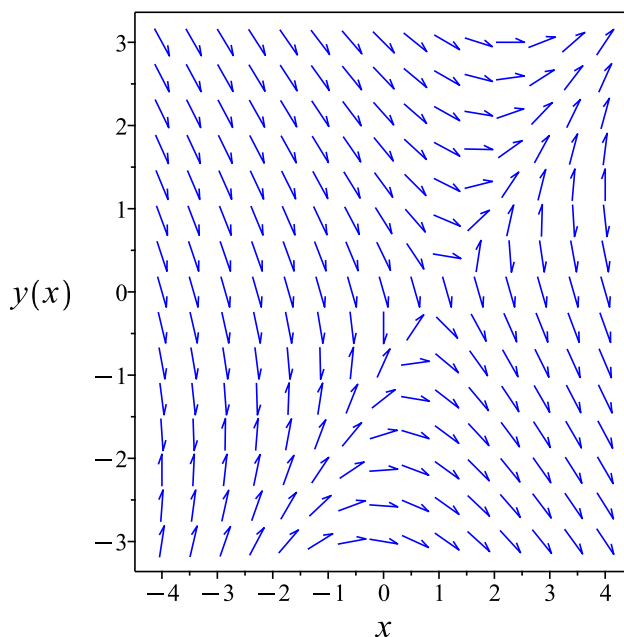


Figure 2.112: Slope field $3y - 7x + 7 + (7y - 3x + 3)y' = 0$

Summary of solutions found

$$y = 1 - x$$

$$y = x - 1$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.905 (sec)

Solve

$$3y - 7x + 7 + (7y - 3x + 3)y' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{3Y(X) + 3y_0 - 7X - 7x_0 + 7}{7Y(X) + 7y_0 - 3X - 3x_0 + 3}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 1$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{3Y(X) - 7X}{7Y(X) - 3X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{3Y - 7X}{7Y - 3X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = 3Y - 7X$ and $N = -7Y + 3X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= \frac{-3u + 7}{7u - 3} \\ \frac{du}{dX} &= \frac{\frac{-3u(X)+7}{7u(X)-3} - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{-3u(X)+7}{7u(X)-3} - u(X)}{X} = 0$$

Or

$$7\left(\frac{d}{dX}u(X)\right)Xu(X) + 7u(X)^2 - 3\left(\frac{d}{dX}u(X)\right)X - 7 = 0$$

Or

$$-7 + X(7u(X) - 3)\left(\frac{d}{dX}u(X)\right) + 7u(X)^2 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{7(u(X) - 1)(u(X) + 1)}{X(7u(X) - 3)} \quad (2.52)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{7(u(X) - 1)(u(X) + 1)}{X(7u(X) - 3)} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{7}{X} \\ g(u) &= \frac{(u - 1)(u + 1)}{7u - 3} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{7u - 3}{(u - 1)(u + 1)} du &= \int -\frac{7}{X} dX \end{aligned}$$

$$5 \ln(u(X) + 1) + 2 \ln(u(X) - 1) = \ln\left(\frac{1}{X^7}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{(u - 1)(u + 1)}{7u - 3} = 0$$

for $u(X)$ gives

$$u(X) = -1$$

$$u(X) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$5 \ln(u(X) + 1) + 2 \ln(u(X) - 1) = \ln\left(\frac{1}{X^7}\right) + c_1$$

$$u(X) = -1$$

$$u(X) = 1$$

Converting $5 \ln(u(X) + 1) + 2 \ln(u(X) - 1) = \ln\left(\frac{1}{X^7}\right) + c_1$ back to $Y(X)$ gives

$$5 \ln\left(\frac{Y(X) + X}{X}\right) + 2 \ln\left(\frac{Y(X) - X}{X}\right) = \ln\left(\frac{1}{X^7}\right) + c_1$$

Converting $u(X) = -1$ back to $Y(X)$ gives

$$Y(X) = -X$$

Converting $u(X) = 1$ back to $Y(X)$ gives

$$Y(X) = X$$

Using the solution for $Y(X)$

$$5 \ln\left(\frac{Y(X) + X}{X}\right) + 2 \ln\left(\frac{Y(X) - X}{X}\right) = \ln\left(\frac{1}{X^7}\right) + c_1 \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$5 \ln\left(\frac{y + x - 1}{x - 1}\right) + 2 \ln\left(\frac{-x + y + 1}{x - 1}\right) = \ln\left(\frac{1}{(x - 1)^7}\right) + c_1$$

Using the solution for $Y(X)$

$$Y(X) = -X \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$y = 1 - x$$

Using the solution for $Y(X)$

$$Y(X) = X \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

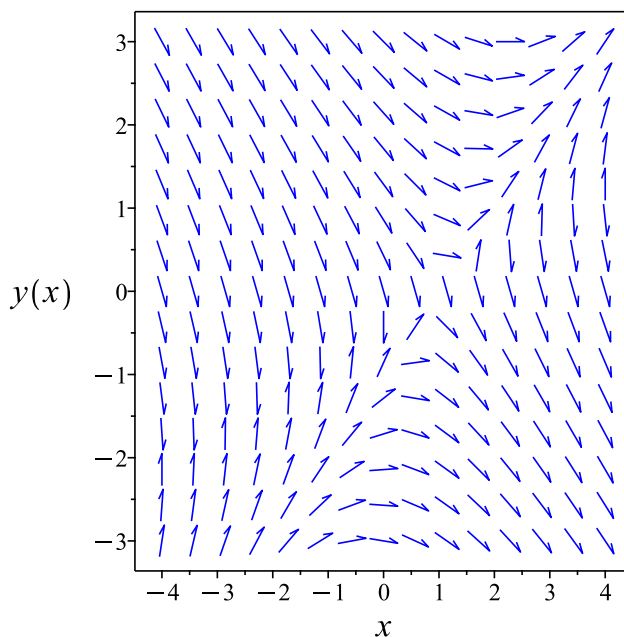
Or

$$Y = y$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$y = x - 1$$

Figure 2.113: Slope field $3y - 7x + 7 + (7y - 3x + 3)y' = 0$ Summary of solutions found

$$5 \ln \left(\frac{y + x - 1}{x - 1} \right) + 2 \ln \left(\frac{-x + y + 1}{x - 1} \right) = \ln \left(\frac{1}{(x - 1)^7} \right) + c_1$$

$$y = 1 - x$$

$$y = x - 1$$

Solve

$$3y - 7x + 7 + (7y - 3x + 3)y' = 0$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$3xu(x) - 7x + 7 + (7xu(x) - 3x + 3)(u(x) + xu'(x)) = 0$$

Which is now solved Unknown ode type.

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 51.658 (sec)

This is Abel second kind ODE, it has the form

$$(y(x) + g) y'(x) = f_0(x) + f_1(x)y(x) + f_2(x)y(x)^2 + f_3(x)y(x)^3$$

Comparing the above to given ODE which is

$$3y(x) - 7x + 7 + (7y(x) - 3x + 3) y'(x) = 0 \quad (1)$$

Shows that

$$g = -\frac{3x}{7} + \frac{3}{7}$$

$$f_0 = x - 1$$

$$f_1 = -\frac{3}{7}$$

$$f_2 = 0$$

$$f_3 = 0$$

Applying transformation

$$y(x) = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = \left(-\frac{40x}{49} + \frac{40}{49} \right) u(x)^3 + \frac{6u(x)^2}{7}$$

Which is now solved.

Solve This is Abel first kind ODE, it has the form

$$u'(x) = f_0(x) + f_1(x)u(x) + f_2(x)u(x)^2 + f_3(x)u(x)^3$$

Comparing the above to given ODE which is

$$u'(x) = \left(-\frac{40x}{49} + \frac{40}{49} \right) u(x)^3 + \frac{6u(x)^2}{7} \quad (1)$$

Therefore

$$f_0 = 0$$

$$f_1 = 0$$

$$f_2 = \frac{6}{7}$$

$$f_3 = -\frac{40x}{49} + \frac{40}{49}$$

Hence

$$\begin{aligned} f'_0 &= 0 \\ f'_3 &= -\frac{40}{49} \end{aligned}$$

Since $f_2(x) = \frac{6}{7}$ is not zero, then the following transformation is used to remove f_2 . Let $u(x) = u(x) - \frac{f_2}{3f_3}$ or

$$\begin{aligned} u(x) &= u(x) - \left(\frac{\frac{6}{7}}{-\frac{120x}{49} + \frac{120}{49}} \right) \\ &= u(x) + \frac{7}{20x - 20} \end{aligned}$$

The above transformation applied to (1) gives a new ODE as

$$u'(x) = -\frac{(2000x^3 - 6000x^2 + 6000x - 2000)u(x)^3}{2450(x-1)^2} - \frac{(-735x + 735)u(x)}{2450(x-1)^2} + \frac{21}{50(x-1)^2} \quad (2)$$

The above ODE (2) can now be solved.

Solve Writing the ode as

$$\begin{aligned} u'(x) &= -\frac{2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029}{2450(x-1)^2} \\ u'(x) &= \omega(x, u(x)) \end{aligned}$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_{u(x)} - \xi_x) - \omega^2 \xi_{u(x)} - \omega_x \xi - \omega_{u(x)} \eta = 0 \quad (A)$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = ua_3 + xa_2 + a_1 \quad (1E)$$

$$\eta = ub_3 + xb_2 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned}
 & b_2 - \frac{(2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029)(b_3 - a_2)}{2450(x-1)^2} \\
 & - \frac{(2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029)^2 a_3}{6002500(x-1)^4} \\
 & - \left(-\frac{6000u^3x^2 - 12000u^3x + 6000u^3 - 735u}{2450(x-1)^2} \right. \\
 & + \frac{2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029}{1225(x-1)^3} \left. \right) (ua_3 + xa_2 + a_1) \\
 & + \frac{(6000u^2x^3 - 18000u^2x^2 + 18000u^2x - 6000u^2 - 735x + 735)(ub_3 + xb_2 + b_1)}{2450(x-1)^2} = 0
 \end{aligned} \tag{5E}$$

Putting the above in normal form gives

$$\begin{aligned}
 & -4000000u^6x^6a_3 - 24000000u^6x^5a_3 + 60000000u^6x^4a_3 - 80000000u^6x^3a_3 + 60000000u^6x^2a_3 - 7840000u^4x^4a_3 \\
 & = 0
 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned}
 & -4000000u^6x^6a_3 + 24000000u^6x^5a_3 - 60000000u^6x^4a_3 \\
 & + 80000000u^6x^3a_3 - 60000000u^6x^2a_3 + 7840000u^4x^4a_3 \\
 & + 9800000u^3x^5a_2 + 9800000u^3x^5b_3 + 14700000u^2x^6b_2 \\
 & + 24000000u^6xa_3 - 31360000u^4x^3a_3 + 4900000u^3x^4a_1 \\
 & - 44100000u^3x^4a_2 - 49000000u^3x^4b_3 + 14700000u^2x^5b_1 \\
 & - 73500000u^2x^5b_2 - 4000000u^6a_3 + 47040000u^4x^2a_3 \\
 & - 19600000u^3x^3a_1 + 78400000u^3x^3a_2 + 4116000u^3x^3a_3 \\
 & + 98000000u^3x^3b_3 - 73500000u^2x^4b_1 + 147000000u^2x^4b_2 \\
 & - 31360000u^4xa_3 + 29400000u^3x^2a_1 - 68600000u^3x^2a_2 \\
 & - 12348000u^3x^2a_3 - 98000000u^3x^2b_3 + 147000000u^2x^3b_1 \\
 & - 147000000u^2x^3b_2 + 7840000u^4a_3 - 19600000u^3xa_1 \\
 & + 29400000u^3xa_2 + 12348000u^3xa_3 + 49000000u^3xb_3 \\
 & + 1260525u^2x^2a_3 - 147000000u^2x^2b_1 + 73500000u^2x^2b_2 \\
 & + 4201750b_2x^4 + 4900000u^3a_1 - 4900000u^3a_2 - 4116000u^3a_3 \\
 & - 9800000u^3b_3 - 2521050u^2xa_3 + 73500000u^2xb_1 \\
 & - 14700000u^2xb_2 + 1800750u^2xa_1 + 1800750u^2xa_2 \\
 & - 1800750x^3b_1 - 18607750b_2x^3 + 1260525u^2a_3 - 14700000u^2b_1 \\
 & - 3601500uxa_1 - 3601500uxa_2 + 3529470uxa_3 + 2521050x^2a_2 \\
 & + 5402250x^2b_1 + 30612750b_2x^2 + 2521050x^2b_3 + 1800750ua_1 \\
 & + 1800750ua_2 - 3529470ua_3 + 5042100xa_1 - 5402250xb_1 \\
 & - 22209250xb_2 - 5042100xb_3 - 5042100a_1 - 2521050a_2 \\
 & - 1058841a_3 + 1800750b_1 + 6002500b_2 + 2521050b_3 = 0
 \end{aligned} \tag{6E}$$

Looking at the above PDE shows the following are all the terms with $\{u, x\}$ in them.

$$\{u, x\}$$

The following substitution is now made to be able to collect on all terms with $\{u, x\}$ in them

$$\{u = v_1, x = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
& -4000000a_3v_1^6v_2^6 + 24000000a_3v_1^6v_2^5 - 60000000a_3v_1^6v_2^4 \\
& + 80000000a_3v_1^6v_2^3 + 98000000a_2v_1^3v_2^5 - 60000000a_3v_1^6v_2^2 \\
& + 78400000a_3v_1^4v_2^4 + 147000000b_2v_1^2v_2^6 + 98000000b_3v_1^3v_2^5 \\
& + 49000000a_1v_1^3v_2^4 - 44100000a_2v_1^3v_2^4 + 24000000a_3v_1^6v_2 \\
& - 31360000a_3v_1^4v_2^3 + 147000000b_1v_1^2v_2^5 - 735000000b_2v_1^2v_2^5 \\
& - 490000000b_3v_1^3v_2^4 - 196000000a_1v_1^3v_2^3 + 784000000a_2v_1^3v_2^3 \\
& - 40000000a_3v_1^6 + 470400000a_3v_1^4v_2^2 + 41160000a_3v_1^3v_2^3 \\
& - 735000000b_1v_1^2v_2^4 + 1470000000b_2v_1^2v_2^4 + 980000000b_3v_1^3v_2^3 \\
& + 294000000a_1v_1^3v_2^2 - 686000000a_2v_1^3v_2^2 - 31360000a_3v_1^4v_2 \\
& - 12348000a_3v_1^3v_2^2 + 1470000000b_1v_1^2v_2^3 - 1470000000b_2v_1^2v_2^3 \\
& - 980000000b_3v_1^3v_2^2 - 196000000a_1v_1^3v_2 + 294000000a_2v_1^3v_2 \\
& + 78400000a_3v_1^4 + 12348000a_3v_1^3v_2 + 1260525a_3v_1^2v_2^2 \\
& - 1470000000b_1v_1^2v_2^2 + 735000000b_2v_1^2v_2^2 + 4201750b_2v_2^4 \\
& + 490000000b_3v_1^3v_2 + 49000000a_1v_1^3 + 1800750a_1v_1v_2^2 \\
& - 49000000a_2v_1^3 + 1800750a_2v_1v_2^2 - 4116000a_3v_1^3 \\
& - 2521050a_3v_1^2v_2 + 735000000b_1v_1^2v_2 - 1800750b_1v_2^3 \\
& - 147000000b_2v_1^2v_2 - 18607750b_2v_2^3 - 98000000b_3v_1^3 \\
& - 3601500a_1v_1v_2 - 3601500a_2v_1v_2 + 2521050a_2v_2^2 \\
& + 1260525a_3v_1^2 + 3529470a_3v_1v_2 - 147000000b_1v_1^2 \\
& + 5402250b_1v_2^2 + 30612750b_2v_2^2 + 2521050b_3v_2^2 + 1800750a_1v_1 \\
& + 5042100a_1v_2 + 1800750a_2v_1 - 3529470a_3v_1 - 5402250b_1v_2 \\
& - 22209250b_2v_2 - 5042100b_3v_2 - 5042100a_1 - 2521050a_2 \\
& - 1058841a_3 + 1800750b_1 + 6002500b_2 + 2521050b_3 = 0
\end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned}
& (9800000a_2 + 9800000b_3) v_1^3 v_2^5 \\
& + (4900000a_1 - 44100000a_2 - 49000000b_3) v_1^3 v_2^4 \\
& + (-19600000a_1 + 78400000a_2 + 4116000a_3 + 98000000b_3) v_1^3 v_2^3 \\
& + (29400000a_1 - 68600000a_2 - 12348000a_3 - 98000000b_3) v_1^3 v_2^2 \\
& + (-19600000a_1 + 29400000a_2 + 12348000a_3 + 49000000b_3) v_1^3 v_2 \\
& + (14700000b_1 - 73500000b_2) v_1^2 v_2^5 \\
& + (-73500000b_1 + 147000000b_2) v_1^2 v_2^4 \\
& + (147000000b_1 - 147000000b_2) v_1^2 v_2^3 \\
& + (1260525a_3 - 147000000b_1 + 73500000b_2) v_1^2 v_2^2 \\
& + (-2521050a_3 + 73500000b_1 - 14700000b_2) v_1^2 v_2 \\
& + (1800750a_1 + 1800750a_2) v_1 v_2^2 \\
& + (-3601500a_1 - 3601500a_2 + 3529470a_3) v_1 v_2 \\
& - 4000000a_3 v_1^6 + 7840000a_3 v_1^4 + 4201750b_2 v_2^4 \\
& + (4900000a_1 - 4900000a_2 - 4116000a_3 - 9800000b_3) v_1^3 \\
& + (1260525a_3 - 14700000b_1) v_1^2 \\
& + (1800750a_1 + 1800750a_2 - 3529470a_3) v_1 \\
& + (-1800750b_1 - 18607750b_2) v_2^3 \\
& + (2521050a_2 + 5402250b_1 + 30612750b_2 + 2521050b_3) v_2^2 \\
& + (5042100a_1 - 5402250b_1 - 22209250b_2 - 5042100b_3) v_2 \\
& - 4000000a_3 v_1^6 v_2^6 + 24000000a_3 v_1^6 v_2^5 - 60000000a_3 v_1^6 v_2^4 \\
& + 80000000a_3 v_1^6 v_2^3 - 60000000a_3 v_1^6 v_2^2 + 7840000a_3 v_1^4 v_2^4 \\
& + 14700000b_2 v_1^2 v_2^6 + 24000000a_3 v_1^6 v_2 - 31360000a_3 v_1^4 v_2^3 \\
& + 47040000a_3 v_1^4 v_2^2 - 31360000a_3 v_1^4 v_2 - 1058841a_3 + 1800750b_1 \\
& + 6002500b_2 + 2521050b_3 - 5042100a_1 - 2521050a_2 = 0
\end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned}
 -60000000a_3 &= 0 \\
 -31360000a_3 &= 0 \\
 -4000000a_3 &= 0 \\
 7840000a_3 &= 0 \\
 24000000a_3 &= 0 \\
 47040000a_3 &= 0 \\
 80000000a_3 &= 0 \\
 4201750b_2 &= 0 \\
 14700000b_2 &= 0 \\
 1800750a_1 + 1800750a_2 &= 0 \\
 9800000a_2 + 9800000b_3 &= 0 \\
 1260525a_3 - 14700000b_1 &= 0 \\
 -73500000b_1 + 147000000b_2 &= 0 \\
 -1800750b_1 - 18607750b_2 &= 0 \\
 14700000b_1 - 73500000b_2 &= 0 \\
 147000000b_1 - 147000000b_2 &= 0 \\
 -3601500a_1 - 3601500a_2 + 3529470a_3 &= 0 \\
 1800750a_1 + 1800750a_2 - 3529470a_3 &= 0 \\
 4900000a_1 - 44100000a_2 - 49000000b_3 &= 0 \\
 -2521050a_3 + 73500000b_1 - 14700000b_2 &= 0 \\
 1260525a_3 - 147000000b_1 + 73500000b_2 &= 0 \\
 -19600000a_1 + 29400000a_2 + 12348000a_3 + 49000000b_3 &= 0 \\
 -19600000a_1 + 78400000a_2 + 4116000a_3 + 98000000b_3 &= 0 \\
 4900000a_1 - 4900000a_2 - 4116000a_3 - 9800000b_3 &= 0 \\
 5042100a_1 - 5402250b_1 - 22209250b_2 - 5042100b_3 &= 0 \\
 29400000a_1 - 68600000a_2 - 12348000a_3 - 98000000b_3 &= 0 \\
 2521050a_2 + 5402250b_1 + 30612750b_2 + 2521050b_3 &= 0 \\
 -5042100a_1 - 2521050a_2 - 1058841a_3 + 1800750b_1 + 6002500b_2 + 2521050b_3 &= 0
 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}a_1 &= b_3 \\a_2 &= -b_3 \\a_3 &= 0 \\b_1 &= 0 \\b_2 &= 0 \\b_3 &= b_3\end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}\xi &= 1 - x \\ \eta &= u\end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, u) \xi \\ &= u - \left(-\frac{2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029}{2450(x-1)^2} \right) (1-x) \\ &= \frac{1029 - 2000(x-1)^3u^3 + (3185x - 3185)u}{2450x - 2450} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, u) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{du}{\eta} = dS \quad (1)$$

The above comes from the requirements that $(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial u}) S(x, u) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{1029 - 2000(x-1)^3u^3 + (3185x - 3185)u}{2450x - 2450}} dy\end{aligned}$$

Which results in

$$S = (2450x - 2450) \left(-\frac{\ln(5ux - 5u - 7)}{1715(-5 + 5x)} - \frac{2 \ln(20ux - 20u + 21)}{343(20x - 20)} + \frac{2 \ln(20ux - 20u + 7)}{245(20x - 20)} \right)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, u)S_u}{R_x + \omega(x, u)R_u} \quad (2)$$

Where in the above R_x, R_u, S_x, S_u are all partial derivatives and $\omega(x, u)$ is the right hand side of the original ode given by

$$\omega(x, u) = -\frac{2000u^3x^3 - 6000u^3x^2 + 6000u^3x - 2000u^3 - 735ux + 735u - 1029}{2450(x - 1)^2}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_u &= 0 \\ S_x &= -\frac{2450u}{-1029 + 2000(x - 1)^3u^3 + (-3185x + 3185)u} \\ S_u &= \frac{-2450x + 2450}{-1029 + 2000(x - 1)^3u^3 + (-3185x + 3185)u} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{x - 1} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, u in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R - 1}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{1}{R - 1} dR \\ S(R) &= \ln(R - 1) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, u coordinates. This results in

$$-\frac{2 \ln (-7+(-5+5 x) u(x))}{7}-\frac{5 \ln (21+20(x-1) u(x))}{7}+\ln (7+20(x-1) u(x))=\ln (x-1)+c_2$$

Substituting $u = u(x) + \frac{2}{7(-\frac{40x}{49} + \frac{40}{49})}$ in the above solution gives

$$-\frac{2 \ln \left(-7+(-5+5 x)\left(u(x)+\frac{2}{7\left(-\frac{40 x}{49}+\frac{40}{49}\right)}\right)\right)}{7}-\frac{5 \ln \left(21+20(x-1)\left(u(x)+\frac{2}{7\left(-\frac{40 x}{49}+\frac{40}{49}\right)}\right)\right)}{7}+\ln \left(7+20\left(x\right.\right.$$

Simplifying the above gives

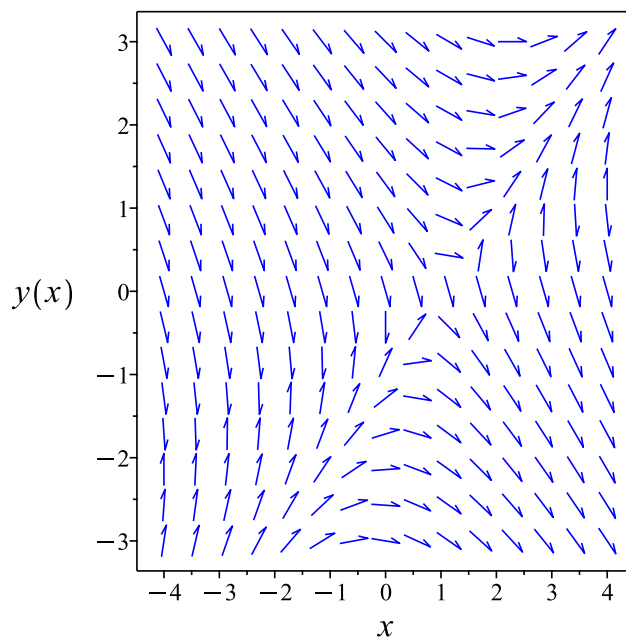
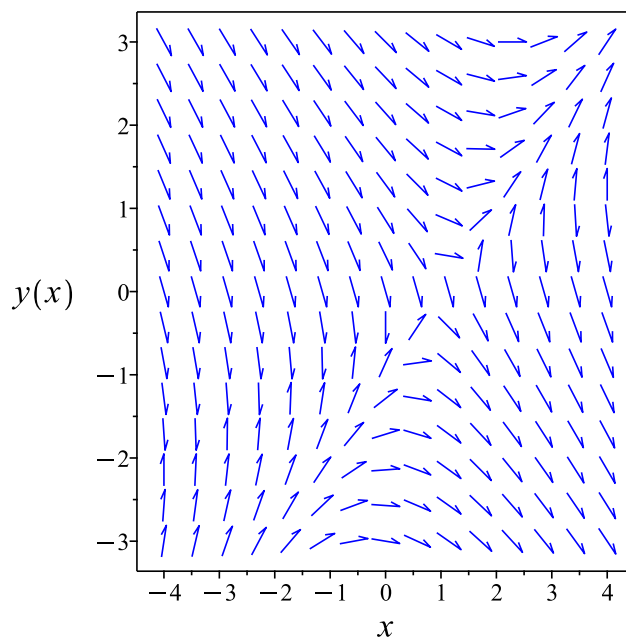
$$\frac{13 \ln (2)}{7}+\frac{5 \ln (5)}{7}-\frac{2 \ln (-7+(4 x-4) u(x))}{7}-\frac{5 \ln (7+10 u(x)(x-1))}{7}+\ln (u(x)(x-1))=\ln (x-1)+c_2$$

Substituting $u(x) = \frac{1}{y(x) - \frac{3x}{7} + \frac{3}{7}}$ in the above solution gives

$$\frac{13 \ln (2)}{7}+\frac{5 \ln (5)}{7}-\frac{2 \ln \left(-7+\frac{4 x-4}{y(x)-\frac{3 x}{7}+\frac{3}{7}}\right)}{7}+\ln \left(\frac{x-1}{y(x)-\frac{3 x}{7}+\frac{3}{7}}\right)-\frac{5 \ln \left(7+\frac{10 x-10}{y(x)-\frac{3 x}{7}+\frac{3}{7}}\right)}{7} \\ =\ln (x-1)+c_2$$

Simplifying the above gives

$$\frac{13 \ln (2)}{7}+\frac{5 \ln (5)}{7}-\ln (7)-\frac{2 \ln \left(\frac{-x+y(x)+1}{-7 y(x)+3 x-3}\right)}{7} \\ +\ln \left(\frac{x-1}{7 y(x)-3 x+3}\right)-\frac{5 \ln \left(\frac{y(x)+x-1}{7 y(x)-3 x+3}\right)}{7}=\ln (x-1)+c_2$$

Figure 2.114: Slope field $3y(x) - 7x + 7 + (7y(x) - 3x + 3)y'(x) = 0$ Figure 2.115: Slope field $3y(x) - 7x + 7 + (7y(x) - 3x + 3)y'(x) = 0$

Summary of solutions found

$$\begin{aligned} & \frac{13 \ln(2)}{7} + \frac{5 \ln(5)}{7} - \ln(7) - \frac{2 \ln\left(\frac{-x+y(x)+1}{-7y(x)+3x-3}\right)}{7} \\ & + \ln\left(\frac{x-1}{7y(x)-3x+3}\right) - \frac{5 \ln\left(\frac{y(x)+x-1}{7y(x)-3x+3}\right)}{7} = \ln(x-1) + c_2 \end{aligned}$$

Solved using first_order_ode_LIE

Time used: 1.664 (sec)

Solve

$$3y(x) - 7x + 7 + (7y(x) - 3x + 3)y'(x) = 0$$

Writing the ode as

$$y'(x) = -\frac{3y - 7x + 7}{7y - 3x + 3}$$

$$y'(x) = \omega(x, y(x))$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_{y(x)} - \xi_x) - \omega^2 \xi_{y(x)} - \omega_x \xi - \omega_{y(x)} \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(3y - 7x + 7)(b_3 - a_2)}{7y - 3x + 3} - \frac{(3y - 7x + 7)^2 a_3}{(7y - 3x + 3)^2}$$

$$- \left(\frac{7}{7y - 3x + 3} - \frac{3(3y - 7x + 7)}{(7y - 3x + 3)^2} \right) (xa_2 + ya_3 + a_1) \quad (5\text{E})$$

$$- \left(-\frac{3}{7y - 3x + 3} + \frac{21y - 49x + 49}{(7y - 3x + 3)^2} \right) (xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$21x^2a_2 - 49x^2a_3 + 49x^2b_2 - 21x^2b_3 - 98xya_2 + 42xya_3 - 42xyb_2 + 98xyb_3 + 21y^2a_2 - 49y^2a_3 + 49y^2b_2 - 21y^2b_3$$

$$= 0$$

Setting the numerator to zero gives

$$\begin{aligned}
 &21x^2a_2 - 49x^2a_3 + 49x^2b_2 - 21x^2b_3 - 98xya_2 + 42xya_3 - 42xyb_2 + 98xyb_3 \\
 &+ 21y^2a_2 - 49y^2a_3 + 49y^2b_2 - 21y^2b_3 - 42xa_2 + 98xa_3 + 40xb_1 - 58xb_2 + 42xb_3 \\
 &- 40ya_1 + 58ya_2 - 42ya_3 + 42yb_2 - 98yb_3 + 21a_2 - 49a_3 - 40b_1 + 9b_2 - 21b_3 = 0
 \end{aligned} \tag{6E}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
 &21a_2v_1^2 - 98a_2v_1v_2 + 21a_2v_2^2 - 49a_3v_1^2 + 42a_3v_1v_2 - 49a_3v_2^2 + 49b_2v_1^2 - 42b_2v_1v_2 \\
 &+ 49b_2v_2^2 - 21b_3v_1^2 + 98b_3v_1v_2 - 21b_3v_2^2 - 40a_1v_2 - 42a_2v_1 + 58a_2v_2 + 98a_3v_1 - 42a_3v_2 \\
 &+ 40b_1v_1 - 58b_2v_1 + 42b_2v_2 + 42b_3v_1 - 98b_3v_2 + 21a_2 - 49a_3 - 40b_1 + 9b_2 - 21b_3 = 0
 \end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned}
 &(21a_2 - 49a_3 + 49b_2 - 21b_3)v_1^2 + (-98a_2 + 42a_3 - 42b_2 + 98b_3)v_1v_2 \\
 &+ (-42a_2 + 98a_3 + 40b_1 - 58b_2 + 42b_3)v_1 + (21a_2 - 49a_3 + 49b_2 - 21b_3)v_2^2 \\
 &+ (-40a_1 + 58a_2 - 42a_3 + 42b_2 - 98b_3)v_2 + 21a_2 - 49a_3 - 40b_1 + 9b_2 - 21b_3 = 0
 \end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned}
 &-98a_2 + 42a_3 - 42b_2 + 98b_3 = 0 \\
 &21a_2 - 49a_3 + 49b_2 - 21b_3 = 0 \\
 &-40a_1 + 58a_2 - 42a_3 + 42b_2 - 98b_3 = 0 \\
 &-42a_2 + 98a_3 + 40b_1 - 58b_2 + 42b_3 = 0 \\
 &21a_2 - 49a_3 - 40b_1 + 9b_2 - 21b_3 = 0
 \end{aligned}$$

Solving the above equations for the unknowns gives

$$a_1 = -b_3$$

$$a_2 = b_3$$

$$a_3 = b_2$$

$$b_1 = -b_2$$

$$b_2 = b_2$$

$$b_3 = b_3$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = y$$

$$\eta = x - 1$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= x - 1 - \left(-\frac{3y - 7x + 7}{7y - 3x + 3} \right) (y) \\ &= \frac{3x^2 - 3y^2 - 6x + 3}{-7y + 3x - 3} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{3x^2 - 3y^2 - 6x + 3}{-7y + 3x - 3}} dy \end{aligned}$$

Which results in

$$S = \frac{5 \ln(y + x - 1)}{3} + \frac{2 \ln(-x + y + 1)}{3}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{3y - 7x + 7}{7y - 3x + 3}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{5}{3y + 3x - 3} + \frac{2}{3x - 3y - 3} \\ S_y &= \frac{5}{3y + 3x - 3} - \frac{2}{3x - 3y - 3} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{5 \ln(y(x) + x - 1)}{3} + \frac{2 \ln(-x + y(x) + 1)}{3} = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

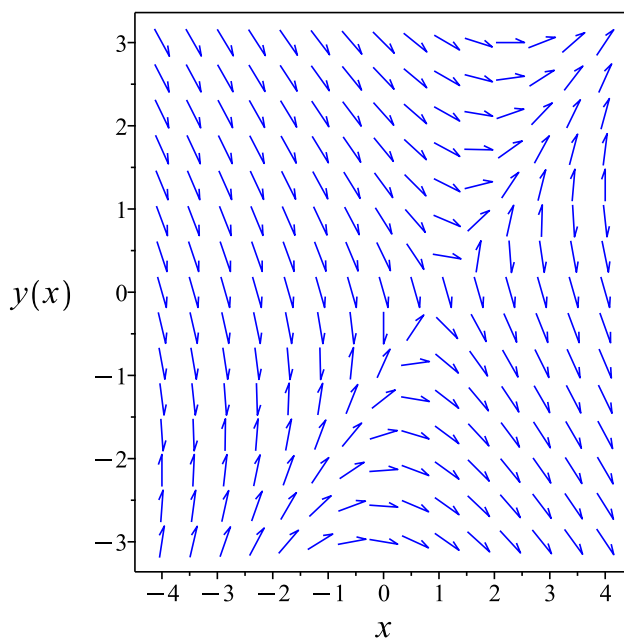
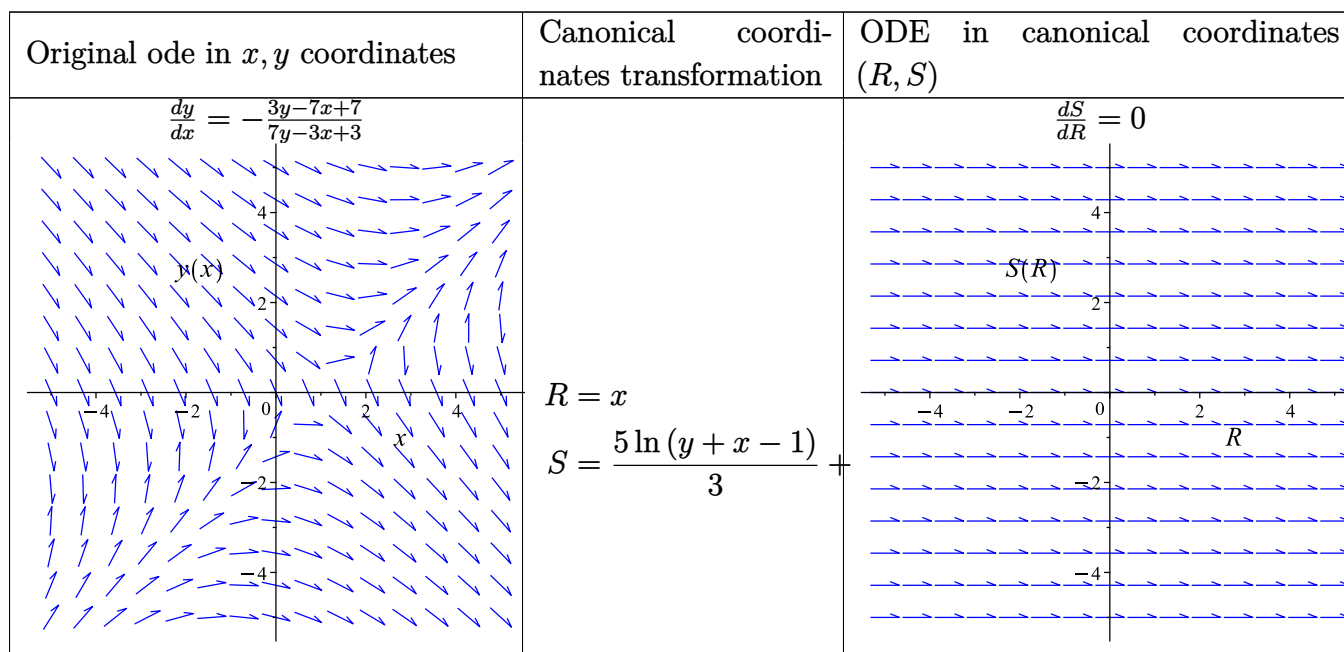


Figure 2.116: Slope field $3y(x) - 7x + 7 + (7y(x) - 3x + 3)y'(x) = 0$

Summary of solutions found

$$\frac{5 \ln(y(x) + x - 1)}{3} + \frac{2 \ln(-x + y(x) + 1)}{3} = c_2$$

✓ **Maple.** Time used: 0.138 (sec). Leaf size: 717

```
ode:=3*y(x)-7*x+7+(7*y(x)-3*x+3)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

Expression too large to display

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous C
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
<- homogeneous successful
```

Maple step by step

Let's solve

$$3y(x) - 7x + 7 + (7y(x) - 3x + 3) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx} y(x)$
- Solve for the highest derivative
 $\frac{d}{dx} y(x) = \frac{-3y(x)+7x-7}{7y(x)-3x+3}$

✓ **Mathematica.** Time used: 60.516 (sec). Leaf size: 7785

```
ode=(3*y[x]-7*x+7)+(7*y[x]-3*x+3)*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Too large to display

✓ Sympy. Time used: 0.745 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-7*x + (-3*x + 7*y(x) + 3)*Derivative(y(x), x) + 3*y(x) + 7, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\log(y(x)) = C_1 - \log\left(\left(\frac{x-1}{y(x)} - 1\right)^{\frac{2}{7}} \left(\frac{x-1}{y(x)} + 1\right)^{\frac{5}{7}}\right)$$

2.2.15 Problem 40

Local contents

Solved using first_order_ode_separable	477
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	479
✓ Maple	483
✓ Mathematica	484
✓ Sympy	484

Internal problem ID [7869]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 40

Date solved : Thursday, November 27, 2025 at 06:25:30 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.177 (sec)

Solve

$$xyy' = (1 + y)(1 - x)$$

The ode

$$y' = -\frac{(1 + y)(x - 1)}{yx} \quad (2.53)$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{(1 + y)(x - 1)}{yx} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{x - 1}{x} \\ g(y) &= \frac{y + 1}{y} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int \frac{y}{y+1} dy = \int -\frac{x-1}{x} dx$$

$$y - \ln(1+y) = -x + \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$\frac{y+1}{y} = 0$$

for y gives

$$y = -1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

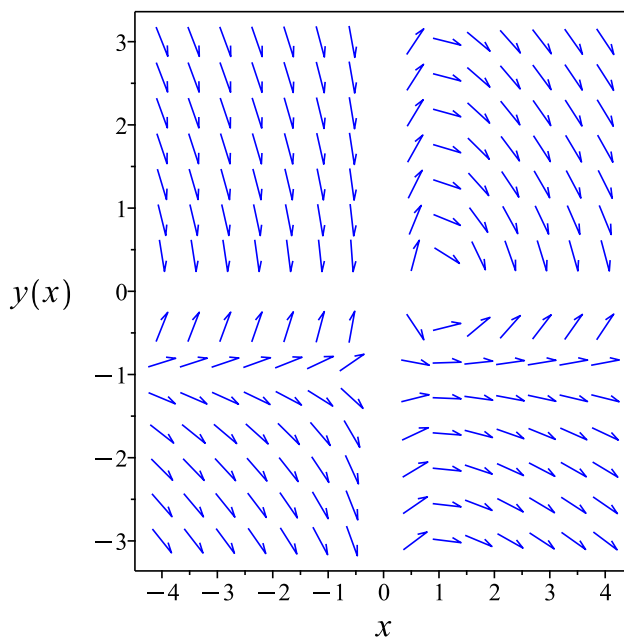
$$y - \ln(1+y) = -x + \ln(x) + c_1$$

$$y = -1$$

Solving for y gives

$$y = -1$$

$$y = -\text{LambertW}\left(-\frac{e^{-c_1+x-1}}{x}\right) - 1$$

Figure 2.117: Slope field $xyy' = (1 + y)(1 - x)$ Summary of solutions found

$$y = -1$$

$$y = -\text{LambertW}\left(-\frac{e^{-c_1+x-1}}{x}\right) - 1$$

Solve

$$xyy' = (1 + y)(1 - x)$$

Solved using**first_order_ode_abel_second_kind_solved_by_converting_to_first_kind**

Time used: 1.185 (sec)

This is Abel second kind ODE, it has the form

$$(y + g)y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$xyy' = (1 + y)(1 - x) \tag{1}$$

Shows that

$$\begin{aligned}g &= 0 \\f_0 &= \frac{1-x}{x} \\f_1 &= \frac{1-x}{x} \\f_2 &= 0 \\f_3 &= 0\end{aligned}$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = \frac{(x-1)u(x)^3}{x} + \frac{(x-1)u(x)^2}{x}$$

Which is now solved.

Solve The ode

$$u'(x) = \frac{u(x)^2(x-1)(u(x)+1)}{x} \quad (2.54)$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= \frac{u(x)^2(x-1)(u(x)+1)}{x} \\&= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= \frac{x-1}{x} \\g(u) &= u^2(u+1)\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u^2(u+1)} du &= \int \frac{x-1}{x} dx\end{aligned}$$

$$\ln(u(x)+1) - \frac{1}{u(x)} - \ln(u(x)) = x + \ln\left(\frac{1}{x}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2(u + 1) = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) + 1) - \frac{1}{u(x)} - \ln(u(x)) = x + \ln\left(\frac{1}{x}\right) + c_1$$

$$u(x) = -1$$

$$u(x) = 0$$

Substituting $u(x) = \frac{1}{y}$ in the above solution gives

$$\ln\left(\frac{1}{y} + 1\right) - y - \ln\left(\frac{1}{y}\right) = x + \ln\left(\frac{1}{x}\right) + c_1$$

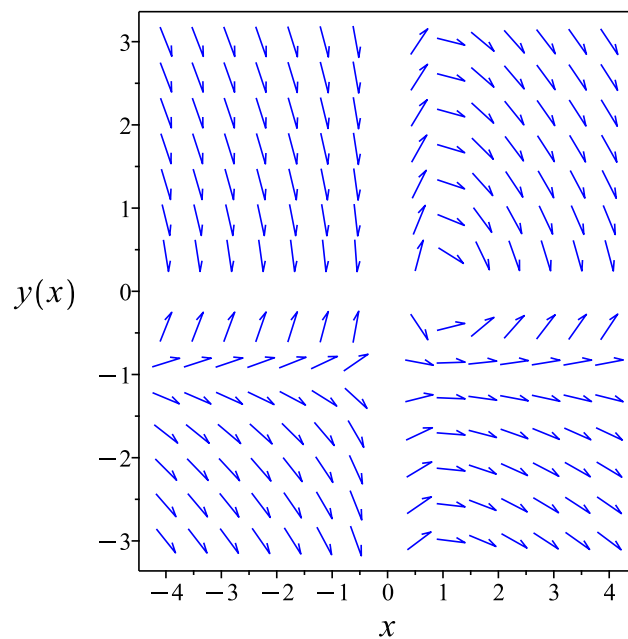
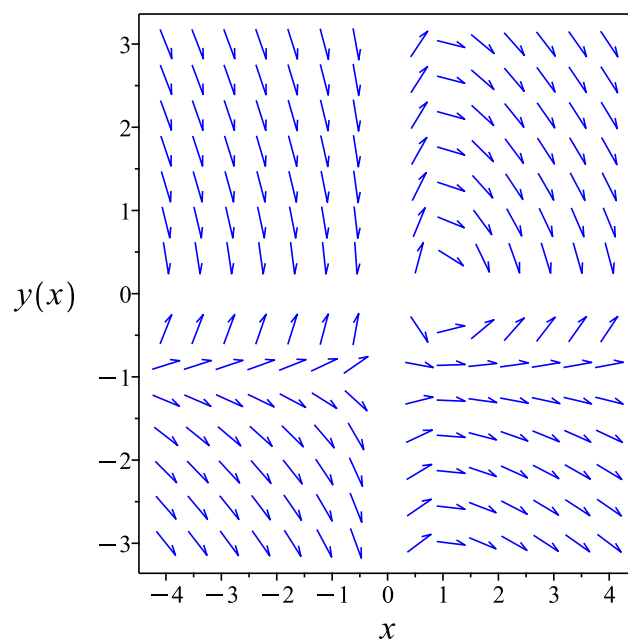
Now we transform the solution $u(x) = -1$ to y using $u(x) = \frac{1}{y}$ which gives

$$y = -1$$

Solving for y gives

$$y = -1$$

$$y = -1 - \text{LambertW}\left(-\frac{e^{c_1+x-1}}{x}\right)$$

Figure 2.118: Slope field $xyy' = (1+y)(1-x)$ Figure 2.119: Slope field $xyy' = (1+y)(1-x)$

Summary of solutions found

$$y = -1$$

$$y = -1 - \text{LambertW}\left(-\frac{e^{c_1+x-1}}{x}\right)$$

✓ Maple. Time used: 0.004 (sec). Leaf size: 19

```
ode:=x*y(x)*diff(y(x),x) = (1+y(x))*(1-x);
dsolve(ode,y(x), singsol=all);
```

$$y = -\text{LambertW}\left(-\frac{c_1 e^{x-1}}{x}\right) - 1$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
<- separable successful
```

Maple step by step

Let's solve

$$xy(x) \left(\frac{d}{dx}y(x)\right) = (y(x) + 1)(1 - x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{(y(x)+1)(1-x)}{xy(x)}$$

- Separate variables

$$\frac{\left(\frac{d}{dx}y(x)\right)y(x)}{y(x)+1} = \frac{1-x}{x}$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx}y(x)\right)y(x)}{y(x)+1} dx = \int \frac{1-x}{x} dx + C1$$

- Evaluate integral

$$y(x) - \ln(y(x) + 1) = -x + \ln(x) + C1$$

- Solve for $y(x)$

$$y(x) = -\text{LambertW}\left(-\frac{e^{-1+x-C1}}{x}\right) - 1$$

- Redefine the integration constant(s)

$$y(x) = -\text{LambertW}\left(\frac{C1 e^x}{x}\right) - 1$$

✓ **Mathematica.** Time used: 0.083 (sec). Leaf size: 37

```
ode=x*y[x]*D[y[x],x]== (y[x]+1)*(1-x);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \text{InverseFunction} \left[\int_1^{\#1} \frac{K[1]}{K[1] + 1} dK[1] \& \right] [-x + \log(x) + c_1]$$

$$y(x) \rightarrow -1$$

✓ **Sympy.** Time used: 0.224 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x)*Derivative(y(x), x) - (1 - x)*(y(x) + 1), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = -W\left(\frac{C_1 e^{x-1}}{x}\right) - 1$$

2.2.16 Problem 41

Local contents

Solved using first_order_ode_bernoulli	485
Solved using first_order_ode_exact	488
Solved using first_order_ode_isobaric	492
Solved using first_order_ode_homog_A	496
Solved using first_order_ode_dAlembert	499
Solved using first_order_ode_homog_type_D2	506
Solved using first_order_ode_homog_type_G	509
Solved using first_order_ode_homog_type_maple_C	510
Solved using first_order_ode_LIE	516
✓ Maple	521
✓ Mathematica	522
✓ Sympy	522

Internal problem ID [7870]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 41

Date solved : Thursday, November 27, 2025 at 06:25:38 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.119 (sec)

Solve

$$y^2 - x^2 + yxy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{-x^2 + y^2}{yx} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{x}\right)y + (x)\frac{1}{y} \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned}f_0 &= -\frac{1}{x} \\f_1 &= x\end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= -\frac{1}{x} \\f_1(x) &= x \\n &= -1\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = -\frac{y^2}{x} + x \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= y^2\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}\frac{v'(x)}{2} &= -\frac{v(x)}{x} + x \\v' &= -\frac{2v}{x} + 2x\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x} \\ p(x) &= 2x \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} \, dx} \\ &= x^2 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu)(2x) \\ \frac{d}{dx}(v x^2) &= (x^2)(2x) \\ d(v x^2) &= (2x^3) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} v x^2 &= \int 2x^3 \, dx \\ &= \frac{x^4}{2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

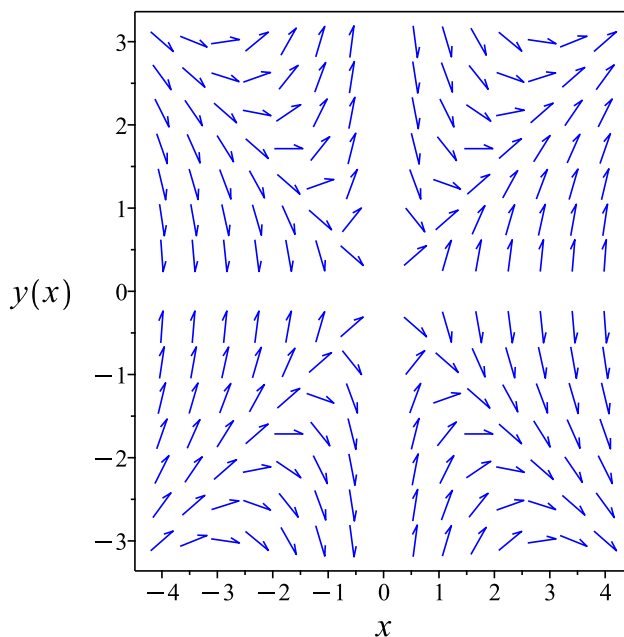
$$v(x) = \frac{x^4 + 2c_1}{2x^2}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^2 = \frac{x^4 + 2c_1}{2x^2}$$

Solving for y gives

$$\begin{aligned} y &= -\frac{\sqrt{2x^4 + 4c_1}}{2x} \\ y &= \frac{\sqrt{2x^4 + 4c_1}}{2x} \end{aligned}$$

Figure 2.120: Slope field $y^2 - x^2 + xy y' = 0$ Summary of solutions found

$$y = -\frac{\sqrt{2x^4 + 4c_1}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 4c_1}}{2x}$$

Solved using first_order_ode_exact

Time used: 0.100 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\frac{\partial \phi}{\partial x} = M$$

$$\frac{\partial \phi}{\partial y} = N$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (yx) dy &= (x^2 - y^2) dx \\ (-x^2 + y^2) dx + (yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x^2 + y^2 \\ N(x, y) &= yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(yx) \\ &= y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{yx} ((2y) - (y)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{1}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(-x^2 + y^2) \\ &= -x^3 + x y^2\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(yx) \\ &= x^2 y\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (-x^3 + x y^2) + (x^2 y) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} \, dx &= \int \overline{M} \, dx \\ \int \frac{\partial \phi}{\partial x} \, dx &= \int -x^3 + x y^2 \, dx \\ \phi &= -\frac{1}{4}x^4 + \frac{1}{2}y^2 x^2 + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x^2 y + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x^2 y$. Therefore equation (4) becomes

$$x^2 y = x^2 y + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{1}{4}x^4 + \frac{1}{2}y^2x^2 + c_1$$

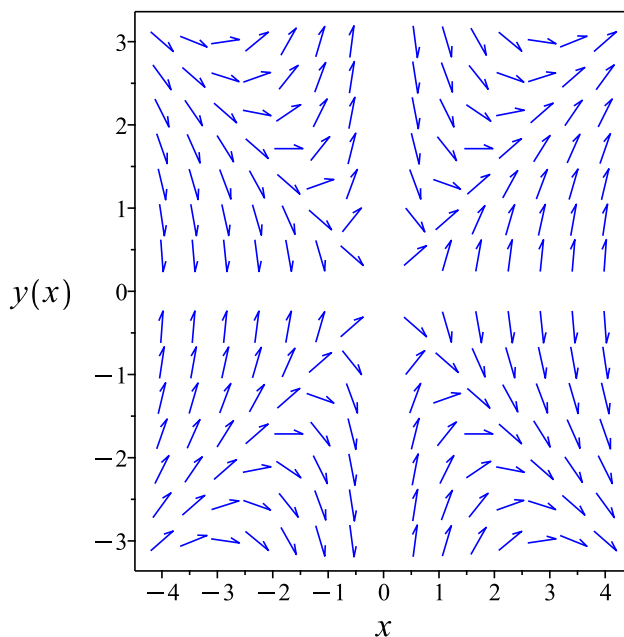
But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{1}{4}x^4 + \frac{1}{2}y^2x^2$$

Solving for y gives

$$y = -\frac{\sqrt{2x^4 + 8c_1}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 8c_1}}{2x}$$

Figure 2.121: Slope field $y^2 - x^2 + xy y' = 0$ Summary of solutions found

$$y = -\frac{\sqrt{2x^4 + 8c_1}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 8c_1}}{2x}$$

Solved using first_order_ode_isobaric

Time used: 0.358 (sec)

Solve

$$y^2 - x^2 + xy y' = 0$$

Solving for y' gives

$$y' = -\frac{y^2 - x^2}{xy} \tag{1}$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \tag{1}$$

Where here

$$f(x, y) = -\frac{y^2 - x^2}{xy} \tag{2}$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x^2u(x)^2 - x^2}{x^2u(x)}$$

The ode

$$u'(x) = -\frac{2u(x)^2 - 1}{u(x)x} \quad (2.55)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 - 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 - 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 - 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(2u(x)^2 - 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 - 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Converting $(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} - 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = -\frac{\sqrt{2}}{2}$$

Converting $u(x) = \frac{\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = \frac{\sqrt{2}}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 - x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$\frac{y}{x} = -\frac{\sqrt{2}}{2}$$

$$\frac{y}{x} = \frac{\sqrt{2}}{2}$$

Solving for y gives

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

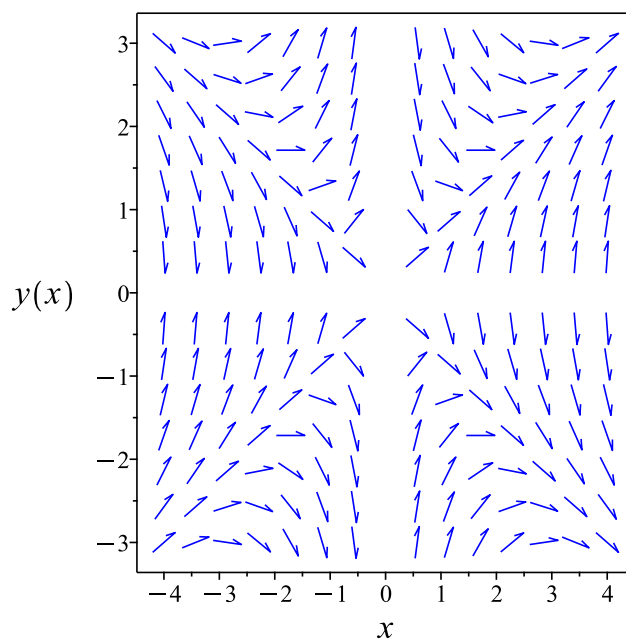


Figure 2.122: Slope field $y^2 - x^2 + xy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

Solved using first_order_ode_homog_A

Time used: 0.669 (sec)

Solve

$$y^2 - x^2 + yxy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{-x^2 + y^2}{yx} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = x^2 - y^2$ and $N = yx$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= \frac{1}{u} - u \\ \frac{du}{dx} &= \frac{\frac{1}{u(x)} - 2u(x)}{x} \end{aligned}$$

Or

$$u'(x) - \frac{\frac{1}{u(x)} - 2u(x)}{x} = 0$$

Or

$$u'(x) u(x) x + 2u(x)^2 - 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{2u(x)^2 - 1}{u(x)x} \quad (2.56)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 - 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(u) = \frac{2u^2 - 1}{u}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{u}{2u^2 - 1} du = \int -\frac{1}{x} dx$$

$$\frac{\ln(2u(x)^2 - 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 - 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Converting $(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(-\frac{x^2 - 2y^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{\sqrt{2}}{2}$ back to y gives

$$y = -\frac{\sqrt{2}x}{2}$$

Converting $u(x) = \frac{\sqrt{2}}{2}$ back to y gives

$$y = \frac{\sqrt{2}x}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 - x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

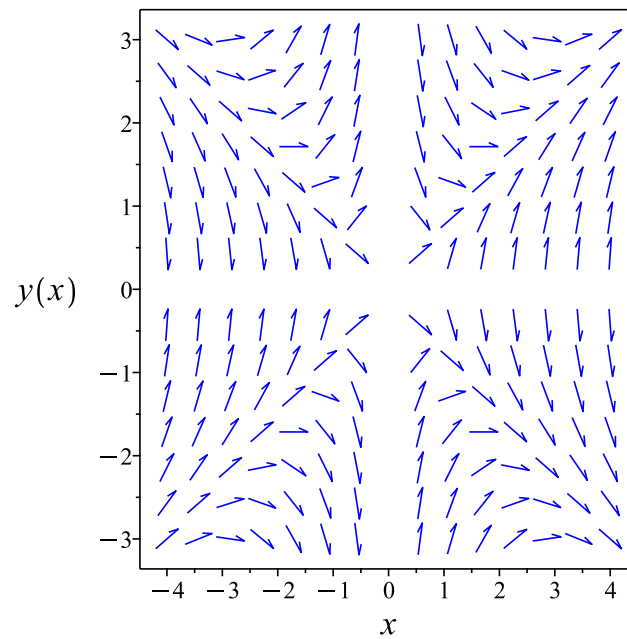
Solving for y gives

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

Figure 2.123: Slope field $y^2 - x^2 + xy y' = 0$ Summary of solutions found

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

Solved using first_order_ode_dAlembert

Time used: 1.040 (sec)

Solve

$$y^2 - x^2 + xy y' = 0$$

Let $p = y'$ the ode becomes

$$yxp - x^2 + y^2 = 0$$

Solving for y from the above results in

$$y = \left(-\frac{p}{2} + \frac{\sqrt{p^2 + 4}}{2} \right) x \quad (1)$$

$$y = \left(-\frac{p}{2} - \frac{\sqrt{p^2 + 4}}{2} \right) x \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{p}{2} + \frac{\sqrt{p^2 + 4}}{2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\frac{3p}{2} - \frac{\sqrt{p^2 + 4}}{2} = \left(-\frac{x}{2} + \frac{xp}{2\sqrt{p^2 + 4}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$\frac{3p}{2} - \frac{\sqrt{p^2 + 4}}{2} = 0$$

Solving the above for p results in

$$p_1 = \frac{\sqrt{2}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{\sqrt{2}x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{\frac{3p(x)}{2} - \frac{\sqrt{p(x)^2+4}}{2}}{-\frac{x}{2} + \frac{xp(x)}{2\sqrt{p(x)^2+4}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{\left(-3p(x) + \sqrt{p(x)^2+4}\right) \sqrt{p(x)^2+4}}{x \left(\sqrt{p(x)^2+4} - p(x)\right)} \quad (2.57)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{\left(-3p(x) + \sqrt{p(x)^2+4}\right) \sqrt{p(x)^2+4}}{x \left(\sqrt{p(x)^2+4} - p(x)\right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(p) &= \frac{(-3p + \sqrt{p^2+4}) \sqrt{p^2+4}}{-p + \sqrt{p^2+4}} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{-p + \sqrt{p^2+4}}{(-3p + \sqrt{p^2+4}) \sqrt{p^2+4}} dp &= \int \frac{1}{x} dx \\ -\frac{\ln(2p(x)^2 - 1)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}+8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}-8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arcsinh}\left(\frac{p(x)}{2}\right)}{4} \\ &= \ln(x) + c_1 \end{aligned}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{(-3p + \sqrt{p^2 + 4})\sqrt{p^2 + 4}}{-p + \sqrt{p^2 + 4}} = 0$$

for $p(x)$ gives

$$p(x) = -2i$$

$$p(x) = 2i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} & -\frac{\ln(2p(x)^2 - 1)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}+8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}-8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arcsinh}\left(\frac{p(x)}{2}\right)}{4} \\ & = \ln(x) + c_1 \end{aligned}$$

$$p(x) = -2i$$

$$p(x) = 2i$$

Substituting the above solution for p in (2A) gives

$$y = ix$$

$$y = -ix$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{p}{2} - \frac{\sqrt{p^2 + 4}}{2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\frac{3p}{2} + \frac{\sqrt{p^2+4}}{2} = \left(-\frac{x}{2} - \frac{xp}{2\sqrt{p^2+4}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$\frac{3p}{2} + \frac{\sqrt{p^2+4}}{2} = 0$$

Solving the above for p results in

$$p_1 = -\frac{\sqrt{2}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{\sqrt{2}x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{\frac{3p(x)}{2} + \frac{\sqrt{p(x)^2+4}}{2}}{-\frac{x}{2} - \frac{xp(x)}{2\sqrt{p(x)^2+4}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{\left(3p(x) + \sqrt{p(x)^2+4} \right) \sqrt{p(x)^2+4}}{x \left(\sqrt{p(x)^2+4} + p(x) \right)} \quad (2.58)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{\left(3p(x) + \sqrt{p(x)^2+4} \right) \sqrt{p(x)^2+4}}{x \left(\sqrt{p(x)^2+4} + p(x) \right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(p) &= \frac{(3p + \sqrt{p^2+4}) \sqrt{p^2+4}}{p + \sqrt{p^2+4}} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{p + \sqrt{p^2 + 4}}{(3p + \sqrt{p^2 + 4}) \sqrt{p^2 + 4}} dp = \int -\frac{1}{x} dx$$

$$\frac{\ln(2p(x)^2 - 1)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}+8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}-8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arcsinh}\left(\frac{p(x)}{2}\right)}{4}$$

$$= \ln\left(\frac{1}{x}\right) + c_2$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{(3p + \sqrt{p^2 + 4}) \sqrt{p^2 + 4}}{p + \sqrt{p^2 + 4}} = 0$$

for $p(x)$ gives

$$p(x) = -2i$$

$$p(x) = 2i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(2p(x)^2 - 1)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}+8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arctanh}\left(\frac{(p(x)\sqrt{2}-8)\sqrt{2}}{6\sqrt{p(x)^2+4}}\right)}{8} + \frac{\operatorname{arcsinh}\left(\frac{p(x)}{2}\right)}{4}$$

$$= \ln\left(\frac{1}{x}\right) + c_2$$

$$p(x) = -2i$$

$$p(x) = 2i$$

Substituting the above solution for p in (2A) gives

$$y = ix$$

$$y = -ix$$

The solution

$$y = -ix$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = ix$$

was found not to satisfy the ode or the IC. Hence it is removed.

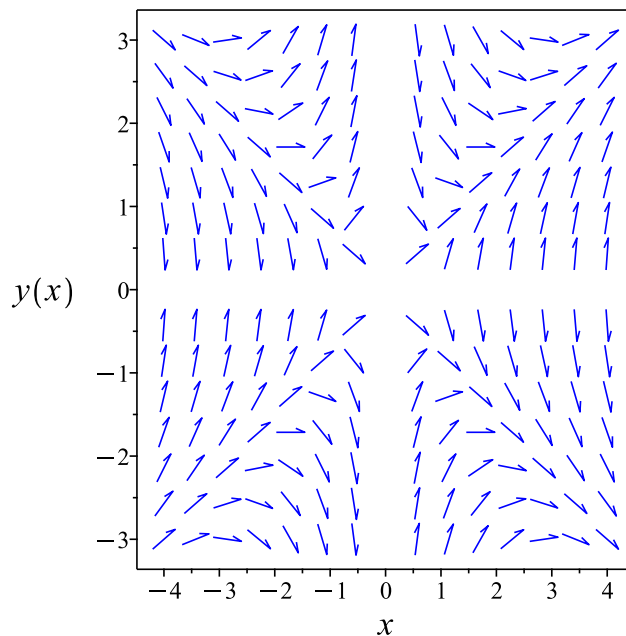


Figure 2.124: Slope field $y^2 - x^2 + yxy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.211 (sec)

Solve

$$y^2 - x^2 + xy y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)^2 x^2 - x^2 + u(x) x^2 (u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{2u(x)^2 - 1}{u(x)x} \quad (2.59)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 - 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 - 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 - 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(2u(x)^2 - 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 - 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{\sqrt{2}}{2}$$

$$u(x) = \frac{\sqrt{2}}{2}$$

Converting $(2u(x)^2 - 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} - 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{\sqrt{2}}{2}$ back to y gives

$$y = -\frac{\sqrt{2}x}{2}$$

Converting $u(x) = \frac{\sqrt{2}}{2}$ back to y gives

$$y = \frac{\sqrt{2}x}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 - x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

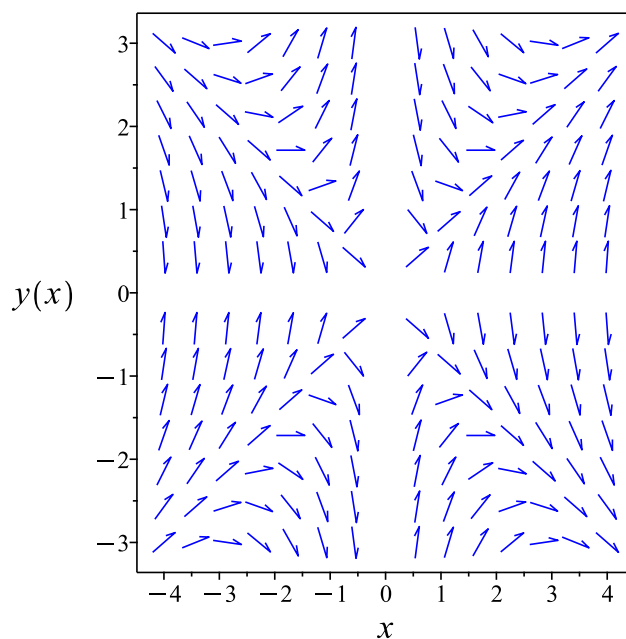


Figure 2.125: Slope field $y^2 - x^2 + xy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.088 (sec)

Solve

$$y^2 - x^2 + yxy' = 0$$

Multiplying the right side of the ode, which is $-\frac{x^2+y^2}{yx}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{-x^2+y^2}{yx} \\ &= \frac{x^2 - y^2}{y^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{2x^2}{y^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{2x^2}{y^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^{-1}} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z^2 - 1}{z^2}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1 + \frac{z^2-1}{z^2})} dz = 0$$

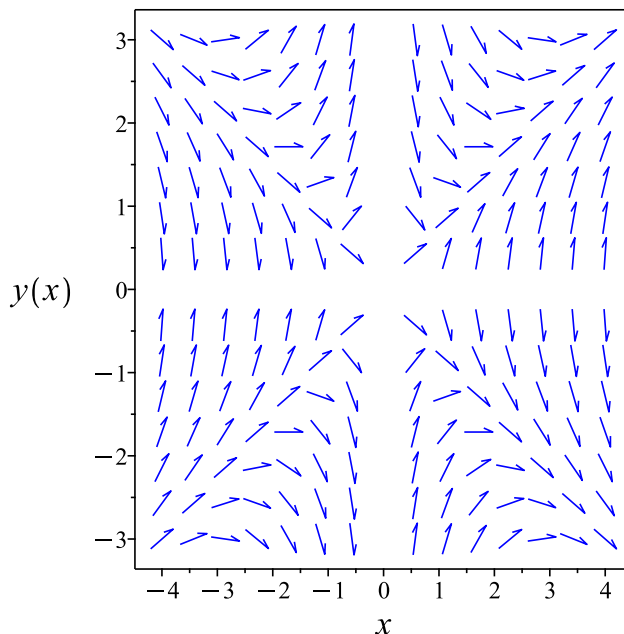


Figure 2.126: Slope field $y^2 - x^2 + xy' = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z(1 + \frac{z^2-1}{z^2})} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.023 (sec)

Solve

$$y^2 - x^2 + xy' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{(Y(X) + y_0)^2 - (X + x_0)^2}{(X + x_0)(Y(X) + y_0)}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$\begin{aligned}x_0 &= 0 \\y_0 &= 0\end{aligned}$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{-X^2 + Y(X)^2}{XY(X)}$$

In canonical form, the ODE is

$$\begin{aligned}Y' &= F(X, Y) \\&= -\frac{-X^2 + Y^2}{XY}\end{aligned}\tag{1}$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = X^2 - Y^2$ and $N = XY$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dX}X + u &= \frac{1}{u} - u \\ \frac{du}{dX} &= \frac{\frac{1}{u(X)} - 2u(X)}{X}\end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{1}{u(X)} - 2u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)u(X)X + 2u(X)^2 - 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{2u(X)^2 - 1}{u(X)X}\tag{2.60}$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= -\frac{2u(X)^2 - 1}{u(X)X} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= -\frac{1}{X} \\ g(u) &= \frac{2u^2 - 1}{u}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{u}{2u^2 - 1} du &= \int -\frac{1}{X} dX\end{aligned}$$

$$\frac{\ln(2u(X)^2 - 1)}{4} = \ln\left(\frac{1}{X}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(X)^2 - 1)^{1/4} = \frac{c_1}{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 - 1}{u} = 0$$

for $u(X)$ gives

$$\begin{aligned}u(X) &= -\frac{\sqrt{2}}{2} \\ u(X) &= \frac{\sqrt{2}}{2}\end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(X)^2 - 1)^{1/4} = \frac{c_1}{X}$$

$$u(X) = -\frac{\sqrt{2}}{2}$$

$$u(X) = \frac{\sqrt{2}}{2}$$

Converting $(2u(X)^2 - 1)^{1/4} = \frac{c_1}{X}$ back to $Y(X)$ gives

$$\left(-\frac{2Y(X)^2 + X^2}{X^2} \right)^{1/4} = \frac{c_1}{X}$$

Converting $u(X) = -\frac{\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = -\frac{X\sqrt{2}}{2}$$

Converting $u(X) = \frac{\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = \frac{X\sqrt{2}}{2}$$

Using the solution for $Y(X)$

$$\left(-\frac{2Y(X)^2 + X^2}{X^2} \right)^{1/4} = \frac{c_1}{X} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\left(-\frac{x^2 - 2y^2}{x^2} \right)^{1/4} = \frac{c_1}{x}$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{X\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{\sqrt{2}x}{2}$$

Using the solution for $Y(X)$

$$Y(X) = \frac{X\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = \frac{\sqrt{2}x}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 - x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

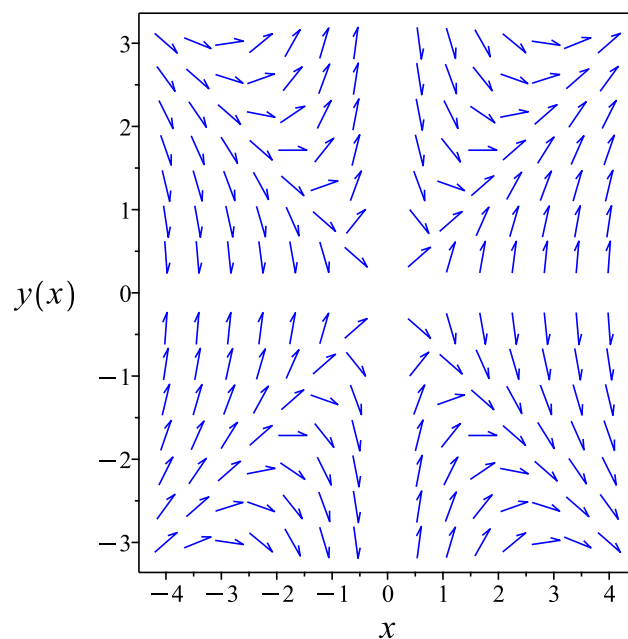


Figure 2.127: Slope field $y^2 - x^2 + xy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{2}x}{2}$$

$$y = \frac{\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 + 2x^4}}{2x}$$

Solved using first_order_ode_LIE

Time used: 0.933 (sec)

Solve

$$y^2 - x^2 + xy' = 0$$

Writing the ode as

$$y' = -\frac{-x^2 + y^2}{yx}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(-x^2 + y^2)(b_3 - a_2)}{yx} - \frac{(-x^2 + y^2)^2 a_3}{y^2 x^2} - \left(\frac{2}{y} + \frac{-x^2 + y^2}{y x^2} \right) (xa_2 + ya_3 + a_1) \quad (5\text{E})$$

$$- \left(-\frac{2}{x} + \frac{-x^2 + y^2}{y^2 x} \right) (xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$\frac{x^4 a_3 - x^4 b_2 + 2x^3 y a_2 - 2x^3 y b_3 - x^2 y^2 a_3 - 2b_2 y^2 x^2 + 2y^4 a_3 - x^3 b_1 + x^2 y a_1 - x y^2 b_1 + y^3 a_1}{y^2 x^2}$$

$$= 0$$

Setting the numerator to zero gives

$$-x^4 a_3 + x^4 b_2 - 2x^3 y a_2 + 2x^3 y b_3 + x^2 y^2 a_3 + 2b_2 y^2 x^2 - 2y^4 a_3 + x^3 b_1 - x^2 y a_1 + x y^2 b_1 - y^3 a_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & -2a_2v_1^3v_2 - a_3v_1^4 + a_3v_1^2v_2^2 - 2a_3v_2^4 + b_2v_1^4 + 2b_2v_1^2v_2^2 \\ & + 2b_3v_1^3v_2 - a_1v_1^2v_2 - a_1v_2^3 + b_1v_1^3 + b_1v_1v_2^2 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-a_3 + b_2)v_1^4 + (-2a_2 + 2b_3)v_1^3v_2 + b_1v_1^3 + (a_3 + 2b_2)v_1^2v_2^2 - a_1v_1^2v_2 + b_1v_1v_2^2 - 2a_3v_2^4 - a_1v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} b_1 &= 0 \\ -a_1 &= 0 \\ -2a_3 &= 0 \\ -2a_2 + 2b_3 &= 0 \\ -a_3 + b_2 &= 0 \\ a_3 + 2b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}\xi &= x \\ \eta &= y\end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{-x^2 + y^2}{yx} \right) (x) \\ &= y + \frac{-x^2 + y^2}{y} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{-x^2 + y^2}{y}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(-x^2 + 2y^2)}{4}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{-x^2 + y^2}{yx}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{x}{2x^2 - 4y^2} \\ S_y &= -\frac{y}{x^2 - 2y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{2R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{2R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

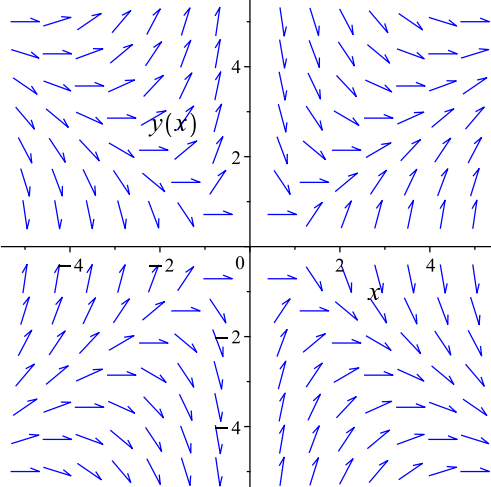
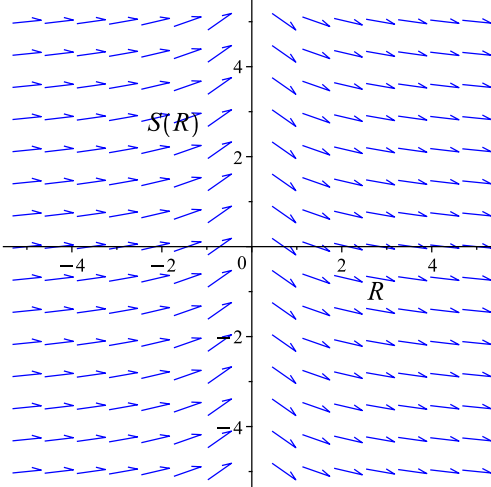
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{2R} dR \\ S(R) &= -\frac{\ln(R)}{2} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(2y^2 - x^2)}{4} = -\frac{\ln(x)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{-x^2+y^2}{yx}$ 	$R = x$ $S = \frac{\ln(-x^2 + 2y^2)}{4}$	$\frac{dS}{dR} = -\frac{1}{2R}$ 

Solving for y gives

$$y = -\frac{\sqrt{2x^4 + 2e^{4c_2}}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 2e^{4c_2}}}{2x}$$

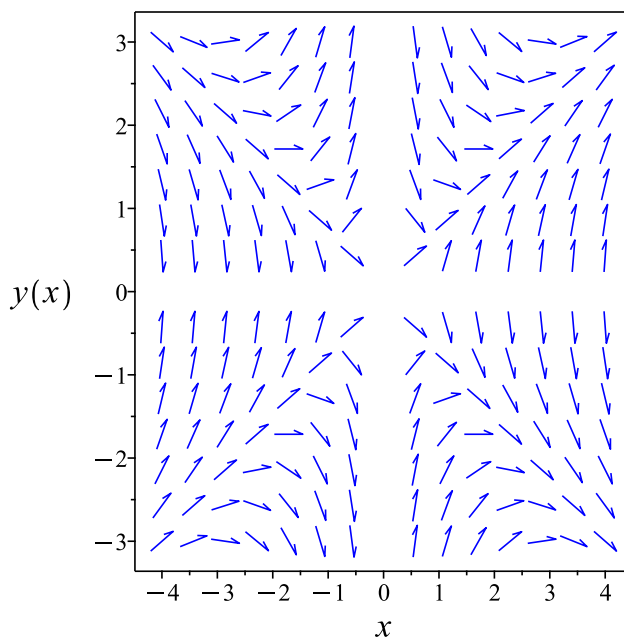


Figure 2.128: Slope field $y^2 - x^2 + xy y' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{2x^4 + 2e^{4c_2}}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 2e^{4c_2}}}{2x}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 39

```
ode:=y(x)^2-x^2+x*y(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{2x^4 + 4c_1}}{2x}$$

$$y = \frac{\sqrt{2x^4 + 4c_1}}{2x}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

<- Bernoulli successful

Maple step by step

Let's solve

$$y(x)^2 - x^2 + xy(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-y(x)^2 + x^2}{xy(x)}$$

✓ **Mathematica.** Time used: 0.137 (sec). Leaf size: 46

```
ode=(y[x]^2-x^2)+x*y[x]*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{\frac{x^4}{2} + c_1}}{x}$$

$$y(x) \rightarrow \frac{\sqrt{\frac{x^4}{2} + c_1}}{x}$$

✓ **Sympy.** Time used: 0.244 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + x*y(x)*Derivative(y(x), x) + y(x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\frac{\sqrt{C_1 + 2x^4}}{2x}, y(x) = \frac{\sqrt{C_1 + 2x^4}}{2x} \right]$$

2.2.17 Problem 42

Local contents

Solved using first_order_ode_exact	523
Solved using first_order_ode_isobaric	528
Solved using first_order_ode_homog_type_G	531
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	533
Solved using first_order_ode_LIE	537
✓ Maple	542
✓ Mathematica	543
✓ Sympy	543

Internal problem ID [7871]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 42

Date solved : Thursday, November 27, 2025 at 06:25:50 AM

CAS classification :

[[_homogeneous, 'class G'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.223 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x(-yx + 1)) dy &= (-y(2yx + 1)) dx \\ (y(2yx + 1)) dx &+ (x(-yx + 1)) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y(2yx + 1) \\ N(x, y) &= x(-yx + 1) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y(2yx + 1)) \\ &= 4yx + 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x(-yx + 1)) \\ &= -2yx + 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{(yx - 1)x} ((4yx + 1) - (-2yx + 1)) \\ &= -\frac{6y}{yx - 1} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{2xy^2 + y} ((-2yx + 1) - (4yx + 1)) \\ &= -\frac{6x}{2yx + 1} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(-2yx + 1) - (4yx + 1)}{x(y(2yx + 1)) - y(x(-yx + 1))} \\ &= -\frac{2}{yx} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = -\frac{2}{t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int (-\frac{2}{t}) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-2 \ln(t)} \\ &= \frac{1}{t^2} \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{y^2 x^2}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{y^2 x^2} (y(2yx + 1)) \\ &= \frac{2yx + 1}{y x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{y^2 x^2} (x(-yx + 1)) \\ &= \frac{-yx + 1}{x y^2}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{2yx + 1}{y x^2} \right) + \left(\frac{-yx + 1}{x y^2} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{2yx + 1}{y x^2} dx \\ \phi &= -\frac{1}{yx} + 2 \ln(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{1}{x y^2} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{-yx+1}{x y^2}$. Therefore equation (4) becomes

$$\frac{-yx+1}{x y^2} = \frac{1}{x y^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{1}{y}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int \left(-\frac{1}{y}\right) dy \\ f(y) &= -\ln(y) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

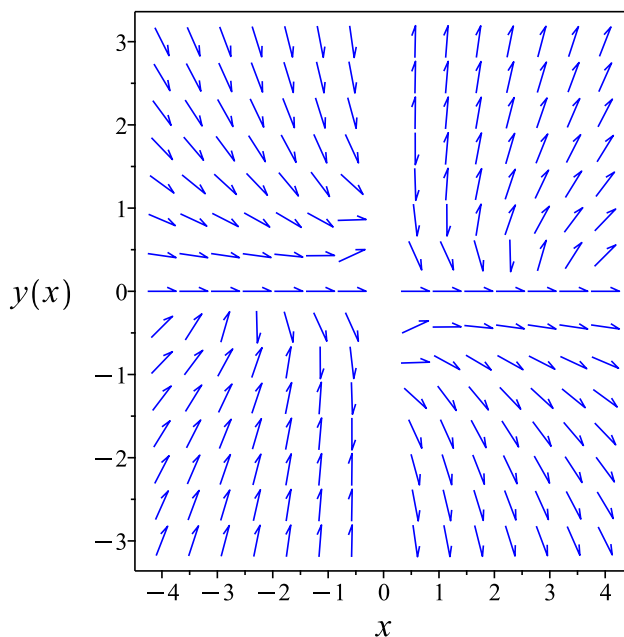
$$\phi = -\frac{1}{yx} + 2 \ln(x) - \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{1}{yx} + 2 \ln(x) - \ln(y)$$

Solving for y gives

$$y = -\frac{1}{x \operatorname{LambertW}\left(-\frac{e^{c_1}}{x^3}\right)}$$

Figure 2.129: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$ Summary of solutions found

$$y = -\frac{1}{x \operatorname{LambertW}\left(-\frac{e^{c_1}}{x^3}\right)}$$

Solved using first_order_ode_isobaric

Time used: 0.490 (sec)

Solve

$$y(2yx + 1) + x(-yx + 1)y' = 0$$

Solving for y' gives

$$y' = \frac{y(2yx + 1)}{x(yx - 1)} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{y(2yx + 1)}{x(yx - 1)} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = -1$$

Since the ode is isobaric of order $m = -1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= \frac{u}{x} \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$-\frac{u(x)}{x^2} + \frac{u'(x)}{x} = \frac{u(x)(2u(x) + 1)}{x^2(u(x) - 1)}$$

The ode

$$u'(x) = \frac{3u(x)^2}{x(u(x) - 1)} \quad (2.61)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{3u(x)^2}{x(u(x) - 1)} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{3}{x} \\ g(u) &= \frac{u^2}{u - 1} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u - 1}{u^2} du &= \int \frac{3}{x} dx \end{aligned}$$

$$\frac{1}{u(x)} + \ln(u(x)) = \ln(x^3) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2}{u - 1} = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{u(x)} + \ln(u(x)) = \ln(x^3) + c_1$$

$$u(x) = 0$$

Converting $\frac{1}{u(x)} + \ln(u(x)) = \ln(x^3) + c_1$ back to y gives

$$\frac{1}{xy} + \ln(yx) = \ln(x^3) + c_1$$

Converting $u(x) = 0$ back to y gives

$$yx = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{1}{\text{LambertW}\left(-\frac{e^{-c_1}}{x^3}\right)x}$$

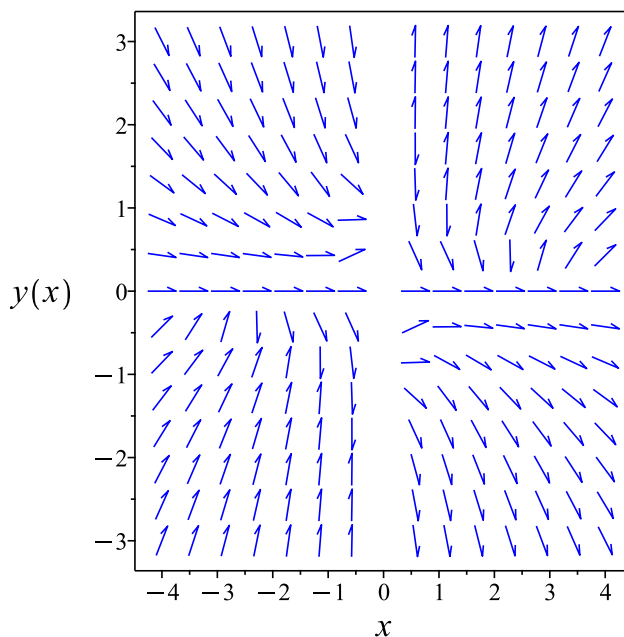


Figure 2.130: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$

Summary of solutions found

$$y = 0$$

$$y = -\frac{1}{\text{LambertW}\left(-\frac{e^{-c_1}}{x^3}\right)x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.091 (sec)

Solve

$$y(2yx + 1) + x(-yx + 1)y' = 0$$

Multiplying the right side of the ode, which is $\frac{y(2yx+1)}{x(yx-1)}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y(2yx+1)}{x(yx-1)} \\ &= \frac{2yx+1}{yx-1} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{3yx}{(yx-1)^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{3yx}{(yx-1)^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= 1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{2z + 1}{z - 1}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{yx} \frac{1}{z(-1 - \frac{2z+1}{z-1})} dz = 0$$

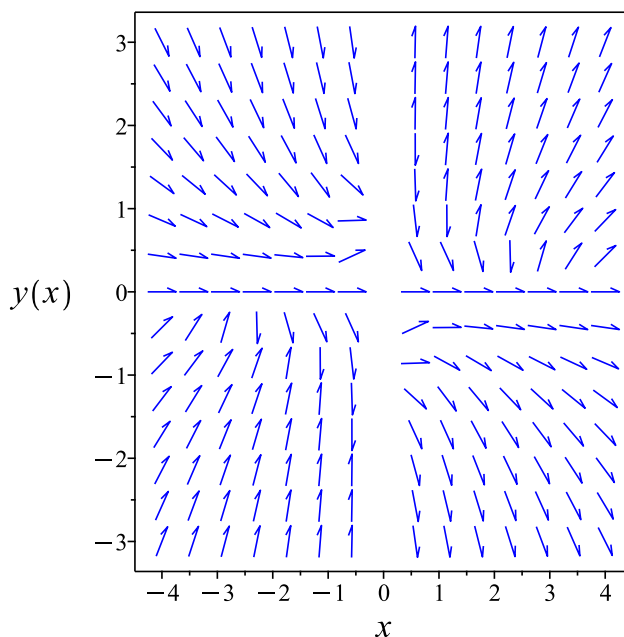


Figure 2.131: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{yx} \frac{1}{z(-1 - \frac{2z+1}{z-1})} dz = 0$$

Solve

$$y(2yx + 1) + x(-yx + 1)y' = 0$$

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 1.959 (sec)

This is Abel second kind ODE, it has the form

$$(y + g) y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$y(2yx + 1) + x(-yx + 1) y' = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= -\frac{1}{x} \\ f_0 &= 0 \\ f_1 &= \frac{1}{x^2} \\ f_2 &= \frac{2}{x} \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = -\frac{3u(x)^3}{x^3} - \frac{6u(x)^2}{x^2} - \frac{2u(x)}{x}$$

Which is now solved.

Solve Solving for u' gives

$$u' = -\frac{u(x) (3u(x)^2 + 6u(x)x + 2x^2)}{x^3} \quad (1)$$

Each of the above ode's is now solved An ode $u'(x) = f(x, u)$ is isobaric if

$$f(tx, t^m u) = t^{m-1} f(x, u) \quad (1)$$

Where here

$$f(x, u) = -\frac{u(x) (3u(x)^2 + 6u(x)x + 2x^2)}{x^3} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} u &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{u(x) (3x^2u(x)^2 + 6x^2u(x) + 2x^2)}{x^2}$$

The ode

$$u'(x) = -\frac{3u(x) (u(x) + 1)^2}{x} \quad (2.62)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{3u(x) (u(x) + 1)^2}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{3}{x} \\ g(u) &= u(u + 1)^2 \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(u + 1)^2} du &= \int -\frac{3}{x} dx \end{aligned}$$

$$\frac{1}{u(x) + 1} - \ln(u(x) + 1) + \ln(u(x)) = \ln\left(\frac{1}{x^3}\right) + c_2$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(u + 1)^2 = 0$$

for $u(x)$ gives

$$u(x) = -1$$

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{u(x)+1} - \ln(u(x)+1) + \ln(u(x)) = \ln\left(\frac{1}{x^3}\right) + c_2$$

$$u(x) = -1$$

$$u(x) = 0$$

Converting $\frac{1}{u(x)+1} - \ln(u(x)+1) + \ln(u(x)) = \ln\left(\frac{1}{x^3}\right) + c_2$ back to $u(x)$ gives

$$\frac{1}{\frac{u(x)}{x}+1} - \ln\left(\frac{u(x)}{x}+1\right) + \ln\left(\frac{u(x)}{x}\right) = \ln\left(\frac{1}{x^3}\right) + c_2$$

Converting $u(x) = -1$ back to $u(x)$ gives

$$\frac{u(x)}{x} = -1$$

Converting $u(x) = 0$ back to $u(x)$ gives

$$\frac{u(x)}{x} = 0$$

Simplifying the above gives

$$\frac{x}{u(x)+x} - \ln\left(\frac{u(x)+x}{x}\right) + \ln\left(\frac{u(x)}{x}\right) = \ln\left(\frac{1}{x^3}\right) + c_2$$

$$\frac{u(x)}{x} = -1$$

$$\frac{u(x)}{x} = 0$$

Solving for $u(x)$ gives

$$\frac{x}{u(x)+x} - \ln\left(\frac{u(x)+x}{x}\right) + \ln\left(\frac{u(x)}{x}\right) = \ln\left(\frac{1}{x^3}\right) + c_2$$

$$u(x) = 0$$

$$u(x) = -x$$

Substituting $u(x) = \frac{1}{y - \frac{1}{x}}$ in the above solution gives

$$\frac{x}{\frac{1}{y - \frac{1}{x}} + x} - \ln \left(\frac{\frac{1}{y - \frac{1}{x}} + x}{x} \right) + \ln \left(\frac{1}{x \left(y - \frac{1}{x} \right)} \right) = \ln \left(\frac{1}{x^3} \right) + c_2$$

Now we transform the solution $u(x) = -x$ to y using $u(x) = \frac{1}{y - \frac{1}{x}}$ which gives

$$y = 0$$

Simplifying the above gives

$$\frac{-\ln \left(\frac{xy}{yx-1} \right) xy + \ln \left(\frac{1}{yx-1} \right) xy + yx - 1}{yx} = \ln \left(\frac{1}{x^3} \right) + c_2$$

$$y = 0$$

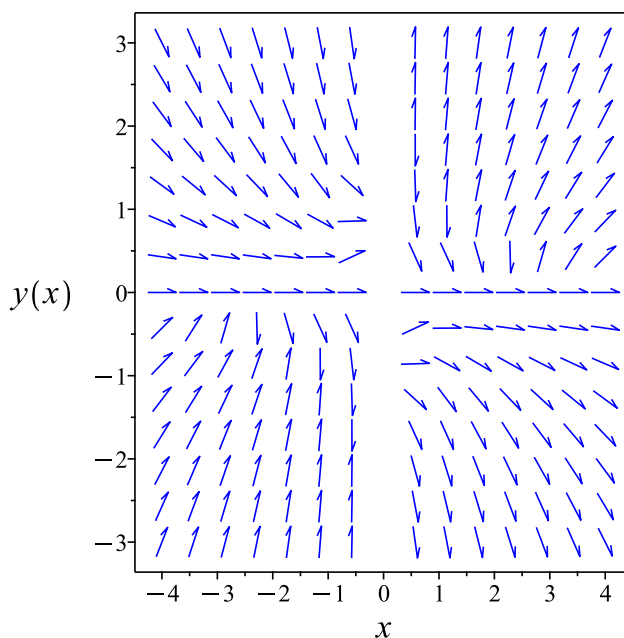
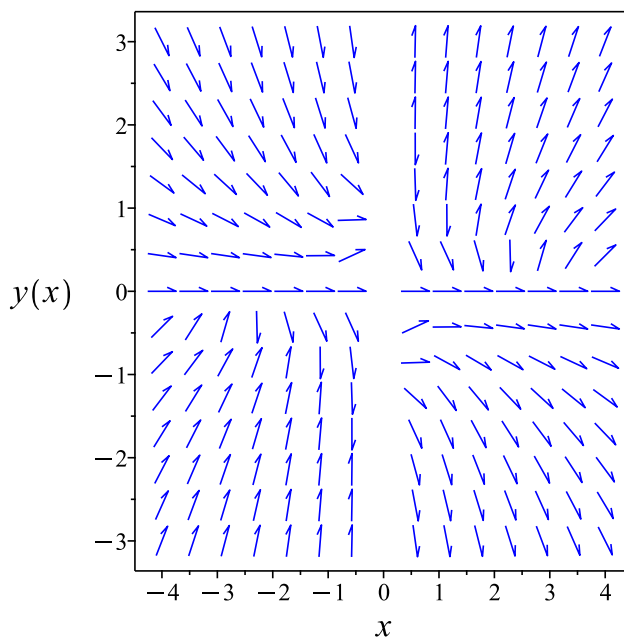


Figure 2.132: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$

Figure 2.133: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$ Summary of solutions found

$$\frac{-\ln\left(\frac{xy}{yx-1}\right)xy + \ln\left(\frac{1}{yx-1}\right)xy + yx - 1}{yx} = \ln\left(\frac{1}{x^3}\right) + c_2$$

$$y = 0$$

Solved using first_order_ode_LIE

Time used: 1.207 (sec)

Solve

$$y(2yx + 1) + x(-yx + 1)y' = 0$$

Writing the ode as

$$y' = \frac{y(2yx + 1)}{x(yx - 1)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 + \frac{y(2yx+1)(b_3-a_2)}{x(yx-1)} - \frac{y^2(2yx+1)^2 a_3}{x^2(yx-1)^2} \\ - \left(\frac{2y^2}{x(yx-1)} - \frac{y(2yx+1)}{x^2(yx-1)} - \frac{y^2(2yx+1)}{x(yx-1)^2} \right) (xa_2 + ya_3 + a_1) \\ - \left(\frac{2yx+1}{x(yx-1)} + \frac{2y}{yx-1} - \frac{(2yx+1)y}{(yx-1)^2} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{x^4 y^2 b_2 + 2x^2 y^4 a_3 + 2x^3 y^2 b_1 - 2x^2 y^3 a_1 - 2x^3 y b_2 - 3x^2 y^2 a_2 - 3x^2 y^2 b_3 + 2x y^3 a_3 - 4x^2 y b_1 - 2x y^2 a_1 - 2b_2 x}{x^2 (yx-1)^2} = 0$$

Setting the numerator to zero gives

$$\begin{aligned} -x^4 y^2 b_2 - 2x^2 y^4 a_3 - 2x^3 y^2 b_1 + 2x^2 y^3 a_1 + 2x^3 y b_2 + 3x^2 y^2 a_2 + 3x^2 y^2 b_3 \\ - 2x y^3 a_3 + 4x^2 y b_1 + 2x y^2 a_1 + 2b_2 x^2 - 2y^2 a_3 + x b_1 - y a_1 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} -2a_3 v_1^2 v_2^4 - b_2 v_1^4 v_2^2 + 2a_1 v_1^2 v_2^3 - 2b_1 v_1^3 v_2^2 + 3a_2 v_1^2 v_2^2 - 2a_3 v_1 v_2^3 + 2b_2 v_1^3 v_2 \\ + 3b_3 v_1^2 v_2^2 + 2a_1 v_1 v_2^2 + 4b_1 v_1^2 v_2 - 2a_3 v_2^2 + 2b_2 v_1^2 - a_1 v_2 + b_1 v_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} & -b_2v_1^4v_2^2 - 2b_1v_1^3v_2^2 + 2b_2v_1^3v_2 - 2a_3v_1^2v_2^4 + 2a_1v_1^2v_2^3 + (3a_2 + 3b_3)v_1^2v_2^2 \\ & + 4b_1v_1^2v_2 + 2b_2v_1^2 - 2a_3v_1v_2^3 + 2a_1v_1v_2^2 + b_1v_1 - 2a_3v_2^2 - a_1v_2 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} b_1 &= 0 \\ -a_1 &= 0 \\ 2a_1 &= 0 \\ -2a_3 &= 0 \\ -2b_1 &= 0 \\ 4b_1 &= 0 \\ -b_2 &= 0 \\ 2b_2 &= 0 \\ 3a_2 + 3b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= -b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= -x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= y - \left(\frac{y(2yx + 1)}{x(yx - 1)} \right) (-x) \\ &= \frac{3y^2x}{yx - 1} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{3y^2x}{yx-1}} dy \end{aligned}$$

Which results in

$$S = \frac{1}{3yx} + \frac{\ln(y)}{3}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y(2yx + 1)}{x(yx - 1)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{1}{3y x^2} \\ S_y &= \frac{yx - 1}{3x y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{2}{3R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{2}{3R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

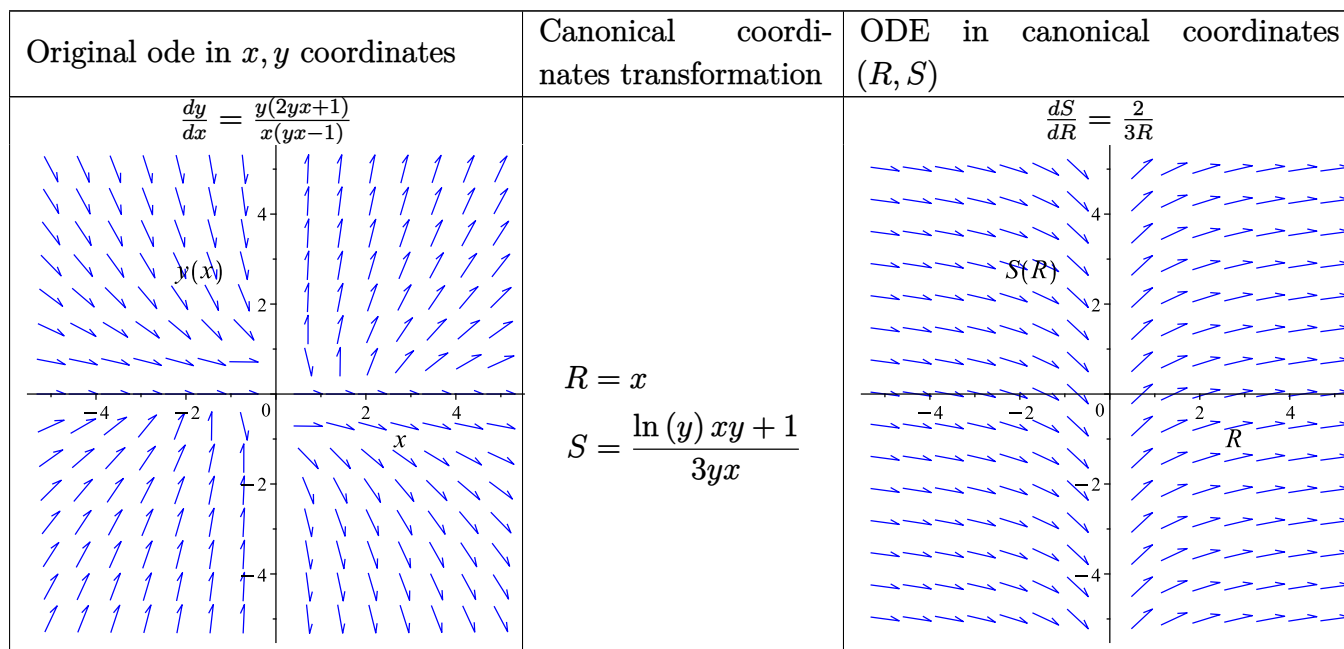
$$\int dS = \int \frac{2}{3R} dR$$

$$S(R) = \frac{2 \ln(R)}{3} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(y)xy + 1}{3yx} = \frac{2 \ln(x)}{3} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{1}{x \operatorname{LambertW}\left(-\frac{e^{-3c_2}}{x^3}\right)}$$

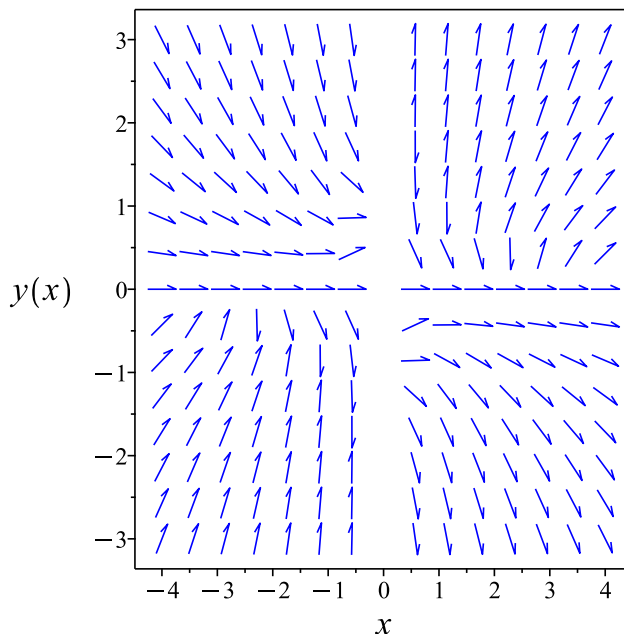


Figure 2.134: Slope field $y(2yx + 1) + x(-yx + 1)y' = 0$

Summary of solutions found

$$y = -\frac{1}{x \operatorname{LambertW}\left(-\frac{e^{-3c_2}}{x^3}\right)}$$

✓ Maple. Time used: 0.011 (sec). Leaf size: 18

```
ode:=y(x)*(1+2*y(x)*x)+x*(-y(x)*x+1)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{1}{\operatorname{LambertW}\left(-\frac{c_1}{x^3}\right) x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
```

```

trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful

```

Maple step by step

Let's solve

$$y(x)(1 + 2xy(x)) + x(1 - xy(x)) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{y(x)(1+2xy(x))}{x(1-xy(x))}$$

✓ **Mathematica.** Time used: 4.92 (sec). Leaf size: 35

```

ode=y[x]*(1+2*x*y[x])+x*(1-x*y[x])*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow -\frac{1}{xW\left(\frac{e^{-1+\frac{9c_1}{2^{2/3}}}}{x^3}\right)}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.529 (sec). Leaf size: 20

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*(-x*y(x) + 1)*Derivative(y(x), x) + (2*x*y(x) + 1)*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)

```

$$y(x) = x^2 e^{C_1 + W\left(-\frac{e^{-C_1}}{x^3}\right)}$$

2.2.18 Problem 43

Local contents

Solved using first_order_ode_separable	544
Solved using first_order_ode_exact	546
Solved using first_order_ode_LIE	550
✓ Maple	556
✓ Mathematica	557
✓ Sympy	557

Internal problem ID [7872]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 43

Date solved : Thursday, November 27, 2025 at 06:26:02 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.135 (sec)

Solve

$$1 + (-x^2 + 1) \cot(y) y' = 0$$

The ode

$$y' = \frac{1}{(x-1)(x+1)\cot(y)} \quad (2.63)$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{1}{(x-1)(x+1)\cot(y)} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{(x-1)(x+1)} \\ g(y) &= \frac{1}{\cot(y)} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int \cot(y) dy = \int \frac{1}{(x-1)(x+1)} dx$$

$$\ln(\sin(y)) = \ln\left(\frac{\sqrt{x-1}}{\sqrt{x+1}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sin(y) = \frac{c_1 \sqrt{x-1}}{\sqrt{x+1}}$$

Solving for y gives

$$y = \arcsin\left(\frac{c_1 \sqrt{x-1}}{\sqrt{x+1}}\right)$$

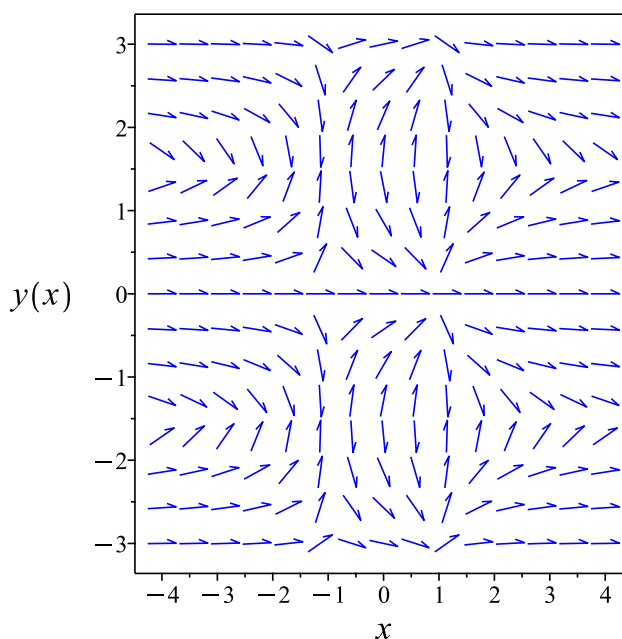


Figure 2.135: Slope field $1 + (-x^2 + 1) \cot(y) y' = 0$

Summary of solutions found

$$y = \arcsin\left(\frac{c_1 \sqrt{x-1}}{\sqrt{x+1}}\right)$$

Solved using first_order_ode_exact

Time used: 0.118 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} ((-x^2 + 1) \cot(y)) dy &= (-1) dx \\ (1) dx + ((-x^2 + 1) \cot(y)) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 1 \\ N(x, y) &= (-x^2 + 1) \cot(y) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(1) \\ &= 0\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}((-x^2 + 1) \cot(y)) \\ &= -2x \cot(y)\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{\tan(y)}{x^2 - 1} ((0) - (-2x \cot(y))) \\ &= -\frac{2x}{x^2 - 1}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{2x}{x^2 - 1} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-\ln(x^2 - 1)} \\ &= \frac{1}{x^2 - 1}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2 - 1}(1) \\ &= \frac{1}{x^2 - 1}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2 - 1}((-x^2 + 1) \cot(y)) \\ &= -\cot(y)\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1}{x^2 - 1}\right) + (-\cot(y)) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{1}{x^2 - 1} dx \\ \phi &= -\operatorname{arctanh}(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 0 + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\cot(y)$. Therefore equation (4) becomes

$$-\cot(y) = 0 + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\cot(y)$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int (-\cot(y)) dy$$

$$f(y) = -\ln(\sin(y)) + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -\operatorname{arctanh}(x) - \ln(\sin(y)) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\operatorname{arctanh}(x) - \ln(\sin(y))$$

Solving for y gives

$$y = \arcsin(e^{-\operatorname{arctanh}(x)-c_1})$$

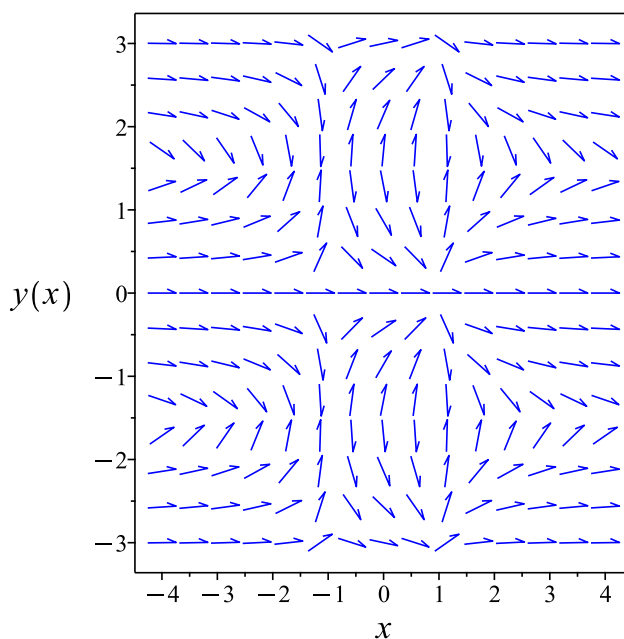


Figure 2.136: Slope field $1 + (-x^2 + 1) \cot(y) y' = 0$

Summary of solutions found

$$y = \arcsin(e^{-\operatorname{arctanh}(x)-c_1})$$

Solved using first_order_ode_LIE

Time used: 1.352 (sec)

Solve

$$1 + (-x^2 + 1) \cot(y) y' = 0$$

Writing the ode as

$$y' = \frac{1}{\cot(y) (x^2 - 1)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 2 to use as anstaz gives

$$\xi = x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (1\text{E})$$

$$\eta = x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3, b_4, b_5, b_6\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 2xb_4 + yb_5 + b_2 + \frac{-2xa_4 + xb_5 - ya_5 + 2yb_6 - a_2 + b_3}{\cot(y) (x^2 - 1)} \\ & - \frac{xa_5 + 2ya_6 + a_3}{\cot(y)^2 (x^2 - 1)^2} + \frac{2x(x^2 a_4 + yxa_5 + y^2 a_6 + xa_2 + ya_3 + a_1)}{\cot(y) (x^2 - 1)^2} \\ & + \frac{(-1 - \cot(y)^2) (x^2 b_4 + yxb_5 + y^2 b_6 + xb_2 + yb_3 + b_1)}{\cot(y)^2 (x^2 - 1)} = 0 \end{aligned} \quad (5\text{E})$$

Putting the above in normal form gives

$$\begin{aligned} & -\cot(y)^2 x^2 y b_3 + 2 \cot(y) x y a_3 + \cot(y)^2 x^4 y b_5 - \cot(y)^2 x^3 y b_5 - \cot(y)^2 x^2 y^2 b_6 - 2 \cot(y)^2 x^2 y b_5 + \cot(y) \\ & = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned}
& -\cot(y)^2 x^2 y b_3 + 2 \cot(y) x y a_3 + \cot(y)^2 x^4 y b_5 - \cot(y)^2 x^3 y b_5 \\
& - \cot(y)^2 x^2 y^2 b_6 - 2 \cot(y)^2 x^2 y b_5 + \cot(y)^2 x y b_5 + \cot(y) x^2 y a_5 \\
& + 2 \cot(y) x^2 y b_6 + 2 \cot(y) x y^2 a_6 - x a_5 - 2 y a_6 - x^4 b_4 + x b_2 + y b_3 + x^2 b_4 \\
& + y^2 b_6 - a_3 + b_1 + y x b_5 - x^3 y b_5 - x^2 y^2 b_6 + 2 \cot(y)^2 x^5 b_4 - \cot(y)^2 x^4 b_4 \\
& - 4 \cot(y)^2 x^3 b_4 + \cot(y)^2 x^2 b_4 + \cot(y)^2 y^2 b_6 + \cot(y) x^3 b_5 - x^3 b_2 \\
& - x^2 b_1 + \cot(y)^2 b_1 + b_2 \cot(y)^2 + \cot(y) a_2 - \cot(y) b_3 + 2 x b_4 \cot(y)^2 \\
& + y b_5 \cot(y)^2 + 2 \cot(y) x a_4 - \cot(y) x b_5 + \cot(y) y a_5 - 2 \cot(y) y b_6 \\
& - x^2 y b_3 + \cot(y)^2 x^4 b_2 - \cot(y)^2 x^3 b_2 - \cot(y)^2 x^2 b_1 - 2 \cot(y)^2 x^2 b_2 \\
& + \cot(y)^2 x b_2 + \cot(y)^2 y b_3 + \cot(y) x^2 a_2 + \cot(y) x^2 b_3 + 2 \cot(y) x a_1 = 0
\end{aligned} \tag{6E}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, \cot(y)\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, \cot(y) = v_3\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
& 2v_3^2 v_1^5 b_4 + v_3^2 v_1^4 v_2 b_5 + v_3^2 v_1^4 b_2 - v_3^2 v_1^4 b_4 - v_3^2 v_1^3 v_2 b_5 - v_3^2 v_1^2 v_2^2 b_6 - v_3^2 v_1^3 b_2 - v_3^2 v_1^2 v_2 b_3 \\
& - 4v_3^2 v_1^3 b_4 - 2v_3^2 v_1^2 v_2 b_5 + v_3 v_1^2 v_2 a_5 + 2v_3 v_1 v_2^2 a_6 - v_3^2 v_1^2 b_1 - 2v_3^2 v_1^2 b_2 - v_1^4 b_4 \\
& + v_3^2 v_1^2 b_4 - v_1^3 v_2 b_5 + v_3 v_1^3 b_5 + v_3^2 v_1 v_2 b_5 - v_1^2 v_2^2 b_6 + 2v_3 v_1^2 v_2 b_6 + v_3^2 v_2^2 b_6 + v_3 v_1^2 a_2 \\
& + 2v_3 v_1 v_2 a_3 - v_1^3 b_2 + v_3^2 v_1 b_2 - v_1^2 v_2 b_3 + v_3 v_1^2 b_3 + v_3^2 v_2 b_3 + 2v_1 b_4 v_3^2 + v_2 b_5 v_3^2 \\
& + 2v_3 v_1 a_1 + 2v_3 v_1 a_4 + v_3 v_2 a_5 - v_1^2 b_1 + v_3^2 b_1 + b_2 v_3^2 + v_1^2 b_4 + v_2 v_1 b_5 - v_3 v_1 b_5 \\
& + v_2^2 b_6 - 2v_3 v_2 b_6 + v_3 a_2 - v_1 a_5 - 2v_2 a_6 + v_1 b_2 + v_2 b_3 - v_3 b_3 - a_3 + b_1 = 0
\end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$\begin{aligned}
& 2v_3 v_1 v_2 a_3 + v_3^2 v_1^4 v_2 b_5 - v_3^2 v_1^3 v_2 b_5 - v_3^2 v_1^2 v_2^2 b_6 + v_3^2 v_1 v_2 b_5 + 2v_3 v_1 v_2^2 a_6 \\
& + (b_2 - b_4) v_1^4 v_3^2 + (-b_2 - 4b_4) v_1^3 v_3^2 + (-b_1 - 2b_2 + b_4) v_1^2 v_3^2 + (a_2 + b_3) v_1^2 v_3 \\
& + (b_2 + 2b_4) v_1 v_3^2 + (2a_1 + 2a_4 - b_5) v_1 v_3 + (b_3 + b_5) v_2 v_3^2 + (a_5 - 2b_6) v_2 v_3 \\
& - a_3 + b_1 + v_2 v_1 b_5 - v_1^3 v_2 b_5 - v_1^2 v_2^2 b_6 + 2v_3^2 v_1^5 b_4 + v_3^2 v_2^2 b_6 + v_3 v_1^3 b_5 - v_1^2 v_2 b_3 \\
& + (-b_3 - 2b_5) v_1^2 v_2 v_3^2 + (a_5 + 2b_6) v_1^2 v_2 v_3 - v_1^4 b_4 + v_2^2 b_6 - v_1^3 b_2 + (-a_5 + b_2) v_1 \\
& + (-2a_6 + b_3) v_2 + (b_1 + b_2) v_3^2 + (-b_3 + a_2) v_3 + (-b_1 + b_4) v_1^2 = 0
\end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$b_5 = 0$$

$$b_6 = 0$$

$$2a_3 = 0$$

$$2a_6 = 0$$

$$-b_2 = 0$$

$$-b_3 = 0$$

$$-b_4 = 0$$

$$2b_4 = 0$$

$$-b_5 = 0$$

$$-b_6 = 0$$

$$a_2 + b_3 = 0$$

$$-a_3 + b_1 = 0$$

$$-a_5 + b_2 = 0$$

$$a_5 - 2b_6 = 0$$

$$a_5 + 2b_6 = 0$$

$$-2a_6 + b_3 = 0$$

$$-b_1 + b_4 = 0$$

$$b_1 + b_2 = 0$$

$$-b_2 - 4b_4 = 0$$

$$b_2 - b_4 = 0$$

$$b_2 + 2b_4 = 0$$

$$-b_3 + a_2 = 0$$

$$-b_3 - 2b_5 = 0$$

$$b_3 + b_5 = 0$$

$$2a_1 + 2a_4 - b_5 = 0$$

$$-b_1 - 2b_2 + b_4 = 0$$

Solving the above equations for the unknowns gives

$$a_1 = -a_4$$

$$a_2 = 0$$

$$a_3 = 0$$

$$a_4 = a_4$$

$$a_5 = 0$$

$$a_6 = 0$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = 0$$

$$b_4 = 0$$

$$b_5 = 0$$

$$b_6 = 0$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x^2 - 1$$

$$\eta = 0$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= 0 - \left(\frac{1}{\cot(y)(x^2 - 1)} \right) (x^2 - 1) \\ &= -\tan(y) \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{-\tan(y)} dy \end{aligned}$$

Which results in

$$S = -\ln(\sin(y))$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{1}{\cot(y)(x^2 - 1)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= -\cot(y) \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{x^2 - 1} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R^2 - 1}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

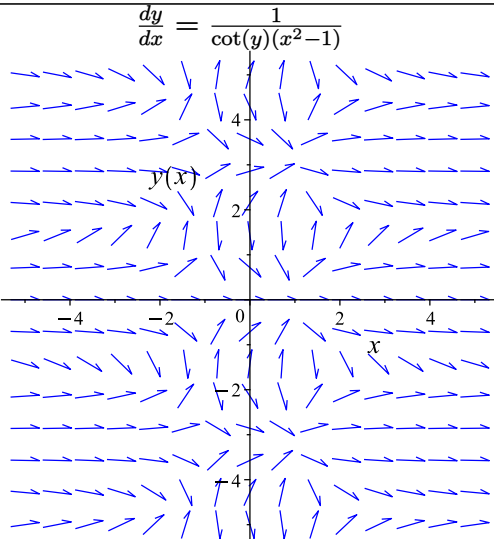
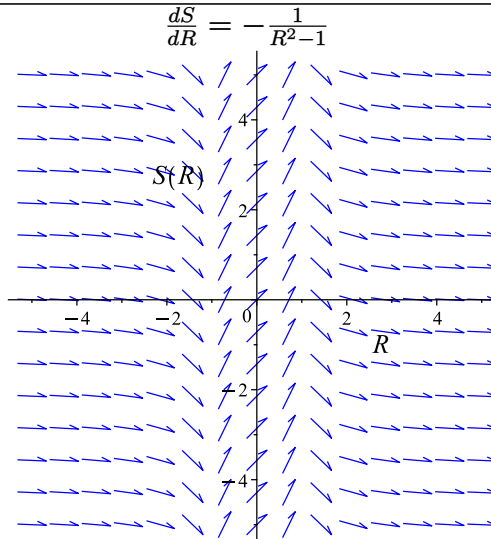
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{R^2 - 1} dR \\ S(R) &= \operatorname{arctanh}(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\ln(\sin(y)) = \operatorname{arctanh}(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{1}{\cot(y)(x^2-1)}$ 	$R = x$ $S = -\ln(\sin(y))$	$\frac{dS}{dR} = -\frac{1}{R^2-1}$ 

Solving for y gives

$$y = \arcsin(e^{-\operatorname{arctanh}(x)-c_2})$$

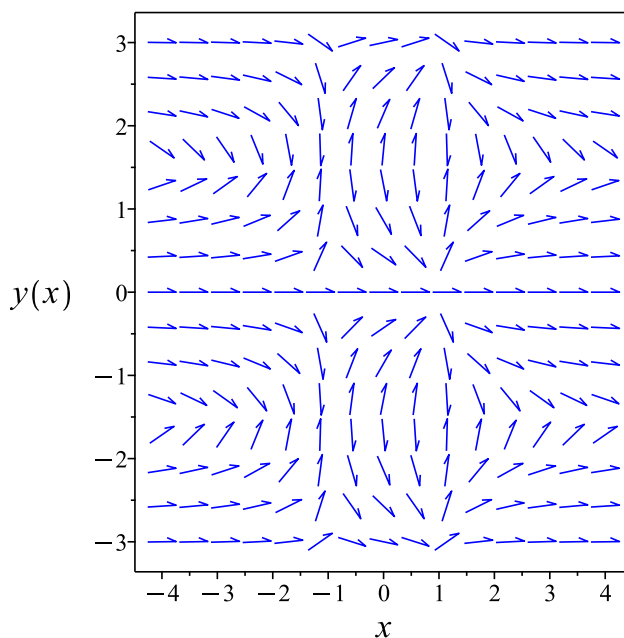


Figure 2.137: Slope field $1 + (-x^2 + 1) \cot(y) y' = 0$

Summary of solutions found

$$y = \arcsin \left(e^{-\operatorname{arctanh}(x) - c_2} \right)$$

✓ Maple. Time used: 0.004 (sec). Leaf size: 21

```
ode:=1+(-x^2+1)*cot(y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \arcsin \left(\frac{\sqrt{-x^2 + 1} c_1}{x + 1} \right)$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

<- separable successful

Maple step by step

Let's solve

$$1 + (-x^2 + 1) \cot(y(x)) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{1}{(-x^2 + 1) \cot(y(x))}$$

- Separate variables

$$\left(\frac{d}{dx} y(x) \right) \cot(y(x)) = -\frac{1}{-x^2 + 1}$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx} y(x) \right) \cot(y(x)) dx = \int -\frac{1}{-x^2 + 1} dx + C1$$

- Evaluate integral

$$\ln(\sin(y(x))) = -\operatorname{arctanh}(x) + C1$$

- Solve for $y(x)$

$$y(x) = \arcsin \left(e^{-\operatorname{arctanh}(x) + C1} \right)$$

- Redefine the integration constant(s)

$$y(x) = \arcsin \left(\frac{C1 \sqrt{-x^2 + 1}}{x + 1} \right)$$

✓ **Mathematica.** Time used: 0.077 (sec). Leaf size: 26

```
ode=1+(1-x^2)*Cot[y[x]]*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \arcsin \left(\exp \left(\int_1^x \frac{1}{K[1]^2 - 1} dK[1] + c_1 \right) \right)$$

✓ **Sympy.** Time used: 0.334 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((1 - x**2)*Derivative(y(x), x)/tan(y(x)) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = \pi - \arcsin \left(\frac{C_1 \sqrt{x-1}}{\sqrt{x+1}} \right), y(x) = \arcsin \left(\frac{C_1 \sqrt{x-1}}{\sqrt{x+1}} \right) \right]$$

2.2.19 Problem 44

Local contents

Solved using first_order_ode_bernoulli	558
Solved using first_order_ode_exact	561
Solved using first_order_ode_isobaric	564
Solved using first_order_ode_homog_A	568
Solved using first_order_ode_dAlembert	571
Solved using first_order_ode_homog_type_D2	582
Solved using first_order_ode_homog_type_G	584
Solved using first_order_ode_homog_type_maple_C	586
Solved using first_order_ode_LIE	591
✓ Maple	597
✓ Mathematica	598
✓ Sympy	599

Internal problem ID [7873]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 44

Date solved : Thursday, November 27, 2025 at 06:26:10 AM

CAS classification :

[[_homogeneous, 'class A'], _exact, _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.115 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^3 + y^3}{3y^2x} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{3x}\right)y + \left(-\frac{x^2}{3}\right)\frac{1}{y^2} \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -\frac{1}{3x} \\ f_1 &= -\frac{x^2}{3} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= -\frac{1}{3x} \\ f_1(x) &= -\frac{x^2}{3} \\ n &= -2 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y^2}$ gives

$$y'y^2 = -\frac{y^3}{3x} - \frac{x^2}{3} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= y^3 \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 3y^2y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(x)}{3} &= -\frac{v(x)}{3x} - \frac{x^2}{3} \\ v' &= -\frac{v}{x} - x^2 \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{1}{x} \\ p(x) &= -x^2 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{1}{x} \, dx} \\ &= x \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) (-x^2) \\ \frac{d}{dx}(vx) &= (x) (-x^2) \\ d(vx) &= (-x^3) \, dx \end{aligned}$$

Integrating gives

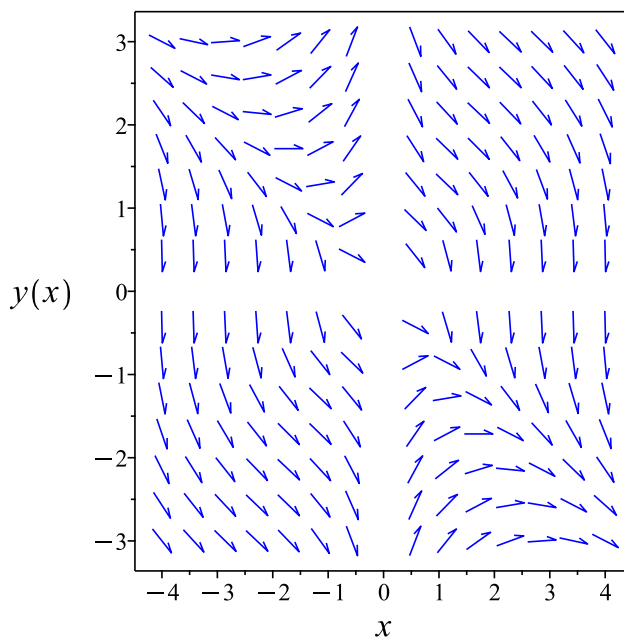
$$\begin{aligned} vx &= \int -x^3 \, dx \\ &= -\frac{x^4}{4} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

$$v(x) = \frac{-\frac{x^4}{4} + c_1}{x}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^3 = \frac{-\frac{x^4}{4} + c_1}{x}$$

Figure 2.138: Slope field $x^3 + y^3 + 3xy^2y' = 0$ Summary of solutions found

$$y^3 = \frac{-\frac{x^4}{4} + c_1}{x}$$

Solved using first_order_ode_exact

Time used: 0.086 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (3x y^2) dy &= (-x^3 - y^3) dx \\ (x^3 + y^3) dx + (3x y^2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^3 + y^3 \\ N(x, y) &= 3x y^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x^3 + y^3) \\ &= 3y^2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (3x y^2) \\ &= 3y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x^3 + y^3 dx \\ \phi &= \frac{1}{4}x^4 + x y^3 + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 3x y^2 + f'(y)\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 3x y^2$. Therefore equation (4) becomes

$$3x y^2 = 3x y^2 + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

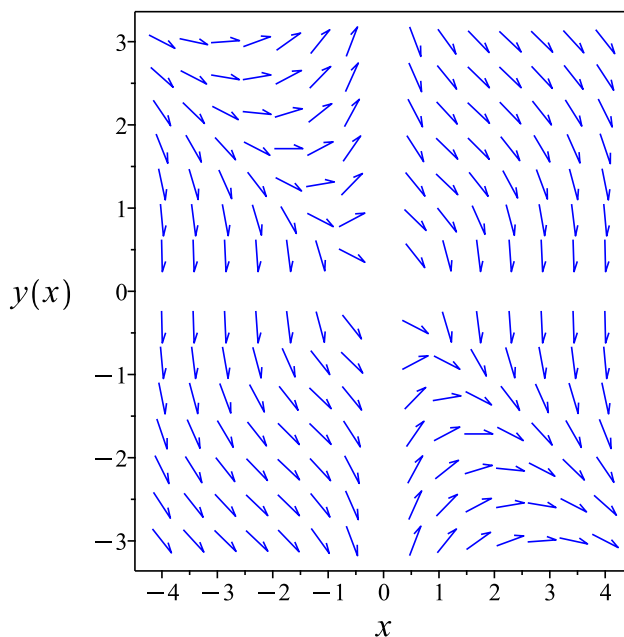
$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{1}{4}x^4 + x y^3 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{1}{4}x^4 + x y^3$$

Figure 2.139: Slope field $x^3 + y^3 + 3xy^2y' = 0$ Summary of solutions found

$$\frac{x^4}{4} + xy^3 = c_1$$

Solved using first_order_ode_isobaric

Time used: 0.346 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Solving for y' gives

$$y' = -\frac{y^3 + x^3}{3xy^2} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{y^3 + x^3}{3xy^2} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x^3u(x)^3 + x^3}{3x^3u(x)^2}$$

The ode

$$u'(x) = -\frac{4u(x)^3 + 1}{3xu(x)^2} \quad (2.64)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{4u(x)^3 + 1}{3xu(x)^2} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{3x} \\ g(u) &= \frac{4u^3 + 1}{u^2} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u^2}{4u^3 + 1} du &= \int -\frac{1}{3x} dx \end{aligned}$$

$$\frac{\ln(4u(x)^3 + 1)}{12} = \ln\left(\frac{1}{x^{1/3}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{4u^3 + 1}{u^2} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Converting $(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$ back to y gives

$$\left(\frac{4y^3}{x^3} + 1\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

Converting $u(x) = -\frac{2^{1/3}}{2}$ back to y gives

$$\frac{y}{x} = -\frac{2^{1/3}}{2}$$

Converting $u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$ back to y gives

$$\frac{y}{x} = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Simplifying the above gives

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$\frac{y}{x} = -\frac{2^{1/3}}{2}$$

$$\frac{y}{x} = -\frac{2^{1/3}(i\sqrt{3} - 1)}{4}$$

Solving for y gives

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

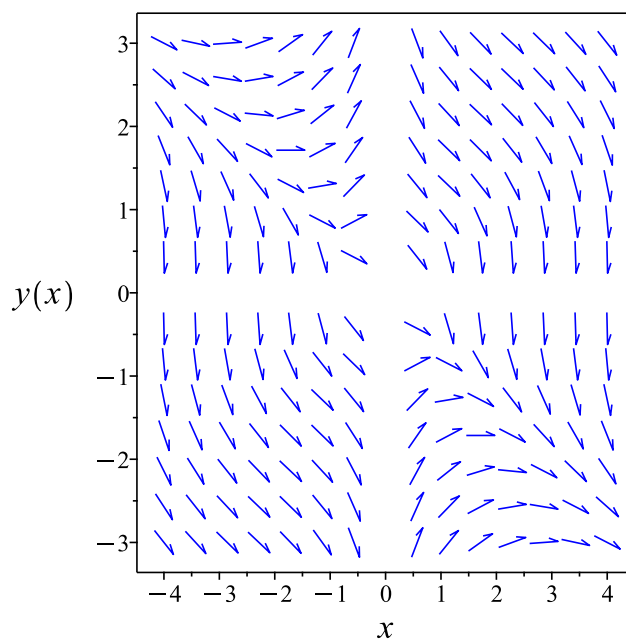


Figure 2.140: Slope field $x^3 + y^3 + 3xy^2y' = 0$

Summary of solutions found

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

Solved using first_order_ode_homog_A

Time used: 0.796 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^3 + y^3}{3y^2x} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x^3 - y^3$ and $N = 3xy^2$ are both homogeneous and of the same order $n = 3$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= -\frac{1}{3u^2} - \frac{u}{3} \\ \frac{du}{dx} &= \frac{-\frac{1}{3u(x)^2} - \frac{4u(x)}{3}}{x} \end{aligned}$$

Or

$$u'(x) - \frac{-\frac{1}{3u(x)^2} - \frac{4u(x)}{3}}{x} = 0$$

Or

$$3u'(x)u(x)^2x + 4u(x)^3 + 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{4u(x)^3 + 1}{3u(x)^2x} \quad (2.65)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{4u(x)^3 + 1}{3u(x)^2x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{1}{3x}$$

$$g(u) = \frac{4u^3 + 1}{u^2}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{u^2}{4u^3 + 1} du = \int -\frac{1}{3x} dx$$

$$\frac{\ln(4u(x)^3 + 1)}{12} = \ln\left(\frac{1}{x^{1/3}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{4u^3 + 1}{u^2} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Converting $(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$ back to y gives

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

Converting $u(x) = -\frac{2^{1/3}}{2}$ back to y gives

$$y = -\frac{2^{1/3}x}{2}$$

Converting $u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$ back to y gives

$$y = x\left(\frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}\right)$$

Simplifying the above gives

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3} - 1)x}{4}$$

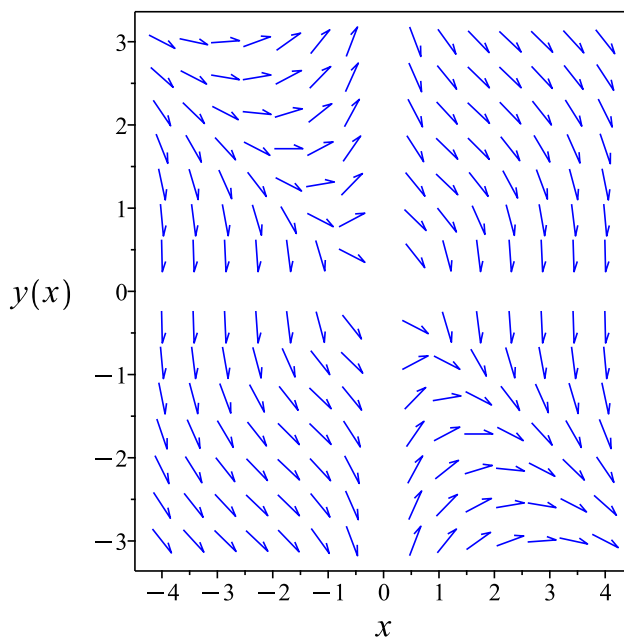


Figure 2.141: Slope field $x^3 + y^3 + 3xy^2y' = 0$

Summary of solutions found

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

Solved using first_order_ode_dAlembert

Time used: 2.641 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Let $p = y'$ the ode becomes

$$3xy^2p + x^3 + y^3 = 0$$

Solving for y from the above results in

$$y = \left(\frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} + \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3} - p} \right) x \quad (1)$$

$$y = \left(-\frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} - \frac{p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3} - p} \right. \\ \left. + \frac{i\sqrt{3} \left(\frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} - \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} \right)}{2} \right) x \quad (2)$$

$$y = \left(-\frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} - \frac{p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} - p - \frac{i\sqrt{3} \left(\frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} - \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} \right)}{2} \right) x \quad (3)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} + \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} - p \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\begin{aligned} 2p - \frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} - \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} &= \left(-\frac{4xp^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{2/3}} + \frac{4xp^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{2/3} \sqrt{4p^3 + 1}} + \dots \right) \end{aligned} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$2p - \frac{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{2} - \frac{2p^2}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} = 0$$

Solving the above for p results in

$$p_1 = \frac{(-2)^{1/3}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{2^{1/3}(1 + i\sqrt{3})x}{4}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$\begin{aligned} p'(x) &= \frac{2p(x) - \frac{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}}{2} - \frac{2p(x)^2}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)}}{\frac{4xp(x)^2}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}} + \frac{4xp(x)^2}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}\sqrt{4p(x)^3+1}} + \frac{4xp(x)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}} + \frac{16xp(x)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)}} \end{aligned} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$\begin{aligned} p'(x) &= \frac{2x \left(8p(x)^5 \sqrt{4p(x)^3+1} - 4p(x)^4 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3} \sqrt{4p(x)^3+1} + 2p(x)^3 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3} \right)}{\left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{2/3} \sqrt{4p(x)^3+1} + 4xp(x)^2 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3}} \end{aligned} \quad (2.66)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{2x \left(8p(x)^5 \sqrt{4p(x)^3+1} - 4p(x)^4 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3} \sqrt{4p(x)^3+1} + 2p(x)^3 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3} \right)}{\left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{2/3} \sqrt{4p(x)^3+1} + 4xp(x)^2 \left(-4 - 8p(x)^3 + 4\sqrt{4p(x)^3+1} \right)^{1/3}} \\ &= f(x)g(p) \end{aligned}$$

Where

$$f(x) = \frac{1}{2x}$$

$$g(p) = \frac{1}{2 \left(-4 - 8p^3 + 4\sqrt{4p^3+1} \right)^{2/3} p^3 \sqrt{4p^3+1} - 4\sqrt{4p^3+1} \left(-4 - 8p^3 + 4\sqrt{4p^3+1} \right)^{1/3} p^4 + 8\sqrt{4p^3+1} p^3}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{2(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{2/3} p^3 \sqrt{4p^3 + 1} - 4\sqrt{4p^3 + 1} (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3} p^4 + 8\sqrt{4p^3 + 1} p^5 - 4}{}$$

$$\int^{p(x)} \frac{2(-4 - 8\tau^3 + 4\sqrt{4\tau^3 + 1})^{2/3} \tau^3 \sqrt{4\tau^3 + 1} - 4\sqrt{4\tau^3 + 1} (-4 - 8\tau^3 + 4\sqrt{4\tau^3 + 1})^{1/3} \tau^4 + 8\sqrt{4\tau^3 + 1} \tau^5 - 4}{}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$2(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{2/3} p^3 \sqrt{4p^3 + 1} - 4\sqrt{4p^3 + 1} (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3} p^4 + 8\sqrt{4p^3 + 1} p^5 - 4(-$$

for $p(x)$ gives

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{2(-4 - 8\tau^3 + 4\sqrt{4\tau^3 + 1})^{2/3} \tau^3 \sqrt{4\tau^3 + 1} - 4\sqrt{4\tau^3 + 1} (-4 - 8\tau^3 + 4\sqrt{4\tau^3 + 1})^{1/3} \tau^4 + 8\sqrt{4\tau^3 + 1} \tau^5 - 4}{}$$

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Substituting the above solution for p in (2A) gives

Expression too large to display

$$y = 2^{1/3}x$$

$$y = \frac{2^{1/3}(i\sqrt{3} - 1)x}{2}$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{(i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} - \frac{p(i\sqrt{3}p + p + (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}$$

$$g = 0$$

Hence (2) becomes

$$\begin{aligned} p - \frac{(i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} \\ + \frac{p(i\sqrt{3}p + p + (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} &= \left(-\frac{2ix\sqrt{3}p}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} + \frac{8ixp^4\sqrt{3}}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^4} \right) \end{aligned} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{(i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} + \frac{p(i\sqrt{3}p + p + (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} = 0$$

Solving the above for p results in

$$p_1 = -\frac{2^{1/3}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{2^{1/3}x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) \tag{3}$$

$$= -\frac{2ix\sqrt{3}p(x)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}} + \frac{8ixp(x)^4\sqrt{3}}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{4/3}\sqrt{4p(x)^3+1}} + \frac{2xp(x)^2}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}} - \frac{(-4-8p(x)^3+4\sqrt{4p(x)^3+1})^{1/3}}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}}$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \tag{2.67}$$

$$-\frac{\left(4i\sqrt{3}p(x)^4\sqrt{4p(x)^3+1}+2i\sqrt{3}\sqrt{4p(x)^3+1}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}\right)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}}$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{\left(4i\sqrt{3}p(x)^4\sqrt{4p(x)^3+1}+2i\sqrt{3}\sqrt{4p(x)^3+1}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}\right)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}} \\ &= f(x)g(p) \end{aligned}$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(p) = \frac{4i\sqrt{3}\sqrt{4p^3+1}p+i\sqrt{3}\sqrt{4p^3+1}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}+4i\sqrt{3}\sqrt{4p^3+1}p^4-i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}}{\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{2/3}}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{4i\sqrt{3}\sqrt{4p^3+1}p+i\sqrt{3}\sqrt{4p^3+1}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}+4i\sqrt{3}\sqrt{4p^3+1}p^4-i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}}{\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{2/3}} dp = \int -\frac{1}{x} dx$$

$$\int^{p(x)} \frac{-4i\sqrt{3}\tau - 12i\sqrt{3}\tau^4 + 4i\sqrt{3}\sqrt{4\tau^3+1}\tau^4 + 2i\sqrt{3}\sqrt{4\tau^3+1}(-4-8\tau^3+4\sqrt{4\tau^3+1})^{1/3}\tau^3 + 2\sqrt{4\tau^3+1}}{}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$4i\sqrt{3}\sqrt{4p^3+1}p + i\sqrt{3}\sqrt{4p^3+1}(-4-8p^3+4\sqrt{4p^3+1})^{1/3} + 4i\sqrt{3}\sqrt{4p^3+1}p^4 - i\sqrt{3}(-4-8p^3+4\sqrt{4p^3+1})^{1/3}$$

for $p(x)$ gives

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{-4i\sqrt{3}\tau - 12i\sqrt{3}\tau^4 + 4i\sqrt{3}\sqrt{4\tau^3+1}\tau^4 + 2i\sqrt{3}\sqrt{4\tau^3+1}(-4-8\tau^3+4\sqrt{4\tau^3+1})^{1/3}\tau^3 + 2\sqrt{4\tau^3+1}}{}$$

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Substituting the above solution for p in (2A) gives

Expression too large to display

$$y = -\frac{2^{1/3}x}{2}$$

$$y = \frac{2^{1/3}(i\sqrt{3}-1)x}{2}$$

Solving ode 3A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{(-i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} + \frac{(i\sqrt{3}p - p - (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})p}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}$$

$$g = 0$$

Hence (2) becomes

$$\begin{aligned} p - \frac{(-i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} \\ - \frac{(i\sqrt{3}p - p - (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})p}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} &= \left(\frac{2ix\sqrt{3}p}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} - \frac{8ixp^4\sqrt{3}}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{4/3}} \right) \end{aligned} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{(-i\sqrt{3} - 1)(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}}{4} - \frac{(i\sqrt{3}p - p - (-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3})p}{(-4 - 8p^3 + 4\sqrt{4p^3 + 1})^{1/3}} = 0$$

Solving the above for p results in

$$p_1 = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{2^{1/3}(i\sqrt{3} - 1)x}{4}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) \quad (3)$$

$$= \frac{2ix\sqrt{3}p(x)}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}} - \frac{8ixp(x)^4\sqrt{3}}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{4/3}\sqrt{4p(x)^3+1}} + \frac{2xp(x)^2}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{2/3}} - \frac{2xp(x)^4}{\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{5/3}}$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) \tag{2.68}$$

$$= \frac{\left(4i\sqrt{3}p(x)^4\sqrt{4p(x)^3+1}+2i\sqrt{3}\sqrt{4p(x)^3+1}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}\right)}{p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}}$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{\left(4i\sqrt{3}p(x)^4\sqrt{4p(x)^3+1}+2i\sqrt{3}\sqrt{4p(x)^3+1}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}\right)}{p(x)^3-4i\sqrt{3}\left(-4-8p(x)^3+4\sqrt{4p(x)^3+1}\right)^{1/3}} \\ &= f(x)g(p) \end{aligned}$$

Where

$$f(x) = \frac{1}{x}$$

$$g(p) = \frac{4i\sqrt{3}\sqrt{4p^3+1}p+i\sqrt{3}\sqrt{4p^3+1}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}+4i\sqrt{3}\sqrt{4p^3+1}p^4-i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}}{p^3-4i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{4i\sqrt{3}\sqrt{4p^3+1}p+i\sqrt{3}\sqrt{4p^3+1}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}+4i\sqrt{3}\sqrt{4p^3+1}p^4-i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}}{p^3-4i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}} dp = \int \frac{1}{x} dx$$

$$\int^{p(x)} \frac{2i\sqrt{3}\sqrt{4\tau^3+1}\left(-4-8\tau^3+4\sqrt{4\tau^3+1}\right)^{1/3}\tau^3-4i\sqrt{3}\tau+i\sqrt{3}\sqrt{4\tau^3+1}\left(-4-8\tau^3+4\sqrt{4\tau^3+1}\right)^{1/3}}{\tau^3-4i\sqrt{3}\left(-4-8\tau^3+4\sqrt{4\tau^3+1}\right)^{1/3}} d\tau = \ln|x| + C$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$4i\sqrt{3}\sqrt{4p^3+1}p+i\sqrt{3}\sqrt{4p^3+1}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3}+4i\sqrt{3}\sqrt{4p^3+1}p^4-i\sqrt{3}\left(-4-8p^3+4\sqrt{4p^3+1}\right)^{1/3} = 0$$

for $p(x)$ gives

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\int^{p(x)} \frac{2i\sqrt{3}\sqrt{4\tau^3+1}(-4-8\tau^3+4\sqrt{4\tau^3+1})^{1/3}\tau^3 - 4i\sqrt{3}\tau + i\sqrt{3}\sqrt{4\tau^3+1}(-4-8\tau^3+4\sqrt{4\tau^3+1})^{1/3}}{d\tau} -$$

$$p(x) = -\frac{2^{1/3}}{2}$$

$$p(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Substituting the above solution for p in (2A) gives

Expression too large to display

$$y = 2^{1/3}x$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

The solution

$$y = 2^{1/3}x$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

Expression too large to display

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Expression too large to display

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$$y = \frac{2^{1/3}(i\sqrt{3} - 1)x}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.

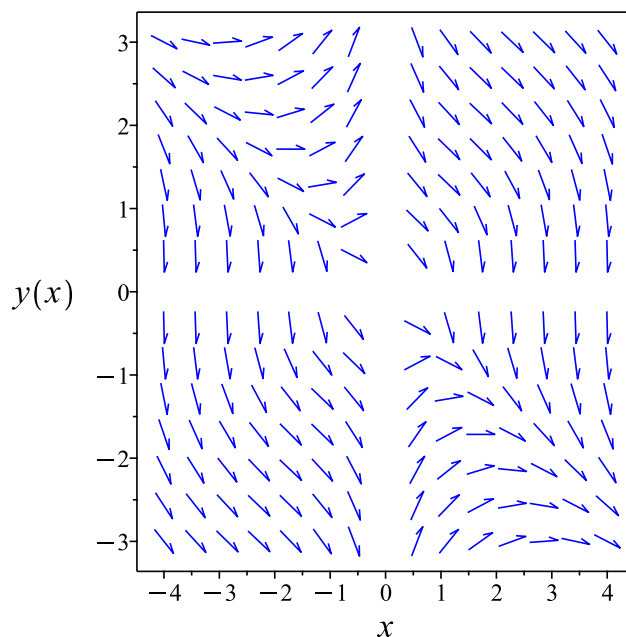


Figure 2.142: Slope field $x^3 + y^3 + 3xy^2y' = 0$

Summary of solutions found

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3} - 1)x}{4}$$

$$y = \frac{2^{1/3}(1 + i\sqrt{3})x}{4}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.191 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$x^3 + u(x)^3 x^3 + 3x^3 u(x)^2 (u'(x)x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{4u(x)^3 + 1}{3u(x)^2 x} \quad (2.69)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{4u(x)^3 + 1}{3u(x)^2 x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{3x} \\ g(u) &= \frac{4u^3 + 1}{u^2} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u^2}{4u^3 + 1} du &= \int -\frac{1}{3x} dx \end{aligned}$$

$$\frac{\ln(4u(x)^3 + 1)}{12} = \ln\left(\frac{1}{x^{1/3}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{4u^3 + 1}{u^2} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$u(x) = -\frac{2^{1/3}}{2}$$

$$u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Converting $(4u(x)^3 + 1)^{1/12} = \frac{c_1}{x^{1/3}}$ back to y gives

$$\left(\frac{4y^3}{x^3} + 1\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

Converting $u(x) = -\frac{2^{1/3}}{2}$ back to y gives

$$y = -\frac{2^{1/3}x}{2}$$

Converting $u(x) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$ back to y gives

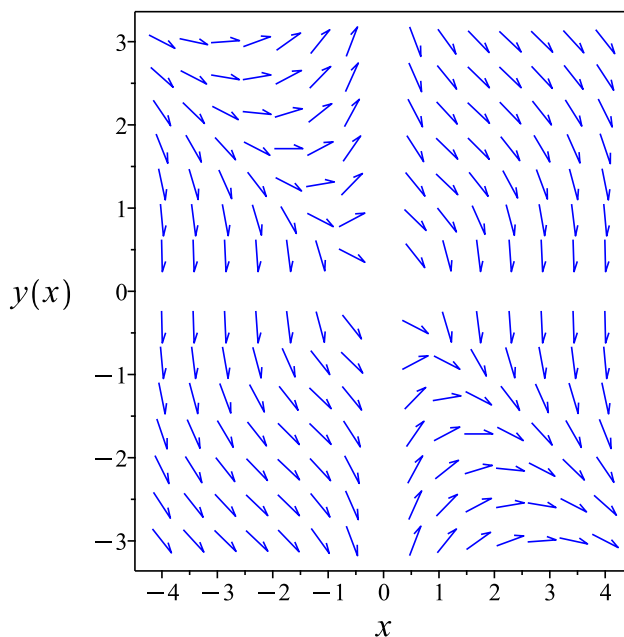
$$y = x \left(\frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4} \right)$$

Simplifying the above gives

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3} - 1)x}{4}$$

Figure 2.143: Slope field $x^3 + y^3 + 3xy^2y' = 0$ Summary of solutions found

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

Solved using first_order_ode_homog_type_G

Time used: 0.085 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Multiplying the right side of the ode, which is $-\frac{x^3+y^3}{3y^2x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x^3 + y^3}{3y^2x} \\ &= -\frac{x^3 + y^3}{3y^3} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x^3}{y^3} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x^3}{y^3} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z^3 + 1}{3z^3}$$

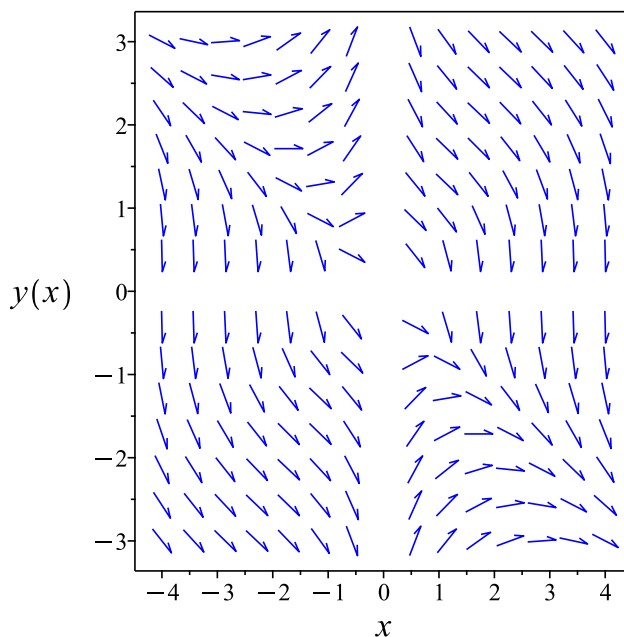
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{z^3 + 1}{3z^3} \right)} dz = 0$$

Figure 2.144: Slope field $x^3 + y^3 + 3xy^2y' = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{z^3+1}{3z^3}\right)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.152 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{(Y(X) + y_0)^3 + (X + x_0)^3}{3(X + x_0)(Y(X) + y_0)^2}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X^3 + Y(X)^3}{3XY(X)^2}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X^3 + Y^3}{3XY^2} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X^3 - Y^3$ and $N = 3XY^2$ are both homogeneous and of the same order $n = 3$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -\frac{1}{3u^2} - \frac{u}{3} \\ \frac{du}{dX} &= \frac{-\frac{1}{3u(X)^2} - \frac{4u(X)}{3}}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-\frac{1}{3u(X)^2} - \frac{4u(X)}{3}}{X} = 0$$

Or

$$3\left(\frac{d}{dX}u(X)\right)u(X)^2X + 4u(X)^3 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{4u(X)^3 + 1}{3u(X)^2X} \quad (2.70)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{4u(X)^3 + 1}{3u(X)^2X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{1}{3X} \\ g(u) &= \frac{4u^3 + 1}{u^2} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{u^2}{4u^3 + 1} du = \int -\frac{1}{3X} dX$$

$$\frac{\ln(4u(X)^3 + 1)}{12} = \ln\left(\frac{1}{X^{1/3}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(4u(X)^3 + 1)^{1/12} = \frac{c_1}{X^{1/3}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{4u^3 + 1}{u^2} = 0$$

for $u(X)$ gives

$$u(X) = -\frac{2^{1/3}}{2}$$

$$u(X) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(4u(X)^3 + 1)^{1/12} = \frac{c_1}{X^{1/3}}$$

$$u(X) = -\frac{2^{1/3}}{2}$$

$$u(X) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$$

Converting $(4u(X)^3 + 1)^{1/12} = \frac{c_1}{X^{1/3}}$ back to $Y(X)$ gives

$$\left(\frac{4Y(X)^3 + X^3}{X^3}\right)^{1/12} = \frac{c_1}{X^{1/3}}$$

Converting $u(X) = -\frac{2^{1/3}}{2}$ back to $Y(X)$ gives

$$Y(X) = -\frac{X2^{1/3}}{2}$$

Converting $u(X) = \frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}$ back to $Y(X)$ gives

$$Y(X) = X\left(\frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4}\right)$$

Using the solution for $Y(X)$

$$\left(\frac{4Y(X)^3 + X^3}{X^3}\right)^{1/12} = \frac{c_1}{X^{1/3}} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{X2^{1/3}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{2^{1/3}x}{2}$$

Using the solution for $Y(X)$

$$Y(X) = X \left(\frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4} \right) \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

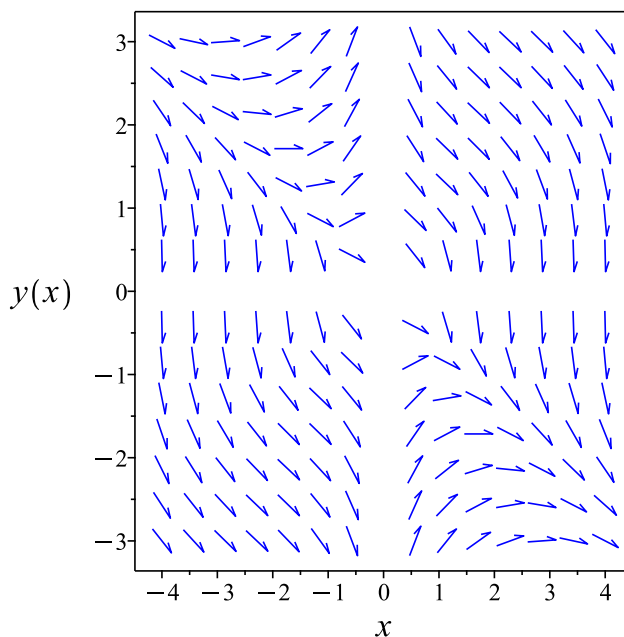
$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = x \left(\frac{2^{1/3}}{4} - \frac{i\sqrt{3}2^{1/3}}{4} \right)$$

Simplifying the above gives

$$\begin{aligned} \left(\frac{4y^3 + x^3}{x^3} \right)^{1/12} &= \frac{c_1}{x^{1/3}} \\ y &= -\frac{2^{1/3}x}{2} \\ y &= -\frac{2^{1/3}(i\sqrt{3}-1)x}{4} \end{aligned}$$

Figure 2.145: Slope field $x^3 + y^3 + 3xy^2y' = 0$ Summary of solutions found

$$\left(\frac{4y^3 + x^3}{x^3}\right)^{1/12} = \frac{c_1}{x^{1/3}}$$

$$y = -\frac{2^{1/3}x}{2}$$

$$y = -\frac{2^{1/3}(i\sqrt{3}-1)x}{4}$$

Solved using first_order_ode_LIE

Time used: 0.990 (sec)

Solve

$$x^3 + y^3 + 3xy^2y' = 0$$

Writing the ode as

$$y' = -\frac{x^3 + y^3}{3y^2x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2\xi_y - \omega_x\xi - \omega_y\eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1E)$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{(x^3 + y^3)(b_3 - a_2)}{3y^2x} - \frac{(x^3 + y^3)^2 a_3}{9y^4x^2} - \left(-\frac{x}{y^2} + \frac{x^3 + y^3}{3y^2x^2} \right) (xa_2 + ya_3 + a_1) \\ - \left(-\frac{1}{x} + \frac{\frac{2x^3}{3} + \frac{2y^3}{3}}{xy^3} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{x^6a_3 + 6x^5yb_2 - 9x^4y^2a_2 + 9x^4y^2b_3 - 4x^3y^3a_3 - 12b_2x^2y^4 + 4y^6a_3 + 6x^4yb_1 - 6x^3y^2a_1 - 3xy^4b_1 + 3y^5a_1}{9x^2y^4} = 0$$

Setting the numerator to zero gives

$$\begin{aligned} -x^6a_3 - 6x^5yb_2 + 9x^4y^2a_2 - 9x^4y^2b_3 + 4x^3y^3a_3 + 12b_2x^2y^4 \\ - 4y^6a_3 - 6x^4yb_1 + 6x^3y^2a_1 + 3xy^4b_1 - 3y^5a_1 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} 9a_2v_1^4v_2^2 - a_3v_1^6 + 4a_3v_1^3v_2^3 - 4a_3v_2^6 - 6b_2v_1^5v_2 + 12b_2v_1^2v_2^4 \\ - 9b_3v_1^4v_2^2 + 6a_1v_1^3v_2^2 - 3a_1v_2^5 - 6b_1v_1^4v_2 + 3b_1v_1v_2^4 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} & -a_3v_1^6 - 6b_2v_1^5v_2 + (9a_2 - 9b_3)v_1^4v_2^2 - 6b_1v_1^4v_2 + 4a_3v_1^3v_2^3 \\ & + 6a_1v_1^3v_2^2 + 12b_2v_1^2v_2^4 + 3b_1v_1v_2^4 - 4a_3v_2^6 - 3a_1v_2^5 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -3a_1 &= 0 \\ 6a_1 &= 0 \\ -4a_3 &= 0 \\ -a_3 &= 0 \\ 4a_3 &= 0 \\ -6b_1 &= 0 \\ 3b_1 &= 0 \\ -6b_2 &= 0 \\ 12b_2 &= 0 \\ 9a_2 - 9b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x^3 + y^3}{3y^2x} \right) (x) \\ &= y + \frac{x^3 + y^3}{3y^2} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x^3 + y^3}{3y^2}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(x^3 + 4y^3)}{4}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x^3 + y^3}{3y^2x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= 1 \\R_y &= 0 \\S_x &= \frac{3x^2}{4x^3 + 16y^3} \\S_y &= \frac{3y^2}{x^3 + 4y^3}\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{4R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{4R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

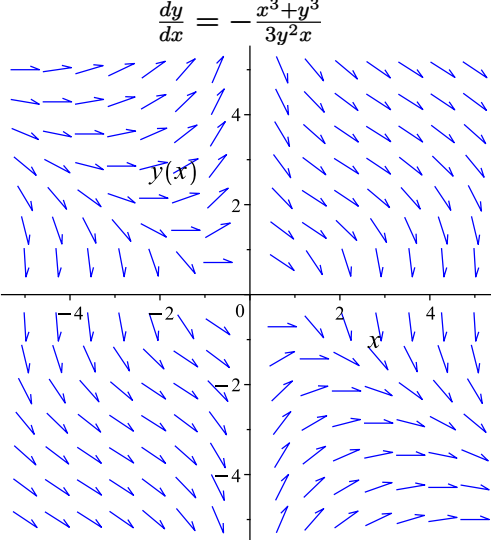
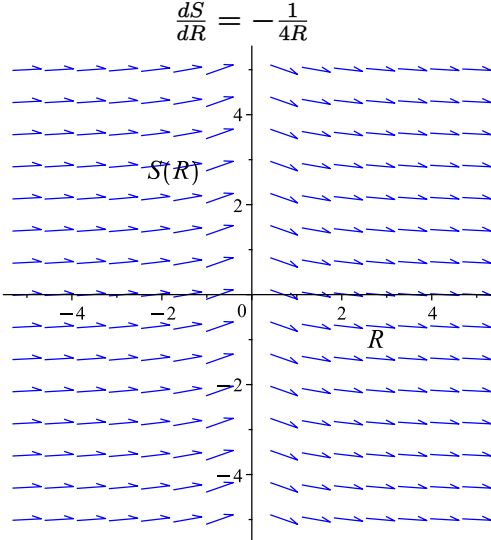
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned}\int dS &= \int -\frac{1}{4R} dR \\S(R) &= -\frac{\ln(R)}{4} + c_2\end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(4y^3 + x^3)}{4} = -\frac{\ln(x)}{4} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x^3+y^3}{3y^2x}$ 	$R = x$ $S = \frac{\ln(x^3 + 4y^3)}{4}$	$\frac{dS}{dR} = -\frac{1}{4R}$ 

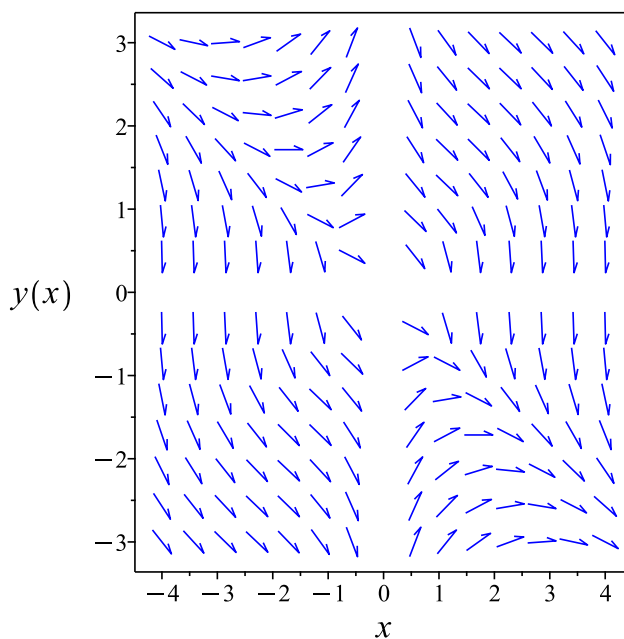


Figure 2.146: Slope field $x^3 + y^3 + 3xy^2y' = 0$

Summary of solutions found

$$\frac{\ln(4y^3 + x^3)}{4} = -\frac{\ln(x)}{4} + c_2$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 92

```
ode:=x^3+y(x)^3+3*x*y(x)^2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{2^{1/3}(-(x^4 - 4c_1)x^2)^{1/3}}{2x}$$

$$y = -\frac{2^{1/3}(-(x^4 - 4c_1)x^2)^{1/3}(1 + i\sqrt{3})}{4x}$$

$$y = \frac{2^{1/3}(-(x^4 - 4c_1)x^2)^{1/3}(i\sqrt{3} - 1)}{4x}$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$x^3 + y(x)^3 + 3xy(x)^2 \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{dx}G(x, y(x)) = 0$
 - Compute derivative of lhs
 - $\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y) \right) \left(\frac{d}{dx}y(x) \right) = 0$
 - Evaluate derivatives
 - $3y^2 = 3y^2$
 - Condition met, ODE is exact
- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y) \right]$$
- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int (x^3 + y^3) dx + _F1(y)$$
- Evaluate integral

$$G(x, y) = \frac{x^4}{4} + y^3x + _F1(y)$$

- Take derivative of $G(x, y)$ with respect to y
 $N(x, y) = \frac{\partial}{\partial y} G(x, y)$
- Compute derivative
 $3y^2x = 3y^2x + \frac{d}{dy} F1(y)$
- Isolate for $\frac{d}{dy} F1(y)$
 $\frac{d}{dy} F1(y) = 0$
- Solve for $F1(y)$
 $F1(y) = 0$
- Substitute $F1(y)$ into equation for $G(x, y)$
 $G(x, y) = \frac{1}{4}x^4 + y^3x$
- Substitute $G(x, y)$ into the solution of the ODE
 $\frac{1}{4}x^4 + y^3x = C1$
- Solve for $y(x)$

$$\left\{ y(x) = \frac{((-2x^4+8C1)x^2)^{1/3}}{2x}, y(x) = -\frac{((-2x^4+8C1)x^2)^{1/3}}{4x} - \frac{1\sqrt{3}((-2x^4+8C1)x^2)^{1/3}}{4x}, y(x) = -\frac{((-2x^4+8C1)x^2)^{1/3}}{4x} \right.$$

✓ **Mathematica.** Time used: 0.149 (sec). Leaf size: 96

```
ode=(x^3+y[x]^3)+3*x*y[x]^2*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\left(-\frac{1}{2}\right)^{2/3} \sqrt[3]{-x^4 + 4c_1}}{\sqrt[3]{x}}$$

$$y(x) \rightarrow \frac{\sqrt[3]{-x^4 + 4c_1}}{2^{2/3} \sqrt[3]{x}}$$

$$y(x) \rightarrow -\frac{\sqrt[3]{-1} \sqrt[3]{-x^4 + 4c_1}}{2^{2/3} \sqrt[3]{x}}$$

✓ Sympy. Time used: 0.988 (sec). Leaf size: 75

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**3 + 3*x*y(x)**2*Derivative(y(x), x) + y(x)**3, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = \frac{\sqrt[3]{2} \sqrt[3]{\frac{C_1}{x} - x^3}}{2}, y(x) = \frac{\sqrt[3]{2}(-1 - \sqrt{3}i) \sqrt[3]{\frac{C_1}{x} - x^3}}{4}, y(x) = \frac{\sqrt[3]{2}(-1 + \sqrt{3}i) \sqrt[3]{\frac{C_1}{x} - x^3}}{4} \right]$$

2.2.20 Problem 45

Local contents

Solved using first_order_ode_dAlembert	600
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	604
Solved using first_order_ode_LIE	606
✓ Maple	611
✓ Mathematica	612
✓ Sympy	613

Internal problem ID [7874]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 45

Date solved : Thursday, November 27, 2025 at 06:26:24 AM

CAS classification :

[[_homogeneous, 'class C'], _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_dAlembert

Time used: 0.458 (sec)

Solve

$$3x + 2y + 1 - (3x + 2y - 1)y' = 0$$

Let $p = y'$ the ode becomes

$$3x + 2y + 1 - (3x + 2y - 1)p = 0$$

Solving for y from the above results in

$$y = -\frac{(3p - 3)x}{2(-1 + p)} - \frac{-p - 1}{2(-1 + p)} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{3}{2} \\ g &= \frac{p+1}{-2+2p} \end{aligned}$$

Hence (2) becomes

$$p + \frac{3}{2} = \left(\frac{1}{-2+2p} - \frac{2p}{(-2+2p)^2} - \frac{2}{(-2+2p)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p + \frac{3}{2} = 0$$

Solving the above for p results in

$$p_1 = -\frac{3}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{1}{10} - \frac{3x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) + \frac{3}{2}}{\frac{1}{-2+2p(x)} - \frac{2p(x)}{(-2+2p(x))^2} - \frac{2}{(-2+2p(x))^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\begin{aligned} \int -\frac{2}{(2p+3)(-1+p)^2} dp &= dx \\ \frac{2}{5(-1+p)} + \frac{4 \ln(-1+p)}{25} - \frac{4 \ln(2p+3)}{25} &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-\frac{(2p+3)(-1+p)^2}{2} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = 1$$

$$p(x) = -\frac{3}{2}$$

Substituting the above solution for p in (2A) gives

$$y = -\frac{3x}{2} + \frac{e^{\text{RootOf}\left(-4\ln\left(\frac{e^{-Z}}{2}-\frac{5}{2}\right)e^{-Z}+25e^{-Z}c_1+4_Ze^{-Z}+25xe^{-Z}+20\ln\left(\frac{e^{-Z}}{2}-\frac{5}{2}\right)-125c_1-20_Z-125x-20\right)} - \frac{1}{2}}{e^{\text{RootOf}\left(-4\ln\left(\frac{e^{-Z}}{2}-\frac{5}{2}\right)e^{-Z}+25e^{-Z}c_1+4_Ze^{-Z}+25xe^{-Z}+20\ln\left(\frac{e^{-Z}}{2}-\frac{5}{2}\right)-125c_1-20_Z-125x-20\right)} - 5}$$

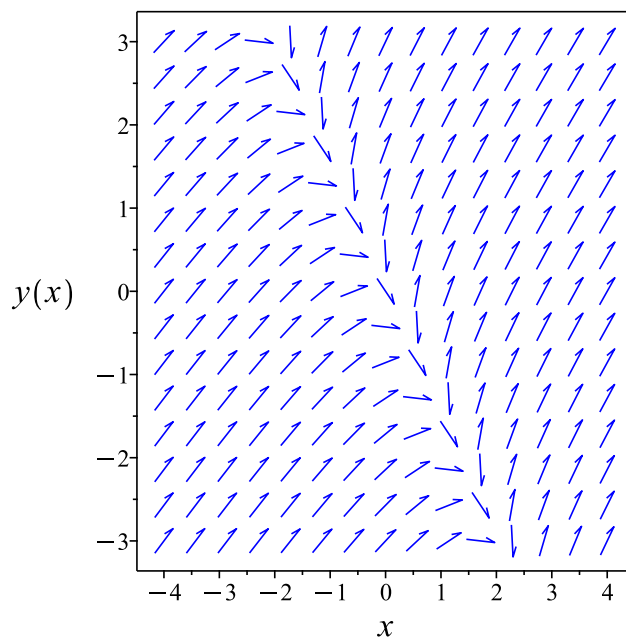
$$y = \frac{1}{10} - \frac{3x}{2}$$

Simplifying the above gives

$$y = \frac{1}{10} - \frac{3x}{2}$$

$$y = \frac{-3xe^{\text{RootOf}\left(4e^{-Z}\ln(2)-4e^{-Z}\ln(e^{-Z}-5)+25e^{-Z}c_1+4_Ze^{-Z}+25xe^{-Z}-20\ln(2)+20\ln(e^{-Z}-5)-125c_1-20_Z-125x-20\right)} + e^{\text{RootOf}\left(4e^{-Z}\ln(2)-4e^{-Z}\ln(e^{-Z}-5)+25e^{-Z}c_1+4_Ze^{-Z}+25xe^{-Z}-20\ln(2)+20\ln(e^{-Z}-5)-125c_1-20_Z-125x-20\right)}}{2e^{\text{RootOf}\left(4e^{-Z}\ln(2)-4e^{-Z}\ln(e^{-Z}-5)+25e^{-Z}c_1+4_Ze^{-Z}+25xe^{-Z}-20\ln(2)+20\ln(e^{-Z}-5)-125c_1-20_Z-125x-20\right)}} + \frac{1}{2}$$

$$y = \frac{1}{10} - \frac{3x}{2}$$

Figure 2.147: Slope field $3x + 2y + 1 - (3x + 2y - 1)y' = 0$

Summary of solutions found

$$y = \frac{-3x e^{\text{RootOf}(4 e^{-Z} \ln(2) - 4 e^{-Z} \ln(e^{-Z} - 5) + 25 e^{-Z} c_1 + 4 Z e^{-Z} + 25 x e^{-Z} - 20 \ln(2) + 20 \ln(e^{-Z} - 5) - 125 c_1 - 20 Z - 125 x - 20)} + e^{\text{RootOf}(4 e^{-Z} \ln(2) - 4 e^{-Z} \ln(e^{-Z} - 5) + 25 e^{-Z} c_1 + 4 Z e^{-Z} + 25 x e^{-Z} - 20 \ln(2) + 20 \ln(e^{-Z} - 5) - 125 c_1 - 20 Z - 125 x - 20)}}{2 e^{\text{RootOf}(4 e^{-Z} \ln(2) - 4 e^{-Z} \ln(e^{-Z} - 5) + 25 e^{-Z} c_1 + 4 Z e^{-Z} + 25 x e^{-Z} - 20 \ln(2) + 20 \ln(e^{-Z} - 5) - 125 c_1 - 20 Z - 125 x - 20)}}$$

$$y = \frac{1}{10} - \frac{3x}{2}$$

Solve

$$3x + 2y + 1 - (3x + 2y - 1)y' = 0$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$3x + 2xu(x) + 1 - (3x + 2xu(x) - 1)(u(x) + xu'(x)) = 0$$

Which is now solved Unknown ode type.

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 0.546 (sec)

This is Abel second kind ODE, it has the form

$$(y(x) + g) y'(x) = f_0(x) + f_1(x)y(x) + f_2(x)y(x)^2 + f_3(x)y(x)^3$$

Comparing the above to given ODE which is

$$3x + 2y(x) + 1 - (3x + 2y(x) - 1) y'(x) = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= \frac{3x}{2} - \frac{1}{2} \\ f_0 &= \frac{1}{2} + \frac{3x}{2} \\ f_1 &= 1 \\ f_2 &= 0 \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y(x) = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = -u(x)^3 - \frac{5u(x)^2}{2}$$

Which is now solved.

Solve Integrating gives

$$\begin{aligned} \int \frac{1}{-u^3 - \frac{5}{2}u^2} du &= dx \\ -\frac{4 \ln(2u + 5)}{25} + \frac{2}{5u} + \frac{4 \ln(u)}{25} &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-u^3 - \frac{5}{2}u^2 = 0$$

for $u(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$\begin{aligned} u(x) &= 0 \\ u(x) &= -\frac{5}{2} \end{aligned}$$

Substituting $u(x) = \frac{1}{\frac{3x}{2} + y(x) - \frac{1}{2}}$ in the above solution gives

$$-\frac{4 \ln \left(\frac{2}{\frac{3x}{2} + y(x) - \frac{1}{2}} + 5 \right)}{25} + \frac{3x}{5} + \frac{2y(x)}{5} - \frac{1}{5} + \frac{4 \ln \left(\frac{1}{\frac{3x}{2} + y(x) - \frac{1}{2}} \right)}{25} = x + c_1$$

Now we transform the solution $u(x) = -\frac{5}{2}$ to $y(x)$ using $u(x) = \frac{1}{\frac{3x}{2} + y(x) - \frac{1}{2}}$ which gives

$$y(x) = \frac{1}{10} - \frac{3x}{2}$$

Simplifying the above gives

$$-\frac{4 \ln \left(\frac{-1+10y(x)+15x}{3x+2y(x)-1} \right)}{25} + \frac{3x}{5} + \frac{2y(x)}{5} - \frac{1}{5} + \frac{4 \ln(2)}{25} + \frac{4 \ln \left(\frac{1}{3x+2y(x)-1} \right)}{25} = x + c_1$$

$$y(x) = \frac{1}{10} - \frac{3x}{2}$$

Solving for $y(x)$ gives

$$y(x) = \frac{1}{10} - \frac{3x}{2}$$

$$y(x) = -\frac{3x}{2} + \frac{1}{10} - \frac{2 \operatorname{LambertW} \left(-\frac{e^{-\frac{25c_1}{4} - \frac{25x}{4}}}{2} \right)}{5}$$

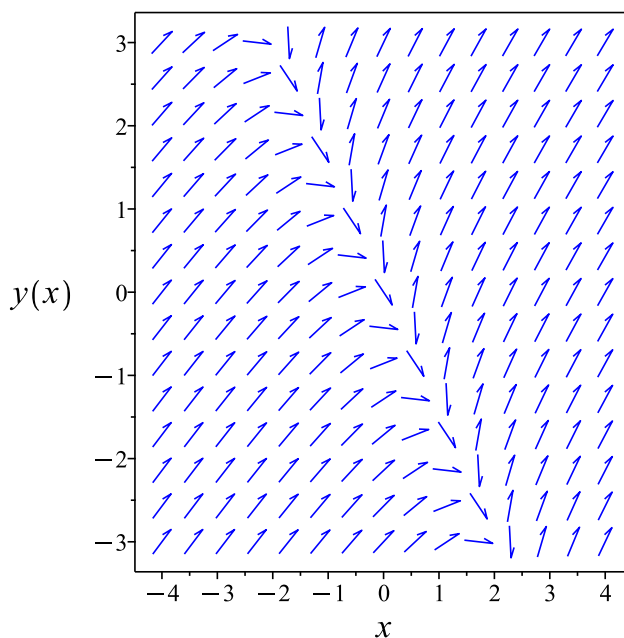
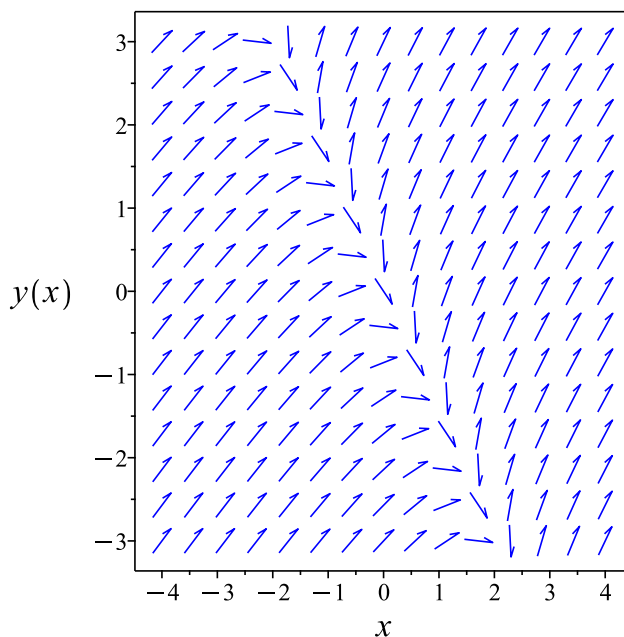


Figure 2.148: Slope field $3x + 2y(x) + 1 - (3x + 2y(x) - 1)y'(x) = 0$

Figure 2.149: Slope field $3x + 2y(x) + 1 - (3x + 2y(x) - 1)y'(x) = 0$ Summary of solutions found

$$y(x) = \frac{1}{10} - \frac{3x}{2}$$

$$y(x) = -\frac{3x}{2} + \frac{1}{10} - \frac{2 \operatorname{LambertW}\left(-\frac{e^{-\frac{25c_1}{4} - \frac{25x}{4} - 1}}{2}\right)}{5}$$

Solved using first_order_ode_LIE

Time used: 0.318 (sec)

Solve

$$3x + 2y(x) + 1 - (3x + 2y(x) - 1)y'(x) = 0$$

Writing the ode as

$$y'(x) = \frac{3x + 2y + 1}{3x + 2y - 1}$$

$$y'(x) = \omega(x, y(x))$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_{y(x)} - \xi_x) - \omega^2 \xi_{y(x)} - \omega_x \xi - \omega_{y(x)} \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1E)$$

$$\eta = \frac{Ax + By}{Cx} \quad (2E)$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & \frac{A}{Cx} - \frac{Ax + By}{Cx^2} + \frac{(3x + 2y + 1)B}{(3x + 2y - 1)Cx} - \frac{3}{3x + 2y - 1} \\ & + \frac{9x + 6y + 3}{(3x + 2y - 1)^2} - \frac{\left(\frac{2}{3x + 2y - 1} - \frac{2(3x + 2y + 1)}{(3x + 2y - 1)^2}\right)(Ax + By)}{Cx} = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{9Bx^3 + 3Bx^2y - 8Bxy^2 - 4By^3 + 4Ax^2 + 10Bxy + 4By^2 + 6Cx^2 - Bx - By}{Cx^2(3x + 2y - 1)^2} = 0$$

Setting the numerator to zero gives

$$9Bx^3 + 3Bx^2y - 8Bxy^2 - 4By^3 + 4Ax^2 + 10Bxy + 4By^2 + 6Cx^2 - Bx - By = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$9Bv_1^3 + 3Bv_1^2v_2 - 8Bv_1v_2^2 - 4Bv_2^3 + 4Av_1^2 + 10Bv_1v_2 + 4Bv_2^2 + 6Cv_1^2 - Bv_1 - Bv_2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$9Bv_1^3 + 3Bv_1^2v_2 + (4A + 6C)v_1^2 - 8Bv_1v_2^2 + 10Bv_1v_2 - Bv_1 - 4Bv_2^3 + 4Bv_2^2 - Bv_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -8B &= 0 \\ -4B &= 0 \\ -B &= 0 \\ 3B &= 0 \\ 4B &= 0 \\ 9B &= 0 \\ 10B &= 0 \\ 4A + 6C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= -\frac{3C}{2} \\ B &= 0 \\ C &= C \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 1 \\ \eta &= -\frac{3}{2} \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical

coordinates, where $S(R)$. Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{-\frac{3}{2}}{1} \\ &= -\frac{3}{2}\end{aligned}$$

This is easily solved to give

$$y(x) = -\frac{3x}{2} + c_1$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = \frac{3x}{2} + y$$

And S is found from

$$\begin{aligned}dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1}\end{aligned}$$

Integrating gives

$$\begin{aligned}S &= \int \frac{dx}{T} \\ &= x\end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{3x + 2y + 1}{3x + 2y - 1}$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= \frac{3}{2} \\ R_y &= 1 \\ S_x &= 1 \\ S_y &= 0\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{4y + 6x - 2}{-1 + 10y + 15x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{4R - 2}{-1 + 10R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

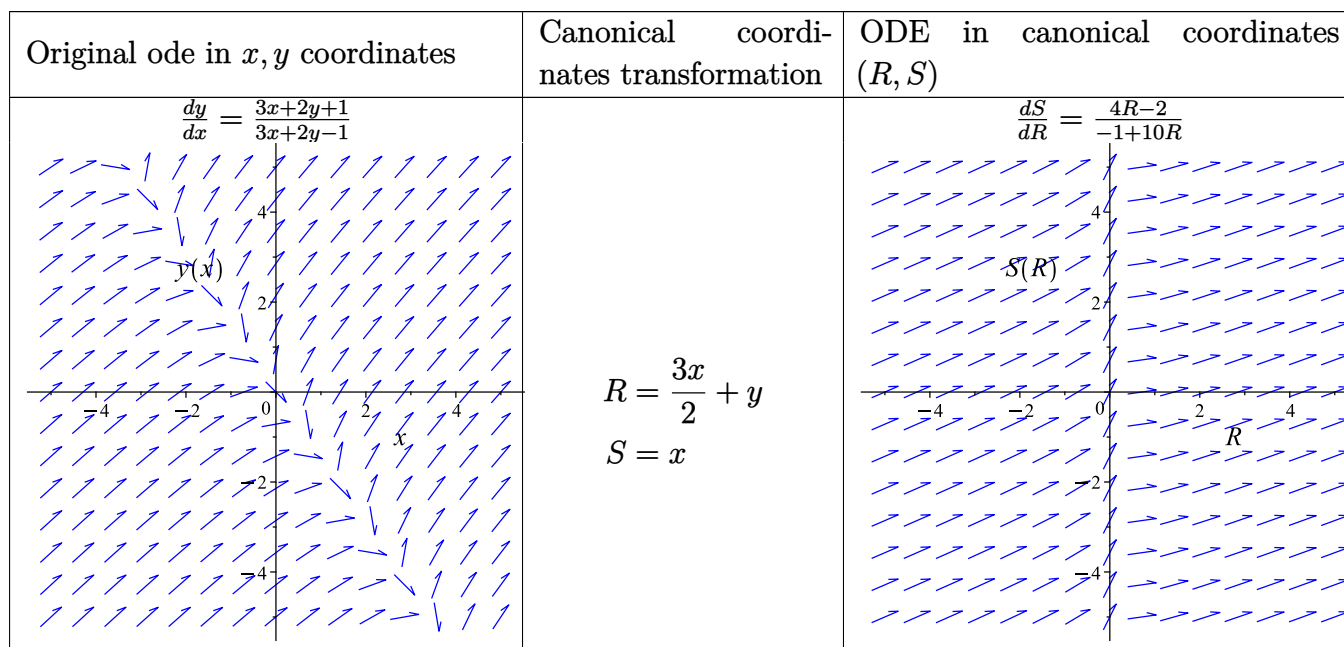
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{4R - 2}{-1 + 10R} dR \\ S(R) &= \frac{2R}{5} - \frac{4 \ln(-1 + 10R)}{25} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = \frac{3x}{5} + \frac{2y(x)}{5} - \frac{4 \ln(-1 + 10y(x) + 15x)}{25} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for $y(x)$ gives

$$y(x) = -\frac{3x}{2} - \frac{2 \operatorname{LambertW}\left(-e^{\frac{1}{4} - \frac{25x}{4} + \frac{25c_2}{4}}\right)}{5} + \frac{1}{10}$$

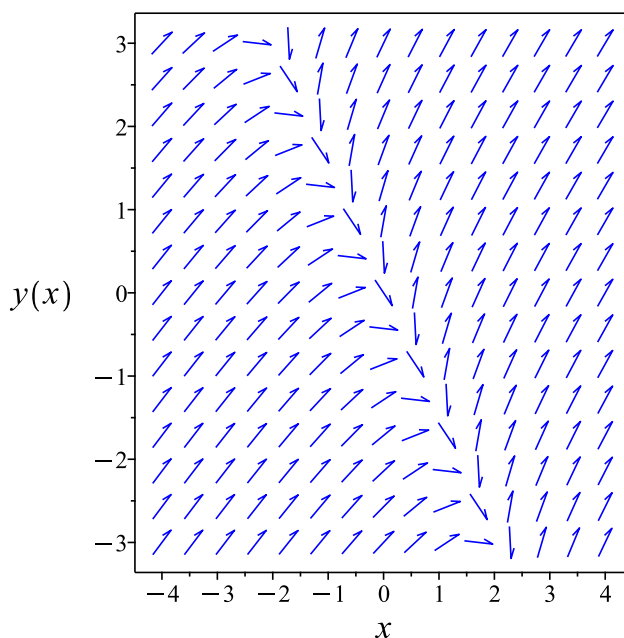


Figure 2.150: Slope field $3x + 2y(x) + 1 - (3x + 2y(x) - 1)y'(x) = 0$

Summary of solutions found

$$y(x) = -\frac{3x}{2} - \frac{2 \operatorname{LambertW}\left(-e^{\frac{1}{4} - \frac{25x}{4} + \frac{25c_2}{4}}\right)}{5} + \frac{1}{10}$$

✓ **Maple.** Time used: 0.011 (sec). Leaf size: 21

```
ode:=3*x+2*y(x)+1-(3*x+2*y(x)-1)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{3x}{2} - \frac{2 \operatorname{LambertW}\left(-c_1 e^{\frac{1}{4} - \frac{25x}{4}}\right)}{5} + \frac{1}{10}$$

Maple trace

Methods for first order ODEs:

```

--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous C
1st order, trying the canonical coordinates of the invariance group
  -> Calling odsolve with the ODE, diff(y(x),x) = -3/2, y(x)
      *** Sublevel 2 ***
      Methods for first order ODEs:
      --- Trying classification methods ---
      trying a quadrature
      trying 1st order linear
      <- 1st order linear successful
<- 1st order, canonical coordinates successful
<- homogeneous successful

```

Maple step by step

Let's solve

$$3x + 2y(x) + 1 - (3x + 2y(x) - 1) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-3x-2y(x)-1}{3x+2y(x)-1}$$

✓ **Mathematica.** Time used: 2.497 (sec). Leaf size: 43

```

ode=(3*x+2*y[x]+1)-(3*x+2*y[x]-1)*D[y[x],x]== 0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow \frac{1}{10} \left(-4W \left(-e^{-\frac{25x}{4}-1+c_1} \right) - 15x + 1 \right)$$

$$y(x) \rightarrow \frac{1}{10} - \frac{3x}{2}$$

✓ Sympy. Time used: 2.703 (sec). Leaf size: 136

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x - (3*x + 2*y(x) - 1)*Derivative(y(x), x) + 2*y(x) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\begin{aligned} y(x) &= -\frac{3x}{2} - \frac{2W\left(-\frac{\sqrt[4]{C_1}e^{-25x}e^{\frac{1}{4}}}{4}\right)}{5} + \frac{1}{10}, \quad y(x) = -\frac{3x}{2} \\ &- \frac{2W\left(\frac{\sqrt[4]{C_1}e^{-25x}e^{\frac{1}{4}}}{4}\right)}{5} + \frac{1}{10}, \quad y(x) = -\frac{3x}{2} - \frac{2W\left(-\frac{i\sqrt[4]{C_1}e^{-25x}e^{\frac{1}{4}}}{4}\right)}{5} \\ &+ \frac{1}{10}, \quad y(x) = -\frac{3x}{2} - \frac{2W\left(\frac{i\sqrt[4]{C_1}e^{-25x}e^{\frac{1}{4}}}{4}\right)}{5} + \frac{1}{10} \end{aligned} \right]$$

2.2.21 Problem 46

Local contents

Existence and uniqueness analysis	614
Solved using first_order_ode_linear	615
Solved using first_order_ode_separable	616
Solved using first_order_ode_exact	618
Solved using first_order_ode_homog_A	622
Solved using first_order_ode_homog_type_D2	625
Solved using first_order_ode_homog_type_G	627
Solved using first_order_ode_homog_type_maple_C	629
Solved using first_order_ode_LIE	633
✓ Maple	638
✓ Mathematica	639
✓ Sympy	639

Internal problem ID [7875]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 46

Date solved : Thursday, November 27, 2025 at 06:26:35 AM

CAS classification : [_separable]

Existence and uniqueness analysis

$$2y + y'x = 0$$

$$y(2) = 1$$

This is a linear ODE. In canonical form it is written as

$$y' + q(x)y = p(x)$$

Where here

$$q(x) = \frac{2}{x}$$

$$p(x) = 0$$

Hence the ode is

$$y' + \frac{2y}{x} = 0$$

The domain of $q(x) = \frac{2}{x}$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 2$ is inside this domain. Hence solution exists and is unique.

Solved using first_order_ode_linear

Time used: 0.036 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = \frac{2}{x}$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} dx} \\ &= x^2\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}(x^2 y) &= 0\end{aligned}$$

Integrating gives

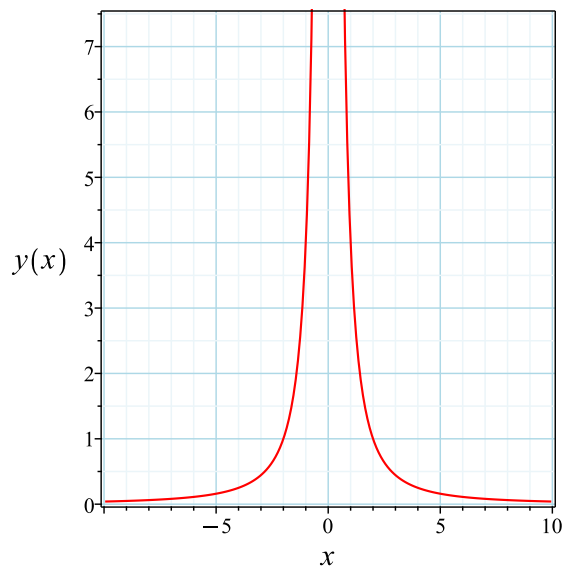
$$\begin{aligned}x^2 y &= \int 0 \, dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

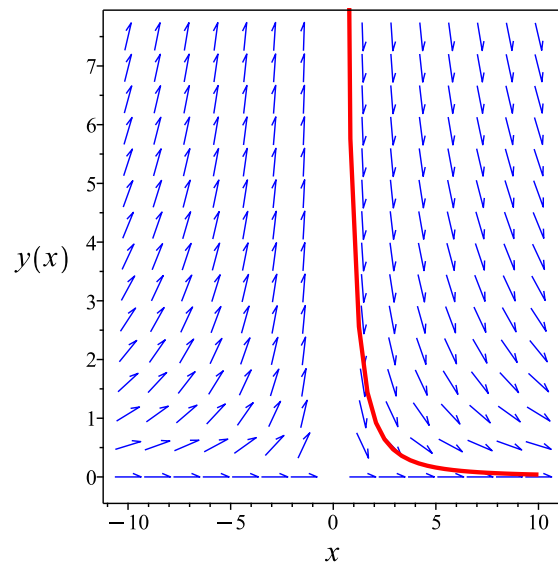
$$y = \frac{c_1}{x^2}$$

Solving for initial conditions the solution is

$$y = \frac{4}{x^2}$$



(a) Solution plot



(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_separable

Time used: 0.119 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

The ode

$$y' = -\frac{2y}{x} \tag{2.71}$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{2y}{x} \\ &= f(x)g(y) \end{aligned}$$

Where

$$f(x) = -\frac{2}{x}$$
$$g(y) = y$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$
$$\int \frac{1}{y} dy = \int -\frac{2}{x} dx$$

$$\ln(y) = \ln\left(\frac{1}{x^2}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$y = \frac{c_1}{x^2}$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y = 0$$

for y gives

$$y = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$y = \frac{c_1}{x^2}$$
$$y = 0$$

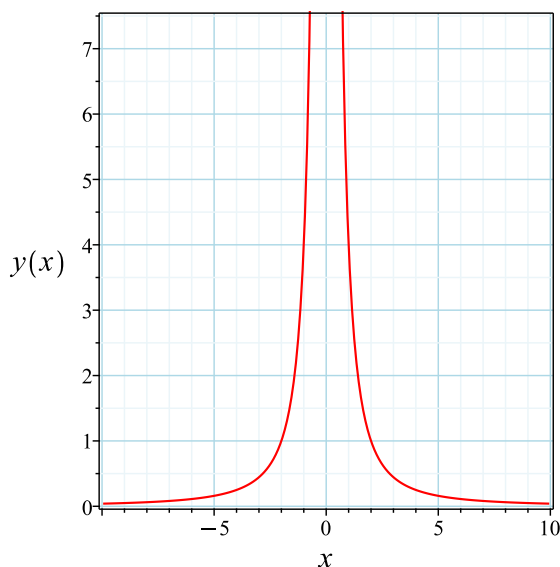
Solving for initial conditions the solution is

$$y = \frac{4}{x^2}$$
$$y = 0$$

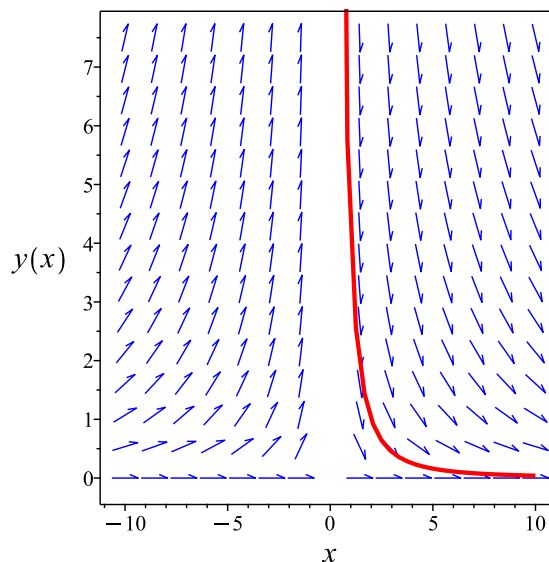
The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_exact

Time used: 0.102 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (-2y) dx \\ (2y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 2y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(2y) \\ &= 2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((2) - (1)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{1}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(2y) \\ &= 2yx\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(x) \\ &= x^2\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (2yx) + (x^2) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} \, dy &= \int \overline{N} \, dy \\ \int \frac{\partial \phi}{\partial y} \, dy &= \int x^2 \, dy \\ \phi &= x^2 y + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = 2yx + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = 2yx$. Therefore equation (4) becomes

$$2yx = 2yx + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

$$\phi = x^2y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

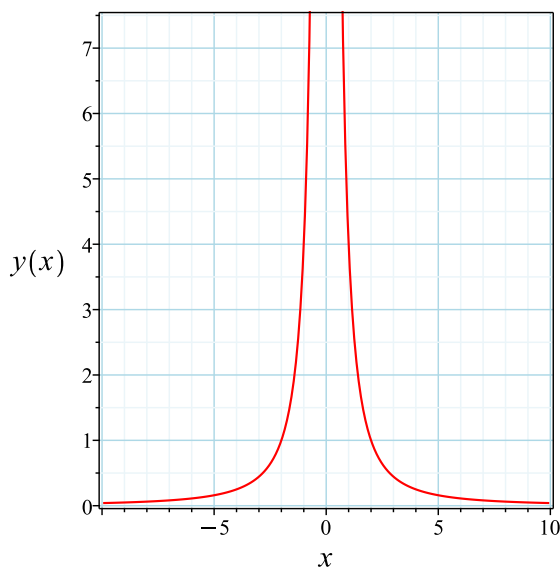
$$c_1 = x^2y$$

Solving for initial conditions the solution is

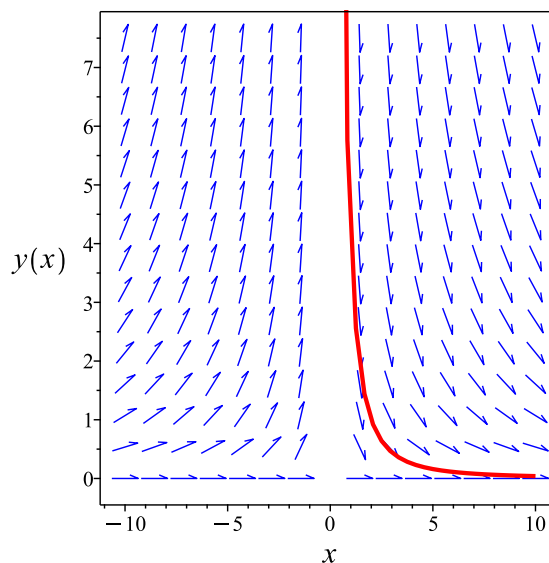
$$x^2y = 4$$

Solving for y gives

$$y = \frac{4}{x^2}$$



(a) Solution plot

(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_homog_A

Time used: 0.420 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{2y}{x} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -2y$ and $N = x$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= -2u \\ \frac{du}{dx} &= -\frac{3u(x)}{x}\end{aligned}$$

Or

$$u'(x) + \frac{3u(x)}{x} = 0$$

Or

$$u'(x)x + 3u(x) = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{3u(x)}{x} \tag{2.72}$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= -\frac{3u(x)}{x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= -\frac{3}{x} \\ g(u) &= u\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u} du &= \int -\frac{3}{x} dx\end{aligned}$$

$$\ln(u(x)) = \ln\left(\frac{1}{x^3}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = \frac{c_1}{x^3}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = \frac{c_1}{x^3}$$

$$u(x) = 0$$

Converting $u(x) = \frac{c_1}{x^3}$ back to y gives

$$y = \frac{c_1}{x^2}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for initial conditions the solution is

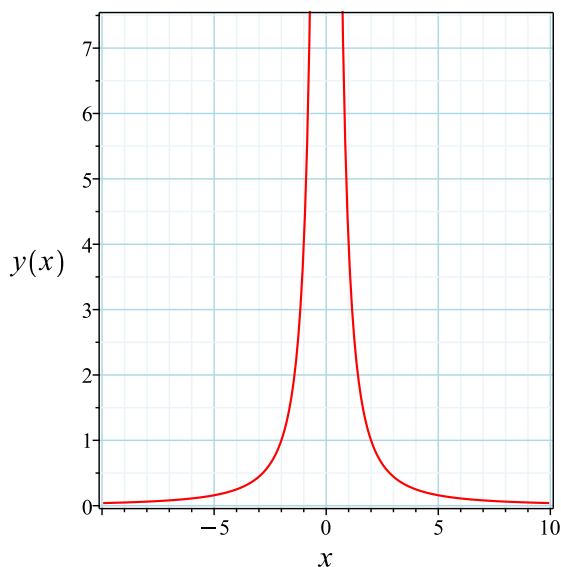
$$y = \frac{4}{x^2}$$

$$y = 0$$

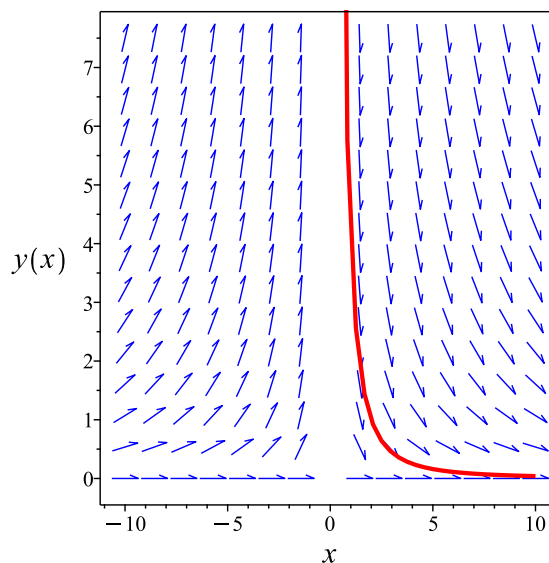
The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot

(b) Slope field $2y + y'x = 0$ Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.152 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$2u(x)x + (u'(x)x + u(x))x = 0$$

Which is now solved The ode

$$u'(x) = -\frac{3u(x)}{x} \tag{2.73}$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{3u(x)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{3}{x}$$

$$g(u) = u$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u} du = \int -\frac{3}{x} dx$$

$$\ln(u(x)) = \ln\left(\frac{1}{x^3}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(x) = \frac{c_1}{x^3}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(x) = \frac{c_1}{x^3}$$

$$u(x) = 0$$

Converting $u(x) = \frac{c_1}{x^3}$ back to y gives

$$y = \frac{c_1}{x^2}$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for initial conditions the solution is

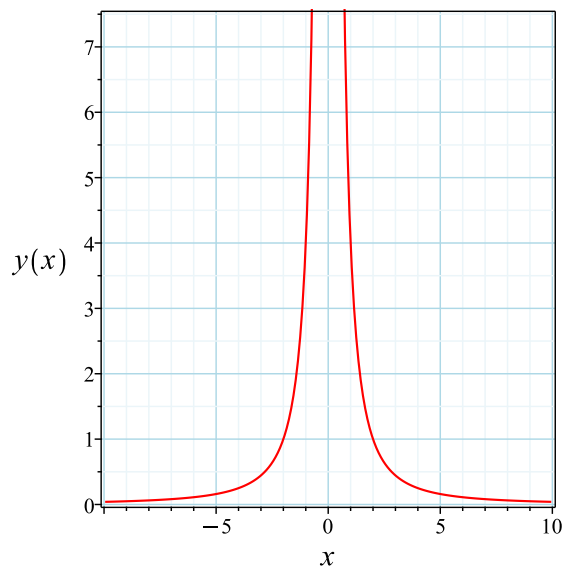
$$y = \frac{4}{x^2}$$

$$y = 0$$

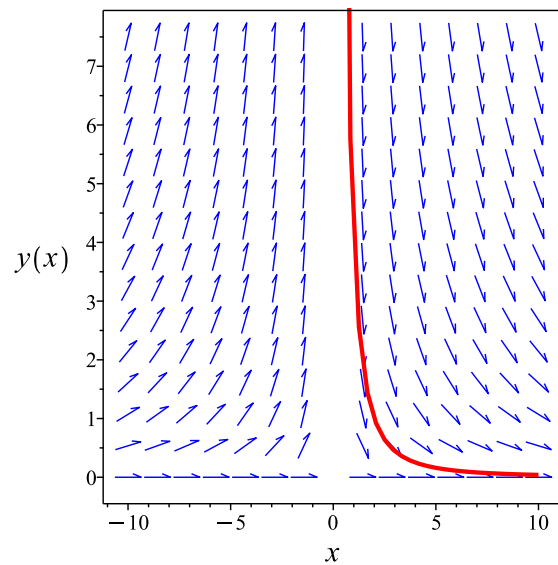
The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_homog_type_G

Time used: 0.144 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

Multiplying the right side of the ode, which is $-\frac{2y}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{2y}{x} \\ &= -2 \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ does not have y and does not have x then

$$\alpha = 0$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -2$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

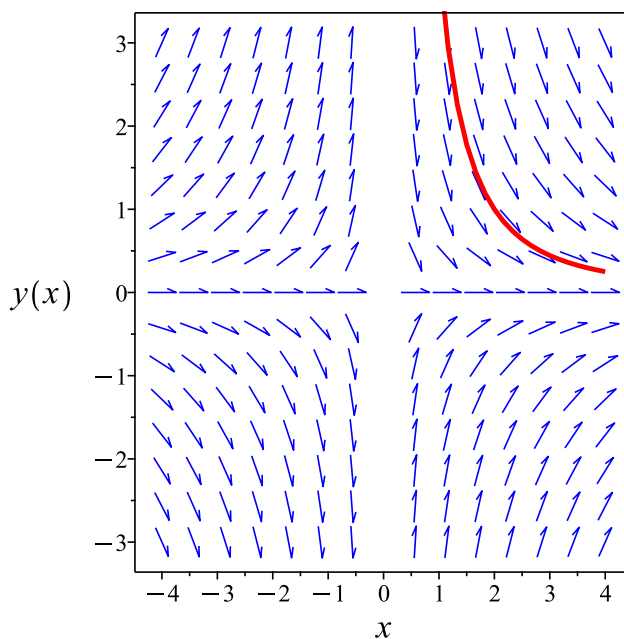
$$\ln(x) - c_1 + \int^y \frac{1}{2z} dz = 0$$

Solving for initial conditions the solution is

$$\ln(x) - \ln(2) - \frac{\int^1 \frac{1}{z} dz}{2} + \frac{\int^y \frac{1}{z} dz}{2} = 0$$

Solving for y gives

$$y = \text{RootOf} \left(-2 \ln(x) + 2 \ln(2) + \int^1 \frac{1}{z} dz - \int^y \frac{1}{z} dz \right)$$

Figure 2.156: Slope field $2y + y'x = 0$ Summary of solutions found

$$y = \text{RootOf} \left(-2 \ln(x) + 2 \ln(2) + \int^1 \frac{1}{z} dz - \int^{-z} \frac{1}{z} dz \right)$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.562 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX} Y(X) = -\frac{2(Y(X) + y_0)}{X + x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{2Y(X)}{X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{2Y}{X} \end{aligned} \tag{1}$$

An ode of the form $Y' = \frac{M(X,Y)}{N(X,Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -2Y$ and $N = X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -2u \\ \frac{du}{dX} &= -\frac{3u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) + \frac{3u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) X + 3u(X) = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{3u(X)}{X} \tag{2.74}$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{3u(X)}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$f(X) = -\frac{3}{X}$$

$$g(u) = u$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{1}{u} du = \int -\frac{3}{X} dX$$

$$\ln(u(X)) = \ln\left(\frac{1}{X^3}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$u(X) = \frac{c_1}{X^3}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$u(X) = \frac{c_1}{X^3}$$

$$u(X) = 0$$

Converting $u(X) = \frac{c_1}{X^3}$ back to $Y(X)$ gives

$$Y(X) = \frac{c_1}{X^2}$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$Y(X) = \frac{c_1}{X^2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = \frac{c_1}{x^2}$$

Using the solution for $Y(X)$

$$Y(X) = 0 \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

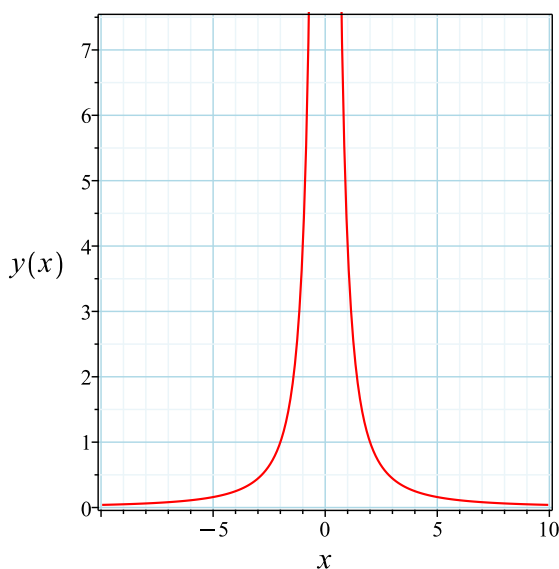
Solving for initial conditions the solution is

$$\begin{aligned} y &= \frac{4}{x^2} \\ y &= 0 \end{aligned}$$

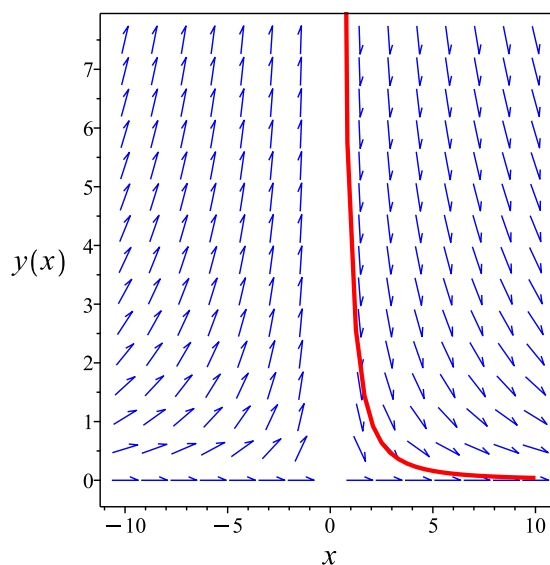
The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

Solved using first_order_ode_LIE

Time used: 0.733 (sec)

Solve

$$2y + y'x = 0$$

$$y(2) = 1$$

Writing the ode as

$$y' = -\frac{2y}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{2y(b_3 - a_2)}{x} - \frac{4y^2a_3}{x^2} - \frac{2y(xa_2 + ya_3 + a_1)}{x^2} + \frac{2xb_2 + 2yb_3 + 2b_1}{x} = 0 \quad (5E)$$

Putting the above in normal form gives

$$\frac{3b_2x^2 - 6y^2a_3 + 2xb_1 - 2ya_1}{x^2} = 0$$

Setting the numerator to zero gives

$$3b_2x^2 - 6y^2a_3 + 2xb_1 - 2ya_1 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-6a_3v_2^2 + 3b_2v_1^2 - 2a_1v_2 + 2b_1v_1 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-6a_3v_2^2 + 3b_2v_1^2 - 2a_1v_2 + 2b_1v_1 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$-2a_1 = 0$$

$$-6a_3 = 0$$

$$2b_1 = 0$$

$$3b_2 = 0$$

Solving the above equations for the unknowns gives

$$a_1 = 0$$

$$a_2 = a_2$$

$$a_3 = 0$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = b_3$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 0$$

$$\eta = y$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y} dy \end{aligned}$$

Which results in

$$S = \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{2y}{x}$$

Evaluating all the partial derivatives gives

$$R_x = 1$$

$$R_y = 0$$

$$S_x = 0$$

$$S_y = \frac{1}{y}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{2}{x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{2}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

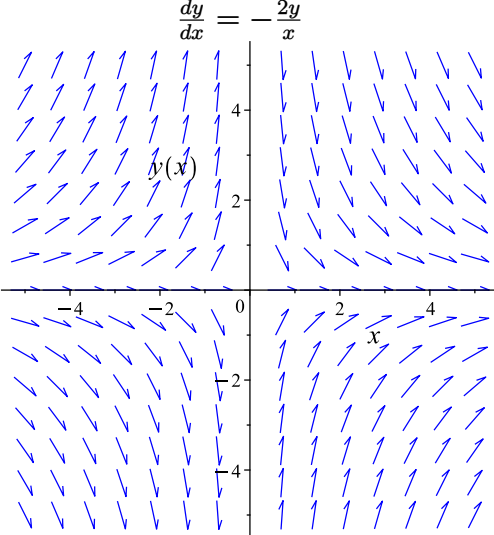
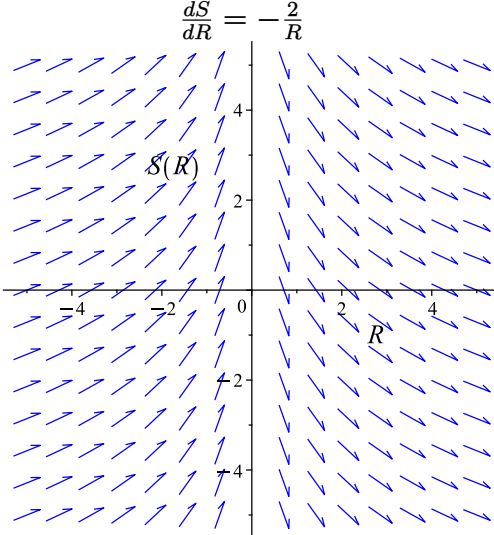
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{2}{R} dR \\ S(R) &= -2 \ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\ln(y) = -2 \ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

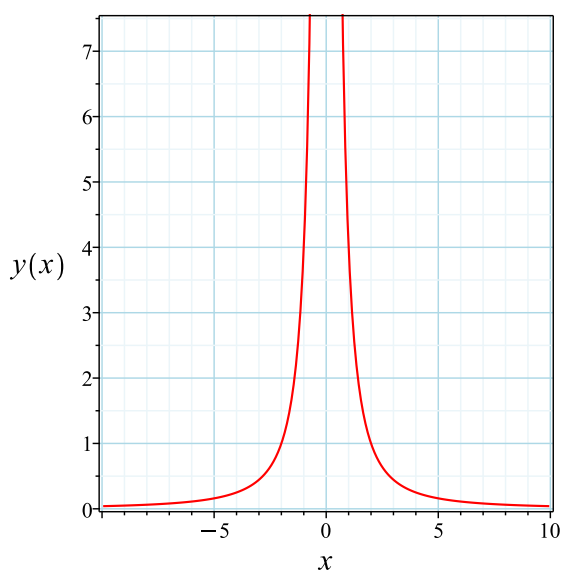
Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{2y}{x}$ 	$R = x$ $S = \ln(y)$	$\frac{dS}{dR} = -\frac{2}{R}$ 

Solving for initial conditions the solution is

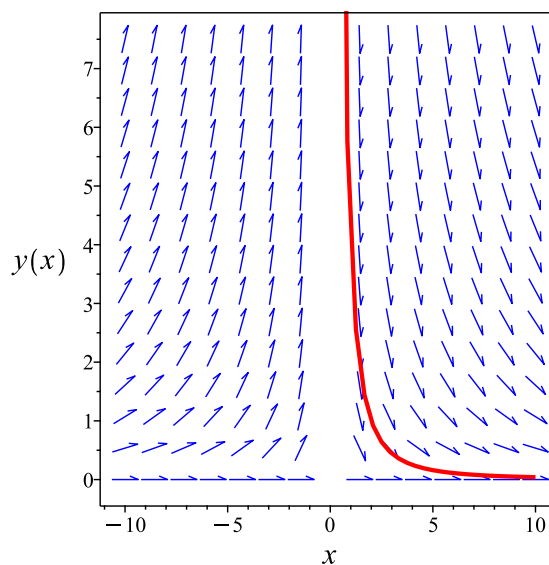
$$\ln(y) = -2\ln(x) + 2\ln(2)$$

Solving for y gives

$$y = \frac{4}{x^2}$$



(a) Solution plot



(b) Slope field $2y + y'x = 0$

Summary of solutions found

$$y = \frac{4}{x^2}$$

✓ **Maple.** Time used: 0.009 (sec). Leaf size: 9

```
ode:=diff(y(x),x)*x+2*y(x) = 0;
ic:=[y(2) = 1];
dsolve([ode,op(ic)],y(x), singsol=all);
```

$$y = \frac{4}{x^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\left[x \left(\frac{d}{dx} y(x) \right) + 2y(x) = 0, y(2) = 1 \right]$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{2y(x)}{x}$$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{y(x)} = -\frac{2}{x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{y(x)} dx = \int -\frac{2}{x} dx + C1$$

- Evaluate integral

$$\ln(y(x)) = -2 \ln(x) + C1$$

- Solve for $y(x)$

$$y(x) = \frac{e^{C1}}{x^2}$$

- Redefine the integration constant(s)

$$y(x) = \frac{C1}{x^2}$$

- Use initial condition $y(2) = 1$

$$1 = \frac{C1}{4}$$

- Solve for $_C1$
 $C1 = 4$
- Substitute $_C1 = 4$ into general solution and simplify
 $y(x) = \frac{4}{x^2}$
- Solution to the IVP
 $y(x) = \frac{4}{x^2}$

✓ **Mathematica.** Time used: 0.015 (sec). Leaf size: 10

```
ode=x*D[y[x],x]+2*y[x]==0;
ic={y[2]==1};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{4}{x^2}$$

✓ **Sympy.** Time used: 0.063 (sec). Leaf size: 7

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) + 2*y(x),0)
ics = {y(2): 1}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{4}{x^2}$$

2.2.22 Problem 47

Local contents

Existence and uniqueness analysis	640
Solved using first_order_ode_bernoulli	641
Solved using first_order_ode_exact	644
Solved using first_order_ode_isobaric	648
Solved using first_order_ode_homog_A	652
Solved using first_order_ode_homog_type_D2	656
Solved using first_order_ode_homog_type_G	659
Solved using first_order_ode_homog_type_maple_C	661
Solved using first_order_ode_LIE	667
✓ Maple	672
✓ Mathematica	672
✓ Sympy	672

Internal problem ID [7876]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 47

Date solved : Thursday, November 27, 2025 at 06:26:43 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Existence and uniqueness analysis

$$\begin{aligned}
 xy' + x^2 + y^2 &= 0 \\
 y(1) &= -1
 \end{aligned}$$

This is non linear first order ODE. In canonical form it is written as

$$\begin{aligned}
 y' &= f(x, y) \\
 &= -\frac{x^2 + y^2}{xy}
 \end{aligned}$$

The x domain of $f(x, y)$ when $y = -1$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 1$ is inside this domain. The y domain of $f(x, y)$ when $x = 1$ is

$$\{y < 0 \vee 0 < y\}$$

And the point $y_0 = -1$ is inside this domain. Now we will look at the continuity of

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} \left(-\frac{x^2 + y^2}{xy} \right) \\ &= -\frac{2}{x} + \frac{x^2 + y^2}{xy^2}\end{aligned}$$

The x domain of $\frac{\partial f}{\partial y}$ when $y = -1$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 1$ is inside this domain. The y domain of $\frac{\partial f}{\partial y}$ when $x = 1$ is

$$\{y < 0 \vee 0 < y\}$$

And the point $y_0 = -1$ is inside this domain. Therefore solution exists and is unique.

Solved using first_order_ode_bernoulli

Time used: 0.112 (sec)

Solve

$$\begin{aligned}xyy' + x^2 + y^2 &= 0 \\ y(1) &= -1\end{aligned}$$

In canonical form, the ODE is

$$\begin{aligned}y' &= F(x, y) \\ &= -\frac{x^2 + y^2}{xy}\end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{x}\right)y + (-x)\frac{1}{y} \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned}f_0 &= -\frac{1}{x} \\f_1 &= -x\end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= -\frac{1}{x} \\f_1(x) &= -x \\n &= -1\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = -\frac{y^2}{x} - x \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= y^2\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}\frac{v'(x)}{2} &= -\frac{v(x)}{x} - x \\v' &= -\frac{2v}{x} - 2x\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x} \\ p(x) &= -2x \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} \, dx} \\ &= x^2 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) (-2x) \\ \frac{d}{dx}(v x^2) &= (x^2) (-2x) \\ d(v x^2) &= (-2x^3) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} v x^2 &= \int -2x^3 \, dx \\ &= -\frac{x^4}{2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

$$v(x) = \frac{-\frac{x^4}{2} + c_1}{x^2}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

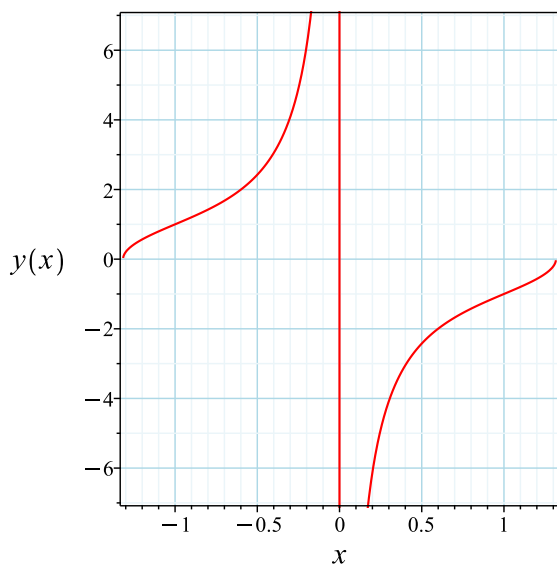
$$y^2 = \frac{-\frac{x^4}{2} + c_1}{x^2}$$

Solving for initial conditions the solution is

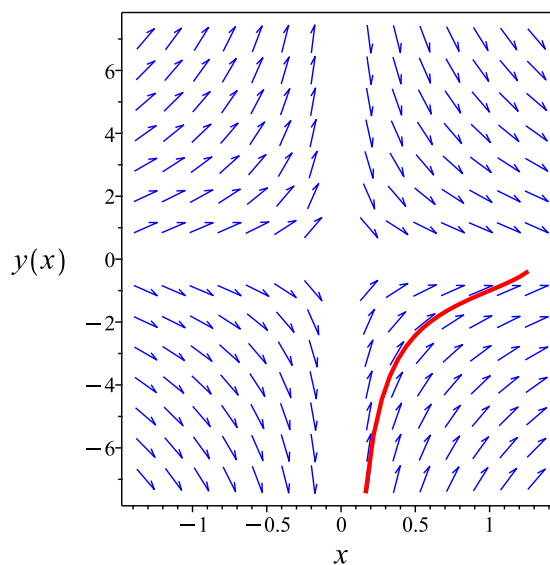
$$y^2 = \frac{-x^4 + 3}{2x^2}$$

Solving for y gives

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$



(a) Solution plot

(b) Slope field $xyy' + x^2 + y^2 = 0$ Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_exact

Time used: 0.122 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (yx) dy &= (-x^2 - y^2) dx \\ (x^2 + y^2) dx + (yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^2 + y^2 \\ N(x, y) &= yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(yx) \\ &= y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{yx} ((2y) - (y)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int \frac{1}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new $\bar{M}$ and new $\bar{N}$ which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= x(x^2 + y^2) \\ &= x(x^2 + y^2)\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= x(yx) \\ &= x^2y\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ (x(x^2 + y^2)) + (x^2y) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \bar{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x(x^2 + y^2) dx \\ \phi &= \frac{(x^2 + y^2)^2}{4} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = (x^2 + y^2) y + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x^2 y$. Therefore equation (4) becomes

$$x^2 y = (x^2 + y^2) y + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -y^3$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (-y^3) dy \\ f(y) &= -\frac{y^4}{4} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{(x^2 + y^2)^2}{4} - \frac{y^4}{4} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{(x^2 + y^2)^2}{4} - \frac{y^4}{4}$$

Simplifying the above gives

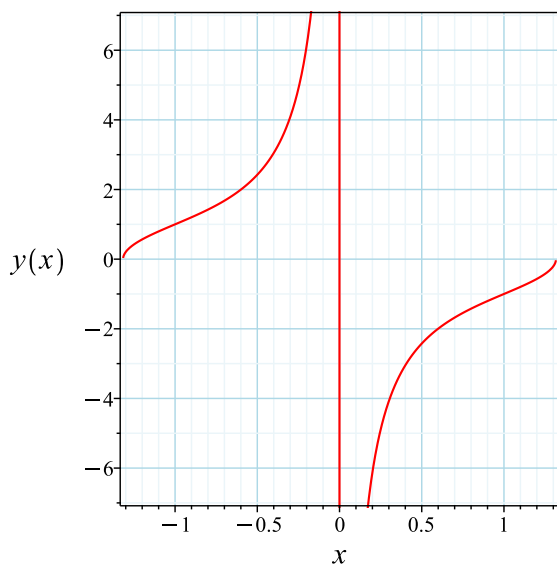
$$\frac{y^2 x^2}{2} + \frac{x^4}{4} = c_1$$

Solving for initial conditions the solution is

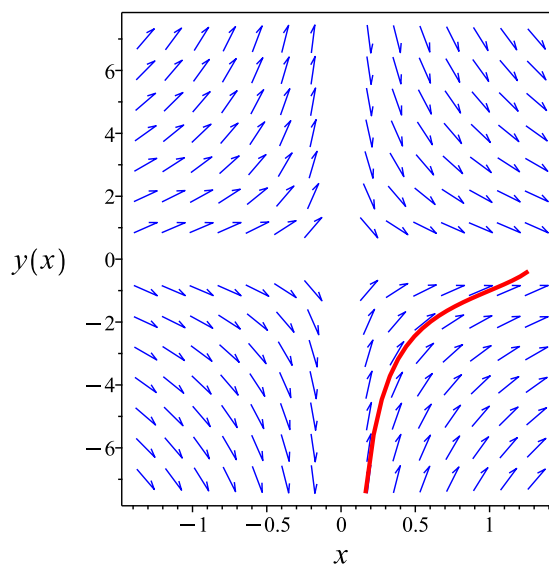
$$\frac{y^2 x^2}{2} + \frac{x^4}{4} = \frac{3}{4}$$

Solving for y gives

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$



(a) Solution plot

(b) Slope field $xyy' + x^2 + y^2 = 0$ Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_isobaric

Time used: 0.362 (sec)

Solve

$$xyy' + x^2 + y^2 = 0$$

$$y(1) = -1$$

Solving for y' gives

$$y' = -\frac{x^2 + y^2}{xy} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{x^2 + y^2}{xy} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x^2 + x^2u(x)^2}{x^2u(x)}$$

The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.75)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} + 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

Solving for initial conditions the solution is

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{3^{1/4}}{x}$$

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

Solving for y gives

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

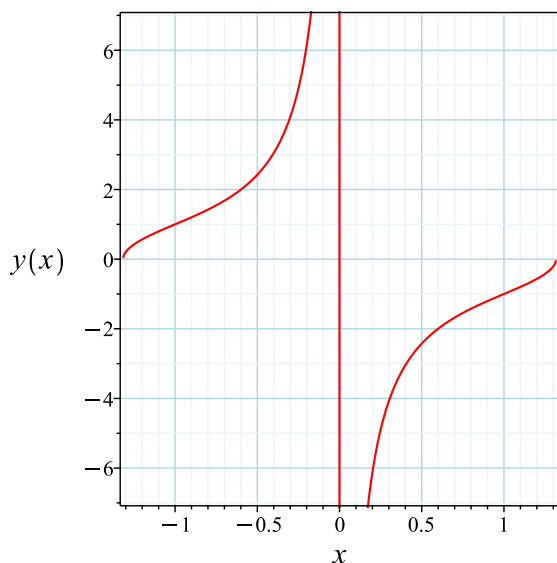
The solution

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

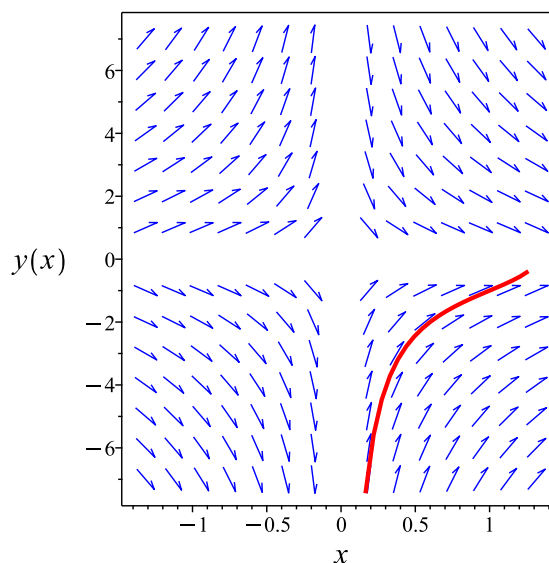
was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot

(b) Slope field $xyy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_homog_A

Time used: 0.671 (sec)

Solve

$$\begin{aligned}xyy' + x^2 + y^2 &= 0 \\y(1) &= -1\end{aligned}$$

In canonical form, the ODE is

$$\begin{aligned}y' &= F(x, y) \\&= -\frac{x^2 + y^2}{xy}\end{aligned}\tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x,y)$ and $N(x,y)$ are both homogeneous functions and of the same order. Recall that a function $f(x,y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x^2 - y^2$ and $N = xy$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= -\frac{1}{u} - u \\ \frac{du}{dx} &= \frac{-\frac{1}{u(x)} - 2u(x)}{x}\end{aligned}$$

Or

$$u'(x) - \frac{-\frac{1}{u(x)} - 2u(x)}{x} = 0$$

Or

$$u'(x)u(x)x + 2u(x)^2 + 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.76)$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= -\frac{2u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 + 1}{u}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{x} dx\end{aligned}$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for initial conditions the solution is

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{3^{1/4}}{x}$$

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

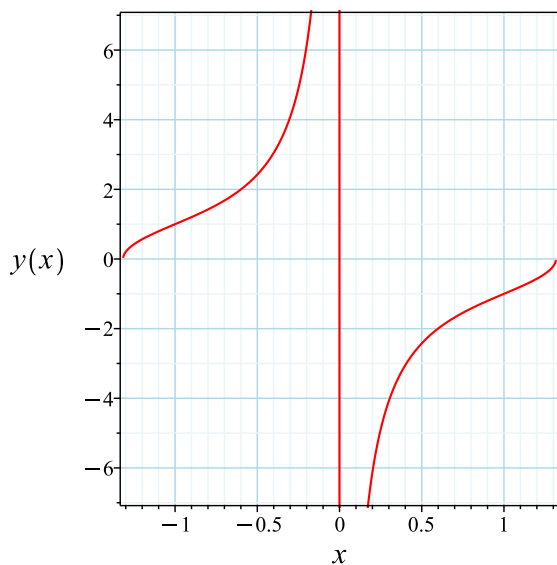
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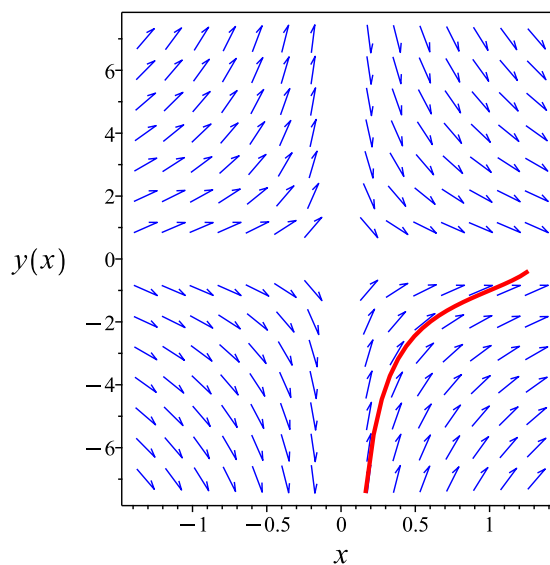
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$$y = \frac{i\sqrt{2}x}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $xyy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.213 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

$$y(1) = -1$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)x^2(u'(x)x + u(x)) + x^2 + u(x)^2x^2 = 0$$

Which is now solved The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.77)$$

is separable as it can be written as

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x}$$

$$= f(x)g(u)$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(u) = \frac{2u^2 + 1}{u}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{u}{2u^2 + 1} du = \int -\frac{1}{x} dx$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} + 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$y = \frac{i\sqrt{2}x}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for initial conditions the solution is

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{3^{1/4}}{x}$$

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

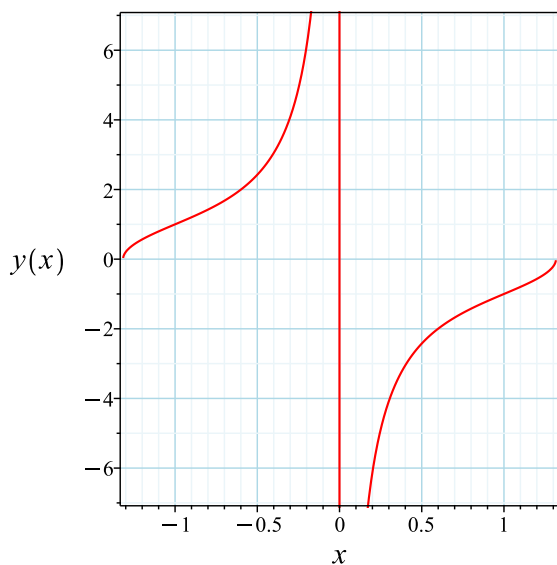
The solution

$$y = -\frac{i\sqrt{2}x}{2}$$

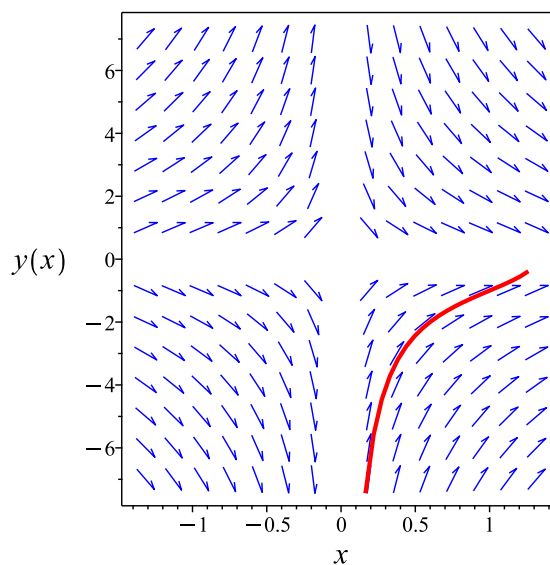
was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = \frac{i\sqrt{2}x}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot

(b) Slope field $xyy' + x^2 + y^2 = 0$ Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.178 (sec)

Solve

$$xyy' + x^2 + y^2 = 0$$

$$y(1) = -1$$

Multiplying the right side of the ode, which is $-\frac{x^2+y^2}{xy}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x^2 + y^2}{xy} \\ &= -\frac{x^2 + y^2}{y^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{2x^2}{y^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{2x^2}{y^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{\frac{1}{x}} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z^2 + 1}{z^2}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{z^2 + 1}{z^2} \right)} dz = 0$$

Solving for initial conditions the solution is

$$\ln(x) - \int^{-1} \frac{z}{2z^2 + 1} dz + \int^{\frac{y}{x}} \frac{z}{2z^2 + 1} dz = 0$$

Solving for y gives

$$y = \text{RootOf} \left(\ln(x) - \int^{-1} \frac{z}{2z^2 + 1} dz + \int^{\frac{z}{x}} \frac{z}{2z^2 + 1} dz \right)$$

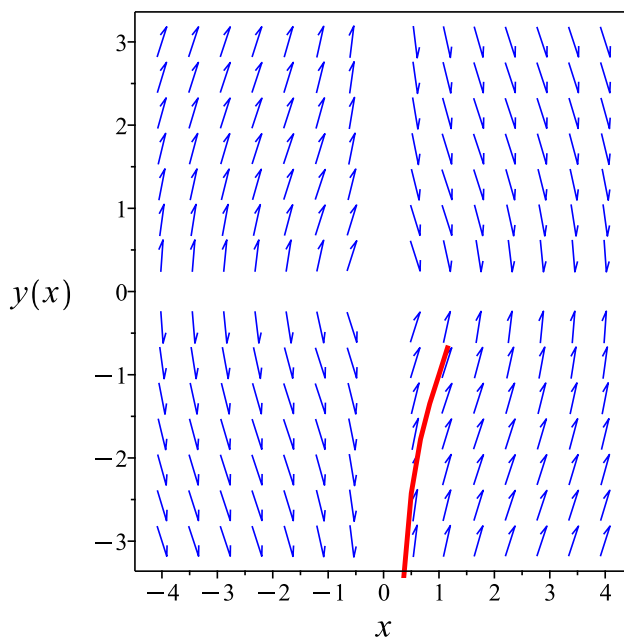


Figure 2.164: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = \text{RootOf} \left(\ln(x) - \int^{-1} \frac{z}{2z^2 + 1} dz + \int^{\frac{z}{x}} \frac{z}{2z^2 + 1} dz \right)$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.032 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

$$y(1) = -1$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX} Y(X) = -\frac{(X + x_0)^2 + (Y(X) + y_0)^2}{(X + x_0)(Y(X) + y_0)}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X^2 + Y(X)^2}{XY(X)}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X^2 + Y^2}{XY} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X^2 - Y^2$ and $N = XY$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -\frac{1}{u} - u \\ \frac{du}{dX} &= \frac{-\frac{1}{u(X)} - 2u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-\frac{1}{u(X)} - 2u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)u(X)X + 2u(X)^2 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{2u(X)^2 + 1}{u(X)X} \quad (2.78)$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= -\frac{2u(X)^2 + 1}{u(X)X} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= -\frac{1}{X} \\ g(u) &= \frac{2u^2 + 1}{u}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{X} dX\end{aligned}$$

$$\frac{\ln(2u(X)^2 + 1)}{4} = \ln\left(\frac{1}{X}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(X)$ gives

$$\begin{aligned}u(X) &= -\frac{i\sqrt{2}}{2} \\ u(X) &= \frac{i\sqrt{2}}{2}\end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$$

$$u(X) = -\frac{i\sqrt{2}}{2}$$

$$u(X) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$ back to $Y(X)$ gives

$$\left(\frac{2Y(X)^2 + X^2}{X^2} \right)^{1/4} = \frac{c_1}{X}$$

Converting $u(X) = -\frac{i\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = -\frac{iX\sqrt{2}}{2}$$

Converting $u(X) = \frac{i\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = \frac{iX\sqrt{2}}{2}$$

Using the solution for $Y(X)$

$$\left(\frac{2Y(X)^2 + X^2}{X^2} \right)^{1/4} = \frac{c_1}{X} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\left(\frac{2y^2 + x^2}{x^2} \right)^{1/4} = \frac{c_1}{x}$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{iX\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{i\sqrt{2}x}{2}$$

Using the solution for $Y(X)$

$$Y(X) = \frac{iX\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for initial conditions the solution is

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{3^{1/4}}{x}$$

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

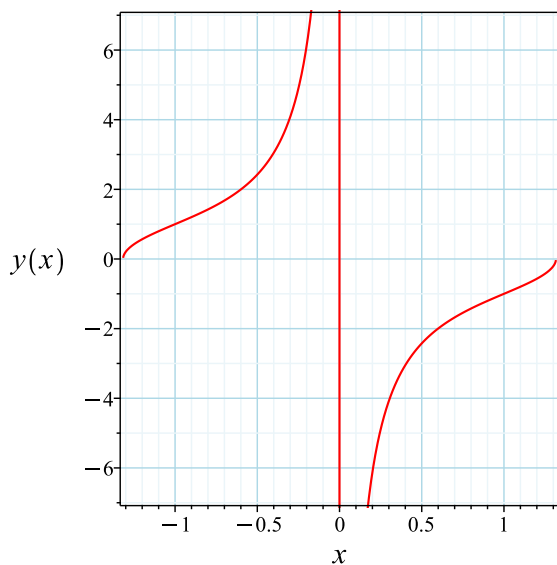
The solution

$$y = -\frac{i\sqrt{2}x}{2}$$

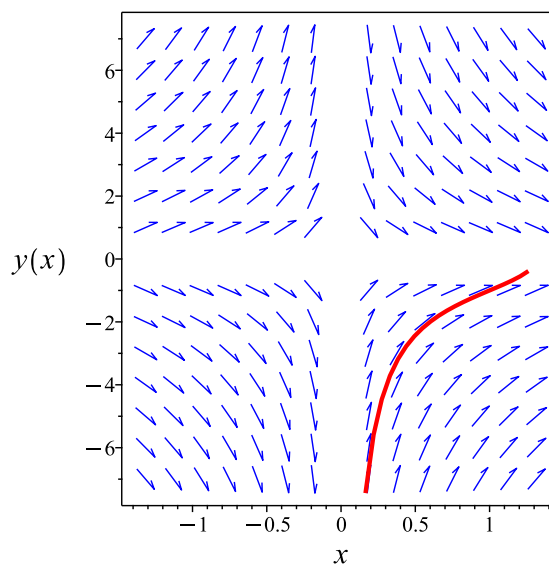
was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = \frac{i\sqrt{2}x}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $xyy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Solved using first_order_ode_LIE

Time used: 0.930 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

$$y(1) = -1$$

Writing the ode as

$$y' = -\frac{x^2 + y^2}{xy}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(x^2 + y^2)(b_3 - a_2)}{xy} - \frac{(x^2 + y^2)^2 a_3}{x^2 y^2} - \left(-\frac{2}{y} + \frac{x^2 + y^2}{x^2 y}\right)(xa_2 + ya_3 + a_1) \quad (5\text{E})$$

$$- \left(-\frac{2}{x} + \frac{x^2 + y^2}{x y^2}\right)(xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$\frac{x^4 a_3 + x^4 b_2 - 2x^3 y a_2 + 2x^3 y b_3 + x^2 y^2 a_3 - 2b_2 y^2 x^2 + 2y^4 a_3 + x^3 b_1 - x^2 y a_1 - x y^2 b_1 + y^3 a_1}{y^2 x^2}$$

$$= 0$$

Setting the numerator to zero gives

$$-x^4 a_3 - x^4 b_2 + 2x^3 y a_2 - 2x^3 y b_3 - x^2 y^2 a_3 + 2b_2 y^2 x^2 - 2y^4 a_3 - x^3 b_1 + x^2 y a_1 + x y^2 b_1 - y^3 a_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$2a_2v_1^3v_2 - a_3v_1^4 - a_3v_1^2v_2^2 - 2a_3v_2^4 - b_2v_1^4 + 2b_2v_1^2v_2^2 - 2b_3v_1^3v_2 + a_1v_1^2v_2 - a_1v_2^3 - b_1v_1^3 + b_1v_1v_2^2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-a_3 - b_2)v_1^4 + (2a_2 - 2b_3)v_1^3v_2 - b_1v_1^3 + (-a_3 + 2b_2)v_1^2v_2^2 + a_1v_1^2v_2 + b_1v_1v_2^2 - 2a_3v_2^4 - a_1v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 &= 0 \\ b_1 &= 0 \\ -a_1 &= 0 \\ -2a_3 &= 0 \\ -b_1 &= 0 \\ 2a_2 - 2b_3 &= 0 \\ -a_3 - b_2 &= 0 \\ -a_3 + 2b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = y$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x^2 + y^2}{xy} \right) (x) \\ &= y + \frac{x^2 + y^2}{y} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x^2 + y^2}{y}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(x^2 + 2y^2)}{4}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x^2 + y^2}{xy}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{x}{2x^2 + 4y^2} \\ S_y &= \frac{y}{x^2 + 2y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{2R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{2R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

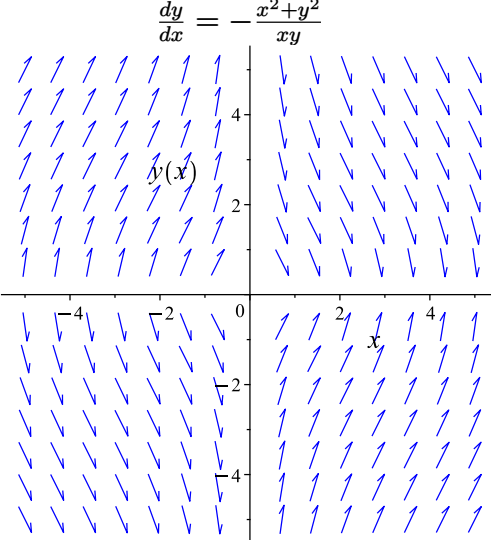
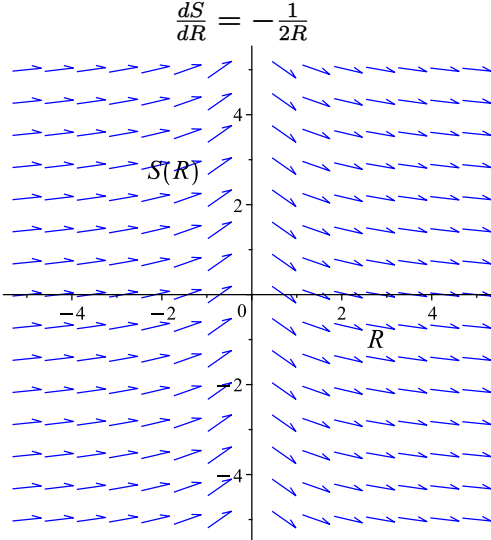
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{2R} dR \\ S(R) &= -\frac{\ln(R)}{2} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(2y^2 + x^2)}{4} = -\frac{\ln(x)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

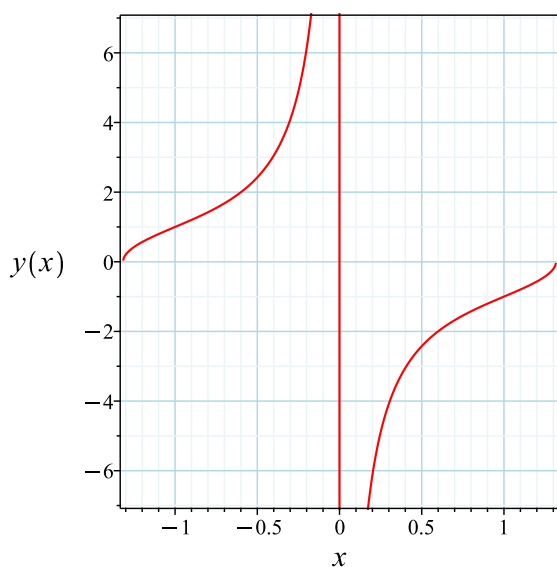
Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x^2+y^2}{xy}$ 	$R = x$ $S = \frac{\ln(x^2 + 2y^2)}{4}$	$\frac{dS}{dR} = -\frac{1}{2R}$ 

Solving for initial conditions the solution is

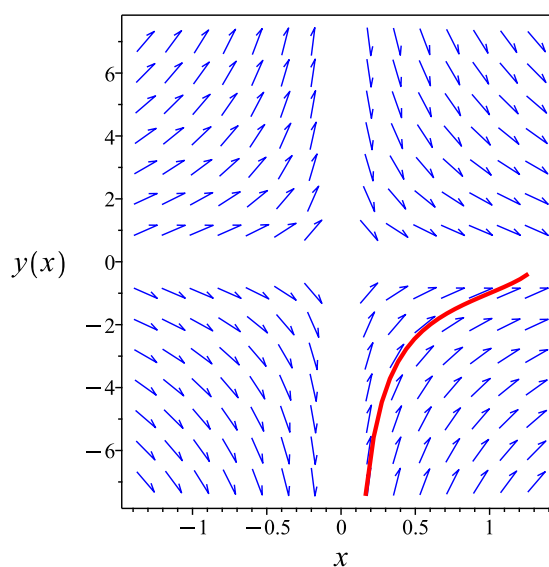
$$\frac{\ln(2y^2 + x^2)}{4} = -\frac{\ln(x)}{2} + \frac{\ln(3)}{4}$$

Solving for y gives

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$



(a) Solution plot



(b) Slope field $xyy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

✓ **Maple.** Time used: 0.058 (sec). Leaf size: 18

```
ode:=x*y(x)*diff(y(x),x)+x^2+y(x)^2 = 0;
ic:=[y(1) = -1];
dsolve([ode,op(ic)],y(x), singsol=all);
```

$$y = -\frac{\sqrt{-2x^4 + 6}}{2x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

✓ **Mathematica.** Time used: 0.136 (sec). Leaf size: 26

```
ode=(x^2+y[x]^2)+x*y[x]*D[y[x],x]==0;
ic={y[1]==-1};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{3-x^4}}{\sqrt{2}x}$$

✓ **Sympy.** Time used: 0.280 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2 + x*y(x)*Derivative(y(x), x) + y(x)**2,0)
ics = {y(1): -1}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = -\frac{\sqrt{6-2x^4}}{2x}$$

2.2.23 Problem 48

Local contents

Existence and uniqueness analysis	673
Solved using first_order_ode_separable	674
Solved using first_order_ode_exact	676
✓ Maple	680
✓ Mathematica	682
✓ Sympy	682

Internal problem ID [7877]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 48

Date solved : Thursday, November 27, 2025 at 06:26:58 AM

CAS classification : [_separable]

Existence and uniqueness analysis

$$\cos(y) + (e^{-x} + 1) \sin(y) y' = 0$$

$$y(0) = \frac{\pi}{4}$$

This is non linear first order ODE. In canonical form it is written as

$$\begin{aligned} y' &= f(x, y) \\ &= -\frac{\cos(y)}{(e^{-x} + 1) \sin(y)} \end{aligned}$$

The x domain of $f(x, y)$ when $y = \frac{\pi}{4}$ is

$$\{-2i\pi_Z765 - i\pi < x\}$$

But the point $x_0 = 0$ is not inside this domain. Hence existence and uniqueness theorem does not apply. There could be infinite number of solutions, or one solution or no solution at all.

Solved using first_order_ode_separable

Time used: 0.300 (sec)

Solve

$$\cos(y) + (e^{-x} + 1) \sin(y) y' = 0$$

$$y(0) = \frac{\pi}{4}$$

The ode

$$y' = -\frac{\cos(y)}{(e^{-x} + 1) \sin(y)} \quad (2.79)$$

is separable as it can be written as

$$\begin{aligned} y' &= -\frac{\cos(y)}{(e^{-x} + 1) \sin(y)} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{e^{-x} + 1} \\ g(y) &= \frac{\cos(y)}{\sin(y)} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{\sin(y)}{\cos(y)} dy &= \int -\frac{1}{e^{-x} + 1} dx \end{aligned}$$

$$-\ln(\cos(y)) = \ln\left(\frac{1}{e^{-x} + 1}\right) - x + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$\frac{\cos(y)}{\sin(y)} = 0$$

for y gives

$$y = -\frac{\pi}{2}$$

$$y = \frac{\pi}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\ln(\cos(y)) = \ln\left(\frac{1}{e^{-x} + 1}\right) - x + c_1$$

$$y = -\frac{\pi}{2}$$

$$y = \frac{\pi}{2}$$

Solving for initial conditions the solution is

$$-\ln(\cos(y)) = \ln\left(\frac{1}{e^{-x} + 1}\right) - x + \frac{3\ln(2)}{2}$$

$$y = -\frac{\pi}{2}$$

$$y = \frac{\pi}{2}$$

Solving for y gives

$$y = -\frac{\pi}{2}$$

$$y = \frac{\pi}{2}$$

$$y = \arccos\left(e^{x - \frac{3\ln(2)}{2}}(1 + e^x)e^{-x}\right)$$

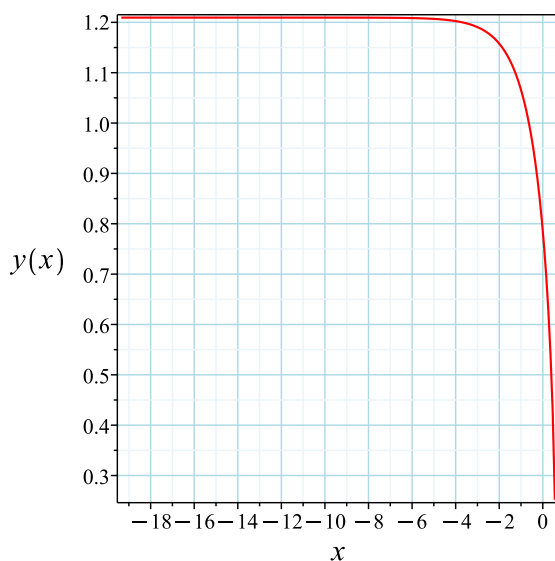
The solution

$$y = -\frac{\pi}{2}$$

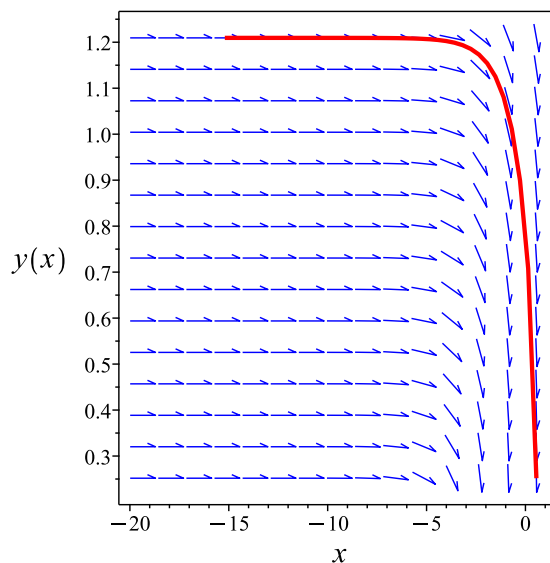
was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = \frac{\pi}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.



(a) Solution plot



(b) Slope field $\cos(y) + (e^{-x} + 1) \sin(y) y' = 0$

Summary of solutions found

$$y = \arccos\left(e^{x - \frac{3 \ln(2)}{2}} (1 + e^x) e^{-x}\right)$$

Solved using first_order_ode_exact

Time used: 0.168 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} ((e^{-x} + 1) \sin(y)) dy &= (-\cos(y)) dx \\ (\cos(y)) dx + ((e^{-x} + 1) \sin(y)) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \cos(y) \\ N(x, y) &= (e^{-x} + 1) \sin(y) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(\cos(y)) \\ &= -\sin(y) \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}((e^{-x} + 1) \sin(y)) \\ &= -e^{-x} \sin(y) \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{\csc(y)}{e^{-x} + 1} ((-\sin(y)) - (-e^{-x} \sin(y))) \\ &= -\tanh\left(\frac{x}{2}\right) \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\tanh(\frac{x}{2}) \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2 \ln(\cosh(\frac{x}{2}))} \\ &= \operatorname{sech}\left(\frac{x}{2}\right)^2\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \operatorname{sech}\left(\frac{x}{2}\right)^2 (\cos(y)) \\ &= \cos(y) \operatorname{sech}\left(\frac{x}{2}\right)^2\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \operatorname{sech}\left(\frac{x}{2}\right)^2 ((e^{-x} + 1) \sin(y)) \\ &= \frac{4 \sin(y)}{1 + e^x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\cos(y) \operatorname{sech}\left(\frac{x}{2}\right)^2 \right) + \left(\frac{4 \sin(y)}{1 + e^x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} \, dx &= \int \overline{M} \, dx \\ \int \frac{\partial \phi}{\partial x} \, dx &= \int \cos(y) \operatorname{sech}\left(\frac{x}{2}\right)^2 \, dx \\ \phi &= 2 \cos(y) \tanh\left(\frac{x}{2}\right) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -2 \sin(y) \tanh\left(\frac{x}{2}\right) + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{4 \sin(y)}{1+e^x}$. Therefore equation (4) becomes

$$\frac{4 \sin(y)}{1+e^x} = -2 \sin(y) \tanh\left(\frac{x}{2}\right) + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$\begin{aligned} f'(y) &= \frac{2 \sin(y) \left(\tanh\left(\frac{x}{2}\right) e^x + \tanh\left(\frac{x}{2}\right) + 2 \right)}{1+e^x} \\ &= 2 \sin(y) \end{aligned}$$

Integrating the above w.r.t y results in

$$\begin{aligned} \int f'(y) dy &= \int (2 \sin(y)) dy \\ f(y) &= -2 \cos(y) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = 2 \cos(y) \tanh\left(\frac{x}{2}\right) - 2 \cos(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = 2 \cos(y) \tanh\left(\frac{x}{2}\right) - 2 \cos(y)$$

Simplifying the above gives

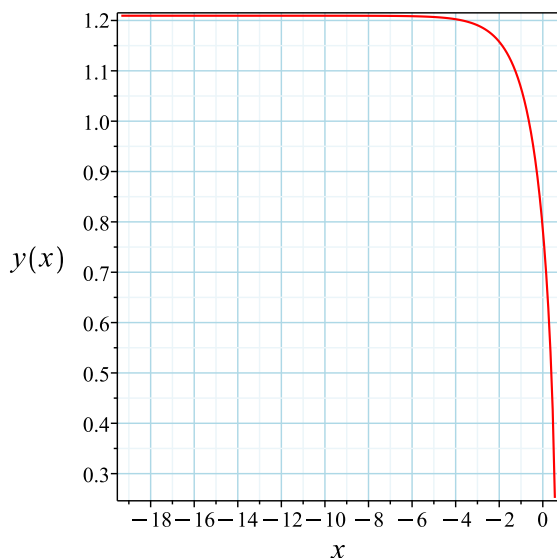
$$2 \cos(y) \left(-1 + \tanh\left(\frac{x}{2}\right) \right) = c_1$$

Solving for initial conditions the solution is

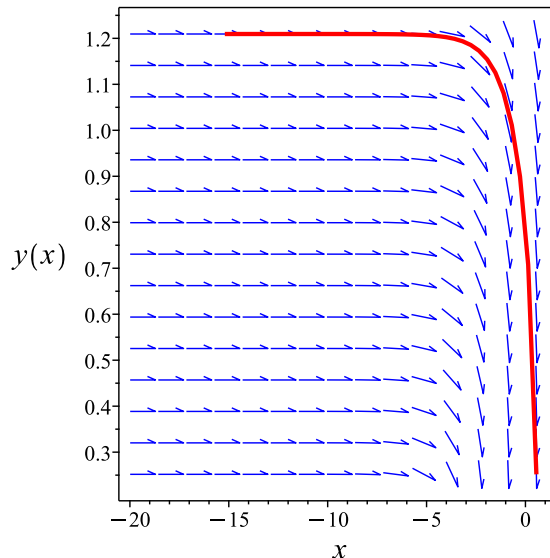
$$2 \cos(y) \left(-1 + \tanh\left(\frac{x}{2}\right) \right) = -\sqrt{2}$$

Solving for y gives

$$y = \pi - \arccos\left(\frac{\sqrt{2}}{2 \tanh\left(\frac{x}{2}\right) - 2}\right)$$



(a) Solution plot



(b) Slope field $\cos(y) + (e^{-x} + 1) \sin(y) y' = 0$

Summary of solutions found

$$y = \pi - \arccos\left(\frac{\sqrt{2}}{2 \tanh\left(\frac{x}{2}\right) - 2}\right)$$

✓ **Maple.** Time used: 0.187 (sec). Leaf size: 14

```
ode:=cos(y(x))+(1+exp(-x))*sin(y(x))*diff(y(x),x) = 0;
ic:=[y(0) = 1/4*Pi];
dsolve([ode,op(ic)],y(x), singsol=all);
```

$$y = \arccos\left(\frac{\sqrt{2}(e^x + 1)}{4}\right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
```

```
trying Bernoulli
trying separable
<- separable successful
```

Maple step by step

Let's solve

$$\left[\cos(y(x)) + (1 + e^{-x}) \sin(y(x)) \left(\frac{d}{dx} y(x) \right) = 0, y(0) = \frac{\pi}{4} \right]$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx} y(x)$
- Solve for the highest derivative
 $\frac{d}{dx} y(x) = -\frac{\cos(y(x))}{(1+e^{-x}) \sin(y(x))}$
- Separate variables
 $\frac{\left(\frac{d}{dx} y(x) \right) \sin(y(x))}{\cos(y(x))} = -\frac{1}{1+e^{-x}}$
- Integrate both sides with respect to x
 $\int \frac{\left(\frac{d}{dx} y(x) \right) \sin(y(x))}{\cos(y(x))} dx = \int -\frac{1}{1+e^{-x}} dx + C1$
- Evaluate integral
 $-\ln(\cos(y(x))) = \ln(e^{-x}) - \ln(1 + e^{-x}) + C1$
- Solve for $y(x)$
 $y(x) = \arccos\left(\frac{e^{x-C1}(e^x+1)}{e^x}\right)$
- Simplify
 $y(x) = \arccos(e^{-C1}(e^x + 1))$
- Redefine the integration constant(s)
 $y(x) = \arccos(C1(e^x + 1))$
- Use initial condition $y(0) = \frac{\pi}{4}$
 $\frac{\pi}{4} = \arccos(2C1)$
- Solve for $_C1$
 $C1 = \frac{\sqrt{2}}{4}$
- Substitute $_C1 = \frac{\sqrt{2}}{4}$ into general solution and simplify
 $y(x) = \arccos\left(\frac{\sqrt{2}(e^x+1)}{4}\right)$
- Solution to the IVP
 $y(x) = \arccos\left(\frac{\sqrt{2}(e^x+1)}{4}\right)$

✓ **Mathematica.** Time used: 59.595 (sec). Leaf size: 20

```
ode=Cos[y[x]]+(1+Exp[-x])*Sin[y[x]]*D[y[x],x]== 0;
ic={y[0]==Pi/4};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \arccos\left(\frac{e^x + 1}{2\sqrt{2}}\right)$$

✓ **Sympy.** Time used: 0.386 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((1 + exp(-x))*sin(y(x))*Derivative(y(x), x) + cos(y(x)),0)
ics = {y(0): pi/4}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \arccos\left(\frac{\sqrt{2}e^x}{4} + \frac{\sqrt{2}}{4}\right)$$

2.2.24 Problem 49

Local contents

Existence and uniqueness analysis	683
Solved using first_order_ode_bernoulli	684
✓ Maple	687
✓ Mathematica	688
✓ Sympy	688

Internal problem ID [7878]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 49

Date solved : Thursday, November 27, 2025 at 06:27:06 AM

CAS classification : [_rational, _Bernoulli]

Existence and uniqueness analysis

$$y^2 + yx - y'x = 0$$

$$y(1) = 1$$

This is non linear first order ODE. In canonical form it is written as

$$\begin{aligned} y' &= f(x, y) \\ &= \frac{y(x+y)}{x} \end{aligned}$$

The x domain of $f(x, y)$ when $y = 1$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 1$ is inside this domain. The y domain of $f(x, y)$ when $x = 1$ is

$$\{-\infty < y < \infty\}$$

And the point $y_0 = 1$ is inside this domain. Now we will look at the continuity of

$$\begin{aligned} \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{y(x+y)}{x} \right) \\ &= \frac{x+y}{x} + \frac{y}{x} \end{aligned}$$

The x domain of $\frac{\partial f}{\partial y}$ when $y = 1$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 1$ is inside this domain. The y domain of $\frac{\partial f}{\partial y}$ when $x = 1$ is

$$\{-\infty < y < \infty\}$$

And the point $y_0 = 1$ is inside this domain. Therefore solution exists and is unique.

Solved using first_order_ode_bernoulli

Time used: 0.121 (sec)

Solve

$$\begin{aligned} y^2 + yx - y'x &= 0 \\ y(1) &= 1 \end{aligned}$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y(x + y)}{x} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = (1)y + \left(\frac{1}{x}\right)y^2 \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= 1 \\ f_1 &= \frac{1}{x} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \tag{3}$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= 1 \\f_1(x) &= \frac{1}{x} \\n &= 2\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^2$ gives

$$y' \frac{1}{y^2} = \frac{1}{y} + \frac{1}{x} \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= \frac{1}{y}\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{1}{y^2} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}-v'(x) &= v(x) + \frac{1}{x} \\v' &= -v - \frac{1}{x}\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= 1 \\p(x) &= -\frac{1}{x}\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\&= e^{\int 1 \, dx} \\&= e^x\end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu v) = \mu p$$

$$\frac{d}{dx}(\mu v) = (\mu) \left(-\frac{1}{x} \right)$$

$$\frac{d}{dx}(v e^x) = (e^x) \left(-\frac{1}{x} \right)$$

$$d(v e^x) = \left(-\frac{e^x}{x} \right) dx$$

Integrating gives

$$\begin{aligned} v e^x &= \int -\frac{e^x}{x} dx \\ &= \text{Ei}_1(-x) + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^x gives the final solution

$$v(x) = e^{-x}(\text{Ei}_1(-x) + c_1)$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

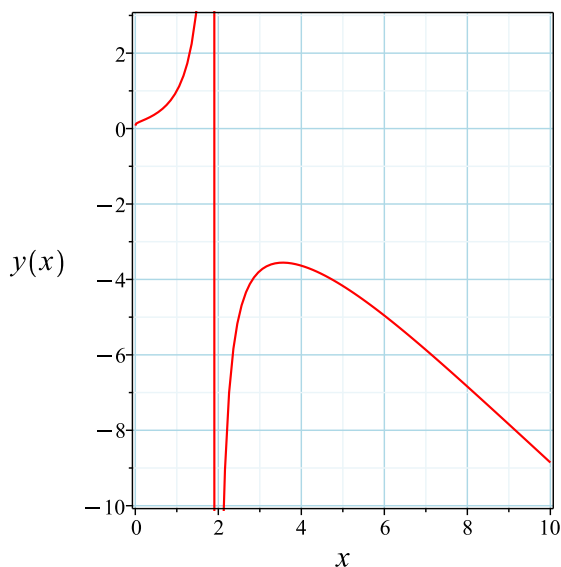
$$\frac{1}{y} = e^{-x}(\text{Ei}_1(-x) + c_1)$$

Solving for initial conditions the solution is

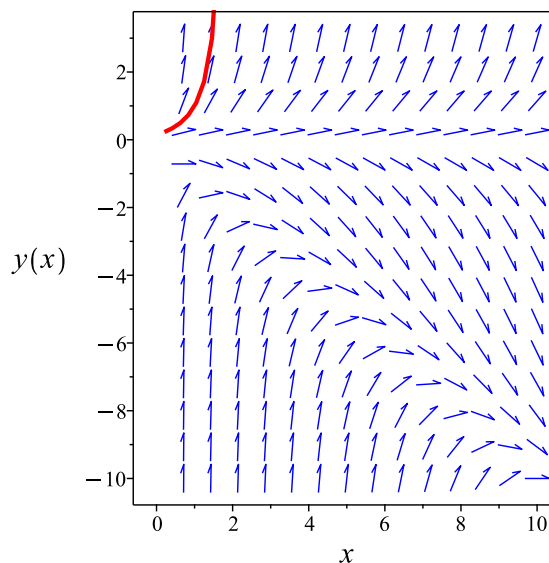
$$\frac{1}{y} = e^{-x}(\text{Ei}_1(-x) - \text{Ei}_1(-1) + e)$$

Solving for y gives

$$y = \frac{e^x}{\text{Ei}_1(-x) - \text{Ei}_1(-1) + e}$$



(a) Solution plot

(b) Slope field $y^2 + yx - y'x = 0$

Summary of solutions found

$$y = \frac{e^x}{\text{Ei}_1(-x) - \text{Ei}_1(-1) + e}$$

✓ Maple. Time used: 0.081 (sec). Leaf size: 22

```
ode:=y(x)^2+y(x)*x-diff(y(x),x)*x = 0;
ic:=[y(1) = 1];
dsolve([ode,op(ic)],y(x), singsol=all);
```

$$y = \frac{e^x}{\text{Ei}_1(-x) + e - \text{Ei}_1(-1)}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

✓ **Mathematica.** Time used: 0.115 (sec). Leaf size: 31

```
ode=(y[x]^2+x*y[x])-x*D[y[x],x]== 0;
ic={y[1]==1};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{e^x}{e - \int_1^x \frac{e^{K[1]}}{K[1]} dK[1]}$$

✓ **Sympy.** Time used: 0.267 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) - x*Derivative(y(x), x) + y(x)**2,0)
ics = {y(1): 1}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{e^x}{- \operatorname{Ei}(x) + \operatorname{Ei}(1) + e}$$

2.2.25 Problem 51

Local contents

Solved using first_order_ode_dAlembert	689
Solved using first_order_ode_LIE	694
Solved using first_order_ode_riccati	699
Solved using first_order_ode_riccati_by_guessing_particular_solution	704
✓ Maple	706
✓ Mathematica	707
✓ Sympy	707

Internal problem ID [7879]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 51

Date solved : Thursday, November 27, 2025 at 06:27:13 AM

CAS classification : [[_homogeneous, 'class C'], _Riccati]

Solved using first_order_ode_dAlembert

Time used: 0.737 (sec)

Solve

$$y' = -2(3y + 2x)^2$$

Let $p = y'$ the ode becomes

$$p = -2(3y + 2x)^2$$

Solving for y from the above results in

$$y = -\frac{2x}{3} + \frac{\sqrt{-2p}}{6} \quad (1)$$

$$y = -\frac{2x}{3} - \frac{\sqrt{-2p}}{6} \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{2}{3} \\ g &= \frac{\sqrt{2}\sqrt{-p}}{6} \end{aligned}$$

Hence (2) becomes

$$p + \frac{2}{3} = -\frac{\sqrt{2}p'(x)}{12\sqrt{-p}} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p + \frac{2}{3} = 0$$

Solving the above for p results in

$$p_1 = -\frac{2}{3}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{2x}{3} + \frac{\sqrt{3}}{9}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = -6 \left(p(x) + \frac{2}{3} \right) \sqrt{2} \sqrt{-p(x)} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\begin{aligned} \int -\frac{\sqrt{2}}{4\sqrt{-p} (3p+2)} dp &= dx \\ \frac{\sqrt{2}\sqrt{6} \operatorname{arctanh}\left(\frac{\sqrt{6}\sqrt{-p}}{2}\right)}{12} &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-2\sqrt{2}\sqrt{-p}(3p+2)=0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$\begin{aligned} p(x) &= 0 \\ p(x) &= -\frac{2}{3} \end{aligned}$$

Substituting the above solution for p in (2A) gives

$$\begin{aligned} y &= -\frac{2x}{3} + \frac{\sqrt{3}\sqrt{\tanh((x+c_1)\sqrt{12})^2}}{9} \\ y &= -\frac{2x}{3} \\ y &= -\frac{2x}{3} + \frac{\sqrt{3}}{9} \end{aligned}$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{2}{3} \\ g &= -\frac{\sqrt{2}\sqrt{-p}}{6} \end{aligned}$$

Hence (2) becomes

$$p + \frac{2}{3} = \frac{\sqrt{2}p'(x)}{12\sqrt{-p}} \tag{2A}$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p + \frac{2}{3} = 0$$

Solving the above for p results in

$$p_1 = -\frac{2}{3}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{2x}{3} - \frac{\sqrt{3}}{9}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = 6 \left(p(x) + \frac{2}{3} \right) \sqrt{2} \sqrt{-p(x)} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\int \frac{\sqrt{2}}{4\sqrt{-p}(3p+2)} dp = dx$$

$$-\frac{\sqrt{2}\sqrt{6} \operatorname{arctanh}\left(\frac{\sqrt{6}\sqrt{-p}}{2}\right)}{12} = x + c_2$$

Singular solutions are found by solving

$$2\sqrt{2}\sqrt{-p}(3p+2) = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = 0$$

$$p(x) = -\frac{2}{3}$$

Substituting the above solution for p in (2A) gives

$$y = -\frac{2x}{3} - \frac{\sqrt{3} \sqrt{\tanh((x+c_2)\sqrt{12})^2}}{9}$$

$$y = -\frac{2x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3}}{9}$$

Simplifying the above gives

$$y = -\frac{2x}{3} + \frac{\sqrt{3}}{9}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3} \sqrt{\tanh(2(x+c_1)\sqrt{3})^2}}{9}$$

$$y = -\frac{2x}{3}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3}}{9}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3}}{9}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3} \sqrt{\tanh(2\sqrt{3}(x+c_2))^2}}{9}$$

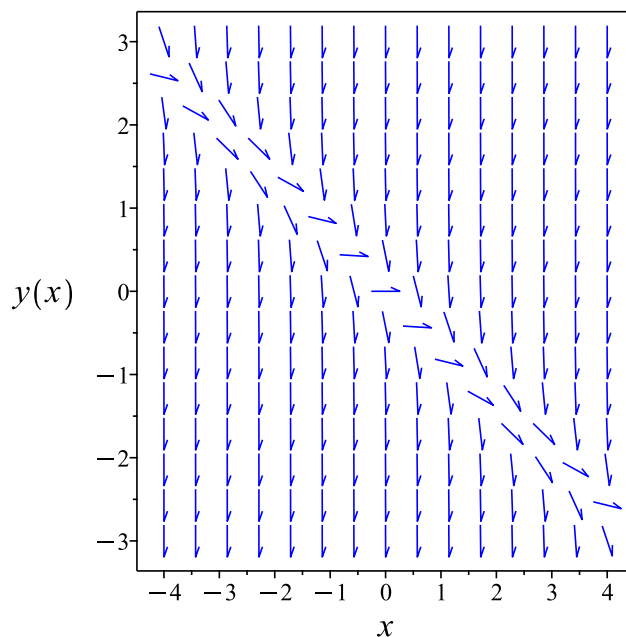
$$y = -\frac{2x}{3}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3}}{9}$$

The solution

$$y = -\frac{2x}{3}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Figure 2.170: Slope field $y' = -2(3y + 2x)^2$ Summary of solutions found

$$y = -\frac{2x}{3} - \frac{\sqrt{3}}{9}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3}}{9}$$

$$y = -\frac{2x}{3} - \frac{\sqrt{3} \sqrt{\tanh(2\sqrt{3}(x + c_2))^2}}{9}$$

$$y = -\frac{2x}{3} + \frac{\sqrt{3} \sqrt{\tanh(2(x + c_1)\sqrt{3})^2}}{9}$$

Solved using first_order_ode_LIE

Time used: 0.294 (sec)

Solve

$$y' = -2(3y + 2x)^2$$

Writing the ode as

$$y' = -2(3y + 2x)^2$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1\text{E})$$

$$\eta = \frac{Ax + By}{Cx} \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\frac{A}{Cx} - \frac{Ax + By}{Cx^2} - \frac{2(3y + 2x)^2 B}{Cx} + 16x + 24y - \frac{(-36y - 24x)(Ax + By)}{Cx} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{24Ax^3 + 36Ax^2y - 8Bx^3 + 18Bxy^2 + 16Cx^3 + 24yCx^2 - By}{Cx^2} = 0$$

Setting the numerator to zero gives

$$24Ax^3 + 36Ax^2y - 8Bx^3 + 18Bxy^2 + 16Cx^3 + 24yCx^2 - By = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$24Av_1^3 + 36Av_1^2v_2 - 8Bv_1^3 + 18Bv_1v_2^2 + 16Cv_1^3 + 24Cv_1^2v_2 - Bv_2 = 0 \quad (7\text{E})$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(24A - 8B + 16C)v_1^3 + (36A + 24C)v_1^2v_2 + 18Bv_1v_2^2 - Bv_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -B &= 0 \\ 18B &= 0 \\ 36A + 24C &= 0 \\ 24A - 8B + 16C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= A \\ B &= 0 \\ C &= -\frac{3A}{2} \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 1 \\ \eta &= -\frac{2}{3} \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{-\frac{2}{3}}{1} \\ &= -\frac{2}{3} \end{aligned}$$

This is easily solved to give

$$y = -\frac{2x}{3} + c_1$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = y + \frac{2x}{3}$$

And S is found from

$$\begin{aligned} dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1} \end{aligned}$$

Integrating gives

$$\begin{aligned} S &= \int \frac{dx}{T} \\ &= x \end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -2(3y + 2x)^2$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= \frac{2}{3} \\ R_y &= 1 \\ S_x &= 1 \\ S_y &= 0 \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{\frac{2}{3} - 2(3y + 2x)^2} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{\frac{2}{3} - 18R^2}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

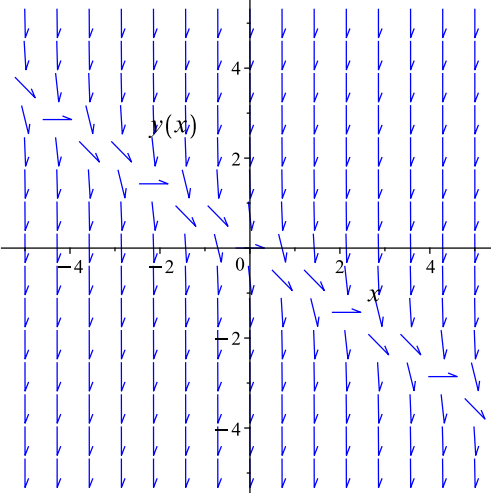
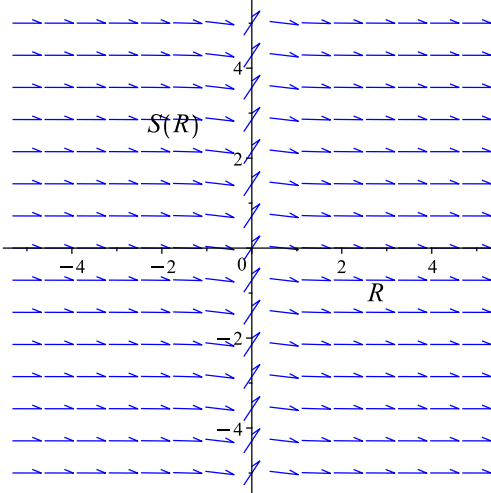
$$\int dS = \int -\frac{3}{2(27R^2 - 1)} dR$$

$$S(R) = \frac{\sqrt{3} \operatorname{arctanh}(3\sqrt{3}R)}{6} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = \frac{\sqrt{3} \operatorname{arctanh}(3\sqrt{3}(y + \frac{2x}{3}))}{6} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

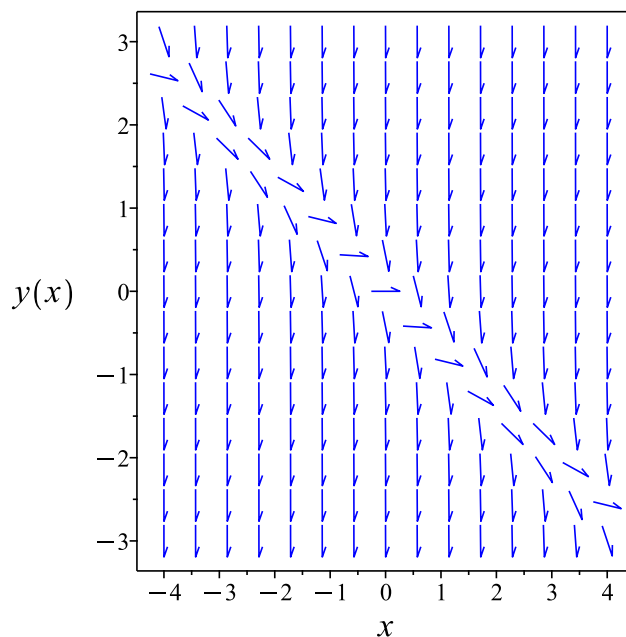
Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -2(3y + 2x)^2$ 	$R = y + \frac{2x}{3}$ $S = x$	$\frac{dS}{dR} = \frac{1}{\frac{2}{3} - 18R^2}$ 

Simplifying the above gives

$$x = \frac{\operatorname{arctanh}(\sqrt{3}(3y + 2x))\sqrt{3}}{6} + c_2$$

Solving for y gives

$$y = -\frac{(2\sqrt{3}x + \tanh(2\sqrt{3}(-x + c_2)))\sqrt{3}}{9}$$

Figure 2.171: Slope field $y' = -2(3y + 2x)^2$ Summary of solutions found

$$y = -\frac{(2\sqrt{3}x + \tanh(2\sqrt{3}(-x + c_2)))\sqrt{3}}{9}$$

Solved using first_order_ode_riccati

Time used: 0.381 (sec)

Solve

$$y' = -2(3y + 2x)^2$$

In canonical form the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -2(3y + 2x)^2 \end{aligned}$$

This is a Riccati ODE. Comparing the ODE to solve

$$y' = -8x^2 - 24yx - 18y^2$$

With Riccati ODE standard form

$$y' = f_0(x) + f_1(x)y + f_2(x)y^2$$

Shows that $f_0(x) = -8x^2$, $f_1(x) = -24x$ and $f_2(x) = -18$. Let

$$\begin{aligned} y &= \frac{-u'}{f_2 u} \\ &= \frac{-u'}{-18u} \end{aligned} \quad (1)$$

Using the above substitution in the given ODE results (after some simplification) in a second order ODE to solve for $u(x)$ which is

$$f_2 u''(x) - (f_2' + f_1 f_2) u'(x) + f_2^2 f_0 u(x) = 0 \quad (2)$$

But

$$\begin{aligned} f_2' &= 0 \\ f_1 f_2 &= 432x \\ f_2^2 f_0 &= -2592x^2 \end{aligned}$$

Substituting the above terms back in equation (2) gives

$$-18u''(x) - 432xu'(x) - 2592x^2u(x) = 0$$

In normal form the given ode is written as

$$\frac{d^2u}{dx^2} + p(x) \left(\frac{du}{dx} \right) + q(x) u = 0 \quad (2)$$

Where

$$\begin{aligned} p(x) &= 24x \\ q(x) &= 144x^2 \end{aligned}$$

Calculating the Liouville ode invariant Q given by

$$\begin{aligned} Q &= q - \frac{p'}{2} - \frac{p^2}{4} \\ &= 144x^2 - \frac{(24x)'}{2} - \frac{(24x)^2}{4} \\ &= 144x^2 - \frac{(24)}{2} - \frac{(576x^2)}{4} \\ &= 144x^2 - (12) - 144x^2 \\ &= -12 \end{aligned}$$

Since the Liouville ode invariant does not depend on the independent variable x then the transformation

$$u = v(x) z(x) \quad (3)$$

is used to change the original ode to a constant coefficients ode in v . In (3) the term $z(x)$ is given by

$$\begin{aligned} z(x) &= e^{-\int \frac{p(x)}{2} dx} \\ &= e^{-\int \frac{24x}{2} dx} \\ &= e^{-6x^2} \end{aligned} \tag{5}$$

Hence (3) becomes

$$u = v(x) e^{-6x^2} \tag{4}$$

Applying this change of variable to the original ode results in

$$\left(-18 \frac{d^2}{dx^2} v(x) + 216v(x) \right) e^{-6x^2} = 0$$

Which is now solved for $v(x)$.

The above ode can be simplified to

$$-18 \frac{d^2}{dx^2} v(x) + 216v(x) = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Av''(x) + Bv'(x) + Cv(x) = 0$$

Where in the above $A = -18, B = 0, C = 216$. Let the solution be $v(x) = e^{\lambda x}$. Substituting this into the ODE gives

$$-18\lambda^2 e^{x\lambda} + 216 e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$-18\lambda^2 + 216 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = -18, B = 0, C = 216$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(-18)} \pm \frac{1}{(2)(-18)} \sqrt{0^2 - (4)(-18)(216)} \\ &= \pm -2\sqrt{3} \end{aligned}$$

Hence

$$\lambda_1 = + - 2\sqrt{3}$$

$$\lambda_2 = - - 2\sqrt{3}$$

Which simplifies to

$$\lambda_1 = -2\sqrt{3}$$

$$\lambda_2 = 2\sqrt{3}$$

Since roots are distinct, then the solution is

$$v(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$v(x) = c_1 e^{(-2\sqrt{3})x} + c_2 e^{(2\sqrt{3})x}$$

Or

$$v(x) = c_1 e^{-2\sqrt{3}x} + c_2 e^{2\sqrt{3}x}$$

Now that $v(x)$ is known, then

$$\begin{aligned} u &= v(x) z(x) \\ &= (c_1 e^{-2\sqrt{3}x} + c_2 e^{2\sqrt{3}x}) (z(x)) \end{aligned} \quad (7)$$

But from (5)

$$z(x) = e^{-6x^2}$$

Hence (7) becomes

$$u = (c_1 e^{-2\sqrt{3}x} + c_2 e^{2\sqrt{3}x}) e^{-6x^2}$$

Taking derivative gives

$$u'(x) = (-2c_1\sqrt{3}e^{-2\sqrt{3}x} + 2c_2\sqrt{3}e^{2\sqrt{3}x})e^{-6x^2} - 12(c_1 e^{-2\sqrt{3}x} + c_2 e^{2\sqrt{3}x})x e^{-6x^2} \quad (4)$$

Substituting equations (3,4) into (1) results in

$$y = \frac{-u'}{f_2 u}$$

$$y = \frac{-u'}{-18u}$$

$$y = \frac{-6c_2\left(x - \frac{\sqrt{3}}{6}\right)e^{4\sqrt{3}x} - 6\left(x + \frac{\sqrt{3}}{6}\right)c_1}{9c_2 e^{4\sqrt{3}x} + 9c_1}$$

Doing change of constants, the above solution becomes

$$y = \frac{\left(\left(-2\sqrt{3}e^{-2\sqrt{3}x} + 2c_3\sqrt{3}e^{2\sqrt{3}x} \right) e^{-6x^2} - 12 \left(e^{-2\sqrt{3}x} + c_3 e^{2\sqrt{3}x} \right) x e^{-6x^2} \right) e^{6x^2}}{18e^{-2\sqrt{3}x} + 18c_3 e^{2\sqrt{3}x}}$$

Simplifying the above gives

$$y = \frac{-6 \left(x - \frac{\sqrt{3}}{6} \right) c_3 e^{4\sqrt{3}x} - 6x - \sqrt{3}}{9e^{4\sqrt{3}x}c_3 + 9}$$

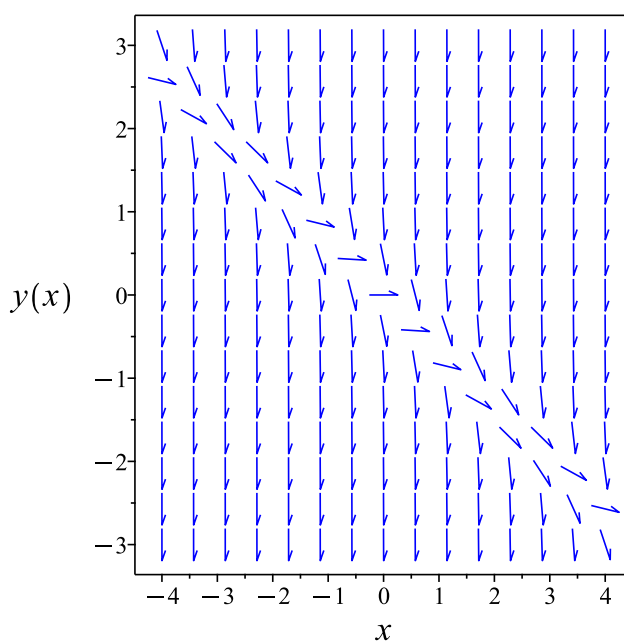


Figure 2.172: Slope field $y' = -2(3y + 2x)^2$

Summary of solutions found

$$y = \frac{-6 \left(x - \frac{\sqrt{3}}{6} \right) c_3 e^{4\sqrt{3}x} - 6x - \sqrt{3}}{9e^{4\sqrt{3}x}c_3 + 9}$$

Solved using first_order_ode_riccati_by_guessing_particular_solution

Time used: 0.159 (sec)

Solve

$$y' = -2(3y + 2x)^2$$

This is a Riccati ODE. Comparing the above ODE to solve with the Riccati standard form

$$y' = f_0(x) + f_1(x)y + f_2(x)y^2$$

Shows that

$$f_0(x) = -8x^2$$

$$f_1(x) = -24x$$

$$f_2(x) = -18$$

Using trial and error, the following particular solution was found

$$y_p = -\frac{2x}{3} + \frac{\sqrt{3}}{9}$$

Since a particular solution is known, then the general solution is given by

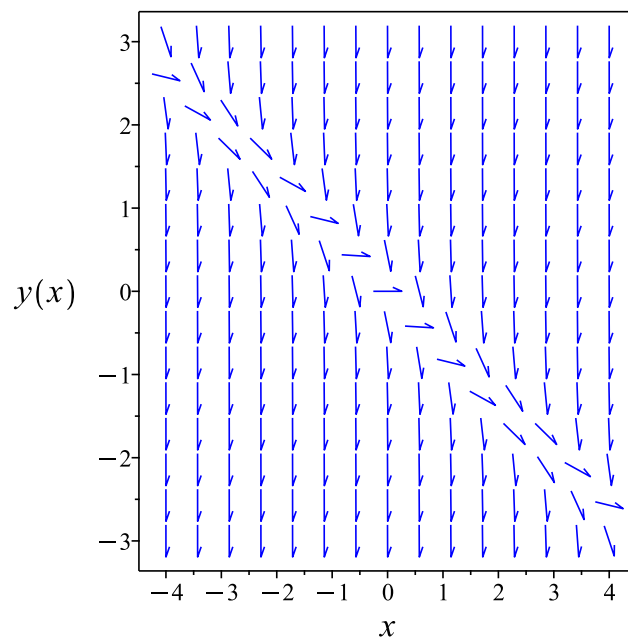
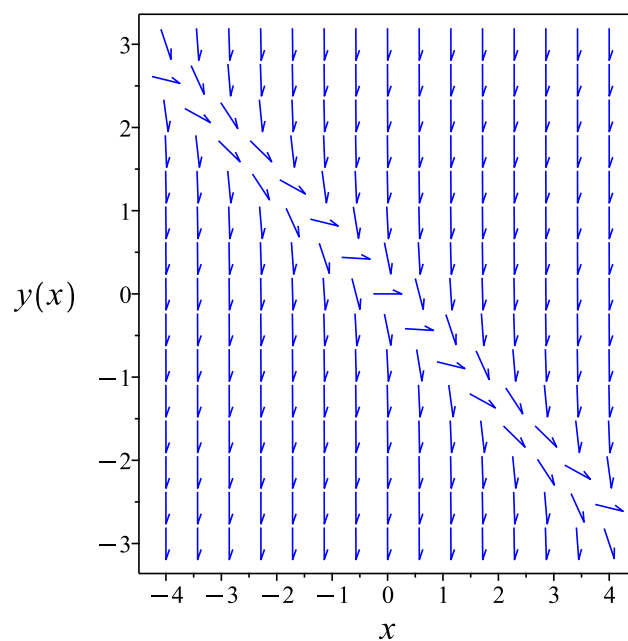
$$y = y_p + \frac{\phi(x)}{c_1 - \int \phi(x) f_2 dx}$$

Where

$$\phi(x) = e^{\int 2f_2 y_p + f_1 dx}$$

Evaluating the above gives the general solution as

$$y = \frac{(18\sqrt{3}x + 9)e^{-4\sqrt{3}x} - 12\left(x - \frac{\sqrt{3}}{6}\right)c_1}{-27\sqrt{3}e^{-4\sqrt{3}x} + 18c_1}$$

Figure 2.173: Slope field $y' = -2(3y + 2x)^2$ Figure 2.174: Slope field $y' = -2(3y + 2x)^2$

Summary of solutions found

$$y = \frac{(18\sqrt{3}x + 9)e^{-4\sqrt{3}x} - 12\left(x - \frac{\sqrt{3}}{6}\right)c_1}{-27\sqrt{3}e^{-4\sqrt{3}x} + 18c_1}$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 24

```
ode:=diff(y(x),x) = -2*(2*x+3*y(x))^2;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{2x}{3} - \frac{\sqrt{3} \tanh(2(c_1 - x) \sqrt{3})}{9}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous C
1st order, trying the canonical coordinates of the invariance group
  -> Calling odsolve with the ODE, diff(y(x),x) = -2/3, y(x)
      *** Sublevel 2 ***
      Methods for first order ODEs:
      --- Trying classification methods ---
      trying a quadrature
      trying 1st order linear
      <- 1st order linear successful
<- 1st order, canonical coordinates successful
<- homogeneous successful
```

Maple step by step

Let's solve

$$\frac{d}{dx}y(x) = -2(2x + 3y(x))^2$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -2(2x + 3y(x))^2$$

✓ **Mathematica.** Time used: 0.096 (sec). Leaf size: 59

```
ode=D[y[x],x]==-2*(2*x+3*y[x])^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{9} \left(-6x - \frac{6}{\sqrt{3} + 12c_1 e^{4\sqrt{3}x}} + \sqrt{3} \right)$$

$$y(x) \rightarrow \frac{1}{9} (\sqrt{3} - 6x)$$

✓ **Sympy.** Time used: 0.210 (sec). Leaf size: 58

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*(2*x + 3*y(x))**2 + Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{-6C_1x - \sqrt{3}C_1 + 6xe^{4\sqrt{3}x} - \sqrt{3}e^{4\sqrt{3}x}}{9(C_1 - e^{4\sqrt{3}x})}$$

2.2.26 Problem 52

Local contents

Solved using first_order_ode_flip_role	708
✓ Maple	712
✓ Mathematica	713
✓ Sympy	713

Internal problem ID [7880]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 4. Equations of first order and first degree (Variable separable). Supplementary problems. Page 22

Problem number : 52

Date solved : Thursday, November 27, 2025 at 06:27:28 AM

CAS classification : [[_1st_order, '_with_symmetry_[F(x),G(y)]']]

Solved using first_order_ode_flip_role

Time used: 3.185 (sec)

Solve

$$x - 2 \sin(y) + 3 + (2x - 4 \sin(y) - 3) \cos(y) y' = 0$$

reversing the roles of the dependent and independent variables, the ode becomes

$$\frac{d}{dy}x(y) = -\frac{\cos(y)(4 \sin(y) - 2x(y) + 3)}{-3 + 2 \sin(y) - x(y)}$$

Which is now solved for $x(y)$ instead for y Solve This is Abel second kind ODE, it has the form

$$(x(y) + g) \frac{d}{dy}x(y) = f_0(y) + f_1(y)x(y) + f_2(y)x(y)^2 + f_3(y)x(y)^3$$

Comparing the above to given ODE which is

$$\frac{d}{dy}x(y) = -\frac{\cos(y)(4 \sin(y) - 2x(y) + 3)}{-3 + 2 \sin(y) - x(y)} \quad (1)$$

Shows that

$$g = -2 \sin(y) + 3$$

$$f_0 = (4 \sin(y) + 3) \cos(y)$$

$$f_1 = -2 \cos(y)$$

$$f_2 = 0$$

$$f_3 = 0$$

Applying transformation

$$x(y) = \frac{1}{u(y)} - g$$

Results in the new ode which is Abel first kind

$$\frac{d}{dy}u(y) = -9 \cos(y) u(y)^3 + 4 \cos(y) u(y)^2$$

Which is now solved.

Solve The ode

$$\frac{d}{dy}u(y) = -\cos(y) u(y)^2 (9u(y) - 4) \quad (2.80)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dy}u(y) &= -\cos(y) u(y)^2 (9u(y) - 4) \\ &= f(y)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(y) &= -\cos(y) \\ g(u) &= u^2(9u - 4) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(y) dy \\ \int \frac{1}{u^2(9u - 4)} du &= \int -\cos(y) dy \end{aligned}$$

$$\frac{9 \ln(9u(y) - 4)}{16} + \frac{1}{4u(y)} - \frac{9 \ln(u(y))}{16} = -\sin(y) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2(9u - 4) = 0$$

for $u(y)$ gives

$$\begin{aligned} u(y) &= 0 \\ u(y) &= \frac{4}{9} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{9 \ln(9u(y) - 4)}{16} + \frac{1}{4u(y)} - \frac{9 \ln(u(y))}{16} = -\sin(y) + c_1$$

$$u(y) = 0$$

$$u(y) = \frac{4}{9}$$

Substituting $u(y) = \frac{1}{-2 \sin(y) + x(y) + 3}$ in the above solution gives

$$\frac{9 \ln\left(\frac{9}{-2 \sin(y) + x(y) + 3} - 4\right)}{16} - \frac{\sin(y)}{2} + \frac{x(y)}{4} + \frac{3}{4} - \frac{9 \ln\left(\frac{1}{-2 \sin(y) + x(y) + 3}\right)}{16} = -\sin(y) + c_1$$

Now we transform the solution $u(y) = \frac{4}{9}$ to $x(y)$ using $u(y) = \frac{1}{-2 \sin(y) + x(y) + 3}$ which gives

$$x(y) = -\frac{3}{4} + 2 \sin(y)$$

Solving for $x(y)$ gives

$$x(y) = -\frac{3}{4} + 2 \sin(y)$$

$$x(y) = 2 \sin(y) - \frac{3}{4} + \frac{9 \operatorname{LambertW}\left(-\frac{e^{\frac{16c_1}{9} - \frac{16 \sin(y)}{9}} - 1}{9}\right)}{4}$$

Now that the reversed roles ode was solved, we will change back to the original roles. This results in the above solution becoming the following.

$$x = -\frac{3}{4} + 2 \sin(y)$$

$$x = 2 \sin(y) - \frac{3}{4} + \frac{9 \operatorname{LambertW}\left(-\frac{e^{\frac{16c_1}{9} - \frac{16 \sin(y)}{9}} - 1}{9}\right)}{4}$$

Solving for y gives

$$y = \arcsin\left(\frac{x}{2} + \frac{3}{8}\right)$$

$$y = \arcsin\left(\frac{x}{2} + \frac{3}{8} + \frac{9 \operatorname{LambertW}\left(\frac{e^{-\frac{5}{3} + \frac{16c_1}{9} - \frac{8x}{9}}}{9}\right)}{8}\right)$$

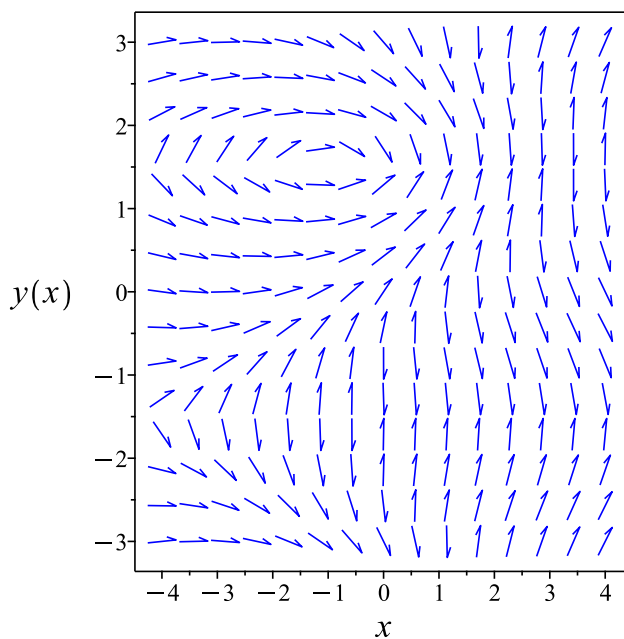


Figure 2.175: Slope field $x - 2 \sin(y) + 3 + (2x - 4 \sin(y) - 3) \cos(y) y' = 0$

Summary of solutions found

$$y = \arcsin\left(\frac{x}{2} + \frac{3}{8}\right)$$

$$y = \arcsin\left(\frac{x}{2} + \frac{3}{8} + \frac{9 \operatorname{LambertW}\left(\frac{e^{-\frac{5}{3} + \frac{16c_1}{9} - \frac{8x}{9}}}{9}\right)}{8}\right)$$

✓ **Maple.** Time used: 0.024 (sec). Leaf size: 22

```
ode:=x-2*sin(y(x))+3+(2*x-4*sin(y(x))-3)*cos(y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \arcsin \left(\frac{9 \operatorname{LambertW} \left(\frac{c_1 e^{-\frac{1}{3} - \frac{8x}{9}}}{9} \right)}{8} + \frac{3}{8} + \frac{x}{2} \right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
Looking for potential symmetries
trying inverse_Riccati
trying an equivalence to an Abel ODE
differential order: 1; trying a linearization to 2nd order
--- trying a change of variables {x -> y(x), y(x) -> x}
differential order: 1; trying a linearization to 2nd order
trying 1st order ODE linearizable_by_differentiation
--- Trying Lie symmetry methods, 1st order ---
-> Computing symmetries using: way = 3
-> Computing symmetries using: way = 4
-> Computing symmetries using: way = 5
[0, (-3+8*sin(y)-4*x)/(-2*x+4*sin(y)+3)/cos(y)]
<- successful computation of symmetries.
1st order, trying the canonical coordinates of the invariance group
<- 1st order, canonical coordinates successful
```

Maple step by step

Let's solve

- $x - 2 \sin(y(x)) + 3 + (2x - 4 \sin(y(x)) - 3) \cos(y(x)) \left(\frac{d}{dx}y(x)\right) = 0$
 - Highest derivative means the order of the ODE is 1
 - $\frac{d}{dx}y(x)$
 - Solve for the highest derivative
- $$\frac{d}{dx}y(x) = \frac{-x + 2 \sin(y(x)) - 3}{(2x - 4 \sin(y(x)) - 3) \cos(y(x))}$$

✓ **Mathematica.** Time used: 1.208 (sec). Leaf size: 795

```
ode=(x-2*Sin[y[x]]+3)+(2*x-4*Sin[y[x]]-3)*Cos[y[x]]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Solution too large to show

✓ **Sympy.** Time used: 86.359 (sec). Leaf size: 364

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x + (2*x - 4*sin(y(x)) - 3)*cos(y(x))*Derivative(y(x), x) - 2*sin(y(x))
        + 3, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = \operatorname{asin} \left(\frac{x}{2} + \frac{9W \left(\frac{\sqrt[9]{C_1} e^{-8x}}{9e^{\frac{1}{3}}} \right)}{8} + \frac{3}{8} \right), y(x) = \operatorname{asin} \left(\frac{x}{2} + \frac{9W \left(\frac{\sqrt[9]{C_1} e^{-8x} e^{-\frac{1}{3} - \frac{2i\pi}{9}}}{9} \right)}{8} + \frac{3}{8} \right), y(x) = \operatorname{asin} \right.$$

2.3 Chapter 5. Equations of first order and first degree (Exact equations). Supplemetary problems.

Page 33

Local contents

2.3.1	Problem 23 (a)	716
2.3.2	Problem 23 (d)	722
2.3.3	Problem 23 (e)	728
2.3.4	Problem 23 (h)	734
2.3.5	Problem 23 (i)	740
2.3.6	Problem 23 (j)	746
2.3.7	Problem 23 (k)	753
2.3.8	Problem 23 (m)	759
2.3.9	Problem 23 (o)	765
2.3.10	Problem 23 (p)	771
2.3.11	Problem 24 (p)	777
2.3.12	Problem 24 (c)	805
2.3.13	Problem 24 (d)	842
2.3.14	Problem 24 (g)	879
2.3.15	Problem 24 (L)	885
2.3.16	Problem 25 (a)	910
2.3.17	Problem 25 (b)	916
2.3.18	Problem 25 (c)	923
2.3.19	Problem 25 (d)	929
2.3.20	Problem 25 (e)	936
2.3.21	Problem 25 (f)	943
2.3.22	Problem 25 (g)	949
2.3.23	Problem 25 (h)	956
2.3.24	Problem 25 (L)	963
2.3.25	Problem 25 (j)	970
2.3.26	Problem 25 (k)	977
2.3.27	Problem 26 (a)	984
2.3.28	Problem 26 (b)	991
2.3.29	Problem 26 (c)	1008
2.3.30	Problem 26 (d)	1015
2.3.31	Problem 26 (e)	1022
2.3.32	Problem 26 (f)	1029
2.3.33	Problem 26 (g)	1035
2.3.34	Problem 26 (h)	1042

2.3.35 Problem 26 (i)	1048
2.3.36 Problem 27	1055

2.3.1 Problem 23 (a)

Local contents

Solved using first_order_ode_exact	716
✓ Maple	719
✓ Mathematica	720
✓ Sympy	721

Internal problem ID [7881]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (a)

Date solved : Thursday, November 27, 2025 at 06:27:41 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.103 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x) dy &= (-x^2 + y) dx \\ (x^2 - y) dx + (-x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^2 - y \\ N(x, y) &= -x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x^2 - y) \\ &= -1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x) \\ &= -1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial y} dy &= \int N dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int -x dy \\ \phi &= -yx + f(x) \end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -y + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = x^2 - y$. Therefore equation (4) becomes

$$x^2 - y = -y + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = x^2$$

Integrating the above w.r.t x gives

$$\begin{aligned} \int f'(x) dx &= \int (x^2) dx \\ f(x) &= \frac{x^3}{3} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

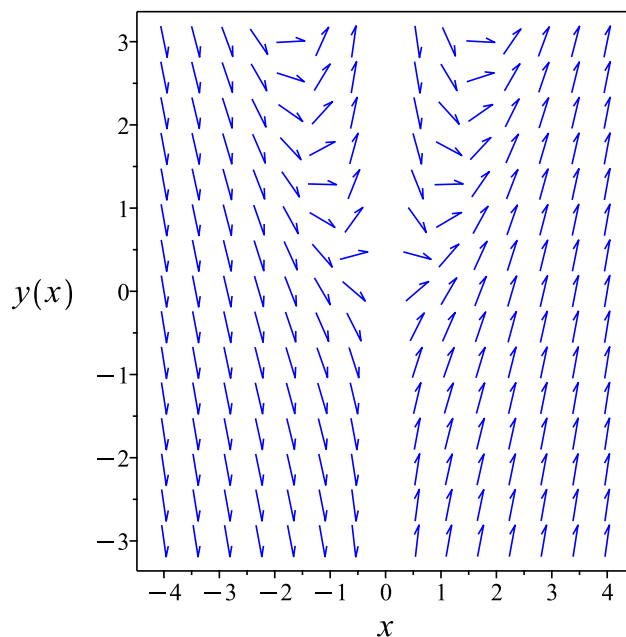
$$\phi = -yx + \frac{1}{3}x^3 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -yx + \frac{1}{3}x^3$$

Solving for y gives

$$y = -\frac{-x^3 + 3c_1}{3x}$$

Figure 2.176: Slope field $x^2 - y - y'x = 0$ Summary of solutions found

$$y = -\frac{-x^3 + 3c_1}{3x}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 16

```
ode:=x^2-y(x)-diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^3 + 3c_1}{3x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$x^2 - y(x) - x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Isolate the derivative
 $\frac{d}{dx}y(x) = -\frac{y(x)}{x} + x$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx}y(x) + \frac{y(x)}{x} = x$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{y(x)}{x} \right) = \mu(x) x$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x) \mu(x))$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$
- Isolate $\frac{d}{dx}\mu(x)$
 $\frac{d}{dx}\mu(x) = \frac{\mu(x)}{x}$
- Solve to find the integrating factor
 $\mu(x) = x$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}(y(x) \mu(x)) \right) dx = \int \mu(x) x dx + C1$
- Evaluate the integral on the lhs
 $y(x) \mu(x) = \int \mu(x) x dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int \mu(x) x dx + C1}{\mu(x)}$
- Substitute $\mu(x) = x$
 $y(x) = \frac{\int x^2 dx + C1}{x}$
- Evaluate the integrals on the rhs
 $y(x) = \frac{\frac{x^3}{3} + C1}{x}$
- Simplify
 $y(x) = \frac{x^3 + 3C1}{3x}$

✓ **Mathematica.** Time used: 0.019 (sec). Leaf size: 19

```
ode=(x^2-y[x])-x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^2}{3} + \frac{c_1}{x}$$

✓ Sympy. Time used: 0.109 (sec). Leaf size: 10

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(x**2 - x*Derivative(y(x), x) - y(x), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1 + \frac{x^3}{3}}{x}$$

2.3.2 Problem 23 (d)

Local contents

Solved using first_order_ode_exact	722
✓ Maple	725
✓ Mathematica	727
✓ Sympy	727

Internal problem ID [7882]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (d)

Date solved : Thursday, November 27, 2025 at 06:27:46 AM

CAS classification :

[[_homogeneous, 'class A'], _exact, _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.077 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2yx) dy &= (-x^2 - y^2) dx \\ (x^2 + y^2) dx + (2yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^2 + y^2 \\ N(x, y) &= 2yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(2yx) \\ &= 2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x^2 + y^2 dx \\ \phi &= \frac{1}{3}x^3 + xy^2 + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 2yx + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 2yx$. Therefore equation (4) becomes

$$2yx = 2yx + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{1}{3}x^3 + xy^2 + c_1$$

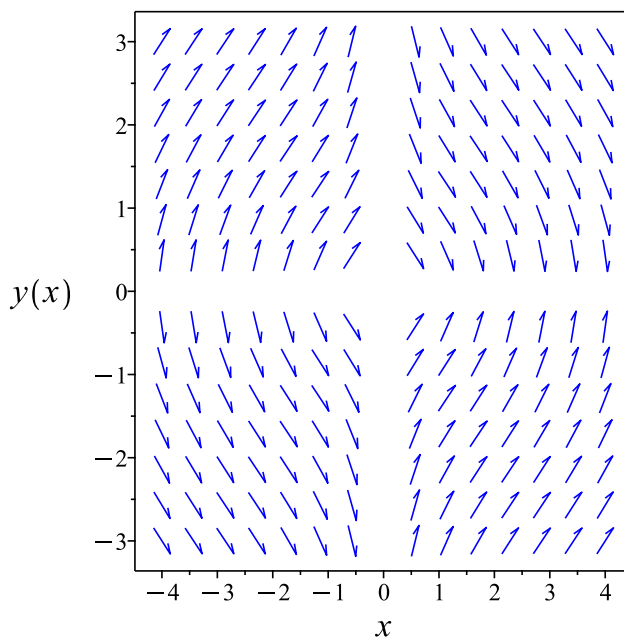
But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{1}{3}x^3 + xy^2$$

Solving for y gives

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

Figure 2.177: Slope field $x^2 + y^2 + 2xyy' = 0$ Summary of solutions found

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

✓ Maple. Time used: 0.003 (sec). Leaf size: 47

```
ode:=x^2+y(x)^2+2*x*y(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{3} \sqrt{-x(x^3 - 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{-x(x^3 - 3c_1)}}{3x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
```

<- Bernoulli successful

Maple step by step

Let's solve

$$x^2 + y(x)^2 + 2xy(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - Compute derivative of lhs
 - Evaluate derivatives
 - Condition met, ODE is exact
- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x} G(x, y), N(x, y) = \frac{\partial}{\partial y} G(x, y) \right]$$
- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int (x^2 + y^2) dx + _F1(y)$$
- Evaluate integral

$$G(x, y) = \frac{x^3}{3} + y^2 x + _F1(y)$$
- Take derivative of $G(x, y)$ with respect to y

$$N(x, y) = \frac{\partial}{\partial y} G(x, y)$$
- Compute derivative

$$2yx = 2yx + \frac{d}{dy} _F1(y)$$
- Isolate for $\frac{d}{dy} _F1(y)$

$$\frac{d}{dy} _F1(y) = 0$$
- Solve for $_F1(y)$

$$_F1(y) = 0$$
- Substitute $_F1(y)$ into equation for $G(x, y)$

$$G(x, y) = \frac{1}{3} x^3 + y^2 x$$
- Substitute $G(x, y)$ into the solution of the ODE

$$\frac{1}{3} x^3 + y^2 x = C1$$
- Solve for $y(x)$

$$\left\{ y(x) = -\frac{\sqrt{3} \sqrt{x(-x^3+3C1)}}{3x}, y(x) = \frac{\sqrt{3} \sqrt{x(-x^3+3C1)}}{3x} \right\}$$

✓ **Mathematica.** Time used: 0.145 (sec). Leaf size: 60

```
ode=(x^2+y[x]^2)+2*x*y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{-x^3+3c_1}}{\sqrt{3}\sqrt{x}}$$

$$y(x) \rightarrow \frac{\sqrt{-x^3+3c_1}}{\sqrt{3}\sqrt{x}}$$

✓ **Sympy.** Time used: 0.314 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2 + 2*x*y(x)*Derivative(y(x), x) + y(x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\frac{\sqrt{3}\sqrt{\frac{C_1}{x} - x^2}}{3}, y(x) = \frac{\sqrt{3}\sqrt{\frac{C_1}{x} - x^2}}{3} \right]$$

2.3.3 Problem 23 (e)

Local contents

Solved using first_order_ode_exact	728
✓ Maple	731
✓ Mathematica	732
✓ Sympy	733

Internal problem ID [7883]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (e)

Date solved : Thursday, November 27, 2025 at 06:27:51 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.076 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (\sin(x)) dy &= (-x - \cos(x)y) dx \\ (x + \cos(x)y) dx + (\sin(x)) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x + \cos(x)y \\ N(x, y) &= \sin(x) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x + \cos(x)y) \\ &= \cos(x) \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(\sin(x)) \\ &= \cos(x) \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial y} dy &= \int N dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \sin(x) dy \\ \phi &= \sin(x)y + f(x) \end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = \cos(x) y + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = x + \cos(x) y$. Therefore equation (4) becomes

$$x + \cos(x) y = \cos(x) y + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = x$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int (x) dx$$

$$f(x) = \frac{x^2}{2} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

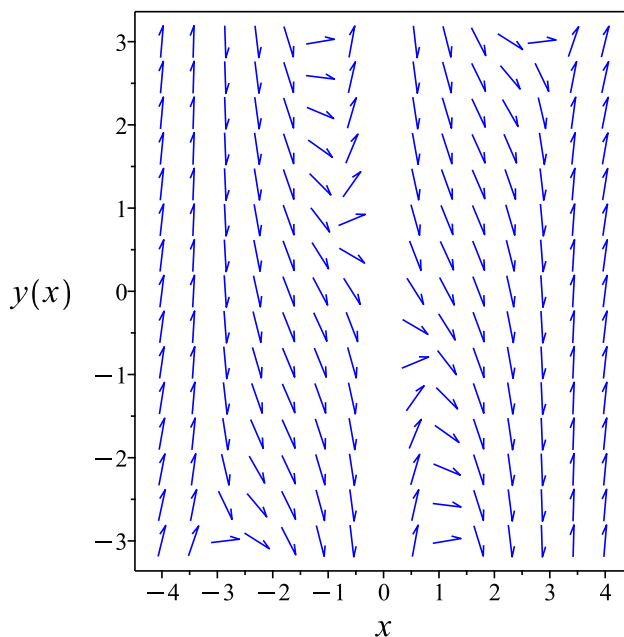
$$\phi = \sin(x) y + \frac{x^2}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \sin(x) y + \frac{x^2}{2}$$

Solving for y gives

$$y = \frac{-x^2 + 2c_1}{2 \sin(x)}$$

Figure 2.178: Slope field $x + \cos(x)y + \sin(x)y' = 0$ Summary of solutions found

$$y = \frac{-x^2 + 2c_1}{2\sin(x)}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 15

```
ode:=x+y(x)*cos(x)+sin(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{(x^2 - 2c_1)\csc(x)}{2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$x + y(x)\cos(x) + \sin(x)\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Isolate the derivative
 $\frac{d}{dx}y(x) = -\frac{\cos(x)y(x)}{\sin(x)} - \frac{x}{\sin(x)}$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx}y(x) + \frac{\cos(x)y(x)}{\sin(x)} = -\frac{x}{\sin(x)}$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{\cos(x)y(x)}{\sin(x)} \right) = -\frac{\mu(x)x}{\sin(x)}$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{\cos(x)y(x)}{\sin(x)} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$
- Isolate $\frac{d}{dx}\mu(x)$
 $\frac{d}{dx}\mu(x) = \frac{\mu(x)\cos(x)}{\sin(x)}$
- Solve to find the integrating factor
 $\mu(x) = \sin(x)$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int -\frac{\mu(x)x}{\sin(x)} dx + C1$
- Evaluate the integral on the lhs
 $y(x)\mu(x) = \int -\frac{\mu(x)x}{\sin(x)} dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int -\frac{\mu(x)x}{\sin(x)} dx + C1}{\mu(x)}$
- Substitute $\mu(x) = \sin(x)$
 $y(x) = \frac{\int -x dx + C1}{\sin(x)}$
- Evaluate the integrals on the rhs
 $y(x) = \frac{-\frac{x^2}{2} + C1}{\sin(x)}$
- Simplify
 $y(x) = -\frac{(x^2 - 2C1) \csc(x)}{2}$

✓ **Mathematica.** Time used: 0.021 (sec). Leaf size: 19

```
ode=(x+y[x]*Cos[x])+Sin[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{1}{2}(x^2 - 2c_1) \csc(x)$$

✓ Sympy. Time used: 0.176 (sec). Leaf size: 12

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(x + y(x)*cos(x) + sin(x)*Derivative(y(x), x), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1 - \frac{x^2}{2}}{\sin(x)}$$

2.3.4 Problem 23 (h)

Local contents

Solved using first_order_ode_exact	734
✓ Maple	737
✓ Mathematica	739
✓ Sympy	739

Internal problem ID [7884]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (h)

Date solved : Thursday, November 27, 2025 at 06:27:56 AM

CAS classification :

[[_homogeneous, 'class C'], _exact, _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_exact

Time used: 0.083 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (3x + 4y + 5) dy &= (-2x - 3y - 4) dx \\ (2x + 3y + 4) dx + (3x + 4y + 5) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 2x + 3y + 4 \\ N(x, y) &= 3x + 4y + 5 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(2x + 3y + 4) \\ &= 3 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(3x + 4y + 5) \\ &= 3 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int 2x + 3y + 4 dx \\ \phi &= x(x + 3y + 4) + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 3x + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 3x + 4y + 5$. Therefore equation (4) becomes

$$3x + 4y + 5 = 3x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 4y + 5$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (4y + 5) dy \\ f(y) &= 2y^2 + 5y + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x(x + 3y + 4) + 2y^2 + 5y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

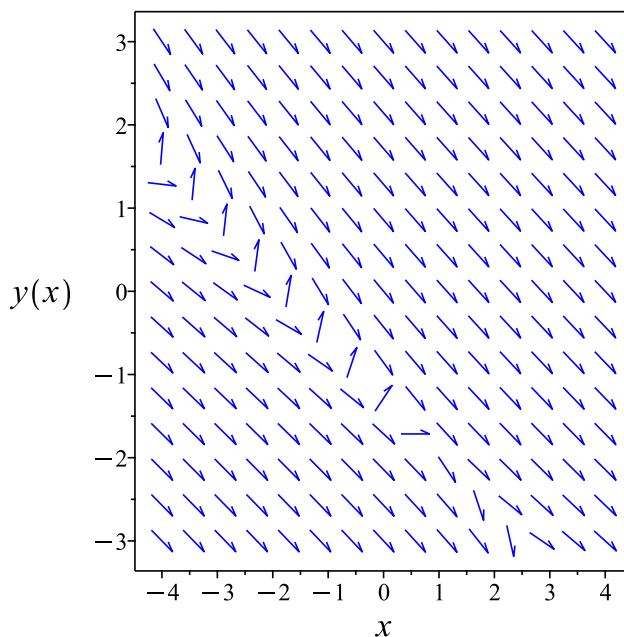
$$c_1 = x(x + 3y + 4) + 2y^2 + 5y$$

Simplifying the above gives

$$2y^2 + (3x + 5)y + x^2 + 4x = c_1$$

Solving for y gives

$$\begin{aligned} y &= -\frac{3x}{4} - \frac{5}{4} - \frac{\sqrt{x^2 + 8c_1 - 2x + 25}}{4} \\ y &= -\frac{3x}{4} - \frac{5}{4} + \frac{\sqrt{x^2 + 8c_1 - 2x + 25}}{4} \end{aligned}$$

Figure 2.179: Slope field $2x + 3y + 4 + (3x + 4y + 5)y' = 0$ Summary of solutions found

$$y = -\frac{3x}{4} - \frac{5}{4} - \frac{\sqrt{x^2 + 8c_1 - 2x + 25}}{4}$$

$$y = -\frac{3x}{4} - \frac{5}{4} + \frac{\sqrt{x^2 + 8c_1 - 2x + 25}}{4}$$

✓ Maple. Time used: 0.065 (sec). Leaf size: 32

```
ode:=2*x+3*y(x)+4+(3*x+4*y(x)+5)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{-\sqrt{(-1+x)^2 c_1^2 + 8} + (-3x - 5) c_1}{4c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
```

```

trying inverse linear
trying homogeneous types:
trying homogeneous C
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
<- homogeneous successful

```

Maple step by step

Let's solve

$$2x + 3y(x) + 4 + (3x + 4y(x) + 5) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx} y(x)$
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 $\frac{d}{dx} G(x, y(x)) = 0$
 - Compute derivative of lhs
 $\frac{\partial}{\partial x} G(x, y) + \left(\frac{\partial}{\partial y} G(x, y) \right) \left(\frac{d}{dx} y(x) \right) = 0$
 - Evaluate derivatives
 $3 = 3$
 - Condition met, ODE is exact
- Exact ODE implies solution will be of this form
 $\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x} G(x, y), N(x, y) = \frac{\partial}{\partial y} G(x, y) \right]$
- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x
 $G(x, y) = \int (2x + 3y + 4) dx + _F1(y)$
- Evaluate integral
 $G(x, y) = x^2 + 3yx + 4x + _F1(y)$
- Take derivative of $G(x, y)$ with respect to y
 $N(x, y) = \frac{\partial}{\partial y} G(x, y)$
- Compute derivative
 $3x + 4y + 5 = 3x + \frac{d}{dy} _F1(y)$
- Isolate for $\frac{d}{dy} _F1(y)$
 $\frac{d}{dy} _F1(y) = 4y + 5$
- Solve for $_F1(y)$
 $_F1(y) = 2y^2 + 5y$
- Substitute $_F1(y)$ into equation for $G(x, y)$
 $G(x, y) = x^2 + 3yx + 2y^2 + 4x + 5y$
- Substitute $G(x, y)$ into the solution of the ODE
 $x^2 + 3yx + 2y^2 + 4x + 5y = C1$

- Solve for $y(x)$

$$\left\{ y(x) = -\frac{3x}{4} - \frac{5}{4} - \frac{\sqrt{x^2+8C_1-2x+25}}{4}, y(x) = -\frac{3x}{4} - \frac{5}{4} + \frac{\sqrt{x^2+8C_1-2x+25}}{4} \right\}$$

✓ **Mathematica.** Time used: 0.079 (sec). Leaf size: 61

```
ode=(2*x+3*y[x]+4)+(3*x+4*y[x]+5)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{4} \left(-\sqrt{x^2 - 2x + 25 + 16c_1} - 3x - 5 \right)$$

$$y(x) \rightarrow \frac{1}{4} \left(\sqrt{x^2 - 2x + 25 + 16c_1} - 3x - 5 \right)$$

✓ **Sympy.** Time used: 1.463 (sec). Leaf size: 49

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x + (3*x + 4*y(x) + 5)*Derivative(y(x), x) + 3*y(x) + 4, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\frac{3x}{4} - \frac{\sqrt{C_1 + x^2 - 2x}}{4} - \frac{5}{4}, y(x) = -\frac{3x}{4} + \frac{\sqrt{C_1 + x^2 - 2x}}{4} - \frac{5}{4} \right]$$

2.3.5 Problem 23 (i)

Local contents

Solved using first_order_ode_exact	740
✓ Maple	743
✓ Mathematica	745
✓ Sympy	745

Internal problem ID [7885]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (i)

Date solved : Thursday, November 27, 2025 at 06:28:01 AM

CAS classification : [[_homogeneous, 'class G'], _exact, _rational]

Solved using first_order_ode_exact

Time used: 0.114 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(3x^4y^2 - \frac{1}{y}\right) dy &= \left(-4x^3y^3 - \frac{1}{x}\right) dx \\ \left(4x^3y^3 + \frac{1}{x}\right) dx + \left(3x^4y^2 - \frac{1}{y}\right) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 4x^3y^3 + \frac{1}{x} \\ N(x, y) &= 3x^4y^2 - \frac{1}{y} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(4x^3y^3 + \frac{1}{x}\right) \\ &= 12x^3y^2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(3x^4y^2 - \frac{1}{y}\right) \\ &= 12x^3y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int 4x^3y^3 + \frac{1}{x} dx \\ \phi &= x^4y^3 + \ln(x) + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 3x^4y^2 + f'(y)\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 3x^4y^2 - \frac{1}{y}$. Therefore equation (4) becomes

$$3x^4y^2 - \frac{1}{y} = 3x^4y^2 + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{1}{y}$$

Integrating the above w.r.t y gives

$$\begin{aligned}\int f'(y) dy &= \int \left(-\frac{1}{y}\right) dy \\ f(y) &= -\ln(y) + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x^4y^3 + \ln(x) - \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x^4y^3 + \ln(x) - \ln(y)$$

Solving for y gives

$$y = \frac{x}{\left(-\frac{3x^7}{\text{LambertW}(-3x^7 e^{-3c_1})}\right)^{1/3}}$$

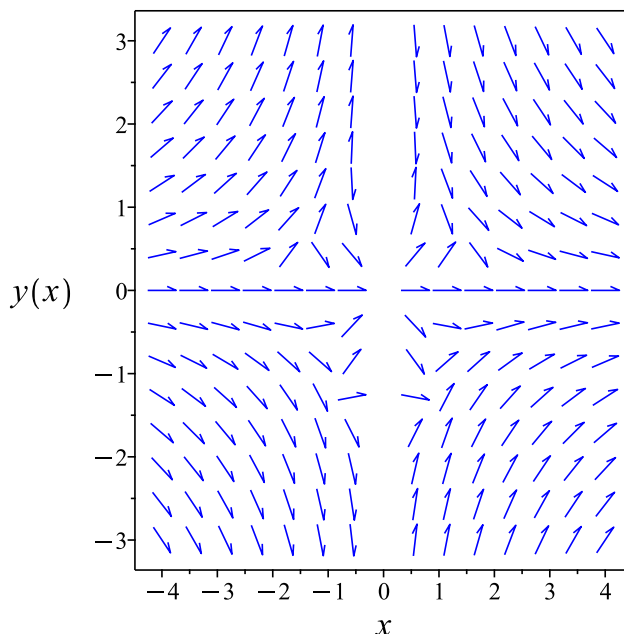


Figure 2.180: Slope field $4x^3y^3 + \frac{1}{x} + \left(3y^2x^4 - \frac{1}{y}\right)y' = 0$

Summary of solutions found

$$y = \frac{x}{\left(-\frac{3x^7}{\text{LambertW}(-3x^7 e^{-3c_1})}\right)^{1/3}}$$

✓ **Maple.** Time used: 0.011 (sec). Leaf size: 25

```
ode:=4*x^3*y(x)^3+1/x+(3*x^4*y(x)^2-1/y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{3^{2/3}}{3 \left(-\frac{x^4}{\text{LambertW}(-3c_1 x^7)}\right)^{1/3}}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
```

```

trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful

```

Maple step by step

Let's solve

$$4x^3y(x)^3 + \frac{1}{x} + \left(3x^4y(x)^2 - \frac{1}{y(x)}\right) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 $\frac{d}{dx}G(x, y(x)) = 0$
 - Compute derivative of lhs
 $\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y)\right) \left(\frac{d}{dx}y(x)\right) = 0$
 - Evaluate derivatives
 $12x^3y^2 = 12x^3y^2$
 - Condition met, ODE is exact
- Exact ODE implies solution will be of this form
 $\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y)\right]$
- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x
 $G(x, y) = \int \left(4x^3y^3 + \frac{1}{x}\right) dx + _F1(y)$
- Evaluate integral
 $G(x, y) = x^4y^3 + \ln(x) + _F1(y)$
- Take derivative of $G(x, y)$ with respect to y
 $N(x, y) = \frac{\partial}{\partial y}G(x, y)$
- Compute derivative
 $3x^4y^2 - \frac{1}{y} = 3x^4y^2 + \frac{d}{dy}_F1(y)$
- Isolate for $\frac{d}{dy}_F1(y)$
 $\frac{d}{dy}_F1(y) = -\frac{1}{y}$
- Solve for $_F1(y)$
 $_F1(y) = -\ln(y)$
- Substitute $_F1(y)$ into equation for $G(x, y)$
 $G(x, y) = x^4y^3 + \ln(x) - \ln(y)$
- Substitute $G(x, y)$ into the solution of the ODE

- $x^4 y^3 + \ln(x) - \ln(y) = C_1$
 Solve for $y(x)$

$$y(x) = \frac{\frac{x}{e^{\frac{\text{LambertW}\left(-3x^7 e^{-3C_1}\right)} + C_1}}}{3}$$

✓ **Mathematica.** Time used: 2.126 (sec). Leaf size: 108

```
ode=(4*x^3*y[x]^3+1/x)+(3*x^4*y[x]^2-1/y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\sqrt[3]{-\frac{1}{3}} \sqrt[3]{W(-3e^{-3c_1}x^7)}}{x^{4/3}}$$

$$y(x) \rightarrow -\frac{\sqrt[3]{W(-3e^{-3c_1}x^7)}}{\sqrt[3]{3}x^{4/3}}$$

$$y(x) \rightarrow -\frac{(-1)^{2/3} \sqrt[3]{W(-3e^{-3c_1}x^7)}}{\sqrt[3]{3}x^{4/3}}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.464 (sec). Leaf size: 29

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x**3*y(x)**3 + (3*x**4*y(x)**2 - 1/y(x))*Derivative(y(x), x) + 1/x, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$-\frac{3x^4 y^3(x)}{7} - \log(x) + \frac{3 \log\left(x^{\frac{4}{3}} y(x)\right)}{7} = C_1$$

2.3.6 Problem 23 (j)

Local contents

Solved using first_order_ode_exact	746
✓ Maple	750
✓ Mathematica	752
✗ Sympy	752

Internal problem ID [7886]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (j)

Date solved : Thursday, November 27, 2025 at 06:28:07 AM

CAS classification :

[[_homogeneous, 'class A'], _exact, _rational, _dAlembert]

Solved using first_order_ode_exact

Time used: 0.115 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(u, v) du + N(u, v) dv = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (u^2 + v^2) dv &= (-2u^2 - 2uv) du \\ (2u^2 + 2uv) du + (u^2 + v^2) dv &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(u, v) &= 2u^2 + 2uv \\ N(u, v) &= u^2 + v^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial v} = \frac{\partial N}{\partial u}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial v} &= \frac{\partial}{\partial v} (2u^2 + 2uv) \\ &= 2u \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial u} &= \frac{\partial}{\partial u} (u^2 + v^2) \\ &= 2u \end{aligned}$$

Since $\frac{\partial M}{\partial v} = \frac{\partial N}{\partial u}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(u, v)$

$$\frac{\partial \phi}{\partial u} = M \quad (1)$$

$$\frac{\partial \phi}{\partial v} = N \quad (2)$$

Integrating (1) w.r.t. u gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial u} du &= \int M du \\ \int \frac{\partial \phi}{\partial u} du &= \int 2u^2 + 2uv du \\ \phi &= \frac{u^2(2u + 3v)}{3} + f(v) \end{aligned} \quad (3)$$

Where $f(v)$ is used for the constant of integration since ϕ is a function of both u and v . Taking derivative of equation (3) w.r.t v gives

$$\frac{\partial \phi}{\partial v} = u^2 + f'(v) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial v} = u^2 + v^2$. Therefore equation (4) becomes

$$u^2 + v^2 = u^2 + f'(v) \quad (5)$$

Solving equation (5) for $f'(v)$ gives

$$f'(v) = v^2$$

Integrating the above w.r.t v gives

$$\begin{aligned} \int f'(v) dv &= \int (v^2) dv \\ f(v) &= \frac{v^3}{3} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(v)$ into equation (3) gives ϕ

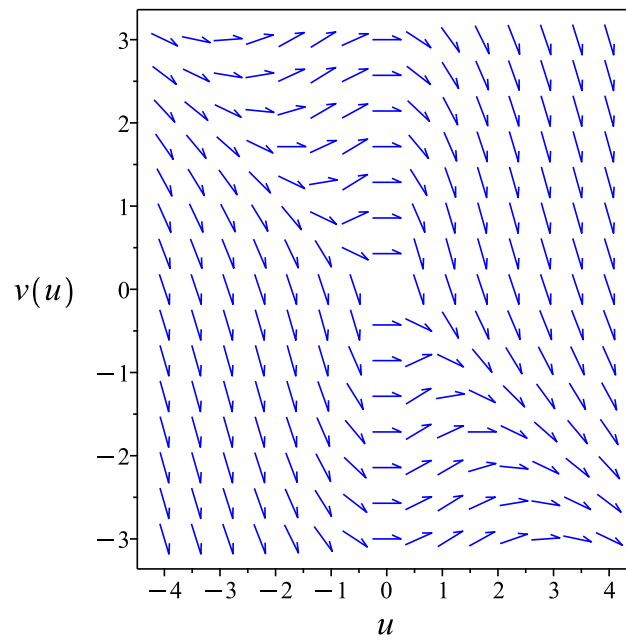
$$\phi = \frac{u^2(2u + 3v)}{3} + \frac{v^3}{3} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{u^2(2u + 3v)}{3} + \frac{v^3}{3}$$

Simplifying the above gives

$$u^2v + \frac{2u^3}{3} + \frac{v^3}{3} = c_1$$

Figure 2.181: Slope field $2u^2 + 2uv + (u^2 + v^2)v' = 0$ Summary of solutions found

$$u^2v + \frac{2u^3}{3} + \frac{v^3}{3} = c_1$$

✓ **Maple.** Time used: 0.015 (sec). Leaf size: 321

```
ode:=2*u^2+2*u*v(u)+(u^2+v(u)^2)*diff(v(u),u) = 0;
dsolve(ode,v(u), singsol=all);
```

$$v = -\frac{2\left(u^2c_1 - \frac{\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{2/3}}{4}\right)}{\sqrt{c_1}\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{1/3}}$$

$$v = -\frac{(1+i\sqrt{3})\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{1/3}}{4\sqrt{c_1}}$$

$$- \frac{(i\sqrt{3}-1)\sqrt{c_1}u^2}{\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{1/3}}$$

$$v = \frac{4i\sqrt{3}c_1u^2 + i\sqrt{3}\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{2/3} + 4u^2c_1 - \left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{1/3}}{4\left(4-8u^3c_1^{3/2}+4\sqrt{8u^6c_1^3-4u^3c_1^{3/2}+1}\right)^{1/3}\sqrt{c_1}}$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
```

Maple step by step

Let's solve

$$2u^2 + 2uv(u) + (u^2 + v(u)^2) \left(\frac{d}{du}v(u)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{du}v(u)$

- Check if ODE is exact
- ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{du}G(u, v(u)) = 0$
 - Compute derivative of lhs
 - $\frac{\partial}{\partial u}G(u, v) + \left(\frac{\partial}{\partial v}G(u, v)\right)\left(\frac{d}{du}v(u)\right) = 0$
 - Evaluate derivatives

$$2u = 2u$$

- Condition met, ODE is exact
- Exact ODE implies solution will be of this form
- $[G(u, v) = C1, M(u, v) = \frac{\partial}{\partial u}G(u, v), N(u, v) = \frac{\partial}{\partial v}G(u, v)]$
- Solve for $G(u, v)$ by integrating $M(u, v)$ with respect to u
- $G(u, v) = \int (2u^2 + 2uv) du + _F1(v)$
- Evaluate integral
- $G(u, v) = \frac{2u^3}{3} + u^2v + _F1(v)$
- Take derivative of $G(u, v)$ with respect to v
- $N(u, v) = \frac{\partial}{\partial v}G(u, v)$
- Compute derivative
- $u^2 + v^2 = u^2 + \frac{d}{dv}_F1(v)$
- Isolate for $\frac{d}{dv}_F1(v)$
- $\frac{d}{dv}_F1(v) = v^2$
- Solve for $_F1(v)$
- $_F1(v) = \frac{v^3}{3}$
- Substitute $_F1(v)$ into equation for $G(u, v)$
- $G(u, v) = \frac{2}{3}u^3 + u^2v + \frac{1}{3}v^3$
- Substitute $G(u, v)$ into the solution of the ODE
- $\frac{2}{3}u^3 + u^2v + \frac{1}{3}v^3 = C1$
- Solve for $v(u)$

$$\left\{ v(u) = \frac{\left(-8u^3+12C1+4\sqrt{8u^6-12C1u^3+9C1^2}\right)^{1/3}}{2} - \frac{2u^2}{\left(-8u^3+12C1+4\sqrt{8u^6-12C1u^3+9C1^2}\right)^{1/3}}, v(u) = -\frac{\left(-8u^3+12C1+4\sqrt{8u^6-12C1u^3+9C1^2}\right)^{1/3}}{2} + \frac{2u^2}{\left(-8u^3+12C1+4\sqrt{8u^6-12C1u^3+9C1^2}\right)^{1/3}} \right\}$$

✓ **Mathematica.** Time used: 25.358 (sec). Leaf size: 622

```
ode=2*(u^2+u*v[u])+(u^2+v[u]^2)*D[v[u],u]==0;
ic={};
DSolve[{ode,ic},v[u],u,IncludeSingularSolutions->True]
```

$$v(u) \rightarrow \frac{\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}}{\sqrt[3]{2}} - \frac{\sqrt[3]{2}u^2}{\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}}$$

$$v(u) \rightarrow \frac{(1+i\sqrt{3})u^2}{2^{2/3}\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}} - \frac{(1-i\sqrt{3})\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}}{2\sqrt[3]{2}}$$

$$v(u) \rightarrow \frac{(1-i\sqrt{3})u^2}{2^{2/3}\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}} - \frac{(1+i\sqrt{3})\sqrt[3]{-2u^3 + \sqrt{8u^6 - 4e^{3c_1}u^3 + e^{6c_1}} + e^{3c_1}}}{2\sqrt[3]{2}}$$

$$v(u) \rightarrow \frac{\left(\sqrt{2}\sqrt{u^6 - u^3}\right)^{2/3} - u^2}{\sqrt[3]{\sqrt{2}\sqrt{u^6 - u^3}}}$$

$$v(u) \rightarrow \frac{i\sqrt{3}u^2 + u^2 + i\sqrt{3}\left(\sqrt{2}\sqrt{u^6 - u^3}\right)^{2/3} - \left(\sqrt{2}\sqrt{u^6 - u^3}\right)^{2/3}}{2\sqrt[3]{\sqrt{2}\sqrt{u^6 - u^3}}}$$

$$v(u) \rightarrow \frac{2\sqrt[3]{2}\left(\sqrt{2}\sqrt{u^6 - u^3}\right)^{2/3}\text{Root}\left[\#1^3 - 32\&, 2\right] + 2^{2/3}u^2\text{Root}\left[\#1^3 + 128\&, 2\right]}{8\sqrt[3]{\sqrt{2}\sqrt{u^6 - u^3}}}$$

✗ **Sympy**

```
from sympy import *
u = symbols("u")
v = Function("v")
ode = Eq(2*u**2 + 2*u*v(u) + (u**2 + v(u)**2)*Derivative(v(u), u), 0)
ics = {}
dsolve(ode, func=v(u), ics=ics)
```

Timed Out

2.3.7 Problem 23 (k)

Local contents

Solved using first_order_ode_exact	753
✓ Maple	756
✓ Mathematica	758
✗ Sympy	758

Internal problem ID [7887]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (k)

Date solved : Thursday, November 27, 2025 at 06:28:12 AM

CAS classification : [_exact]

Solved using first_order_ode_exact

Time used: 0.109 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} & (\sqrt{x^2 + y^2} y - x) dy = (-x\sqrt{x^2 + y^2} + y) dx \\ & (x\sqrt{x^2 + y^2} - y) dx + (\sqrt{x^2 + y^2} y - x) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x\sqrt{x^2 + y^2} - y \\ N(x, y) &= \sqrt{x^2 + y^2} y - x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x\sqrt{x^2 + y^2} - y) \\ &= \frac{xy}{\sqrt{x^2 + y^2}} - 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (\sqrt{x^2 + y^2} y - x) \\ &= \frac{xy}{\sqrt{x^2 + y^2}} - 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x\sqrt{x^2 + y^2} - y dx \\ \phi &= \frac{(x^2 + y^2)^{3/2}}{3} - yx + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \sqrt{x^2 + y^2} y - x + f'(y)\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \sqrt{x^2 + y^2} y - x$. Therefore equation (4) becomes

$$\sqrt{x^2 + y^2} y - x = \sqrt{x^2 + y^2} y - x + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{(x^2 + y^2)^{3/2}}{3} - yx + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{(x^2 + y^2)^{3/2}}{3} - yx$$

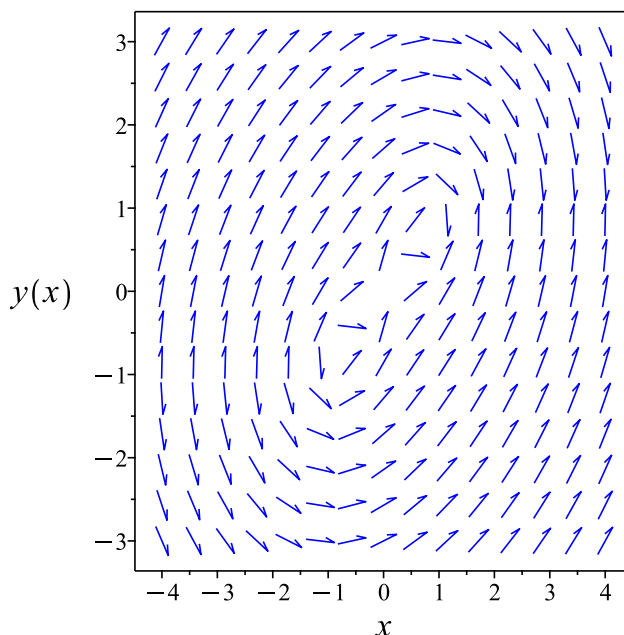


Figure 2.182: Slope field $x\sqrt{x^2+y^2}-y+(y\sqrt{x^2+y^2}-x)y' = 0$

Summary of solutions found

$$\frac{(x^2 + y^2)^{3/2}}{3} - yx = c_1$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 22

```
ode:=x*(x^2+y(x)^2)^(1/2)-y(x)+(y(x)*(x^2+y(x)^2)^(1/2)-x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$\frac{(x^2 + y^2)^{3/2}}{3} - xy + c_1 = 0$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying homogeneous types:
differential order: 1; looking for linear symmetries
trying exact
<- exact successful
```

Maple step by step

Let's solve

$$x\sqrt{x^2 + y(x)^2} - y(x) + \left(y(x)\sqrt{x^2 + y(x)^2} - x\right) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{dx}G(x, y(x)) = 0$
 - Compute derivative of lhs
 - $\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y)\right) \left(\frac{d}{dx}y(x)\right) = 0$
 - Evaluate derivatives
 - $\frac{xy}{\sqrt{x^2+y^2}} - 1 = \frac{xy}{\sqrt{x^2+y^2}} - 1$
 - Condition met, ODE is exact
 - Exact ODE implies solution will be of this form
 - $\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y)\right]$
 - Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x
 - $G(x, y) = \int (x\sqrt{x^2 + y^2} - y) dx + _F1(y)$
 - Evaluate integral
 - $G(x, y) = \frac{(x^2+y^2)^{3/2}}{3} - xy + _F1(y)$
 - Take derivative of $G(x, y)$ with respect to y
 - $N(x, y) = \frac{\partial}{\partial y}G(x, y)$
 - Compute derivative
 - $y\sqrt{x^2 + y^2} - x = y\sqrt{x^2 + y^2} - x + \frac{d}{dy}_F1(y)$
 - Isolate for $\frac{d}{dy}_F1(y)$
 - $\frac{d}{dy}_F1(y) = 0$
 - Solve for $_F1(y)$
 - $_F1(y) = 0$
 - Substitute $_F1(y)$ into equation for $G(x, y)$
 - $G(x, y) = \frac{(x^2+y^2)^{3/2}}{3} - xy$
 - Substitute $G(x, y)$ into the solution of the ODE
 - $\frac{(x^2+y^2)^{3/2}}{3} - xy = C1$
 - Solve for $y(x)$
 - $y(x) = \frac{\text{RootOf}\left(_Z^6 - 6C1_Z^3 - 9x^2_Z^2 + 9x^4 + 9C1^2\right)^3 - 3C1}{3x}$

✓ **Mathematica.** Time used: 36.8 (sec). Leaf size: 319

```
ode=(x*Sqrt[x^2+y[x]^2]-y[x])+(y[x]*Sqrt[x^2+y[x]^2]-x)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 1\right]$$

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 2\right]$$

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 3\right]$$

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 4\right]$$

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 5\right]$$

$$y(x) \rightarrow \text{Root}\left[\#1^6 + 3\#1^4x^2 + \#1^2(3x^4 - 9x^2) - 18\#1c_1x + x^6 - 9c_1^2\&, 6\right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*sqrt(x**2 + y(x)**2) + (-x + sqrt(x**2 + y(x)**2)*y(x))*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.3.8 Problem 23 (m)

Local contents

Solved using first_order_ode_exact	759
✓ Maple	762
✓ Mathematica	764
✓ Sympy	764

Internal problem ID [7888]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (m)

Date solved : Thursday, November 27, 2025 at 06:28:17 AM

CAS classification :

[[_homogeneous, 'class C'], _exact, _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_exact

Time used: 0.083 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-y + x - 3) dy &= (-1 - y - x) dx \\ (x + y + 1) dx + (-y + x - 3) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x + y + 1 \\ N(x, y) &= -y + x - 3 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x + y + 1) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-y + x - 3) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x + y + 1 dx \\ \phi &= \frac{x(x + 2y + 2)}{2} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -y + x - 3$. Therefore equation (4) becomes

$$-y + x - 3 = x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -y - 3$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) \, dy &= \int (-y - 3) \, dy \\ f(y) &= -\frac{1}{2}y^2 - 3y + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x(x + 2y + 2)}{2} - \frac{y^2}{2} - 3y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

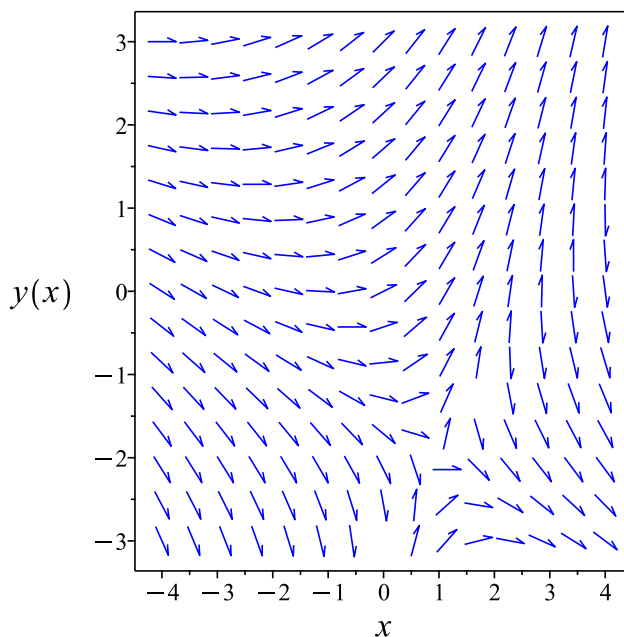
$$c_1 = \frac{x(x + 2y + 2)}{2} - \frac{y^2}{2} - 3y$$

Simplifying the above gives

$$-\frac{y^2}{2} + (x - 3)y + \frac{x^2}{2} + x = c_1$$

Solving for y gives

$$\begin{aligned} y &= x - 3 - \sqrt{2x^2 - 2c_1 - 4x + 9} \\ y &= x - 3 + \sqrt{2x^2 - 2c_1 - 4x + 9} \end{aligned}$$

Figure 2.183: Slope field $x + y + 1 - (y - x + 3)y' = 0$ Summary of solutions found

$$y = x - 3 - \sqrt{2x^2 - 2c_1 - 4x + 9}$$

$$y = x - 3 + \sqrt{2x^2 - 2c_1 - 4x + 9}$$

✓ **Maple.** Time used: 0.067 (sec). Leaf size: 30

```
ode:=x+y(x)+1-(3-x+y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{-\sqrt{2(x-1)^2 c_1^2 + 1} + (x-3)c_1}{c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
```

```

trying homogeneous C
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
<- homogeneous successful

```

Maple step by step

Let's solve

$$x + y(x) + 1 - (y(x) - x + 3) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Check if ODE is exact

- ODE is exact if the lhs is the total derivative of a C^2 function

$$\frac{d}{dx} G(x, y(x)) = 0$$

- Compute derivative of lhs

$$\frac{\partial}{\partial x} G(x, y) + \left(\frac{\partial}{\partial y} G(x, y) \right) \left(\frac{d}{dx} y(x) \right) = 0$$

- Evaluate derivatives

$$1 = 1$$

- Condition met, ODE is exact

- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x} G(x, y), N(x, y) = \frac{\partial}{\partial y} G(x, y) \right]$$

- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int (x + y + 1) dx + _F1(y)$$

- Evaluate integral

$$G(x, y) = \frac{x^2}{2} + yx + x + _F1(y)$$

- Take derivative of $G(x, y)$ with respect to y

$$N(x, y) = \frac{\partial}{\partial y} G(x, y)$$

- Compute derivative

$$-y + x - 3 = x + \frac{d}{dy} _F1(y)$$

- Isolate for $\frac{d}{dy} _F1(y)$

$$\frac{d}{dy} _F1(y) = -y - 3$$

- Solve for $_F1(y)$

$$_F1(y) = -\frac{1}{2}y^2 - 3y$$

- Substitute $_F1(y)$ into equation for $G(x, y)$

$$G(x, y) = \frac{1}{2}x^2 + yx + x - \frac{1}{2}y^2 - 3y$$

- Substitute $G(x, y)$ into the solution of the ODE

$$\frac{1}{2}x^2 + yx + x - \frac{1}{2}y^2 - 3y = C1$$

- Solve for $y(x)$

$$\{y(x) = x - 3 - \sqrt{2x^2 - 2C1 - 4x + 9}, y(x) = x - 3 + \sqrt{2x^2 - 2C1 - 4x + 9}\}$$

✓ **Mathematica.** Time used: 0.085 (sec). Leaf size: 59

```
ode=(x+y[x]+1)-(y[x]-x+3)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -i\sqrt{-2x^2 + 4x - 9 - c_1} + x - 3$$

$$y(x) \rightarrow i\sqrt{-2x^2 + 4x - 9 - c_1} + x - 3$$

✓ **Sympy.** Time used: 1.253 (sec). Leaf size: 37

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x - (-x + y(x) + 3)*Derivative(y(x), x) + y(x) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = x - \sqrt{C_1 + 2x^2 - 4x} - 3, \quad y(x) = x + \sqrt{C_1 + 2x^2 - 4x} - 3 \right]$$

2.3.9 Problem 23 (o)

Local contents

Solved using first_order_ode_exact	765
✓ Maple	768
✓ Mathematica	770
✗ Sympy	770

Internal problem ID [7889]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (o)

Date solved : Thursday, November 27, 2025 at 06:28:22 AM

CAS classification : [_exact, _rational]

Solved using first_order_ode_exact

Time used: 0.111 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{1}{x+y} + 2(x+1)y \right) dy &= \left(-y^2 + \frac{y}{x(x+y)} - 2 \right) dx \\ \left(y^2 - \frac{y}{x(x+y)} + 2 \right) dx &+ \left(\frac{1}{x+y} + 2(x+1)y \right) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y^2 - \frac{y}{x(x+y)} + 2 \\ N(x, y) &= \frac{1}{x+y} + 2(x+1)y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(y^2 - \frac{y}{x(x+y)} + 2 \right) \\ &= 2y - \frac{1}{x(x+y)} + \frac{y}{x(x+y)^2} \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{x+y} + 2(x+1)y \right) \\ &= -\frac{1}{(x+y)^2} + 2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int y^2 - \frac{y}{x(x+y)} + 2 dx \\ \phi &= x y^2 + 2x + \ln(x+y) - \ln(x) + f(y)\end{aligned}\quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 2yx + \frac{1}{x+y} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{1}{x+y} + 2(x+1)y$. Therefore equation (4) becomes

$$\frac{1}{x+y} + 2(x+1)y = 2yx + \frac{1}{x+y} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 2y$$

Integrating the above w.r.t y gives

$$\begin{aligned}\int f'(y) dy &= \int (2y) dy \\ f(y) &= y^2 + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x y^2 + 2x + \ln(x+y) - \ln(x) + y^2 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x y^2 + 2x + \ln(x+y) - \ln(x) + y^2$$

Simplifying the above gives

$$\ln(x+y) + (x+1)y^2 + 2x - \ln(x) = c_1$$

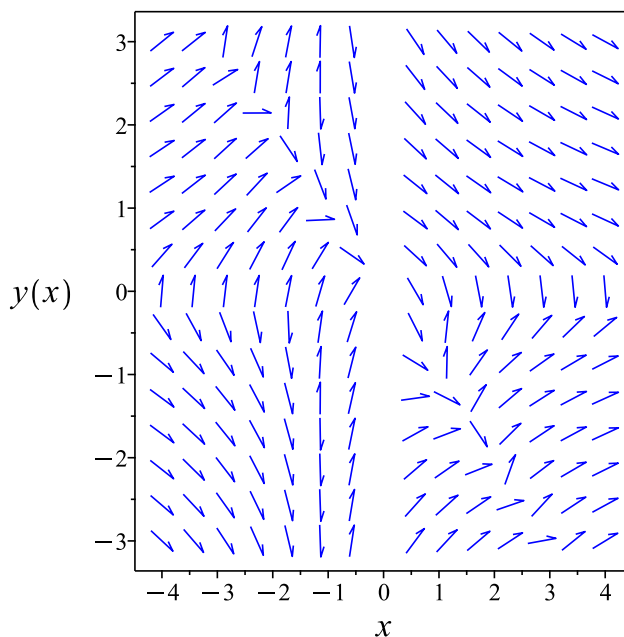


Figure 2.184: Slope field $y^2 - \frac{y}{x+y} + 2 + \left(\frac{1}{x+y} + 2(x+1)y\right)y' = 0$

Summary of solutions found

$$\ln(x+y) + (x+1)y^2 + 2x - \ln(x) = c_1$$

✓ **Maple.** Time used: 0.017 (sec). Leaf size: 69

```
ode:=y(x)^2-y(x)/x/(x+y(x))+2+(1/(x+y(x))+2*(1+x)*y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = x \left(-1 + e^{-\text{RootOf}(x^3 e^{2-Z} + x^2 e^{2-Z} - 2x^3 e^{-Z} + c_1 e^{2-Z} - Z e^{2-Z} + 2x e^{2-Z} - 2x^2 e^{-Z} + x^3 + x^2)} \right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
```

```
trying exact
<- exact successful
```

Maple step by step

Let's solve

$$y(x)^2 - \frac{y(x)}{x(x+y(x))} + 2 + \left(\frac{1}{x+y(x)} + 2y(x)(1+x) \right) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Check if ODE is exact

- ODE is exact if the lhs is the total derivative of a C^2 function

$$\frac{d}{dx}G(x, y(x)) = 0$$

- Compute derivative of lhs

$$\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y) \right) \left(\frac{d}{dx}y(x) \right) = 0$$

- Evaluate derivatives

$$2y - \frac{1}{(x+y)x} + \frac{y}{x(x+y)^2} = -\frac{1}{(x+y)^2} + 2y$$

- Simplify

$$2y - \frac{1}{(x+y)x} + \frac{y}{x(x+y)^2} = -\frac{1}{(x+y)^2} + 2y$$

- Condition met, ODE is exact

- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y) \right]$$

- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int \left(y^2 - \frac{y}{x(x+y)} + 2 \right) dx + _F1(y)$$

- Evaluate integral

$$G(x, y) = y^2x + 2x - \ln(x) + \ln(x+y) + _F1(y)$$

- Take derivative of $G(x, y)$ with respect to y

$$N(x, y) = \frac{\partial}{\partial y}G(x, y)$$

- Compute derivative

$$\frac{1}{x+y} + 2y(1+x) = 2yx + \frac{1}{x+y} + \frac{d}{dy}_F1(y)$$

- Isolate for $\frac{d}{dy}_F1(y)$

$$\frac{d}{dy}_F1(y) = 2y(1+x) - 2yx$$

- Solve for $_F1(y)$

$$_F1(y) = y^2$$

- Substitute $_F1(y)$ into equation for $G(x, y)$

$$G(x, y) = y^2x + 2x - \ln(x) + \ln(x+y) + y^2$$

- Substitute $G(x, y)$ into the solution of the ODE

$$y^2x + 2x - \ln(x) + \ln(x+y) + y^2 = C1$$

- Solve for $y(x)$

$$y(x) = \frac{-xe^{\frac{\text{RootOf}\left(-x^3(e^{-Z})^2 - x^2(e^{-Z})^2 + 2x^3e^{-Z} + C1(e^{-Z})^2 + _Z(e^{-Z})^2 - 2x(e^{-Z})^2 + 2x^2e^{-Z} - x^3 - x^2\right)}{e}}}{\text{RootOf}\left(-x^3(e^{-Z})^2 - x^2(e^{-Z})^2 + 2x^3e^{-Z} + C1(e^{-Z})^2 + _Z(e^{-Z})^2 - 2x(e^{-Z})^2 + 2x^2e^{-Z} - x^3 - x^2\right)} + x$$

✓ **Mathematica.** Time used: 0.331 (sec). Leaf size: 29

```
ode=(y[x]^2- y[x]/(x*(x+y[x]))+2)+( 1/(x+y[x]) + 2*y[x]*(1+x))*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve}\left[xy(x)^2 + y(x)^2 + \log(y(x) + x) + 2x - \log(x) = c_1, y(x)\right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(((2*x + 2)*y(x) + 1/(x + y(x)))*Derivative(y(x), x) + y(x)**2 + 2 - y(x)
        /(x*(x + y(x))),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.3.10 Problem 23 (p)

Local contents

Solved using first_order_ode_exact	771
✓ Maple	774
✓ Mathematica	776
✗ Sympy	776

Internal problem ID [7890]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 23 (p)

Date solved : Thursday, November 27, 2025 at 06:28:27 AM

CAS classification : [_exact]

Solved using first_order_ode_exact

Time used: 0.135 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x^2 e^{x^2 y} + 2xy e^{x y^2} - 2y) dy &= (-2xy e^{x^2 y} - y^2 e^{x y^2} - 1) dx \\ (2xy e^{x^2 y} + y^2 e^{x y^2} + 1) dx &+ (x^2 e^{x^2 y} + 2xy e^{x y^2} - 2y) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 2xy e^{x^2 y} + y^2 e^{x y^2} + 1 \\ N(x, y) &= x^2 e^{x^2 y} + 2xy e^{x y^2} - 2y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (2xy e^{x^2 y} + y^2 e^{x y^2} + 1) \\ &= (2y x^3 + 2x) e^{x^2 y} + 2 e^{x y^2} y (x y^2 + 1) \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x^2 e^{x^2 y} + 2xy e^{x y^2} - 2y) \\ &= (2y x^3 + 2x) e^{x^2 y} + 2 e^{x y^2} y (x y^2 + 1) \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int 2xy e^{x^2 y} + y^2 e^{x y^2} + 1 dx \\ \phi &= x + e^{x y^2} + e^{x^2 y} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 2xy e^{xy^2} + x^2 e^{x^2y} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x^2 e^{x^2y} + 2xy e^{xy^2} - 2y$. Therefore equation (4) becomes

$$x^2 e^{x^2y} + 2xy e^{xy^2} - 2y = 2xy e^{xy^2} + x^2 e^{x^2y} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -2y$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (-2y) dy \\ f(y) &= -y^2 + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x + e^{xy^2} + e^{x^2y} - y^2 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x + e^{xy^2} + e^{x^2y} - y^2$$

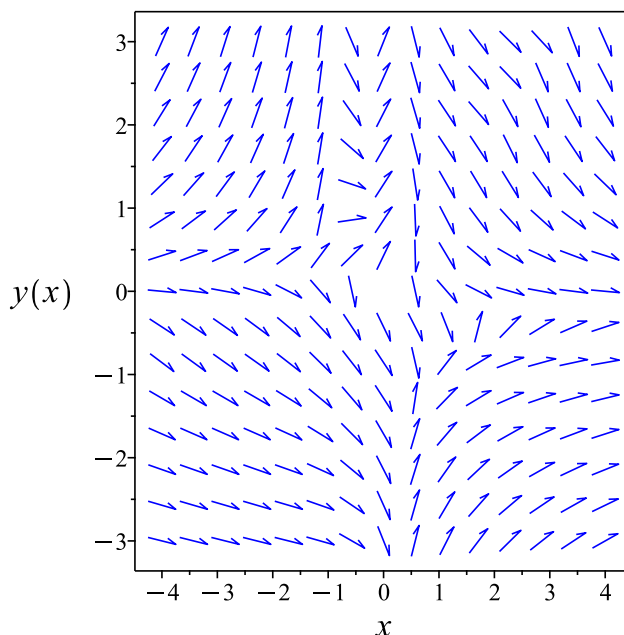


Figure 2.185: Slope field $2xy e^{x^2y} + y^2 e^{y^2x} + 1 + (x^2 e^{x^2y} + 2xy e^{y^2x} - 2y) y' = 0$

Summary of solutions found

$$x + e^{y^2x} + e^{x^2y} - y^2 = c_1$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 41

```
ode:=2*x*y(x)*exp(x^2*y(x))+y(x)^2*exp(x*y(x)^2)+1+(x^2*exp(x^2*y(x))+2*x*y(x)*
  exp(x*y(x)^2)-2*y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\text{RootOf}\left(e^{-Z}x^4 + e^{\frac{Z^2}{x^3}}x^4 + c_1x^4 + x^5 - _Z^2\right)}{x^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
```

```

differential order: 1; looking for linear symmetries
trying exact
<- exact successful

```

Maple step by step

Let's solve

$$2xy(x) e^{x^2y(x)} + y(x)^2 e^{xy(x)^2} + 1 + \left(x^2 e^{x^2y(x)} + 2xy(x) e^{xy(x)^2} - 2y(x) \right) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Check if ODE is exact

- ODE is exact if the lhs is the total derivative of a C^2 function

$$\frac{d}{dx} G(x, y(x)) = 0$$

- Compute derivative of lhs

$$\frac{\partial}{\partial x} G(x, y) + \left(\frac{\partial}{\partial y} G(x, y) \right) \left(\frac{d}{dx} y(x) \right) = 0$$

- Evaluate derivatives

$$2x e^{x^2y} + 2x^3 y e^{x^2y} + 2y e^{xy^2} + 2y^3 x e^{xy^2} = 2x e^{x^2y} + 2x^3 y e^{x^2y} + 2y e^{xy^2} + 2y^3 x e^{xy^2}$$

- Condition met, ODE is exact

- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x} G(x, y), N(x, y) = \frac{\partial}{\partial y} G(x, y) \right]$$

- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int \left(2xy e^{x^2y} + y^2 e^{xy^2} + 1 \right) dx + _F1(y)$$

- Evaluate integral

$$G(x, y) = x + e^{xy^2} + e^{x^2y} + _F1(y)$$

- Take derivative of $G(x, y)$ with respect to y

$$N(x, y) = \frac{\partial}{\partial y} G(x, y)$$

- Compute derivative

$$x^2 e^{x^2y} + 2xy e^{xy^2} - 2y = 2xy e^{xy^2} + x^2 e^{x^2y} + \frac{d}{dy} _F1(y)$$

- Isolate for $\frac{d}{dy} _F1(y)$

$$\frac{d}{dy} _F1(y) = -2y$$

- Solve for $_F1(y)$

$$_F1(y) = -y^2$$

- Substitute $_F1(y)$ into equation for $G(x, y)$

$$G(x, y) = x + e^{xy^2} + e^{x^2y} - y^2$$

- Substitute $G(x, y)$ into the solution of the ODE

$$x + e^{xy^2} + e^{x^2y} - y^2 = C1$$

- Solve for $y(x)$

$$y(x) = \frac{\text{RootOf}\left(-e^{-Zx^4} - e^{-\frac{Z^2}{x^3}} x^4 + C1 x^4 - x^5 + _Z^2\right)}{x^2}$$

✓ **Mathematica.** Time used: 0.274 (sec). Leaf size: 30

```
ode=(2*x*y[x]*Exp[x^2*y[x]]+ y[x]^2*Exp[x*y[x]^2]+1)+(x^2*Exp[x^2*y[x]]+ 2*x*y[x]
]*Exp[x*y[x]^2]-2*y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve}\left[e^{x^2y(x)} - y(x)^2 + e^{xy(x)^2} + x = c_1, y(x)\right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*y(x)*exp(x**2*y(x)) + (x**2*exp(x**2*y(x)) + 2*x*y(x)*exp(x*y(x)**2)
- 2*y(x))*Derivative(y(x), x) + y(x)**2*exp(x*y(x)**2) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.3.11 Problem 24 (p)

Local contents

Solved using first_order_ode_bernoulli	777
Solved using first_order_ode_exact	780
Solved using first_order_ode_isobaric	785
Solved using first_order_ode_homog_A	787
Solved using first_order_ode_homog_type_D2	790
Solved using first_order_ode_homog_type_G	792
Solved using first_order_ode_homog_type_maple_C	794
Solved using first_order_ode_LIE	798
✓ Maple	803
✓ Mathematica	804
✓ Sympy	804

Internal problem ID [7891]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 24 (p)

Date solved : Thursday, November 27, 2025 at 06:28:32 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.112 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{y(-x + 2y)}{x^2} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(\frac{1}{x}\right)y + \left(-\frac{2}{x^2}\right)y^2 \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1}{x} \\ f_1 &= -\frac{2}{x^2} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= \frac{1}{x} \\ f_1(x) &= -\frac{2}{x^2} \\ n &= 2 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^2$ gives

$$y' \frac{1}{y^2} = \frac{1}{yx} - \frac{2}{x^2} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= \frac{1}{y} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{1}{y^2} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -v'(x) &= \frac{v(x)}{x} - \frac{2}{x^2} \\ v' &= -\frac{v}{x} + \frac{2}{x^2} \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{1}{x} \\ p(x) &= \frac{2}{x^2} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{1}{x} \, dx} \\ &= x \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) \left(\frac{2}{x^2} \right) \\ \frac{d}{dx}(vx) &= (x) \left(\frac{2}{x^2} \right) \\ d(vx) &= \left(\frac{2}{x} \right) dx \end{aligned}$$

Integrating gives

$$\begin{aligned} vx &= \int \frac{2}{x} \, dx \\ &= 2 \ln(x) + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

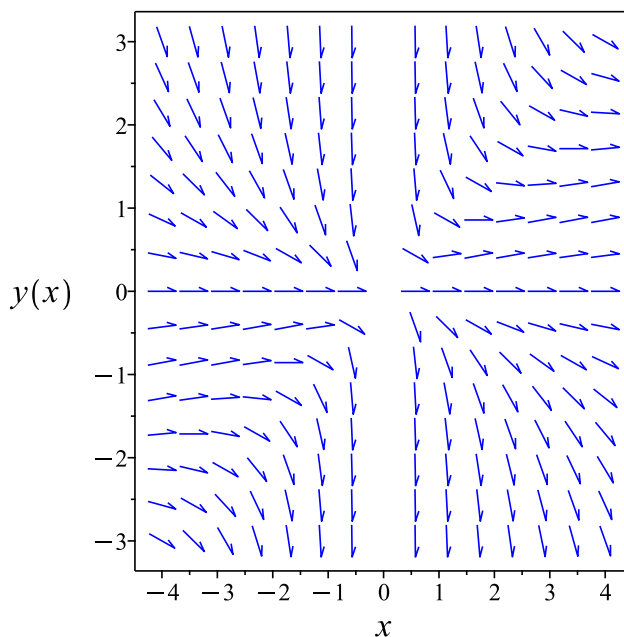
$$v(x) = \frac{2 \ln(x) + c_1}{x}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y} = \frac{2 \ln(x) + c_1}{x}$$

Solving for y gives

$$y = \frac{x}{2 \ln(x) + c_1}$$

Figure 2.186: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$y = \frac{x}{2 \ln(x) + c_1}$$

Solved using first_order_ode_exact

Time used: 0.116 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x^2) dy &= (-y(x - 2y)) dx \\ (y(x - 2y)) dx + (-x^2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y(x - 2y) \\ N(x, y) &= -x^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y(x - 2y)) \\ &= x - 4y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x^2) \\ &= -2x \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{xy^2}$ is an integrating factor. Therefore by multiplying $M = y(x - 2y)$ and $N = -x^2$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{x - 2y}{xy} \\ N &= -\frac{x}{y^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} \left(-\frac{x}{y^2}\right) dy &= \left(-\frac{x-2y}{yx}\right) dx \\ \left(\frac{x-2y}{yx}\right) dx + \left(-\frac{x}{y^2}\right) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{x-2y}{yx} \\ N(x, y) &= -\frac{x}{y^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{x-2y}{yx} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(-\frac{x}{y^2} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{x-2y}{yx} dx \\ \phi &= -2 \ln(x) + \frac{x}{y} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{x}{y^2} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{x}{y^2}$. Therefore equation (4) becomes

$$-\frac{x}{y^2} = -\frac{x}{y^2} + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -2 \ln(x) + \frac{x}{y} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -2 \ln(x) + \frac{x}{y}$$

Solving for y gives

$$y = \frac{x}{2 \ln(x) + c_1}$$

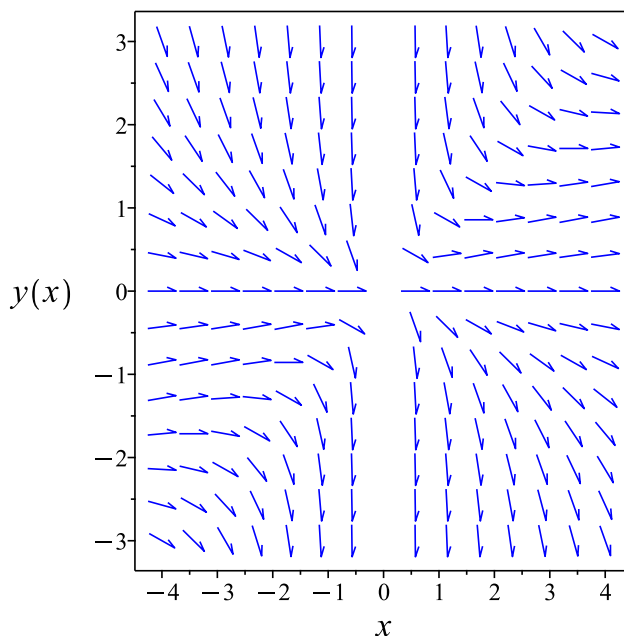


Figure 2.187: Slope field $y(x - 2y) - y'x^2 = 0$

Summary of solutions found

$$y = \frac{x}{2 \ln(x) + c_1}$$

Solved using first_order_ode_isobaric

Time used: 0.324 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

Solving for y' gives

$$y' = -\frac{y(-x + 2y)}{x^2} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{y(-x + 2y)}{x^2} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{u(x)(-x + 2xu(x))}{x}$$

The ode

$$u'(x) = -\frac{2u(x)^2}{x} \quad (2.81)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{2}{x} \\ g(u) &= u^2 \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u^2} du = \int -\frac{2}{x} dx$$

$$-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2 = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$$

$$u(x) = 0$$

Converting $-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$ back to y gives

$$-\frac{x}{y} = \ln\left(\frac{1}{x^2}\right) + c_1$$

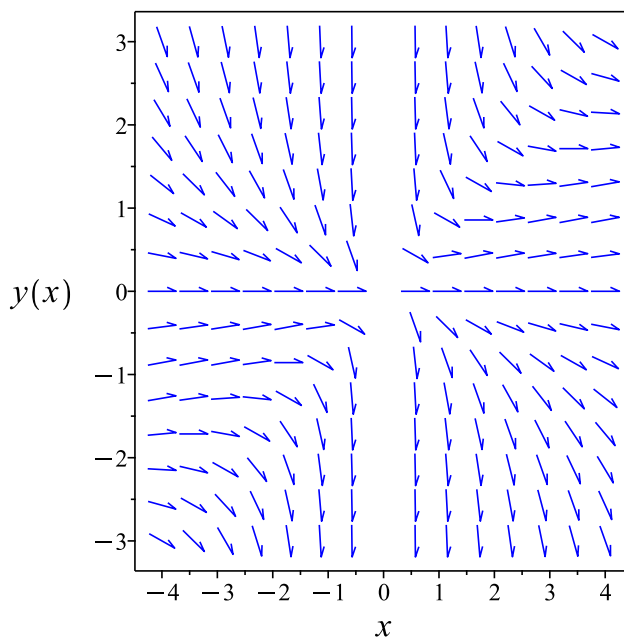
Converting $u(x) = 0$ back to y gives

$$\frac{y}{x} = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Figure 2.188: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Solved using first_order_ode_homog_A

Time used: 0.598 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{y(-x + 2y)}{x^2} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x, y)}{N(x, y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = y(x - 2y)$ and $N = x^2$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous,

it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dx}x + u &= -2u^2 + u \\ \frac{du}{dx} &= -\frac{2u(x)^2}{x}\end{aligned}$$

Or

$$u'(x) + \frac{2u(x)^2}{x} = 0$$

Or

$$2u(x)^2 + u'(x)x = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{2u(x)^2}{x} \tag{2.82}$$

is separable as it can be written as

$$\begin{aligned}u'(x) &= -\frac{2u(x)^2}{x} \\ &= f(x)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(x) &= -\frac{2}{x} \\ g(u) &= u^2\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u^2} du &= \int -\frac{2}{x} dx\end{aligned}$$

$$-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2 = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$$

$$u(x) = 0$$

Converting $-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$ back to y gives

$$-\frac{x}{y} = \ln\left(\frac{1}{x^2}\right) + c_1$$

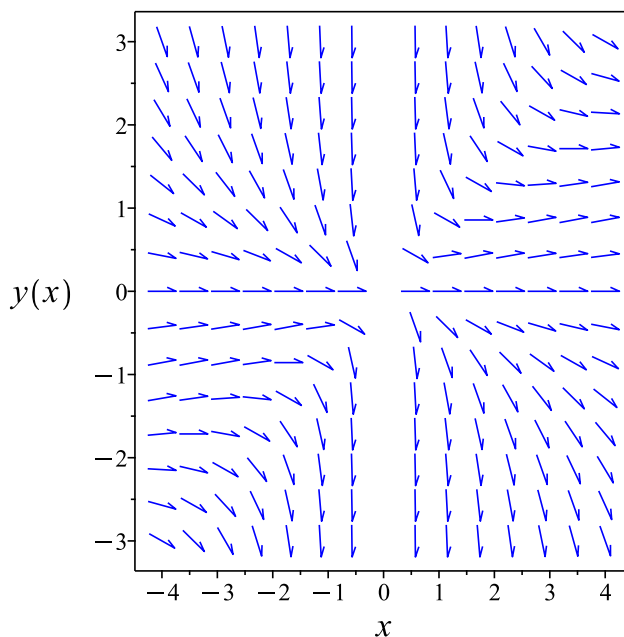
Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Figure 2.189: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.155 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)x(x - 2u(x)x) - (u'(x)x + u(x))x^2 = 0$$

Which is now solved The ode

$$u'(x) = -\frac{2u(x)^2}{x} \tag{2.83}$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{2}{x} \\ g(u) &= u^2 \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u^2} du &= \int -\frac{2}{x} dx \end{aligned}$$

$$-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2 = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} -\frac{1}{u(x)} &= \ln\left(\frac{1}{x^2}\right) + c_1 \\ u(x) &= 0 \end{aligned}$$

Converting $-\frac{1}{u(x)} = \ln\left(\frac{1}{x^2}\right) + c_1$ back to y gives

$$-\frac{x}{y} = \ln\left(\frac{1}{x^2}\right) + c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

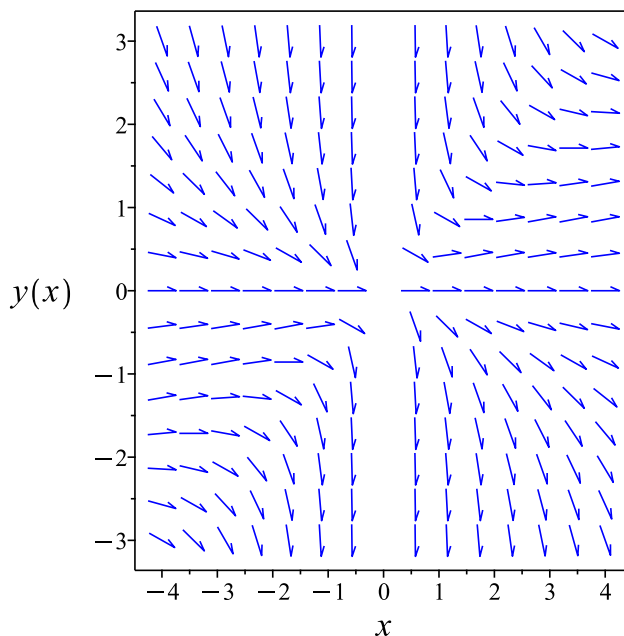


Figure 2.190: Slope field $y(x - 2y) - y'x^2 = 0$

Summary of solutions found

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.081 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

Multiplying the right side of the ode, which is $-\frac{y(-x+2y)}{x^2}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{y(-x+2y)}{x^2} \\ &= \frac{x-2y}{x} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{2y}{x} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{2y}{x} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -2z + 1$$

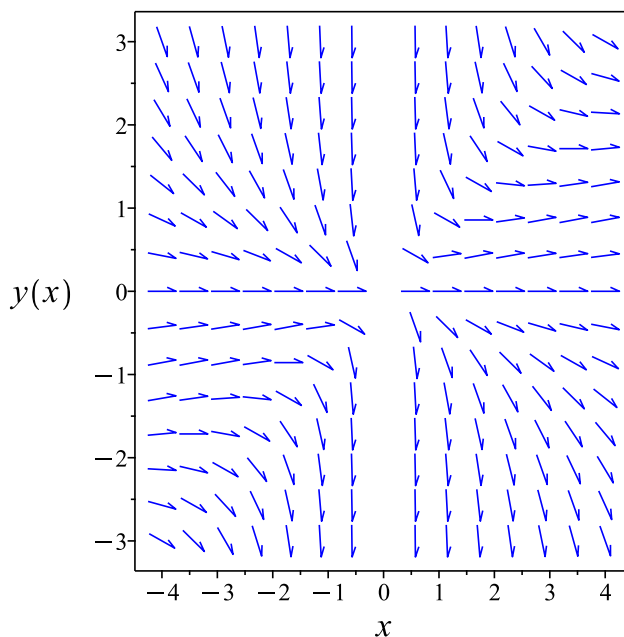
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{2z^2} dz = 0$$

Figure 2.191: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{2z^2} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.901 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{(Y(X) + y_0)(-X - x_0 + 2Y(X) + 2y_0)}{(X + x_0)^2}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{-Y(X)X + 2Y(X)^2}{X^2}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{Y(-X + 2Y)}{X^2} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = Y(X - 2Y)$ and $N = X^2$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -2u^2 + u \\ \frac{du}{dX} &= -\frac{2u(X)^2}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) + \frac{2u(X)^2}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X) \right) X + 2u(X)^2 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{2u(X)^2}{X} \quad (2.84)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{2u(X)^2}{X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{2}{X} \\ g(u) &= u^2 \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{1}{u^2} du = \int -\frac{2}{X} dX$$

$$-\frac{1}{u(X)} = \ln \left(\frac{1}{X^2} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2 = 0$$

for $u(X)$ gives

$$u(X) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\frac{1}{u(X)} = \ln \left(\frac{1}{X^2} \right) + c_1$$

$$u(X) = 0$$

Converting $-\frac{1}{u(X)} = \ln \left(\frac{1}{X^2} \right) + c_1$ back to $Y(X)$ gives

$$-\frac{X}{Y(X)} = \ln \left(\frac{1}{X^2} \right) + c_1$$

Converting $u(X) = 0$ back to $Y(X)$ gives

$$Y(X) = 0$$

Using the solution for $Y(X)$

$$-\frac{X}{Y(X)} = \ln \left(\frac{1}{X^2} \right) + c_1 \tag{A}$$

And replacing back terms in the above solution using

$$\begin{aligned}Y &= y + y_0 \\X &= x_0 + x\end{aligned}$$

Or

$$\begin{aligned}Y &= y \\X &= x\end{aligned}$$

Then the solution in y becomes using EQ (A)

$$-\frac{x}{y} = \ln\left(\frac{1}{x^2}\right) + c_1$$

Using the solution for $Y(X)$

$$Y(X) = 0 \tag{A}$$

And replacing back terms in the above solution using

$$\begin{aligned}Y &= y + y_0 \\X &= x_0 + x\end{aligned}$$

Or

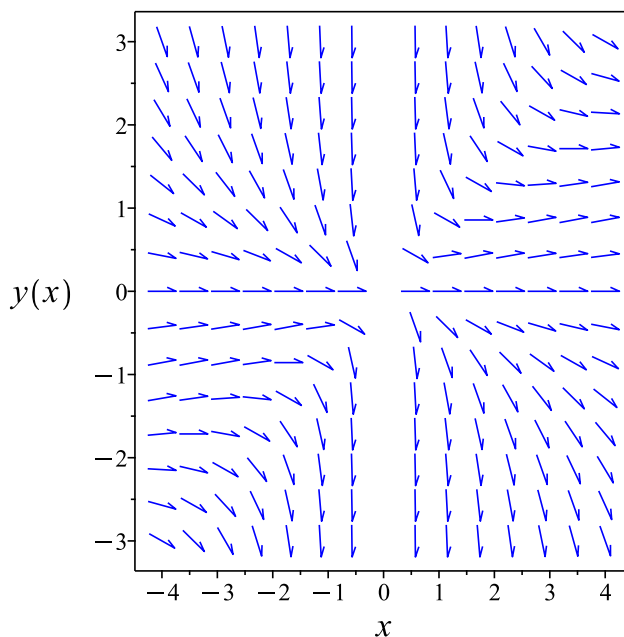
$$\begin{aligned}Y &= y \\X &= x\end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = 0$$

Solving for y gives

$$\begin{aligned}y &= 0 \\y &= -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}\end{aligned}$$

Figure 2.192: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$y = 0$$

$$y = -\frac{x}{\ln\left(\frac{1}{x^2}\right) + c_1}$$

Solved using first_order_ode_LIE

Time used: 0.694 (sec)

Solve

$$y(x - 2y) - y'x^2 = 0$$

Writing the ode as

$$y' = -\frac{y(-x + 2y)}{x^2}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{y(-x+2y)(b_3-a_2)}{x^2} - \frac{y^2(-x+2y)^2 a_3}{x^4} \\ - \left(\frac{y}{x^2} + \frac{2y(-x+2y)}{x^3} \right) (xa_2 + ya_3 + a_1) - \left(-\frac{-x+2y}{x^2} - \frac{2y}{x^2} \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{4x^3 y b_2 - 2x^2 y^2 a_2 + 2x^2 y^2 b_3 - 4y^4 a_3 - x^3 b_1 + x^2 y a_1 + 4x^2 y b_1 - 4x y^2 a_1}{x^4} = 0$$

Setting the numerator to zero gives

$$4x^3 y b_2 - 2x^2 y^2 a_2 + 2x^2 y^2 b_3 - 4y^4 a_3 - x^3 b_1 + x^2 y a_1 + 4x^2 y b_1 - 4x y^2 a_1 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-2a_2 v_1^2 v_2^2 - 4a_3 v_2^4 + 4b_2 v_1^3 v_2 + 2b_3 v_1^2 v_2^2 + a_1 v_1^2 v_2 - 4a_1 v_1 v_2^2 - b_1 v_1^3 + 4b_1 v_1^2 v_2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$4b_2 v_1^3 v_2 - b_1 v_1^3 + (-2a_2 + 2b_3) v_1^2 v_2^2 + (a_1 + 4b_1) v_1^2 v_2 - 4a_1 v_1 v_2^2 - 4a_3 v_2^4 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -4a_1 &= 0 \\ -4a_3 &= 0 \\ -b_1 &= 0 \\ 4b_2 &= 0 \\ a_1 + 4b_1 &= 0 \\ -2a_2 + 2b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{y(-x + 2y)}{x^2} \right) (x) \\ &= \frac{2y^2}{x} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \tag{1}$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{2y^2}{x}} dy \end{aligned}$$

Which results in

$$S = -\frac{x}{2y}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{y(-x + 2y)}{x^2}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{1}{2y} \\ S_y &= \frac{x}{2y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

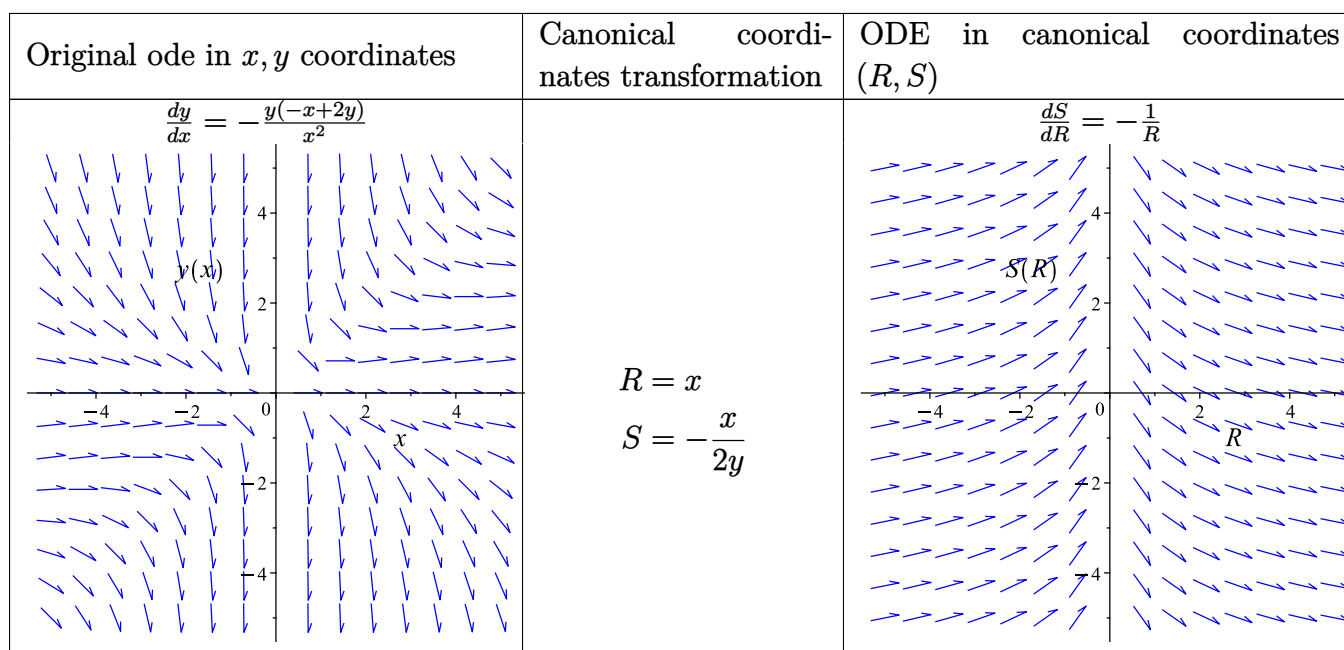
$$\int dS = \int -\frac{1}{R} dR$$

$$S(R) = -\ln(R) + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

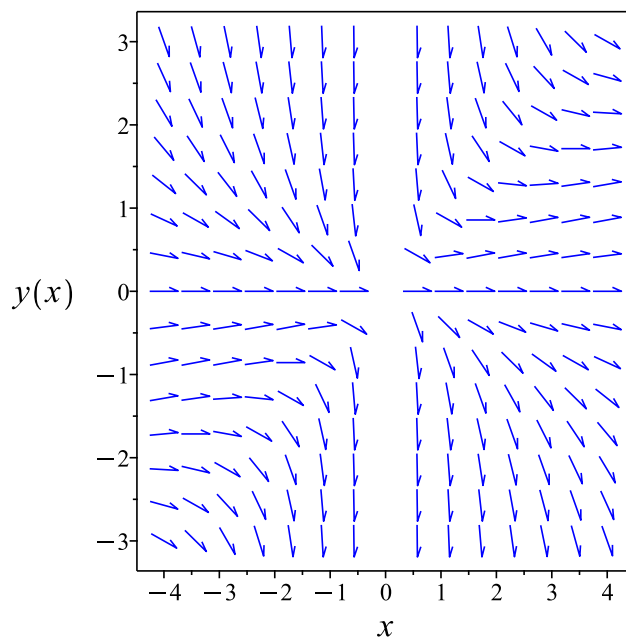
$$-\frac{x}{2y} = -\ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = \frac{x}{2\ln(x) - 2c_2}$$

Figure 2.193: Slope field $y(x - 2y) - y'x^2 = 0$ Summary of solutions found

$$y = \frac{x}{2 \ln(x) - 2c_2}$$

✓ Maple. Time used: 0.001 (sec). Leaf size: 14

```
ode:=y(x)*(x-2*y(x))-x^2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x}{2 \ln(x) + c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

- $y(x)(x - 2y(x)) - x^2\left(\frac{d}{dx}y(x)\right) = 0$
- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative
- $\frac{d}{dx}y(x) = \frac{y(x)(x-2y(x))}{x^2}$

✓ **Mathematica.** Time used: 0.087 (sec). Leaf size: 21

```
ode=y[x]*(x-2*y[x])-x^2*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x}{2\log(x) + c_1}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.119 (sec). Leaf size: 10

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*Derivative(y(x), x) + (x - 2*y(x))*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x}{C_1 + 2\log(x)}$$

2.3.12 Problem 24 (c)

Local contents

Solved using first_order_ode_bernoulli	805
Solved using first_order_ode_exact	808
Solved using first_order_ode_isobaric	812
Solved using first_order_ode_homog_A	816
Solved using first_order_ode_dAlembert	819
Solved using first_order_ode_homog_type_D2	825
Solved using first_order_ode_homog_type_G	828
Solved using first_order_ode_homog_type_maple_C	830
Solved using first_order_ode_LIE	835
✓ Maple	840
✓ Mathematica	841
✓ Sympy	841

Internal problem ID [7892]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 24 (c)

Date solved : Thursday, November 27, 2025 at 06:28:41 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.113 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^2 + y^2}{xy} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{x}\right)y + (-x)\frac{1}{y} \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned}f_0 &= -\frac{1}{x} \\f_1 &= -x\end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= -\frac{1}{x} \\f_1(x) &= -x \\n &= -1\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = -\frac{y^2}{x} - x \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= y^2\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}\frac{v'(x)}{2} &= -\frac{v(x)}{x} - x \\v' &= -\frac{2v}{x} - 2x\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x} \\ p(x) &= -2x \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} \, dx} \\ &= x^2 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu)(-2x) \\ \frac{d}{dx}(v x^2) &= (x^2)(-2x) \\ d(v x^2) &= (-2x^3) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} v x^2 &= \int -2x^3 \, dx \\ &= -\frac{x^4}{2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

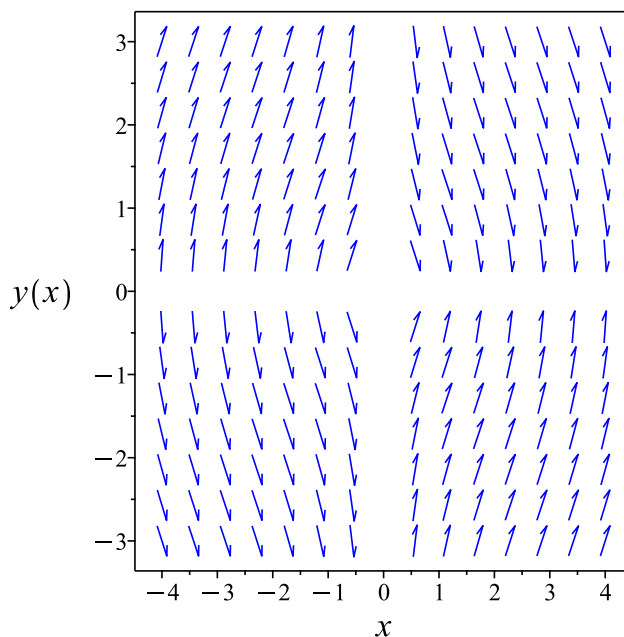
$$v(x) = \frac{-\frac{x^4}{2} + c_1}{x^2}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^2 = \frac{-\frac{x^4}{2} + c_1}{x^2}$$

Solving for y gives

$$\begin{aligned} y &= -\frac{\sqrt{-2x^4 + 4c_1}}{2x} \\ y &= \frac{\sqrt{-2x^4 + 4c_1}}{2x} \end{aligned}$$

Figure 2.194: Slope field $xy' + x^2 + y^2 = 0$ Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 4c_1}}{2x}$$

$$y = \frac{\sqrt{-2x^4 + 4c_1}}{2x}$$

Solved using first_order_ode_exact

Time used: 0.106 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\frac{\partial \phi}{\partial x} = M$$

$$\frac{\partial \phi}{\partial y} = N$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (yx) dy &= (-x^2 - y^2) dx \\ (x^2 + y^2) dx + (yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^2 + y^2 \\ N(x, y) &= yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(yx) \\ &= y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{yx} ((2y) - (y)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{1}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(x^2 + y^2) \\ &= x(x^2 + y^2)\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(yx) \\ &= x^2y\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (x(x^2 + y^2)) + (x^2y) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} \, dx &= \int \overline{M} \, dx \\ \int \frac{\partial \phi}{\partial x} \, dx &= \int x(x^2 + y^2) \, dx \\ \phi &= \frac{(x^2 + y^2)^2}{4} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = (x^2 + y^2) y + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x^2 y$. Therefore equation (4) becomes

$$x^2 y = (x^2 + y^2) y + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -y^3$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (-y^3) dy \\ f(y) &= -\frac{y^4}{4} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{(x^2 + y^2)^2}{4} - \frac{y^4}{4} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

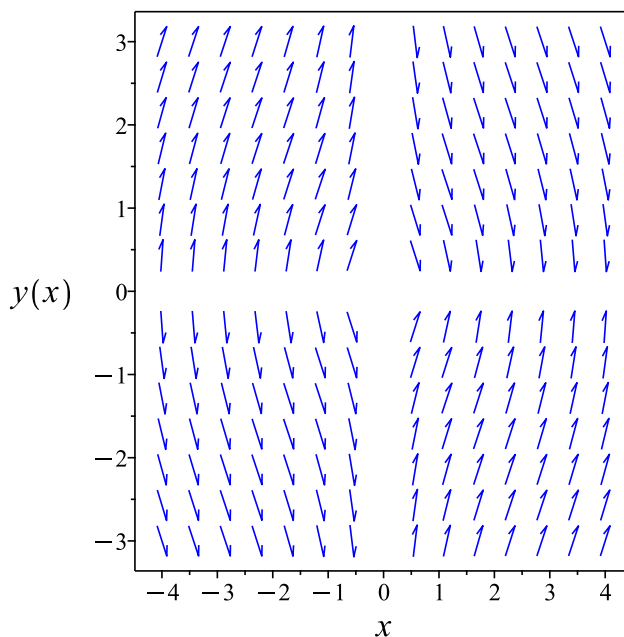
$$c_1 = \frac{(x^2 + y^2)^2}{4} - \frac{y^4}{4}$$

Simplifying the above gives

$$\frac{y^2 x^2}{2} + \frac{x^4}{4} = c_1$$

Solving for y gives

$$\begin{aligned} y &= -\frac{\sqrt{-2x^4 + 8c_1}}{2x} \\ y &= \frac{\sqrt{-2x^4 + 8c_1}}{2x} \end{aligned}$$

Figure 2.195: Slope field $yxy' + x^2 + y^2 = 0$ Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 8c_1}}{2x}$$

$$y = \frac{\sqrt{-2x^4 + 8c_1}}{2x}$$

Solved using first_order_ode_isobaric

Time used: 0.366 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

Solving for y' gives

$$y' = -\frac{x^2 + y^2}{xy} \tag{1}$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \tag{1}$$

Where here

$$f(x, y) = -\frac{x^2 + y^2}{xy} \tag{2}$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x^2 + x^2u(x)^2}{x^2u(x)}$$

The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.85)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} + 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$\frac{y}{x} = -\frac{i\sqrt{2}}{2}$$

$$\frac{y}{x} = \frac{i\sqrt{2}}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

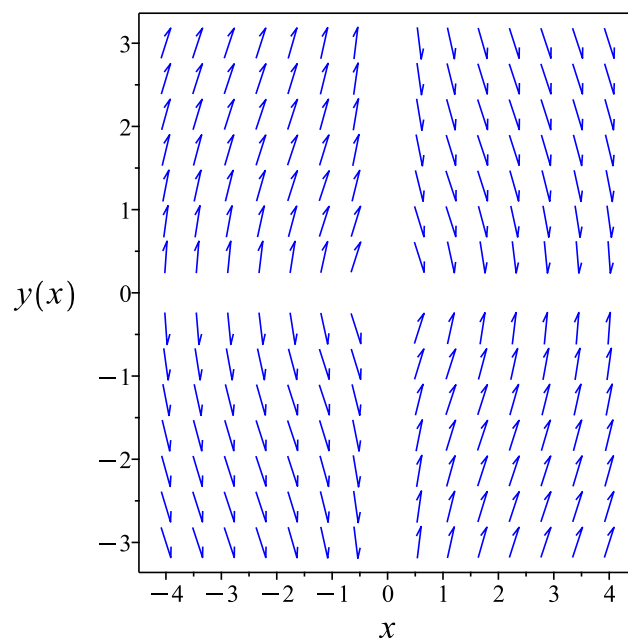


Figure 2.196: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

Solved using first_order_ode_homog_A

Time used: 0.669 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^2 + y^2}{xy} \end{aligned} \quad (1)$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x^2 - y^2$ and $N = yx$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= -\frac{1}{u} - u \\ \frac{du}{dx} &= \frac{-\frac{1}{u(x)} - 2u(x)}{x} \end{aligned}$$

Or

$$u'(x) - \frac{-\frac{1}{u(x)} - 2u(x)}{x} = 0$$

Or

$$u'(x)u(x)x + 2u(x)^2 + 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.86)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(u) = \frac{2u^2 + 1}{u}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{u}{2u^2 + 1} du = \int -\frac{1}{x} dx$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

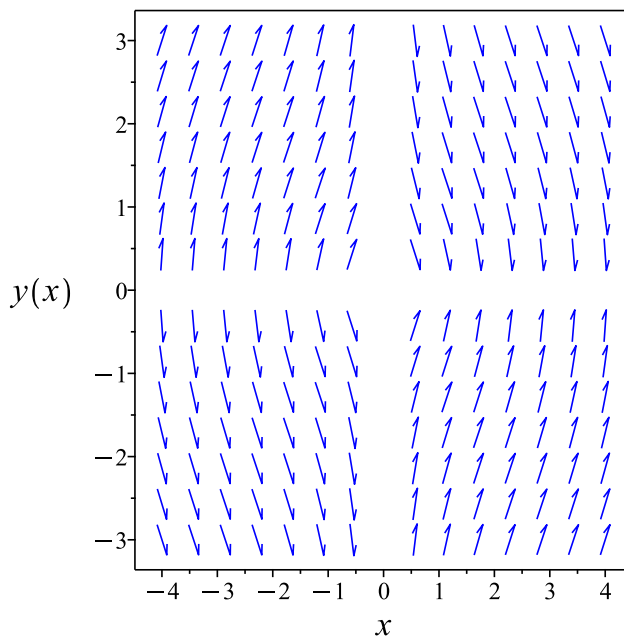


Figure 2.197: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

Solved using first_order_ode_dAlembert

Time used: 1.124 (sec)

Solve

$$xyy' + x^2 + y^2 = 0$$

Let $p = y'$ the ode becomes

$$yxp + x^2 + y^2 = 0$$

Solving for y from the above results in

$$y = \left(-\frac{p}{2} + \frac{\sqrt{p^2 - 4}}{2} \right) x \quad (1)$$

$$y = \left(-\frac{p}{2} - \frac{\sqrt{p^2 - 4}}{2} \right) x \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is dAlembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{p}{2} + \frac{\sqrt{p^2 - 4}}{2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\frac{3p}{2} - \frac{\sqrt{p^2 - 4}}{2} = \left(-\frac{x}{2} + \frac{xp}{2\sqrt{p^2 - 4}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$\frac{3p}{2} - \frac{\sqrt{p^2 - 4}}{2} = 0$$

Solving the above for p results in

$$p_1 = \frac{i\sqrt{2}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{i\sqrt{2}x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{\frac{3p(x)}{2} - \frac{\sqrt{p(x)^2 - 4}}{2}}{-\frac{x}{2} + \frac{xp(x)}{2\sqrt{p(x)^2 - 4}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{\left(-3p(x) + \sqrt{(p(x) - 2)(p(x) + 2)} \right) \sqrt{(p(x) - 2)(p(x) + 2)}}{x \left(\sqrt{(p(x) - 2)(p(x) + 2)} - p(x) \right)} \quad (2.87)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{\left(-3p(x) + \sqrt{(p(x) - 2)(p(x) + 2)} \right) \sqrt{(p(x) - 2)(p(x) + 2)}}{x \left(\sqrt{(p(x) - 2)(p(x) + 2)} - p(x) \right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$f(x) = \frac{1}{x}$$

$$g(p) = \frac{\left(-3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}}{\sqrt{(p-2)(p+2)} - p}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{\sqrt{(p-2)(p+2)} - p}{\left(-3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}} dp = \int \frac{1}{x} dx$$

$$\ln \left(\frac{\left(p(x) + \sqrt{p(x)^2 - 4}\right)^{1/4}}{(2p(x)^2 + 1)^{1/8}} \right) + \frac{\operatorname{arctanh} \left(\frac{3p(x)}{\sqrt{p(x)^2 - 4}} \right)}{4} = \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\left(-3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}}{\sqrt{(p-2)(p+2)} - p} = 0$$

for $p(x)$ gives

$$p(x) = -2$$

$$p(x) = 2$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln \left(\frac{\left(p(x) + \sqrt{p(x)^2 - 4}\right)^{1/4}}{(2p(x)^2 + 1)^{1/8}} \right) + \frac{\operatorname{arctanh} \left(\frac{3p(x)}{\sqrt{p(x)^2 - 4}} \right)}{4} = \ln(x) + c_1$$

$$p(x) = -2$$

$$p(x) = 2$$

Substituting the above solution for p in (2A) gives

$$y = x$$

$$y = -x$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -\frac{p}{2} - \frac{\sqrt{p^2 - 4}}{2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\frac{3p}{2} + \frac{\sqrt{p^2 - 4}}{2} = \left(-\frac{x}{2} - \frac{xp}{2\sqrt{p^2 - 4}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$\frac{3p}{2} + \frac{\sqrt{p^2 - 4}}{2} = 0$$

Solving the above for p results in

$$p_1 = -\frac{i\sqrt{2}}{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{i\sqrt{2}x}{2}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{\frac{3p(x)}{2} + \frac{\sqrt{p(x)^2 - 4}}{2}}{-\frac{x}{2} - \frac{xp(x)}{2\sqrt{p(x)^2 - 4}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{\left(3p(x) + \sqrt{(p(x)-2)(p(x)+2)}\right) \sqrt{(p(x)-2)(p(x)+2)}}{x \left(\sqrt{(p(x)-2)(p(x)+2)} + p(x)\right)} \quad (2.88)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{\left(3p(x) + \sqrt{(p(x)-2)(p(x)+2)}\right) \sqrt{(p(x)-2)(p(x)+2)}}{x \left(\sqrt{(p(x)-2)(p(x)+2)} + p(x)\right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(p) &= \frac{\left(3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}}{p + \sqrt{(p-2)(p+2)}} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{p + \sqrt{(p-2)(p+2)}}{\left(3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}} dp &= \int -\frac{1}{x} dx \end{aligned}$$

$$\ln \left(\left(p(x) + \sqrt{p(x)^2 - 4} \right)^{1/4} (2p(x)^2 + 1)^{1/8} \right) + \frac{\operatorname{arctanh} \left(\frac{3p(x)}{\sqrt{p(x)^2 - 4}} \right)}{4} = \ln \left(\frac{1}{x} \right) + c_2$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\left(3p + \sqrt{(p-2)(p+2)}\right) \sqrt{(p-2)(p+2)}}{p + \sqrt{(p-2)(p+2)}} = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= -2 \\ p(x) &= 2 \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln \left(\left(p(x) + \sqrt{p(x)^2 - 4} \right)^{1/4} (2p(x)^2 + 1)^{1/8} \right) + \frac{\operatorname{arctanh} \left(\frac{3p(x)}{\sqrt{p(x)^2 - 4}} \right)}{4} = \ln \left(\frac{1}{x} \right) + c_2$$

$$p(x) = -2$$

$$p(x) = 2$$

Substituting the above solution for p in (2A) gives

$$y = x$$

$$y = -x$$

The solution

$$y = x$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = -x$$

was found not to satisfy the ode or the IC. Hence it is removed.

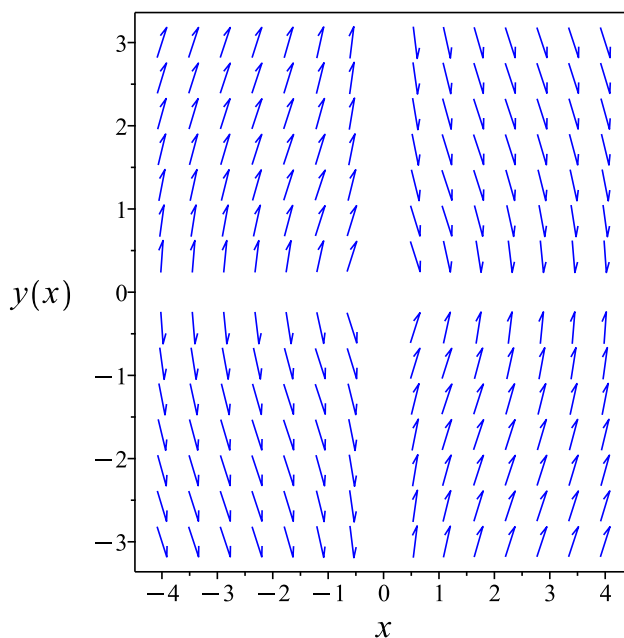


Figure 2.198: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.216 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u(x)x^2(u'(x)x + u(x)) + x^2 + u(x)^2x^2 = 0$$

Which is now solved The ode

$$u'(x) = -\frac{2u(x)^2 + 1}{u(x)x} \quad (2.89)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{2u(x)^2 + 1}{u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(u) &= \frac{2u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{x} dx \end{aligned}$$

$$\frac{\ln(2u(x)^2 + 1)}{4} = \ln\left(\frac{1}{x}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$$

$$u(x) = -\frac{i\sqrt{2}}{2}$$

$$u(x) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(x)^2 + 1)^{1/4} = \frac{c_1}{x}$ back to y gives

$$\left(\frac{2y^2}{x^2} + 1\right)^{1/4} = \frac{c_1}{x}$$

Converting $u(x) = -\frac{i\sqrt{2}}{2}$ back to y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

Converting $u(x) = \frac{i\sqrt{2}}{2}$ back to y gives

$$y = \frac{i\sqrt{2}x}{2}$$

Simplifying the above gives

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

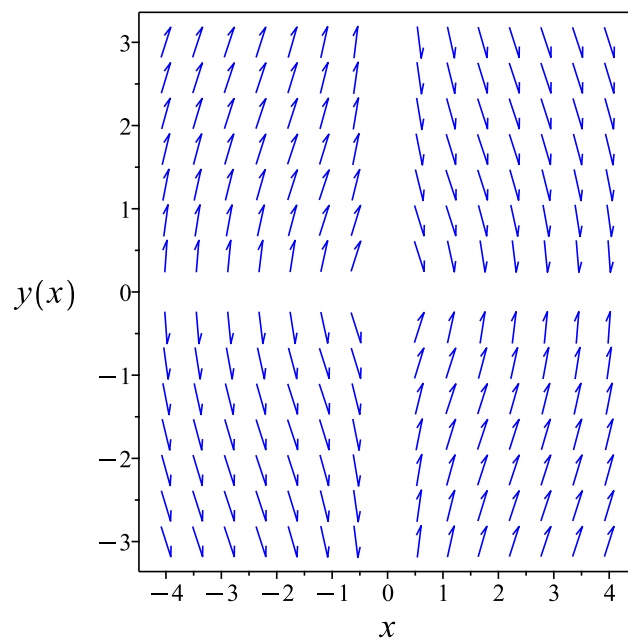


Figure 2.199: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.098 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

Multiplying the right side of the ode, which is $-\frac{x^2+y^2}{xy}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x^2 + y^2}{xy} \\ &= -\frac{x^2 + y^2}{y^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{2x^2}{y^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{2x^2}{y^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{\frac{z}{x}}{\frac{1}{x}} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z^2 + 1}{z^2}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z\left(1 + \frac{z^2+1}{z^2}\right)} dz = 0$$

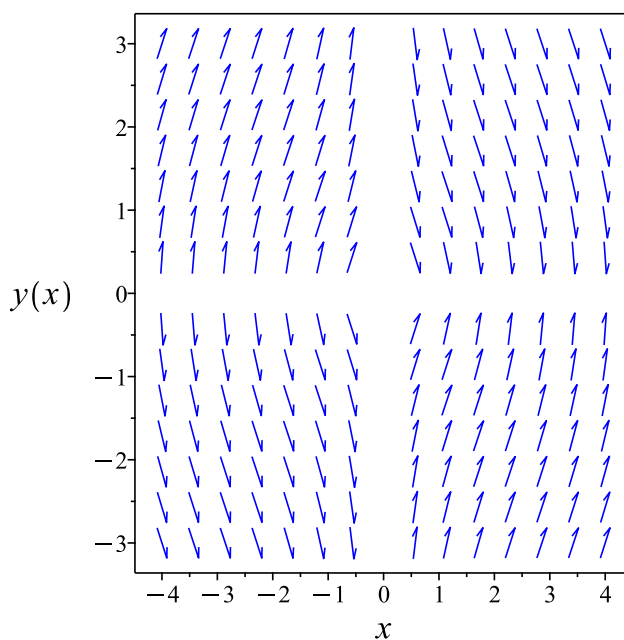


Figure 2.200: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z\left(1 + \frac{z^2+1}{z^2}\right)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.042 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{(X+x_0)^2 + (Y(X)+y_0)^2}{(X+x_0)(Y(X)+y_0)}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X^2 + Y(X)^2}{XY(X)}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X^2 + Y^2}{XY} \end{aligned} \tag{1}$$

An ode of the form $Y' = \frac{M(X,Y)}{N(X,Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X^2 - Y^2$ and $N = XY$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -\frac{1}{u} - u \\ \frac{du}{dX} &= \frac{-\frac{1}{u(X)} - 2u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-\frac{1}{u(X)} - 2u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)u(X)X + 2u(X)^2 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{2u(X)^2 + 1}{u(X)X} \quad (2.90)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{2u(X)^2 + 1}{u(X)X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{1}{X} \\ g(u) &= \frac{2u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{u}{2u^2 + 1} du &= \int -\frac{1}{X} dX \end{aligned}$$

$$\frac{\ln(2u(X)^2 + 1)}{4} = \ln\left(\frac{1}{X}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{2u^2 + 1}{u} = 0$$

for $u(X)$ gives

$$u(X) = -\frac{i\sqrt{2}}{2}$$

$$u(X) = \frac{i\sqrt{2}}{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$$

$$u(X) = -\frac{i\sqrt{2}}{2}$$

$$u(X) = \frac{i\sqrt{2}}{2}$$

Converting $(2u(X)^2 + 1)^{1/4} = \frac{c_1}{X}$ back to $Y(X)$ gives

$$\left(\frac{2Y(X)^2 + X^2}{X^2}\right)^{1/4} = \frac{c_1}{X}$$

Converting $u(X) = -\frac{i\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = -\frac{iX\sqrt{2}}{2}$$

Converting $u(X) = \frac{i\sqrt{2}}{2}$ back to $Y(X)$ gives

$$Y(X) = \frac{iX\sqrt{2}}{2}$$

Using the solution for $Y(X)$

$$\left(\frac{2Y(X)^2 + X^2}{X^2}\right)^{1/4} = \frac{c_1}{X} \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\left(\frac{2y^2 + x^2}{x^2}\right)^{1/4} = \frac{c_1}{x}$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{iX\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{i\sqrt{2}x}{2}$$

Using the solution for $Y(X)$

$$Y(X) = \frac{iX\sqrt{2}}{2} \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y = \frac{i\sqrt{2}x}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

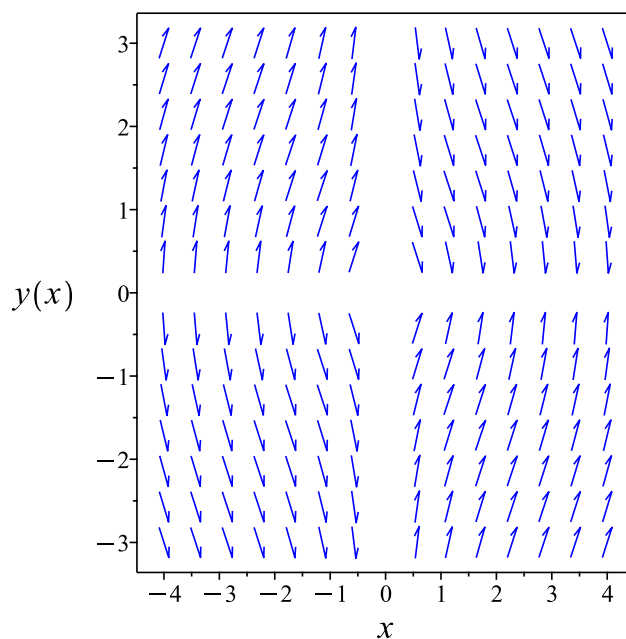


Figure 2.201: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{2}x}{2}$$

$$y = \frac{i\sqrt{2}x}{2}$$

$$y = -\frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

$$y = \frac{\sqrt{2c_1^4 - 2x^4}}{2x}$$

Solved using first_order_ode_LIE

Time used: 0.931 (sec)

Solve

$$yxy' + x^2 + y^2 = 0$$

Writing the ode as

$$y' = -\frac{x^2 + y^2}{xy}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(x^2 + y^2)(b_3 - a_2)}{xy} - \frac{(x^2 + y^2)^2 a_3}{x^2 y^2} - \left(-\frac{2}{y} + \frac{x^2 + y^2}{x^2 y}\right)(xa_2 + ya_3 + a_1) \quad (5\text{E})$$

$$- \left(-\frac{2}{x} + \frac{x^2 + y^2}{x y^2}\right)(xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$\frac{x^4 a_3 + x^4 b_2 - 2x^3 y a_2 + 2x^3 y b_3 + x^2 y^2 a_3 - 2b_2 y^2 x^2 + 2y^4 a_3 + x^3 b_1 - x^2 y a_1 - x y^2 b_1 + y^3 a_1}{y^2 x^2}$$

$$= 0$$

Setting the numerator to zero gives

$$-x^4 a_3 - x^4 b_2 + 2x^3 y a_2 - 2x^3 y b_3 - x^2 y^2 a_3 + 2b_2 y^2 x^2 - 2y^4 a_3 - x^3 b_1 + x^2 y a_1 + x y^2 b_1 - y^3 a_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$2a_2v_1^3v_2 - a_3v_1^4 - a_3v_1^2v_2^2 - 2a_3v_2^4 - b_2v_1^4 + 2b_2v_1^2v_2^2 - 2b_3v_1^3v_2 + a_1v_1^2v_2 - a_1v_2^3 - b_1v_1^3 + b_1v_1v_2^2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-a_3 - b_2)v_1^4 + (2a_2 - 2b_3)v_1^3v_2 - b_1v_1^3 + (-a_3 + 2b_2)v_1^2v_2^2 + a_1v_1^2v_2 + b_1v_1v_2^2 - 2a_3v_2^4 - a_1v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_1 &= 0 \\ b_1 &= 0 \\ -a_1 &= 0 \\ -2a_3 &= 0 \\ -b_1 &= 0 \\ 2a_2 - 2b_3 &= 0 \\ -a_3 - b_2 &= 0 \\ -a_3 + 2b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = y$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x^2 + y^2}{xy} \right) (x) \\ &= y + \frac{x^2 + y^2}{y} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x^2 + y^2}{y}} dy\end{aligned}$$

Which results in

$$S = \frac{\ln(x^2 + 2y^2)}{4}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x^2 + y^2}{xy}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{x}{2x^2 + 4y^2} \\ S_y &= \frac{y}{x^2 + 2y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{2R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{2R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

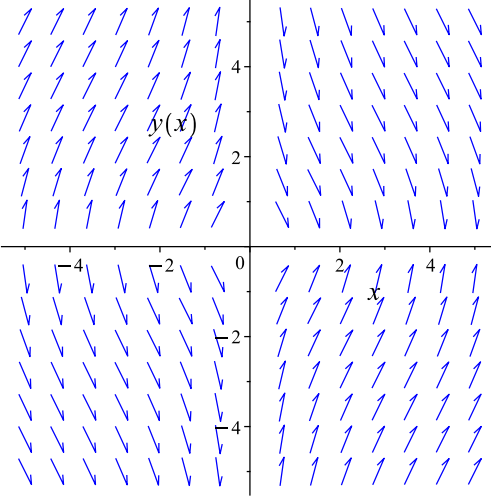
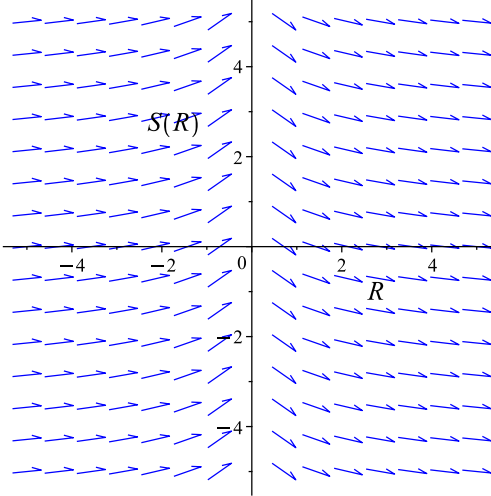
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{2R} dR \\ S(R) &= -\frac{\ln(R)}{2} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(2y^2 + x^2)}{4} = -\frac{\ln(x)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x^2+y^2}{xy}$ 	$R = x$ $S = \frac{\ln(x^2 + 2y^2)}{4}$	$\frac{dS}{dR} = -\frac{1}{2R}$ 

Solving for y gives

$$y = -\frac{\sqrt{-2x^4 + 2e^{4c_2}}}{2x}$$

$$y = \frac{\sqrt{-2x^4 + 2e^{4c_2}}}{2x}$$

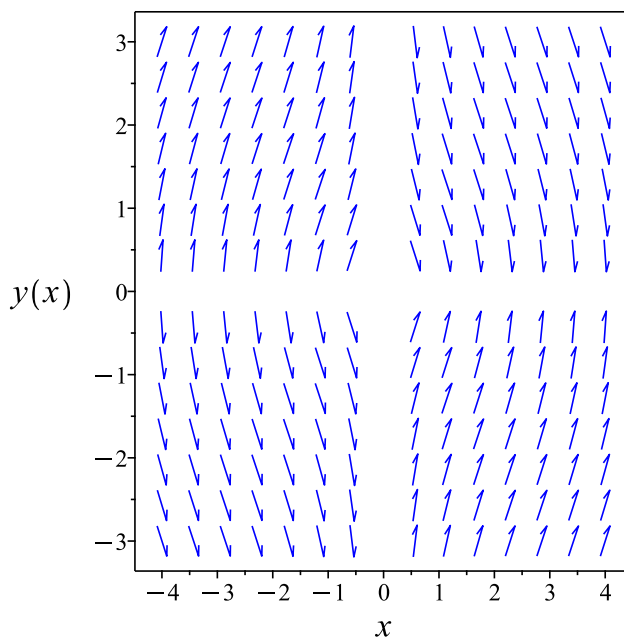


Figure 2.202: Slope field $yxy' + x^2 + y^2 = 0$

Summary of solutions found

$$y = -\frac{\sqrt{-2x^4 + 2e^{4c_2}}}{2x}$$

$$y = \frac{\sqrt{-2x^4 + 2e^{4c_2}}}{2x}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 39

```
ode:=x*y(x)*diff(y(x),x)+x^2+y(x)^2 = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{-2x^4 + 4c_1}}{2x}$$

$$y = \frac{\sqrt{-2x^4 + 4c_1}}{2x}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

<- Bernoulli successful

Maple step by step

Let's solve

$$x^2 + y(x)^2 + xy(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{x^2 + y(x)^2}{xy(x)}$$

✓ **Mathematica.** Time used: 0.075 (sec). Leaf size: 46

```
ode=(x^2+y[x]^2)+x*y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{-\frac{x^4}{2} + c_1}}{x}$$

$$y(x) \rightarrow \frac{\sqrt{-\frac{x^4}{2} + c_1}}{x}$$

✓ **Sympy.** Time used: 0.250 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2 + x*y(x)*Derivative(y(x), x) + y(x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\frac{\sqrt{C_1 - 2x^4}}{2x}, y(x) = \frac{\sqrt{C_1 - 2x^4}}{2x} \right]$$

2.3.13 Problem 24 (d)

Local contents

Solved using first_order_ode_bernoulli	842
Solved using first_order_ode_exact	845
Solved using first_order_ode_isobaric	848
Solved using first_order_ode_homog_A	852
Solved using first_order_ode_dAlembert	855
Solved using first_order_ode_homog_type_D2	861
Solved using first_order_ode_homog_type_G	864
Solved using first_order_ode_homog_type_maple_C	866
Solved using first_order_ode_LIE	871
✓ Maple	876
✓ Mathematica	878
✓ Sympy	878

Internal problem ID [7893]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 24 (d)

Date solved : Thursday, November 27, 2025 at 06:28:53 AM

CAS classification :

[[_homogeneous, 'class A'], _exact, _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.111 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^2 + y^2}{2xy} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{2x}\right)y + \left(-\frac{x}{2}\right)\frac{1}{y} \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -\frac{1}{2x} \\ f_1 &= -\frac{x}{2} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= -\frac{1}{2x} \\ f_1(x) &= -\frac{x}{2} \\ n &= -1 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = -\frac{y^2}{2x} - \frac{x}{2} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= y^2 \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(x)}{2} &= -\frac{v(x)}{2x} - \frac{x}{2} \\ v' &= -\frac{v}{x} - x \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{1}{x} \\ p(x) &= -x \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{1}{x} \, dx} \\ &= x \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu)(-x) \\ \frac{d}{dx}(vx) &= (x)(-x) \\ d(vx) &= (-x^2) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} vx &= \int -x^2 \, dx \\ &= -\frac{x^3}{3} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

$$v(x) = \frac{-\frac{x^3}{3} + c_1}{x}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^2 = \frac{-\frac{x^3}{3} + c_1}{x}$$

Solving for y gives

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

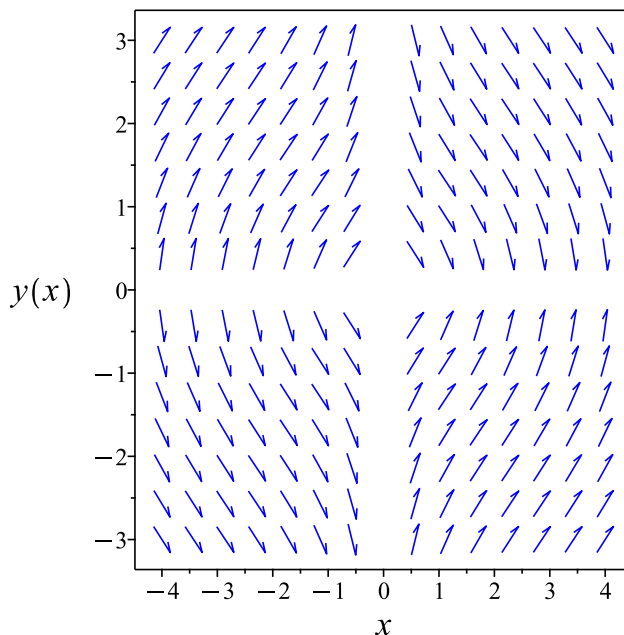


Figure 2.203: Slope field $x^2 + y^2 + 2xyy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

Solved using first_order_ode_exact

Time used: 0.070 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \tag{A}$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2yx) dy &= (-x^2 - y^2) dx \\ (x^2 + y^2) dx + (2yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x^2 + y^2 \\ N(x, y) &= 2yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(2yx) \\ &= 2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int x^2 + y^2 dx \\ \phi &= \frac{1}{3}x^3 + xy^2 + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 2yx + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 2yx$. Therefore equation (4) becomes

$$2yx = 2yx + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{1}{3}x^3 + xy^2 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{1}{3}x^3 + xy^2$$

Solving for y gives

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

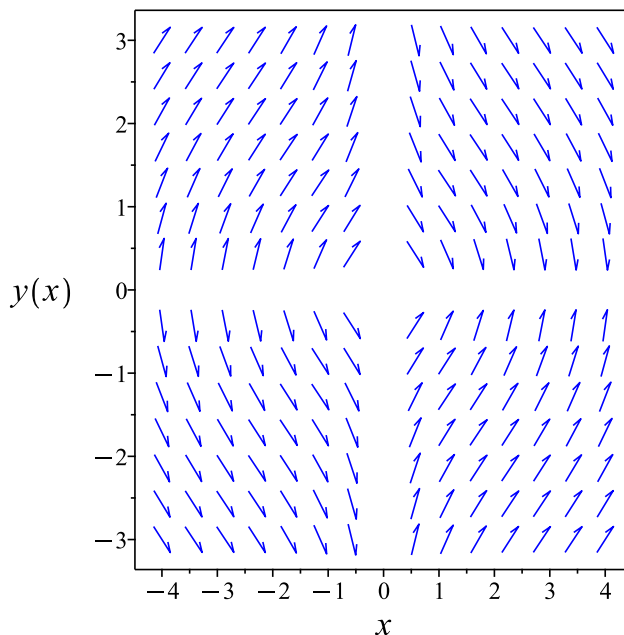


Figure 2.204: Slope field $x^2 + y^2 + 2xyy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(-x^3 + 3c_1)}}{3x}$$

Solved using first_order_ode_isobaric

Time used: 0.332 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Solving for y' gives

$$y' = -\frac{x^2 + y^2}{2xy} \tag{1}$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{x^2 + y^2}{2xy} \quad (2)$$

m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = 1$$

Since the ode is isobaric of order $m = 1$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= ux \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$u(x) + xu'(x) = -\frac{x^2 + x^2 u(x)^2}{2x^2 u(x)}$$

The ode

$$u'(x) = -\frac{3u(x)^2 + 1}{2u(x)x} \quad (2.91)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{3u(x)^2 + 1}{2u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{2x} \\ g(u) &= \frac{3u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{3u^2 + 1} du &= \int -\frac{1}{2x} dx \end{aligned}$$

$$\frac{\ln(3u(x)^2 + 1)}{6} = \ln\left(\frac{1}{\sqrt{x}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{3u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{3}}{3}$$

$$u(x) = \frac{i\sqrt{3}}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

$$u(x) = -\frac{i\sqrt{3}}{3}$$

$$u(x) = \frac{i\sqrt{3}}{3}$$

Converting $(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$ back to y gives

$$\left(\frac{3y^2}{x^2} + 1\right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

Converting $u(x) = -\frac{i\sqrt{3}}{3}$ back to y gives

$$\frac{y}{x} = -\frac{i\sqrt{3}}{3}$$

Converting $u(x) = \frac{i\sqrt{3}}{3}$ back to y gives

$$\frac{y}{x} = \frac{i\sqrt{3}}{3}$$

Simplifying the above gives

$$\left(\frac{x^2 + 3y^2}{x^2}\right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

$$\frac{y}{x} = -\frac{i\sqrt{3}}{3}$$

$$\frac{y}{x} = \frac{i\sqrt{3}}{3}$$

Solving for y gives

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

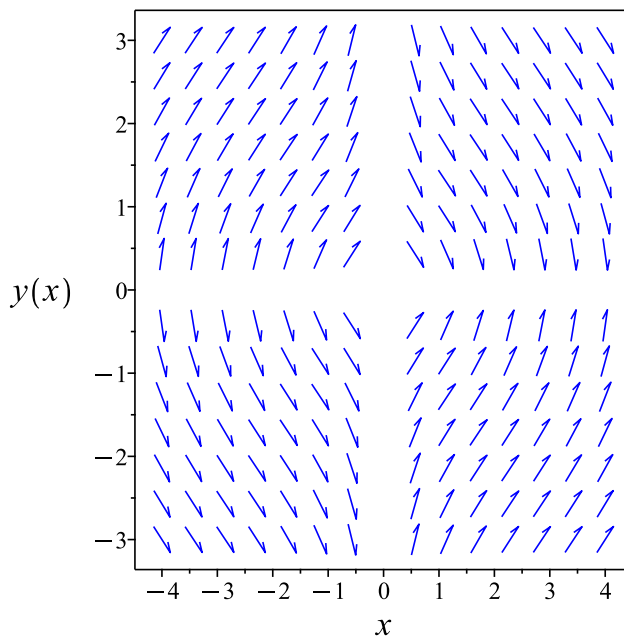


Figure 2.205: Slope field $x^2 + y^2 + 2xyy' = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

Solved using first_order_ode_homog_A

Time used: 0.642 (sec)

Solve

$$x^2 + y^2 + 2yxy' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{x^2 + y^2}{2xy} \end{aligned} \tag{1}$$

An ode of the form $y' = \frac{M(x,y)}{N(x,y)}$ is called homogeneous if the functions $M(x, y)$ and $N(x, y)$ are both homogeneous functions and of the same order. Recall that a function $f(x, y)$ is homogeneous of order n if

$$f(t^n x, t^n y) = t^n f(x, y)$$

In this case, it can be seen that both $M = -x^2 - y^2$ and $N = 2yx$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{y}{x}$, or $y = ux$. Hence

$$\frac{dy}{dx} = \frac{du}{dx}x + u$$

Applying the transformation $y = ux$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dx}x + u &= -\frac{1}{2u} - \frac{u}{2} \\ \frac{du}{dx} &= \frac{-\frac{1}{2u(x)} - \frac{3u(x)}{2}}{x} \end{aligned}$$

Or

$$u'(x) - \frac{-\frac{1}{2u(x)} - \frac{3u(x)}{2}}{x} = 0$$

Or

$$2u'(x)u(x)x + 3u(x)^2 + 1 = 0$$

Which is now solved as separable in $u(x)$.

The ode

$$u'(x) = -\frac{3u(x)^2 + 1}{2u(x)x} \quad (2.92)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{3u(x)^2 + 1}{2u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{2x} \\ g(u) &= \frac{3u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{3u^2 + 1} du &= \int -\frac{1}{2x} dx \end{aligned}$$

$$\frac{\ln(3u(x)^2 + 1)}{6} = \ln\left(\frac{1}{\sqrt{x}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{3u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$u(x) = -\frac{i\sqrt{3}}{3}$$

$$u(x) = \frac{i\sqrt{3}}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

$$u(x) = -\frac{i\sqrt{3}}{3}$$

$$u(x) = \frac{i\sqrt{3}}{3}$$

Converting $(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$ back to y gives

$$\left(\frac{x^2 + 3y^2}{x^2}\right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

Converting $u(x) = -\frac{i\sqrt{3}}{3}$ back to y gives

$$y = -\frac{i\sqrt{3}x}{3}$$

Converting $u(x) = \frac{i\sqrt{3}}{3}$ back to y gives

$$y = \frac{i\sqrt{3}x}{3}$$

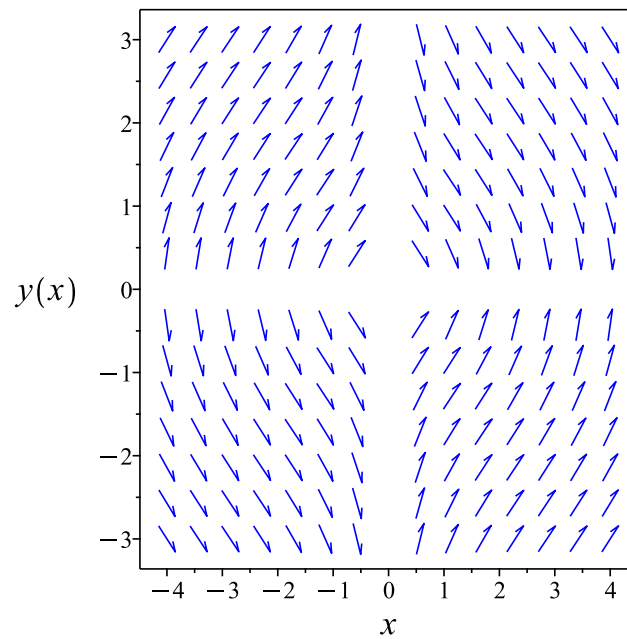
Solving for y gives

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

Figure 2.206: Slope field $x^2 + y^2 + 2xyy' = 0$ Summary of solutions found

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

Solved using first_order_ode_dAlembert

Time used: 1.039 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Let $p = y'$ the ode becomes

$$2yxp + x^2 + y^2 = 0$$

Solving for y from the above results in

$$y = (-p + \sqrt{p^2 - 1})x \quad (1)$$

$$y = (-p - \sqrt{p^2 - 1})x \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -p + \sqrt{p^2 - 1} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$2p - \sqrt{p^2 - 1} = \left(-x + \frac{xp}{\sqrt{p^2 - 1}}\right)p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$2p - \sqrt{p^2 - 1} = 0$$

Solving the above for p results in

$$p_1 = \frac{i\sqrt{3}}{3}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{i\sqrt{3}x}{3}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{2p(x) - \sqrt{p(x)^2 - 1}}{-x + \frac{xp(x)}{\sqrt{p(x)^2 - 1}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{\left(-2p(x) + \sqrt{(p(x) - 1)(p(x) + 1)}\right) \sqrt{(p(x) - 1)(p(x) + 1)}}{x \left(\sqrt{(p(x) - 1)(p(x) + 1)} - p(x)\right)} \quad (2.93)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{\left(-2p(x) + \sqrt{(p(x) - 1)(p(x) + 1)}\right) \sqrt{(p(x) - 1)(p(x) + 1)}}{x \left(\sqrt{(p(x) - 1)(p(x) + 1)} - p(x)\right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(p) &= \frac{\left(-2p + \sqrt{(p - 1)(p + 1)}\right) \sqrt{(p - 1)(p + 1)}}{\sqrt{(p - 1)(p + 1)} - p} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{\sqrt{(p - 1)(p + 1)} - p}{\left(-2p + \sqrt{(p - 1)(p + 1)}\right) \sqrt{(p - 1)(p + 1)}} dp &= \int \frac{1}{x} dx \\ \ln \left(\frac{\left(p(x) + \sqrt{p(x)^2 - 1}\right)^{1/3}}{(3p(x)^2 + 1)^{1/6}} \right) + \frac{\operatorname{arctanh} \left(\frac{2p(x)}{\sqrt{p(x)^2 - 1}} \right)}{3} &= \ln(x) + c_1 \end{aligned}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{(-2p + \sqrt{(p-1)(p+1)}) \sqrt{(p-1)(p+1)}}{\sqrt{(p-1)(p+1)} - p} = 0$$

for $p(x)$ gives

$$p(x) = -1$$

$$p(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln \left(\frac{(p(x) + \sqrt{p(x)^2 - 1})^{1/3}}{(3p(x)^2 + 1)^{1/6}} \right) + \frac{\operatorname{arctanh} \left(\frac{2p(x)}{\sqrt{p(x)^2 - 1}} \right)}{3} = \ln(x) + c_1$$

$$p(x) = -1$$

$$p(x) = 1$$

Substituting the above solution for p in (2A) gives

$$y = x$$

$$y = -x$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = -p - \sqrt{p^2 - 1}$$

$$g = 0$$

Hence (2) becomes

$$2p + \sqrt{p^2 - 1} = \left(-x - \frac{xp}{\sqrt{p^2 - 1}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$2p + \sqrt{p^2 - 1} = 0$$

Solving the above for p results in

$$p_1 = -\frac{i\sqrt{3}}{3}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{i\sqrt{3}x}{3}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{2p(x) + \sqrt{p(x)^2 - 1}}{-x - \frac{xp(x)}{\sqrt{p(x)^2 - 1}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{\left(2p(x) + \sqrt{(p(x) - 1)(p(x) + 1)} \right) \sqrt{(p(x) - 1)(p(x) + 1)}}{x \left(\sqrt{(p(x) - 1)(p(x) + 1)} + p(x) \right)} \quad (2.94)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{\left(2p(x) + \sqrt{(p(x) - 1)(p(x) + 1)} \right) \sqrt{(p(x) - 1)(p(x) + 1)}}{x \left(\sqrt{(p(x) - 1)(p(x) + 1)} + p(x) \right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{x} \\ g(p) &= \frac{\left(2p + \sqrt{(p - 1)(p + 1)} \right) \sqrt{(p - 1)(p + 1)}}{p + \sqrt{(p - 1)(p + 1)}} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{p + \sqrt{(p-1)(p+1)}}{(2p + \sqrt{(p-1)(p+1)}) \sqrt{(p-1)(p+1)}} dp = \int -\frac{1}{x} dx$$

$$\ln \left(\left(p(x) + \sqrt{p(x)^2 - 1} \right)^{1/3} (3p(x)^2 + 1)^{1/6} \right) + \frac{\operatorname{arctanh} \left(\frac{2p(x)}{\sqrt{p(x)^2 - 1}} \right)}{3} = \ln \left(\frac{1}{x} \right) + c_2$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{(2p + \sqrt{(p-1)(p+1)}) \sqrt{(p-1)(p+1)}}{p + \sqrt{(p-1)(p+1)}} = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= -1 \\ p(x) &= 1 \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln \left(\left(p(x) + \sqrt{p(x)^2 - 1} \right)^{1/3} (3p(x)^2 + 1)^{1/6} \right) + \frac{\operatorname{arctanh} \left(\frac{2p(x)}{\sqrt{p(x)^2 - 1}} \right)}{3} = \ln \left(\frac{1}{x} \right) + c_2$$

$$p(x) = -1$$

$$p(x) = 1$$

Substituting the above solution for p in (2A) gives

$$y = x$$

$$y = -x$$

The solution

$$y = x$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = -x$$

was found not to satisfy the ode or the IC. Hence it is removed.

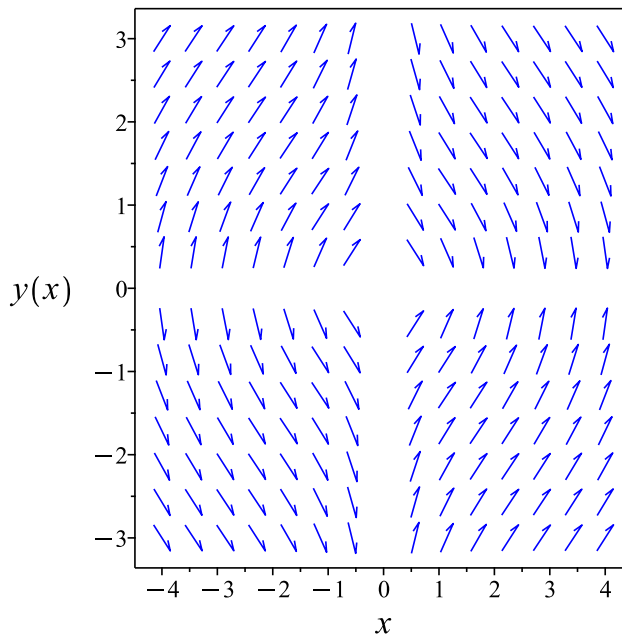


Figure 2.207: Slope field $x^2 + y^2 + 2xyy' = 0$

Summary of solutions found

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

Solved using first_order_ode_homog_type_D2

Time used: 0.184 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$x^2 + u(x)^2 x^2 + 2u(x) x^2 (u'(x) x + u(x)) = 0$$

Which is now solved The ode

$$u'(x) = -\frac{3u(x)^2 + 1}{2u(x)x} \quad (2.95)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{3u(x)^2 + 1}{2u(x)x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{2x} \\ g(u) &= \frac{3u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{u}{3u^2 + 1} du &= \int -\frac{1}{2x} dx \end{aligned}$$

$$\frac{\ln(3u(x)^2 + 1)}{6} = \ln\left(\frac{1}{\sqrt{x}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{3u^2 + 1}{u} = 0$$

for $u(x)$ gives

$$\begin{aligned} u(x) &= -\frac{i\sqrt{3}}{3} \\ u(x) &= \frac{i\sqrt{3}}{3} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$$

$$u(x) = -\frac{i\sqrt{3}}{3}$$

$$u(x) = \frac{i\sqrt{3}}{3}$$

Converting $(3u(x)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{x}}$ back to y gives

$$\left(\frac{3y^2}{x^2} + 1\right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

Converting $u(x) = -\frac{i\sqrt{3}}{3}$ back to y gives

$$y = -\frac{i\sqrt{3}x}{3}$$

Converting $u(x) = \frac{i\sqrt{3}}{3}$ back to y gives

$$y = \frac{i\sqrt{3}x}{3}$$

Simplifying the above gives

$$\left(\frac{x^2 + 3y^2}{x^2}\right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

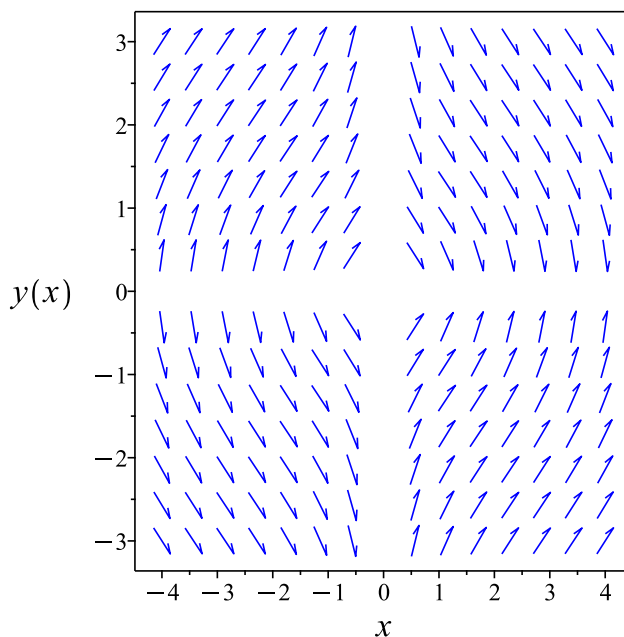
Solving for y gives

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

Figure 2.208: Slope field $x^2 + y^2 + 2xyy' = 0$ Summary of solutions found

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3} \sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(c_1^6 - x^3)}}{3x}$$

Solved using first_order_ode_homog_type_G

Time used: 0.088 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Multiplying the right side of the ode, which is $-\frac{x^2+y^2}{2xy}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{x^2 + y^2}{2xy} \\ &= -\frac{x^2 + y^2}{2y^2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x^2}{y^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x^2}{y^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{z^2 + 1}{2z^2}$$

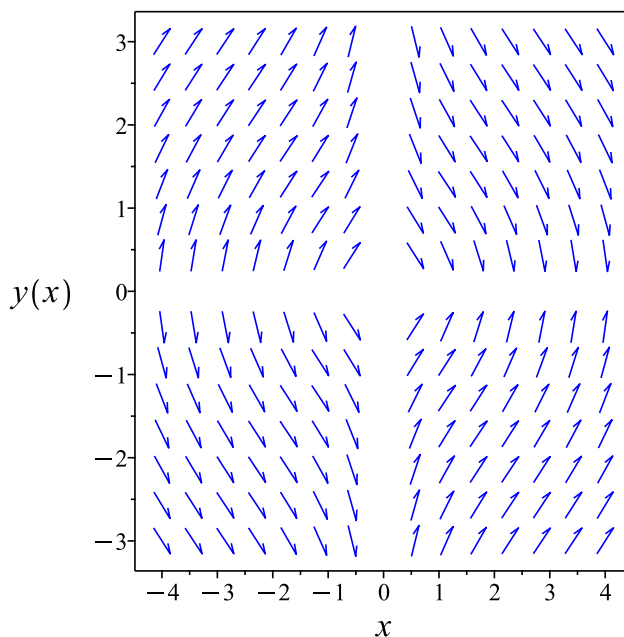
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{z^2 + 1}{2z^2} \right)} dz = 0$$

Figure 2.209: Slope field $x^2 + y^2 + 2xyy' = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 + \frac{z^2+1}{2z^2}\right)} dz = 0$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.004 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{(X+x_0)^2 + (Y(X)+y_0)^2}{2(X+x_0)(Y(X)+y_0)}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 0$$

$$y_0 = 0$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X^2 + Y(X)^2}{2XY(X)}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X^2 + Y^2}{2XY} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X, Y)}{N(X, Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X^2 - Y^2$ and $N = 2XY$ are both homogeneous and of the same order $n = 2$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= -\frac{1}{2u} - \frac{u}{2} \\ \frac{du}{dX} &= \frac{-\frac{1}{2u(X)} - \frac{3u(X)}{2}}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-\frac{1}{2u(X)} - \frac{3u(X)}{2}}{X} = 0$$

Or

$$2\left(\frac{d}{dX}u(X)\right)u(X)X + 3u(X)^2 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{3u(X)^2 + 1}{2u(X)X} \quad (2.96)$$

is separable as it can be written as

$$\begin{aligned} \frac{d}{dX}u(X) &= -\frac{3u(X)^2 + 1}{2u(X)X} \\ &= f(X)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(X) &= -\frac{1}{2X} \\ g(u) &= \frac{3u^2 + 1}{u} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(X) dX$$

$$\int \frac{u}{3u^2 + 1} du = \int -\frac{1}{2X} dX$$

$$\frac{\ln(3u(X)^2 + 1)}{6} = \ln\left(\frac{1}{\sqrt{X}}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$(3u(X)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{X}}$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{3u^2 + 1}{u} = 0$$

for $u(X)$ gives

$$u(X) = -\frac{i\sqrt{3}}{3}$$

$$u(X) = \frac{i\sqrt{3}}{3}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$(3u(X)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{X}}$$

$$u(X) = -\frac{i\sqrt{3}}{3}$$

$$u(X) = \frac{i\sqrt{3}}{3}$$

Converting $(3u(X)^2 + 1)^{1/6} = \frac{c_1}{\sqrt{X}}$ back to $Y(X)$ gives

$$\left(\frac{3Y(X)^2 + X^2}{X^2}\right)^{1/6} = \frac{c_1}{\sqrt{X}}$$

Converting $u(X) = -\frac{i\sqrt{3}}{3}$ back to $Y(X)$ gives

$$Y(X) = -\frac{iX\sqrt{3}}{3}$$

Converting $u(X) = \frac{i\sqrt{3}}{3}$ back to $Y(X)$ gives

$$Y(X) = \frac{iX\sqrt{3}}{3}$$

Using the solution for $Y(X)$

$$\left(\frac{3Y(X)^2 + X^2}{X^2} \right)^{1/6} = \frac{c_1}{\sqrt{X}} \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$\left(\frac{x^2 + 3y^2}{x^2} \right)^{1/6} = \frac{c_1}{\sqrt{x}}$$

Using the solution for $Y(X)$

$$Y(X) = -\frac{iX\sqrt{3}}{3} \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = -\frac{i\sqrt{3}x}{3}$$

Using the solution for $Y(X)$

$$Y(X) = \frac{iX\sqrt{3}}{3} \quad (\text{A})$$

And replacing back terms in the above solution using

$$\begin{aligned} Y &= y + y_0 \\ X &= x_0 + x \end{aligned}$$

Or

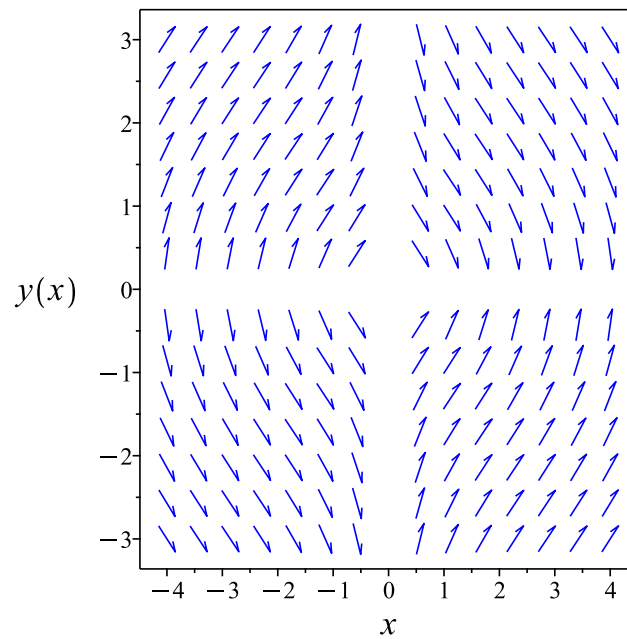
$$\begin{aligned} Y &= y \\ X &= x \end{aligned}$$

Then the solution in y becomes using EQ (A)

$$y = \frac{i\sqrt{3}x}{3}$$

Solving for y gives

$$\begin{aligned} y &= -\frac{i\sqrt{3}x}{3} \\ y &= \frac{i\sqrt{3}x}{3} \\ y &= -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x} \\ y &= \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x} \end{aligned}$$

Figure 2.210: Slope field $x^2 + y^2 + 2xyy' = 0$ Summary of solutions found

$$y = -\frac{i\sqrt{3}x}{3}$$

$$y = \frac{i\sqrt{3}x}{3}$$

$$y = -\frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

$$y = \frac{\sqrt{3}\sqrt{x(c_1^6 - x^3)}}{3x}$$

Solved using first_order_ode_LIE

Time used: 0.942 (sec)

Solve

$$x^2 + y^2 + 2xyy' = 0$$

Writing the ode as

$$y' = -\frac{x^2 + y^2}{2xy}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 - \frac{(x^2 + y^2)(b_3 - a_2)}{2xy} - \frac{(x^2 + y^2)^2 a_3}{4x^2 y^2} - \left(-\frac{1}{y} + \frac{x^2 + y^2}{2x^2 y}\right)(xa_2 + ya_3 + a_1) \\ - \left(-\frac{1}{x} + \frac{x^2 + y^2}{2x y^2}\right)(xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5\text{E})$$

Putting the above in normal form gives

$$-\frac{x^4 a_3 + 2x^4 b_2 - 4x^3 y a_2 + 4x^3 y b_3 - 6b_2 y^2 x^2 + 3y^4 a_3 + 2x^3 b_1 - 2x^2 y a_1 - 2x y^2 b_1 + 2y^3 a_1}{4y^2 x^2} = 0$$

Setting the numerator to zero gives

$$-x^4 a_3 - 2x^4 b_2 + 4x^3 y a_2 - 4x^3 y b_3 + 6b_2 y^2 x^2 - 3y^4 a_3 - 2x^3 b_1 + 2x^2 y a_1 + 2x y^2 b_1 - 2y^3 a_1 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$4a_2 v_1^3 v_2 - a_3 v_1^4 - 3a_3 v_2^4 - 2b_2 v_1^4 + 6b_2 v_1^2 v_2^2 - 4b_3 v_1^3 v_2 + 2a_1 v_1^2 v_2 - 2a_1 v_2^3 - 2b_1 v_1^3 + 2b_1 v_1 v_2^2 = 0 \quad (7\text{E})$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(-a_3 - 2b_2)v_1^4 + (4a_2 - 4b_3)v_1^3v_2 - 2b_1v_1^3 + 6b_2v_1^2v_2^2 + 2a_1v_1^2v_2 + 2b_1v_1v_2^2 - 3a_3v_2^4 - 2a_1v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -2a_1 &= 0 \\ 2a_1 &= 0 \\ -3a_3 &= 0 \\ -2b_1 &= 0 \\ 2b_1 &= 0 \\ 6b_2 &= 0 \\ 4a_2 - 4b_3 &= 0 \\ -a_3 - 2b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{x^2 + y^2}{2xy} \right) (x) \\ &= y + \frac{x^2 + y^2}{2y} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y + \frac{x^2+y^2}{2y}} dy \end{aligned}$$

Which results in

$$S = \frac{\ln(x^2 + 3y^2)}{3}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x^2 + y^2}{2xy}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{2x}{3x^2 + 9y^2} \\ S_y &= \frac{2y}{x^2 + 3y^2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{3R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{3R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

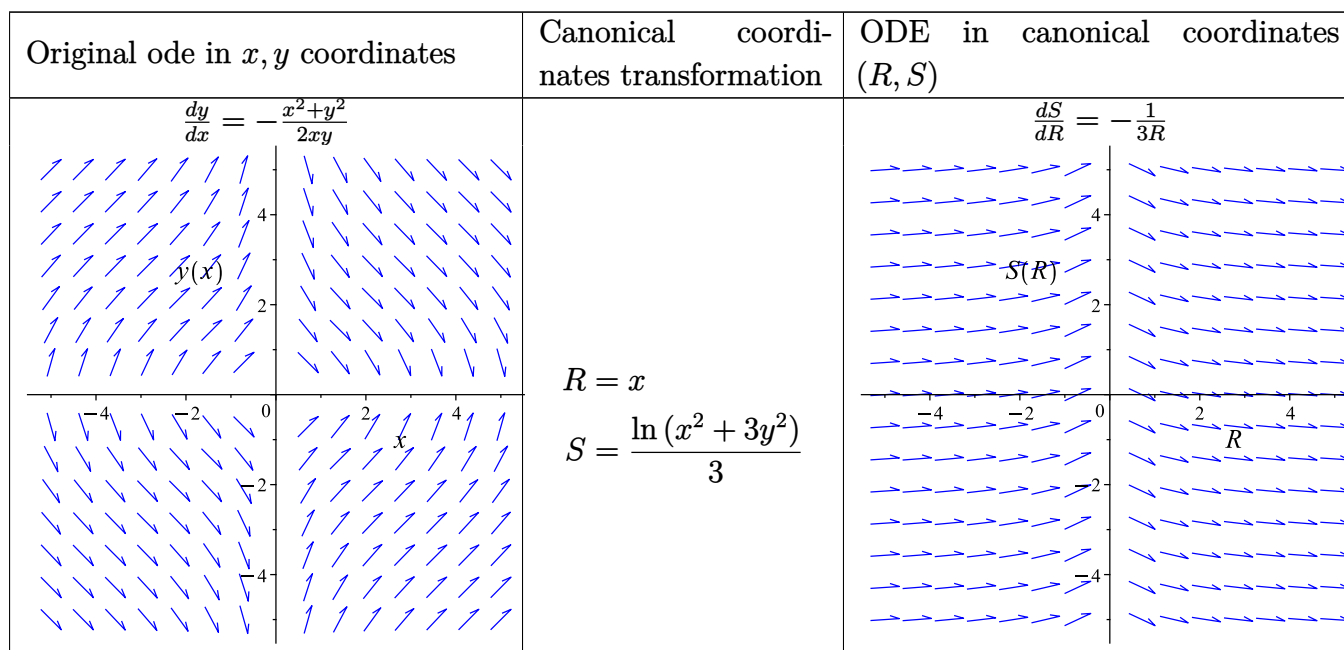
$$\int dS = \int -\frac{1}{3R} dR$$

$$S(R) = -\frac{\ln(R)}{3} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{\ln(x^2 + 3y^2)}{3} = -\frac{\ln(x)}{3} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{\sqrt{3} \sqrt{x(e^{3c_2} - x^3)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(e^{3c_2} - x^3)}}{3x}$$

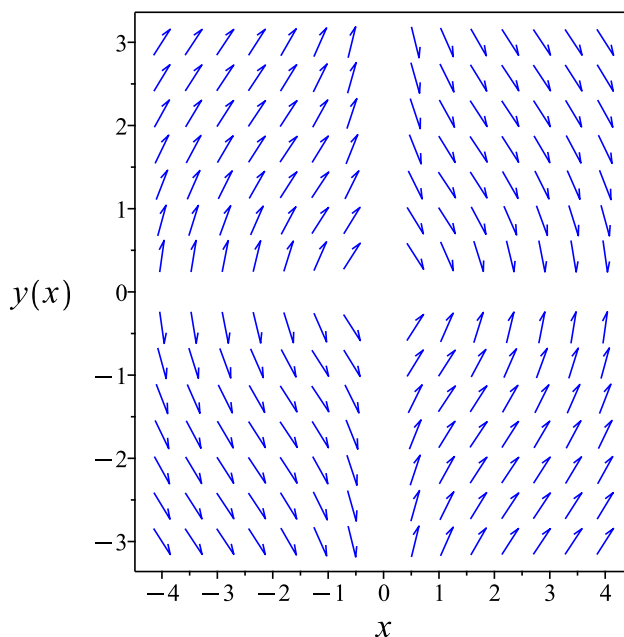


Figure 2.211: Slope field $x^2 + y^2 + 2xyy' = 0$

Summary of solutions found

$$y = -\frac{\sqrt{3} \sqrt{x(e^{3c_2} - x^3)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{x(e^{3c_2} - x^3)}}{3x}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 47

```
ode:=x^2+y(x)^2+2*x*y(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{3} \sqrt{-x(x^3 - 3c_1)}}{3x}$$

$$y = \frac{\sqrt{3} \sqrt{-x(x^3 - 3c_1)}}{3x}$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$x^2 + y(x)^2 + 2xy(x) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$ Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{dx}G(x, y(x)) = 0$
 - Compute derivative of lhs
 - $\frac{\partial}{\partial x}G(x, y) + \left(\frac{\partial}{\partial y}G(x, y) \right) \left(\frac{d}{dx}y(x) \right) = 0$
 - Evaluate derivatives
 - $2y = 2y$
 - Condition met, ODE is exact
- Exact ODE implies solution will be of this form

$$\left[G(x, y) = C1, M(x, y) = \frac{\partial}{\partial x}G(x, y), N(x, y) = \frac{\partial}{\partial y}G(x, y) \right]$$
- Solve for $G(x, y)$ by integrating $M(x, y)$ with respect to x

$$G(x, y) = \int (x^2 + y^2) dx + _F1(y)$$
- Evaluate integral

$$G(x, y) = \frac{x^3}{3} + y^2x + _F1(y)$$
- Take derivative of $G(x, y)$ with respect to y

$$N(x, y) = \frac{\partial}{\partial y}G(x, y)$$
- Compute derivative

$$2yx = 2yx + \frac{d}{dy}_F1(y)$$
- Isolate for $\frac{d}{dy}_F1(y)$

$$\frac{d}{dy}_F1(y) = 0$$
- Solve for $_F1(y)$

$$_F1(y) = 0$$
- Substitute $_F1(y)$ into equation for $G(x, y)$

$$G(x, y) = \frac{1}{3}x^3 + y^2x$$
- Substitute $G(x, y)$ into the solution of the ODE

$$\frac{1}{3}x^3 + y^2x = C1$$

- Solve for $y(x)$

$$\left\{ y(x) = -\frac{\sqrt{3}\sqrt{x(-x^3+3C1)}}{3x}, y(x) = \frac{\sqrt{3}\sqrt{x(-x^3+3C1)}}{3x} \right\}$$

✓ **Mathematica.** Time used: 0.138 (sec). Leaf size: 60

```
ode=(x^2+y[x]^2)+2*x*y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{-x^3+3c_1}}{\sqrt{3}\sqrt{x}}$$

$$y(x) \rightarrow \frac{\sqrt{-x^3+3c_1}}{\sqrt{3}\sqrt{x}}$$

✓ **Sympy.** Time used: 0.303 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2 + 2*x*y(x)*Derivative(y(x), x) + y(x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\frac{\sqrt{3}\sqrt{\frac{C_1}{x} - x^2}}{3}, y(x) = \frac{\sqrt{3}\sqrt{\frac{C_1}{x} - x^2}}{3} \right]$$

2.3.14 Problem 24 (g)

Local contents

Solved using first_order_ode_quadrature	879
Solved using first_order_ode_exact	880
✓ Maple	883
✓ Mathematica	884
✓ Sympy	884

Internal problem ID [7894]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 24 (g)

Date solved : Thursday, November 27, 2025 at 06:29:04 AM

CAS classification : [_quadrature]

Solved using first_order_ode_quadrature

Time used: 0.114 (sec)

Solve

$$1 - \sqrt{a^2 - x^2} y' = 0$$

Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \frac{1}{\sqrt{a^2 - x^2}} dx$$

$$y = \arctan \left(\frac{x}{\sqrt{a^2 - x^2}} \right) + c_1$$

Summary of solutions found

$$y = \arctan \left(\frac{x}{\sqrt{a^2 - x^2}} \right) + c_1$$

Solved using first_order_ode_exact

Time used: 0.154 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} & \left(-\sqrt{a^2 - x^2} \right) dy = (-1) dx \\ (1) dx + & \left(-\sqrt{a^2 - x^2} \right) dy = 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 1 \\ N(x, y) &= -\sqrt{a^2 - x^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(1) \\ &= 0\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}\left(-\sqrt{a^2 - x^2}\right) \\ &= \frac{x}{\sqrt{a^2 - x^2}}\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) \\ &= -\frac{1}{\sqrt{a^2 - x^2}}\left(0 - \left(\frac{x}{\sqrt{a^2 - x^2}}\right)\right) \\ &= \frac{x}{a^2 - x^2}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{x}{a^2 - x^2} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-\frac{\ln(a^2 - x^2)}{2}} \\ &= \frac{1}{\sqrt{a^2 - x^2}}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= \frac{1}{\sqrt{a^2 - x^2}}(1) \\ &= \frac{1}{\sqrt{a^2 - x^2}}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{\sqrt{a^2 - x^2}} \left(-\sqrt{a^2 - x^2} \right) \\ &= -1\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1}{\sqrt{a^2 - x^2}} \right) + (-1) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \bar{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{1}{\sqrt{a^2 - x^2}} dx \\ \phi &= \arctan \left(\frac{x}{\sqrt{a^2 - x^2}} \right) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 0 + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -1$. Therefore equation (4) becomes

$$-1 = 0 + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -1$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int (-1) dy$$

$$f(y) = -y + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right) - y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right) - y$$

Solving for y gives

$$y = -c_1 + \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right)$$

Summary of solutions found

$$y = -c_1 + \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right)$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 20

```
ode:=1-(a^2-x^2)^(1/2)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right) + c_1$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
<- quadrature successful
```

Maple step by step

Let's solve

$$1 - \sqrt{a^2 - x^2} \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{1}{\sqrt{a^2 - x^2}}$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx} y(x) \right) dx = \int \frac{1}{\sqrt{a^2 - x^2}} dx + C1$$

- Evaluate integral

$$y(x) = \arctan \left(\frac{x}{\sqrt{a^2 - x^2}} \right) + C1$$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 24

```
ode=1-(Sqrt[a^2-x^2])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \arctan \left(\frac{x}{\sqrt{a^2 - x^2}} \right) + c_1$$

✓ **Sympy.** Time used: 0.437 (sec). Leaf size: 41

```
from sympy import *
x = symbols("x")
a = symbols("a")
y = Function("y")
ode = Eq(-sqrt(a**2 - x**2)*Derivative(y(x), x) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 + \begin{cases} -i \log(-2x + 2i\sqrt{a^2 - x^2}) & \text{for } a^2 \neq 0 \\ \frac{x \log(x)}{\sqrt{-x^2}} & \text{otherwise} \end{cases}$$

2.3.15 Problem 24 (L)

Local contents

Solved using first_order_ode_dAlembert	885
Solved using first_order_ode_homog_type_maple_C	888
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	894
Solved using first_order_ode_LIE	903
✓ Maple	908
✓ Mathematica	908
✓ Sympy	909

Internal problem ID [7895]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 24 (L)

Date solved : Thursday, November 27, 2025 at 06:29:09 AM

CAS classification :

[[_homogeneous, 'class C'], _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_dAlembert

Time used: 0.424 (sec)

Solve

$$x + y + 1 - (-y + x - 3)y' = 0$$

Let $p = y'$ the ode becomes

$$x + y + 1 - (-y + x - 3)p = 0$$

Solving for y from the above results in

$$y = \frac{(p-1)x}{1+p} + \frac{-3p-1}{1+p} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{p-1}{1+p} \\ g &= \frac{-3p-1}{1+p} \end{aligned}$$

Hence (2) becomes

$$p - \frac{p-1}{1+p} = \left(\frac{x}{1+p} - \frac{xp}{(1+p)^2} + \frac{x}{(1+p)^2} - \frac{3}{1+p} + \frac{3p}{(1+p)^2} + \frac{1}{(1+p)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p-1}{1+p} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= i \\ p_2 &= -i \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= ix - i - 2 \\ y &= -ix + i - 2 \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)-1}{1+p(x)}}{\frac{x}{1+p(x)} - \frac{xp(x)}{(1+p(x))^2} + \frac{x}{(1+p(x))^2} - \frac{3}{1+p(x)} + \frac{3p(x)}{(1+p(x))^2} + \frac{1}{(1+p(x))^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{(1+p(x))(p(x)^2+1)}{2x-2} \quad (2.97)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{(1+p(x))(p(x)^2+1)}{2x-2} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{2x-2} \\ g(p) &= (1+p)(p^2+1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{(1+p)(p^2+1)} dp &= \int \frac{1}{2x-2} dx \end{aligned}$$

$$\ln \left(\frac{\sqrt{1+p(x)}}{(p(x)^2+1)^{1/4}} \right) + \frac{\arctan(p(x))}{2} = \ln(\sqrt{x-1}) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$(1+p)(p^2+1) = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= -1 \\ p(x) &= -i \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \ln \left(\frac{\sqrt{1+p(x)}}{(p(x)^2+1)^{1/4}} \right) + \frac{\arctan(p(x))}{2} &= \ln(\sqrt{x-1}) + c_1 \\ p(x) &= -1 \\ p(x) &= -i \end{aligned}$$

Substituting the above solution for p in (2A) gives

$$y = -i(x - 1 - 2i)$$

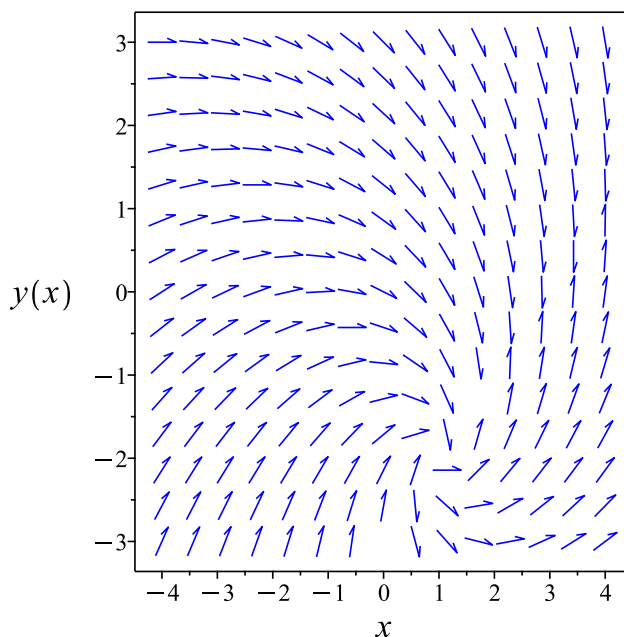


Figure 2.212: Slope field $x + y + 1 - (-y + x - 3)y' = 0$

Summary of solutions found

$$y = -ix + i - 2$$

$$y = ix - i - 2$$

Solved using first_order_ode_homog_type_maple_C

Time used: 1.000 (sec)

Solve

$$x + y + 1 - (-y + x - 3)y' = 0$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = -\frac{X + x_0 + Y(X) + y_0 + 1}{Y(X) + y_0 - X - x_0 + 3}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$x_0 = 1$$

$$y_0 = -2$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = -\frac{X+Y(X)}{Y(X)-X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= -\frac{X+Y}{Y-X} \end{aligned} \quad (1)$$

An ode of the form $Y' = \frac{M(X,Y)}{N(X,Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = X+Y$ and $N = -Y+X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned} \frac{du}{dX}X + u &= \frac{-u-1}{u-1} \\ \frac{du}{dX} &= \frac{\frac{-u(X)-1}{u(X)-1} - u(X)}{X} \end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{\frac{-u(X)-1}{u(X)-1} - u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)Xu(X) - \left(\frac{d}{dX}u(X)\right)X + u(X)^2 + 1 = 0$$

Or

$$X(u(X) - 1) \left(\frac{d}{dX}u(X)\right) + u(X)^2 + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = -\frac{u(X)^2 + 1}{X(u(X) - 1)} \quad (2.98)$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= -\frac{u(X)^2 + 1}{X(u(X) - 1)} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= -\frac{1}{X} \\ g(u) &= \frac{u^2 + 1}{u - 1}\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{u - 1}{u^2 + 1} du &= \int -\frac{1}{X} dX\end{aligned}$$

$$\frac{\ln(u(X)^2 + 1)}{2} - \arctan(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$\frac{u^2 + 1}{u - 1} = 0$$

for $u(X)$ gives

$$\begin{aligned}u(X) &= -i \\ u(X) &= i\end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(u(X)^2 + 1)}{2} - \arctan(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$$

$$u(X) = -i$$

$$u(X) = i$$

Converting $\frac{\ln(u(X)^2+1)}{2} - \arctan(u(X)) = \ln\left(\frac{1}{X}\right) + c_1$ back to $Y(X)$ gives

$$\frac{\ln\left(\frac{Y(X)^2+X^2}{X^2}\right)}{2} - \arctan\left(\frac{Y(X)}{X}\right) = \ln\left(\frac{1}{X}\right) + c_1$$

Converting $u(X) = -i$ back to $Y(X)$ gives

$$Y(X) = -iX$$

Converting $u(X) = i$ back to $Y(X)$ gives

$$Y(X) = iX$$

Using the solution for $Y(X)$

$$\frac{\ln\left(\frac{Y(X)^2+X^2}{X^2}\right)}{2} - \arctan\left(\frac{Y(X)}{X}\right) = \ln\left(\frac{1}{X}\right) + c_1 \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - 2$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$\frac{\ln\left(\frac{(y+2)^2+(x-1)^2}{(x-1)^2}\right)}{2} - \arctan\left(\frac{y+2}{x-1}\right) = \ln\left(\frac{1}{x-1}\right) + c_1$$

Using the solution for $Y(X)$

$$Y(X) = -iX \quad (\text{A})$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - 2$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$y + 2 = -i(x - 1)$$

Using the solution for $Y(X)$

$$Y(X) = iX \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - 2$$

$$X = x + 1$$

Then the solution in y becomes using EQ (A)

$$y + 2 = i(x - 1)$$

Simplifying the above gives

$$\frac{\ln \left(\frac{(y+2)^2 + (x-1)^2}{(x-1)^2} \right)}{2} - \arctan \left(\frac{y+2}{x-1} \right) = \ln \left(\frac{1}{x-1} \right) + c_1$$

$$y + 2 = -ix + i$$

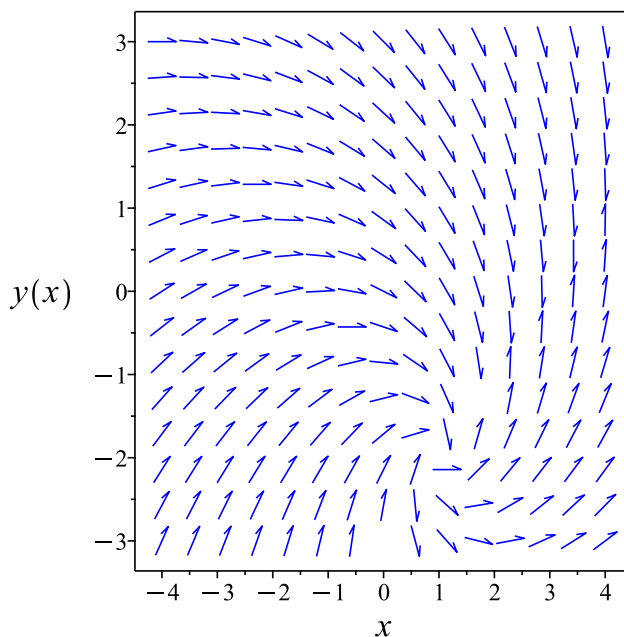
$$y + 2 = i(x - 1)$$

Solving for y gives

$$\frac{\ln \left(\frac{(y+2)^2 + (x-1)^2}{(x-1)^2} \right)}{2} - \arctan \left(\frac{y+2}{x-1} \right) = \ln \left(\frac{1}{x-1} \right) + c_1$$

$$y = -ix + i - 2$$

$$y = ix - i - 2$$

Figure 2.213: Slope field $x + y + 1 - (-y + x - 3)y' = 0$ Summary of solutions found

$$\frac{\ln \left(\frac{(y+2)^2 + (x-1)^2}{(x-1)^2} \right)}{2} - \arctan \left(\frac{y+2}{x-1} \right) = \ln \left(\frac{1}{x-1} \right) + c_1$$

$$y = -ix + i - 2$$

$$y = ix - i - 2$$

Solve

$$x + y + 1 - (-y + x - 3)y' = 0$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$x + xu(x) + 1 - (-xu(x) + x - 3)(u(x) + xu'(x)) = 0$$

Which is now solved Unknown ode type.

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 6.108 (sec)

This is Abel second kind ODE, it has the form

$$(y(x) + g) y'(x) = f_0(x) + f_1(x)y(x) + f_2(x)y(x)^2 + f_3(x)y(x)^3$$

Comparing the above to given ODE which is

$$x + y(x) + 1 - (-y(x) + x - 3) y'(x) = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= -x + 3 \\ f_0 &= -1 - x \\ f_1 &= -1 \\ f_2 &= 0 \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y(x) = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = (2x - 2) u(x)^3 + 2u(x)^2$$

Which is now solved.

Solve This is Abel first kind ODE, it has the form

$$u'(x) = f_0(x) + f_1(x)u(x) + f_2(x)u(x)^2 + f_3(x)u(x)^3$$

Comparing the above to given ODE which is

$$u'(x) = (2x - 2) u(x)^3 + 2u(x)^2 \quad (1)$$

Therefore

$$\begin{aligned} f_0 &= 0 \\ f_1 &= 0 \\ f_2 &= 2 \\ f_3 &= 2x - 2 \end{aligned}$$

Hence

$$\begin{aligned} f'_0 &= 0 \\ f'_3 &= 2 \end{aligned}$$

Since $f_2(x) = 2$ is not zero, then the following transformation is used to remove f_2 . Let $u(x) = u(x) - \frac{f_2}{3f_3}$ or

$$\begin{aligned} u(x) &= u(x) - \left(\frac{2}{6x-6} \right) \\ &= u(x) - \frac{1}{3x-3} \end{aligned}$$

The above transformation applied to (1) gives a new ODE as

$$u'(x) = \frac{(54x^3 - 162x^2 + 162x - 54)u(x)^3}{27(x-1)^2} + \frac{(-18x + 18)u(x)}{27(x-1)^2} - \frac{5}{27(x-1)^2} \quad (2)$$

The above ODE (2) can now be solved.

Solve Writing the ode as

$$\begin{aligned} u'(x) &= \frac{54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5}{27(x-1)^2} \\ u'(x) &= \omega(x, u(x)) \end{aligned}$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_{u(x)} - \xi_x) - \omega^2 \xi_{u(x)} - \omega_x \xi - \omega_{u(x)} \eta = 0 \quad (A)$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = ua_3 + xa_2 + a_1 \quad (1E)$$

$$\eta = ub_3 + xb_2 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned}
 & b_2 + \frac{(54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5)(b_3 - a_2)}{27(x-1)^2} \\
 & - \frac{(54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5)^2 a_3}{729(x-1)^4} \\
 & - \left(\frac{162u^3x^2 - 324u^3x + 162u^3 - 18u}{27(x-1)^2} \right. \\
 & \left. - \frac{2(54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5)}{27(x-1)^3} \right) (ua_3 + xa_2 + a_1) \\
 & - \frac{(162u^2x^3 - 486u^2x^2 + 486u^2x - 162u^2 - 18x + 18)(ub_3 + xb_2 + b_1)}{27(x-1)^2} = 0
 \end{aligned} \tag{5E}$$

Putting the above in normal form gives

$$\begin{aligned}
 & \frac{2916u^6x^6a_3 - 17496u^6x^5a_3 + 43740u^6x^4a_3 - 58320u^6x^3a_3 + 43740u^6x^2a_3 - 486u^4x^4a_3 + 2916u^3x^5a_2 + 2916u^3x^4a_1}{27(x-1)^2} \\
 & = 0
 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned}
 & -2916u^6x^6a_3 + 17496u^6x^5a_3 - 43740u^6x^4a_3 + 58320u^6x^3a_3 \\
 & - 43740u^6x^2a_3 + 486u^4x^4a_3 - 2916u^3x^5a_2 - 2916u^3x^4a_1 \\
 & - 4374u^2x^6b_2 + 17496u^6xa_3 - 1944u^4x^3a_3 - 1458u^3x^4a_1 \\
 & + 13122u^3x^4a_2 + 14580u^3x^4b_3 - 4374u^2x^5b_1 + 21870u^2x^5b_2 \\
 & - 2916u^6a_3 + 2916u^4x^2a_3 + 5832u^3x^3a_1 - 23328u^3x^3a_2 + 540u^3x^3a_3 \\
 & - 29160u^3x^3b_3 + 21870u^2x^4b_1 - 43740u^2x^4b_2 - 1944u^4xa_3 \\
 & - 8748u^3x^2a_1 + 20412u^3x^2a_2 - 1620u^3x^2a_3 + 29160u^3x^2b_3 \\
 & - 43740u^2x^3b_1 + 43740u^2x^3b_2 + 486u^4a_3 + 5832u^3xa_1 - 8748u^3xa_2 \\
 & + 1620u^3xa_3 - 14580u^3xb_3 - 810u^2x^2a_3 + 43740u^2x^2b_1 \\
 & - 21870u^2x^2b_2 + 1215b_2x^4 - 1458u^3a_1 + 1458u^3a_2 - 540u^3a_3 \\
 & + 2916u^3b_3 + 1620u^2xa_3 - 21870u^2xb_1 + 4374u^2xb_2 - 486u^2xa_1 \\
 & - 486u^2xa_2 + 486x^3b_1 - 4374b_2x^3 - 810u^2a_3 + 4374u^2b_1 + 972uxa_1 \\
 & + 972uxa_2 - 450uxa_3 - 135x^2a_2 - 1458x^2b_1 + 5832b_2x^2 - 135x^2b_3 \\
 & - 486ua_1 - 486ua_2 + 450ua_3 - 270xa_1 + 1458xb_1 - 3402xb_2 \\
 & + 270xb_3 + 270a_1 + 135a_2 - 25a_3 - 486b_1 + 729b_2 - 135b_3 = 0
 \end{aligned} \tag{6E}$$

Looking at the above PDE shows the following are all the terms with $\{u, x\}$ in them.

$$\{u, x\}$$

The following substitution is now made to be able to collect on all terms with $\{u, x\}$ in them

$$\{u = v_1, x = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
& -2916a_3v_1^6v_2^6 + 17496a_3v_1^6v_2^5 - 43740a_3v_1^6v_2^4 + 58320a_3v_1^6v_2^3 \\
& - 2916a_2v_1^3v_2^5 - 43740a_3v_1^6v_2^2 + 486a_3v_1^4v_2^4 - 4374b_2v_1^2v_2^6 \\
& - 2916b_3v_1^3v_2^5 - 1458a_1v_1^3v_2^4 + 13122a_2v_1^3v_2^4 + 17496a_3v_1^6v_2 \\
& - 1944a_3v_1^4v_2^3 - 4374b_1v_1^2v_2^5 + 21870b_2v_1^2v_2^5 + 14580b_3v_1^3v_2^4 \\
& + 5832a_1v_1^3v_2^3 - 23328a_2v_1^3v_2^3 - 2916a_3v_1^6 + 2916a_3v_1^4v_2^2 + 540a_3v_1^3v_2^3 \\
& + 21870b_1v_1^2v_2^4 - 43740b_2v_1^2v_2^4 - 29160b_3v_1^3v_2^3 - 8748a_1v_1^3v_2^2 \\
& + 20412a_2v_1^3v_2^2 - 1944a_3v_1^4v_2 - 1620a_3v_1^3v_2^2 - 43740b_1v_1^2v_2^3 \\
& + 43740b_2v_1^2v_2^3 + 29160b_3v_1^3v_2^2 + 5832a_1v_1^3v_2 - 8748a_2v_1^3v_2 + 486a_3v_1^4 \\
& + 1620a_3v_1^3v_2 - 810a_3v_1^2v_2^2 + 43740b_1v_1^2v_2^2 - 21870b_2v_1^2v_2^2 + 1215b_2v_2^4 \\
& - 14580b_3v_1^3v_2 - 1458a_1v_1^3 - 486a_1v_1v_2^2 + 1458a_2v_1^3 - 486a_2v_1v_2^2 \\
& - 540a_3v_1^3 + 1620a_3v_1^2v_2 - 21870b_1v_1^2v_2 + 486b_1v_2^3 + 4374b_2v_1^2v_2 \\
& - 4374b_2v_2^3 + 2916b_3v_1^3 + 972a_1v_1v_2 + 972a_2v_1v_2 - 135a_2v_2^2 \\
& - 810a_3v_1^2 - 450a_3v_1v_2 + 4374b_1v_1^2 - 1458b_1v_2^2 + 5832b_2v_2^2 - 135b_3v_2^2 \\
& - 486a_1v_1 - 270a_1v_2 - 486a_2v_1 + 450a_3v_1 + 1458b_1v_2 - 3402b_2v_2 \\
& + 270b_3v_2 + 270a_1 + 135a_2 - 25a_3 - 486b_1 + 729b_2 - 135b_3 = 0
\end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned}
& (-2916a_2 - 2916b_3)v_1^3v_2^5 + (-1458a_1 + 13122a_2 + 14580b_3)v_1^3v_2^4 \\
& + (5832a_1 - 23328a_2 + 540a_3 - 29160b_3)v_1^3v_2^3 + (-8748a_1 + 20412a_2 - 1620a_3 + 29160b_3)v_1^3v_2^2 \\
& + (5832a_1 - 8748a_2 + 1620a_3 - 14580b_3)v_1^3v_2 + (-4374b_1 + 21870b_2)v_1^2v_2^5 \\
& + (21870b_1 - 43740b_2)v_1^2v_2^4 + (-43740b_1 + 43740b_2)v_1^2v_2^3 + (-810a_3 + 43740b_1 - 21870b_2)v_1^2v_2^2 \\
& + (1620a_3 - 21870b_1 + 4374b_2)v_1^2v_2 + (-486a_1 - 486a_2)v_1v_2^2 \\
& + (972a_1 + 972a_2 - 450a_3)v_1v_2 - 2916a_3v_1^6 + 486a_3v_1^4 + 1215b_2v_2^4 + 270a_1 \\
& + 135a_2 - 25a_3 - 486b_1 + 729b_2 - 135b_3 + (-1458a_1 + 1458a_2 - 540a_3 + 2916b_3)v_1^3 \\
& + (-810a_3 + 4374b_1)v_1^2 + (-486a_1 - 486a_2 + 450a_3)v_1 + (486b_1 - 4374b_2)v_2^3 \\
& + (-135a_2 - 1458b_1 + 5832b_2 - 135b_3)v_2^2 + (-270a_1 + 1458b_1 - 3402b_2 + 270b_3)v_2 \\
& - 2916a_3v_1^6v_2^6 + 17496a_3v_1^6v_2^5 - 43740a_3v_1^6v_2^4 + 58320a_3v_1^6v_2^3 - 43740a_3v_1^6v_2^2 + 486a_3v_1^4v_2^4 \\
& - 4374b_2v_1^2v_2^6 + 17496a_3v_1^6v_2 - 1944a_3v_1^4v_2^3 + 2916a_3v_1^4v_2^2 - 1944a_3v_1^4v_2 = 0
\end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -43740a_3 &= 0 \\ -2916a_3 &= 0 \\ -1944a_3 &= 0 \\ 486a_3 &= 0 \\ 2916a_3 &= 0 \\ 17496a_3 &= 0 \\ 58320a_3 &= 0 \\ -4374b_2 &= 0 \\ 1215b_2 &= 0 \\ -486a_1 - 486a_2 &= 0 \\ -2916a_2 - 2916b_3 &= 0 \\ -810a_3 + 4374b_1 &= 0 \\ -43740b_1 + 43740b_2 &= 0 \\ -4374b_1 + 21870b_2 &= 0 \\ 486b_1 - 4374b_2 &= 0 \\ 21870b_1 - 43740b_2 &= 0 \\ -1458a_1 + 13122a_2 + 14580b_3 &= 0 \\ -486a_1 - 486a_2 + 450a_3 &= 0 \\ 972a_1 + 972a_2 - 450a_3 &= 0 \\ -810a_3 + 43740b_1 - 21870b_2 &= 0 \\ 1620a_3 - 21870b_1 + 4374b_2 &= 0 \\ -8748a_1 + 20412a_2 - 1620a_3 + 29160b_3 &= 0 \\ -1458a_1 + 1458a_2 - 540a_3 + 2916b_3 &= 0 \\ -270a_1 + 1458b_1 - 3402b_2 + 270b_3 &= 0 \\ 5832a_1 - 23328a_2 + 540a_3 - 29160b_3 &= 0 \\ 5832a_1 - 8748a_2 + 1620a_3 - 14580b_3 &= 0 \\ -135a_2 - 1458b_1 + 5832b_2 - 135b_3 &= 0 \\ 270a_1 + 135a_2 - 25a_3 - 486b_1 + 729b_2 - 135b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}a_1 &= b_3 \\a_2 &= -b_3 \\a_3 &= 0 \\b_1 &= 0 \\b_2 &= 0 \\b_3 &= b_3\end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}\xi &= 1 - x \\ \eta &= u\end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, u) \xi \\ &= u - \left(\frac{54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5}{27(x-1)^2} \right) (1-x) \\ &= \frac{-5 + 54(x-1)^3u^3 + (9x-9)u}{27x-27} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, u) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{du}{\eta} = dS \quad (1)$$

The above comes from the requirements that $(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial u}) S(x, u) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{-5 + 54(x-1)^3u^3 + (9x-9)u}{27x-27}} dy\end{aligned}$$

Which results in

$$S = (27x - 27) \left(\frac{(-6x + 6) \ln(18u^2x^2 - 36u^2x + 18u^2 + 6ux - 6u + 5)}{324x^2 - 648x + 324} + \frac{2 \left(-4 - \frac{(-6x+6)(6x-6)}{2(18x^2-36x+18)} \right) \arctan \left(\frac{u}{x-1} \right)}{9(18x - 18)} \right)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, u)S_u}{R_x + \omega(x, u)R_u} \quad (2)$$

Where in the above R_x, R_u, S_x, S_u are all partial derivatives and $\omega(x, u)$ is the right hand side of the original ode given by

$$\omega(x, u) = \frac{54u^3x^3 - 162u^3x^2 + 162u^3x - 54u^3 - 18ux + 18u - 5}{27(x-1)^2}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_u &= 0 \\ S_x &= \frac{u}{2 \left(-\frac{1}{3} + u(x-1) \right) \left(\frac{5}{18} + (x-1)^2 u^2 + \frac{u(x-1)}{3} \right)} \\ S_u &= \frac{x-1}{2 \left(-\frac{1}{3} + u(x-1) \right) \left(\frac{5}{18} + (x-1)^2 u^2 + \frac{u(x-1)}{3} \right)} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{x-1} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, u in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R-1}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{1}{R-1} dR \\ S(R) &= \ln(R-1) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, u coordinates. This results in

$$-\arctan\left(\frac{1}{3} + 2u(x)(x-1)\right) - \frac{\ln(5 + 18(x-1)^2 u(x)^2 + (6x-6)u(x))}{2} + \ln(u(x)(3x-3)-1) = \ln(x)$$

Simplifying the above gives

$$-\arctan\left(\frac{1}{3} + 2u(x)(x-1)\right) - \frac{\ln(5 + 18(x-1)^2 u(x)^2 + 6u(x)(x-1))}{2} + \ln(-1 + 3u(x)(x-1)) = \ln(x-1) + c_2$$

Substituting $u = u(x) + \frac{2}{3(2x-2)}$ in the above solution gives

$$-\arctan\left(\frac{1}{3} + 2\left(u(x) + \frac{2}{3(2x-2)}\right)(x-1)\right) - \frac{\ln\left(5 + 18(x-1)^2 \left(u(x) + \frac{2}{3(2x-2)}\right)^2 + 6\left(u(x) + \frac{2}{3(2x-2)}\right)(x-1)\right)}{2}$$

Simplifying the above gives

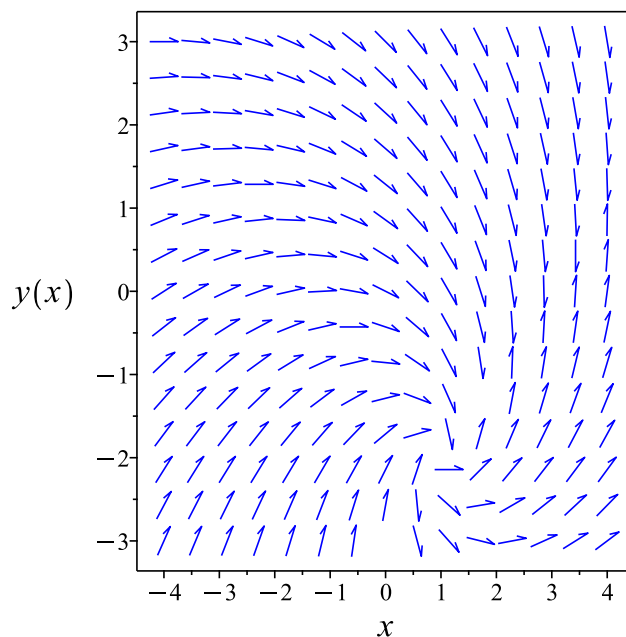
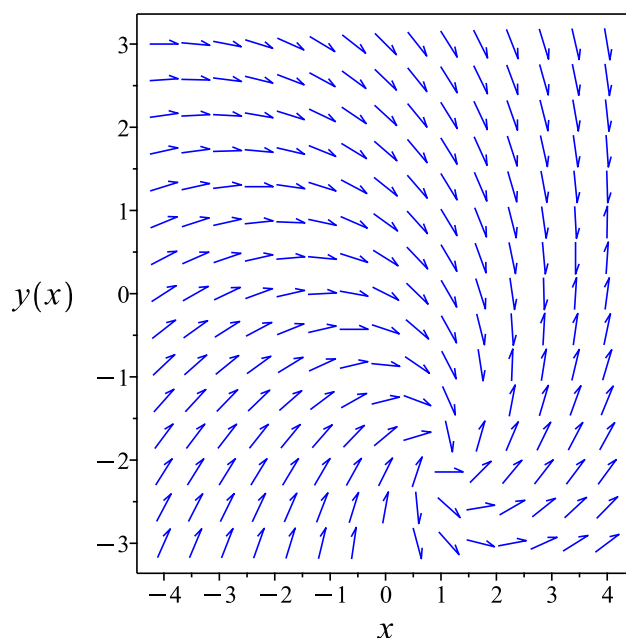
$$-\arctan(1 + (2x-2)u(x)) - \frac{\ln(1 + 2(x-1)^2 u(x)^2 + (2x-2)u(x))}{2} + \ln(u(x)(x-1)) = \ln(x-1) + c_2$$

Substituting $u(x) = \frac{1}{y(x)-x+3}$ in the above solution gives

$$-\arctan\left(1 + \frac{2x-2}{y(x)-x+3}\right) - \frac{\ln\left(1 + \frac{2(x-1)^2}{(y(x)-x+3)^2} + \frac{2x-2}{y(x)-x+3}\right)}{2} + \ln\left(\frac{x-1}{y(x)-x+3}\right) = \ln(x-1) + c_2$$

Simplifying the above gives

$$-\arctan\left(\frac{-1-y(x)-x}{-y(x)+x-3}\right) - \frac{\ln\left(\frac{y(x)^2+x^2+4y(x)-2x+5}{(y(x)-x+3)^2}\right)}{2} + \ln\left(\frac{x-1}{y(x)-x+3}\right) = \ln(x-1) + c_2$$

Figure 2.214: Slope field $x + y(x) + 1 - (-y(x) + x - 3)y'(x) = 0$ Figure 2.215: Slope field $x + y(x) + 1 - (-y(x) + x - 3)y'(x) = 0$

Summary of solutions found

$$-\arctan\left(\frac{-1 - y(x) - x}{-y(x) + x - 3}\right) - \frac{\ln\left(\frac{y(x)^2 + x^2 + 4y(x) - 2x + 5}{(y(x) - x + 3)^2}\right)}{2} + \ln\left(\frac{x - 1}{y(x) - x + 3}\right) = \ln(x - 1) + c_2$$

Solved using first_order_ode_LIE

Time used: 6.536 (sec)

Solve

$$x + y(x) + 1 - (-y(x) + x - 3) y'(x) = 0$$

Writing the ode as

$$y'(x) = -\frac{x + y + 1}{y - x + 3}$$

$$y'(x) = \omega(x, y(x))$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_{y(x)} - \xi_x) - \omega^2 \xi_{y(x)} - \omega_x \xi - \omega_{y(x)} \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{(x + y + 1)(b_3 - a_2)}{y - x + 3} - \frac{(x + y + 1)^2 a_3}{(y - x + 3)^2} - \left(-\frac{1}{y - x + 3} - \frac{x + y + 1}{(y - x + 3)^2} \right) (xa_2 + ya_3 + a_1) - \left(-\frac{1}{y - x + 3} + \frac{x + y + 1}{(y - x + 3)^2} \right) (xb_2 + yb_3 + b_1) = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{x^2 a_2 + x^2 a_3 + x^2 b_2 - x^2 b_3 - 2xya_2 + 2xya_3 + 2xyb_2 + 2xyb_3 - y^2 a_2 - y^2 a_3 - y^2 b_2 + y^2 b_3 - 6xa_2 + 2xa_3 + 2yb_2 - 2yb_3 - 6xb_2 + 2xb_3 - 6a_1 + 6b_1}{(-y + x - 3)} = 0$$

Setting the numerator to zero gives

$$\begin{aligned} & -x^2a_2 - x^2a_3 - x^2b_2 + x^2b_3 + 2xya_2 - 2xya_3 - 2xyb_2 - 2xyb_3 + y^2a_2 \\ & + y^2a_3 + y^2b_2 - y^2b_3 + 6xa_2 - 2xa_3 - 2xb_1 - 4xb_2 - 2xb_3 + 2ya_1 \\ & + 4ya_2 + 2ya_3 + 6yb_2 - 2yb_3 + 4a_1 + 3a_2 - a_3 + 2b_1 + 9b_2 - 3b_3 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & -a_2v_1^2 + 2a_2v_1v_2 + a_2v_2^2 - a_3v_1^2 - 2a_3v_1v_2 + a_3v_2^2 - b_2v_1^2 - 2b_2v_1v_2 + b_2v_2^2 \\ & + b_3v_1^2 - 2b_3v_1v_2 - b_3v_2^2 + 2a_1v_2 + 6a_2v_1 + 4a_2v_2 - 2a_3v_1 + 2a_3v_2 - 2b_1v_1 \\ & - 4b_2v_1 + 6b_2v_2 - 2b_3v_1 - 2b_3v_2 + 4a_1 + 3a_2 - a_3 + 2b_1 + 9b_2 - 3b_3 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} & (-a_2 - a_3 - b_2 + b_3)v_1^2 + (2a_2 - 2a_3 - 2b_2 - 2b_3)v_1v_2 \\ & + (6a_2 - 2a_3 - 2b_1 - 4b_2 - 2b_3)v_1 + (a_2 + a_3 + b_2 - b_3)v_2^2 \\ & + (2a_1 + 4a_2 + 2a_3 + 6b_2 - 2b_3)v_2 + 4a_1 + 3a_2 - a_3 + 2b_1 + 9b_2 - 3b_3 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} & -a_2 - a_3 - b_2 + b_3 = 0 \\ & a_2 + a_3 + b_2 - b_3 = 0 \\ & 2a_2 - 2a_3 - 2b_2 - 2b_3 = 0 \\ & 2a_1 + 4a_2 + 2a_3 + 6b_2 - 2b_3 = 0 \\ & 6a_2 - 2a_3 - 2b_1 - 4b_2 - 2b_3 = 0 \\ & 4a_1 + 3a_2 - a_3 + 2b_1 + 9b_2 - 3b_3 = 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}a_1 &= -2b_2 - b_3 \\a_2 &= b_3 \\a_3 &= -b_2 \\b_1 &= -b_2 + 2b_3 \\b_2 &= b_2 \\b_3 &= b_3\end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned}\xi &= -2 - y \\ \eta &= x - 1\end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= x - 1 - \left(-\frac{x + y + 1}{y - x + 3} \right) (-2 - y) \\ &= \frac{x^2 + y^2 - 2x + 4y + 5}{-y + x - 3} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{x^2 + y^2 - 2x + 4y + 5}{-y + x - 3}} dy\end{aligned}$$

Which results in

$$S = -\frac{\ln(x^2 + y^2 - 2x + 4y + 5)}{2} + \frac{2(x-1) \arctan\left(\frac{2y+4}{2x-2}\right)}{2x-2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{x + y + 1}{y - x + 3}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{-1 - y - x}{x^2 + y^2 - 2x + 4y + 5} \\ S_y &= \frac{-y + x - 3}{x^2 + y^2 - 2x + 4y + 5} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

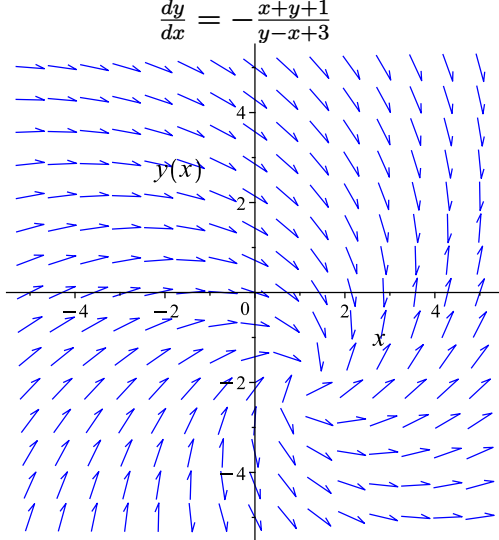
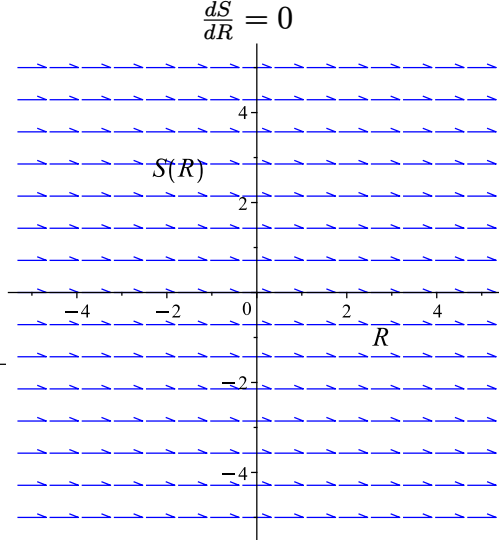
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\frac{\ln(y(x)^2 + x^2 + 4y(x) - 2x + 5)}{2} + \arctan\left(\frac{y(x) + 2}{x - 1}\right) = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{x+y+1}{y-x+3}$ 	$R = x$ $S = -\frac{\ln(x^2 + y^2 - 2x)}{2}$	$\frac{dS}{dR} = 0$ 

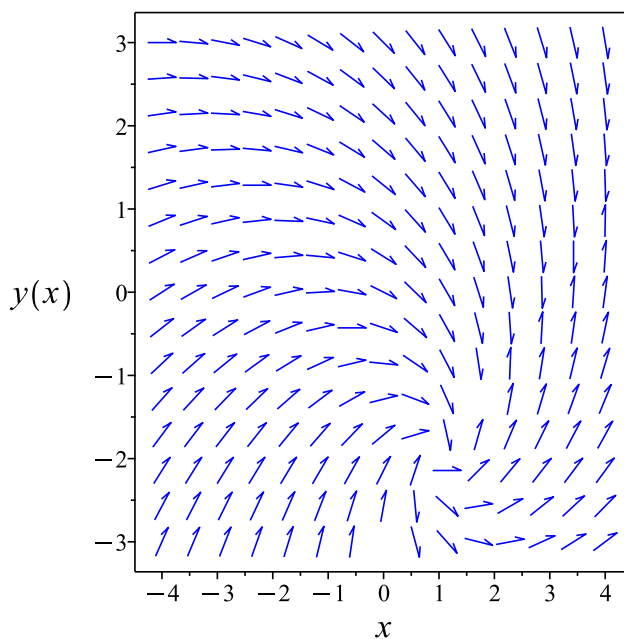


Figure 2.216: Slope field $x + y(x) + 1 - (-y(x) + x - 3)y'(x) = 0$

Summary of solutions found

$$-\frac{\ln(y(x)^2 + x^2 + 4y(x) - 2x + 5)}{2} + \arctan\left(\frac{y(x) + 2}{x - 1}\right) = c_2$$

✓ **Maple.** Time used: 0.013 (sec). Leaf size: 31

```
ode:=x+y(x)+1-(-3+x-y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -2 - \tan \left(\text{RootOf} \left(2_Z + \ln \left(\sec \left(_Z \right)^2 \right) + 2 \ln (x - 1) + 2c_1 \right) \right) (x - 1)$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous C
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
<- homogeneous successful
```

Maple step by step

Let's solve

$$x + y(x) + 1 - (x - y(x) - 3) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-x-y(x)-1}{x-y(x)-3}$$

✓ **Mathematica.** Time used: 0.034 (sec). Leaf size: 58

```
ode=(x+y[x]+1)-(x-y[x]-3)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[2 \arctan \left(\frac{y(x) + x + 1}{y(x) - x + 3} \right) + \log \left(\frac{x^2 + y(x)^2 + 4y(x) - 2x + 5}{2(x - 1)^2} \right) + 2 \log(x - 1) + c_1 = 0, y(x) \right]$$

✓ Sympy. Time used: 2.269 (sec). Leaf size: 34

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x - (x - y(x) - 3)*Derivative(y(x), x) + y(x) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\log(x-1) = C_1 - \log\left(\sqrt{1 + \frac{(y(x)+2)^2}{(x-1)^2}}\right) + \operatorname{atan}\left(\frac{y(x)+2}{x-1}\right)$$

2.3.16 Problem 25 (a)

Local contents

Solved using first_order_ode_exact	910
✓ Maple	914
✓ Mathematica	914
✓ Sympy	915

Internal problem ID [7896]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (a)

Date solved : Thursday, November 27, 2025 at 06:29:34 AM

CAS classification : [[_homogeneous, 'class D'], _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.125 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (y) dy &= (x^2 + y^2 - x) dx \\ (-x^2 - y^2 + x) dx + (y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x^2 - y^2 + x \\ N(x, y) &= y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-x^2 - y^2 + x) \\ &= -2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(y) \\ &= 0 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{y} ((-2y) - (0)) \\ &= -2 \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -2 dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2x} \\ &= e^{-2x}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= e^{-2x}(-x^2 - y^2 + x) \\ &= (-x^2 - y^2 + x) e^{-2x}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= e^{-2x}(y) \\ &= e^{-2x}y\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ ((-x^2 - y^2 + x) e^{-2x}) + (e^{-2x}y) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int (-x^2 - y^2 + x) e^{-2x} dx \\ \phi &= \frac{e^{-2x}(x^2 + y^2)}{2} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = e^{-2x}y + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = e^{-2x}y$. Therefore equation (4) becomes

$$e^{-2x}y = e^{-2x}y + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{e^{-2x}(x^2 + y^2)}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{e^{-2x}(x^2 + y^2)}{2}$$

Solving for y gives

$$y = e^{2x} \sqrt{-e^{-2x}(x^2 e^{-2x} - 2c_1)}$$

$$y = -e^{2x} \sqrt{-e^{-2x}(x^2 e^{-2x} - 2c_1)}$$

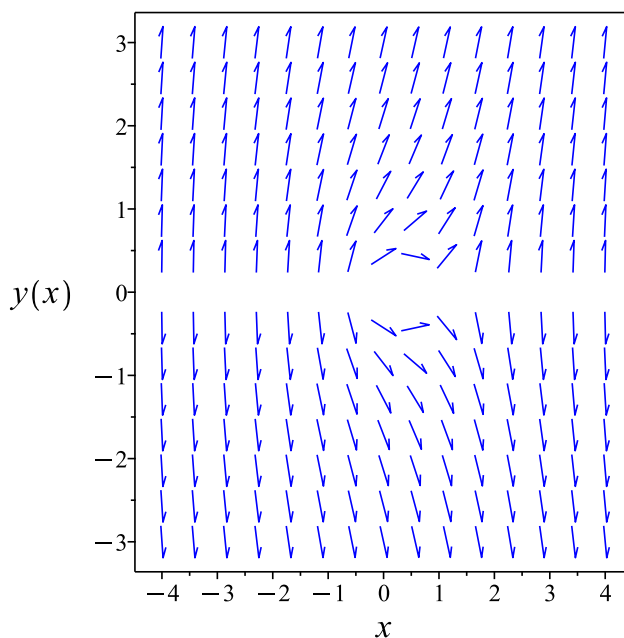


Figure 2.217: Slope field $x - x^2 - y^2 + yy' = 0$

Summary of solutions found

$$y = e^{2x} \sqrt{-e^{-2x} (x^2 e^{-2x} - 2c_1)}$$

$$y = -e^{2x} \sqrt{-e^{-2x} (x^2 e^{-2x} - 2c_1)}$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 37

```
ode:=x-x^2-y(x)^2+diff(y(x),x)*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{e^{2x}c_1 - x^2}$$

$$y = -\sqrt{e^{2x}c_1 - x^2}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

<- Bernoulli successful

Maple step by step

Let's solve

$$x - x^2 - y(x)^2 + y(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-x+x^2+y(x)^2}{y(x)}$$

✓ **Mathematica.** Time used: 3.578 (sec). Leaf size: 47

```
ode=(x-x^2-y[x]^2)+y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{-x^2 + c_1 e^{2x}}$$

$$y(x) \rightarrow \sqrt{-x^2 + c_1 e^{2x}}$$

✓ Sympy. Time used: 0.326 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + x - y(x)**2 + y(x)*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\sqrt{C_1 e^{2x} - x^2}, \quad y(x) = \sqrt{C_1 e^{2x} - x^2} \right]$$

2.3.17 Problem 25 (b)

Local contents

Solved using first_order_ode_exact	916
✓ Maple	920
✓ Mathematica	921
✓ Sympy	922

Internal problem ID [7897]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (b)

Date solved : Thursday, November 27, 2025 at 06:29:39 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.131 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (3x - 2y) dx \\ (-3x + 2y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -3x + 2y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-3x + 2y) \\ &= 2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((2) - (1)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int \frac{1}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(-3x + 2y) \\ &= (-3x + 2y)x\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(x) \\ &= x^2\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ ((-3x + 2y)x) + (x^2) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int x^2 dy \\ \phi &= x^2 y + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = 2yx + f'(x) \tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = (-3x + 2y)x$. Therefore equation (4) becomes

$$(-3x + 2y)x = 2yx + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -3x^2$$

Integrating the above w.r.t x gives

$$\begin{aligned} \int f'(x) dx &= \int (-3x^2) dx \\ f(x) &= -x^3 + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = -x^3 + x^2y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

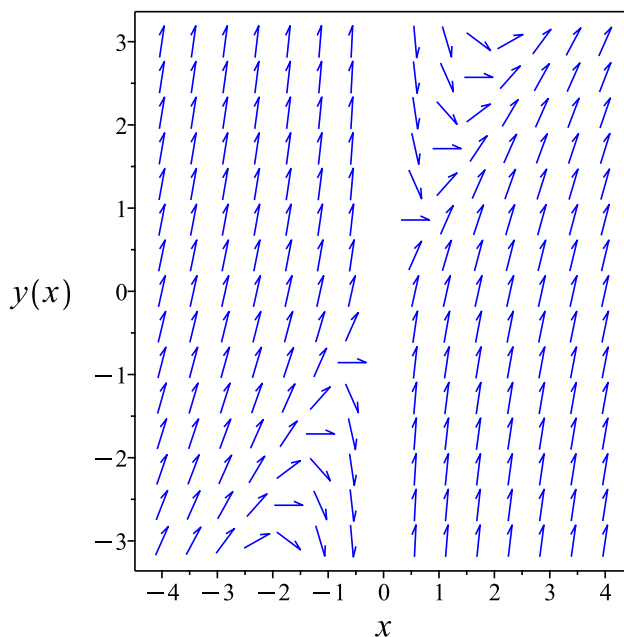
$$c_1 = -x^3 + x^2y$$

Simplifying the above gives

$$x^2(-x + y) = c_1$$

Solving for y gives

$$y = \frac{x^3 + c_1}{x^2}$$

Figure 2.218: Slope field $2y - 3x + y'x = 0$ Summary of solutions found

$$y = \frac{x^3 + c_1}{x^2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 11

```
ode:=2*y(x)-3*x+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = x + \frac{c_1}{x^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$2y(x) - 3x + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Solve for the highest derivative
 $\frac{d}{dx}y(x) = \frac{-2y(x)+3x}{x}$
- Collect w.r.t. $y(x)$ and simplify
 $\frac{d}{dx}y(x) = 3 - \frac{2y(x)}{x}$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx}y(x) + \frac{2y(x)}{x} = 3$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{2y(x)}{x} \right) = 3\mu(x)$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{2y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$
- Isolate $\frac{d}{dx}\mu(x)$
 $\frac{d}{dx}\mu(x) = \frac{2\mu(x)}{x}$
- Solve to find the integrating factor
 $\mu(x) = x^2$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int 3\mu(x) dx + C1$
- Evaluate the integral on the lhs
 $y(x)\mu(x) = \int 3\mu(x) dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int 3\mu(x)dx + C1}{\mu(x)}$
- Substitute $\mu(x) = x^2$
 $y(x) = \frac{\int 3x^2dx + C1}{x^2}$
- Evaluate the integrals on the rhs
 $y(x) = \frac{x^3 + C1}{x^2}$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 13

```
ode=(2*y[x]-3*x)+x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x + \frac{C_1}{x^2}$$

✓ Sympy. Time used: 0.094 (sec). Leaf size: 8

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(x*Derivative(y(x), x) - 3*x + 2*y(x), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1}{x^2} + x$$

2.3.18 Problem 25 (c)

Local contents

Solved using first_order_ode_exact	923
✓ Maple	927
✓ Mathematica	928
✓ Sympy	928

Internal problem ID [7898]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (c)

Date solved : Thursday, November 27, 2025 at 06:29:44 AM

CAS classification : [[_homogeneous, 'class G'], _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.108 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2yx) dy &= (y^2 - x) dx \\ (-y^2 + x) dx + (2yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -y^2 + x \\ N(x, y) &= 2yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-y^2 + x) \\ &= -2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(2yx) \\ &= 2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{2yx} ((-2y) - (2y)) \\ &= -\frac{2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{2}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2\ln(x)} \\ &= \frac{1}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2}(-y^2 + x) \\ &= \frac{-y^2 + x}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2}(2yx) \\ &= \frac{2y}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-y^2 + x}{x^2} \right) + \left(\frac{2y}{x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-y^2 + x}{x^2} dx \\ \phi &= \frac{y^2}{x} + \ln(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{2y}{x} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{2y}{x}$. Therefore equation (4) becomes

$$\frac{2y}{x} = \frac{2y}{x} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{y^2}{x} + \ln(x) + c_1$$

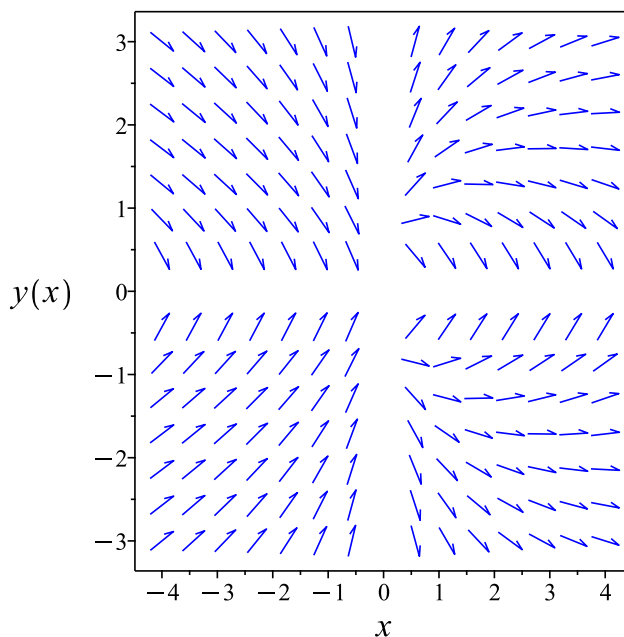
But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y^2}{x} + \ln(x)$$

Solving for y gives

$$y = \sqrt{-x \ln(x) + c_1 x}$$

$$y = -\sqrt{-x \ln(x) + c_1 x}$$

Figure 2.219: Slope field $x - y^2 + 2xyy' = 0$ Summary of solutions found

$$y = \sqrt{-x \ln(x) + c_1 x}$$

$$y = -\sqrt{-x \ln(x) + c_1 x}$$

✓ Maple. Time used: 0.004 (sec). Leaf size: 30

```
ode:=x-y(x)^2+2*x*y(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{-x (\ln(x) - c_1)}$$

$$y = -\sqrt{(-\ln(x) + c_1) x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$x - y(x)^2 + 2xy(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-x + y(x)^2}{2xy(x)}$$

✓ **Mathematica.** Time used: 0.134 (sec). Leaf size: 44

```
ode=(x-y[x]^2)+2*x*y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{x} \sqrt{-\log(x) + c_1}$$

$$y(x) \rightarrow \sqrt{x} \sqrt{-\log(x) + c_1}$$

✓ **Sympy.** Time used: 0.285 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*y(x)*Derivative(y(x), x) + x - y(x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\sqrt{x(C_1 - \log(x))}, y(x) = \sqrt{x(C_1 - \log(x))} \right]$$

2.3.19 Problem 25 (d)

Local contents

Solved using first_order_ode_exact	929
✓ Maple	934
✓ Mathematica	934
✓ Sympy	935

Internal problem ID [7899]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (d)

Date solved : Thursday, November 27, 2025 at 06:29:50 AM

CAS classification : [[_homogeneous, 'class D'], _rational, _Riccati]

Solved using first_order_ode_exact

Time used: 0.135 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (y + 3x^2(x^2 + y^2)) dx \\ (-y - 3x^2(x^2 + y^2)) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -y - 3x^2(x^2 + y^2) \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-y - 3x^2(x^2 + y^2)) \\ &= -6x^2y - 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{x^2+y^2}$ is an integrating factor. Therefore by multiplying $M = -y - 3x^2(x^2 + y^2)$ and $N = x$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} \\ N &= \frac{x}{x^2 + y^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{x}{x^2 + y^2} \right) dy &= \left(-\frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} \right) dx \\ \left(\frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} \right) dx &+ \left(\frac{x}{x^2 + y^2} \right) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} \\ N(x, y) &= \frac{x}{x^2 + y^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} \right) \\ &= \frac{-x^2 + y^2}{(x^2 + y^2)^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) \\ &= \frac{-x^2 + y^2}{(x^2 + y^2)^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-y - 3x^2(x^2 + y^2)}{x^2 + y^2} dx \\ \phi &= -x^3 - \arctan\left(\frac{x}{y}\right) + f(y)\end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= \frac{x}{y^2 \left(\frac{x^2}{y^2} + 1 \right)} + f'(y) \\ &= \frac{x}{x^2 + y^2} + f'(y)\end{aligned} \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{x}{x^2 + y^2}$. Therefore equation (4) becomes

$$\frac{x}{x^2 + y^2} = \frac{x}{x^2 + y^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -x^3 - \arctan\left(\frac{x}{y}\right) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -x^3 - \arctan\left(\frac{x}{y}\right)$$

Solving for y gives

$$y = -\frac{x}{\tan(x^3 + c_1)}$$

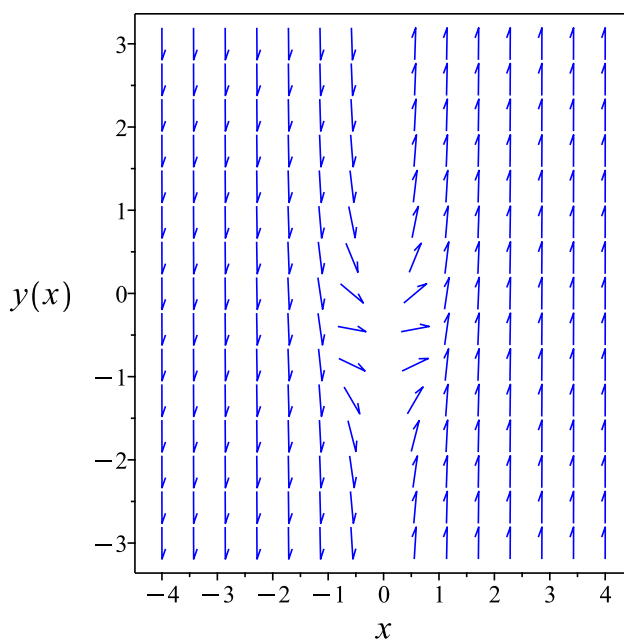


Figure 2.220: Slope field $-y - 3x^2(x^2 + y^2) + y'x = 0$

Summary of solutions found

$$y = -\frac{x}{\tan(x^3 + c_1)}$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 14

```
ode:=-y(x)-3*(x^2+y(x)^2)*x^2+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \tan(x^3 + 3c_1)x$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

trying homogeneous types:

trying homogeneous D

<- homogeneous successful

Maple step by step

Let's solve

$$-y(x) - 3x^2(x^2 + y(x)^2) + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x) + 3x^2(x^2 + y(x)^2)}{x}$$

✓ **Mathematica.** Time used: 0.049 (sec). Leaf size: 30

```
ode=(-y[x]-3*x^2*(x^2+y[x]^2))+x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\int_1^{\frac{y(x)}{x}} \frac{1}{K[1]^2 + 1} dK[1] = x^3 + c_1, y(x) \right]$$

✓ Sympy. Time used: 0.188 (sec). Leaf size: 27

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-3*x**2*(x**2 + y(x)**2) + x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x(iC_1 + ie^{2ix^3})}{C_1 - e^{2ix^3}}$$

2.3.20 Problem 25 (e)

Local contents

Solved using first_order_ode_exact	936
✓ Maple	940
✓ Mathematica	941
✓ Sympy	942

Internal problem ID [7900]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (e)

Date solved : Thursday, November 27, 2025 at 06:29:55 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.129 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x) dy &= (-y + \ln(x)) dx \\ (y - \ln(x)) dx + (-x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y - \ln(x) \\ N(x, y) &= -x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y - \ln(x)) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x) \\ &= -1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{x} ((1) - (-1)) \\ &= -\frac{2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{2}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2\ln(x)} \\ &= \frac{1}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2}(y - \ln(x)) \\ &= \frac{y - \ln(x)}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2}(-x) \\ &= -\frac{1}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{y - \ln(x)}{x^2} \right) + \left(-\frac{1}{x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int -\frac{1}{x} dy \\ \phi &= -\frac{y}{x} + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = \frac{y}{x^2} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{y - \ln(x)}{x^2}$. Therefore equation (4) becomes

$$\frac{y - \ln(x)}{x^2} = \frac{y}{x^2} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -\frac{\ln(x)}{x^2}$$

Integrating the above w.r.t x gives

$$\begin{aligned} \int f'(x) dx &= \int \left(-\frac{\ln(x)}{x^2} \right) dx \\ f(x) &= \frac{\ln(x)}{x} + \frac{1}{x} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = -\frac{y}{x} + \frac{\ln(x)}{x} + \frac{1}{x} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

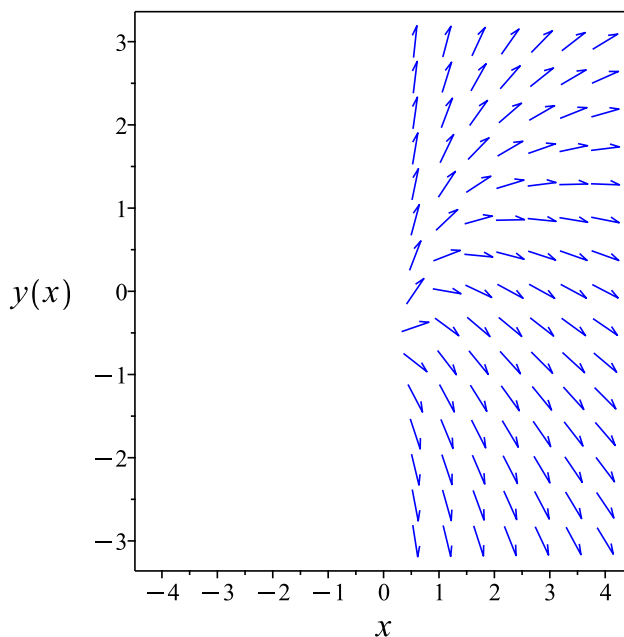
$$c_1 = -\frac{y}{x} + \frac{\ln(x)}{x} + \frac{1}{x}$$

Simplifying the above gives

$$\frac{-y + \ln(x) + 1}{x} = c_1$$

Solving for y gives

$$y = -c_1 x + \ln(x) + 1$$

Figure 2.221: Slope field $y - \ln(x) - y'x = 0$ Summary of solutions found

$$y = -c_1x + \ln(x) + 1$$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 11

```
ode:=y(x)-ln(x)-diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1x + \ln(x) + 1$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$y(x) - \ln(x) - x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{-y(x)+\ln(x)}{x}$$
- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx}y(x) = \frac{y(x)}{x} - \frac{\ln(x)}{x}$$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - \frac{y(x)}{x} = -\frac{\ln(x)}{x}$$
- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x} \right) = -\frac{\mu(x)\ln(x)}{x}$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$
- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{\mu(x)}{x}$$
- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x}$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int -\frac{\mu(x)\ln(x)}{x} dx + C1$$
- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int -\frac{\mu(x)\ln(x)}{x} dx + C1$$
- Solve for $y(x)$

$$y(x) = \frac{\int -\frac{\mu(x)\ln(x)}{x} dx + C1}{\mu(x)}$$
- Substitute $\mu(x) = \frac{1}{x}$

$$y(x) = x \left(\int -\frac{\ln(x)}{x^2} dx + C1 \right)$$
- Evaluate the integrals on the rhs

$$y(x) = x \left(\frac{\ln(x)}{x} + \frac{1}{x} + C1 \right)$$
- Simplify

$$y(x) = C1x + \ln(x) + 1$$

✓ **Mathematica.** Time used: 0.019 (sec). Leaf size: 13

```
ode=(y[x]-Log[x])-x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \log(x) + c_1x + 1$$

✓ Sympy. Time used: 0.138 (sec). Leaf size: 10

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-x*Derivative(y(x), x) + y(x) - log(x), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1x + \log(x) + 1$$

2.3.21 Problem 25 (f)

Local contents

Solved using first_order_ode_exact	943
✓ Maple	947
✓ Mathematica	948
✓ Sympy	948

Internal problem ID [7901]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (f)

Date solved : Thursday, November 27, 2025 at 06:30:00 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.109 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-2yx) dy &= (-3x^2 - y^2) dx \\ (3x^2 + y^2) dx + (-2yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 3x^2 + y^2 \\ N(x, y) &= -2yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(3x^2 + y^2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-2yx) \\ &= -2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{2yx} ((2y) - (-2y)) \\ &= -\frac{2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{2}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2\ln(x)} \\ &= \frac{1}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2}(3x^2 + y^2) \\ &= \frac{3x^2 + y^2}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2}(-2yx) \\ &= -\frac{2y}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{3x^2 + y^2}{x^2} \right) + \left(-\frac{2y}{x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{3x^2 + y^2}{x^2} dx \\ \phi &= 3x - \frac{y^2}{x} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{2y}{x} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{2y}{x}$. Therefore equation (4) becomes

$$-\frac{2y}{x} = -\frac{2y}{x} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = 3x - \frac{y^2}{x} + c_1$$

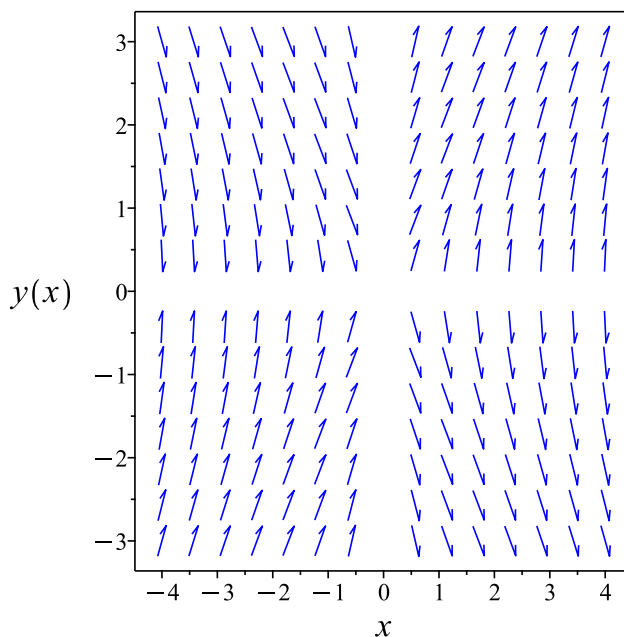
But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = 3x - \frac{y^2}{x}$$

Solving for y gives

$$y = \sqrt{-c_1 x + 3x^2}$$

$$y = -\sqrt{-c_1 x + 3x^2}$$

Figure 2.222: Slope field $3x^2 + y^2 - 2xyy' = 0$ Summary of solutions found

$$y = \sqrt{-c_1x + 3x^2}$$

$$y = -\sqrt{-c_1x + 3x^2}$$

✓ Maple. Time used: 0.002 (sec). Leaf size: 27

```
ode:=3*x^2+y(x)^2-2*x*y(x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{(3x + c_1)x}$$

$$y = -\sqrt{(3x + c_1)x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$3x^2 + y(x)^2 - 2xy(x) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{3x^2 - y(x)^2}{2xy(x)}$$

✓ **Mathematica.** Time used: 0.12 (sec). Leaf size: 42

```
ode=(3*x^2+y[x]^2)-2*x*y[x]*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{x}\sqrt{3x + c_1}$$

$$y(x) \rightarrow \sqrt{x}\sqrt{3x + c_1}$$

✓ **Sympy.** Time used: 0.269 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x**2 - 2*x*y(x)*Derivative(y(x), x) + y(x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\sqrt{x(C_1 + 3x)}, \quad y(x) = \sqrt{x(C_1 + 3x)} \right]$$

2.3.22 Problem 25 (g)

Local contents

Solved using first_order_ode_exact	949
✓ Maple	954
✓ Mathematica	954
✓ Sympy	955

Internal problem ID [7902]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (g)

Date solved : Thursday, November 27, 2025 at 06:30:05 AM

CAS classification :

[[_homogeneous, 'class A'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.163 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x^2 + 3yx) dy &= (-yx + 2y^2) dx \\ (yx - 2y^2) dx + (-x^2 + 3yx) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= yx - 2y^2 \\ N(x, y) &= -x^2 + 3yx \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(yx - 2y^2) \\ &= x - 4y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x^2 + 3yx) \\ &= -2x + 3y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{xy^2}$ is an integrating factor. Therefore by multiplying $M = -2y^2 + yx$ and $N = -x^2 + 3yx$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{-2y^2 + yx}{xy^2} \\ N &= \frac{-x^2 + 3yx}{xy^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{-x^2 + 3yx}{x y^2} \right) dy &= \left(-\frac{yx - 2y^2}{x y^2} \right) dx \\ \left(\frac{yx - 2y^2}{x y^2} \right) dx + \left(\frac{-x^2 + 3yx}{x y^2} \right) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{yx - 2y^2}{x y^2} \\ N(x, y) &= \frac{-x^2 + 3yx}{x y^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{yx - 2y^2}{x y^2} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{-x^2 + 3yx}{x y^2} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{yx - 2y^2}{x y^2} dx \\ \phi &= -2 \ln(x) + \frac{x}{y} + f(y)\end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{x}{y^2} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{-x^2 + 3yx}{x y^2}$. Therefore equation (4) becomes

$$\frac{-x^2 + 3yx}{x y^2} = -\frac{x}{y^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = \frac{3}{y}$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int \left(\frac{3}{y}\right) dy$$

$$f(y) = 3 \ln(y) + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -2 \ln(x) + \frac{x}{y} + 3 \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -2 \ln(x) + \frac{x}{y} + 3 \ln(y)$$

Solving for y gives

$$y = e^{\text{LambertW}\left(-\frac{x e^{-\frac{2 \ln(x)}{3} - \frac{c_1}{3}}}{3}\right) + \frac{2 \ln(x)}{3} + \frac{c_1}{3}}$$

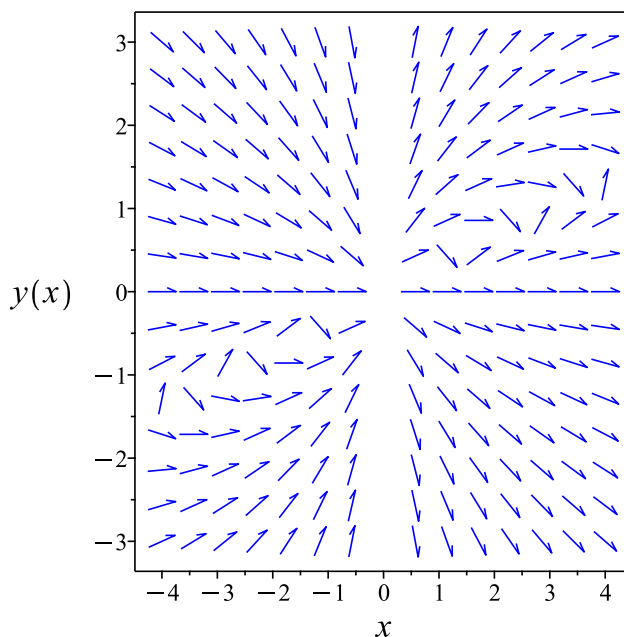


Figure 2.223: Slope field $yx - 2y^2 - (x^2 - 3yx)y' = 0$

Summary of solutions found

$$y = e^{\text{LambertW}\left(-\frac{x e^{-\frac{2 \ln(x)}{3} - \frac{c_1}{3}}}{3}\right) + \frac{2 \ln(x)}{3} + \frac{c_1}{3}}$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 19

```
ode:=y(x)*x-2*y(x)^2-(x^2-3*y(x)*x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{x}{3 \operatorname{LambertW}\left(-e^{\frac{c_1}{3}} x^{1/3}\right)}$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
```

Maple step by step

Let's solve

$$xy(x) - 2y(x)^2 - (x^2 - 3xy(x)) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{-xy(x)+2y(x)^2}{x^2-3xy(x)}$$

✓ **Mathematica.** Time used: 3.39 (sec). Leaf size: 37

```
ode=(x*y[x]-2*y[x]^2)-(x^2-3*x*y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{x}{3W\left(-\frac{1}{3}e^{-1-\frac{c_1}{3}}\sqrt[3]{x}\right)}$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 9.981 (sec). Leaf size: 90

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) - (x**2 - 3*x*y(x))*Derivative(y(x), x) - 2*y(x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = x^{\frac{2}{3}} e^{C_1 + W\left(\frac{\sqrt[3]{-x}(-1 + \sqrt{3}i)e^{-C_1}}{6}\right)}, y(x) = x^{\frac{2}{3}} e^{C_1 + W\left(\frac{\sqrt[3]{-x}(-1 - \sqrt{3}i)e^{-C_1}}{6}\right)}, y(x) = x^{\frac{2}{3}} e^{C_1 + W\left(\frac{\sqrt[3]{-x}e^{-C_1}}{3}\right)} \right]$$

2.3.23 Problem 25 (h)

Local contents

Solved using first_order_ode_exact	956
✓ Maple	961
✓ Mathematica	961
✓ Sympy	962

Internal problem ID [7903]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (h)

Date solved : Thursday, November 27, 2025 at 06:30:10 AM

CAS classification :

[[_homogeneous, 'class A'], _rational, [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_exact

Time used: 0.125 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x + y) dy &= (-x - y) dx \\ (x + y) dx + (-x + y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x + y \\ N(x, y) &= -x + y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(x + y) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x + y) \\ &= -1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{x^2 + y^2}$ is an integrating factor. Therefore by multiplying $M = x + y$ and $N = -x + y$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{x + y}{x^2 + y^2} \\ N &= \frac{-x + y}{x^2 + y^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

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$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{-x+y}{x^2+y^2} \right) dy &= \left(-\frac{x+y}{x^2+y^2} \right) dx \\ \left(\frac{x+y}{x^2+y^2} \right) dx + \left(\frac{-x+y}{x^2+y^2} \right) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{x+y}{x^2+y^2} \\ N(x, y) &= \frac{-x+y}{x^2+y^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{x+y}{x^2+y^2} \right) \\ &= \frac{x^2 - 2yx - y^2}{(x^2+y^2)^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{-x+y}{x^2+y^2} \right) \\ &= \frac{x^2 - 2yx - y^2}{(x^2+y^2)^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{x+y}{x^2+y^2} dx \\ \phi &= \frac{\ln(x^2+y^2)}{2} + \arctan\left(\frac{x}{y}\right) + f(y)\end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= \frac{y}{x^2+y^2} - \frac{x}{y^2 \left(\frac{x^2}{y^2} + 1 \right)} + f'(y) \\ &= \frac{-x+y}{x^2+y^2} + f'(y)\end{aligned} \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{-x+y}{x^2+y^2}$. Therefore equation (4) becomes

$$\frac{-x+y}{x^2+y^2} = \frac{-x+y}{x^2+y^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{\ln(x^2 + y^2)}{2} + \arctan\left(\frac{x}{y}\right) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{\ln(x^2 + y^2)}{2} + \arctan\left(\frac{x}{y}\right)$$

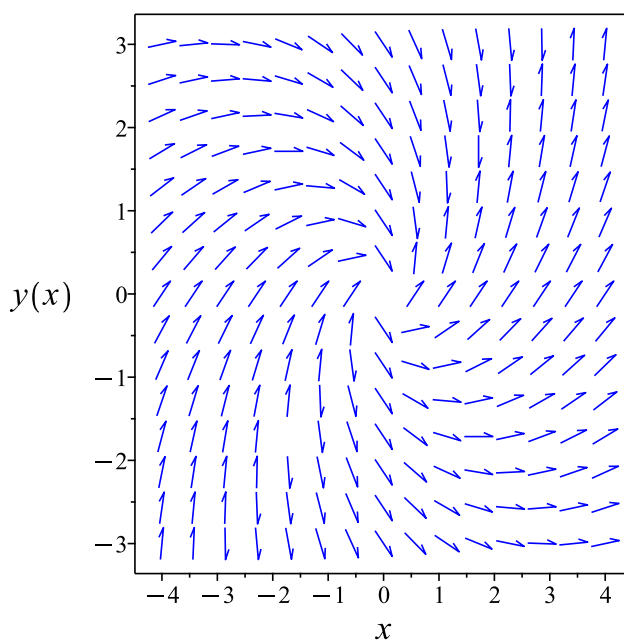


Figure 2.224: Slope field $x + y - (x - y)y' = 0$

Summary of solutions found

$$\frac{\ln(x^2 + y^2)}{2} + \arctan\left(\frac{x}{y}\right) = c_1$$

✓ **Maple.** Time used: 0.008 (sec). Leaf size: 24

```
ode:=x+y(x)-(x-y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \tan \left(\text{RootOf} \left(-2_Z + \ln \left(\sec \left(_Z \right)^2 \right) + 2 \ln (x) + 2c_1 \right) \right) x$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

trying homogeneous types:

trying homogeneous D

<- homogeneous successful

Maple step by step

Let's solve

$$x + y(x) - (x - y(x)) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{x + y(x)}{x - y(x)}$$

✓ **Mathematica.** Time used: 0.02 (sec). Leaf size: 36

```
ode=(x+y[x])-(x-y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\int_1^{\frac{y(x)}{x}} \frac{K[1] - 1}{K[1]^2 + 1} dK[1] = -\log(x) + c_1, y(x) \right]$$

✓ Sympy. Time used: 0.902 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x - (x - y(x))*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\log(x) = C_1 - \log\left(\sqrt{1 + \frac{y^2(x)}{x^2}}\right) + \operatorname{atan}\left(\frac{y(x)}{x}\right)$$

2.3.24 Problem 25 (L)

Local contents

Solved using first_order_ode_exact	963
✓ Maple	968
✓ Mathematica	969
✓ Sympy	969

Internal problem ID [7904]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (L)

Date solved : Thursday, November 27, 2025 at 06:30:15 AM

CAS classification : [[_homogeneous, 'class G'], _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.209 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x) dy &= (3x y^2 - 2y) dx \\ (-3x y^2 + 2y) dx + (-x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -3x y^2 + 2y \\ N(x, y) &= -x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-3x y^2 + 2y) \\ &= -6yx + 2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x) \\ &= -1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{x} ((-6yx + 2) - (-1)) \\ &= \frac{6yx - 3}{x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{-3xy^2 + 2y} ((-1) - (-6yx + 2)) \\ &= \frac{-6yx + 3}{3xy^2 - 2y} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(-1) - (-6yx + 2)}{x(-3xy^2 + 2y) - y(-x)} \\ &= \frac{-2yx + 1}{(yx - 1)xy} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = \frac{-2t + 1}{(t - 1)t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int \left(\frac{-2t+1}{(t-1)t} \right) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\ln((t-1)t)} \\ &= \frac{1}{(t-1)t} \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{(yx - 1)yx}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{(yx-1)yx}(-3xy^2+2y) \\ &= \frac{-3yx+2}{x(yx-1)}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{(yx-1)yx}(-x) \\ &= -\frac{1}{y(yx-1)}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-3yx+2}{x(yx-1)} \right) + \left(-\frac{1}{y(yx-1)} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-3yx+2}{x(yx-1)} dx \\ \phi &= -\ln(yx-1) - 2\ln(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{x}{yx-1} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{1}{y(yx-1)}$. Therefore equation (4) becomes

$$-\frac{1}{y(yx-1)} = -\frac{x}{yx-1} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = \frac{1}{y}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) \, dy &= \int \left(\frac{1}{y} \right) \, dy \\ f(y) &= \ln(y) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

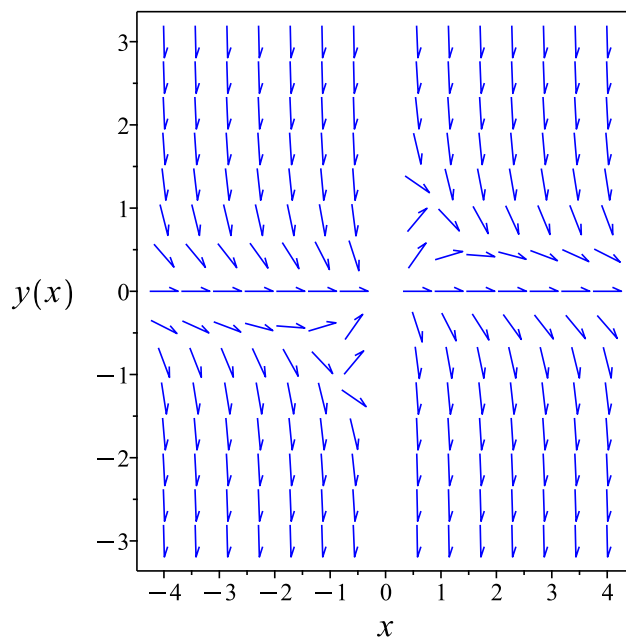
$$\phi = -\ln(yx-1) - 2\ln(x) + \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\ln(yx-1) - 2\ln(x) + \ln(y)$$

Solving for y gives

$$y = \frac{e^{c_1} x^2}{e^{c_1} x^3 - 1}$$

Figure 2.225: Slope field $2y - 3y^2x - y'x = 0$ Summary of solutions found

$$y = \frac{e^{c_1} x^2}{e^{c_1} x^3 - 1}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 15

```
ode:=2*y(x)-3*x*y(x)^2-diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2}{x^3 + c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$2y(x) - 3xy(x)^2 - x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{-2y(x)+3xy(x)^2}{x}$$

✓ **Mathematica.** Time used: 0.106 (sec). Leaf size: 22

```
ode=(2*y[x]-3*x*y[x]^2)-x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^2}{x^3 + c_1}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.121 (sec). Leaf size: 10

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-3*x*y(x)**2 - x*Derivative(y(x), x) + 2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x^2}{C_1 + x^3}$$

2.3.25 Problem 25 (j)

Local contents

Solved using first_order_ode_exact	970
✓ Maple	975
✓ Mathematica	975
✗ Sympy	976

Internal problem ID [7905]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (j)

Date solved : Thursday, November 27, 2025 at 06:30:21 AM

CAS classification :

[[_homogeneous, 'class G'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.128 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x(x^2y - 1)) dy &= (-y) dx \\ (y) dx + (x(x^2y - 1)) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y \\ N(x, y) &= x(x^2y - 1) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x(x^2y - 1)) \\ &= 3x^2y - 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{y}{x^3}$ is an integrating factor. Therefore by multiplying $M = y$ and $N = x(x^2y - 1)$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{y^2}{x^3} \\ N &= \frac{y(x^2y - 1)}{x^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{y(x^2y - 1)}{x^2} \right) dy &= \left(-\frac{y^2}{x^3} \right) dx \\ \left(\frac{y^2}{x^3} \right) dx + \left(\frac{y(x^2y - 1)}{x^2} \right) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{y^2}{x^3} \\ N(x, y) &= \frac{y(x^2y - 1)}{x^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{y^2}{x^3} \right) \\ &= \frac{2y}{x^3}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{y(x^2y - 1)}{x^2} \right) \\ &= \frac{2y}{x^3}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{y^2}{x^3} dx \\ \phi &= -\frac{y^2}{2x^2} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{y}{x^2} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{y(x^2y - 1)}{x^2}$. Therefore equation (4) becomes

$$\frac{y(x^2y - 1)}{x^2} = -\frac{y}{x^2} + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = y^2$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int (y^2) dy$$

$$f(y) = \frac{y^3}{3} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{y^2}{2x^2} + \frac{y^3}{3} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{y^2}{2x^2} + \frac{y^3}{3}$$

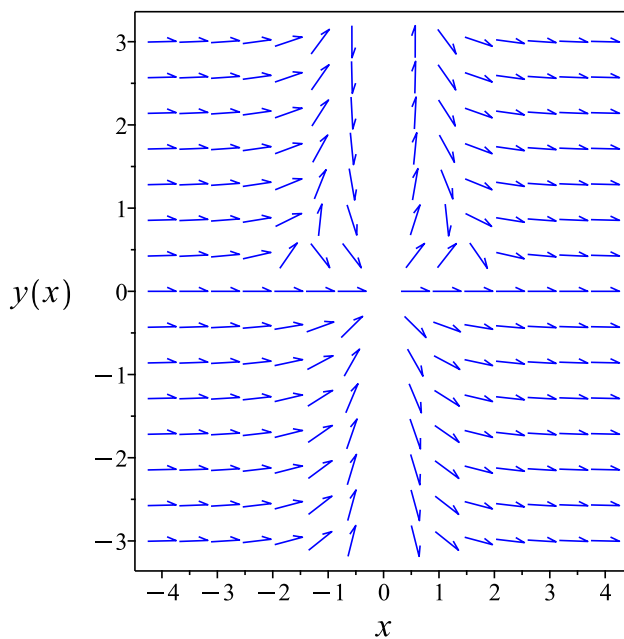


Figure 2.226: Slope field $y + x(x^2y - 1)y' = 0$

Summary of solutions found

$$-\frac{y^2}{2x^2} + \frac{y^3}{3} = c_1$$

✓ **Maple.** Time used: 0.159 (sec). Leaf size: 685

```
ode:=y(x)+x*(x^2*y(x)-1)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

Solution too large to show

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful
```

Maple step by step

Let's solve

$$y(x) + x(x^2y(x) - 1) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{y(x)}{x(x^2y(x)-1)}$$

✓ **Mathematica.** Time used: 5.824 (sec). Leaf size: 89

```
ode=y[x]+x*(x^2*y[x]-1)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\int_1^{\frac{(-1)^{2/3}(x^2y(x)-4)}{2^{2/3}\sqrt[3]{5}(x^2y(x)-1)}} \frac{1}{K[1]^3 + \frac{21\sqrt[3]{-\frac{1}{2}K[1]}}{2\sqrt[5]{2/3}} + 1} dK[1] = \frac{2}{9}(-5)^{2/3}\sqrt[3]{2}\log(x) + c_1, y(x) \right]$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*(x**2*y(x) - 1)*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.3.26 Problem 25 (k)

Local contents

Solved using first_order_ode_exact	977
✓ Maple	982
✓ Mathematica	983
✗ Sympy	983

Internal problem ID [7906]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 25 (k)

Date solved : Thursday, November 27, 2025 at 06:30:27 AM

CAS classification : [_rational]

Solved using first_order_ode_exact

Time used: 0.227 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (4x y^4 + 8y^3 + x) dy &= (-y x^3 - 2x^2 - y) dx \\ (y x^3 + 2x^2 + y) dx &+ (4x y^4 + 8y^3 + x) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y x^3 + 2x^2 + y \\ N(x, y) &= 4x y^4 + 8y^3 + x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (y x^3 + 2x^2 + y) \\ &= x^3 + 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (4x y^4 + 8y^3 + x) \\ &= 4y^4 + 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{4x y^4 + 8y^3 + x} ((x^3 + 1) - (4y^4 + 1)) \\ &= \frac{-4y^4 + x^3}{4x y^4 + 8y^3 + x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y x^3 + 2x^2 + y} ((4y^4 + 1) - (x^3 + 1)) \\ &= \frac{4y^4 - x^3}{y x^3 + 2x^2 + y} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(4y^4 + 1) - (x^3 + 1)}{x(y x^3 + 2x^2 + y) - y(4x y^4 + 8y^3 + x)} \\ &= -\frac{1}{yx + 2} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = -\frac{1}{t + 2}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int \left(-\frac{1}{t+2}\right) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\ln(t+2)} \\ &= \frac{1}{t + 2} \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{yx + 2}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{yx+2}(y x^3 + 2x^2 + y) \\ &= \frac{y x^3 + 2x^2 + y}{yx+2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{yx+2}(4x y^4 + 8y^3 + x) \\ &= \frac{4x y^4 + 8y^3 + x}{yx+2}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{y x^3 + 2x^2 + y}{yx+2} \right) + \left(\frac{4x y^4 + 8y^3 + x}{yx+2} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{y x^3 + 2x^2 + y}{yx+2} dx \\ \phi &= \frac{x^3}{3} + \ln(yx+2) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{x}{yx+2} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{4xy^4 + 8y^3 + x}{yx+2}$. Therefore equation (4) becomes

$$\frac{4xy^4 + 8y^3 + x}{yx + 2} = \frac{x}{yx + 2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 4y^3$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (4y^3) dy \\ f(y) &= y^4 + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x^3}{3} + \ln(yx + 2) + y^4 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x^3}{3} + \ln(yx + 2) + y^4$$

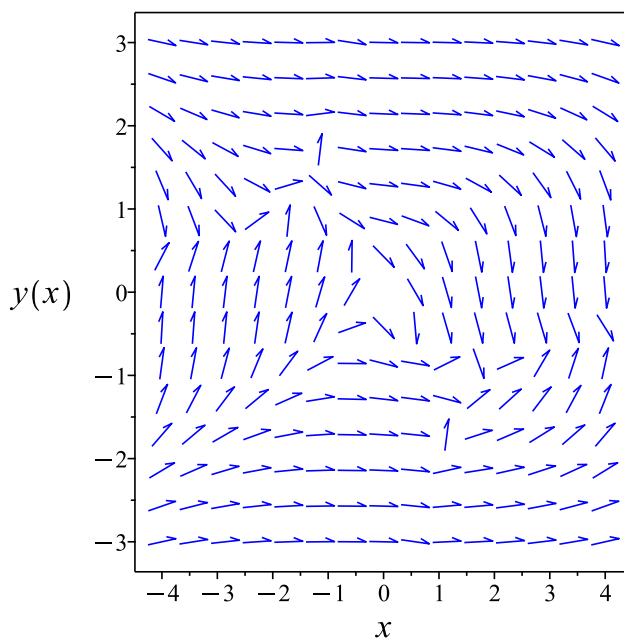


Figure 2.227: Slope field $y + yx^3 + 2x^2 + (x + 4y^4x + 8y^3)y' = 0$

Summary of solutions found

$$\frac{x^3}{3} + \ln(yx + 2) + y^4 = c_1$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 25

```
ode:=y(x)+x^3*y(x)+2*x^2+(x+4*x*y(x)^4+8*y(x)^3)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$-\frac{x^3}{3} - \ln(xy + 2) - y^4 + c_1 = 0$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
Looking for potential symmetries
trying inverse_Riccati
trying an equivalence to an Abel ODE
differential order: 1; trying a linearization to 2nd order
--- trying a change of variables {x -> y(x), y(x) -> x}
differential order: 1; trying a linearization to 2nd order
trying 1st order ODE linearizable_by_differentiation
--- Trying Lie symmetry methods, 1st order ---
-> Computing symmetries using: way = 2
[0, (x*y+2)/(4*x*y^4+8*y^3+x)]
<- successful computation of symmetries.
trying an integrating factor from the invariance group
<- integrating factor successful
```

Maple step by step

Let's solve

$$y(x) + x^3 y(x) + 2x^2 + (x + 4xy(x)^4 + 8y(x)^3) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-y(x) - x^3 y(x) - 2x^2}{x + 4xy(x)^4 + 8y(x)^3}$$

✓ **Mathematica.** Time used: 0.119 (sec). Leaf size: 25

```
ode=(y[x]+x^3*y[x]+2*x^2)+(x+4*x*y[x]^4+8*y[x]^3)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\frac{x^3}{3} + y(x)^4 + \log(xy(x) + 2) = c_1, y(x) \right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**3*y(x) + 2*x**2 + (4*x*y(x)**4 + x + 8*y(x)**3)*Derivative(y(x), x) +
        y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.3.27 Problem 26 (a)

Local contents

Solved using first_order_ode_exact	984
✓ Maple	988
✓ Mathematica	989
✓ Sympy	990

Internal problem ID [7907]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (a)

Date solved : Thursday, November 27, 2025 at 06:30:32 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.118 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (x^2 e^x + y) dx \\ (-x^2 e^x - y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x^2 e^x - y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (-x^2 e^x - y) \\ &= -1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((-1) - (1)) \\ &= -\frac{2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{2}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2\ln(x)} \\ &= \frac{1}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2}(-x^2 e^x - y) \\ &= \frac{-x^2 e^x - y}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2}(x) \\ &= \frac{1}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-x^2 e^x - y}{x^2} \right) + \left(\frac{1}{x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{1}{x} dy \\ \phi &= \frac{y}{x} + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{y}{x^2} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{-x^2 e^x - y}{x^2}$. Therefore equation (4) becomes

$$\frac{-x^2 e^x - y}{x^2} = -\frac{y}{x^2} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -e^x$$

Integrating the above w.r.t x gives

$$\begin{aligned} \int f'(x) dx &= \int (-e^x) dx \\ f(x) &= -e^x + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

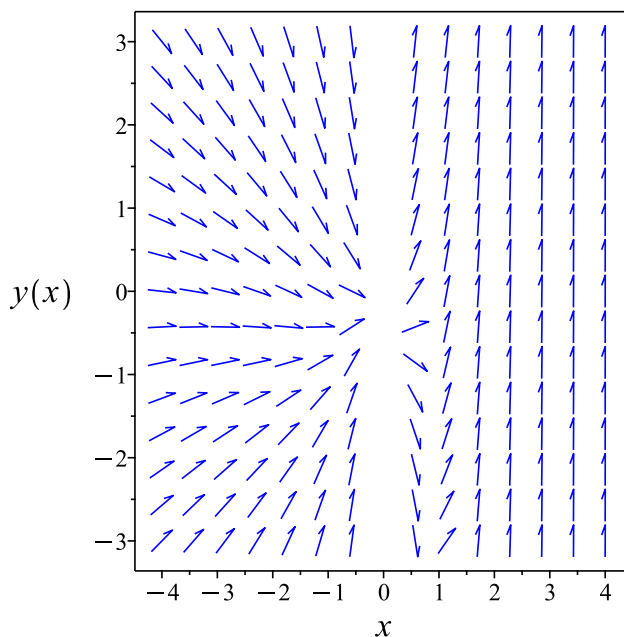
$$\phi = \frac{y}{x} - e^x + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y}{x} - e^x$$

Solving for y gives

$$y = x(e^x + c_1)$$

Figure 2.228: Slope field $-y - x^2e^x + y'x = 0$ Summary of solutions found

$$y = x(e^x + c_1)$$

✓ Maple. Time used: 0.001 (sec). Leaf size: 10

```
ode:=-y(x)-exp(x)*x^2+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (e^x + c_1)x$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$-y(x) - x^2e^x + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x)+x^2e^x}{x}$$
- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx}y(x) = \frac{y(x)}{x} + xe^x$$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - \frac{y(x)}{x} = xe^x$$
- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x} \right) = \mu(x) xe^x$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$
- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{\mu(x)}{x}$$
- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x}$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int \mu(x) xe^x dx + C1$$
- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int \mu(x) xe^x dx + C1$$
- Solve for $y(x)$

$$y(x) = \frac{\int \mu(x) xe^x dx + C1}{\mu(x)}$$
- Substitute $\mu(x) = \frac{1}{x}$

$$y(x) = x \left(\int e^x dx + C1 \right)$$
- Evaluate the integrals on the rhs

$$y(x) = x(e^x + C1)$$

✓ **Mathematica.** Time used: 0.031 (sec). Leaf size: 13

```
ode=(-y[x]-x^2*Exp[x])+x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x(e^x + c_1)$$

✓ Sympy. Time used: 0.128 (sec). Leaf size: 8

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*exp(x) + x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = x(C_1 + e^x)$$

2.3.28 Problem 26 (b)

Local contents

Solved using first_order_ode_separable	991
Solved using first_order_ode_LIE	993
Solved using first_order_ode_riccati	1000
Solved using first_order_ode_riccati_by_guessing_particular_solution	1004
✓ Maple	1006
✓ Mathematica	1007
✓ Sympy	1007

Internal problem ID [7908]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (b)

Date solved : Thursday, November 27, 2025 at 06:30:37 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.116 (sec)

Solve

$$1 + y^2 = (x^2 + x) y'$$

The ode

$$y' = \frac{1 + y^2}{x(x + 1)} \quad (2.99)$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{1 + y^2}{x(x + 1)} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x(x + 1)} \\ g(y) &= y^2 + 1 \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int \frac{1}{y^2 + 1} dy = \int \frac{1}{x(x+1)} dx$$

$$\arctan(y) = \ln\left(\frac{x}{x+1}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y^2 + 1 = 0$$

for y gives

$$y = -i$$

$$y = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\arctan(y) = \ln\left(\frac{x}{x+1}\right) + c_1$$

$$y = -i$$

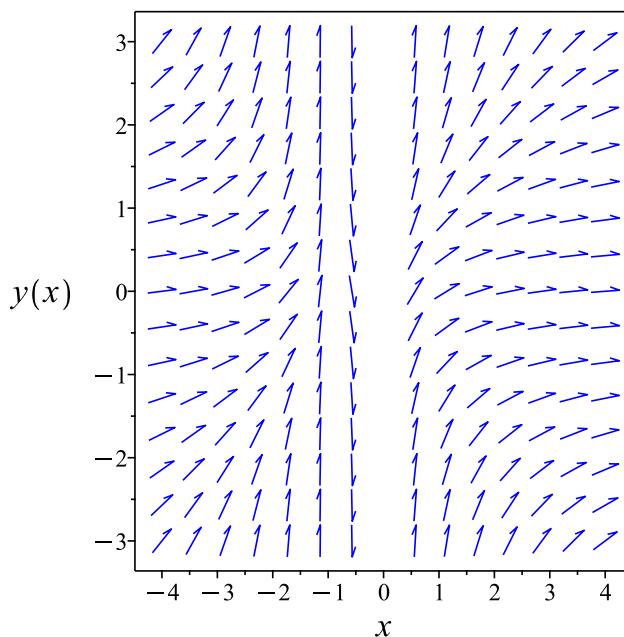
$$y = i$$

Solving for y gives

$$y = -i$$

$$y = i$$

$$y = \tan\left(\ln\left(\frac{x}{x+1}\right) + c_1\right)$$

Figure 2.229: Slope field $1 + y^2 = (x^2 + x)y'$ Summary of solutions found

$$y = -i$$

$$y = i$$

$$y = \tan \left(\ln \left(\frac{x}{x+1} \right) + c_1 \right)$$

Solved using first_order_ode_LIE

Time used: 1.066 (sec)

Solve

$$1 + y^2 = (x^2 + x)y'$$

Writing the ode as

$$y' = \frac{y^2 + 1}{x(x+1)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 2 to use as anstaz gives

$$\xi = x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (1E)$$

$$\eta = x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3, b_4, b_5, b_6\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 2xb_4 + yb_5 + b_2 + \frac{(y^2 + 1)(-2xa_4 + xb_5 - ya_5 + 2yb_6 - a_2 + b_3)}{x(x+1)} \\ & - \frac{(y^2 + 1)^2(xa_5 + 2ya_6 + a_3)}{x^2(x+1)^2} \\ & - \left(-\frac{y^2 + 1}{x^2(x+1)} - \frac{y^2 + 1}{x(x+1)^2} \right) (x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1) \\ & - \frac{2y(x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1)}{x(x+1)} = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\begin{aligned} & \frac{2x^5 b_4 - 2x^4 y b_4 + x^4 y b_5 - x^3 y^2 b_5 + x^2 y^3 a_5 - x y^4 a_5 + 2x y^4 a_6 - 2y^5 a_6 + x^4 b_2 + 4x^4 b_4 - 2x^3 y b_2 - 2x^3 y b_4 + 2x^3 y b_5 + x^2 y^2 a_2 - x^2 y^2 a_4 - x^2 y^2 b_3}{1} \\ & - x^2 y^2 b_5 + 2x y^3 a_3 - y^4 a_3 + y^4 a_6 + 2x^3 b_2 + 2x^3 b_4 + x^3 b_5 + x^2 y a_5 - 2x^2 y b_1 \\ & - 2x^2 y b_2 + y b_5 x^2 + 2x^2 y b_6 + 2x y^2 a_1 - 2x y^2 a_5 + 2x y^2 a_6 - x y^2 b_3 + y^3 a_3 \\ & - 4y^3 a_6 + x^2 a_2 - x^2 a_4 + b_2 x^2 + x^2 b_3 + x^2 b_5 + 2x y a_3 - 2x y b_1 + 2x y b_6 \\ & + y^2 a_1 - 2y^2 a_3 + y^2 a_6 + 2x a_1 - x a_5 + x b_3 + y a_3 - 2y a_6 + a_1 - a_3 = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned} & 2x^5 b_4 - 2x^4 y b_4 + x^4 y b_5 - x^3 y^2 b_5 + x^2 y^3 a_5 - x y^4 a_5 + 2x y^4 a_6 - 2y^5 a_6 \\ & + x^4 b_2 + 4x^4 b_4 - 2x^3 y b_2 - 2x^3 y b_4 + 2x^3 y b_5 + x^2 y^2 a_2 - x^2 y^2 a_4 - x^2 y^2 b_3 \\ & - x^2 y^2 b_5 + 2x y^3 a_3 - y^4 a_3 + y^4 a_6 + 2x^3 b_2 + 2x^3 b_4 + x^3 b_5 + x^2 y a_5 - 2x^2 y b_1 \\ & - 2x^2 y b_2 + y b_5 x^2 + 2x^2 y b_6 + 2x y^2 a_1 - 2x y^2 a_5 + 2x y^2 a_6 - x y^2 b_3 + y^3 a_3 \\ & - 4y^3 a_6 + x^2 a_2 - x^2 a_4 + b_2 x^2 + x^2 b_3 + x^2 b_5 + 2x y a_3 - 2x y b_1 + 2x y b_6 \\ & + y^2 a_1 - 2y^2 a_3 + y^2 a_6 + 2x a_1 - x a_5 + x b_3 + y a_3 - 2y a_6 + a_1 - a_3 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & a_5 v_1^2 v_2^3 - a_5 v_1 v_2^4 + 2a_6 v_1 v_2^4 - 2a_6 v_2^5 + 2b_4 v_1^5 - 2b_4 v_1^4 v_2 + b_5 v_1^4 v_2 - b_5 v_1^3 v_2^2 \\ & + a_2 v_1^2 v_2^2 + 2a_3 v_1 v_2^3 - a_3 v_2^4 - a_4 v_1^2 v_2^2 + a_6 v_2^4 + b_2 v_1^4 - 2b_2 v_1^3 v_2 - b_3 v_1^2 v_2^2 + 4b_4 v_1^4 \\ & - 2b_4 v_1^3 v_2 + 2b_5 v_1^3 v_2 - b_5 v_1^2 v_2^2 + 2a_1 v_1 v_2^2 + a_3 v_2^3 + a_5 v_1^2 v_2 - 2a_5 v_1 v_2^2 + 2a_6 v_1 v_2^2 \quad (7E) \\ & - 4a_6 v_2^3 - 2b_1 v_1^2 v_2 + 2b_2 v_1^3 - 2b_2 v_1^2 v_2 - b_3 v_1 v_2^2 + 2b_4 v_1^3 + b_5 v_1^3 + b_5 v_1^2 v_2 \\ & + 2b_6 v_1^2 v_2 + a_1 v_2^2 + a_2 v_1^2 + 2a_3 v_1 v_2 - 2a_3 v_2^2 - a_4 v_1^2 + a_6 v_2^2 - 2b_1 v_1 v_2 + b_2 v_1^2 \\ & + b_3 v_1^2 + b_5 v_1^2 + 2b_6 v_1 v_2 + 2a_1 v_1 + a_3 v_2 - a_5 v_1 - 2a_6 v_2 + b_3 v_1 + a_1 - a_3 = 0 \end{aligned}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} & 2b_4 v_1^5 + (-2b_4 + b_5) v_1^4 v_2 + (b_2 + 4b_4) v_1^4 - b_5 v_1^3 v_2^2 + (-2b_2 - 2b_4 + 2b_5) v_1^3 v_2 \\ & + (2b_2 + 2b_4 + b_5) v_1^3 + a_5 v_1^2 v_2^3 + (a_2 - a_4 - b_3 - b_5) v_1^2 v_2^2 \\ & + (a_5 - 2b_1 - 2b_2 + b_5 + 2b_6) v_1^2 v_2 + (a_2 - a_4 + b_2 + b_3 + b_5) v_1^2 \quad (8E) \\ & + (-a_5 + 2a_6) v_1 v_2^4 + 2a_3 v_1 v_2^3 + (2a_1 - 2a_5 + 2a_6 - b_3) v_1 v_2^2 \\ & + (2a_3 - 2b_1 + 2b_6) v_1 v_2 + (2a_1 - a_5 + b_3) v_1 - 2a_6 v_2^5 + (-a_3 + a_6) v_2^4 \\ & + (a_3 - 4a_6) v_2^3 + (a_1 - 2a_3 + a_6) v_2^2 + (a_3 - 2a_6) v_2 + a_1 - a_3 = 0 \end{aligned}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$a_5 = 0$$

$$2a_3 = 0$$

$$-2a_6 = 0$$

$$2b_4 = 0$$

$$-b_5 = 0$$

$$a_1 - a_3 = 0$$

$$-a_3 + a_6 = 0$$

$$a_3 - 4a_6 = 0$$

$$a_3 - 2a_6 = 0$$

$$-a_5 + 2a_6 = 0$$

$$b_2 + 4b_4 = 0$$

$$-2b_4 + b_5 = 0$$

$$a_1 - 2a_3 + a_6 = 0$$

$$2a_1 - a_5 + b_3 = 0$$

$$2a_3 - 2b_1 + 2b_6 = 0$$

$$-2b_2 - 2b_4 + 2b_5 = 0$$

$$2b_2 + 2b_4 + b_5 = 0$$

$$2a_1 - 2a_5 + 2a_6 - b_3 = 0$$

$$a_2 - a_4 - b_3 - b_5 = 0$$

$$a_2 - a_4 + b_2 + b_3 + b_5 = 0$$

$$a_5 - 2b_1 - 2b_2 + b_5 + 2b_6 = 0$$

Solving the above equations for the unknowns gives

$$a_1 = 0$$

$$a_2 = a_4$$

$$a_3 = 0$$

$$a_4 = a_4$$

$$a_5 = 0$$

$$a_6 = 0$$

$$b_1 = b_6$$

$$b_2 = 0$$

$$b_3 = 0$$

$$b_4 = 0$$

$$b_5 = 0$$

$$b_6 = b_6$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 0$$

$$\eta = y^2 + 1$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y^2 + 1} dy \end{aligned}$$

Which results in

$$S = \arctan(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y^2 + 1}{x(x + 1)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{y^2 + 1} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{x(x + 1)} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R(R + 1)}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

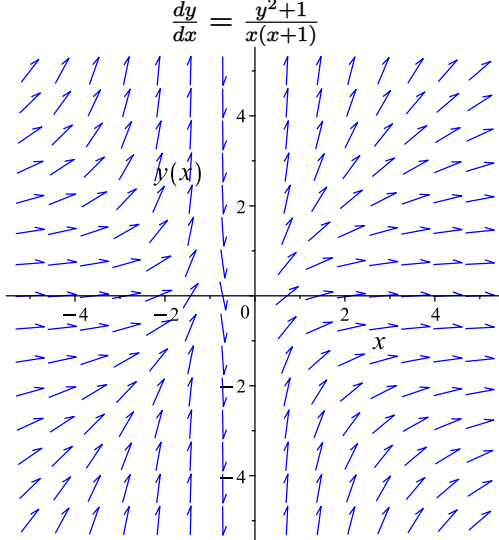
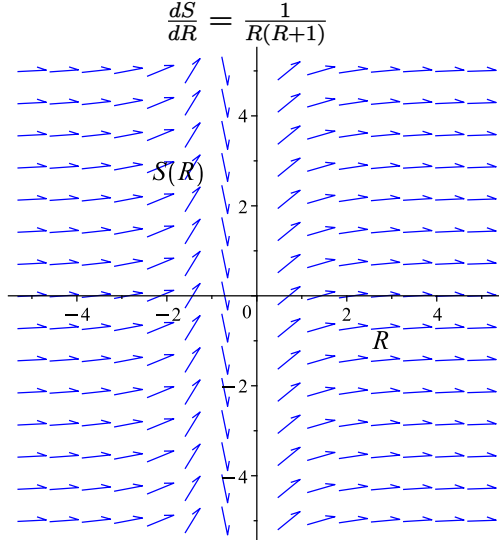
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{1}{R(R + 1)} dR \\ S(R) &= -\ln(R + 1) + \ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\arctan(y) = -\ln(x + 1) + \ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{y^2+1}{x(x+1)}$ 	$R = x$ $S = \arctan(y)$	$\frac{dS}{dR} = \frac{1}{R(R+1)}$ 

Solving for y gives

$$y = \tan(-\ln(x+1) + \ln(x) + c_2)$$

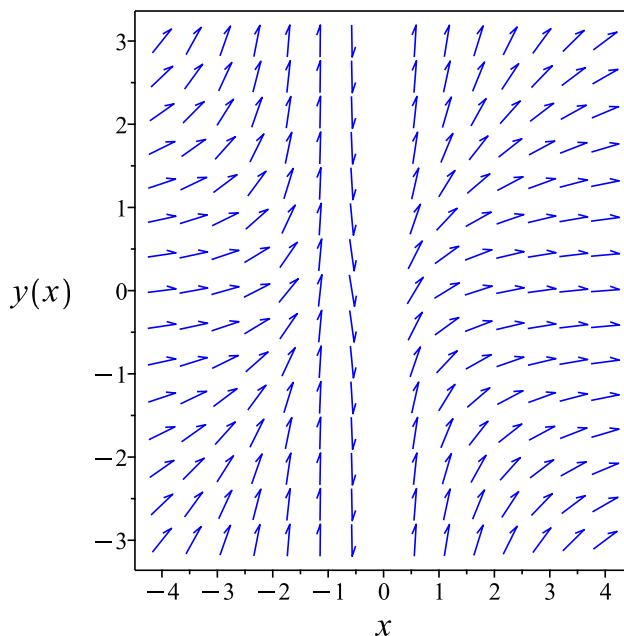


Figure 2.230: Slope field $1 + y^2 = (x^2 + x)y'$

Summary of solutions found

$$y = \tan(-\ln(x+1) + \ln(x) + c_2)$$

Solved using first_order_ode_riccati

Time used: 0.760 (sec)

Solve

$$1 + y^2 = (x^2 + x) y'$$

In canonical form the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y^2 + 1}{x(x+1)} \end{aligned}$$

This is a Riccati ODE. Comparing the ODE to solve

$$y' = \frac{y^2}{x(x+1)} + \frac{1}{x(x+1)}$$

With Riccati ODE standard form

$$y' = f_0(x) + f_1(x)y + f_2(x)y^2$$

Shows that $f_0(x) = \frac{1}{x(x+1)}$, $f_1(x) = 0$ and $f_2(x) = \frac{1}{x(x+1)}$. Let

$$\begin{aligned} y &= \frac{-u'}{f_2 u} \\ &= \frac{-u'}{\frac{u}{x(x+1)}} \end{aligned} \tag{1}$$

Using the above substitution in the given ODE results (after some simplification) in a second order ODE to solve for $u(x)$ which is

$$f_2 u''(x) - (f_2' + f_1 f_2) u'(x) + f_2^2 f_0 u(x) = 0 \tag{2}$$

But

$$\begin{aligned} f_2' &= -\frac{1}{x^2(x+1)} - \frac{1}{x(x+1)^2} \\ f_1 f_2 &= 0 \\ f_2^2 f_0 &= \frac{1}{x^3(x+1)^3} \end{aligned}$$

Substituting the above terms back in equation (2) gives

$$\frac{u''(x)}{x(x+1)} - \left(-\frac{1}{x^2(x+1)} - \frac{1}{x(x+1)^2} \right) u'(x) + \frac{u(x)}{x^3(x+1)^3} = 0$$

In normal form the ode

$$\frac{\frac{d^2u}{dx^2}}{x(x+1)} - \left(-\frac{1}{x^2(x+1)} - \frac{1}{x(x+1)^2} \right) \left(\frac{du}{dx} \right) + \frac{u}{x^3(x+1)^3} = 0 \quad (1)$$

Becomes

$$\frac{d^2u}{dx^2} + p(x) \left(\frac{du}{dx} \right) + q(x) u = 0 \quad (2)$$

Where

$$p(x) = \frac{2x+1}{x(x+1)}$$

$$q(x) = \frac{1}{x^2(x+1)^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2} u(\tau) + p_1 \left(\frac{d}{d\tau} u(\tau) \right) + q_1 u(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\frac{d^2}{dx^2} \tau(x) + p(x) \left(\frac{d}{dx} \tau(x) \right)}{\left(\frac{d}{dx} \tau(x) \right)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\left(\frac{d}{dx} \tau(x) \right)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\frac{d^2}{dx^2} \tau(x) + p(x) \left(\frac{d}{dx} \tau(x) \right) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x) dx} dx \\ &= \int e^{-\int \frac{2x+1}{x(x+1)} dx} dx \\ &= \int e^{-\ln(x(x+1))} dx \\ &= \int \frac{1}{x(x+1)} dx \\ &= -\ln(x+1) + \ln(x) \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\left(\frac{d}{dx}\tau(x)\right)^2} \\ &= \frac{1}{\frac{x^2(x+1)^2}{1}} \\ &= 1 \end{aligned} \tag{7}$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2}u(\tau) + q_1u(\tau) &= 0 \\ \frac{d^2}{d\tau^2}u(\tau) + u(\tau) &= 0 \end{aligned}$$

The above ode is now solved for $u(\tau)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Au''(\tau) + Bu'(\tau) + Cu(\tau) = 0$$

Where in the above $A = 1, B = 0, C = 1$. Let the solution be $u(\tau) = e^{\lambda\tau}$. Substituting this into the ODE gives

$$\lambda^2 e^{\lambda\tau} + e^{\lambda\tau} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda\tau}$ gives

$$\lambda^2 + 1 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(1)} \\ &= \pm i \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= +i \\ \lambda_2 &= -i \end{aligned}$$

Which simplifies to

$$\begin{aligned}\lambda_1 &= i \\ \lambda_2 &= -i\end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$u(\tau) = e^{\alpha\tau}(c_1 \cos(\beta\tau) + c_2 \sin(\beta\tau))$$

Which becomes

$$u(\tau) = e^0(c_1 \cos(\tau) + c_2 \sin(\tau))$$

Or

$$u(\tau) = c_1 \cos(\tau) + c_2 \sin(\tau)$$

The above solution is now transformed back to u using (6) which results in

$$u = c_1 \cos(-\ln(x+1) + \ln(x)) + c_2 \sin(-\ln(x+1) + \ln(x))$$

Taking derivative gives

$$\begin{aligned}u'(x) &= -c_1 \left(-\frac{1}{x+1} + \frac{1}{x} \right) \sin(-\ln(x+1) + \ln(x)) \\ &\quad + c_2 \left(-\frac{1}{x+1} + \frac{1}{x} \right) \cos(-\ln(x+1) + \ln(x))\end{aligned}\tag{4}$$

Substituting equations (3,4) into (1) results in

$$\begin{aligned}y &= \frac{-u'}{f_2 u} \\ y &= \frac{-u'}{\frac{u}{x(x+1)}} \\ y &= \frac{c_1 \sin(-\ln(x+1) + \ln(x)) - c_2 \cos(-\ln(x+1) + \ln(x))}{c_1 \cos(-\ln(x+1) + \ln(x)) + c_2 \sin(-\ln(x+1) + \ln(x))}\end{aligned}$$

Doing change of constants, the above solution becomes

$$y = - \frac{\left(-\left(-\frac{1}{x+1} + \frac{1}{x}\right) \sin(-\ln(x+1) + \ln(x)) + c_3\left(-\frac{1}{x+1} + \frac{1}{x}\right) \cos(-\ln(x+1) + \ln(x))\right) x(x+1)}{\cos(-\ln(x+1) + \ln(x)) + c_3 \sin(-\ln(x+1) + \ln(x))}$$

Simplifying the above gives

$$y = \frac{\sin(-\ln(x+1) + \ln(x)) - c_3 \cos(-\ln(x+1) + \ln(x))}{\cos(-\ln(x+1) + \ln(x)) + c_3 \sin(-\ln(x+1) + \ln(x))}$$

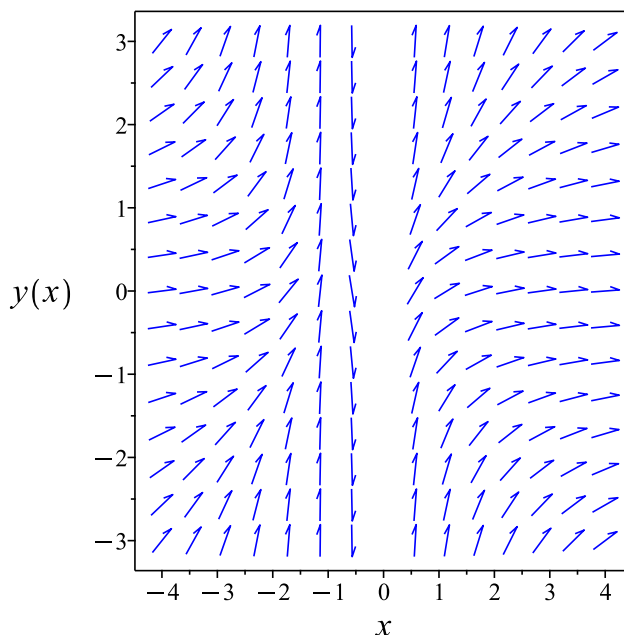


Figure 2.231: Slope field $1 + y^2 = (x^2 + x) y'$

Summary of solutions found

$$y = \frac{\sin(-\ln(x+1) + \ln(x)) - c_3 \cos(-\ln(x+1) + \ln(x))}{\cos(-\ln(x+1) + \ln(x)) + c_3 \sin(-\ln(x+1) + \ln(x))}$$

Solved using first_order_ode_riccati_by_guessing_particular_solution

Time used: 0.059 (sec)

Solve

$$1 + y^2 = (x^2 + x) y'$$

This is a Riccati ODE. Comparing the above ODE to solve with the Riccati standard form

$$y' = f_0(x) + f_1(x)y + f_2(x)y^2$$

Shows that

$$f_0(x) = \frac{1}{x(x+1)}$$

$$f_1(x) = 0$$

$$f_2(x) = \frac{1}{x(x+1)}$$

Using trial and error, the following particular solution was found

$$y_p = i$$

Since a particular solution is known, then the general solution is given by

$$y = y_p + \frac{\phi(x)}{c_1 - \int \phi(x) f_2 dx}$$

Where

$$\phi(x) = e^{\int 2f_2 y_p + f_1 dx}$$

Evaluating the above gives the general solution as

$$y = \frac{2ic_1(x+1)^{2i} + x^{2i}}{ix^{2i} + 2(x+1)^{2i}c_1}$$

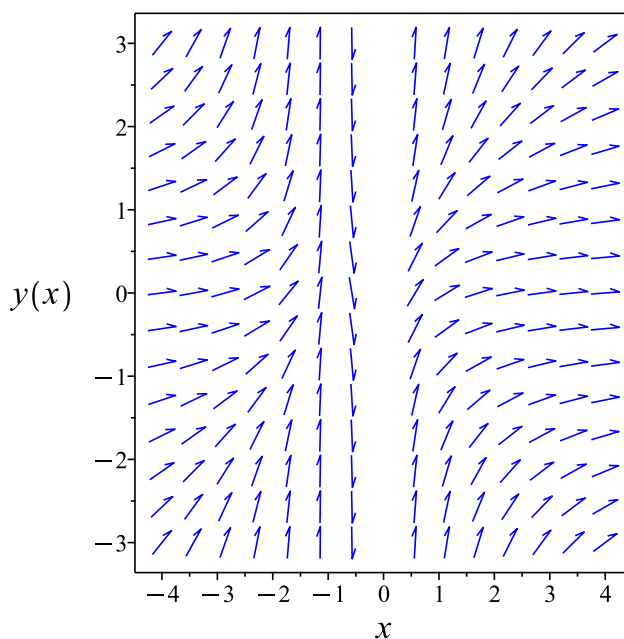
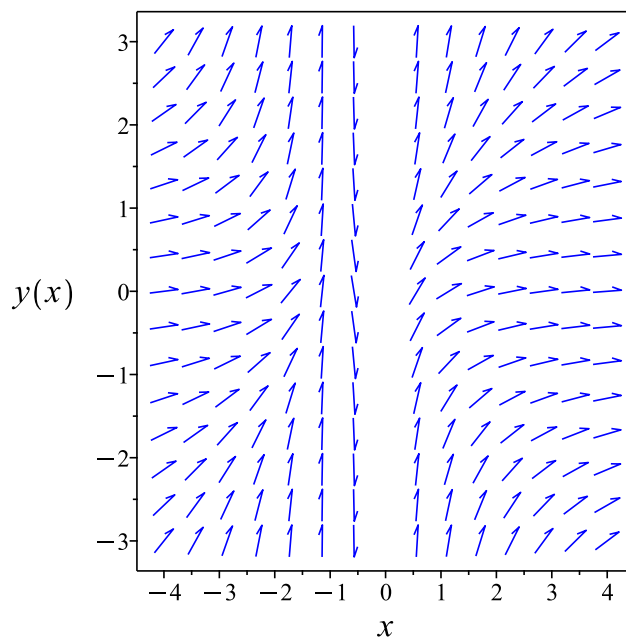


Figure 2.232: Slope field $1 + y^2 = (x^2 + x)y'$

Figure 2.233: Slope field $1 + y^2 = (x^2 + x)y'$ Summary of solutions found

$$y = \frac{2ic_1(x+1)^{2i} + x^{2i}}{ix^{2i} + 2(x+1)^{2i}c_1}$$

✓ Maple. Time used: 0.005 (sec). Leaf size: 15

```
ode:=1+y(x)^2 = (x^2+x)*diff(y(x),x);
dsolve(ode,y(x), singsol=all);
```

$$y = \tan(-\ln(x+1) + \ln(x) + c_1)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
<- separable successful
```

Maple step by step

Let's solve

$$1 + y(x)^2 = (x^2 + x) \left(\frac{d}{dx} y(x) \right)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{1-y(x)^2}{x^2+x}$$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{-1-y(x)^2} = -\frac{1}{x^2+x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{-1-y(x)^2} dx = \int -\frac{1}{x^2+x} dx + C1$$

- Evaluate integral

$$-\arctan(y(x)) = -\ln(x) + \ln(x+1) + C1$$

- Solve for $y(x)$

$$y(x) = -\tan(-\ln(x) + \ln(x+1) + C1)$$

✓ **Mathematica.** Time used: 0.167 (sec). Leaf size: 56

```
ode=1+y[x]^2==(x+x^2)*D[y[x],x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \text{InverseFunction} \left[\int_1^{\#1} \frac{1}{K[1]^2 + 1} dK[1] \& \right] \left[\int_1^x \frac{1}{K[2]^2 + K[2]} dK[2] + c_1 \right]$$

$$y(x) \rightarrow -i$$

$$y(x) \rightarrow i$$

✓ **Sympy.** Time used: 0.379 (sec). Leaf size: 14

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-x**2 - x)*Derivative(y(x), x) + y(x)**2 + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \tan(C_1 + \log(x) - \log(x+1))$$

2.3.29 Problem 26 (c)

Local contents

Solved using first_order_ode_exact	1008
✓ Maple	1012
✓ Mathematica	1013
✓ Sympy	1014

Internal problem ID [7909]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (c)

Date solved : Thursday, November 27, 2025 at 06:30:45 AM

CAS classification : [_linear]

Solved using first_order_ode_exact

Time used: 0.102 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (x^3 - 2y) dx \\ (-x^3 + 2y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x^3 + 2y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-x^3 + 2y) \\ &= 2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((2) - (1)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int \frac{1}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(-x^3 + 2y) \\ &= -(x^3 - 2y)x\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(x) \\ &= x^2\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (- (x^3 - 2y)x) + (x^2) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int x^2 dy \\ \phi &= x^2 y + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = 2yx + f'(x) \tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = -(x^3 - 2y)x$. Therefore equation (4) becomes

$$-(x^3 - 2y)x = 2yx + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -x^4$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int (-x^4) dx$$

$$f(x) = -\frac{x^5}{5} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = x^2y - \frac{1}{5}x^5 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

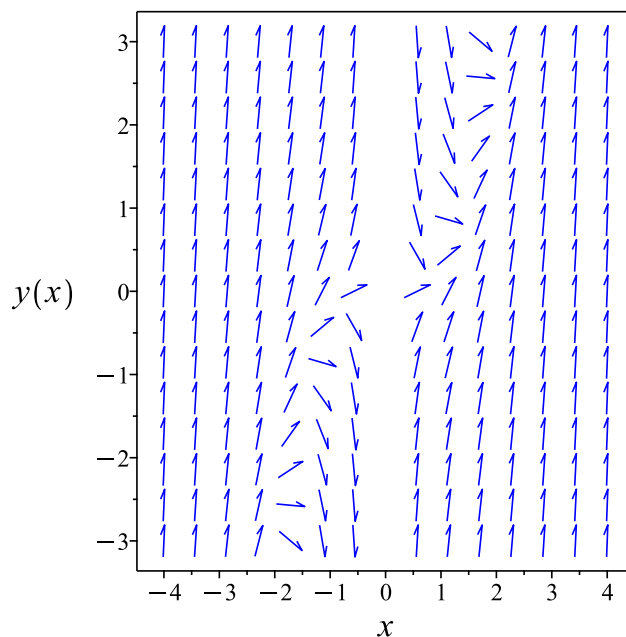
$$c_1 = x^2y - \frac{1}{5}x^5$$

Simplifying the above gives

$$-\frac{x^2(x^3 - 5y)}{5} = c_1$$

Solving for y gives

$$y = \frac{x^5 + 5c_1}{5x^2}$$

Figure 2.234: Slope field $2y - x^3 + y'/x = 0$ Summary of solutions found

$$y = \frac{x^5 + 5c_1}{5x^2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 16

```
ode:=2*y(x)-x^3+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^5 + 5c_1}{5x^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$2y(x) - x^3 + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx}y(x)$
- Solve for the highest derivative
 $\frac{d}{dx}y(x) = \frac{-2y(x)+x^3}{x}$
- Collect w.r.t. $y(x)$ and simplify
 $\frac{d}{dx}y(x) = -\frac{2y(x)}{x} + x^2$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx}y(x) + \frac{2y(x)}{x} = x^2$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{2y(x)}{x} \right) = \mu(x) x^2$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x) \mu(x))$
 $\mu(x) \left(\frac{d}{dx}y(x) + \frac{2y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$
- Isolate $\frac{d}{dx}\mu(x)$
 $\frac{d}{dx}\mu(x) = \frac{2\mu(x)}{x}$
- Solve to find the integrating factor
 $\mu(x) = x^2$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}(y(x) \mu(x)) \right) dx = \int \mu(x) x^2 dx + C1$
- Evaluate the integral on the lhs
 $y(x) \mu(x) = \int \mu(x) x^2 dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int \mu(x) x^2 dx + C1}{\mu(x)}$
- Substitute $\mu(x) = x^2$
 $y(x) = \frac{\int x^4 dx + C1}{x^2}$
- Evaluate the integrals on the rhs
 $y(x) = \frac{\frac{x^5}{5} + C1}{x^2}$
- Simplify
 $y(x) = \frac{x^5 + 5C1}{5x^2}$

✓ **Mathematica.** Time used: 0.019 (sec). Leaf size: 19

```
ode=(2*y[x]-x^3)+x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^3}{5} + \frac{c_1}{x^2}$$

✓ Sympy. Time used: 0.110 (sec). Leaf size: 12

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-x**3 + x*Derivative(y(x), x) + 2*y(x), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1 + \frac{x^5}{5}}{x^2}$$

2.3.30 Problem 26 (d)

Local contents

Solved using first_order_ode_exact	1015
✓ Maple	1020
✓ Mathematica	1020
✓ Sympy	1021

Internal problem ID [7910]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (d)

Date solved : Thursday, November 27, 2025 at 06:30:50 AM

CAS classification : [[_homogeneous, 'class G'], _rational]

Solved using first_order_ode_exact

Time used: 0.108 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (y^2 - x) dy &= (-y) dx \\ (y) dx + (y^2 - x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y \\ N(x, y) &= y^2 - x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(y^2 - x) \\ &= -1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{y^2 - x} ((1) - (-1)) \\ &= -\frac{2}{-y^2 + x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y} ((-1) - (1)) \\ &= -\frac{2}{y} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int -\frac{2}{y} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-2 \ln(y)} \\ &= \frac{1}{y^2} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{1}{y^2} (y) \\ &= \frac{1}{y} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{1}{y^2} (y^2 - x) \\ &= \frac{y^2 - x}{y^2} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1}{y} \right) + \left(\frac{y^2 - x}{y^2} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{1}{y} dx \\ \phi &= \frac{x}{y} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{x}{y^2} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{y^2 - x}{y^2}$. Therefore equation (4) becomes

$$\frac{y^2 - x}{y^2} = -\frac{x}{y^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 1$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (1) dy \\ f(y) &= y + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x}{y} + y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x}{y} + y$$

Solving for y gives

$$y = \frac{c_1}{2} - \frac{\sqrt{c_1^2 - 4x}}{2}$$

$$y = \frac{c_1}{2} + \frac{\sqrt{c_1^2 - 4x}}{2}$$

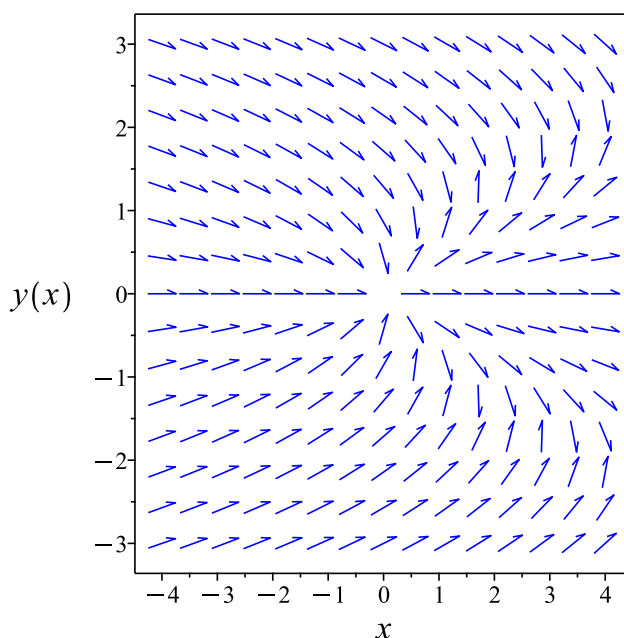


Figure 2.235: Slope field $y + (-x + y^2)y' = 0$

Summary of solutions found

$$y = \frac{c_1}{2} - \frac{\sqrt{c_1^2 - 4x}}{2}$$

$$y = \frac{c_1}{2} + \frac{\sqrt{c_1^2 - 4x}}{2}$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 37

```
ode:=y(x)+(y(x)^2-x)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{c_1}{2} - \frac{\sqrt{c_1^2 - 4x}}{2}$$

$$y = \frac{c_1}{2} + \frac{\sqrt{c_1^2 - 4x}}{2}$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
<- 1st order linear successful
<- inverse linear successful
```

Maple step by step

Let's solve

$$y(x) + (-x + y(x)^2) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{y(x)}{-x + y(x)^2}$$

✓ **Mathematica.** Time used: 0.175 (sec). Leaf size: 54

```
ode=y[x]+(-x+y[x]^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2} \left(c_1 - \sqrt{-4x + c_1^2} \right)$$

$$y(x) \rightarrow \frac{1}{2} \left(\sqrt{-4x + c_1^2} + c_1 \right)$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 0.450 (sec). Leaf size: 36

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-x + y(x)**2)*Derivative(y(x), x) + y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\frac{C_1}{2} - \frac{\sqrt{C_1^2 - 4x}}{2}, y(x) = -\frac{C_1}{2} + \frac{\sqrt{C_1^2 - 4x}}{2} \right]$$

2.3.31 Problem 26 (e)

Local contents

Solved using first_order_ode_exact	1022
✓ Maple	1027
✓ Mathematica	1028
✓ Sympy	1028

Internal problem ID [7911]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (e)

Date solved : Thursday, November 27, 2025 at 06:30:56 AM

CAS classification : [[_homogeneous, 'class G'], _rational]

Solved using first_order_ode_exact

Time used: 0.153 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-6xy^2 - x^2) dy &= (-3y^3 + yx) dx \\ (3y^3 - yx) dx + (-6xy^2 - x^2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 3y^3 - yx \\ N(x, y) &= -6xy^2 - x^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (3y^3 - yx) \\ &= 9y^2 - x \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (-6xy^2 - x^2) \\ &= -6y^2 - 2x \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{x^2y}$ is an integrating factor. Therefore by multiplying $M = 3y^3 - yx$ and $N = -x^2 - 6y^2x$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{3y^3 - yx}{x^2y} \\ N &= \frac{-x^2 - 6y^2x}{x^2y} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(\frac{-6xy^2 - x^2}{x^2y} \right) dy &= \left(-\frac{3y^3 - yx}{x^2y} \right) dx \\ \left(\frac{3y^3 - yx}{x^2y} \right) dx &+ \left(\frac{-6xy^2 - x^2}{x^2y} \right) dy = 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{3y^3 - yx}{x^2y} \\ N(x, y) &= \frac{-6xy^2 - x^2}{x^2y} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{3y^3 - yx}{x^2y} \right) \\ &= \frac{6y}{x^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{-6xy^2 - x^2}{x^2y} \right) \\ &= \frac{6y}{x^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{3y^3 - yx}{x^2y} dx \\ \phi &= -\frac{3y^2}{x} - \ln(x) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{6y}{x} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{-6xy^2 - x^2}{x^2y}$. Therefore equation (4) becomes

$$\frac{-6xy^2 - x^2}{x^2y} = -\frac{6y}{x} + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{1}{y}$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int \left(-\frac{1}{y}\right) dy$$

$$f(y) = -\ln(y) + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{3y^2}{x} - \ln(x) - \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{3y^2}{x} - \ln(x) - \ln(y)$$

Solving for y gives

$$y = \frac{e^{-\frac{\text{LambertW}\left(\frac{6e^{-2c_1}}{x^3}\right)}{2} - c_1}}{x}$$

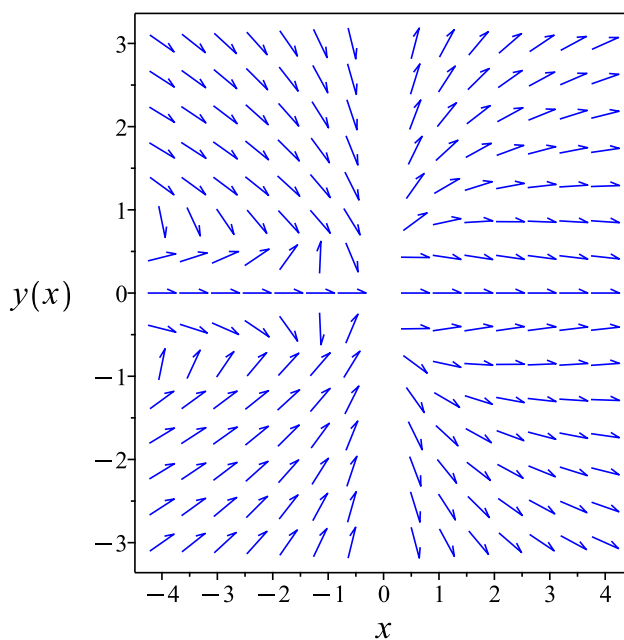


Figure 2.236: Slope field $3y^3 - yx - (x^2 + 6y^2x)y' = 0$

Summary of solutions found

$$y = \frac{e^{-\frac{\text{LambertW}\left(\frac{6e^{-2c_1}}{x^3}\right)}{2} - c_1}}{x}$$

✓ **Maple.** Time used: 0.016 (sec). Leaf size: 38

```
ode:=3*y(x)^3-y(x)*x-(x^2+6*x*y(x)^2)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{e^{\frac{3c_1}{2}} \sqrt{6}}{6x \sqrt{\frac{e^{3c_1}}{x^3 \text{LambertW}\left(\frac{6e^{3c_1}}{x^3}\right)}}}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

trying homogeneous types:

trying homogeneous G

<- homogeneous successful

Maple step by step

Let's solve

$$3y(x)^3 - xy(x) - (x^2 + 6xy(x)^2) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{3y(x)^3 + xy(x)}{x^2 + 6xy(x)^2}$$

✓ **Mathematica.** Time used: 2.144 (sec). Leaf size: 73

```
ode=(3*y[x]^3-x*y[x])-(x^2+6*x*y[x]^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{x}\sqrt{W\left(\frac{6e^{2+3c_1}}{x^3}\right)}}{\sqrt{6}}$$

$$y(x) \rightarrow \frac{\sqrt{x}\sqrt{W\left(\frac{6e^{2+3c_1}}{x^3}\right)}}{\sqrt{6}}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 6.267 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*y(x) - (x**2 + 6*x*y(x)**2)*Derivative(y(x), x) + 3*y(x)**3,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{e^{-3C_1 - \frac{W\left(\frac{6e^{-6C_1}}{x^3}\right)}}{x}}$$

2.3.32 Problem 26 (f)

Local contents

Solved using first_order_ode_exact	1029
✓ Maple	1033
✓ Mathematica	1034
✗ Sympy	1034

Internal problem ID [7912]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (f)

Date solved : Thursday, November 27, 2025 at 06:31:01 AM

CAS classification :

[[_homogeneous, 'class G'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.153 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (4y x^3 - 12) dy &= (-3y^2 x^2) dx \\ (3y^2 x^2) dx + (4y x^3 - 12) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 3y^2 x^2 \\ N(x, y) &= 4y x^3 - 12 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (3y^2 x^2) \\ &= 6x^2 y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (4y x^3 - 12) \\ &= 12x^2 y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{4y x^3 - 12} ((6x^2 y) - (12x^2 y)) \\ &= -\frac{3x^2 y}{2y x^3 - 6} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{3y^2x^2} ((12x^2y) - (6x^2y)) \\ &= \frac{2}{y} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int \frac{2}{y} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{2 \ln(y)} \\ &= y^2 \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= y^2 (3y^2x^2) \\ &= 3x^2y^4 \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= y^2 (4y x^3 - 12) \\ &= 4(y x^3 - 3) y^2 \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (3x^2y^4) + (4(y x^3 - 3) y^2) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int 3x^2 y^4 dx \\ \phi &= x^3 y^4 + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 4x^3 y^3 + f'(y)\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 4(y x^3 - 3) y^2$. Therefore equation (4) becomes

$$4(y x^3 - 3) y^2 = 4x^3 y^3 + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -12y^2$$

Integrating the above w.r.t y gives

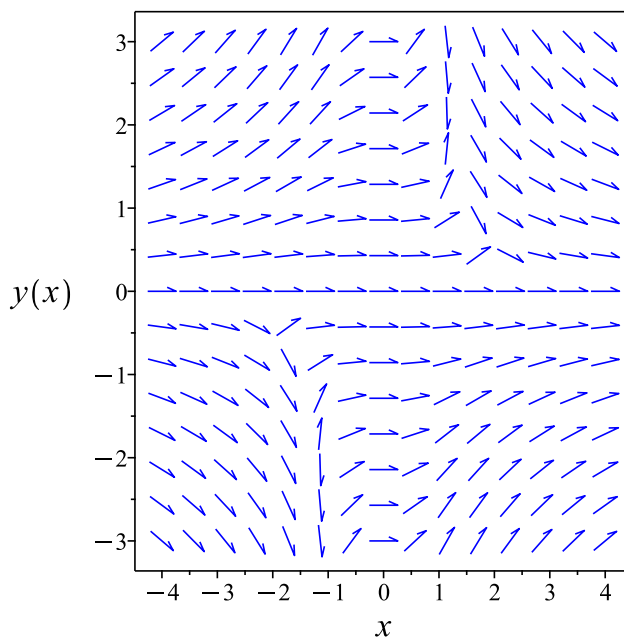
$$\begin{aligned}\int f'(y) dy &= \int (-12y^2) dy \\ f(y) &= -4y^3 + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x^3 y^4 - 4y^3 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x^3 y^4 - 4y^3$$

Figure 2.237: Slope field $3y^2x^2 + 4(yx^3 - 3)y' = 0$ Summary of solutions found

$$x^3y^4 - 4y^3 = c_1$$

✓ Maple. Time used: 0.436 (sec). Leaf size: 30

```
ode:=3*x^2*y(x)^2+4*(x^3*y(x)-3)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\text{RootOf}(c_1_Z^{12} + 4c_1_Z^3 - x^3)^9 + 4}{x^3}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful
```

Maple step by step

Let's solve

$$3x^2y(x)^2 + 4(x^3y(x) - 3) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{3x^2y(x)^2}{4(x^3y(x)-3)}$$

✓ **Mathematica.** Time used: 0.199 (sec). Leaf size: 96

```
ode=(3*x^2*y[x]^2)+4*(x^3*y[x]-3)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[70^{2/3} \left(-\frac{1}{x^3} \right)^{2/3} x^2 \log(x) + 12c_1 = 12 \int_1^x \frac{\left(-\frac{1}{x^3} \right)^{2/3} x^2 (2x^3y(x)-15)}{\sqrt[3]{70(x^3y(x)-3)}} \frac{1}{K[1]^3 + \frac{39\sqrt[3]{-1K[1]}}{70^{2/3}} + 1} dK[1], y(x) \right]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x**2*y(x)**2 + (4*x**3*y(x) - 12)*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.3.33 Problem 26 (g)

Local contents

Solved using first_order_ode_exact	1035
✓ Maple	1040
✓ Mathematica	1040
✓ Sympy	1041

Internal problem ID [7913]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (g)

Date solved : Thursday, November 27, 2025 at 06:31:06 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _Bernoulli]

Solved using first_order_ode_exact

Time used: 0.109 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-x^2) dy &= (-y(x + y)) dx \\ (y(x + y)) dx + (-x^2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y(x + y) \\ N(x, y) &= -x^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y(x + y)) \\ &= x + 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-x^2) \\ &= -2x \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. By inspection $\frac{1}{xy^2}$ is an integrating factor. Therefore by multiplying $M = y(x + y)$ and $N = -x^2$ by this integrating factor the ode becomes exact. The new M, N are

$$\begin{aligned} M &= \frac{x + y}{xy} \\ N &= -\frac{x}{y^2} \end{aligned}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (A)$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} \left(-\frac{x}{y^2}\right) dy &= \left(-\frac{x+y}{xy}\right) dx \\ \left(\frac{x+y}{xy}\right) dx + \left(-\frac{x}{y^2}\right) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= \frac{x+y}{xy} \\ N(x, y) &= -\frac{x}{y^2} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{x+y}{xy} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} \left(-\frac{x}{y^2} \right) \\ &= -\frac{1}{y^2}\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{x+y}{xy} dx \\ \phi &= \ln(x) + \frac{x}{y} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{x}{y^2} + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -\frac{x}{y^2}$. Therefore equation (4) becomes

$$-\frac{x}{y^2} = -\frac{x}{y^2} + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \ln(x) + \frac{x}{y} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \ln(x) + \frac{x}{y}$$

Solving for y gives

$$y = -\frac{x}{\ln(x) - c_1}$$

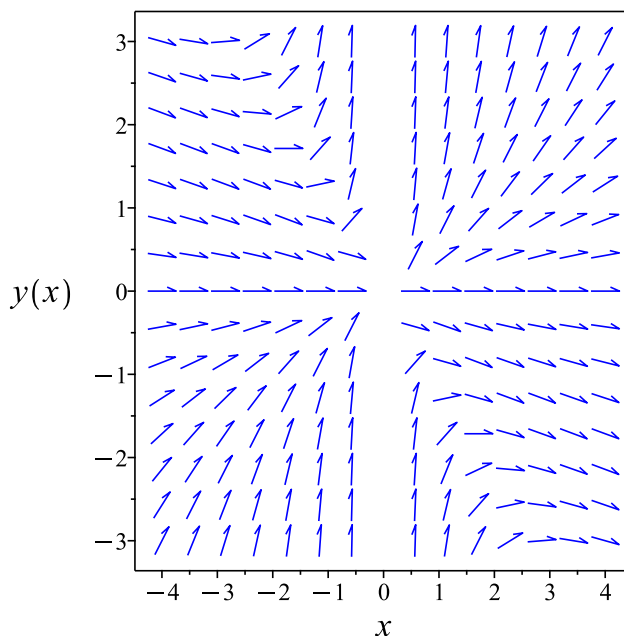


Figure 2.238: Slope field $y(x+y) - y'x^2 = 0$

Summary of solutions found

$$y = -\frac{x}{\ln(x) - c_1}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 14

```
ode:=y(x)*(x+y(x))-x^2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x}{-\ln(x) + c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$y(x)(x + y(x)) - x^2\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx}y(x)$$

$$\frac{d}{dx}y(x) = \frac{y(x)(x+y(x))}{x^2}$$

✓ **Mathematica.** Time used: 0.092 (sec). Leaf size: 21

```
ode=y[x]*(x+y[x])-x^2*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x}{-\log(x) + c_1}$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 0.120 (sec). Leaf size: 8

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*Derivative(y(x), x) + (x + y(x))*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x}{C_1 - \log(x)}$$

2.3.34 Problem 26 (h)

Local contents

Solved using first_order_ode_exact	1042
✓ Maple	1046
✓ Mathematica	1047
✓ Sympy	1047

Internal problem ID [7914]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (h)

Date solved : Thursday, November 27, 2025 at 06:31:11 AM

CAS classification :

[[_homogeneous, 'class G'], _rational, [_Abel, '2nd type', 'class B']]

Solved using first_order_ode_exact

Time used: 0.135 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2x^2y + x) dy &= (-3xy^2 - 2y) dx \\ (3xy^2 + 2y) dx + (2x^2y + x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 3xy^2 + 2y \\ N(x, y) &= 2x^2y + x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (3xy^2 + 2y) \\ &= 6yx + 2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (2x^2y + x) \\ &= 4yx + 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{2x^2y + x} ((6yx + 2) - (4yx + 1)) \\ &= \frac{1}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int \frac{1}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\ln(x)} \\ &= x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= x(3x y^2 + 2y) \\ &= 3y^2 x^2 + 2yx\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= x(2x^2 y + x) \\ &= 2y x^3 + x^2\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (3y^2 x^2 + 2yx) + (2y x^3 + x^2) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int 3y^2 x^2 + 2yx dx \\ \phi &= y x^2(yx + 1) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= x^2(yx + 1) + y x^3 + f'(y) \\ &= 2y x^3 + x^2 + f'(y)\end{aligned} \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = 2y x^3 + x^2$. Therefore equation (4) becomes

$$2y x^3 + x^2 = 2y x^3 + x^2 + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = y x^2 (yx + 1) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = y x^2 (yx + 1)$$

Solving for y gives

$$y = \frac{-x + \sqrt{4c_1 x + x^2}}{2x^2}$$

$$y = -\frac{x + \sqrt{4c_1 x + x^2}}{2x^2}$$

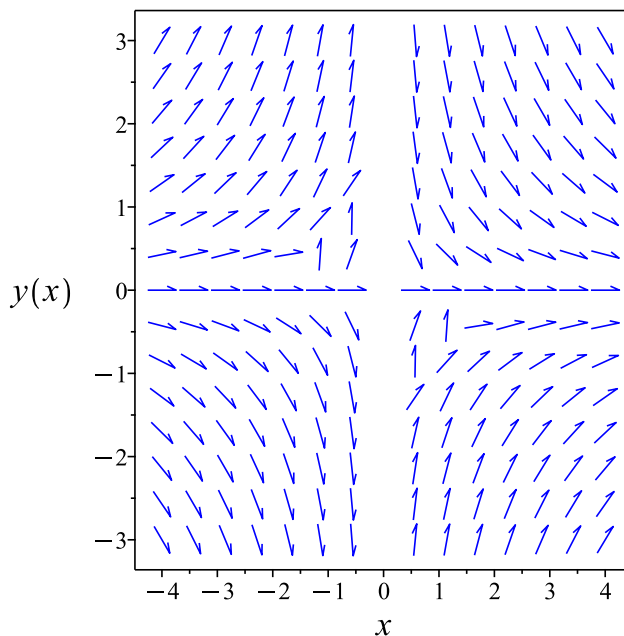


Figure 2.239: Slope field $2y + 3y^2x + (x + 2x^2y) y' = 0$

Summary of solutions found

$$y = \frac{-x + \sqrt{4c_1x + x^2}}{2x^2}$$

$$y = -\frac{x + \sqrt{4c_1x + x^2}}{2x^2}$$

✓ Maple. Time used: 0.005 (sec). Leaf size: 45

```
ode:=2*y(x)+3*x*y(x)^2+(x+2*x^2*y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{-x + \sqrt{x(4c_1 + x)}}{2x^2}$$

$$y = \frac{-x - \sqrt{x(4c_1 + x)}}{2x^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying homogeneous G
<- homogeneous successful
```

Maple step by step

Let's solve

$$2y(x) + 3xy(x)^2 + (x + 2x^2y(x)) \left(\frac{d}{dx}y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-2y(x) - 3xy(x)^2}{x + 2x^2y(x)}$$

✓ **Mathematica.** Time used: 0.463 (sec). Leaf size: 69

```
ode=(2*y[x]+3*x*y[x]^2)+(x+2*x^2*y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{x^{3/2} + \sqrt{x^2(x + 4c_1)}}{2x^{5/2}}$$

$$y(x) \rightarrow \frac{-x^{3/2} + \sqrt{x^2(x + 4c_1)}}{2x^{5/2}}$$

✓ **Sympy.** Time used: 0.632 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x*y(x)**2 + (2*x**2*y(x) + x)*Derivative(y(x), x) + 2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = \frac{-\sqrt{\frac{C_1}{x} + 1} - 1}{2x}, y(x) = \frac{\sqrt{\frac{C_1}{x} + 1} - 1}{2x} \right]$$

2.3.35 Problem 26 (i)

Local contents

Solved using first_order_ode_exact	1048
✓ Maple	1053
✓ Mathematica	1054
✓ Sympy	1054

Internal problem ID [7915]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 26 (i)

Date solved : Thursday, November 27, 2025 at 06:31:16 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_exact

Time used: 0.207 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x(-x^2 + 2y^2)) dy &= (-y(-2x^2 + y^2)) dx \\ (y(-2x^2 + y^2)) dx + (x(-x^2 + 2y^2)) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y(-2x^2 + y^2) \\ N(x, y) &= x(-x^2 + 2y^2) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (y(-2x^2 + y^2)) \\ &= -2x^2 + 3y^2 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x(-x^2 + 2y^2)) \\ &= -3x^2 + 2y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{x(x^2 - 2y^2)} ((-2x^2 + 3y^2) - (-3x^2 + 2y^2)) \\ &= \frac{-x^2 - y^2}{x(x^2 - 2y^2)} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y(-2x^2 + y^2)} ((-3x^2 + 2y^2) - (-2x^2 + 3y^2)) \\ &= \frac{x^2 + y^2}{2x^2y - y^3} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(-3x^2 + 2y^2) - (-2x^2 + 3y^2)}{x(y(-2x^2 + y^2)) - y(x(-x^2 + 2y^2))} \\ &= \frac{1}{yx} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = \frac{1}{t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int (\frac{1}{t}) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{\ln(t)} \\ &= t \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = yx$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= yx(y(-2x^2 + y^2)) \\ &= y^2(-2x^2 + y^2)x\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= yx(x(-x^2 + 2y^2)) \\ &= -x^2(x^2 - 2y^2)y\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (y^2(-2x^2 + y^2)x) + (-x^2(x^2 - 2y^2)y) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int y^2(-2x^2 + y^2)x dx \\ \phi &= -\frac{y^2(2x^2 - y^2)^2}{8} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= -\frac{y(2x^2 - y^2)^2}{4} + \frac{y^3(2x^2 - y^2)}{2} + f'(y) \\ &= -x^4y + 2y^3x^2 - \frac{3}{4}y^5 + f'(y)\end{aligned} \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = -x^2(x^2 - 2y^2)y$. Therefore equation (4) becomes

$$-x^2(x^2 - 2y^2)y = -x^4y + 2y^3x^2 - \frac{3}{4}y^5 + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = \frac{3y^5}{4}$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int \left(\frac{3y^5}{4} \right) dy$$

$$f(y) = \frac{y^6}{8} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

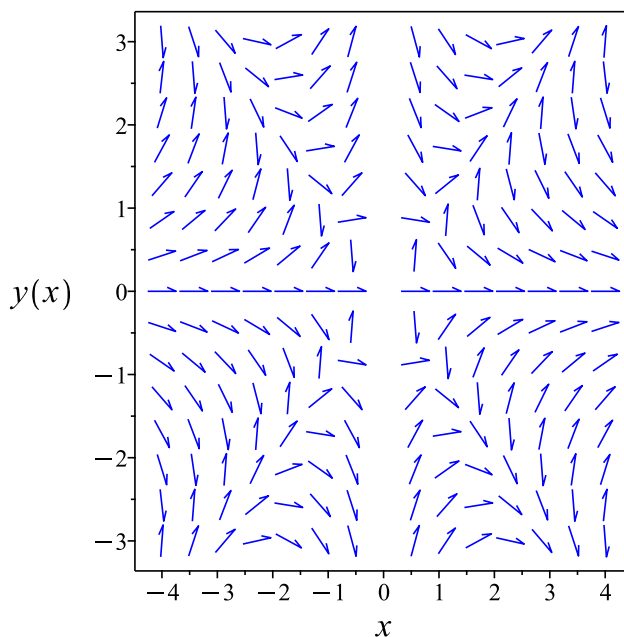
$$\phi = -\frac{y^2(2x^2 - y^2)^2}{8} + \frac{y^6}{8} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{y^2(2x^2 - y^2)^2}{8} + \frac{y^6}{8}$$

Simplifying the above gives

$$\frac{y^4x^2}{2} - \frac{y^2x^4}{2} = c_1$$

Figure 2.240: Slope field $y(y^2 - 2x^2) + x(2y^2 - x^2)y' = 0$ Summary of solutions found

$$\frac{y^4 x^2}{2} - \frac{y^2 x^4}{2} = c_1$$

✓ Maple. Time used: 0.053 (sec). Leaf size: 71

```
ode:=y(x)*(y(x)^2-2*x^2)+x*(2*y(x)^2-x^2)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\sqrt{\frac{2c_1 x^3 - 2\sqrt{c_1^2 x^6 + 4}}{c_1 x^3}} x}{2}$$

$$y = \frac{\sqrt{2} \sqrt{\frac{c_1 x^3 + \sqrt{c_1^2 x^6 + 4}}{c_1 x^3}} x}{2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
```

```
trying inverse linear
trying homogeneous types:
trying homogeneous D
<- homogeneous successful
```

Maple step by step

Let's solve

$$y(x) (y(x)^2 - 2x^2) + x(2y(x)^2 - x^2) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{y(x)(y(x)^2 - 2x^2)}{x(2y(x)^2 - x^2)}$$

✓ **Mathematica.** Time used: 0.084 (sec). Leaf size: 48

```
ode=y[x]*(y[x]^2-2*x^2)+x*(2*y[x]^2-x^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\int_1^{\frac{y(x)}{x}} \frac{2K[1]^2 - 1}{(K[1] - 1)K[1](K[1] + 1)} dK[1] = -3 \log(x) + c_1, y(x) \right]$$

✓ **Sympy.** Time used: 3.787 (sec). Leaf size: 105

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*(-x**2 + 2*y(x)**2)*Derivative(y(x), x) + (-2*x**2 + y(x)**2)*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\frac{\sqrt{2}\sqrt{x^2 - \frac{\sqrt{C_1+x^6}}{x}}}{2}, y(x) = \frac{\sqrt{2}\sqrt{x^2 - \frac{\sqrt{C_1+x^6}}{x}}}{2}, y(x) = -\frac{\sqrt{2}\sqrt{x^2 + \frac{\sqrt{C_1+x^6}}{x}}}{2}, y(x) = \frac{\sqrt{2}\sqrt{x^2 + \frac{\sqrt{C_1+x^6}}{x}}}{2} \right]$$

2.3.36 Problem 27

Local contents

Solved using first_order_ode_exact	1055
✓ Maple	1059
✓ Mathematica	1060
✓ Sympy	1060

Internal problem ID [7916]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 5. Equations of first order and first degree (Exact equations). Supplementary problems. Page 33

Problem number : 27

Date solved : Thursday, November 27, 2025 at 06:31:22 AM

CAS classification : [_separable]

Solved using first_order_ode_exact

Time used: 0.080 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (y) dx \\ (-y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-y) \\ &= -1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((-1) - (1)) \\ &= -\frac{2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{2}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-2\ln(x)} \\ &= \frac{1}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^2}(-y) \\ &= -\frac{y}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^2}(x) \\ &= \frac{1}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(-\frac{y}{x^2}\right) + \left(\frac{1}{x}\right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{1}{x} dy \\ \phi &= \frac{y}{x} + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{y}{x^2} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = -\frac{y}{x^2}$. Therefore equation (4) becomes

$$-\frac{y}{x^2} = -\frac{y}{x^2} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{y}{x} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y}{x}$$

Solving for y gives

$$y = c_1 x$$

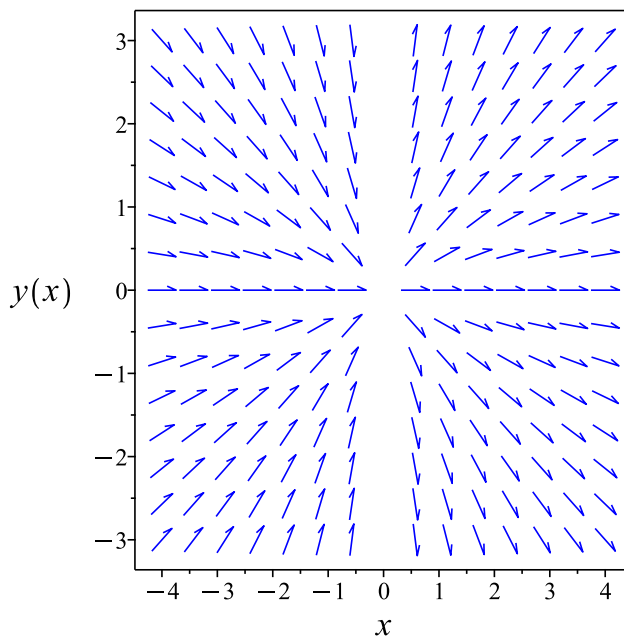


Figure 2.241: Slope field $y'x - y = 0$

Summary of solutions found

$$y = c_1 x$$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 7

```
ode:=-y(x)+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$-y(x) + x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x)}{x}$$

- Separate variables

$$\frac{\frac{d}{dx}y(x)}{y(x)} = \frac{1}{x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int \frac{1}{x} dx + C1$$

- Evaluate integral

$$\ln(y(x)) = \ln(x) + C1$$

- Solve for $y(x)$

$$y(x) = x e^{C1}$$

- Redefine the integration constant(s)

$$y(x) = C1x$$

✓ **Mathematica.** Time used: 0.015 (sec). Leaf size: 14

```
ode=-y[x]+x*D[y[x],x]==0;  
ic={};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 x$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.061 (sec). Leaf size: 5

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(x*Derivative(y(x), x) - y(x),0)  
ics = {}  
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 x$$

2.4 Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Local contents

2.4.1	Problem 19 (a)	1062
2.4.2	Problem 19 (c)	1077
2.4.3	Problem 19 (d)	1093
2.4.4	Problem 19 (e)	1109
2.4.5	Problem 19 (f)	1117
2.4.6	Problem 19 (g)	1126
2.4.7	Problem 19 (h)	1136
2.4.8	Problem 19 (i)	1144
2.4.9	Problem 19 (j)	1164
2.4.10	Problem 19 (k)	1172
2.4.11	Problem 19 (L)	1178
2.4.12	Problem 19 (m)	1190
2.4.13	Problem 19 (o)	1197
2.4.14	Problem 19 (p)	1207
2.4.15	Problem 19 (q)	1215
2.4.16	Problem 19 (r)	1228
2.4.17	Problem 19 (s)	1241
2.4.18	Problem 19 (t)	1257
2.4.19	Problem 22 (a)	1263
2.4.20	Problem 22 (b)	1278
2.4.21	Problem 23 (a)	1287
2.4.22	Problem 23 (b)	1293
2.4.23	Problem 23 (c)	1296
2.4.24	Problem 23 (d)	1305
2.4.25	Problem 23 (e)	1312

2.4.1 Problem 19 (a)

Local contents

Solved using first_order_ode_linear	1062
Solved using first_order_ode_exact	1064
Solved using first_order_ode_homog_type_D2	1068
Solved using first_order_ode_LIE	1070
✓ Maple	1074
✓ Mathematica	1075
✓ Sympy	1076

Internal problem ID [7917]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (a)

Date solved : Thursday, November 27, 2025 at 06:31:27 AM

CAS classification : `[_linear, 'class A']`

Solved using first_order_ode_linear

Time used: 0.042 (sec)

Solve

$$y' + y = 2x + 2$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = 1$$

$$p(x) = 2x + 2$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int 1 \, dx} \\ &= e^x\end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu)(2x + 2)$$

$$\frac{d}{dx}(e^x y) = (e^x)(2x + 2)$$

$$d(e^x y) = ((2x + 2)e^x) dx$$

Integrating gives

$$\begin{aligned} e^x y &= \int (2x + 2)e^x dx \\ &= 2xe^x + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^x gives the final solution

$$y = e^{-x}c_1 + 2x$$

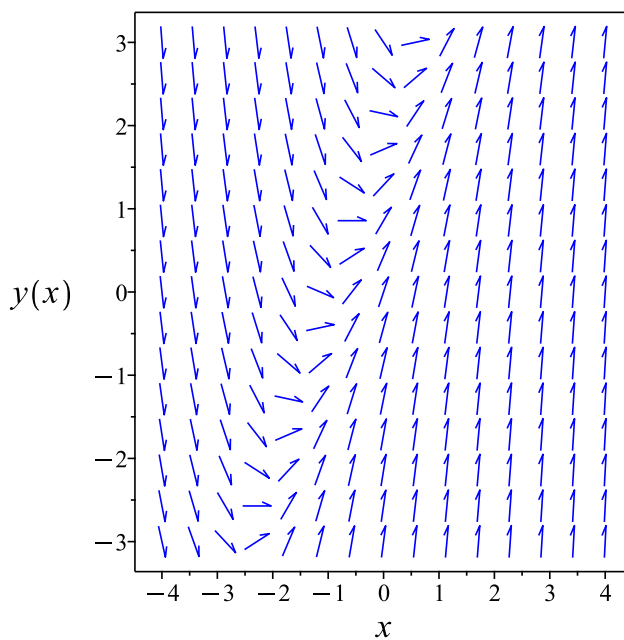


Figure 2.242: Slope field $y' + y = 2x + 2$

Summary of solutions found

$$y = e^{-x}c_1 + 2x$$

Solved using first_order_ode_exact

Time used: 0.099 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} dy &= (-y + 2x + 2) dx \\ (y - 2x - 2) dx + dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y - 2x - 2 \\ N(x, y) &= 1 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y - 2x - 2) \\ &= 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(1) \\ &= 0\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= 1((1) - (0)) \\ &= 1\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int 1 \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^x \\ &= e^x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= e^x(y - 2x - 2) \\ &= (y - 2x - 2)e^x\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= e^x(1) \\ &= e^x\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ ((y - 2x - 2)e^x) + (e^x) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int e^x dy \\ \phi &= e^x y + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = e^x y + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = (y - 2x - 2)e^x$. Therefore equation (4) becomes

$$(y - 2x - 2)e^x = e^x y + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$\begin{aligned}f'(x) &= -2x e^x - 2e^x \\ &= -2e^x(x + 1)\end{aligned}$$

Integrating the above w.r.t x results in

$$\begin{aligned}\int f'(x) dx &= \int (-2e^x(x + 1)) dx \\ f(x) &= -2x e^x + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = e^x y - 2x e^x + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = e^x y - 2x e^x$$

Simplifying the above gives

$$e^x(y - 2x) = c_1$$

Solving for y gives

$$y = e^{-x}(2x e^x + c_1)$$

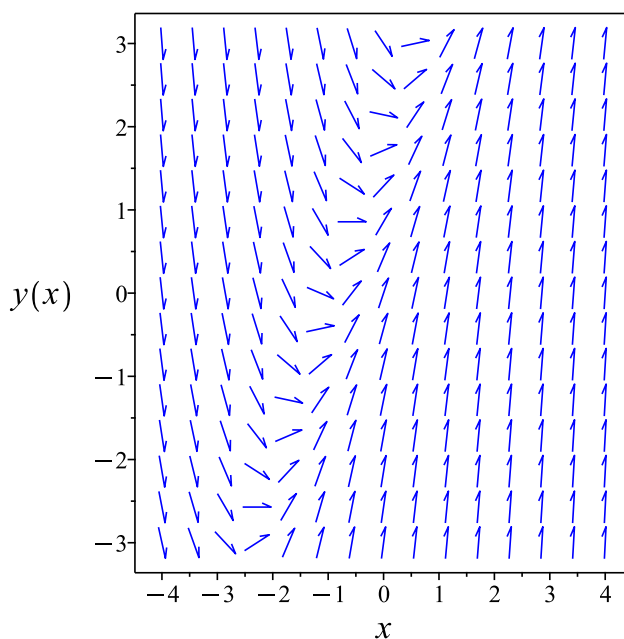


Figure 2.243: Slope field $y' + y = 2x + 2$

Summary of solutions found

$$y = e^{-x}(2x e^x + c_1)$$

Solved using first_order_ode_homog_type_D2

Time used: 0.185 (sec)

Solve

$$y' + y = 2x + 2$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u'(x)x + u(x) + u(x)x = 2x + 2$$

Which is now solved The ode

$$u'(x) = -\frac{(x+1)(u(x)-2)}{x} \quad (2.100)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{(x+1)(u(x)-2)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{x+1}{x} \\ g(u) &= -u+2 \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{-u+2} du &= \int \frac{x+1}{x} dx \end{aligned}$$

$$-\ln(-u(x)+2) = x + \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$-u+2=0$$

for $u(x)$ gives

$$u(x) = 2$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\ln(-u(x) + 2) = x + \ln(x) + c_1$$

$$u(x) = 2$$

Converting $-\ln(-u(x) + 2) = x + \ln(x) + c_1$ back to y gives

$$-\ln\left(-\frac{y}{x} + 2\right) = x + \ln(x) + c_1$$

Converting $u(x) = 2$ back to y gives

$$y = 2x$$

Solving for y gives

$$y = 2x$$

$$y = -e^{-x-c_1} + 2x$$

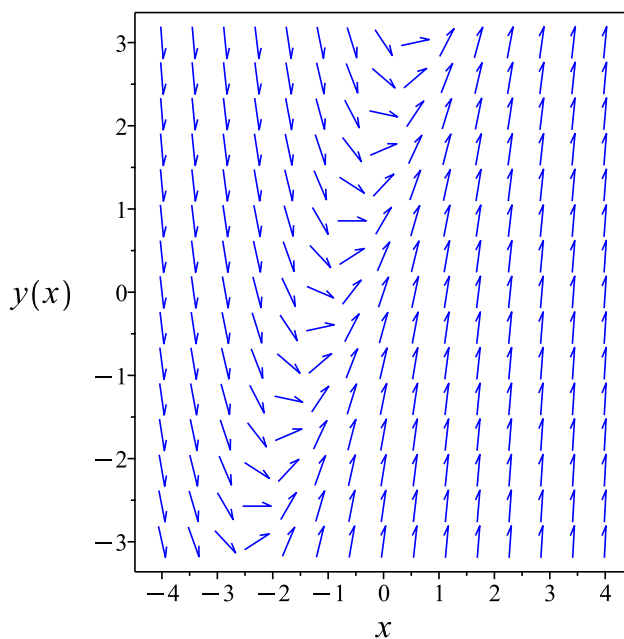


Figure 2.244: Slope field $y' + y = 2x + 2$

Summary of solutions found

$$y = 2x$$

$$y = -e^{-x-c_1} + 2x$$

Solved using first_order_ode_LIE

Time used: 0.319 (sec)

Solve

$$y' + y = 2x + 2$$

Writing the ode as

$$y' = -y + 2x + 2$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1\text{E})$$

$$\eta = \frac{Ax + By}{Cx} \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\frac{A}{Cx} - \frac{Ax + By}{Cx^2} + \frac{(-y + 2x + 2)B}{Cx} - 2 + \frac{Ax + By}{Cx} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{Ax^2 + 2Bx^2 - 2Cx^2 + 2Bx - By}{Cx^2} = 0$$

Setting the numerator to zero gives

$$Ax^2 + 2Bx^2 - 2Cx^2 + 2Bx - By = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$Av_1^2 + 2Bv_1^2 - 2Cv_1^2 + 2Bv_1 - Bv_2 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(A + 2B - 2C)v_1^2 + 2Bv_1 - Bv_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$-B = 0$$

$$2B = 0$$

$$A + 2B - 2C = 0$$

Solving the above equations for the unknowns gives

$$A = 2C$$

$$B = 0$$

$$C = C$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 1$$

$$\eta = 2$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{2}{1} \\ &= 2\end{aligned}$$

This is easily solved to give

$$y = c_1 + 2x$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = -2x + y$$

And S is found from

$$\begin{aligned}dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1}\end{aligned}$$

Integrating gives

$$\begin{aligned}S &= \int \frac{dx}{T} \\ &= x\end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -y + 2x + 2$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= -2 \\ R_y &= 1 \\ S_x &= 1 \\ S_y &= 0\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{2x - y} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

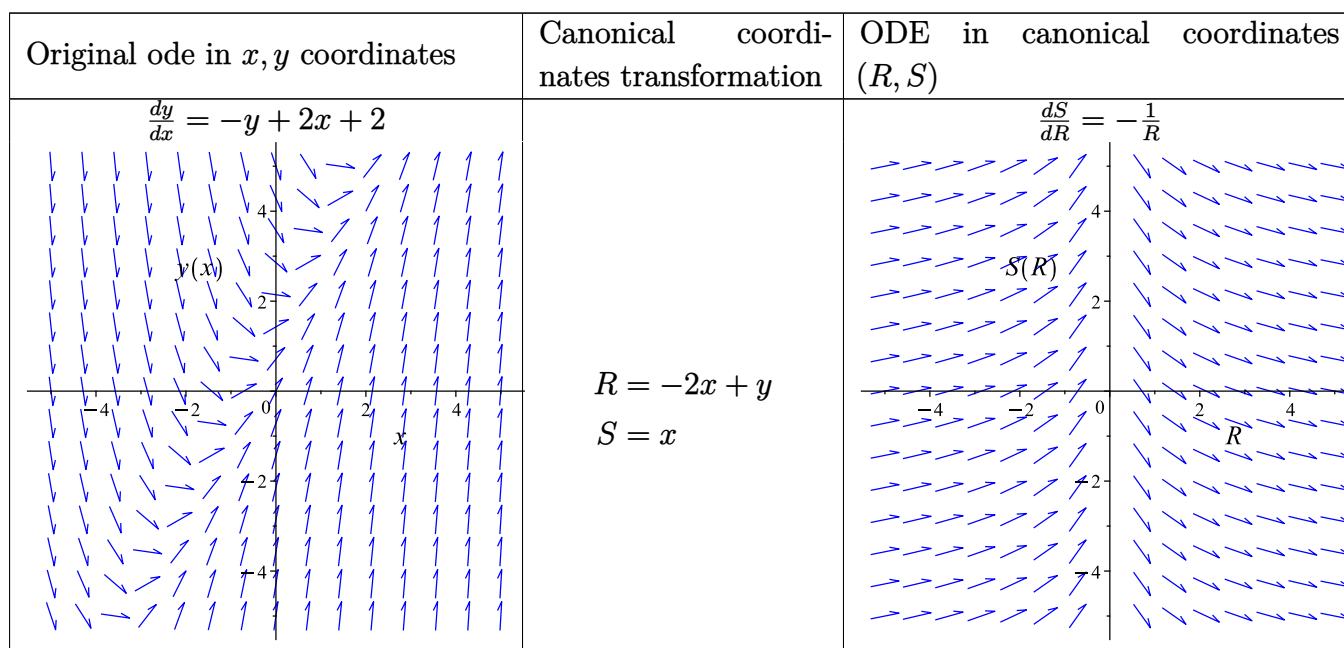
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{R} dR \\ S(R) &= -\ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = -\ln(y - 2x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = e^{-x+c_2} + 2x$$

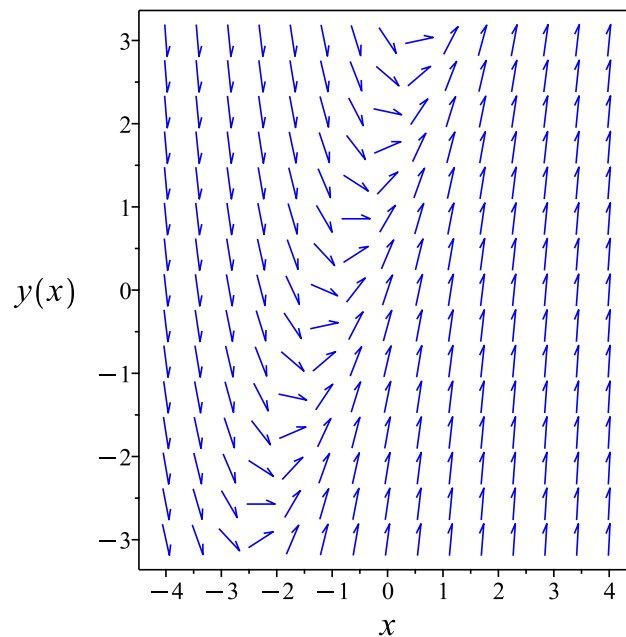


Figure 2.245: Slope field $y' + y = 2x + 2$

Summary of solutions found

$$y = e^{-x+c_2} + 2x$$

✓ Maple. Time used: 0.000 (sec). Leaf size: 14

```
ode:=diff(y(x),x)+y(x) = 2*x+2;
dsolve(ode,y(x), singsol=all);
```

$$y = 2x + e^{-x}c_1$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\frac{d}{dx}y(x) + y(x) = 2 + 2x$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -y(x) + 2 + 2x$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) + y(x) = 2 + 2x$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) + y(x) \right) = \mu(x) (2 + 2x)$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) + y(x) \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$

- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = \mu(x)$$

- Solve to find the integrating factor

$$\mu(x) = e^x$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int \mu(x) (2 + 2x) dx + C1$$

- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int \mu(x) (2 + 2x) dx + C1$$

- Solve for $y(x)$

$$y(x) = \frac{\int \mu(x)(2+2x)dx + C1}{\mu(x)}$$

- Substitute $\mu(x) = e^x$

$$y(x) = \frac{\int e^x(2+2x)dx + C1}{e^x}$$

- Evaluate the integrals on the rhs

$$y(x) = \frac{2e^x x + C1}{e^x}$$

- Simplify

$$y(x) = e^{-x} C1 + 2x$$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 17

```
ode=D[y[x],x]+y[x]==2+2*x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow 2x + c_1 e^{-x}$$

✓ Sympy. Time used: 0.066 (sec). Leaf size: 10

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-2*x + y(x) + Derivative(y(x), x) - 2, 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-x} + 2x$$

2.4.2 Problem 19 (c)

Local contents

Solved using first_order_ode_linear	1077
Solved using first_order_ode_separable	1079
Solved using first_order_ode_exact	1081
Solved using first_order_ode_homog_type_D2	1084
Solved using first_order_ode_LIE	1086
✓ Maple	1091
✓ Mathematica	1092
✓ Sympy	1092

Internal problem ID [7918]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (c)

Date solved : Thursday, November 27, 2025 at 06:31:33 AM

CAS classification : [_separable]

Solved using first_order_ode_linear

Time used: 0.042 (sec)

Solve

$$y' - y = yx$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -1 - x$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int (-1-x) \, dx} \\ &= e^{-\frac{x(x+2)}{2}}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}\left(y e^{-\frac{x(x+2)}{2}}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}y e^{-\frac{x(x+2)}{2}} &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $e^{-\frac{x(x+2)}{2}}$ gives the final solution

$$y = e^{\frac{x(x+2)}{2}} c_1$$

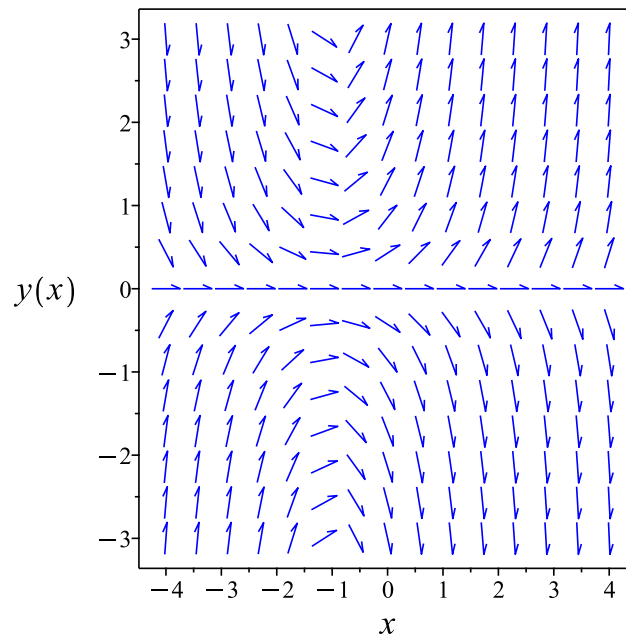


Figure 2.246: Slope field $y' - y = yx$

Summary of solutions found

$$y = e^{\frac{x(x+2)}{2}} c_1$$

Solved using first_order_ode_separable

Time used: 0.072 (sec)

Solve

$$y' - y = yx$$

The ode

$$y' = (x + 1) y \tag{2.101}$$

is separable as it can be written as

$$\begin{aligned} y' &= (x + 1) y \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= x + 1 \\ g(y) &= y \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(y)} dy &= \int f(x) dx \\ \int \frac{1}{y} dy &= \int x + 1 dx \end{aligned}$$

$$\ln(y) = \frac{1}{2}x^2 + x + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$y = 0$$

for y gives

$$y = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(y) = \frac{1}{2}x^2 + x + c_1$$

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = e^{\frac{1}{2}x^2 + x + c_1}$$

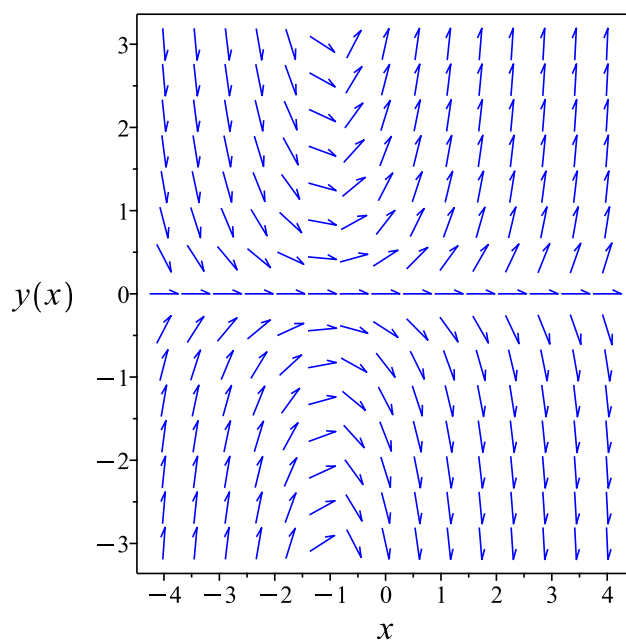


Figure 2.247: Slope field $y' - y = yx$

Summary of solutions found

$$y = 0$$

$$y = e^{\frac{1}{2}x^2 + x + c_1}$$

Solved using first_order_ode_exact

Time used: 0.109 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} dy &= (yx + y) dx \\ (-yx - y) dx + dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -yx - y \\ N(x, y) &= 1 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-yx - y) \\ &= -1 - x\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(1) \\ &= 0\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= 1((-1 - x) - (0)) \\ &= -1 - x\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -1-x \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-x - \frac{1}{2}x^2} \\ &= e^{-\frac{x(x+2)}{2}}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= e^{-\frac{x(x+2)}{2}}(-yx - y) \\ &= -(x+1)y e^{-\frac{x(x+2)}{2}}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= e^{-\frac{x(x+2)}{2}}(1) \\ &= e^{-\frac{x(x+2)}{2}}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(-(x+1)y e^{-\frac{x(x+2)}{2}} \right) + \left(e^{-\frac{x(x+2)}{2}} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int e^{-\frac{x(x+2)}{2}} dy \\ \phi &= y e^{-\frac{x(x+2)}{2}} + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = y(-1-x) e^{-\frac{x(x+2)}{2}} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = -(x+1)y e^{-\frac{x(x+2)}{2}}$. Therefore equation (4) becomes

$$-(x+1)y e^{-\frac{x(x+2)}{2}} = y(-1-x) e^{-\frac{x(x+2)}{2}} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

$$\phi = y e^{-\frac{x(x+2)}{2}} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = y e^{-\frac{x(x+2)}{2}}$$

Solving for y gives

$$y = e^{\frac{x(x+2)}{2}} c_1$$

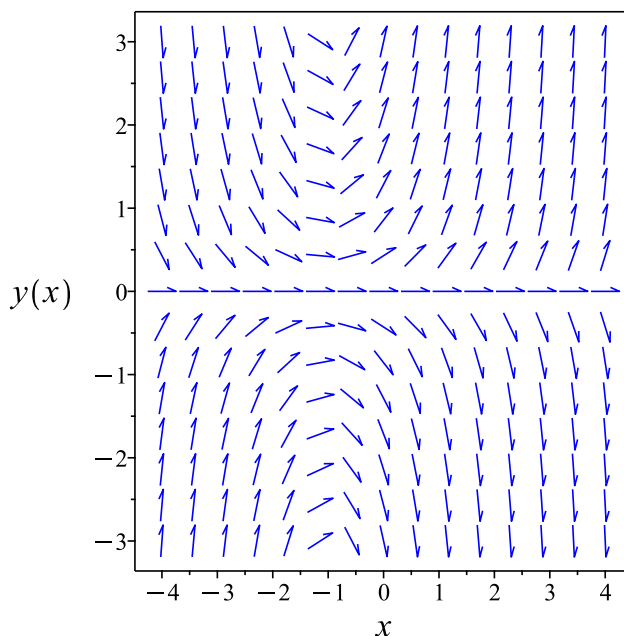


Figure 2.248: Slope field $y' - y = yx$

Summary of solutions found

$$y = e^{\frac{x(x+2)}{2}} c_1$$

Solved using first_order_ode_homog_type_D2

Time used: 0.158 (sec)

Solve

$$y' - y = yx$$

Applying change of variables $y = u(x)x$, then the ode becomes

$$u'(x)x + u(x) - u(x)x = u(x)x^2$$

Which is now solved The ode

$$u'(x) = \frac{u(x)(x^2 + x - 1)}{x} \tag{2.102}$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(x^2 + x - 1)}{x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{x^2 + x - 1}{x} \\ g(u) &= u \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u} du &= \int \frac{x^2 + x - 1}{x} dx \end{aligned}$$

$$\ln(u(x)) = x + \frac{x^2}{2} + \ln\left(\frac{1}{x}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u = 0$$

for $u(x)$ gives

$$u(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned} \ln(u(x)) &= x + \frac{x^2}{2} + \ln\left(\frac{1}{x}\right) + c_1 \\ u(x) &= 0 \end{aligned}$$

Converting $\ln(u(x)) = x + \frac{x^2}{2} + \ln\left(\frac{1}{x}\right) + c_1$ back to y gives

$$\ln\left(\frac{y}{x}\right) = x + \frac{x^2}{2} + \ln\left(\frac{1}{x}\right) + c_1$$

Converting $u(x) = 0$ back to y gives

$$y = 0$$

Solving for y gives

$$y = 0$$

$$y = e^{\frac{1}{2}x^2 + x + c_1}$$

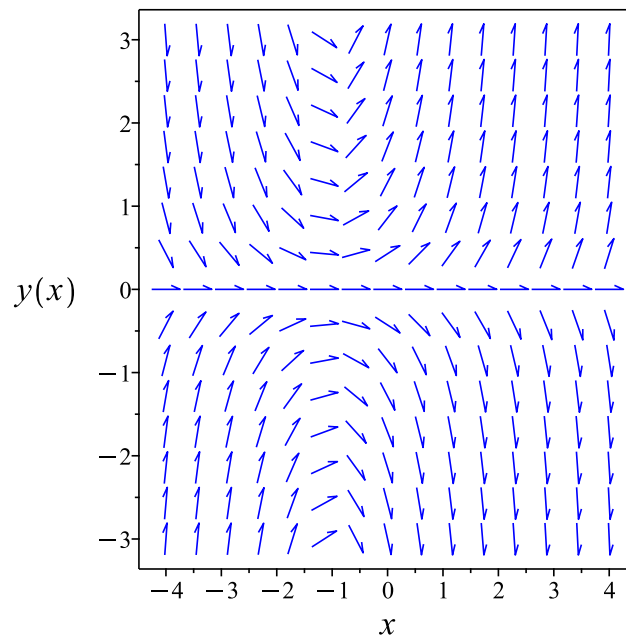


Figure 2.249: Slope field $y' - y = yx$

Summary of solutions found

$$y = 0$$

$$y = e^{\frac{1}{2}x^2 + x + c_1}$$

Solved using first_order_ode_LIE

Time used: 0.727 (sec)

Solve

$$y' - y = yx$$

Writing the ode as

$$y' = yx + y$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 + (yx + y)(b_3 - a_2) - (yx + y)^2 a_3 - y(xa_2 + ya_3 + a_1) - (x + 1)(xb_2 + yb_3 + b_1) = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-x^2 y^2 a_3 - 2x y^2 a_3 - x^2 b_2 - 2xya_2 - 2y^2 a_3 - xb_1 - xb_2 - ya_1 - ya_2 - b_1 + b_2 = 0$$

Setting the numerator to zero gives

$$-x^2 y^2 a_3 - 2x y^2 a_3 - x^2 b_2 - 2xya_2 - 2y^2 a_3 - xb_1 - xb_2 - ya_1 - ya_2 - b_1 + b_2 = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-a_3 v_1^2 v_2^2 - 2a_3 v_1 v_2^2 - 2a_2 v_1 v_2 - 2a_3 v_2^2 - b_2 v_1^2 - a_1 v_2 - a_2 v_2 - b_1 v_1 - b_2 v_1 - b_1 + b_2 = 0 \quad (7\text{E})$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-a_3v_1^2v_2^2 - b_2v_1^2 - 2a_3v_1v_2^2 - 2a_2v_1v_2 + (-b_1 - b_2)v_1 - 2a_3v_2^2 + (-a_1 - a_2)v_2 - b_1 + b_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -2a_2 &= 0 \\ -2a_3 &= 0 \\ -a_3 &= 0 \\ -b_2 &= 0 \\ -a_1 - a_2 &= 0 \\ -b_1 - b_2 &= 0 \\ -b_1 + b_2 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= 0 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= 0 \\ \eta &= y \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{y} dy \end{aligned}$$

Which results in

$$S = \ln(y)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = yx + y$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= 0 \\ S_y &= \frac{1}{y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = R + 1 \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = R + 1$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

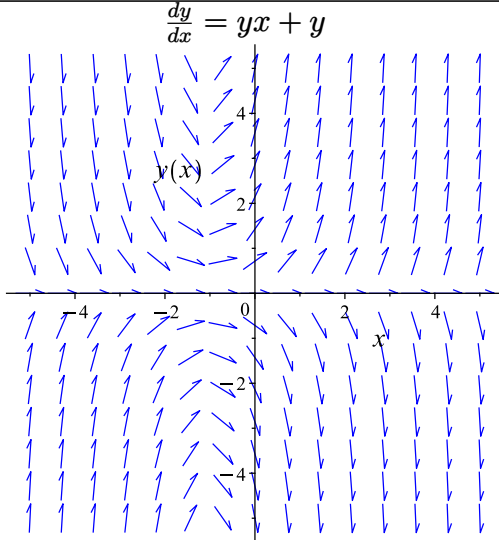
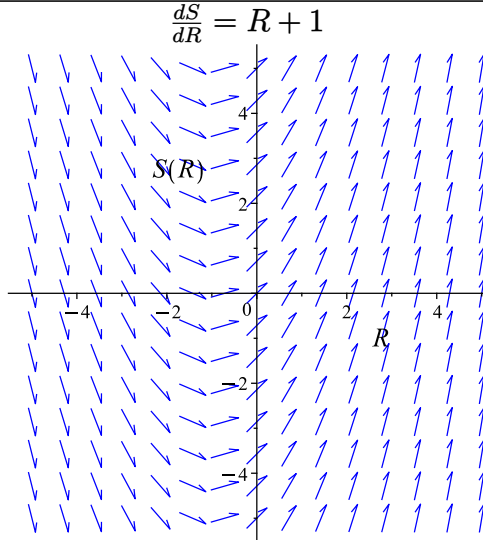
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int R + 1 dR \\ S(R) &= \frac{1}{2}R^2 + R + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\ln(y) = \frac{1}{2}x^2 + x + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
 $\frac{dy}{dx} = yx + y$	$R = x$ $S = \ln(y)$	 $\frac{dS}{dR} = R + 1$

Solving for y gives

$$y = e^{\frac{1}{2}x^2 + x + c_2}$$

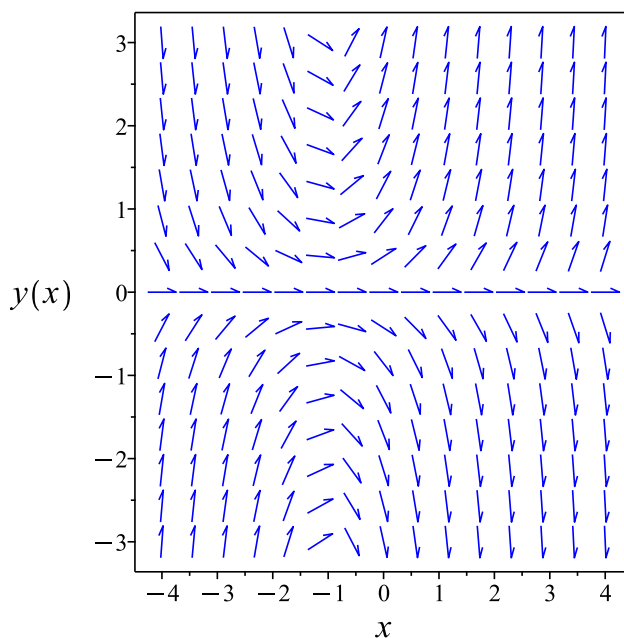


Figure 2.250: Slope field $y' - y = yx$

Summary of solutions found

$$y = e^{\frac{1}{2}x^2+x+c_2}$$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 13

```
ode:=diff(y(x),x)-y(x) = y(x)*x;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 e^{\frac{x(x+2)}{2}}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\frac{d}{dx}y(x) - y(x) = xy(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = y(x) + xy(x)$$

- Separate variables

$$\frac{\frac{d}{dx}y(x)}{y(x)} = x + 1$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int (x + 1) dx + C1$$

- Evaluate integral

$$\ln(y(x)) = \frac{1}{2}x^2 + x + C1$$

- Solve for $y(x)$

$$y(x) = e^{\frac{1}{2}x^2+x+C1}$$

- Redefine the integration constant(s)

$$y(x) = C1 e^{\frac{1}{2}x^2+x}$$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 23

```
ode=D[y[x],x]-y[x]==x*y[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 e^{\frac{1}{2}x(x+2)}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.167 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*y(x) - y(x) + Derivative(y(x), x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 e^{x(\frac{x}{2}+1)}$$

2.4.3 Problem 19 (d)

Local contents

Solved using first_order_ode_linear	1093
Solved using first_order_ode_exact	1095
Solved using first_order_ode_LIE	1099
✓ Maple	1106
✓ Mathematica	1107
✓ Sympy	1108

Internal problem ID [7919]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (d)

Date solved : Thursday, November 27, 2025 at 06:31:40 AM

CAS classification : [_linear]

Solved using first_order_ode_linear

Time used: 0.069 (sec)

Solve

$$-3y - (x - 2)e^x + y'x = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{3}{x}$$

$$p(x) = \frac{e^x(x - 2)}{x}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{3}{x} \, dx} \\ &= \frac{1}{x^3}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{e^x(x-2)}{x} \right) \\ \frac{d}{dx} \left(\frac{y}{x^3} \right) &= \left(\frac{1}{x^3} \right) \left(\frac{e^x(x-2)}{x} \right) \\ d \left(\frac{y}{x^3} \right) &= \left(\frac{e^x(x-2)}{x^4} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^3} &= \int \frac{e^x(x-2)}{x^4} dx \\ &= \frac{-\text{Ei}_1(-x)x^3 - e^x(x^2 + x - 4)}{6x^3} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^3}$ gives the final solution

$$y = -\frac{\text{Ei}_1(-x)x^3}{6} + \frac{(-x^2 - x + 4)e^x}{6} + c_1 x^3$$

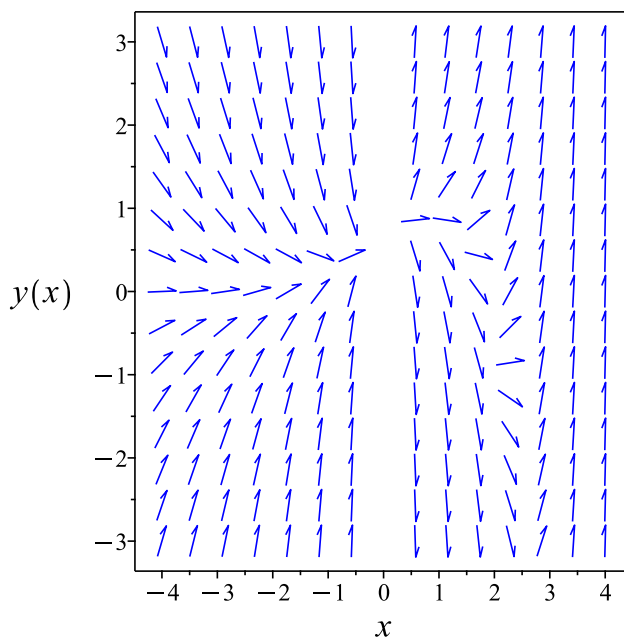


Figure 2.251: Slope field $-3y - (x-2)e^x + y'x = 0$

Summary of solutions found

$$y = -\frac{\text{Ei}_1(-x)x^3}{6} + \frac{(-x^2 - x + 4)e^x}{6} + c_1 x^3$$

Solved using first_order_ode_exact

Time used: 0.125 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (x) dy &= (3y + (x - 2) e^x) dx \\ (-3y - (x - 2) e^x) dx + (x) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -3y - (x - 2) e^x \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-3y - (x-2)e^x) \\ &= -3\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x}((-3) - (1)) \\ &= -\frac{4}{x}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{4}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-4 \ln(x)} \\ &= \frac{1}{x^4}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^4}(-3y - (x-2)e^x) \\ &= \frac{(-x+2)e^x - 3y}{x^4}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{x^4}(x) \\ &= \frac{1}{x^3}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{(-x+2)e^x - 3y}{x^4} \right) + \left(\frac{1}{x^3} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{1}{x^3} dy \\ \phi &= \frac{y}{x^3} + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{3y}{x^4} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{(-x+2)e^x - 3y}{x^4}$. Therefore equation (4) becomes

$$\frac{(-x+2)e^x - 3y}{x^4} = -\frac{3y}{x^4} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -\frac{e^x(x-2)}{x^4}$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int \left(-\frac{e^x(x-2)}{x^4} \right) dx$$

$$f(x) = \frac{e^x}{6x^2} + \frac{e^x}{6x} + \frac{\text{Ei}_1(-x)}{6} - \frac{2e^x}{3x^3} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{y}{x^3} + \frac{e^x}{6x^2} + \frac{e^x}{6x} + \frac{\text{Ei}_1(-x)}{6} - \frac{2e^x}{3x^3} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y}{x^3} + \frac{e^x}{6x^2} + \frac{e^x}{6x} + \frac{\text{Ei}_1(-x)}{6} - \frac{2e^x}{3x^3}$$

Simplifying the above gives

$$\frac{\text{Ei}_1(-x)x^3 + e^x(x^2 + x - 4) + 6y}{6x^3} = c_1$$

Solving for y gives

$$y = -\frac{\text{Ei}_1(-x)x^3}{6} - \frac{x^2 e^x}{6} - \frac{x e^x}{6} + \frac{2e^x}{3} + c_1 x^3$$

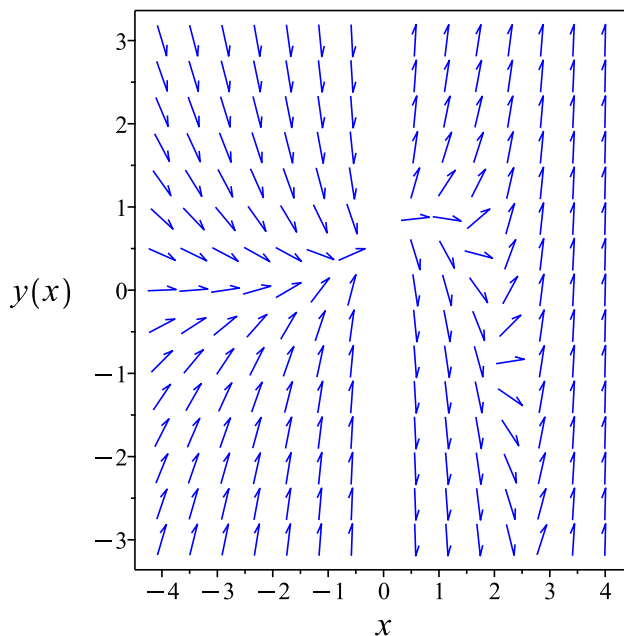


Figure 2.252: Slope field $-3y - (x-2)e^x + y'x = 0$

Summary of solutions found

$$y = -\frac{\text{Ei}_1(-x) x^3}{6} - \frac{x^2 e^x}{6} - \frac{x e^x}{6} + \frac{2 e^x}{3} + c_1 x^3$$

Solved using first_order_ode_LIE

Time used: 1.449 (sec)

Solve

$$-3y - (x - 2) e^x + y' x = 0$$

Writing the ode as

$$y' = \frac{x e^x - 2 e^x + 3y}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 3 to use as anstaz gives

$$\xi = x^3 a_7 + x^2 y a_8 + x y^2 a_9 + y^3 a_{10} + x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (1\text{E})$$

$$\eta = x^3 b_7 + x^2 y b_8 + x y^2 b_9 + y^3 b_{10} + x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9, b_{10}\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 3x^2 b_7 + 2xy b_8 + y^2 b_9 + 2xb_4 + yb_5 + b_2 \quad (5\text{E}) \\ & + \frac{(x e^x - 2 e^x + 3y) (-3x^2 a_7 + x^2 b_8 - 2xy a_8 + 2xy b_9 - y^2 a_9 + 3y^2 b_{10} - 2xa_4 + xb_5 - ya_5 + 2yb_6 - a_2 + b_3)}{x} \\ & - \frac{(x e^x - 2 e^x + 3y)^2 (x^2 a_8 + 2xy a_9 + 3y^2 a_{10} + xa_5 + 2ya_6 + a_3)}{x^2} - \left(\frac{x e^x - e^x}{x} \right. \\ & \left. - \frac{x e^x - 2 e^x + 3y}{x^2} \right) (x^3 a_7 + x^2 y a_8 + x y^2 a_9 + y^3 a_{10} + x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1) \\ & - \frac{3(x^3 b_7 + x^2 y b_8 + x y^2 b_9 + y^3 b_{10} + x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1)}{x} = 0 \end{aligned}$$

Putting the above in normal form gives

$$\frac{e^x x^2 y^2 a_6 + 5 e^x x^2 y a_5 - 2 e^x x^2 y b_6 + 10 e^x x y^2 a_6 - 12 e^x x y a_5 + 4 e^x x y b_6 + e^x x^3 y a_5 + e^x x^2 y a_3 + 4 e^x x y a_3 -}{= 0}$$

Setting the numerator to zero gives

$$\begin{aligned} & -e^x x^2 y^2 a_6 - 5 e^x x^2 y a_5 + 2 e^x x^2 y b_6 - 10 e^x x y^2 a_6 + 12 e^x x y a_5 \\ & - 4 e^x x y b_6 - e^x x^3 y a_5 - e^x x^2 y a_3 - 4 e^x x y a_3 + 8 e^{2x} x y a_6 - 2 e^{2x} x^2 y a_6 \\ & - 6 e^x x^3 y a_8 + 2 e^x x^3 y b_9 - 11 e^x x^2 y^2 a_9 + 3 e^x x^2 y^2 b_{10} + 14 e^x x^2 y a_8 \\ & - 4 e^x x^2 y b_9 + 24 e^x x y^2 a_9 - 6 e^x x y^2 b_{10} - 16 e^x x y^3 a_{10} - e^x x^4 y a_8 \\ & - e^x x^3 y^2 a_9 - e^x x^2 y^3 a_{10} - 2 e^{2x} x^3 y a_9 - 3 e^{2x} x^2 y^2 a_{10} + 8 e^{2x} x^2 y a_9 \\ & + 12 e^{2x} x y^2 a_{10} - 8 e^{2x} x y a_9 + 2 e^x x^2 a_4 - 2 e^x x^2 b_5 + 22 e^x y^2 a_6 \\ & + 2 x^3 y b_8 + 4 y^2 b_9 x^2 - e^{2x} x^2 a_3 + 4 e^{2x} x a_3 - 4 e^{2x} a_3 + y b_5 x^2 - 3 x^2 y a_4 \\ & - 9 x y^2 a_5 + 3 x y^2 b_6 - e^x x^4 a_4 + e^x x^3 b_5 - e^{2x} x^4 a_8 + 4 e^{2x} x^3 a_8 \\ & - 4 e^{2x} x^2 a_8 - 12 e^{2x} y^2 a_{10} - e^{2x} x^3 a_5 + 4 e^{2x} x^2 a_5 - 4 e^{2x} x a_5 \\ & - 8 e^{2x} y a_6 - x^3 b_4 - 15 y^3 a_6 - e^x x^3 a_2 - e^x x^2 a_1 + e^x x^2 a_2 + e^x x^2 b_3 \\ & + 2 e^x x a_1 - 2 e^x x b_3 + 10 e^x y a_3 - 2 b_2 x^2 - 6 y^2 a_3 - 3 x b_1 + 3 y a_1 \\ & - 2 e^x a_1 - 24 y^4 a_{10} - 6 x^3 y a_7 - 12 x^2 y^2 a_8 - 18 x y^3 a_9 + 6 x y^3 b_{10} \\ & - e^x x^4 a_7 + e^x x^4 b_8 + 4 e^x x^3 a_7 - 2 e^x x^3 b_8 + 34 e^x y^3 a_{10} - e^x x^5 a_7 = 0 \end{aligned} \quad (6E)$$

Simplifying the above gives

$$\begin{aligned} & -e^x x^2 y^2 a_6 - 5 e^x x^2 y a_5 + 2 e^x x^2 y b_6 - 10 e^x x y^2 a_6 + 12 e^x x y a_5 \\ & - 4 e^x x y b_6 - e^x x^3 y a_5 - e^x x^2 y a_3 - 4 e^x x y a_3 + 8 e^{2x} x y a_6 - 2 e^{2x} x^2 y a_6 \\ & - 6 e^x x^3 y a_8 + 2 e^x x^3 y b_9 - 11 e^x x^2 y^2 a_9 + 3 e^x x^2 y^2 b_{10} + 14 e^x x^2 y a_8 \\ & - 4 e^x x^2 y b_9 + 24 e^x x y^2 a_9 - 6 e^x x y^2 b_{10} - 16 e^x x y^3 a_{10} - e^x x^4 y a_8 \\ & - e^x x^3 y^2 a_9 - e^x x^2 y^3 a_{10} - 2 e^{2x} x^3 y a_9 - 3 e^{2x} x^2 y^2 a_{10} + 8 e^{2x} x^2 y a_9 \\ & + 12 e^{2x} x y^2 a_{10} - 8 e^{2x} x y a_9 + 2 e^x x^2 a_4 - 2 e^x x^2 b_5 + 22 e^x y^2 a_6 \\ & + 2 x^3 y b_8 + 4 y^2 b_9 x^2 - e^{2x} x^2 a_3 + 4 e^{2x} x a_3 - 4 e^{2x} a_3 + y b_5 x^2 - 3 x^2 y a_4 \\ & - 9 x y^2 a_5 + 3 x y^2 b_6 - e^x x^4 a_4 + e^x x^3 b_5 - e^{2x} x^4 a_8 + 4 e^{2x} x^3 a_8 \\ & - 4 e^{2x} x^2 a_8 - 12 e^{2x} y^2 a_{10} - e^{2x} x^3 a_5 + 4 e^{2x} x^2 a_5 - 4 e^{2x} x a_5 \\ & - 8 e^{2x} y a_6 - x^3 b_4 - 15 y^3 a_6 - e^x x^3 a_2 - e^x x^2 a_1 + e^x x^2 a_2 + e^x x^2 b_3 \\ & + 2 e^x x a_1 - 2 e^x x b_3 + 10 e^x y a_3 - 2 b_2 x^2 - 6 y^2 a_3 - 3 x b_1 + 3 y a_1 \\ & - 2 e^x a_1 - 24 y^4 a_{10} - 6 x^3 y a_7 - 12 x^2 y^2 a_8 - 18 x y^3 a_9 + 6 x y^3 b_{10} \\ & - e^x x^4 a_7 + e^x x^4 b_8 + 4 e^x x^3 a_7 - 2 e^x x^3 b_8 + 34 e^x y^3 a_{10} - e^x x^5 a_7 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, e^x, e^{2x}\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, e^x = v_3, e^{2x} = v_4\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & -v_3v_1^2v_2^2a_6 - 5v_3v_1^2v_2a_5 + 2v_3v_1^2v_2b_6 - 10v_3v_1v_2^2a_6 + 12v_3v_1v_2a_5 - 4v_3v_1v_2b_6 - v_3v_1^3v_2a_5 - v_3v_1^2v_2a_3 \\ & - 4v_3v_1v_2a_3 + 8v_4v_1v_2a_6 - 2v_4v_1^2v_2a_6 - 6v_3v_1^3v_2a_8 + 2v_3v_1^3v_2b_9 - 11v_3v_1^2v_2^2a_9 + 3v_3v_1^2v_2^2b_{10} \\ & + 14v_3v_1^2v_2a_8 - 4v_3v_1^2v_2b_9 + 2v_3v_1^2a_4 - 2v_3v_1^2b_5 + 22v_3v_2^2a_6 + 2v_1^3v_2b_8 + 4v_2^2b_9v_1^2 - v_4v_1^2a_3 \\ & + 4v_4v_1a_3 + v_2b_5v_1^2 - 3v_1^2v_2a_4 - 9v_1v_2^2a_5 + 3v_1v_2^2b_6 - v_3v_1^4a_4 + v_3v_1^3b_5 + 24v_3v_1v_2^2a_9 - 6v_3v_1v_2^2b_{10} \\ & - 16v_3v_1v_2^3a_{10} - v_3v_1^4v_2a_8 - v_3v_1^3v_2^2a_9 - v_3v_1^2v_2^3a_{10} - 2v_4v_1^3v_2a_9 - 3v_4v_1^2v_2^2a_{10} + 8v_4v_1^2v_2a_9 \\ & + 12v_4v_1v_2^2a_{10} - 8v_4v_1v_2a_9 - 4v_4a_3 - v_1^3b_4 - 15v_2^3a_6 - 2b_2v_1^2 - 6v_2^2a_3 - 3v_1b_1 + 3v_2a_1 - 2v_3a_1 \\ & - 24v_2^4a_{10} - v_4v_1^4a_8 + 4v_4v_1^3a_8 - 4v_4v_1^2a_8 - 12v_4v_2^2a_{10} - v_4v_1^3a_5 + 4v_4v_1^2a_5 - 4v_4v_1a_5 - 8v_4v_2a_6 \\ & - v_3v_1^3a_2 - v_3v_1^2a_1 + v_3v_1^2a_2 + v_3v_1^2b_3 + 2v_3v_1a_1 - 2v_3v_1b_3 + 10v_3v_2a_3 - 6v_1^3v_2a_7 - 12v_1^2v_2^2a_8 \\ & - 18v_1v_2^3a_9 + 6v_1v_2^3b_{10} - v_3v_1^4a_7 + v_3v_1^4b_8 + 4v_3v_1^3a_7 - 2v_3v_1^3b_8 + 34v_3v_2^3a_{10} - v_3v_1^5a_7 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3, v_4\}$$

Equation (7E) now becomes

$$\begin{aligned} & (-18a_9 + 6b_{10})v_2^3v_1 + (-a_1 + a_2 + 2a_4 + b_3 - 2b_5)v_1^2v_3 + (2b_8 - 6a_7)v_2v_1^3 \\ & + (4b_9 - 12a_8)v_1^2v_2^2 + (-a_3 - 4a_8 + 4a_5)v_1^2v_4 + (4a_3 - 4a_5)v_1v_4 \\ & + (-3a_4 + b_5)v_1^2v_2 + (-9a_5 + 3b_6)v_2^2v_1 + (-a_4 - a_7 + b_8)v_1^4v_3 \\ & + (b_5 - a_2 + 4a_7 - 2b_8)v_1^3v_3 + (4a_8 - a_5)v_1^3v_4 + (2a_1 - 2b_3)v_1v_3 \\ & + 22v_3v_2^2a_6 - 16v_3v_1v_2^3a_{10} - v_3v_1^4v_2a_8 - v_3v_1^3v_2^2a_9 - v_3v_1^2v_2^3a_{10} - 2v_4v_1^3v_2a_9 \\ & - 3v_4v_1^2v_2^2a_{10} + 12v_4v_1v_2^2a_{10} - 4v_4a_3 - v_1^3b_4 - 15v_2^3a_6 - 2b_2v_1^2 - 6v_2^2a_3 \\ & - 3v_1b_1 + 3v_2a_1 - 2v_3a_1 - 24v_2^4a_{10} - v_4v_1^4a_8 - 12v_4v_2^2a_{10} - 8v_4v_2a_6 \\ & + 10v_3v_2a_3 + 34v_3v_2^3a_{10} - v_3v_1^5a_7 + (-5a_5 + 2b_6 - a_3 + 14a_8 - 4b_9)v_2v_1^2v_3 \\ & + (-10a_6 + 24a_9 - 6b_{10})v_2^2v_1v_3 + (-4a_3 + 12a_5 - 4b_6)v_2v_1v_3 \\ & + (-a_5 - 6a_8 + 2b_9)v_2v_1^3v_3 + (8a_6 - 8a_9)v_2v_1v_4 \\ & + (-2a_6 + 8a_9)v_2v_1^2v_4 + (-a_6 - 11a_9 + 3b_{10})v_2^2v_1^2v_3 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$-2a_1 = 0$$

$$3a_1 = 0$$

$$-6a_3 = 0$$

$$-4a_3 = 0$$

$$10a_3 = 0$$

$$-15a_6 = 0$$

$$-8a_6 = 0$$

$$22a_6 = 0$$

$$-a_7 = 0$$

$$-a_8 = 0$$

$$-2a_9 = 0$$

$$-a_9 = 0$$

$$-24a_{10} = 0$$

$$-16a_{10} = 0$$

$$-12a_{10} = 0$$

$$-3a_{10} = 0$$

$$-a_{10} = 0$$

$$12a_{10} = 0$$

$$34a_{10} = 0$$

$$-3b_1 = 0$$

$$-2b_2 = 0$$

$$-b_4 = 0$$

$$2a_1 - 2b_3 = 0$$

$$4a_3 - 4a_5 = 0$$

$$-3a_4 + b_5 = 0$$

$$-9a_5 + 3b_6 = 0$$

$$-2a_6 + 8a_9 = 0$$

$$8a_6 - 8a_9 = 0$$

$$4a_8 - a_5 = 0$$

$$-18a_9 + 6b_{10} = 0$$

$$2b_8 - 6a_7 = 0$$

$$4b_9 - 12a_8 = 0$$

$$-4a_3 + 12a_5 - 4b_6 = 0$$

$$-a_3 - 4a_8 + 4a_5 = 0$$

$$-a_4 - a_7 + b_8 = 0$$

$$-a_5 - 6a_8 + 2b_9 = 0$$

$$-10a_6 + 24a_9 - 6b_{10} = 0$$

$$11a_7 + 2b_{10} = 0$$

Solving the above equations for the unknowns gives

$$a_1 = 0$$

$$a_2 = 0$$

$$a_3 = 0$$

$$a_4 = 0$$

$$a_5 = 0$$

$$a_6 = 0$$

$$a_7 = 0$$

$$a_8 = 0$$

$$a_9 = 0$$

$$a_{10} = 0$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = 0$$

$$b_4 = 0$$

$$b_5 = 0$$

$$b_6 = 0$$

$$b_7 = b_7$$

$$b_8 = 0$$

$$b_9 = 0$$

$$b_{10} = 0$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 0$$

$$\eta = x^3$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical

coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{x^3} dy \end{aligned}$$

Which results in

$$S = \frac{y}{x^3}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{x e^x - 2 e^x + 3y}{x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{3y}{x^4} \\ S_y &= \frac{1}{x^3} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{e^x(x-2)}{x^4} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{e^R(R-2)}{R^4}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\int dS = \int \frac{e^R(R-2)}{R^4} dR$$

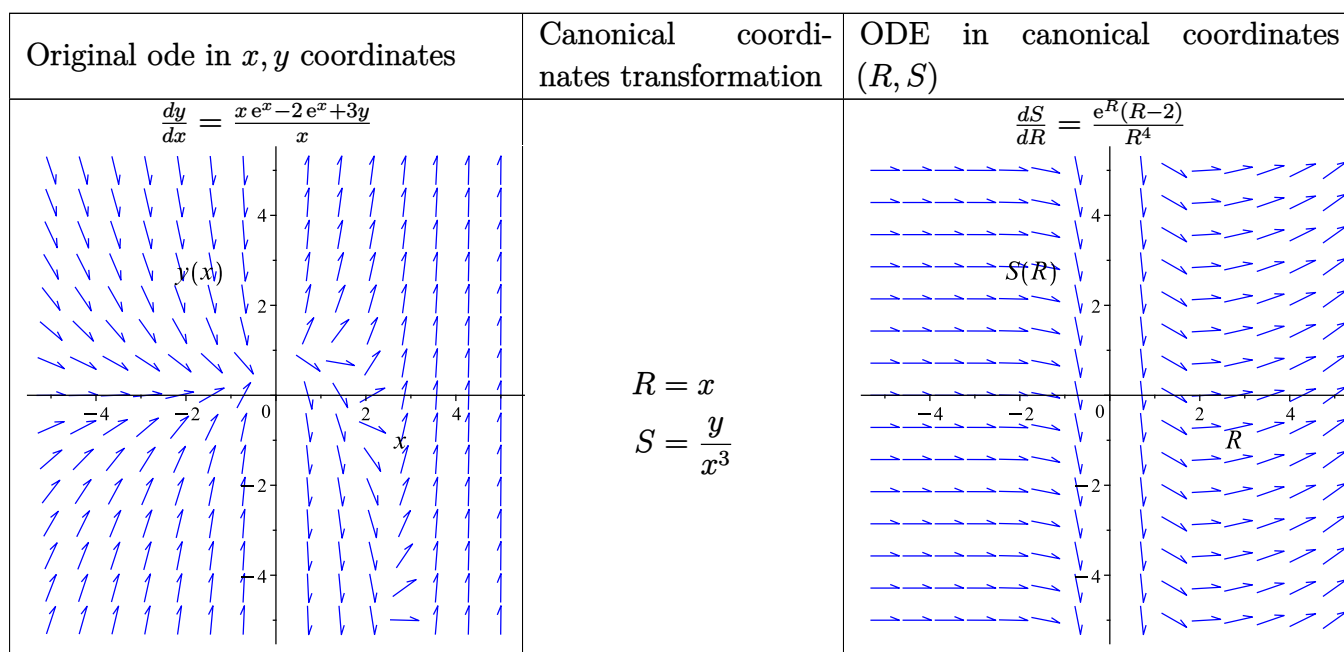
$$S(R) = -\frac{e^R}{6R^2} - \frac{e^R}{6R} - \frac{\text{Ei}_1(-R)}{6} + \frac{2e^R}{3R^3} + c_2$$

$$S(R) = \frac{-\text{Ei}_1(-R)R^3 + (-R^2 - R + 4)e^R + 6c_2R^3}{6R^3}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

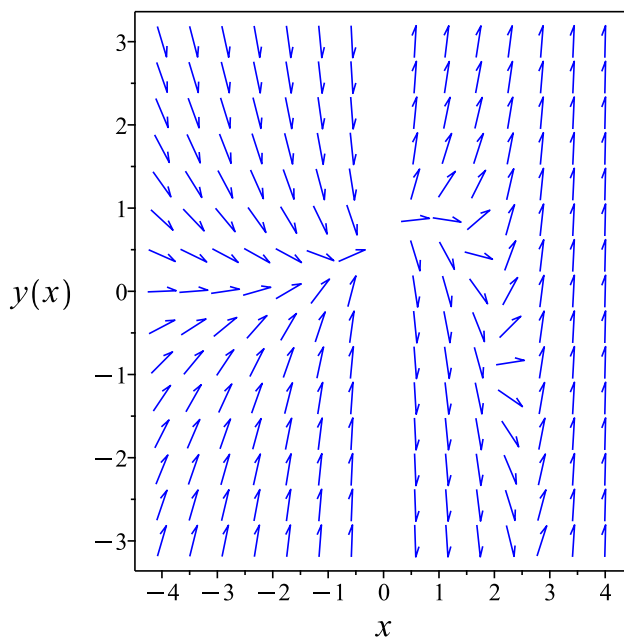
$$\frac{y}{x^3} = \frac{-\text{Ei}_1(-x)x^3 + (-x^2 - x + 4)e^x + 6c_2x^3}{6x^3}$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{\text{Ei}_1(-x)x^3}{6} - \frac{x^2e^x}{6} - \frac{xe^x}{6} + \frac{2e^x}{3} + c_2x^3$$

Figure 2.253: Slope field $-3y - (x - 2)e^x + y'x = 0$ Summary of solutions found

$$y = -\frac{\text{Ei}_1(-x)x^3}{6} - \frac{x^2 e^x}{6} - \frac{x e^x}{6} + \frac{2 e^x}{3} + c_2 x^3$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 34

```
ode:=-3*y(x)-(x-2)*exp(x)+diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{x^3 \text{Ei}_1(-x)}{6} + \frac{(-x^2 - x + 4)e^x}{6} + c_1 x^3$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

- $$-3y(x) - (x-2)e^x + x\left(\frac{d}{dx}y(x)\right) = 0$$
- Highest derivative means the order of the ODE is 1
- $$\frac{d}{dx}y(x)$$
- Solve for the highest derivative
- $$\frac{d}{dx}y(x) = \frac{3y(x) + (x-2)e^x}{x}$$
- Collect w.r.t. $y(x)$ and simplify
- $$\frac{d}{dx}y(x) = \frac{3y(x)}{x} + \frac{(x-2)e^x}{x}$$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
- $$\frac{d}{dx}y(x) - \frac{3y(x)}{x} = \frac{(x-2)e^x}{x}$$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
- $$\mu(x) \left(\frac{d}{dx}y(x) - \frac{3y(x)}{x} \right) = \frac{\mu(x)(x-2)e^x}{x}$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$
- $$\mu(x) \left(\frac{d}{dx}y(x) - \frac{3y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$
- Isolate $\frac{d}{dx}\mu(x)$
- $$\frac{d}{dx}\mu(x) = -\frac{3\mu(x)}{x}$$
- Solve to find the integrating factor
- $$\mu(x) = \frac{1}{x^3}$$
- Integrate both sides with respect to x
- $$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int \frac{\mu(x)(x-2)e^x}{x} dx + C1$$
- Evaluate the integral on the lhs
- $$y(x)\mu(x) = \int \frac{\mu(x)(x-2)e^x}{x} dx + C1$$
- Solve for $y(x)$
- $$y(x) = \frac{\int \frac{\mu(x)(x-2)e^x}{x} dx + C1}{\mu(x)}$$
- Substitute $\mu(x) = \frac{1}{x^3}$
- $$y(x) = x^3 \left(\int \frac{(x-2)e^x}{x^4} dx + C1 \right)$$
- Evaluate the integrals on the rhs
- $$y(x) = x^3 \left(-\frac{e^x}{6x^2} - \frac{e^x}{6x} - \frac{\text{Ei}_1(-x)}{6} + \frac{2e^x}{3x^3} + C1 \right)$$
- Simplify
- $$y(x) = -\frac{\text{Ei}_1(-x)x^3}{6} + \frac{(-x^2-x+4)e^x}{6} + C1 x^3$$

✓ **Mathematica.** Time used: 0.1 (sec). Leaf size: 32

```
ode=(-2*y[x]-(x-2)*Exp[x])+x*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x^3 \left(\int_1^x \frac{e^{K[1]}(K[1]-2)}{K[1]^4} dK[1] + c_1 \right)$$

✓ Sympy. Time used: 0.515 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) - (x - 2)*exp(x) - 3*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 x^3 - x E_3(-x) + 2 E_4(-x)$$

2.4.4 Problem 19 (e)

Local contents

Solved using first_order_ode_linear	1109
Solved using first_order_ode_exact	1111
✓ Maple	1115
✓ Mathematica	1116
✓ Sympy	1116

Internal problem ID [7920]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (e)

Date solved : Thursday, November 27, 2025 at 06:31:49 AM

CAS classification : [[_linear, 'class A']]

Solved using first_order_ode_linear

Time used: 0.101 (sec)

Solve

$$i' - 6i = 10 \sin(2t)$$

In canonical form a linear first order is

$$i' + q(t)i = p(t)$$

Comparing the above to the given ode shows that

$$q(t) = -6$$

$$p(t) = 10 \sin(2t)$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dt} \\ &= e^{\int (-6) dt} \\ &= e^{-6t} \end{aligned}$$

The ode becomes

$$\frac{d}{dt}(\mu i) = \mu p$$

$$\frac{d}{dt}(\mu i) = (\mu) (10 \sin(2t))$$

$$\frac{d}{dt}(i e^{-6t}) = (e^{-6t}) (10 \sin(2t))$$

$$d(i e^{-6t}) = (10 \sin(2t) e^{-6t}) dt$$

Integrating gives

$$\begin{aligned} i e^{-6t} &= \int 10 \sin(2t) e^{-6t} dt \\ &= -\frac{e^{-6t}(\cos(2t) + 3 \sin(2t))}{2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^{-6t} gives the final solution

$$i = e^{6t} c_1 - \frac{\cos(2t)}{2} - \frac{3 \sin(2t)}{2}$$

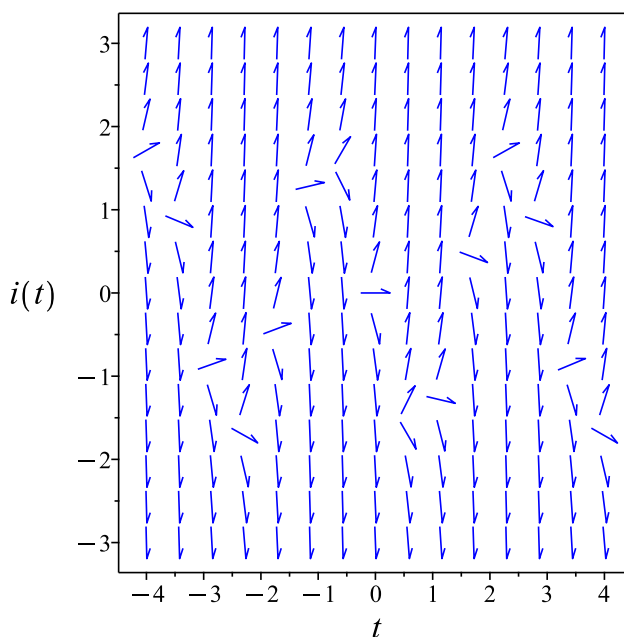


Figure 2.254: Slope field $i' - 6i = 10 \sin(2t)$

Summary of solutions found

$$i = e^{6t} c_1 - \frac{\cos(2t)}{2} - \frac{3 \sin(2t)}{2}$$

Solved using first_order_ode_exact

Time used: 0.138 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(t, i) dt + N(t, i) di = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} di &= (6i + 10 \sin(2t)) dt \\ (-6i - 10 \sin(2t)) dt + di &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(t, i) &= -6i - 10 \sin(2t) \\ N(t, i) &= 1 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial i} = \frac{\partial N}{\partial t}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial i} &= \frac{\partial}{\partial i}(-6i - 10 \sin(2t)) \\ &= -6\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial t} &= \frac{\partial}{\partial t}(1) \\ &= 0\end{aligned}$$

Since $\frac{\partial M}{\partial i} \neq \frac{\partial N}{\partial t}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial i} - \frac{\partial N}{\partial t} \right) \\ &= 1((-6) - (0)) \\ &= -6\end{aligned}$$

Since A does not depend on i , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dt} \\ &= e^{\int -6 \, dt}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-6t} \\ &= e^{-6t}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= e^{-6t}(-6i - 10 \sin(2t)) \\ &= (-6i - 10 \sin(2t)) e^{-6t}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= e^{-6t}(1) \\ &= e^{-6t}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{di}{dt} &= 0 \\ ((-6i - 10 \sin(2t)) e^{-6t}) + (e^{-6t}) \frac{di}{dt} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(t, i)$

$$\frac{\partial \phi}{\partial t} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial i} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. i gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial i} di &= \int \overline{N} di \\ \int \frac{\partial \phi}{\partial i} di &= \int e^{-6t} di \\ \phi &= i e^{-6t} + f(t)\end{aligned} \quad (3)$$

Where $f(t)$ is used for the constant of integration since ϕ is a function of both t and i . Taking derivative of equation (3) w.r.t t gives

$$\frac{\partial \phi}{\partial t} = -6i e^{-6t} + f'(t) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial t} = (-6i - 10 \sin(2t)) e^{-6t}$. Therefore equation (4) becomes

$$(-6i - 10 \sin(2t)) e^{-6t} = -6i e^{-6t} + f'(t) \quad (5)$$

Solving equation (5) for $f'(t)$ gives

$$f'(t) = -10 \sin(2t) e^{-6t}$$

Integrating the above w.r.t t gives

$$\begin{aligned}\int f'(t) dt &= \int (-10 \sin(2t) e^{-6t}) dt \\ f(t) &= \frac{\cos(2t) e^{-6t}}{2} + \frac{3 \sin(2t) e^{-6t}}{2} + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(t)$ into equation (3) gives ϕ

$$\phi = i e^{-6t} + \frac{\cos(2t) e^{-6t}}{2} + \frac{3 \sin(2t) e^{-6t}}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = i e^{-6t} + \frac{\cos(2t) e^{-6t}}{2} + \frac{3 \sin(2t) e^{-6t}}{2}$$

Simplifying the above gives

$$\frac{e^{-6t}(2i + \cos(2t) + 3 \sin(2t))}{2} = c_1$$

Solving for i gives

$$i = -\frac{(3 \sin(2t) e^{-6t} + \cos(2t) e^{-6t} - 2c_1) e^{6t}}{2}$$

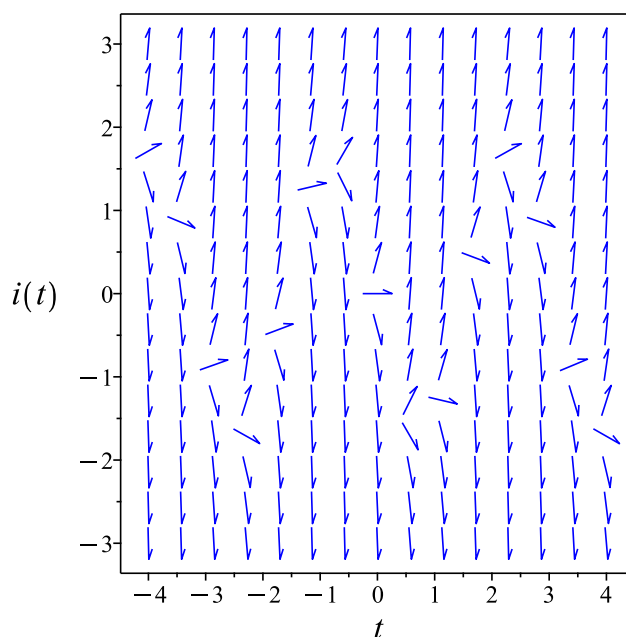


Figure 2.255: Slope field $i' - 6i = 10 \sin(2t)$

Summary of solutions found

$$i = -\frac{(3 \sin(2t) e^{-6t} + \cos(2t) e^{-6t} - 2c_1) e^{6t}}{2}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 23

```
ode:=diff(i(t),t)-6*i(t) = 10*sin(2*t);
dsolve(ode,i(t), singsol=all);
```

$$i = -\frac{\cos(2t)}{2} - \frac{3 \sin(2t)}{2} + e^{6t} c_1$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

Maple step by step

Let's solve

$$\frac{d}{dt}i(t) - 6i(t) = 10 \sin(2t)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dt}i(t)$$

- Solve for the highest derivative

$$\frac{d}{dt}i(t) = 6i(t) + 10 \sin(2t)$$

- Group terms with $i(t)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dt}i(t) - 6i(t) = 10 \sin(2t)$$

- The ODE is linear; multiply by an integrating factor $\mu(t)$

$$\mu(t) \left(\frac{d}{dt}i(t) - 6i(t) \right) = 10\mu(t) \sin(2t)$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dt}(i(t) \mu(t))$

$$\mu(t) \left(\frac{d}{dt}i(t) - 6i(t) \right) = \left(\frac{d}{dt}i(t) \right) \mu(t) + i(t) \left(\frac{d}{dt}\mu(t) \right)$$

- Isolate $\frac{d}{dt}\mu(t)$

$$\frac{d}{dt}\mu(t) = -6\mu(t)$$

- Solve to find the integrating factor

$$\mu(t) = e^{-6t}$$

- Integrate both sides with respect to t

$$\int \left(\frac{d}{dt}(i(t) \mu(t)) \right) dt = \int 10\mu(t) \sin(2t) dt + C1$$

- Evaluate the integral on the lhs

$$i(t) \mu(t) = \int 10\mu(t) \sin(2t) dt + C1$$

- Solve for $i(t)$

$$i(t) = \frac{\int 10\mu(t) \sin(2t) dt + C1}{\mu(t)}$$

- Substitute $\mu(t) = e^{-6t}$

$$i(t) = \frac{\int 10 e^{-6t} \sin(2t) dt + C1}{e^{-6t}}$$

- Evaluate the integrals on the rhs

$$i(t) = \frac{-\frac{e^{-6t} \cos(2t)}{2} - \frac{3e^{-6t} \sin(2t)}{2} + C1}{e^{-6t}}$$

- Simplify

$$i(t) = -\frac{\cos(2t)}{2} - \frac{3\sin(2t)}{2} + e^{6t} C1$$

✓ **Mathematica.** Time used: 0.061 (sec). Leaf size: 34

```
ode=D[i[t],t]-6*i[t]==10*Sin[2*t];
ic={};
DSolve[{ode,ic},i[t],t,IncludeSingularSolutions->True]
```

$$i(t) \rightarrow e^{6t} \left(\int_1^t 10e^{-6K[1]} \sin(2K[1]) dK[1] + c_1 \right)$$

✓ **Sympy.** Time used: 0.093 (sec). Leaf size: 24

```
from sympy import *
t = symbols("t")
i = Function("i")
ode = Eq(-6*i(t) - 10*sin(2*t) + Derivative(i(t), t), 0)
ics = {}
dsolve(ode, func=i(t), ics=ics)
```

$$i(t) = C_1 e^{6t} - \frac{3 \sin(2t)}{2} - \frac{\cos(2t)}{2}$$

2.4.5 Problem 19 (f)

Local contents

Solved using first_order_ode_bernoulli	1117
Solved using first_order_ode_LIE	1120
✓ Maple	1124
✓ Mathematica	1125
✓ Sympy	1125

Internal problem ID [7921]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (f)

Date solved : Thursday, November 27, 2025 at 06:31:55 AM

CAS classification : [[_1st_order, _with_linear_symmetries], _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.111 (sec)

Solve

$$y' + y = e^x y^2$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -y + e^x y^2 \end{aligned}$$

This is a Bernoulli ODE.

$$y' = (-1)y + (e^x)y^2 \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -1 \\ f_1 &= e^x \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \tag{3}$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= -1 \\f_1(x) &= e^x \\n &= 2\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^2$ gives

$$y' \frac{1}{y^2} = -\frac{1}{y} + e^x \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= \frac{1}{y}\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{1}{y^2} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}-v'(x) &= -v(x) + e^x \\v' &= v - e^x\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -1 \\p(x) &= -e^x\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\&= e^{\int (-1) \, dx} \\&= e^{-x}\end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu v) = \mu p$$

$$\frac{d}{dx}(\mu v) = (\mu)(-e^x)$$

$$\frac{d}{dx}(v e^{-x}) = (e^{-x})(-e^x)$$

$$d(v e^{-x}) = (-e^x e^{-x}) dx$$

Integrating gives

$$\begin{aligned} v e^{-x} &= \int -e^x e^{-x} dx \\ &= -x + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^{-x} gives the final solution

$$v(x) = (c_1 - x) e^x$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y} = (c_1 - x) e^x$$

Solving for y gives

$$y = \frac{e^{-x}}{c_1 - x}$$

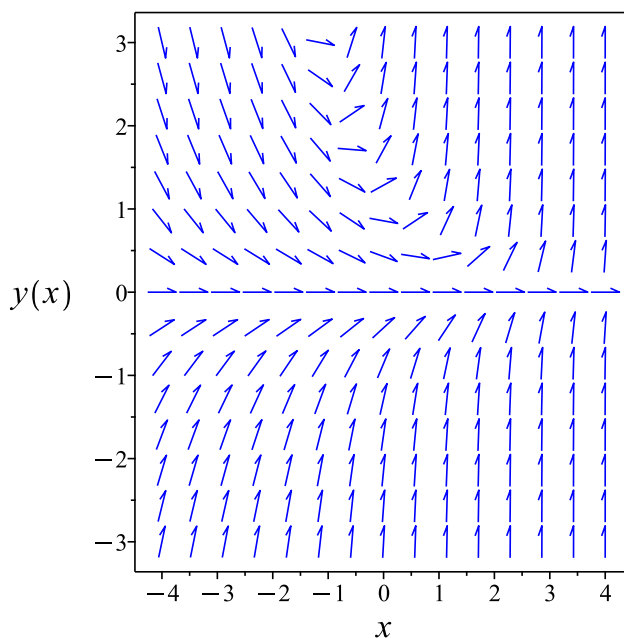


Figure 2.256: Slope field $y' + y = e^x y^2$

Summary of solutions found

$$y = \frac{e^{-x}}{c_1 - x}$$

Solved using first_order_ode_LIE

Time used: 0.427 (sec)

Solve

$$y' + y = e^x y^2$$

Writing the ode as

$$\begin{aligned} y' &= -y + e^x y^2 \\ y' &= \omega(x, y) \end{aligned}$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = 1 \quad (1\text{E})$$

$$\eta = Ax + By \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$A + (-y + e^x y^2) B - e^x y^2 - (2e^x y - 1)(Ax + By) = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$-2A e^x x y - B e^x y^2 - e^x y^2 + Ax + A = 0$$

Setting the numerator to zero gives

$$-2A e^x x y - B e^x y^2 - e^x y^2 + Ax + A = 0 \quad (6\text{E})$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, e^x\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, e^x = v_3\}$$

The above PDE (6E) now becomes

$$-2Av_3v_1v_2 - Bv_3v_2^2 - v_3v_2^2 + Av_1 + A = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$-2Av_3v_1v_2 + Av_1 + (-B - 1)v_3v_2^2 + A = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$A = 0$$

$$-2A = 0$$

$$-B - 1 = 0$$

Solving the above equations for the unknowns gives

$$A = 0$$

$$B = -1$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 1$$

$$\eta = -y$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{-y}{1} \\ &= -y\end{aligned}$$

This is easily solved to give

$$y = e^{-x} c_1$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = e^x y$$

And S is found from

$$\begin{aligned}dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1}\end{aligned}$$

Integrating gives

$$\begin{aligned}S &= \int \frac{dx}{T} \\ &= x\end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -y + e^x y^2$$

Evaluating all the partial derivatives gives

$$\begin{aligned}R_x &= e^x y \\ R_y &= e^x \\ S_x &= 1 \\ S_y &= 0\end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{e^{-2x}}{y^2} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{R^2}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

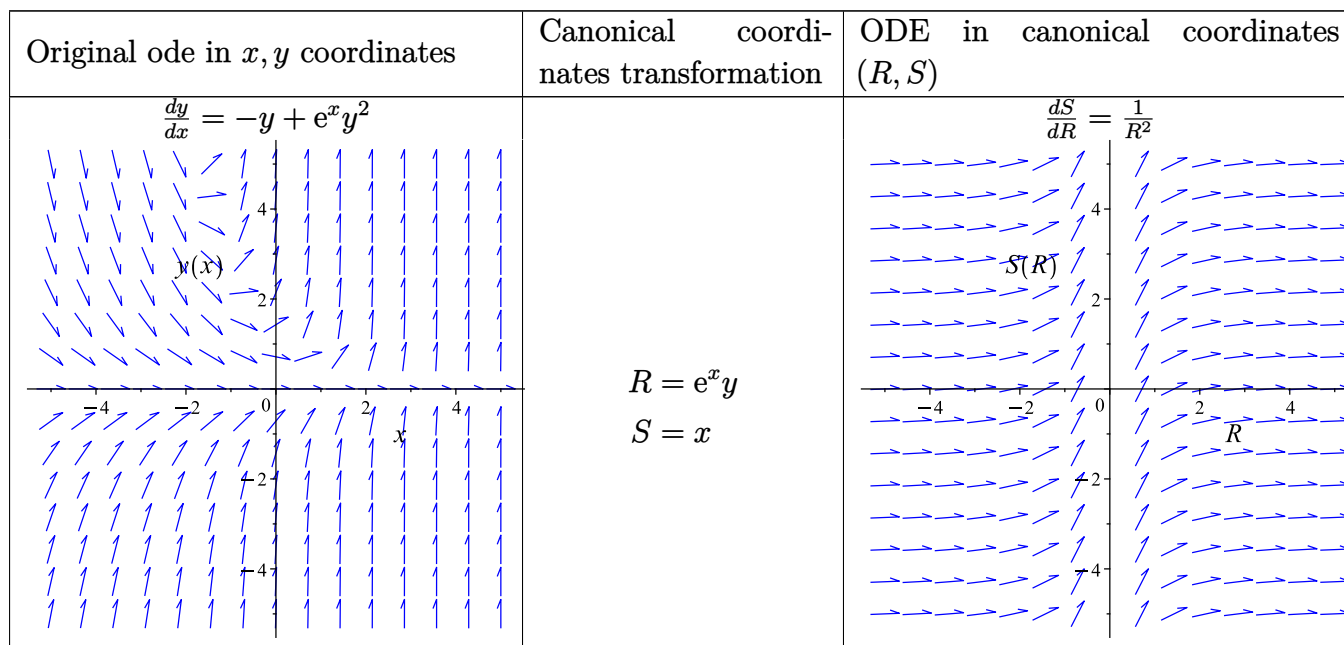
$$\int dS = \int \frac{1}{R^2} dR$$

$$S(R) = -\frac{1}{R} + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = -\frac{e^{-x}}{y} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = \frac{e^{-x}}{-x + c_2}$$

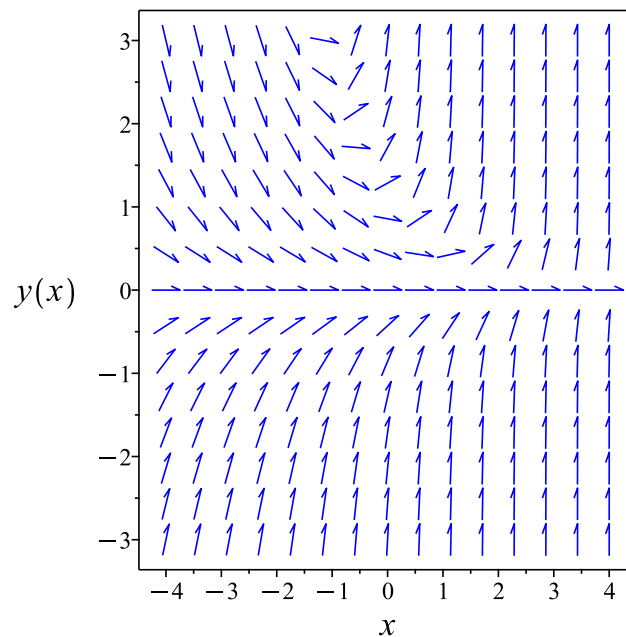


Figure 2.257: Slope field $y' + y = e^x y^2$

Summary of solutions found

$$y = \frac{e^{-x}}{-x + c_2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 16

```
ode:=diff(y(x),x)+y(x) = y(x)^2*exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{e^{-x}}{-x + c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```


Maple step by step

Let's solve

$$\frac{d}{dx}y(x) + y(x) = y(x)^2 e^x$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -y(x) + y(x)^2 e^x$$

✓ **Mathematica.** Time used: 0.134 (sec). Leaf size: 25

```
ode=D[y[x],x]+y[x]==y[x]^2*Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{e^{-x}}{x - c_1}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 0.130 (sec). Leaf size: 10

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x)**2*exp(x) + y(x) + Derivative(y(x), x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{e^{-x}}{C_1 - x}$$

2.4.6 Problem 19 (g)

Local contents

Solved using first_order_ode_exact	1126
Solved using first_order_ode_abel_second_kind_solved_by_convert- ing_to_first_kind	1130
✓ Maple	1134
✓ Mathematica	1135
✓ Sympy	1135

Internal problem ID [7922]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (g)

Date solved : Thursday, November 27, 2025 at 06:32:02 AM

CAS classification :

[_rational, [_1st_order, ‘_with_symmetry_[F(x)*G(y),0]’], [_Abel, ‘2nd type’, ‘class B’]

Solved using first_order_ode_exact

Time used: 0.118 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (yx + x - 3y) dy &= (-y) dx \\ (y) dx + (yx + x - 3y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y \\ N(x, y) &= yx + x - 3y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(yx + x - 3y) \\ &= y + 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{(x-3)y+x} ((1) - (y+1)) \\ &= -\frac{y}{(x-3)y+x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y} ((y+1) - (1)) \\ &= 1 \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int 1 \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^y \\ &= e^y \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \bar{M} &= \mu M \\ &= e^y(y) \\ &= e^y y \end{aligned}$$

And

$$\begin{aligned} \bar{N} &= \mu N \\ &= e^y(yx + x - 3y) \\ &= (yx + x - 3y) e^y \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ (e^y y) + ((yx + x - 3y) e^y) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int e^y y dx \\ \phi &= e^y yx + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= e^y yx + x e^y + f'(y) \\ &= x(y + 1) e^y + f'(y)\end{aligned}\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = (yx + x - 3y) e^y$. Therefore equation (4) becomes

$$(yx + x - 3y) e^y = x(y + 1) e^y + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -3 e^y y$$

Integrating the above w.r.t y gives

$$\begin{aligned}\int f'(y) dy &= \int (-3 e^y y) dy \\ f(y) &= -3(-1 + y) e^y + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = e^y yx - 3(-1 + y) e^y + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = e^y yx - 3(-1 + y) e^y$$

Simplifying the above gives

$$(3 + (x - 3) y) e^y = c_1$$

Solving for y gives

$$y = \frac{\text{LambertW}\left(\frac{c_1 e^{\frac{3}{x-3}}}{x-3}\right) x - 3 \text{LambertW}\left(\frac{c_1 e^{\frac{3}{x-3}}}{x-3}\right) - 3}{x-3}$$

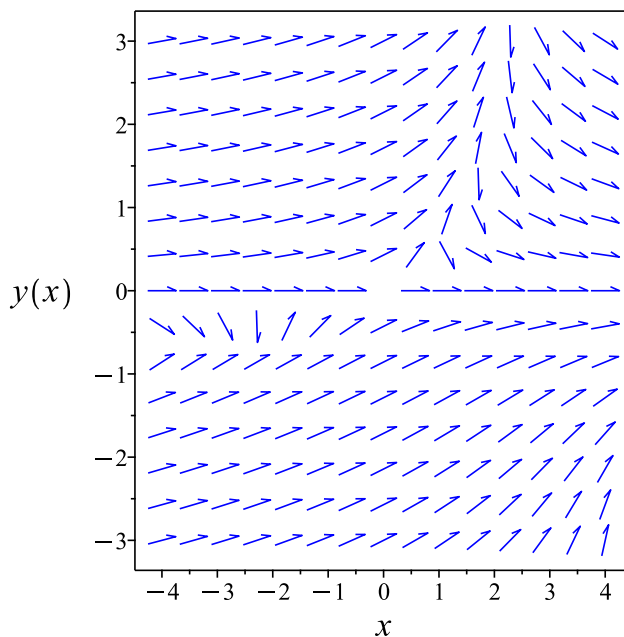


Figure 2.258: Slope field $y + (yx + x - 3y)y' = 0$

Summary of solutions found

$$y = \frac{\text{LambertW}\left(\frac{c_1 e^{\frac{3}{x-3}}}{x-3}\right) x - 3 \text{LambertW}\left(\frac{c_1 e^{\frac{3}{x-3}}}{x-3}\right) - 3}{x-3}$$

Solve

$$y + (yx + x - 3y)y' = 0$$

Solved using

first_order_ode_abel_second_kind_solved_by_converting_to_first_kind

Time used: 1.530 (sec)

This is Abel second kind ODE, it has the form

$$(y + g)y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$y + (yx + x - 3y) y' = 0 \quad (1)$$

Shows that

$$\begin{aligned} g &= \frac{x}{x-3} \\ f_0 &= 0 \\ f_1 &= -\frac{1}{x-3} \\ f_2 &= 0 \\ f_3 &= 0 \end{aligned}$$

Applying transformation

$$y = \frac{1}{u(x)} - g$$

Results in the new ode which is Abel first kind

$$u'(x) = -\frac{xu(x)^3}{x^2 - 6x + 9} + \frac{xu(x)^2}{x^2 - 6x + 9}$$

Which is now solved.

Solve The ode

$$u'(x) = -\frac{xu(x)^2(u(x) - 1)}{(x - 3)^2} \quad (2.103)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{xu(x)^2(u(x) - 1)}{(x - 3)^2} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{x}{(x - 3)^2} \\ g(u) &= u^2(u - 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u^2(u - 1)} du &= \int -\frac{x}{(x - 3)^2} dx \end{aligned}$$

$$\ln(u(x) - 1) + \frac{1}{u(x)} - \ln(u(x)) = \frac{3}{x-3} - \ln(x-3) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u^2(u-1) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\ln(u(x) - 1) + \frac{1}{u(x)} - \ln(u(x)) = \frac{3}{x-3} - \ln(x-3) + c_1$$

$$u(x) = 0$$

$$u(x) = 1$$

Substituting $u(x) = \frac{1}{y + \frac{x}{x-3}}$ in the above solution gives

$$\ln\left(\frac{1}{y + \frac{x}{x-3}} - 1\right) + y + \frac{x}{x-3} - \ln\left(\frac{1}{y + \frac{x}{x-3}}\right) = \frac{3}{x-3} - \ln(x-3) + c_1$$

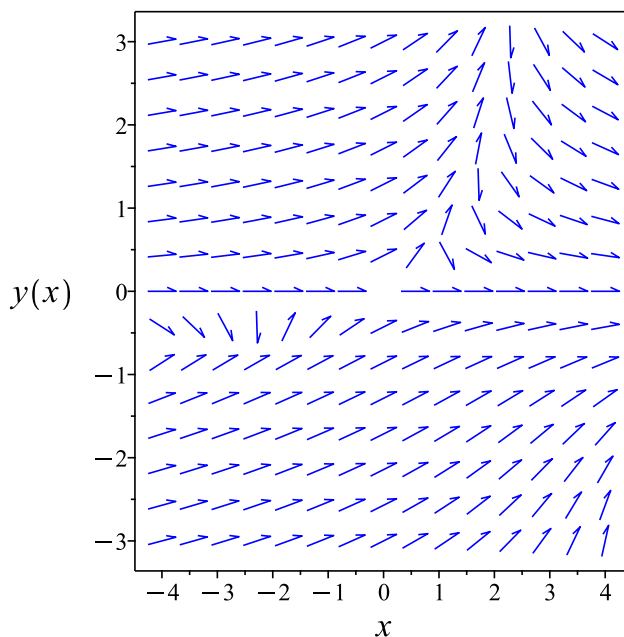
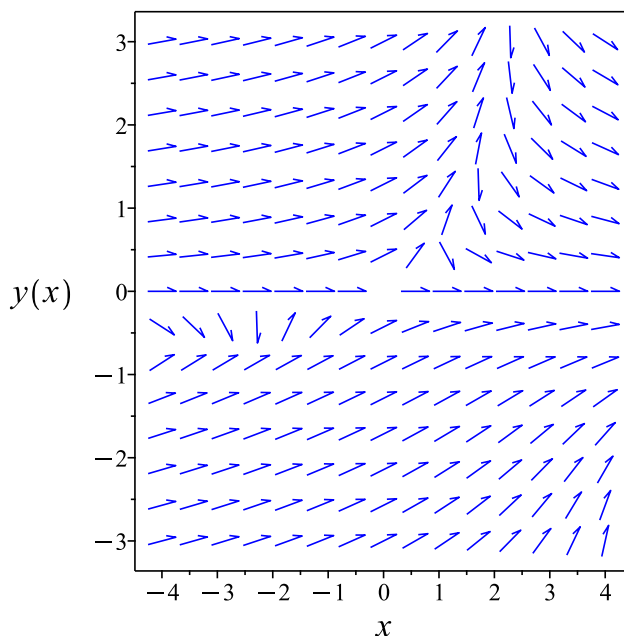
Now we transform the solution $u(x) = 1$ to y using $u(x) = \frac{1}{y + \frac{x}{x-3}}$ which gives

$$y = -\frac{3}{x-3}$$

Simplifying the above gives

$$\ln\left(-\frac{yx - 3y + 3}{yx + x - 3y}\right) + y + \frac{x}{x-3} - \ln\left(\frac{x-3}{(x-3)y + x}\right) = \frac{3}{x-3} - \ln(x-3) + c_1$$

$$y = -\frac{3}{x-3}$$

Figure 2.259: Slope field $y + (yx + x - 3y)y' = 0$ Figure 2.260: Slope field $y + (yx + x - 3y)y' = 0$

Summary of solutions found

$$\ln \left(-\frac{yx - 3y + 3}{yx + x - 3y} \right) + y + \frac{x}{x - 3} - \ln \left(\frac{x - 3}{(x - 3)y + x} \right) = \frac{3}{x - 3} - \ln(x - 3) + c_1$$

$$y = -\frac{3}{x - 3}$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 30

```
ode:=y(x)+(y(x)*x+x-3*y(x))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \text{LambertW}\left(\frac{e^{\frac{3}{x-3}}}{c_1(x-3)}\right) - \frac{3}{x-3}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

-> Trying a Liouvillian solution using Kovacic's algorithm

A Liouvillian solution exists

Reducible group (found an exponential solution)

Reducible group (found another exponential solution)

<- Kovacic's algorithm successful

<- Abel AIR successful: ODE belongs to the 1F1 2-parameter class

<- inverse linear successful

Maple step by step

Let's solve

$$y(x) + (xy(x) + x - 3y(x)) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{y(x)}{xy(x) + x - 3y(x)}$$

✓ **Mathematica.** Time used: 0.066 (sec). Leaf size: 52

```
ode=y[x]+(x*y[x]+x-3*y[x])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[x = \frac{e^{-y(x)-1} \int_1^{y(x)} 3e^{K[1]+1} K[1] dK[1]}{y(x)} + \frac{c_1 e^{-y(x)-1}}{y(x)}, y(x) \right]$$

✓ **Sympy.** Time used: 1.009 (sec). Leaf size: 27

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x*y(x) + x - 3*y(x))*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{(x-3) W\left(\frac{e^{\frac{C_1(x-3)+3}{x-3}}}{x-3}\right) - 3}{x-3}$$

2.4.7 Problem 19 (h)

Local contents

Solved using first_order_ode_exact	1136
Solved using first_order_ode_abel_second_kind_case_5	1140
✓ Maple	1141
✓ Mathematica	1142
✗ Sympy	1143

Internal problem ID [7923]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (h)

Date solved : Thursday, November 27, 2025 at 06:32:10 AM

CAS classification :

[_exact, [_1st_order, '_with_symmetry_[F(x),G(x)]'], [_Abel, '2nd type', 'class A']]

Solved using first_order_ode_exact

Time used: 0.141 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(t, s) dt + N(t, s) ds = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (2s - e^{2t}) ds &= (2s e^{2t} - 2 \cos(2t)) dt \\ (-2s e^{2t} + 2 \cos(2t)) dt + (2s - e^{2t}) ds &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(t, s) &= -2s e^{2t} + 2 \cos(2t) \\ N(t, s) &= 2s - e^{2t} \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial s} = \frac{\partial N}{\partial t}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial s} &= \frac{\partial}{\partial s} (-2s e^{2t} + 2 \cos(2t)) \\ &= -2 e^{2t} \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial t} &= \frac{\partial}{\partial t} (2s - e^{2t}) \\ &= -2 e^{2t} \end{aligned}$$

Since $\frac{\partial M}{\partial s} = \frac{\partial N}{\partial t}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(t, s)$

$$\frac{\partial \phi}{\partial t} = M \quad (1)$$

$$\frac{\partial \phi}{\partial s} = N \quad (2)$$

Integrating (1) w.r.t. t gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial t} dt &= \int M dt \\ \int \frac{\partial \phi}{\partial t} dt &= \int -2s e^{2t} + 2 \cos(2t) dt \\ \phi &= -s e^{2t} + \sin(2t) + f(s) \end{aligned} \quad (3)$$

Where $f(s)$ is used for the constant of integration since ϕ is a function of both t and s . Taking derivative of equation (3) w.r.t s gives

$$\frac{\partial \phi}{\partial s} = -e^{2t} + f'(s) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial s} = 2s - e^{2t}$. Therefore equation (4) becomes

$$2s - e^{2t} = -e^{2t} + f'(s) \quad (5)$$

Solving equation (5) for $f'(s)$ gives

$$f'(s) = 2s$$

Integrating the above w.r.t s gives

$$\begin{aligned} \int f'(s) \, ds &= \int (2s) \, ds \\ f(s) &= s^2 + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(s)$ into equation (3) gives ϕ

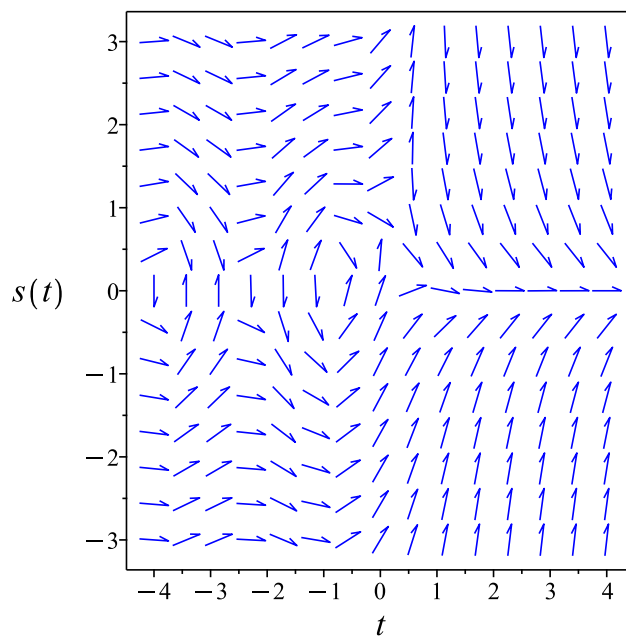
$$\phi = -s e^{2t} + \sin(2t) + s^2 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -s e^{2t} + \sin(2t) + s^2$$

Solving for s gives

$$\begin{aligned} s &= \frac{e^{2t}}{2} - \frac{\sqrt{e^{4t} - 4 \sin(2t) + 4c_1}}{2} \\ s &= \frac{e^{2t}}{2} + \frac{\sqrt{e^{4t} - 4 \sin(2t) + 4c_1}}{2} \end{aligned}$$

Figure 2.261: Slope field $(2s - e^{2t}) s' = 2s e^{2t} - 2 \cos(2t)$ Summary of solutions found

$$s = \frac{e^{2t}}{2} - \frac{\sqrt{e^{4t} - 4 \sin(2t) + 4c_1}}{2}$$

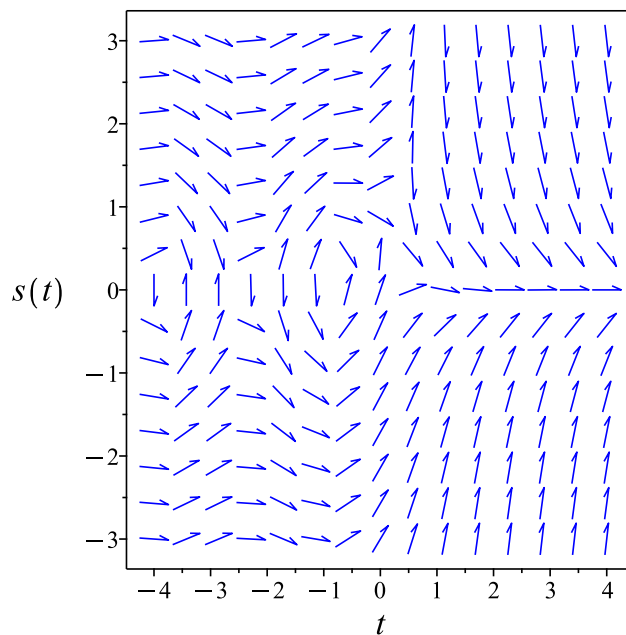
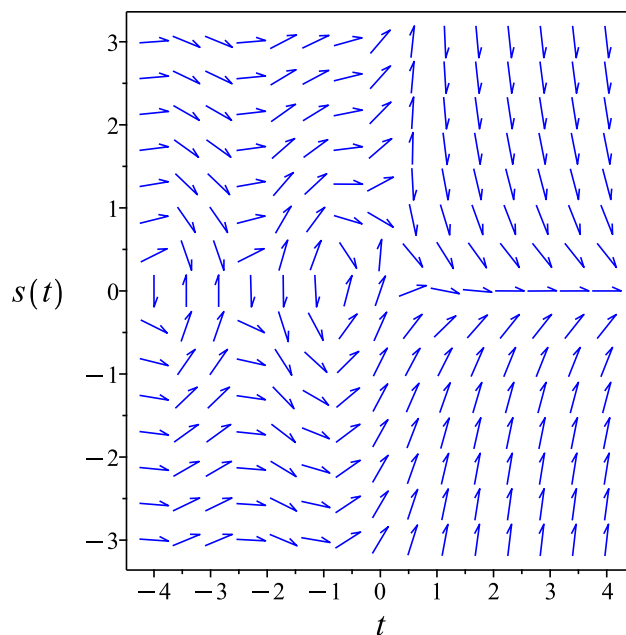
$$s = \frac{e^{2t}}{2} + \frac{\sqrt{e^{4t} - 4 \sin(2t) + 4c_1}}{2}$$

Solve

$$(2s - e^{2t}) s' = 2s e^{2t} - 2 \cos(2t)$$

Solved using first_order_ode_abel_second_kind_case_5

Time used: 0.166 (sec)

Figure 2.262: Slope field $(2s - e^{2t})s' = 2se^{2t} - 2\cos(2t)$ Figure 2.263: Slope field $(2s - e^{2t})s' = 2se^{2t} - 2\cos(2t)$ Summary of solutions found

$$s = \frac{e^{2t}}{2} - \frac{\sqrt{e^{4t} - 4\sin(2t) + 4c_1}}{2}$$

$$s = \frac{e^{2t}}{2} + \frac{\sqrt{e^{4t} - 4 \sin(2t) + 4c_1}}{2}$$

✓ Maple. Time used: 0.006 (sec). Leaf size: 57

```
ode:=(2*s(t)-exp(2*t))*diff(s(t),t) = 2*s(t)*exp(2*t)-2*cos(2*t);
dsolve(ode,s(t), singsol=all);
```

$$s = \frac{e^{2t}}{2} - \frac{\sqrt{e^{4t} - 4 \sin(2t) - 4c_1}}{2}$$

$$s = \frac{e^{2t}}{2} + \frac{\sqrt{e^{4t} - 4 \sin(2t) - 4c_1}}{2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
<- exact successful
```

Maple step by step

Let's solve

$$(2s(t) - e^{2t}) \left(\frac{d}{dt} s(t) \right) = 2s(t) e^{2t} - 2 \cos(2t)$$

- Highest derivative means the order of the ODE is 1
- Check if ODE is exact
 - ODE is exact if the lhs is the total derivative of a C^2 function
 - $\frac{d}{dt} G(t, s(t)) = 0$
 - Compute derivative of lhs
 - $\frac{\partial}{\partial t} G(t, s) + \left(\frac{\partial}{\partial s} G(t, s) \right) \left(\frac{d}{dt} s(t) \right) = 0$
 - Evaluate derivatives
 - $-2e^{2t} = -2e^{2t}$
 - Condition met, ODE is exact

- Exact ODE implies solution will be of this form
 $[G(t, s) = C1, M(t, s) = \frac{\partial}{\partial t}G(t, s), N(t, s) = \frac{\partial}{\partial s}G(t, s)]$
- Solve for $G(t, s)$ by integrating $M(t, s)$ with respect to t
 $G(t, s) = \int (-2s e^{2t} + 2 \cos(2t)) dt + _F1(s)$
- Evaluate integral
 $G(t, s) = -s e^{2t} + \sin(2t) + _F1(s)$
- Take derivative of $G(t, s)$ with respect to s
 $N(t, s) = \frac{\partial}{\partial s}G(t, s)$
- Compute derivative
 $2s - e^{2t} = -e^{2t} + \frac{d}{ds}_F1(s)$
- Isolate for $\frac{d}{ds}_F1(s)$
 $\frac{d}{ds}_F1(s) = 2s$
- Solve for $_F1(s)$
 $_F1(s) = s^2$
- Substitute $_F1(s)$ into equation for $G(t, s)$
 $G(t, s) = -s e^{2t} + \sin(2t) + s^2$
- Substitute $G(t, s)$ into the solution of the ODE
 $-s e^{2t} + \sin(2t) + s^2 = C1$
- Solve for $s(t)$
 $\left\{ s(t) = \frac{e^{2t}}{2} - \frac{\sqrt{(e^{2t})^2 - 4 \sin(2t) + 4 C1}}{2}, s(t) = \frac{e^{2t}}{2} + \frac{\sqrt{(e^{2t})^2 - 4 \sin(2t) + 4 C1}}{2} \right\}$

✓ **Mathematica.** Time used: 3.317 (sec). Leaf size: 101

```
ode=(2*s[t]-Exp[2*t])*D[s[t],t]==2*(s[t]*Exp[2*t]-Cos[2*t]);
ic={};
DSolve[{ode,ic},s[t],t,IncludeSingularSolutions->True]
```

$$s(t) \rightarrow \frac{1}{2} \left(e^{2t} - i \sqrt{-8 \int_1^t -\cos(2K[1]) dK[1] - e^{4t} - 4c_1} \right)$$

$$s(t) \rightarrow \frac{1}{2} \left(e^{2t} + i \sqrt{-8 \int_1^t -\cos(2K[1]) dK[1] - e^{4t} - 4c_1} \right)$$

Sympy

```
from sympy import *
t = symbols("t")
s = Function("s")
ode = Eq((2*s(t) - exp(2*t))*Derivative(s(t), t) - 2*s(t)*exp(2*t) + 2*cos(2*t)
,0)
ics = {}
dsolve(ode,func=s(t),ics=ics)
```

Timed Out

2.4.8 Problem 19 (i)

Local contents

Solved using first_order_ode_bernoulli	1144
Solved using first_order_ode_exact	1147
Solved using first_order_ode_isobaric	1152
Solved using first_order_ode_homog_type_G	1155
Solved using first_order_ode_LIE	1156
✓ Maple	1162
✓ Mathematica	1163
✓ Sympy	1163

Internal problem ID [7924]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (i)

Date solved : Thursday, November 27, 2025 at 06:32:22 AM

CAS classification : [[_homogeneous, 'class G'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.125 (sec)

Solve

$$y'x + y - y^6x^3 = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{y(x^3y^5 - 1)}{x} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(-\frac{1}{x}\right)y + (x^2)y^6 \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -\frac{1}{x} \\ f_1 &= x^2 \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= -\frac{1}{x} \\ f_1(x) &= x^2 \\ n &= 6 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^6$ gives

$$y' \frac{1}{y^6} = -\frac{1}{x y^5} + x^2 \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= \frac{1}{y^5} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{5}{y^6} y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -\frac{v'(x)}{5} &= -\frac{v(x)}{x} + x^2 \\ v' &= \frac{5v}{x} - 5x^2 \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{5}{x} \\ p(x) &= -5x^2 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{5}{x} \, dx} \\ &= \frac{1}{x^5}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) (-5x^2) \\ \frac{d}{dx}\left(\frac{v}{x^5}\right) &= \left(\frac{1}{x^5}\right) (-5x^2) \\ d\left(\frac{v}{x^5}\right) &= \left(-\frac{5}{x^3}\right) dx\end{aligned}$$

Integrating gives

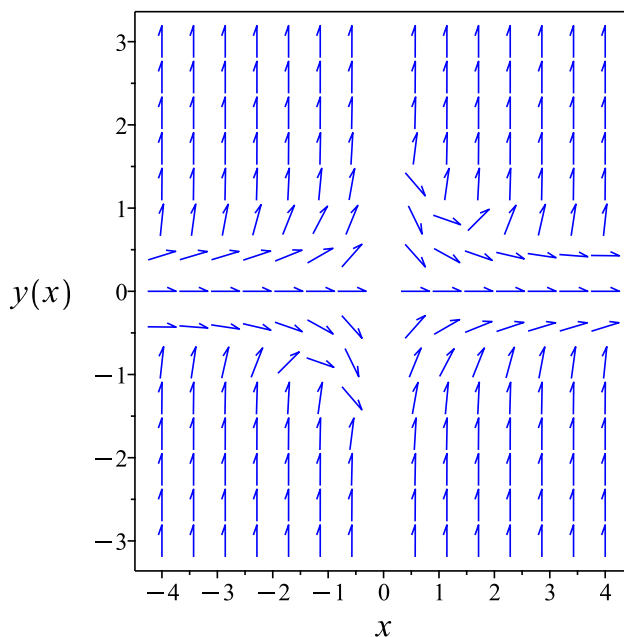
$$\begin{aligned}\frac{v}{x^5} &= \int -\frac{5}{x^3} \, dx \\ &= \frac{5}{2x^2} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^5}$ gives the final solution

$$v(x) = x^5 \left(\frac{5}{2x^2} + c_1 \right)$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y^5} = x^5 \left(\frac{5}{2x^2} + c_1 \right)$$

Figure 2.264: Slope field $y'x + y - y^6x^3 = 0$ Summary of solutions found

$$\frac{1}{y^5} = c_1 x^5 + \frac{5}{2}x^3$$

Solved using first_order_ode_exact

Time used: 0.181 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x) dy &= (x^3 y^6 - y) dx \\ (-x^3 y^6 + y) dx + (x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x^3 y^6 + y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (-x^3 y^6 + y) \\ &= -6x^3 y^5 + 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((-6x^3 y^5 + 1) - (1)) \\ &= -6x^2 y^5 \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{-x^3y^6 + y} ((1) - (-6x^3y^5 + 1)) \\ &= -\frac{6x^3y^4}{x^3y^5 - 1} \end{aligned}$$

Since B depends on x , it can not be used to obtain an integrating factor. We will now try a third method to find an integrating factor. Let

$$R = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN}$$

R is now checked to see if it is a function of only $t = xy$. Therefore

$$\begin{aligned} R &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{xM - yN} \\ &= \frac{(1) - (-6x^3y^5 + 1)}{x(-x^3y^6 + y) - y(x)} \\ &= -\frac{6}{yx} \end{aligned}$$

Replacing all powers of terms xy by t gives

$$R = -\frac{6}{t}$$

Since R depends on t only, then it can be used to find an integrating factor. Let the integrating factor be μ then

$$\begin{aligned} \mu &= e^{\int R dt} \\ &= e^{\int (-\frac{6}{t}) dt} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-6 \ln(t)} \\ &= \frac{1}{t^6} \end{aligned}$$

Now t is replaced back with xy giving

$$\mu = \frac{1}{y^6 x^6}$$

Multiplying M and N by this integrating factor gives new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{y^6 x^6} (-x^3 y^6 + y) \\ &= \frac{-x^3 y^5 + 1}{y^5 x^6}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{y^6 x^6} (x) \\ &= \frac{1}{x^5 y^6}\end{aligned}$$

A modified ODE is now obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-x^3 y^5 + 1}{y^5 x^6} \right) + \left(\frac{1}{x^5 y^6} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-x^3 y^5 + 1}{y^5 x^6} dx \\ \phi &= \frac{5x^3 y^5 - 2}{10x^5 y^5} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= \frac{5}{2y x^2} - \frac{5x^3 y^5 - 2}{2x^5 y^6} + f'(y) \\ &= \frac{1}{x^5 y^6} + f'(y)\end{aligned} \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{1}{x^5 y^6}$. Therefore equation (4) becomes

$$\frac{1}{x^5 y^6} = \frac{1}{x^5 y^6} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{5x^3 y^5 - 2}{10x^5 y^5} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{5x^3 y^5 - 2}{10x^5 y^5}$$

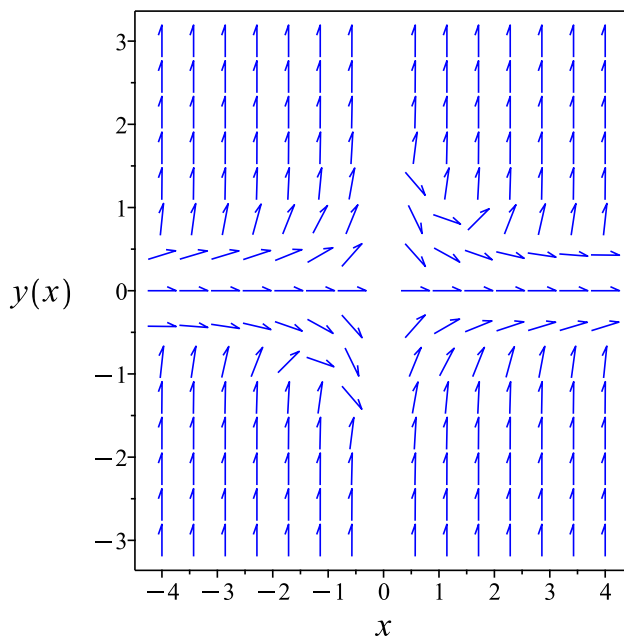


Figure 2.265: Slope field $y'x + y - y^6x^3 = 0$

Summary of solutions found

$$\frac{5x^3 y^5 - 2}{10x^5 y^5} = c_1$$

Solved using first_order_ode_isobaric

Time used: 0.565 (sec)

Solve

$$y'x + y - y^6x^3 = 0$$

Solving for y' gives

$$y' = \frac{y(x^3y^5 - 1)}{x} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = \frac{y(x^3y^5 - 1)}{x} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = -\frac{3}{5}$$

Since the ode is isobaric of order $m = -\frac{3}{5}$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= \frac{u}{x^{3/5}} \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$-\frac{3u(x)}{5x^{8/5}} + \frac{u'(x)}{x^{3/5}} = \frac{u(x)(u(x)^5 - 1)}{x^{8/5}}$$

The ode

$$u'(x) = \frac{u(x)(5u(x)^5 - 2)}{5x} \quad (2.104)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= \frac{u(x)(5u(x)^5 - 2)}{5x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{5x} \\ g(u) &= u(5u^5 - 2) \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(u)} du = \int f(x) dx$$

$$\int \frac{1}{u(5u^5 - 2)} du = \int \frac{1}{5x} dx$$

$$\frac{\ln(5u(x)^5 - 2)}{10} - \frac{\ln(u(x))}{2} = \ln(x^{1/5}) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(5u^5 - 2) = 0$$

for $u(x)$ gives

$$u(x) = 0$$

$$u(x) = \frac{2^{1/5}5^{4/5}}{5}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(5u(x)^5 - 2)}{10} - \frac{\ln(u(x))}{2} = \ln(x^{1/5}) + c_1$$

$$u(x) = 0$$

$$u(x) = \frac{2^{1/5}5^{4/5}}{5}$$

Converting $\frac{\ln(5u(x)^5 - 2)}{10} - \frac{\ln(u(x))}{2} = \ln(x^{1/5}) + c_1$ back to y gives

$$\frac{\ln(5x^3y^5 - 2)}{10} - \frac{\ln(yx^{3/5})}{2} = \ln(x^{1/5}) + c_1$$

Converting $u(x) = 0$ back to y gives

$$yx^{3/5} = 0$$

Converting $u(x) = \frac{2^{1/5}5^{4/5}}{5}$ back to y gives

$$yx^{3/5} = \frac{2^{1/5}5^{4/5}}{5}$$

Solving for y gives

$$\frac{\ln(5x^3y^5 - 2)}{10} - \frac{\ln(yx^{3/5})}{2} = \frac{\ln(x)}{5} + c_1$$

$$y = 0$$

$$y = \frac{2^{1/5}5^{4/5}}{5x^{3/5}}$$

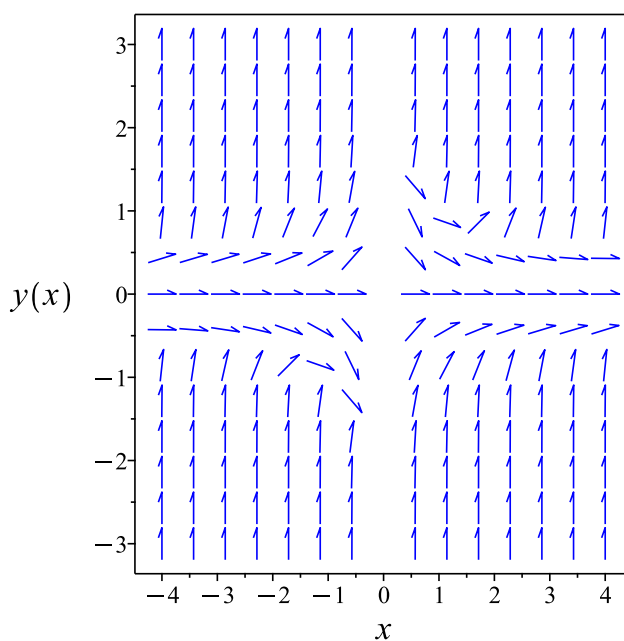


Figure 2.266: Slope field $y'x + y - y^6x^3 = 0$

Summary of solutions found

$$\frac{\ln(5x^3y^5 - 2)}{10} - \frac{\ln(yx^{3/5})}{2} = \frac{\ln(x)}{5} + c_1$$

$$y = 0$$

$$y = \frac{2^{1/5}5^{4/5}}{5x^{3/5}}$$

Solved using first_order_ode_homog_type_G

Time used: 0.079 (sec)

Solve

$$y'x + y - y^6x^3 = 0$$

Multiplying the right side of the ode, which is $\frac{y(x^3y^5-1)}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y(x^3y^5-1)}{x} \\ &= x^3y^5 - 1 \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= 3x^3y^5 \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= 5x^3y^5 \\ \alpha &= \frac{f_x}{f_y} \\ &= \frac{3}{5} \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^{3/5}} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = z^5 - 1$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{yx^{3/5}} \frac{1}{z \left(\frac{2}{5} - z^5 \right)} dz = 0$$

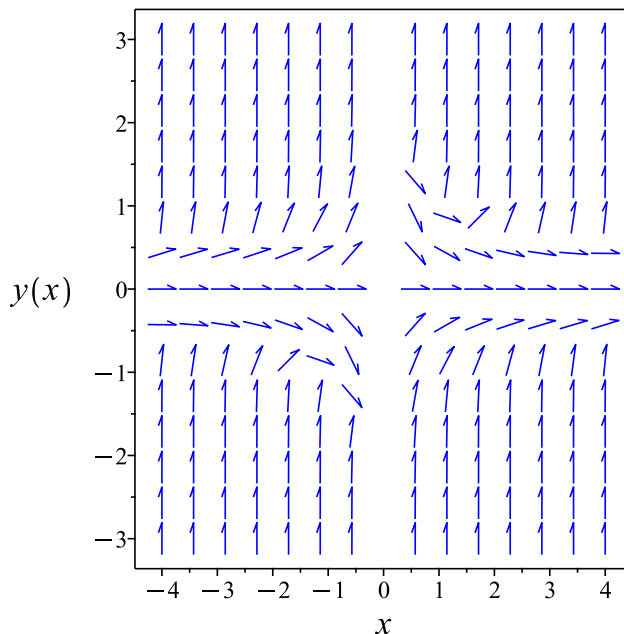


Figure 2.267: Slope field $y'x + y - y^6x^3 = 0$

Summary of solutions found

$$\ln(x) - c_1 + \int^{yx^{3/5}} \frac{1}{z \left(\frac{2}{5} - z^5 \right)} dz = 0$$

Solved using first_order_ode_LIE

Time used: 1.670 (sec)

Solve

$$y'x + y - y^6x^3 = 0$$

Writing the ode as

$$y' = \frac{y(x^3y^5 - 1)}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1E)$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2E)$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} b_2 + \frac{y(x^3y^5 - 1)(b_3 - a_2)}{x} - \frac{y^2(x^3y^5 - 1)^2 a_3}{x^2} \\ - \left(3xy^6 - \frac{y(x^3y^5 - 1)}{x^2} \right) (xa_2 + ya_3 + a_1) - \left(\frac{x^3y^5 - 1}{x} + 5x^2y^5 \right) (xb_2 + yb_3 + b_1) = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\begin{aligned} & \frac{-x^6y^{12}a_3 + 6x^5y^5b_2 + 3x^4y^6a_2 + 5x^4y^6b_3 + 6x^4y^5b_1 + 2x^3y^6a_1 - 2b_2x^2 + 2y^2a_3 - xb_1 + ya_1}{x^2} \\ & = 0 \end{aligned}$$

Setting the numerator to zero gives

$$-x^6y^{12}a_3 - 6x^5y^5b_2 - 3x^4y^6a_2 - 5x^4y^6b_3 - 6x^4y^5b_1 - 2x^3y^6a_1 + 2b_2x^2 - 2y^2a_3 + xb_1 - ya_1 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$-a_3v_1^6v_2^{12} - 3a_2v_1^4v_2^6 - 6b_2v_1^5v_2^5 - 5b_3v_1^4v_2^6 - 2a_1v_1^3v_2^6 - 6b_1v_1^4v_2^5 - 2a_3v_2^2 + 2b_2v_1^2 - a_1v_2 + b_1v_1 = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$-a_3 v_1^6 v_2^{12} - 6b_2 v_1^5 v_2^5 + (-3a_2 - 5b_3) v_1^4 v_2^6 - 6b_1 v_1^4 v_2^5 - 2a_1 v_1^3 v_2^6 + 2b_2 v_1^2 + b_1 v_1 - 2a_3 v_2^2 - a_1 v_2 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} b_1 &= 0 \\ -2a_1 &= 0 \\ -a_1 &= 0 \\ -2a_3 &= 0 \\ -a_3 &= 0 \\ -6b_1 &= 0 \\ -6b_2 &= 0 \\ 2b_2 &= 0 \\ -3a_2 - 5b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= -\frac{5b_3}{3} \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= -\frac{5x}{3} \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= y - \left(\frac{y(x^3 y^5 - 1)}{x} \right) \left(-\frac{5x}{3} \right) \\ &= \frac{5}{3} x^3 y^6 - \frac{2}{3} y \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{5}{3} x^3 y^6 - \frac{2}{3} y} dy\end{aligned}$$

Which results in

$$S = \frac{3 \ln(5x^3 y^5 - 2)}{10} - \frac{3 \ln(y)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y(x^3 y^5 - 1)}{x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{9x^2y^5}{10x^3y^5 - 4} \\ S_y &= \frac{3}{5x^3y^6 - 2y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{3}{2x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{3}{2R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int \frac{3}{2R} dR \\ S(R) &= \frac{3 \ln(R)}{2} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{3 \ln(5x^3y^5 - 2)}{10} - \frac{3 \ln(y)}{2} = \frac{3 \ln(x)}{2} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

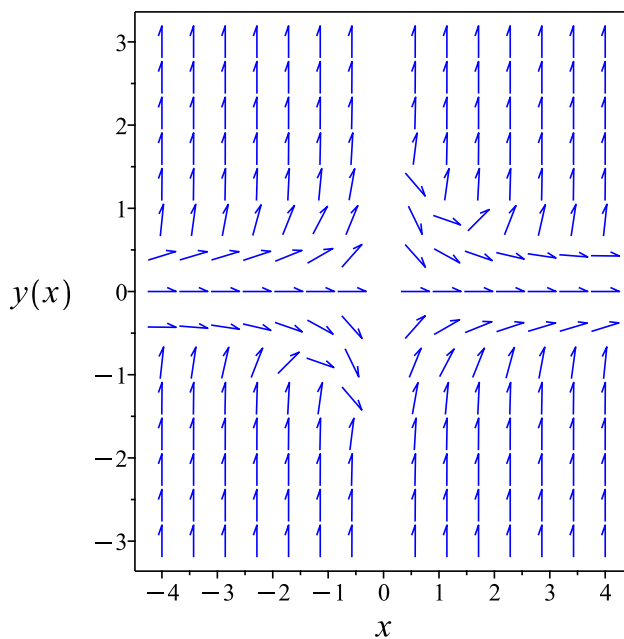
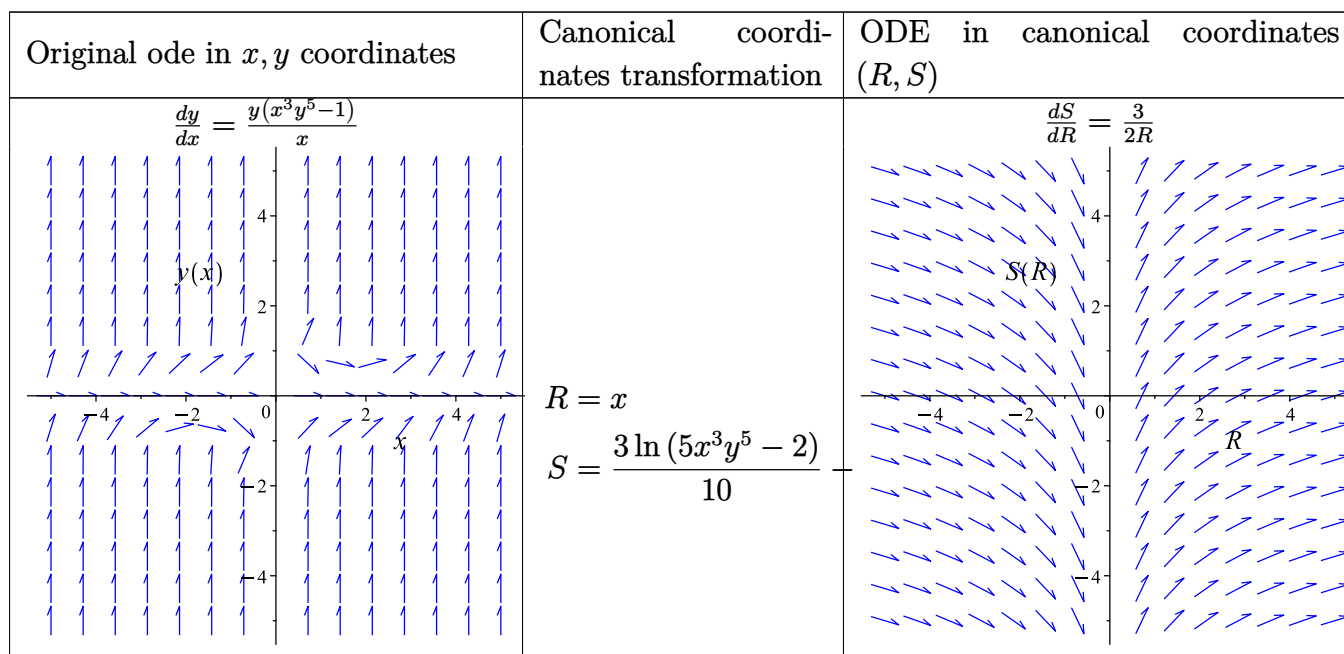


Figure 2.268: Slope field $y'x + y - y^6x^3 = 0$

Summary of solutions found

$$\frac{3 \ln(5x^3y^5 - 2)}{10} - \frac{3 \ln(y)}{2} = \frac{3 \ln(x)}{2} + c_2$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 258

```
ode:=diff(y(x),x)*x+y(x)-x^3*y(x)^6 = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{2^{1/5} \left(x^2 (2c_1 x^2 + 5)^4 \right)^{1/5}}{2c_1 x^3 + 5x}$$

$$y = - \frac{\left(i\sqrt{2} \sqrt{5 - \sqrt{5}} + \sqrt{5} + 1 \right) 2^{1/5} \left(x^2 (2c_1 x^2 + 5)^4 \right)^{1/5}}{8c_1 x^3 + 20x}$$

$$y = - \frac{\left(i\sqrt{2} \sqrt{5 - \sqrt{5}} - \sqrt{5} - 1 \right) 2^{1/5} \left(x^2 (2c_1 x^2 + 5)^4 \right)^{1/5}}{8c_1 x^3 + 20x}$$

$$y = - \frac{\left(i\sqrt{2} \sqrt{5 + \sqrt{5}} - \sqrt{5} + 1 \right) 2^{1/5} \left(x^2 (2c_1 x^2 + 5)^4 \right)^{1/5}}{8c_1 x^3 + 20x}$$

$$y = \frac{\left(i\sqrt{2} \sqrt{5 + \sqrt{5}} + \sqrt{5} - 1 \right) 2^{1/5} \left(x^2 (2c_1 x^2 + 5)^4 \right)^{1/5}}{8c_1 x^3 + 20x}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$x \left(\frac{d}{dx} y(x) \right) + y(x) - x^3 y(x)^6 = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{-y(x) + x^3 y(x)^6}{x}$$

✓ **Mathematica.** Time used: 0.75 (sec). Leaf size: 141

```
ode=x*D[y[x],x]+(y[x]-x^3*y[x]^6)==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt[5]{-2}}{\sqrt[5]{x^3(5+2c_1x^2)}}$$

$$y(x) \rightarrow \frac{1}{\sqrt[5]{\frac{5x^3}{2} + c_1x^5}}$$

$$y(x) \rightarrow \frac{(-1)^{2/5}}{\sqrt[5]{\frac{5x^3}{2} + c_1x^5}}$$

$$y(x) \rightarrow -\frac{(-1)^{3/5}}{\sqrt[5]{\frac{5x^3}{2} + c_1x^5}}$$

$$y(x) \rightarrow \frac{(-1)^{4/5}}{\sqrt[5]{\frac{5x^3}{2} + c_1x^5}}$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 8.792 (sec). Leaf size: 228

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**3*y(x)**6 + x*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = \sqrt[5]{2} \sqrt[5]{\frac{1}{x^3(C_1x^2 + 5)}}, y(x) = \frac{\sqrt[5]{\frac{1}{x^3(C_1x^2 + 5)}} \left(-\sqrt[5]{2} + \sqrt[5]{2}\sqrt{5} - 2^{\frac{7}{10}}i\sqrt{\sqrt{5} + 5} \right)}{4}, y(x) = \frac{\sqrt[5]{\frac{1}{x^3(C_1x^2 + 5)}}}{4} \right]$$

2.4.9 Problem 19 (j)

Local contents

Solved using first_order_ode_linear	1164
Solved using first_order_ode_exact	1166
✓ Maple	1170
✓ Mathematica	1171
✓ Sympy	1171

Internal problem ID [7925]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (j)

Date solved : Thursday, November 27, 2025 at 06:32:32 AM

CAS classification : [_linear]

Solved using first_order_ode_linear

Time used: 0.159 (sec)

Solve

$$r' + 2r \cos(\theta) + \sin(2\theta) = 0$$

In canonical form a linear first order is

$$r' + q(\theta)r = p(\theta)$$

Comparing the above to the given ode shows that

$$q(\theta) = 2 \cos(\theta)$$

$$p(\theta) = -\sin(2\theta)$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\theta} \\ &= e^{\int 2 \cos(\theta) d\theta} \\ &= e^{2 \sin(\theta)} \end{aligned}$$

The ode becomes

$$\frac{d}{d\theta}(\mu r) = \mu p$$

$$\frac{d}{d\theta}(\mu r) = (\mu) (-\sin(2\theta))$$

$$\frac{d}{d\theta}(r e^{2\sin(\theta)}) = (e^{2\sin(\theta)}) (-\sin(2\theta))$$

$$d(r e^{2\sin(\theta)}) = (-\sin(2\theta) e^{2\sin(\theta)}) d\theta$$

Integrating gives

$$\begin{aligned} r e^{2\sin(\theta)} &= \int -\sin(2\theta) e^{2\sin(\theta)} d\theta \\ &= e^{2\sin(\theta)} \left(\frac{1}{2} - \sin(\theta) \right) + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $e^{2\sin(\theta)}$ gives the final solution

$$r = \frac{1}{2} - \sin(\theta) + e^{-2\sin(\theta)} c_1$$

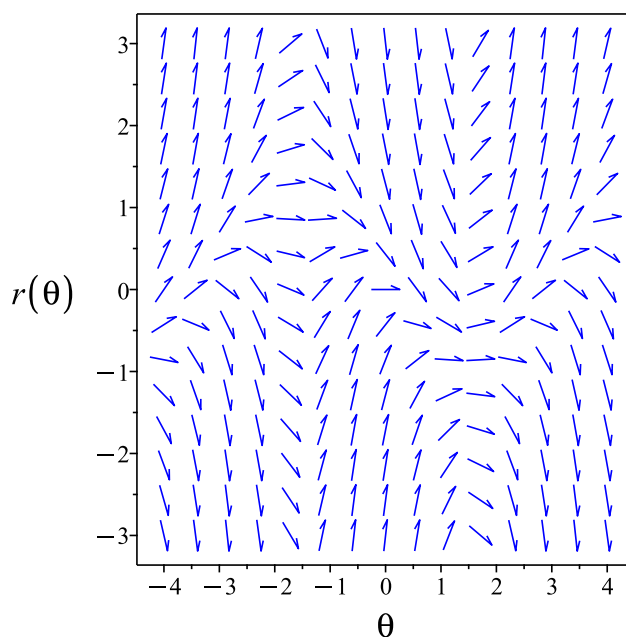


Figure 2.269: Slope field $r' + 2r \cos(\theta) + \sin(2\theta) = 0$

Summary of solutions found

$$r = \frac{1}{2} - \sin(\theta) + e^{-2\sin(\theta)} c_1$$

Solved using first_order_ode_exact

Time used: 0.156 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(\theta, r) d\theta + N(\theta, r) dr = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} dr &= (-2r \cos(\theta) - \sin(2\theta)) d\theta \\ (2r \cos(\theta) + \sin(2\theta)) d\theta + dr &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(\theta, r) &= 2r \cos(\theta) + \sin(2\theta) \\ N(\theta, r) &= 1 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial r} = \frac{\partial N}{\partial \theta}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial r} &= \frac{\partial}{\partial r}(2r \cos(\theta) + \sin(2\theta)) \\ &= 2 \cos(\theta)\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial \theta} &= \frac{\partial}{\partial \theta}(1) \\ &= 0\end{aligned}$$

Since $\frac{\partial M}{\partial r} \neq \frac{\partial N}{\partial \theta}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial r} - \frac{\partial N}{\partial \theta} \right) \\ &= 1((2 \cos(\theta)) - (0)) \\ &= 2 \cos(\theta)\end{aligned}$$

Since A does not depend on r , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, d\theta} \\ &= e^{\int 2 \cos(\theta) \, d\theta}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{2 \sin(\theta)} \\ &= e^{2 \sin(\theta)}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\bar{M}$ and $\bar{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\bar{M} &= \mu M \\ &= e^{2 \sin(\theta)}(2r \cos(\theta) + \sin(2\theta)) \\ &= 2 \cos(\theta)(r + \sin(\theta)) e^{2 \sin(\theta)}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= e^{2 \sin(\theta)}(1) \\ &= e^{2 \sin(\theta)}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dr}{d\theta} &= 0 \\ (2 \cos(\theta) (r + \sin(\theta)) e^{2 \sin(\theta)}) + (e^{2 \sin(\theta)}) \frac{dr}{d\theta} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(\theta, r)$

$$\frac{\partial \phi}{\partial \theta} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial r} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. r gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial r} dr &= \int \overline{N} dr \\ \int \frac{\partial \phi}{\partial r} dr &= \int e^{2 \sin(\theta)} dr \\ \phi &= r e^{2 \sin(\theta)} + f(\theta)\end{aligned} \quad (3)$$

Where $f(\theta)$ is used for the constant of integration since ϕ is a function of both θ and r . Taking derivative of equation (3) w.r.t θ gives

$$\frac{\partial \phi}{\partial \theta} = 2 \cos(\theta) e^{2 \sin(\theta)} r + f'(\theta) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial \theta} = 2 \cos(\theta) (r + \sin(\theta)) e^{2 \sin(\theta)}$. Therefore equation (4) becomes

$$2 \cos(\theta) (r + \sin(\theta)) e^{2 \sin(\theta)} = 2 \cos(\theta) e^{2 \sin(\theta)} r + f'(\theta) \quad (5)$$

Solving equation (5) for $f'(\theta)$ gives

$$f'(\theta) = 2 \cos(\theta) e^{2 \sin(\theta)} \sin(\theta)$$

Integrating the above w.r.t θ gives

$$\begin{aligned}\int f'(\theta) d\theta &= \int (2 \cos(\theta) e^{2 \sin(\theta)} \sin(\theta)) d\theta \\ f(\theta) &= \sin(\theta) e^{2 \sin(\theta)} - \frac{e^{2 \sin(\theta)}}{2} + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(\theta)$ into equation (3) gives ϕ

$$\phi = r e^{2 \sin(\theta)} + \sin(\theta) e^{2 \sin(\theta)} - \frac{e^{2 \sin(\theta)}}{2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = r e^{2 \sin(\theta)} + \sin(\theta) e^{2 \sin(\theta)} - \frac{e^{2 \sin(\theta)}}{2}$$

Simplifying the above gives

$$e^{2 \sin(\theta)} \left(-\frac{1}{2} + r + \sin(\theta) \right) = c_1$$

Solving for r gives

$$r = -\frac{(2 \sin(\theta) e^{2 \sin(\theta)} - e^{2 \sin(\theta)} - 2c_1) e^{-2 \sin(\theta)}}{2}$$

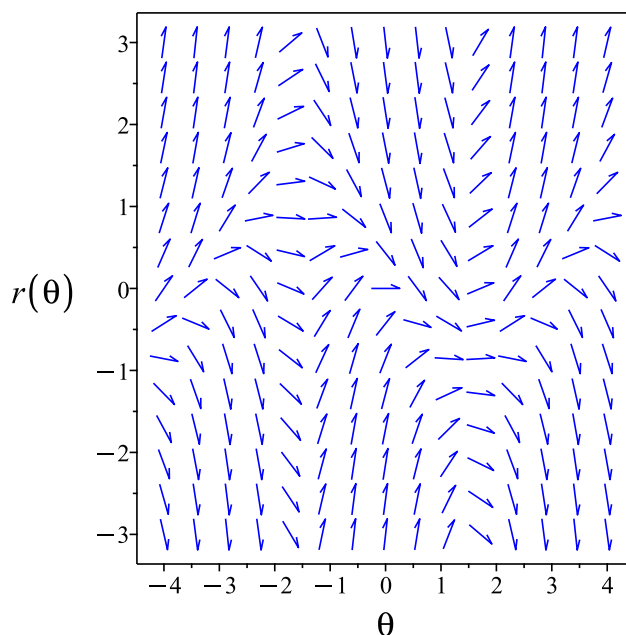


Figure 2.270: Slope field $r' + 2r \cos(\theta) + \sin(2\theta) = 0$

Summary of solutions found

$$r = -\frac{(2 \sin(\theta) e^{2 \sin(\theta)} - e^{2 \sin(\theta)} - 2c_1) e^{-2 \sin(\theta)}}{2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 17

```
ode:=diff(r(theta),theta)+2*r(theta)*cos(theta)+sin(2*theta) = 0;
dsolve(ode,r(theta), singsol=all);
```

$$r = -\sin(\theta) + \frac{1}{2} + e^{-2\sin(\theta)} c_1$$

Maple trace

Methods for first order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\frac{d}{d\theta} r(\theta) + 2r(\theta) \cos(\theta) + \sin(2\theta) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{d\theta} r(\theta)$$

- Solve for the highest derivative

$$\frac{d}{d\theta} r(\theta) = -2r(\theta) \cos(\theta) - \sin(2\theta)$$

- Group terms with $r(\theta)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{d\theta} r(\theta) + 2r(\theta) \cos(\theta) = -\sin(2\theta)$$

- The ODE is linear; multiply by an integrating factor $\mu(\theta)$

$$\mu(\theta) \left(\frac{d}{d\theta} r(\theta) + 2r(\theta) \cos(\theta) \right) = -\mu(\theta) \sin(2\theta)$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{d\theta} (r(\theta) \mu(\theta))$

$$\mu(\theta) \left(\frac{d}{d\theta} r(\theta) + 2r(\theta) \cos(\theta) \right) = \left(\frac{d}{d\theta} r(\theta) \right) \mu(\theta) + r(\theta) \left(\frac{d}{d\theta} \mu(\theta) \right)$$

- Isolate $\frac{d}{d\theta} \mu(\theta)$

$$\frac{d}{d\theta} \mu(\theta) = 2\mu(\theta) \cos(\theta)$$

- Solve to find the integrating factor

$$\mu(\theta) = e^{2\sin(\theta)}$$

- Integrate both sides with respect to θ

$$\int \left(\frac{d}{d\theta} (r(\theta) \mu(\theta)) \right) d\theta = \int -\mu(\theta) \sin(2\theta) d\theta + C_1$$

- Evaluate the integral on the lhs

$$r(\theta) \mu(\theta) = \int -\mu(\theta) \sin(2\theta) d\theta + C_1$$

- Solve for $r(\theta)$

$$r(\theta) = \frac{\int -\mu(\theta) \sin(2\theta) d\theta + C_1}{\mu(\theta)}$$

- Substitute $\mu(\theta) = e^{2\sin(\theta)}$

$$r(\theta) = \frac{\int -e^{2\sin(\theta)} \sin(2\theta) d\theta + C_1}{e^{2\sin(\theta)}}$$

- Evaluate the integrals on the rhs

$$r(\theta) = \frac{-e^{2\sin(\theta)} \sin(\theta) + \frac{e^{2\sin(\theta)}}{2} + C_1}{e^{2\sin(\theta)}}$$

- Simplify

$$r(\theta) = e^{-2\sin(\theta)} C_1 - \sin(\theta) + \frac{1}{2}$$

✓ **Mathematica.** Time used: 0.046 (sec). Leaf size: 54

```
ode=D[r[t],t]+(2*r[t]*Cos[t]+Sin[2*t])==0;
ic={};
DSolve[{ode,ic},r[t],t,IncludeSingularSolutions->True]
```

$$r(t) \rightarrow \exp\left(\int_1^t -2\cos(K[1])dK[1]\right) \left(\int_1^t -\exp\left(-\int_1^{K[2]} -2\cos(K[1])dK[1]\right) \sin(2K[2])dK[2] + c_1\right)$$

✓ **Sympy.** Time used: 0.312 (sec). Leaf size: 17

```
from sympy import *
theta = symbols("theta")
r = Function("r")
ode = Eq(2*r(theta)*cos(theta) + sin(2*theta) + Derivative(r(theta), theta), 0)
ics = {}
dsolve(ode, func=r(theta), ics=ics)
```

$$r(\theta) = C_1 e^{-2\sin(\theta)} - \sin(\theta) + \frac{1}{2}$$

2.4.10 Problem 19 (k)

Local contents

Solved using first_order_ode_exact	1172
✓ Maple	1176
✓ Mathematica	1177
✓ Sympy	1177

Internal problem ID [7926]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (k)

Date solved : Thursday, November 27, 2025 at 06:32:39 AM

CAS classification : [_rational, [_1st_order, ['_with_symmetry_[F(x)*G(y),0]']]

Solved using first_order_ode_exact

Time used: 0.164 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (4x y^2 - 2) dy &= (-y(y^2 + 1)) dx \\ (y(y^2 + 1)) dx + (4x y^2 - 2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y(y^2 + 1) \\ N(x, y) &= 4x y^2 - 2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (y(y^2 + 1)) \\ &= 3y^2 + 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (4x y^2 - 2) \\ &= 4y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{4x y^2 - 2} ((3y^2 + 1) - (4y^2)) \\ &= \frac{-y^2 + 1}{4x y^2 - 2} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y(y^2 + 1)} ((4y^2) - (3y^2 + 1)) \\ &= \frac{y^2 - 1}{y(y^2 + 1)} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int \frac{y^2 - 1}{y(y^2 + 1)} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{\ln(y^2 + 1) - \ln(y)} \\ &= \frac{y^2 + 1}{y} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{y^2 + 1}{y} (y(y^2 + 1)) \\ &= (y^2 + 1)^2 \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{y^2 + 1}{y} (4xy^2 - 2) \\ &= \frac{(4xy^2 - 2)(y^2 + 1)}{y} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left((y^2 + 1)^2 \right) + \left(\frac{(4xy^2 - 2)(y^2 + 1)}{y} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int (y^2 + 1)^2 dx \\ \phi &= (y^2 + 1)^2 x + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = 4(y^2 + 1)xy + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{(4xy^2 - 2)(y^2 + 1)}{y}$. Therefore equation (4) becomes

$$\frac{(4xy^2 - 2)(y^2 + 1)}{y} = 4(y^2 + 1)xy + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{2(y^2 + 1)}{y}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int \left(\frac{-2y^2 - 2}{y} \right) dy \\ f(y) &= -y^2 - 2 \ln(y) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = (y^2 + 1)^2 x - y^2 - 2 \ln(y) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = (y^2 + 1)^2 x - y^2 - 2 \ln(y)$$

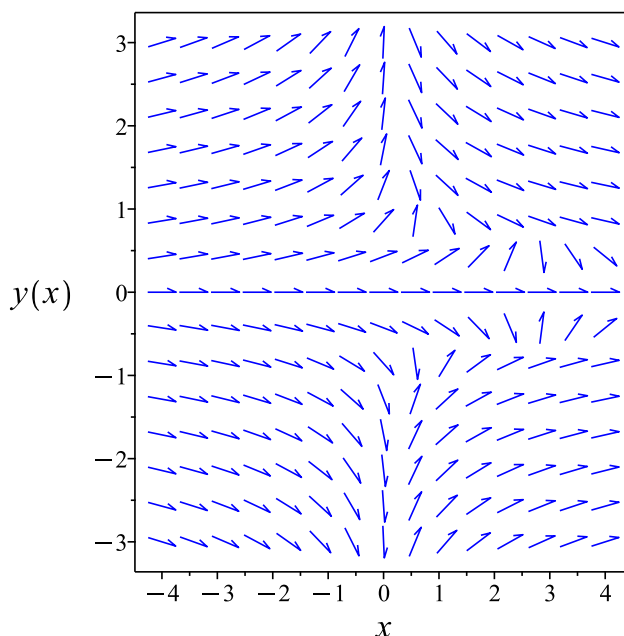


Figure 2.271: Slope field $(1 + y^2)y = 2(1 - 2y^2x)y'$

Summary of solutions found

$$(1 + y^2)^2 x - y^2 - 2 \ln(y) = c_1$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 32

```
ode:=y(x)*(1+y(x)^2) = 2*(1-2*x*y(x)^2)*diff(y(x),x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{\text{RootOf}(-x e^{4-Z} - 2x e^{2-Z} + e^{2-Z} + c_1 + 2_Z - x)}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
```

```
trying separable
trying inverse linear
<- 1st order linear successful
<- inverse linear successful
```

Maple step by step

Let's solve

$$y(x) (1 + y(x)^2) = 2(1 - 2xy(x)^2) \left(\frac{d}{dx}y(x)\right)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x)(1+y(x)^2)}{2(1-2xy(x)^2)}$$

✓ **Mathematica.** Time used: 0.122 (sec). Leaf size: 36

```
ode=y[x]*(1+y[x]^2)==2*(1-2*x*y[x]^2)*D[y[x],x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[x = \frac{y(x)^2 + 2 \log(y(x))}{(y(x)^2 + 1)^2} + \frac{c_1}{(y(x)^2 + 1)^2}, y(x) \right]$$

✓ **Sympy.** Time used: 0.591 (sec). Leaf size: 29

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((4*x*y(x)**2 - 2)*Derivative(y(x), x) + (y(x)**2 + 1)*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$C_1 + x(y^4(x) + 2y^2(x) + 1) - y^2(x) - 2 \log(y(x)) = 0$$

2.4.11 Problem 19 (L)

Local contents

Solved using first_order_ode_separable	1178
Solved using first_order_ode_bernoulli	1180
Solved using first_order_ode_exact	1183
✓ Maple	1187
✓ Mathematica	1188
✓ Sympy	1189

Internal problem ID [7927]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (L)

Date solved : Thursday, November 27, 2025 at 06:32:46 AM

CAS classification : [_separable]

Solved using first_order_ode_separable

Time used: 0.081 (sec)

Solve

$$yy' - y^2x + x = 0$$

The ode

$$y' = \frac{x(-1+y)(1+y)}{y} \quad (2.105)$$

is separable as it can be written as

$$\begin{aligned} y' &= \frac{x(-1+y)(1+y)}{y} \\ &= f(x)g(y) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= x \\ g(y) &= \frac{(-1+y)(y+1)}{y} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int \frac{y}{(-1+y)(y+1)} dy = \int x dx$$

$$\frac{\ln(y^2 - 1)}{2} = \frac{x^2}{2} + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(y)$ is zero, since we had to divide by this above. Solving $g(y) = 0$ or

$$\frac{(-1+y)(y+1)}{y} = 0$$

for y gives

$$y = -1$$

$$y = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(y^2 - 1)}{2} = \frac{x^2}{2} + c_1$$

$$y = -1$$

$$y = 1$$

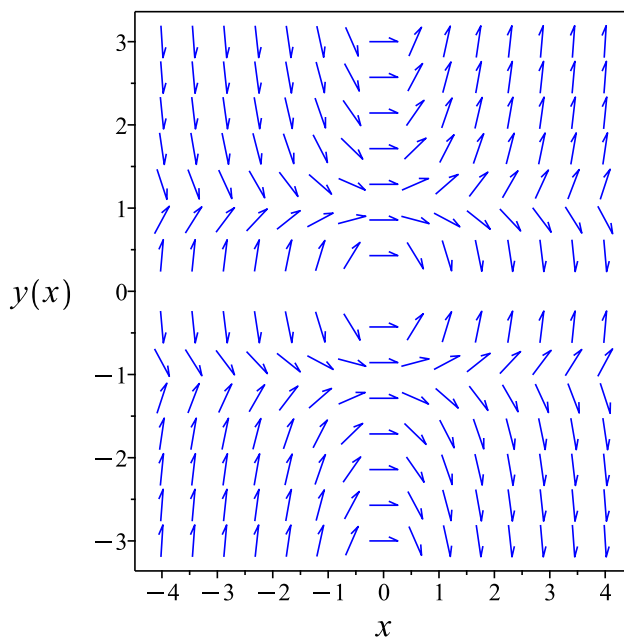
Solving for y gives

$$y = -1$$

$$y = 1$$

$$y = \sqrt{1 + e^{x^2+2c_1}}$$

$$y = -\sqrt{1 + e^{x^2+2c_1}}$$

Figure 2.272: Slope field $yy' - y^2x + x = 0$ Summary of solutions found

$$y = -1$$

$$y = 1$$

$$y = \sqrt{1 + e^{x^2+2c_1}}$$

$$y = -\sqrt{1 + e^{x^2+2c_1}}$$

Solved using first_order_ode_bernoulli

Time used: 0.144 (sec)

Solve

$$yy' - y^2x + x = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{x(y^2 - 1)}{y} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = (x)y + (-x)\frac{1}{y} \tag{1}$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned}f_0 &= x \\f_1 &= -x\end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned}f_0(x) &= x \\f_1(x) &= -x \\n &= -1\end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y}$ gives

$$y'y = x y^2 - x \quad (4)$$

Let

$$\begin{aligned}v &= y^{1-n} \\&= y^2\end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 2yy' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned}\frac{v'(x)}{2} &= v(x)x - x \\v' &= 2xv - 2x\end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -2x \\p(x) &= -2x\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\&= e^{\int -2x \, dx} \\&= e^{-x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu)(-2x) \\ \frac{d}{dx}(v e^{-x^2}) &= (e^{-x^2})(-2x) \\ d(v e^{-x^2}) &= (-2x e^{-x^2}) \, dx\end{aligned}$$

Integrating gives

$$\begin{aligned}v e^{-x^2} &= \int -2x e^{-x^2} \, dx \\&= e^{-x^2} + c_1\end{aligned}$$

Dividing throughout by the integrating factor e^{-x^2} gives the final solution

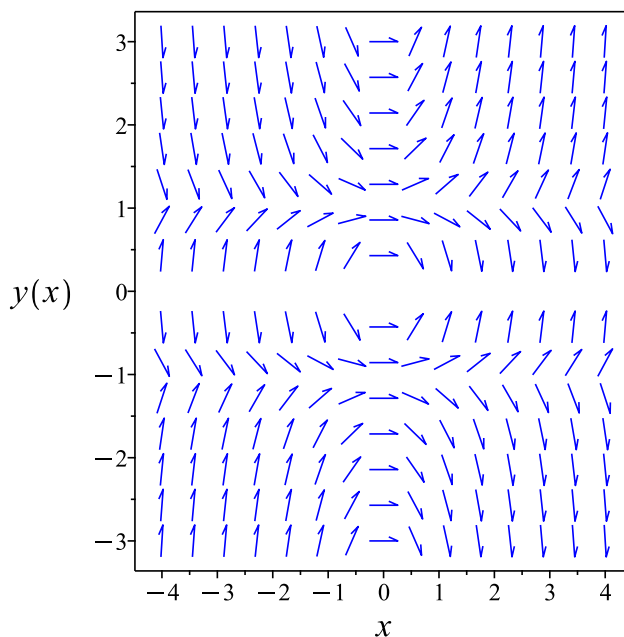
$$v(x) = c_1 e^{x^2} + 1$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^2 = c_1 e^{x^2} + 1$$

Solving for y gives

$$\begin{aligned}y &= \sqrt{c_1 e^{x^2} + 1} \\y &= -\sqrt{c_1 e^{x^2} + 1}\end{aligned}$$

Figure 2.273: Slope field $yy' - y^2x + x = 0$ Summary of solutions found

$$y = \sqrt{c_1 e^{x^2} + 1}$$

$$y = -\sqrt{c_1 e^{x^2} + 1}$$

Solved using first_order_ode_exact

Time used: 0.108 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (y) dy &= (x y^2 - x) dx \\ (-x y^2 + x) dx + (y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -x y^2 + x \\ N(x, y) &= y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-x y^2 + x) \\ &= -2yx \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(y) \\ &= 0 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{y} ((-2yx) - (0)) \\ &= -2x \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -2x \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-x^2} \\ &= e^{-x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= e^{-x^2}(-x y^2 + x) \\ &= x(-y^2 + 1) e^{-x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= e^{-x^2}(y) \\ &= e^{-x^2}y\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(x(-y^2 + 1) e^{-x^2}\right) + \left(e^{-x^2}y\right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} \, dx &= \int \overline{M} \, dx \\ \int \frac{\partial \phi}{\partial x} \, dx &= \int x(-y^2 + 1) e^{-x^2} \, dx \\ \phi &= \frac{(y^2 - 1) e^{-x^2}}{2} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = e^{-x^2} y + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = e^{-x^2} y$. Therefore equation (4) becomes

$$e^{-x^2} y = e^{-x^2} y + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{(y^2 - 1) e^{-x^2}}{2} + c_1$$

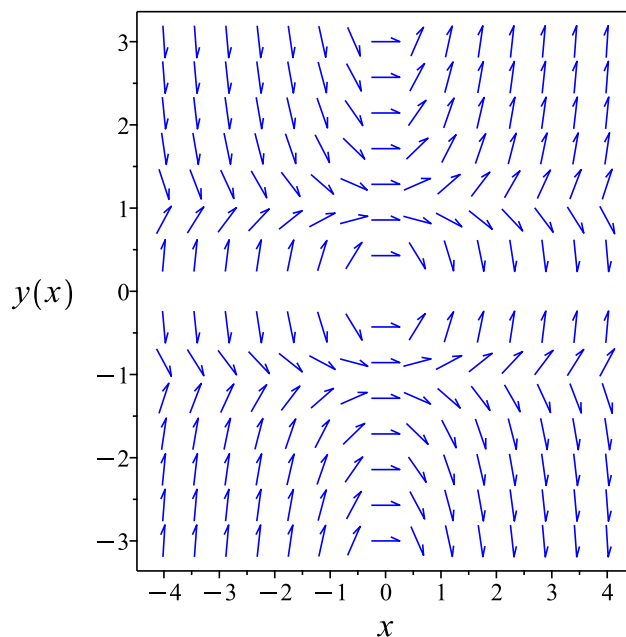
But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{(y^2 - 1) e^{-x^2}}{2}$$

Solving for y gives

$$y = e^{x^2} \sqrt{e^{-x^2} (e^{-x^2} + 2c_1)}$$

$$y = -e^{x^2} \sqrt{e^{-x^2} (e^{-x^2} + 2c_1)}$$

Figure 2.274: Slope field $yy' - y^2x + x = 0$ Summary of solutions found

$$y = e^{x^2} \sqrt{e^{-x^2} (e^{-x^2} + 2c_1)}$$

$$y = -e^{x^2} \sqrt{e^{-x^2} (e^{-x^2} + 2c_1)}$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 29

```
ode:=diff(y(x),x)*y(x)-x*y(x)^2+x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{e^{x^2} c_1 + 1}$$

$$y = -\sqrt{e^{x^2} c_1 + 1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$y(x) \left(\frac{d}{dx} y(x) \right) - xy(x)^2 + x = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{xy(x)^2 - x}{y(x)}$$

- Separate variables

$$\frac{\left(\frac{d}{dx} y(x) \right) y(x)}{(y(x)-1)(y(x)+1)} = x$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx} y(x) \right) y(x)}{(y(x)-1)(y(x)+1)} dx = \int x dx + C1$$

- Evaluate integral

$$\frac{\ln((y(x)-1)(y(x)+1))}{2} = \frac{x^2}{2} + C1$$

- Solve for $y(x)$

$$\left\{ y(x) = \sqrt{1 + e^{x^2+2C1}}, y(x) = -\sqrt{1 + e^{x^2+2C1}} \right\}$$

- Redefine the integration constant(s)

$$\left\{ y(x) = \sqrt{1 + C1 e^{x^2}}, y(x) = -\sqrt{1 + C1 e^{x^2}} \right\}$$

✓ **Mathematica.** Time used: 1.773 (sec). Leaf size: 53

```
ode=y[x]*D[y[x],x]-x*y[x]^2+x==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{1 + e^{x^2+2c_1}}$$

$$y(x) \rightarrow \sqrt{1 + e^{x^2+2c_1}}$$

$$y(x) \rightarrow -1$$

$$y(x) \rightarrow 1$$

✓ Sympy. Time used: 0.348 (sec). Leaf size: 29

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*y(x)**2 + x + y(x)*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\sqrt{C_1 e^{x^2} + 1}, y(x) = \sqrt{C_1 e^{x^2} + 1} \right]$$

2.4.12 Problem 19 (m)

Local contents

Solved using first_order_ode_LIE	1190
✓ Maple	1195
✓ Mathematica	1196
✗ Sympy	1196

Internal problem ID [7928]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (m)

Date solved : Thursday, November 27, 2025 at 06:32:54 AM

CAS classification : ['y=_G(x,y)']

Solved using first_order_ode_LIE

Time used: 0.836 (sec)

Solve

$$(x - x\sqrt{x^2 - y^2})y' - y = 0$$

Writing the ode as

$$y' = -\frac{y}{x(\sqrt{x^2 - y^2} - 1)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Using these anstaz

$$\xi = \frac{Ax + By}{Cy} \quad (1\text{E})$$

$$\eta = 1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{A, B, C\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & \frac{A}{x(\sqrt{x^2-y^2}-1)C} - \frac{y^2\left(\frac{B}{Cy} - \frac{Ax+By}{Cy^2}\right)}{x^2(\sqrt{x^2-y^2}-1)^2} \\ & - \frac{\left(\frac{y}{x^2(\sqrt{x^2-y^2}-1)} + \frac{y}{(\sqrt{x^2-y^2}-1)^2\sqrt{x^2-y^2}}\right)(Ax+By)}{Cy} \\ & + \frac{1}{x(\sqrt{x^2-y^2}-1)} + \frac{y^2}{x(\sqrt{x^2-y^2}-1)^2\sqrt{x^2-y^2}} = 0 \end{aligned} \quad (5E)$$

Putting the above in normal form gives

$$\frac{-Ax^3 - 2Bx^2y + By^3 + Cx^3 + Ax\sqrt{x^2-y^2} + By\sqrt{x^2-y^2} - Cx\sqrt{x^2-y^2}}{x^2(\sqrt{x^2-y^2}-1)^2\sqrt{x^2-y^2}C} = 0$$

Setting the numerator to zero gives

$$-Ax^3 - 2Bx^2y + By^3 + Cx^3 + Ax\sqrt{x^2-y^2} + By\sqrt{x^2-y^2} - Cx\sqrt{x^2-y^2} = 0 \quad (6E)$$

Simplifying the above gives

$$\begin{aligned} & y^2\left(-Ax^3 - By(x^2-y^2) - Bx^2y + Cx(x^2-y^2) + y^2Cx\right. \\ & \left.+ Ax\sqrt{x^2-y^2} + By\sqrt{x^2-y^2} - Cx\sqrt{x^2-y^2}\right) = 0 \end{aligned} \quad (6E)$$

Since the PDE has radicals, simplifying gives

$$y^2\left(-Ax^3 - 2Bx^2y + By^3 + Cx^3 + Ax\sqrt{x^2-y^2} + By\sqrt{x^2-y^2} - Cx\sqrt{x^2-y^2}\right) = 0$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, \sqrt{x^2-y^2}\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, \sqrt{x^2-y^2} = v_3\}$$

The above PDE (6E) now becomes

$$v_2^2(-Av_1^3 - 2Bv_1^2v_2 + Bv_2^3 + Cv_1^3 + Av_1v_3 + Bv_2v_3 - Cv_1v_3) = 0 \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$(-A + C)v_2^2v_1^3 - 2Bv_2^3v_1^2 + (A - C)v_3v_1v_2^2 + Bv_2^5 + Bv_3v_2^3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} B &= 0 \\ -2B &= 0 \\ -A + C &= 0 \\ A - C &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} A &= C \\ B &= 0 \\ C &= C \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= \frac{x}{y} \\ \eta &= 1 \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= 1 - \left(-\frac{y}{x(\sqrt{x^2 - y^2} - 1)} \right) \left(\frac{x}{y} \right) \\ &= \frac{\sqrt{x^2 - y^2}}{\sqrt{x^2 - y^2} - 1} \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{\sqrt{x^2 - y^2}}{\sqrt{x^2 - y^2} - 1}} dy \end{aligned}$$

Which results in

$$S = y - \arctan\left(\frac{y}{\sqrt{x^2 - y^2}}\right)$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{y}{x(\sqrt{x^2 - y^2} - 1)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{y}{x\sqrt{x^2 - y^2}} \\ S_y &= \frac{\sqrt{x^2 - y^2} - 1}{\sqrt{x^2 - y^2}} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

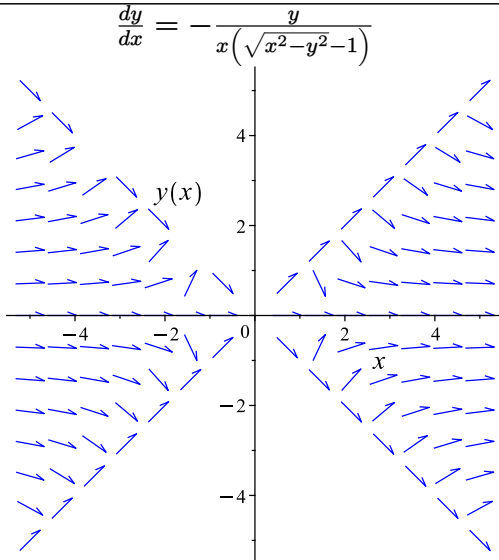
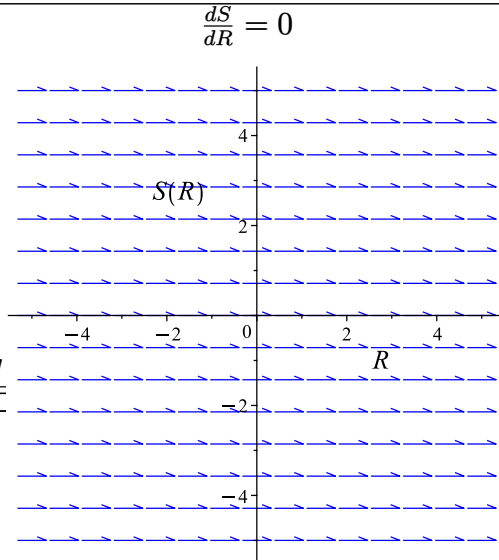
$$\int dS = \int 0 dR + c_2$$

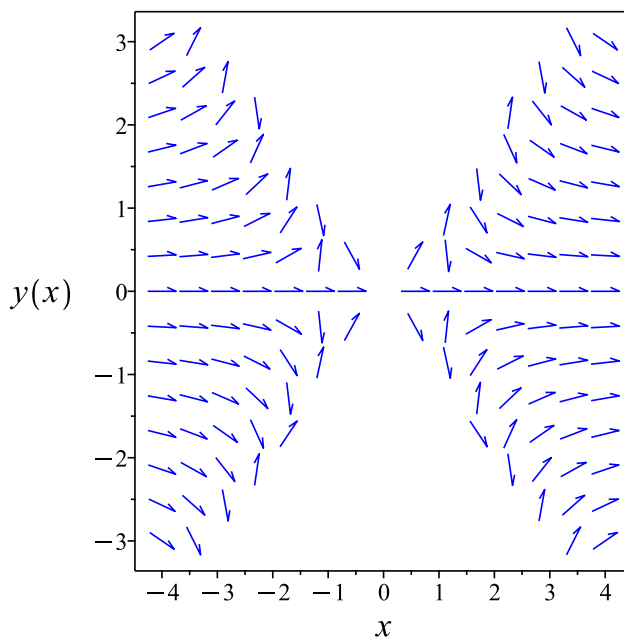
$$S(R) = c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$y - \arctan\left(\frac{y}{\sqrt{x^2 - y^2}}\right) = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = -\frac{y}{x(\sqrt{x^2 - y^2} - 1)}$ 	$R = x$ $S = y - \arctan\left(\frac{y}{\sqrt{x^2 - y^2}}\right)$	$\frac{dS}{dR} = 0$ 

Figure 2.275: Slope field $(x - x\sqrt{x^2 - y^2})y' - y = 0$ Summary of solutions found

$$y - \arctan\left(\frac{y}{\sqrt{x^2 - y^2}}\right) = c_2$$

✓ Maple. Time used: 0.052 (sec). Leaf size: 27

```
ode:=(x-x*(x^2-y(x)^2)^(1/2))*diff(y(x),x)-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y - \arctan\left(\frac{y}{\sqrt{x^2 - y^2}}\right) - c_1 = 0$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying homogeneous types:
differential order: 1; looking for linear symmetries
trying exact
Looking for potential symmetries
trying an equivalence to an Abel ODE
trying 1st order ODE linearizable_by_differentiation
--- Trying Lie symmetry methods, 1st order ---
```

```

-> Computing symmetries using: way = 3
-> Computing symmetries using: way = 5
[0, (x^2-y^2)^(1/2)/((x^2-y^2)^(1/2)-1)]
<- successful computation of symmetries.
1st order, trying the canonical coordinates of the invariance group
<- 1st order, canonical coordinates successful

```

Maple step by step

Let's solve

$$\left(x - x\sqrt{x^2 - y(x)^2}\right) \left(\frac{d}{dx}y(x)\right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{y(x)}{x - x\sqrt{x^2 - y(x)^2}}$$

✓ **Mathematica.** Time used: 0.316 (sec). Leaf size: 29

```

ode=(x-x*Sqrt[x^2-y[x]^2])*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$\text{Solve} \left[\arctan \left(\frac{\sqrt{x^2 - y(x)^2}}{y(x)} \right) + y(x) = c_1, y(x) \right]$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-x*sqrt(x**2 - y(x)**2) + x)*Derivative(y(x), x) - y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)

```

Timed Out

2.4.13 Problem 19 (o)

Local contents

Solved using first_order_ode_bernoulli	1197
Solved using first_order_ode_exact	1200
✓ Maple	1205
✓ Mathematica	1206
✓ Sympy	1206

Internal problem ID [7929]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (o)

Date solved : Thursday, November 27, 2025 at 06:33:04 AM

CAS classification : [_Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.153 (sec)

Solve

$$2x' - \frac{x}{y} + x^3 \cos(y) = 0$$

In canonical form, the ODE is

$$\begin{aligned} x' &= F(y, x) \\ &= -\frac{x(x^2 \cos(y) y - 1)}{2y} \end{aligned}$$

This is a Bernoulli ODE.

$$x' = \left(\frac{1}{2y}\right)x + \left(-\frac{\cos(y)}{2}\right)x^3 \quad (1)$$

The standard Bernoulli ODE has the form

$$x' = f_0(y)x + f_1(y)x^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1}{2y} \\ f_1 &= -\frac{\cos(y)}{2} \end{aligned}$$

The first step is to divide the above equation by x^n which gives

$$\frac{x'}{x^n} = f_0(y)x^{1-n} + f_1(y) \quad (3)$$

The next step is use the substitution $v = x^{1-n}$ in equation (3) which generates a new ODE in $v(y)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $x(y)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(y) &= \frac{1}{2y} \\ f_1(y) &= -\frac{\cos(y)}{2} \\ n &= 3 \end{aligned}$$

Dividing both sides of ODE (1) by $x^n = x^3$ gives

$$x' \frac{1}{x^3} = \frac{1}{2y x^2} - \frac{\cos(y)}{2} \quad (4)$$

Let

$$\begin{aligned} v &= x^{1-n} \\ &= \frac{1}{x^2} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t y gives

$$v' = -\frac{2}{x^3} x' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -\frac{v'(y)}{2} &= \frac{v(y)}{2y} - \frac{\cos(y)}{2} \\ v' &= -\frac{v}{y} + \cos(y) \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(y)$ which is now solved.

In canonical form a linear first order is

$$v'(y) + q(y)v(y) = p(y)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(y) &= \frac{1}{y} \\ p(y) &= \cos(y) \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dy} \\ &= e^{\int \frac{1}{y} \, dy} \\ &= y\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dy}(\mu v) &= \mu p \\ \frac{d}{dy}(\mu v) &= (\mu) (\cos(y)) \\ \frac{d}{dy}(vy) &= (y) (\cos(y)) \\ d(vy) &= (\cos(y) y) \, dy\end{aligned}$$

Integrating gives

$$\begin{aligned}vy &= \int \cos(y) y \, dy \\ &= \cos(y) + \sin(y) y + c_1\end{aligned}$$

Dividing throughout by the integrating factor y gives the final solution

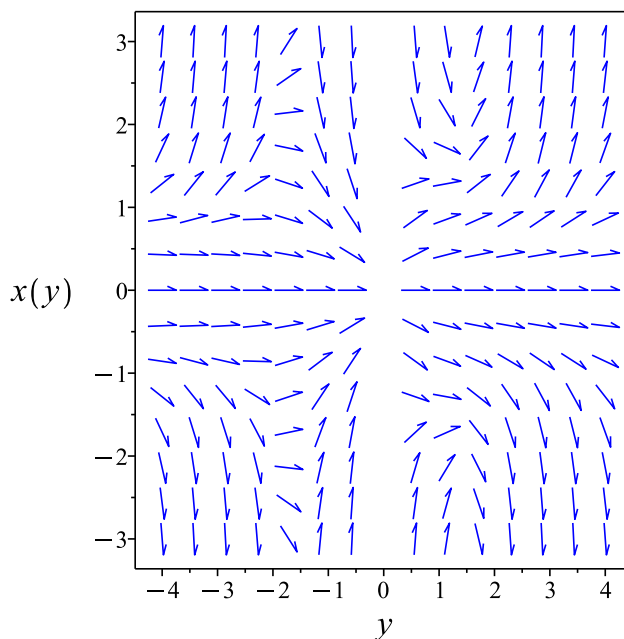
$$v(y) = \frac{\cos(y) + \sin(y) y + c_1}{y}$$

The substitution $v = x^{1-n}$ is now used to convert the above solution back to x which results in

$$\frac{1}{x^2} = \frac{\cos(y) + \sin(y) y + c_1}{y}$$

Solving for x gives

$$\begin{aligned}x &= \frac{\sqrt{(\cos(y) + \sin(y) y + c_1) y}}{\cos(y) + \sin(y) y + c_1} \\ x &= -\frac{\sqrt{(\cos(y) + \sin(y) y + c_1) y}}{\cos(y) + \sin(y) y + c_1}\end{aligned}$$

Figure 2.276: Slope field $2x' - \frac{x}{y} + x^3 \cos(y) = 0$ Summary of solutions found

$$x = \frac{\sqrt{(\cos(y) + \sin(y)y + c_1)y}}{\cos(y) + \sin(y)y + c_1}$$

$$x = -\frac{\sqrt{(\cos(y) + \sin(y)y + c_1)y}}{\cos(y) + \sin(y)y + c_1}$$

Solved using first_order_ode_exact

Time used: 0.142 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N\end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(y, x) dy + N(y, x) dx = 0 \quad (1A)$$

Therefore

$$\begin{aligned}(2y) dx &= (-x(x^2 \cos(y) y - 1)) dy \\ (x(x^2 \cos(y) y - 1)) dy &+ (2y) dx = 0\end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(y, x) &= x(x^2 \cos(y) y - 1) \\ N(y, x) &= 2y\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial x} &= \frac{\partial}{\partial x}(x(x^2 \cos(y) y - 1)) \\ &= 3x^2 \cos(y) y - 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial y} &= \frac{\partial}{\partial y}(2y) \\ &= 2\end{aligned}$$

Since $\frac{\partial M}{\partial x} \neq \frac{\partial N}{\partial y}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) \\ &= \frac{1}{2y} ((3x^2 \cos(y)y - 1) - (2)) \\ &= \frac{3x^2 \cos(y)y - 3}{2y} \end{aligned}$$

Since A depends on x , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} \right) \\ &= \frac{1}{x(x^2 \cos(y)y - 1)} ((2) - (3x^2 \cos(y)y - 1)) \\ &= -\frac{3}{x} \end{aligned}$$

Since B does not depend on y , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dx} \\ &= e^{\int -\frac{3}{x} \, dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-3 \ln(x)} \\ &= \frac{1}{x^3} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{1}{x^3} (x(x^2 \cos(y)y - 1)) \\ &= \frac{x^2 \cos(y)y - 1}{x^2} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{1}{x^3} (2y) \\ &= \frac{2y}{x^3} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dx}{dy} &= 0 \\ \left(\frac{x^2 \cos(y) y - 1}{x^2} \right) + \left(\frac{2y}{x^3} \right) \frac{dx}{dy} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(y, x)$

$$\frac{\partial \phi}{\partial y} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial x} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \overline{M} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{x^2 \cos(y) y - 1}{x^2} dy \\ \phi &= \frac{x^2 \sin(y) y + x^2 \cos(y) - y}{x^2} + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both y and x . Taking derivative of equation (3) w.r.t x gives

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= -\frac{2(x^2 \sin(y) y + x^2 \cos(y) - y)}{x^3} + \frac{2x \sin(y) y + 2 \cos(y) x}{x^2} + f'(x) \\ &= \frac{2y}{x^3} + f'(x)\end{aligned} \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial x} = \frac{2y}{x^3}$. Therefore equation (4) becomes

$$\frac{2y}{x^3} = \frac{2y}{x^3} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = 0$$

Therefore

$$f(x) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{x^2 \sin(y) y + x^2 \cos(y) - y}{x^2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x^2 \sin(y) y + x^2 \cos(y) - y}{x^2}$$

Simplifying the above gives

$$\frac{-y + x^2(\cos(y) + \sin(y) y)}{x^2} = c_1$$

Solving for x gives

$$x = \frac{\sqrt{(\sin(y) y + \cos(y) - c_1) y}}{\sin(y) y + \cos(y) - c_1}$$

$$x = -\frac{\sqrt{(\sin(y) y + \cos(y) - c_1) y}}{\sin(y) y + \cos(y) - c_1}$$

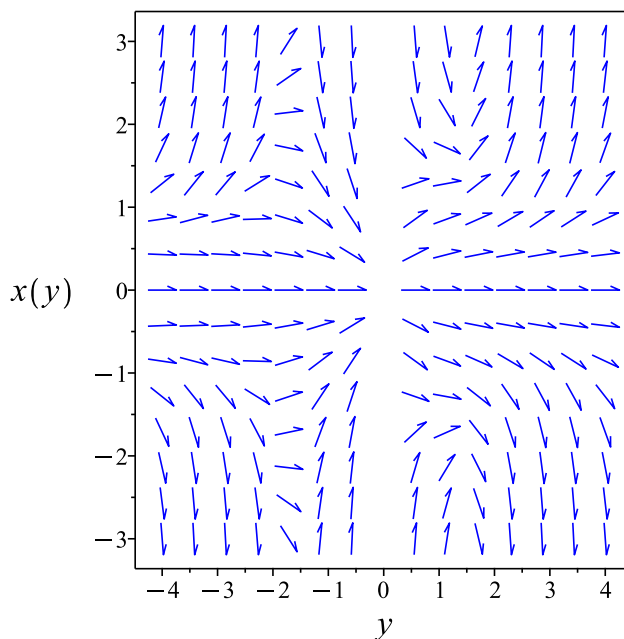


Figure 2.277: Slope field $2x' - \frac{x}{y} + x^3 \cos(y) = 0$

Summary of solutions found

$$x = \frac{\sqrt{(\sin(y)y + \cos(y) - c_1)y}}{\sin(y)y + \cos(y) - c_1}$$

$$x = -\frac{\sqrt{(\sin(y)y + \cos(y) - c_1)y}}{\sin(y)y + \cos(y) - c_1}$$

✓ Maple. Time used: 0.010 (sec). Leaf size: 54

```
ode:=2*diff(x(y),y)-x(y)/y+x(y)^3*cos(y) = 0;
dsolve(ode,x(y), singsol=all);
```

$$x = \frac{\sqrt{(\cos(y) + y \sin(y) + c_1)y}}{\cos(y) + y \sin(y) + c_1}$$

$$x = -\frac{\sqrt{(\cos(y) + y \sin(y) + c_1)y}}{\cos(y) + y \sin(y) + c_1}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$2\frac{d}{dy}x(y) - \frac{x(y)}{y} + x(y)^3 \cos(y) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dy}x(y)$$

- Solve for the highest derivative

$$\frac{d}{dy}x(y) = \frac{x(y)}{2y} - \frac{x(y)^3 \cos(y)}{2}$$

✓ **Mathematica.** Time used: 0.138 (sec). Leaf size: 77

```
ode=2*D[x[y],y]-x[y]/y+x[y]^3*Cos[y]==0;
ic={};
DSolve[{ode,ic},x[y],y,IncludeSingularSolutions->True]
```

$$x(y) \rightarrow -\frac{\sqrt{y}}{\sqrt{-2 \int_1^y -\frac{1}{2} \cos(K[1])K[1]dK[1] + c_1}}$$

$$x(y) \rightarrow \frac{\sqrt{y}}{\sqrt{-2 \int_1^y -\frac{1}{2} \cos(K[1])K[1]dK[1] + c_1}}$$

$$x(y) \rightarrow 0$$

✓ **Sympy.** Time used: 0.970 (sec). Leaf size: 36

```
from sympy import *
y = symbols("y")
x = Function("x")
ode = Eq(x(y)**3*cos(y) + 2*Derivative(x(y), y) - x(y)/y,0)
ics = {}
dsolve(ode,func=x(y),ics=ics)
```

$$\left[x(y) = -\sqrt{\frac{y}{C_1 + y \sin(y) + \cos(y)}}, x(y) = \sqrt{\frac{y}{C_1 + y \sin(y) + \cos(y)}} \right]$$

2.4.14 Problem 19 (p)

Local contents

Solved using first_order_ode_linear	1207
Solved using first_order_ode_exact	1209
✓ Maple	1213
✓ Mathematica	1214
✓ Sympy	1214

Internal problem ID [7930]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (p)

Date solved : Thursday, November 27, 2025 at 06:33:16 AM

CAS classification : [_linear]

Solved using first_order_ode_linear

Time used: 0.096 (sec)

Solve

$$y'x = y(1 - \tan(x)x) + \cos(x)x^2$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1 - \tan(x)x}{x}$$

$$p(x) = x \cos(x)$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1 - \tan(x)x}{x} dx} \\ &= \frac{1}{\cos(x)x}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) (x \cos(x)) \\ \frac{d}{dx} \left(\frac{y}{\cos(x) x} \right) &= \left(\frac{1}{\cos(x) x} \right) (x \cos(x)) \\ d \left(\frac{y}{\cos(x) x} \right) &= dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{\cos(x) x} &= \int 1 dx \\ &= x + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{\cos(x)x}$ gives the final solution

$$y = \cos(x) x(x + c_1)$$

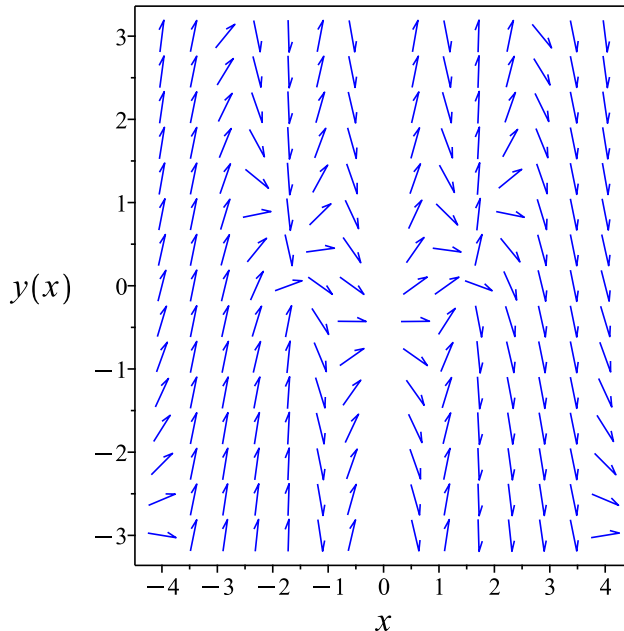


Figure 2.278: Slope field $y'x = y(1 - \tan(x)x) + \cos(x)x^2$

Summary of solutions found

$$y = \cos(x) x(x + c_1)$$

Solved using first_order_ode_exact

Time used: 0.216 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (x) dy &= (y(1 - \tan(x)x) + \cos(x)x^2) dx \\ (-y(1 - \tan(x)x) - \cos(x)x^2) dx &+ (x) dy = 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -y(1 - \tan(x)x) - \cos(x)x^2 \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-y(1 - \tan(x)x) - \cos(x)x^2) \\ &= \tan(x)x - 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((\tan(x)x - 1) - (1)) \\ &= \frac{\tan(x)x - 2}{x}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{\tan(x)x - 2}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-\ln(\cos(x)) - 2\ln(x)} \\ &= \frac{1}{\cos(x)x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{\cos(x)x^2} (-y(1 - \tan(x)x) - \cos(x)x^2) \\ &= \frac{-x^2 + y \sec(x)(\tan(x)x - 1)}{x^2}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{\cos(x) x^2}(x) \\ &= \frac{\sec(x)}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-x^2 + y \sec(x) (\tan(x) x - 1)}{x^2} \right) + \left(\frac{\sec(x)}{x} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{\sec(x)}{x} dy \\ \phi &= \frac{\sec(x) y}{x} + f(x)\end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= -\frac{\sec(x) y}{x^2} + \frac{\tan(x) \sec(x) y}{x} + f'(x) \\ &= \frac{y \sec(x) (\tan(x) x - 1)}{x^2} + f'(x)\end{aligned} \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{-x^2 + y \sec(x) (\tan(x) x - 1)}{x^2}$. Therefore equation (4) becomes

$$\frac{-x^2 + y \sec(x) (\tan(x) x - 1)}{x^2} = \frac{y \sec(x) (\tan(x) x - 1)}{x^2} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -1$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int (-1) dx$$

$$f(x) = -x + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{\sec(x)y}{x} - x + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{\sec(x)y}{x} - x$$

Solving for y gives

$$y = \frac{x(x + c_1)}{\sec(x)}$$

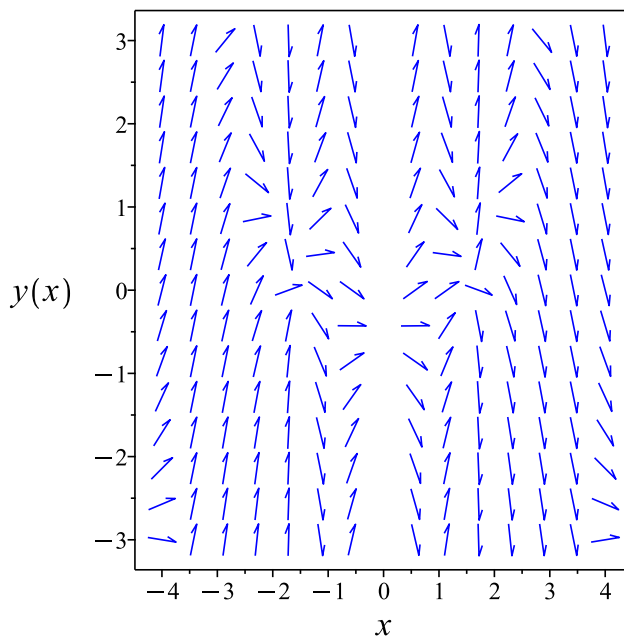


Figure 2.279: Slope field $y'x = y(1 - \tan(x)x) + \cos(x)x^2$

Summary of solutions found

$$y = \frac{x(x + c_1)}{\sec(x)}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 11

```
ode:=diff(y(x),x)*x = y(x)*(1-x*tan(x))+x^2*cos(x);
dsolve(ode,y(x), singsol=all);
```

$$y = (x + c_1) \cos(x) x$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

Maple step by step

Let's solve

$$x \left(\frac{d}{dx} y(x) \right) = y(x) (1 - x \tan(x)) + x^2 \cos(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{y(x)(1-x \tan(x)) + x^2 \cos(x)}{x}$$

- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx} y(x) = -\frac{(x \tan(x) - 1)y(x)}{x} + x \cos(x)$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx} y(x) + \frac{(x \tan(x) - 1)y(x)}{x} = x \cos(x)$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx} y(x) + \frac{(x \tan(x) - 1)y(x)}{x} \right) = \mu(x) x \cos(x)$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx} (y(x) \mu(x))$

$$\mu(x) \left(\frac{d}{dx} y(x) + \frac{(x \tan(x) - 1)y(x)}{x} \right) = \left(\frac{d}{dx} y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx} \mu(x) \right)$$

- Isolate $\frac{d}{dx} \mu(x)$

$$\frac{d}{dx} \mu(x) = \frac{\mu(x)(x \tan(x) - 1)}{x}$$

- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x \cos(x)}$$

- Integrate both sides with respect to x

- $\int \left(\frac{d}{dx}(y(x) \mu(x)) \right) dx = \int \mu(x) x \cos(x) dx + C1$
- Evaluate the integral on the lhs
 $y(x) \mu(x) = \int \mu(x) x \cos(x) dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int \mu(x) x \cos(x) dx + C1}{\mu(x)}$
- Substitute $\mu(x) = \frac{1}{x \cos(x)}$
 $y(x) = x \cos(x) \left(\int 1 dx + C1 \right)$
- Evaluate the integrals on the rhs
 $y(x) = x \cos(x) (x + C1)$

✓ **Mathematica.** Time used: 0.047 (sec). Leaf size: 13

```
ode=x*D[y[x],x]==y[x]*(1-x*Tan[x])+x^2*Cos[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x(x + c_1) \cos(x)$$

✓ **Sympy.** Time used: 4.121 (sec). Leaf size: 54

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*cos(x) + x*Derivative(y(x), x) - (-x*tan(x) + 1)*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$= C_1 - \int \sqrt{\tan^2(x) + 1} \cos(x) dx - \int \frac{\sqrt{\tan^2(x) + 1} y(x)}{x^2} dx + \int \frac{\sqrt{\tan^2(x) + 1} y(x) \tan(x)}{x} dx$$

2.4.15 Problem 19 (q)

Local contents

Solved using first_order_ode_exact	1215
Solved using first_order_ode_LIE	1219
✓ Maple	1226
✓ Mathematica	1227
✓ Sympy	1227

Internal problem ID [7931]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (q)

Date solved : Thursday, November 27, 2025 at 06:33:36 AM

CAS classification : [_rational, [_1st_order, ['_with_symmetry_[F(x)*G(y),0]']]

Solved using first_order_ode_exact

Time used: 0.476 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-y^3 - yx - 2y) dy &= (-y^2 - 2) dx \\ (y^2 + 2) dx + (-y^3 - yx - 2y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y^2 + 2 \\ N(x, y) &= -y^3 - yx - 2y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y^2 + 2) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-y^3 - yx - 2y) \\ &= -y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= -\frac{1}{y(y^2 + x + 2)} ((2y) - (-y)) \\ &= -\frac{3}{y^2 + x + 2} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y^2 + 2} ((-y) - (2y)) \\ &= -\frac{3y}{y^2 + 2} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int -\frac{3y}{y^2+2} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\frac{3 \ln(y^2+2)}{2}} \\ &= \frac{1}{(y^2 + 2)^{3/2}} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{1}{(y^2 + 2)^{3/2}} (y^2 + 2) \\ &= \frac{1}{\sqrt{y^2 + 2}} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{1}{(y^2 + 2)^{3/2}} (-y^3 - yx - 2y) \\ &= \frac{(-y^2 - x - 2)y}{(y^2 + 2)^{3/2}} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{1}{\sqrt{y^2 + 2}} \right) + \left(\frac{(-y^2 - x - 2)y}{(y^2 + 2)^{3/2}} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{1}{\sqrt{y^2 + 2}} dx \\ \phi &= \frac{x}{\sqrt{y^2 + 2}} + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = -\frac{xy}{(y^2 + 2)^{3/2}} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{(-y^2 - x - 2)y}{(y^2 + 2)^{3/2}}$. Therefore equation (4) becomes

$$\frac{(-y^2 - x - 2)y}{(y^2 + 2)^{3/2}} = -\frac{xy}{(y^2 + 2)^{3/2}} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{y}{\sqrt{y^2 + 2}}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int \left(-\frac{y}{\sqrt{y^2 + 2}} \right) dy \\ f(y) &= -\sqrt{y^2 + 2} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = \frac{x}{\sqrt{y^2 + 2}} - \sqrt{y^2 + 2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{x}{\sqrt{y^2 + 2}} - \sqrt{y^2 + 2}$$

Simplifying the above gives

$$\frac{-y^2 - 2 + x}{\sqrt{y^2 + 2}} = c_1$$

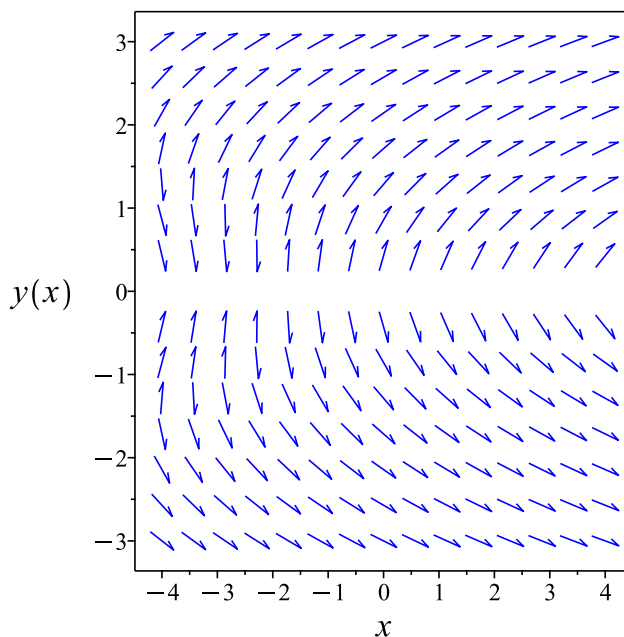


Figure 2.280: Slope field $2 + y^2 - (yx + 2y + y^3) y' = 0$

Summary of solutions found

$$\frac{-y^2 - 2 + x}{\sqrt{y^2 + 2}} = c_1$$

Solved using first_order_ode_LIE

Time used: 6.941 (sec)

Solve

$$2 + y^2 - (yx + 2y + y^3) y' = 0$$

Writing the ode as

$$y' = \frac{y^2 + 2}{y(y^2 + x + 2)}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 2 to use as anstaz gives

$$\xi = x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (1\text{E})$$

$$\eta = x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3, b_4, b_5, b_6\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 2x b_4 + y b_5 + b_2 + \frac{(y^2 + 2)(-2x a_4 + x b_5 - y a_5 + 2y b_6 - a_2 + b_3)}{y(y^2 + x + 2)} \\ & - \frac{(y^2 + 2)^2(x a_5 + 2y a_6 + a_3)}{y^2(y^2 + x + 2)^2} + \frac{(y^2 + 2)(x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1)}{y(y^2 + x + 2)^2} \quad (5\text{E}) \\ & - \left(\frac{2}{y^2 + x + 2} - \frac{y^2 + 2}{y^2(y^2 + x + 2)} - \frac{2(y^2 + 2)}{(y^2 + x + 2)^2} \right) (x^2 b_4 \\ & + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1) = 0 \end{aligned}$$

Putting the above in normal form gives

$$\begin{aligned} & 2x y^6 b_4 + y^7 b_5 + 5x^2 y^4 b_4 - 2x y^5 a_4 + 4x y^5 b_5 - y^6 a_5 + y^6 b_2 + 3y^6 b_6 + x^3 y^2 b_4 - x^2 y^3 a_4 + x^2 y^3 b_5 - x y^4 a_5 + 3 \\ & = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned} & 2x y^6 b_4 + y^7 b_5 + 5x^2 y^4 b_4 - 2x y^5 a_4 + 4x y^5 b_5 - y^6 a_5 + y^6 b_2 + 3y^6 b_6 \\ & + x^3 y^2 b_4 - x^2 y^3 a_4 + x^2 y^3 b_5 - x y^4 a_5 + 3x y^4 b_2 + 8x y^4 b_4 + x y^4 b_6 - y^5 a_2 \\ & - y^5 a_6 + 2y^5 b_3 + 4y^5 b_5 + 12x^2 y^2 b_4 - 8x y^3 a_4 + 12x y^3 b_5 - 4y^4 a_5 + y^4 b_1 \quad (6\text{E}) \\ & + 4y^4 b_2 + 12y^4 b_6 + 2x^3 b_4 - 2x^2 y a_4 + 4x^2 y b_5 - 4x y^2 a_5 - x y^2 b_1 + 8x y^2 b_2 \\ & + 8x b_4 y^2 + 6x y^2 b_6 + y^3 a_1 - 4y^3 a_2 - 6y^3 a_6 + 8y^3 b_3 + 4y^3 b_5 + 2x^2 b_2 + 4x^2 b_4 \\ & - 8x y a_4 + 4x y b_3 + 8y x b_5 - 2y^2 a_3 - 4y^2 a_5 + 4y^2 b_1 + 4b_2 y^2 + 12y^2 b_6 \\ & - 4x a_5 + 2x b_1 + 4x b_2 + 2y a_1 - 4y a_2 - 8y a_6 + 8y b_3 - 4a_3 + 4b_1 = 0 \end{aligned}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & 2b_4v_1v_2^6 + b_5v_2^7 - 2a_4v_1v_2^5 - a_5v_2^6 + b_2v_2^6 + 5b_4v_1^2v_2^4 + 4b_5v_1v_2^5 + 3b_6v_2^6 - a_2v_2^5 \\ & - a_4v_1^2v_2^3 - a_5v_1v_2^4 - a_6v_2^5 + 3b_2v_1v_2^4 + 2b_3v_2^5 + b_4v_1^3v_2^2 + 8b_4v_1v_2^4 + b_5v_1^2v_2^3 \\ & + 4b_5v_2^5 + b_6v_1v_2^4 - 8a_4v_1v_2^3 - 4a_5v_2^4 + b_1v_2^4 + 4b_2v_2^4 + 12b_4v_1^2v_2^2 + 12b_5v_1v_2^3 \\ & + 12b_6v_2^4 + a_1v_2^3 - 4a_2v_2^3 - 2a_4v_1^2v_2 - 4a_5v_1v_2^2 - 6a_6v_2^3 - b_1v_1v_2^2 + 8b_2v_1v_2^2 \\ & + 8b_3v_2^3 + 2b_4v_1^3 + 8b_4v_1v_2^2 + 4b_5v_1^2v_2 + 4b_5v_2^3 + 6b_6v_1v_2^2 - 2a_3v_2^2 - 8a_4v_1v_2 \\ & - 4a_5v_2^2 + 4b_1v_2^2 + 2b_2v_1^2 + 4b_2v_2^2 + 4b_3v_1v_2 + 4b_4v_1^2 + 8b_5v_1v_2 + 12b_6v_2^2 \\ & + 2a_1v_2 - 4a_2v_2 - 4a_5v_1 - 8a_6v_2 + 2b_1v_1 + 4b_2v_1 + 8b_3v_2 - 4a_3 + 4b_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} & b_4v_1^3v_2^2 + 2b_4v_1^3 + 5b_4v_1^2v_2^4 + (-a_4 + b_5)v_1^2v_2^3 + 12b_4v_1^2v_2^2 + (-2a_4 + 4b_5)v_1^2v_2 \\ & + (2b_2 + 4b_4)v_1^2 + 2b_4v_1v_2^6 + (-2a_4 + 4b_5)v_1v_2^5 + (-a_5 + 3b_2 + 8b_4 + b_6)v_1v_2^4 \\ & + (-8a_4 + 12b_5)v_1v_2^3 + (-4a_5 - b_1 + 8b_2 + 8b_4 + 6b_6)v_1v_2^2 \\ & + (-8a_4 + 4b_3 + 8b_5)v_1v_2 + (-4a_5 + 2b_1 + 4b_2)v_1 + b_5v_2^7 \\ & + (-a_5 + b_2 + 3b_6)v_2^6 + (-a_2 - a_6 + 2b_3 + 4b_5)v_2^5 \\ & + (-4a_5 + b_1 + 4b_2 + 12b_6)v_2^4 + (a_1 - 4a_2 - 6a_6 + 8b_3 + 4b_5)v_2^3 \\ & + (-2a_3 - 4a_5 + 4b_1 + 4b_2 + 12b_6)v_2^2 + (2a_1 - 4a_2 - 8a_6 + 8b_3)v_2 - 4a_3 + 4b_1 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned}b_4 &= 0 \\b_5 &= 0 \\2b_4 &= 0 \\5b_4 &= 0 \\12b_4 &= 0 \\-4a_3 + 4b_1 &= 0 \\-8a_4 + 12b_5 &= 0 \\-2a_4 + 4b_5 &= 0 \\-a_4 + b_5 &= 0 \\2b_2 + 4b_4 &= 0 \\-8a_4 + 4b_3 + 8b_5 &= 0 \\-4a_5 + 2b_1 + 4b_2 &= 0 \\-a_5 + b_2 + 3b_6 &= 0 \\2a_1 - 4a_2 - 8a_6 + 8b_3 &= 0 \\-a_2 - a_6 + 2b_3 + 4b_5 &= 0 \\-4a_5 + b_1 + 4b_2 + 12b_6 &= 0 \\-a_5 + 3b_2 + 8b_4 + b_6 &= 0 \\a_1 - 4a_2 - 6a_6 + 8b_3 + 4b_5 &= 0 \\-2a_3 - 4a_5 + 4b_1 + 4b_2 + 12b_6 &= 0 \\-4a_5 - b_1 + 8b_2 + 8b_4 + 6b_6 &= 0\end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned}a_1 &= 2a_6 \\a_2 &= -a_6 \\a_3 &= 0 \\a_4 &= 0 \\a_5 &= 0 \\a_6 &= a_6 \\b_1 &= 0 \\b_2 &= 0 \\b_3 &= 0 \\b_4 &= 0 \\b_5 &= 0 \\b_6 &= 0\end{aligned}$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = y^2 - x + 2$$

$$\eta = 0$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= 0 - \left(\frac{y^2 + 2}{y(y^2 + x + 2)} \right) (y^2 - x + 2) \\ &= \frac{(y^2 + 2)(-y^2 + x - 2)}{y(y^2 + x + 2)} \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{\frac{(y^2+2)(-y^2+x-2)}{y(y^2+x+2)}} dy\end{aligned}$$

Which results in

$$S = -\ln(y^2 - x + 2) + \frac{\ln(y^2 + 2)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y) S_y}{R_x + \omega(x, y) R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y^2 + 2}{y(y^2 + x + 2)}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= \frac{1}{y^2 - x + 2} \\ S_y &= -\frac{2y}{y^2 - x + 2} + \frac{y}{y^2 + 2} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = 0 \tag{2A}$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = 0$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

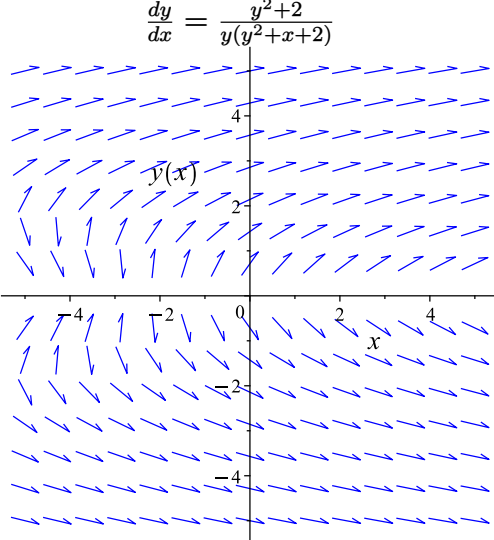
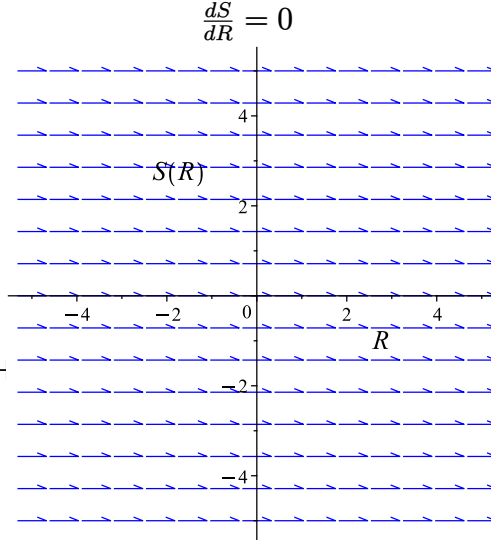
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int 0 dR + c_2 \\ S(R) &= c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\ln(y^2 + 2 - x) + \frac{\ln(y^2 + 2)}{2} = c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{y^2+2}{y(y^2+x+2)}$ 	$R = x$ $S = -\ln(y^2 - x + 2)$	$\frac{dS}{dR} = 0$ 

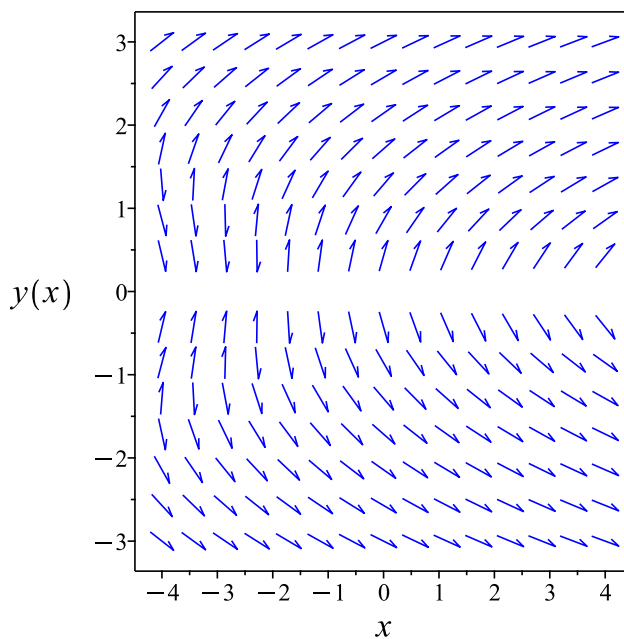


Figure 2.281: Slope field $2 + y^2 - (yx + 2y + y^3) y' = 0$

Summary of solutions found

$$-\ln(y^2 + 2 - x) + \frac{\ln(y^2 + 2)}{2} = c_2$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 23

```
ode:=2+y(x)^2-(y(x)*x+2*y(x)+y(x)^3)*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$x - y^2 - 2 - \sqrt{y^2 + 2} c_1 = 0$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

trying separable

trying inverse linear

<- 1st order linear successful

<- inverse linear successful

Maple step by step

Let's solve

$$2 + y(x)^2 - (xy(x) + 2y(x) + y(x)^3) \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-2 - y(x)^2}{xy(x) + 2y(x) + y(x)^3}$$

✓ **Mathematica.** Time used: 5.282 (sec). Leaf size: 189

```
ode=(2+y[x]^2)-(x*y[x]+2*y[x]+y[x]^3)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{2x - \sqrt{4c_1^2x + c_1^4} - 4 + c_1^2}}{\sqrt{2}}$$

$$y(x) \rightarrow \frac{\sqrt{2x - \sqrt{4c_1^2x + c_1^4} - 4 + c_1^2}}{\sqrt{2}}$$

$$y(x) \rightarrow -\frac{\sqrt{2x + \sqrt{4c_1^2x + c_1^4} - 4 + c_1^2}}{\sqrt{2}}$$

$$y(x) \rightarrow \frac{\sqrt{2x + \sqrt{4c_1^2x + c_1^4} - 4 + c_1^2}}{\sqrt{2}}$$

$$y(x) \rightarrow -i\sqrt{2}$$

$$y(x) \rightarrow i\sqrt{2}$$

✓ **Sympy.** Time used: 31.156 (sec). Leaf size: 112

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-x*y(x) - y(x)**3 - 2*y(x))*Derivative(y(x), x) + y(x)**2 + 2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = -\sqrt{\frac{C_1^2}{2} - \frac{C_1\sqrt{C_1^2 + 4x}}{2}} + x - 2, y(x) = \sqrt{\frac{C_1^2}{2} - \frac{C_1\sqrt{C_1^2 + 4x}}{2}} + x - 2, y(x) = \right. \\ \left. -\sqrt{\frac{C_1^2}{2} + \frac{C_1\sqrt{C_1^2 + 4x}}{2}} + x - 2, y(x) = \sqrt{\frac{C_1^2}{2} + \frac{C_1\sqrt{C_1^2 + 4x}}{2}} + x - 2 \right]$$

2.4.16 Problem 19 (r)

Local contents

Solved using first_order_ode_exact	1228
Solved using first_order_ode_LIE	1232
✓ Maple	1239
✓ Mathematica	1240
✓ Sympy	1240

Internal problem ID [7932]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (r)

Date solved : Thursday, November 27, 2025 at 06:33:53 AM

CAS classification : `[[_1st_order, '_with_symmetry_[F(x),G(y)]']]`

Solved using first_order_ode_exact

Time used: 0.216 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-\arctan(y) + x) dy &= (-y^2 - 1) dx \\ (y^2 + 1) dx + (-\arctan(y) + x) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y^2 + 1 \\ N(x, y) &= -\arctan(y) + x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y^2 + 1) \\ &= 2y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-\arctan(y) + x) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{-\arctan(y) + x} ((2y) - (1)) \\ &= \frac{1 - 2y}{\arctan(y) - x} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{y^2 + 1} ((1) - (2y)) \\ &= \frac{1 - 2y}{y^2 + 1} \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int \frac{1-2y}{y^2+1} \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\ln(y^2+1) + \arctan(y)} \\ &= \frac{e^{\arctan(y)}}{y^2 + 1} \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{e^{\arctan(y)}}{y^2 + 1} (y^2 + 1) \\ &= e^{\arctan(y)} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{e^{\arctan(y)}}{y^2 + 1} (-\arctan(y) + x) \\ &= \frac{(-\arctan(y) + x) e^{\arctan(y)}}{y^2 + 1} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (e^{\arctan(y)}) + \left(\frac{(-\arctan(y) + x) e^{\arctan(y)}}{y^2 + 1} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int e^{\arctan(y)} dx \\ \phi &= e^{\arctan(y)} x + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{e^{\arctan(y)} x}{y^2 + 1} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{(-\arctan(y) + x)e^{\arctan(y)}}{y^2 + 1}$. Therefore equation (4) becomes

$$\frac{(-\arctan(y) + x)e^{\arctan(y)}}{y^2 + 1} = \frac{e^{\arctan(y)} x}{y^2 + 1} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -\frac{e^{\arctan(y)} \arctan(y)}{y^2 + 1}$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int \left(-\frac{e^{\arctan(y)} \arctan(y)}{y^2 + 1} \right) dy \\ f(y) &= -e^{\arctan(y)} \arctan(y) + e^{\arctan(y)} + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = e^{\arctan(y)} x - e^{\arctan(y)} \arctan(y) + e^{\arctan(y)} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = e^{\arctan(y)} x - e^{\arctan(y)} \arctan(y) + e^{\arctan(y)}$$

Simplifying the above gives

$$e^{\arctan(y)}(x - \arctan(y) + 1) = c_1$$

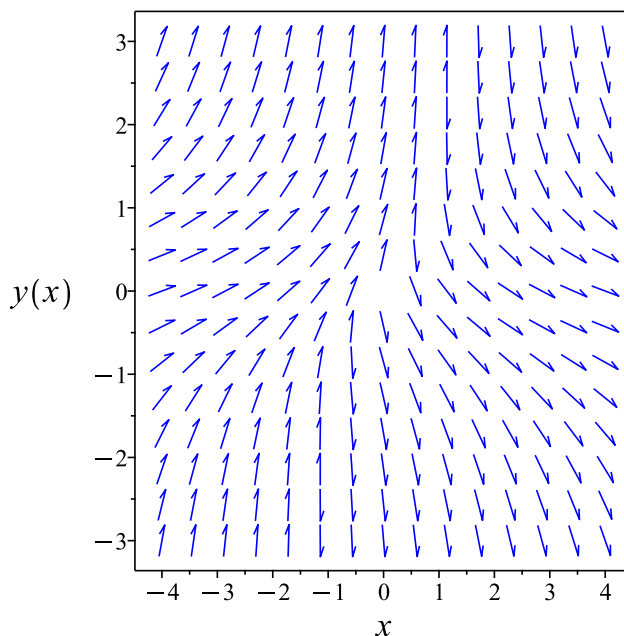


Figure 2.282: Slope field $1 + y^2 = (\arctan(y) - x)y'$

Summary of solutions found

$$e^{\arctan(y)}(x - \arctan(y) + 1) = c_1$$

Solved using first_order_ode_LIE

Time used: 1.224 (sec)

Solve

$$1 + y^2 = (\arctan(y) - x)y'$$

Writing the ode as

$$y' = \frac{y^2 + 1}{\arctan(y) - x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 2 to use as anstaz gives

$$\xi = x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (\text{1E})$$

$$\eta = x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (\text{2E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3, b_4, b_5, b_6\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 2xb_4 + yb_5 + b_2 + \frac{(y^2 + 1)(-2xa_4 + xb_5 - ya_5 + 2yb_6 - a_2 + b_3)}{\arctan(y) - x} \\ & - \frac{(y^2 + 1)^2(xa_5 + 2ya_6 + a_3)}{(\arctan(y) - x)^2} - \frac{(y^2 + 1)(x^2a_4 + yxa_5 + y^2a_6 + xa_2 + ya_3 + a_1)}{(\arctan(y) - x)^2} \quad (\text{5E}) \\ & - \left(\frac{2y}{\arctan(y) - x} - \frac{1}{(\arctan(y) - x)^2} \right) (x^2b_4 + yxb_5 + y^2b_6 + xb_2 + yb_3 + b_1) = 0 \end{aligned}$$

Putting the above in normal form gives

$$\begin{aligned} & yxb_5 - y^4a_3 - y^3a_3 + x^2b_2 - y^2a_1 - 2y^2a_3 - xb_3 + \arctan(y)^2 b_2 - \arctan(y) a_2 + \arctan(y) b_3 - 2 \arctan(y) \\ & = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned} & yxb_5 - y^4a_3 - y^3a_3 + x^2b_2 - y^2a_1 - 2y^2a_3 - xb_3 + \arctan(y)^2 b_2 \\ & - \arctan(y) a_2 + \arctan(y) b_3 - 2 \arctan(y) xyb_2 + x^2a_4 - y^2a_6 \\ & + x^2b_4 + y^2b_6 - a_1 - a_3 + b_1 - ya_3 + xb_2 + yb_3 - 2 \arctan(y) x^2yb_4 \\ & - 2 \arctan(y) x y^2a_4 - \arctan(y) x y^2b_5 - 2 \arctan(y) xyb_5 - 2y^5a_6 \quad (\text{6E}) \\ & - y^4a_6 + 2x^3b_4 - 4y^3a_6 - x^2b_5 - xa_5 - 2ya_6 + 2x^2yb_2 + x y^2b_3 + 2xyb_1 \\ & - \arctan(y) y^2a_2 - \arctan(y) y^2b_3 - 2 \arctan(y) xb_2 - 2 \arctan(y) yb_1 \\ & - x y^4a_5 + 2x^3yb_4 + x^2y^2a_4 + x^2y^2b_5 + x^2yb_5 - 2x y^2a_5 - 2xyb_6 \\ & - \arctan(y) y^3a_5 + 2 \arctan(y)^2 xb_4 + \arctan(y)^2 yb_5 - 4 \arctan(y) x^2b_4 \\ & - 2 \arctan(y) xa_4 + \arctan(y) xb_5 - \arctan(y) ya_5 + 2 \arctan(y) yb_6 = 0 \end{aligned}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, \arctan(y)\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, \arctan(y) = v_3\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & -v_1v_2^4a_5 - 2v_2^5a_6 - v_2^4a_3 + v_1^2v_2^2a_4 - 2v_3v_1v_2^2a_4 - v_3v_2^3a_5 - v_2^4a_6 + 2v_1^3v_2b_4 \\ & - 2v_3v_1^2v_2b_4 + v_1^2v_2^2b_5 - v_3v_1v_2^2b_5 - v_3v_2^2a_2 - v_2^3a_3 - 2v_1v_2^2a_5 - 4v_2^3a_6 + 2v_1^2v_2b_2 \\ & - 2v_3v_1v_2b_2 + v_1v_2^2b_3 - v_3v_2^2b_3 + 2v_1^3b_4 - 4v_3v_1^2b_4 + 2v_3^2v_1b_4 + v_1^2v_2b_5 - 2v_3v_1v_2b_5 \\ & + v_3^2v_2b_5 - v_2^2a_1 - 2v_2^2a_3 + v_1^2a_4 - 2v_3v_1a_4 - v_3v_2a_5 - v_2^2a_6 + 2v_1v_2b_1 - 2v_3v_2b_1 \\ & + v_1^2b_2 - 2v_3v_1b_2 + v_3^2b_2 + v_1^2b_4 - v_1^2b_5 + v_2v_1b_5 + v_3v_1b_5 - 2v_1v_2b_6 + v_2^2b_6 + 2v_3v_2b_6 \\ & - v_3a_2 - v_2a_3 - v_1a_5 - 2v_2a_6 + v_1b_2 - v_1b_3 + v_2b_3 + v_3b_3 - a_1 - a_3 + b_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3\}$$

Equation (7E) now becomes

$$\begin{aligned} & -a_1 - a_3 + b_1 - v_1v_2^4a_5 + 2v_1^3v_2b_4 - v_3v_2^3a_5 + 2v_3^2v_1b_4 + v_3^2v_2b_5 - 4v_3v_1^2b_4 \\ & + (-2a_4 - b_5)v_1v_2^2v_3 + (-2b_2 - 2b_5)v_1v_2v_3 + (a_4 + b_2 + b_4 - b_5)v_1^2 \\ & + (-a_5 + b_2 - b_3)v_1 + (-a_3 - a_6)v_2^4 + (-a_3 - 4a_6)v_2^3 + (-a_1 - 2a_3 - a_6 + b_6)v_2^2 \\ & + (-a_3 - 2a_6 + b_3)v_2 + (b_3 - a_2)v_3 - 2v_3v_1^2v_2b_4 + (-a_5 - 2b_1 + 2b_6)v_2v_3 \\ & + (a_4 + b_5)v_1^2v_2^2 + (2b_2 + b_5)v_1^2v_2 + (-2a_5 + b_3)v_1v_2^2 + (2b_1 + b_5 - 2b_6)v_1v_2 \\ & + (-2a_4 - 2b_2 + b_5)v_1v_3 + (-a_2 - b_3)v_2^2v_3 + v_3^2b_2 - 2v_2^5a_6 + 2v_1^3b_4 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$b_2 = 0$$

$$b_5 = 0$$

$$-a_5 = 0$$

$$-2a_6 = 0$$

$$-4b_4 = 0$$

$$-2b_4 = 0$$

$$2b_4 = 0$$

$$-a_2 - b_3 = 0$$

$$-a_3 - 4a_6 = 0$$

$$-a_3 - a_6 = 0$$

$$-2a_4 - b_5 = 0$$

$$a_4 + b_5 = 0$$

$$-2a_5 + b_3 = 0$$

$$-2b_2 - 2b_5 = 0$$

$$2b_2 + b_5 = 0$$

$$b_3 - a_2 = 0$$

$$-a_1 - a_3 + b_1 = 0$$

$$-a_3 - 2a_6 + b_3 = 0$$

$$-2a_4 - 2b_2 + b_5 = 0$$

$$-a_5 - 2b_1 + 2b_6 = 0$$

$$-a_5 + b_2 - b_3 = 0$$

$$2b_1 + b_5 - 2b_6 = 0$$

$$-a_1 - 2a_3 - a_6 + b_6 = 0$$

$$a_4 + b_2 + b_4 - b_5 = 0$$

Solving the above equations for the unknowns gives

$$a_1 = b_6$$

$$a_2 = 0$$

$$a_3 = 0$$

$$a_4 = 0$$

$$a_5 = 0$$

$$a_6 = 0$$

$$b_1 = b_6$$

$$b_2 = 0$$

$$b_3 = 0$$

$$b_4 = 0$$

$$b_5 = 0$$

$$b_6 = b_6$$

Substituting the above solution in the ansatz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = 1$$

$$\eta = y^2 + 1$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{\eta}{\xi} \\ &= \frac{y^2 + 1}{1} \\ &= y^2 + 1 \end{aligned}$$

This is easily solved to give

$$y = \tan(x + c_1)$$

Where now the coordinate R is taken as the constant of integration. Hence

$$R = \arctan(y) - x$$

And S is found from

$$\begin{aligned} dS &= \frac{dx}{\xi} \\ &= \frac{dx}{1} \end{aligned}$$

Integrating gives

$$\begin{aligned} S &= \int \frac{dx}{T} \\ &= x \end{aligned}$$

Where the constant of integration is set to zero as we just need one solution. Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{y^2 + 1}{\arctan(y) - x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= -1 \\ R_y &= \frac{1}{y^2 + 1} \\ S_x &= 1 \\ S_y &= 0 \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{-\arctan(y) + x}{-1 + \arctan(y) - x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{R}{-1 + R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

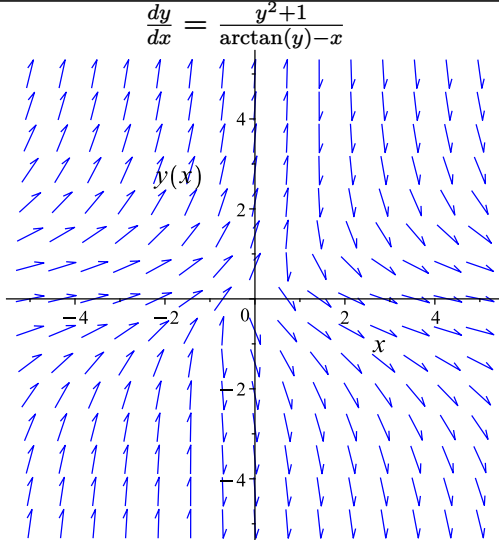
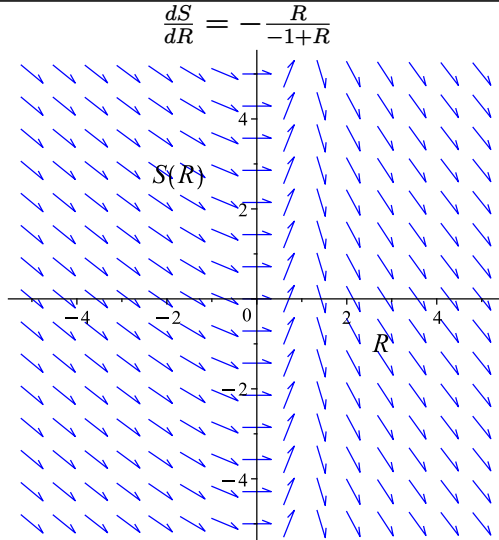
$$\int dS = \int -\frac{R}{-1+R} dR$$

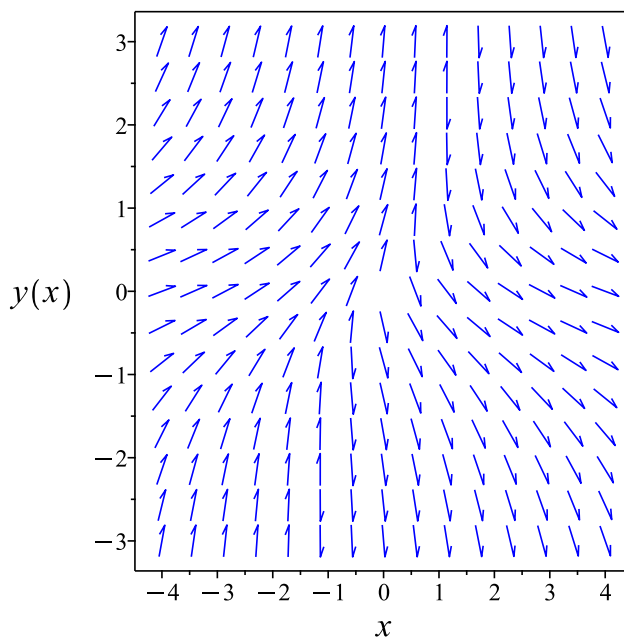
$$S(R) = -R - \ln(-1+R) + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$x = -\arctan(y) + x - \ln(-x + \arctan(y) - 1) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

Original ode in x, y coordinates	Canonical coordinates transformation	ODE in canonical coordinates (R, S)
$\frac{dy}{dx} = \frac{y^2+1}{\arctan(y)-x}$ 	$R = \arctan(y) - x$ $S = x$	$\frac{dS}{dR} = -\frac{R}{-1+R}$ 

Figure 2.283: Slope field $1 + y^2 = (\arctan(y) - x)y'$ Summary of solutions found

$$x = -\arctan(y) + x - \ln(-x + \arctan(y) - 1) + c_2$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 18

```
ode:=1+y(x)^2 = (arctan(y(x))-x)*diff(y(x),x);
dsolve(ode,y(x), singsol=all);
```

$$y = \tan(\text{LambertW}(-c_1 e^{-x-1}) + x + 1)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
<- 1st order linear successful
<- inverse linear successful
```

Maple step by step

Let's solve

$$1 + y(x)^2 = (\arctan(y(x)) - x) \left(\frac{d}{dx} y(x) \right)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-1-y(x)^2}{\arctan(y(x))-x}$$

✓ **Mathematica.** Time used: 0.151 (sec). Leaf size: 95

```
ode=(1+y[x]^2)==(ArcTan[y[x]]-x)*D[y[x],x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[x = \exp \left(\int_1^{y(x)} \frac{1}{-K[1]^2 - 1} dK[1] \right) \int_1^{y(x)} \exp \left(- \int_1^{K[2]} \frac{1}{-K[1]^2 - 1} dK[1] \right) \arctan(K[2]) - \frac{\arctan(K[2])}{-K[2]^2 - 1} dK[2] + c_1 \exp \left(\int_1^{y(x)} \frac{1}{-K[1]^2 - 1} dK[1] \right), y(x) \right]$$

✓ **Sympy.** Time used: 1.925 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x - atan(y(x)))*Derivative(y(x), x) + y(x)**2 + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \tan(x + W(C_1 e^{-x-1}) + 1)$$

2.4.17 Problem 19 (s)

Local contents

Solved using first_order_ode_bernoulli	1241
Solved using first_order_ode_isobaric	1244
Solved using first_order_ode_homog_type_G	1247
Solved using first_order_ode_LIE	1249
✓ Maple	1254
✓ Mathematica	1255
✓ Sympy	1256

Internal problem ID [7933]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (s)

Date solved : Thursday, November 27, 2025 at 06:34:05 AM

CAS classification : [[_homogeneous, 'class G'], _rational, _Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.131 (sec)

Solve

$$2xy^5 - y + 2y'x = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= -\frac{y(2xy^4 - 1)}{2x} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(\frac{1}{2x}\right)y + (-1)y^5 \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1}{2x} \\ f_1 &= -1 \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= \frac{1}{2x} \\ f_1(x) &= -1 \\ n &= 5 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = y^5$ gives

$$y' \frac{1}{y^5} = \frac{1}{2y^4x} - 1 \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= \frac{1}{y^4} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{4}{y^5}y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -\frac{v'(x)}{4} &= \frac{v(x)}{2x} - 1 \\ v' &= -\frac{2v}{x} + 4 \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x} \\ p(x) &= 4 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} \, dx} \\ &= x^2\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) (4) \\ \frac{d}{dx}(v x^2) &= (x^2) (4) \\ d(v x^2) &= (4x^2) \, dx\end{aligned}$$

Integrating gives

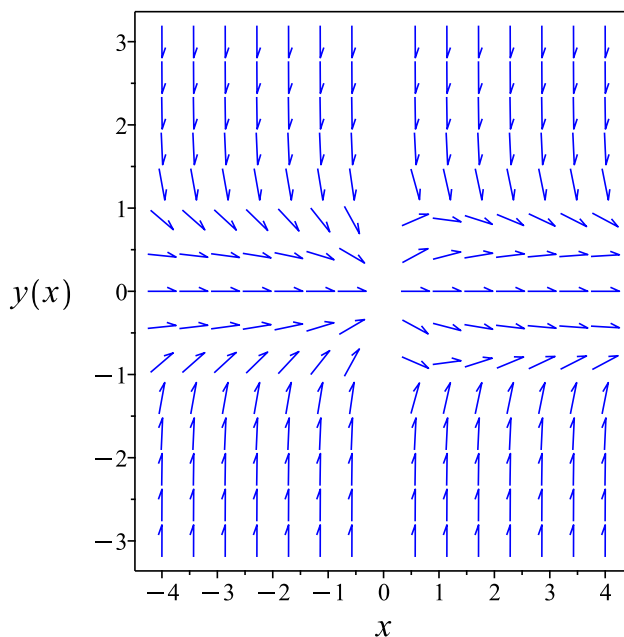
$$\begin{aligned}v x^2 &= \int 4x^2 \, dx \\ &= \frac{4x^3}{3} + c_1\end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

$$v(x) = \frac{\frac{4x^3}{3} + c_1}{x^2}$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$\frac{1}{y^4} = \frac{\frac{4x^3}{3} + c_1}{x^2}$$

Figure 2.284: Slope field $2xy^5 - y + 2y'x = 0$ Summary of solutions found

$$\frac{1}{y^4} = \frac{\frac{4x^3}{3} + c_1}{x^2}$$

Solved using first_order_ode_isobaric

Time used: 0.635 (sec)

Solve

$$2xy^5 - y + 2y'x = 0$$

Solving for y' gives

$$y' = -\frac{y(2y^4x - 1)}{2x} \quad (1)$$

Each of the above ode's is now solved An ode $y' = f(x, y)$ is isobaric if

$$f(tx, t^m y) = t^{m-1} f(x, y) \quad (1)$$

Where here

$$f(x, y) = -\frac{y(2y^4x - 1)}{2x} \quad (2)$$

 m is the order of isobaric. Substituting (2) into (1) and solving for m gives

$$m = -\frac{1}{4}$$

Since the ode is isobaric of order $m = -\frac{1}{4}$, then the substitution

$$\begin{aligned} y &= ux^m \\ &= \frac{u}{x^{1/4}} \end{aligned}$$

Converts the ODE to a separable in $u(x)$. Performing this substitution gives

$$-\frac{u(x)}{4x^{5/4}} + \frac{u'(x)}{x^{1/4}} = -\frac{u(x)(2u(x)^4 - 1)}{2x^{5/4}}$$

The ode

$$u'(x) = -\frac{u(x)(4u(x)^4 - 3)}{4x} \quad (2.106)$$

is separable as it can be written as

$$\begin{aligned} u'(x) &= -\frac{u(x)(4u(x)^4 - 3)}{4x} \\ &= f(x)g(u) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{4x} \\ g(u) &= u(4u^4 - 3) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(u)} du &= \int f(x) dx \\ \int \frac{1}{u(4u^4 - 3)} du &= \int -\frac{1}{4x} dx \end{aligned}$$

$$\frac{\ln(4u(x)^4 - 3)}{12} - \frac{\ln(u(x))}{3} = \ln\left(\frac{1}{x^{1/4}}\right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$u(4u^4 - 3) = 0$$

for $u(x)$ gives

$$\begin{aligned} u(x) &= 0 \\ u(x) &= -\frac{i3^{1/4}\sqrt{2}}{2} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{\ln(4u(x)^4 - 3)}{12} - \frac{\ln(u(x))}{3} = \ln\left(\frac{1}{x^{1/4}}\right) + c_1$$

$$u(x) = 0$$

$$u(x) = -\frac{i3^{1/4}\sqrt{2}}{2}$$

Converting $\frac{\ln(4u(x)^4 - 3)}{12} - \frac{\ln(u(x))}{3} = \ln\left(\frac{1}{x^{1/4}}\right) + c_1$ back to y gives

$$\frac{\ln(4y^4x - 3)}{12} - \frac{\ln(yx^{1/4})}{3} = \ln\left(\frac{1}{x^{1/4}}\right) + c_1$$

Converting $u(x) = 0$ back to y gives

$$yx^{1/4} = 0$$

Converting $u(x) = -\frac{i3^{1/4}\sqrt{2}}{2}$ back to y gives

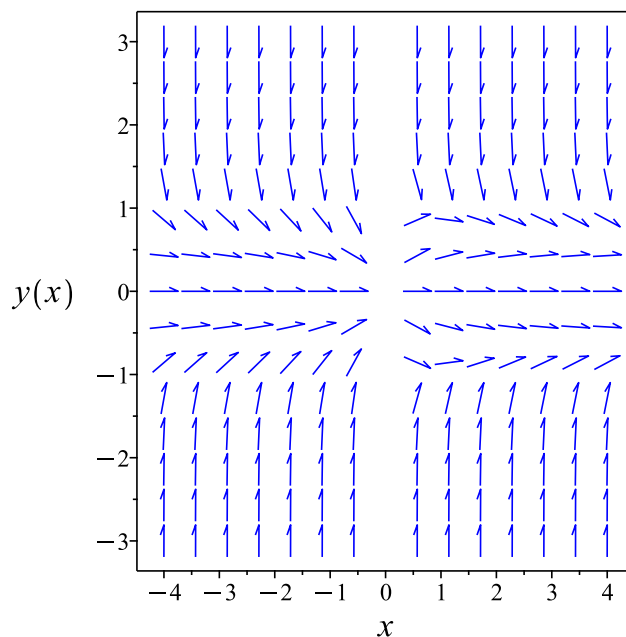
$$yx^{1/4} = -\frac{i3^{1/4}\sqrt{2}}{2}$$

Solving for y gives

$$\frac{\ln(4y^4x - 3)}{12} - \frac{\ln(yx^{1/4})}{3} = -\frac{\ln(x)}{4} + c_1$$

$$y = 0$$

$$y = -\frac{i3^{1/4}\sqrt{2}}{2x^{1/4}}$$

Figure 2.285: Slope field $2xy^5 - y + 2y'x = 0$ Summary of solutions found

$$\frac{\ln(4y^4x - 3)}{12} - \frac{\ln(yx^{1/4})}{3} = -\frac{\ln(x)}{4} + c_1$$

$$y = 0$$

$$y = -\frac{i3^{1/4}\sqrt{2}}{2x^{1/4}}$$

Solved using first_order_ode_homog_type_G

Time used: 0.091 (sec)

Solve

$$2xy^5 - y + 2y'x = 0$$

Multiplying the right side of the ode, which is $-\frac{y(2xy^4-1)}{2x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{y(2xy^4-1)}{2x} \\ &= -xy^4 + \frac{1}{2} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -x y^4 \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -4x y^4 \\ \alpha &= \frac{f_x}{f_y} \\ &= \frac{1}{4} \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^{1/4}} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -z^4 + \frac{1}{2}$$

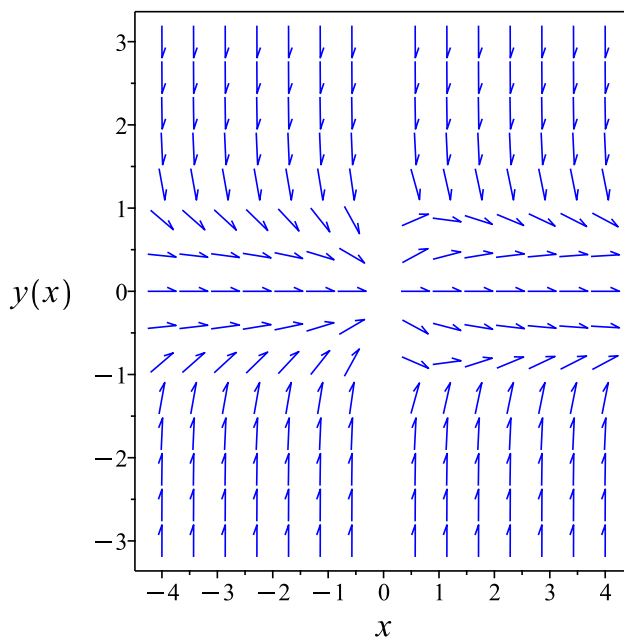
We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{yx^{1/4}} \frac{1}{z(-\frac{3}{4} + z^4)} dz = 0$$

Figure 2.286: Slope field $2xy^5 - y + 2y'x = 0$ Summary of solutions found

$$\ln(x) - c_1 + \int^{yx^{1/4}} \frac{1}{z(-\frac{3}{4} + z^4)} dz = 0$$

Solved using first_order_ode_LIE

Time used: 1.818 (sec)

Solve

$$2xy^5 - y + 2y'x = 0$$

Writing the ode as

$$y' = -\frac{y(2xy^4 - 1)}{2x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 - \frac{y(2x y^4 - 1)(b_3 - a_2)}{2x} - \frac{y^2(2x y^4 - 1)^2 a_3}{4x^2} \quad (5E)$$

$$- \left(-\frac{y^5}{x} + \frac{y(2x y^4 - 1)}{2x^2} \right) (xa_2 + ya_3 + a_1) - \left(-\frac{2x y^4 - 1}{2x} - 4y^4 \right) (xb_2 + yb_3 + b_1) = 0$$

Putting the above in normal form gives

$$\frac{-4x^2 y^{10} a_3 + 20x^3 y^4 b_2 + 4x^2 y^5 a_2 + 16x^2 y^5 b_3 + 4x y^6 a_3 + 20x^2 y^4 b_1 + 2b_2 x^2 + y^2 a_3 - 2xb_1 + 2ya_1}{4x^2} = 0$$

Setting the numerator to zero gives

$$\begin{aligned} & -4x^2 y^{10} a_3 + 20x^3 y^4 b_2 + 4x^2 y^5 a_2 + 16x^2 y^5 b_3 + 4x y^6 a_3 \\ & + 20x^2 y^4 b_1 + 2b_2 x^2 + y^2 a_3 - 2xb_1 + 2ya_1 = 0 \end{aligned} \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} & -4a_3 v_1^2 v_2^{10} + 4a_2 v_1^2 v_2^5 + 4a_3 v_1 v_2^6 + 20b_2 v_1^3 v_2^4 + 16b_3 v_1^2 v_2^5 \\ & + 20b_1 v_1^2 v_2^4 + a_3 v_2^2 + 2b_2 v_1^2 + 2a_1 v_2 - 2b_1 v_1 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$\begin{aligned} 20b_2v_1^3v_2^4 - 4a_3v_1^2v_2^{10} + (4a_2 + 16b_3)v_1^2v_2^5 + 20b_1v_1^2v_2^4 \\ + 2b_2v_1^2 + 4a_3v_1v_2^6 - 2b_1v_1 + a_3v_2^2 + 2a_1v_2 = 0 \end{aligned} \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} a_3 &= 0 \\ 2a_1 &= 0 \\ -4a_3 &= 0 \\ 4a_3 &= 0 \\ -2b_1 &= 0 \\ 20b_1 &= 0 \\ 2b_2 &= 0 \\ 20b_2 &= 0 \\ 4a_2 + 16b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= -4b_3 \\ a_3 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= b_3 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\begin{aligned} \xi &= -4x \\ \eta &= y \end{aligned}$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned} \eta &= \eta - \omega(x, y) \xi \\ &= y - \left(-\frac{y(2xy^4 - 1)}{2x} \right) (-4x) \\ &= -4xy^5 + 3y \\ \xi &= 0 \end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{-4x y^5 + 3y} dy \end{aligned}$$

Which results in

$$S = -\frac{\ln(4x y^4 - 3)}{12} + \frac{\ln(y)}{3}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = -\frac{y(2x y^4 - 1)}{2x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{y^4}{12x y^4 - 9} \\ S_y &= -\frac{1}{4x y^5 - 3y} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = \frac{1}{6R} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = \frac{1}{6R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

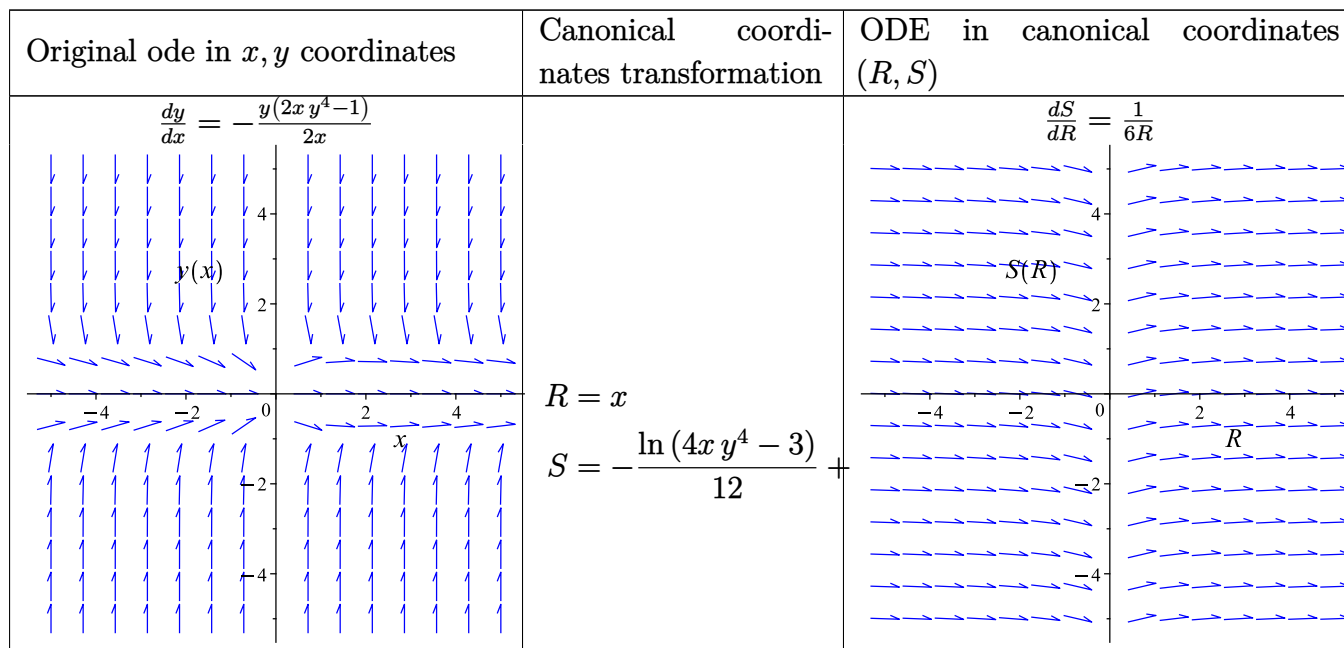
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

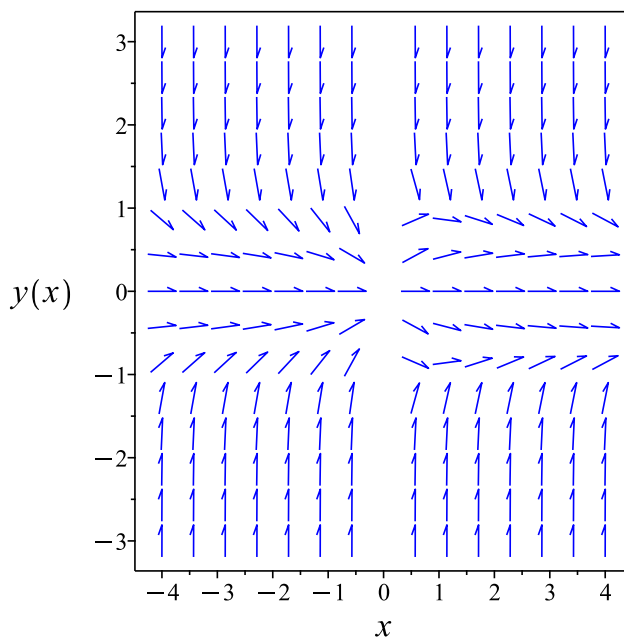
$$\begin{aligned} \int dS &= \int \frac{1}{6R} dR \\ S(R) &= \frac{\ln(R)}{6} + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\frac{\ln(4y^4x - 3)}{12} + \frac{\ln(y)}{3} = \frac{\ln(x)}{6} + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Figure 2.287: Slope field $2xy^5 - y + 2y'/x = 0$ Summary of solutions found

$$-\frac{\ln(4y^4x - 3)}{12} + \frac{\ln(y)}{3} = \frac{\ln(x)}{6} + c_2$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 137

```
ode:=2*y(x)^5*x-y(x)+2*diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{3^{1/4} \sqrt{-\sqrt{4x^3 + 3c_1}} x}{\sqrt{4x^3 + 3c_1}}$$

$$y = \frac{3^{1/4} \sqrt{\sqrt{4x^3 + 3c_1}} x}{\sqrt{4x^3 + 3c_1}}$$

$$y = -\frac{3^{1/4} \sqrt{-\sqrt{4x^3 + 3c_1}} x}{\sqrt{4x^3 + 3c_1}}$$

$$y = -\frac{3^{1/4} \sqrt{\sqrt{4x^3 + 3c_1}} x}{\sqrt{4x^3 + 3c_1}}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

```
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$2xy(x)^5 - y(x) + 2x\left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-2xy(x)^5 + y(x)}{2x}$$

✓ **Mathematica.** Time used: 0.139 (sec). Leaf size: 109

```
ode=(2*x*y[x]^5-y[x])+2*x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{x}}{\sqrt[4]{\frac{4x^3}{3} + c_1}}$$

$$y(x) \rightarrow -\frac{i\sqrt{x}}{\sqrt[4]{\frac{4x^3}{3} + c_1}}$$

$$y(x) \rightarrow \frac{i\sqrt{x}}{\sqrt[4]{\frac{4x^3}{3} + c_1}}$$

$$y(x) \rightarrow \frac{\sqrt{x}}{\sqrt[4]{\frac{4x^3}{3} + c_1}}$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 1.418 (sec). Leaf size: 88

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*y(x)**5 + 2*x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = -\sqrt[4]{3}\sqrt[4]{\frac{x^2}{C_1 + 4x^3}}, y(x) = \sqrt[4]{3}\sqrt[4]{\frac{x^2}{C_1 + 4x^3}}, y(x) = -\sqrt[4]{3}i\sqrt[4]{\frac{x^2}{C_1 + 4x^3}}, y(x) = \sqrt[4]{3}i\sqrt[4]{\frac{x^2}{C_1 + 4x^3}} \right]$$

2.4.18 Problem 19 (t)

Local contents

Solved using first_order_ode_exact	1257
✓ Maple	1261
✓ Mathematica	1262
✗ Sympy	1262

Internal problem ID [7934]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 19 (t)

Date solved : Thursday, November 27, 2025 at 06:34:15 AM

CAS classification : `[[_1st_order, '_with_symmetry_[F(x)*G(y),0]']]`

Solved using first_order_ode_exact

Time used: 0.194 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (-2 \cos(y) y + x(\sec(y) + \tan(y))) dy &= (-1 - \sin(y)) dx \\ (1 + \sin(y)) dx + (-2 \cos(y) y + x(\sec(y) + \tan(y))) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= 1 + \sin(y) \\ N(x, y) &= -2 \cos(y) y + x(\sec(y) + \tan(y)) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(1 + \sin(y)) \\ &= \cos(y) \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(-2 \cos(y) y + x(\sec(y) + \tan(y))) \\ &= \sec(y) + \tan(y) \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{\cos(y)}{-2y \cos(y)^2 + x \sin(y) + x} ((\cos(y)) - (\sec(y) + \tan(y))) \\ &= -\frac{\sin(y)}{2 \sin(y) y + x - 2y} \end{aligned}$$

Since A depends on y , it can not be used to obtain an integrating factor. We will now try a second method to find an integrating factor. Let

$$\begin{aligned} B &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{1 + \sin(y)} ((\sec(y) + \tan(y)) - (\cos(y))) \\ &= \tan(y) \end{aligned}$$

Since B does not depend on x , it can be used to obtain an integrating factor. Let the integrating factor be μ . Then

$$\begin{aligned} \mu &= e^{\int B \, dy} \\ &= e^{\int \tan(y) \, dy} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-\ln(\cos(y))} \\ &= \sec(y) \end{aligned}$$

M and N are now multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \sec(y) (1 + \sin(y)) \\ &= \sec(y) + \tan(y) \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \sec(y) (-2 \cos(y) y + x(\sec(y) + \tan(y))) \\ &= \frac{-2 \sin(y) y - x + 2y}{\sin(y) - 1} \end{aligned}$$

So now a modified ODE is obtained from the original ODE which will be exact and can be solved using the standard method. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (\sec(y) + \tan(y)) + \left(\frac{-2 \sin(y) y - x + 2y}{\sin(y) - 1} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \sec(y) + \tan(y) dx \\ \phi &= x(\sec(y) + \tan(y)) + f(y)\end{aligned}\tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\begin{aligned}\frac{\partial \phi}{\partial y} &= x(\tan(y) \sec(y) + 1 + \tan(y)^2) + f'(y) \\ &= -\frac{x}{\sin(y) - 1} + f'(y)\end{aligned}\tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{-2 \sin(y)y - x + 2y}{\sin(y) - 1}$. Therefore equation (4) becomes

$$\frac{-2 \sin(y)y - x + 2y}{\sin(y) - 1} = -\frac{x}{\sin(y) - 1} + f'(y)\tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = -2y$$

Integrating the above w.r.t y gives

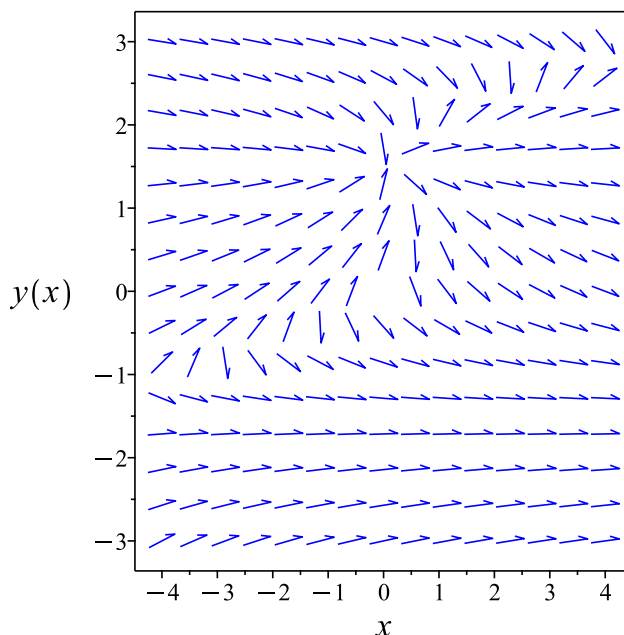
$$\begin{aligned}\int f'(y) dy &= \int (-2y) dy \\ f(y) &= -y^2 + c_1\end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = x(\sec(y) + \tan(y)) - y^2 + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = x(\sec(y) + \tan(y)) - y^2$$

Figure 2.288: Slope field $1 + \sin(y) = (2y \cos(y) - x(\sec(y) + \tan(y))) y'$ Summary of solutions found

$$x(\sec(y) + \tan(y)) - y^2 = c_1$$

✓ Maple. Time used: 0.029 (sec). Leaf size: 25

```
ode:=1+sin(y(x)) = (2*y(x)*cos(y(x))-x*(sec(y(x))+tan(y(x))))*diff(y(x),x);
dsolve(ode,y(x), singsol=all);
```

$$x + \frac{-y^2 - c_1}{\sec(y) + \tan(y)} = 0$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
<- 1st order linear successful
<- inverse linear successful
```

Maple step by step

Let's solve

$$1 + \sin(y(x)) = (2y(x) \cos(y(x)) - x(\sec(y(x)) + \tan(y(x)))) \left(\frac{d}{dx} y(x) \right)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = -\frac{-1 - \sin(y(x))}{2y(x) \cos(y(x)) - x(\sec(y(x)) + \tan(y(x)))}$$

✓ **Mathematica.** Time used: 0.692 (sec). Leaf size: 66

```
ode=(1+Sin[y[x]])==(2*y[x]*Cos[y[x]]-x*(Sec[y[x]]+Tan[y[x]]))*D[y[x],x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[x = y(x)^2 e^{-2 \operatorname{arctanh} \left(\tan \left(\frac{y(x)}{2} \right) \right)} + c_1 e^{-2 \operatorname{arctanh} \left(\tan \left(\frac{y(x)}{2} \right) \right)}, y(x) \right]$$

$$y(x) \rightarrow -\frac{\pi}{2}$$

$$y(x) \rightarrow \frac{3\pi}{2}$$

$$y(x) \rightarrow -\frac{\pi}{2}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x*(tan(y(x)) + 1/cos(y(x))) - 2*y(x)*cos(y(x)))*Derivative(y(x), x) +
        sin(y(x)) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.4.19 Problem 22 (a)**Local contents**

Existence and uniqueness analysis	1263
Solved using first_order_ode_linear	1264
Solved using first_order_ode_exact	1265
Solved using first_order_ode_LIE	1269
✓ Maple	1275
✓ Mathematica	1277
✓ Sympy	1277

Internal problem ID [7935]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 22 (a)

Date solved : Thursday, November 27, 2025 at 06:34:25 AM

CAS classification : [_linear]

Existence and uniqueness analysis

$$y'x = 2y + x^3e^x$$

$$y(1) = 0$$

This is a linear ODE. In canonical form it is written as

$$y' + q(x)y = p(x)$$

Where here

$$q(x) = -\frac{2}{x}$$

$$p(x) = x^2e^x$$

Hence the ode is

$$y' - \frac{2y}{x} = x^2e^x$$

The domain of $q(x) = -\frac{2}{x}$ is

$$\{x < 0 \vee 0 < x\}$$

And the point $x_0 = 1$ is inside this domain. The domain of $p(x) = x^2 e^x$ is

$$\{-\infty < x < \infty\}$$

And the point $x_0 = 1$ is also inside this domain. Hence solution exists and is unique.

Solved using first_order_ode_linear

Time used: 0.046 (sec)

Solve

$$y'x = 2y + x^3 e^x$$

$$y(1) = 0$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{2}{x}$$

$$p(x) = x^2 e^x$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) (x^2 e^x) \\ \frac{d}{dx}\left(\frac{y}{x^2}\right) &= \left(\frac{1}{x^2}\right) (x^2 e^x) \\ d\left(\frac{y}{x^2}\right) &= e^x \, dx\end{aligned}$$

Integrating gives

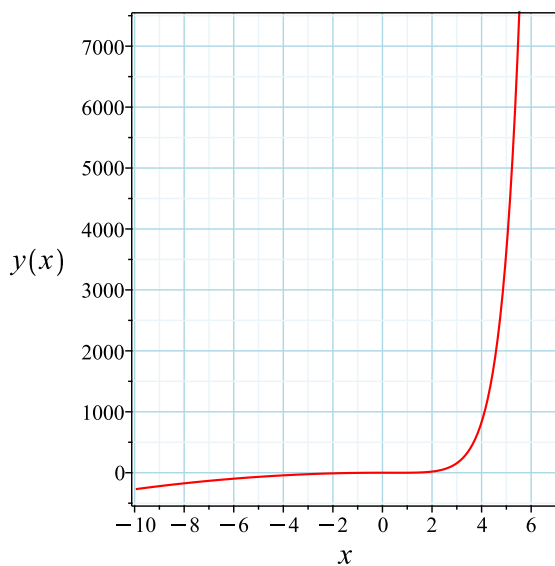
$$\begin{aligned}\frac{y}{x^2} &= \int e^x \, dx \\ &= e^x + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

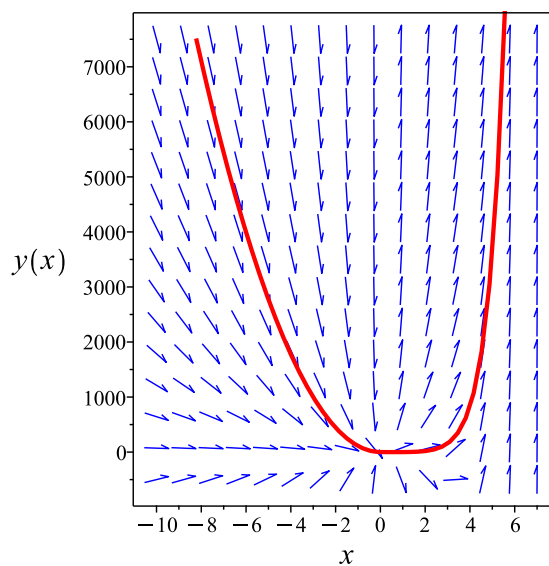
$$y = x^2(e^x + c_1)$$

Solving for initial conditions the solution is

$$y = x^2(e^x - e)$$



(a) Solution plot



(b) Slope field $y'x = 2y + x^3e^x$

Summary of solutions found

$$y = x^2(e^x - e)$$

Solved using first_order_ode_exact

Time used: 0.134 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N\end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned}(x) dy &= (2y + x^3 e^x) dx \\ (-2y - x^3 e^x) dx + (x) dy &= 0\end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= -2y - x^3 e^x \\ N(x, y) &= x\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-2y - x^3 e^x) \\ &= -2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x} ((-2) - (1)) \\ &= -\frac{3}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{3}{x} \, dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned} \mu &= e^{-3 \ln(x)} \\ &= \frac{1}{x^3} \end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned} \overline{M} &= \mu M \\ &= \frac{1}{x^3} (-2y - x^3 e^x) \\ &= \frac{-2y - x^3 e^x}{x^3} \end{aligned}$$

And

$$\begin{aligned} \overline{N} &= \mu N \\ &= \frac{1}{x^3} (x) \\ &= \frac{1}{x^2} \end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned} \overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-2y - x^3 e^x}{x^3} \right) + \left(\frac{1}{x^2} \right) \frac{dy}{dx} &= 0 \end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \quad (1)$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \quad (2)$$

Integrating (2) w.r.t. y gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial y} dy &= \int \overline{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{1}{x^2} dy \\ \phi &= \frac{y}{x^2} + f(x) \end{aligned} \quad (3)$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{2y}{x^3} + f'(x) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{-2y - x^3 e^x}{x^3}$. Therefore equation (4) becomes

$$\frac{-2y - x^3 e^x}{x^3} = -\frac{2y}{x^3} + f'(x) \quad (5)$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = -e^x$$

Integrating the above w.r.t x gives

$$\begin{aligned} \int f'(x) dx &= \int (-e^x) dx \\ f(x) &= -e^x + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{y}{x^2} - e^x + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

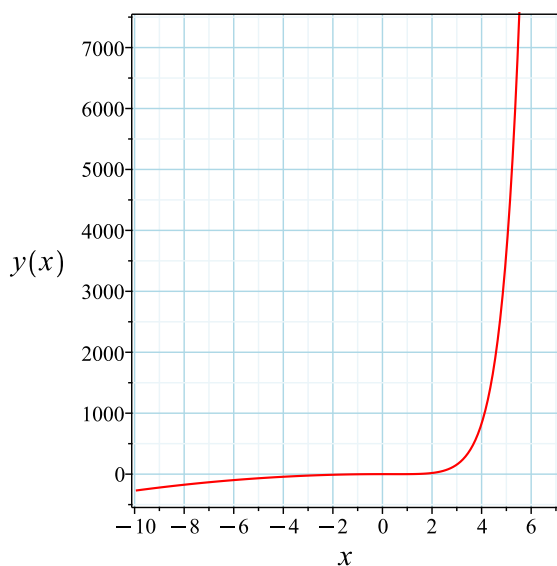
$$c_1 = \frac{y}{x^2} - e^x$$

Solving for initial conditions the solution is

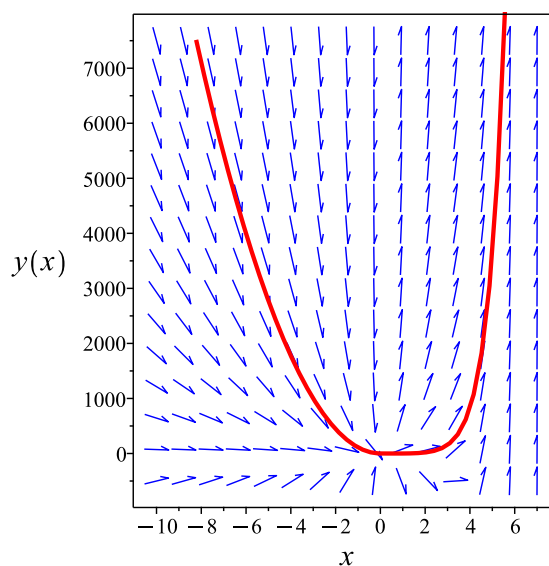
$$\frac{y}{x^2} - e^x = -e$$

Solving for y gives

$$y = x^2(e^x - e)$$



(a) Solution plot



(b) Slope field $y'x = 2y + x^3e^x$

Summary of solutions found

$$y = x^2(e^x - e)$$

Solved using first_order_ode_LIE

Time used: 0.902 (sec)

Solve

$$y'x = 2y + x^3e^x$$

$$y(1) = 0$$

Writing the ode as

$$y' = \frac{2y + x^3 e^x}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 2 to use as anstaz gives

$$\xi = x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1 \quad (\text{1E})$$

$$\eta = x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1 \quad (\text{2E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3, b_4, b_5, b_6\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$\begin{aligned} & 2xb_4 + yb_5 + b_2 + \frac{(2y + x^3 e^x)(-2xa_4 + xb_5 - ya_5 + 2yb_6 - a_2 + b_3)}{x} \\ & - \frac{(2y + x^3 e^x)^2 (xa_5 + 2ya_6 + a_3)}{x^2} \\ & - \left(\frac{3x^2 e^x + x^3 e^x}{x} - \frac{2y + x^3 e^x}{x^2} \right) (x^2 a_4 + y x a_5 + y^2 a_6 + x a_2 + y a_3 + a_1) \\ & - \frac{2(x^2 b_4 + y x b_5 + y^2 b_6 + x b_2 + y b_3 + b_1)}{x} = 0 \end{aligned} \quad (\text{5E})$$

Putting the above in normal form gives

$$\begin{aligned} & \underline{b_2 x^2 + 2y^2 a_3 + 2xb_1 - 2ya_1 + e^x x^4 y a_3 + 6e^x x^3 y a_3 + 7e^x x^4 y a_5 - 2e^x x^4 y b_6 + 10e^x x^3 y^2 a_6 + e^x x^5 y a_5 + 2e^x x^4 y a_4} \\ & = 0 \end{aligned}$$

Setting the numerator to zero gives

$$\begin{aligned} & -b_2 x^2 - 2y^2 a_3 - 2xb_1 + 2ya_1 - e^x x^4 y a_3 - 6e^x x^3 y a_3 - 7e^x x^4 y a_5 + 2e^x x^4 y b_6 \\ & - 10e^x x^3 y^2 a_6 - e^x x^5 y a_5 - 2e^{2x} x^6 y a_6 - e^x x^4 y^2 a_6 + y b_5 x^2 - 2x^2 y a_4 \\ & - 4x y^2 a_5 + 2x y^2 b_6 - e^x x^6 a_4 - 4e^x x^5 a_4 + e^x x^5 b_5 - e^x x^5 a_2 - e^x x^4 a_1 \\ & - 3e^x x^4 a_2 + e^x x^4 b_3 - 2e^x x^3 a_1 - e^{2x} x^6 a_3 - 6y^3 a_6 - e^{2x} x^7 a_5 = 0 \end{aligned} \quad (\text{6E})$$

Simplifying the above gives

$$\begin{aligned}
 & -b_2x^2 - 2y^2a_3 - 2xb_1 + 2ya_1 - e^xx^4ya_3 - 6e^xx^3ya_3 - 7e^xx^4ya_5 + 2e^xx^4yb_6 \\
 & - 10e^xx^3y^2a_6 - e^xx^5ya_5 - 2e^{2x}x^6ya_6 - e^xx^4y^2a_6 + yb_5x^2 - 2x^2ya_4 \\
 & - 4xy^2a_5 + 2xy^2b_6 - e^xx^6a_4 - 4e^xx^5a_4 + e^xx^5b_5 - e^xx^5a_2 - e^xx^4a_1 \\
 & - 3e^xx^4a_2 + e^xx^4b_3 - 2e^xx^3a_1 - e^{2x}x^6a_3 - 6y^3a_6 - e^{2x}x^7a_5 = 0
 \end{aligned} \tag{6E}$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y, e^x, e^{2x}\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2, e^x = v_3, e^{2x} = v_4\}$$

The above PDE (6E) now becomes

$$\begin{aligned}
 & -v_4v_1^7a_5 - 2v_4v_1^6v_2a_6 - v_4v_1^6a_3 - v_3v_1^6a_4 - v_3v_1^5v_2a_5 - v_3v_1^4v_2^2a_6 - v_3v_1^5a_2 \\
 & - v_3v_1^4v_2a_3 - 4v_3v_1^5a_4 - 7v_3v_1^4v_2a_5 - 10v_3v_1^3v_2^2a_6 + v_3v_1^5b_5 + 2v_3v_1^4v_2b_6 \\
 & - v_3v_1^4a_1 - 3v_3v_1^4a_2 - 6v_3v_1^3v_2a_3 + v_3v_1^4b_3 - 2v_3v_1^3a_1 - 2v_1^2v_2a_4 - 4v_1v_2^2a_5 \\
 & - 6v_2^3a_6 + v_2b_5v_1^2 + 2v_1v_2^2b_6 - 2v_2^2a_3 - b_2v_1^2 + 2v_2a_1 - 2v_1b_1 = 0
 \end{aligned} \tag{7E}$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2, v_3, v_4\}$$

Equation (7E) now becomes

$$\begin{aligned}
 & -v_4v_1^7a_5 - 2v_4v_1^6v_2a_6 - v_3v_1^6a_4 - v_4v_1^6a_3 - v_3v_1^5v_2a_5 + (-a_2 - 4a_4 + b_5)v_1^5v_3 \\
 & - v_3v_1^4v_2^2a_6 + (-a_3 - 7a_5 + 2b_6)v_1^4v_2v_3 + (-a_1 - 3a_2 + b_3)v_1^4v_3 \\
 & - 10v_3v_1^3v_2^2a_6 - 6v_3v_1^3v_2a_3 - 2v_3v_1^3a_1 + (-2a_4 + b_5)v_1^2v_2 - b_2v_1^2 \\
 & + (-4a_5 + 2b_6)v_1v_2^2 - 2v_1b_1 - 6v_2^3a_6 - 2v_2^2a_3 + 2v_2a_1 = 0
 \end{aligned} \tag{8E}$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -2a_1 &= 0 \\ 2a_1 &= 0 \\ -6a_3 &= 0 \\ -2a_3 &= 0 \\ -a_3 &= 0 \\ -a_4 &= 0 \\ -a_5 &= 0 \\ -10a_6 &= 0 \\ -6a_6 &= 0 \\ -2a_6 &= 0 \\ -a_6 &= 0 \\ -2b_1 &= 0 \\ -b_2 &= 0 \\ -2a_4 + b_5 &= 0 \\ -4a_5 + 2b_6 &= 0 \\ -a_1 - 3a_2 + b_3 &= 0 \\ -a_2 - 4a_4 + b_5 &= 0 \\ -a_3 - 7a_5 + 2b_6 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= 0 \\ a_3 &= 0 \\ a_4 &= 0 \\ a_5 &= 0 \\ a_6 &= 0 \\ b_1 &= 0 \\ b_2 &= 0 \\ b_3 &= 0 \\ b_4 &= b_4 \\ b_5 &= 0 \\ b_6 &= 0 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any

unknown in the RHS) gives

$$\xi = 0$$

$$\eta = x^2$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}\right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned} S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{x^2} dy \end{aligned}$$

Which results in

$$S = \frac{y}{x^2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{2y + x^3 e^x}{x}$$

Evaluating all the partial derivatives gives

$$R_x = 1$$

$$R_y = 0$$

$$S_x = -\frac{2y}{x^3}$$

$$S_y = \frac{1}{x^2}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = e^x \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = e^R$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

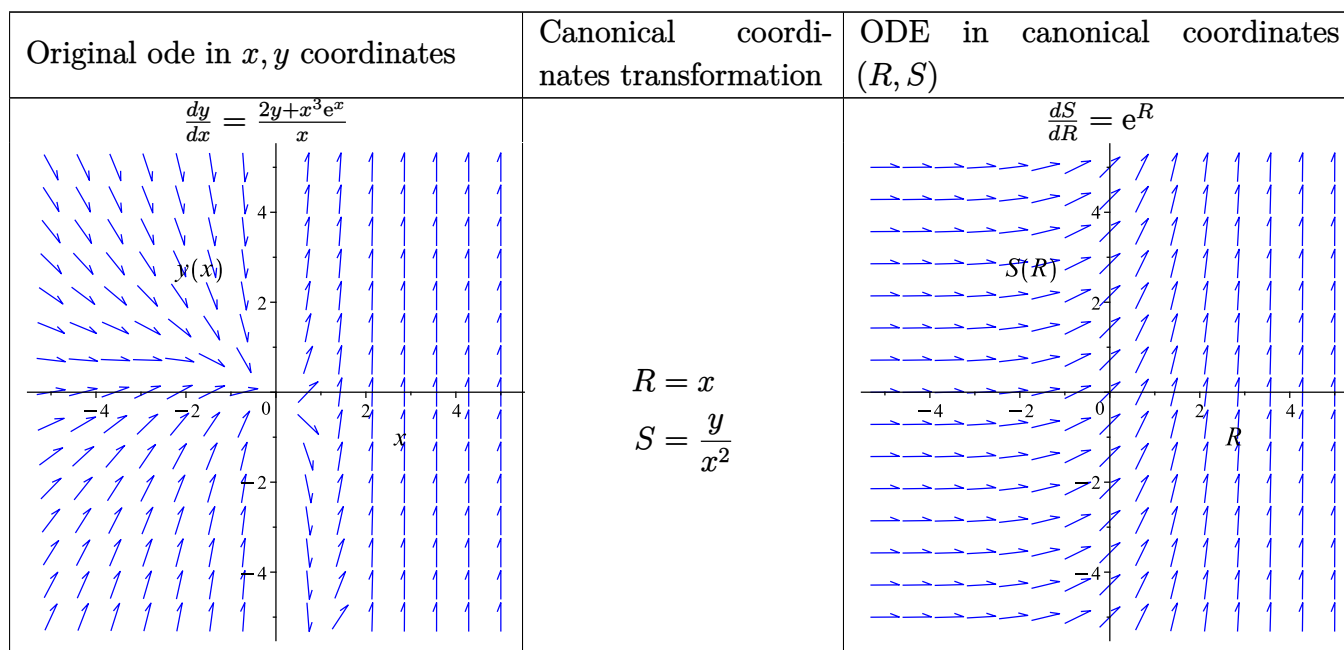
$$\int dS = \int e^R dR$$

$$S(R) = e^R + c_2$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$\frac{y}{x^2} = e^x + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.

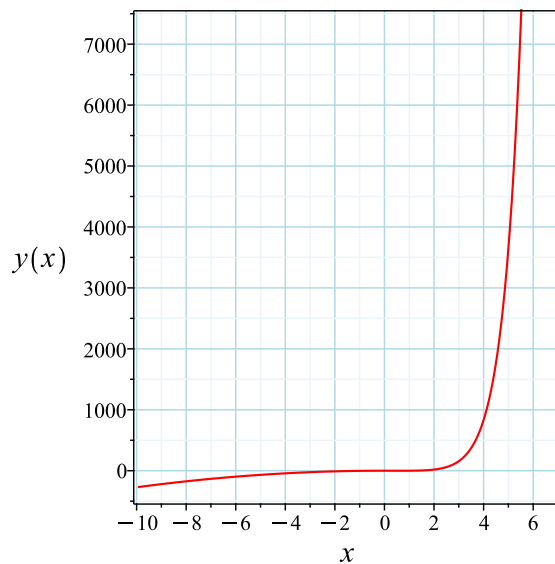


Solving for initial conditions the solution is

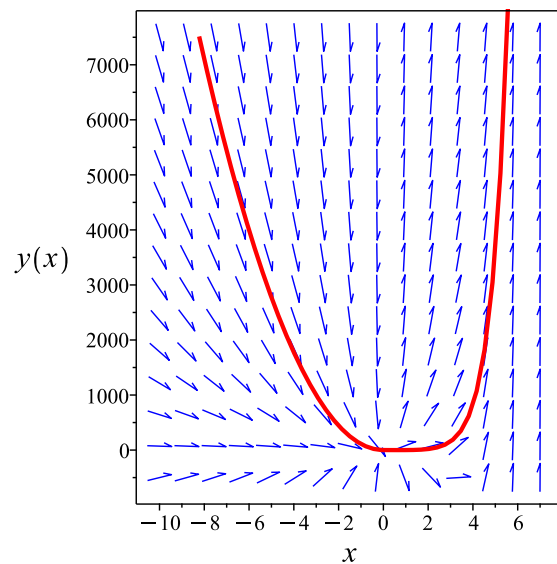
$$\frac{y}{x^2} = e^x - e$$

Solving for y gives

$$y = x^2(e^x - e)$$



(a) Solution plot



(b) Slope field $y'x = 2y + x^3e^x$

Summary of solutions found

$$y = x^2(e^x - e)$$

✓ Maple. Time used: 0.028 (sec). Leaf size: 15

```
ode:=diff(y(x),x)*x = 2*y(x)+exp(x)*x^3;
ic:=[y(1) = 0];
dsolve([ode,op(ic)],y(x), singsol=all);
```

$$y = (e^x - e)x^2$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\left[x \left(\frac{d}{dx} y(x) \right) = 2y(x) + x^3 e^x, y(1) = 0 \right]$$

- Highest derivative means the order of the ODE is 1
 $\frac{d}{dx} y(x)$
- Solve for the highest derivative
 $\frac{d}{dx} y(x) = \frac{2y(x) + x^3 e^x}{x}$
- Collect w.r.t. $y(x)$ and simplify
 $\frac{d}{dx} y(x) = \frac{2y(x)}{x} + x^2 e^x$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx} y(x) - \frac{2y(x)}{x} = x^2 e^x$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx} y(x) - \frac{2y(x)}{x} \right) = \mu(x) x^2 e^x$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx} (y(x) \mu(x))$
 $\mu(x) \left(\frac{d}{dx} y(x) - \frac{2y(x)}{x} \right) = \left(\frac{d}{dx} y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx} \mu(x) \right)$
- Isolate $\frac{d}{dx} \mu(x)$
 $\frac{d}{dx} \mu(x) = -\frac{2\mu(x)}{x}$
- Solve to find the integrating factor
 $\mu(x) = \frac{1}{x^2}$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx} (y(x) \mu(x)) \right) dx = \int \mu(x) x^2 e^x dx + C1$
- Evaluate the integral on the lhs
 $y(x) \mu(x) = \int \mu(x) x^2 e^x dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int \mu(x) x^2 e^x dx + C1}{\mu(x)}$
- Substitute $\mu(x) = \frac{1}{x^2}$
 $y(x) = x^2 \left(\int e^x dx + C1 \right)$
- Evaluate the integrals on the rhs
 $y(x) = x^2 (e^x + C1)$
- Use initial condition $y(1) = 0$
 $0 = e + C1$
- Solve for $C1$
 $C1 = -e$
- Substitute $C1 = -e$ into general solution and simplify
 $y(x) = x^2 (e^x - e)$
- Solution to the IVP
 $y(x) = x^2 (e^x - e)$

✓ **Mathematica.** Time used: 0.033 (sec). Leaf size: 16

```
ode=x*D[y[x],x]==2*y[x]+x^3*Exp[x];  
ic={y[1]==0};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow (e^x - e) x^2$$

✓ **Sympy.** Time used: 0.144 (sec). Leaf size: 12

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-x**3*exp(x) + x*Derivative(y(x), x) - 2*y(x),0)  
ics = {y(1): 0}  
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x^2(e^x - e)$$

2.4.20 Problem 22 (b)

Local contents

Existence and uniqueness analysis	1278
Solved using first_order_ode_linear	1279
Solved using first_order_ode_exact	1280
✓ Maple	1284
✓ Mathematica	1286
✓ Sympy	1286

Internal problem ID [7936]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 22 (b)

Date solved : Thursday, November 27, 2025 at 06:34:32 AM

CAS classification : `[_linear, 'class A']`

Existence and uniqueness analysis

$$Li' + Ri = E \sin(2t)$$

$$i(0) = 0$$

This is a linear ODE. In canonical form it is written as

$$i' + q(t)i = p(t)$$

Where here

$$q(t) = \frac{R}{L}$$

$$p(t) = \frac{E \sin(2t)}{L}$$

Hence the ode is

$$i' + \frac{Ri}{L} = \frac{E \sin(2t)}{L}$$

The domain of $q(t) = \frac{R}{L}$ is

$$\{-\infty < t < \infty\}$$

And the point $t_0 = 0$ is inside this domain. The domain of $p(t) = \frac{E \sin(2t)}{L}$ is

$$\{-\infty < t < \infty\}$$

And the point $t_0 = 0$ is also inside this domain. Hence solution exists and is unique.

Solved using first_order_ode_linear

Time used: 0.215 (sec)

Solve

$$Li' + Ri = E \sin(2t)$$

$$i(0) = 0$$

In canonical form a linear first order is

$$i' + q(t)i = p(t)$$

Comparing the above to the given ode shows that

$$q(t) = \frac{R}{L}$$

$$p(t) = \frac{E \sin(2t)}{L}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dt} \\ &= e^{\int \frac{R}{L} dt} \\ &= e^{\frac{Rt}{L}}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dt}(\mu i) &= \mu p \\ \frac{d}{dt}(\mu i) &= (\mu) \left(\frac{E \sin(2t)}{L} \right) \\ \frac{d}{dt} \left(i e^{\frac{Rt}{L}} \right) &= \left(e^{\frac{Rt}{L}} \right) \left(\frac{E \sin(2t)}{L} \right) \\ d \left(i e^{\frac{Rt}{L}} \right) &= \left(\frac{E \sin(2t) e^{\frac{Rt}{L}}}{L} \right) dt\end{aligned}$$

Integrating gives

$$\begin{aligned} i e^{\frac{Rt}{L}} &= \int \frac{E \sin(2t) e^{\frac{Rt}{L}}}{L} dt \\ &= -\frac{2 e^{\frac{Rt}{L}} E \left(L \cos(2t) - \frac{\sin(2t)R}{2} \right)}{4L^2 + R^2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $e^{\frac{Rt}{L}}$ gives the final solution

$$i = -\frac{2E \left(L \cos(2t) - \frac{\sin(2t)R}{2} \right)}{4L^2 + R^2} + e^{-\frac{Rt}{L}} c_1$$

Solving for initial conditions the solution is

$$i = \frac{E \left(2L e^{-\frac{Rt}{L}} - 2L \cos(2t) + \sin(2t) R \right)}{4L^2 + R^2}$$

Summary of solutions found

$$i = \frac{E \left(2L e^{-\frac{Rt}{L}} - 2L \cos(2t) + \sin(2t) R \right)}{4L^2 + R^2}$$

Solved using first_order_ode_exact

Time used: 0.163 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \tag{A}$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \tag{B}$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(t, i) dt + N(t, i) di = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (L) di &= (-Ri + E \sin(2t)) dt \\ (Ri - E \sin(2t)) dt + (L) di &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(t, i) &= Ri - E \sin(2t) \\ N(t, i) &= L \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial i} = \frac{\partial N}{\partial t}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial i} &= \frac{\partial}{\partial i}(Ri - E \sin(2t)) \\ &= R \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial t} &= \frac{\partial}{\partial t}(L) \\ &= 0 \end{aligned}$$

Since $\frac{\partial M}{\partial i} \neq \frac{\partial N}{\partial t}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial i} - \frac{\partial N}{\partial t} \right) \\ &= \frac{1}{L} ((R) - (0)) \\ &= \frac{R}{L} \end{aligned}$$

Since A does not depend on i , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A dt} \\ &= e^{\int \frac{R}{L} dt}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{\frac{Rt}{L}} \\ &= e^{\frac{Rt}{L}}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= e^{\frac{Rt}{L}} (Ri - E \sin(2t)) \\ &= (Ri - E \sin(2t)) e^{\frac{Rt}{L}}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= e^{\frac{Rt}{L}} (L) \\ &= L e^{\frac{Rt}{L}}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{di}{dt} &= 0 \\ \left((Ri - E \sin(2t)) e^{\frac{Rt}{L}} \right) + \left(L e^{\frac{Rt}{L}} \right) \frac{di}{dt} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(t, i)$

$$\frac{\partial \phi}{\partial t} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial i} = \overline{N} \tag{2}$$

Integrating (2) w.r.t. i gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial i} di &= \int \overline{N} di \\ \int \frac{\partial \phi}{\partial i} di &= \int L e^{\frac{Rt}{L}} di \\ \phi &= L e^{\frac{Rt}{L}} i + f(t)\end{aligned} \tag{3}$$

Where $f(t)$ is used for the constant of integration since ϕ is a function of both t and i . Taking derivative of equation (3) w.r.t t gives

$$\frac{\partial \phi}{\partial t} = R e^{\frac{Rt}{L}} i + f'(t) \quad (4)$$

But equation (1) says that $\frac{\partial \phi}{\partial t} = (Ri - E \sin(2t)) e^{\frac{Rt}{L}}$. Therefore equation (4) becomes

$$(Ri - E \sin(2t)) e^{\frac{Rt}{L}} = R e^{\frac{Rt}{L}} i + f'(t) \quad (5)$$

Solving equation (5) for $f'(t)$ gives

$$f'(t) = -E \sin(2t) e^{\frac{Rt}{L}}$$

Integrating the above w.r.t t gives

$$\begin{aligned} \int f'(t) dt &= \int \left(-E \sin(2t) e^{\frac{Rt}{L}} \right) dt \\ f(t) &= -E \left(-\frac{2 e^{\frac{Rt}{L}} \cos(2t)}{\frac{R^2}{L^2} + 4} + \frac{R e^{\frac{Rt}{L}} \sin(2t)}{L \left(\frac{R^2}{L^2} + 4 \right)} \right) + c_1 \end{aligned}$$

Where c_1 is constant of integration. Substituting result found above for $f(t)$ into equation (3) gives ϕ

$$\phi = L e^{\frac{Rt}{L}} i - E \left(-\frac{2 e^{\frac{Rt}{L}} \cos(2t)}{\frac{R^2}{L^2} + 4} + \frac{R e^{\frac{Rt}{L}} \sin(2t)}{L \left(\frac{R^2}{L^2} + 4 \right)} \right) + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = L e^{\frac{Rt}{L}} i - E \left(-\frac{2 e^{\frac{Rt}{L}} \cos(2t)}{\frac{R^2}{L^2} + 4} + \frac{R e^{\frac{Rt}{L}} \sin(2t)}{L \left(\frac{R^2}{L^2} + 4 \right)} \right)$$

Simplifying the above gives

$$e^{\frac{Rt}{L}} \left(Li - \frac{E(-2L^2 \cos(2t) + LR \sin(2t))}{4L^2 + R^2} \right) = c_1$$

Solving for initial conditions the solution is

$$e^{\frac{Rt}{L}} \left(Li - \frac{E(-2L^2 \cos(2t) + LR \sin(2t))}{4L^2 + R^2} \right) = \frac{2L^2 E}{4L^2 + R^2}$$

Solving for i gives

$$i = -\frac{E\left(2\cos(2t)e^{\frac{Rt}{L}}L - Re^{\frac{Rt}{L}}\sin(2t) - 2L\right)e^{-\frac{Rt}{L}}}{4L^2 + R^2}$$

Summary of solutions found

$$i = -\frac{E\left(2\cos(2t)e^{\frac{Rt}{L}}L - Re^{\frac{Rt}{L}}\sin(2t) - 2L\right)e^{-\frac{Rt}{L}}}{4L^2 + R^2}$$

✓ **Maple.** Time used: 0.019 (sec). Leaf size: 42

```
ode:=L*diff(i(t),t)+R*i(t) = E*sin(2*t);
ic:=[i(0) = 0];
dsolve([ode,op(ic)],i(t), singsol=all);
```

$$i = \frac{E\left(2Le^{-\frac{Rt}{L}} - 2L\cos(2t) + \sin(2t)R\right)}{4L^2 + R^2}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\left[L\left(\frac{d}{dt}i(t)\right) + Ri(t) = E\sin(2t), i(0) = 0\right]$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dt}i(t)$$

- Solve for the highest derivative

$$\frac{d}{dt}i(t) = \frac{-Ri(t) + E\sin(2t)}{L}$$

- Collect w.r.t. $i(t)$ and simplify

$$\frac{d}{dt}i(t) = -\frac{Ri(t)}{L} + \frac{E\sin(2t)}{L}$$

- Group terms with $i(t)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dt}i(t) + \frac{Ri(t)}{L} = \frac{E\sin(2t)}{L}$$

- The ODE is linear; multiply by an integrating factor $\mu(t)$

- $$\mu(t) \left(\frac{d}{dt} i(t) + \frac{Ri(t)}{L} \right) = \frac{\mu(t)E \sin(2t)}{L}$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dt}(i(t) \mu(t))$

$$\mu(t) \left(\frac{d}{dt} i(t) + \frac{Ri(t)}{L} \right) = \left(\frac{d}{dt} i(t) \right) \mu(t) + i(t) \left(\frac{d}{dt} \mu(t) \right)$$
- Isolate $\frac{d}{dt} \mu(t)$

$$\frac{d}{dt} \mu(t) = \frac{\mu(t)R}{L}$$
- Solve to find the integrating factor

$$\mu(t) = e^{\frac{Rt}{L}}$$
- Integrate both sides with respect to t

$$\int \left(\frac{d}{dt} (i(t) \mu(t)) \right) dt = \int \frac{\mu(t)E \sin(2t)}{L} dt + C1$$
- Evaluate the integral on the lhs

$$i(t) \mu(t) = \int \frac{\mu(t)E \sin(2t)}{L} dt + C1$$
- Solve for $i(t)$

$$i(t) = \frac{\int \frac{\mu(t)E \sin(2t)}{L} dt + C1}{\mu(t)}$$
- Substitute $\mu(t) = e^{\frac{Rt}{L}}$

$$i(t) = \frac{\int \frac{e^{\frac{Rt}{L}} E \sin(2t)}{L} dt + C1}{e^{\frac{Rt}{L}}}$$
- Evaluate the integrals on the rhs

$$i(t) = \frac{E \left(-\frac{2e^{\frac{Rt}{L}} \cos(2t)}{\frac{R^2}{L^2} + 4} + \frac{R e^{\frac{Rt}{L}} \sin(2t)}{L \left(\frac{R^2}{L^2} + 4 \right)} \right)}{\frac{L}{e^{\frac{Rt}{L}}}} + C1$$
- Simplify

$$i(t) = e^{-\frac{Rt}{L}} C1 + \frac{E(-2L^2 \cos(2t) + LR \sin(2t))}{(4L^2 + R^2)L}$$
- Use initial condition $i(0) = 0$

$$0 = C1 - \frac{2EL}{4L^2 + R^2}$$
- Solve for $C1$

$$C1 = \frac{2EL}{4L^2 + R^2}$$
- Substitute $C1 = \frac{2EL}{4L^2 + R^2}$ into general solution and simplify

$$i(t) = -\frac{2 \left(L \cos(2t) - L e^{-\frac{Rt}{L}} - \frac{\sin(2t)R}{2} \right) E}{4L^2 + R^2}$$
- Solution to the IVP

$$i(t) = -\frac{2 \left(L \cos(2t) - L e^{-\frac{Rt}{L}} - \frac{\sin(2t)R}{2} \right) E}{4L^2 + R^2}$$

✓ **Mathematica.** Time used: 0.08 (sec). Leaf size: 41

```
ode=L*D[i[t],t]+R*i[t]==e*Sin[2*t];
ic={i[0]==0};
DSolve[{ode,ic},i[t],t,IncludeSingularSolutions->True]
```

$$i(t) \rightarrow e^{-\frac{Rt}{L}} \int_0^t \frac{ee^{\frac{RK[1]}{L}} \sin(2K[1])}{L} dK[1]$$

✓ **Sympy.** Time used: 0.159 (sec). Leaf size: 63

```
from sympy import *
t = symbols("t")
E = symbols("E")
L = symbols("L")
R = symbols("R")
i = Function("i")
ode = Eq(L*Derivative(i(t), t) + R*i(t) - E*sin(2*t),0)
ics = {i(0): 0}
dsolve(ode,func=i(t),ics=ics)
```

$$i(t) = -\frac{2eL \cos(2t)}{4L^2 + R^2} + \frac{2eLe^{-\frac{Rt}{L}}}{4L^2 + R^2} + \frac{eR \sin(2t)}{4L^2 + R^2}$$

2.4.21 Problem 23 (a)

Local contents

Solved using first_order_ode_exact	1287
✓ Maple	1291
✓ Mathematica	1292
✗ Sympy	1292

Internal problem ID [7937]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 23 (a)

Date solved : Thursday, November 27, 2025 at 06:34:40 AM

CAS classification : ['y=_G(x,y)']

Solved using first_order_ode_exact

Time used: 0.125 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x^2 \cos(y)) dy &= (2x \sin(y) - 1) dx \\ (-2x \sin(y) + 1) dx + (x^2 \cos(y)) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -2x \sin(y) + 1 \\ N(x, y) &= x^2 \cos(y) \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-2x \sin(y) + 1) \\ &= -2 \cos(y) x \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x^2 \cos(y)) \\ &= 2 \cos(y) x \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{\sec(y)}{x^2} ((-2 \cos(y) x) - (2 \cos(y) x)) \\ &= -\frac{4}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int -\frac{4}{x} dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-4 \ln(x)} \\ &= \frac{1}{x^4}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^4}(-2x \sin(y) + 1) \\ &= \frac{-2x \sin(y) + 1}{x^4}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{1}{x^4}(x^2 \cos(y)) \\ &= \frac{\cos(y)}{x^2}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-2x \sin(y) + 1}{x^4} \right) + \left(\frac{\cos(y)}{x^2} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int \frac{-2x \sin(y) + 1}{x^4} dx \\ \phi &= \frac{3x \sin(y) - 1}{3x^3} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{\cos(y)}{x^2} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{\cos(y)}{x^2}$. Therefore equation (4) becomes

$$\frac{\cos(y)}{x^2} = \frac{\cos(y)}{x^2} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

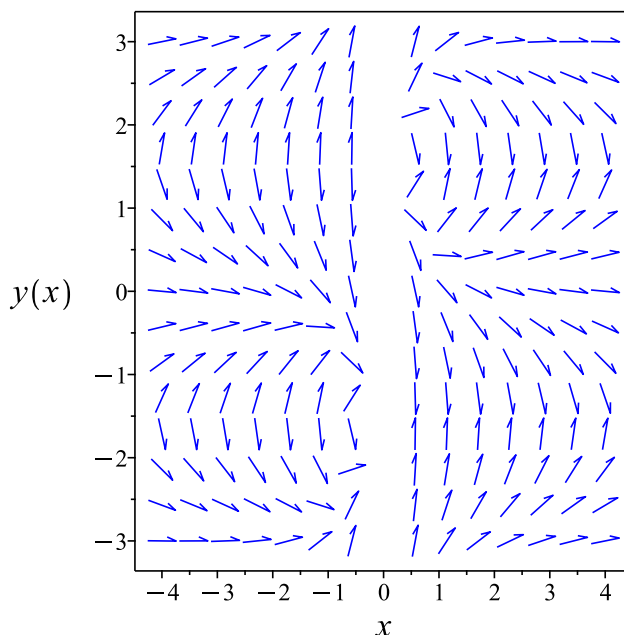
$$\phi = \frac{3x \sin(y) - 1}{3x^3} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{3x \sin(y) - 1}{3x^3}$$

Solving for y gives

$$y = \arcsin\left(\frac{3c_1 x^3 + 1}{3x}\right)$$

Figure 2.292: Slope field $x^2 \cos(y) y' = 2x \sin(y) - 1$ Summary of solutions found

$$y = \arcsin\left(\frac{3c_1 x^3 + 1}{3x}\right)$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 20

```
ode:=x^2*cos(y(x))*diff(y(x),x) = 2*x*sin(y(x))-1;
dsolve(ode,y(x), singsol=all);
```

$$y = -\arcsin\left(\frac{3c_1 x^3 - 1}{3x}\right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
```

```
trying exact
<- exact successful
```

Maple step by step

Let's solve

$$x^2 \cos(y(x)) \left(\frac{d}{dx} y(x) \right) = 2x \sin(y(x)) - 1$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx} y(x) = \frac{2x \sin(y(x)) - 1}{x^2 \cos(y(x))}$$

✓ **Mathematica.** Time used: 8.535 (sec). Leaf size: 21

```
ode=x^2*Cos[y[x]]*D[y[x],x]==2*x*Sin[y[x]]-1;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \arcsin\left(\frac{1}{3x} + 2c_1 x^2\right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*cos(y(x))*Derivative(y(x), x) - 2*x*sin(y(x)) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (2*x*sin(y(x)) - 1)/(x
**2*cos(y(x))) cannot be solved by the factorable group method
```


2.4.22 Problem 23 (b)

Local contents

✓ Maple	1293
✓ Mathematica	1295
✓ Sympy	1295

Internal problem ID [7938]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 23 (b)

Date solved : Thursday, November 27, 2025 at 06:34:48 AM

CAS classification : [_rational]

$$4x^2yy' = 3x(3y^2 + 2) + 2(3y^2 + 2)^3$$

Unknown ode type.

✓ **Maple.** Time used: 0.009 (sec). Leaf size: 175

```
ode:=4*x^2*y(x)*diff(y(x),x) = 3*x*(3*y(x)^2+2)+2*(3*y(x)^2+2)^3;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{6} \sqrt{\frac{-3c_1 x^8 + \sqrt{-3(c_1 x^8 - \frac{1}{3})c_1 x^9 + 1}}{3c_1 x^8 - 1}}}{3}$$

$$y = \frac{\sqrt{6} \sqrt{\frac{-3c_1 x^8 + \sqrt{-3(c_1 x^8 - \frac{1}{3})c_1 x^9 + 1}}{3c_1 x^8 - 1}}}{3}$$

$$y = -\frac{\sqrt{\frac{-18c_1 x^8 - 6\sqrt{-3(c_1 x^8 - \frac{1}{3})c_1 x^9 + 6}}{3c_1 x^8 - 1}}}{3}$$

$$y = \frac{\sqrt{\frac{-18c_1 x^8 - 6\sqrt{-3(c_1 x^8 - \frac{1}{3})c_1 x^9 + 6}}{3c_1 x^8 - 1}}}{3}$$

Maple trace

Methods for first order ODEs:

```

--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
Looking for potential symmetries
trying inverse_Riccati
trying an equivalence to an Abel ODE
differential order: 1; trying a linearization to 2nd order
--- trying a change of variables {x -> y(x), y(x) -> x}
differential order: 1; trying a linearization to 2nd order
trying 1st order ODE linearizable_by_differentiation
--- Trying Lie symmetry methods, 1st order ---
-> Computing symmetries using: way = 2
[0, (4/3*x*y^2+4/27*x^2*y^2+2*x*y^4+x*y^6+8/27*x+8/81*x^2)/x^2/y]
<- successful computation of symmetries.
trying an integrating factor from the invariance group
<- integrating factor successful

```

Maple step by step

Let's solve

$$4x^2y(x) \left(\frac{d}{dx}y(x) \right) = 3x(3y(x)^2 + 2) + 2(3y(x)^2 + 2)^3$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{3x(3y(x)^2+2)+2(3y(x)^2+2)^3}{4x^2y(x)}$$

✓ **Mathematica.** Time used: 15.893 (sec). Leaf size: 277

```
ode=4*x^2*y[x]*D[y[x],x]==3*x*(3*y[x]^2+2)+2*(3*y[x]^2+2)^3;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{1}{3}\sqrt{2}\sqrt{-\frac{3x^8 + \sqrt{3}\sqrt{-x^9(x^8 + 72c_1)} + 216c_1}{x^8 + 72c_1}}$$

$$y(x) \rightarrow \frac{1}{3}\sqrt{2}\sqrt{-\frac{3x^8 + \sqrt{3}\sqrt{-x^9(x^8 + 72c_1)} + 216c_1}{x^8 + 72c_1}}$$

$$y(x) \rightarrow -\frac{1}{3}\sqrt{2}\sqrt{\frac{-3x^8 + \sqrt{3}\sqrt{-x^9(x^8 + 72c_1)} - 216c_1}{x^8 + 72c_1}}$$

$$y(x) \rightarrow \frac{1}{3}\sqrt{2}\sqrt{\frac{-3x^8 + \sqrt{3}\sqrt{-x^9(x^8 + 72c_1)} - 216c_1}{x^8 + 72c_1}}$$

$$y(x) \rightarrow -i\sqrt{\frac{2}{3}}$$

$$y(x) \rightarrow i\sqrt{\frac{2}{3}}$$

✓ **Sympy.** Time used: 8.845 (sec). Leaf size: 194

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x**2*y(x)*Derivative(y(x), x) - 3*x*(3*y(x)**2 + 2) - 2*(3*y(x)**2 +
2)**3, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\begin{array}{l} y(x) = \\ -\frac{\sqrt{2}\sqrt{-\frac{48C_1-3x^8-\sqrt{3}\sqrt{x^9(16C_1-x^8)}}{16C_1-x^8}}}{3}, y(x) = \frac{\sqrt{2}\sqrt{-\frac{48C_1-3x^8-\sqrt{3}\sqrt{x^9(16C_1-x^8)}}{16C_1-x^8}}}{3}, y(x) = \\ -\frac{\sqrt{2}\sqrt{-\frac{48C_1-3x^8+\sqrt{3}\sqrt{x^9(16C_1-x^8)}}{16C_1-x^8}}}{3}, y(x) = \frac{\sqrt{2}\sqrt{-\frac{48C_1-3x^8+\sqrt{3}\sqrt{x^9(16C_1-x^8)}}{16C_1-x^8}}}{3} \end{array} \right]$$

2.4.23 Problem 23 (c)

Local contents

Solved using first_order_ode_bernoulli	1296
Solved using first_order_ode_exact	1299
✓ Maple	1303
✓ Mathematica	1304
✓ Sympy	1304

Internal problem ID [7939]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 23 (c)

Date solved : Thursday, November 27, 2025 at 06:35:06 AM

CAS classification : [_Bernoulli]

Solved using first_order_ode_bernoulli

Time used: 0.163 (sec)

Solve

$$xy^3 - y^3 - x^2e^x + 3xy^2y' = 0$$

In canonical form, the ODE is

$$\begin{aligned} y' &= F(x, y) \\ &= \frac{-xy^3 + y^3 + x^2e^x}{3xy^2} \end{aligned}$$

This is a Bernoulli ODE.

$$y' = \left(\frac{1-x}{3x} \right) y + \left(\frac{x e^x}{3} \right) \frac{1}{y^2} \quad (1)$$

The standard Bernoulli ODE has the form

$$y' = f_0(x)y + f_1(x)y^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1-x}{3x} \\ f_1 &= \frac{x e^x}{3} \end{aligned}$$

The first step is to divide the above equation by y^n which gives

$$\frac{y'}{y^n} = f_0(x)y^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = y^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $y(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= \frac{1-x}{3x} \\ f_1(x) &= \frac{x e^x}{3} \\ n &= -2 \end{aligned}$$

Dividing both sides of ODE (1) by $y^n = \frac{1}{y^2}$ gives

$$y'y^2 = \frac{(1-x)y^3}{3x} + \frac{x e^x}{3} \quad (4)$$

Let

$$\begin{aligned} v &= y^{1-n} \\ &= y^3 \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = 3y^2 y' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(x)}{3} &= \frac{(1-x)v(x)}{3x} + \frac{x e^x}{3} \\ v' &= \frac{(1-x)v}{x} + x e^x \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{1-x}{x} \\ p(x) &= x e^x \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1-x}{x} dx} \\ &= \frac{e^x}{x}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu)(x e^x) \\ \frac{d}{dx}\left(\frac{v e^x}{x}\right) &= \left(\frac{e^x}{x}\right)(x e^x) \\ d\left(\frac{v e^x}{x}\right) &= e^{2x} dx\end{aligned}$$

Integrating gives

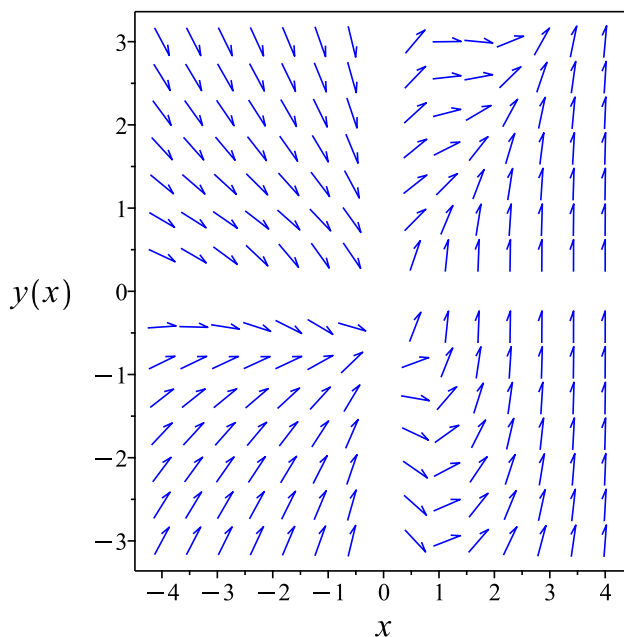
$$\begin{aligned}\frac{v e^x}{x} &= \int e^{2x} dx \\ &= \frac{e^{2x}}{2} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{e^x}{x}$ gives the final solution

$$v(x) = x\left(\frac{e^x}{2} + e^{-x}c_1\right)$$

The substitution $v = y^{1-n}$ is now used to convert the above solution back to y which results in

$$y^3 = x\left(\frac{e^x}{2} + e^{-x}c_1\right)$$

Figure 2.293: Slope field $xy^3 - y^3 - x^2e^x + 3xy^2y' = 0$ Summary of solutions found

$$y^3 = x \left(\frac{e^x}{2} + e^{-x}c_1 \right)$$

Solved using first_order_ode_exact

Time used: 0.141 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (3x y^2) dy &= (-x y^3 + y^3 + x^2 e^x) dx \\ (x y^3 - y^3 - x^2 e^x) dx + (3x y^2) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= x y^3 - y^3 - x^2 e^x \\ N(x, y) &= 3x y^2 \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x y^3 - y^3 - x^2 e^x) \\ &= 3y^2(x - 1) \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (3x y^2) \\ &= 3y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{3x y^2} ((3x y^2 - 3y^2) - (3y^2)) \\ &= \frac{x - 2}{x} \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int \frac{x-2}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{x-2\ln(x)} \\ &= \frac{e^x}{x^2}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{e^x}{x^2}(xy^3 - y^3 - x^2e^x) \\ &= -\frac{(x^2e^x - y^3(x-1))e^x}{x^2}\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= \frac{e^x}{x^2}(3xy^2) \\ &= \frac{3y^2e^x}{x}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ \left(-\frac{(x^2e^x - y^3(x-1))e^x}{x^2}\right) + \left(\frac{3y^2e^x}{x}\right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int -\frac{(x^2 e^x - y^3(x-1)) e^x}{x^2} dx \\ \phi &= -\frac{e^x(-2y^3 + x e^x)}{2x} + f(y)\end{aligned}\quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = \frac{3y^2 e^x}{x} + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = \frac{3y^2 e^x}{x}$. Therefore equation (4) becomes

$$\frac{3y^2 e^x}{x} = \frac{3y^2 e^x}{x} + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

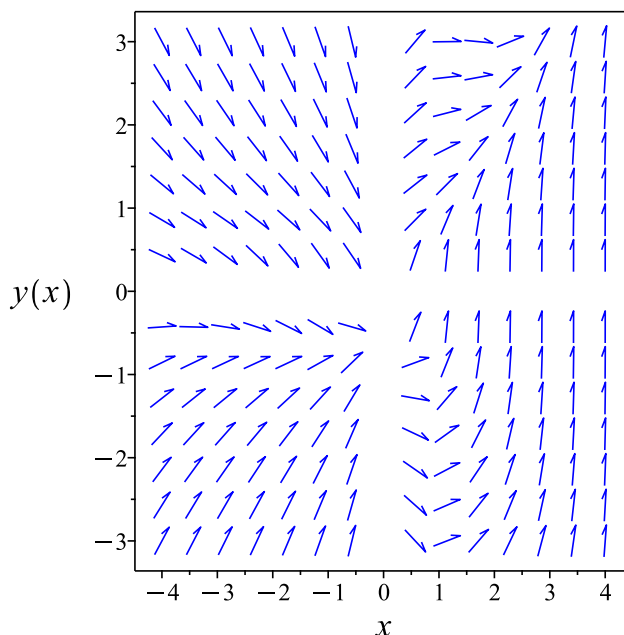
$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -\frac{e^x(-2y^3 + x e^x)}{2x} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -\frac{e^x(-2y^3 + x e^x)}{2x}$$

Figure 2.294: Slope field $xy^3 - y^3 - x^2e^x + 3xy^2y' = 0$ Summary of solutions found

$$-\frac{e^x(-2y^3 + x e^x)}{2x} = c_1$$

✓ Maple. Time used: 0.004 (sec). Leaf size: 101

```
ode:=x*y(x)^3-y(x)^3-exp(x)*x^2+3*x*y(x)^2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{2^{2/3}((e^{2x} + 2c_1)x e^{2x})^{1/3} e^{-x}}{2}$$

$$y = -\frac{2^{2/3}((e^{2x} + 2c_1)x e^{2x})^{1/3} (1 + i\sqrt{3}) e^{-x}}{4}$$

$$y = \frac{2^{2/3}((e^{2x} + 2c_1)x e^{2x})^{1/3} (i\sqrt{3} - 1) e^{-x}}{4}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
```

Maple step by step

Let's solve

$$xy(x)^3 - y(x)^3 - x^2e^x + 3xy(x)^2 \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-xy(x)^3 + y(x)^3 + x^2e^x}{3xy(x)^2}$$

✓ **Mathematica.** Time used: 0.676 (sec). Leaf size: 117

```
ode=(x*y[x]^3-y[x]^3-x^2*Exp[x])+(3*x*y[x]^2)*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt[3]{-\frac{1}{2}e^{-x/3}\sqrt[3]{x}\sqrt[3]{e^{2x}+2c_1}}$$

$$y(x) \rightarrow \frac{e^{-x/3}\sqrt[3]{x}\sqrt[3]{e^{2x}+2c_1}}{\sqrt[3]{2}}$$

$$y(x) \rightarrow \frac{(-1)^{2/3}e^{-x/3}\sqrt[3]{x}\sqrt[3]{e^{2x}+2c_1}}{\sqrt[3]{2}}$$

✓ **Sympy.** Time used: 1.329 (sec). Leaf size: 85

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*exp(x) + x*y(x)**3 + 3*x*y(x)**2*Derivative(y(x), x) - y(x)**3, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = \frac{2^{\frac{2}{3}}\sqrt[3]{x}(C_1e^{-x}+e^x)}{2}, y(x) = \frac{2^{\frac{2}{3}}\sqrt[3]{x}(C_1e^{-x}+e^x)(-1-\sqrt{3}i)}{4}, y(x) = \frac{2^{\frac{2}{3}}\sqrt[3]{x}(C_1e^{-x}+e^x)(-1+\sqrt{3}i)}{4} \right]$$

2.4.24 Problem 23 (d)

Local contents

Solved using first_order_ode_abel_first_kind	1305
✓ Maple	1309
✓ Mathematica	1310
✓ Sympy	1311

Internal problem ID [7940]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 23 (d)

Date solved : Thursday, November 27, 2025 at 06:35:14 AM

CAS classification : [_Abel]

Solved using first_order_ode_abel_first_kind

Time used: 1.053 (sec)

Solve

$$y' + x(x + y) = x^3(x + y)^3 - 1$$

This is Abel first kind ODE, it has the form

$$y' = f_0(x) + f_1(x)y + f_2(x)y^2 + f_3(x)y^3$$

Comparing the above to given ODE which is

$$y' = x^3y^3 + 3y^2x^4 + (3x^5 - x)y + x^6 - x^2 - 1 \quad (1)$$

Therefore

$$f_0 = x^6 - x^2 - 1$$

$$f_1 = 3x^5 - x$$

$$f_2 = 3x^4$$

$$f_3 = x^3$$

Hence

$$f'_0 = 6x^5 - 2x$$

$$f'_3 = 3x^2$$

Since $f_2(x) = 3x^4$ is not zero, then the following transformation is used to remove f_2 . Let $y = u(x) - \frac{f_2}{3f_3}$ or

$$\begin{aligned} y &= u(x) - \left(\frac{3x^4}{3x^3} \right) \\ &= u(x) - x \end{aligned}$$

The above transformation applied to (1) gives a new ODE as

$$u'(x) = x^3 u(x)^3 - xu(x) \quad (2)$$

The above ODE (2) can now be solved.

Solve

In canonical form, the ODE is

$$\begin{aligned} u' &= F(x, u) \\ &= x^3 u^3 - xu \end{aligned}$$

This is a Bernoulli ODE.

$$u' = (-x)u(x) + (x^3)u^3 \quad (1)$$

The standard Bernoulli ODE has the form

$$u' = f_0(x)u + f_1(x)u^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -x \\ f_1 &= x^3 \end{aligned}$$

The first step is to divide the above equation by u^n which gives

$$\frac{u'}{u^n} = f_0(x)u^{1-n} + f_1(x) \quad (3)$$

The next step is use the substitution $v = u^{1-n}$ in equation (3) which generates a new ODE in $v(x)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $u(x)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(x) &= -x \\ f_1(x) &= x^3 \\ n &= 3 \end{aligned}$$

Dividing both sides of ODE (1) by $u^n = u^3$ gives

$$u' \frac{1}{u^3} = -\frac{x}{u^2} + x^3 \quad (4)$$

Let

$$\begin{aligned} v &= u^{1-n} \\ &= \frac{1}{u^2} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t x gives

$$v' = -\frac{2}{u^3} u' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -\frac{v'(x)}{2} &= -v(x)x + x^3 \\ v' &= -2x^3 + 2xv \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(x)$ which is now solved.

In canonical form a linear first order is

$$v'(x) + q(x)v(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -2x \\ p(x) &= -2x^3 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int -2x \, dx} \\ &= e^{-x^2} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu v) &= \mu p \\ \frac{d}{dx}(\mu v) &= (\mu) (-2x^3) \\ \frac{d}{dx}(v e^{-x^2}) &= (e^{-x^2}) (-2x^3) \\ d(v e^{-x^2}) &= (-2x^3 e^{-x^2}) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} v e^{-x^2} &= \int -2x^3 e^{-x^2} dx \\ &= (x^2 + 1) e^{-x^2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^{-x^2} gives the final solution

$$v(x) = c_1 e^{x^2} + x^2 + 1$$

The substitution $v = u^{1-n}$ is now used to convert the above solution back to $u(x)$ which results in

$$\frac{1}{u(x)^2} = c_1 e^{x^2} + x^2 + 1$$

Solving for $u(x)$ gives

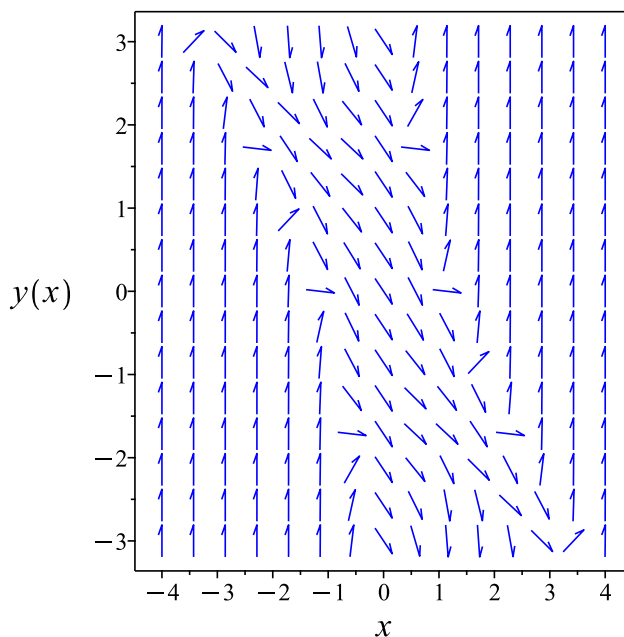
$$\begin{aligned} u(x) &= \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} \\ u(x) &= -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} \end{aligned}$$

Now we transform the solution $u(x) = \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}}$ to y using $y = u(x) - \frac{f_2}{3f_3}$, which gives

$$\begin{aligned} y &= \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - (x) \\ &= \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x \\ &= \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x \end{aligned}$$

Now we transform the solution $u(x) = -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}}$ to y using $y = u(x) - \frac{f_2}{3f_3}$, which gives

$$\begin{aligned} y &= -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - (x) \\ &= -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x \\ &= -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x \end{aligned}$$

Figure 2.295: Slope field $y' + x(x+y) = x^3(x+y)^3 - 1$ Summary of solutions found

$$y = -\frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x$$

$$y = \frac{1}{\sqrt{c_1 e^{x^2} + x^2 + 1}} - x$$

✓ Maple. Time used: 0.004 (sec). Leaf size: 76

```
ode:=diff(y(x),x)+x*(x+y(x)) = x^3*(x+y(x))^3-1;
dsolve(ode,y(x), singsol=all);
```

$$y = -x$$

$$y = -\frac{e^{-\frac{x^2}{2}}}{\sqrt{e^{-x^2}x^2 + e^{-x^2} + c_1}} - x$$

$$y = \frac{e^{-\frac{x^2}{2}}}{\sqrt{e^{-x^2}x^2 + e^{-x^2} + c_1}} - x$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

```

trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
trying Abel
<- Abel successful

```

Maple step by step

Let's solve

- $\frac{d}{dx}y(x) + x(x + y(x)) = x^3(x + y(x))^3 - 1$
Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
Solve for the highest derivative
 $\frac{d}{dx}y(x) = -x(x + y(x)) + x^3(x + y(x))^3 - 1$

✓ **Mathematica.** Time used: 9.746 (sec). Leaf size: 85

```

ode=D[y[x],x]+x*(x+y[x])==x^3*(x+y[x])^3-1;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow -x - \frac{e^{-\frac{x^2}{2}}}{\sqrt{e^{-x^2}(x^2 + 1) + c_1}}$$

$$y(x) \rightarrow -x + \frac{e^{-\frac{x^2}{2}}}{\sqrt{e^{-x^2}(x^2 + 1) + c_1}}$$

$$y(x) \rightarrow -x$$

✓ Sympy. Time used: 5.831 (sec). Leaf size: 134

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**3*(x + y(x))**3 + x*(x + y(x)) + Derivative(y(x), x) + 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = \frac{-x \left(x^2 e^{2C_1} + e^{2C_1} - e^{x^2} \right) - \sqrt{x^2 e^{2C_1} + e^{2C_1} - e^{x^2}} e^{C_1}}{x^2 e^{2C_1} + e^{2C_1} - e^{x^2}}, y(x) = \frac{-x \left(x^2 e^{2C_1} + e^{2C_1} - e^{x^2} \right) + \sqrt{x^2 e^{2C_1} + e^{2C_1} - e^{x^2}} e^{C_1}}{x^2 e^{2C_1} + e^{2C_1} - e^{x^2}} \right]$$

2.4.25 Problem 23 (e)

Local contents

Solved using first_order_ode_exact	1312
✓ Maple	1316
✓ Mathematica	1317
✓ Sympy	1317

Internal problem ID [7941]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 6. Equations of first order and first degree (Linear equations). Supplementary problems. Page 39

Problem number : 23 (e)

Date solved : Thursday, November 27, 2025 at 06:35:23 AM

CAS classification : ['y=_G(x,y)']

Solved using first_order_ode_exact

Time used: 0.107 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (1 + e^y) dy &= (-y - e^y + e^{-x}) dx \\ (y + e^y - e^{-x}) dx + (1 + e^y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y + e^y - e^{-x} \\ N(x, y) &= 1 + e^y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (y + e^y - e^{-x}) \\ &= 1 + e^y \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (1 + e^y) \\ &= 0 \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned} A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{1 + e^y} ((1 + e^y) - (0)) \\ &= 1 \end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int A dx} \\ &= e^{\int 1 dx} \end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^x \\ &= e^x\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= e^x(y + e^y - e^{-x}) \\ &= e^{x+y} - 1 + e^x y\end{aligned}$$

And

$$\begin{aligned}\overline{N} &= \mu N \\ &= e^x(1 + e^y) \\ &= (1 + e^y)e^x\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\overline{M} + \overline{N} \frac{dy}{dx} &= 0 \\ (e^{x+y} - 1 + e^x y) + ((1 + e^y)e^x) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \overline{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \overline{N} \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int \overline{M} dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int e^{x+y} - 1 + e^x y dx \\ \phi &= -x + e^x y + e^{x+y} + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = e^{x+y} + e^x + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = (1 + e^y) e^x$. Therefore equation (4) becomes

$$(1 + e^y) e^x = (1 + e^y) e^x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 0$$

Therefore

$$f(y) = c_1$$

Where c_1 is constant of integration. Substituting this result for $f(y)$ into equation (3) gives ϕ

$$\phi = -x + e^x y + e^{x+y} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = -x + e^x y + e^{x+y}$$

Solving for y gives

$$y = -\left(\text{LambertW}\left(e^{-x+(x e^x + c_1 + x)e^{-x}}\right) e^x - c_1 - x\right) e^{-x}$$

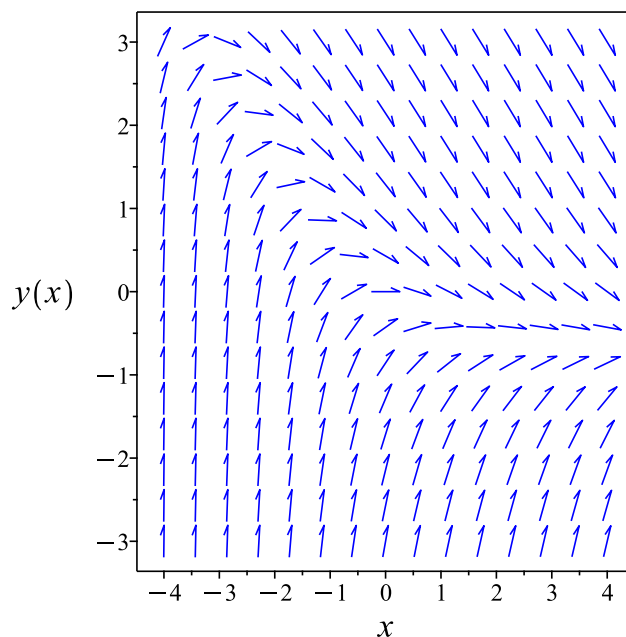


Figure 2.296: Slope field $y + e^y - e^{-x} + (1 + e^y) y' = 0$

Summary of solutions found

$$y = -\left(\text{LambertW}\left(e^{-x+(x e^x+c_1+x)e^{-x}}\right) e^x - c_1 - x\right) e^{-x}$$

✓ **Maple.** Time used: 0.020 (sec). Leaf size: 29

```
ode:=y(x)+exp(y(x))-exp(-x)+(1+exp(y(x)))*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x}(-c_1 + x) - \text{LambertW}\left(e^{e^{-x}(-c_1+x)}\right)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
trying separable
trying inverse linear
trying homogeneous types:
trying Chini
differential order: 1; looking for linear symmetries
trying exact
<- exact successful
```

Maple step by step

Let's solve

$$y(x) + e^{y(x)} - e^{-x} + (1 + e^{y(x)}) \left(\frac{d}{dx}y(x)\right) = 0$$

- Highest derivative means the order of the ODE is 1
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{-y(x) - e^{y(x)} + e^{-x}}{1 + e^{y(x)}}$$

✓ **Mathematica.** Time used: 5.908 (sec). Leaf size: 33

```
ode=(y[x]+Exp[y[x]]-Exp[-x])+(1+Exp[y[x]])*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(-e^x W \left(e^{e^{-x}(x+c_1)} \right) + x + c_1 \right)$$

✓ **Sympy.** Time used: 1.048 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((exp(y(x)) + 1)*Derivative(y(x), x) + y(x) + exp(y(x)) - exp(-x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \left(-C_1 + x - e^x W \left(e^{-(C_1-x)e^{-x}} \right) \right) e^{-x}$$

2.5 Chapter 9. Equations of first order and higher degree. Supplemetary problems. Page 65

Local contents

2.5.1	Problem 17	1319
2.5.2	Problem 18	1325
2.5.3	Problem 19	1330
2.5.4	Problem 20	1339
2.5.5	Problem 21	1346
2.5.6	Problem 22	1355
2.5.7	Problem 23	1360
2.5.8	Problem 24	1364
2.5.9	Problem 25	1369
2.5.10	Problem 26	1376
2.5.11	Problem 27	1385
2.5.12	Problem 28	1390
2.5.13	Problem 29	1397
2.5.14	Problem 30	1403
2.5.15	Problem 31	1408

2.5.1 Problem 17

Local contents

Solved by factoring the differential equation	1319
✓ Maple	1322
✓ Mathematica	1323
✓ Sympy	1324

Internal problem ID [7942]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:35:31 AM

CAS classification : [_separable]

Solved by factoring the differential equation

Time used: 0.053 (sec)

Solve

$$y'^2 x^2 + xy y' - 6y^2 = 0$$

Writing the ode as

$$(y'x + 3y)(-y'x + 2y) = 0$$

Therefore we need to solve the following equations

$$y'x + 3y = 0 \tag{1}$$

$$-y'x + 2y = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = \frac{3}{x}$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int \frac{3}{x} dx} \\ &= x^3\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu y &= 0 \\ \frac{d}{dx}(y x^3) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}y x^3 &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor x^3 gives the final solution

$$y = \frac{c_1}{x^3}$$

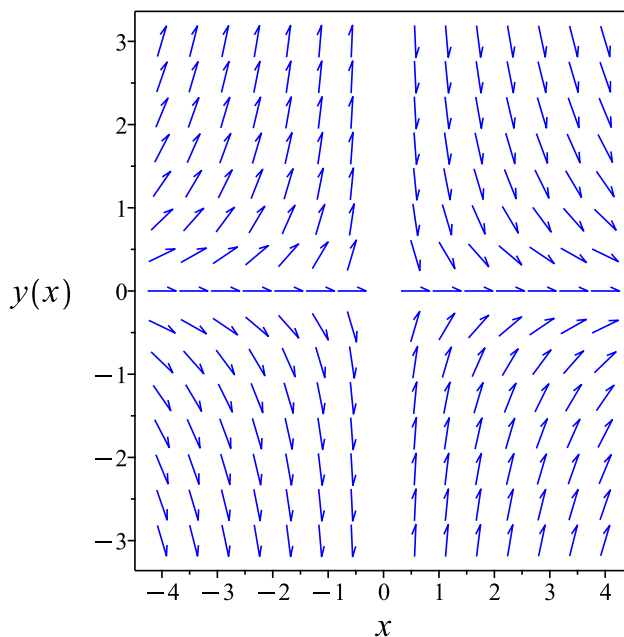


Figure 2.297: Slope field $y'x + 3y = 0$

Solving equation (2)

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{2}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x^2} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu y &= 0 \\ \frac{d}{dx} \left(\frac{y}{x^2} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{y}{x^2} &= \int 0 \, dx + c_2 \\ &= c_2 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_2 x^2$$

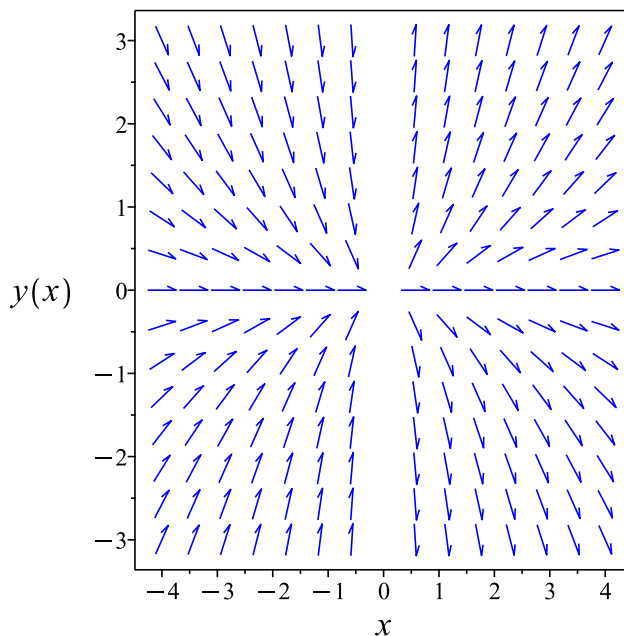


Figure 2.298: Slope field $-y'x + 2y = 0$

Summary of solutions found

$$y = \frac{c_1}{x^3}$$

$$y = c_2 x^2$$

✓ Maple. Time used: 0.000 (sec). Leaf size: 17

```
ode:=diff(y(x),x)^2*x^2+x*y(x)*diff(y(x),x)-6*y(x)^2 = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x^2$$

$$y = \frac{c_1}{x^3}$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$x^2 \left(\frac{d}{dx} y(x) \right)^2 + xy(x) \left(\frac{d}{dx} y(x) \right) - 6y(x)^2 = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = -\frac{3y(x)}{x}, \frac{d}{dx} y(x) = \frac{2y(x)}{x} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = -\frac{3y(x)}{x}$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{y(x)} = -\frac{3}{x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{y(x)} dx = \int -\frac{3}{x} dx + _C1$$

- Evaluate integral
 $\ln(y(x)) = -3 \ln(x) + _C1$
- Solve for $y(x)$
 $y(x) = \frac{e^{-C1}}{x^3}$
- Redefine the integration constant(s)
 $y(x) = \frac{C1}{x^3}$
- Solve the equation $\frac{d}{dx}y(x) = \frac{2y(x)}{x}$
- Separate variables
 $\frac{\frac{d}{dx}y(x)}{y(x)} = \frac{2}{x}$
- Integrate both sides with respect to x
 $\int \frac{\frac{d}{dx}y(x)}{y(x)} dx = \int \frac{2}{x} dx + _C1$
- Evaluate integral
 $\ln(y(x)) = 2 \ln(x) + _C1$
- Solve for $y(x)$
 $y(x) = e^{-C1} x^2$
- Redefine the integration constant(s)
 $y(x) = _C1 x^2$
- Set of solutions
 $\{y(x) = \frac{C1}{x^3}, y(x) = C1 x^2\}$

✓ **Mathematica.** Time used: 0.028 (sec). Leaf size: 26

```
ode=x^2*(D[y[x],x])^2+x*y[x]*D[y[x],x]-6*y[x]^2==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{c_1}{x^3}$$

$$y(x) \rightarrow c_1 x^2$$

$$y(x) \rightarrow 0$$

✓ Sympy. Time used: 0.099 (sec). Leaf size: 14

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), x)**2 + x*y(x)*Derivative(y(x), x) - 6*y(x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = C_1 x^2, \quad y(x) = \frac{C_1}{x^3} \right]$$

2.5.2 Problem 18

Local contents

Solved by factoring the differential equation	1325
✓ Maple	1327
✓ Mathematica	1329
✓ Sympy	1329

Internal problem ID [7943]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:35:36 AM

CAS classification : [_quadrature]

Solved by factoring the differential equation

Time used: 0.103 (sec)

Solve

$$xy'^2 + (-x^2 + y - 1)y' - (-1 + y)x = 0$$

Writing the ode as

$$(y' - x)(y'x + y - 1) = 0$$

Therefore we need to solve the following equations

$$y' - x = 0 \tag{1}$$

$$y'x + y - 1 = 0 \tag{2}$$

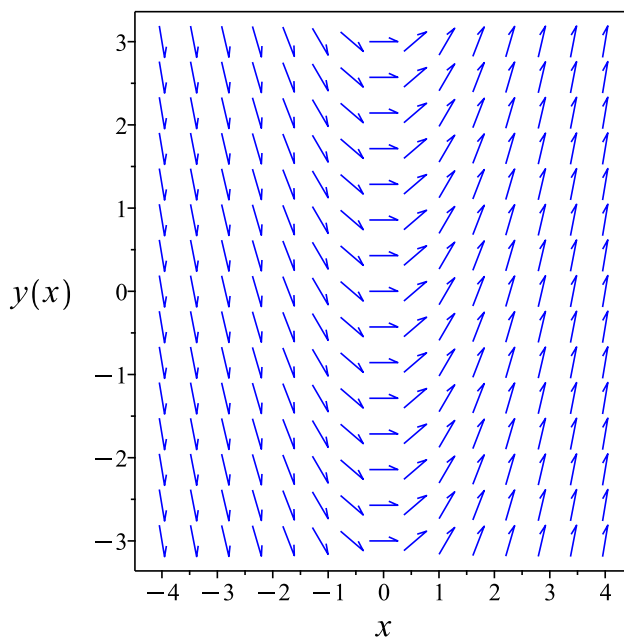
Now each of the above equations is solved in turn.

Solving equation (1)

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int x dx$$

$$y = \frac{x^2}{2} + c_1$$

Figure 2.299: Slope field $y' - x = 0$ Solving equation (2)

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = \frac{1}{x}$$

$$p(x) = \frac{1}{x}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int \frac{1}{x} \, dx} \\ &= x\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{1}{x} \right) \\ \frac{d}{dx}(yx) &= (x) \left(\frac{1}{x} \right) \\ d(yx) &= dx\end{aligned}$$

Integrating gives

$$\begin{aligned} yx &= \int 1 \, dx \\ &= x + c_2 \end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

$$y = \frac{x + c_2}{x}$$

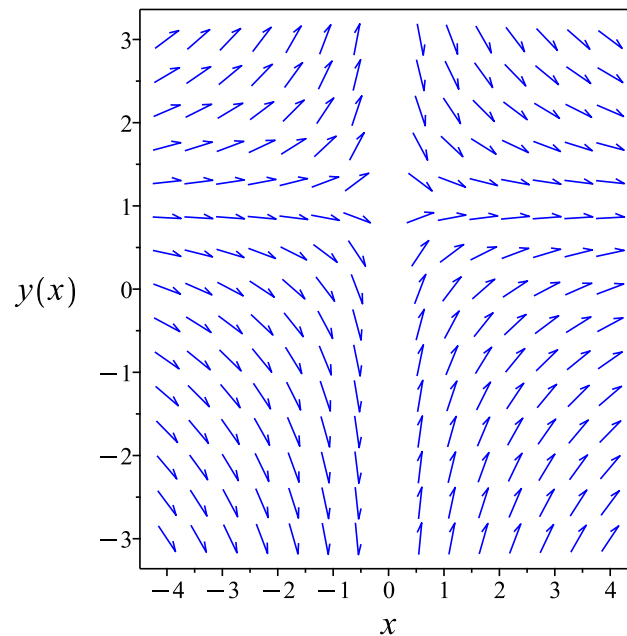


Figure 2.300: Slope field $y'x + y - 1 = 0$

Summary of solutions found

$$y = \frac{x + c_2}{x}$$

$$y = \frac{x^2}{2} + c_1$$

✓ Maple. Time used: 0.002 (sec). Leaf size: 21

```
ode:=diff(y(x),x)^2*x+(y(x)-1-x^2)*diff(y(x),x)-(y(x)-1)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2}{2} + c_1$$

$$y = \frac{x + c_1}{x}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

<- quadrature successful

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^2 + (y(x) - 1 - x^2)\left(\frac{d}{dx}y(x)\right) - x(y(x) - 1) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = x, \frac{d}{dx}y(x) = -\frac{y(x)-1}{x}\right]$$

- Solve the equation $\frac{d}{dx}y(x) = x$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}y(x)\right) dx = \int x dx + _C1$$

- Evaluate integral

$$y(x) = \frac{x^2}{2} + _C1$$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{y(x)-1}{x}$

- Separate variables

$$\frac{\frac{d}{dx}y(x)}{y(x)-1} = -\frac{1}{x}$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}y(x)}{y(x)-1} dx = \int -\frac{1}{x} dx + _C1$$

- Evaluate integral

$$\ln(y(x) - 1) = -\ln(x) + _C1$$

- Solve for $y(x)$

$$y(x) = \frac{e^{-C1+x}}{x}$$

- Redefine the integration constant(s)

$$y(x) = \frac{C1+x}{x}$$

- Set of solutions

$$\left\{y(x) = \frac{C1+x}{x}, y(x) = \frac{x^2}{2} + C1\right\}$$

✓ **Mathematica.** Time used: 0.026 (sec). Leaf size: 32

```
ode=x*(D[y[x],x])^2+(y[x]-1-x^2)*D[y[x],x]-x*(y[x]-1)==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^2}{2} + c_1$$

$$y(x) \rightarrow \frac{x + c_1}{x}$$

$$y(x) \rightarrow 1$$

✓ **Sympy.** Time used: 0.135 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*(y(x) - 1) + x*Derivative(y(x), x)**2 + (-x**2 + y(x) - 1)*Derivative
        (y(x), x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[y(x) = C_1 + \frac{x^2}{2}, \quad y(x) = \frac{C_1 + x}{x} \right]$$

2.5.3 Problem 19

Local contents

Solved using first_order_ode_dAlembert	1330
Solved using first_order_ode_homog_type_G	1333
Solved using first_order_ode_parametric method	1335
✓ Maple	1337
✓ Mathematica	1338
✓ Sympy	1338

Internal problem ID [7944]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:35:41 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.162 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Let $p = y'$ the ode becomes

$$x p^2 - 2yp + 4x = 0$$

Solving for y from the above results in

$$y = \frac{x(p^2 + 4)}{2p} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{p^2 + 4}{2p}$$

$$g = 0$$

Hence (2) becomes

$$p - \frac{p^2 + 4}{2p} = \left(\frac{x}{2} - \frac{2x}{p^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p^2 + 4}{2p} = 0$$

Solving the above for p results in

$$p_1 = 2$$

$$p_2 = -2$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 2x$$

$$y = -2x$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)^2 + 4}{2p(x)}}{\frac{x}{2} - \frac{2x}{p(x)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{p(x)}{x} \quad (2.107)$$

is separable as it can be written as

$$p'(x) = \frac{p(x)}{x}$$

$$= f(x)g(p)$$

Where

$$f(x) = \frac{1}{x}$$

$$g(p) = p$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{1}{p} dp = \int \frac{1}{x} dx$$

$$\ln(p(x)) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$p(x) = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p = 0$$

for $p(x)$ gives

$$p(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$p(x) = c_1 x$$

$$p(x) = 0$$

Substituting the above solution for p in (2A) gives

$$y = \frac{c_1^2 x^2 + 4}{2c_1}$$

Summary of solutions found

$$y = -2x$$

$$y = 2x$$

$$y = \frac{c_1^2 x^2 + 4}{2c_1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.310 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Multiplying the right side of the ode, which is $\frac{y+\sqrt{-4x^2+y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y + \sqrt{-4x^2 + y^2}}{x} \\ &= \frac{y + \sqrt{-4x^2 + y^2}}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{4x^2}{y\sqrt{-4x^2 + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{(y + \sqrt{-4x^2 + y^2})(\sqrt{-4x^2 + y^2} - y)}{y\sqrt{-4x^2 + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{\sqrt{z^2 - 4} + z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2 - 4} + z}{z}\right)} dz = 0$$

Multiplying the right side of the ode, which is $\frac{y - \sqrt{-4x^2 + y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y - \sqrt{-4x^2 + y^2}}{x} \\ &= -\frac{\sqrt{-4x^2 + y^2} - y}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{4x^2}{y\sqrt{-4x^2 + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{(y + \sqrt{-4x^2 + y^2})(\sqrt{-4x^2 + y^2} - y)}{y\sqrt{-4x^2 + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{-\sqrt{z^2 - 4} + z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

Solved using first_order_ode_parametric method

Time used: 0.110 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Let y' be a parameter λ . The ode becomes

$$x\lambda^2 - 2y\lambda + 4x = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{x(\lambda^2 + 4)}{2\lambda} \\ &= \frac{x(\lambda^2 + 4)}{2\lambda} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{x - \frac{x(\lambda^2+4)}{2\lambda^2}}{\lambda - \frac{\lambda^2+4}{2\lambda}} \\ &= \frac{x(\lambda)}{\lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{1}{\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{1}{\lambda} d\lambda} \\ &= \frac{1}{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x}{\lambda}\right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{x}{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{\lambda}$ gives the final solution

$$x(\lambda) = \lambda c_1$$

Now that we found solution x we have two equations with parameter λ . They are

$$y = \frac{x(\lambda^2 + 4)}{2\lambda}$$

$$x = \lambda c_1$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \frac{4c_1^2 + x^2}{2c_1}$$

✓ **Maple.** Time used: 0.027 (sec). Leaf size: 30

```
ode:=4*x-2*diff(y(x),x)*y(x)+diff(y(x),x)^2*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -2x$$

$$y = 2x$$

$$y = \frac{4c_1^2 + x^2}{2c_1}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

<- dAlembert successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^2 - 2y(x)\left(\frac{d}{dx}y(x)\right) + 4x = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x^2}}{x}, \frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 - 4x^2}}{x} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x^2}}{x}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 - 4x^2}}{x}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.178 (sec). Leaf size: 43

```
ode=x*(D[y[x],x])^2-2*y[x]*D[y[x],x]+4*x==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -2x \cosh(-\log(x) + c_1)$$

$$y(x) \rightarrow -2x \cosh(\log(x) + c_1)$$

$$y(x) \rightarrow -2x$$

$$y(x) \rightarrow 2x$$

✓ **Sympy.** Time used: 1.423 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x)**2 + 4*x - 2*y(x)*Derivative(y(x), x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = 2x^2e^{-C_1} + \frac{e^{C_1}}{2}$$

2.5.4 Problem 20

Local contents

Solved using first_order_ode_homog_type_G	1339
Solved using first_order_ode_parametric method	1342
✓ Maple	1343
✓ Mathematica	1345
✓ Sympy	1345

Internal problem ID [7945]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 20

Date solved : Thursday, November 27, 2025 at 06:35:47 AM

CAS classification : [[_homogeneous, 'class G'], _rational]

Solved using first_order_ode_homog_type_G

Time used: 0.186 (sec)

Solve

$$3x^4y'^2 - y'x - y = 0$$

Multiplying the right side of the ode, which is $\frac{1+\sqrt{12x^2y+1}}{6x^3}$ by $\frac{x}{y}$ gives

$$\begin{aligned}
 y' &= \left(\frac{x}{y}\right) \frac{1 + \sqrt{12x^2y + 1}}{6x^3} \\
 &= \frac{1 + \sqrt{12x^2y + 1}}{6x^2y} \\
 &= F(x, y)
 \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{6x^2y + 1 + \sqrt{12x^2y + 1}}{3x^2y\sqrt{12x^2y + 1}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{6x^2y + 1 + \sqrt{12x^2y + 1}}{6x^2y\sqrt{12x^2y + 1}} \\ \alpha &= \frac{f_x}{f_y} \\ &= 2 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^2} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{1 + \sqrt{12z + 1}}{6z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{x^2y} \frac{1}{z \left(-2 - \frac{1 + \sqrt{12z + 1}}{6z} \right)} dz = 0$$

Multiplying the right side of the ode, which is $-\frac{-1 + \sqrt{12x^2y + 1}}{6x^3}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y} \right) - \frac{-1 + \sqrt{12x^2y + 1}}{6x^3} \\ &= -\frac{-1 + \sqrt{12x^2y + 1}}{6x^2y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{-6x^2y + \sqrt{12x^2y + 1} - 1}{3x^2y\sqrt{12x^2y + 1}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{-6x^2y + \sqrt{12x^2y + 1} - 1}{6x^2y\sqrt{12x^2y + 1}} \\ \alpha &= \frac{f_x}{f_y} \\ &= 2 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^2} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{-1 + \sqrt{12z + 1}}{6z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{x^2y} \frac{1}{z \left(-2 + \frac{-1 + \sqrt{12z + 1}}{6z} \right)} dz = 0$$

Summary of solutions found

$$\begin{aligned} \ln(x) - c_1 + \int^{x^2y} \frac{1}{z \left(-2 - \frac{1 + \sqrt{12z + 1}}{6z} \right)} dz &= 0 \\ \ln(x) - c_2 + \int^{x^2y} \frac{1}{z \left(-2 + \frac{-1 + \sqrt{12z + 1}}{6z} \right)} dz &= 0 \end{aligned}$$

Solved using first_order_ode_parametric method

Time used: 0.345 (sec)

Solve

$$3x^4y'^2 - y'x - y = 0$$

Let y' be a parameter λ . The ode becomes

$$3x^4\lambda^2 - \lambda x - y = 0$$

Isolating y gives

$$\begin{aligned} y &= 3x^4\lambda^2 - \lambda x \\ &= 3x^4\lambda^2 - \lambda x \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{6\lambda x^4 - x}{-12\lambda^2 x^3 + 2\lambda} \\ &= -\frac{x(\lambda)}{2\lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{2\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{2\lambda} d\lambda} \\ &= \sqrt{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}(x\sqrt{\lambda}) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} x\sqrt{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{\lambda}$ gives the final solution

$$x(\lambda) = \frac{c_1}{\sqrt{\lambda}}$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned} y &= 3x^4\lambda^2 - \lambda x \\ x &= \frac{c_1}{\sqrt{\lambda}} \end{aligned}$$

Eliminating λ gives the solution for y . Simplifying the above gives

$$y = \frac{3x c_1^4 - c_1^2}{x}$$

Summary of solutions found

$$y = \frac{3x c_1^4 - c_1^2}{x}$$

✓ **Maple.** Time used: 0.153 (sec). Leaf size: 97

```
ode:=3*x^4*diff(y(x),x)^2-diff(y(x),x)*x-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$\begin{aligned} y &= -\frac{1}{12x^2} \\ y &= \frac{ic_1\sqrt{3} - 3x}{3x c_1^2} \\ y &= \frac{ic_1\sqrt{3} - 3x}{3x c_1^2} \\ y &= \frac{-ic_1\sqrt{3} - 3x}{3x c_1^2} \\ y &= \frac{-ic_1\sqrt{3} - 3x}{3x c_1^2} \end{aligned}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

trying simple symmetries for implicit equations

Successful isolation of dy/dx: 2 solutions were found. Trying to solve each \ resulting ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

trying homogeneous G

<- homogeneous successful

* Tackling next ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

trying homogeneous G

<- homogeneous successful

Maple step by step

Let's solve

$$3x^4 \left(\frac{d}{dx} y(x) \right)^2 - x \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = \frac{1 + \sqrt{1 + 12x^2 y(x)}}{6x^3}, \frac{d}{dx} y(x) = -\frac{-1 + \sqrt{1 + 12x^2 y(x)}}{6x^3} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = \frac{1 + \sqrt{1 + 12x^2 y(x)}}{6x^3}$
- Solve the equation $\frac{d}{dx} y(x) = -\frac{-1 + \sqrt{1 + 12x^2 y(x)}}{6x^3}$

- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.327 (sec). Leaf size: 123

```
ode=3*x^4*D[y[x],x]^2-x*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[-\frac{x\sqrt{12x^2y(x)+1}\operatorname{arctanh}\left(\sqrt{12x^2y(x)+1}\right)}{\sqrt{12x^4y(x)+x^2}} - \frac{1}{2}\log(y(x)) = c_1, y(x) \right]$$

$$\text{Solve} \left[\frac{x\sqrt{12x^2y(x)+1}\operatorname{arctanh}\left(\sqrt{12x^2y(x)+1}\right)}{\sqrt{12x^4y(x)+x^2}} - \frac{1}{2}\log(y(x)) = c_1, y(x) \right]$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 2.542 (sec). Leaf size: 10

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x**4*Derivative(y(x), x)**2 - x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1(C_1 + \frac{2}{x})}{12}$$

2.5.5 Problem 21

Local contents

Solved using first_order_ode_dAlembert	1346
Solved using first_order_ode_parametric method	1350
✓ Maple	1352
✓ Mathematica	1353
✓ Sympy	1354

Internal problem ID [7946]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 21

Date solved : Thursday, November 27, 2025 at 06:35:52 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.325 (sec)

Solve

$$8y'^2y - 2y'x + y = 0$$

Let $p = y'$ the ode becomes

$$8p^2y - 2px + y = 0$$

Solving for y from the above results in

$$y = \frac{2px}{8p^2 + 1} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{2p}{8p^2 + 1} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{2p}{8p^2 + 1} = \left(\frac{2x}{8p^2 + 1} - \frac{32xp^2}{(8p^2 + 1)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{2p}{8p^2 + 1} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= 0 \\ p_2 &= \frac{\sqrt{2}}{4} \\ p_3 &= -\frac{\sqrt{2}}{4} \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= 0 \\ y &= \frac{\sqrt{2}x}{4} \\ y &= -\frac{\sqrt{2}x}{4} \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{2p(x)}{8p(x)^2 + 1}}{\frac{2x}{8p(x)^2 + 1} - \frac{32xp(x)^2}{(8p(x)^2 + 1)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{p(x)(8p(x)^2 + 1)}{2x} \quad (2.108)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{p(x)(8p(x)^2 + 1)}{2x} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= -\frac{1}{2x} \\ g(p) &= p(8p^2 + 1) \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{p(8p^2 + 1)} dp &= \int -\frac{1}{2x} dx \end{aligned}$$

$$\ln \left(\frac{p(x)}{\sqrt{8p(x)^2 + 1}} \right) = \ln \left(\frac{1}{\sqrt{x}} \right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{p(x)}{\sqrt{8p(x)^2 + 1}} = \frac{c_1}{\sqrt{x}}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p(8p^2 + 1) = 0$$

for $p(x)$ gives

$$\begin{aligned} p(x) &= 0 \\ p(x) &= -\frac{i\sqrt{2}}{4} \end{aligned}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{p(x)}{\sqrt{8p(x)^2 + 1}} = \frac{c_1}{\sqrt{x}}$$

$$p(x) = 0$$

$$p(x) = -\frac{i\sqrt{2}}{4}$$

Substituting the above solution for p in (2A) gives

$$y = -\frac{2c_1x}{\sqrt{-8c_1^2 + x} \left(\frac{8c_1^2}{-8c_1^2 + x} + 1 \right)}$$

$$y = 0$$

Simplifying the above gives

$$y = 0$$

$$y = \frac{\sqrt{2}x}{4}$$

$$y = -\frac{\sqrt{2}x}{4}$$

$$y = -2\sqrt{-8c_1^2 + x} c_1$$

$$y = 0$$

Summary of solutions found

$$y = 0$$

$$y = -\frac{\sqrt{2}x}{4}$$

$$y = \frac{\sqrt{2}x}{4}$$

$$y = -2\sqrt{-8c_1^2 + x} c_1$$

Solved using first_order_ode_parametric method

Time used: 0.144 (sec)

Solve

$$8y'^2y - 2y'x + y = 0$$

Let y' be a parameter λ . The ode becomes

$$8\lambda^2y - 2\lambda x + y = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{2\lambda x}{8\lambda^2 + 1} \\ &= \frac{2\lambda x}{8\lambda^2 + 1} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{\frac{2x}{8\lambda^2+1} - \frac{32\lambda^2x}{(8\lambda^2+1)^2}}{\lambda - \frac{2\lambda}{8\lambda^2+1}} \\ &= -\frac{2x(\lambda)}{8\lambda^3 + \lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{2}{8\lambda^3 + \lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{2}{8\lambda^3 + \lambda} d\lambda} \\ &= \frac{\lambda^2}{8\lambda^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x\lambda^2}{8\lambda^2+1}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{x\lambda^2}{8\lambda^2+1} &= \int 0 d\lambda + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{\lambda^2}{8\lambda^2+1}$ gives the final solution

$$x(\lambda) = \frac{(8\lambda^2+1)c_1}{\lambda^2}$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned}y &= \frac{2\lambda x}{8\lambda^2+1} \\ x &= \frac{(8\lambda^2+1)c_1}{\lambda^2}\end{aligned}$$

Eliminating λ gives the solution for y . Simplifying the above gives

$$\begin{aligned}y &= -2\sqrt{c_1(-8c_1+x)} \\ y &= 2\sqrt{c_1(-8c_1+x)}\end{aligned}$$

Summary of solutions found

$$\begin{aligned}y &= -2\sqrt{c_1(-8c_1+x)} \\ y &= 2\sqrt{c_1(-8c_1+x)}\end{aligned}$$

✓ **Maple.** Time used: 0.030 (sec). Leaf size: 95

```
ode:=8*y(x)*diff(y(x),x)^2-2*diff(y(x),x)*x+y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{2}x}{4}$$

$$y = \frac{\sqrt{2}x}{4}$$

$$y = 0$$

$$\ln(x) + \operatorname{arctanh}\left(\frac{1}{\sqrt{-\frac{8y^2-x^2}{x^2}}}\right) + \ln\left(\frac{y}{x}\right) - c_1 = 0$$

$$\ln(x) - \operatorname{arctanh}\left(\frac{1}{\sqrt{-\frac{8y^2-x^2}{x^2}}}\right) + \ln\left(\frac{y}{x}\right) - c_1 = 0$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying simple symmetries for implicit equations

<- symmetries for implicit equations successful

Maple step by step

Let's solve

$$8y(x) \left(\frac{d}{dx}y(x)\right)^2 - 2x\left(\frac{d}{dx}y(x)\right) + y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = -\frac{-x + \sqrt{x^2 - 8y(x)^2}}{8y(x)}, \frac{d}{dx}y(x) = \frac{x + \sqrt{x^2 - 8y(x)^2}}{8y(x)} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{-x + \sqrt{x^2 - 8y(x)^2}}{8y(x)}$

- Solve the equation $\frac{d}{dx}y(x) = \frac{x + \sqrt{x^2 - 8y(x)^2}}{8y(x)}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.217 (sec). Leaf size: 174

```
ode=8*y[x]*D[y[x],x]^2-2*x*D[y[x],x]+y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{e^{4c_1}\sqrt{e^{8c_1}-2ix}}{2\sqrt{2}}$$

$$y(x) \rightarrow \frac{e^{4c_1}\sqrt{e^{8c_1}-2ix}}{2\sqrt{2}}$$

$$y(x) \rightarrow -\frac{e^{4c_1}\sqrt{2ix+e^{8c_1}}}{2\sqrt{2}}$$

$$y(x) \rightarrow \frac{e^{4c_1}\sqrt{2ix+e^{8c_1}}}{2\sqrt{2}}$$

$$y(x) \rightarrow 0$$

$$y(x) \rightarrow -\frac{x}{2\sqrt{2}}$$

$$y(x) \rightarrow \frac{x}{2\sqrt{2}}$$

✓ Sympy. Time used: 135.995 (sec). Leaf size: 388

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*Derivative(y(x), x) + 8*y(x)*Derivative(y(x), x)**2 + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\begin{aligned} y(x) &= -\frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \\ &-\frac{\sqrt{C_1 + 2\sqrt{2}x\sqrt{-C_1}}}{4}, y(x) = \frac{\sqrt{C_1 + 2\sqrt{2}x\sqrt{-C_1}}}{4}, y(x) = \\ &-\frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \\ &-\frac{\sqrt{2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \\ &-\frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \\ &-\frac{\sqrt{C_1 + 2\sqrt{2}x\sqrt{-C_1}}}{4}, y(x) = \frac{\sqrt{C_1 + 2\sqrt{2}x\sqrt{-C_1}}}{4}, y(x) = \\ &-\frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{-2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \\ &-\frac{\sqrt{2\sqrt{2}\sqrt{C_1}x - C_1}}{4}, y(x) = \frac{\sqrt{2\sqrt{2}\sqrt{C_1}x - C_1}}{4} \end{aligned} \right]$$

2.5.6 Problem 22

Local contents

Solved using first_order_ode_parametric method	1355
✓ Maple	1357
✓ Mathematica	1359
✗ Sympy	1359

Internal problem ID [7947]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 22

Date solved : Thursday, November 27, 2025 at 06:35:58 AM

CAS classification : `[[_1st_order, _with_linear_symmetries], _rational]`

Solved using first_order_ode_parametric method

Time used: 0.579 (sec)

Solve

$$y'^2 y^2 + 3y'x - y = 0$$

Let y' be a parameter λ . The ode becomes

$$\lambda^2 y^2 + 3\lambda x - y = 0$$

Isolating x gives

$$x = -\frac{y(\lambda^2 y - 1)}{3\lambda}$$

$$x = F(y, \lambda)$$

Now we generate an ode in $y(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda} y(\lambda) &= \frac{\lambda \frac{\partial F}{\partial \lambda}}{1 - \frac{\partial F}{\partial y}} \\ &= \frac{\lambda \left(-\frac{2y^2}{3} + \frac{y(\lambda^2 y - 1)}{3\lambda^2} \right)}{1 - \lambda \left(-\frac{\lambda^2 y - 1}{3\lambda} - \frac{y\lambda}{3} \right)} \\ &= -\frac{y(\lambda)}{2\lambda} \end{aligned}$$

Which is now solved for y .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}y(\lambda) + q(\lambda)y(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{2\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{2\lambda} d\lambda} \\ &= \sqrt{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu y &= 0 \\ \frac{d}{d\lambda}(y\sqrt{\lambda}) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} y\sqrt{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{\lambda}$ gives the final solution

$$y(\lambda) = \frac{c_1}{\sqrt{\lambda}}$$

Now that we have found solution y , we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{c_1}{\sqrt{\lambda}} \\ x &= -\frac{y(\lambda^2 y - 1)}{3\lambda} \end{aligned}$$

Eliminating λ gives the solution for y .

$$y\sqrt{\text{RootOf}(y^2_Z^2 + 3_Zx - y)} - c_1$$

Which can be written as

$$\frac{c_1^4}{y^2} + \frac{3c_1^2 x}{y^2} - y = 0$$

Simplifying the above gives

$$\frac{c_1^4 + 3c_1^2x - y^3}{y^2} = 0$$

Summary of solutions found

$$\frac{c_1^4 + 3c_1^2x - y^3}{y^2} = 0$$

✓ **Maple.** Time used: 0.062 (sec). Leaf size: 125

```
ode:=y(x)^2*diff(y(x),x)^2+3*diff(y(x),x)*x-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{18^{1/3}(-x^2)^{1/3}}{2}$$

$$y = -\frac{2^{1/3}(-x^2)^{1/3}(3i3^{1/6} + 3^{2/3})}{4}$$

$$y = \frac{2^{1/3}(-x^2)^{1/3}(-3^{2/3} + 3i3^{1/6})}{4}$$

$$y = 0$$

$$y = \text{RootOf} \left(-2\ln(x) - 3 \int^{-x} \frac{4a^3 + 3\sqrt{4a^3 + 9} + 9}{-a(4a^3 + 9)} da + 2c_1 \right) x^{2/3}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying simple symmetries for implicit equations

Successful isolation of dy/dx: 2 solutions were found. Trying to solve each \ resulting ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

```

trying exact
Looking for potential symmetries
trying an equivalence to an Abel ODE
trying 1st order ODE linearizable_by_differentiation
-> Solving 1st order ODE of high degree, Lie methods, 1st trial
    -> Computing symmetries using: way = 3
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
    <- successful computation of symmetries.
1st order, trying reduction of order with given symmetries:
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
1st order, trying the canonical coordinates of the invariance group
    -> Calling odsolve with the ODE, diff(y(x),x) = 2/3*y(x)/x, y(x)
    *** Sublevel 3 ***
    Methods for first order ODEs:
    --- Trying classification methods ---
    trying a quadrature
    trying 1st order linear
    <- 1st order linear successful
<- 1st order, canonical coordinates successful
1st order, trying reduction of order with given symmetries:
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
1st order, trying the canonical coordinates of the invariance group
<- 1st order, canonical coordinates successful

```

Maple step by step

Let's solve

$$y(x)^2 \left(\frac{d}{dx} y(x) \right)^2 + 3x \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = -\frac{3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}, \frac{d}{dx} y(x) = \frac{-3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = -\frac{3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}$
- Solve the equation $\frac{d}{dx} y(x) = \frac{-3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}$

- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.324 (sec). Leaf size: 204

```
ode=y[x]^2*D[y[x],x]^2+3*x*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\frac{3}{2} \log(y(x)) - \frac{2 \sqrt{\frac{9x^2}{4y(x)^3} + 1} \operatorname{arcsinh}\left(\frac{3}{2}x \sqrt{\frac{1}{y(x)^3}}\right)}{\sqrt{\frac{1}{y(x)^3}} \sqrt{9x^2 + 4y(x)^3}} = c_1, y(x) \right]$$

$$\text{Solve} \left[\frac{2 \sqrt{\frac{9x^2}{4y(x)^3} + 1} \operatorname{arcsinh}\left(\frac{3}{2}x \sqrt{\frac{1}{y(x)^3}}\right)}{\sqrt{\frac{1}{y(x)^3}} \sqrt{9x^2 + 4y(x)^3}} + \frac{3}{2} \log(y(x)) = c_1, y(x) \right]$$

$$y(x) \rightarrow -\left(-\frac{3}{2}\right)^{2/3} x^{2/3}$$

$$y(x) \rightarrow -\left(\frac{3}{2}\right)^{2/3} x^{2/3}$$

$$y(x) \rightarrow \frac{\sqrt[3]{-13} 2^{2/3} x^{2/3}}{2^{2/3}}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x*Derivative(y(x), x) + y(x)**2*Derivative(y(x), x)**2 - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE -(-3*x + sqrt(9*x**2 + 4*y(x)**3))/(2*y(x)
**2) + Derivative(y(x), x) cannot be solved by the factorable group method
```

2.5.7 Problem 23

Local contents

Solved using first_order_ode_clairaut	1360
✓ Maple	1362
✓ Mathematica	1363
✓ Sympy	1363

Internal problem ID [7948]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 23

Date solved : Thursday, November 27, 2025 at 06:36:04 AM

CAS classification : `[_1st_order, _with_linear_symmetries], _Clairaut]`

Solved using first_order_ode_clairaut

Time used: 0.047 (sec)

Solve

$$y'^2 - y'x + y = 0$$

This is Clairaut ODE. It has the form

$$y = y'x + g(y')$$

Where g is function of $y'(x)$. Let $p = y'$ the ode becomes

$$p^2 - px + y = 0$$

Solving for y from the above results in

$$y = -p^2 + px \tag{1A}$$

The above ode is a Clairaut ode which is now solved.

Writing the equation (1A) as

$$y = px + g(p)$$

We now write $g \equiv g(p)$ to make notation simpler but we should always remember that g is function of p which in turn is function of x . Hence the above becomes

$$y = px + g \tag{1}$$

Where

$$g = -p^2$$

Taking derivative of (1) w.r.t. x gives

$$\begin{aligned} p &= \frac{d}{dx}(xp + g) \\ p &= \left(p + x \frac{dp}{dx}\right) + \left(g' \frac{dp}{dx}\right) \\ p &= p + (x + g') \frac{dp}{dx} \\ 0 &= (x + g') \frac{dp}{dx} \end{aligned}$$

Where g' is derivative of $g(p)$ w.r.t. p .

The general solution is given by

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ p &= c_1 \end{aligned}$$

Substituting this in (1) gives the general solution as

$$y = -c_1^2 + c_1x$$

The singular solution is found from solving for p from

$$x + g'(p) = 0$$

And substituting the result back in (1). Since we found above that $g = -p^2$, then the above equation becomes

$$\begin{aligned} x + g'(p) &= x - 2p \\ &= 0 \end{aligned}$$

Solving the above for p results in

$$p_1 = y = \frac{x^2}{4}$$

Substituting the above back in (1) results in

$$y = \frac{x^2}{4}$$

Simplifying the above gives

$$y = c_1(-c_1 + x)$$

$$y = \frac{x^2}{4}$$

Summary of solutions found

$$y = c_1(-c_1 + x)$$

$$y = \frac{x^2}{4}$$

✓ **Maple.** Time used: 0.017 (sec). Leaf size: 19

```
ode:=diff(y(x),x)^2-diff(y(x),x)*x+y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2}{4}$$

$$y = c_1(-c_1 + x)$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

<- dAlembert successful

Maple step by step

Let's solve

$$\left(\frac{d}{dx}y(x)\right)^2 - x\left(\frac{d}{dx}y(x)\right) + y(x) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{x}{2} - \frac{\sqrt{x^2-4y(x)}}{2}, \frac{d}{dx}y(x) = \frac{x}{2} + \frac{\sqrt{x^2-4y(x)}}{2} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{2} - \frac{\sqrt{x^2-4y(x)}}{2}$
- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{2} + \frac{\sqrt{x^2-4y(x)}}{2}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.005 (sec). Leaf size: 25

```
ode=D[y[x],x]^2-x*D[y[x],x]+y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1(x - c_1)$$

$$y(x) \rightarrow \frac{x^2}{4}$$

✓ **Sympy.** Time used: 1.163 (sec). Leaf size: 14

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) + y(x) + Derivative(y(x), x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \frac{x^2}{4} - \frac{(C_1 + x)^2}{4}$$

2.5.8 Problem 24

Local contents

Solved using first_order_ode_parametric method	1364
✓ Maple	1366
✓ Mathematica	1368
✗ Sympy	1368

Internal problem ID [7949]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 24

Date solved : Thursday, November 27, 2025 at 06:36:09 AM

CAS classification : `[_1st_order, _with_linear_symmetries], _rational]`

Solved using first_order_ode_parametric method

Time used: 0.389 (sec)

Solve

$$16y^3y'^2 - 4y'x + y = 0$$

Let y' be a parameter λ . The ode becomes

$$16y^3\lambda^2 - 4\lambda x + y = 0$$

Isolating x gives

$$x = \frac{y(16\lambda^2y^2 + 1)}{4\lambda}$$

$$x = F(y, \lambda)$$

Now we generate an ode in $y(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}y(\lambda) &= \frac{\lambda \frac{\partial F}{\partial \lambda}}{1 - \frac{\partial F}{\partial y}} \\ &= \frac{\lambda \left(8y^3 - \frac{y(16\lambda^2y^2+1)}{4\lambda^2} \right)}{1 - \lambda \left(\frac{16\lambda^2y^2+1}{4\lambda} + 8y^2\lambda \right)} \\ &= -\frac{y(\lambda)}{3\lambda} \end{aligned}$$

Which is now solved for y .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}y(\lambda) + q(\lambda)y(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{3\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{3\lambda} d\lambda} \\ &= \lambda^{1/3} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu y &= 0 \\ \frac{d}{d\lambda}(y \lambda^{1/3}) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} y \lambda^{1/3} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\lambda^{1/3}$ gives the final solution

$$y(\lambda) = \frac{c_1}{\lambda^{1/3}}$$

Now that we have found solution y , we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{c_1}{\lambda^{1/3}} \\ x &= \frac{16y^3\lambda^2 + y}{4\lambda} \end{aligned}$$

Eliminating λ gives the solution for y .

$$y4^{2/3} \text{RootOf}(y^3_Z^2 - _Zx + y)^{1/3} - 4c_1$$

Which can be written as

$$\frac{16c_1^6}{y^3} - \frac{4c_1^3x}{y^3} + y = 0$$

Simplifying the above gives

$$\frac{16c_1^6 + y^4 - 4x c_1^3}{y^3} = 0$$

Summary of solutions found

$$\frac{16c_1^6 + y^4 - 4x c_1^3}{y^3} = 0$$

✓ **Maple.** Time used: 0.118 (sec). Leaf size: 105

```
ode:=16*y(x)^3*diff(y(x),x)^2-4*diff(y(x),x)*x+y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{\sqrt{2}\sqrt{-x}}{2}$$

$$y = \frac{\sqrt{2}\sqrt{-x}}{2}$$

$$y = -\frac{\sqrt{2}\sqrt{x}}{2}$$

$$y = \frac{\sqrt{2}\sqrt{x}}{2}$$

$$y = 0$$

$$y = \text{RootOf} \left(-\ln(x) + 2 \int^{-z} -\frac{4_a^4 - \sqrt{-4_a^4 + 1} - 1}{_a(4_a^4 - 1)} d_a + c_1 \right) \sqrt{x}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying simple symmetries for implicit equations

Successful isolation of dy/dx: 2 solutions were found. Trying to solve each \ resulting ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

```

--- Trying classification methods ---
trying homogeneous types:
trying exact
Looking for potential symmetries
trying an equivalence to an Abel ODE
trying 1st order ODE linearizable_by_differentiation
-> Solving 1st order ODE of high degree, Lie methods, 1st trial
-> Computing symmetries using: way = 3
[2*x, y]
<- successful computation of symmetries.
1st order, trying reduction of order with given symmetries:
[2*x, y]
1st order, trying the canonical coordinates of the invariance group
-> Calling odsolve with the ODE, diff(y(x),x) = 1/2*y(x)/x, y(x)
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- 1st order, canonical coordinates successful
1st order, trying reduction of order with given symmetries:
[2*x, y]
1st order, trying the canonical coordinates of the invariance group
<- 1st order, canonical coordinates successful

```

Maple step by step

Let's solve

$$16y(x)^3 \left(\frac{d}{dx}y(x)\right)^2 - 4x\left(\frac{d}{dx}y(x)\right) + y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = -\frac{-x + \sqrt{x^2 - 4y(x)^4}}{8y(x)^3}, \frac{d}{dx}y(x) = \frac{x + \sqrt{x^2 - 4y(x)^4}}{8y(x)^3} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{-x + \sqrt{x^2 - 4y(x)^4}}{8y(x)^3}$

- Solve the equation $\frac{d}{dx}y(x) = \frac{x + \sqrt{x^2 - 4y(x)^4}}{8y(x)^3}$

- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.328 (sec). Leaf size: 307

```
ode=16*y[x]^3*D[y[x],x]^2-4*x*D[y[x],x]+y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\begin{aligned}
 y(x) &\rightarrow -\sqrt[4]{2}\sqrt[4]{e^{c_1}(x-2e^{c_1})} \\
 y(x) &\rightarrow -i\sqrt[4]{2}\sqrt[4]{e^{c_1}(x-2e^{c_1})} \\
 y(x) &\rightarrow i\sqrt[4]{2}\sqrt[4]{e^{c_1}(x-2e^{c_1})} \\
 y(x) &\rightarrow \sqrt[4]{2}\sqrt[4]{e^{c_1}(x-2e^{c_1})} \\
 y(x) &\rightarrow -\frac{\sqrt[4]{-e^{c_1}(-2x+e^{c_1})}}{\sqrt{2}} \\
 y(x) &\rightarrow -\frac{i\sqrt[4]{-e^{c_1}(-2x+e^{c_1})}}{\sqrt{2}} \\
 y(x) &\rightarrow \frac{i\sqrt[4]{-e^{c_1}(-2x+e^{c_1})}}{\sqrt{2}} \\
 y(x) &\rightarrow \frac{\sqrt[4]{-e^{c_1}(-2x+e^{c_1})}}{\sqrt{2}} \\
 y(x) &\rightarrow 0 \\
 y(x) &\rightarrow -\frac{\sqrt{x}}{\sqrt{2}} \\
 y(x) &\rightarrow -\frac{i\sqrt{x}}{\sqrt{2}} \\
 y(x) &\rightarrow \frac{i\sqrt{x}}{\sqrt{2}} \\
 y(x) &\rightarrow \frac{\sqrt{x}}{\sqrt{2}}
 \end{aligned}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*x*Derivative(y(x), x) + 16*y(x)**3*Derivative(y(x), x)**2 + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

NotImplementedError : The given ODE $-(x + \sqrt{x^2 - 4y(x)^4})/(8y(x)^3) + \text{Derivative}(y(x), x)$ cannot be solved by the factorable group method

2.5.9 Problem 25

Local contents

Solved by factoring the differential equation	1369
✓ Maple	1373
✓ Mathematica	1375
✗ Sympy	1375

Internal problem ID [7950]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 25

Date solved : Thursday, November 27, 2025 at 06:36:14 AM

CAS classification : `[_homogeneous, 'class G', _rational, _Clairaut]`

Solved by factoring the differential equation

Time used: 0.132 (sec)

Solve

$$xy'^5 - yy'^4 + (x^2 + 1)y'^3 - 2xyy'^2 + (x + y^2)y' - y = 0$$

Writing the ode as

$$\left(-y'^3 - y'x + y\right)\left(-xy'^2 + yy' - 1\right) = 0$$

Therefore we need to solve the following equations

$$-y'^3 - y'x + y = 0 \tag{1}$$

$$-xy'^2 + yy' - 1 = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)

Solve This is Clairaut ODE. It has the form

$$y = y'x + g(y')$$

Where g is function of $y'(x)$. Let $p = y'$ the ode becomes

$$-p^3 - px + y = 0$$

Solving for y from the above results in

$$y = p^3 + px \quad (1A)$$

The above ode is a Clairaut ode which is now solved.

Writing the equation (1A) as

$$y = px + g(p)$$

We now write $g \equiv g(p)$ to make notation simpler but we should always remember that g is function of p which in turn is function of x . Hence the above becomes

$$y = px + g \quad (1)$$

Where

$$g = p^3$$

Taking derivative of (1) w.r.t. x gives

$$\begin{aligned} p &= \frac{d}{dx}(xp + g) \\ p &= \left(p + x \frac{dp}{dx}\right) + \left(g' \frac{dp}{dx}\right) \\ p &= p + (x + g') \frac{dp}{dx} \\ 0 &= (x + g') \frac{dp}{dx} \end{aligned}$$

Where g' is derivative of $g(p)$ w.r.t. p .

The general solution is given by

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ p &= c_1 \end{aligned}$$

Substituting this in (1) gives the general solution as

$$y = c_1^3 + c_1 x$$

The singular solution is found from solving for p from

$$x + g'(p) = 0$$

And substituting the result back in (1). Since we found above that $g = p^3$, then the above equation becomes

$$\begin{aligned} x + g'(p) &= 3p^2 + x \\ &= 0 \end{aligned}$$

Solving the above for p results in

$$\begin{aligned} p_1 = y &= -\frac{2\sqrt{3}(-x)^{3/2}}{9} \\ p_2 = y &= \frac{2\sqrt{3}(-x)^{3/2}}{9} \end{aligned}$$

Substituting the above back in (1) results in

$$\begin{aligned} y &= -\frac{2\sqrt{3}(-x)^{3/2}}{9} \\ y &= \frac{2\sqrt{3}(-x)^{3/2}}{9} \end{aligned}$$

Solving equation (2)

Solve This is Clairaut ODE. It has the form

$$y = y'x + g(y')$$

Where g is function of $y'(x)$. Let $p = y'$ the ode becomes

$$-xp^2 + yp - 1 = 0$$

Solving for y from the above results in

$$y = \frac{xp^2 + 1}{p} \tag{1A}$$

The above ode is a Clairaut ode which is now solved.

Writing the equation (1A) as

$$y = px + g(p)$$

We now write $g \equiv g(p)$ to make notation simpler but we should always remember that g is function of p which in turn is function of x . Hence the above becomes

$$y = px + g \tag{1}$$

Where

$$g = \frac{1}{p}$$

Taking derivative of (1) w.r.t. x gives

$$\begin{aligned} p &= \frac{d}{dx}(xp + g) \\ p &= \left(p + x \frac{dp}{dx}\right) + \left(g' \frac{dp}{dx}\right) \\ p &= p + (x + g') \frac{dp}{dx} \\ 0 &= (x + g') \frac{dp}{dx} \end{aligned}$$

Where g' is derivative of $g(p)$ w.r.t. p .

The general solution is given by

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ p &= c_1 \end{aligned}$$

Substituting this in (1) gives the general solution as

$$y = c_2 x + \frac{1}{c_2}$$

The singular solution is found from solving for p from

$$x + g'(p) = 0$$

And substituting the result back in (1). Since we found above that $g = \frac{1}{p}$, then the above equation becomes

$$\begin{aligned} x + g'(p) &= x - \frac{1}{p^2} \\ &= 0 \end{aligned}$$

Solving the above for p results in

$$\begin{aligned} p_1 &= y = 2\sqrt{x} \\ p_2 &= y = -2\sqrt{x} \end{aligned}$$

Substituting the above back in (1) results in

$$\begin{aligned} y &= 2\sqrt{x} \\ y &= -2\sqrt{x} \end{aligned}$$

Summary of solutions found

$$y = c_1(c_1^2 + x)$$

$$y = -2\sqrt{x}$$

$$y = 2\sqrt{x}$$

$$y = -\frac{2\sqrt{3}(-x)^{3/2}}{9}$$

$$y = \frac{2\sqrt{3}(-x)^{3/2}}{9}$$

$$y = c_2x + \frac{1}{c_2}$$

✓ **Maple.** Time used: 0.053 (sec). Leaf size: 63

```
ode:=x*diff(y(x),x)^5-y(x)*diff(y(x),x)^4+(x^2+1)*diff(y(x),x)^3-2*x*y(x)*diff(y(x),x)^2+(x+y(x)^2)*diff(y(x),x)-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{2\sqrt{3}(-x)^{3/2}}{9}$$

$$y = -\frac{2\sqrt{3}(-x)^{3/2}}{9}$$

$$y = c_1(c_1^2 + x)$$

$$y = -2\sqrt{x}$$

$$y = 2\sqrt{x}$$

$$y = c_1x + \frac{1}{c_1}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

```

trying dAlembert
<- dAlembert successful
Methods for first order ODEs:
*** Sublevel 2 ***
Methods for first order ODEs:
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful

```

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^5 - y(x)\left(\frac{d}{dx}y(x)\right)^4 + (x^2 + 1)\left(\frac{d}{dx}y(x)\right)^3 - 2xy(x)\left(\frac{d}{dx}y(x)\right)^2 + (x + y(x)^2)\left(\frac{d}{dx}y(x)\right) - y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{y(x) - \sqrt{y(x)^2 - 4x}}{2x}, \frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x}}{2x}, \frac{d}{dx}y(x) = \frac{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}{6} - \frac{1}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) - \sqrt{y(x)^2 - 4x}}{2x}$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x}}{2x}$

- Solve the equation $\frac{d}{dx}y(x) = \frac{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}{6} - \frac{2x}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}{12} + \frac{x}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}} - \frac{I\sqrt{3}\left(\frac{108y(x) - \sqrt{12x^3 + 81y(x)^2}}{12}\right)^{1/3}}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}{12} + \frac{x}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}} + \frac{I\sqrt{3}\left(\frac{108y(x) - \sqrt{12x^3 + 81y(x)^2}}{12}\right)^{1/3}}{\left(108y(x) + 12\sqrt{12x^3 + 81y(x)^2}\right)^{1/3}}$

- Set of solutions

$$\{workingODE, workingODE, workingODE, workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.027 (sec). Leaf size: 142

```
ode=x*D[y[x],x]^5-y[x]*D[y[x],x]^4+(1+x^2)*D[y[x],x]^3-2*x*y[x]*D[y[x],x]^2+(x+y[x]^2)*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 x + \frac{1}{c_1}$$

$$y(x) \rightarrow c_1 (x + c_1^2)$$

$$y(x) \rightarrow \text{Indeterminate}$$

$$y(x) \rightarrow -x - 1$$

$$y(x) \rightarrow -2\sqrt{x}$$

$$y(x) \rightarrow 2\sqrt{x}$$

$$y(x) \rightarrow -\frac{2ix^{3/2}}{3\sqrt{3}}$$

$$y(x) \rightarrow \frac{2ix^{3/2}}{3\sqrt{3}}$$

$$y(x) \rightarrow x + 1$$

$$y(x) \rightarrow -\sqrt{-(x-1)^2}$$

$$y(x) \rightarrow \sqrt{-(x-1)^2}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*y(x)*Derivative(y(x), x)**2 + x*Derivative(y(x), x)**5 + (x + y(x)
**2)*Derivative(y(x), x) + (x**2 + 1)*Derivative(y(x), x)**3 - y(x)*Derivative
(y(x), x)**4 - y(x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.5.10 Problem 26

Local contents

Solved using first_order_ode_dAlembert	1376
Solved using first_order_ode_homog_type_G	1379
Solved using first_order_ode_parametric method	1381
✓ Maple	1383
✓ Mathematica	1384
✗ Sympy	1384

Internal problem ID [7951]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 26

Date solved : Thursday, November 27, 2025 at 06:36:20 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.414 (sec)

Solve

$$xy'^2 - yy' - y = 0$$

Let $p = y'$ the ode becomes

$$x p^2 - yp - y = 0$$

Solving for y from the above results in

$$y = \frac{x p^2}{p + 1} \quad (1)$$

This has the form

$$y = x f(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of $(*)$ w.r.t. x gives

$$\begin{aligned} p &= f + (x f' + g') \frac{dp}{dx} \\ p - f &= (x f' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{p^2}{p+1} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{p^2}{p+1} = \left(\frac{2xp}{p+1} - \frac{xp^2}{(p+1)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p^2}{p+1} = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 0$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)^2}{p(x)+1}}{\frac{2xp(x)}{p(x)+1} - \frac{xp(x)^2}{(p(x)+1)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{p(x) + 1}{x(p(x) + 2)} \quad (2.109)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{p(x) + 1}{x(p(x) + 2)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(p) &= \frac{p+1}{p+2} \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{p+2}{p+1} dp = \int \frac{1}{x} dx$$

$$p(x) + \ln(p(x) + 1) = \ln(x) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{p+1}{p+2} = 0$$

for $p(x)$ gives

$$p(x) = -1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$p(x) + \ln(p(x) + 1) = \ln(x) + c_1$$

$$p(x) = -1$$

Substituting the above solution for p in (2A) gives

$$y = \frac{x(\text{LambertW}(x e^{c_1+1}) - 1)^2}{\text{LambertW}(x e^{c_1+1})}$$

Summary of solutions found

$$y = 0$$

$$y = \frac{x(\text{LambertW}(x e^{c_1+1}) - 1)^2}{\text{LambertW}(x e^{c_1+1})}$$

Solved using first_order_ode_homog_type_G

Time used: 0.330 (sec)

Solve

$$xy'^2 - yy' - y = 0$$

Multiplying the right side of the ode, which is $\frac{y+\sqrt{4yx+y^2}}{2x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y + \sqrt{4yx + y^2}}{2x} \\ &= \frac{y + \sqrt{4yx + y^2}}{2y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{x}{\sqrt{4yx + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{x}{\sqrt{4yx + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{z + \sqrt{z(z+4)}}{2z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{z + \sqrt{z(z+4)}}{2z}\right)} dz = 0$$

Multiplying the right side of the ode, which is $\frac{y - \sqrt{4yx + y^2}}{2x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y - \sqrt{4yx + y^2}}{2x} \\ &= -\frac{-y + \sqrt{4yx + y^2}}{2y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x}{\sqrt{4yx + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x}{\sqrt{4yx + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{z - \sqrt{z(z+4)}}{2z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{z - \sqrt{z(z+4)}}{2z}\right)} dz = 0$$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{z + \sqrt{z(z+4)}}{2z}\right)} dz = 0$$

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{z - \sqrt{z(z+4)}}{2z}\right)} dz = 0$$

Solved using first_order_ode_parametric method

Time used: 0.162 (sec)

Solve

$$xy'^2 - yy' - y = 0$$

Let y' be a parameter λ . The ode becomes

$$x\lambda^2 - y\lambda - y = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{x\lambda^2}{\lambda + 1} \\ &= \frac{x\lambda^2}{\lambda + 1} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda} x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{\frac{2x\lambda}{\lambda+1} - \frac{x\lambda^2}{(\lambda+1)^2}}{\lambda - \frac{\lambda^2}{\lambda+1}} \\ &= \frac{(\lambda+2)x(\lambda)}{\lambda+1} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{\lambda+2}{\lambda+1} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{\lambda+2}{\lambda+1} d\lambda} \\ &= \frac{e^{-\lambda}}{\lambda+1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x e^{-\lambda}}{\lambda+1}\right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{x e^{-\lambda}}{\lambda+1} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{e^{-\lambda}}{\lambda+1}$ gives the final solution

$$x(\lambda) = (\lambda+1) e^{\lambda} c_1$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{x \lambda^2}{\lambda+1} \\ x &= (\lambda+1) e^{\lambda} c_1 \end{aligned}$$

Eliminating λ gives the solution for y .

$$e^{\text{RootOf}(x_Z^2 - y_Z - y)} c_1 \text{RootOf}(x_Z^2 - y_Z - y) + e^{\text{RootOf}(x_Z^2 - y_Z - y)} c_1 - x$$

Which can be written as

$$x \left(\text{LambertW} \left(\frac{x e}{c_1} \right) - 1 \right)^2 - y \left(\text{LambertW} \left(\frac{x e}{c_1} \right) - 1 \right) - y = 0$$

Simplifying the above gives

$$y = \frac{x \left(\text{LambertW} \left(\frac{x e}{c_1} \right) - 1 \right)^2}{\text{LambertW} \left(\frac{x e}{c_1} \right)}$$

Summary of solutions found

$$y = \frac{x \left(\text{LambertW} \left(\frac{x e}{c_1} \right) - 1 \right)^2}{\text{LambertW} \left(\frac{x e}{c_1} \right)}$$

✓ **Maple.** Time used: 0.020 (sec). Leaf size: 32

```
ode:=diff(y(x),x)^2*x-diff(y(x),x)*y(x)-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = \frac{x \left(\text{LambertW} \left(\frac{x e}{c_1} \right) - 1 \right)^2}{\text{LambertW} \left(\frac{x e}{c_1} \right)}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

<- dAlembert successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^2 - y(x)\left(\frac{d}{dx}y(x)\right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 + 4xy(x)}}{2x}, \frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 + 4xy(x)}}{2x} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 + 4xy(x)}}{2x}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 + 4xy(x)}}{2x}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 1.887 (sec). Leaf size: 161

```
ode=x*D[y[x],x]^2-y[x]*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\begin{aligned} \text{Solve} \left[\operatorname{arctanh} \left(\frac{\sqrt{\frac{y(x)}{x} + 4}}{\sqrt{\frac{y(x)}{x}}} \right) + \frac{\sqrt{\frac{y(x)}{x} + 4}}{\sqrt{\frac{y(x)}{x} + 4} - \sqrt{\frac{y(x)}{x}}} = \frac{\log(x)}{2} + c_1, y(x) \right] \\ \text{Solve} \left[\operatorname{arctanh} \left(\frac{\sqrt{\frac{y(x)}{x} + 4}}{\sqrt{\frac{y(x)}{x}}} \right) - \frac{\sqrt{\frac{y(x)}{x} + 4}}{\sqrt{\frac{y(x)}{x} + 4} + \sqrt{\frac{y(x)}{x}}} = -\frac{\log(x)}{2} + c_1, y(x) \right] \\ y(x) \rightarrow 0 \end{aligned}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x)**2 - y(x)*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

RecursionError : maximum recursion depth exceeded

2.5.11 Problem 27

Local contents

Solved using first_order_ode_parametric method	1385
✓ Maple	1387
✓ Mathematica	1389
✗ Sympy	1389

Internal problem ID [7952]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 27

Date solved : Thursday, November 27, 2025 at 06:36:26 AM

CAS classification : [[_1st_order, _with_linear_symmetries]]

Solved using first_order_ode_parametric method

Time used: 0.125 (sec)

Solve

$$y = 2y'x + y^2y'^3$$

Let y' be a parameter λ . The ode becomes

$$-y^2\lambda^3 - 2\lambda x + y = 0$$

Isolating x gives

$$x = -\frac{y(\lambda^3y - 1)}{2\lambda}$$

$$x = F(y, \lambda)$$

Now we generate an ode in $y(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}y(\lambda) &= \frac{\lambda \frac{\partial F}{\partial \lambda}}{1 - \frac{\partial F}{\partial y}} \\ &= \frac{\lambda \left(-\frac{3y^2\lambda}{2} + \frac{y(\lambda^3y-1)}{2\lambda^2} \right)}{1 - \lambda \left(-\frac{\lambda^3y-1}{2\lambda} - \frac{y\lambda^2}{2} \right)} \\ &= -\frac{y(\lambda)}{\lambda} \end{aligned}$$

Which is now solved for y .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}y(\lambda) + q(\lambda)y(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{\lambda} d\lambda} \\ &= \lambda \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu y &= 0 \\ \frac{d}{d\lambda}(y\lambda) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} y\lambda &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor λ gives the final solution

$$y(\lambda) = \frac{c_1}{\lambda}$$

Now that we have found solution y , we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{c_1}{\lambda} \\ x &= -\frac{y(\lambda^3 y - 1)}{2\lambda} \end{aligned}$$

Eliminating λ gives the solution for y . Solving for y gives

$$\begin{aligned} y &= \sqrt{c_1^3 + 2c_1 x} \\ y &= -\sqrt{c_1^3 + 2c_1 x} \end{aligned}$$

Simplifying the above gives

$$y = \sqrt{c_1 (c_1^2 + 2x)}$$

$$y = -\sqrt{c_1 (c_1^2 + 2x)}$$

Summary of solutions found

$$y = \sqrt{c_1 (c_1^2 + 2x)}$$

$$y = -\sqrt{c_1 (c_1^2 + 2x)}$$

✓ **Maple.** Time used: 0.181 (sec). Leaf size: 97

```
ode:=y(x) = 2*diff(y(x),x)*x+y(x)^2*diff(y(x),x)^3;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{2(-x^3)^{1/4} 6^{1/4}}{3}$$

$$y = \frac{2(-x^3)^{1/4} 6^{1/4}}{3}$$

$$y = -\frac{2i(-x^3)^{1/4} 6^{1/4}}{3}$$

$$y = \frac{2i(-x^3)^{1/4} 6^{1/4}}{3}$$

$$y = 0$$

$$y = \sqrt{c_1 (c_1^2 + 2x)}$$

$$y = -\sqrt{c_1 (c_1^2 + 2x)}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

```

trying simple symmetries for implicit equations
Successful isolation of dy/dx: 3 solutions were found. Trying to solve each \
resulting ODE.
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying homogeneous types:
trying exact
Looking for potential symmetries
trying an equivalence to an Abel ODE
trying 1st order ODE linearizable_by_differentiation
-> Solving 1st order ODE of high degree, Lie methods, 1st trial
-> Computing symmetries using: way = 2
-> Computing symmetries using: way = 2
-> Solving 1st order ODE of high degree, 2nd attempt. Trying parametric meth\
ods
trying dAlembert
-> Calling odsolve with the ODE, diff(y(x),x) = (-2*y(x)^2*x^3-y(x))/(2*y(x)
*x^4+x), y(x)
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- 1st order, parametric methods successful

```

Maple step by step

Let's solve

$$y(x) = 2x\left(\frac{d}{dx}y(x)\right) + y(x)^2\left(\frac{d}{dx}y(x)\right)^3$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}{6y(x)} - \frac{4x}{y(x)\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}, \frac{d}{dx}y(x) = -\frac{\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}{6y(x)} + \frac{4x}{y(x)\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}{6y(x)} - \frac{4x}{y(x)\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}{12y(x)} + \frac{2x}{y(x)\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}} - \frac{1}{y(x)}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}}{12y(x)} + \frac{2x}{y(x)\left(12\sqrt{3}\sqrt{27y(x)^4+32x^3+108y(x)^2}\right)^{1/3}} + \frac{1}{y(x)}$
- Set of solutions
 $\{workingODE, workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.074 (sec). Leaf size: 119

```
ode=y[x]==2*x*D[y[x],x]+y[x]^2*D[y[x],x]^3;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\sqrt{2c_1x + c_1^3}$$

$$y(x) \rightarrow \sqrt{2c_1x + c_1^3}$$

$$y(x) \rightarrow (-1-i)\left(\frac{2}{3}\right)^{3/4}x^{3/4}$$

$$y(x) \rightarrow (1-i)\left(\frac{2}{3}\right)^{3/4}x^{3/4}$$

$$y(x) \rightarrow (-1+i)\left(\frac{2}{3}\right)^{3/4}x^{3/4}$$

$$y(x) \rightarrow (1+i)\left(\frac{2}{3}\right)^{3/4}x^{3/4}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*Derivative(y(x), x) - y(x)**2*Derivative(y(x), x)**3 + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

IndexError : list index out of range

2.5.12 Problem 28

Local contents

Solved using first_order_ode_dAlembert	1390
Solved using first_order_ode_parametric method	1392
✓ Maple	1395
✓ Mathematica	1396
✗ Sympy	1396

Internal problem ID [7953]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 28

Date solved : Thursday, November 27, 2025 at 06:36:31 AM

CAS classification : [[_1st_order, _with_linear_symmetries], _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.134 (sec)

Solve

$$y'^2 - y'x - y = 0$$

Let $p = y'$ the ode becomes

$$p^2 - px - y = 0$$

Solving for y from the above results in

$$y = p^2 - px \tag{1}$$

This has the form

$$y = xf(p) + g(p) \tag{*}$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= -p \\ g &= p^2 \end{aligned}$$

Hence (2) becomes

$$2p = (-x + 2p) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$2p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 0$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{2p(x)}{-x + 2p(x)} \quad (3)$$

Inverting the above ode gives

$$\frac{d}{dp}x(p) = \frac{-x(p) + 2p}{2p} \quad (4)$$

This ODE is now solved for $x(p)$. The integrating factor is

$$\begin{aligned} \mu &= e^{\int \frac{1}{2p} dp} \\ \mu &= \sqrt{p} \\ \mu &= \sqrt{p} \end{aligned} \quad (5)$$

Integrating gives

$$\begin{aligned} x(p) &= \frac{1}{\mu} \left(\int \mu(1) dp + c_1 \right) \\ &= \frac{1}{\mu} \left(\frac{2p^{3/2} + 3c_1}{3\sqrt{p}} + c_1 \right) \\ &= \frac{2p^{3/2} + 3c_1}{3\sqrt{p}} \end{aligned} \quad (5)$$

Now we need to eliminate p between the above solution and (1A). The first method is to solve for p from Eq. (1A) and substitute the result into Eq. (5). The Second method is to solve for p from Eq. (5) and substitute the result into (1A).

Eliminating p from the following two equations

$$x = \frac{2p^{3/2} + 3c_1}{3\sqrt{p}}$$

$$y = p^2 - px$$

results in

$$p = \text{RootOf} \left(2_Z^3 - 3_Zx + 3c_1 \right)^2$$

Substituting the above into Eq (1A) and simplifying gives

$$y = \frac{\text{RootOf} \left(2_Z^3 - 3_Zx + 3c_1 \right) \left(x \text{RootOf} \left(2_Z^3 - 3_Zx + 3c_1 \right) - 3c_1 \right)}{2}$$

Summary of solutions found

$$y = 0$$

$$y = \frac{\text{RootOf} \left(2_Z^3 - 3_Zx + 3c_1 \right) \left(x \text{RootOf} \left(2_Z^3 - 3_Zx + 3c_1 \right) - 3c_1 \right)}{2}$$

Solved using first_order_ode_parametric method

Time used: 0.463 (sec)

Solve

$$y'^2 - y'x - y = 0$$

Let y' be a parameter λ . The ode becomes

$$\lambda^2 - \lambda x - y = 0$$

Isolating y gives

$$\begin{aligned} y &= \lambda^2 - \lambda x \\ &= \lambda(\lambda - x) \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned}\frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{2\lambda - x}{2\lambda} \\ &= \frac{\lambda - \frac{x(\lambda)}{2}}{\lambda}\end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(\lambda) &= \frac{1}{2\lambda} \\ p(\lambda) &= 1\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{2\lambda} d\lambda} \\ &= \sqrt{\lambda}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{d\lambda}(\mu x) &= \mu \\ \frac{d}{d\lambda}(x\sqrt{\lambda}) &= \sqrt{\lambda} \\ d(x\sqrt{\lambda}) &= \sqrt{\lambda}d\lambda\end{aligned}$$

Integrating gives

$$\begin{aligned}x\sqrt{\lambda} &= \int \sqrt{\lambda} d\lambda \\ &= \frac{2\lambda^{3/2}}{3} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{\lambda}$ gives the final solution

$$x(\lambda) = \frac{2\lambda^{3/2} + 3c_1}{3\sqrt{\lambda}}$$

Now that we found solution x we have two equations with parameter λ . They are

$$y = \lambda(\lambda - x)$$

$$x = \frac{2\lambda^{3/2} + 3c_1}{3\sqrt{\lambda}}$$

Eliminating λ gives the solution for y .

$$2\text{RootOf}(_Z^2 - _Zx - y)^{3/2} - 3x\sqrt{\text{RootOf}(_Z^2 - _Zx - y)} + 3c_1$$

Which can be written as

$$\left(\frac{\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}}{2} + \frac{x}{\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}} \right)^4$$

$$- \left(\frac{\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}}{2} + \frac{x}{\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}} \right)^2 x - y = 0$$

Simplifying the above gives

$$y$$

$$= \frac{\left(\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{2/3} + 2x \right)^2 \left(-3 \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3} c_1 + 2x^2 + \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3} \right)}{16 \left(-3c_1 + \sqrt{-2x^3 + 9c_1^2} \right) \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}}$$

Summary of solutions found

$$y$$

$$= \frac{\left(\left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{2/3} + 2x \right)^2 \left(-3 \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3} c_1 + 2x^2 + \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3} \right)}{16 \left(-3c_1 + \sqrt{-2x^3 + 9c_1^2} \right) \left(-6c_1 + 2\sqrt{-2x^3 + 9c_1^2} \right)^{1/3}}$$

✓ **Maple.** Time used: 0.017 (sec). Leaf size: 77

```
ode:=diff(y(x),x)^2-diff(y(x),x)*x-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$\frac{c_1}{\sqrt{2x - 2\sqrt{x^2 + 4y}}} + \frac{2x}{3} + \frac{\sqrt{x^2 + 4y}}{3} = 0$$

$$\frac{c_1}{\sqrt{2x + 2\sqrt{x^2 + 4y}}} + \frac{2x}{3} - \frac{\sqrt{x^2 + 4y}}{3} = 0$$

Maple trace

Methods for first order ODEs:

```
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful
```

Maple step by step

Let's solve

$$\left(\frac{d}{dx}y(x)\right)^2 - x\left(\frac{d}{dx}y(x)\right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{x}{2} - \frac{\sqrt{x^2+4y(x)}}{2}, \frac{d}{dx}y(x) = \frac{x}{2} + \frac{\sqrt{x^2+4y(x)}}{2}\right]$$
- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{2} - \frac{\sqrt{x^2+4y(x)}}{2}$
- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{2} + \frac{\sqrt{x^2+4y(x)}}{2}$
- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 60.069 (sec). Leaf size: 1003

```
ode=D[y[x],x]^2-x*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\left(x^2 + \sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}}\right)^2 + 8e^{3c_1}x}{4\sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}}}$$

$$y(x) \rightarrow \frac{1}{8} \left(4x^2 - \frac{i(\sqrt{3} - i)x(x^3 + 8e^{3c_1})}{\sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}}} + i(\sqrt{3} + i)\sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}} \right)$$

$$y(x) \rightarrow \frac{1}{8} \left(4x^2 + \frac{i(\sqrt{3} + i)x(x^3 + 8e^{3c_1})}{\sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}}} - (1 + i\sqrt{3})\sqrt[3]{-x^6 + 20e^{3c_1}x^3 + 8\sqrt{e^{3c_1}(-x^3 + e^{3c_1})^3 + 8e^{6c_1}}} \right)$$

$$y(x) \rightarrow \frac{2\sqrt[3]{2}x^4 + 2^{2/3}(-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}})^{2/3} + 4x^2\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}}}{8\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}}}$$

$$y(x) \rightarrow \frac{1}{16} \left(8x^2 + \frac{2\sqrt[3]{2}(1 + i\sqrt{3})x(-x^3 + 2e^{3c_1})}{\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}}} + i2^{2/3}(\sqrt{3} + i)\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}} \right)$$

$$y(x) \rightarrow \frac{1}{16} \left(8x^2 + \frac{2i\sqrt[3]{2}(\sqrt{3} + i)x(x^3 - 2e^{3c_1})}{\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}}} - 2^{2/3}(1 + i\sqrt{3})\sqrt[3]{-2x^6 - 10e^{3c_1}x^3 + \sqrt{e^{3c_1}(4x^3 + e^{3c_1})^3 + e^{6c_1}}} \right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) - y(x) + Derivative(y(x), x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE -x/2 - sqrt(x**2 + 4*y(x))/2 + Derivative(y(x), x) cannot be solved by the factorable group method
```


2.5.13 Problem 29

Local contents

Solved using first_order_ode_dAlembert	1397
Solved using first_order_ode_parametric method	1399
✓ Maple	1401
✓ Mathematica	1402
✓ Sympy	1402

Internal problem ID [7954]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 29

Date solved : Thursday, November 27, 2025 at 06:36:37 AM

CAS classification : `[[_1st_order, _with_linear_symmetries], _dAlembert]`

Solved using first_order_ode_dAlembert

Time used: 0.105 (sec)

Solve

$$y = (1 + y')x + y'^2$$

Let $p = y'$ the ode becomes

$$y = (1 + p)x + p^2$$

Solving for y from the above results in

$$y = (1 + p)x + p^2 \tag{1}$$

This has the form

$$y = xf(p) + g(p) \tag{*}$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= 1 + p \\ g &= p^2 \end{aligned}$$

Hence (2) becomes

$$-1 = (x + 2p)p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$-1 = 0$$

Unable to solve for p from the above. No singular solution can be found.

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = -\frac{1}{x + 2p(x)} \quad (3)$$

Inverting the above ode gives

$$\frac{d}{dp}x(p) = -x(p) - 2p \quad (4)$$

This ODE is now solved for $x(p)$. The integrating factor is

$$\begin{aligned} \mu &= e^{\int 1 dp} \\ \mu &= e^p \\ \mu &= e^p \end{aligned} \quad (5)$$

Integrating gives

$$\begin{aligned} x(p) &= \frac{1}{\mu} \left(\int \mu(-2p) dp + c_1 \right) \\ &= \frac{1}{\mu} (-2p + 2 + e^{-p}c_1 + c_1) \\ &= -2p + 2 + e^{-p}c_1 \end{aligned} \quad (5)$$

Now we need to eliminate p between the above solution and (1A). The first method is to solve for p from Eq. (1A) and substitute the result into Eq. (5). The Second method is to solve for p from Eq. (5) and substitute the result into (1A).

Eliminating p from the following two equations

$$\begin{aligned} x &= -2p + 2 + e^{-p}c_1 \\ y &= p^2 + xp + x \end{aligned}$$

results in

$$p = \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right) - \frac{x}{2} + 1$$

Substituting the above into Eq (1A) and simplifying gives

$$y = \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right)^2 + 2 \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right) - \frac{x^2}{4} + x + 1$$

Summary of solutions found

$$y = \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right)^2 + 2 \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right) - \frac{x^2}{4} + x + 1$$

Solved using first_order_ode_parametric method

Time used: 0.944 (sec)

Solve

$$y = (1 + y')x + y'^2$$

Let y' be a parameter λ . The ode becomes

$$y - (1 + \lambda)x - \lambda^2 = 0$$

Isolating y gives

$$\begin{aligned} y &= \lambda^2 + x\lambda + x \\ &= (1 + \lambda)x + \lambda^2 \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= -2\lambda - x \\ &= -2\lambda - x(\lambda) \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= 1 \\ p(\lambda) &= -2\lambda \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int 1 d\lambda} \\ &= e^{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}(\mu x) &= \mu p \\ \frac{d}{d\lambda}(\mu x) &= (\mu)(-2\lambda) \\ \frac{d}{d\lambda}(x e^{\lambda}) &= (e^{\lambda})(-2\lambda) \\ d(x e^{\lambda}) &= (-2\lambda e^{\lambda}) d\lambda \end{aligned}$$

Integrating gives

$$\begin{aligned} x e^{\lambda} &= \int -2\lambda e^{\lambda} d\lambda \\ &= -2(\lambda - 1)e^{\lambda} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^{λ} gives the final solution

$$x(\lambda) = e^{-\lambda}c_1 - 2\lambda + 2$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned} y &= (1 + \lambda)x + \lambda^2 \\ x &= e^{-\lambda}c_1 - 2\lambda + 2 \end{aligned}$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right)^2 + 2 \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right) - \frac{x^2}{4} + x + 1$$

✓ Maple. Time used: 0.017 (sec). Leaf size: 36

```
ode:=y(x) = x*(diff(y(x),x)+1)+diff(y(x),x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = x - \frac{x^2}{4} + \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right)^2 + 2 \text{LambertW}\left(\frac{c_1 e^{\frac{x}{2}-1}}{2}\right) + 1$$

Maple trace

Methods for first order ODEs:

```
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful
```

Maple step by step

Let's solve

$$y(x) = \left(1 + \frac{d}{dx}y(x)\right)x + \left(\frac{d}{dx}y(x)\right)^2$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = -\frac{x}{2} - \frac{\sqrt{x^2+4y(x)-4x}}{2}, \frac{d}{dx}y(x) = -\frac{x}{2} + \frac{\sqrt{x^2+4y(x)-4x}}{2} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{x}{2} - \frac{\sqrt{x^2+4y(x)-4x}}{2}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{x}{2} + \frac{\sqrt{x^2+4y(x)-4x}}{2}$
- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 1.863 (sec). Leaf size: 177

```
ode=y[x]==(1+D[y[x],x])*x+D[y[x],x]^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve}\left[-\sqrt{x^2+4y(x)-4x}+2\log\left(\sqrt{x^2+4y(x)-4x}-x+2\right)-2\log\left(-x\sqrt{x^2+4y(x)-4x}+x^2+4y(x)\right)\right. \\ \left.\text{Solve}\left[-4\operatorname{arctanh}\left(\frac{(x-5)\sqrt{x^2+4y(x)-4x}-x^2-4y(x)+7x-6}{(x-3)\sqrt{x^2+4y(x)-4x}-x^2-4y(x)+5x-2}\right)+\sqrt{x^2+4y(x)-4x}\right]\right]$$

✓ **Sympy.** Time used: 2.545 (sec). Leaf size: 75

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*(Derivative(y(x), x) + 1) + y(x) - Derivative(y(x), x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$\left[C_1 + x + 2\sqrt{\frac{x^2}{4} - x + y(x)} - 2\log\left(\sqrt{\frac{x^2}{4} - x + y(x)} + 1\right) = 0, C_1 \right. \\ \left. + x - 2\sqrt{\frac{x^2}{4} - x + y(x)} - 2\log\left(\sqrt{\frac{x^2}{4} - x + y(x)} - 1\right) = 0 \right]$$

2.5.14 Problem 30

Local contents

Solved using first_order_ode_dAlembert	1403
✓ Maple	1405
✓ Mathematica	1407
✓ Sympy	1407

Internal problem ID [7955]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 30

Date solved : Thursday, November 27, 2025 at 06:36:43 AM

CAS classification : [_quadrature]

Solved using first_order_ode_dAlembert

Time used: 0.188 (sec)

Solve

$$y = 2y' + \sqrt{1 + y'^2}$$

Let $p = y'$ the ode becomes

$$y = 2p + \sqrt{p^2 + 1}$$

Solving for y from the above results in

$$y = 2p + \sqrt{p^2 + 1} \tag{1}$$

This has the form

$$y = xf(p) + g(p) \tag{*}$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= 0 \\ g &= 2p + \sqrt{p^2 + 1} \end{aligned}$$

Hence (2) becomes

$$p = \left(2 + \frac{p}{\sqrt{p^2 + 1}} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 1$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x)}{2 + \frac{p(x)}{\sqrt{p(x)^2 + 1}}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\begin{aligned} \int \frac{2\sqrt{p^2 + 1} + p}{p\sqrt{p^2 + 1}} dp &= dx \\ 2 \ln(p) + \operatorname{arcsinh}(p) &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$\frac{p\sqrt{p^2 + 1}}{2\sqrt{p^2 + 1} + p} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$\begin{aligned} p(x) &= 0 \\ p(x) &= -i \end{aligned}$$

Substituting the above solution for p in (2A) gives

$$y = 2e^{\text{RootOf}(-2_Z - \text{arcsinh}(e^{-Z}) + x + c_1)} + \sqrt{e^{2\text{RootOf}(-2_Z - \text{arcsinh}(e^{-Z}) + x + c_1)} + 1}$$

$$y = 1$$

$$y = -2i$$

The solution

$$y = -2i$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = 2e^{\text{RootOf}(-2_Z - \text{arcsinh}(e^{-Z}) + x + c_1)} + \sqrt{e^{2\text{RootOf}(-2_Z - \text{arcsinh}(e^{-Z}) + x + c_1)} + 1}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = 1$$

✓ **Maple.** Time used: 0.027 (sec). Leaf size: 125

```
ode:=y(x) = 2*diff(y(x),x)+(1+diff(y(x),x)^2)^(1/2);
dsolve(ode,y(x), singsol=all);
```

$$\begin{aligned} & x - \text{arcsinh}\left(\frac{\sqrt{3}y}{3}\right) + \text{arctanh}\left(\frac{3+y}{2\sqrt{y^2+3}}\right) \\ & + \text{arctanh}\left(\frac{-3+y}{2\sqrt{y^2+3}}\right) - \ln(y-1) - \ln(y+1) - c_1 = 0 \\ & x + \text{arcsinh}\left(\frac{\sqrt{3}y}{3}\right) - \text{arctanh}\left(\frac{3+y}{2\sqrt{y^2+3}}\right) \\ & - \text{arctanh}\left(\frac{-3+y}{2\sqrt{y^2+3}}\right) - \ln(y-1) - \ln(y+1) - c_1 = 0 \end{aligned}$$

Maple trace

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

```
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
<- differential order: 1; missing x successful
```

Maple step by step

Let's solve

$$y(x) = 2 \frac{d}{dx} y(x) + \sqrt{1 + \left(\frac{d}{dx} y(x) \right)^2}$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = \frac{2y(x)}{3} - \frac{\sqrt{y(x)^2+3}}{3}, \frac{d}{dx} y(x) = \frac{2y(x)}{3} + \frac{\sqrt{y(x)^2+3}}{3} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = \frac{2y(x)}{3} - \frac{\sqrt{y(x)^2+3}}{3}$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{\frac{2y(x)}{3} - \frac{\sqrt{y(x)^2+3}}{3}} = 1$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{\frac{2y(x)}{3} - \frac{\sqrt{y(x)^2+3}}{3}} dx = \int 1 dx + _C1$$

- Evaluate integral

$$\frac{\sqrt{(y(x)-1)^2+2y(x)+2}}{2} + \operatorname{arcsinh}\left(\frac{\sqrt{3}y(x)}{3}\right) - \operatorname{arctanh}\left(\frac{6+2y(x)}{4\sqrt{(y(x)-1)^2+2y(x)+2}}\right) - \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2} + \operatorname{arctanh}\left(\frac{6+2y(x)}{4\sqrt{(y(x)+1)^2-2y(x)+2}}\right)$$

- Convert *arctanh* to 'ln'

$$\frac{\sqrt{(y(x)-1)^2+2y(x)+2}}{2} + \ln\left(\frac{\sqrt{3}y(x)}{3} + \frac{\sqrt{9+3y(x)^2}}{3}\right) - \frac{\ln\left(\frac{3+y(x)}{2\sqrt{y(x)^2+3}} + 1\right)}{2} + \frac{\ln\left(1 - \frac{3+y(x)}{2\sqrt{y(x)^2+3}}\right)}{2} - \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2} - \ln\left(\frac{\sqrt{3}y(x)}{3} + \frac{\sqrt{9+3y(x)^2}}{3}\right) + \frac{\ln\left(\frac{3+y(x)}{2\sqrt{y(x)^2+3}} + 1\right)}{2} - \frac{\ln\left(1 - \frac{3+y(x)}{2\sqrt{y(x)^2+3}}\right)}{2} + \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2}$$

- Solve the equation $\frac{d}{dx} y(x) = \frac{2y(x)}{3} + \frac{\sqrt{y(x)^2+3}}{3}$

- Separate variables

$$\frac{\frac{d}{dx} y(x)}{\frac{2y(x)}{3} + \frac{\sqrt{y(x)^2+3}}{3}} = 1$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx} y(x)}{\frac{2y(x)}{3} + \frac{\sqrt{y(x)^2+3}}{3}} dx = \int 1 dx + _C1$$

- Evaluate integral

$$-\frac{\sqrt{(y(x)-1)^2+2y(x)+2}}{2} - \operatorname{arcsinh}\left(\frac{\sqrt{3}y(x)}{3}\right) + \operatorname{arctanh}\left(\frac{6+2y(x)}{4\sqrt{(y(x)-1)^2+2y(x)+2}}\right) + \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2} - \operatorname{arctanh}\left(\frac{6+2y(x)}{4\sqrt{(y(x)+1)^2-2y(x)+2}}\right)$$

- Convert *arctanh* to 'ln'

$$-\frac{\sqrt{(y(x)-1)^2+2y(x)+2}}{2} - \ln\left(\frac{\sqrt{3}y(x)}{3} + \frac{\sqrt{9+3y(x)^2}}{3}\right) + \frac{\ln\left(\frac{3+y(x)}{2\sqrt{y(x)^2+3}} + 1\right)}{2} - \frac{\ln\left(1 - \frac{3+y(x)}{2\sqrt{y(x)^2+3}}\right)}{2} + \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2} - \ln\left(\frac{\sqrt{3}y(x)}{3} + \frac{\sqrt{9+3y(x)^2}}{3}\right) + \frac{\ln\left(\frac{3+y(x)}{2\sqrt{y(x)^2+3}} + 1\right)}{2} - \frac{\ln\left(1 - \frac{3+y(x)}{2\sqrt{y(x)^2+3}}\right)}{2} + \frac{\sqrt{(y(x)+1)^2-2y(x)+2}}{2}$$

- Set of solutions

$\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 60.218 (sec). Leaf size: 4821

```
ode=y[x]==2*D[y[x],x]+Sqrt[1+D[y[x],x]^2];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Too large to display

✓ **Sympy.** Time used: 0.799 (sec). Leaf size: 65

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-sqrt(Derivative(y(x), x)**2 + 1) + y(x) - 2*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\begin{aligned} x - 2 \log \left(\sqrt{y^2(x) + 3} - 2y(x) \right) - \operatorname{asinh} \left(\frac{\sqrt{3}y(x)}{3} \right) &= C_1, \quad x \\ - 2 \log \left(\sqrt{y^2(x) + 3} + 2y(x) \right) + \operatorname{asinh} \left(\frac{\sqrt{3}y(x)}{3} \right) &= C_1 \end{aligned} \right]$$

2.5.15 Problem 31

Local contents

Solved using first_order_ode_dAlembert	1408
Solved using first_order_ode_parametric method	1412
✓ Maple	1414
✓ Mathematica	1415
✓ Sympy	1415

Internal problem ID [7956]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 9. Equations of first order and higher degree. Supplementary problems. Page 65

Problem number : 31

Date solved : Thursday, November 27, 2025 at 06:36:58 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.558 (sec)

Solve

$$y'^2 y - y' x + 3y = 0$$

Let $p = y'$ the ode becomes

$$p^2 y - px + 3y = 0$$

Solving for y from the above results in

$$y = \frac{px}{p^2 + 3} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= \frac{p}{p^2 + 3} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$p - \frac{p}{p^2 + 3} = \left(\frac{x}{p^2 + 3} - \frac{2xp^2}{(p^2 + 3)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p}{p^2 + 3} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= 0 \\ p_2 &= i\sqrt{2} \\ p_3 &= -i\sqrt{2} \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= 0 \\ y &= i\sqrt{2}x \\ y &= -i\sqrt{2}x \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)}{p(x)^2 + 3}}{\frac{x}{p(x)^2 + 3} - \frac{2xp(x)^2}{(p(x)^2 + 3)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = -\frac{(p(x)^2 + 3)(p(x)^2 + 2)p(x)}{x(p(x)^2 - 3)} \quad (2.110)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= -\frac{(p(x)^2 + 3)(p(x)^2 + 2)p(x)}{x(p(x)^2 - 3)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(p) = \frac{(p^2 + 3)(p^2 + 2)p}{p^2 - 3}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{p^2 - 3}{(p^2 + 3)(p^2 + 2)p} dp = \int -\frac{1}{x} dx$$

$$\ln \left(\frac{(p(x)^2 + 2)^{5/4}}{(p(x)^2 + 3) \sqrt{p(x)}} \right) = \ln \left(\frac{1}{x} \right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{(p(x)^2 + 2)^{5/4}}{(p(x)^2 + 3) \sqrt{p(x)}} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{(p^2 + 3)(p^2 + 2)p}{p^2 - 3} = 0$$

for $p(x)$ gives

$$p(x) = 0$$

$$p(x) = -i\sqrt{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{(p(x)^2 + 2)^{5/4}}{(p(x)^2 + 3) \sqrt{p(x)}} = \frac{c_1}{x}$$

$$p(x) = 0$$

$$p(x) = -i\sqrt{2}$$

Substituting the above solution for p in (2A) gives

$$y = \frac{\text{RootOf}((c_1^4 - x^4)Z^{20} + 2c_1^4Z^{16} - 2c_1^4Z^{12} - 8c_1^4Z^8 - 7c_1^4Z^4)}{c_1^2 \left(\text{RootOf}((c_1^4 - x^4)Z^{20} + 2c_1^4Z^{16} - 2c_1^4Z^{12} - 8c_1^4Z^8 - 7c_1^4Z^4 - 2c_1^4) + 1 \right)^2 \left(\frac{\text{RootOf}((c_1^4 - x^4)Z^2)}{c_1^4 \left(\text{RootOf}((c_1^4 - x^4)Z^{20} + 2c_1^4Z^{16} - 2c_1^4Z^{12} - 8c_1^4Z^8 - 7c_1^4Z^4 - 2c_1^4) + 1 \right)^2} \right)}$$

$$y = 0$$

$$y = -i\sqrt{2}x$$

Simplifying the above gives

$$y = 0$$

$$y = i\sqrt{2}x$$

$$y = -i\sqrt{2}x$$

Expression too large to display

$$y = 0$$

$$y = -i\sqrt{2}x$$

The solution

Expression too large to display

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = 0$$

$$y = -i\sqrt{2}x$$

$$y = i\sqrt{2}x$$

Solved using first_order_ode_parametric method

Time used: 0.332 (sec)

Solve

$$y'^2 y - y' x + 3y = 0$$

Let y' be a parameter λ . The ode becomes

$$\lambda^2 y - \lambda x + 3y = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{\lambda x}{\lambda^2 + 3} \\ &= \frac{\lambda x}{\lambda^2 + 3} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda} x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{\frac{x}{\lambda^2 + 3} - \frac{2\lambda^2 x}{(\lambda^2 + 3)^2}}{\lambda - \frac{\lambda}{\lambda^2 + 3}} \\ &= -\frac{(\lambda^2 - 3)x(\lambda)}{(\lambda^2 + 2)\lambda(\lambda^2 + 3)} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda} x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{-\lambda^2 + 3}{(\lambda^2 + 2)\lambda(\lambda^2 + 3)} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{-\lambda^2 + 3}{(\lambda^2 + 2)\lambda(\lambda^2 + 3)} d\lambda} \\ &= \frac{(\lambda^2 + 2)^{5/4}}{(\lambda^2 + 3)\sqrt{\lambda}} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x(\lambda^2+2)^{5/4}}{(\lambda^2+3)\sqrt{\lambda}}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{x(\lambda^2+2)^{5/4}}{(\lambda^2+3)\sqrt{\lambda}} &= \int 0 d\lambda + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{(\lambda^2+2)^{5/4}}{(\lambda^2+3)\sqrt{\lambda}}$ gives the final solution

$$x(\lambda) = \frac{(\lambda^2+3)\sqrt{\lambda}c_1}{(\lambda^2+2)^{5/4}}$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned}y &= \frac{\lambda x}{\lambda^2+3} \\ x &= \frac{(\lambda^2+3)\sqrt{\lambda}c_1}{(\lambda^2+2)^{5/4}}\end{aligned}$$

Eliminating λ gives the solution for y .

$$\begin{aligned}x\left(\frac{x\text{RootOf}(_Z^2y - _Zx + 3y) - y}{y}\right)^{5/4} \\ - \text{RootOf}(_Z^2y - _Zx + 3y)^{5/2}c_1 - 3\sqrt{\text{RootOf}(_Z^2y - _Zx + 3y)}c_1\end{aligned}$$

Which can be written as

$$x\left(\frac{\frac{x(x+\sqrt{x^2-12y^2})}{2y} - y}{y}\right)^{5/4} - \frac{\sqrt{2}\left(\frac{x+\sqrt{x^2-12y^2}}{y}\right)^{5/2}c_1}{8} - \frac{3\sqrt{2}\sqrt{\frac{x+\sqrt{x^2-12y^2}}{y}}c_1}{2} = 0$$

Simplifying the above gives

$$\frac{\left((-2y^2 + x\sqrt{x^2-12y^2} + x^2)\left(\frac{-4y^2+2x\sqrt{x^2-12y^2}+2x^2}{y^2}\right)^{1/4} - c_1\sqrt{\frac{x+\sqrt{x^2-12y^2}}{y}}(x+\sqrt{x^2-12y^2})\right)x\sqrt{2}}{4y^2} = 0$$

Summary of solutions found

$$\frac{\left((-2y^2 + x\sqrt{x^2 - 12y^2} + x^2) \left(\frac{-4y^2 + 2x\sqrt{x^2 - 12y^2} + 2x^2}{y^2} \right)^{1/4} - c_1 \sqrt{\frac{x + \sqrt{x^2 - 12y^2}}{y}} (x + \sqrt{x^2 - 12y^2}) \right) x\sqrt{2}}{4y^2} = 0$$

✓ **Maple.** Time used: 0.238 (sec). Leaf size: 159

```
ode:=y(x)*diff(y(x),x)^2-diff(y(x),x)*x+3*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$\ln(x) - \frac{\operatorname{arctanh}\left(\frac{1}{\sqrt{-\frac{12y^2-x^2}{x^2}}}\right)}{4} + \frac{5 \operatorname{arctanh}\left(\frac{\sqrt{\frac{x^2-12y^2}{x^2}}}{5}\right)}{4} - \frac{\ln\left(\frac{y}{x}\right)}{4} + \frac{5 \ln\left(\frac{2x^2+y^2}{x^2}\right)}{8} - c_1 = 0$$

$$\ln(x) + \frac{\operatorname{arctanh}\left(\frac{1}{\sqrt{-\frac{12y^2-x^2}{x^2}}}\right)}{4} - \frac{5 \operatorname{arctanh}\left(\frac{\sqrt{\frac{x^2-12y^2}{x^2}}}{5}\right)}{4} - \frac{\ln\left(\frac{y}{x}\right)}{4} + \frac{5 \ln\left(\frac{2x^2+y^2}{x^2}\right)}{8} - c_1 = 0$$

Maple trace

Methods for first order ODEs:

```
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful
```

Maple step by step

Let's solve

$$y(x) \left(\frac{d}{dx} y(x) \right)^2 - x \left(\frac{d}{dx} y(x) \right) + 3y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = -\frac{-x + \sqrt{x^2 - 12y(x)^2}}{2y(x)}, \frac{d}{dx} y(x) = \frac{x + \sqrt{x^2 - 12y(x)^2}}{2y(x)} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = -\frac{-x + \sqrt{x^2 - 12y(x)^2}}{2y(x)}$

- Solve the equation $\frac{d}{dx}y(x) = \frac{x + \sqrt{x^2 - 12y(x)^2}}{2y(x)}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 60.171 (sec). Leaf size: 1131

```
ode=y[x]*D[y[x],x]^2-x*D[y[x],x]+3*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Solution too large to show

✓ **Sympy.** Time used: 17.139 (sec). Leaf size: 87

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) + y(x)*Derivative(y(x), x)**2 + 3*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\log(x) = C_1 + \log \left(\frac{\sqrt{2} \left(\sqrt{1 - \frac{12y^2(x)}{x^2}} - 1 \right)^{\frac{3}{4}}}{2 \left(\sqrt{1 - \frac{12y^2(x)}{x^2}} - 5 \right)^{\frac{3}{4}} \sqrt{\sqrt{1 - \frac{12y^2(x)}{x^2}} - 1 + \frac{2y^2(x)}{x^2}}} \right), y(x) = 0 \right]$$

2.6 Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Local contents

2.6.1	Problem 10	1417
2.6.2	Problem 11	1421
2.6.3	Problem 12	1426
2.6.4	Problem 13	1435
2.6.5	Problem 14	1445
2.6.6	Problem 15	1452
2.6.7	Problem 16	1459
2.6.8	Problem 17	1465
2.6.9	Problem 18	1470
2.6.10	Problem 19	1475

2.6.1 Problem 10

Local contents

Solved using first_order_ode_clairaut	1417
✓ Maple	1419
✓ Mathematica	1420
✓ Sympy	1420

Internal problem ID [7957]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplemetary problems. Page 74

Problem number : 10

Date solved : Thursday, November 27, 2025 at 06:37:04 AM

CAS classification : `[_1st_order, _with_linear_symmetries], _Clairaut]`

Solved using first_order_ode_clairaut

Time used: 0.050 (sec)

Solve

$$y = y'x - 2y'^2$$

This is Clairaut ODE. It has the form

$$y = y'x + g(y')$$

Where g is function of $y'(x)$. Let $p = y'$ the ode becomes

$$y = -2p^2 + px$$

Solving for y from the above results in

$$y = -2p^2 + px \tag{1A}$$

The above ode is a Clairaut ode which is now solved.

Writing the equation (1A) as

$$y = px + g(p)$$

We now write $g \equiv g(p)$ to make notation simpler but we should always remember that g is function of p which in turn is function of x . Hence the above becomes

$$y = px + g \tag{1}$$

Where

$$g = -2p^2$$

Taking derivative of (1) w.r.t. x gives

$$\begin{aligned} p &= \frac{d}{dx}(xp + g) \\ p &= \left(p + x \frac{dp}{dx}\right) + \left(g' \frac{dp}{dx}\right) \\ p &= p + (x + g') \frac{dp}{dx} \\ 0 &= (x + g') \frac{dp}{dx} \end{aligned}$$

Where g' is derivative of $g(p)$ w.r.t. p .

The general solution is given by

$$\begin{aligned} \frac{dp}{dx} &= 0 \\ p &= c_1 \end{aligned}$$

Substituting this in (1) gives the general solution as

$$y = -2c_1^2 + c_1x$$

The singular solution is found from solving for p from

$$x + g'(p) = 0$$

And substituting the result back in (1). Since we found above that $g = -2p^2$, then the above equation becomes

$$\begin{aligned} x + g'(p) &= x - 4p \\ &= 0 \end{aligned}$$

Solving the above for p results in

$$p_1 = y = \frac{x^2}{8}$$

Substituting the above back in (1) results in

$$y = \frac{x^2}{8}$$

Simplifying the above gives

$$y = c_1(-2c_1 + x)$$

$$y = \frac{x^2}{8}$$

Summary of solutions found

$$y = c_1(-2c_1 + x)$$

$$y = \frac{x^2}{8}$$

✓ **Maple.** Time used: 0.018 (sec). Leaf size: 19

```
ode:=y(x) = diff(y(x),x)*x-2*diff(y(x),x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2}{8}$$

$$y = c_1(-2c_1 + x)$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

<- dAlembert successful

Maple step by step

Let's solve

$$y(x) = \left(\frac{d}{dx}y(x)\right)x - 2\left(\frac{d}{dx}y(x)\right)^2$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{x}{4} - \frac{\sqrt{x^2-8y(x)}}{4}, \frac{d}{dx}y(x) = \frac{x}{4} + \frac{\sqrt{x^2-8y(x)}}{4} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{4} - \frac{\sqrt{x^2-8y(x)}}{4}$
- Solve the equation $\frac{d}{dx}y(x) = \frac{x}{4} + \frac{\sqrt{x^2-8y(x)}}{4}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.005 (sec). Leaf size: 25

```
ode=y[x]==D[y[x],x]*x-2*D[y[x],x]^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1(x - 2c_1)$$

$$y(x) \rightarrow \frac{x^2}{8}$$

✓ **Sympy.** Time used: 1.083 (sec). Leaf size: 14

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*Derivative(y(x), x) + y(x) + 2*Derivative(y(x), x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x^2}{8} - \frac{(C_1 + x)^2}{8}$$

2.6.2 Problem 11

Local contents

Solved using first_order_ode_parametric method	1421
✓ Maple	1423
✓ Mathematica	1425
✗ Sympy	1425

Internal problem ID [7958]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 11

Date solved : Thursday, November 27, 2025 at 06:37:10 AM

CAS classification : `[_1st_order, _with_linear_symmetries], _rational]`

Solved using first_order_ode_parametric method

Time used: 0.537 (sec)

Solve

$$y'^2 y^2 + 3y'x - y = 0$$

Let y' be a parameter λ . The ode becomes

$$\lambda^2 y^2 + 3\lambda x - y = 0$$

Isolating x gives

$$x = -\frac{y(\lambda^2 y - 1)}{3\lambda}$$

$$x = F(y, \lambda)$$

Now we generate an ode in $y(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda} y(\lambda) &= \frac{\lambda \frac{\partial F}{\partial \lambda}}{1 - \frac{\partial F}{\partial y}} \\ &= \frac{\lambda \left(-\frac{2y^2}{3} + \frac{y(\lambda^2 y - 1)}{3\lambda^2} \right)}{1 - \lambda \left(-\frac{\lambda^2 y - 1}{3\lambda} - \frac{y\lambda}{3} \right)} \\ &= -\frac{y(\lambda)}{2\lambda} \end{aligned}$$

Which is now solved for y .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}y(\lambda) + q(\lambda)y(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{2\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{2\lambda} d\lambda} \\ &= \sqrt{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu y &= 0 \\ \frac{d}{d\lambda}(y\sqrt{\lambda}) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} y\sqrt{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{\lambda}$ gives the final solution

$$y(\lambda) = \frac{c_1}{\sqrt{\lambda}}$$

Now that we have found solution y , we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{c_1}{\sqrt{\lambda}} \\ x &= -\frac{y(\lambda^2 y - 1)}{3\lambda} \end{aligned}$$

Eliminating λ gives the solution for y .

$$y\sqrt{\text{RootOf}(y^2_Z^2 + 3_Zx - y)} - c_1$$

Which can be written as

$$\frac{c_1^4}{y^2} + \frac{3c_1^2 x}{y^2} - y = 0$$

Simplifying the above gives

$$\frac{c_1^4 + 3c_1^2x - y^3}{y^2} = 0$$

Summary of solutions found

$$\frac{c_1^4 + 3c_1^2x - y^3}{y^2} = 0$$

✓ **Maple.** Time used: 0.158 (sec). Leaf size: 125

```
ode:=y(x)^2*diff(y(x),x)^2+3*diff(y(x),x)*x-y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{18^{1/3}(-x^2)^{1/3}}{2}$$

$$y = -\frac{2^{1/3}(-x^2)^{1/3}(3i3^{1/6} + 3^{2/3})}{4}$$

$$y = \frac{2^{1/3}(-x^2)^{1/3}(-3^{2/3} + 3i3^{1/6})}{4}$$

$$y = 0$$

$$y = \text{RootOf} \left(-2\ln(x) - 3 \int^{-x} \frac{4a^3 + 3\sqrt{4a^3 + 9} + 9}{-a(4a^3 + 9)} da + 2c_1 \right) x^{2/3}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying simple symmetries for implicit equations

Successful isolation of dy/dx: 2 solutions were found. Trying to solve each \ resulting ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

```

trying exact
Looking for potential symmetries
trying an equivalence to an Abel ODE
trying 1st order ODE linearizable_by_differentiation
-> Solving 1st order ODE of high degree, Lie methods, 1st trial
    -> Computing symmetries using: way = 3
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
    <- successful computation of symmetries.
1st order, trying reduction of order with given symmetries:
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
1st order, trying the canonical coordinates of the invariance group
    -> Calling odsolve with the ODE, diff(y(x),x) = 2/3*y(x)/x, y(x)
    *** Sublevel 3 ***
    Methods for first order ODEs:
    --- Trying classification methods ---
    trying a quadrature
    trying 1st order linear
    <- 1st order linear successful
<- 1st order, canonical coordinates successful
1st order, trying reduction of order with given symmetries:
[3/2*x, y], [2/3*y^3+3*x^2, y*x]
1st order, trying the canonical coordinates of the invariance group
<- 1st order, canonical coordinates successful

```

Maple step by step

Let's solve

$$y(x)^2 \left(\frac{d}{dx} y(x) \right)^2 + 3x \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = -\frac{3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}, \frac{d}{dx} y(x) = \frac{-3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = -\frac{3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}$
- Solve the equation $\frac{d}{dx} y(x) = \frac{-3x + \sqrt{9x^2 + 4y(x)^3}}{2y(x)^2}$

- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.326 (sec). Leaf size: 204

```
ode=y[x]^2*D[y[x],x]^2+3*x*D[y[x],x]-y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[\frac{3}{2} \log(y(x)) - \frac{2 \sqrt{\frac{9x^2}{4y(x)^3} + 1} \operatorname{arcsinh}\left(\frac{3}{2}x \sqrt{\frac{1}{y(x)^3}}\right)}{\sqrt{\frac{1}{y(x)^3}} \sqrt{9x^2 + 4y(x)^3}} = c_1, y(x) \right]$$

$$\text{Solve} \left[\frac{2 \sqrt{\frac{9x^2}{4y(x)^3} + 1} \operatorname{arcsinh}\left(\frac{3}{2}x \sqrt{\frac{1}{y(x)^3}}\right)}{\sqrt{\frac{1}{y(x)^3}} \sqrt{9x^2 + 4y(x)^3}} + \frac{3}{2} \log(y(x)) = c_1, y(x) \right]$$

$$y(x) \rightarrow -\left(-\frac{3}{2}\right)^{2/3} x^{2/3}$$

$$y(x) \rightarrow -\left(\frac{3}{2}\right)^{2/3} x^{2/3}$$

$$y(x) \rightarrow \frac{\sqrt[3]{-13} 2^{2/3} x^{2/3}}{2^{2/3}}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x*Derivative(y(x), x) + y(x)**2*Derivative(y(x), x)**2 - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE -(-3*x + sqrt(9*x**2 + 4*y(x)**3))/(2*y(x)
**2) + Derivative(y(x), x) cannot be solved by the factorable group method
```

2.6.3 Problem 12

Local contents

Solved using first_order_ode_dAlembert	1426
Solved using first_order_ode_homog_type_G	1429
Solved using first_order_ode_parametric method	1431
✓ Maple	1433
✓ Mathematica	1434
✓ Sympy	1434

Internal problem ID [7959]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 12

Date solved : Thursday, November 27, 2025 at 06:37:15 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.164 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Let $p = y'$ the ode becomes

$$x p^2 - 2yp + 4x = 0$$

Solving for y from the above results in

$$y = \frac{x(p^2 + 4)}{2p} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of $(*)$ w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{p^2 + 4}{2p}$$

$$g = 0$$

Hence (2) becomes

$$p - \frac{p^2 + 4}{2p} = \left(\frac{x}{2} - \frac{2x}{p^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p^2 + 4}{2p} = 0$$

Solving the above for p results in

$$p_1 = 2$$

$$p_2 = -2$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 2x$$

$$y = -2x$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)^2 + 4}{2p(x)}}{\frac{x}{2} - \frac{2x}{p(x)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{p(x)}{x} \quad (2.111)$$

is separable as it can be written as

$$p'(x) = \frac{p(x)}{x}$$

$$= f(x)g(p)$$

Where

$$f(x) = \frac{1}{x}$$

$$g(p) = p$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{1}{p} dp = \int \frac{1}{x} dx$$

$$\ln(p(x)) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$p(x) = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p = 0$$

for $p(x)$ gives

$$p(x) = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$p(x) = c_1 x$$

$$p(x) = 0$$

Substituting the above solution for p in (2A) gives

$$y = \frac{c_1^2 x^2 + 4}{2c_1}$$

Summary of solutions found

$$y = -2x$$

$$y = 2x$$

$$y = \frac{c_1^2 x^2 + 4}{2c_1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.315 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Multiplying the right side of the ode, which is $\frac{y+\sqrt{-4x^2+y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y + \sqrt{-4x^2 + y^2}}{x} \\ &= \frac{y + \sqrt{-4x^2 + y^2}}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{4x^2}{y\sqrt{-4x^2 + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{(y + \sqrt{-4x^2 + y^2})(\sqrt{-4x^2 + y^2} - y)}{y\sqrt{-4x^2 + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{\sqrt{z^2 - 4} + z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2 - 4} + z}{z}\right)} dz = 0$$

Multiplying the right side of the ode, which is $\frac{y - \sqrt{-4x^2 + y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y - \sqrt{-4x^2 + y^2}}{x} \\ &= -\frac{\sqrt{-4x^2 + y^2} - y}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{4x^2}{y\sqrt{-4x^2 + y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{(y + \sqrt{-4x^2 + y^2})(\sqrt{-4x^2 + y^2} - y)}{y\sqrt{-4x^2 + y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{-\sqrt{z^2 - 4} + z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-4}+z}{z}\right)} dz = 0$$

Solved using first_order_ode_parametric method

Time used: 0.114 (sec)

Solve

$$xy'^2 - 2yy' + 4x = 0$$

Let y' be a parameter λ . The ode becomes

$$x\lambda^2 - 2y\lambda + 4x = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{x(\lambda^2 + 4)}{2\lambda} \\ &= \frac{x(\lambda^2 + 4)}{2\lambda} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{x - \frac{x(\lambda^2+4)}{2\lambda^2}}{\lambda - \frac{\lambda^2+4}{2\lambda}} \\ &= \frac{x(\lambda)}{\lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{1}{\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{1}{\lambda} d\lambda} \\ &= \frac{1}{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x}{\lambda}\right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{x}{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{\lambda}$ gives the final solution

$$x(\lambda) = \lambda c_1$$

Now that we found solution x we have two equations with parameter λ . They are

$$y = \frac{x(\lambda^2 + 4)}{2\lambda}$$

$$x = \lambda c_1$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \frac{4c_1^2 + x^2}{2c_1}$$

✓ **Maple.** Time used: 0.026 (sec). Leaf size: 30

```
ode:=4*x-2*diff(y(x),x)*y(x)+diff(y(x),x)^2*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -2x$$

$$y = 2x$$

$$y = \frac{4c_1^2 + x^2}{2c_1}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

<- dAlembert successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^2 - 2y(x)\left(\frac{d}{dx}y(x)\right) + 4x = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x^2}}{x}, \frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 - 4x^2}}{x} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 4x^2}}{x}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{-y(x) + \sqrt{y(x)^2 - 4x^2}}{x}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.092 (sec). Leaf size: 43

```
ode=x*D[y[x],x]^2-2*y[x]*D[y[x],x]+4*x==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -2x \cosh(-\log(x) + c_1)$$

$$y(x) \rightarrow -2x \cosh(\log(x) + c_1)$$

$$y(x) \rightarrow -2x$$

$$y(x) \rightarrow 2x$$

✓ **Sympy.** Time used: 1.445 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x)**2 + 4*x - 2*y(x)*Derivative(y(x), x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = 2x^2e^{-C_1} + \frac{e^{C_1}}{2}$$

2.6.4 Problem 13

Local contents

Solved using first_order_ode_dAlembert	1435
Solved using first_order_ode_homog_type_G	1438
Solved using first_order_ode_parametric method	1441
✓ Maple	1442
✓ Mathematica	1443
✓ Sympy	1444

Internal problem ID [7960]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 13

Date solved : Thursday, November 27, 2025 at 06:37:21 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.191 (sec)

Solve

$$xy'^2 - 2yy' + x + 2y = 0$$

Let $p = y'$ the ode becomes

$$x p^2 - 2yp + x + 2y = 0$$

Solving for y from the above results in

$$y = \frac{x(p^2 + 1)}{2p - 2} \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = \frac{p^2 + 1}{2p - 2}$$

$$g = 0$$

Hence (2) becomes

$$p - \frac{p^2 + 1}{2p - 2} = \left(\frac{2xp}{2p - 2} - \frac{2xp^2}{(2p - 2)^2} - \frac{2x}{(2p - 2)^2} \right) p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p - \frac{p^2 + 1}{2p - 2} = 0$$

Solving the above for p results in

$$p_1 = 1 + \sqrt{2}$$

$$p_2 = 1 - \sqrt{2}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = (1 + \sqrt{2})x$$

$$y = -(\sqrt{2} - 1)x$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = \frac{p(x) - \frac{p(x)^2 + 1}{2p(x) - 2}}{\frac{2xp(x)}{2p(x) - 2} - \frac{2xp(x)^2}{(2p(x) - 2)^2} - \frac{2x}{(2p(x) - 2)^2}} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{p(x) - 1}{x} \quad (2.112)$$

is separable as it can be written as

$$p'(x) = \frac{p(x) - 1}{x}$$

$$= f(x)g(p)$$

Where

$$\begin{aligned}f(x) &= \frac{1}{x} \\g(p) &= p - 1\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{1}{p-1} dp &= \int \frac{1}{x} dx\end{aligned}$$

$$\ln(p(x) - 1) = \ln(x) + c_1$$

Taking the exponential of both sides the solution becomes

$$p(x) - 1 = c_1 x$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p - 1 = 0$$

for $p(x)$ gives

$$p(x) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\begin{aligned}p(x) - 1 &= c_1 x \\ p(x) &= 1\end{aligned}$$

Substituting the above solution for p in (2A) gives

$$y = \frac{(c_1 x + 1)^2 + 1}{2c_1}$$

Simplifying the above gives

$$y = (1 + \sqrt{2})x$$

$$y = -(\sqrt{2} - 1)x$$

$$y = \frac{c_1^2 x^2 + 2c_1 x + 2}{2c_1}$$

Summary of solutions found

$$y = (1 + \sqrt{2})x$$

$$y = -(\sqrt{2} - 1)x$$

$$y = \frac{c_1^2 x^2 + 2c_1 x + 2}{2c_1}$$

Solved using first_order_ode_homog_type_G

Time used: 0.336 (sec)

Solve

$$xy'^2 - 2yy' + x + 2y = 0$$

Multiplying the right side of the ode, which is $\frac{y + \sqrt{-x^2 - 2yx + y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) \frac{y + \sqrt{-x^2 - 2yx + y^2}}{x} \\ &= \frac{y + \sqrt{-x^2 - 2yx + y^2}}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x(x+y)}{y\sqrt{-x^2-2yx+y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x(x+y)}{y\sqrt{-x^2-2yx+y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{1} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{\sqrt{z^2 - 2z - 1} + z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2 - 2z - 1} + z}{z} \right)} dz = 0$$

Multiplying the right side of the ode, which is $\frac{y - \sqrt{-x^2 - 2yx + y^2}}{x}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y} \right) \frac{y - \sqrt{-x^2 - 2yx + y^2}}{x} \\ &= -\frac{-y + \sqrt{-x^2 - 2yx + y^2}}{y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= \frac{x(x+y)}{y\sqrt{-x^2-2yx+y^2}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{x(x+y)}{y\sqrt{-x^2-2yx+y^2}} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{-\sqrt{z^2-2z-1}+z}{z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-2z-1}+z}{z} \right)} dz = 0$$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{z^2-2z-1}}{z} \right)} dz = 0$$

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z^2-2z-1}+z}{z} \right)} dz = 0$$

Solved using first_order_ode_parametric method

Time used: 0.123 (sec)

Solve

$$xy'^2 - 2yy' + x + 2y = 0$$

Let y' be a parameter λ . The ode becomes

$$x\lambda^2 - 2y\lambda + x + 2y = 0$$

Isolating y gives

$$\begin{aligned} y &= \frac{x(\lambda^2 + 1)}{2\lambda - 2} \\ &= \frac{x(\lambda^2 + 1)}{2\lambda - 2} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{\frac{x\lambda}{\lambda-1} - \frac{x(\lambda^2+1)}{2(\lambda-1)^2}}{\lambda - \frac{\lambda^2+1}{2(\lambda-1)}} \\ &= \frac{x(\lambda)}{\lambda - 1} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{1}{\lambda - 1} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{1}{\lambda-1} d\lambda} \\ &= \frac{1}{\lambda - 1} \end{aligned}$$

The ode becomes

$$\frac{d}{d\lambda} \mu x = 0$$

$$\frac{d}{d\lambda} \left(\frac{x}{\lambda - 1} \right) = 0$$

Integrating gives

$$\frac{x}{\lambda - 1} = \int 0 d\lambda + c_1$$

$$= c_1$$

Dividing throughout by the integrating factor $\frac{1}{\lambda-1}$ gives the final solution

$$x(\lambda) = (\lambda - 1) c_1$$

Now that we found solution x we have two equations with parameter λ . They are

$$y = \frac{x(\lambda^2 + 1)}{2\lambda - 2}$$

$$x = (\lambda - 1) c_1$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \frac{2c_1^2 + 2c_1x + x^2}{2c_1}$$

✓ **Maple.** Time used: 0.041 (sec). Leaf size: 44

```
ode:=diff(y(x),x)^2*x-2*diff(y(x),x)*y(x)+x+2*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (1 - \sqrt{2}) x$$

$$y = (1 + \sqrt{2}) x$$

$$y = \frac{2c_1^2 + 2c_1x + x^2}{2c_1}$$

Maple trace

```
Methods for first order ODEs:
*** Sublevel 2 ***
```

Methods for first order ODEs:

```
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful
```

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right)^2 - 2y(x)\left(\frac{d}{dx}y(x)\right) + x + 2y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{y(x) - \sqrt{y(x)^2 - 2xy(x) - x^2}}{x}, \frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 2xy(x) - x^2}}{x} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) - \sqrt{y(x)^2 - 2xy(x) - x^2}}{x}$
- Solve the equation $\frac{d}{dx}y(x) = \frac{y(x) + \sqrt{y(x)^2 - 2xy(x) - x^2}}{x}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 0.107 (sec). Leaf size: 78

```
ode=x*D[y[x],x]^2-2*y[x]*D[y[x],x]+x+2*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{1}{2}e^{-c_1}x^2 + x - e^{c_1}$$

$$y(x) \rightarrow -e^{c_1}x^2 + x - \frac{e^{-c_1}}{2}$$

$$y(x) \rightarrow x - \sqrt{2}x$$

$$y(x) \rightarrow (1 + \sqrt{2})x$$

✓ Sympy. Time used: 2.032 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x)**2 + x - 2*y(x)*Derivative(y(x), x) + 2*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = 2x^2e^{-C_1} + x + \frac{e^{C_1}}{4}$$

2.6.5 Problem 14

Local contents

✓ Maple	1448
✓ Mathematica	1451
✗ Sympy	1451

Internal problem ID [7961]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 14

Date solved : Thursday, November 27, 2025 at 06:37:27 AM

CAS classification : [_quadrature]

Solve

$$(-1 + 3y)^2 y'^2 = 4y$$

Solving for the derivative gives these ODE's to solve

$$y' = -\frac{2\sqrt{y}}{-1 + 3y} \quad (1)$$

$$y' = \frac{2\sqrt{y}}{-1 + 3y} \quad (2)$$

Now each of the above is solved separately.

Solving Eq. (1)

Solve Integrating gives

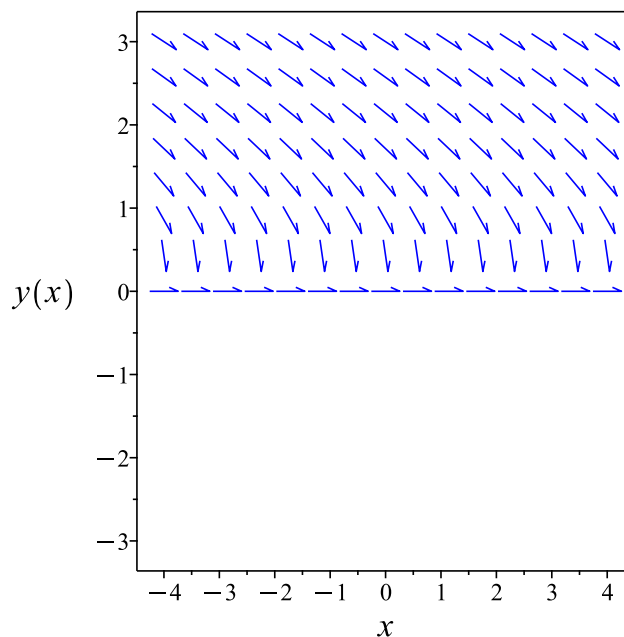
$$\begin{aligned} \int -\frac{-1 + 3y}{2\sqrt{y}} dy &= dx \\ -\sqrt{y}(-1 + y) &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-\frac{2\sqrt{y}}{-1 + 3y} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = 0$$

Figure 2.301: Slope field $y' = -\frac{2\sqrt{y}}{-1+3y}$

Solving Eq. (2)

Solve Integrating gives

$$\int \frac{-1+3y}{2\sqrt{y}} dy = dx$$

$$\sqrt{y}(-1+y) = x + c_2$$

Singular solutions are found by solving

$$\frac{2\sqrt{y}}{-1+3y} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = 0$$

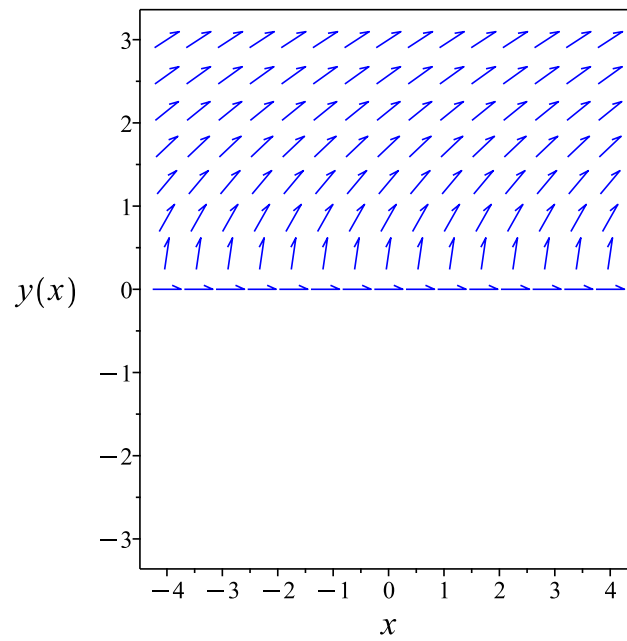


Figure 2.302: Slope field $y' = \frac{2\sqrt{y}}{-1+3y}$

✓ **Maple.** Time used: 0.025 (sec). Leaf size: 519

```
ode:=(3*y(x)-1)^2*diff(y(x),x)^2 = 4*y(x);
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = \frac{\left(\left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} + 12 \right)^2}{36 \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

$$y$$

$$= \frac{\left(i\sqrt{3} \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} - 12i\sqrt{3} + \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} \right)^2}{144 \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

$$y = \frac{\left((i\sqrt{3} - 1) \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} - 12i\sqrt{3} - 12 \right)^2}{144 \left(108c_1 - 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

$$y = \frac{\left(\left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} + 12 \right)^2}{36 \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

$$y$$

$$= \frac{\left(i \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} \sqrt{3} - 12i\sqrt{3} + \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} \right)^2}{144 \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

$$y = \frac{\left((i\sqrt{3} - 1) \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3} - 12i\sqrt{3} - 12 \right)^2}{144 \left(-108c_1 + 108x + 12\sqrt{81c_1^2 - 162c_1x + 81x^2 - 12} \right)^{2/3}}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

```
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
<- differential order: 1; missing x successful
```

Maple step by step

Let's solve

$$(3y(x) - 1)^2 \left(\frac{d}{dx}y(x)\right)^2 = 4y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = -\frac{2\sqrt{y(x)}}{3y(x)-1}, \frac{d}{dx}y(x) = \frac{2\sqrt{y(x)}}{3y(x)-1} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = -\frac{2\sqrt{y(x)}}{3y(x)-1}$

- Separate variables

$$\frac{\left(\frac{d}{dx}y(x)\right)(3y(x)-1)}{\sqrt{y(x)}} = -2$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx}y(x)\right)(3y(x)-1)}{\sqrt{y(x)}} dx = \int (-2) dx + _C1$$

- Evaluate integral

$$2y(x)^{3/2} - 2\sqrt{y(x)} = -2x + _C1$$

- Solve for $y(x)$

$$\left\{ y(x) = \left(\frac{\left(54_C1 - 108x + 6\sqrt{81_C1^2 - 324_C1x + 324x^2 - 48} \right)^{1/3}}{6} + \frac{2}{\left(54_C1 - 108x + 6\sqrt{81_C1^2 - 324_C1x + 324x^2} \right)} \right.$$

- Simplify

$$\left\{ y(x) = \left(\frac{\left(54_C1 - 108x + 6\sqrt{81_C1^2 - 324_C1x + 324x^2 - 48} \right)^{1/3}}{6} + \frac{2}{\left(54_C1 - 108x + 6\sqrt{81_C1^2 - 324_C1x + 324x^2} \right)} \right.$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{2\sqrt{y(x)}}{3y(x)-1}$

- Separate variables

$$\frac{\left(\frac{d}{dx}y(x)\right)(3y(x)-1)}{\sqrt{y(x)}} = 2$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx}y(x)\right)(3y(x)-1)}{\sqrt{y(x)}} dx = \int 2 dx + _C1$$

- Evaluate integral

$$2y(x)^{3/2} - 2\sqrt{y(x)} = 2x + _C1$$

- Solve for $y(x)$

$$\left\{ y(x) = \left(\frac{\left(54_C1+108x+6\sqrt{81_CI^2+324_C1x+324x^2-48} \right)^{1/3}}{6} + \frac{2}{\left(54_C1+108x+6\sqrt{81_CI^2+324_C1x+324x^2-48} \right)^{1/3}} \right) \right.$$

- Simplify

$$\left\{ y(x) = \left(\frac{\left(54_C1+108x+6\sqrt{81_CI^2+324_C1x+324x^2-48} \right)^{1/3}}{6} + \frac{2}{\left(54_C1+108x+6\sqrt{81_CI^2+324_C1x+324x^2-48} \right)^{1/3}} \right) \right.$$

- Set of solutions

$$\left\{ y(x) = \left(\frac{\left(54C1-108x+6\sqrt{81CI^2-324C1x+324x^2-48} \right)^{1/3}}{6} + \frac{2}{\left(54C1-108x+6\sqrt{81CI^2-324C1x+324x^2-48} \right)^{1/3}} \right)^2, y(x) \right.$$

✓ **Mathematica.** Time used: 3.537 (sec). Leaf size: 892

```
ode=(3*y[x]-1)^2*D[y[x],x]^2==4*y[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\left(2 + \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(-2x + c_1)^2(108x^2 - 108c_1x - 16 + 27c_1^2)} - 108c_1x - 8 + 27c_1^2}\right)^2}{6\sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(-2x + c_1)^2(108x^2 - 108c_1x - 16 + 27c_1^2)} - 108c_1x - 8 + 27c_1^2}}$$

$$y(x) \rightarrow \frac{1}{24} \left(2i(\sqrt{3} + i) \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(-2x + c_1)^2(108x^2 - 108c_1x - 16 + 27c_1^2)} - 108c_1x - 8 + 27c_1^2} \right)^2$$

$$y(x) \rightarrow \frac{1}{24} \left(-2(1 + i\sqrt{3}) \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(-2x + c_1)^2(108x^2 - 108c_1x - 16 + 27c_1^2)} - 108c_1x - 8 + 27c_1^2} \right)^2$$

$$y(x) \rightarrow \frac{\left(2 + \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(2x + c_1)^2(108x^2 + 108c_1x - 16 + 27c_1^2)} + 108c_1x - 8 + 27c_1^2}\right)^2}{6\sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(2x + c_1)^2(108x^2 + 108c_1x - 16 + 27c_1^2)} + 108c_1x - 8 + 27c_1^2}}$$

$$y(x) \rightarrow \frac{1}{24} \left(2i(\sqrt{3} + i) \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(2x + c_1)^2(108x^2 + 108c_1x - 16 + 27c_1^2)} + 108c_1x - 8 + 27c_1^2} \right)^2$$

$$y(x) \rightarrow \frac{1}{24} \left(-2(1 + i\sqrt{3}) \sqrt[3]{108x^2 + 3\sqrt{3}\sqrt{(2x + c_1)^2(108x^2 + 108c_1x - 16 + 27c_1^2)} + 108c_1x - 8 + 27c_1^2} \right)^2$$

$$y(x) \rightarrow 0$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((3*y(x) - 1)**2*Derivative(y(x), x)**2 - 4*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.6.6 Problem 15

Local contents

Solved using first_order_ode_homog_type_G	1452
Solved using first_order_ode_parametric method	1455
✓ Maple	1456
✓ Mathematica	1458
✓ Sympy	1458

Internal problem ID [7962]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 15

Date solved : Thursday, November 27, 2025 at 06:37:32 AM

CAS classification : [[_homogeneous, 'class G'], _rational]

Solved using first_order_ode_homog_type_G

Time used: 0.184 (sec)

Solve

$$y = -y'x + x^4y'^2$$

Multiplying the right side of the ode, which is $\frac{1+\sqrt{4x^2y+1}}{2x^3}$ by $\frac{x}{y}$ gives

$$\begin{aligned}
 y' &= \left(\frac{x}{y}\right) \frac{1+\sqrt{4x^2y+1}}{2x^3} \\
 &= \frac{1+\sqrt{4x^2y+1}}{2x^2y} \\
 &= F(x, y)
 \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{2x^2y + 1 + \sqrt{4x^2y + 1}}{x^2y\sqrt{4x^2y + 1}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{2x^2y + 1 + \sqrt{4x^2y + 1}}{2x^2y\sqrt{4x^2y + 1}} \\ \alpha &= \frac{f_x}{f_y} \\ &= 2 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^2} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{1 + \sqrt{4z + 1}}{2z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{x^2y} \frac{1}{z \left(-2 - \frac{1 + \sqrt{4z + 1}}{2z} \right)} dz = 0$$

Multiplying the right side of the ode, which is $-\frac{-1 + \sqrt{4x^2y + 1}}{2x^3}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y} \right) - \frac{-1 + \sqrt{4x^2y + 1}}{2x^3} \\ &= -\frac{-1 + \sqrt{4x^2y + 1}}{2x^2y} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{-2x^2y + \sqrt{4x^2y + 1} - 1}{x^2y\sqrt{4x^2y + 1}} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= -\frac{-2x^2y + \sqrt{4x^2y + 1} - 1}{2x^2y\sqrt{4x^2y + 1}} \\ \alpha &= \frac{f_x}{f_y} \\ &= 2 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x^2} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = -\frac{-1 + \sqrt{4z + 1}}{2z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{x^2y} \frac{1}{z \left(-2 + \frac{-1 + \sqrt{4z + 1}}{2z} \right)} dz = 0$$

Summary of solutions found

$$\begin{aligned} \ln(x) - c_1 + \int^{x^2y} \frac{1}{z \left(-2 - \frac{1 + \sqrt{4z + 1}}{2z} \right)} dz &= 0 \\ \ln(x) - c_2 + \int^{x^2y} \frac{1}{z \left(-2 + \frac{-1 + \sqrt{4z + 1}}{2z} \right)} dz &= 0 \end{aligned}$$

Solved using first_order_ode_parametric method

Time used: 0.339 (sec)

Solve

$$y = -y'x + x^4y'^2$$

Let y' be a parameter λ . The ode becomes

$$-x^4\lambda^2 + \lambda x + y = 0$$

Isolating y gives

$$\begin{aligned} y &= x^4\lambda^2 - \lambda x \\ &= x^4\lambda^2 - \lambda x \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{2\lambda x^4 - x}{-4\lambda^2 x^3 + 2\lambda} \\ &= -\frac{x(\lambda)}{2\lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= \frac{1}{2\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{1}{2\lambda} d\lambda} \\ &= \sqrt{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}(x\sqrt{\lambda}) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} x\sqrt{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\sqrt{\lambda}$ gives the final solution

$$x(\lambda) = \frac{c_1}{\sqrt{\lambda}}$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned} y &= x^4 \lambda^2 - \lambda x \\ x &= \frac{c_1}{\sqrt{\lambda}} \end{aligned}$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \frac{c_1^2(c_1^2 x - 1)}{x}$$

✓ **Maple.** Time used: 0.154 (sec). Leaf size: 81

```
ode:=y(x) = -diff(y(x),x)*x+x^4*diff(y(x),x)^2;
dsolve(ode,y(x), singsol=all);
```

$$\begin{aligned} y &= -\frac{1}{4x^2} \\ y &= \frac{ic_1 - x}{x c_1^2} \\ y &= \frac{ic_1 - x}{x c_1^2} \\ y &= \frac{-ic_1 - x}{x c_1^2} \\ y &= \frac{-ic_1 - x}{x c_1^2} \end{aligned}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

```

-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
trying simple symmetries for implicit equations
Successful isolation of dy/dx: 2 solutions were found. Trying to solve each \
resulting ODE.
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying homogeneous types:
trying homogeneous G
<- homogeneous successful
-----
* Tackling next ODE.
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying homogeneous types:
trying homogeneous G
<- homogeneous successful

```

Maple step by step

Let's solve

$$y(x) = -x\left(\frac{d}{dx}y(x)\right) + x^4\left(\frac{d}{dx}y(x)\right)^2$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{1+\sqrt{1+4x^2y(x)}}{2x^3}, \frac{d}{dx}y(x) = -\frac{-1+\sqrt{1+4x^2y(x)}}{2x^3} \right]$$
- Solve the equation $\frac{d}{dx}y(x) = \frac{1+\sqrt{1+4x^2y(x)}}{2x^3}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{-1+\sqrt{1+4x^2y(x)}}{2x^3}$
- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 0.319 (sec). Leaf size: 123

```
ode=y[x]==-x*D[y[x],x]+x^4*D[y[x],x]^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\text{Solve} \left[-\frac{x\sqrt{4x^2y(x)+1}\operatorname{arctanh}\left(\sqrt{4x^2y(x)+1}\right)}{\sqrt{4x^4y(x)+x^2}} - \frac{1}{2}\log(y(x)) = c_1, y(x) \right]$$

$$\text{Solve} \left[\frac{x\sqrt{4x^2y(x)+1}\operatorname{arctanh}\left(\sqrt{4x^2y(x)+1}\right)}{\sqrt{4x^4y(x)+x^2}} - \frac{1}{2}\log(y(x)) = c_1, y(x) \right]$$

$$y(x) \rightarrow 0$$

✓ **Sympy.** Time used: 2.457 (sec). Leaf size: 10

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**4*Derivative(y(x), x)**2 + x*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{C_1(C_1 + \frac{2}{x})}{4}$$

2.6.7 Problem 16

Local contents

Solved using first_order_ode_dAlembert	1459
Solved using first_order_ode_parametric method	1461
✓ Maple	1463
✓ Mathematica	1464
✗ Sympy	1464

Internal problem ID [7963]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 16

Date solved : Thursday, November 27, 2025 at 06:37:38 AM

CAS classification : [[_1st_order, _with_linear_symmetries], _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 0.108 (sec)

Solve

$$2y = y'^2 + 4y'x$$

Let $p = y'$ the ode becomes

$$2y = p^2 + 4px$$

Solving for y from the above results in

$$y = \frac{1}{2}p^2 + 2px \quad (1)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. The above ode is dAlembert ode which is now solved.

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g')\frac{dp}{dx} \\ p - f &= (xf' + g')\frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= 2p \\ g &= \frac{p^2}{2} \end{aligned}$$

Hence (2) becomes

$$-p = (2x + p)p'(x) \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$-p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = 0$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = -\frac{p(x)}{2x + p(x)} \quad (3)$$

Inverting the above ode gives

$$\frac{d}{dp}x(p) = -\frac{2x(p) + p}{p} \quad (4)$$

This ODE is now solved for $x(p)$. The integrating factor is

$$\begin{aligned} \mu &= e^{\int \frac{2}{p} dp} \\ \mu &= p^2 \\ \mu &= p^2 \end{aligned} \quad (5)$$

Integrating gives

$$\begin{aligned} x(p) &= \frac{1}{\mu} \left(\int \mu(-1) dp + c_1 \right) \\ &= \frac{1}{\mu} \left(\frac{-p^3}{3} + c_1 \right) \\ &= \frac{-\frac{p^3}{3} + c_1}{p^2} \end{aligned} \quad (5)$$

Now we need to eliminate p between the above solution and (1A). The first method is to solve for p from Eq. (1A) and substitute the result into Eq. (5). The Second method is to solve for p from Eq. (5) and substitute the result into (1A).

Eliminating p from the following two equations

$$x = \frac{-\frac{p^3}{3} + c_1}{p^2}$$

$$y = \frac{1}{2}p^2 + 2px$$

results in

$$p = \text{RootOf}(_Z^3 + 3x_Z^2 - 3c_1)$$

Substituting the above into Eq (1A) and simplifying gives

$$y = \frac{\text{RootOf}(_Z^3 + 3x_Z^2 - 3c_1) (\text{RootOf}(_Z^3 + 3x_Z^2 - 3c_1) + 4x)}{2}$$

Summary of solutions found

$$y = 0$$

$$y = \frac{\text{RootOf}(_Z^3 + 3x_Z^2 - 3c_1) (\text{RootOf}(_Z^3 + 3x_Z^2 - 3c_1) + 4x)}{2}$$

Solved using first_order_ode_parametric method

Time used: 0.188 (sec)

Solve

$$2y = y'^2 + 4y'x$$

Let y' be a parameter λ . The ode becomes

$$-\lambda^2 - 4\lambda x + 2y = 0$$

Isolating y gives

$$y = \frac{1}{2}\lambda^2 + 2\lambda x$$

$$= \frac{\lambda(\lambda + 4x)}{2}$$

$$= F(x, \lambda)$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned}\frac{d}{d\lambda}x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= -\frac{\lambda + 2x}{\lambda} \\ &= \frac{-\lambda - 2x(\lambda)}{\lambda}\end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(\lambda) &= \frac{2}{\lambda} \\ p(\lambda) &= -1\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q d\lambda} \\ &= e^{\int \frac{2}{\lambda} d\lambda} \\ &= \lambda^2\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{d\lambda}(\mu x) &= \mu p \\ \frac{d}{d\lambda}(\mu x) &= (\mu)(-1) \\ \frac{d}{d\lambda}(x \lambda^2) &= (\lambda^2)(-1) \\ d(x \lambda^2) &= (-\lambda^2) d\lambda\end{aligned}$$

Integrating gives

$$\begin{aligned}x \lambda^2 &= \int -\lambda^2 d\lambda \\ &= -\frac{\lambda^3}{3} + c_1\end{aligned}$$

Dividing throughout by the integrating factor λ^2 gives the final solution

$$x(\lambda) = \frac{-\frac{\lambda^3}{3} + c_1}{\lambda^2}$$

Now that we found solution x we have two equations with parameter λ . They are

$$y = \frac{\lambda(\lambda + 4x)}{2}$$

$$x = \frac{-\frac{\lambda^3}{3} + c_1}{\lambda^2}$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$48c_1 x^3 - 12y^2 x^2 + 36c_1 xy - 8y^3 + 9c_1^2 = 0$$

✓ **Maple.** Time used: 0.018 (sec). Leaf size: 700

```
ode:=2*y(x) = diff(y(x),x)^2+4*diff(y(x),x)*x;
dsolve(ode,y(x), singsol=all);
```

Solution too large to show

Maple trace

```
Methods for first order ODEs:
-> Solving 1st order ODE of high degree, 1st attempt
trying 1st order WeierstrassP solution for high degree ODE
trying 1st order WeierstrassPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying dAlembert
<- dAlembert successful
```

Maple step by step

Let's solve

$$2y(x) = \left(\frac{d}{dx}y(x)\right)^2 + 4x\left(\frac{d}{dx}y(x)\right)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = -2x - \sqrt{4x^2 + 2y(x)}, \frac{d}{dx}y(x) = -2x + \sqrt{4x^2 + 2y(x)} \right]$$

- Solve the equation $\frac{d}{dx}y(x) = -2x - \sqrt{4x^2 + 2y(x)}$

- Solve the equation $\frac{d}{dx}y(x) = -2x + \sqrt{4x^2 + 2y(x)}$
- Set of solutions
 $\{workingODE, workingODE\}$

✓ **Mathematica.** Time used: 60.16 (sec). Leaf size: 1344

```
ode=2*y[x]==D[y[x],x]^2+4*x*D[y[x],x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Solution too large to show

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*x*Derivative(y(x), x) + 2*y(x) - Derivative(y(x), x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE 2*x - sqrt(4*x**2 + 2*y(x)) + Derivative(y(x),
x) cannot be solved by the factorable group method
```

2.6.8 Problem 17

Local contents

✓ Maple	1467
✓ Mathematica	1468
✓ Sympy	1469

Internal problem ID [7964]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:37:44 AM

CAS classification : [_quadrature]

Solve

$$y(3 - 4y)^2 y'^2 = 4 - 4y$$

Solving for the derivative gives these ODE's to solve

$$y' = -\frac{2\sqrt{-(-1+y)y}}{y(4y-3)} \quad (1)$$

$$y' = \frac{2\sqrt{-(-1+y)y}}{y(4y-3)} \quad (2)$$

Now each of the above is solved separately.

Solving Eq. (1)

Solve Integrating gives

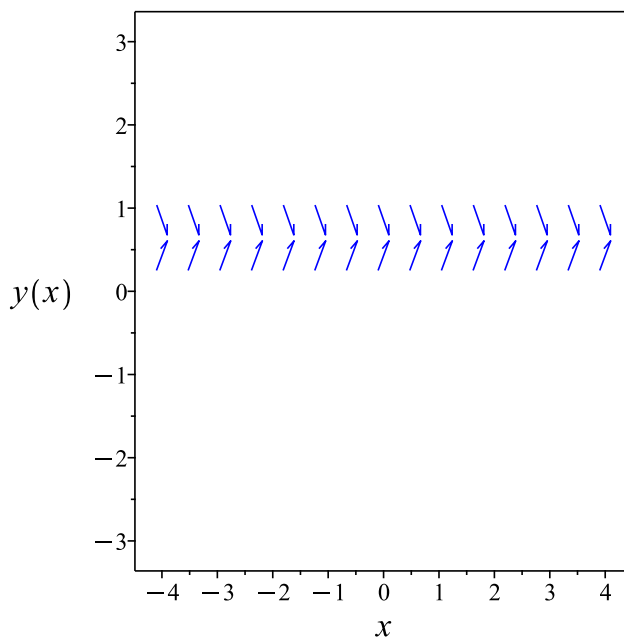
$$\begin{aligned} \int -\frac{y(4y-3)}{2\sqrt{-(-1+y)y}} dy &= dx \\ -\frac{(-1+y)y^2}{\sqrt{-(-1+y)y}} &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-\frac{2\sqrt{-(-1+y)y}}{y(4y-3)} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = 1$$

Figure 2.303: Slope field $y' = -\frac{2\sqrt{-(-1+y)y}}{y(4y-3)}$ Solving Eq. (2)

Solve Integrating gives

$$\int \frac{y(4y-3)}{2\sqrt{-(-1+y)y}} dy = dx$$

$$\frac{(-1+y)y^2}{\sqrt{-(-1+y)y}} = x + c_2$$

Singular solutions are found by solving

$$\frac{2\sqrt{-(-1+y)y}}{y(4y-3)} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = 1$$

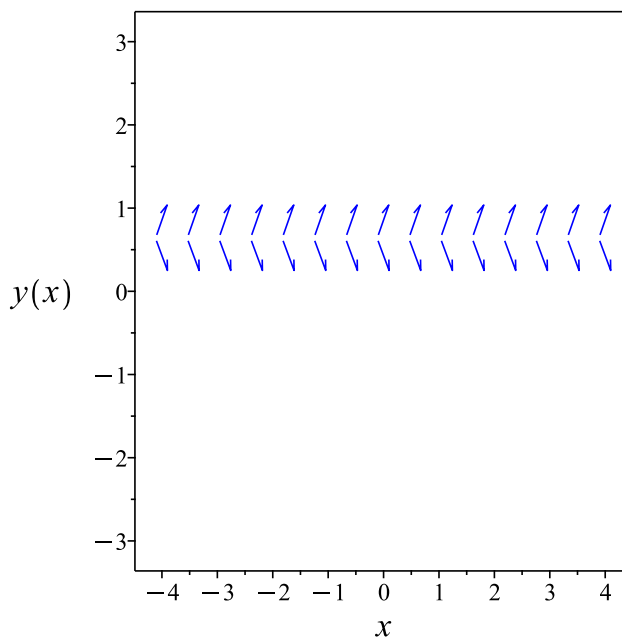


Figure 2.304: Slope field $y' = \frac{2\sqrt{-1+y}y}{y(4y-3)}$

✓ **Maple.** Time used: 0.030 (sec). Leaf size: 58

```
ode:=y(x)*(3-4*y(x))^2*diff(y(x),x)^2 = 4-4*y(x);
dsolve(ode,y(x), singsol=all);
```

$$y = 1$$

$$x + \frac{y^2(y-1)}{\sqrt{-y(y-1)}} - c_1 = 0$$

$$x - \frac{y^2(y-1)}{\sqrt{-y(y-1)}} - c_1 = 0$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

<- differential order: 1; missing x successful

Maple step by step

Let's solve

$$y(x) (3 - 4y(x))^2 \left(\frac{d}{dx} y(x) \right)^2 = 4 - 4y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx} y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx} y(x) = -\frac{2\sqrt{-y(x)(y(x)-1)}}{y(x)(4y(x)-3)}, \frac{d}{dx} y(x) = \frac{2\sqrt{-y(x)(y(x)-1)}}{y(x)(4y(x)-3)} \right]$$

- Solve the equation $\frac{d}{dx} y(x) = -\frac{2\sqrt{-y(x)(y(x)-1)}}{y(x)(4y(x)-3)}$

- Separate variables

$$\frac{\left(\frac{d}{dx} y(x) \right) y(x)(4y(x)-3)}{\sqrt{-y(x)(y(x)-1)}} = -2$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx} y(x) \right) y(x)(4y(x)-3)}{\sqrt{-y(x)(y(x)-1)}} dx = \int (-2) dx + _C1$$

- Evaluate integral

$$-2y(x) \sqrt{-y(x)^2 + y(x)} = -2x + _C1$$

- Solve the equation $\frac{d}{dx} y(x) = \frac{2\sqrt{-y(x)(y(x)-1)}}{y(x)(4y(x)-3)}$

- Separate variables

$$\frac{\left(\frac{d}{dx} y(x) \right) y(x)(4y(x)-3)}{\sqrt{-y(x)(y(x)-1)}} = 2$$

- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx} y(x) \right) y(x)(4y(x)-3)}{\sqrt{-y(x)(y(x)-1)}} dx = \int 2 dx + _C1$$

- Evaluate integral

$$-2y(x) \sqrt{-y(x)^2 + y(x)} = 2x + _C1$$

- Set of solutions

$$\left\{ -2y(x) \sqrt{-y(x)^2 + y(x)} = -2x + C1, -2y(x) \sqrt{-y(x)^2 + y(x)} = 2x + C1 \right\}$$

✓ **Mathematica.** Time used: 60.303 (sec). Leaf size: 3751

```
ode=y[x]*(3-4*y[x])^2*D[y[x],x]^2==4*(1-y[x]);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Too large to display

✓ Sympy. Time used: 2.968 (sec). Leaf size: 228

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((3 - 4*y(x))*2*y(x)*Derivative(y(x), x)**2 + 4*y(x) - 4, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[\begin{aligned} & \left\{ \begin{aligned} & 3i\sqrt{y(x)-1}\sqrt{y(x)} - \frac{2iy^{\frac{5}{2}}(x)}{\sqrt{y(x)-1}} - \frac{iy^{\frac{3}{2}}(x)}{\sqrt{y(x)-1}} + \frac{3i\sqrt{y(x)}}{\sqrt{y(x)-1}} && \text{for } |y(x)| > 1 \\ & \frac{2y^{\frac{5}{2}}(x)}{\sqrt{1-y(x)}} - \frac{2y^{\frac{3}{2}}(x)}{\sqrt{1-y(x)}} && \text{otherwise} \end{aligned} \right. &= C_1 \\ & - 2x, \left\{ \begin{aligned} & 3i\sqrt{y(x)-1}\sqrt{y(x)} - \frac{2iy^{\frac{5}{2}}(x)}{\sqrt{y(x)-1}} - \frac{iy^{\frac{3}{2}}(x)}{\sqrt{y(x)-1}} + \frac{3i\sqrt{y(x)}}{\sqrt{y(x)-1}} && \text{for } |y(x)| > 1 \\ & \frac{2y^{\frac{5}{2}}(x)}{\sqrt{1-y(x)}} - \frac{2y^{\frac{3}{2}}(x)}{\sqrt{1-y(x)}} && \text{otherwise} \end{aligned} \right. &= C_1 + 2x \end{aligned} \right]$$

2.6.9 Problem 18

Local contents

Solved using first_order_ode_parametric method	1470
✓ Maple	1472
✗ Mathematica	1473
✗ Sympy	1474

Internal problem ID [7965]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:37:50 AM

CAS classification : [[_1st_order, _with_linear_symmetries]]

Solved using first_order_ode_parametric method

Time used: 0.124 (sec)

Solve

$$y'^3 - 4x^4 y' + 8yx^3 = 0$$

Let y' be a parameter λ . The ode becomes

$$-4x^4 \lambda + 8y x^3 + \lambda^3 = 0$$

Isolating y gives

$$\begin{aligned} y &= -\frac{\lambda(-4x^4 + \lambda^2)}{8x^3} \\ &= \frac{4x^4 \lambda - \lambda^3}{8x^3} \\ &= F(x, \lambda) \end{aligned}$$

Now we generate an ode in $x(\lambda)$ using

$$\begin{aligned} \frac{d}{d\lambda} x(\lambda) &= \frac{\frac{\partial F}{\partial \lambda}}{\lambda - \frac{\partial F}{\partial x}} \\ &= \frac{-\frac{4x^4 + \lambda^2}{8x^3} - \frac{\lambda^2}{4x^3}}{-\lambda - \frac{3\lambda(-4x^4 + \lambda^2)}{8x^4}} \\ &= \frac{x(\lambda)}{\lambda} \end{aligned}$$

Which is now solved for x .

Solve In canonical form a linear first order is

$$\frac{d}{d\lambda}x(\lambda) + q(\lambda)x(\lambda) = p(\lambda)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(\lambda) &= -\frac{1}{\lambda} \\ p(\lambda) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q d\lambda} \\ &= e^{\int -\frac{1}{\lambda} d\lambda} \\ &= \frac{1}{\lambda} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{d\lambda}\mu x &= 0 \\ \frac{d}{d\lambda}\left(\frac{x}{\lambda}\right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{x}{\lambda} &= \int 0 d\lambda + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{\lambda}$ gives the final solution

$$x(\lambda) = \lambda c_1$$

Now that we found solution x we have two equations with parameter λ . They are

$$\begin{aligned} y &= \frac{4x^4\lambda - \lambda^3}{8x^3} \\ x &= \lambda c_1 \end{aligned}$$

Eliminating λ gives the solution for y .

Summary of solutions found

$$y = \frac{4c_1^2x^2 - 1}{8c_1^3}$$

✓ **Maple.** Time used: 0.166 (sec). Leaf size: 40

```
ode:=diff(y(x),x)^3-4*x^4*diff(y(x),x)+8*x^3*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{2\sqrt{3}x^3}{9}$$

$$y = \frac{2\sqrt{3}x^3}{9}$$

$$y = \frac{x^2}{2c_1} - \frac{1}{8c_1^3}$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

trying 1st order WeierstrassPPrime solution for high degree ODE

trying 1st order JacobiSN solution for high degree ODE

trying 1st order ODE linearizable_by_differentiation

trying differential order: 1; missing variables

trying dAlembert

trying simple symmetries for implicit equations

Successful isolation of dy/dx: 3 solutions were found. Trying to solve each \ resulting ODE.

*** Sublevel 3 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying homogeneous types:

trying exact

Looking for potential symmetries

trying an equivalence to an Abel ODE

trying 1st order ODE linearizable_by_differentiation

-> Solving 1st order ODE of high degree, Lie methods, 1st trial

-> Computing symmetries using: way = 2

-> Computing symmetries using: way = 2

-> Solving 1st order ODE of high degree, 2nd attempt. Trying parametric meth\ ods

-> Calling odsolve with the ODE, diff(y(x),x) = (-4*y(x)^5+3*y(x)*x^2)/(-4*x

```

*y(x)^4+3*x^3), y(x)
*** Sublevel 3 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- 1st order, parametric methods successful

```

Maple step by step

Let's solve

$$\left(\frac{d}{dx}y(x)\right)^3 - 4x^4\left(\frac{d}{dx}y(x)\right) + 8x^3y(x) = 0$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = 2 \left(\frac{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}}{6} + \frac{2x^2}{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}} \right) x, \frac{d}{dx}y(x) = 2 \left(-\frac{(-108y(x)+12\sqrt{-12x^6+81y(x)^2})^{1/3}}{6} + \frac{2x^2}{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}} \right) x \right]$$

- Solve the equation $\frac{d}{dx}y(x) = 2 \left(\frac{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}}{6} + \frac{2x^2}{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}} \right) x$

- Solve the equation $\frac{d}{dx}y(x) = 2 \left(-\frac{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}}{12} - \frac{x^2}{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}} - \frac{I\sqrt{3}}{12} \right) x$

- Solve the equation $\frac{d}{dx}y(x) = 2 \left(-\frac{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}}{12} - \frac{x^2}{\left(-108y(x)+12\sqrt{-12x^6+81y(x)^2}\right)^{1/3}} + \frac{I\sqrt{3}}{12} \right) x$

- Set of solutions
 $\{workingODE, workingODE, workingODE\}$

✗ Mathematica

```

ode=D[y[x],x]^3-4*x^4*D[y[x],x]+8*x^3*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

Timed out

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*x**4*Derivative(y(x), x) + 8*x**3*y(x) + Derivative(y(x), x)**3, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

Timed Out

2.6.10 Problem 19

Local contents

Solved using first_order_ode_dAlembert	1475
Solved using first_order_ode_homog_type_G	1483
✓ Maple	1485
✓ Mathematica	1486
✗ Sympy	1487

Internal problem ID [7966]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 10. Singular solutions, Extraneous loci. Supplementary problems. Page 74

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:37:55 AM

CAS classification : [[_homogeneous, 'class A'], _rational, _dAlembert]

Solved using first_order_ode_dAlembert

Time used: 2.023 (sec)

Solve

$$(1 + y'^2)(x - y)^2 = (x + yy')^2$$

Let $p = y'$ the ode becomes

$$(p^2 + 1)(x - y)^2 = (yp + x)^2$$

Solving for y from the above results in

$$y = (p^2 + p + 1 + \sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1})x \quad (1)$$

$$y = (p^2 + p + 1 - \sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1})x \quad (2)$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is dAlembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= p^2 + p + 1 + \sqrt{(p^2 + 1)(p + 1)^2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\begin{aligned} -p^2 - 1 - \sqrt{(p^2 + 1)(p + 1)^2} &= \left(2px + x + \frac{2xp^3}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \right. \\ &\quad + \frac{3xp^2}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} + \frac{2xp}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \\ &\quad \left. + \frac{x}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \right) p'(x) \end{aligned} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$-p^2 - 1 - \sqrt{(p^2 + 1)(p + 1)^2} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= i \\ p_2 &= -i \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= ix \\ y &= -ix \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$\begin{aligned} p'(x) &= \frac{-p(x)^2 - 1 - \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{2p(x)x + x + \frac{2xp(x)^3}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} + \frac{3xp(x)^2}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} + \frac{2xp(x)}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} + \frac{x}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}}} \end{aligned} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = - \frac{\left(p(x)^2 + \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} \right) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{x \left(2p(x)^3 + 3p(x)^2 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} + 2p(x) + \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} \right)} \quad (2.113)$$

is separable as it can be written as

$$p'(x) = - \frac{\left(p(x)^2 + \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} \right) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{x \left(2p(x)^3 + 3p(x)^2 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} + 2p(x) + \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1} \right)}$$

$$= f(x)g(p)$$

Where

$$f(x) = -\frac{1}{x}$$

$$g(p) = \frac{\left(p^2 + 1 + \sqrt{(p^2 + 1)(p + 1)^2 + 1} \right) \sqrt{(p^2 + 1)(p + 1)^2}}{2p^3 + 3p^2 + 2p \sqrt{(p^2 + 1)(p + 1)^2 + 1} + 2p + \sqrt{(p^2 + 1)(p + 1)^2 + 1}}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(x) dx$$

$$\int \frac{2p^3 + 3p^2 + 2p \sqrt{(p^2 + 1)(p + 1)^2 + 1} + 2p + \sqrt{(p^2 + 1)(p + 1)^2 + 1}}{\left(p^2 + 1 + \sqrt{(p^2 + 1)(p + 1)^2 + 1} \right) \sqrt{(p^2 + 1)(p + 1)^2}} dp = \int -\frac{1}{x} dx$$

$$-\frac{\ln \left(\frac{-2p(x)^3 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 - 2p(x)^2 - p(x) - 1}}{(p(x)^2 + 1)(p(x) + 1)} \right)}{2} = \ln \left(\frac{1}{x} \right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\frac{1}{\sqrt{\frac{-2p(x)^3 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 - 2p(x)^2 - p(x) - 1}}{(p(x)^2 + 1)(p(x) + 1)}}} = \frac{c_1}{x}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\left(p^2 + 1 + \sqrt{(p^2 + 1)(p + 1)^2}\right) \sqrt{(p^2 + 1)(p + 1)^2}}{2p^3 + 3p^2 + 2p\sqrt{(p^2 + 1)(p + 1)^2} + 2p + \sqrt{(p^2 + 1)(p + 1)^2} + 1} = 0$$

for $p(x)$ gives

$$p(x) = -i$$

$$p(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\frac{1}{\sqrt{\frac{-2p(x)^3 + 2p(x)\sqrt{(p(x)^2 + 1)(p(x) + 1)^2} - 2p(x)^2 - p(x) - 1}{(p(x)^2 + 1)(p(x) + 1)}}} = \frac{c_1}{x}$$

$$p(x) = -i$$

$$p(x) = i$$

Substituting the above solution for p in (2A) gives

$$y = x \left(\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)} + \sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)} + 1} \right) + \sqrt{\left(\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)} + 1 \right) \left(\sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)} + 1} \right)^2}$$

$$y = -ix$$

$$y = ix$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= p^2 + p + 1 - \sqrt{(p^2 + 1)(p + 1)^2} \\ g &= 0 \end{aligned}$$

Hence (2) becomes

$$\begin{aligned} -p^2 - 1 + \sqrt{(p^2 + 1)(p + 1)^2} &= \left(2px + x - \frac{2xp^3}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \right. \\ &\quad - \frac{3xp^2}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} - \frac{2xp}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \\ &\quad \left. - \frac{x}{\sqrt{p^4 + 2p^3 + 2p^2 + 2p + 1}} \right) p'(x) \end{aligned} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$-p^2 - 1 + \sqrt{(p^2 + 1)(p + 1)^2} = 0$$

Solving the above for p results in

$$\begin{aligned} p_1 &= 0 \\ p_2 &= i \\ p_3 &= -i \end{aligned}$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$\begin{aligned} y &= 0 \\ y &= ix \\ y &= -ix \end{aligned}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$\begin{aligned} p'(x) &= \frac{-p(x)^2 - 1 + \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{2p(x)x + x - \frac{2xp(x)^3}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} - \frac{3xp(x)^2}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} - \frac{2xp(x)}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}} - \frac{x}{\sqrt{p(x)^4 + 2p(x)^3 + 2p(x)^2 + 2p(x) + 1}}} \end{aligned} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

The ode

$$p'(x) = \frac{\left(p(x)^2 - \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1}\right) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{x \left(2p(x)^3 + 3p(x)^2 - 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 2p(x) - \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1}}\right)} \quad (2.114)$$

is separable as it can be written as

$$\begin{aligned} p'(x) &= \frac{\left(p(x)^2 - \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1}\right) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2}}{x \left(2p(x)^3 + 3p(x)^2 - 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 2p(x) - \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 1}}\right)} \\ &= f(x)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(x) &= \frac{1}{x} \\ g(p) &= \frac{\left(p^2 + 1 - \sqrt{(p^2 + 1)(p + 1)^2}\right) \sqrt{(p^2 + 1)(p + 1)^2}}{2p^3 + 3p^2 - 2p \sqrt{(p^2 + 1)(p + 1)^2 + 2p - \sqrt{(p^2 + 1)(p + 1)^2 + 1}}} \end{aligned}$$

Integrating gives

$$\begin{aligned} \int \frac{1}{g(p)} dp &= \int f(x) dx \\ \int \frac{2p^3 + 3p^2 - 2p \sqrt{(p^2 + 1)(p + 1)^2 + 2p - \sqrt{(p^2 + 1)(p + 1)^2 + 1}}}{\left(p^2 + 1 - \sqrt{(p^2 + 1)(p + 1)^2}\right) \sqrt{(p^2 + 1)(p + 1)^2}} dp &= \int \frac{1}{x} dx \end{aligned}$$

$$\frac{\ln \left(\frac{2p(x)^3 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 2p(x)^2 + p(x) + 1}}{(p(x)^2 + 1)(p(x) + 1)} \right)}{2} = \ln(x) + c_2$$

Taking the exponential of both sides the solution becomes

$$\sqrt{\frac{2p(x)^3 + 2p(x) \sqrt{(p(x)^2 + 1)(p(x) + 1)^2 + 2p(x)^2 + p(x) + 1}}{(p(x)^2 + 1)(p(x) + 1)}} = c_2 x$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{\left(p^2 + 1 - \sqrt{(p^2 + 1)(p + 1)^2}\right) \sqrt{(p^2 + 1)(p + 1)^2}}{2p^3 + 3p^2 - 2p\sqrt{(p^2 + 1)(p + 1)^2} + 2p - \sqrt{(p^2 + 1)(p + 1)^2} + 1} = 0$$

for $p(x)$ gives

$$p(x) = -i$$

$$p(x) = i$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{\frac{2p(x)^3 + 2p(x)\sqrt{(p(x)^2 + 1)(p(x) + 1)^2} + 2p(x)^2 + p(x) + 1}{(p(x)^2 + 1)(p(x) + 1)}} = c_2x$$

$$p(x) = -i$$

$$p(x) = i$$

Substituting the above solution for p in (2A) gives

$$y = x \left(\frac{(c_2x + 1)^2}{-c_2^2x^2 - 2c_2x} - \frac{c_2x + 1}{\sqrt{-c_2^2x^2 - 2c_2x}} + 1 - \sqrt{\left(\frac{(c_2x + 1)^2}{-c_2^2x^2 - 2c_2x} + 1\right) \left(-\frac{c_2x + 1}{\sqrt{-c_2^2x^2 - 2c_2x}} + 1\right)^2} \right)$$

$$y = -ix$$

$$y = ix$$

Simplifying the above gives

$$y = ix$$

$$y = -ix$$

$$y = \frac{(4c_1^2x + x^3) \sqrt{\frac{c_1^2 \left(\sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)}} + 1 \right)^2 (2ic_1 + x)}{x(4c_1^2 + x^2)}} + (4c_1^2x + x^3) \sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)}} + 2ic_1^3 + c_1^2x}{4c_1^2 + x^2}$$

$$y = -ix$$

$$y = ix$$

$$y = 0$$

$$y = ix$$

$$y = -ix$$

$$y =$$

$$- \frac{xc_2 \sqrt{-(c_2x + 2)xc_2} (c_2x + 2) \sqrt{\frac{(c_2x - \sqrt{-(c_2x + 2)xc_2} + 1)^2}{(c_2x + 2)^2 x^2 c_2^2}} + c_2^3 x^3 + 3c_2^2 x^2 + 2c_2 x + \sqrt{-(c_2x + 2)xc_2}}{\sqrt{-(c_2x + 2)xc_2} (c_2x + 2) c_2}$$

$$y = -ix$$

$$y = ix$$

The solution

$$y = \frac{(4c_1^2x + x^3) \sqrt{\frac{c_1^2 \left(\sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)}} + 1 \right)^2 (2ic_1 + x)}{x(4c_1^2 + x^2)}} + (4c_1^2x + x^3) \sqrt{\frac{2ic_1^3 - 3c_1^2x - x^3}{x(4c_1^2 + x^2)}} + 2ic_1^3 + c_1^2x}{4c_1^2 + x^2}$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y =$$

$$- \frac{xc_2 \sqrt{-(c_2x + 2)xc_2} (c_2x + 2) \sqrt{\frac{(c_2x - \sqrt{-(c_2x + 2)xc_2} + 1)^2}{(c_2x + 2)^2 x^2 c_2^2}} + c_2^3 x^3 + 3c_2^2 x^2 + 2c_2 x + \sqrt{-(c_2x + 2)xc_2}}{\sqrt{-(c_2x + 2)xc_2} (c_2x + 2) c_2}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = 0$$

$$y = -ix$$

$$y = ix$$

Solved using first_order_ode_homog_type_G

Time used: 0.384 (sec)

Solve

$$(1 + y'^2)(x - y)^2 = (x + yy')^2$$

Multiplying the right side of the ode, which is $-\frac{yx + \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{x(-x+2y)}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y}\right) - \frac{yx + \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{x(-x+2y)} \\ &= \frac{yx + \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{y(x-2y)} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x(-x^3 + 6x^2y - 7xy^2 + 2y^3 + 2y\sqrt{2yx^3 - 4y^2x^2 + 2xy^3})}{\sqrt{2yx^3 - 4y^2x^2 + 2xy^3} (x-2y)^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x(-x^3 + 6x^2y - 7xy^2 + 2y^3 + 2y\sqrt{2yx^3 - 4y^2x^2 + 2xy^3})}{\sqrt{2yx^3 - 4y^2x^2 + 2xy^3} (x-2y)^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{-\sqrt{z(z-1)^2}\sqrt{2} - z}{2z^2 - z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z(z-1)^2 - 2z}}{2z^2 - z} \right)} dz = 0$$

Multiplying the right side of the ode, which is $-\frac{yx - \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{x(-x+2y)}$ by $\frac{x}{y}$ gives

$$\begin{aligned} y' &= \left(\frac{x}{y} \right) - \frac{yx - \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{x(-x+2y)} \\ &= -\frac{-yx + \sqrt{2yx^3 - 4y^2x^2 + 2xy^3}}{y(x-2y)} \\ &= F(x, y) \end{aligned}$$

Since $F(x, y)$ has y , then let

$$\begin{aligned} f_x &= x \left(\frac{\partial}{\partial x} F(x, y) \right) \\ &= -\frac{x(x^3 - 6x^2y + 7xy^2 - 2y^3 + 2y\sqrt{2yx^3 - 4y^2x^2 + 2xy^3})}{\sqrt{2yx^3 - 4y^2x^2 + 2xy^3} (x-2y)^2} \\ f_y &= y \left(\frac{\partial}{\partial y} F(x, y) \right) \\ &= \frac{x(x^3 - 6x^2y + 7xy^2 - 2y^3 + 2y\sqrt{2yx^3 - 4y^2x^2 + 2xy^3})}{\sqrt{2yx^3 - 4y^2x^2 + 2xy^3} (x-2y)^2} \\ \alpha &= \frac{f_x}{f_y} \\ &= -1 \end{aligned}$$

Since α is independent of x, y then this is Homogeneous type G.

Let

$$\begin{aligned} y &= \frac{z}{x^\alpha} \\ &= \frac{z}{x} \end{aligned}$$

Substituting the above back into $F(x, y)$ gives

$$F(z) = \frac{\sqrt{2} \sqrt{z(z-1)^2} - z}{2z^2 - z}$$

We see that $F(z)$ does not depend on x nor on y . If this was not the case, then this method will not work.

Therefore, the implicit solution is given by

$$\ln(x) - c_1 - \int^{yx^\alpha} \frac{1}{z(\alpha + F(z))} dz = 0$$

Which gives

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{2} \sqrt{z(z-1)^2 - z}}{2z^2 - z} \right)} dz = 0$$

Summary of solutions found

$$\ln(x) - c_1 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{-\sqrt{z(z-1)^2 \sqrt{2} - z}}{2z^2 - z} \right)} dz = 0$$

$$\ln(x) - c_2 + \int^{\frac{y}{x}} \frac{1}{z \left(1 - \frac{\sqrt{2} \sqrt{z(z-1)^2 - z}}{2z^2 - z} \right)} dz = 0$$

✓ **Maple.** Time used: 0.042 (sec). Leaf size: 103

```
ode:=(1+diff(y(x),x)^2)*(x-y(x))^2 = (diff(y(x),x)*y(x)+x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = \text{RootOf} \left(-2 \ln(x) - \int^{-z} \frac{2_a^2 + \sqrt{2} \sqrt{-a(-a-1)^2}}{-a(-a^2+1)} d_a + 2c_1 \right) x$$

$$y = \text{RootOf} \left(-2 \ln(x) + \int^{-z} \frac{\sqrt{2} \sqrt{-a(-a-1)^2} - 2_a^2}{-a(-a^2+1)} d_a + 2c_1 \right) x$$

Maple trace

Methods for first order ODEs:

*** Sublevel 2 ***

Methods for first order ODEs:

-> Solving 1st order ODE of high degree, 1st attempt

trying 1st order WeierstrassP solution for high degree ODE

```

trying 1st order WeierstrassPPPrime solution for high degree ODE
trying 1st order JacobiSN solution for high degree ODE
trying 1st order ODE linearizable_by_differentiation
trying differential order: 1; missing variables
trying simple symmetries for implicit equations
<- symmetries for implicit equations successful

```

Maple step by step

Let's solve

$$\left(\left(\frac{d}{dx}y(x)\right)^2 + 1\right)(x - y(x))^2 = \left(x + y(x)\left(\frac{d}{dx}y(x)\right)\right)^2$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\left[\frac{d}{dx}y(x) = \frac{-xy(x) + \sqrt{2y(x)^3x - 4x^2y(x)^2 + 2y(x)x^3}}{x(2y(x) - x)}, \frac{d}{dx}y(x) = -\frac{xy(x) + \sqrt{2y(x)^3x - 4x^2y(x)^2 + 2y(x)x^3}}{x(2y(x) - x)}\right]$$

- Solve the equation $\frac{d}{dx}y(x) = \frac{-xy(x) + \sqrt{2y(x)^3x - 4x^2y(x)^2 + 2y(x)x^3}}{x(2y(x) - x)}$
- Solve the equation $\frac{d}{dx}y(x) = -\frac{xy(x) + \sqrt{2y(x)^3x - 4x^2y(x)^2 + 2y(x)x^3}}{x(2y(x) - x)}$
- Set of solutions

$$\{workingODE, workingODE\}$$

✓ **Mathematica.** Time used: 2.284 (sec). Leaf size: 167

```

ode=(D[y[x],x]^2+1)*(x-y[x])^2==(x+y[x]*D[y[x],x])^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow -\sqrt{-x\left(x + 2e^{\frac{c_1}{2}}\right)} - e^{\frac{c_1}{2}}$$

$$y(x) \rightarrow \sqrt{-x\left(x + 2e^{\frac{c_1}{2}}\right)} - e^{\frac{c_1}{2}}$$

$$y(x) \rightarrow e^{\frac{c_1}{2}} - \sqrt{x\left(-x + 2e^{\frac{c_1}{2}}\right)}$$

$$y(x) \rightarrow \sqrt{x\left(-x + 2e^{\frac{c_1}{2}}\right)} + e^{\frac{c_1}{2}}$$

$$y(x) \rightarrow -\sqrt{-x^2}$$

$$y(x) \rightarrow \sqrt{-x^2}$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-(x + y(x)*Derivative(y(x), x))**2 + (x - y(x))**2*(Derivative(y(x), x)
    **2 + 1),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.7 Chapter 12. Linear equations of order n . Supplementary problems. Page 81

Local contents

2.7.1	Problem 10	1489
2.7.2	Problem 11	1496
2.7.3	Problem 12	1499
2.7.4	Problem 13	1509
2.7.5	Problem 14	1519
2.7.6	Problem 15	1534
2.7.7	Problem 16	1540
2.7.8	Problem 17	1553
2.7.9	Problem 18	1561
2.7.10	Problem 19	1575

2.7.1 Problem 10

Local contents

Solved as second order linear constant coeff ode	1489
Solved as second order ode using Kovacic algorithm	1491
✓ Maple	1494
✓ Mathematica	1495
✓ Sympy	1495

Internal problem ID [7967]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 10

Date solved : Thursday, November 27, 2025 at 06:39:10 AM

CAS classification : [[_2nd_order, _missing_x]]

Solved as second order linear constant coeff ode

Time used: 0.059 (sec)

Solve

$$y'' + y' - 6y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 1, C = -6$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + \lambda e^{x\lambda} - 6 e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + \lambda - 6 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 1, C = -6$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-1}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{1^2 - (4)(1)(-6)} \\ &= -\frac{1}{2} \pm \frac{5}{2} \end{aligned}$$

Hence

$$\lambda_1 = -\frac{1}{2} + \frac{5}{2}$$

$$\lambda_2 = -\frac{1}{2} - \frac{5}{2}$$

Which simplifies to

$$\lambda_1 = 2$$

$$\lambda_2 = -3$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(2)x} + c_2 e^{(-3)x}$$

Or

$$y = e^{2x} c_1 + c_2 e^{-3x}$$

Summary of solutions found

$$y = e^{2x} c_1 + c_2 e^{-3x}$$

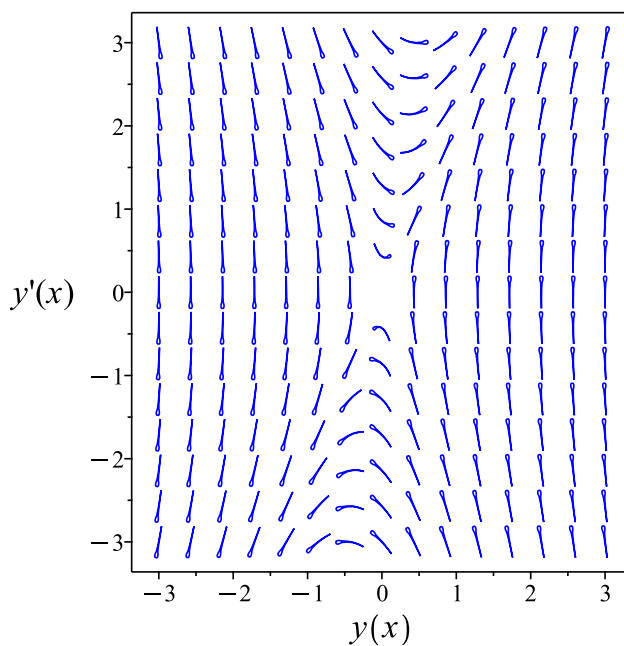


Figure 2.305: Slope field $y'' + y' - 6y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.062 (sec)

Solve

$$y'' + y' - 6y = 0$$

Writing the ode as

$$y'' + y' - 6y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 1 \\ C &= -6 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{25}{4} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 25 \\ t &= 4 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \frac{25z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.166: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{25}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{5x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{1}{1} dx} \\ &= z_1 e^{-\frac{x}{2}} \\ &= z_1 \left(e^{-\frac{x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

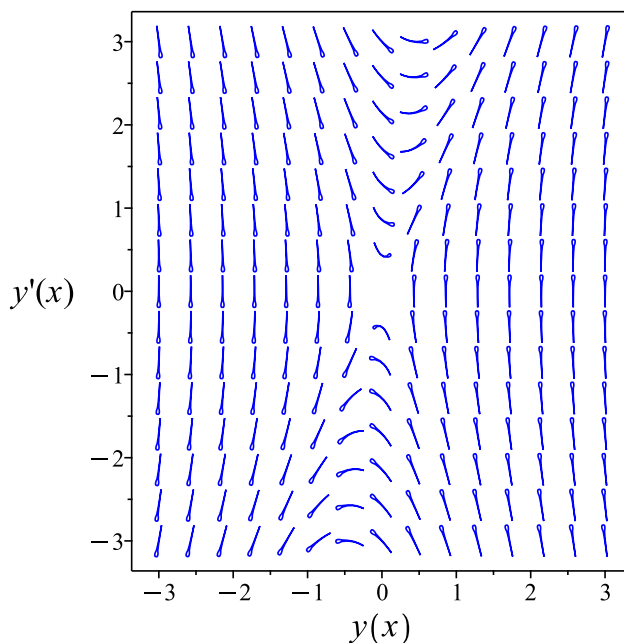
$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{1}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{-x} e^{6x}}{5} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-3x}) + c_2 \left(e^{-3x} \left(\frac{e^{-x} e^{6x}}{5} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = e^{-3x} c_1 + \frac{e^{2x} c_2}{5}$$

Figure 2.306: Slope field $y'' + y' - 6y = 0$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 17

```
ode:=diff(diff(y(x),x),x)+diff(y(x),x)-6*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (c_1 e^{5x} + c_2) e^{-3x}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + \frac{d}{dx}y(x) - 6y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 + r - 6 = 0$
- Factor the characteristic polynomial

- $(r + 3)(r - 2) = 0$
 - Roots of the characteristic polynomial
 $r = (-3, 2)$
 - 1st solution of the ODE
 $y_1(x) = e^{-3x}$
 - 2nd solution of the ODE
 $y_2(x) = e^{2x}$
 - General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x)$
 - Substitute in solutions
 $y(x) = C_1 e^{-3x} + C_2 e^{2x}$

✓ **Mathematica.** Time used: 0.01 (sec). Leaf size: 22

```
ode=D[y[x],{x,2}]+D[y[x],x]-6*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-3x}(c_2 e^{5x} + c_1)$$

✓ **Sympy.** Time used: 0.074 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-6*y(x) + Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-3x} + C_2 e^{2x}$$

2.7.2 Problem 11

Local contents

Solved as higher order constant coeff ode	1496
✓ Maple	1497
✓ Mathematica	1498
✓ Sympy	1498

Internal problem ID [7968]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 11

Date solved : Thursday, November 27, 2025 at 06:39:15 AM

CAS classification : `[[_3rd_order, _missing_x]]`

Solve

$$-8y + 12y' - 6y'' + y''' = 0$$

Solved as higher order constant coeff ode

Time used: 0.033 (sec)

The characteristic equation is

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

The roots of the above equation are

$$\lambda_1 = 2$$

$$\lambda_2 = 2$$

$$\lambda_3 = 2$$

Therefore the homogeneous solution is

$$y_h(x) = e^{2x}c_1 + x e^{2x}c_2 + x^2 e^{2x}c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = e^{2x}$$

$$y_2 = x e^{2x}$$

$$y_3 = e^{2x}x^2$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 19

```
ode:=diff(diff(diff(y(x),x),x),x)-6*diff(diff(y(x),x),x)+12*diff(y(x),x)-8*y(x) =
0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{2x}(c_3 x^2 + c_2 x + c_1)$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) - 6\frac{d^2}{dx^2}y(x) + 12\frac{d}{dx}y(x) - 8y(x) = 0$$

- Highest derivative means the order of the ODE is 3
- $\frac{d^3}{dx^3}y(x)$
- Characteristic polynomial of ODE

$$r^3 - 6r^2 + 12r - 8 = 0$$
- Roots of the characteristic polynomial and corresponding multiplicities

$$r = [[2, 3]]$$
- 1st solution from $r = 2$

$$y_1(x) = e^{2x}$$
- 2nd solution from $r = 2$

$$y_2(x) = x e^{2x}$$
- 3rd solution from $r = 2$

$$y_3(x) = x^2 e^{2x}$$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x)$$
- Substitute in solutions and simplify

$$y(x) = e^{2x}(C_3 x^2 + C_2 x + C_1)$$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 23

```
ode=D[y[x],{x,3}]-6*D[y[x],{x,2}]+12*D[y[x],x]-8*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{2x}(x(c_3x + c_2) + c_1)$$

✓ **Sympy.** Time used: 0.102 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-8*y(x) + 12*Derivative(y(x), x) - 6*Derivative(y(x), (x, 2)) +
        Derivative(y(x), (x, 3)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + x(C_2 + C_3x))e^{2x}$$

2.7.3 Problem 12

Local contents

Solved as second order linear constant coeff ode	1499
Solved as second order ode using Kovacic algorithm	1502
✓ Maple	1506
✓ Mathematica	1508
✓ Sympy	1508

Internal problem ID [7969]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 12

Date solved : Thursday, November 27, 2025 at 06:39:20 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.129 (sec)

Solve

$$y'' - 3y' + 2y = e^{5x}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -3, C = 2, f(x) = e^{5x}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -3, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 3\lambda e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 3\lambda + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -3, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{3}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-3^2 - (4)(1)(2)} \\ &= \frac{3}{2} \pm \frac{1}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= \frac{3}{2} + \frac{1}{2} \\ \lambda_2 &= \frac{3}{2} - \frac{1}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= 2 \\ \lambda_2 &= 1 \end{aligned}$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(2)x} + c_2 e^{(1)x} \end{aligned}$$

Or

$$y = e^{2x} c_1 + c_2 e^x$$

Therefore the homogeneous solution y_h is

$$y_h = e^{2x} c_1 + c_2 e^x$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{5x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{5x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{2x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{5x}$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$12A_1 e^{5x} = e^{5x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{12} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

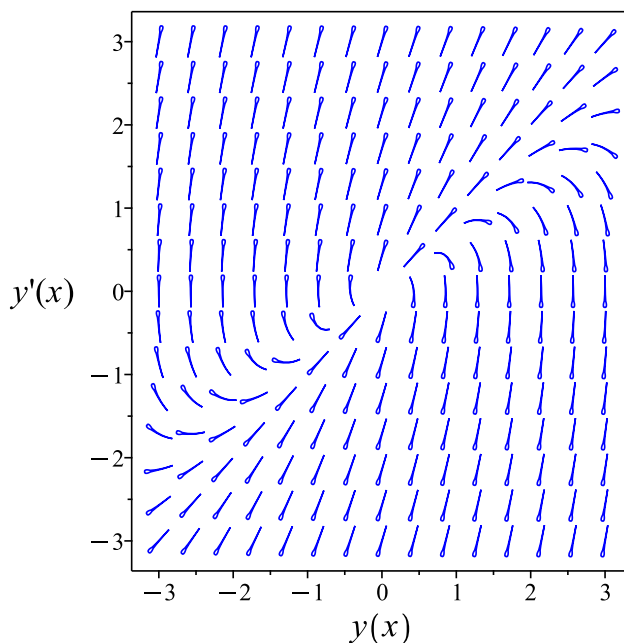
$$y_p = \frac{e^{5x}}{12}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x} c_1 + c_2 e^x) + \left(\frac{e^{5x}}{12} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{e^{5x}}{12} + e^{2x} c_1 + c_2 e^x$$

Figure 2.307: Slope field $y'' - 3y' + 2y = e^{5x}$

Solved as second order ode using Kovacic algorithm

Time used: 0.106 (sec)

Solve

$$y'' - 3y' + 2y = e^{5x}$$

Writing the ode as

$$y'' - 3y' + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -3 \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{4} \quad (6)$$

Comparing the above to (5) shows that

$$s = 1$$

$$t = 4$$

Therefore eq. (4) becomes

$$z''(x) = \frac{z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.169: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{1}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3}{1} dx} \\ &= z_1 e^{\frac{3x}{2}} \\ &= z_1 \left(e^{\frac{3x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3x}}{(y_1)^2} dx \\ &= y_1 (e^{3x} e^{-2x}) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 (e^x (e^{3x} e^{-2x})) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^x c_1 + e^{2x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{5x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{5x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{2x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{5x}$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$12A_1 e^{5x} = e^{5x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{12} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{e^{5x}}{12}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{2x} c_2) + \left(\frac{e^{5x}}{12} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{e^{5x}}{12} + e^x c_1 + e^{2x} c_2$$

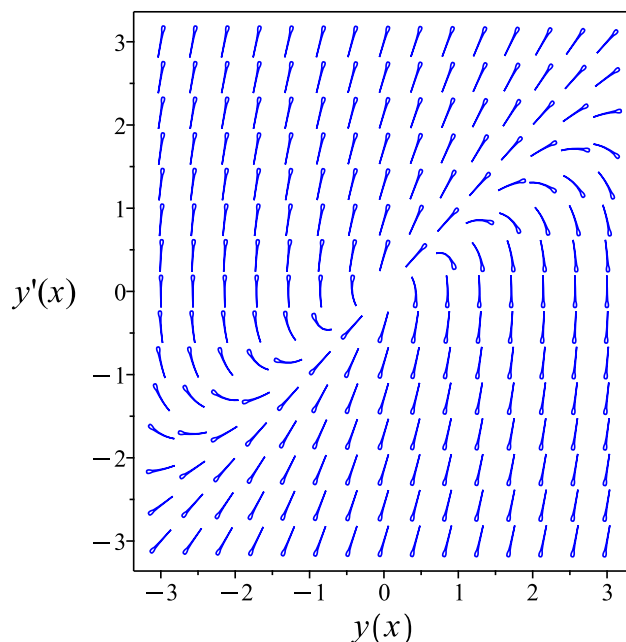


Figure 2.308: Slope field $y'' - 3y' + 2y = e^{5x}$

✓ Maple. Time used: 0.001 (sec). Leaf size: 19

```
ode:=2*y(x)-3*diff(y(x),x)+diff(diff(y(x),x),x) = exp(5*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \left(\frac{e^{4x}}{12} + e^x c_1 + c_2 \right) e^x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
```

```

trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
<- double symmetry of the form [xi=0, eta=F(x)] successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 3\frac{d}{dx}y(x) + 2y(x) = e^{5x}$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^2 - 3r + 2 = 0$$
- Factor the characteristic polynomial

$$(r - 1)(r - 2) = 0$$
- Roots of the characteristic polynomial

$$r = (1, 2)$$
- 1st solution of the homogeneous ODE

$$y_1(x) = e^x$$
- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{2x}$$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^x + C_2 e^{2x} + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^{5x} \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{3x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = e^x \left(- \int e^{4x} dx + e^x \int e^{3x} dx \right)$$
 - Compute integrals

$$y_p(x) = \frac{e^{5x}}{12}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^x + C_2 e^{2x} + \frac{e^{5x}}{12}$$

✓ **Mathematica.** Time used: 0.014 (sec). Leaf size: 29

```
ode=D[y[x],{x,2}]-3*D[y[x],x]+2*y[x]==Exp[5*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{e^{5x}}{12} + c_1 e^x + c_2 e^{2x}$$

✓ **Sympy.** Time used: 0.114 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*y(x) - exp(5*x) - 3*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \left(C_1 + C_2 e^x + \frac{e^{4x}}{12} \right) e^x$$

2.7.4 Problem 13

Local contents

Solved as second order linear constant coeff ode	1509
Solved as second order ode using Kovacic algorithm	1512
✓ Maple	1516
✓ Mathematica	1518
✓ Sympy	1518

Internal problem ID [7970]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 13

Date solved : Thursday, November 27, 2025 at 06:39:25 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.220 (sec)

Solve

$$y'' + 9y = x \cos(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 9, f(x) = x \cos(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 9y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 9$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 9e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 9 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 9$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(9)} \\ &= \pm 3i \end{aligned}$$

Hence

$$\lambda_1 = +3i$$

$$\lambda_2 = -3i$$

Which simplifies to

$$\lambda_1 = 3i$$

$$\lambda_2 = -3i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 3$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(3x) + c_2 \sin(3x))$$

Or

$$y = c_1 \cos(3x) + c_2 \sin(3x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(3x) + c_2 \sin(3x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x \cos(x), x \sin(x), \cos(x), \sin(x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{\cos(3x), \sin(3x)\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 x \cos(x) + A_2 x \sin(x) + A_3 \cos(x) + A_4 \sin(x)$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_1 \sin(x) + 8A_1 x \cos(x) + 2A_2 \cos(x) + 8A_2 x \sin(x) + 8A_3 \cos(x) + 8A_4 \sin(x) = x \cos(x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{8}, A_2 = 0, A_3 = 0, A_4 = \frac{1}{32} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

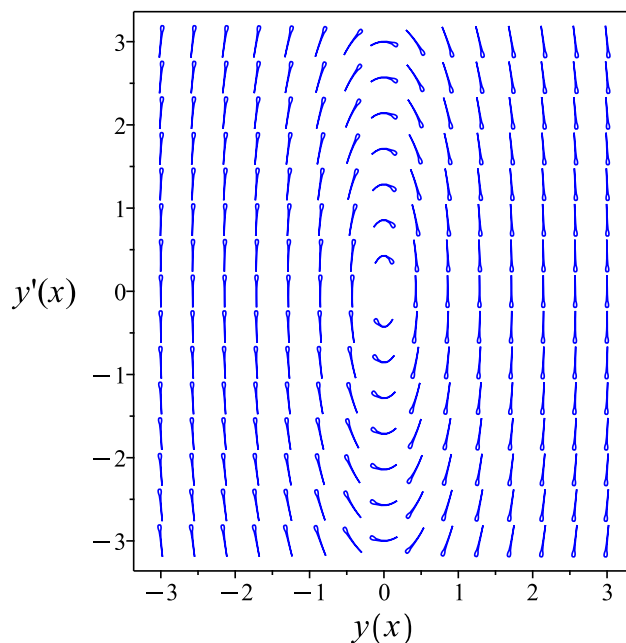
$$y_p = \frac{x \cos(x)}{8} + \frac{\sin(x)}{32}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 \cos(3x) + c_2 \sin(3x)) + \left(\frac{x \cos(x)}{8} + \frac{\sin(x)}{32} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x \cos(x)}{8} + \frac{\sin(x)}{32} + c_1 \cos(3x) + c_2 \sin(3x)$$

Figure 2.309: Slope field $y'' + 9y = x \cos(x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.191 (sec)

Solve

$$y'' + 9y = x \cos(x)$$

Writing the ode as

$$y'' + 9y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 9 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-9}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -9 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -9z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.171: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -9$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(3x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(3x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(3x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(3x) \int \frac{1}{\cos(3x)^2} dx \\ &= \cos(3x) \left(\frac{\tan(3x)}{3} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(3x)) + c_2 \left(\cos(3x) \left(\frac{\tan(3x)}{3} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 9y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(3x) + \frac{c_2 \sin(3x)}{3}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x \cos(x), x \sin(x), \cos(x), \sin(x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{\sin(3x)}{3}, \cos(3x) \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 x \cos(x) + A_2 x \sin(x) + A_3 \cos(x) + A_4 \sin(x)$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_1 \sin(x) + 8A_1 x \cos(x) + 2A_2 \cos(x) + 8A_2 x \sin(x) + 8A_3 \cos(x) + 8A_4 \sin(x) = x \cos(x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{8}, A_2 = 0, A_3 = 0, A_4 = \frac{1}{32} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{x \cos(x)}{8} + \frac{\sin(x)}{32}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(3x) + \frac{c_2 \sin(3x)}{3} \right) + \left(\frac{x \cos(x)}{8} + \frac{\sin(x)}{32} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x \cos(x)}{8} + \frac{\sin(x)}{32} + c_1 \cos(3x) + \frac{c_2 \sin(3x)}{3}$$

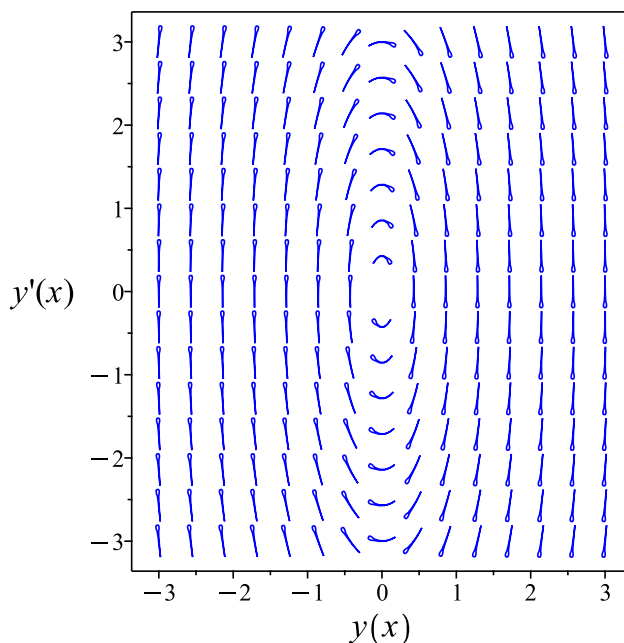


Figure 2.310: Slope field $y'' + 9y = x \cos(x)$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 26

```
ode:=diff(diff(y(x),x),x)+9*y(x) = cos(x)*x;
dsolve(ode,y(x), singsol=all);
```

$$y = \sin(3x) c_2 + \cos(3x) c_1 + \frac{\sin(x)}{32} + \frac{x \cos(x)}{8}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 9y(x) = x \cos(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 9 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-36})}{2}$$

- Roots of the characteristic polynomial

$$r = (-3i, 3i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(3x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(3x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(3x) + C_2 \sin(3x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = x \cos(x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(3x) & \sin(3x) \\ -3\sin(3x) & 3\cos(3x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 3$$

- Substitute functions into equation for $y_p(x)$

- $$y_p(x) = -\frac{\cos(3x) \int \sin(3x)x \cos(x) dx}{3} + \frac{\sin(3x) \int \cos(3x)x \cos(x) dx}{3}$$
- Compute integrals

$$y_p(x) = \frac{x \cos(x)}{8} + \frac{\sin(x)}{32}$$
 - Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(3x) + C_2 \sin(3x) + \frac{x \cos(x)}{8} + \frac{\sin(x)}{32}$$

✓ **Mathematica.** Time used: 0.014 (sec). Leaf size: 32

```
ode=D[y[x],{x,2}]+9*y[x]==x*Cos[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{32}(\sin(x) + 4x \cos(x)) + c_1 \cos(3x) + c_2 \sin(3x)$$

✓ **Sympy.** Time used: 0.056 (sec). Leaf size: 27

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*cos(x) + 9*y(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 \sin(3x) + C_2 \cos(3x) + \frac{x \cos(x)}{8} + \frac{\sin(x)}{32}$$

2.7.5 Problem 14

Local contents

Solved as second order Euler type ode	1519
Solved as second order ode using change of variable on x method 2 .	1520
Solved as second order ode using change of variable on x method 1 .	1522
Solved as second order ode using change of variable on y method 2 .	1524
Solved as second order ode using Kovacic algorithm	1527
✓ Maple	1531
✓ Mathematica	1533
✓ Sympy	1533

Internal problem ID [7971]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 14

Date solved : Thursday, November 27, 2025 at 06:39:31 AM

CAS classification :

[[_Emden, _Fowler], [_2nd_order, _linear, '_with_symmetry_[0,F(x)]']]

Solved as second order Euler type ode

Time used: 0.066 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = 0$$

This is Euler second order ODE. Let the solution be $y = x^r$, then $y' = rx^{r-1}$ and $y'' = r(r-1)x^{r-2}$. Substituting these back into the given ODE gives

$$x^2(r(r-1))x^{r-2} - 3rx^{r-1} + 4x^r = 0$$

Simplifying gives

$$r(r-1)x^r - 3rx^r + 4x^r = 0$$

Since $x^r \neq 0$ then dividing throughout by x^r gives

$$r(r-1) - 3r + 4 = 0$$

Or

$$r^2 - 4r + 4 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = 2$$

$$r_2 = 2$$

Since the roots are equal, then the general solution is

$$y = c_1 y_1 + c_2 y_2$$

Where $y_1 = x^r$ and $y_2 = x^r \ln(x)$. Hence

$$y = c_1 x^2 + c_2 \ln(x) x^2$$

Summary of solutions found

$$y = c_1 x^2 + c_2 \ln(x) x^2$$

Solved as second order ode using change of variable on x method 2

Time used: 0.244 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2 y'' - 3y'x + 4y = 0 \quad (1)$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{4}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2} y(\tau) + p_1 \left(\frac{d}{d\tau} y(\tau) \right) + q_1 y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x) dx} dx \\ &= \int e^{-\int -\frac{3}{x} dx} dx \\ &= \int e^{3 \ln(x)} dx \\ &= \int x^3 dx \\ &= \frac{x^4}{4} \end{aligned} \tag{6}$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{\frac{4}{x^2}}{x^6} \\ &= \frac{4}{x^8} \end{aligned} \tag{7}$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2} y(\tau) + q_1 y(\tau) &= 0 \\ \frac{d^2}{d\tau^2} y(\tau) + \frac{4y(\tau)}{x^8} &= 0 \end{aligned}$$

But in terms of τ

$$\frac{4}{x^8} = \frac{1}{4\tau^2}$$

Hence the above ode becomes

$$\frac{d^2}{d\tau^2} y(\tau) + \frac{y(\tau)}{4\tau^2} = 0$$

The above ode is now solved for $y(\tau)$. This is Euler second order ODE. Let the solution be $y(\tau) = \tau^r$, then $y' = r\tau^{r-1}$ and $y'' = r(r-1)\tau^{r-2}$. Substituting these back into the given ODE gives

$$4\tau^2(r(r-1))\tau^{r-2} + 0r\tau^{r-1} + \tau^r = 0$$

Simplifying gives

$$4r(r-1)\tau^r + 0\tau^r + \tau^r = 0$$

Since $\tau^r \neq 0$ then dividing throughout by τ^r gives

$$4r(r-1) + 0 + 1 = 0$$

Or

$$4r^2 - 4r + 1 = 0 \quad (1)$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = \frac{1}{2}$$

$$r_2 = \frac{1}{2}$$

Since the roots are equal, then the general solution is

$$y(\tau) = c_1 y_1 + c_2 y_2$$

Where $y_1 = \tau^r$ and $y_2 = \tau^r \ln(\tau)$. Hence

$$y(\tau) = c_1 \sqrt{\tau} + c_2 \sqrt{\tau} \ln(\tau)$$

The above solution is now transformed back to y using (6) which results in

$$y = \frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4}$$

Summary of solutions found

$$y = \frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4}$$

Solved as second order ode using change of variable on x method 1

Time used: 0.325 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2 y'' - 3y'x + 4y = 0 \quad (1)$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \quad (2)$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{4}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) results

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x)\tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $q_1 = c^2$ where c is some constant. Therefore from (5)

$$\begin{aligned} \tau' &= \frac{1}{c}\sqrt{q} \\ &= \frac{2\sqrt{\frac{1}{x^2}}}{c} \\ \tau'' &= -\frac{2}{c\sqrt{\frac{1}{x^2}}x^3} \end{aligned} \quad (6)$$

Substituting the above into (4) results in

$$\begin{aligned} p_1(\tau) &= \frac{\tau''(x) + p(x)\tau'(x)}{\tau'(x)^2} \\ &= \frac{-\frac{2}{c\sqrt{\frac{1}{x^2}}x^3} - \frac{3}{x}\frac{2\sqrt{\frac{1}{x^2}}}{c}}{\left(\frac{2\sqrt{\frac{1}{x^2}}}{c}\right)^2} \\ &= -2c \end{aligned}$$

Therefore ode (3) now becomes

$$\begin{aligned} y(\tau)'' + p_1y(\tau)' + q_1y(\tau) &= 0 \\ \frac{d^2}{d\tau^2}y(\tau) - 2c\left(\frac{d}{d\tau}y(\tau)\right) + c^2y(\tau) &= 0 \end{aligned} \quad (7)$$

The above ode is now solved for $y(\tau)$. Since the ode is now constant coefficients, it can be easily solved to give

$$y(\tau) = c_1 e^{c\tau} + c_2 \tau e^{c\tau}$$

Now from (6)

$$\begin{aligned}\tau &= \int \frac{1}{c} \sqrt{q} dx \\ &= \frac{\int 2\sqrt{\frac{1}{x^2}} dx}{c} \\ &= \frac{2 \ln(x)}{c}\end{aligned}$$

Substituting the above into the solution obtained gives

$$y = c_1 x^2 + \frac{2c_2 \ln(x) x^2}{c}$$

Summary of solutions found

$$y = c_1 x^2 + \frac{2c_2 \ln(x) x^2}{c}$$

Solved as second order ode using change of variable on y method 2

Time used: 0.231 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2 y'' - 3y'x + 4y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \tag{2}$$

Where

$$\begin{aligned}p(x) &= -\frac{3}{x} \\ q(x) &= \frac{4}{x^2}\end{aligned}$$

Applying change of variables on the dependent variable $y = v(x) x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right) v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right) v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{3n}{x^2} + \frac{4}{x^2} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 2 \quad (6)$$

Substituting this value in (3) gives

$$\begin{aligned} v''(x) + \frac{v'(x)}{x} &= 0 \\ v''(x) + \frac{v'(x)}{x} &= 0 \end{aligned} \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{u(x)}{x} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{1}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int \frac{1}{x} dx} \\ &= x\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu u &= 0 \\ \frac{d}{dx}(ux) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}ux &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

$$u(x) = \frac{c_1}{x}$$

Now that $u(x)$ is known, then

$$\begin{aligned}v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= c_1 \ln(x) + c_2\end{aligned}$$

Hence

$$\begin{aligned}y &= v(x) x^n \\ &= (c_1 \ln(x) + c_2) x^2 \\ &= (c_1 \ln(x) + c_2) x^2\end{aligned}$$

Summary of solutions found

$$y = (c_1 \ln(x) + c_2) x^2$$

Solved as second order ode using Kovacic algorithm

Time used: 0.109 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = 0$$

Writing the ode as

$$x^2 y'' - 3y'x + 4y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 \\ B &= -3x \\ C &= 4 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{1}{4x^2}\right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.173: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -\frac{1}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = -\frac{1}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{1}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{1}{2}$	$\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{1}{2}$	$\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = \frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= \frac{1}{2} - \left(\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= \frac{1}{2x} + (-)(0) \\ &= \frac{1}{2x} \\ &= \frac{1}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(\frac{1}{2x}\right)(0) + \left(\left(-\frac{1}{2x^2}\right) + \left(\frac{1}{2x}\right)^2 - \left(-\frac{1}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \frac{1}{2x} dx} \\ &= \sqrt{x} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3x}{x^2} dx} \\ &= z_1 e^{\frac{3 \ln(x)}{2}} \\ &= z_1 (x^{3/2}) \end{aligned}$$

Which simplifies to

$$y_1 = x^2$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3\ln(x)}}{(y_1)^2} dx \\ &= y_1 (\ln(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (x^2) + c_2 (x^2 (\ln(x))) \end{aligned}$$

Summary of solutions found

$$y = c_1 x^2 + c_2 \ln(x) x^2$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 14

```
ode:=x^2*diff(diff(y(x),x),x)-3*diff(y(x),x)*x+4*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = x^2(c_1 + c_2 \ln(x))$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
<- LODE of Euler type successful
```

Maple step by step

Let's solve

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - 3x \left(\frac{d}{dx} y(x) \right) + 4y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{4y(x)}{x^2} + \frac{3 \left(\frac{d}{dx} y(x) \right)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{3 \left(\frac{d}{dx} y(x) \right)}{x} + \frac{4y(x)}{x^2} = 0$$

- Multiply by denominators of the ODE

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - 3x \left(\frac{d}{dx} y(x) \right) + 4y(x) = 0$$

- Make a change of variables

$$t = \ln(x)$$

- Substitute the change of variables back into the ODE

- Calculate the 1st derivative of y with respect to x , using the chain rule

$$\frac{d}{dx} y(x) = \left(\frac{d}{dt} y(t) \right) \left(\frac{d}{dx} t(x) \right)$$

- Compute derivative

$$\frac{d}{dx} y(x) = \frac{\frac{d}{dt} y(t)}{x}$$

- Calculate the 2nd derivative of y with respect to x , using the chain rule

$$\frac{d^2}{dx^2} y(x) = \left(\frac{d^2}{dt^2} y(t) \right) \left(\frac{d}{dx} t(x) \right)^2 + \left(\frac{d^2}{dx^2} t(x) \right) \left(\frac{d}{dt} y(t) \right)$$

- Compute derivative

$$\frac{d^2}{dx^2} y(x) = \frac{\frac{d^2}{dt^2} y(t)}{x^2} - \frac{\frac{d}{dt} y(t)}{x^2}$$

Substitute the change of variables back into the ODE

$$x^2 \left(\frac{\frac{d^2}{dt^2} y(t)}{x^2} - \frac{\frac{d}{dt} y(t)}{x^2} \right) - 3 \frac{d}{dt} y(t) + 4y(t) = 0$$

- Simplify

$$\frac{d^2}{dt^2} y(t) - 4 \frac{d}{dt} y(t) + 4y(t) = 0$$

- Characteristic polynomial of ODE

$$r^2 - 4r + 4 = 0$$

- Factor the characteristic polynomial

$$(r - 2)^2 = 0$$

- Root of the characteristic polynomial

$$r = 2$$

- 1st solution of the ODE

$$y_1(t) = e^{2t}$$

- Repeated root, multiply $y_1(t)$ by t to ensure linear independence

$$y_2(t) = t e^{2t}$$

- General solution of the ODE

- $y(t) = C_1 y_1(t) + C_2 y_2(t)$
- Substitute in solutions
 $y(t) = C_1 e^{2t} + C_2 t e^{2t}$
- Change variables back using $t = \ln(x)$
 $y(x) = C_2 \ln(x) x^2 + C_1 x^2$
- Simplify
 $y(x) = x^2(C_2 \ln(x) + C_1)$

✓ **Mathematica.** Time used: 0.012 (sec). Leaf size: 18

```
ode=x^2*D[y[x],{x,2}]-3*x*D[y[x],x]+4*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x^2(2c_2 \log(x) + c_1)$$

✓ **Sympy.** Time used: 0.093 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) - 3*x*Derivative(y(x), x) + 4*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = x^2(C_1 + C_2 \log(x))$$

2.7.6 Problem 15

Local contents

Solved as higher order Euler type ode	1534
✓ Maple	1538
✓ Mathematica	1539
✓ Sympy	1539

Internal problem ID [7972]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 15

Date solved : Thursday, November 27, 2025 at 06:39:37 AM

CAS classification : `[_3rd_order, _with_linear_symmetries]`

Solve

$$-y + y'x + x^3y''' = 3x^4$$

Solved as higher order Euler type ode

Time used: 0.257 (sec)

This is Euler ODE of higher order. Let $y = x^\lambda$. Hence

$$\begin{aligned} y' &= \lambda x^{\lambda-1} \\ y'' &= \lambda(\lambda-1) x^{\lambda-2} \\ y''' &= \lambda(\lambda-1)(\lambda-2) x^{\lambda-3} \end{aligned}$$

Substituting these back into

$$-y + y'x + x^3y''' = 3x^4$$

gives

$$x\lambda x^{\lambda-1} + x^3\lambda(\lambda-1)(\lambda-2)x^{\lambda-3} - x^\lambda = 0$$

Which simplifies to

$$\lambda x^\lambda + \lambda(\lambda-1)(\lambda-2)x^\lambda - x^\lambda = 0$$

And since $x^\lambda \neq 0$ then dividing through by x^λ , the above becomes

$$\lambda + \lambda(\lambda-1)(\lambda-2) - 1 = 0$$

Simplifying gives the characteristic equation as

$$(\lambda - 1)^3 = 0$$

Solving the above gives the following roots

$$\lambda_1 = 1$$

$$\lambda_2 = 1$$

$$\lambda_3 = 1$$

This table summarises the result

root	multiplicity	type of root
1	3	real root

The solution is generated by going over the above table. For each real root λ of multiplicity one generates a $c_1 x^\lambda$ basis solution. Each real root of multiplicity two, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ basis solutions. Each real root of multiplicity three, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ and $c_3 x^\lambda \ln(x)^2$ basis solutions, and so on. Each complex root $\alpha \pm i\beta$ of multiplicity one generates $x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity two generates $\ln(x) x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity three generates $\ln(x)^2 x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And so on. Using the above show that the solution is

$$y = c_1 x + c_2 \ln(x) x + c_3 \ln(x)^2 x$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = x$$

$$y_2 = x \ln(x)$$

$$y_3 = \ln(x)^2 x$$

This is higher order nonhomogeneous Euler type ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous Euler ODE And y_p is a particular solution to the nonhomogeneous Euler ODE. y_h is the solution to

$$-y + y'x + x^3 y''' = 0$$

Now the particular solution to the given ODE is found

$$-y + y'x + x^3 y''' = 3x^4$$

Let the particular solution be

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Where y_i are the basis solutions found above for the homogeneous solution y_h and $U_i(x)$ are functions to be determined as follows

$$U_i = (-1)^{n-i} \int \frac{F(x)W_i(x)}{aW(x)} dx$$

Where $W(x)$ is the Wronskian and $W_i(x)$ is the Wronskian that results after deleting the last row and the i -th column of the determinant and n is the order of the ODE or equivalently, the number of basis solutions, and a is the coefficient of the leading derivative in the ODE, and $F(x)$ is the RHS of the ODE. Therefore, the first step is to find the Wronskian $W(x)$. This is given by

$$W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Substituting the fundamental set of solutions y_i found above in the Wronskian gives

$$W = \begin{bmatrix} x & x \ln(x) & \ln(x)^2 x \\ 1 & \ln(x) + 1 & \ln(x)(\ln(x) + 2) \\ 0 & \frac{1}{x} & \frac{2+2\ln(x)}{x} \end{bmatrix}$$

$$|W| = 2$$

The determinant simplifies to

$$|W| = 2$$

Now we determine W_i for each U_i .

$$W_1(x) = \det \begin{bmatrix} x \ln(x) & \ln(x)^2 x \\ \ln(x) + 1 & \ln(x)(\ln(x) + 2) \end{bmatrix}$$

$$= \ln(x)^2 x$$

$$W_2(x) = \det \begin{bmatrix} x & \ln(x)^2 x \\ 1 & \ln(x)(\ln(x) + 2) \end{bmatrix}$$

$$= 2x \ln(x)$$

$$W_3(x) = \det \begin{bmatrix} x & x \ln(x) \\ 1 & \ln(x) + 1 \end{bmatrix}$$

$$= x$$

Now we are ready to evaluate each $U_i(x)$.

$$\begin{aligned}
 U_1 &= (-1)^{3-1} \int \frac{F(x)W_1(x)}{aW(x)} dx \\
 &= (-1)^2 \int \frac{(3x^4)(\ln(x)^2 x)}{(x^3)(2)} dx \\
 &= \int \frac{3x^5 \ln(x)^2}{2x^3} dx \\
 &= \int \left(\frac{3x^2 \ln(x)^2}{2} \right) dx \\
 &= \frac{\ln(x)^2 x^3}{2} - \frac{x^3 \ln(x)}{3} + \frac{x^3}{9}
 \end{aligned}$$

$$\begin{aligned}
 U_2 &= (-1)^{3-2} \int \frac{F(x)W_2(x)}{aW(x)} dx \\
 &= (-1)^1 \int \frac{(3x^4)(2x \ln(x))}{(x^3)(2)} dx \\
 &= - \int \frac{6x^5 \ln(x)}{2x^3} dx \\
 &= - \int (3x^2 \ln(x)) dx \\
 &= -x^3 \ln(x) + \frac{x^3}{3}
 \end{aligned}$$

$$\begin{aligned}
 U_3 &= (-1)^{3-3} \int \frac{F(x)W_3(x)}{aW(x)} dx \\
 &= (-1)^0 \int \frac{(3x^4)(x)}{(x^3)(2)} dx \\
 &= \int \frac{3x^5}{2x^3} dx \\
 &= \int \left(\frac{3x^2}{2} \right) dx \\
 &= \frac{x^3}{2}
 \end{aligned}$$

Now that all the U_i functions have been determined, the particular solution is found from

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Hence

$$\begin{aligned}
 y_p &= \left(\frac{\ln(x)^2 x^3}{2} - \frac{x^3 \ln(x)}{3} + \frac{x^3}{9} \right) (x) \\
 &\quad + \left(-x^3 \ln(x) + \frac{x^3}{3} \right) (x \ln(x)) \\
 &\quad + \left(\frac{x^3}{2} \right) (\ln(x)^2 x)
 \end{aligned}$$

Therefore the particular solution is

$$y_p = \frac{x^4}{9}$$

Therefore the general solution is

$$\begin{aligned}
 y &= y_h + y_p \\
 &= (c_1 x + c_2 \ln(x) x + c_3 \ln(x)^2 x) + \left(\frac{x^4}{9} \right)
 \end{aligned}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 26

```
ode:=-y(x)+diff(y(x),x)*x+x^3*diff(diff(diff(y(x),x),x),x) = 3*x^4;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x(9c_3 \ln(x)^2 + x^3 + 9c_2 \ln(x) + 9c_1)}{9}$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 3; linear nonhomogeneous with symmetry [0,1]
trying high order linear exact nonhomogeneous
trying differential order: 3; missing the dependent variable
checking if the LODE is of Euler type
<- LODE of Euler type successful
Euler equation successful
```

Maple step by step

Let's solve

$$x^3 \left(\frac{d^3}{dx^3} y(x) \right) + x \left(\frac{d}{dx} y(x) \right) - y(x) = 3x^4$$

- Highest derivative means the order of the ODE is 3
 $\frac{d^3}{dx^3} y(x)$

✓ **Mathematica.** Time used: 0.006 (sec). Leaf size: 31

```
ode=x^3*D[y[x],{x,3}]+x*D[y[x],x]-y[x]==3*x^4;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^4}{9} + c_1 x + c_3 x \log^2(x) + c_2 x \log(x)$$

✓ **Sympy.** Time used: 0.167 (sec). Leaf size: 22

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-3*x**4 + x**3*Derivative(y(x), (x, 3)) + x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x \left(C_1 + C_2 \log(x) + C_3 \log(x)^2 + \frac{x^3}{9} \right)$$

2.7.7 Problem 16

Local contents

Solved as second order ode using change of variable on x method 2	1540
Solved as second order Bessel ode	1543
Solved as second order ode using Kovacic algorithm	1544
✓ Maple	1550
✓ Mathematica	1552
✓ Sympy	1552

Internal problem ID [7973]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 16

Date solved : Thursday, November 27, 2025 at 06:39:42 AM

CAS classification :

[[_Emden, _Fowler], [_2nd_order, _linear, '_with_symmetry_[0,F(x)]']]

Solved as second order ode using change of variable on x method 2

Time used: 0.228 (sec)

Solve

$$y''x - y' + 4yx^3 = 0$$

In normal form the ode

$$y''x - y' + 4yx^3 = 0 \tag{1}$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$\begin{aligned} p(x) &= -\frac{1}{x} \\ q(x) &= 4x^2 \end{aligned}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \tag{3}$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x) dx} dx \\ &= \int e^{-\int -\frac{1}{x} dx} dx \\ &= \int e^{\ln(x)} dx \\ &= \int x dx \\ &= \frac{x^2}{2} \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{4x^2}{x^2} \\ &= 4 \end{aligned} \quad (7)$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2} y(\tau) + q_1 y(\tau) &= 0 \\ \frac{d^2}{d\tau^2} y(\tau) + 4y(\tau) &= 0 \end{aligned}$$

The above ode is now solved for $y(\tau)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(\tau) + By'(\tau) + Cy(\tau) = 0$$

Where in the above $A = 1, B = 0, C = 4$. Let the solution be $y(\tau) = e^{\lambda\tau}$. Substituting this into the ODE gives

$$\lambda^2 e^{\tau\lambda} + 4e^{\tau\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda\tau}$ gives

$$\lambda^2 + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(4)} \\ &= \pm 2i \end{aligned}$$

Hence

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

Which simplifies to

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 2$. Therefore the final solution, when using Euler relation, can be written as

$$y(\tau) = e^{\alpha\tau} (c_1 \cos(\beta\tau) + c_2 \sin(\beta\tau))$$

Which becomes

$$y(\tau) = e^0 (c_1 \cos(2\tau) + c_2 \sin(2\tau))$$

Or

$$y(\tau) = c_1 \cos(2\tau) + c_2 \sin(2\tau)$$

The above solution is now transformed back to y using (6) which results in

$$y = c_1 \cos(x^2) + c_2 \sin(x^2)$$

Summary of solutions found

$$y = c_1 \cos(x^2) + c_2 \sin(x^2)$$

Solved as second order Bessel ode

Time used: 0.117 (sec)

Solve

$$y''x - y' + 4yx^3 = 0$$

Writing the ode as

$$x^2y'' - y'x + 4yx^4 = 0 \quad (1)$$

Bessel ode has the form

$$x^2y'' + y'x + (-n^2 + x^2)y = 0 \quad (2)$$

The generalized form of Bessel ode is given by Bowman (1958) as the following

$$x^2y'' + (1 - 2\alpha)xy' + (\beta^2\gamma^2x^{2\gamma} - n^2\gamma^2 + \alpha^2)y = 0 \quad (3)$$

With the standard solution

$$y = x^\alpha(c_3 \text{BesselJ}(n, \beta x^\gamma) + c_4 \text{BesselY}(n, \beta x^\gamma)) \quad (4)$$

Comparing (3) to (1) and solving for α, β, n, γ gives

$$\alpha = 1$$

$$\beta = 1$$

$$n = \frac{1}{2}$$

$$\gamma = 2$$

Substituting all the above into (4) gives the solution as

$$y = \frac{c_3x\sqrt{2} \sin(x^2)}{\sqrt{\pi} \sqrt{x^2}} - \frac{c_4x\sqrt{2} \cos(x^2)}{\sqrt{\pi} \sqrt{x^2}}$$

Summary of solutions found

$$y = \frac{c_3x\sqrt{2} \sin(x^2)}{\sqrt{\pi} \sqrt{x^2}} - \frac{c_4x\sqrt{2} \cos(x^2)}{\sqrt{\pi} \sqrt{x^2}}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.651 (sec)

Solve

$$y''x - y' + 4yx^3 = 0$$

Writing the ode as

$$y''x - y' + 4yx^3 = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= -1 \\ C &= 4x^3 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-16x^4 + 3}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -16x^4 + 3 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{-16x^4 + 3}{4x^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.176: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 4 \\ &= -2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is -2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Therefore

$$L = [1, 2]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -4x^2 + \frac{3}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r

given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $O_r(\infty) = -2$ then

$$v = \frac{-O_r(\infty)}{2} = \frac{2}{2} = 1$$

$[\sqrt{r}]_\infty$ is the sum of terms involving x^i for $0 \leq i \leq v$ in the Laurent series for $\sqrt{r}$ at ∞ . Therefore

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^v a_i x^i \\ &= \sum_{i=0}^1 a_i x^i \end{aligned} \tag{8}$$

Let a be the coefficient of $x^v = x^1$ in the above sum. The Laurent series of $\sqrt{r}$ at ∞ is

$$\begin{aligned} \sqrt{r} \approx 2ix &- \frac{3i}{16x^3} - \frac{9i}{1024x^7} - \frac{27i}{32768x^{11}} - \frac{405i}{4194304x^{15}} \\ &- \frac{1701i}{134217728x^{19}} - \frac{15309i}{8589934592x^{23}} - \frac{72171i}{274877906944x^{27}} + \dots \end{aligned} \tag{9}$$

Comparing Eq. (9) with Eq. (8) shows that

$$a = 2i$$

From Eq. (9) the sum up to $v = 1$ gives

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^1 a_i x^i \\ &= 2ix \end{aligned} \tag{10}$$

Now we need to find b , where b be the coefficient of $x^{v-1} = x^0 = 1$ in r minus the coefficient of same term but in $([\sqrt{r}]_\infty)^2$ where $[\sqrt{r}]_\infty$ was found above in Eq (10). Hence

$$([\sqrt{r}]_\infty)^2 = -4x^2$$

This shows that the coefficient of 1 in the above is 0. Now we need to find the coefficient of 1 in r . How this is done depends on if $v = 0$ or not. Since $v = 1$ which is not zero, then starting $r = \frac{s}{t}$, we do long division and write this in the form

$$r = Q + \frac{R}{t}$$

Where Q is the quotient and R is the remainder. Then the coefficient of 1 in r will be the coefficient this term in the quotient. Doing long division gives

$$\begin{aligned}
 r &= \frac{s}{t} \\
 &= \frac{-16x^4 + 3}{4x^2} \\
 &= Q + \frac{R}{4x^2} \\
 &= (-4x^2) + \left(\frac{3}{4x^2}\right) \\
 &= -4x^2 + \frac{3}{4x^2}
 \end{aligned}$$

We see that the coefficient of the term x in the quotient is 0. Now b can be found.

$$\begin{aligned}
 b &= (0) - (0) \\
 &= 0
 \end{aligned}$$

Hence

$$\begin{aligned}
 [\sqrt{r}]_{\infty} &= 2ix \\
 \alpha_{\infty}^{+} &= \frac{1}{2} \left(\frac{b}{a} - v \right) = \frac{1}{2} \left(\frac{0}{2i} - 1 \right) = -\frac{1}{2} \\
 \alpha_{\infty}^{-} &= \frac{1}{2} \left(-\frac{b}{a} - v \right) = \frac{1}{2} \left(-\frac{0}{2i} - 1 \right) = -\frac{1}{2}
 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{-16x^4 + 3}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^{+}	α_c^{-}
0	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_{\infty}$	α_{∞}^{+}	α_{∞}^{-}
-2	$2ix$	$-\frac{1}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^{\pm}$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_{\infty}$ and its associated $\alpha_{\infty}^{\pm}$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_{\infty}^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_{\infty}^{\pm}$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying

$\alpha_{\infty}^{-} = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_{\infty}^{-} - (\alpha_{c_1}^{-}) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_{\infty}$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^{-}}{x - c_1} \right) + (-)[\sqrt{r}]_{\infty} \\ &= -\frac{1}{2x} + (-)(2ix) \\ &= -\frac{1}{2x} - 2ix \\ &= -\frac{1}{2x} - 2ix \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(-\frac{1}{2x} - 2ix\right)(0) + \left(\left(\frac{1}{2x^2} - 2i\right) + \left(-\frac{1}{2x} - 2ix\right)^2 - \left(\frac{-16x^4 + 3}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2x} - 2ix\right) dx} \\ &= \frac{e^{-ix^2}}{\sqrt{x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-1}{x} dx} \\ &= z_1 e^{\frac{\ln(x)}{2}} \\ &= z_1 (\sqrt{x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-ix^2}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-1}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{ie^{2ix^2}}{4} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-ix^2}) + c_2 \left(e^{-ix^2} \left(-\frac{ie^{2ix^2}}{4} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^{-ix^2} - \frac{ic_2 e^{ix^2}}{4}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 17

```
ode:=x*diff(diff(y(x),x),x)-diff(y(x),x)+4*x^3*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 \sin(x^2) + c_2 \cos(x^2)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

<- linear_1 successful

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) - \frac{d}{dx} y(x) + 4x^3 y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -4x^2 y(x) + \frac{\frac{d}{dx} y(x)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{\frac{d}{dx} y(x)}{x} + 4x^2 y(x) = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- Define functions

$$[P_2(x) = -\frac{1}{x}, P_3(x) = 4x^2]$$

- $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -1$$

- $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$

- $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$x \left(\frac{d^2}{dx^2} y(x) \right) - \frac{d}{dx} y(x) + 4x^3 y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $x^3 \cdot y(x)$ to series expansion

$$x^3 \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+3}$$

- Shift index using $k \rightarrow k-3$

$$x^3 \cdot y(x) = \sum_{k=3}^{\infty} a_{k-3} x^{k+r}$$

- Convert $\frac{d}{dx}y(x)$ to series expansion

$$\frac{d}{dx}y(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}$$

- Shift index using $k \rightarrow k+1$

$$\frac{d}{dx}y(x) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) x^{k+r}$$

- Convert $x \cdot \left(\frac{d^2}{dx^2}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) x^{k+r-1}$$

- Shift index using $k \rightarrow k+1$

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) (k+r) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0 r(-2+r) x^{-1+r} + a_1 (1+r) (-1+r) x^r + a_2 (2+r) r x^{1+r} + a_3 (3+r) (1+r) x^{2+r} + \left(\sum_{k=3}^{\infty} (a_{k+1} (k+1+r) (k+r) - a_k (k+r) (k+r-1)) x^{k+r} \right)$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$r(-2+r) = 0$$

- Values of r that satisfy the indicial equation

$$r \in \{0, 2\}$$

- The coefficients of each power of x must be 0

$$[a_1 (1+r) (-1+r) = 0, a_2 (2+r) r = 0, a_3 (3+r) (1+r) = 0]$$

- Solve for the dependent coefficient(s)

$$\{a_1 = 0, a_2 = 0, a_3 = 0\}$$

- Each term in the series must be 0, giving the recursion relation

$$a_{k+1} (k+1+r) (k+r-1) + 4a_{k-3} = 0$$

- Shift index using $k \rightarrow k+3$

$$a_{k+4} (k+4+r) (k+2+r) + 4a_k = 0$$

- Recursion relation that defines series solution to ODE

$$a_{k+4} = -\frac{4a_k}{(k+4+r)(k+2+r)}$$

- Recursion relation for $r = 0$

$$a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}$$

- Solution for $r = 0$

- $$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}, a_1 = 0, a_2 = 0, a_3 = 0 \right]$$
- Recursion relation for $r = 2$

$$a_{k+4} = -\frac{4a_k}{(k+6)(k+4)}$$
- Solution for $r = 2$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+2}, a_{k+4} = -\frac{4a_k}{(k+6)(k+4)}, a_1 = 0, a_2 = 0, a_3 = 0 \right]$$
- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^k \right) + \left(\sum_{k=0}^{\infty} b_k x^{k+2} \right), a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}, a_1 = 0, a_2 = 0, a_3 = 0, b_{k+4} = -\frac{4b_k}{(k+6)(k+4)} \right]$$

✓ **Mathematica.** Time used: 0.012 (sec). Leaf size: 20

```
ode=x*D[y[x],{x,2}]-D[y[x],x]+4*x^3*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 \cos(x^2) + c_2 \sin(x^2)$$

✓ **Sympy.** Time used: 0.107 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x**3*y(x) + x*Derivative(y(x), (x, 2)) - Derivative(y(x), x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x \left(C_1 J_{\frac{1}{2}}(x^2) + C_2 Y_{\frac{1}{2}}(x^2) \right)$$

2.7.8 Problem 17

Local contents

Solved as second order missing x ode	1553
Solved as second order missing y ode	1556
✓ Maple	1559
✓ Mathematica	1560
✓ Sympy	1560

Internal problem ID [7974]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:39:49 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_xy]]

Solved as second order missing x ode

Time used: 6.958 (sec)

Solve

$$y'' + y'^2 + 1 = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$p(y) \left(\frac{d}{dy} p(y) \right) + p(y)^2 + 1 = 0$$

Which is now solved as first order ode for $p(y)$.

Solve Integrating gives

$$\int -\frac{p}{p^2 + 1} dp = dy$$

$$-\frac{\ln(p^2 + 1)}{2} = c_1 + y$$

Singular solutions are found by solving

$$-\frac{p^2 + 1}{p} = 0$$

for p . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p = -i$$

$$p = i$$

Solving for p gives

$$p = -i$$

$$p = i$$

$$p = \sqrt{-1 + e^{-2c_1 - 2y}}$$

$$p = -\sqrt{-1 + e^{-2c_1 - 2y}}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -i dx$$

$$y = -ix + c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int i dx$$

$$y = ix + c_3$$

For solution (3) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \sqrt{-1 + e^{-2c_1 - 2y}}$$

Solve Integrating gives

$$\int \frac{1}{\sqrt{-1 + e^{-2c_1 - 2y}}} dy = dx$$

$$- \arctan \left(\sqrt{-1 + e^{-2c_1 - 2y}} \right) = x + c_4$$

Singular solutions are found by solving

$$\sqrt{-1 + e^{-2c_1 - 2y}} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -c_1$$

Simplifying the above gives

$$- \arctan \left(\sqrt{-e^{-2c_1 - 2y} (e^{2c_1 + 2y} - 1)} \right) = x + c_4$$

$$y = -c_1$$

For solution (4) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\sqrt{-1 + e^{-2c_1 - 2y}}$$

Solve Integrating gives

$$\int -\frac{1}{\sqrt{-1 + e^{-2c_1 - 2y}}} dy = dx$$

$$\arctan \left(\sqrt{-1 + e^{-2c_1 - 2y}} \right) = x + c_5$$

Singular solutions are found by solving

$$-\sqrt{-1 + e^{-2c_1 - 2y}} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -c_1$$

Simplifying the above gives

$$\arctan \left(\sqrt{-e^{-2c_1-2y} (e^{2c_1+2y} - 1)} \right) = x + c_5$$

$$y = -c_1$$

Solving for y from the above solution(s) gives (after possible removing of solutions that do not verify)

$$y = -c_1$$

$$y = -c_1 - \frac{\ln (\tan (x + c_4)^2 + 1)}{2}$$

$$y = -c_1 - \frac{\ln (\tan (x + c_5)^2 + 1)}{2}$$

$$y = -ix + c_2$$

$$y = ix + c_3$$

The solution

$$y = -c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = -c_1 - \frac{\ln (\tan (x + c_4)^2 + 1)}{2}$$

$$y = -c_1 - \frac{\ln (\tan (x + c_5)^2 + 1)}{2}$$

$$y = -ix + c_2$$

$$y = ix + c_3$$

Solved as second order missing y ode

Time used: 0.588 (sec)

Solve

$$y'' + y'^2 + 1 = 0$$

This is second order ode with missing dependent variable y . Let

$$u(x) = y'$$

Then

$$u'(x) = y''$$

Hence the ode becomes

$$u'(x) + u(x)^2 + 1 = 0$$

Which is now solved for $u(x)$ as first order ode.

Solve Integrating gives

$$\begin{aligned} \int \frac{1}{-u^2 - 1} du &= dx \\ -\arctan(u) &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-u^2 - 1 = 0$$

for $u(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \end{aligned}$$

Solving for $u(x)$ gives

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \\ u(x) &= -\tan(x + c_1) \end{aligned}$$

In summary, these are the solution found for y

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \\ u(x) &= -\tan(x + c_1) \end{aligned}$$

For solution $u(x) = -i$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = -i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dy &= \int -i dx \\ y &= -ix + c_2 \end{aligned}$$

In summary, these are the solution found for (y)

$$y = -ix + c_2$$

For solution $u(x) = i$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int i dx$$

$$y = ix + c_3$$

In summary, these are the solution found for (y)

$$y = ix + c_3$$

For solution $u(x) = -\tan(x + c_1)$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = -\tan(x + c_1)$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -\tan(x + c_1) dx$$

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

In summary, these are the solution found for (y)

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

Summary of solutions found

$$y = -ix + c_2$$

$$y = ix + c_3$$

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

✓ **Maple.** Time used: 0.011 (sec). Leaf size: 15

```
ode:=diff(diff(y(x),x),x)+diff(y(x),x)^2+1 = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \ln(-c_1 \sin(x) + c_2 \cos(x))$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying 2nd order Liouville

trying 2nd order WeierstrassP

trying 2nd order JacobiSN

differential order: 2; trying a linearization to 3rd order

trying 2nd order ODE linearizable_by_differentiation

trying 2nd order, 2 integrating factors of the form $\mu(x,y)$

trying a quadrature

checking if the LODE has constant coefficients

<- constant coefficients successful

<- 2nd order, 2 integrating factors of the form $\mu(x,y)$ successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + \left(\frac{d}{dx}y(x)\right)^2 + 1 = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Make substitution $u = \frac{d}{dx}y(x)$ to reduce order of ODE

$$\frac{d}{dx}u(x) + u(x)^2 + 1 = 0$$

- Solve for the highest derivative

$$\frac{d}{dx}u(x) = -u(x)^2 - 1$$

- Separate variables

$$\frac{\frac{d}{dx}u(x)}{-u(x)^2-1} = 1$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}u(x)}{-u(x)^2-1} dx = \int 1 dx + C1$$

- Evaluate integral

$$-\arctan(u(x)) = x + C1$$

- Solve for $u(x)$

$$u(x) = -\tan(x + C1)$$

- Solve 1st ODE for $u(x)$

- $u(x) = -\tan(x + C1)$
- Make substitution $u = \frac{d}{dx}y(x)$
 $\frac{d}{dx}y(x) = -\tan(x + C1)$
- Integrate both sides to solve for $y(x)$
 $\int \left(\frac{d}{dx}y(x)\right) dx = \int -\tan(x + C1) dx + C2$
- Compute integrals
 $y(x) = -\frac{\ln(1+\tan(x+C1)^2)}{2} + C2$

✓ **Mathematica.** Time used: 0.423 (sec). Leaf size: 40

```
ode=D[y[x],{x,2}]+D[y[x],x]^2+1==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x \text{InverseFunction} \left[\int_1^{\#1} \frac{1}{K[1]^2 + 1} dK[1] \& \right] [c_1 - K[2]] dK[2] + c_2$$

✓ **Sympy.** Time used: 0.727 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(Derivative(y(x), x)**2 + Derivative(y(x), (x, 2)) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = C_1 - \frac{\log(\tan^2(C_2 - x) + 1)}{2}, \quad y(x) = C_1 - \frac{\log(\tan^2(C_2 - x) + 1)}{2} \right]$$

2.7.9 Problem 18

Local contents

Solved as second order nonlinear exact ode	1561
Solved as second order missing x ode	1563
Solved as second order integrable as is ode	1568
Solved as second order integrable as is ode (ABC method)	1569
✓ Maple	1571
✓ Mathematica	1573
✗ Sympy	1574

Internal problem ID [7975]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:40:02 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _exact, _nonlinear], [_2nd_order, _reducible, _m

Solved as second order nonlinear exact ode

Time used: 0.215 (sec)

Solve

$$yy'' + y'^2 = 2$$

An exact non-linear second order ode has the form

$$a_2(x, y, y') y'' + a_1(x, y, y') y' + a_0(x, y, y') = 0$$

Where the following conditions are satisfied

$$\begin{aligned}\frac{\partial a_2}{\partial y} &= \frac{\partial a_1}{\partial y'} \\ \frac{\partial a_2}{\partial x} &= \frac{\partial a_0}{\partial y'} \\ \frac{\partial a_1}{\partial x} &= \frac{\partial a_0}{\partial y}\end{aligned}$$

Looking at the the ode given we see that

$$\begin{aligned}a_2 &= y \\ a_1 &= y' \\ a_0 &= -2\end{aligned}$$

Applying the conditions to the above shows this is a nonlinear exact second order ode. Therefore it can be reduced to first order ode given by

$$\int a_2 dy' + \int a_1 dy + \int a_0 dx = c_1$$

$$\int y dy' + \int y' dy + \int -2 dx = c_1$$

Which results in

$$yy' - 2x = c_1$$

Which is now solved.

Solve The ode

$$y' = \frac{c_1 + 2x}{y} \quad (2.115)$$

is separable as it can be written as

$$y' = \frac{c_1 + 2x}{y}$$

$$= f(x)g(y)$$

Where

$$f(x) = c_1 + 2x$$

$$g(y) = \frac{1}{y}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int y dy = \int c_1 + 2x dx$$

$$\frac{y^2}{2} = x(x + c_1) + c_2$$

Simplifying the above gives

$$\frac{y^2}{2} = c_1x + x^2 + c_2$$

Solving for y gives

$$y = \sqrt{2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1x + 2x^2 + 2c_2}$$

Summary of solutions found

$$y = \sqrt{2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1x + 2x^2 + 2c_2}$$

Solved as second order missing x ode

Time used: 2.225 (sec)

Solve

$$yy'' + y'^2 = 2$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$yp(y) \left(\frac{d}{dy} p(y) \right) + p(y)^2 = 2$$

Which is now solved as first order ode for $p(y)$.

Solve The ode

$$p' = -\frac{p^2 - 2}{yp} \tag{2.116}$$

is separable as it can be written as

$$\begin{aligned} p' &= -\frac{p^2 - 2}{yp} \\ &= f(y)g(p) \end{aligned}$$

Where

$$f(y) = -\frac{1}{y}$$

$$g(p) = \frac{p^2 - 2}{p}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(y) dy$$

$$\int \frac{p}{p^2 - 2} dp = \int -\frac{1}{y} dy$$

$$\frac{\ln(p^2 - 2)}{2} = \ln\left(\frac{1}{y}\right) + c_1$$

Taking the exponential of both sides the solution becomes

$$\sqrt{p^2 - 2} = \frac{c_1}{y}$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$\frac{p^2 - 2}{p} = 0$$

for p gives

$$p = \sqrt{2}$$

$$p = -\sqrt{2}$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$\sqrt{p^2 - 2} = \frac{c_1}{y}$$

$$p = \sqrt{2}$$

$$p = -\sqrt{2}$$

Solving for p gives

$$p = \sqrt{2}$$

$$p = \frac{\sqrt{c_1^2 + 2y^2}}{y}$$

$$p = -\sqrt{2}$$

$$p = -\frac{\sqrt{c_1^2 + 2y^2}}{y}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \sqrt{2}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \sqrt{2} dx$$

$$y = \sqrt{2}x + c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{\sqrt{c_1^2 + 2y^2}}{y}$$

Solve Integrating gives

$$\int \frac{y}{\sqrt{c_1^2 + 2y^2}} dy = dx$$

$$\frac{\sqrt{c_1^2 + 2y^2}}{2} = x + c_3$$

Singular solutions are found by solving

$$\frac{\sqrt{c_1^2 + 2y^2}}{y} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -\frac{i\sqrt{2}c_1}{2}$$

$$y = \frac{i\sqrt{2}c_1}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}c_1}{2}$$

$$y = \frac{i\sqrt{2}c_1}{2}$$

$$y = -\frac{\sqrt{-2c_1^2 + 8c_3^2 + 16c_3x + 8x^2}}{2}$$

$$y = \frac{\sqrt{-2c_1^2 + 8c_3^2 + 16c_3x + 8x^2}}{2}$$

For solution (3) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\sqrt{2}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -\sqrt{2} dx$$

$$y = -\sqrt{2}x + c_4$$

For solution (4) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\frac{\sqrt{c_1^2 + 2y^2}}{y}$$

Solve Integrating gives

$$\int -\frac{y}{\sqrt{c_1^2 + 2y^2}} dy = dx$$

$$-\frac{\sqrt{c_1^2 + 2y^2}}{2} = x + c_5$$

Singular solutions are found by solving

$$-\frac{\sqrt{c_1^2 + 2y^2}}{y} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -\frac{i\sqrt{2}c_1}{2}$$

$$y = \frac{i\sqrt{2}c_1}{2}$$

Solving for y gives

$$y = -\frac{i\sqrt{2}c_1}{2}$$

$$y = \frac{i\sqrt{2}c_1}{2}$$

$$y = -\frac{\sqrt{-2c_1^2 + 8c_5^2 + 16c_5x + 8x^2}}{2}$$

$$y = \frac{\sqrt{-2c_1^2 + 8c_5^2 + 16c_5x + 8x^2}}{2}$$

The solution

$$y = -\frac{i\sqrt{2}c_1}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = \frac{i\sqrt{2}c_1}{2}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = -\frac{\sqrt{-2c_1^2 + 8c_3^2 + 16c_3x + 8x^2}}{2}$$

$$y = \frac{\sqrt{-2c_1^2 + 8c_3^2 + 16c_3x + 8x^2}}{2}$$

$$y = -\frac{\sqrt{-2c_1^2 + 8c_5^2 + 16c_5x + 8x^2}}{2}$$

$$y = \frac{\sqrt{-2c_1^2 + 8c_5^2 + 16c_5x + 8x^2}}{2}$$

$$y = -\sqrt{2}x + c_4$$

$$y = \sqrt{2}x + c_2$$

Solved as second order integrable as is ode

Time used: 0.119 (sec)

Solve

$$yy'' + y'^2 = 2$$

Integrating both sides of the ODE w.r.t x gives

$$\int (yy'' + y'^2) dx = \int 2dx$$
$$yy' = 2x + c_1$$

Which is now solved for y . Solve The ode

$$y' = \frac{c_1 + 2x}{y} \tag{2.117}$$

is separable as it can be written as

$$y' = \frac{c_1 + 2x}{y}$$
$$= f(x)g(y)$$

Where

$$f(x) = c_1 + 2x$$
$$g(y) = \frac{1}{y}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$
$$\int y dy = \int c_1 + 2x dx$$

$$\frac{y^2}{2} = x(x + c_1) + c_2$$

Simplifying the above gives

$$\frac{y^2}{2} = c_1x + x^2 + c_2$$

Solving for y gives

$$y = \sqrt{2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1x + 2x^2 + 2c_2}$$

Summary of solutions found

$$y = \sqrt{2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1x + 2x^2 + 2c_2}$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.111 (sec)

Solve

$$yy'' + y'^2 = 2$$

Writing the ode as

$$yy'' + y'^2 = 2$$

Integrating both sides of the ODE w.r.t x gives

$$\int (yy'' + y'^2) dx = \int 2dx$$

$$yy' = 2x + c_1$$

Which is now solved for y . Solve The ode

$$y' = \frac{c_1 + 2x}{y} \tag{2.118}$$

is separable as it can be written as

$$y' = \frac{c_1 + 2x}{y}$$

$$= f(x)g(y)$$

Where

$$f(x) = c_1 + 2x$$

$$g(y) = \frac{1}{y}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int y dy = \int c_1 + 2x dx$$

$$\frac{y^2}{2} = x(x + c_1) + c_2$$

Simplifying the above gives

$$\frac{y^2}{2} = c_1 x + x^2 + c_2$$

Solving for y gives

$$y = \sqrt{2c_1 x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1 x + 2x^2 + 2c_2}$$

Solve The ode

$$y' = \frac{c_1 + 2x}{y} \tag{2.119}$$

is separable as it can be written as

$$y' = \frac{c_1 + 2x}{y}$$

$$= f(x)g(y)$$

Where

$$f(x) = c_1 + 2x$$

$$g(y) = \frac{1}{y}$$

Integrating gives

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$\int y dy = \int c_1 + 2x dx$$

$$\frac{y^2}{2} = x(x + c_1) + c_2$$

Simplifying the above gives

$$\frac{y^2}{2} = c_1x + x^2 + c_2$$

Solving for y gives

$$y = \sqrt{2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{2c_1x + 2x^2 + 2c_2}$$

✓ **Maple.** Time used: 0.014 (sec). Leaf size: 39

```
ode:=diff(y(x),x)^2+y(x)*diff(diff(y(x),x),x) = 2;
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{-2c_1x + 2x^2 + 2c_2}$$

$$y = -\sqrt{-2c_1x + 2x^2 + 2c_2}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying 2nd order Liouville

trying 2nd order WeierstrassP

trying 2nd order JacobiSN

differential order: 2; trying a linearization to 3rd order

trying 2nd order ODE linearizable_by_differentiation

trying 2nd order, 2 integrating factors of the form $\mu(x,y)$

trying a quadrature

<- quadrature successful

<- 2nd order, 2 integrating factors of the form $\mu(x,y)$ successful

Maple step by step

Let's solve

$$y(x) \left(\frac{d^2}{dx^2} y(x) \right) + \left(\frac{d}{dx} y(x) \right)^2 = 2$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Define new dependent variable u

$$u(x) = \frac{d}{dx} y(x)$$

- Compute $\frac{d^2}{dx^2} y(x)$

$$\frac{d}{dx} u(x) = \frac{d^2}{dx^2} y(x)$$

- Use chain rule on the lhs

$$\left(\frac{d}{dx}y(x)\right)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Substitute in the definition of u

$$u(y)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Make substitutions $\frac{d}{dx}y(x) = u(y)$, $\frac{d^2}{dx^2}y(x) = u(y)\left(\frac{d}{dy}u(y)\right)$ to reduce order of ODE

$$yu(y)\left(\frac{d}{dy}u(y)\right) + u(y)^2 = 2$$
- Solve for the highest derivative

$$\frac{d}{dy}u(y) = \frac{-u(y)^2+2}{yu(y)}$$
- Separate variables

$$\frac{\left(\frac{d}{dy}u(y)\right)u(y)}{-u(y)^2+2} = \frac{1}{y}$$
- Integrate both sides with respect to y

$$\int \frac{\left(\frac{d}{dy}u(y)\right)u(y)}{-u(y)^2+2} dy = \int \frac{1}{y} dy + C1$$
- Evaluate integral

$$-\frac{\ln(u(y)^2-2)}{2} = \ln(y) + C1$$
- Solve for $u(y)$

$$\left\{u(y) = \frac{\sqrt{2y^2(e^{C1})^2+1}}{e^{C1}y}, u(y) = -\frac{\sqrt{2y^2(e^{C1})^2+1}}{e^{C1}y}\right\}$$
- Redefine the integration constant(s)

$$\left\{u(y) = \frac{\sqrt{2C1^2y^2+1}}{C1y}, u(y) = -\frac{\sqrt{2C1^2y^2+1}}{C1y}\right\}$$
- Solve 1st ODE for $u(y)$

$$u(y) = \frac{\sqrt{2C1^2y^2+1}}{C1y}$$
- Revert to original variables with substitution $u(y) = \frac{d}{dx}y(x)$, $y = y(x)$

$$\frac{d}{dx}y(x) = \frac{\sqrt{2y(x)^2C1^2+1}}{C1y(x)}$$
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{\sqrt{2y(x)^2C1^2+1}}{C1y(x)}$$
- Separate variables

$$\frac{\left(\frac{d}{dx}y(x)\right)y(x)}{\sqrt{2y(x)^2C1^2+1}} = \frac{1}{C1}$$
- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx}y(x)\right)y(x)}{\sqrt{2y(x)^2C1^2+1}} dx = \int \frac{1}{C1} dx + C2$$
- Evaluate integral

$$\frac{\sqrt{2y(x)^2C1^2+1}}{2C1^2} = \frac{x}{C1} + C2$$
- Solve for $y(x)$

$$\left\{y(x) = -\frac{\sqrt{8C1^4C2^2+16C1^3C2x+8C1^2x^2-2}}{2C1}, y(x) = \frac{\sqrt{8C1^4C2^2+16C1^3C2x+8C1^2x^2-2}}{2C1}\right\}$$
- Solve 2nd ODE for $u(y)$

- $$u(y) = -\frac{\sqrt{2C1^2y^2+1}}{C1y}$$
- Revert to original variables with substitution $u(y) = \frac{d}{dx}y(x), y = y(x)$

$$\frac{d}{dx}y(x) = -\frac{\sqrt{2y(x)^2C1^2+1}}{C1y(x)}$$
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = -\frac{\sqrt{2y(x)^2C1^2+1}}{C1y(x)}$$
- Separate variables

$$\frac{\left(\frac{d}{dx}y(x)\right)y(x)}{\sqrt{2y(x)^2C1^2+1}} = -\frac{1}{C1}$$
- Integrate both sides with respect to x

$$\int \frac{\left(\frac{d}{dx}y(x)\right)y(x)}{\sqrt{2y(x)^2C1^2+1}} dx = \int -\frac{1}{C1} dx + C2$$
- Evaluate integral

$$\frac{\sqrt{2y(x)^2C1^2+1}}{2C1^2} = -\frac{x}{C1} + C2$$
- Solve for $y(x)$

$$\left\{ y(x) = -\frac{\sqrt{8C1^4C2^2-16C1^3C2x+8C1^2x^2-2}}{2C1}, y(x) = \frac{\sqrt{8C1^4C2^2-16C1^3C2x+8C1^2x^2-2}}{2C1} \right\}$$

✓ **Mathematica.** Time used: 6.211 (sec). Leaf size: 101

```
ode=y[x]*D[y[x],{x,2}]+D[y[x],x]^2==2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{\sqrt{4(x+c_2)^2-e^{2c_1}}}{\sqrt{2}}$$

$$y(x) \rightarrow \sqrt{2(x+c_2)^2-\frac{e^{2c_1}}{2}}$$

$$y(x) \rightarrow -\sqrt{2}\sqrt{(x+c_2)^2}$$

$$y(x) \rightarrow \sqrt{2}\sqrt{(x+c_2)^2}$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x)*Derivative(y(x), (x, 2)) + Derivative(y(x), x)**2 - 2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE -sqrt(-y(x)*Derivative(y(x), (x, 2)) + 2) +
Derivative(y(x), x) cannot be solved by the factorable group method
```

2.7.10 Problem 19

Local contents

Solved as second order missing x ode	1575
Solved by factoring the differential equation	1576
✓ Maple	1579
✓ Mathematica	1581
✗ Sympy	1581

Internal problem ID [7976]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 12. Linear equations of order n. Supplementary problems. Page 81

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:40:09 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_x_y1], [_2nd_order, _reducible,

Solved as second order missing x ode

Time used: 0.773 (sec)

Solve

$$y'^3 + yy'' = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$p(y)^3 + yp(y) \left(\frac{d}{dy} p(y) \right) = 0$$

Which is now solved as first order ode for $p(y)$.

Solved by factoring the differential equation

Time used: 0.130 (sec)

Solve

$$p^3 + ypp' = 0$$

Writing the ode as

$$(p) (p'y + p^2) = 0$$

Therefore we need to solve the following equations

$$p = 0 \tag{1}$$

$$p'y + p^2 = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)Solving for p from

$$p = 0$$

Solving gives

$$p = 0$$

Solving equation (2)

Solve The ode

$$p' = -\frac{p^2}{y} \tag{2.120}$$

is separable as it can be written as

$$\begin{aligned} p' &= -\frac{p^2}{y} \\ &= f(y)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(y) &= -\frac{1}{y} \\ g(p) &= p^2 \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(y) dy$$

$$\int \frac{1}{p^2} dp = \int -\frac{1}{y} dy$$

$$-\frac{1}{p} = \ln \left(\frac{1}{y} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p^2 = 0$$

for p gives

$$p = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\frac{1}{p} = \ln \left(\frac{1}{y} \right) + c_1$$

$$p = 0$$

Solving for p gives

$$p = 0$$

$$p = -\frac{1}{\ln \left(\frac{1}{y} \right) + c_1}$$

Summary of solutions found

$$p = 0$$

$$p = -\frac{1}{\ln \left(\frac{1}{y} \right) + c_1}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = 0$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int 0 dx + c_2$$

$$y = c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\frac{1}{\ln\left(\frac{1}{y}\right) + c_1}$$

Solve Integrating gives

$$\int \left(-\ln\left(\frac{1}{y}\right) - c_1 \right) dy = dx$$

$$-c_1 y - \ln\left(\frac{1}{y}\right) y - y = x + c_3$$

Simplifying the above gives

$$-y \left(\ln\left(\frac{1}{y}\right) + c_1 + 1 \right) = x + c_3$$

Solving for y from the above solution(s) gives (after possible removing of solutions that do not verify)

$$y = c_2$$

$$y = \frac{x + c_3}{\text{LambertW}((x + c_3) e^{-c_1 - 1})}$$

Summary of solutions found

$$y = c_2$$

$$y = \frac{x + c_3}{\text{LambertW}((x + c_3) e^{-c_1 - 1})}$$

✓ **Maple.** Time used: 0.016 (sec). Leaf size: 27

```
ode:=diff(y(x),x)^3+y(x)*diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = c_1$$

$$y = \frac{c_2 + x}{\text{LambertW}((c_2 + x)e^{c_1-1})}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying 2nd order Liouville

trying 2nd order WeierstrassP

trying 2nd order JacobiSN

differential order: 2; trying a linearization to 3rd order

trying 2nd order ODE linearizable_by_differentiation

trying 2nd order, 2 integrating factors of the form $\mu(x,y)$

trying differential order: 2; missing variables

-> Calling odsolve with the ODE, $\text{diff}(_b(_a),_a)*_b(_a)+_b(_a)^3/_a = 0$, $_b(_a)$

*** Sublevel 2 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

trying Bernoulli

<- Bernoulli successful

<- differential order: 2; canonical coordinates successful

<- differential order 2; missing variables successful

Maple step by step

Let's solve

$$y(x) \left(\frac{d^2}{dx^2} y(x) \right) + \left(\frac{d}{dx} y(x) \right)^3 = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Define new dependent variable u

$$u(x) = \frac{d}{dx} y(x)$$

- Compute $\frac{d^2}{dx^2} y(x)$

$$\frac{d}{dx} u(x) = \frac{d^2}{dx^2} y(x)$$

- Use chain rule on the lhs

$$\left(\frac{d}{dx}y(x)\right)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Substitute in the definition of u

$$u(y)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Make substitutions $\frac{d}{dx}y(x) = u(y)$, $\frac{d^2}{dx^2}y(x) = u(y)\left(\frac{d}{dy}u(y)\right)$ to reduce order of ODE

$$yu(y)\left(\frac{d}{dy}u(y)\right) + u(y)^3 = 0$$
- Solve for the highest derivative

$$\frac{d}{dy}u(y) = -\frac{u(y)^2}{y}$$
- Separate variables

$$\frac{\frac{d}{dy}u(y)}{u(y)^2} = -\frac{1}{y}$$
- Integrate both sides with respect to y

$$\int \frac{\frac{d}{dy}u(y)}{u(y)^2} dy = \int -\frac{1}{y} dy + C1$$
- Evaluate integral

$$-\frac{1}{u(y)} = -\ln(y) + C1$$
- Solve for $u(y)$

$$u(y) = \frac{1}{\ln(y) - C1}$$
- Redefine the integration constant(s)

$$u(y) = \frac{1}{\ln(y) + C1}$$
- Solve 1st ODE for $u(y)$

$$u(y) = \frac{1}{\ln(y) + C1}$$
- Revert to original variables with substitution $u(y) = \frac{d}{dx}y(x)$, $y = y(x)$

$$\frac{d}{dx}y(x) = \frac{1}{\ln(y(x)) + C1}$$
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{1}{\ln(y(x)) + C1}$$
- Separate variables

$$\left(\frac{d}{dx}y(x)\right)(\ln(y(x)) + C1) = 1$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}y(x)\right)(\ln(y(x)) + C1) dx = \int 1 dx + C2$$
- Evaluate integral

$$y(x) C1 + y(x) \ln(y(x)) - y(x) = x + C2$$
- Solve for $y(x)$

$$y(x) = e^{\text{Lambert } W((x+C2)e^{C1-1}) - C1 + 1}$$
- Simplify

$$y(x) = \frac{x+C2}{\text{Lambert } W((x+C2)e^{C1-1})}$$
- Redefine the integration constant(s)

$$y(x) = \frac{x+C2}{\text{Lambert } W((x+C2)C1)}$$

✓ **Mathematica.** Time used: 60.065 (sec). Leaf size: 26

```
ode=y[x]*D[y[x],{x,2}]+D[y[x],x]^3==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x + c_2}{W(e^{-1-c_1}(x + c_2))}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x)*Derivative(y(x), (x, 2)) + Derivative(y(x), x)**3,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE (-y(x)*Derivative(y(x), (x, 2))**(1/3)/2 -
sqrt(3)*I*(-y(x)*Derivative(y(x), (x, 2))**(1/3)/2 + Derivative(y(x), x)
cannot be solved by the factorable group method
```

2.8 Chapter 13. Homogeneous Linear equations with constant coefficients. Supplemetary problems.

Page 86

Local contents

2.8.1	Problem 16	1583
2.8.2	Problem 17	1590
2.8.3	Problem 18	1593
2.8.4	Problem 19	1601
2.8.5	Problem 20	1604
2.8.6	Problem 21	1611
2.8.7	Problem 22	1623
2.8.8	Problem 23	1626
2.8.9	Problem 24	1629
2.8.10	Problem 25	1632

2.8.1 Problem 16

Local contents

Solved as second order linear constant coeff ode	1583
Solved as second order ode using Kovacic algorithm	1585
✓ Maple	1588
✓ Mathematica	1589
✓ Sympy	1589

Internal problem ID [7977]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 16

Date solved : Thursday, November 27, 2025 at 06:40:15 AM

CAS classification : `[_2nd_order, _missing_x]`

Solved as second order linear constant coeff ode

Time used: 0.056 (sec)

Solve

$$y'' + 2y' - 15y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 2, C = -15$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 2\lambda e^{x\lambda} - 15e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 2\lambda - 15 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 2, C = -15$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-2}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{2^2 - (4)(1)(-15)} \\ &= -1 \pm 4 \end{aligned}$$

Hence

$$\lambda_1 = -1 + 4$$

$$\lambda_2 = -1 - 4$$

Which simplifies to

$$\lambda_1 = 3$$

$$\lambda_2 = -5$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(3)x} + c_2 e^{(-5)x}$$

Or

$$y = e^{3x} c_1 + c_2 e^{-5x}$$

Summary of solutions found

$$y = e^{3x} c_1 + c_2 e^{-5x}$$

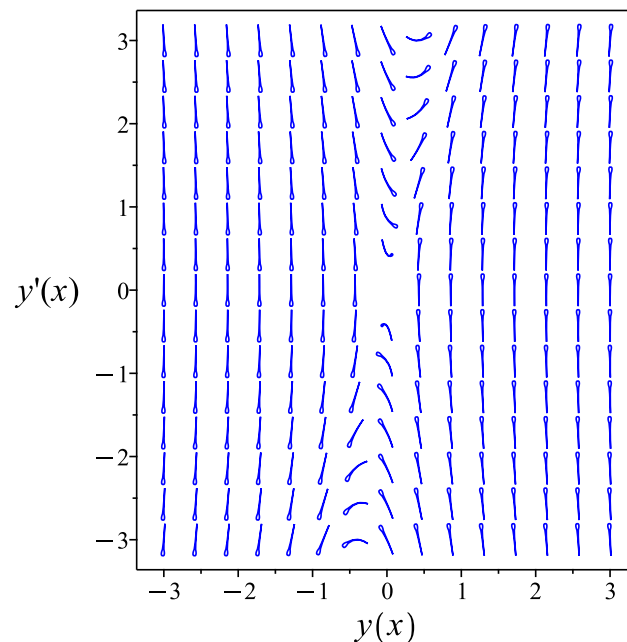


Figure 2.311: Slope field $y'' + 2y' - 15y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.062 (sec)

Solve

$$y'' + 2y' - 15y = 0$$

Writing the ode as

$$y'' + 2y' - 15y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 2 \\ C &= -15 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{16}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 16 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 16z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.181: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 16$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-4x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{2}{1} dx} \\ &= z_1 e^{-x} \\ &= z_1 (e^{-x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-5x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

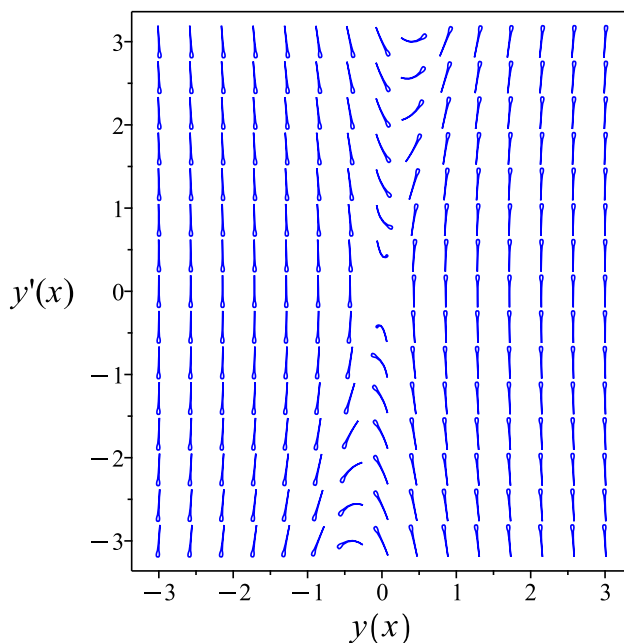
$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{2}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-2x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{-2x} e^{10x}}{8} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-5x}) + c_2 \left(e^{-5x} \left(\frac{e^{-2x} e^{10x}}{8} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^{-5x} + \frac{e^{3x} c_2}{8}$$

Figure 2.312: Slope field $y'' + 2y' - 15y = 0$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 17

```
ode:=diff(diff(y(x),x),x)+2*diff(y(x),x)-15*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (c_1 e^{8x} + c_2) e^{-5x}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 2\frac{d}{dx}y(x) - 15y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 + 2r - 15 = 0$
- Factor the characteristic polynomial

- $(r + 5)(r - 3) = 0$
 - Roots of the characteristic polynomial
 $r = (-5, 3)$
 - 1st solution of the ODE
 $y_1(x) = e^{-5x}$
 - 2nd solution of the ODE
 $y_2(x) = e^{3x}$
 - General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x)$
 - Substitute in solutions
 $y(x) = C_1 e^{-5x} + C_2 e^{3x}$

✓ **Mathematica.** Time used: 0.01 (sec). Leaf size: 22

```
ode=D[y[x],{x,2}]+2*D[y[x],x]-15*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-5x}(c_2 e^{8x} + c_1)$$

✓ **Sympy.** Time used: 0.090 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-15*y(x) + 2*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-5x} + C_2 e^{3x}$$

2.8.2 Problem 17

Local contents

Solved as higher order constant coeff ode	1590
✓ Maple	1591
✓ Mathematica	1592
✓ Sympy	1592

Internal problem ID [7978]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:40:20 AM

CAS classification : `[_3rd_order, _missing_x]`

Solve

$$y''' + y'' - 2y' = 0$$

Solved as higher order constant coeff ode

Time used: 0.027 (sec)

The characteristic equation is

$$\lambda^3 + \lambda^2 - 2\lambda = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = 1$$

$$\lambda_3 = -2$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 + c_2 e^x + e^{-2x} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = e^x$$

$$y_3 = e^{-2x}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 16

```
ode:=diff(diff(diff(y(x),x),x),x)+diff(diff(y(x),x),x)-2*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 + c_2 e^x + c_3 e^{-2x}$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

<- constant coefficients successful

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) + \frac{d^2}{dx^2}y(x) - 2\frac{d}{dx}y(x) = 0$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of ODE

$$r^3 + r^2 - 2r = 0$$

- Roots of the characteristic polynomial

$$r = [0, 1, -2]$$

- Solution from $r = 0$

$$y_1(x) = 1$$

- Solution from $r = 1$

$$y_2(x) = e^x$$

- Solution from $r = -2$

$$y_3(x) = e^{-2x}$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + C3y_3(x)$$

- Substitute in solutions

$$y(x) = C1 + C2e^x + C3e^{-2x}$$

✓ **Mathematica.** Time used: 0.016 (sec). Leaf size: 25

```
ode=D[y[x],{x,3}]+D[y[x],{x,2}]-2*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{1}{2}c_1e^{-2x} + c_2e^x + c_3$$

✓ **Sympy.** Time used: 0.090 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*Derivative(y(x), x) + Derivative(y(x), (x, 2)) + Derivative(y(x), (x,
    3)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 + C_2e^{-2x} + C_3e^x$$

2.8.3 Problem 18

Local contents

Solved as second order linear constant coeff ode	1593
Solved as second order solved by an integrating factor	1594
Solved as second order ode using Kovacic algorithm	1596
✓ Maple	1599
✓ Mathematica	1600
✓ Sympy	1600

Internal problem ID [7979]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:40:25 AM

CAS classification : `[[_2nd_order, _missing_x]]`

Solved as second order linear constant coeff ode

Time used: 0.050 (sec)

Solve

$$y'' + 6y' + 9y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 6, C = 9$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 6\lambda e^{x\lambda} + 9e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 6\lambda + 9 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 6, C = 9$ into the above gives

$$\begin{aligned}\lambda_{1,2} &= \frac{-6}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{(6)^2 - (4)(1)(9)} \\ &= -3\end{aligned}$$

Hence this is the case of a double root $\lambda_{1,2} = 3$. Therefore the solution is

$$y = c_1 e^{-3x} + c_2 x e^{-3x} \quad (1)$$

Summary of solutions found

$$y = e^{-3x} c_1 + c_2 x e^{-3x}$$

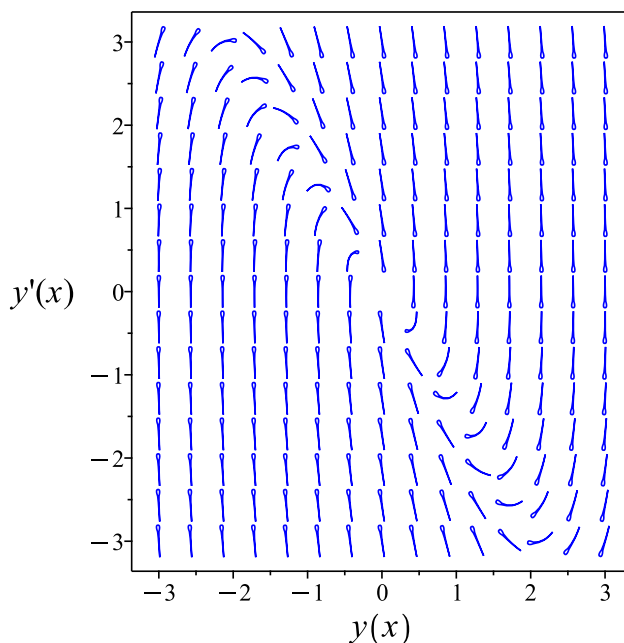


Figure 2.313: Slope field $y'' + 6y' + 9y = 0$

Solved as second order solved by an integrating factor

Time used: 0.072 (sec)

Solve

$$y'' + 6y' + 9y = 0$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x))^2 + p'(x)}{2}y = f(x)$$

Where $p(x) = 6$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int 6 \, dx} \\ &= e^{3x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$\begin{aligned} (M(x)y)'' &= 0 \\ (e^{3x}y)'' &= 0 \end{aligned}$$

Integrating once gives

$$(e^{3x}y)' = c_1$$

Integrating again gives

$$(e^{3x}y) = c_1x + c_2$$

Hence the solution is

$$y = \frac{c_1x + c_2}{e^{3x}}$$

Or

$$y = c_1x e^{-3x} + c_2 e^{-3x}$$

Summary of solutions found

$$y = c_1x e^{-3x} + c_2 e^{-3x}$$

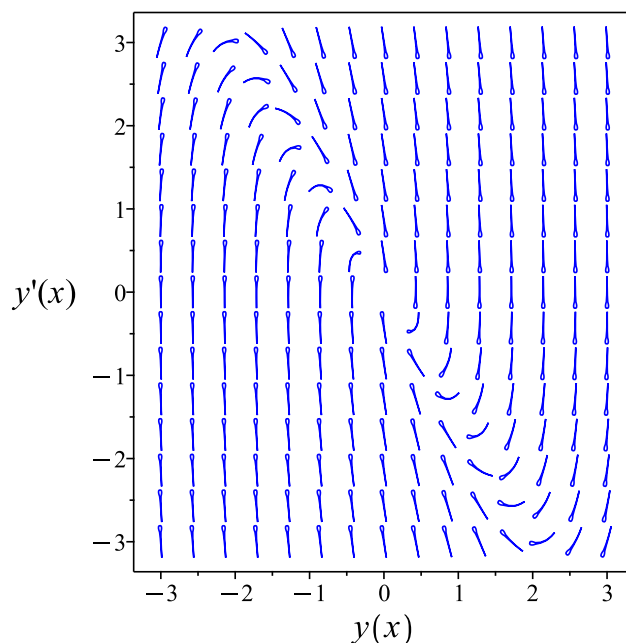


Figure 2.314: Slope field $y'' + 6y' + 9y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.055 (sec)

Solve

$$y'' + 6y' + 9y = 0$$

Writing the ode as

$$y'' + 6y' + 9y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 6 \\ C &= 9 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 0 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.184: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{6}{1} dx} \\ &= z_1 e^{-3x} \\ &= z_1 (e^{-3x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{6}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-6x}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-3x}) + c_2 (e^{-3x}(x)) \end{aligned}$$

Summary of solutions found

$$y = e^{-3x} c_1 + c_2 x e^{-3x}$$

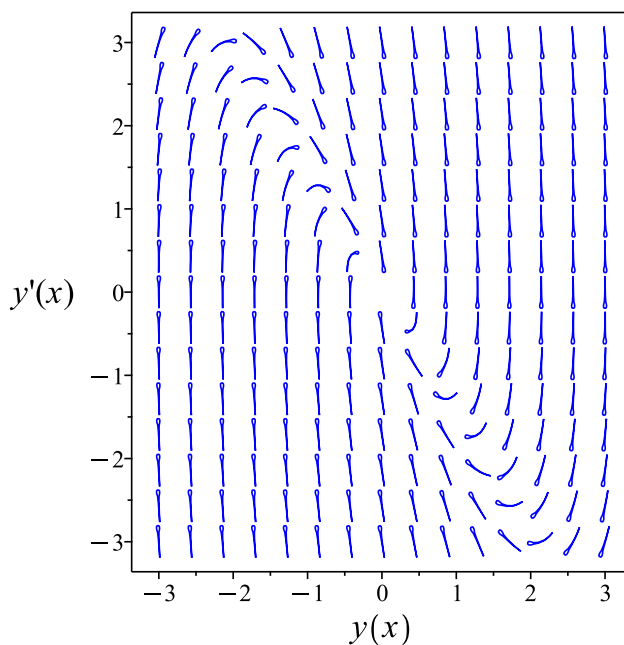


Figure 2.315: Slope field $y'' + 6y' + 9y = 0$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 14

```
ode:=diff(diff(y(x),x),x)+6*diff(y(x),x)+9*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-3x}(c_2x + c_1)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 6\frac{d}{dx}y(x) + 9y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 + 6r + 9 = 0$
- Factor the characteristic polynomial
 $(r + 3)^2 = 0$
- Root of the characteristic polynomial
 $r = -3$
- 1st solution of the ODE
 $y_1(x) = e^{-3x}$
- Repeated root, multiply $y_1(x)$ by x to ensure linear independence
 $y_2(x) = x e^{-3x}$
- General solution of the ODE
 $y(x) = C1y_1(x) + C2y_2(x)$
- Substitute in solutions
 $y(x) = C1 e^{-3x} + C2x e^{-3x}$

✓ **Mathematica.** Time used: 0.009 (sec). Leaf size: 18

```
ode=D[y[x],{x,2}]+6*D[y[x],x]+9*y[x]==0;  
ic={};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-3x}(c_2x + c_1)$$

✓ **Sympy.** Time used: 0.083 (sec). Leaf size: 12

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(9*y(x) + 6*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + C_2x)e^{-3x}$$

2.8.4 Problem 19

Local contents

Solved as higher order constant coeff ode	1601
✓ Maple	1602
✓ Mathematica	1603
✓ Sympy	1603

Internal problem ID [7980]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:40:30 AM

CAS classification : `[[_high_order, _missing_x]]`

Solve

$$y'''' - 6y''' + 12y'' - 8y' = 0$$

Solved as higher order constant coeff ode

Time used: 0.033 (sec)

The characteristic equation is

$$\lambda^4 - 6\lambda^3 + 12\lambda^2 - 8\lambda = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = 2$$

$$\lambda_3 = 2$$

$$\lambda_4 = 2$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 + e^{2x}c_2 + x e^{2x}c_3 + x^2 e^{2x}c_4$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = e^{2x}$$

$$y_3 = x e^{2x}$$

$$y_4 = e^{2x} x^2$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 21

```
ode:=diff(diff(diff(diff(y(x),x),x),x),x)-6*diff(diff(diff(y(x),x),x),x)+12*diff(
    diff(y(x),x),x)-8*diff(y(x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (c_4 x^2 + c_3 x + c_2) e^{2x} + c_1$$

Maple trace

```
Methods for high order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^4}{dx^4}y(x) - 6\frac{d^3}{dx^3}y(x) + 12\frac{d^2}{dx^2}y(x) - 8\frac{d}{dx}y(x) = 0$$

- Highest derivative means the order of the ODE is 4
 $\frac{d^4}{dx^4}y(x)$
- Characteristic polynomial of ODE
 $r^4 - 6r^3 + 12r^2 - 8r = 0$
- Roots of the characteristic polynomial and corresponding multiplicities
 $r = [[0, 1], [2, 3]]$
- Solution from $r = 0$
 $y_1(x) = 1$
- 1st solution from $r = 2$
 $y_2(x) = e^{2x}$
- 2nd solution from $r = 2$
 $y_3(x) = x e^{2x}$
- 3rd solution from $r = 2$
 $y_4(x) = x^2 e^{2x}$
- General solution of the ODE
 $y(x) = C1 y_1(x) + C2 y_2(x) + C3 y_3(x) + C4 y_4(x)$
- Substitute in solutions and simplify
 $y(x) = (C4 x^2 + C3 x + C2) e^{2x} + C1$

✓ **Mathematica.** Time used: 0.02 (sec). Leaf size: 36

```
ode=D[y[x],{x,4}]-6*D[y[x],{x,3}]+12*D[y[x],{x,2}]-8*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x e^{2K[1]}(c_1 + K[1](c_2 + c_3 K[1]))dK[1] + c_4$$

✓ **Sympy.** Time used: 0.105 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-8*Derivative(y(x), x) + 12*Derivative(y(x), (x, 2)) - 6*Derivative(y(x),
    (x, 3)) + Derivative(y(x), (x, 4)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 + (C_2 + x(C_3 + C_4 x)) e^{2x}$$

2.8.5 Problem 20

Local contents

Solved as second order linear constant coeff ode	1604
Solved as second order ode using Kovacic algorithm	1606
✓ Maple	1609
✓ Mathematica	1610
✓ Sympy	1610

Internal problem ID [7981]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 20

Date solved : Thursday, November 27, 2025 at 06:40:36 AM

CAS classification : `[_2nd_order, _missing_x]`

Solved as second order linear constant coeff ode

Time used: 0.085 (sec)

Solve

$$y'' - 4y' + 13y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 13$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} + 13e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda + 13 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 13$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-4^2 - (4)(1)(13)} \\ &= 2 \pm 3i \end{aligned}$$

Hence

$$\lambda_1 = 2 + 3i$$

$$\lambda_2 = 2 - 3i$$

Which simplifies to

$$\lambda_1 = 2 + 3i$$

$$\lambda_2 = 2 - 3i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 2$ and $\beta = 3$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^{2x}(c_1 \cos(3x) + c_2 \sin(3x))$$

Summary of solutions found

$$y = e^{2x}(c_1 \cos(3x) + c_2 \sin(3x))$$

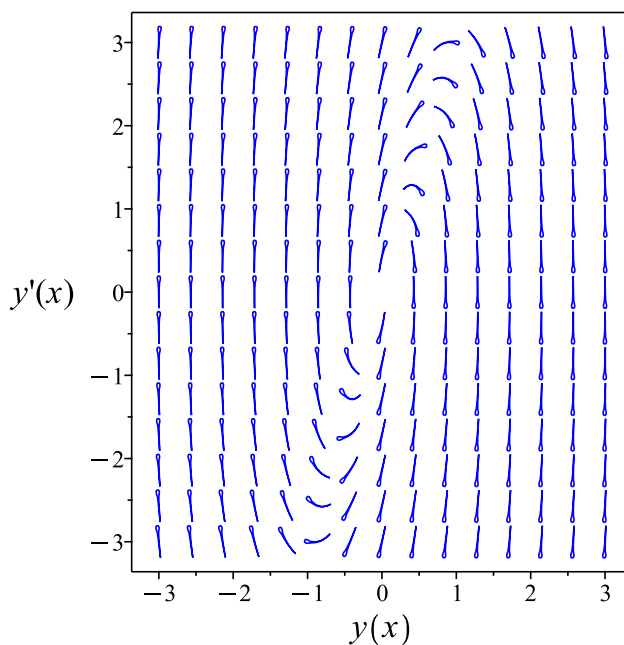


Figure 2.316: Slope field $y'' - 4y' + 13y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.122 (sec)

Solve

$$y'' - 4y' + 13y = 0$$

Writing the ode as

$$y'' - 4y' + 13y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 13 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-9}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -9 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -9z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.187: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -9$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(3x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{2x} \cos(3x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

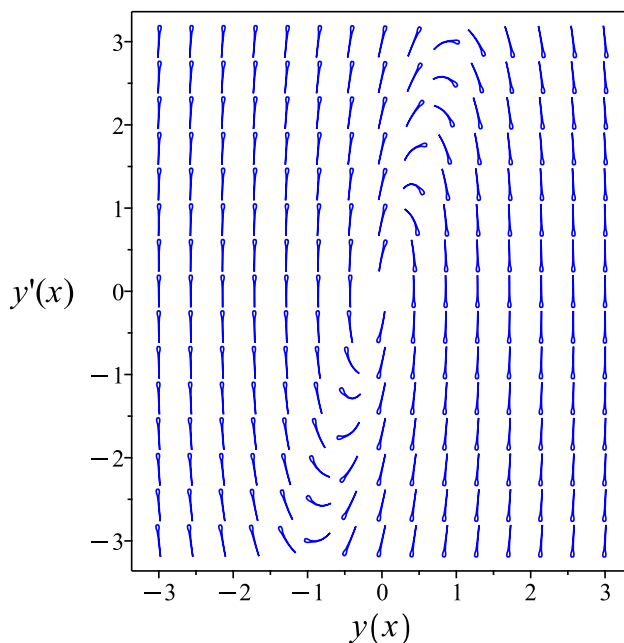
$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{\tan(3x)}{3} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{2x} \cos(3x)) + c_2 \left(e^{2x} \cos(3x) \left(\frac{\tan(3x)}{3} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^{2x} \cos(3x) + \frac{c_2 e^{2x} \sin(3x)}{3}$$

Figure 2.317: Slope field $y'' - 4y' + 13y = 0$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 22

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x)+13*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{2x}(c_1 \sin(3x) + c_2 \cos(3x))$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 13y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 - 4r + 13 = 0$
- Use quadratic formula to solve for r

- $r = \frac{4 \pm (\sqrt{-36})}{2}$
 - Roots of the characteristic polynomial
 $r = (2 - 3I, 2 + 3I)$
 - 1st solution of the ODE
 $y_1(x) = e^{2x} \cos(3x)$
 - 2nd solution of the ODE
 $y_2(x) = e^{2x} \sin(3x)$
 - General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x)$
 - Substitute in solutions
 $y(x) = C_1 e^{2x} \cos(3x) + C_2 e^{2x} \sin(3x)$

✓ **Mathematica.** Time used: 0.011 (sec). Leaf size: 26

```
ode=D[y[x],{x,2}]-4*D[y[x],x]+13*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{2x}(c_2 \cos(3x) + c_1 \sin(3x))$$

✓ **Sympy.** Time used: 0.094 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(13*y(x) - 4*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 \sin(3x) + C_2 \cos(3x)) e^{2x}$$

2.8.6 Problem 21

Local contents

Solved as second order linear constant coeff ode	1611
Solved as second order can be made integrable	1613
Solved as second order ode using Kovacic algorithm	1618
✓ Maple	1621
✓ Mathematica	1622
✓ Sympy	1622

Internal problem ID [7982]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 21

Date solved : Thursday, November 27, 2025 at 06:40:41 AM

CAS classification : `[[_2nd_order, _missing_x]]`

Solved as second order linear constant coeff ode

Time used: 0.083 (sec)

Solve

$$y'' + 25y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 25$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 25 e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 25 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 25$ into the above gives

$$\begin{aligned}\lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(25)} \\ &= \pm 5i\end{aligned}$$

Hence

$$\lambda_1 = +5i$$

$$\lambda_2 = -5i$$

Which simplifies to

$$\lambda_1 = 5i$$

$$\lambda_2 = -5i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 5$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

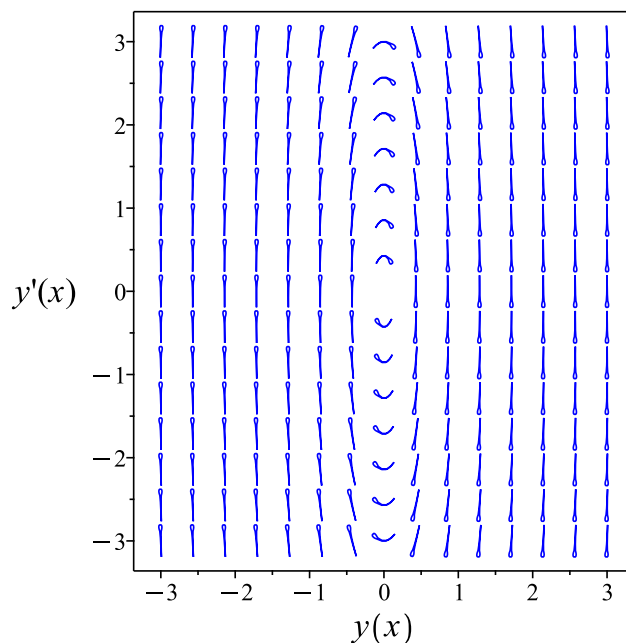
$$y = e^0(c_1 \cos(5x) + c_2 \sin(5x))$$

Or

$$y = c_1 \cos(5x) + c_2 \sin(5x)$$

Summary of solutions found

$$y = c_1 \cos(5x) + c_2 \sin(5x)$$

Figure 2.318: Slope field $y'' + 25y = 0$

Solved as second order can be made integrable

Time used: 0.891 (sec)

Solve

$$y'' + 25y = 0$$

Multiplying the ode by y' gives

$$y'y'' + 25yy' = 0$$

Integrating the above w.r.t x gives

$$\int (y'y'' + 25yy') dx = 0$$

$$\frac{y'^2}{2} + \frac{25y^2}{2} = c_1$$

Which is now solved for y . Solve Let $p = y'$ the ode becomes

$$\frac{p^2}{2} + \frac{25y^2}{2} = c_1$$

Solving for y from the above results in

$$y = \frac{\sqrt{-p^2 + 2c_1}}{5} \tag{1}$$

$$y = -\frac{\sqrt{-p^2 + 2c_1}}{5} \tag{2}$$

This has the form

$$y = xf(p) + g(p) \quad (*)$$

Where f, g are functions of $p = y'(x)$. Each of the above ode's is d'Alembert ode which is now solved.

Solving ode 1A

Taking derivative of (*) w.r.t. x gives

$$\begin{aligned} p &= f + (xf' + g') \frac{dp}{dx} \\ p - f &= (xf' + g') \frac{dp}{dx} \end{aligned} \quad (2)$$

Comparing the form $y = xf + g$ to (1A) shows that

$$\begin{aligned} f &= 0 \\ g &= \frac{\sqrt{-p^2 + 2c_1}}{5} \end{aligned}$$

Hence (2) becomes

$$p = -\frac{pp'(x)}{5\sqrt{-p^2 + 2c_1}} \quad (2A)$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = \frac{\sqrt{2}\sqrt{c_1}}{5}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = -5\sqrt{-p(x)^2 + 2c_1} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\int -\frac{1}{5\sqrt{-p^2 + 2c_1}} dp = dx$$

$$-\frac{\arctan\left(\frac{p}{\sqrt{-p^2 + 2c_1}}\right)}{5} = x + c_2$$

Singular solutions are found by solving

$$-5\sqrt{-p^2 + 2c_1} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = \sqrt{2} \sqrt{c_1}$$

$$p(x) = -\sqrt{2} \sqrt{c_1}$$

Substituting the above solution for p in (2A) gives

$$y = \frac{\sqrt{-\frac{2 \tan(5x+5c_2)^2 c_1}{\tan(5x+5c_2)^2 + 1} + 2c_1}}{5}$$

$$y = 0$$

$$y = 0$$

Solving ode 2A

Taking derivative of (*) w.r.t. x gives

$$p = f + (xf' + g') \frac{dp}{dx}$$

$$p - f = (xf' + g') \frac{dp}{dx} \tag{2}$$

Comparing the form $y = xf + g$ to (1A) shows that

$$f = 0$$

$$g = -\frac{\sqrt{-p^2 + 2c_1}}{5}$$

Hence (2) becomes

$$p = \frac{pp'(x)}{5\sqrt{-p^2 + 2c_1}} \tag{2A}$$

The singular solution is found by setting $\frac{dp}{dx} = 0$ in the above which gives

$$p = 0$$

Solving the above for p results in

$$p_1 = 0$$

Substituting these in (1A) and keeping singular solution that verifies the ode gives

$$y = -\frac{\sqrt{2}\sqrt{c_1}}{5}$$

The general solution is found when $\frac{dp}{dx} \neq 0$. From eq. (2A). This results in

$$p'(x) = 5\sqrt{-p(x)^2 + 2c_1} \quad (3)$$

This ODE is now solved for $p(x)$. No inversion is needed.

Integrating gives

$$\int \frac{1}{5\sqrt{-p^2 + 2c_1}} dp = dx$$

$$\frac{\arctan\left(\frac{p}{\sqrt{-p^2 + 2c_1}}\right)}{5} = x + c_3$$

Singular solutions are found by solving

$$5\sqrt{-p^2 + 2c_1} = 0$$

for $p(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p(x) = \sqrt{2}\sqrt{c_1}$$

$$p(x) = -\sqrt{2}\sqrt{c_1}$$

Substituting the above solution for p in (2A) gives

$$y = -\frac{\sqrt{-\frac{2 \tan(5x+5c_3)^2 c_1}{\tan(5x+5c_3)^2 + 1} + 2c_1}}{5}$$

$$y = 0$$

$$y = 0$$

Simplifying the above gives

$$y = \frac{\sqrt{2}\sqrt{c_1}}{5}$$

$$y = \frac{\sqrt{2}\sqrt{c_1 \cos(5x + 5c_2)^2}}{5}$$

$$y = 0$$

$$y = 0$$

$$y = -\frac{\sqrt{2}\sqrt{c_1}}{5}$$

$$y = -\frac{\sqrt{2}\sqrt{c_1 \cos(5x + 5c_3)^2}}{5}$$

$$y = 0$$

$$y = 0$$

The solution

$$y = 0$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = -\frac{\sqrt{2}\sqrt{c_1}}{5}$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

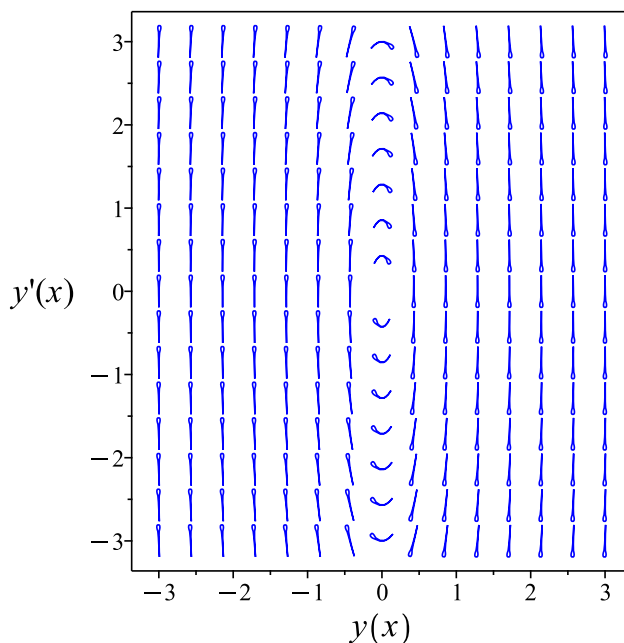
$$y = \frac{\sqrt{2}\sqrt{c_1}}{5}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = \frac{\sqrt{2}\sqrt{c_1 \cos(5x + 5c_2)^2}}{5}$$

$$y = -\frac{\sqrt{2}\sqrt{c_1 \cos(5x + 5c_3)^2}}{5}$$

Figure 2.319: Slope field $y'' + 25y = 0$

Solved as second order ode using Kovacic algorithm

Time used: 0.059 (sec)

Solve

$$y'' + 25y = 0$$

Writing the ode as

$$y'' + 25y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 25 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-25}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -25 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -25z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.189: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -25$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(5x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(5x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(5x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

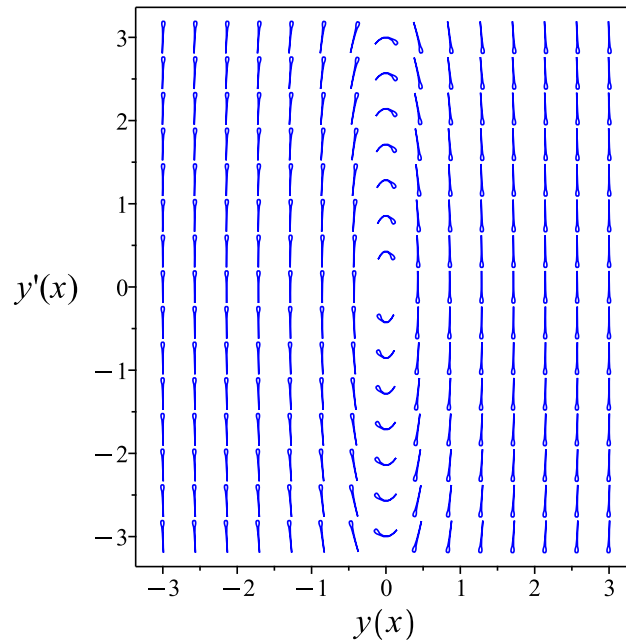
$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(5x) \int \frac{1}{\cos^2(5x)} dx \\ &= \cos(5x) \left(\frac{\tan(5x)}{5} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(5x)) + c_2 \left(\cos(5x) \left(\frac{\tan(5x)}{5} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 \cos(5x) + \frac{c_2 \sin(5x)}{5}$$

Figure 2.320: Slope field $y'' + 25y = 0$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 17

```
ode:=diff(diff(y(x),x),x)+25*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 \sin(5x) + c_2 \cos(5x)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 25y(x) = 0$$

- Highest derivative means the order of the ODE is 2
 $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of ODE
 $r^2 + 25 = 0$
- Use quadratic formula to solve for r
 $r = \frac{0 \pm (\sqrt{-100})}{2}$
- Roots of the characteristic polynomial
 $r = (-5I, 5I)$
- 1st solution of the ODE
 $y_1(x) = \cos(5x)$
- 2nd solution of the ODE
 $y_2(x) = \sin(5x)$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x)$
- Substitute in solutions
 $y(x) = C_1 \cos(5x) + C_2 \sin(5x)$

✓ **Mathematica.** Time used: 0.009 (sec). Leaf size: 20

```
ode=D[y[x],{x,2}]+25*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 \cos(5x) + c_2 \sin(5x)$$

✓ **Sympy.** Time used: 0.025 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(25*y(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 \sin(5x) + C_2 \cos(5x)$$

2.8.7 Problem 22

Local contents

Solved as higher order constant coeff ode	1623
✓ Maple	1624
✓ Mathematica	1624
✓ Sympy	1625

Internal problem ID [7983]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 22

Date solved : Thursday, November 27, 2025 at 06:40:47 AM

CAS classification : `[_3rd_order, _missing_x]`

Solve

$$y''' - y'' + 9y' - 9y = 0$$

Solved as higher order constant coeff ode

Time used: 0.036 (sec)

The characteristic equation is

$$\lambda^3 - \lambda^2 + 9\lambda - 9 = 0$$

The roots of the above equation are

$$\lambda_1 = 1$$

$$\lambda_2 = 3i$$

$$\lambda_3 = -3i$$

Therefore the homogeneous solution is

$$y_h(x) = e^x c_1 + e^{-3ix} c_2 + e^{3ix} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = e^x$$

$$y_2 = e^{-3ix}$$

$$y_3 = e^{3ix}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 21

```
ode:=diff(diff(diff(y(x),x),x),x)-diff(diff(y(x),x),x)+9*diff(y(x),x)-9*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 e^x + c_2 \sin(3x) + c_3 \cos(3x)$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) - \frac{d^2}{dx^2}y(x) + 9\frac{d}{dx}y(x) - 9y(x) = 0$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of ODE

$$r^3 - r^2 + 9r - 9 = 0$$

- Roots of the characteristic polynomial

$$r = [1, 3I, -3I]$$

- Solution from $r = 1$

$$y_1(x) = e^x$$

- Solutions from $r = 3I$ and $r = -3I$

$$[y_2(x) = \sin(3x), y_3(x) = \cos(3x)]$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + C3y_3(x)$$

- Substitute in solutions

$$y(x) = C1 e^x + C2 \sin(3x) + C3 \cos(3x)$$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 26

```
ode=D[y[x],{x,3}]-D[y[x],{x,2}]+9*D[y[x],x]-9*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_3 e^x + c_1 \cos(3x) + c_2 \sin(3x)$$

✓ Sympy. Time used: 0.092 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-9*y(x) + 9*Derivative(y(x), x) - Derivative(y(x), (x, 2)) + Derivative(
    y(x), (x, 3)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^x + C_2 \sin(3x) + C_3 \cos(3x)$$

2.8.8 Problem 23

Local contents

Solved as higher order constant coeff ode	1626
✓ Maple	1627
✓ Mathematica	1628
✓ Sympy	1628

Internal problem ID [7984]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 23

Date solved : Thursday, November 27, 2025 at 06:40:52 AM

CAS classification : `[_high_order, _missing_x]`

Solve

$$y'''' + 4y'' = 0$$

Solved as higher order constant coeff ode

Time used: 0.034 (sec)

The characteristic equation is

$$\lambda^4 + 4\lambda^2 = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = 0$$

$$\lambda_3 = 2i$$

$$\lambda_4 = -2i$$

Therefore the homogeneous solution is

$$y_h(x) = c_2x + c_1 + e^{2ix}c_3 + e^{-2ix}c_4$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = x$$

$$y_3 = e^{2ix}$$

$$y_4 = e^{-2ix}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 21

```
ode:=diff(diff(diff(diff(y(x),x),x),x),x)+4*diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 + c_2x + c_3 \sin(2x) + c_4 \cos(2x)$$

Maple trace

```
Methods for high order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^4}{dx^4}y(x) + 4\frac{d^2}{dx^2}y(x) = 0$$

- Highest derivative means the order of the ODE is 4
- $\frac{d^4}{dx^4}y(x)$
- Characteristic polynomial of ODE
 $r^4 + 4r^2 = 0$
- Roots of the characteristic polynomial and corresponding multiplicities
 $r = [[0, 2], [2I, 1], [-2I, 1]]$
- 1st solution from $r = 0$
 $y_1(x) = 1$
- 2nd solution from $r = 0$
 $y_2(x) = x$
- Solutions from $r = 2I$ and $r = -2I$
 $[y_3(x) = \sin(2x), y_4(x) = \cos(2x)]$
- General solution of the ODE
 $y(x) = C1y_1(x) + C2y_2(x) + C3y_3(x) + C4y_4(x)$
- Substitute in solutions
 $y(x) = C1 + C2x + C3 \sin(2x) + C4 \cos(2x)$

✓ **Mathematica.** Time used: 60.009 (sec). Leaf size: 44

```
ode=D[y[x],{x,4}]+4*D[y[x],{x,2}]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x \int_1^{K[2]} (c_1 \cos(2K[1]) + c_2 \sin(2K[1])) dK[1] dK[2] + c_4 x + c_3$$

✓ **Sympy.** Time used: 0.029 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*Derivative(y(x), (x, 2)) + Derivative(y(x), (x, 4)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 + C_2 x + C_3 \sin(2x) + C_4 \cos(2x)$$

2.8.9 Problem 24

Local contents

Solved as higher order constant coeff ode	1629
✓ Maple	1630
✓ Mathematica	1631
✓ Sympy	1631

Internal problem ID [7985]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 24

Date solved : Thursday, November 27, 2025 at 06:40:57 AM

CAS classification : `[[_high_order, _missing_x]]`

Solve

$$y'''' - 6y''' + 13y'' - 12y' + 4y = 0$$

Solved as higher order constant coeff ode

Time used: 0.036 (sec)

The characteristic equation is

$$\lambda^4 - 6\lambda^3 + 13\lambda^2 - 12\lambda + 4 = 0$$

The roots of the above equation are

$$\lambda_1 = 1$$

$$\lambda_2 = 1$$

$$\lambda_3 = 2$$

$$\lambda_4 = 2$$

Therefore the homogeneous solution is

$$y_h(x) = e^x c_1 + x e^x c_2 + e^{2x} c_3 + x e^{2x} c_4$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = e^x$$

$$y_2 = x e^x$$

$$y_3 = e^{2x}$$

$$y_4 = x e^{2x}$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 20

```
ode:=diff(diff(diff(diff(y(x),x),x),x),x)-6*diff(diff(diff(y(x),x),x),x)+13*diff(
    diff(y(x),x),x)-12*diff(y(x),x)+4*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x((c_2x + c_1)e^x + c_4x + c_3)$$

Maple trace

```
Methods for high order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^4}{dx^4}y(x) - 6\frac{d^3}{dx^3}y(x) + 13\frac{d^2}{dx^2}y(x) - 12\frac{d}{dx}y(x) + 4y(x) = 0$$

- Highest derivative means the order of the ODE is 4
 $\frac{d^4}{dx^4}y(x)$
- Characteristic polynomial of ODE
 $r^4 - 6r^3 + 13r^2 - 12r + 4 = 0$
- Roots of the characteristic polynomial and corresponding multiplicities
 $r = [[1, 2], [2, 2]]$
- 1st solution from $r = 1$
 $y_1(x) = e^x$
- 2nd solution from $r = 1$
 $y_2(x) = x e^x$
- 1st solution from $r = 2$
 $y_3(x) = e^{2x}$
- 2nd solution from $r = 2$
 $y_4(x) = x e^{2x}$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + C_4 y_4(x)$
- Substitute in solutions and simplify
 $y(x) = e^x((C_4 x + C_3) e^x + C_2 x + C_1)$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 29

```
ode=D[y[x],{x,4}]-6*D[y[x],{x,3}]+13*D[y[x],{x,2}]-12*D[y[x],x]+4*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x(c_3e^x + x(c_4e^x + c_2) + c_1)$$

✓ **Sympy.** Time used: 0.127 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*y(x) - 12*Derivative(y(x), x) + 13*Derivative(y(x), (x, 2)) - 6*
        Derivative(y(x), (x, 3)) + Derivative(y(x), (x, 4)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = (C_1 + C_4e^x + x(C_2 + C_3e^x))e^x$$

2.8.10 Problem 25

Local contents

Solved as higher order constant coeff ode	1632
✓ Maple	1633
✓ Mathematica	1633
✓ Sympy	1634

Internal problem ID [7986]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 13. Homogeneous Linear equations with constant coefficients. Supplementary problems. Page 86

Problem number : 25

Date solved : Thursday, November 27, 2025 at 06:41:02 AM

CAS classification : `[_high_order, _missing_x]`

Solve

$$y^{(6)} + 9y'''' + 24y'' + 16y = 0$$

Solved as higher order constant coeff ode

Time used: 0.052 (sec)

The characteristic equation is

$$\lambda^6 + 9\lambda^4 + 24\lambda^2 + 16 = 0$$

The roots of the above equation are

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

$$\lambda_3 = 2i$$

$$\lambda_4 = -2i$$

$$\lambda_5 = 2i$$

$$\lambda_6 = -2i$$

Therefore the homogeneous solution is

$$y_h(x) = e^{ix} c_1 + e^{2ix} c_2 + x e^{2ix} c_3 + e^{-ix} c_4 + e^{-2ix} c_5 + x e^{-2ix} c_6$$

The fundamental set of solutions for the homogeneous solution are the following

$$\begin{aligned}y_1 &= e^{ix} \\y_2 &= e^{2ix} \\y_3 &= e^{2ix} x \\y_4 &= e^{-ix} \\y_5 &= e^{-2ix} \\y_6 &= x e^{-2ix}\end{aligned}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 33

```
ode:=diff(diff(diff(diff(diff(diff(y(x),x),x),x),x),x),x)+9*diff(diff(diff(y
(x),x),x),x),x)+24*diff(diff(y(x),x),x)+16*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = (c_6x + c_4) \cos(2x) + (c_5x + c_3) \sin(2x) + c_1 \sin(x) + c_2 \cos(x)$$

Maple trace

```
Methods for high order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 40

```
ode=D[y[x],{x,6}]+9*D[y[x],{x,4}]+24*D[y[x],{x,2}]+16*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow (c_2x + c_1) \cos(2x) + c_6 \sin(x) + \cos(x)(2(c_4x + c_3) \sin(x) + c_5)$$

✓ Sympy. Time used: 0.074 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(16*y(x) + 24*Derivative(y(x), (x, 2)) + 9*Derivative(y(x), (x, 4)) +
        Derivative(y(x), (x, 6)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_5 \sin(x) + C_6 \cos(x) + (C_1 + C_2 x) \sin(2x) + (C_3 + C_4 x) \cos(2x)$$

2.9 Chapter 14. Linear equations with constant coefficients. Supplemetary problems. Page 92

Local contents

2.9.1	Problem 11	1636
2.9.2	Problem 12	1646
2.9.3	Problem 13	1665
2.9.4	Problem 14	1670
2.9.5	Problem 15	1677
2.9.6	Problem 16	1683
2.9.7	Problem 17	1694
2.9.8	Problem 18	1704
2.9.9	Problem 19	1715
2.9.10	Problem 20	1725
2.9.11	Problem 21	1737
2.9.12	Problem 22	1749

2.9.1 Problem 11

Local contents

Solved as second order linear constant coeff ode	1636
Solved as second order ode using Kovacic algorithm	1639
✓ Maple	1643
✓ Mathematica	1645
✓ Sympy	1645

Internal problem ID [7987]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 11

Date solved : Thursday, November 27, 2025 at 06:41:07 AM

CAS classification : `[[_2nd_order, _missing_x]]`

Solved as second order linear constant coeff ode

Time used: 0.122 (sec)

Solve

$$y'' - 4y' + 3y = 1$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -4, C = 3, f(x) = 1$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 3y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 3$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} + 3e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda + 3 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 3$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-4^2 - (4)(1)(3)} \\ &= 2 \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = 2 + 1$$

$$\lambda_2 = 2 - 1$$

Which simplifies to

$$\lambda_1 = 3$$

$$\lambda_2 = 1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(3)x} + c_2 e^{(1)x}$$

Or

$$y = e^{3x} c_1 + c_2 e^x$$

Therefore the homogeneous solution y_h is

$$y_h = e^{3x} c_1 + c_2 e^x$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{3x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_1 = 1$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{3} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

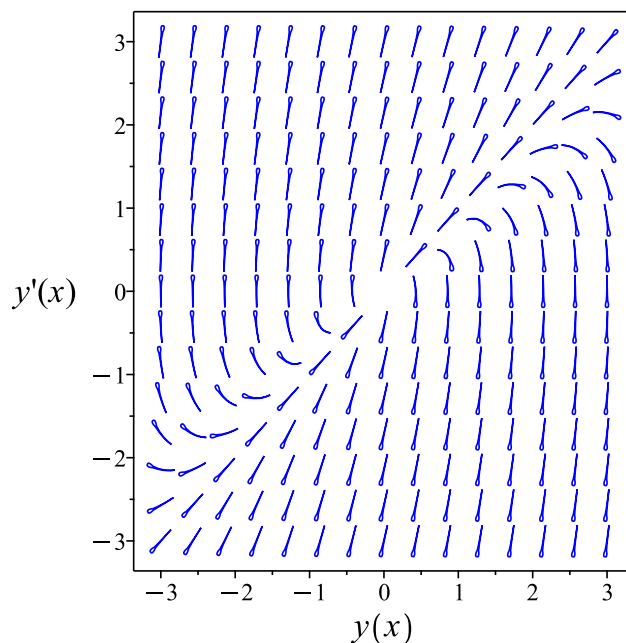
$$y_p = \frac{1}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{3x} c_1 + c_2 e^x) + \left(\frac{1}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{1}{3} + e^{3x} c_1 + c_2 e^x$$

Figure 2.321: Slope field $y'' - 4y' + 3y = 1$

Solved as second order ode using Kovacic algorithm

Time used: 0.102 (sec)

Solve

$$y'' - 4y' + 3y = 1$$

Writing the ode as

$$y'' - 4y' + 3y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 3 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.194: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{4x} e^{-2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 \left(e^x \left(\frac{e^{4x} e^{-2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 3y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^x c_1 + \frac{e^{3x} c_2}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^{3x}}{2}, e^x \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_1 = 1$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{3} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{1}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^x c_1 + \frac{e^{3x} c_2}{2} \right) + \left(\frac{1}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{1}{3} + e^x c_1 + \frac{e^{3x} c_2}{2}$$

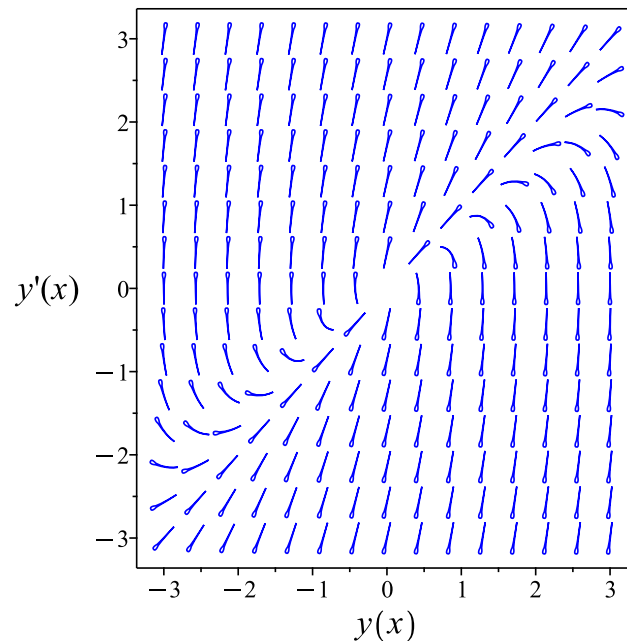


Figure 2.322: Slope field $y'' - 4y' + 3y = 1$

✓ Maple. Time used: 0.002 (sec). Leaf size: 16

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x)+3*y(x) = 1;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{3x} c_2 + e^x c_1 + \frac{1}{3}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
```

```

trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 3y(x) = 1$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^2 - 4r + 3 = 0$$
- Factor the characteristic polynomial

$$(r - 1)(r - 3) = 0$$
- Roots of the characteristic polynomial

$$r = (1, 3)$$
- 1st solution of the homogeneous ODE

$$y_1(x) = e^x$$
- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{3x}$$
- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^x + C2 e^{3x} + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = 1 \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = 2e^{4x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^x(-\int e^{-3x} dx e^{2x} + \int e^{-x} dx)}{2}$$
 - Compute integrals

$$y_p(x) = \frac{1}{3}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^x + C_2 e^{3x} + \frac{1}{3}$$

✓ **Mathematica.** Time used: 0.009 (sec). Leaf size: 23

```
ode=D[y[x],{x,2}]-4*D[y[x],x]+3*y[x]==1;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 e^x + c_2 e^{3x} + \frac{1}{3}$$

✓ **Sympy.** Time used: 0.096 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*y(x) - 4*Derivative(y(x), x) + Derivative(y(x), (x, 2)) - 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^x + C_2 e^{3x} + \frac{1}{3}$$

2.9.2 Problem 12

Local contents

Solved as second order linear constant coeff ode	1646
Solved as second order linear exact ode	1649
Solved as second order missing y ode	1651
Solved as second order integrable as is ode	1654
Solved as second order integrable as is ode (ABC method)	1656
Solved as second order ode using Kovacic algorithm	1658
✓ Maple	1662
✓ Mathematica	1664
✓ Sympy	1664

Internal problem ID [7988]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 12

Date solved : Thursday, November 27, 2025 at 06:41:13 AM

CAS classification : [[_2nd_order, _missing_x]]

Solved as second order linear constant coeff ode

Time used: 0.122 (sec)

Solve

$$y'' - 4y' = 5$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -4, C = 0, f(x) = 5$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 0$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 0$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-4^2 - (4)(1)(0)} \\ &= 2 \pm 2 \end{aligned}$$

Hence

$$\lambda_1 = 2 + 2$$

$$\lambda_2 = 2 - 2$$

Which simplifies to

$$\lambda_1 = 4$$

$$\lambda_2 = 0$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(4)x} + c_2 e^{(0)x}$$

Or

$$y = c_1 e^{4x} + c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^{4x} + c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, e^{4x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 = 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{5}{4} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

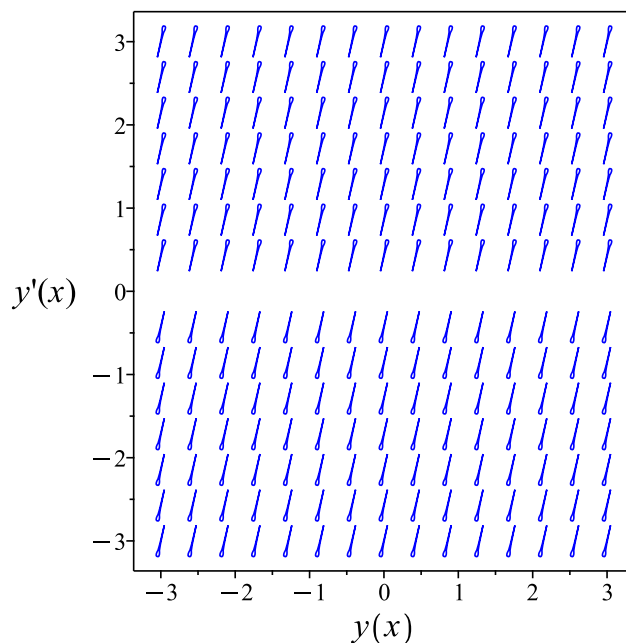
$$y_p = -\frac{5x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^{4x} + c_2) + \left(-\frac{5x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{5x}{4} + c_1 e^{4x} + c_2$$

Figure 2.323: Slope field $y'' - 4y' = 5$

Solved as second order linear exact ode

Time used: 0.191 (sec)

Solve

$$y'' - 4y' = 5$$

An ode of the form

$$p(x) y'' + q(x) y' + r(x) y = s(x)$$

is exact if

$$p''(x) - q'(x) + r(x) = 0 \tag{1}$$

For the given ode we have

$$\begin{aligned} p(x) &= 1 \\ q(x) &= -4 \\ r(x) &= 0 \\ s(x) &= 5 \end{aligned}$$

Hence

$$\begin{aligned} p''(x) &= 0 \\ q'(x) &= 0 \end{aligned}$$

Therefore (1) becomes

$$0 - (0) + (0) = 0$$

Hence the ode is exact. Since we now know the ode is exact, it can be written as

$$(p(x) y' + (q(x) - p'(x)) y)' = s(x)$$

Integrating gives

$$p(x) y' + (q(x) - p'(x)) y = \int s(x) dx$$

Substituting the above values for p, q, r, s gives

$$y' - 4y = \int 5 dx$$

We now have a first order ode to solve which is

$$y' - 4y = 5x + c_1$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -4 \\ p(x) &= 5x + c_1 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int (-4) dx} \\ &= e^{-4x} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) (5x + c_1) \\ \frac{d}{dx}(y e^{-4x}) &= (e^{-4x}) (5x + c_1) \\ d(y e^{-4x}) &= ((5x + c_1) e^{-4x}) dx \end{aligned}$$

Integrating gives

$$\begin{aligned} y e^{-4x} &= \int (5x + c_1) e^{-4x} dx \\ &= -\frac{(20x + 4c_1 + 5) e^{-4x}}{16} + c_2 \end{aligned}$$

Dividing throughout by the integrating factor e^{-4x} gives the final solution

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

Summary of solutions found

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

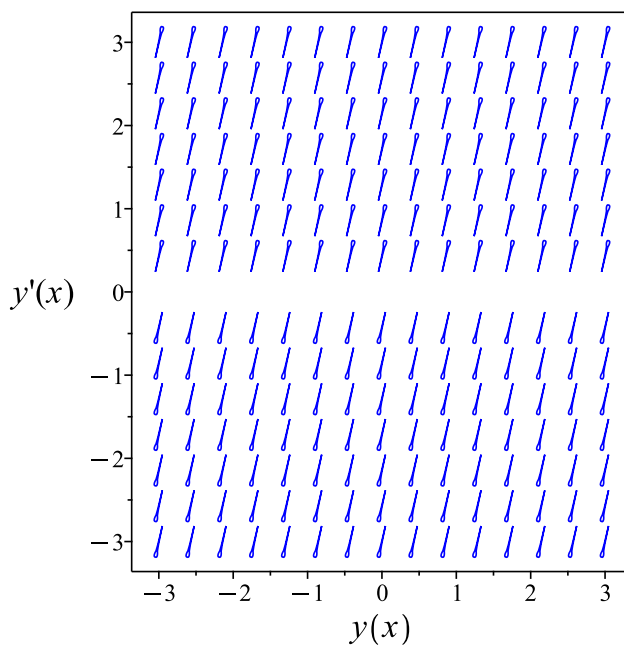


Figure 2.324: Slope field $y'' - 4y' = 5$

Solved as second order missing y ode

Time used: 0.499 (sec)

Solve

$$y'' - 4y' = 5$$

This is second order ode with missing dependent variable y . Let

$$u(x) = y'$$

Then

$$u'(x) = y''$$

Hence the ode becomes

$$u'(x) - 4u(x) - 5 = 0$$

Which is now solved for $u(x)$ as first order ode.

Solve Integrating gives

$$\int \frac{1}{4u+5} du = dx$$
$$\frac{\ln(4u+5)}{4} = x + c_1$$

Singular solutions are found by solving

$$4u + 5 = 0$$

for $u(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$u(x) = -\frac{5}{4}$$

Solving for $u(x)$ gives

$$u(x) = -\frac{5}{4}$$
$$u(x) = \frac{e^{4x+4c_1}}{4} - \frac{5}{4}$$

In summary, these are the solution found for y

$$u(x) = -\frac{5}{4}$$
$$u(x) = \frac{e^{4x+4c_1}}{4} - \frac{5}{4}$$

For solution $u(x) = -\frac{5}{4}$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = -\frac{5}{4}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -\frac{5}{4} dx$$

$$y = -\frac{5x}{4} + c_2$$

In summary, these are the solution found for (y)

$$y = -\frac{5x}{4} + c_2$$

For solution $u(x) = \frac{e^{4x+4c_1}}{4} - \frac{5}{4}$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{e^{4x+4c_1}}{4} - \frac{5}{4}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \frac{e^{4x+4c_1}}{4} - \frac{5}{4} dx$$

$$y = -\frac{5x}{4} + \frac{e^{4x+4c_1}}{16} + c_3$$

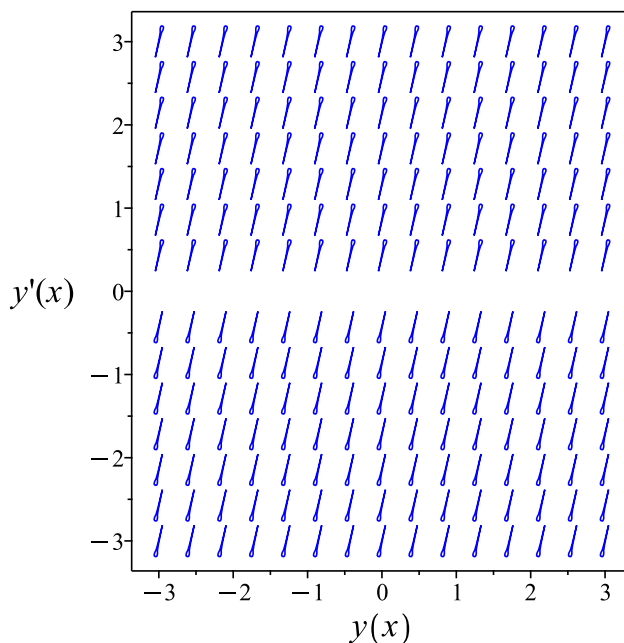
In summary, these are the solution found for (y)

$$y = -\frac{5x}{4} + \frac{e^{4x+4c_1}}{16} + c_3$$

Summary of solutions found

$$y = -\frac{5x}{4} + c_2$$

$$y = -\frac{5x}{4} + \frac{e^{4x+4c_1}}{16} + c_3$$

Figure 2.325: Slope field $y'' - 4y' = 5$

Solved as second order integrable as is ode

Time used: 0.077 (sec)

Solve

$$y'' - 4y' = 5$$

Integrating both sides of the ODE w.r.t x gives

$$\begin{aligned} \int (y'' - 4y') dx &= \int 5 dx \\ y' - 4y &= 5x + c_1 \end{aligned}$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -4 \\ p(x) &= 5x + c_1 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int (-4) dx} \\ &= e^{-4x} \end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu) (5x + c_1)$$

$$\frac{d}{dx}(y e^{-4x}) = (e^{-4x}) (5x + c_1)$$

$$d(y e^{-4x}) = ((5x + c_1) e^{-4x}) dx$$

Integrating gives

$$\begin{aligned} y e^{-4x} &= \int (5x + c_1) e^{-4x} dx \\ &= -\frac{(20x + 4c_1 + 5) e^{-4x}}{16} + c_2 \end{aligned}$$

Dividing throughout by the integrating factor e^{-4x} gives the final solution

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

Summary of solutions found

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

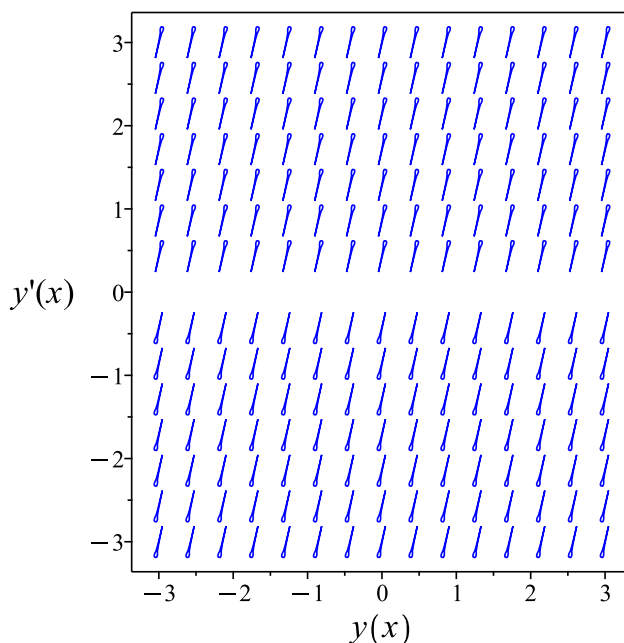


Figure 2.326: Slope field $y'' - 4y' = 5$

Solved as second order integrable as is ode (ABC method)

Time used: 0.076 (sec)

Solve

$$y'' - 4y' = 5$$

Writing the ode as

$$y'' - 4y' = 5$$

Integrating both sides of the ODE w.r.t x gives

$$\int (y'' - 4y') dx = \int 5 dx$$

$$y' - 4y = 5x + c_1$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -4$$

$$p(x) = 5x + c_1$$

The integrating factor μ is

$$\mu = e^{\int q dx}$$

$$= e^{\int (-4) dx}$$

$$= e^{-4x}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu)(5x + c_1)$$

$$\frac{d}{dx}(y e^{-4x}) = (e^{-4x})(5x + c_1)$$

$$d(y e^{-4x}) = ((5x + c_1) e^{-4x}) dx$$

Integrating gives

$$y e^{-4x} = \int (5x + c_1) e^{-4x} dx$$

$$= -\frac{(20x + 4c_1 + 5) e^{-4x}}{16} + c_2$$

Dividing throughout by the integrating factor e^{-4x} gives the final solution

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -4 \\ p(x) &= 5x + c_1 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int (-4) \, dx} \\ &= e^{-4x} \end{aligned}$$

The ode becomes

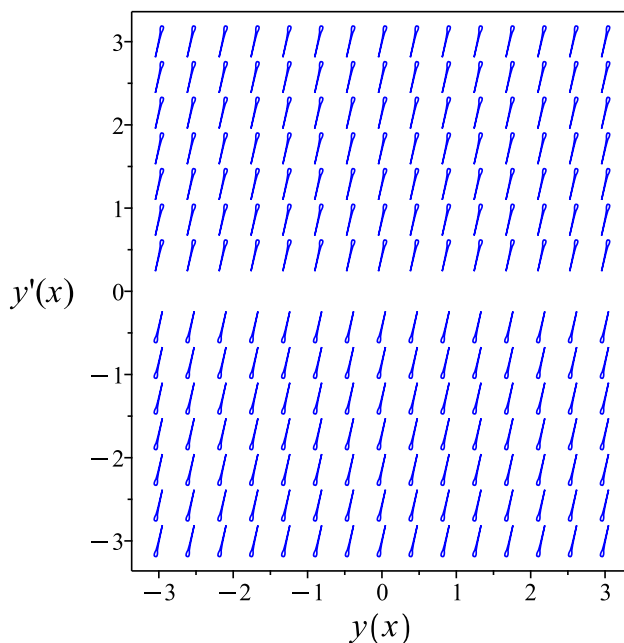
$$\begin{aligned} \frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu)(5x + c_1) \\ \frac{d}{dx}(y e^{-4x}) &= (e^{-4x})(5x + c_1) \\ d(y e^{-4x}) &= ((5x + c_1) e^{-4x}) \, dx \end{aligned}$$

Integrating gives

$$\begin{aligned} y e^{-4x} &= \int (5x + c_1) e^{-4x} \, dx \\ &= -\frac{(20x + 4c_1 + 5) e^{-4x}}{16} + c_2 \end{aligned}$$

Dividing throughout by the integrating factor e^{-4x} gives the final solution

$$y = -\frac{c_1}{4} - \frac{5x}{4} - \frac{5}{16} + c_2 e^{4x}$$

Figure 2.327: Slope field $y'' - 4y' = 5$

Solved as second order ode using Kovacic algorithm

Time used: 0.101 (sec)

Solve

$$y'' - 4y' = 5$$

Writing the ode as

$$y'' - 4y' = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 0 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{4}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 4 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 4z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.196: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 4$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-2x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = 1$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{4x}}{4} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (1) + c_2 \left(1 \left(\frac{e^{4x}}{4} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 + \frac{c_2 e^{4x}}{4}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ 1, \frac{e^{4x}}{4} \right\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 = 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{5}{4} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{5x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 + \frac{c_2 e^{4x}}{4} \right) + \left(-\frac{5x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{5x}{4} + c_1 + \frac{c_2 e^{4x}}{4}$$

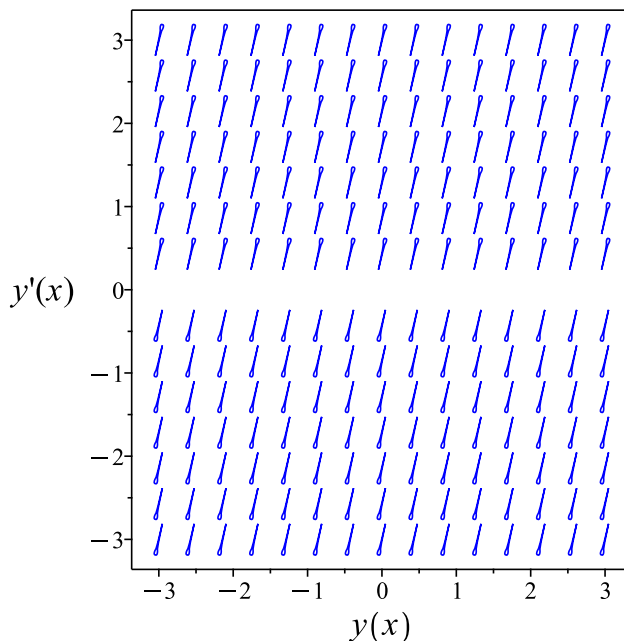


Figure 2.328: Slope field $y'' - 4y' = 5$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 16

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x) = 5;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{e^{4x}c_1}{4} - \frac{5x}{4} + c_2$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

-> Calling odsolve with the ODE, $\text{diff}(_b(_a),_a) = 4*_b(_a)+5, _b(_a)$

*** Sublevel 2 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

<- high order exact linear fully integrable successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) = 5$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE
 $r^2 - 4r = 0$
- Factor the characteristic polynomial
 $r(r - 4) = 0$
- Roots of the characteristic polynomial
 $r = (0, 4)$
- 1st solution of the homogeneous ODE
 $y_1(x) = 1$
- 2nd solution of the homogeneous ODE
 $y_2(x) = e^{4x}$
- General solution of the ODE
 $y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C1 + C2e^{4x} + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))}dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))}dx, f(x) = 5 \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} 1 & e^{4x} \\ 0 & 4e^{4x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = 4e^{4x}$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{5 \int 1 dx}{4} + \frac{5e^{4x} \int e^{-4x} dx}{4}$$

- Compute integrals

$$y_p(x) = -\frac{5x}{4} - \frac{5}{16}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 + C_2 e^{4x} - \frac{5x}{4} - \frac{5}{16}$$

✓ **Mathematica.** Time used: 0.008 (sec). Leaf size: 24

```
ode=D[y[x],{x,2}]-4*D[y[x],x]==5;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{5x}{4} + \frac{1}{4}c_1 e^{4x} + c_2$$

✓ **Sympy.** Time used: 0.085 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*Derivative(y(x), x) + Derivative(y(x), (x, 2)) - 5, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2 e^{4x} - \frac{5x}{4}$$

2.9.3 Problem 13

Local contents

Solved as higher order constant coeff ode	1665
✓ Maple	1667
✓ Mathematica	1669
✓ Sympy	1669

Internal problem ID [7989]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 13

Date solved : Thursday, November 27, 2025 at 06:41:19 AM

CAS classification : `[_3rd_order, _missing_x]`

Solve

$$y''' - 4y'' = 5$$

Solved as higher order constant coeff ode

Time used: 0.085 (sec)

The characteristic equation is

$$\lambda^3 - 4\lambda^2 = 0$$

The roots of the above equation are

$$\lambda_1 = 4$$

$$\lambda_2 = 0$$

$$\lambda_3 = 0$$

Therefore the homogeneous solution is

$$y_h(x) = c_2x + c_1 + e^{4x}c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = x$$

$$y_3 = e^{4x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' - 4y'' = 0$$

Now the particular solution to the given ODE is found

$$y''' - 4y'' = 5$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, x, e^{4x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x\}]$$

Since x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x^2$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-8A_1 = 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{5}{8} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{5x^2}{8}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_2x + c_1 + e^{4x}c_3) + \left(-\frac{5x^2}{8}\right) \end{aligned}$$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 21

```
ode:=diff(diff(diff(y(x),x),x),x)-4*diff(diff(y(x),x),x) = 5;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{e^{4x}c_1}{16} - \frac{5x^2}{8} + c_2x + c_3$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
-> Calling odsolve with the ODE, diff(_b(_a),_a) = 4*_b(_a)+5, _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- high order exact linear fully integrable successful
```

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) - 4\frac{d^2}{dx^2}y(x) = 5$$

- Highest derivative means the order of the ODE is 3
- $\frac{d^3}{dx^3}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^3 - 4r^2 = 0$$

- Roots of the characteristic polynomial and corresponding multiplicities

$$r = [[4, 1], [0, 2]]$$

- Homogeneous solution from $r = 4$

$$y_1(x) = e^{4x}$$

- 1st homogeneous solution from $r = 0$

$$y_2(x) = 1$$

- 2nd homogeneous solution from $r = 0$

$$y_3(x) = x$$

- General solution of the ODE

$$y(x) = C1 y_1(x) + C2 y_2(x) + C3 y_3(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{4x} + C2 + C3 x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Define the forcing function of the ODE

$$f(x) = 5$$

- Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$

- Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$

- Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$

- The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^3}{dx^3} y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$

- Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$

- Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} -\frac{5}{64}e^{4x} \\ \frac{5}{8}x^2 - \frac{5}{16}x \\ -\frac{5x}{4} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = -\frac{5}{64} - \frac{5}{8}x^2 - \frac{5}{16}x$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{4x} + C_2 + C_3 x - \frac{5}{64} - \frac{5x^2}{8} - \frac{5x}{16}$$

✓ **Mathematica.** Time used: 0.02 (sec). Leaf size: 30

```
ode=D[y[x],{x,3}]-4*D[y[x],{x,2}]==5;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{5x^2}{8} + c_3 x + \frac{1}{16} c_1 e^{4x} + c_2$$

✓ **Sympy.** Time used: 0.039 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*Derivative(y(x), (x, 2)) + Derivative(y(x), (x, 3)) - 5, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2 x + C_3 e^{4x} - \frac{5x^2}{8}$$

2.9.4 Problem 14

Local contents

Solved as higher order constant coeff ode	1670
✓ Maple	1672
✓ Mathematica	1675
✓ Sympy	1676

Internal problem ID [7990]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 14

Date solved : Thursday, November 27, 2025 at 06:41:24 AM

CAS classification : `[_high_order, _missing_x]`

Solve

$$y^{(5)} - 4y''' = 5$$

Solved as higher order constant coeff ode

Time used: 0.106 (sec)

The characteristic equation is

$$\lambda^5 - 4\lambda^3 = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = 0$$

$$\lambda_3 = 0$$

$$\lambda_4 = 2$$

$$\lambda_5 = -2$$

Therefore the homogeneous solution is

$$y_h(x) = c_3 x^2 + c_2 x + c_1 + e^{-2x} c_4 + e^{2x} c_5$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = x$$

$$y_3 = x^2$$

$$y_4 = e^{-2x}$$

$$y_5 = e^{2x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y^{(5)} - 4y''' = 0$$

Now the particular solution to the given ODE is found

$$y^{(5)} - 4y''' = 5$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, x, x^2, e^{-2x}, e^{2x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x\}]$$

Since x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2\}]$$

Since x^2 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^3\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set .

$$y_p = A_1 x^3$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-24A_1 = 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{5}{24} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{5x^3}{24}$$

Therefore the general solution is

$$y = y_h + y_p \\ = (c_3 x^2 + c_2 x + c_1 + e^{-2x} c_4 + e^{2x} c_5) + \left(-\frac{5x^3}{24} \right)$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 34

```
ode:=diff(diff(diff(diff(diff(y(x),x),x),x),x),x)-4*diff(diff(diff(y(x),x),x),x)
= 5;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{5x^3}{24} + \frac{c_3 x^2}{2} - \frac{e^{-2x} c_2}{8} + \frac{e^{2x} c_1}{8} + c_4 x + c_5$$

Maple trace

```
Methods for high order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 5; linear nonhomogeneous with symmetry [0,1]
```

```

-> Calling odsolve with the ODE, diff(diff(_b(_a),_a),_a) = 4*_b(_a)+5, _b(_a)
*** Sublevel 2 ***
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
    <- solving first the homogeneous part of the ODE successful
<- differential order: 5; linear nonhomogeneous with symmetry [0,1] successful

```

Maple step by step

Let's solve

$$\frac{d^5}{dx^5}y(x) - 4\frac{d^3}{dx^3}y(x) = 5$$

- Highest derivative means the order of the ODE is 5
- $\frac{d^5}{dx^5}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^5 - 4r^3 = 0$$
- Roots of the characteristic polynomial and corresponding multiplicities

$$r = [[0, 3], [2, 1], [-2, 1]]$$
- 1st homogeneous solution from $r = 0$

$$y_1(x) = 1$$
- 2nd homogeneous solution from $r = 0$

$$y_2(x) = x$$
- 3rd homogeneous solution from $r = 0$

$$y_3(x) = x^2$$
- Homogeneous solution from $r = 2$

$$y_4(x) = e^{2x}$$
- Homogeneous solution from $r = -2$

$$y_5(x) = e^{-2x}$$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + C_4 y_4(x) + C_5 y_5(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 + C_2 x + C_3 x^2 + C_4 e^{2x} + C_5 e^{-2x} + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Define the forcing function of the ODE

$$f(x) = 5$$

- Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^5 u_i(x) y_i(x)$$

- Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^5 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$

- Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^5 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$

- Calculate the 3rd derivative of $y_p(x)$

$$\frac{d^3}{dx^3} y_p(x) = \sum_{i=1}^5 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = 0$$

- Calculate the 4th derivative of $y_p(x)$

$$\frac{d^4}{dx^4} y_p(x) = \sum_{i=1}^5 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) + u_i(x) \left(\frac{d^4}{dx^4} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) = 0$$

- The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^5}{dx^5} y(x) + \left(\sum_{i=0}^4 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$

- Substitute $y_p(x) = \sum_{i=1}^5 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^4 P_j(x) \left(\sum_{i=1}^5 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^5 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^4}{dx^4} y_i(x) \right) + u_i(x) \left(\frac{d^5}{dx^5} y_i(x) \right) \right) = f(x)$$

- Rearrange the ODE

$$\sum_{i=1}^5 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^4 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^5}{dx^5} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^4}{dx^4} y_i(x) \right) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^4}{dx^4} y_i(x) \right) = f(x)$$

- We have now made a system of 5 equations in 5 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = 0, \sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) = 0, \sum_{i=1}^5 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^4}{dx^4} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) & y_4(x) & y_5(x) \\ \frac{d}{dx}y_1(x) & \frac{d}{dx}y_2(x) & \frac{d}{dx}y_3(x) & \frac{d}{dx}y_4(x) & \frac{d}{dx}y_5(x) \\ \frac{d^2}{dx^2}y_1(x) & \frac{d^2}{dx^2}y_2(x) & \frac{d^2}{dx^2}y_3(x) & \frac{d^2}{dx^2}y_4(x) & \frac{d^2}{dx^2}y_5(x) \\ \frac{d^3}{dx^3}y_1(x) & \frac{d^3}{dx^3}y_2(x) & \frac{d^3}{dx^3}y_3(x) & \frac{d^3}{dx^3}y_4(x) & \frac{d^3}{dx^3}y_5(x) \\ \frac{d^4}{dx^4}y_1(x) & \frac{d^4}{dx^4}y_2(x) & \frac{d^4}{dx^4}y_3(x) & \frac{d^4}{dx^4}y_4(x) & \frac{d^4}{dx^4}y_5(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx}u_1(x) \\ \frac{d}{dx}u_2(x) \\ \frac{d}{dx}u_3(x) \\ \frac{d}{dx}u_4(x) \\ \frac{d}{dx}u_5(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \\ u_4(x) \\ u_5(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \\ u_4(x) \\ u_5(x) \end{bmatrix} = \begin{bmatrix} -\frac{5}{16}x - \frac{5}{24}x^3 \\ \frac{5x^2}{8} \\ -\frac{5x}{8} \\ -\frac{5}{64e^{2x}} \\ \frac{5}{64e^{-2x}} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = -\frac{5}{16}x - \frac{5}{24}x^3$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 + C_2x + C_3x^2 + C_4e^{2x} + C_5e^{-2x} - \frac{5x}{16} - \frac{5x^3}{24}$$

✓ **Mathematica.** Time used: 0.048 (sec). Leaf size: 47

```
ode=D[y[x],{x,5}]-4*D[y[x],{x,3}]==5;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{5x^3}{24} + c_5x^2 + c_4x + \frac{1}{8}c_1e^{2x} - \frac{1}{8}c_2e^{-2x} + c_3$$

✓ Sympy. Time used: 0.069 (sec). Leaf size: 32

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-4*Derivative(y(x), (x, 3)) + Derivative(y(x), (x, 5)) - 5, 0)  
ics = {}  
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2x + C_3x^2 + C_4e^{-2x} + C_5e^{2x} - \frac{5x^3}{24}$$

2.9.5 Problem 15

Local contents

Solved as higher order constant coeff ode	1677
✓ Maple	1679
✓ Mathematica	1681
✓ Sympy	1682

Internal problem ID [7991]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 15

Date solved : Thursday, November 27, 2025 at 06:41:29 AM

CAS classification : `[_3rd_order, _missing_y]`

Solve

$$-4y' + y''' = x$$

Solved as higher order constant coeff ode

Time used: 0.093 (sec)

The characteristic equation is

$$\lambda^3 - 4\lambda = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = 2$$

$$\lambda_3 = -2$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 + e^{-2x}c_2 + e^{2x}c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = e^{-2x}$$

$$y_3 = e^{2x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$-4y' + y''' = 0$$

Now the particular solution to the given ODE is found

$$-4y' + y''' = x$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$x$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, e^{-2x}, e^{2x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x, x^2\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_2x^2 + A_1x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-8xA_2 - 4A_1 = x$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 0, A_2 = -\frac{1}{8} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{x^2}{8}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= (c_1 + e^{-2x}c_2 + e^{2x}c_3) + \left(-\frac{x^2}{8}\right)$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 25

```
ode:=-4*diff(y(x),x)+diff(diff(diff(y(x),x),x),x) = x;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{x^2}{8} - \frac{e^{-2x}c_1}{2} + \frac{e^{2x}c_2}{2} + c_3$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 3; linear nonhomogeneous with symmetry [0,1]

-> Calling odsolve with the ODE, diff(diff(_b(_a),_a),_a) = 4*_b(_a)+_a, _b(_a)

*** Sublevel 2 ***

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

<- differential order: 3; linear nonhomogeneous with symmetry [0,1] successful

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) - 4\frac{d}{dx}y(x) = x$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of homogeneous ODE
 $r^3 - 4r = 0$
- Roots of the characteristic polynomial
 $r = [0, 2, -2]$
- Homogeneous solution from $r = 0$
 $y_1(x) = 1$
- Homogeneous solution from $r = 2$
 $y_2(x) = e^{2x}$
- Homogeneous solution from $r = -2$
 $y_3(x) = e^{-2x}$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C_1 + C_2 e^{2x} + C_3 e^{-2x} + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Define the forcing function of the ODE
 $f(x) = x$
 - Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$
 - Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$
 - Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$
 - The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^3}{dx^3} y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$
 - Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$
 - Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} -\frac{x^2}{8} \\ -\frac{x}{16e^{2x}} - \frac{1}{32e^{2x}} \\ \frac{x}{16e^{-2x}} - \frac{1}{32e^{-2x}} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = -\frac{x^2}{8} - \frac{1}{16}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 + C_2 e^{2x} + C_3 e^{-2x} - \frac{x^2}{8} - \frac{1}{16}$$

✓ **Mathematica.** Time used: 0.021 (sec). Leaf size: 37

```
ode=D[y[x],{x,3}]-4*D[y[x],x]==x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -\frac{x^2}{8} + \frac{1}{2}c_1e^{2x} - \frac{1}{2}c_2e^{-2x} + c_3$$

✓ Sympy. Time used: 0.108 (sec). Leaf size: 22

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x - 4*Derivative(y(x), x) + Derivative(y(x), (x, 3)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2 e^{-2x} + C_3 e^{2x} - \frac{x^2}{8}$$

2.9.6 Problem 16

Local contents

Solved as second order linear constant coeff ode	1683
Solved as second order solved by an integrating factor	1686
Solved as second order ode using Kovacic algorithm	1687
✓ Maple	1691
✓ Mathematica	1693
✓ Sympy	1693

Internal problem ID [7992]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 16

Date solved : Thursday, November 27, 2025 at 06:41:34 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.124 (sec)

Solve

$$y'' - 6y' + 9y = e^{2x}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -6, C = 9, f(x) = e^{2x}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 6y' + 9y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -6, C = 9$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 6\lambda e^{x\lambda} + 9e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 6\lambda + 9 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -6, C = 9$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{6}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{(-6)^2 - (4)(1)(9)} \\ &= 3 \end{aligned}$$

Hence this is the case of a double root $\lambda_{1,2} = -3$. Therefore the solution is

$$y = c_1 e^{3x} + c_2 x e^{3x} \quad (1)$$

Therefore the homogeneous solution y_h is

$$y_h = e^{3x} c_1 + c_2 x e^{3x}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{2x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{2x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{x e^{3x}, e^{3x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{2x}$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$A_1 e^{2x} = e^{2x}$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = e^{2x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{3x}c_1 + c_2x e^{3x}) + (e^{2x}) \end{aligned}$$

Summary of solutions found

$$y = e^{2x} + e^{3x}c_1 + c_2x e^{3x}$$

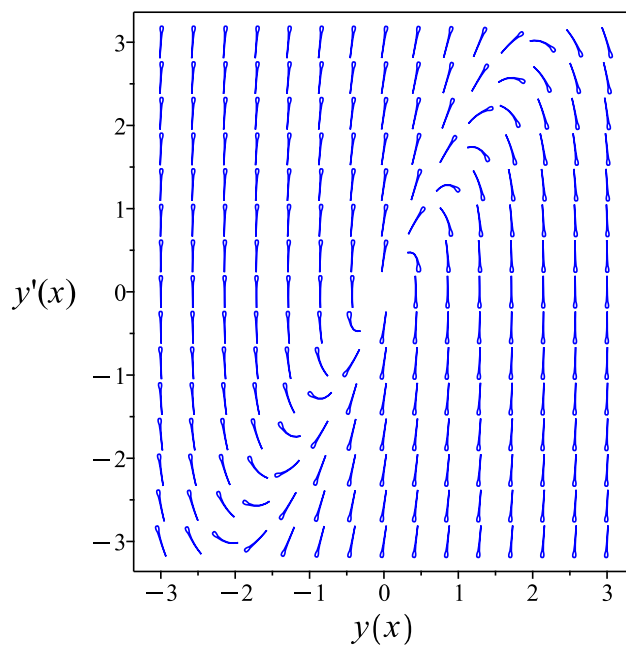


Figure 2.329: Slope field $y'' - 6y' + 9y = e^{2x}$

Solved as second order solved by an integrating factor

Time used: 0.069 (sec)

Solve

$$y'' - 6y' + 9y = e^{2x}$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x))^2 + p'(x)}{2}y = f(x)$$

Where $p(x) = -6$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int -6 \, dx} \\ &= e^{-3x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$(M(x)y)'' = e^{-3x}e^{2x}$$

$$(ye^{-3x})'' = e^{-3x}e^{2x}$$

Integrating once gives

$$(ye^{-3x})' = -e^{-x} + c_1$$

Integrating again gives

$$(ye^{-3x}) = c_1x + e^{-x} + c_2$$

Hence the solution is

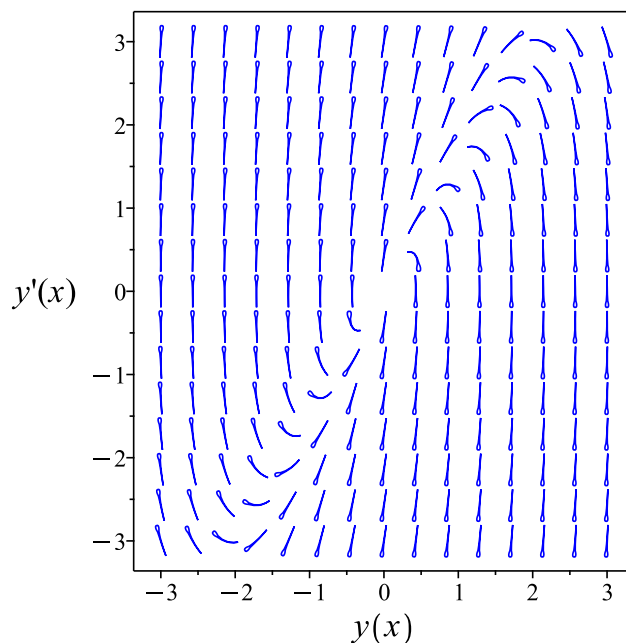
$$y = \frac{c_1x + e^{-x} + c_2}{e^{-3x}}$$

Or

$$y = c_1xe^{3x} + e^{3x}c_2 + e^{2x}$$

Summary of solutions found

$$y = c_1xe^{3x} + e^{3x}c_2 + e^{2x}$$

Figure 2.330: Slope field $y'' - 6y' + 9y = e^{2x}$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.104 (sec)

Solve

$$y'' - 6y' + 9y = e^{2x}$$

Writing the ode as

$$y'' - 6y' + 9y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -6 \\ C &= 9 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 0 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.201: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-6}{1} dx} \\ &= z_1 e^{3x} \\ &= z_1 (e^{3x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-6}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{6x}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{3x}) + c_2 (e^{3x}(x)) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 6y' + 9y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{3x}c_1 + c_2x e^{3x}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{2x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{2x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{x e^{3x}, e^{3x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{2x}$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$A_1 e^{2x} = e^{2x}$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = e^{2x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{3x}c_1 + c_2x e^{3x}) + (e^{2x}) \end{aligned}$$

Summary of solutions found

$$y = e^{2x} + e^{3x}c_1 + c_2x e^{3x}$$

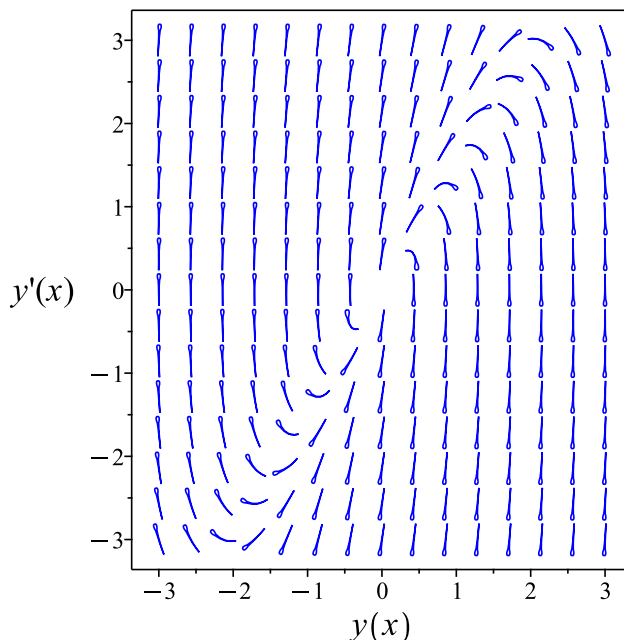


Figure 2.331: Slope field $y'' - 6y' + 9y = e^{2x}$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 19

```
ode:=diff(diff(y(x),x),x)-6*diff(y(x),x)+9*y(x) = exp(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{2x} + e^{3x}(c_1x + c_2)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE
 checking if the LODE has constant coefficients
 <- constant coefficients successful
 <- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 6\frac{d}{dx}y(x) + 9y(x) = e^{2x}$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE
 $r^2 - 6r + 9 = 0$
- Factor the characteristic polynomial
 $(r - 3)^2 = 0$
- Root of the characteristic polynomial
 $r = 3$
- 1st solution of the homogeneous ODE
 $y_1(x) = e^{3x}$
- Repeated root, multiply $y_1(x)$ by x to ensure linear independence
 $y_2(x) = x e^{3x}$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C_1 e^{3x} + C_2 x e^{3x} + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = e^{2x} \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{3x} & x e^{3x} \\ 3 e^{3x} & e^{3x} + 3x e^{3x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{6x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = e^{3x} \left(x \int e^{-x} dx - \int x e^{-x} dx \right)$$
 - Compute integrals

$$y_p(x) = e^{2x}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{3x} + C_2 x e^{3x} + e^{2x}$$

✓ **Mathematica.** Time used: 0.023 (sec). Leaf size: 49

```
ode=D[y[x],{x,2}]-6*D[y[x],x]+9*y[x]==Exp[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{2x} \left(e^x \int_1^x -e^{-K[1]} K[1] dK[1] + c_1 e^x + x(-1 + c_2 e^x) \right)$$

✓ **Sympy.** Time used: 0.115 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(9*y(x) - exp(2*x) - 6*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = ((C_1 + C_2 x) e^x + 1) e^{2x}$$

2.9.7 Problem 17

Local contents

Solved as second order linear constant coeff ode	1694
Solved as second order ode using Kovacic algorithm	1697
✓ Maple	1701
✓ Mathematica	1703
✓ Sympy	1703

Internal problem ID [7993]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 17

Date solved : Thursday, November 27, 2025 at 06:41:40 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.127 (sec)

Solve

$$y'' + y' - 2y = -2x^2 + 2x + 2$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 1, C = -2, f(x) = -2x^2 + 2x + 2$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y' - 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 1, C = -2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + \lambda e^{x\lambda} - 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + \lambda - 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 1, C = -2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-1}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{1^2 - (4)(1)(-2)} \\ &= -\frac{1}{2} \pm \frac{3}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} + \frac{3}{2} \\ \lambda_2 &= -\frac{1}{2} - \frac{3}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= 1 \\ \lambda_2 &= -2 \end{aligned}$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(1)x} + c_2 e^{(-2)x} \end{aligned}$$

Or

$$y = e^x c_1 + e^{-2x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = e^x c_1 + e^{-2x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2 + x + 1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-2x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_3x^2 + A_2x + A_1$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_3x^2 - 2A_2x + 2xA_3 - 2A_1 + A_2 + 2A_3 = -2x^2 + 2x + 2$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 0, A_2 = 0, A_3 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

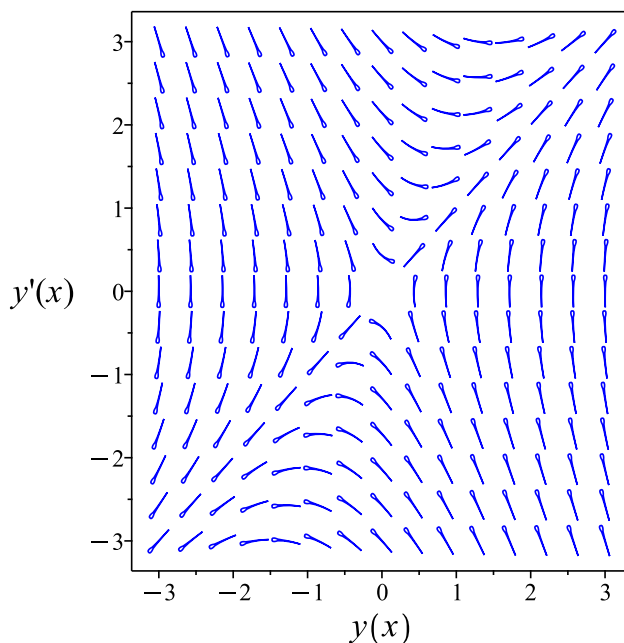
$$y_p = x^2$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{-2x} c_2) + (x^2) \end{aligned}$$

Summary of solutions found

$$y = x^2 + e^x c_1 + e^{-2x} c_2$$

Figure 2.332: Slope field $y'' + y' - 2y = -2x^2 + 2x + 2$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.116 (sec)

Solve

$$y'' + y' - 2y = -2x^2 + 2x + 2$$

Writing the ode as

$$y'' + y' - 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 1 \\ C &= -2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{9}{4} \quad (6)$$

Comparing the above to (5) shows that

$$s = 9$$

$$t = 4$$

Therefore eq. (4) becomes

$$z''(x) = \frac{9z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.203: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{9}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{3x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{1}{1} dx} \\ &= z_1 e^{-\frac{x}{2}} \\ &= z_1 (e^{-\frac{x}{2}}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-2x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{1}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{-x} e^{4x}}{3} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-2x}) + c_2 \left(e^{-2x} \left(\frac{e^{-x} e^{4x}}{3} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y' - 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{-2x}c_1 + \frac{c_2 e^x}{3}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2 + x + 1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{3}, e^{-2x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_3x^2 + A_2x + A_1$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_3x^2 - 2A_2x + 2xA_3 - 2A_1 + A_2 + 2A_3 = -2x^2 + 2x + 2$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 0, A_2 = 0, A_3 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = x^2$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-2x} c_1 + \frac{c_2 e^x}{3} \right) + (x^2) \end{aligned}$$

Summary of solutions found

$$y = x^2 + e^{-2x} c_1 + \frac{c_2 e^x}{3}$$

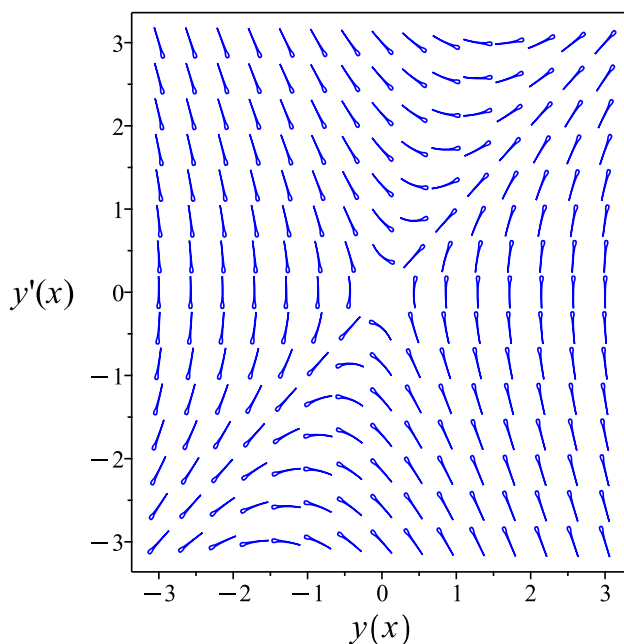


Figure 2.333: Slope field $y'' + y' - 2y = -2x^2 + 2x + 2$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 18

```
ode:=diff(diff(y(x),x),x)+diff(y(x),x)-2*y(x) = -2*x^2+2*x+2;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x c_2 + e^{-2x} c_1 + x^2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
```

```

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + \frac{d}{dx}y(x) - 2y(x) = -2x^2 + 2x + 2$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + r - 2 = 0$$

- Factor the characteristic polynomial

$$(r + 2)(r - 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-2, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-2x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-2x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = -2x^2 + 2x + 2 \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-2x} & e^x \\ -2e^{-2x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 3e^{-x}$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{2(\int e^{-x}(x^2-x-1)dx e^{3x} - \int e^{2x}(x^2-x-1)dx) e^{-2x}}{3}$$

- Compute integrals

$$y_p(x) = x^2$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-2x} + C_2 e^x + x^2$$

✓ **Mathematica.** Time used: 0.01 (sec). Leaf size: 23

```
ode=D[y[x],{x,2}]+D[y[x],x]-2*y[x]==2*(1+x-x^2);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x^2 + c_1 e^{-2x} + c_2 e^x$$

✓ **Sympy.** Time used: 0.084 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x**2 - 2*x - 2*y(x) + Derivative(y(x), x) + Derivative(y(x), (x, 2)) -
        2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 e^{-2x} + C_2 e^x + x^2$$

2.9.8 Problem 18

Local contents

Solved as second order linear constant coeff ode	1704
Solved as second order ode using Kovacic algorithm	1707
✓ Maple	1712
✓ Mathematica	1714
✓ Sympy	1714

Internal problem ID [7994]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 18

Date solved : Thursday, November 27, 2025 at 06:41:45 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.138 (sec)

Solve

$$y'' - y = 4x e^x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = 4x e^x$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = e^x c_1 + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = e^x c_1 + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$4x e^x$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x e^x, e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since e^x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2e^x, xe^x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1x^2e^x + A_2xe^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$2A_1e^x + 4A_1xe^x + 2A_2e^x = 4xe^x$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 1, A_2 = -1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

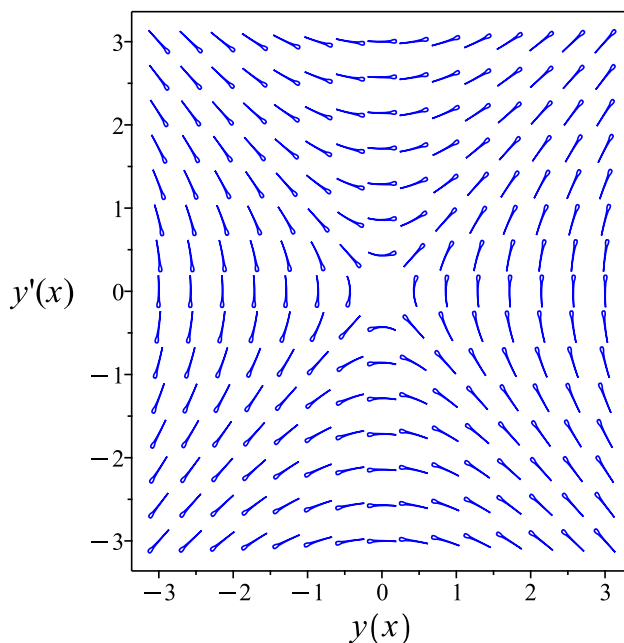
$$y_p = x^2e^x - xe^x$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{-x} c_2) + (x^2e^x - xe^x) \end{aligned}$$

Summary of solutions found

$$y = -xe^x + x^2e^x + e^x c_1 + e^{-x} c_2$$

Figure 2.334: Slope field $y'' - y = 4xe^x$

Solved as second order ode using Kovacic algorithm

Time used: 0.174 (sec)

Solve

$$y'' - y = 4xe^x$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.205: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{-x}c_1 + \frac{c_2 e^x}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-x}$$

$$y_2 = \frac{e^x}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ \frac{d}{dx}(e^{-x}) & \frac{d}{dx}\left(\frac{e^x}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ -e^{-x} & \frac{e^x}{2} \end{vmatrix}$$

Therefore

$$W = (e^{-x}) \left(\frac{e^x}{2} \right) - \left(\frac{e^x}{2} \right) (-e^{-x})$$

Which simplifies to

$$W = e^x e^{-x}$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{2x e^{2x}}{1} dx$$

Which simplifies to

$$u_1 = - \int 2x e^{2x} dx$$

Hence

$$u_1 = - \frac{(2x - 1) e^{2x}}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{4 e^{-x} e^x x}{1} dx$$

Which simplifies to

$$u_2 = \int 4x dx$$

Hence

$$u_2 = 2x^2$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = - \frac{(2x - 1) e^{2x} e^{-x}}{2} + x^2 e^x$$

Which simplifies to

$$y_p(x) = e^x \left(-x + \frac{1}{2} + x^2 \right)$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-x} c_1 + \frac{c_2 e^x}{2} \right) + \left(e^x \left(-x + \frac{1}{2} + x^2 \right) \right) \end{aligned}$$

Summary of solutions found

$$y = e^x \left(-x + \frac{1}{2} + x^2 \right) + e^{-x} c_1 + \frac{c_2 e^x}{2}$$

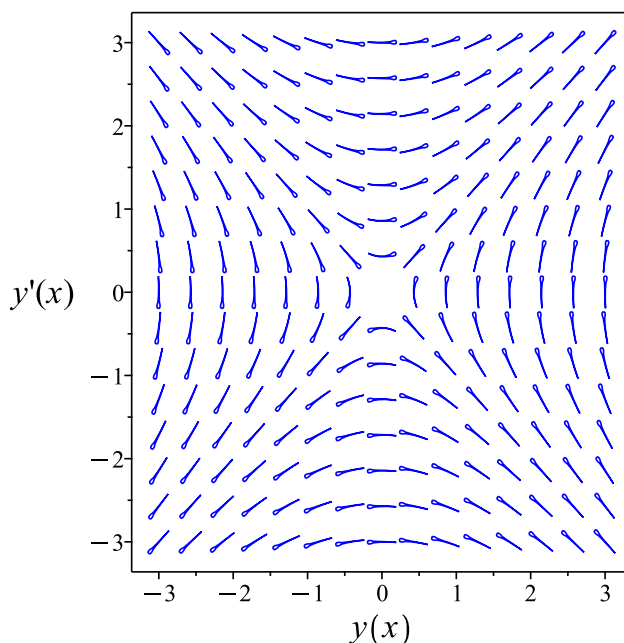


Figure 2.335: Slope field $y'' - y = 4x e^x$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 22

```
ode:=diff(diff(y(x),x),x)-y(x) = 4*x*exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x} c_1 + e^x (x^2 + c_2 - x)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = 4x e^x$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = 4x e^x \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

- $y_p(x) = -2e^{-x} \int e^{2x} x dx + 2e^x \int x dx$
 - Compute integrals
 - $y_p(x) = e^x \left(-x + \frac{1}{2} + x^2\right)$
 - Substitute particular solution into general solution to ODE
- $$y(x) = C_1 e^{-x} + C_2 e^x + e^x \left(-x + \frac{1}{2} + x^2\right)$$

✓ **Mathematica.** Time used: 0.019 (sec). Leaf size: 43

```
ode=D[y[x],{x,2}]-y[x]==4*x*Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(\int_1^x -2e^{2K[1]} K[1] dK[1] + e^{2x} (x^2 + c_1) + c_2 \right)$$

✓ **Sympy.** Time used: 0.060 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*x*exp(x) - y(x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_2 e^{-x} + (C_1 + x^2 - x) e^x$$

2.9.9 Problem 19

Local contents

Solved as second order linear constant coeff ode	1715
Solved as second order ode using Kovacic algorithm	1718
✓ Maple	1722
✓ Mathematica	1724
✓ Sympy	1724

Internal problem ID [7995]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 19

Date solved : Thursday, November 27, 2025 at 06:41:50 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.187 (sec)

Solve

$$y'' - y = \sin(x)^2$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = \sin(x)^2$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = e^x c_1 + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = e^x c_1 + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\sin(x)^2$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 \cos(2x) + A_3 \sin(2x)$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-5A_2 \cos(2x) - 5A_3 \sin(2x) - A_1 = \sin(x)^2$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{2}, A_2 = \frac{1}{10}, A_3 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

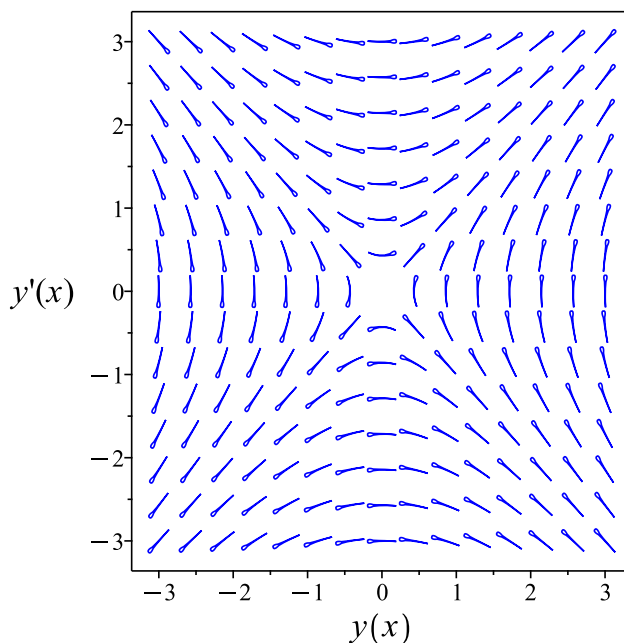
$$y_p = -\frac{1}{2} + \frac{\cos(2x)}{10}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{-x} c_2) + \left(-\frac{1}{2} + \frac{\cos(2x)}{10} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{1}{2} + \frac{\cos(2x)}{10} + e^x c_1 + e^{-x} c_2$$

Figure 2.336: Slope field $y'' - y = \sin(x)^2$

Solved as second order ode using Kovacic algorithm

Time used: 0.163 (sec)

Solve

$$y'' - y = \sin(x)^2$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.207: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{-x}c_1 + \frac{c_2 e^x}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\sin(x)^2$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{2}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 \cos(2x) + A_3 \sin(2x)$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-5A_2 \cos(2x) - 5A_3 \sin(2x) - A_1 = \sin(x)^2$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{2}, A_2 = \frac{1}{10}, A_3 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{1}{2} + \frac{\cos(2x)}{10}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-x} c_1 + \frac{c_2 e^x}{2} \right) + \left(-\frac{1}{2} + \frac{\cos(2x)}{10} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{1}{2} + \frac{\cos(2x)}{10} + e^{-x} c_1 + \frac{c_2 e^x}{2}$$

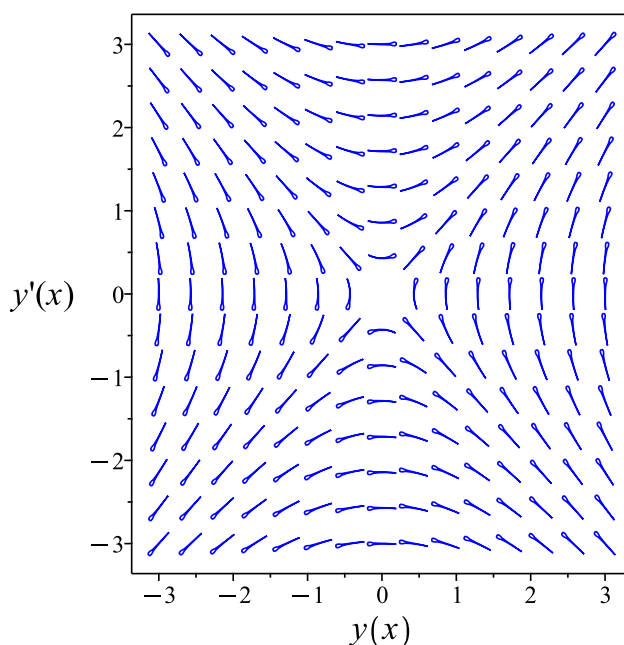


Figure 2.337: Slope field $y'' - y = \sin(x)^2$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 22

```
ode:=diff(diff(y(x),x),x)-y(x) = sin(x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x c_2 + e^{-x} c_1 + \frac{\cos(x)^2}{5} - \frac{3}{5}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
```

```

trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = \sin(x)^2$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$
- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$
- Roots of the characteristic polynomial

$$r = (-1, 1)$$
- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$
- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \sin(x)^2 \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^{-x} \int e^x \sin(x)^2 dx}{2} + \frac{e^x \int e^{-x} \sin(x)^2 dx}{2}$$
 - Compute integrals

$$y_p(x) = -\frac{1}{2} + \frac{\cos(2x)}{10}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x - \frac{1}{2} + \frac{\cos(2x)}{10}$$

✓ **Mathematica.** Time used: 0.047 (sec). Leaf size: 70

```
ode=D[y[x],{x,2}]-y[x]==Sin[x]^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(e^{2x} \int_1^x \frac{1}{2} e^{-K[1]} \sin^2(K[1]) dK[1] + \int_1^x -\frac{1}{2} e^{K[2]} \sin^2(K[2]) dK[2] + c_1 e^{2x} + c_2 \right)$$

✓ **Sympy.** Time used: 0.293 (sec). Leaf size: 22

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - sin(x)**2 + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x + \frac{\cos(2x)}{10} - \frac{1}{2}$$

2.9.10 Problem 20

Local contents

Solved as second order linear constant coeff ode	1725
Solved as second order ode using Kovacic algorithm	1729
✓ Maple	1734
✓ Mathematica	1736
✓ Sympy	1736

Internal problem ID [7996]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 20

Date solved : Thursday, November 27, 2025 at 06:41:56 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.288 (sec)

Solve

$$y'' - y = \frac{1}{(e^{-x} + 1)^2}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = \frac{1}{(e^{-x}+1)^2}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = e^x c_1 + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = e^x c_1 + e^{-x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = e^{-x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & e^{-x} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}(e^{-x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

Therefore

$$W = (e^x)(-e^{-x}) - (e^{-x})(e^x)$$

Which simplifies to

$$W = -2e^x e^{-x}$$

Which simplifies to

$$W = -2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^{-x}}{(e^{-x}+1)^2}}{-2} dx$$

Which simplifies to

$$u_1 = - \int -\frac{\operatorname{sech}\left(\frac{x}{2}\right)^2}{8} dx$$

Hence

$$u_1 = \frac{\tanh\left(\frac{x}{2}\right)}{4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^x}{(e^{-x}+1)^2}}{-2} dx$$

Which simplifies to

$$u_2 = \int -\frac{e^{2x} \operatorname{sech}\left(\frac{x}{2}\right)^2}{8} dx$$

Hence

$$u_2 = -\frac{e^x}{2} + \frac{1}{2+2e^x} + \ln(1+e^x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\tanh\left(\frac{x}{2}\right) e^x}{4} + e^{-x} \left(-\frac{e^x}{2} + \frac{1}{2+2e^x} + \ln(1+e^x) \right)$$

Which simplifies to

$$y_p(x) = -1 + \left(\ln(1+e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{-x} c_2) + \left(-1 + \left(\ln(1+e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -1 + \left(\ln(1+e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} + e^x c_1 + e^{-x} c_2$$

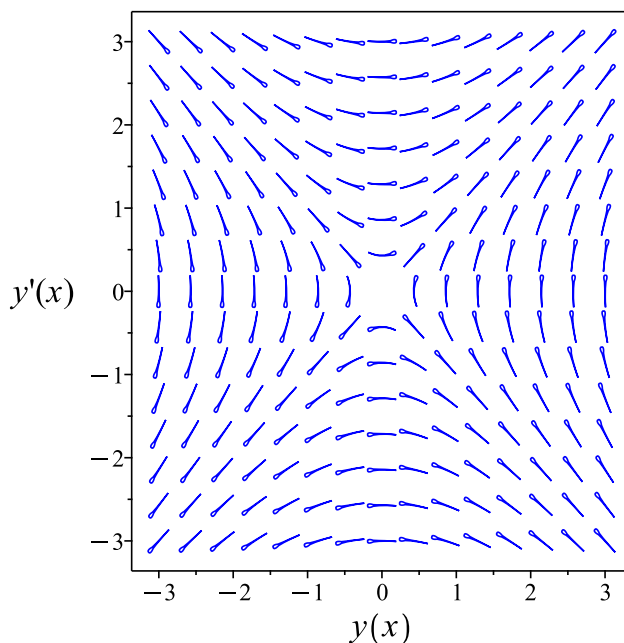


Figure 2.338: Slope field $y'' - y = \frac{1}{(e^{-x}+1)^2}$

Solved as second order ode using Kovacic algorithm

Time used: 0.151 (sec)

Solve

$$y'' - y = \frac{1}{(e^{-x} + 1)^2}$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.209: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{-x}c_1 + \frac{c_2 e^x}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-x}$$

$$y_2 = \frac{e^x}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ \frac{d}{dx}(e^{-x}) & \frac{d}{dx}\left(\frac{e^x}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ -e^{-x} & \frac{e^x}{2} \end{vmatrix}$$

Therefore

$$W = (e^{-x}) \left(\frac{e^x}{2} \right) - \left(\frac{e^x}{2} \right) (-e^{-x})$$

Which simplifies to

$$W = e^x e^{-x}$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^x}{2(e^{-x}+1)^2}}{1} dx$$

Which simplifies to

$$u_1 = - \int \frac{e^{2x} \operatorname{sech} \left(\frac{x}{2} \right)^2}{8} dx$$

Hence

$$u_1 = -\frac{e^x}{2} + \frac{1}{2+2e^x} + \ln(1+e^x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^{-x}}{(e^{-x}+1)^2}}{1} dx$$

Which simplifies to

$$u_2 = \int \frac{\operatorname{sech} \left(\frac{x}{2} \right)^2}{4} dx$$

Hence

$$u_2 = \frac{\tanh \left(\frac{x}{2} \right)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\tanh \left(\frac{x}{2} \right) e^x}{4} + e^{-x} \left(-\frac{e^x}{2} + \frac{1}{2+2e^x} + \ln(1+e^x) \right)$$

Which simplifies to

$$y_p(x) = -1 + \left(\ln(1 + e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-x} c_1 + \frac{c_2 e^x}{2} \right) + \left(-1 + \left(\ln(1 + e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -1 + \left(\ln(1 + e^x) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} + e^{-x} c_1 + \frac{c_2 e^x}{2}$$

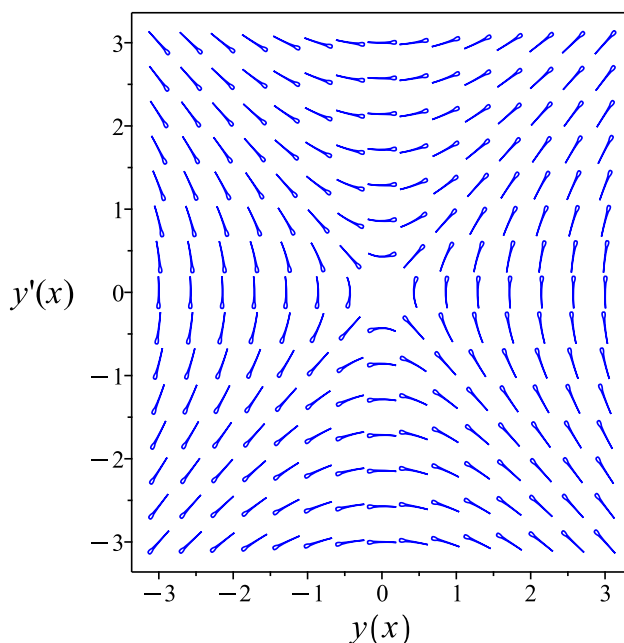


Figure 2.339: Slope field $y'' - y = \frac{1}{(e^{-x} + 1)^2}$

✓ **Maple.** Time used: 0.071 (sec). Leaf size: 41

```
ode:=diff(diff(y(x),x),x)-y(x) = 1/(1+exp(-x))^2;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{e^{-x} \left(4 \ln \left(-\frac{1}{e^x + 1} \right) + 4 \ln(2) - 4c_1 + 1 \right)}{4} - \frac{e^x (-4c_2 - 1)}{4} - 1$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = \frac{1}{(1+e^{-x})^2}$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \frac{1}{(1+e^{-x})^2} \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

- $$y_p(x) = -\frac{e^{-x} \int \frac{e^{2x}}{\cosh\left(\frac{x}{2}\right)^2} dx}{8} + \frac{e^x \int \frac{1}{\cosh\left(\frac{x}{2}\right)^2} dx}{8}$$
- Compute integrals
 - $y_p(x) = -1 - \frac{e^{-x} \left(4 \ln(2) + 4 \ln\left(-\frac{1}{e^x+1}\right) + 1\right)}{4} + \frac{e^x}{4}$
 - Substitute particular solution into general solution to ODE
- $$y(x) = C_1 e^{-x} + C_2 e^x - 1 - \frac{e^{-x} \left(4 \ln(2) + 4 \ln\left(-\frac{1}{e^x+1}\right) + 1\right)}{4} + \frac{e^x}{4}$$

✓ **Mathematica.** Time used: 0.2 (sec). Leaf size: 42

```
ode=D[y[x],{x,2}]-y[x]==(1+Exp[-x])^(-2);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2} e^{-x} (-2e^x + 2 \log(e^x + 1) + 2c_1 e^{2x} + 1 + 2c_2)$$

✓ **Sympy.** Time used: 0.244 (sec). Leaf size: 42

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) + Derivative(y(x), (x, 2)) - 1/(1 + exp(-x))**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \left(C_1 - \frac{1}{2(e^x + 1)}\right) e^x + (C_2 + \log(e^x + 1)) e^{-x} - \frac{1}{2} + \frac{e^{-x}}{2(e^x + 1)}$$

2.9.11 Problem 21

Local contents

Solved as second order linear constant coeff ode	1737
Solved as second order ode using Kovacic algorithm	1741
✓ Maple	1746
✓ Mathematica	1748
✓ Sympy	1748

Internal problem ID [7997]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 21

Date solved : Thursday, November 27, 2025 at 06:42:01 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.168 (sec)

Solve

$$y'' + y = \csc(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 1, f(x) = \csc(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(1)} \\ &= \pm i \end{aligned}$$

Hence

$$\lambda_1 = +i$$

$$\lambda_2 = -i$$

Which simplifies to

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (\cos(x) c_1 + c_2 \sin(x))$$

Or

$$y = \cos(x) c_1 + c_2 \sin(x)$$

Therefore the homogeneous solution y_h is

$$y_h = \cos(x) c_1 + c_2 \sin(x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(x)$$

$$y_2 = \sin(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ \frac{d}{dx}(\cos(x)) & \frac{d}{dx}(\sin(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix}$$

Therefore

$$W = (\cos(x))(\cos(x)) - (\sin(x))(-\sin(x))$$

Which simplifies to

$$W = \cos(x)^2 + \sin(x)^2$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\sin(x) \csc(x)}{1} dx$$

Which simplifies to

$$u_1 = - \int 1 dx$$

Hence

$$u_1 = -x$$

And Eq. (3) becomes

$$u_2 = \int \frac{\csc(x) \cos(x)}{1} dx$$

Which simplifies to

$$u_2 = \int \cot(x) dx$$

Hence

$$u_2 = \ln(\sin(x))$$

Therefore the particular solution, from equation (1) is

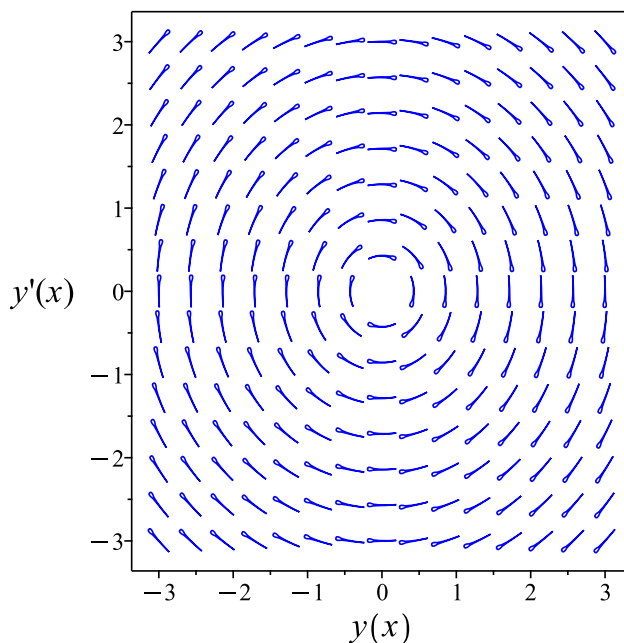
$$y_p(x) = -x \cos(x) + \sin(x) \ln(\sin(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (\cos(x) c_1 + c_2 \sin(x)) + (-x \cos(x) + \sin(x) \ln(\sin(x))) \end{aligned}$$

Summary of solutions found

$$y = -x \cos(x) + \sin(x) \ln(\sin(x)) + \cos(x) c_1 + c_2 \sin(x)$$

Figure 2.340: Slope field $y'' + y = \csc(x)$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.119 (sec)

Solve

$$y'' + y = \csc(x)$$

Writing the ode as

$$y'' + y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.211: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(x) \int \frac{1}{\cos(x)^2} dx \\ &= \cos(x) (\tan(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(x)) + c_2 (\cos(x) (\tan(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \cos(x) c_1 + c_2 \sin(x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(x)$$

$$y_2 = \sin(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ \frac{d}{dx}(\cos(x)) & \frac{d}{dx}(\sin(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix}$$

Therefore

$$W = (\cos(x))(\cos(x)) - (\sin(x))(-\sin(x))$$

Which simplifies to

$$W = \cos(x)^2 + \sin(x)^2$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\sin(x) \csc(x)}{1} dx$$

Which simplifies to

$$u_1 = - \int 1 dx$$

Hence

$$u_1 = -x$$

And Eq. (3) becomes

$$u_2 = \int \frac{\csc(x) \cos(x)}{1} dx$$

Which simplifies to

$$u_2 = \int \cot(x) dx$$

Hence

$$u_2 = \ln(\sin(x))$$

Therefore the particular solution, from equation (1) is

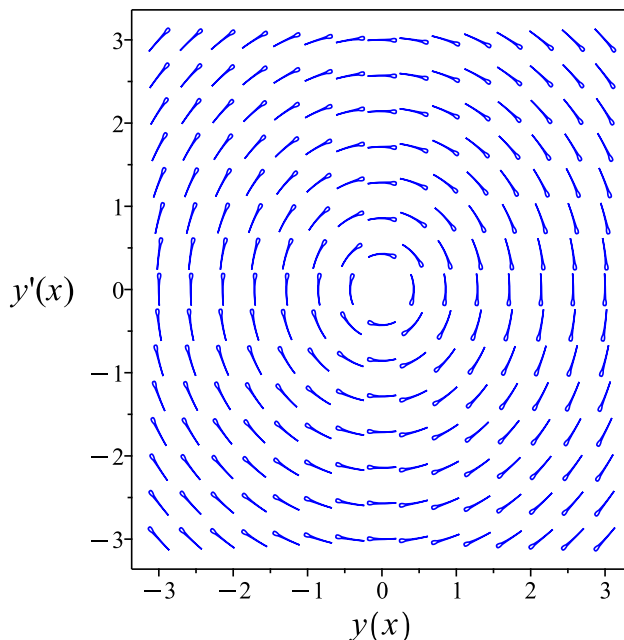
$$y_p(x) = -x \cos(x) + \sin(x) \ln(\sin(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (\cos(x) c_1 + c_2 \sin(x)) + (-x \cos(x) + \sin(x) \ln(\sin(x))) \end{aligned}$$

Summary of solutions found

$$y = -x \cos(x) + \sin(x) \ln(\sin(x)) + \cos(x) c_1 + c_2 \sin(x)$$

Figure 2.341: Slope field $y'' + y = \csc(x)$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 24

```
ode:=diff(diff(y(x),x),x)+y(x) = csc(x);
dsolve(ode,y(x), singsol=all);
```

$$y = -\ln(\csc(x)) \sin(x) + (-x + c_1) \cos(x) + \sin(x) c_2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + y(x) = \csc(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 1 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-4})}{2}$$

- Roots of the characteristic polynomial

$$r = (-i, i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \csc(x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 1$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\cos(x) \int 1 dx + \sin(x) \int \cot(x) dx$$

- Compute integrals

$$y_p(x) = -\cos(x)x + \sin(x) \ln(\sin(x))$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) - \cos(x)x + \sin(x) \ln(\sin(x))$$

✓ **Mathematica.** Time used: 0.061 (sec). Leaf size: 66

```
ode=D[y[x],{x,2}]-y[x]==Csc[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(e^{2x} \int_1^x \frac{1}{2} e^{-K[1]} \csc(K[1]) dK[1] + \int_1^x -\frac{1}{2} e^{K[2]} \csc(K[2]) dK[2] + c_1 e^{2x} + c_2 \right)$$

✓ **Sympy.** Time used: 0.135 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x) + Derivative(y(x), (x, 2)) - 1/sin(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 - x) \cos(x) + (C_2 + \log(\sin(x))) \sin(x)$$

2.9.12 Problem 22

Local contents

Solved as second order linear constant coeff ode	1749
Solved as second order ode using Kovacic algorithm	1753
✓ Maple	1758
✓ Mathematica	1760
✓ Sympy	1760

Internal problem ID [7998]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 14. Linear equations with constant coefficients. Supplementary problems. Page 92

Problem number : 22

Date solved : Thursday, November 27, 2025 at 06:42:07 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.250 (sec)

Solve

$$y'' - 3y' + 2y = \sin(e^{-x})$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -3, C = 2, f(x) = \sin(e^{-x})$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -3, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 3\lambda e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 3\lambda + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -3, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{3}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-3^2 - (4)(1)(2)} \\ &= \frac{3}{2} \pm \frac{1}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= \frac{3}{2} + \frac{1}{2} \\ \lambda_2 &= \frac{3}{2} - \frac{1}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= 2 \\ \lambda_2 &= 1 \end{aligned}$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(2)x} + c_2 e^{(1)x} \end{aligned}$$

Or

$$y = e^{2x} c_1 + c_2 e^x$$

Therefore the homogeneous solution y_h is

$$y_h = e^{2x} c_1 + c_2 e^x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{2x}$$

$$y_2 = e^x$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{2x} & e^x \\ \frac{d}{dx}(e^{2x}) & \frac{d}{dx}(e^x) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{2x} & e^x \\ 2e^{2x} & e^x \end{vmatrix}$$

Therefore

$$W = (e^{2x})(e^x) - (e^x)(2e^{2x})$$

Which simplifies to

$$W = -e^x e^{2x}$$

Which simplifies to

$$W = -e^{3x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^x \sin(e^{-x})}{-e^{3x}} dx$$

Which simplifies to

$$u_1 = - \int -\sin(e^{-x}) e^{-2x} dx$$

Hence

$$u_1 = -\frac{e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - e^{-x} + 2 \tan\left(\frac{e^{-x}}{2}\right)}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{2x} \sin(e^{-x})}{-e^{3x}} dx$$

Which simplifies to

$$u_2 = \int -e^{-x} \sin(e^{-x}) dx$$

Hence

$$u_2 = -\cos(e^{-x})$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\frac{\left(e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - e^{-x} + 2 \tan\left(\frac{e^{-x}}{2}\right)\right) e^{2x}}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2} - e^x \cos(e^{-x})$$

Which simplifies to

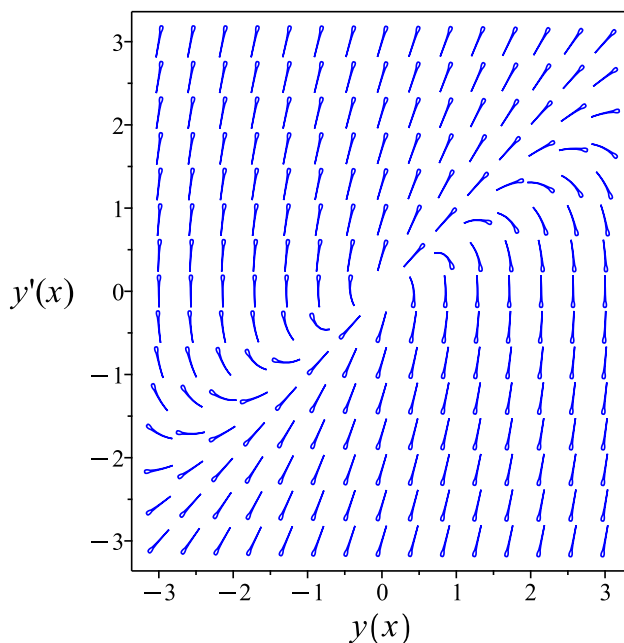
$$y_p(x) = -e^{2x} \sin(e^{-x})$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x} c_1 + c_2 e^x) + (-e^{2x} \sin(e^{-x})) \end{aligned}$$

Summary of solutions found

$$y = -e^{2x} \sin(e^{-x}) + e^{2x} c_1 + c_2 e^x$$

Figure 2.342: Slope field $y'' - 3y' + 2y = \sin(e^{-x})$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.162 (sec)

Solve

$$y'' - 3y' + 2y = \sin(e^{-x})$$

Writing the ode as

$$y'' - 3y' + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -3 \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{4} \quad (6)$$

Comparing the above to (5) shows that

$$s = 1$$

$$t = 4$$

Therefore eq. (4) becomes

$$z''(x) = \frac{z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.213: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{1}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3}{1} dx} \\ &= z_1 e^{\frac{3x}{2}} \\ &= z_1 \left(e^{\frac{3x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3x}}{(y_1)^2} dx \\ &= y_1 (e^{3x} e^{-2x}) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 (e^x (e^{3x} e^{-2x})) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 3y' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^x c_1 + e^{2x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = e^{2x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & e^{2x} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}(e^{2x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix}$$

Therefore

$$W = (e^x) (2 e^{2x}) - (e^{2x}) (e^x)$$

Which simplifies to

$$W = e^x e^{2x}$$

Which simplifies to

$$W = e^{3x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{2x} \sin(e^{-x})}{e^{3x}} dx$$

Which simplifies to

$$u_1 = - \int e^{-x} \sin(e^{-x}) dx$$

Hence

$$u_1 = - \cos(e^{-x})$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^x \sin(e^{-x})}{e^{3x}} dx$$

Which simplifies to

$$u_2 = \int \sin(e^{-x}) e^{-2x} dx$$

Hence

$$u_2 = \frac{-e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 + e^{-x} - 2 \tan\left(\frac{e^{-x}}{2}\right)}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -e^x \cos(e^{-x}) + \frac{e^{2x} \left(-e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 + e^{-x} - 2 \tan\left(\frac{e^{-x}}{2}\right) \right)}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2}$$

Which simplifies to

$$y_p(x) = -e^{2x} \sin(e^{-x})$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^x c_1 + e^{2x} c_2) + (-e^{2x} \sin(e^{-x})) \end{aligned}$$

Summary of solutions found

$$y = -e^{2x} \sin(e^{-x}) + e^x c_1 + e^{2x} c_2$$

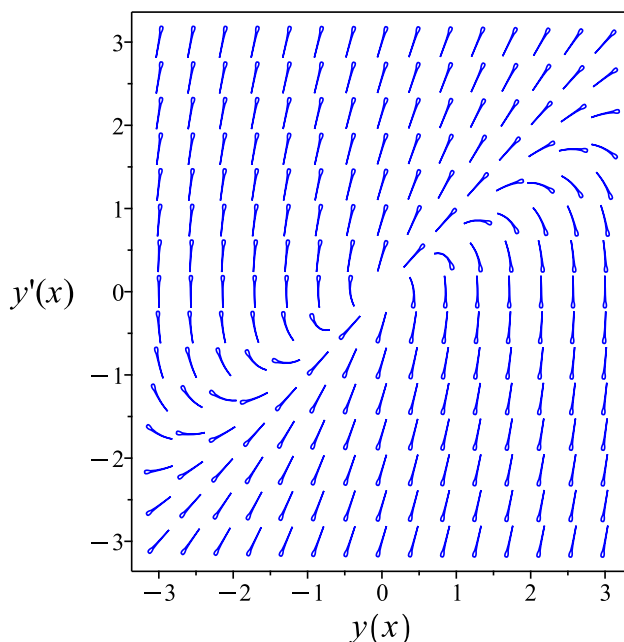


Figure 2.343: Slope field $y'' - 3y' + 2y = \sin(e^{-x})$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 22

```
ode:=2*y(x)-3*diff(y(x),x)+diff(diff(y(x),x),x) = sin(exp(-x));
dsolve(ode,y(x), singsol=all);
```

$$y = (e^x c_1 - e^x \sin(e^{-x}) + c_2) e^x$$

Maple trace

Methods for second order ODEs:

```

--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
<- double symmetry of the form [xi=0, eta=F(x)] successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 3\frac{d}{dx}y(x) + 2y(x) = \sin(e^{-x})$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^2 - 3r + 2 = 0$$
- Factor the characteristic polynomial

$$(r - 1)(r - 2) = 0$$
- Roots of the characteristic polynomial

$$r = (1, 2)$$
- 1st solution of the homogeneous ODE

$$y_1(x) = e^x$$
- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{2x}$$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^x + C_2 e^{2x} + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \sin(e^{-x}) \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{3x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = e^x \left(- \int e^{-x} \sin(e^{-x}) dx + e^x \int \sin(e^{-x}) e^{-2x} dx \right)$$
 - Compute integrals

$$y_p(x) = -e^{2x} \sin(e^{-x})$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^x + C_2 e^{2x} - e^{2x} \sin(e^{-x})$$

✓ **Mathematica.** Time used: 0.037 (sec). Leaf size: 65

```
ode=D[y[x],{x,2}]-3*D[y[x],x]+2*y[x]==Sin[Exp[-x]];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x \left(\int_1^x -e^{-K[1]} \sin(e^{-K[1]}) dK[1] + e^x \int_1^x e^{-2K[2]} \sin(e^{-K[2]}) dK[2] + c_2 e^x + c_1 \right)$$

✓ **Sympy.** Time used: 2.753 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*y(x) - sin(exp(-x)) - 3*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = (C_1 + (C_2 - \sin(e^{-x})) e^x) e^x$$

2.10 Chapter 15. Linear equations with constant coefficients (Variation of parameters). Supplementary problems. Page 98

Local contents

2.10.1 Problem 10	1762
2.10.2 Problem 11	1774
2.10.3 Problem 12	1786
2.10.4 Problem 13	1797
2.10.5 Problem 14	1809
2.10.6 Problem 15	1821
2.10.7 Problem 16	1831
2.10.8 Problem 17	1841
2.10.9 Problem 18	1851
2.10.10 Problem 19	1861
2.10.11 Problem 20	1867
2.10.12 Problem 21	1877

2.10.1 Problem 10

Local contents

Solved as second order linear constant coeff ode	1762
Solved as second order ode using Kovacic algorithm	1766
✓ Maple	1771
✓ Mathematica	1773
✓ Sympy	1773

Internal problem ID [7999]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 10

Date solved : Thursday, November 27, 2025 at 06:42:12 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.169 (sec)

Solve

$$y'' + y = \csc(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 1, f(x) = \csc(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(1)} \\ &= \pm i \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= +i \\ \lambda_2 &= -i \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= i \\ \lambda_2 &= -i \end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (\cos(x) c_1 + c_2 \sin(x))$$

Or

$$y = \cos(x) c_1 + c_2 \sin(x)$$

Therefore the homogeneous solution y_h is

$$y_h = \cos(x) c_1 + c_2 \sin(x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(x)$$

$$y_2 = \sin(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ \frac{d}{dx}(\cos(x)) & \frac{d}{dx}(\sin(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix}$$

Therefore

$$W = (\cos(x))(\cos(x)) - (\sin(x))(-\sin(x))$$

Which simplifies to

$$W = \cos(x)^2 + \sin(x)^2$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\sin(x) \csc(x)}{1} dx$$

Which simplifies to

$$u_1 = - \int 1 dx$$

Hence

$$u_1 = -x$$

And Eq. (3) becomes

$$u_2 = \int \frac{\csc(x) \cos(x)}{1} dx$$

Which simplifies to

$$u_2 = \int \cot(x) dx$$

Hence

$$u_2 = \ln(\sin(x))$$

Therefore the particular solution, from equation (1) is

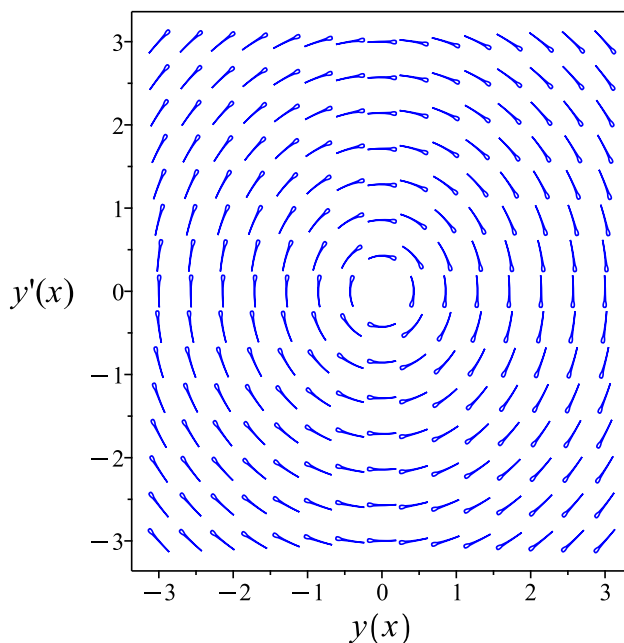
$$y_p(x) = -x \cos(x) + \sin(x) \ln(\sin(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (\cos(x) c_1 + c_2 \sin(x)) + (-x \cos(x) + \sin(x) \ln(\sin(x))) \end{aligned}$$

Summary of solutions found

$$y = -x \cos(x) + \sin(x) \ln(\sin(x)) + \cos(x) c_1 + c_2 \sin(x)$$

Figure 2.344: Slope field $y'' + y = \csc(x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.119 (sec)

Solve

$$y'' + y = \csc(x)$$

Writing the ode as

$$y'' + y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.215: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(x) \int \frac{1}{\cos(x)^2} dx \\ &= \cos(x) (\tan(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(x)) + c_2 (\cos(x) (\tan(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \cos(x) c_1 + c_2 \sin(x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(x)$$

$$y_2 = \sin(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ \frac{d}{dx}(\cos(x)) & \frac{d}{dx}(\sin(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix}$$

Therefore

$$W = (\cos(x))(\cos(x)) - (\sin(x))(-\sin(x))$$

Which simplifies to

$$W = \cos(x)^2 + \sin(x)^2$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\sin(x) \csc(x)}{1} dx$$

Which simplifies to

$$u_1 = - \int 1 dx$$

Hence

$$u_1 = -x$$

And Eq. (3) becomes

$$u_2 = \int \frac{\csc(x) \cos(x)}{1} dx$$

Which simplifies to

$$u_2 = \int \cot(x) dx$$

Hence

$$u_2 = \ln(\sin(x))$$

Therefore the particular solution, from equation (1) is

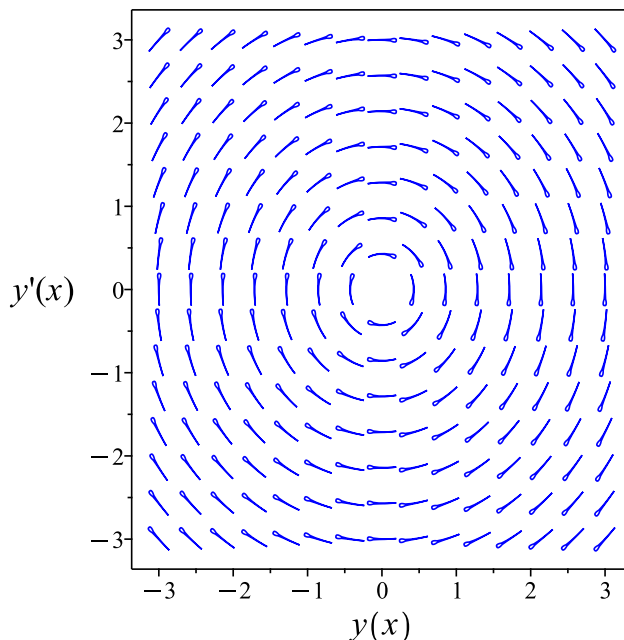
$$y_p(x) = -x \cos(x) + \sin(x) \ln(\sin(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (\cos(x) c_1 + c_2 \sin(x)) + (-x \cos(x) + \sin(x) \ln(\sin(x))) \end{aligned}$$

Summary of solutions found

$$y = -x \cos(x) + \sin(x) \ln(\sin(x)) + \cos(x) c_1 + c_2 \sin(x)$$

Figure 2.345: Slope field $y'' + y = \csc(x)$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 24

```
ode:=diff(diff(y(x),x),x)+y(x) = csc(x);
dsolve(ode,y(x), singsol=all);
```

$$y = -\ln(\csc(x)) \sin(x) + (c_1 - x) \cos(x) + \sin(x) c_2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + y(x) = \csc(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 1 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-4})}{2}$$

- Roots of the characteristic polynomial

$$r = (-i, i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \csc(x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 1$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\cos(x) \int 1 dx + \sin(x) \int \cot(x) dx$$

- Compute integrals

$$y_p(x) = -\cos(x)x + \sin(x) \ln(\sin(x))$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) - \cos(x)x + \sin(x) \ln(\sin(x))$$

✓ **Mathematica.** Time used: 0.013 (sec). Leaf size: 24

```
ode=D[y[x],{x,2}]+y[x]==Csc[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow (-x + c_1) \cos(x) + \sin(x)(\log(\sin(x)) + c_2)$$

✓ **Sympy.** Time used: 0.138 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x) + Derivative(y(x), (x, 2)) - 1/sin(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 - x) \cos(x) + (C_2 + \log(\sin(x))) \sin(x)$$

2.10.2 Problem 11

Local contents

Solved as second order linear constant coeff ode	1774
Solved as second order ode using Kovacic algorithm	1778
✓ Maple	1783
✓ Mathematica	1785
✓ Sympy	1785

Internal problem ID [8000]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 11

Date solved : Thursday, November 27, 2025 at 06:42:18 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.260 (sec)

Solve

$$y'' + 4y = 4 \sec(x)^2$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 4, f(x) = 4 \sec(x)^2$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 4y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 4$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 4e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(4)} \\ &= \pm 2i \end{aligned}$$

Hence

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

Which simplifies to

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 2$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(2x) + c_2 \sin(2x))$$

Or

$$y = c_1 \cos(2x) + c_2 \sin(2x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(2x) + c_2 \sin(2x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(2x)$$

$$y_2 = \sin(2x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(2x) & \sin(2x) \\ \frac{d}{dx}(\cos(2x)) & \frac{d}{dx}(\sin(2x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix}$$

Therefore

$$W = (\cos(2x))(2\cos(2x)) - (\sin(2x))(-2\sin(2x))$$

Which simplifies to

$$W = 2\cos(2x)^2 + 2\sin(2x)^2$$

Which simplifies to

$$W = 2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{4\sin(2x)\sec(x)^2}{2} dx$$

Which simplifies to

$$u_1 = - \int 4 \tan (x) dx$$

Hence

$$u_1 = 4 \ln (\cos (x))$$

And Eq. (3) becomes

$$u_2 = \int \frac{4 \cos (2x) \sec (x)^2}{2} dx$$

Which simplifies to

$$u_2 = \int 2 \cos (2x) \sec (x)^2 dx$$

Hence

$$u_2 = 4x - 2 \tan (x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = 4 \ln (\cos (x)) \cos (2x) + \sin (2x) (4x - 2 \tan (x))$$

Which simplifies to

$$y_p(x) = (8 \cos (x)^2 - 4) \ln (\cos (x)) + 8 \sin (x) \cos (x) x - 4 \sin (x)^2$$

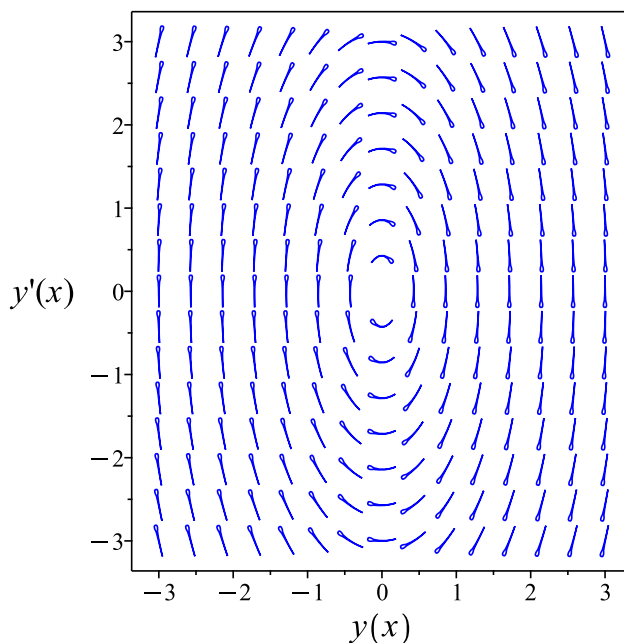
Therefore the general solution is

$$y = y_h + y_p$$

$$= (c_1 \cos (2x) + c_2 \sin (2x)) + ((8 \cos (x)^2 - 4) \ln (\cos (x)) + 8 \sin (x) \cos (x) x - 4 \sin (x)^2)$$

Summary of solutions found

$$y = (8 \cos (x)^2 - 4) \ln (\cos (x)) + 8 \sin (x) \cos (x) x - 4 \sin (x)^2 + c_1 \cos (2x) + c_2 \sin (2x)$$

Figure 2.346: Slope field $y'' + 4y = 4 \sec(x)^2$

Solved as second order ode using Kovacic algorithm

Time used: 0.161 (sec)

Solve

$$y'' + 4y = 4 \sec(x)^2$$

Writing the ode as

$$y'' + 4y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 4 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-4}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -4 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -4z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.217: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -4$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(2x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(2x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(2x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(2x) \int \frac{1}{\cos^2(2x)} dx \\ &= \cos(2x) \left(\frac{\tan(2x)}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(2x)) + c_2 \left(\cos(2x) \left(\frac{\tan(2x)}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 4y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(2x)$$

$$y_2 = \frac{\sin(2x)}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(2x) & \frac{\sin(2x)}{2} \\ \frac{d}{dx}(\cos(2x)) & \frac{d}{dx}\left(\frac{\sin(2x)}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(2x) & \frac{\sin(2x)}{2} \\ -2\sin(2x) & \cos(2x) \end{vmatrix}$$

Therefore

$$W = (\cos(2x))(\cos(2x)) - \left(\frac{\sin(2x)}{2}\right)(-2\sin(2x))$$

Which simplifies to

$$W = \cos(2x)^2 + \sin(2x)^2$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{2\sin(2x)\sec(x)^2}{1} dx$$

Which simplifies to

$$u_1 = - \int 4\tan(x) dx$$

Hence

$$u_1 = 4\ln(\cos(x))$$

And Eq. (3) becomes

$$u_2 = \int \frac{4\cos(2x)\sec(x)^2}{1} dx$$

Which simplifies to

$$u_2 = \int 4\cos(2x)\sec(x)^2 dx$$

Hence

$$u_2 = 8x - 4\tan(x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = 4\ln(\cos(x))\cos(2x) + \frac{\sin(2x)(8x - 4\tan(x))}{2}$$

Which simplifies to

$$y_p(x) = (8 \cos(x)^2 - 4) \ln(\cos(x)) + 8 \sin(x) \cos(x) x - 4 \sin(x)^2$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2} \right) + ((8 \cos(x)^2 - 4) \ln(\cos(x)) + 8 \sin(x) \cos(x) x - 4 \sin(x)^2)$$

Summary of solutions found

$$y = (8 \cos(x)^2 - 4) \ln(\cos(x)) + 8 \sin(x) \cos(x) x - 4 \sin(x)^2 + c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2}$$

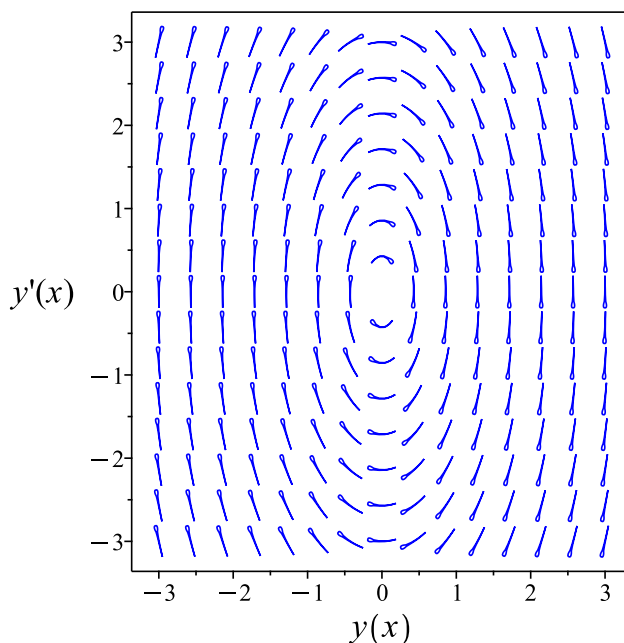


Figure 2.347: Slope field $y'' + 4y = 4 \sec(x)^2$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 44

```
ode:=4*y(x)+diff(diff(y(x),x),x) = 4*sec(x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = (-8 \cos(x)^2 + 4) \ln(\sec(x)) + 2c_1 \cos(x)^2 + 8 \sin(x) \left(x + \frac{c_2}{4} \right) \cos(x) - 4 \sin(x)^2 - c_1$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 4y(x) = 4\sec(x)^2$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 4 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-16})}{2}$$

- Roots of the characteristic polynomial

$$r = (-2i, 2i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(2x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(2x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(2x) + C_2 \sin(2x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = 4\sec(x)^2 \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -4 \cos(2x) \int \tan(x) dx + 2 \sin(2x) \int \cos(2x) \sec(x)^2 dx$$

- Compute integrals

$$y_p(x) = (8 \cos(x)^2 - 4) \ln(\cos(x)) + 8 \sin(x) \cos(x) x - 4 \sin(x)^2$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(2x) + C_2 \sin(2x) + (8 \cos(x)^2 - 4) \ln(\cos(x)) + 8 \sin(x) \cos(x) x - 4 \sin(x)^2$$

✓ **Mathematica.** Time used: 0.041 (sec). Leaf size: 44

```
ode=D[y[x],{x,2}]+4*y[x]==4*Sec[x]^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow 2 \sin(2x) \arctan(\tan(x)) + 2x \sin(2x) + c_2 \sin(2x) + \cos(2x)(4 \log(\cos(x)) + 2 + c_1) - 2$$

✓ **Sympy.** Time used: 0.219 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*y(x) + Derivative(y(x), (x, 2)) - 4/cos(x)**2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = (C_1 + 4 \log(\cos(x))) \cos(2x) + (C_2 + 4x - 2 \tan(x)) \sin(2x)$$

2.10.3 Problem 12

Local contents

Solved as second order linear constant coeff ode	1786
Solved as second order ode using Kovacic algorithm	1790
✓ Maple	1795
✓ Mathematica	1796
✗ Sympy	1796

Internal problem ID [8001]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 12

Date solved : Thursday, November 27, 2025 at 07:32:14 AM

CAS classification : [[_2nd_order, _linear, _nonhomogeneous]]

Solved as second order linear constant coeff ode

Time used: 0.246 (sec)

Solve

$$y'' - 4y' + 3y = \frac{1}{1 + e^{-x}}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -4, C = 3, f(x) = \frac{1}{1+e^{-x}}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 3y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 3$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} + 3e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda + 3 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 3$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-4^2 - (4)(1)(3)} \\ &= 2 \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = 2 + 1$$

$$\lambda_2 = 2 - 1$$

Which simplifies to

$$\lambda_1 = 3$$

$$\lambda_2 = 1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(3)x} + c_2 e^{(1)x}$$

Or

$$y = c_1 e^{3x} + c_2 e^x$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^{3x} + c_2 e^x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{3x}$$

$$y_2 = e^x$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{3x} & e^x \\ \frac{d}{dx}(e^{3x}) & \frac{d}{dx}(e^x) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{3x} & e^x \\ 3e^{3x} & e^x \end{vmatrix}$$

Therefore

$$W = (e^{3x})(e^x) - (e^x)(3e^{3x})$$

Which simplifies to

$$W = -2e^{3x}e^x$$

Which simplifies to

$$W = -2e^{4x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^x}{-2e^{4x}} dx$$

Which simplifies to

$$u_1 = - \int -\frac{e^{-3x}}{2 + 2e^{-x}} dx$$

Hence

$$u_1 = -\frac{\ln(e^x + 1)}{2} - \frac{e^{-2x}}{4} + \frac{\ln(e^x)}{2} + \frac{e^{-x}}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^{3x}}{1+e^{-x}}}{-2e^{4x}} dx$$

Which simplifies to

$$u_2 = \int -\frac{1}{2e^x + 2} dx$$

Hence

$$u_2 = \frac{\ln(e^x + 1)}{2} - \frac{\ln(e^x)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{\ln(e^x + 1)}{2} - \frac{e^{-2x}}{4} + \frac{\ln(e^x)}{2} + \frac{e^{-x}}{2} \right) e^{3x} + e^x \left(\frac{\ln(e^x + 1)}{2} - \frac{\ln(e^x)}{2} \right)$$

Which simplifies to

$$y_p(x) = -\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= (c_1 e^{3x} + c_2 e^x) + \left(-\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4} \right)$$

Summary of solutions found

$$y = -\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4} + c_1 e^{3x} + c_2 e^x$$

Solved as second order ode using Kovacic algorithm

Time used: 0.174 (sec)

Solve

$$y'' - 4y' + 3y = \frac{1}{1 + e^{-x}}$$

Writing the ode as

$$y'' - 4y' + 3y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 3 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.219: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{4x} e^{-2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 \left(e^x \left(\frac{e^{4x} e^{-2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 3y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^x + \frac{c_2 e^{3x}}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = \frac{e^{3x}}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & \frac{e^{3x}}{2} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}\left(\frac{e^{3x}}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^x & \frac{e^{3x}}{2} \\ e^x & \frac{3e^{3x}}{2} \end{vmatrix}$$

Therefore

$$W = (e^x) \left(\frac{3e^{3x}}{2} \right) - \left(\frac{e^{3x}}{2} \right) (e^x)$$

Which simplifies to

$$W = e^{3x} e^x$$

Which simplifies to

$$W = e^{4x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^{3x}}{2+2e^{-x}}}{e^{4x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{1}{2e^x + 2} dx$$

Hence

$$u_1 = \frac{\ln(e^x + 1)}{2} - \frac{\ln(e^x)}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^x}{1+e^{-x}}}{e^{4x}} dx$$

Which simplifies to

$$u_2 = \int \frac{e^{-2x}}{e^x + 1} dx$$

Hence

$$u_2 = -\ln(e^x + 1) - \frac{e^{-2x}}{2} + \ln(e^x) + e^{-x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = e^x \left(\frac{\ln(e^x + 1)}{2} - \frac{\ln(e^x)}{2} \right) + \frac{e^{3x} \left(-\ln(e^x + 1) - \frac{e^{-2x}}{2} + \ln(e^x) + e^{-x} \right)}{2}$$

Which simplifies to

$$y_p(x) = -\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(c_1 e^x + \frac{c_2 e^{3x}}{2} \right) + \left(-\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4} \right)$$

Summary of solutions found

$$y = -\frac{e^x(2(\ln(e^x + 1) - \ln(e^x))e^{2x} - 2e^x - 2\ln(e^x + 1) + 2\ln(e^x) + 1)}{4} + c_1 e^x + \frac{c_2 e^{3x}}{2}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 45

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x)+3*y(x) = 1/(1+exp(-x));
dsolve(ode,y(x), singsol=all);
```

$$y = e^x \left(c_2 - \frac{1}{4} - \frac{\ln(e^x)}{2} + \frac{\ln(e^x + 1)}{2} + \frac{e^x}{2} + \left(c_1 + \frac{\ln(e^x)}{2} - \frac{\ln(e^x + 1)}{2} \right) e^{2x} \right)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 3y(x) = \frac{1}{1+e^{-x}}$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 4r + 3 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r - 3) = 0$$

- Roots of the characteristic polynomial

$$r = (1, 3)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^x$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{3x}$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^x + C2 e^{3x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = \frac{1}{1+e^{-x}} \right]$$
- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{bmatrix}$$
- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2e^{4x}$$
- Substitute functions into equation for $y_p(x)$

$$y_p(x) = \frac{e^x \left(\int \frac{e^{-2x}}{e^x+1} dx e^{2x} - \int \frac{1}{e^x+1} dx \right)}{2}$$
- Compute integrals

$$y_p(x) = \frac{e^x \left(-\frac{1}{2} - \ln(e^x) + \ln(e^x+1) + e^x + (\ln(e^x) - \ln(e^x+1))e^{2x} \right)}{2}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^x + C_2 e^{3x} + \frac{e^x \left(-\frac{1}{2} - \ln(e^x) + \ln(e^x+1) + e^x + (\ln(e^x) - \ln(e^x+1))e^{2x} \right)}{2}$$

✓ **Mathematica.** Time used: 0.196 (sec). Leaf size: 49

```
ode=D[y[x],{x,2}]-4*D[y[x],x]+3*y[x]==1/(1+Exp[-x]);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{4}e^x(-4(e^{2x}-1)\operatorname{arctanh}(2e^x+1)+2e^x+4c_2e^{2x}-1+4c_1)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*y(x) - 4*Derivative(y(x), x) + Derivative(y(x), (x, 2)) - 1/(1 + exp(-x)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (3*y(x)*exp(x) + 3*y(x)
+ exp(x)*Derivative(y(x), (x, 2)) - exp(x) + Derivative(y(x), (x, 2)))/(4*(
exp(x) + 1)) cannot be solved by the factorable group method
```


2.10.4 Problem 13

Local contents

Solved as second order linear constant coeff ode	1797
Solved as second order ode using Kovacic algorithm	1801
✓ Maple	1807
✓ Mathematica	1808
✓ Sympy	1808

Internal problem ID [8002]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 13

Date solved : Thursday, November 27, 2025 at 07:32:14 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.284 (sec)

Solve

$$y'' - y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = e^{-x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & e^{-x} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}(e^{-x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

Therefore

$$W = (e^x)(-e^{-x}) - (e^{-x})(e^x)$$

Which simplifies to

$$W = -2e^x e^{-x}$$

Which simplifies to

$$W = -2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{-x}(e^{-x} \sin(e^{-x}) + \cos(e^{-x}))}{-2} dx$$

Which simplifies to

$$u_1 = - \int \left(-\frac{e^{-x} \cos(e^{-x})}{2} - \frac{e^{-2x} \sin(e^{-x})}{2} \right) dx$$

Hence

$$u_1 = -\frac{e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - e^{-x} + 2 \tan\left(\frac{e^{-x}}{2}\right)}{2\left(1 + \tan\left(\frac{e^{-x}}{2}\right)^2\right)} - \frac{\sin(e^{-x})}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^x(e^{-x} \sin(e^{-x}) + \cos(e^{-x}))}{-2} dx$$

Which simplifies to

$$u_2 = \int \left(-\frac{\cos(e^{-x}) e^x}{2} - \frac{\sin(e^{-x})}{2} \right) dx$$

Hence

$$u_2 = -\frac{e^{x-ie^{-x}}}{4} - \frac{e^{x+ie^{-x}}}{4}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - e^{-x} + 2 \tan\left(\frac{e^{-x}}{2}\right)}{2\left(1 + \tan\left(\frac{e^{-x}}{2}\right)^2\right)} - \frac{\sin(e^{-x})}{2} \right) e^x + e^{-x} \left(-\frac{e^{x-ie^{-x}}}{4} - \frac{e^{x+ie^{-x}}}{4} \right)$$

Which simplifies to

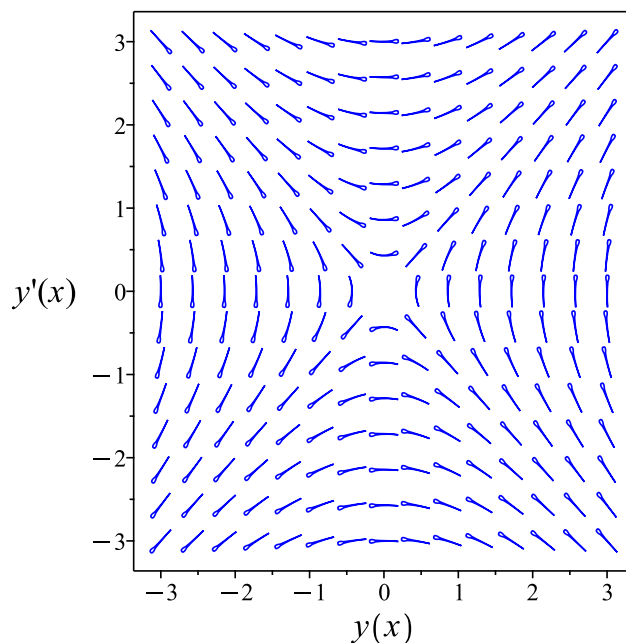
$$y_p(x) = -e^x \sin(e^{-x})$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x} c_2) + (-e^x \sin(e^{-x})) \end{aligned}$$

Summary of solutions found

$$y = -e^x \sin(e^{-x}) + c_1 e^x + e^{-x} c_2$$

Figure 2.348: Slope field $y'' - y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$

Solved as second order ode using Kovacic algorithm

Time used: 0.188 (sec)

Solve

$$y'' - y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.221: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-x}$$

$$y_2 = \frac{e^x}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ \frac{d}{dx}(e^{-x}) & \frac{d}{dx}\left(\frac{e^x}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ -e^{-x} & \frac{e^x}{2} \end{vmatrix}$$

Therefore

$$W = (e^{-x}) \left(\frac{e^x}{2} \right) - \left(\frac{e^x}{2} \right) (-e^{-x})$$

Which simplifies to

$$W = e^x e^{-x}$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^x (e^{-x} \sin(e^{-x}) + \cos(e^{-x}))}{2}}{1} dx$$

Which simplifies to

$$u_1 = - \int \left(\frac{\cos(e^{-x}) e^x}{2} + \frac{\sin(e^{-x})}{2} \right) dx$$

Hence

$$u_1 = -\frac{e^{x-ie^{-x}}}{4} - \frac{e^{x+ie^{-x}}}{4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{-x} (e^{-x} \sin(e^{-x}) + \cos(e^{-x}))}{1} dx$$

Which simplifies to

$$u_2 = \int (e^{-x} \cos(e^{-x}) + e^{-2x} \sin(e^{-x})) dx$$

Hence

$$u_2 = \frac{-e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - 2 \tan\left(\frac{e^{-x}}{2}\right) + e^{-x}}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2} - \sin(e^{-x})$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = e^{-x} \left(-\frac{e^{x-ie^{-x}}}{4} - \frac{e^{x+ie^{-x}}}{4} \right) + \frac{e^x \left(\frac{-e^{-x} \tan\left(\frac{e^{-x}}{2}\right)^2 - 2 \tan\left(\frac{e^{-x}}{2}\right) + e^{-x}}{1 + \tan\left(\frac{e^{-x}}{2}\right)^2} - \sin(e^{-x}) \right)}{2}$$

Which simplifies to

$$y_p(x) = -e^x \sin(e^{-x})$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + (-e^x \sin(e^{-x})) \end{aligned}$$

Summary of solutions found

$$y = -e^x \sin(e^{-x}) + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

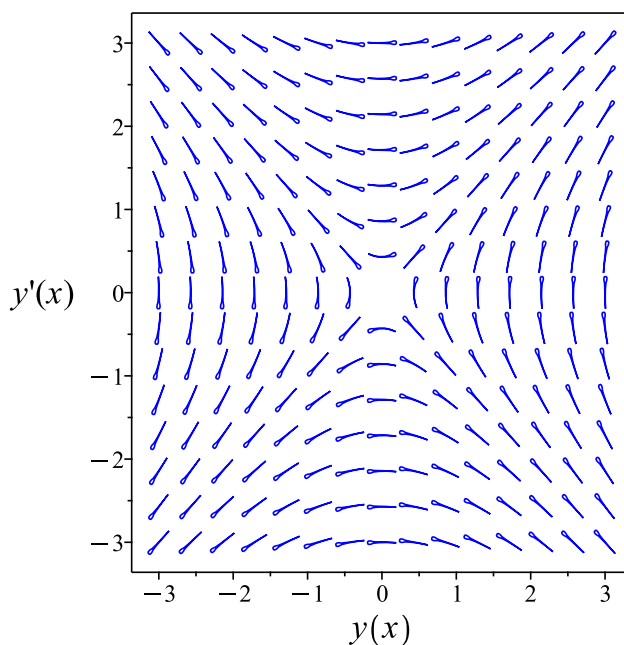


Figure 2.349: Slope field $y'' - y = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 24

```
ode:=diff(diff(y(x),x),x)-y(x) = exp(-x)*sin(exp(-x))+cos(exp(-x));
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x}c_2 + e^x c_1 - e^x \sin(e^{-x})$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = e^{-x} \sin(e^{-x}) + \cos(e^{-x})$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

- $$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = e^{-x} \sin(e^{-x}) + \cos(e^{-x}) \right]$$
- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = \frac{e^x \int (\cos(e^{-x})e^x + \sin(e^{-x}))e^{-2x} dx}{2} - \frac{e^{-x} \int (\cos(e^{-x})e^x + \sin(e^{-x})) dx}{2}$$
 - Compute integrals

$$y_p(x) = -e^x \sin(e^{-x})$$
 - Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x - e^x \sin(e^{-x})$$

✓ **Mathematica.** Time used: 0.086 (sec). Leaf size: 81

```
ode=D[y[x],{x,2}]-y[x]==Exp[-x]*Sin[Exp[-x]]+Cos[Exp[-x]];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \int_1^x \frac{1}{2} (-e^{K[1]} \cos(e^{-K[1]}) - \sin(e^{-K[1]})) dK[1] - e^x \sin(e^{-x}) + \frac{1}{2} \cos(e^{-x}) + c_1 e^x + c_2 e^{-x}$$

✓ **Sympy.** Time used: 1.214 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - cos(exp(-x)) + Derivative(y(x), (x, 2)) - exp(-x)*sin(exp(-x)), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_2 e^{-x} + (C_1 - \sin(e^{-x})) e^x$$

2.10.5 Problem 14

Local contents

Solved as second order linear constant coeff ode	1809
Solved as second order ode using Kovacic algorithm	1813
✓ Maple	1818
✓ Mathematica	1820
✓ Sympy	1820

Internal problem ID [8003]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 14

Date solved : Thursday, November 27, 2025 at 07:32:16 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.260 (sec)

Solve

$$y'' - y = \frac{1}{(1 + e^{-x})^2}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = \frac{1}{(1+e^{-x})^2}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = e^{-x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & e^{-x} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}(e^{-x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

Therefore

$$W = (e^x)(-e^{-x}) - (e^{-x})(e^x)$$

Which simplifies to

$$W = -2e^x e^{-x}$$

Which simplifies to

$$W = -2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^{-x}}{(1+e^{-x})^2}}{-2} dx$$

Which simplifies to

$$u_1 = - \int -\frac{\operatorname{sech}\left(\frac{x}{2}\right)^2}{8} dx$$

Hence

$$u_1 = \frac{\tanh\left(\frac{x}{2}\right)}{4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^x}{(1+e^{-x})^2}}{-2} dx$$

Which simplifies to

$$u_2 = \int -\frac{e^{2x} \operatorname{sech}\left(\frac{x}{2}\right)^2}{8} dx$$

Hence

$$u_2 = -\frac{e^x}{2} + \frac{1}{2e^x + 2} + \ln(e^x + 1)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\tanh\left(\frac{x}{2}\right) e^x}{4} + e^{-x} \left(-\frac{e^x}{2} + \frac{1}{2e^x + 2} + \ln(e^x + 1) \right)$$

Which simplifies to

$$y_p(x) = -1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x} c_2) + \left(-1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} + c_1 e^x + e^{-x} c_2$$

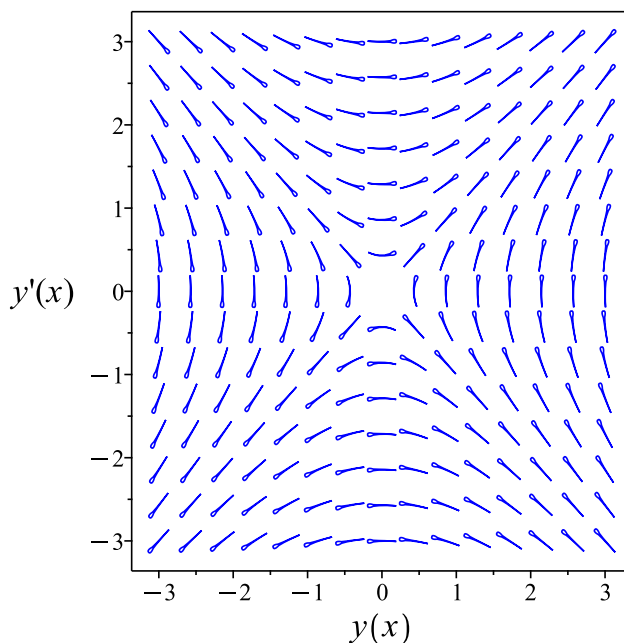


Figure 2.350: Slope field $y'' - y = \frac{1}{(1+e^{-x})^2}$

Solved as second order ode using Kovacic algorithm

Time used: 0.138 (sec)

Solve

$$y'' - y = \frac{1}{(1 + e^{-x})^2}$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.223: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-x}$$

$$y_2 = \frac{e^x}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ \frac{d}{dx}(e^{-x}) & \frac{d}{dx}\left(\frac{e^x}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ -e^{-x} & \frac{e^x}{2} \end{vmatrix}$$

Therefore

$$W = (e^{-x}) \left(\frac{e^x}{2} \right) - \left(\frac{e^x}{2} \right) (-e^{-x})$$

Which simplifies to

$$W = e^x e^{-x}$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^x}{2(1+e^{-x})^2}}{1} dx$$

Which simplifies to

$$u_1 = - \int \frac{e^{2x} \operatorname{sech} \left(\frac{x}{2} \right)^2}{8} dx$$

Hence

$$u_1 = -\frac{e^x}{2} + \frac{1}{2e^x + 2} + \ln(e^x + 1)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^{-x}}{(1+e^{-x})^2}}{1} dx$$

Which simplifies to

$$u_2 = \int \frac{\operatorname{sech} \left(\frac{x}{2} \right)^2}{4} dx$$

Hence

$$u_2 = \frac{\tanh \left(\frac{x}{2} \right)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\tanh \left(\frac{x}{2} \right) e^x}{4} + e^{-x} \left(-\frac{e^x}{2} + \frac{1}{2e^x + 2} + \ln(e^x + 1) \right)$$

Which simplifies to

$$y_p(x) = -1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + \left(-1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -1 + \left(\ln(e^x + 1) + \frac{1}{2} \right) e^{-x} + \frac{e^x}{4} + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

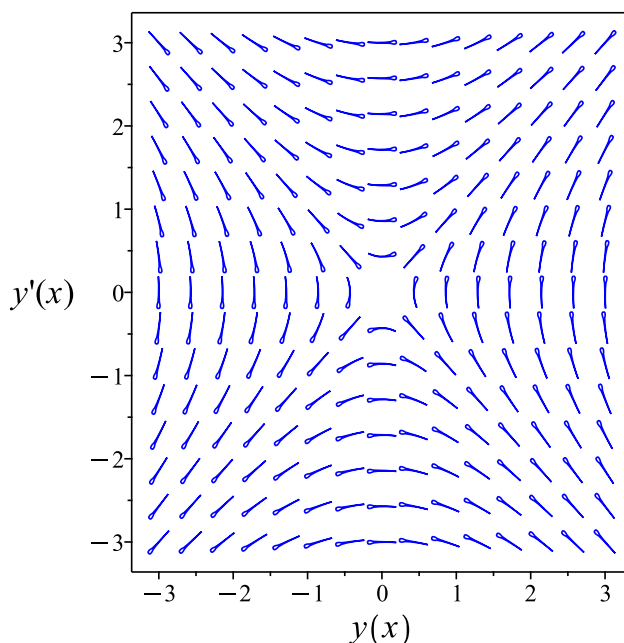


Figure 2.351: Slope field $y'' - y = \frac{1}{(1+e^{-x})^2}$

✓ Maple. Time used: 0.016 (sec). Leaf size: 41

```
ode:=diff(diff(y(x),x),x)-y(x) = 1/(1+exp(-x))^2;
dsolve(ode,y(x), singsol=all);
```

$$y = -1 - \frac{e^x(-4c_2 - 1)}{4} - \frac{e^{-x}(4 \ln(-\frac{1}{e^x+1}) + 4 \ln(2) - 4c_1 + 1)}{4}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = \frac{1}{(1+e^{-x})^2}$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \frac{1}{(1+e^{-x})^2} \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

- $$y_p(x) = -\frac{e^{-x} \int \frac{e^{2x}}{\cosh\left(\frac{x}{2}\right)^2} dx}{8} + \frac{e^x \int \frac{1}{\cosh\left(\frac{x}{2}\right)^2} dx}{8}$$
- Compute integrals
 - $y_p(x) = \frac{e^{-x}(-4\ln(2)-4\ln(-\frac{1}{1+e^x})-1)}{4} - 1 + \frac{e^x}{4}$
 - Substitute particular solution into general solution to ODE
- $$y(x) = C_1 e^{-x} + C_2 e^x + \frac{e^{-x}(-4\ln(2)-4\ln(-\frac{1}{1+e^x})-1)}{4} - 1 + \frac{e^x}{4}$$

✓ **Mathematica.** Time used: 0.021 (sec). Leaf size: 42

```
ode=D[y[x],{x,2}]-y[x]==1/(1+Exp[-x])^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}e^{-x}(-2e^x + 2\log(e^x + 1) + 2c_1 e^{2x} + 1 + 2c_2)$$

✓ **Sympy.** Time used: 0.247 (sec). Leaf size: 42

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) + Derivative(y(x), (x, 2)) - 1/(1 + exp(-x))**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \left(C_1 - \frac{1}{2(e^x + 1)}\right)e^x + (C_2 + \log(e^x + 1))e^{-x} - \frac{1}{2} + \frac{e^{-x}}{2(e^x + 1)}$$

2.10.6 Problem 15

Local contents

Solved as second order linear constant coeff ode	1821
Solved as second order ode using Kovacic algorithm	1824
✓ Maple	1829
✓ Mathematica	1830
✓ Sympy	1830

Internal problem ID [8004]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 15

Date solved : Thursday, November 27, 2025 at 07:32:16 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.115 (sec)

Solve

$$y'' + 2y = 2 + e^x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 2, f(x) = 2 + e^x$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(2)} \\ &= \pm i\sqrt{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= +i\sqrt{2} \\ \lambda_2 &= -i\sqrt{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= i\sqrt{2} \\ \lambda_2 &= -i\sqrt{2} \end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = \sqrt{2}$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x))$$

Or

$$y = c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$2 + e^x$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \cos(\sqrt{2}x), \sin(\sqrt{2}x) \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 e^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_2 e^x + 2A_1 = 2 + e^x$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 1, A_2 = \frac{1}{3} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

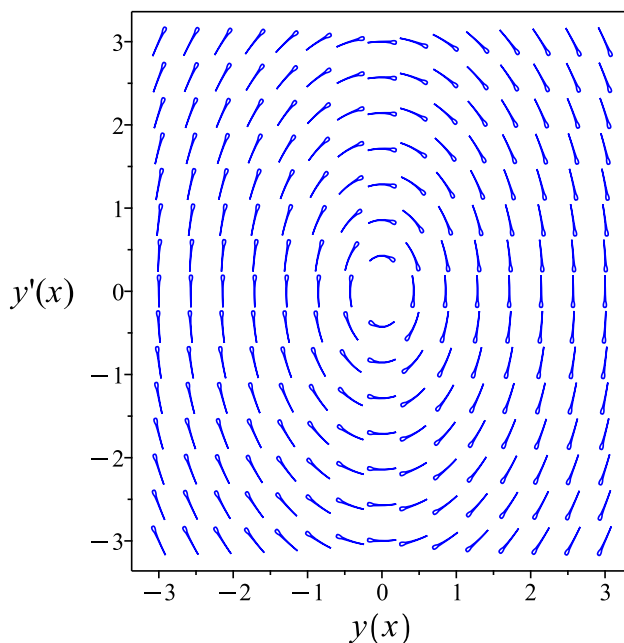
$$y_p = 1 + \frac{e^x}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x) \right) + \left(1 + \frac{e^x}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = 1 + \frac{e^x}{3} + c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

Figure 2.352: Slope field $y'' + 2y = 2 + e^x$

Solved as second order ode using Kovacic algorithm

Time used: 0.138 (sec)

Solve

$$y'' + 2y = 2 + e^x$$

Writing the ode as

$$y'' + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-2}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -2 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -2z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.225: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -2$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(\sqrt{2}x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(\sqrt{2}x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(\sqrt{2}x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(\sqrt{2}x) \int \frac{1}{\cos^2(\sqrt{2}x)} dx \\ &= \cos(\sqrt{2}x) \left(\frac{\sqrt{2} \tan(\sqrt{2}x)}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned}
y &= c_1 y_1 + c_2 y_2 \\
&= c_1 \left(\cos \left(\sqrt{2} x \right) \right) + c_2 \left(\cos \left(\sqrt{2} x \right) \left(\frac{\sqrt{2} \tan \left(\sqrt{2} x \right)}{2} \right) \right)
\end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos \left(\sqrt{2} x \right) + \frac{c_2 \sqrt{2} \sin \left(\sqrt{2} x \right)}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$2 + e^x$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{\sqrt{2} \sin \left(\sqrt{2} x \right)}{2}, \cos \left(\sqrt{2} x \right) \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 e^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_2 e^x + 2A_1 = 2 + e^x$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 1, A_2 = \frac{1}{3} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = 1 + \frac{e^x}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(\sqrt{2}x) + \frac{c_2 \sqrt{2} \sin(\sqrt{2}x)}{2} \right) + \left(1 + \frac{e^x}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = 1 + \frac{e^x}{3} + c_1 \cos(\sqrt{2}x) + \frac{c_2 \sqrt{2} \sin(\sqrt{2}x)}{2}$$

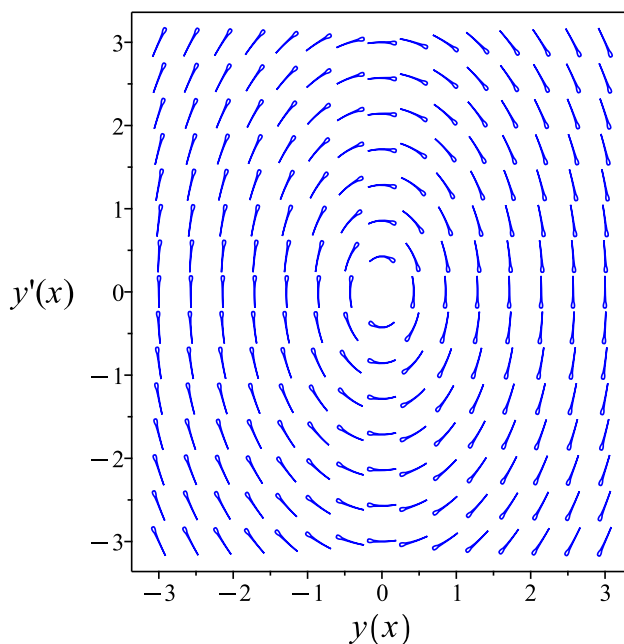


Figure 2.353: Slope field $y'' + 2y = 2 + e^x$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 26

```
ode:=diff(diff(y(x),x),x)+2*y(x) = exp(x)+2;
dsolve(ode,y(x), singsol=all);
```

$$y = \sin(\sqrt{2}x) c_2 + \cos(\sqrt{2}x) c_1 + 1 + \frac{e^x}{3}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 2y(x) = 2 + e^x$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 2 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-8})}{2}$$

- Roots of the characteristic polynomial

$$r = (-I\sqrt{2}, I\sqrt{2})$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(\sqrt{2}x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(\sqrt{2}x)$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 \cos(\sqrt{2}x) + C2 \sin(\sqrt{2}x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = 2 + e^x \right]$$
- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(\sqrt{2}x) & \sin(\sqrt{2}x) \\ -\sqrt{2} \sin(\sqrt{2}x) & \sqrt{2} \cos(\sqrt{2}x) \end{bmatrix}$$
- Compute Wronskian

$$W(y_1(x), y_2(x)) = \sqrt{2}$$
- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{\sqrt{2} \left(\int \sin(\sqrt{2}x) (2+e^x) dx \cos(\sqrt{2}x) - \int \cos(\sqrt{2}x) (2+e^x) dx \sin(\sqrt{2}x) \right)}{2}$$
- Compute integrals

$$y_p(x) = \frac{e^x}{3} + 1$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(\sqrt{2}x) + C_2 \sin(\sqrt{2}x) + \frac{e^x}{3} + 1$$

✓ **Mathematica.** Time used: 0.055 (sec). Leaf size: 103

```
ode=D[y[x],{x,2}]+2*y[x]==2+Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \cos(\sqrt{2}x) \int_1^x -\frac{(2+e^{K[1]}) \sin(\sqrt{2}K[1])}{\sqrt{2}} dK[1] + \sin(\sqrt{2}x) \int_1^x \frac{(2+e^{K[2]}) \cos(\sqrt{2}K[2])}{\sqrt{2}} dK[2] + c$$

✓ **Sympy.** Time used: 0.043 (sec). Leaf size: 29

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*y(x) - exp(x) + Derivative(y(x), (x, 2)) - 2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 \sin(\sqrt{2}x) + C_2 \cos(\sqrt{2}x) + \frac{e^x}{3} + 1$$

2.10.7 Problem 16

Local contents

Solved as second order linear constant coeff ode	1831
Solved as second order ode using Kovacic algorithm	1834
✓ Maple	1838
✓ Mathematica	1840
✓ Sympy	1840

Internal problem ID [8005]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 16

Date solved : Thursday, November 27, 2025 at 07:32:17 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.112 (sec)

Solve

$$y'' - y = e^x \sin(2x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = e^x \sin(2x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x \sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x \cos(2x), e^x \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^x \cos(2x) + A_2 e^x \sin(2x)$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 e^x \cos(2x) - 4A_1 e^x \sin(2x) - 4A_2 e^x \sin(2x) + 4A_2 e^x \cos(2x) = e^x \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{8}, A_2 = -\frac{1}{8} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

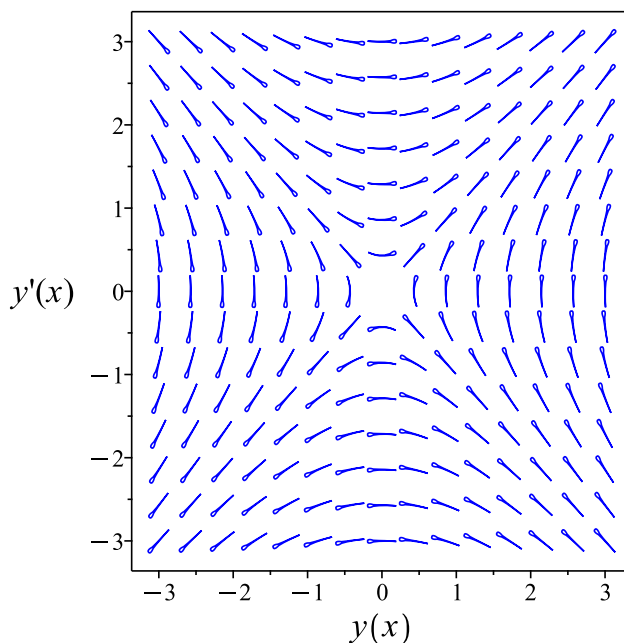
$$y_p = -\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x} c_2) + \left(-\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8} + c_1 e^x + e^{-x} c_2$$

Figure 2.354: Slope field $y'' - y = e^x \sin(2x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.098 (sec)

Solve

$$y'' - y = e^x \sin(2x)$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.227: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x \sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x \cos(2x), e^x \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{2}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^x \cos(2x) + A_2 e^x \sin(2x)$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 e^x \cos(2x) - 4A_1 e^x \sin(2x) - 4A_2 e^x \sin(2x) + 4A_2 e^x \cos(2x) = e^x \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{8}, A_2 = -\frac{1}{8} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8}$$

Therefore the general solution is

$$y = y_h + y_p \\ = \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + \left(-\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8} \right)$$

Summary of solutions found

$$y = -\frac{e^x \cos(2x)}{8} - \frac{e^x \sin(2x)}{8} + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

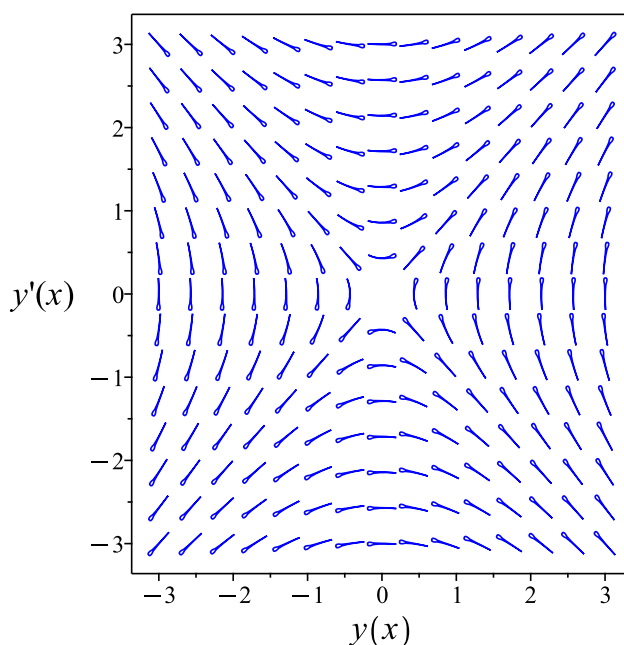


Figure 2.355: Slope field $y'' - y = e^x \sin(2x)$

✓ Maple. Time used: 0.003 (sec). Leaf size: 30

```
ode:=diff(diff(y(x),x),x)-y(x) = exp(x)*sin(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{e^x \cos(2x)}{8} + e^{-x} c_1 + e^x \left(c_2 - \frac{\sin(2x)}{8} \right)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
```

```

trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = e^x \sin(2x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C1 y_1(x) + C2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-x} + C2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^x \sin(2x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^{-x} \int e^{2x} \sin(2x) dx}{2} + \frac{e^x \int \sin(2x) dx}{2}$$

- Compute integrals

$$y_p(x) = -\frac{e^x (\sin(2x) + \cos(2x))}{8}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x - \frac{e^x (\sin(2x) + \cos(2x))}{8}$$

✓ **Mathematica.** Time used: 0.045 (sec). Leaf size: 58

```
ode=D[y[x],{x,2}]-y[x]==Exp[x]*Sin[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \int_1^x -e^{2K[1]} \cos(K[1]) \sin(K[1]) dK[1] - \frac{1}{2} e^x \cos^2(x) + c_1 e^x + c_2 e^{-x}$$

✓ **Sympy.** Time used: 0.074 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - exp(x)*sin(2*x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_2 e^{-x} + \left(C_1 - \frac{\sin(2x)}{8} - \frac{\cos(2x)}{8} \right) e^x$$

2.10.8 Problem 17

Local contents

Solved as second order linear constant coeff ode	1841
Solved as second order ode using Kovacic algorithm	1844
✓ Maple	1848
✓ Mathematica	1850
✓ Sympy	1850

Internal problem ID [8006]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 17

Date solved : Thursday, November 27, 2025 at 07:32:17 AM

CAS classification : [[_2nd_order, _linear, _nonhomogeneous]]

Solved as second order linear constant coeff ode

Time used: 0.144 (sec)

Solve

$$y'' + 2y' + 2y = x^2 + \sin(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 2, C = 2, f(x) = x^2 + \sin(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 2, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 2\lambda e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 2\lambda + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 2, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-2}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{2^2 - (4)(1)(2)} \\ &= -1 \pm i \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= -1 + i \\ \lambda_2 &= -1 - i \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= -1 + i \\ \lambda_2 &= -1 - i \end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = -1$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^{-x} (c_1 \cos(x) + c_2 \sin(x))$$

Therefore the homogeneous solution y_h is

$$y_h = e^{-x} (c_1 \cos(x) + c_2 \sin(x))$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2 + \sin(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(x), \sin(x)\}, \{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^{-x} \cos(x), e^{-x} \sin(x)\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 \cos(x) + A_2 \sin(x) + A_3 + A_4 x + A_5 x^2$$

The unknowns $\{A_1, A_2, A_3, A_4, A_5\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$\begin{aligned} A_1 \cos(x) + A_2 \sin(x) + 2A_5 - 2A_1 \sin(x) + 2A_2 \cos(x) \\ + 2A_4 + 4A_5 x + 2A_3 + 2A_4 x + 2A_5 x^2 = x^2 + \sin(x) \end{aligned}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{2}{5}, A_2 = \frac{1}{5}, A_3 = \frac{1}{2}, A_4 = -1, A_5 = \frac{1}{2} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

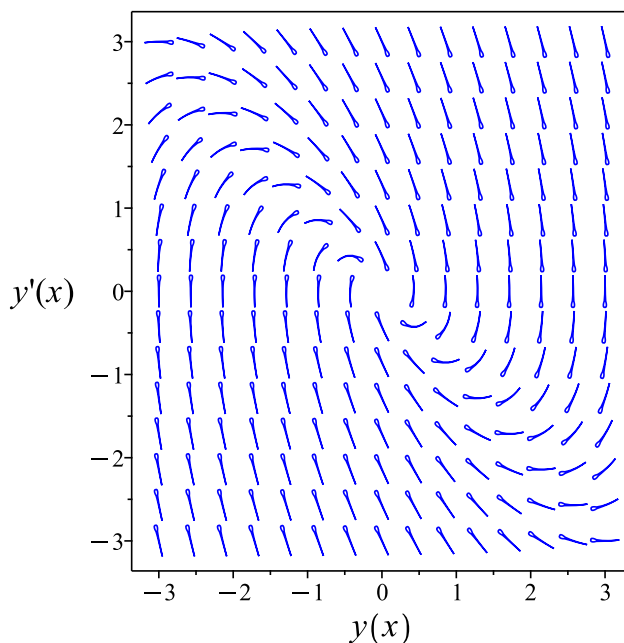
$$y_p = -\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{-x}(c_1 \cos(x) + c_2 \sin(x))) + \left(-\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2} + e^{-x}(c_1 \cos(x) + c_2 \sin(x))$$

Figure 2.356: Slope field $y'' + 2y' + 2y = x^2 + \sin(x)$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.141 (sec)

Solve

$$y'' + 2y' + 2y = x^2 + \sin(x)$$

Writing the ode as

$$y'' + 2y' + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$A = 1$$

$$B = 2 \quad (3)$$

$$C = 2$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.229: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{2}{1} dx} \\ &= z_1 e^{-x} \\ &= z_1 (e^{-x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x} \cos(x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{2}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-2x}}{(y_1)^2} dx \\ &= y_1 (\tan(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x} \cos(x)) + c_2 (e^{-x} \cos(x) (\tan(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} \cos(x) + c_2 e^{-x} \sin(x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2 + \sin(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(x), \sin(x)\}, \{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^{-x} \cos(x), e^{-x} \sin(x)\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 \cos(x) + A_2 \sin(x) + A_3 + A_4 x + A_5 x^2$$

The unknowns $\{A_1, A_2, A_3, A_4, A_5\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$\begin{aligned} A_1 \cos(x) + A_2 \sin(x) + 2A_5 - 2A_1 \sin(x) + 2A_2 \cos(x) \\ + 2A_4 + 4A_5 x + 2A_3 + 2A_4 x + 2A_5 x^2 = x^2 + \sin(x) \end{aligned}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{2}{5}, A_2 = \frac{1}{5}, A_3 = \frac{1}{2}, A_4 = -1, A_5 = \frac{1}{2} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2}$$

Therefore the general solution is

$$y = y_h + y_p \\ = (c_1 e^{-x} \cos(x) + c_2 e^{-x} \sin(x)) + \left(-\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2} \right)$$

Summary of solutions found

$$y = -\frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} + \frac{1}{2} - x + \frac{x^2}{2} + c_1 e^{-x} \cos(x) + c_2 e^{-x} \sin(x)$$

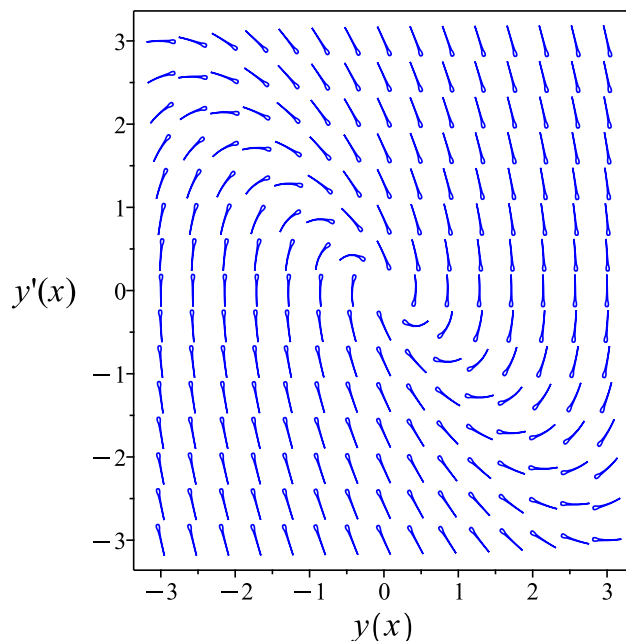


Figure 2.357: Slope field $y'' + 2y' + 2y = x^2 + \sin(x)$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 38

```
ode:=diff(diff(y(x),x),x)+2*diff(y(x),x)+2*y(x) = x^2+sin(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x} \sin(x) c_2 + e^{-x} \cos(x) c_1 + \frac{x^2}{2} - x + \frac{1}{2} - \frac{2 \cos(x)}{5} + \frac{\sin(x)}{5}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form $[x_i=0, \eta=F(x)]$

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 2\frac{d}{dx}y(x) + 2y(x) = x^2 + \sin(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 2r + 2 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{(-2) \pm (\sqrt{-4})}{2}$$

- Roots of the characteristic polynomial

$$r = (-1 - i, -1 + i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x} \cos(x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{-x} \sin(x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} \cos(x) + C_2 e^{-x} \sin(x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = x^2 + \sin(x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} \cos(x) & e^{-x} \sin(x) \\ -e^{-x} \cos(x) - e^{-x} \sin(x) & -e^{-x} \sin(x) + e^{-x} \cos(x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{-2x}$$

- Substitute functions into equation for $y_p(x)$

- $$y_p(x) = e^{-x} \left(- \int \sin(x) e^x (x^2 + \sin(x)) dx \cos(x) + \int \cos(x) e^x (x^2 + \sin(x)) dx \sin(x) \right)$$
- Compute integrals

$$y_p(x) = \frac{x^2}{2} - \frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} - x + \frac{1}{2}$$
 - Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} \cos(x) + C_2 e^{-x} \sin(x) + \frac{x^2}{2} - \frac{2 \cos(x)}{5} + \frac{\sin(x)}{5} - x + \frac{1}{2}$$

✓ **Mathematica.** Time used: 0.103 (sec). Leaf size: 75

```
ode=D[y[x],{x,2}]+2*D[y[x],x]+2*y[x]==x^2+Sin[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(\cos(x) \int_1^x -e^{K[2]} \sin(K[2]) (K[2]^2 + \sin(K[2])) dK[2] + \sin(x) \int_1^x e^{K[1]} \cos(K[1]) (K[1]^2 + \sin(K[1])) dK[1] \right)$$

✓ **Sympy.** Time used: 0.155 (sec). Leaf size: 37

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + 2*y(x) - sin(x) + 2*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x^2}{2} - x + (C_1 \sin(x) + C_2 \cos(x)) e^{-x} + \frac{\sin(x)}{5} - \frac{2 \cos(x)}{5} + \frac{1}{2}$$

2.10.9 Problem 18

Local contents

Solved as second order linear constant coeff ode	1851
Solved as second order ode using Kovacic algorithm	1854
✓ Maple	1858
✓ Mathematica	1860
✓ Sympy	1860

Internal problem ID [8007]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 18

Date solved : Thursday, November 27, 2025 at 07:32:17 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.171 (sec)

Solve

$$y'' - 9y = x + e^{2x} - \sin(2x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -9, f(x) = x + e^{2x} - \sin(2x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 9y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -9$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 9e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 9 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -9$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-9)} \\ &= \pm 3 \end{aligned}$$

Hence

$$\lambda_1 = +3$$

$$\lambda_2 = -3$$

Which simplifies to

$$\lambda_1 = 3$$

$$\lambda_2 = -3$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(3)x} + c_2 e^{(-3)x} \end{aligned}$$

Or

$$y = c_1 e^{3x} + c_2 e^{-3x}$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^{3x} + c_2 e^{-3x}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x + e^{2x} - \sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{2x}\}, \{1, x\}, \{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^{-3x}, e^{3x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{2x} + A_2 + A_3 x + A_4 \cos(2x) + A_5 \sin(2x)$$

The unknowns $\{A_1, A_2, A_3, A_4, A_5\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-5A_1 e^{2x} - 13A_4 \cos(2x) - 13A_5 \sin(2x) - 9A_2 - 9A_3 x = x + e^{2x} - \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{5}, A_2 = 0, A_3 = -\frac{1}{9}, A_4 = 0, A_5 = \frac{1}{13} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

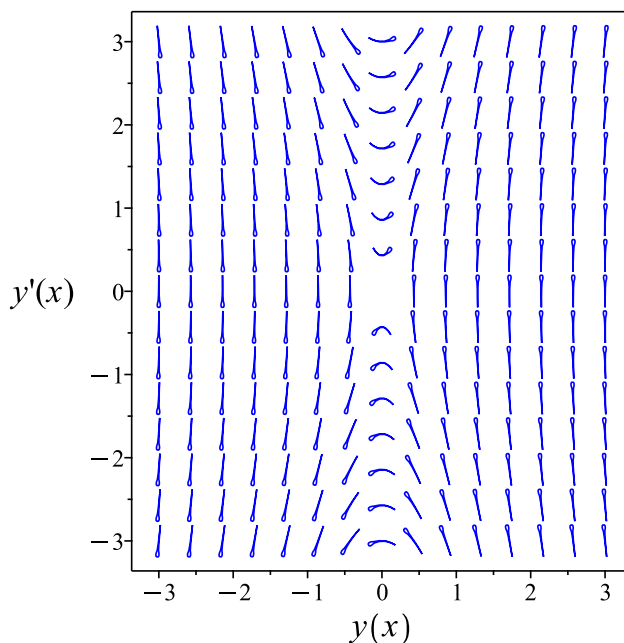
$$y_p = -\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^{3x} + c_2 e^{-3x}) + \left(-\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13} + c_1 e^{3x} + c_2 e^{-3x}$$

Figure 2.358: Slope field $y'' - 9y = x + e^{2x} - \sin(2x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.152 (sec)

Solve

$$y'' - 9y = x + e^{2x} - \sin(2x)$$

Writing the ode as

$$y'' - 9y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -9 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{9}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 9 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 9z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.231: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 9$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-3x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-3x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-3x} \int \frac{1}{e^{-6x}} dx \\ &= e^{-3x} \left(\frac{e^{6x}}{6} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-3x}) + c_2 \left(e^{-3x} \left(\frac{e^{6x}}{6} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 9y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-3x} + \frac{c_2 e^{3x}}{6}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x + e^{2x} - \sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^{2x}\}, \{1, x\}, \{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^{3x}}{6}, e^{-3x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^{2x} + A_2 + A_3 x + A_4 \cos(2x) + A_5 \sin(2x)$$

The unknowns $\{A_1, A_2, A_3, A_4, A_5\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-5A_1 e^{2x} - 13A_4 \cos(2x) - 13A_5 \sin(2x) - 9A_2 - 9A_3 x = x + e^{2x} - \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{5}, A_2 = 0, A_3 = -\frac{1}{9}, A_4 = 0, A_5 = \frac{1}{13} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13}$$

Therefore the general solution is

$$y = y_h + y_p \\ = \left(c_1 e^{-3x} + \frac{c_2 e^{3x}}{6} \right) + \left(-\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13} \right)$$

Summary of solutions found

$$y = -\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13} + c_1 e^{-3x} + \frac{c_2 e^{3x}}{6}$$

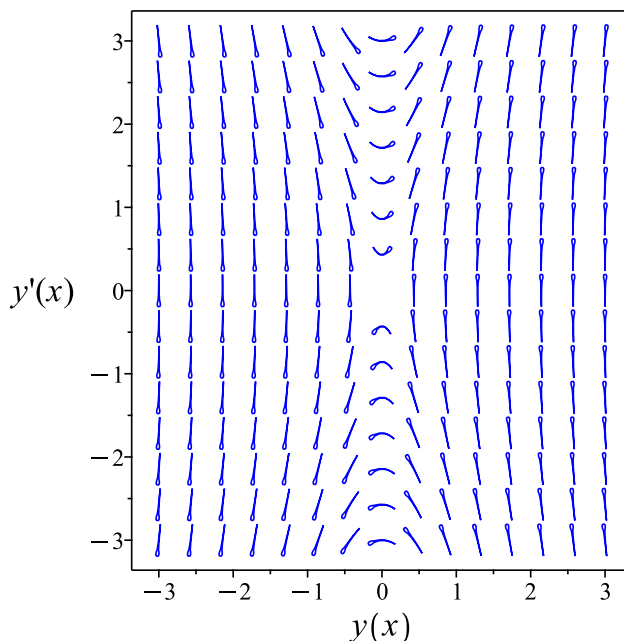


Figure 2.359: Slope field $y'' - 9y = x + e^{2x} - \sin(2x)$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 32

```
ode:=diff(diff(y(x),x),x)-9*y(x) = x+exp(2*x)-sin(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{3x} c_2 + e^{-3x} c_1 - \frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form $[xi=0, eta=F(x)]$

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 9y(x) = x + e^{2x} - \sin(2x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 9 = 0$$

- Factor the characteristic polynomial

$$(r - 3)(r + 3) = 0$$

- Roots of the characteristic polynomial

$$r = (-3, 3)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-3x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{3x}$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-3x} + C2 e^{3x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- o Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = x + e^{2x} - \sin(2x) \right]$$

- o Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-3x} & e^{3x} \\ -3e^{-3x} & 3e^{3x} \end{bmatrix}$$

- o Compute Wronskian

$$W(y_1(x), y_2(x)) = 6$$

- o Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^{-3x} \int e^{3x} (x + e^{2x} - \sin(2x)) dx}{6} + \frac{e^{3x} \int e^{-3x} (x + e^{2x} - \sin(2x)) dx}{6}$$

- o Compute integrals

$$y_p(x) = -\frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-3x} + C_2 e^{3x} - \frac{e^{2x}}{5} - \frac{x}{9} + \frac{\sin(2x)}{13}$$

✓ **Mathematica.** Time used: 0.381 (sec). Leaf size: 94

```
ode=D[y[x],{x,2}]-9*y[x]==x+Exp[2*x]-Sin[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-3x} \left(e^{6x} \int_1^x \frac{1}{6} e^{-3K[1]} (K[1] + e^{2K[1]} - \sin(2K[1])) dK[1] + \int_1^x -\frac{1}{6} e^{3K[2]} (K[2] + e^{2K[2]} - \sin(2K[2])) dK[2] \right)$$

✓ **Sympy.** Time used: 0.064 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x - 9*y(x) - exp(2*x) + sin(2*x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-3x} + C_2 e^{3x} - \frac{x}{9} - \frac{e^{2x}}{5} + \frac{\sin(2x)}{13}$$

2.10.10 Problem 19

Local contents

Solved as higher order constant coeff ode	1861
✓ Maple	1863
✓ Mathematica	1865
✓ Sympy	1866

Internal problem ID [8008]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 19

Date solved : Thursday, November 27, 2025 at 07:32:18 AM

CAS classification : `[_3rd_order, _missing_y]`

Solve

$$y''' + 3y'' + 2y' = x^2 + 4x + 8$$

Solved as higher order constant coeff ode

Time used: 0.063 (sec)

The characteristic equation is

$$\lambda^3 + 3\lambda^2 + 2\lambda = 0$$

The roots of the above equation are

$$\lambda_1 = 0$$

$$\lambda_2 = -1$$

$$\lambda_3 = -2$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 + e^{-x}c_2 + e^{-2x}c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = e^{-x}$$

$$y_3 = e^{-2x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' + 3y'' + 2y' = 0$$

Now the particular solution to the given ODE is found

$$y''' + 3y'' + 2y' = x^2 + 4x + 8$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$x^2 + x + 1$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[1, x, x^2]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, e^{-2x}, e^{-x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[x, x^2, x^3]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_3x^3 + A_2x^2 + A_1x$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$6x^2A_3 + 4xA_2 + 18xA_3 + 2A_1 + 6A_2 + 6A_3 = x^2 + 4x + 8$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{11}{4}, A_2 = \frac{1}{4}, A_3 = \frac{1}{6} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x$$

Therefore the general solution is

$$y = y_h + y_p \\ = (c_1 + e^{-x}c_2 + e^{-2x}c_3) + \left(\frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x\right)$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 33

```
ode:=diff(diff(diff(y(x),x),x),x)+3*diff(diff(y(x),x),x)+2*diff(y(x),x) = x^2+4*x
+8;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2}{4} + \frac{x^3}{6} + \frac{e^{-2x}c_1}{2} - e^{-x}c_2 + \frac{11x}{4} + c_3$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 3; linear nonhomogeneous with symmetry [0,1]
-> Calling odsolve with the ODE, diff(diff(_b(_a),_a),_a) = _a^2-2*_b(_a)-3*
diff(_b(_a),_a)+4*_a+8, _b(_a)
*** Sublevel 2 ***
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
<- double symmetry of the form [xi=0, eta=F(x)] successful
<- differential order: 3; linear nonhomogeneous with symmetry [0,1] successful
```

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) + 3\frac{d^2}{dx^2}y(x) + 2\frac{d}{dx}y(x) = x^2 + 4x + 8$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^3 + 3r^2 + 2r = 0$$

- Roots of the characteristic polynomial

$$r = [0, -1, -2]$$

- Homogeneous solution from $r = 0$

$$y_1(x) = 1$$

- Homogeneous solution from $r = -1$

$$y_2(x) = e^{-x}$$

- Homogeneous solution from $r = -2$

$$y_3(x) = e^{-2x}$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 + C_2 e^{-x} + C_3 e^{-2x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Define the forcing function of the ODE

$$f(x) = x^2 + 4x + 8$$

- Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$

- Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx}y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx}y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx}u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx}u_i(x) \right) y_i(x) = 0$$

- Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2}y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) \left(\frac{d}{dx}y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2}y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx}u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx}u_i(x) \right) \left(\frac{d}{dx}y_i(x) \right) = 0$$

- The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^3}{dx^3}y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i}y(x) \right) \right) = f(x)$$

- Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$

- Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} \frac{1}{6}x^3 + x^2 + 4x \\ -\frac{x^2}{e^{-x}} - \frac{2x}{e^{-x}} - \frac{6}{e^{-x}} \\ \frac{x^2}{4e^{-2x}} + \frac{3x}{4e^{-2x}} + \frac{13}{8e^{-2x}} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x - \frac{35}{8}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 + C_2 e^{-x} + C_3 e^{-2x} + \frac{x^3}{6} + \frac{x^2}{4} + \frac{11x}{4} - \frac{35}{8}$$

✓ **Mathematica.** Time used: 0.091 (sec). Leaf size: 43

```
ode=D[y[x],{x,3}]+3*D[y[x],{x,2}]+2*D[y[x],x]==x^2+4*x+8;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{12}(2x^3 + 3x^2 + 33x - 6e^{-2x}(2c_2e^x + c_1)) + c_3$$

✓ Sympy. Time used: 0.141 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 - 4*x + 2*Derivative(y(x), x) + 3*Derivative(y(x), (x, 2)) +
        Derivative(y(x), (x, 3)) - 8, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2 e^{-2x} + C_3 e^{-x} + \frac{x^3}{6} + \frac{x^2}{4} + \frac{11x}{4}$$

2.10.11 Problem 20

Local contents

Solved as second order linear constant coeff ode	1867
Solved as second order ode using Kovacic algorithm	1870
✓ Maple	1875
✓ Mathematica	1876
✓ Sympy	1876

Internal problem ID [8009]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 20

Date solved : Thursday, November 27, 2025 at 07:32:18 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.122 (sec)

Solve

$$y'' + y = -2 \sin(x) + 4x \cos(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 1, f(x) = -2 \sin(x) + 4x \cos(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(1)} \\ &= \pm i \end{aligned}$$

Hence

$$\lambda_1 = +i$$

$$\lambda_2 = -i$$

Which simplifies to

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(x) + c_2 \sin(x))$$

Or

$$y = c_1 \cos(x) + c_2 \sin(x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(x) + c_2 \sin(x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$-2 \sin(x) + 4x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x \cos(x), x \sin(x), \cos(x), \sin(x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{\cos(x), \sin(x)\}$$

Since $\cos(x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x \cos(x), x \sin(x), x^2 \cos(x), x^2 \sin(x)\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x \cos(x) + A_2 x \sin(x) + A_3 x^2 \cos(x) + A_4 x^2 \sin(x)$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$\begin{aligned} & -2A_1 \sin(x) + 2A_2 \cos(x) + 2A_3 \cos(x) - 4A_3 x \sin(x) + 2A_4 \sin(x) + 4A_4 x \cos(x) \\ & = -2 \sin(x) + 4x \cos(x) \end{aligned}$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 2, A_2 = 0, A_3 = 0, A_4 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

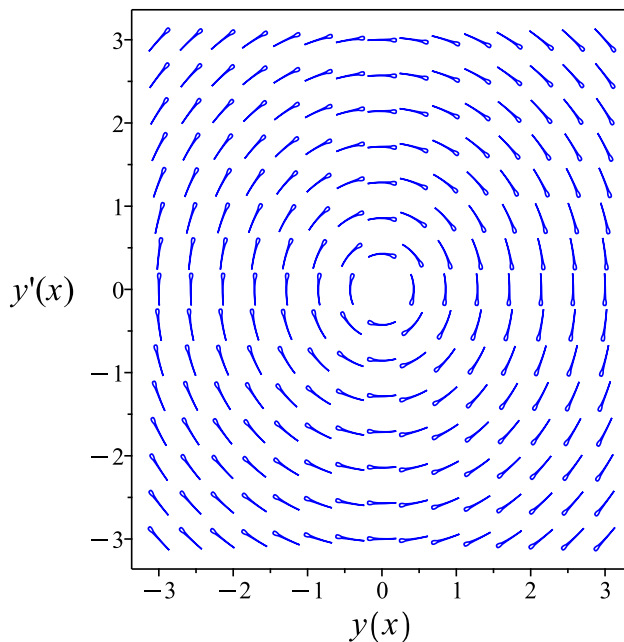
$$y_p = 2x \cos(x) + x^2 \sin(x)$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 \cos(x) + c_2 \sin(x)) + (2x \cos(x) + x^2 \sin(x)) \end{aligned}$$

Summary of solutions found

$$y = 2x \cos(x) + x^2 \sin(x) + c_1 \cos(x) + c_2 \sin(x)$$

Figure 2.360: Slope field $y'' + y = -2 \sin(x) + 4x \cos(x)$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.103 (sec)

Solve

$$y'' + y = -2 \sin(x) + 4x \cos(x)$$

Writing the ode as

$$y'' + y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$A = 1$$

$$B = 0$$

$$C = 1$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.234: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(x) \int \frac{1}{\cos(x)^2} dx \\ &= \cos(x) (\tan(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(\cos(x)) + c_2(\cos(x)(\tan(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(x) + c_2 \sin(x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$-2 \sin(x) + 4x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x \cos(x), x \sin(x), \cos(x), \sin(x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{\cos(x), \sin(x)\}$$

Since $\cos(x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x \cos(x), x \sin(x), x^2 \cos(x), x^2 \sin(x)\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x \cos(x) + A_2 x \sin(x) + A_3 x^2 \cos(x) + A_4 x^2 \sin(x)$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$\begin{aligned} -2A_1 \sin(x) + 2A_2 \cos(x) + 2A_3 \cos(x) - 4A_3 x \sin(x) + 2A_4 \sin(x) + 4A_4 x \cos(x) \\ = -2 \sin(x) + 4x \cos(x) \end{aligned}$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = 2, A_2 = 0, A_3 = 0, A_4 = 1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = 2x \cos(x) + x^2 \sin(x)$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 \cos(x) + c_2 \sin(x)) + (2x \cos(x) + x^2 \sin(x)) \end{aligned}$$

Summary of solutions found

$$y = 2x \cos(x) + x^2 \sin(x) + c_1 \cos(x) + c_2 \sin(x)$$

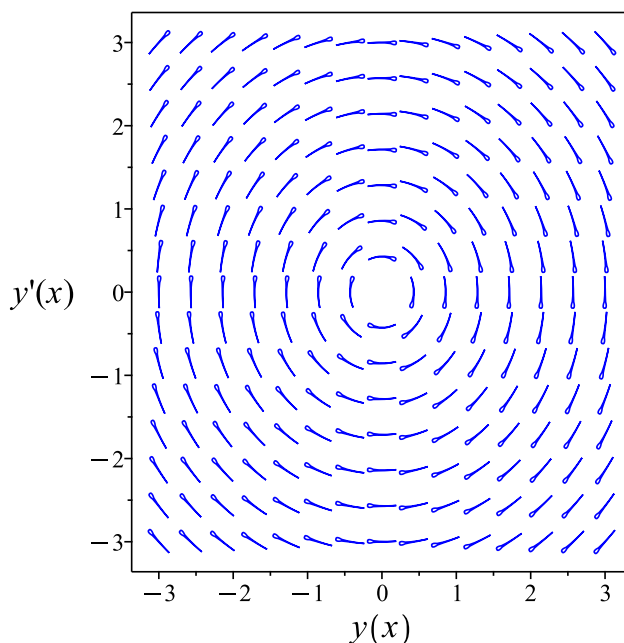


Figure 2.361: Slope field $y'' + y = -2 \sin(x) + 4x \cos(x)$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 23

```
ode:=diff(diff(y(x),x),x)+y(x) = -2*sin(x)+4*cos(x)*x;
dsolve(ode,y(x), singsol=all);
```

$$y = (x^2 + c_2 - 1) \sin(x) + 2\left(x + \frac{c_1}{2}\right) \cos(x)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + y(x) = -2\sin(x) + 4x\cos(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 1 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-4})}{2}$$

- Roots of the characteristic polynomial

$$r = (-i, i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = -2 \sin(x) + 4x \cos(x) \right]$$
- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix}$$
- Compute Wronskian

$$W(y_1(x), y_2(x)) = 1$$
- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\cos(x) \int (2x \sin(2x) - 1 + \cos(2x)) dx + \sin(x) \int (2x \cos(2x) + 2x - \sin(2x)) dx$$
- Compute integrals

$$y_p(x) = \sin(x) x^2 + 2x \cos(x) - \sin(x)$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(x) + C_2 \sin(x) + \sin(x) x^2 + 2x \cos(x) - \sin(x)$$

✓ **Mathematica.** Time used: 0.16 (sec). Leaf size: 70

```
ode=D[y[x],{x,2}]+y[x]==-2*Sin[x]+4*x*Cos[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \sin(x) \int_1^x (4 \cos^2(K[2])K[2] - 2 \cos(K[2]) \sin(K[2])) dK[2] + \cos(x) \int_1^x 2 \sin(K[1])(\sin(K[1]) - 2 \cos(K[1])) dK[1]$$

✓ **Sympy.** Time used: 0.069 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-4*x*cos(x) + y(x) + 2*sin(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = (C_1 + 2x) \cos(x) + (C_2 + x^2) \sin(x)$$

2.10.12 Problem 21**Local contents**

Solved as higher order constant coeff ode	1877
✓ Maple	1880
✓ Mathematica	1883
✓ Sympy	1883

Internal problem ID [8010]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 15. Linear equations with constant coefficients (Variation of parameters).

Supplementary problems. Page 98

Problem number : 21

Date solved : Thursday, November 27, 2025 at 07:32:19 AM

CAS classification : `[[_3rd_order, _linear, _nonhomogeneous]]`

Solve

$$y''' - y'' - 4y' + 4y = 2x^2 - 4x - 1 + 2x^2e^{2x} + 5xe^{2x} + e^{2x}$$

Solved as higher order constant coeff ode

Time used: 0.280 (sec)

The characteristic equation is

$$\lambda^3 - \lambda^2 - 4\lambda + 4 = 0$$

The roots of the above equation are

$$\lambda_1 = 1$$

$$\lambda_2 = 2$$

$$\lambda_3 = -2$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 e^x + e^{-2x} c_2 + e^{2x} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = e^x$$

$$y_2 = e^{-2x}$$

$$y_3 = e^{2x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' - y'' - 4y' + 4y = 0$$

Now the particular solution to the given ODE is found

$$y''' - y'' - 4y' + 4y = -(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1$$

Let the particular solution be

$$y_p = U_1y_1 + U_2y_2 + U_3y_3$$

Where y_i are the basis solutions found above for the homogeneous solution y_h and $U_i(x)$ are functions to be determined as follows

$$U_i = (-1)^{n-i} \int \frac{F(x)W_i(x)}{aW(x)} dx$$

Where $W(x)$ is the Wronskian and $W_i(x)$ is the Wronskian that results after deleting the last row and the i -th column of the determinant and n is the order of the ODE or equivalently, the number of basis solutions, and a is the coefficient of the leading derivative in the ODE, and $F(x)$ is the RHS of the ODE. Therefore, the first step is to find the Wronskian $W(x)$. This is given by

$$W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Substituting the fundamental set of solutions y_i found above in the Wronskian gives

$$W = \begin{bmatrix} e^x & e^{-2x} & e^{2x} \\ e^x & -2e^{-2x} & 2e^{2x} \\ e^x & 4e^{-2x} & 4e^{2x} \end{bmatrix}$$

$$|W| = -12e^xe^{-2x}e^{2x}$$

The determinant simplifies to

$$|W| = -12e^x$$

Now we determine W_i for each U_i .

$$W_1(x) = \det \begin{bmatrix} e^{-2x} & e^{2x} \\ -2e^{-2x} & 2e^{2x} \end{bmatrix}$$

$$= 4$$

$$\begin{aligned} W_2(x) &= \det \begin{bmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{bmatrix} \\ &= e^{3x} \end{aligned}$$

$$\begin{aligned} W_3(x) &= \det \begin{bmatrix} e^x & e^{-2x} \\ e^x & -2e^{-2x} \end{bmatrix} \\ &= -3e^{-x} \end{aligned}$$

Now we are ready to evaluate each $U_i(x)$.

$$\begin{aligned} U_1 &= (-1)^{3-1} \int \frac{F(x)W_1(x)}{aW(x)} dx \\ &= (-1)^2 \int \frac{(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)(4)}{(1)(-12e^x)} dx \\ &= \int \frac{-4(-2x^2 - 5x - 1)e^{2x} + 8x^2 - 16x - 4}{-12e^x} dx \\ &= \int \left(-\frac{(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)e^{-x}}{3} \right) dx \\ &= -\frac{e^{-x}}{3} + \frac{2x^2e^{-x}}{3} - \frac{xe^x}{3} - \frac{2e^xx^2}{3} \end{aligned}$$

$$\begin{aligned} U_2 &= (-1)^{3-2} \int \frac{F(x)W_2(x)}{aW(x)} dx \\ &= (-1)^1 \int \frac{(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)(e^{3x})}{(1)(-12e^x)} dx \\ &= - \int \frac{(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)e^{3x}}{-12e^x} dx \\ &= - \int \left(\frac{(-2x^2 - 5x - 1)e^{4x}}{12} + \frac{(-2x^2 + 4x + 1)e^{2x}}{12} \right) dx \\ &= \frac{e^{4x}x}{12} + \frac{e^{4x}x^2}{24} - \frac{xe^{2x}}{4} + \frac{e^{2x}}{12} + \frac{x^2e^{2x}}{12} \\ &= \frac{e^{4x}x}{12} + \frac{e^{4x}x^2}{24} - \frac{xe^{2x}}{4} + \frac{e^{2x}}{12} + \frac{x^2e^{2x}}{12} \end{aligned}$$

$$\begin{aligned}
U_3 &= (-1)^{3-3} \int \frac{F(x)W_3(x)}{aW(x)} dx \\
&= (-1)^0 \int \frac{(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)(-3e^{-x})}{(1)(-12e^x)} dx \\
&= \int \frac{-3(-(-2x^2 - 5x - 1)e^{2x} + 2x^2 - 4x - 1)e^{-x}}{-12e^x} dx \\
&= \int \left(\frac{x^2}{2} + \frac{5x}{4} + \frac{1}{4} + \frac{e^{-2x}(2x^2 - 4x - 1)}{4} \right) dx \\
&= \frac{x}{4} - \frac{e^{-2x}x^2}{4} + \frac{e^{-2x}x}{4} + \frac{e^{-2x}}{4} + \frac{5x^2}{8} + \frac{x^3}{6} \\
&= \frac{x}{4} - \frac{e^{-2x}x^2}{4} + \frac{e^{-2x}x}{4} + \frac{e^{-2x}}{4} + \frac{5x^2}{8} + \frac{x^3}{6}
\end{aligned}$$

Now that all the U_i functions have been determined, the particular solution is found from

$$y_p = U_1y_1 + U_2y_2 + U_3y_3$$

Hence

$$\begin{aligned}
y_p &= \left(-\frac{e^{-x}}{3} + \frac{2x^2e^{-x}}{3} - \frac{xe^x}{3} - \frac{2e^xx^2}{3} \right) (e^x) \\
&\quad + \left(\frac{e^{4x}x}{12} + \frac{e^{4x}x^2}{24} - \frac{xe^{2x}}{4} + \frac{e^{2x}}{12} + \frac{x^2e^{2x}}{12} \right) (e^{-2x}) \\
&\quad + \left(\frac{x}{4} - \frac{e^{-2x}x^2}{4} + \frac{e^{-2x}x}{4} + \frac{e^{-2x}}{4} + \frac{5x^2}{8} + \frac{x^3}{6} \right) (e^{2x})
\end{aligned}$$

Therefore the particular solution is

$$y_p = \frac{x^2(xe^{2x} + 3)}{6}$$

Therefore the general solution is

$$\begin{aligned}
y &= y_h + y_p \\
&= (c_1e^x + e^{-2x}c_2 + e^{2x}c_3) + \left(\frac{x^2(xe^{2x} + 3)}{6} \right)
\end{aligned}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 33

```
ode:=diff(diff(diff(y(x),x),x),x)-diff(diff(y(x),x),x)-4*diff(y(x),x)+4*y(x) = 2*
x^2-4*x-1+2*x^2*exp(2*x)+5*x*exp(2*x)+exp(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(x^3 + 6c_3)e^{2x}}{6} + \frac{x^2}{2} + c_1e^x + c_2e^{-2x}$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 3; linear nonhomogeneous with symmetry [0,1]

trying high order linear exact nonhomogeneous

trying differential order: 3; missing the dependent variable

checking if the LODE has constant coefficients

<- constant coefficients successful

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) - \frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 4y(x) = 2x^2 - 4x - 1 + 2x^2e^{2x} + 5xe^{2x} + e^{2x}$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^3 - r^2 - 4r + 4 = 0$$

- Roots of the characteristic polynomial

$$r = [1, 2, -2]$$

- Homogeneous solution from $r = 1$

$$y_1(x) = e^x$$

- Homogeneous solution from $r = 2$

$$y_2(x) = e^{2x}$$

- Homogeneous solution from $r = -2$

$$y_3(x) = e^{-2x}$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^x + C_2 e^{2x} + C_3 e^{-2x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- o Define the forcing function of the ODE

$$f(x) = 2x^2 - 4x - 1 + 2x^2e^{2x} + 5xe^{2x} + e^{2x}$$

- o Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$

- o Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx}y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx}y_i(x) \right) \right)$$

- o Choose equation to add to a system of equations in $\frac{d}{dx}u_i(x)$

- $$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$
 - Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$
 - The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the derivative
$$\frac{d^3}{dx^3} y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$
 - Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE
$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$
 - Rearrange the ODE
$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) \right) = f(x)$$
 - Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0
$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$
 - We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$
 - Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$
 - Solve for the varied parameters
$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$
 - Substitute in the homogeneous solutions and forcing function and solve
$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} -\frac{1}{3e^x} + \frac{2x^2}{3e^x} - \frac{e^x x}{3} - \frac{2e^x x^2}{3} \\ \frac{x}{4} + \frac{x^3}{6} - \frac{x^2}{4e^{2x}} + \frac{x}{4e^{2x}} + \frac{5x^2}{8} + \frac{1}{4e^{2x}} \\ \frac{(e^x)^2}{12} - \frac{x(e^x)^2}{4} + \frac{x^2(e^x)^2}{12} + \frac{(e^x)^4 x}{12} + \frac{(e^x)^4 x^2}{24} \end{bmatrix}$$
- Find a particular solution $y_p(x)$ of the ODE
- $$y_p(x) = \frac{x^2(xe^{2x}+3)}{6}$$
- Substitute particular solution into general solution to ODE
$$y(x) = C_1 e^x + C_2 e^{2x} + C_3 e^{-2x} + \frac{x^2(xe^{2x}+3)}{6}$$

✓ **Mathematica.** Time used: 0.027 (sec). Leaf size: 187

```
ode=D[y[x],{x,3}]-D[y[x],{x,2}]-4*D[y[x],x]+4*y[x]==2*x^2-4*x-1+2*x^2*Exp[2*x]+5*
  x*Exp[2*x]+Exp[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-2x} \left(\int_1^x \frac{1}{12} e^{2K[1]} (2K[1]^2 - 4K[1] + e^{2K[1]} (2K[1]^2 + 5K[1] + 1) - 1) dK[1] + e^{3x} \int_1^x \frac{1}{3} e^{-K[2]} (-2K[2] + 1) dK[2] \right) + e^{-2x}$$

✓ **Sympy.** Time used: 0.264 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x**2*exp(2*x) - 2*x**2 - 5*x*exp(2*x) + 4*x + 4*y(x) - exp(2*x) - 4*
  Derivative(y(x), x) - Derivative(y(x), (x, 2)) + Derivative(y(x), (x, 3)) +
  1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_2 e^{-2x} + C_3 e^x + \frac{x^2}{2} + \left(C_1 + \frac{x^3}{6} \right) e^{2x}$$

2.11 Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Local contents

2.11.1 Problem 26	1885
2.11.2 Problem 27	1895
2.11.3 Problem 28	1906
2.11.4 Problem 29	1918
2.11.5 Problem 30	1924
2.11.6 Problem 31	1929
2.11.7 Problem 32	1939
2.11.8 Problem 33	1950
2.11.9 Problem 34	1957
2.11.10 Problem 36	1967
2.11.11 Problem 37	1980
2.11.12 Problem 38	1990
2.11.13 Problem 39	2003
2.11.14 Problem 40	2013

2.11.1 Problem 26

Local contents

Solved as second order linear constant coeff ode	1885
Solved as second order ode using Kovacic algorithm	1888
✓ Maple	1893
✓ Mathematica	1894
✓ Sympy	1894

Internal problem ID [8011]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 26

Date solved : Thursday, November 27, 2025 at 07:32:19 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.146 (sec)

Solve

$$y'' + y' + y = e^{3x} + 6e^x - 3e^{-2x} + 5$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 1, C = 1, f(x) = e^{3x} + 6e^x - 3e^{-2x} + 5$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y' + y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 1, C = 1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + \lambda e^{x\lambda} + e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + \lambda + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 1, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-1}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{1^2 - (4)(1)(1)} \\ &= -\frac{1}{2} \pm \frac{i\sqrt{3}}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} + \frac{i\sqrt{3}}{2} \\ \lambda_2 &= -\frac{1}{2} - \frac{i\sqrt{3}}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} + \frac{i\sqrt{3}}{2} \\ \lambda_2 &= -\frac{1}{2} - \frac{i\sqrt{3}}{2} \end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = -\frac{1}{2}$ and $\beta = \frac{\sqrt{3}}{2}$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^{-\frac{x}{2}} \left(c_1 \cos \left(\frac{\sqrt{3}x}{2} \right) + c_2 \sin \left(\frac{\sqrt{3}x}{2} \right) \right)$$

Therefore the homogeneous solution y_h is

$$y_h = e^{-\frac{x}{2}} \left(c_1 \cos \left(\frac{\sqrt{3}x}{2} \right) + c_2 \sin \left(\frac{\sqrt{3}x}{2} \right) \right)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{3x} + 6e^x - 3e^{-2x} + 5$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{e^x\}, \{e^{-2x}\}, \{e^{3x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right), e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 e^x + A_3 e^{-2x} + A_4 e^{3x}$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_2 e^x + 3A_3 e^{-2x} + 13A_4 e^{3x} + A_1 = e^{3x} + 6e^x - 3e^{-2x} + 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 5, A_2 = 2, A_3 = -1, A_4 = \frac{1}{13} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = 5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(e^{-\frac{x}{2}} \left(c_1 \cos\left(\frac{\sqrt{3}x}{2}\right) + c_2 \sin\left(\frac{\sqrt{3}x}{2}\right) \right) \right) + \left(5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13} \right) \end{aligned}$$

Summary of solutions found

$$y = 5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13} + e^{-\frac{x}{2}} \left(c_1 \cos\left(\frac{\sqrt{3}x}{2}\right) + c_2 \sin\left(\frac{\sqrt{3}x}{2}\right) \right)$$

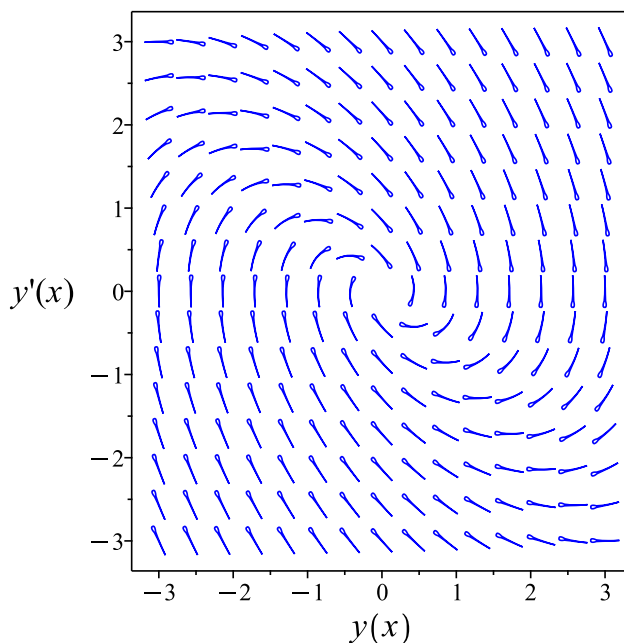


Figure 2.362: Slope field $y'' + y' + y = e^{3x} + 6e^x - 3e^{-2x} + 5$

Solved as second order ode using Kovacic algorithm

Time used: 0.204 (sec)

Solve

$$y'' + y' + y = e^{3x} + 6e^x - 3e^{-2x} + 5$$

Writing the ode as

$$y'' + y' + y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$A = 1$$

$$B = 1 \quad (3)$$

$$C = 1$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-3}{4} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -3 \\ t &= 4 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -\frac{3z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.237: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -\frac{3}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos\left(\frac{\sqrt{3}x}{2}\right)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{1}{1} dx} \\ &= z_1 e^{-\frac{x}{2}} \\ &= z_1 (e^{-\frac{x}{2}}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{1}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{2\sqrt{3} \tan\left(\frac{\sqrt{3}x}{2}\right)}{3} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned}
y &= c_1 y_1 + c_2 y_2 \\
&= c_1 \left(e^{-\frac{x}{2}} \cos \left(\frac{\sqrt{3}x}{2} \right) \right) + c_2 \left(e^{-\frac{x}{2}} \cos \left(\frac{\sqrt{3}x}{2} \right) \left(\frac{2\sqrt{3} \tan \left(\frac{\sqrt{3}x}{2} \right)}{3} \right) \right)
\end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + y' + y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-\frac{x}{2}} \cos \left(\frac{\sqrt{3}x}{2} \right) + \frac{2c_2 e^{-\frac{x}{2}} \sqrt{3} \sin \left(\frac{\sqrt{3}x}{2} \right)}{3}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^{3x} + 6e^x - 3e^{-2x} + 5$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1\}, \{e^x\}, \{e^{-2x}\}, \{e^{3x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ e^{-\frac{x}{2}} \cos \left(\frac{\sqrt{3}x}{2} \right), \frac{2e^{-\frac{x}{2}} \sqrt{3} \sin \left(\frac{\sqrt{3}x}{2} \right)}{3} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 + A_2 e^x + A_3 e^{-2x} + A_4 e^{3x}$$

The unknowns $\{A_1, A_2, A_3, A_4\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$3A_2 e^x + 3A_3 e^{-2x} + 13A_4 e^{3x} + A_1 = e^{3x} + 6e^x - 3e^{-2x} + 5$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 5, A_2 = 2, A_3 = -1, A_4 = \frac{1}{13} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = 5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(c_1 e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{2c_2 e^{-\frac{x}{2}} \sqrt{3} \sin\left(\frac{\sqrt{3}x}{2}\right)}{3} \right) + \left(5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13} \right)$$

Summary of solutions found

$$y = 5 + 2e^x - e^{-2x} + \frac{e^{3x}}{13} + c_1 e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{2c_2 e^{-\frac{x}{2}} \sqrt{3} \sin\left(\frac{\sqrt{3}x}{2}\right)}{3}$$

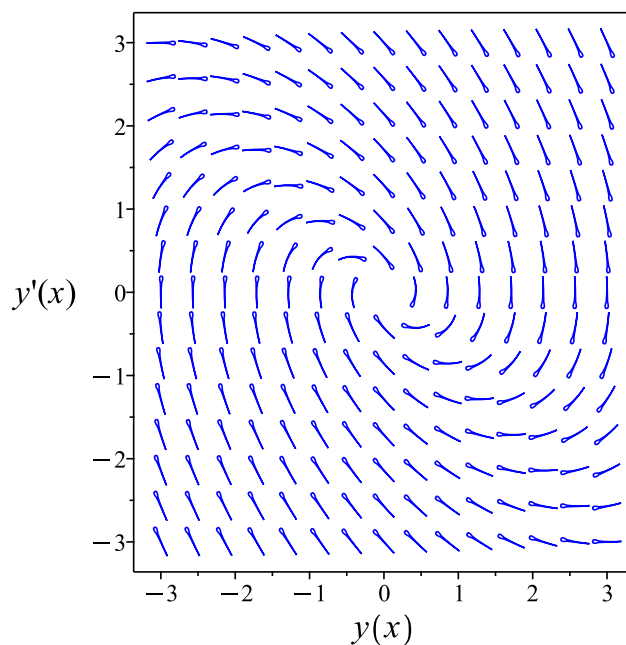


Figure 2.363: Slope field $y'' + y' + y = e^{3x} + 6e^x - 3e^{-2x} + 5$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 48

```
ode:=diff(diff(y(x),x),x)+diff(y(x),x)+y(x) = exp(3*x)+6*exp(x)-3*exp(-2*x)+5;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) c_2 + e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) c_1 + 5 + \frac{e^{3x}}{13} + 2e^x - e^{-2x}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + \frac{d}{dx}y(x) + y(x) = e^{3x} + 6e^x - 3e^{-2x} + 5$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + r + 1 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{(-1) \pm (\sqrt{-3})}{2}$$

- Roots of the characteristic polynomial

$$r = \left(-\frac{1}{2} - \frac{i\sqrt{3}}{2}, -\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + C_2 e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) + y_p(x)$$

□ Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^{3x} + 6e^x - 3e^{-2x} + 5 \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) & e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) \\ -\frac{e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right)}{2} - \frac{e^{-\frac{x}{2}} \sqrt{3} \sin\left(\frac{\sqrt{3}x}{2}\right)}{2} & -\frac{e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right)}{2} + \frac{e^{-\frac{x}{2}} \sqrt{3} \cos\left(\frac{\sqrt{3}x}{2}\right)}{2} \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = \frac{\sqrt{3}e^{-x}}{2}$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{2\sqrt{3}e^{-\frac{x}{2}} \left(\int e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) (e^{3x} + 6e^x - 3e^{-2x} + 5) dx \cos\left(\frac{\sqrt{3}x}{2}\right) - \int e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) (e^{3x} + 6e^x - 3e^{-2x} + 5) dx \sin\left(\frac{\sqrt{3}x}{2}\right) \right)}{3}$$

- Compute integrals

$$y_p(x) = -e^{-2x} + 2e^x + 5 + \frac{e^{3x}}{13}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + C_2 e^{-\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) - e^{-2x} + 2e^x + 5 + \frac{e^{3x}}{13}$$

✓ **Mathematica.** Time used: 1.76 (sec). Leaf size: 182

```
ode=D[y[x],{x,2}]+D[y[x],x]+y[x]==Exp[3*x]+6*Exp[x]-3*Exp[-2*x]+5;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x/2} \left(\cos\left(\frac{\sqrt{3}x}{2}\right) \int_1^x -\frac{2e^{-\frac{3K[2]}{2}} (-3 + 5e^{2K[2]} + 6e^{3K[2]} + e^{5K[2]}) \sin\left(\frac{1}{2}\sqrt{3}K[2]\right)}{\sqrt{3}} dK[2] + \sin\left(\frac{\sqrt{3}x}{2}\right) \right)$$

✓ **Sympy.** Time used: 0.167 (sec). Leaf size: 51

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x) - exp(3*x) - 6*exp(x) + Derivative(y(x), x) + Derivative(y(x), (x,
2)) - 5 + 3*exp(-2*x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \left(C_1 \sin\left(\frac{\sqrt{3}x}{2}\right) + C_2 \cos\left(\frac{\sqrt{3}x}{2}\right) \right) e^{-\frac{x}{2}} + \frac{e^{3x}}{13} + 2e^x + 5 - e^{-2x}$$

2.11.2 Problem 27

Local contents

Solved as second order linear constant coeff ode	1895
Solved as second order ode using Kovacic algorithm	1898
✓ Maple	1903
✓ Mathematica	1905
✓ Sympy	1905

Internal problem ID [8012]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 27

Date solved : Thursday, November 27, 2025 at 07:32:20 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.088 (sec)

Solve

$$y'' - y = e^x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = e^x$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since e^x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x e^x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x e^x$$

The unknowns $\{A_1\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$2A_1 e^x = e^x$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{2} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

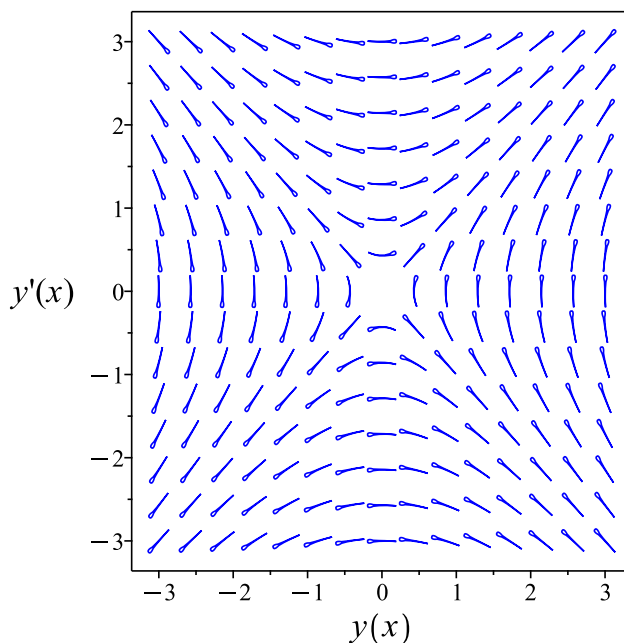
$$y_p = \frac{x e^x}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x} c_2) + \left(\frac{x e^x}{2} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x e^x}{2} + c_1 e^x + e^{-x} c_2$$

Figure 2.364: Slope field $y'' - y = e^x$

Solved as second order ode using Kovacic algorithm

Time used: 0.116 (sec)

Solve

$$y'' - y = e^x$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.239: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-x}$$

$$y_2 = \frac{e^x}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ \frac{d}{dx}(e^{-x}) & \frac{d}{dx}\left(\frac{e^x}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-x} & \frac{e^x}{2} \\ -e^{-x} & \frac{e^x}{2} \end{vmatrix}$$

Therefore

$$W = (e^{-x}) \left(\frac{e^x}{2} \right) - \left(\frac{e^x}{2} \right) (-e^{-x})$$

Which simplifies to

$$W = e^x e^{-x}$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{2x}}{1} dx$$

Which simplifies to

$$u_1 = - \int \frac{e^{2x}}{2} dx$$

Hence

$$u_1 = - \frac{e^{2x}}{4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^x e^{-x}}{1} dx$$

Which simplifies to

$$u_2 = \int 1 dx$$

Hence

$$u_2 = x$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\frac{e^{2x}e^{-x}}{4} + \frac{x e^x}{2}$$

Which simplifies to

$$y_p(x) = \frac{e^x(-1+2x)}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + \left(\frac{e^x(-1+2x)}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{e^x(-1+2x)}{4} + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

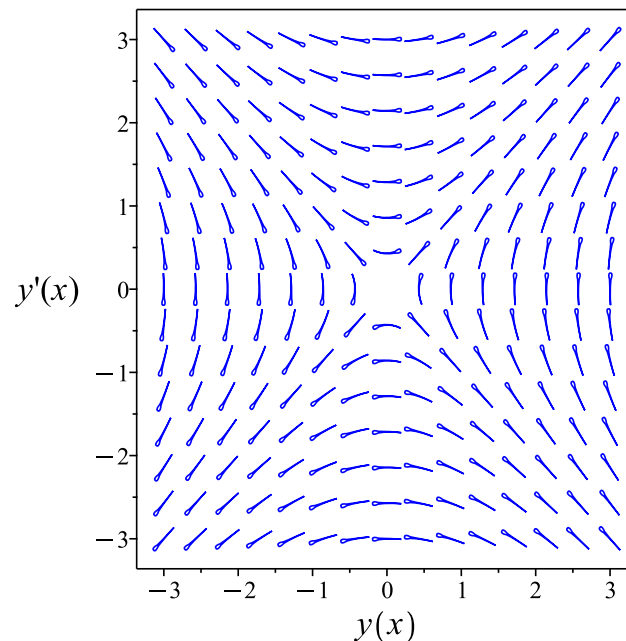


Figure 2.365: Slope field $y'' - y = e^x$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 20

```
ode:=diff(diff(y(x),x),x)-y(x) = exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x}c_2 + \frac{e^x(x+2c_1)}{2}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = e^x$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C1 y_1(x) + C2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-x} + C2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^x \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

- $$y_p(x) = -\frac{e^{-x} \int e^{2x} dx}{2} + \frac{e^x \int 1 dx}{2}$$
- Compute integrals

$$y_p(x) = \frac{e^x(-1+2x)}{4}$$
 - Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + \frac{e^x(-1+2x)}{4}$$

✓ **Mathematica.** Time used: 0.011 (sec). Leaf size: 29

```
ode=D[y[x],{x,2}]-y[x]==Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x \left(\frac{x}{2} - \frac{1}{4} + c_1 \right) + c_2 e^{-x}$$

✓ **Sympy.** Time used: 0.054 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - exp(x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_2 e^{-x} + \left(C_1 + \frac{x}{2} \right) e^x$$

2.11.3 Problem 28

Local contents

Solved as second order linear constant coeff ode	1906
Solved as second order solved by an integrating factor	1909
Solved as second order ode using Kovacic algorithm	1911
✓ Maple	1915
✓ Mathematica	1916
✓ Sympy	1917

Internal problem ID [8013]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 28

Date solved : Thursday, November 27, 2025 at 07:32:20 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.118 (sec)

Solve

$$y'' - 4y' + 4y = e^x + x e^{2x}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -4, C = 4, f(x) = e^x + x e^{2x}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 4y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 4$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} + 4e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{(-4)^2 - (4)(1)(4)} \\ &= 2 \end{aligned}$$

Hence this is the case of a double root $\lambda_{1,2} = -2$. Therefore the solution is

$$y = c_1 e^{2x} + c_2 x e^{2x} \quad (1)$$

Therefore the homogeneous solution y_h is

$$y_h = e^{2x} c_1 + x e^{2x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x + x e^{2x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x\}, \{x e^{2x}, e^{2x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{x e^{2x}, e^{2x}\}$$

Since e^{2x} is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{e^x\}, \{x e^{2x}, x^2 e^{2x}\}]$$

Since $x e^{2x}$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{e^x\}, \{x^2 e^{2x}, x^3 e^{2x}\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 e^x + A_2 x^2 e^{2x} + A_3 x^3 e^{2x}$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$A_1 e^x + 2A_2 e^{2x} + 6A_3 x e^{2x} = e^x + x e^{2x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 1, A_2 = 0, A_3 = \frac{1}{6} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

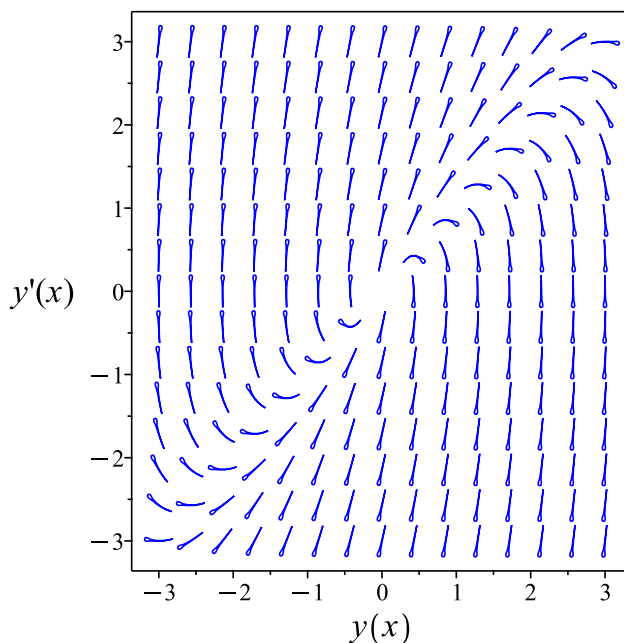
$$y_p = e^x + \frac{x^3 e^{2x}}{6}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x} c_1 + x e^{2x} c_2) + \left(e^x + \frac{x^3 e^{2x}}{6} \right) \end{aligned}$$

Summary of solutions found

$$y = e^x + \frac{x^3 e^{2x}}{6} + e^{2x} c_1 + x e^{2x} c_2$$

Figure 2.366: Slope field $y'' - 4y' + 4y = e^x + x e^{2x}$ **Solved as second order solved by an integrating factor**

Time used: 0.046 (sec)

Solve

$$y'' - 4y' + 4y = e^x + x e^{2x}$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x))^2 + p'(x)}{2}y = f(x)$$

Where $p(x) = -4$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int -4 \, dx} \\ &= e^{-2x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$\begin{aligned} (M(x)y)'' &= e^{-2x}(e^x + x e^{2x}) \\ (e^{-2x}y)'' &= e^{-2x}(e^x + x e^{2x}) \end{aligned}$$

Integrating once gives

$$(e^{-2x}y)' = \frac{x^2}{2} - e^{-x} + c_1$$

Integrating again gives

$$(e^{-2x}y)' = \frac{x^3}{6} + c_1x + e^{-x} + c_2$$

Hence the solution is

$$y = \frac{\frac{x^3}{6} + c_1x + e^{-x} + c_2}{e^{-2x}}$$

Or

$$y = \frac{x^3 e^{2x}}{6} + c_1 x e^{2x} + e^{2x} c_2 + e^x$$

Summary of solutions found

$$y = \frac{x^3 e^{2x}}{6} + c_1 x e^{2x} + e^{2x} c_2 + e^x$$

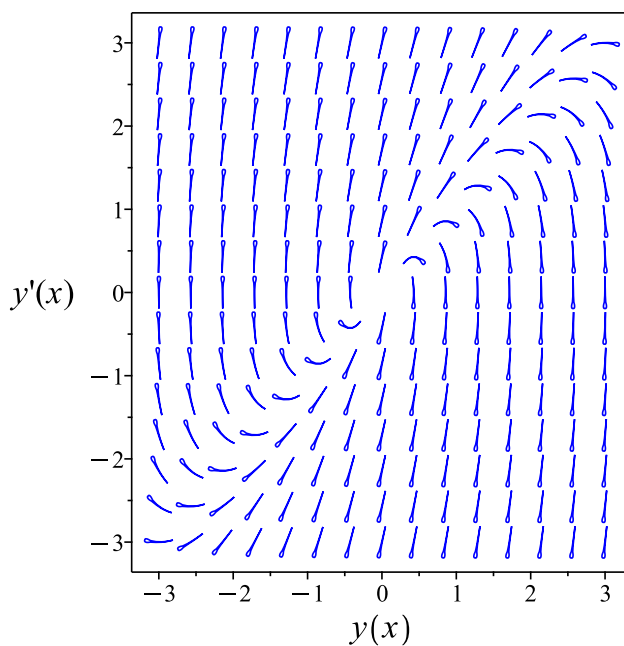


Figure 2.367: Slope field $y'' - 4y' + 4y = e^x + x e^{2x}$

Solved as second order ode using Kovacic algorithm

Time used: 0.105 (sec)

Solve

$$y'' - 4y' + 4y = e^x + x e^{2x}$$

Writing the ode as

$$y'' - 4y' + 4y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 4 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = y e^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = r z(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 0 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.241: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{2x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{2x}) + c_2 (e^{2x}(x)) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 4y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{2x} c_1 + x e^{2x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x + x e^{2x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x\}, \{x e^{2x}, e^{2x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{x e^{2x}, e^{2x}\}$$

Since e^{2x} is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{e^x\}, \{x e^{2x}, x^2 e^{2x}\}]$$

Since $x e^{2x}$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{e^x\}, \{x^2 e^{2x}, x^3 e^{2x}\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 e^x + A_2 x^2 e^{2x} + A_3 x^3 e^{2x}$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$A_1 e^x + 2A_2 e^{2x} + 6A_3 x e^{2x} = e^x + x e^{2x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 1, A_2 = 0, A_3 = \frac{1}{6} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

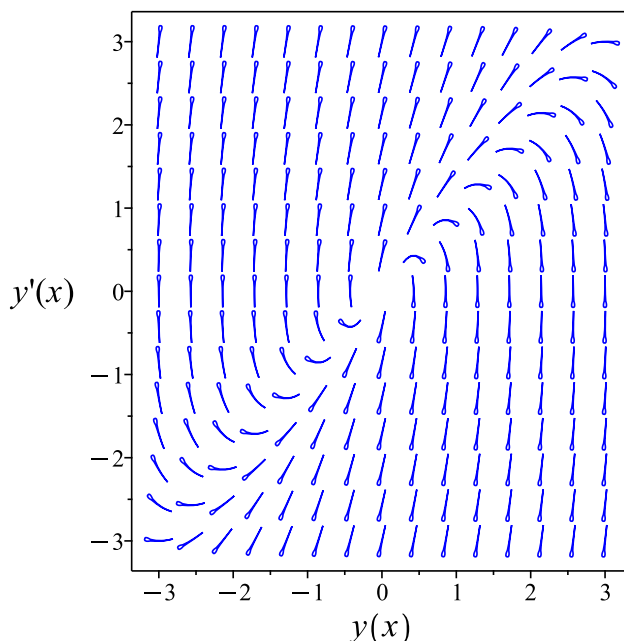
$$y_p = e^x + \frac{x^3 e^{2x}}{6}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x} c_1 + x e^{2x} c_2) + \left(e^x + \frac{x^3 e^{2x}}{6} \right) \end{aligned}$$

Summary of solutions found

$$y = e^x + \frac{x^3 e^{2x}}{6} + e^{2x} c_1 + x e^{2x} c_2$$

Figure 2.368: Slope field $y'' - 4y' + 4y = e^x + x e^{2x}$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 24

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x)+4*y(x) = exp(x)+x*exp(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(x^3 + 6c_1x + 6c_2)e^{2x}}{6} + e^x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful
```

Maple step by step

Let's solve

- $\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 4y(x) = e^x + xe^{2x}$
Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
Characteristic polynomial of homogeneous ODE
 $r^2 - 4r + 4 = 0$
- Factor the characteristic polynomial
 $(r - 2)^2 = 0$
- Root of the characteristic polynomial
 $r = 2$
- 1st solution of the homogeneous ODE
 $y_1(x) = e^{2x}$
- Repeated root, multiply $y_1(x)$ by x to ensure linear independence
 $y_2(x) = xe^{2x}$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C_1 e^{2x} + C_2 x e^{2x} + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^x + xe^{2x} \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & e^{2x} + 2x e^{2x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{4x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = e^{2x} \left(x \int (x + e^{-x}) dx - \int x(x + e^{-x}) dx \right)$$
 - Compute integrals

$$y_p(x) = \frac{e^{2x}x^3}{6} + e^x$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{2x} + C_2 x e^{2x} + \frac{e^{2x}x^3}{6} + e^x$$

✓ **Mathematica.** Time used: 0.133 (sec). Leaf size: 66

```
ode=D[y[x],{x,2}]-4*D[y[x],x]+4*y[x]==Exp[x]+x*Exp[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}e^x \left(2e^x \int_1^x (-K[1] - e^{-K[1]}) K[1] dK[1] + e^x x^3 + 2x(-1 + c_2 e^x) + 2c_1 e^x \right)$$

✓ Sympy. Time used: 0.169 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*exp(2*x) + 4*y(x) - exp(x) - 4*Derivative(y(x), x) + Derivative(y(x),
    (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \left(\left(C_1 + x \left(C_2 + \frac{x^2}{6} \right) \right) e^x + 1 \right) e^x$$

2.11.4 Problem 29

Local contents

Solved as higher order constant coeff ode	1918
✓ Maple	1920
✓ Mathematica	1922
✓ Sympy	1923

Internal problem ID [8014]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 29

Date solved : Thursday, November 27, 2025 at 07:32:21 AM

CAS classification : `[[_high_order, _linear, _nonhomogeneous]]`

Solve

$$y'''' - y = \sin(2x)$$

Solved as higher order constant coeff ode

Time used: 0.069 (sec)

The characteristic equation is

$$\lambda^4 - 1 = 0$$

The roots of the above equation are

$$\begin{aligned}\lambda_1 &= 1 \\ \lambda_2 &= -1 \\ \lambda_3 &= i \\ \lambda_4 &= -i\end{aligned}$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 e^{-x} + c_2 e^x + e^{ix} c_3 + e^{-ix} c_4$$

The fundamental set of solutions for the homogeneous solution are the following

$$\begin{aligned}y_1 &= e^{-x} \\ y_2 &= e^x \\ y_3 &= e^{ix} \\ y_4 &= e^{-ix}\end{aligned}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y'''' - y = 0$$

Now the particular solution to the given ODE is found

$$y'''' - y = \sin(2x)$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$\sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{ix}, e^{-x}, e^{-ix}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 \cos(2x) + A_2 \sin(2x)$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$15A_1 \cos(2x) + 15A_2 \sin(2x) = \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 0, A_2 = \frac{1}{15} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{\sin(2x)}{15}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= (c_1 e^{-x} + c_2 e^x + e^{ix} c_3 + e^{-ix} c_4) + \left(\frac{\sin(2x)}{15} \right)$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 29

```
ode:=diff(diff(diff(diff(y(x),x),x),x),x)-y(x) = sin(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{\sin(2x)}{15} + c_1 \cos(x) + c_2 e^x + c_3 \sin(x) + c_4 e^{-x}$$

Maple trace

Methods for high order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 4; linear nonhomogeneous with symmetry [0,1]

trying high order linear exact nonhomogeneous

trying differential order: 4; missing the dependent variable

checking if the LODE has constant coefficients

<- constant coefficients successful

Maple step by step

Let's solve

$$\frac{d^4}{dx^4} y(x) - y(x) = \sin(2x)$$

- Highest derivative means the order of the ODE is 4

$$\frac{d^4}{dx^4} y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^4 - 1 = 0$$

- Roots of the characteristic polynomial

$$r = [1, -1, i, -i]$$

- Homogeneous solution from $r = 1$

$$y_1(x) = e^x$$

- Homogeneous solution from $r = -1$

$$y_2(x) = e^{-x}$$

- Homogeneous solutions from $r = i$ and $r = -i$

- $[y_3(x) = \sin(x), y_4(x) = \cos(x)]$
- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + C_4 y_4(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^x + C_2 e^{-x} + C_3 \sin(x) + C_4 \cos(x) + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Define the forcing function of the ODE

$$f(x) = \sin(2x)$$
 - Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^4 u_i(x) y_i(x)$$
 - Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^4 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^4 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$
 - Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^4 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^4 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$
 - Calculate the 3rd derivative of $y_p(x)$

$$\frac{d^3}{dx^3} y_p(x) = \sum_{i=1}^4 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^4 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = 0$$
 - The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the derivative

$$\frac{d^4}{dx^4} y(x) + \left(\sum_{i=0}^3 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$
 - Substitute $y_p(x) = \sum_{i=1}^4 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^3 P_j(x) \left(\sum_{i=1}^4 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^4 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) + u_i(x) \left(\frac{d^4}{dx^4} y_i(x) \right) \right) = f(x)$$
 - Rearrange the ODE

$$\sum_{i=1}^4 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^3 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^4}{dx^4} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$
 - Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^4 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^3}{dx^3} y_i(x) \right) = f(x)$$

- We have now made a system of 4 equations in 4 unknowns $(\frac{d}{dx}u_i(x))$

$$\left[\sum_{i=1}^4 \left(\frac{d}{dx}u_i(x) \right) y_i(x) = 0, \sum_{i=1}^4 \left(\frac{d}{dx}u_i(x) \right) \left(\frac{d}{dx}y_i(x) \right) = 0, \sum_{i=1}^4 \left(\frac{d}{dx}u_i(x) \right) \left(\frac{d^2}{dx^2}y_i(x) \right) = 0, \sum_{i=1}^4 \left(\frac{d}{dx}u_i(x) \right) \left(\frac{d^3}{dx^3}y_i(x) \right) = 0 \right]$$
- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) & y_4(x) \\ \frac{d}{dx}y_1(x) & \frac{d}{dx}y_2(x) & \frac{d}{dx}y_3(x) & \frac{d}{dx}y_4(x) \\ \frac{d^2}{dx^2}y_1(x) & \frac{d^2}{dx^2}y_2(x) & \frac{d^2}{dx^2}y_3(x) & \frac{d^2}{dx^2}y_4(x) \\ \frac{d^3}{dx^3}y_1(x) & \frac{d^3}{dx^3}y_2(x) & \frac{d^3}{dx^3}y_3(x) & \frac{d^3}{dx^3}y_4(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx}u_1(x) \\ \frac{d}{dx}u_2(x) \\ \frac{d}{dx}u_3(x) \\ \frac{d}{dx}u_4(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ f(x) \end{bmatrix}$$
- Solve for the varied parameters
$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \\ u_4(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ f(x) \end{bmatrix} dx$$
- Substitute in the homogeneous solutions and forcing function and solve
$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \\ u_4(x) \end{bmatrix} = \begin{bmatrix} -\frac{e^{-x} \cos(2x)}{10} - \frac{e^{-x} \sin(2x)}{20} \\ \frac{e^x \cos(2x)}{10} - \frac{e^x \sin(2x)}{20} \\ \frac{\cos(x)^3}{3} \\ \frac{\sin(x)^3}{3} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = \frac{\sin(2x)}{15}$$

- Substitute particular solution into general solution to ODE
$$y(x) = C_1 e^x + C_2 e^{-x} + C_3 \sin(x) + C_4 \cos(x) + \frac{\sin(2x)}{15}$$

✓ **Mathematica.** Time used: 0.033 (sec). Leaf size: 90

```
ode=D[y[x],{x,4}]-y[x]==Sin[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^x \int_1^x \frac{1}{4} e^{-K[1]} \sin(2K[1]) dK[1] + e^{-x} \int_1^x -\frac{1}{4} e^{K[2]} \sin(2K[2]) dK[2] + c_1 e^x + c_3 e^{-x} + \frac{1}{3} \sin(x) \cos(x)$$

✓ Sympy. Time used: 0.055 (sec). Leaf size: 29

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - sin(2*x) + Derivative(y(x), (x, 4)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x + C_3 \sin(x) + C_4 \cos(x) + \frac{\sin(2x)}{15}$$

2.11.5 Problem 30

Local contents

Solved as higher order constant coeff ode	1924
✓ Maple	1926
✓ Mathematica	1928
✓ Sympy	1928

Internal problem ID [8015]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 30

Date solved : Thursday, November 27, 2025 at 07:32:21 AM

CAS classification : `[[_3rd_order, _linear, _nonhomogeneous]]`

Solve

$$y''' + y = \cos(x)$$

Solved as higher order constant coeff ode

Time used: 0.066 (sec)

The characteristic equation is

$$\lambda^3 + 1 = 0$$

The roots of the above equation are

$$\begin{aligned}\lambda_1 &= -1 \\ \lambda_2 &= \frac{1}{2} - \frac{i\sqrt{3}}{2} \\ \lambda_3 &= \frac{1}{2} + \frac{i\sqrt{3}}{2}\end{aligned}$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 e^{-x} + e^{\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)x} c_2 + e^{\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)x} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$\begin{aligned}y_1 &= e^{-x} \\ y_2 &= e^{\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)x} \\ y_3 &= e^{\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)x}\end{aligned}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' + y = 0$$

Now the particular solution to the given ODE is found

$$y''' + y = \cos(x)$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$\cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(x), \sin(x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ e^{\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)x}, e^{\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)x}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 \cos(x) + A_2 \sin(x)$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$A_1 \sin(x) - A_2 \cos(x) + A_1 \cos(x) + A_2 \sin(x) = \cos(x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{2}, A_2 = -\frac{1}{2} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{\cos(x)}{2} - \frac{\sin(x)}{2}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(c_1 e^{-x} + e^{\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)x} c_2 + e^{\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)x} c_3 \right) + \left(\frac{\cos(x)}{2} - \frac{\sin(x)}{2} \right)$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 45

```
ode:=diff(diff(diff(y(x),x),x),x)+y(x) = cos(x);
dsolve(ode,y(x), singsol=all);
```

$$y = c_2 e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + c_3 e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) + c_1 e^{-x} + \frac{\cos(x)}{2} - \frac{\sin(x)}{2}$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 3; linear nonhomogeneous with symmetry [0,1]
trying high order linear exact nonhomogeneous
trying differential order: 3; missing the dependent variable
checking if the LODE has constant coefficients
<- constant coefficients successful
```

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) + y(x) = \cos(x)$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^3 + 1 = 0$$

- Roots of the characteristic polynomial

$$r = \left[-1, \frac{1}{2} - \frac{i\sqrt{3}}{2}, \frac{1}{2} + \frac{i\sqrt{3}}{2} \right]$$

- Homogeneous solution from $r = -1$

$$y_1(x) = e^{-x}$$

- Homogeneous solutions from $r = \frac{1}{2} - \frac{i\sqrt{3}}{2}$ and $r = \frac{1}{2} + \frac{i\sqrt{3}}{2}$

$$\left[y_2(x) = e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right), y_3(x) = e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) \right]$$

- General solution of the ODE
 $y(x) = C1 y_1(x) + C2 y_2(x) + C3 y_3(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C1 e^{-x} + C2 e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) + C3 e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Define the forcing function of the ODE
 $f(x) = \cos(x)$
 - Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$
 - Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$
 - Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$
 - Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$
 - The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^3}{dx^3} y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$
 - Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$
 - Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) \right) = f(x)$$
 - Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$
 - We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$
 - Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} -\frac{\left(-\frac{\sqrt{3}}{2}-1\right)e^{-\frac{x}{2}}\cos\left(x\left(\frac{\sqrt{3}}{2}+1\right)\right)}{6\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}+1\right)^2\right)} + \frac{e^{-\frac{x}{2}}\sin\left(x\left(\frac{\sqrt{3}}{2}+1\right)\right)}{12\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}+1\right)^2\right)} - \frac{\left(-\frac{\sqrt{3}}{2}+1\right)e^{-\frac{x}{2}}\cos\left(x\left(\frac{\sqrt{3}}{2}-1\right)\right)}{6\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}-1\right)^2\right)} + \frac{e^{-\frac{x}{2}}\sin\left(x\left(\frac{\sqrt{3}}{2}-1\right)\right)}{12\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}-1\right)^2\right)} \\ \frac{e^{-\frac{x}{2}}\cos\left(x\left(\frac{\sqrt{3}}{2}-1\right)\right)}{12\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}-1\right)^2\right)} + \frac{\left(-\frac{\sqrt{3}}{2}+1\right)e^{-\frac{x}{2}}\sin\left(x\left(\frac{\sqrt{3}}{2}-1\right)\right)}{6\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}-1\right)^2\right)} + \frac{e^{-\frac{x}{2}}\cos\left(x\left(\frac{\sqrt{3}}{2}+1\right)\right)}{12\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}+1\right)^2\right)} + \frac{\left(-\frac{\sqrt{3}}{2}-1\right)e^{-\frac{x}{2}}\sin\left(x\left(\frac{\sqrt{3}}{2}+1\right)\right)}{6\left(\frac{1}{4}+\left(\frac{\sqrt{3}}{2}+1\right)^2\right)} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = \frac{\cos(x)}{2} - \frac{\sin(x)}{2}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right) + C_3 e^{\frac{x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{\cos(x)}{2} - \frac{\sin(x)}{2}$$

✓ **Mathematica.** Time used: 0.208 (sec). Leaf size: 232

```
ode=D[y[x],{x,3}]+y[x]==Cos[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-x} \left(\int_1^x \frac{1}{3} e^{K[1]} \cos(K[1]) dK[1] + e^{3x/2} \sin\left(\frac{\sqrt{3}x}{2}\right) \int_1^x -\frac{e^{-\frac{K[2]}{2}} \cos(K[2]) \left(\sqrt{3} \sin\left(\frac{1}{2}\sqrt{3}K[2]\right) - 3 \cos\left(\frac{1}{2}\sqrt{3}K[2]\right)\right)}{3\sqrt{3}} dK[2] \right)$$

✓ **Sympy.** Time used: 0.087 (sec). Leaf size: 46

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x) - cos(x) + Derivative(y(x), (x, 3)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_3 e^{-x} + \left(C_1 \sin\left(\frac{\sqrt{3}x}{2}\right) + C_2 \cos\left(\frac{\sqrt{3}x}{2}\right) \right) e^{\frac{x}{2}} - \frac{\sin(x)}{2} + \frac{\cos(x)}{2}$$

2.11.6 Problem 31

Local contents

Solved as second order linear constant coeff ode	1929
Solved as second order ode using Kovacic algorithm	1932
✓ Maple	1936
✓ Mathematica	1938
✓ Sympy	1938

Internal problem ID [8016]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 31

Date solved : Thursday, November 27, 2025 at 07:32:21 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.104 (sec)

Solve

$$y'' + 4y = \sin(2x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 4, f(x) = \sin(2x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 4y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 4$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 4e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(4)} \\ &= \pm 2i \end{aligned}$$

Hence

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

Which simplifies to

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 2$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(2x) + c_2 \sin(2x))$$

Or

$$y = c_1 \cos(2x) + c_2 \sin(2x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(2x) + c_2 \sin(2x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{\cos(2x), \sin(2x)\}$$

Since $\cos(2x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x \cos(2x), \sin(2x) x\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x \cos(2x) + A_2 \sin(2x) x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 \sin(2x) + 4A_2 \cos(2x) = \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{4}, A_2 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

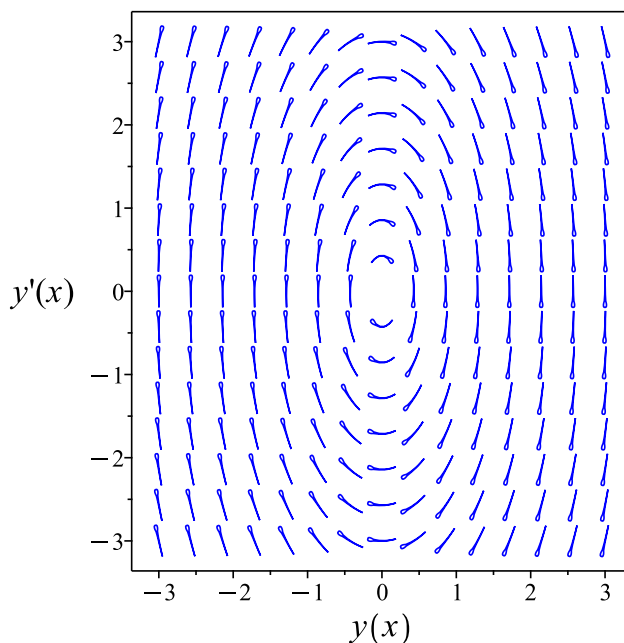
$$y_p = -\frac{x \cos(2x)}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 \cos(2x) + c_2 \sin(2x)) + \left(-\frac{x \cos(2x)}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{x \cos(2x)}{4} + c_1 \cos(2x) + c_2 \sin(2x)$$

Figure 2.369: Slope field $y'' + 4y = \sin(2x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.093 (sec)

Solve

$$y'' + 4y = \sin(2x)$$

Writing the ode as

$$y'' + 4y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 4 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-4}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -4 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -4z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.245: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -4$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(2x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(2x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(2x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(2x) \int \frac{1}{\cos^2(2x)} dx \\ &= \cos(2x) \left(\frac{\tan(2x)}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\cos(2x)) + c_2 \left(\cos(2x) \left(\frac{\tan(2x)}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 4y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\sin(2x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{\cos(2x), \sin(2x)\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{\sin(2x)}{2}, \cos(2x) \right\}$$

Since $\cos(2x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[x \cos(2x), \sin(2x) x]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 x \cos(2x) + A_2 \sin(2x) x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-4A_1 \sin(2x) + 4A_2 \cos(2x) = \sin(2x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{4}, A_2 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{x \cos(2x)}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2} \right) + \left(-\frac{x \cos(2x)}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{x \cos(2x)}{4} + c_1 \cos(2x) + \frac{c_2 \sin(2x)}{2}$$

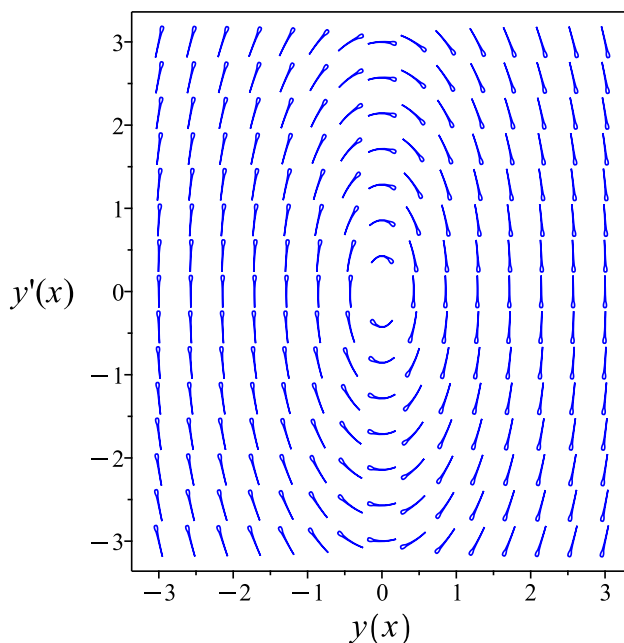


Figure 2.370: Slope field $y'' + 4y = \sin(2x)$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 24

```
ode:=4*y(x)+diff(diff(y(x),x),x) = sin(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(-x + 4c_1) \cos(2x)}{4} + \sin(2x) c_2$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 4y(x) = \sin(2x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 4 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-16})}{2}$$

- Roots of the characteristic polynomial

$$r = (-2i, 2i)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(2x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(2x)$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 \cos(2x) + C_2 \sin(2x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \sin(2x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{\cos(2x) \int \sin(2x)^2 dx}{2} + \frac{\sin(2x) \int \sin(4x) dx}{4}$$

- Compute integrals

$$y_p(x) = \frac{\sin(2x)}{16} - \frac{\cos(2x)x}{4}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(2x) + C_2 \sin(2x) + \frac{\sin(2x)}{16} - \frac{\cos(2x)x}{4}$$

✓ **Mathematica.** Time used: 0.029 (sec). Leaf size: 64

```
ode=D[y[x],{x,2}]+4*y[x]==Sin[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \sin(2x) \int_1^x \frac{1}{4} \sin(4K[2]) dK[2] + \cos(2x) \int_1^x -\frac{1}{2} \sin^2(2K[1]) dK[1] + c_1 \cos(2x) + c_2 \sin(2x)$$

✓ **Sympy.** Time used: 0.058 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*y(x) - sin(2*x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_2 \sin(2x) + \left(C_1 - \frac{x}{4}\right) \cos(2x)$$

2.11.7 Problem 32

Local contents

Solved as second order linear constant coeff ode	1939
Solved as second order ode using Kovacic algorithm	1942
✓ Maple	1947
✓ Mathematica	1949
✓ Sympy	1949

Internal problem ID [8017]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 32

Date solved : Thursday, November 27, 2025 at 07:32:21 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.129 (sec)

Solve

$$y'' + 5y = \cos(\sqrt{5}x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 5, f(x) = \cos(\sqrt{5}x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 5y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 5$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 5 e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 5 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 5$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(5)} \\ &= \pm i\sqrt{5} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= +i\sqrt{5} \\ \lambda_2 &= -i\sqrt{5} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= i\sqrt{5} \\ \lambda_2 &= -i\sqrt{5} \end{aligned}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = \sqrt{5}$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x))$$

Or

$$y = c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x)$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\cos(\sqrt{5}x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$\left[\left\{ \cos(\sqrt{5}x), \sin(\sqrt{5}x) \right\} \right]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \cos(\sqrt{5}x), \sin(\sqrt{5}x) \right\}$$

Since $\cos(\sqrt{5}x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$\left[\left\{ \cos(\sqrt{5}x)x, \sin(\sqrt{5}x)x \right\} \right]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 \cos(\sqrt{5}x)x + A_2 \sin(\sqrt{5}x)x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_1\sqrt{5} \sin(\sqrt{5}x) + 2A_2\sqrt{5} \cos(\sqrt{5}x) = \cos(\sqrt{5}x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 0, A_2 = \frac{\sqrt{5}}{10} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{\sqrt{5} \sin(\sqrt{5}x)x}{10}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x) \right) + \left(\frac{\sqrt{5} \sin(\sqrt{5}x)x}{10} \right)$$

Summary of solutions found

$$y = \frac{\sqrt{5} \sin(\sqrt{5}x)x}{10} + c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x)$$

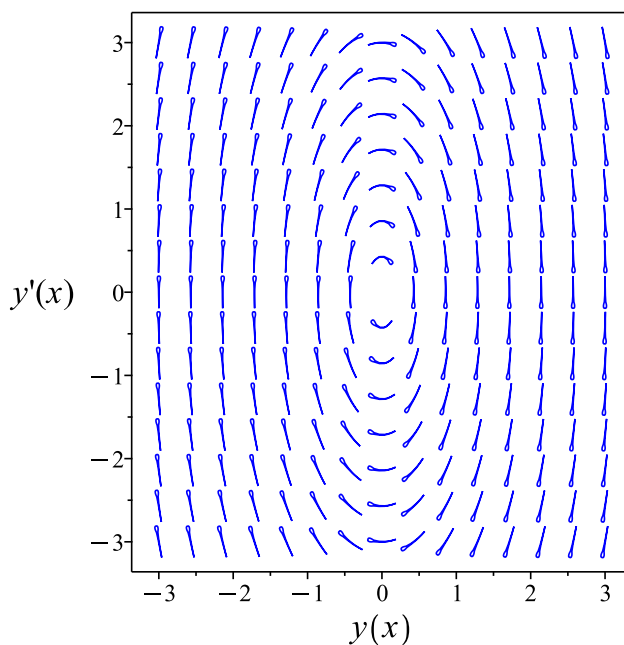


Figure 2.371: Slope field $y'' + 5y = \cos(\sqrt{5}x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.124 (sec)

Solve

$$y'' + 5y = \cos(\sqrt{5}x)$$

Writing the ode as

$$y'' + 5y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= 5 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-5}{1} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -5 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -5z(x) \tag{7}$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.247: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\
 &= 0 - 0 \\
 &= 0
 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -5$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(\sqrt{5}x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned}
 y_1 &= z_1 \\
 &= \cos(\sqrt{5}x)
 \end{aligned}$$

Which simplifies to

$$y_1 = \cos(\sqrt{5}x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(\sqrt{5}x) \int \frac{1}{\cos(\sqrt{5}x)^2} dx \\ &= \cos(\sqrt{5}x) \left(\frac{\sqrt{5} \tan(\sqrt{5}x)}{5} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\cos(\sqrt{5}x) \right) + c_2 \left(\cos(\sqrt{5}x) \left(\frac{\sqrt{5} \tan(\sqrt{5}x)}{5} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 5y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(\sqrt{5}x) + \frac{c_2 \sqrt{5} \sin(\sqrt{5}x)}{5}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$\cos(\sqrt{5}x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$\left[\left\{ \cos(\sqrt{5}x), \sin(\sqrt{5}x) \right\} \right]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{\sqrt{5} \sin(\sqrt{5}x)}{5}, \cos(\sqrt{5}x) \right\}$$

Since $\cos(\sqrt{5}x)$ is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$\left[\left\{ \cos(\sqrt{5}x)x, \sin(\sqrt{5}x)x \right\} \right]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_1 \cos(\sqrt{5}x)x + A_2 \sin(\sqrt{5}x)x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-2A_1\sqrt{5} \sin(\sqrt{5}x) + 2A_2\sqrt{5} \cos(\sqrt{5}x) = \cos(\sqrt{5}x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 0, A_2 = \frac{\sqrt{5}}{10} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

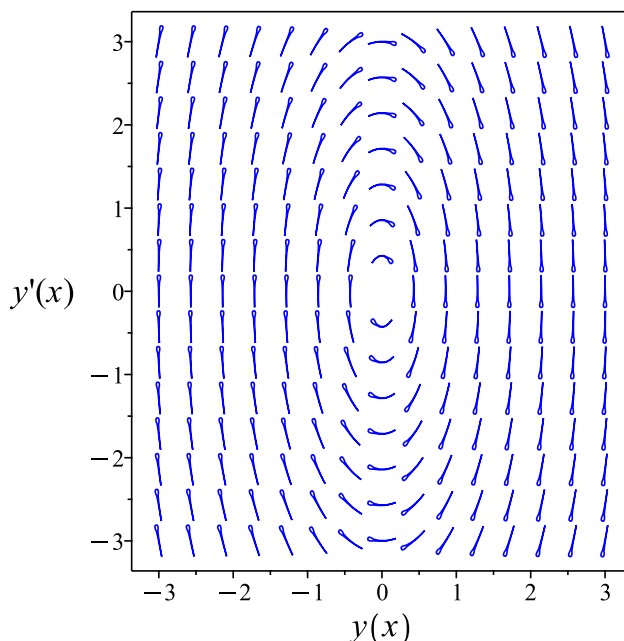
$$y_p = \frac{\sqrt{5} \sin(\sqrt{5}x)x}{10}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(\sqrt{5}x) + \frac{c_2\sqrt{5} \sin(\sqrt{5}x)}{5} \right) + \left(\frac{\sqrt{5} \sin(\sqrt{5}x)x}{10} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{\sqrt{5} \sin(\sqrt{5}x)x}{10} + c_1 \cos(\sqrt{5}x) + \frac{c_2\sqrt{5} \sin(\sqrt{5}x)}{5}$$

Figure 2.372: Slope field $y'' + 5y = \cos(\sqrt{5}x)$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 35

```
ode:=diff(diff(y(x),x),x)+5*y(x) = cos(5^(1/2)*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(10c_1 + 1) \cos(\sqrt{5}x)}{10} + \frac{\sin(\sqrt{5}x) (\sqrt{5}x + 10c_2)}{10}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful
```

Maple step by step

Let's solve

- $\frac{d^2}{dx^2}y(x) + 5y(x) = \cos(\sqrt{5}x)$
- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE
 $r^2 + 5 = 0$
- Use quadratic formula to solve for r
 $r = \frac{0 \pm (\sqrt{-20})}{2}$
- Roots of the characteristic polynomial
 $r = (-i\sqrt{5}, i\sqrt{5})$
- 1st solution of the homogeneous ODE
 $y_1(x) = \cos(\sqrt{5}x)$
- 2nd solution of the homogeneous ODE
 $y_2(x) = \sin(\sqrt{5}x)$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C_1 \cos(\sqrt{5}x) + C_2 \sin(\sqrt{5}x) + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \cos(\sqrt{5}x) \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(\sqrt{5}x) & \sin(\sqrt{5}x) \\ -\sqrt{5} \sin(\sqrt{5}x) & \sqrt{5} \cos(\sqrt{5}x) \end{bmatrix}$$
 - Compute Wronskian
 $W(y_1(x), y_2(x)) = \sqrt{5}$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{\sqrt{5} \left(\int \sin(2\sqrt{5}x) dx \cos(\sqrt{5}x) - 2 \int \cos(\sqrt{5}x)^2 dx \sin(\sqrt{5}x) \right)}{10}$$
 - Compute integrals

$$y_p(x) = \frac{\sin(\sqrt{5}x)\sqrt{5}x}{10} + \frac{\cos(\sqrt{5}x)}{20}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(\sqrt{5}x) + C_2 \sin(\sqrt{5}x) + \frac{\sin(\sqrt{5}x)\sqrt{5}x}{10} + \frac{\cos(\sqrt{5}x)}{20}$$

✓ **Mathematica.** Time used: 0.09 (sec). Leaf size: 96

```
ode=D[y[x],{x,2}]+5*y[x]==Cos[Sqrt[5]*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \sin(\sqrt{5}x) \int_1^x \frac{\cos^2(\sqrt{5}K[2])}{\sqrt{5}} dK[2] + \cos(\sqrt{5}x) \int_1^x -\frac{\sin(2\sqrt{5}K[1])}{2\sqrt{5}} dK[1] + c_1 \cos(\sqrt{5}x) + c_2 \sin(\sqrt{5}x)$$

✓ **Sympy.** Time used: 0.060 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(5*y(x) - cos(sqrt(5)*x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_2 \cos(\sqrt{5}x) + \left(C_1 + \frac{\sqrt{5}x}{10}\right) \sin(\sqrt{5}x)$$

2.11.8 Problem 33

Local contents

Solved as higher order constant coeff ode	1950
✓ Maple	1954
✓ Mathematica	1956
✓ Sympy	1956

Internal problem ID [8018]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 33

Date solved : Thursday, November 27, 2025 at 07:32:22 AM

CAS classification : `[[_3rd_order, _linear, _nonhomogeneous]]`

Solve

$$y''' + y'' + y' + y = e^x + e^{-x} + \sin(x)$$

Solved as higher order constant coeff ode

Time used: 0.799 (sec)

The characteristic equation is

$$\lambda^3 + \lambda^2 + \lambda + 1 = 0$$

The roots of the above equation are

$$\lambda_1 = -1$$

$$\lambda_2 = i$$

$$\lambda_3 = -i$$

Therefore the homogeneous solution is

$$y_h(x) = c_1 e^{-x} + e^{ix} c_2 + e^{-ix} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = e^{-x}$$

$$y_2 = e^{ix}$$

$$y_3 = e^{-ix}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' + y'' + y' + y = 0$$

Now the particular solution to the given ODE is found

$$y''' + y'' + y' + y = \sin(x) + 2 \cosh(x)$$

Let the particular solution be

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Where y_i are the basis solutions found above for the homogeneous solution y_h and $U_i(x)$ are functions to be determined as follows

$$U_i = (-1)^{n-i} \int \frac{F(x)W_i(x)}{aW(x)} dx$$

Where $W(x)$ is the Wronskian and $W_i(x)$ is the Wronskian that results after deleting the last row and the i -th column of the determinant and n is the order of the ODE or equivalently, the number of basis solutions, and a is the coefficient of the leading derivative in the ODE, and $F(x)$ is the RHS of the ODE. Therefore, the first step is to find the Wronskian $W(x)$. This is given by

$$W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Substituting the fundamental set of solutions y_i found above in the Wronskian gives

$$W = \begin{bmatrix} e^{-x} & e^{ix} & e^{-ix} \\ -e^{-x} & ie^{ix} & -ie^{-ix} \\ e^{-x} & -e^{ix} & -e^{-ix} \end{bmatrix}$$

$$|W| = -4ie^{-x}e^{ix}e^{-ix}$$

The determinant simplifies to

$$|W| = -4ie^{-x}$$

Now we determine W_i for each U_i .

$$W_1(x) = \det \begin{bmatrix} e^{ix} & e^{-ix} \\ ie^{ix} & -ie^{-ix} \end{bmatrix}$$

$$= -2i$$

$$\begin{aligned}
 W_2(x) &= \det \begin{bmatrix} e^{-x} & e^{-ix} \\ -e^{-x} & -ie^{-ix} \end{bmatrix} \\
 &= (1-i)e^{(-1-i)x}
 \end{aligned}$$

$$\begin{aligned}
 W_3(x) &= \det \begin{bmatrix} e^{-x} & e^{ix} \\ -e^{-x} & ie^{ix} \end{bmatrix} \\
 &= (1+i)e^{(-1+i)x}
 \end{aligned}$$

Now we are ready to evaluate each $U_i(x)$.

$$\begin{aligned}
 U_1 &= (-1)^{3-1} \int \frac{F(x)W_1(x)}{aW(x)} dx \\
 &= (-1)^2 \int \frac{(\sin(x) + 2 \cosh(x))(-2i)}{(1)(-4ie^{-x})} dx \\
 &= \int \frac{-2i(\sin(x) + 2 \cosh(x))}{-4ie^{-x}} dx \\
 &= \int \left(\left(\frac{\sin(x)}{2} + \cosh(x) \right) e^x \right) dx \\
 &= \frac{x}{2} + \frac{e^{2x}}{4} - \frac{e^x \cos(x)}{4} + \frac{\sin(x) e^x}{4} \\
 &= \frac{x}{2} + \frac{e^{2x}}{4} - \frac{e^x \cos(x)}{4} + \frac{\sin(x) e^x}{4}
 \end{aligned}$$

$$\begin{aligned}
 U_2 &= (-1)^{3-2} \int \frac{F(x)W_2(x)}{aW(x)} dx \\
 &= (-1)^1 \int \frac{(\sin(x) + 2 \cosh(x))((1-i)e^{(-1-i)x})}{(1)(-4ie^{-x})} dx \\
 &= - \int \frac{(1-i)(\sin(x) + 2 \cosh(x))e^{(-1-i)x}}{-4ie^{-x}} dx \\
 &= - \int \left(\left(\frac{1}{4} + \frac{i}{4} \right) (\sin(x) + 2 \cosh(x)) e^{-ix} \right) dx \\
 &= - \int \left(\frac{1}{4} + \frac{i}{4} \right) (\sin(x) + 2 \cosh(x)) e^{-ix} dx
 \end{aligned}$$

$$\begin{aligned}
U_3 &= (-1)^{3-3} \int \frac{F(x)W_3(x)}{aW(x)} dx \\
&= (-1)^0 \int \frac{(\sin(x) + 2 \cosh(x)) ((1+i)e^{(-1+i)x})}{(1)(-4ie^{-x})} dx \\
&= \int \frac{(1+i)(\sin(x) + 2 \cosh(x)) e^{(-1+i)x}}{-4ie^{-x}} dx \\
&= \int \left(\left(-\frac{1}{4} + \frac{i}{4} \right) (\sin(x) + 2 \cosh(x)) e^{ix} \right) dx \\
&= -\frac{x}{8} - \frac{ix}{8} + \frac{e^{2ix}}{16} - \frac{ie^{2ix}}{16} + \frac{i(e^{2x} - i)e^{(-1+i)x}}{4} \\
&= -\frac{x}{8} - \frac{ix}{8} + \frac{e^{2ix}}{16} - \frac{ie^{2ix}}{16} + \frac{i(e^{2x} - i)e^{(-1+i)x}}{4}
\end{aligned}$$

Now that all the U_i functions have been determined, the particular solution is found from

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Hence

$$\begin{aligned}
y_p &= \left(\frac{x}{2} + \frac{e^{2x}}{4} - \frac{e^x \cos(x)}{4} + \frac{\sin(x) e^x}{4} \right) (e^{-x}) \\
&\quad + \left(- \int \left(\frac{1}{4} + \frac{i}{4} \right) (\sin(x) + 2 \cosh(x)) e^{-ix} dx \right) (e^{ix}) \\
&\quad + \left(-\frac{x}{8} - \frac{ix}{8} + \frac{e^{2ix}}{16} - \frac{ie^{2ix}}{16} + \frac{i(e^{2x} - i)e^{(-1+i)x}}{4} \right) (e^{-ix})
\end{aligned}$$

Therefore the particular solution is

$$y_p = \left(-\frac{1}{8} - \frac{i}{4} \right) e^{ix} + \left(-\frac{1}{16} + \frac{3i}{16} \right) e^{-ix} - \frac{x \cos(x)}{4} - \frac{x \sin(x)}{4} + \frac{e^x}{4} + \frac{(1+x)e^{-x}}{2}$$

Which simplifies to

$$y_p = -\frac{(-4e^{2x} + e^x(\cos(x)(3+i+4x) + (-7+i+4x)\sin(x)) - 8x - 8)e^{-x}}{16}$$

Therefore the general solution is

$$\begin{aligned}
y &= y_h + y_p \\
&= (c_1 e^{-x} + e^{ix} c_2 + e^{-ix} c_3) \\
&\quad + \left(-\frac{(-4e^{2x} + e^x(\cos(x)(3+i+4x) + (-7+i+4x)\sin(x)) - 8x - 8)e^{-x}}{16} \right)
\end{aligned}$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 46

```
ode:=diff(diff(diff(y(x),x),x),x)+diff(diff(y(x),x),x)+diff(y(x),x)+y(x) = exp(x)
+exp(-x)+sin(x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(4c_3 + 2x + 2)e^{-x}}{4} + \frac{(-x + 4c_2 + 1)\sin(x)}{4} + \frac{(-x + 4c_1)\cos(x)}{4} + \frac{e^x}{4}$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 3; linear nonhomogeneous with symmetry [0,1]

trying high order linear exact nonhomogeneous

trying differential order: 3; missing the dependent variable

checking if the LODE has constant coefficients

<- constant coefficients successful

Maple step by step

Let's solve

$$\frac{d^3}{dx^3}y(x) + \frac{d^2}{dx^2}y(x) + \frac{d}{dx}y(x) + y(x) = e^x + e^{-x} + \sin(x)$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^3 + r^2 + r + 1 = 0$$

- Roots of the characteristic polynomial

$$r = [-1, I, -I]$$

- Homogeneous solution from $r = -1$

$$y_1(x) = e^{-x}$$

- Homogeneous solutions from $r = I$ and $r = -I$

$$[y_2(x) = \sin(x), y_3(x) = \cos(x)]$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + C3y_3(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-x} + C2 \sin(x) + C3 \cos(x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Define the forcing function of the ODE

$$f(x) = e^x + e^{-x} + \sin(x)$$

- Form of the particular solution to the ODE where the $u_i(x)$ are to be found

$$y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$$

- Calculate the 1st derivative of $y_p(x)$

$$\frac{d}{dx} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0$$

- Calculate the 2nd derivative of $y_p(x)$

$$\frac{d^2}{dx^2} y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2} y_i(x) \right) \right)$$

- Choose equation to add to a system of equations in $\frac{d}{dx} u_i(x)$

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0$$

- The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the der

$$\frac{d^3}{dx^3} y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i} y(x) \right) \right) = f(x)$$

- Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE

$$\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3} y_i(x) \right) \right) = f(x)$$

- Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} \frac{(e^x)^2}{4} + \frac{x}{2} - \frac{e^x \cos(x)}{4} + \frac{e^x \sin(x)}{4} \\ -\frac{x}{4} + \frac{e^x \cos(x)}{2} + \frac{e^{-x} \sin(x)}{2} + \frac{\sin(2x)}{8} - \frac{\cos(2x)}{8} \\ -\frac{x}{4} - \frac{e^x \sin(x)}{2} + \frac{e^{-x} \cos(x)}{2} + \frac{\sin(2x)}{8} + \frac{\cos(2x)}{8} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = \frac{(4x+4)e^{-x}}{8} + \frac{(-2x-1)\cos(x)}{8} + \frac{(-2x+3)\sin(x)}{8} + \frac{e^x}{4}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 \sin(x) + C_3 \cos(x) + \frac{(4x+4)e^{-x}}{8} + \frac{(-2x-1)\cos(x)}{8} + \frac{(-2x+3)\sin(x)}{8} + \frac{e^x}{4}$$

✓ **Mathematica.** Time used: 0.108 (sec). Leaf size: 146

```
ode=D[y[x],{x,3}]+D[y[x],{x,2}]+D[y[x],x]+y[x]==Exp[x]+Exp[-x]+Sin[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \cos(x) \int_1^x -\frac{1}{2}e^{-K[1]}(\cos(K[1]) + \sin(K[1]))(e^{K[1]}\sin(K[1]) + e^{2K[1]} + 1) dK[1] + \sin(x) \int_1^x \frac{1}{2}e^{-K[2]}$$

✓ **Sympy.** Time used: 0.198 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x) - exp(x) - sin(x) + Derivative(y(x), x) + Derivative(y(x), (x, 2))
        + Derivative(y(x), (x, 3)) - exp(-x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \left(C_1 - \frac{x}{4}\right) \sin(x) + \left(C_2 - \frac{x}{4}\right) \cos(x) + \left(C_3 + \frac{x}{2}\right) e^{-x} + \frac{e^x}{4}$$

2.11.9 Problem 34

Local contents

Solved as second order linear constant coeff ode	1957
Solved as second order ode using Kovacic algorithm	1960
✓ Maple	1964
✓ Mathematica	1966
✓ Sympy	1966

Internal problem ID [8019]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 34

Date solved : Thursday, November 27, 2025 at 07:32:23 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order linear constant coeff ode

Time used: 0.082 (sec)

Solve

$$y'' - y = x^2$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = x^2$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_3x^2 + A_2x + A_1$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-A_3x^2 - A_2x - A_1 + 2A_3 = x^2$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = -2, A_2 = 0, A_3 = -1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

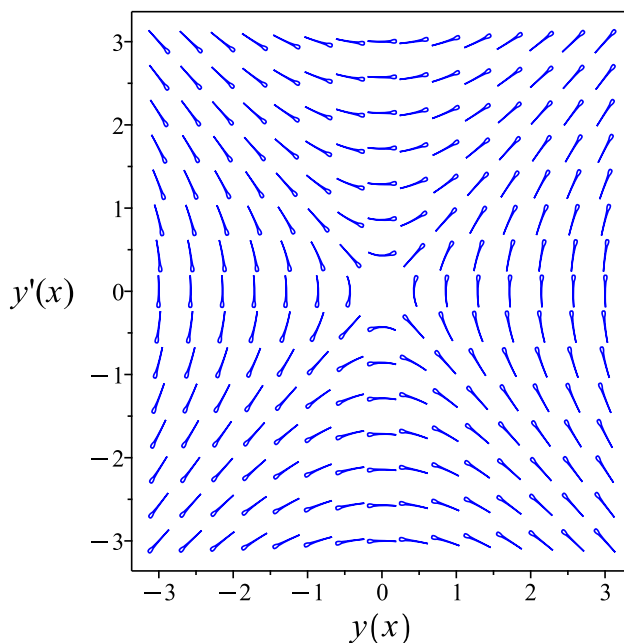
$$y_p = -x^2 - 2$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x}c_2) + (-x^2 - 2) \end{aligned}$$

Summary of solutions found

$$y = -x^2 - 2 + c_1 e^x + e^{-x}c_2$$

Figure 2.373: Slope field $y'' - y = x^2$

Solved as second order ode using Kovacic algorithm

Time used: 0.075 (sec)

Solve

$$y'' - y = x^2$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.250: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x^2$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{2}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_3 x^2 + A_2 x + A_1$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-A_3 x^2 - A_2 x - A_1 + 2A_3 = x^2$$

Solving for the unknowns by comparing coefficients results in

$$[A_1 = -2, A_2 = 0, A_3 = -1]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -x^2 - 2$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + (-x^2 - 2) \end{aligned}$$

Summary of solutions found

$$y = -x^2 - 2 + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

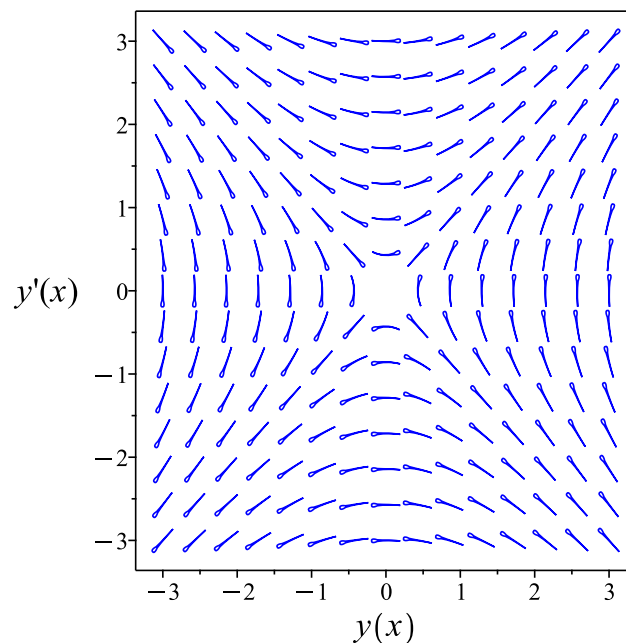


Figure 2.374: Slope field $y'' - y = x^2$

✓ Maple. Time used: 0.002 (sec). Leaf size: 21

```
ode:=diff(diff(y(x),x),x)-y(x) = x^2;
dsolve(ode,y(x), singsol=all);
```

$$y = e^x c_2 + e^{-x} c_1 - x^2 - 2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
```



```

trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = x^2$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$
- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$
- Roots of the characteristic polynomial

$$r = (-1, 1)$$
- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$
- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$
- General solution of the ODE

$$y(x) = C1 y_1(x) + C2 y_2(x) + y_p(x)$$
- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-x} + C2 e^x + y_p(x)$$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = x^2 \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^{-x} \int e^x x^2 dx}{2} + \frac{e^x \int e^{-x} x^2 dx}{2}$$
 - Compute integrals

$$y_p(x) = -x^2 - 2$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x - x^2 - 2$$

✓ **Mathematica.** Time used: 0.009 (sec). Leaf size: 26

```
ode=D[y[x],{x,2}]-y[x]==x^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow -x^2 + c_1 e^x + c_2 e^{-x} - 2$$

✓ **Sympy.** Time used: 0.037 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 - y(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x - x^2 - 2$$

2.11.10 Problem 36

Local contents

Solved as second order linear constant coeff ode	1967
Solved as second order ode using Kovacic algorithm	1972
✓ Maple	1977
✓ Mathematica	1979
✓ Sympy	1979

Internal problem ID [8020]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 36

Date solved : Thursday, November 27, 2025 at 07:32:23 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.365 (sec)

Solve

$$y'' + 2y = x^3 + x^2 + e^{-2x} + \cos(3x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = 2, f(x) = x^3 + x^2 + e^{-2x} + \cos(3x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = 2$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 2e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 2 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 2$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(2)} \\ &= \pm i\sqrt{2} \end{aligned}$$

Hence

$$\lambda_1 = +i\sqrt{2}$$

$$\lambda_2 = -i\sqrt{2}$$

Which simplifies to

$$\lambda_1 = i\sqrt{2}$$

$$\lambda_2 = -i\sqrt{2}$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = \sqrt{2}$. Therefore the final solution, when using Euler relation, can be written as

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$y = e^0 (c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x))$$

Or

$$y = c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(\sqrt{2}x)$$

$$y_2 = \sin(\sqrt{2}x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(\sqrt{2}x) & \sin(\sqrt{2}x) \\ \frac{d}{dx}(\cos(\sqrt{2}x)) & \frac{d}{dx}(\sin(\sqrt{2}x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(\sqrt{2}x) & \sin(\sqrt{2}x) \\ -\sqrt{2} \sin(\sqrt{2}x) & \sqrt{2} \cos(\sqrt{2}x) \end{vmatrix}$$

Therefore

$$W = (\cos(\sqrt{2}x))(\sqrt{2} \cos(\sqrt{2}x)) - (\sin(\sqrt{2}x))(-\sqrt{2} \sin(\sqrt{2}x))$$

Which simplifies to

$$W = \cos^2(\sqrt{2}x) \sqrt{2} + \sin^2(\sqrt{2}x) \sqrt{2}$$

Which simplifies to

$$W = \sqrt{2}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\sin(\sqrt{2}x)(x^3 + x^2 + e^{-2x} + \cos(3x))}{\sqrt{2}} dx$$

Which simplifies to

$$u_1 = - \int \frac{\sin(\sqrt{2}x)(x^3 + x^2 + e^{-2x} + \cos(3x))\sqrt{2}}{2} dx$$

Hence

$$u_1 = \frac{\sqrt{2} \left(\frac{\sqrt{2}(-2\cos(\sqrt{2}x)x^2 + 2\cos(\sqrt{2}x) + 2\sqrt{2}\sin(\sqrt{2}x)x)}{4} - \frac{\sqrt{2}x^3\cos(\sqrt{2}x)}{2} + \frac{3\sin(\sqrt{2}x)x^2}{2} - \frac{3\sin(\sqrt{2}x)}{2} + \frac{3\sqrt{2}x\cos(\sqrt{2}x)}{2} \right)}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\cos(\sqrt{2}x)(x^3 + x^2 + e^{-2x} + \cos(3x))}{\sqrt{2}} dx$$

Which simplifies to

$$u_2 = \int \frac{\cos(\sqrt{2}x)(x^3 + x^2 + e^{-2x} + \cos(3x))\sqrt{2}}{2} dx$$

Hence

$$u_2 = \frac{\sqrt{2} \left(\frac{\sqrt{2}(2\sin(\sqrt{2}x)x^2 - 2\sin(\sqrt{2}x) + 2\sqrt{2}x\cos(\sqrt{2}x))}{4} + \frac{\sin(\sqrt{2}x)\sqrt{2}x^3}{2} + \frac{3\cos(\sqrt{2}x)x^2}{2} - \frac{3\cos(\sqrt{2}x)}{2} - \frac{3\sqrt{2}\sin(\sqrt{2}x)x}{2} - \frac{3\sqrt{2}\cos(\sqrt{2}x)}{2} \right)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\sqrt{2} \left(\frac{\sqrt{2}(-2\cos(\sqrt{2}x)x^2 + 2\cos(\sqrt{2}x) + 2\sqrt{2}\sin(\sqrt{2}x)x)}{4} - \frac{\sqrt{2}x^3\cos(\sqrt{2}x)}{2} + \frac{3\sin(\sqrt{2}x)x^2}{2} - \frac{3\sin(\sqrt{2}x)}{2} + \frac{3\sqrt{2}x\cos(\sqrt{2}x)}{2} \right)}{2} + \frac{\sin(\sqrt{2}x)\sqrt{2} \left(\frac{\sqrt{2}(2\sin(\sqrt{2}x)x^2 - 2\sin(\sqrt{2}x) + 2\sqrt{2}x\cos(\sqrt{2}x))}{4} + \frac{\sin(\sqrt{2}x)\sqrt{2}x^3}{2} + \frac{3\cos(\sqrt{2}x)x^2}{2} - \frac{3\cos(\sqrt{2}x)}{2} - \frac{3\sqrt{2}\sin(\sqrt{2}x)x}{2} - \frac{3\sqrt{2}\cos(\sqrt{2}x)}{2} \right)}{2}$$

Which simplifies to

$$y_p(x) = \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x) \right) + \left(\frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2} + c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x)$$

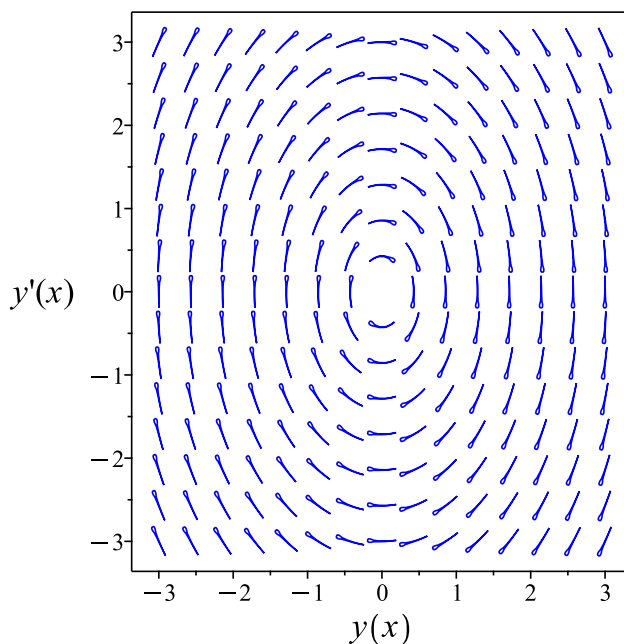


Figure 2.375: Slope field $y'' + 2y = x^3 + x^2 + e^{-2x} + \cos(3x)$

Solved as second order ode using Kovacic algorithm

Time used: 0.328 (sec)

Solve

$$y'' + 2y = x^3 + x^2 + e^{-2x} + \cos(3x)$$

Writing the ode as

$$y'' + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$A = 1$$

$$B = 0 \quad (3)$$

$$C = 2$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-2}{1} \quad (6)$$

Comparing the above to (5) shows that

$$s = -2$$

$$t = 1$$

Therefore eq. (4) becomes

$$z''(x) = -2z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.252: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -2$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(\sqrt{2}x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= \cos(\sqrt{2}x) \end{aligned}$$

Which simplifies to

$$y_1 = \cos(\sqrt{2}x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= \cos(\sqrt{2}x) \int \frac{1}{\cos(\sqrt{2}x)^2} dx \\ &= \cos(\sqrt{2}x) \left(\frac{\sqrt{2} \tan(\sqrt{2}x)}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\cos(\sqrt{2}x) \right) + c_2 \left(\cos(\sqrt{2}x) \left(\frac{\sqrt{2} \tan(\sqrt{2}x)}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 \cos(\sqrt{2}x) + \frac{c_2 \sqrt{2} \sin(\sqrt{2}x)}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos(\sqrt{2}x)$$

$$y_2 = \frac{\sqrt{2} \sin(\sqrt{2}x)}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos(\sqrt{2}x) & \frac{\sqrt{2} \sin(\sqrt{2}x)}{2} \\ \frac{d}{dx}(\cos(\sqrt{2}x)) & \frac{d}{dx}\left(\frac{\sqrt{2} \sin(\sqrt{2}x)}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos(\sqrt{2}x) & \frac{\sqrt{2} \sin(\sqrt{2}x)}{2} \\ -\sqrt{2} \sin(\sqrt{2}x) & \cos(\sqrt{2}x) \end{vmatrix}$$

Therefore

$$W = (\cos(\sqrt{2}x))(\cos(\sqrt{2}x)) - \left(\frac{\sqrt{2} \sin(\sqrt{2}x)}{2}\right)(-\sqrt{2} \sin(\sqrt{2}x))$$

Which simplifies to

$$W = \cos^2(\sqrt{2}x) + \sin^2(\sqrt{2}x)$$

Which simplifies to

$$W = 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{\sin(\sqrt{2}x)(x^3+x^2+e^{-2x}+\cos(3x))\sqrt{2}}{2}}{1} dx$$

Which simplifies to

$$u_1 = - \int \frac{\sin(\sqrt{2}x)(x^3+x^2+e^{-2x}+\cos(3x))\sqrt{2}}{2} dx$$

Hence

$$u_1 = \frac{\sqrt{2} \left(\frac{\sqrt{2}(-2\cos(\sqrt{2}x)x^2+2\cos(\sqrt{2}x)+2\sqrt{2}\sin(\sqrt{2}x)x)}{4} - \frac{\sqrt{2}x^3\cos(\sqrt{2}x)}{2} + \frac{3\sin(\sqrt{2}x)x^2}{2} - \frac{3\sin(\sqrt{2}x)}{2} + \frac{3\sqrt{2}x\cos(\sqrt{2}x)}{2} \right)}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\cos(\sqrt{2}x)(x^3+x^2+e^{-2x}+\cos(3x))}{1} dx$$

Which simplifies to

$$u_2 = \int \cos(\sqrt{2}x)(x^3+x^2+e^{-2x}+\cos(3x)) dx$$

Hence

$$\begin{aligned} u_2 = & \frac{\sqrt{2}(2\sin(\sqrt{2}x)x^2-2\sin(\sqrt{2}x)+2\sqrt{2}x\cos(\sqrt{2}x))}{4} + \frac{\sin(\sqrt{2}x)\sqrt{2}x^3}{2} \\ & + \frac{3\cos(\sqrt{2}x)x^2}{2} - \frac{3\cos(\sqrt{2}x)}{2} - \frac{3\sqrt{2}\sin(\sqrt{2}x)x}{2} - \frac{e^{-2x}\cos(\sqrt{2}x)}{3} \\ & + \frac{e^{-2x}\sqrt{2}\sin(\sqrt{2}x)}{6} + \frac{\sin(x(-3+\sqrt{2}))}{-6+2\sqrt{2}} + \frac{\sin(x(3+\sqrt{2}))}{6+2\sqrt{2}} \end{aligned}$$

Therefore the particular solution, from equation (1) is

$$\begin{aligned} y_p(x) = & \frac{\sqrt{2} \left(\frac{\sqrt{2}(-2\cos(\sqrt{2}x)x^2+2\cos(\sqrt{2}x)+2\sqrt{2}\sin(\sqrt{2}x)x)}{4} - \frac{\sqrt{2}x^3\cos(\sqrt{2}x)}{2} + \frac{3\sin(\sqrt{2}x)x^2}{2} - \frac{3\sin(\sqrt{2}x)}{2} + \frac{3\sqrt{2}x\cos(\sqrt{2}x)}{2} \right)}{2} \\ & + \frac{\sin(\sqrt{2}x)\sqrt{2} \left(\frac{\sqrt{2}(2\sin(\sqrt{2}x)x^2-2\sin(\sqrt{2}x)+2\sqrt{2}x\cos(\sqrt{2}x))}{4} + \frac{\sin(\sqrt{2}x)\sqrt{2}x^3}{2} + \frac{3\cos(\sqrt{2}x)x^2}{2} - \frac{3\cos(\sqrt{2}x)}{2} - \frac{3\sqrt{2}\sin(\sqrt{2}x)x}{2} - \frac{e^{-2x}\cos(\sqrt{2}x)}{3} \right)}{2} \\ & + \frac{e^{-2x}\sqrt{2}\sin(\sqrt{2}x)}{6} + \frac{\sin(x(-3+\sqrt{2}))}{-6+2\sqrt{2}} + \frac{\sin(x(3+\sqrt{2}))}{6+2\sqrt{2}} \end{aligned}$$

Which simplifies to

$$y_p(x) = \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2}$$

Therefore the general solution is

$$y = y_h + y_p \\ = \left(c_1 \cos(\sqrt{2}x) + \frac{c_2 \sqrt{2} \sin(\sqrt{2}x)}{2} \right) + \left(\frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2} \right)$$

Summary of solutions found

$$y = \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2} + c_1 \cos(\sqrt{2}x) + \frac{c_2 \sqrt{2} \sin(\sqrt{2}x)}{2}$$

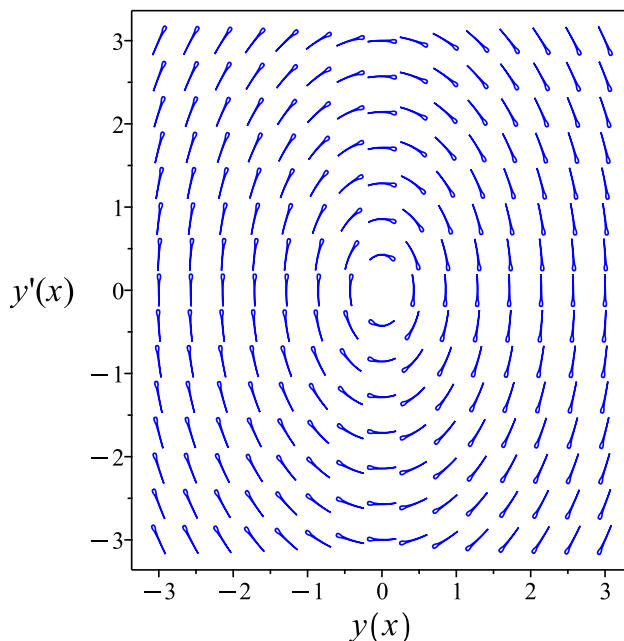


Figure 2.376: Slope field $y'' + 2y = x^3 + x^2 + e^{-2x} + \cos(3x)$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 47

```
ode:=diff(diff(y(x),x),x)+2*y(x) = x^3+x^2+exp(-2*x)+cos(3*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \sin(\sqrt{2}x) c_2 + \cos(\sqrt{2}x) c_1 + \frac{x^3}{2} + \frac{x^2}{2} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2} + \frac{e^{-2x}}{6}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form $[xi=0, eta=F(x)]$

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

<- constant coefficients successful

<- solving first the homogeneous part of the ODE successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 2y(x) = x^3 + x^2 + e^{-2x} + \cos(3x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 2 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{0 \pm (\sqrt{-8})}{2}$$

- Roots of the characteristic polynomial

$$r = (-I\sqrt{2}, I\sqrt{2})$$

- 1st solution of the homogeneous ODE

$$y_1(x) = \cos(\sqrt{2}x)$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = \sin(\sqrt{2}x)$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 \cos(\sqrt{2}x) + C2 \sin(\sqrt{2}x) + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x),y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x),y_2(x))} dx, f(x) = x^3 + x^2 + e^{-2x} + \cos(3x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} \cos(\sqrt{2}x) & \sin(\sqrt{2}x) \\ -\sqrt{2} \sin(\sqrt{2}x) & \sqrt{2} \cos(\sqrt{2}x) \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = \sqrt{2}$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{\sqrt{2} \left(\int \sin(\sqrt{2}x) (x^3 + x^2 + e^{-2x} + \cos(3x)) dx \cos(\sqrt{2}x) - \int \cos(\sqrt{2}x) (x^3 + x^2 + e^{-2x} + \cos(3x)) dx \sin(\sqrt{2}x) \right)}{2}$$

- o Compute integrals

$$y_p(x) = \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 \cos(\sqrt{2}x) + C_2 \sin(\sqrt{2}x) + \frac{x^3}{2} + \frac{x^2}{2} + \frac{e^{-2x}}{6} - \frac{\cos(3x)}{7} - \frac{3x}{2} - \frac{1}{2}$$

✓ **Mathematica.** Time used: 1.085 (sec). Leaf size: 161

```
ode=D[y[x],{x,2}]+2*y[x]==x^3+x^2+Exp[-2*x]+Cos[3*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \cos(\sqrt{2}x) \int_1^x -\frac{e^{-2K[1]}(e^{2K[1]}(K[1]+1)K[1]^2 + e^{2K[1]}\cos(3K[1]) + 1)\sin(\sqrt{2}K[1])}{\sqrt{2}} dK[1] + \sin(\sqrt{2}x) \int_1^x \frac{e^{-2K[1]}(e^{2K[1]}(K[1]+1)K[1]^2 + e^{2K[1]}\cos(3K[1]) + 1)\cos(\sqrt{2}K[1])}{\sqrt{2}} dK[1]$$

✓ **Sympy.** Time used: 0.086 (sec). Leaf size: 56

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**3 - x**2 + 2*y(x) - cos(3*x) + Derivative(y(x), (x, 2)) - exp(-2*x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 \sin(\sqrt{2}x) + C_2 \cos(\sqrt{2}x) + \frac{x^3}{2} + \frac{x^2}{2} - \frac{3x}{2} - \frac{\cos(3x)}{7} - \frac{1}{2} + \frac{e^{-2x}}{6}$$

2.11.11 Problem 37

Local contents

Solved as second order linear constant coeff ode	1980
Solved as second order ode using Kovacic algorithm	1983
✓ Maple	1987
✓ Mathematica	1989
✓ Sympy	1989

Internal problem ID [8021]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 37

Date solved : Thursday, November 27, 2025 at 07:32:24 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.112 (sec)

Solve

$$y'' - 2y' - y = e^x \cos(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -2, C = -1, f(x) = e^x \cos(x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 2y' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -2, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 2\lambda e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 2\lambda - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -2, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{2}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{-2^2 - (4)(1)(-1)} \\ &= 1 \pm \sqrt{2} \end{aligned}$$

Hence

$$\lambda_1 = 1 + \sqrt{2}$$

$$\lambda_2 = 1 - \sqrt{2}$$

Which simplifies to

$$\lambda_1 = 1 + \sqrt{2}$$

$$\lambda_2 = 1 - \sqrt{2}$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1+\sqrt{2})x} + c_2 e^{(1-\sqrt{2})x}$$

Or

$$y = c_1 e^{x(1+\sqrt{2})} + c_2 e^{x(1-\sqrt{2})}$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^{x(1+\sqrt{2})} + c_2 e^{x(1-\sqrt{2})}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x \cos(x), \sin(x) e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ e^{x(1-\sqrt{2})}, e^{x(1+\sqrt{2})} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^x \cos(x) + A_2 \sin(x) e^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-3A_1 e^x \cos(x) - 3A_2 \sin(x) e^x = e^x \cos(x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{3}, A_2 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

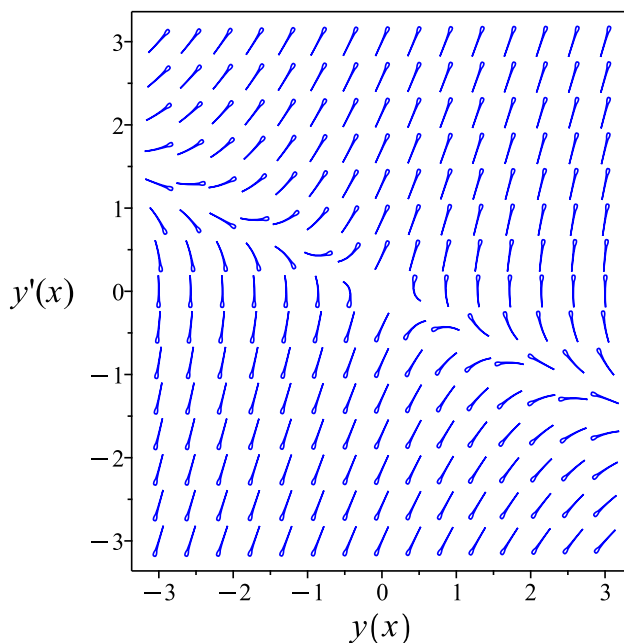
$$y_p = -\frac{e^x \cos(x)}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{x(1+\sqrt{2})} + c_2 e^{x(1-\sqrt{2})} \right) + \left(-\frac{e^x \cos(x)}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{e^x \cos(x)}{3} + c_1 e^{x(1+\sqrt{2})} + c_2 e^{x(1-\sqrt{2})}$$

Figure 2.377: Slope field $y'' - 2y' - y = e^x \cos(x)$ **Solved as second order ode using Kovacic algorithm**

Time used: 0.109 (sec)

Solve

$$y'' - 2y' - y = e^x \cos(x)$$

Writing the ode as

$$y'' - 2y' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -2 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{2}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 2 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 2z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.254: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 2$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\sqrt{2}x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2}{1} dx} \\ &= z_1 e^x \\ &= z_1(e^x) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x(\sqrt{2}-1)}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{2x}}{(y_1)^2} dx \\ &= y_1 \left(\frac{\sqrt{2} e^{2x} e^{2x(\sqrt{2}-1)}}{4} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(e^{-x(\sqrt{2}-1)} \right) + c_2 \left(e^{-x(\sqrt{2}-1)} \left(\frac{\sqrt{2} e^{2x} e^{2x(\sqrt{2}-1)}}{4} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 2y' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x(\sqrt{2}-1)} + \frac{c_2 \sqrt{2} e^{x(1+\sqrt{2})}}{4}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$e^x \cos(x)$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{e^x \cos(x), \sin(x) e^x\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{\sqrt{2} e^{x(1+\sqrt{2})}}{4}, e^{-x(\sqrt{2}-1)} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 e^x \cos(x) + A_2 \sin(x) e^x$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$-3A_1 e^x \cos(x) - 3A_2 \sin(x) e^x = e^x \cos(x)$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = -\frac{1}{3}, A_2 = 0 \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = -\frac{e^x \cos(x)}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x(\sqrt{2}-1)} + \frac{c_2 \sqrt{2} e^{x(1+\sqrt{2})}}{4} \right) + \left(-\frac{e^x \cos(x)}{3} \right) \end{aligned}$$

Summary of solutions found

$$y = -\frac{e^x \cos(x)}{3} + c_1 e^{-x(\sqrt{2}-1)} + \frac{c_2 \sqrt{2} e^{x(1+\sqrt{2})}}{4}$$

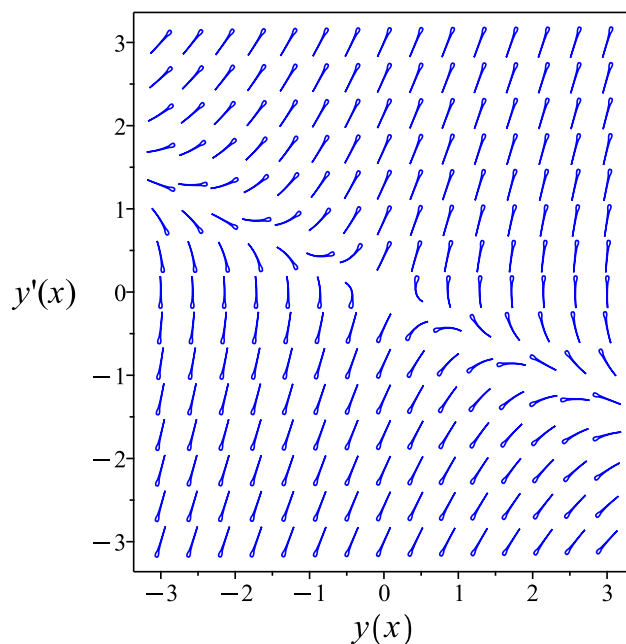


Figure 2.378: Slope field $y'' - 2y' - y = e^x \cos(x)$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 29

```
ode:=diff(diff(y(x),x),x)-2*diff(y(x),x)-y(x) = cos(x)*exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^x \left(e^{x\sqrt{2}} c_2 + e^{-x\sqrt{2}} c_1 - \frac{\cos(x)}{3} \right)$$

Maple trace

Methods for second order ODEs:

```

--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 2\frac{d}{dx}y(x) - y(x) = e^x \cos(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 2r - 1 = 0$$

- Use quadratic formula to solve for r

$$r = \frac{2 \pm (\sqrt{8})}{2}$$

- Roots of the characteristic polynomial

$$r = (1 - \sqrt{2}, 1 + \sqrt{2})$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{(1-\sqrt{2})x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{(1+\sqrt{2})x}$$

- General solution of the ODE

$$y(x) = C1 y_1(x) + C2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{(1-\sqrt{2})x} + C2 e^{(1+\sqrt{2})x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^x \cos(x) \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{(1-\sqrt{2})x} & e^{(1+\sqrt{2})x} \\ (1-\sqrt{2})e^{(1-\sqrt{2})x} & (1+\sqrt{2})e^{(1+\sqrt{2})x} \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2\sqrt{2}e^{2x}$$
- Substitute functions into equation for $y_p(x)$

$$y_p(x) = \frac{\sqrt{2} \left(\int \cos(x)e^{-x\sqrt{2}} dx e^{2x\sqrt{2}} - \int \cos(x)e^{x\sqrt{2}} dx \right) e^{-(\sqrt{2}-1)x}}{4}$$
- Compute integrals

$$y_p(x) = -\frac{e^x \cos(x)}{3}$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{(1-\sqrt{2})x} + C_2 e^{(1+\sqrt{2})x} - \frac{e^x \cos(x)}{3}$$

✓ **Mathematica.** Time used: 0.077 (sec). Leaf size: 104

```
ode=D[y[x],{x,2}]-2*D[y[x],x]-y[x]==Exp[x]*Cos[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{x-\sqrt{2}x} \left(\int_1^x -\frac{e^{\sqrt{2}K[1]} \cos(K[1])}{2\sqrt{2}} dK[1] + e^{2\sqrt{2}x} \int_1^x \frac{e^{-\sqrt{2}K[2]} \cos(K[2])}{2\sqrt{2}} dK[2] + c_2 e^{2\sqrt{2}x} + c_1 \right)$$

✓ **Sympy.** Time used: 0.170 (sec). Leaf size: 34

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x) - exp(x)*cos(x) - 2*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 e^{x(1-\sqrt{2})} + C_2 e^{x(1+\sqrt{2})} - \frac{e^x \cos(x)}{3}$$

2.11.12 Problem 38

Local contents

Solved as second order linear constant coeff ode	1990
Solved as second order solved by an integrating factor	1994
Solved as second order ode using Kovacic algorithm	1995
✓ Maple	2000
✓ Mathematica	2002
✓ Sympy	2002

Internal problem ID [8022]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 38

Date solved : Thursday, November 27, 2025 at 07:32:24 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.125 (sec)

Solve

$$y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = -4, C = 4, f(x) = \frac{e^{2x}}{x^2}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 4y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = -4, C = 4$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 4\lambda e^{x\lambda} + 4e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 4\lambda + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = -4, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{4}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{(-4)^2 - (4)(1)(4)} \\ &= 2 \end{aligned}$$

Hence this is the case of a double root $\lambda_{1,2} = -2$. Therefore the solution is

$$y = c_1 e^{2x} + c_2 x e^{2x} \quad (1)$$

Therefore the homogeneous solution y_h is

$$y_h = e^{2x} c_1 + x e^{2x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{2x}$$

$$y_2 = x e^{2x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ \frac{d}{dx}(e^{2x}) & \frac{d}{dx}(x e^{2x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & e^{2x} + 2x e^{2x} \end{vmatrix}$$

Therefore

$$W = (e^{2x})(e^{2x} + 2x e^{2x}) - (x e^{2x})(2e^{2x})$$

Which simplifies to

$$W = e^{4x}$$

Which simplifies to

$$W = e^{4x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{4x}}{x} dx$$

Which simplifies to

$$u_1 = - \int \frac{1}{x} dx$$

Hence

$$u_1 = - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{4x}}{x^2} dx$$

Which simplifies to

$$u_2 = \int \frac{1}{x^2} dx$$

Hence

$$u_2 = -\frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\ln(x) e^{2x} - e^{2x}$$

Which simplifies to

$$y_p(x) = -e^{2x}(1 + \ln(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x}c_1 + x e^{2x}c_2) + (-e^{2x}(1 + \ln(x))) \end{aligned}$$

Summary of solutions found

$$y = -e^{2x}(1 + \ln(x)) + e^{2x}c_1 + x e^{2x}c_2$$

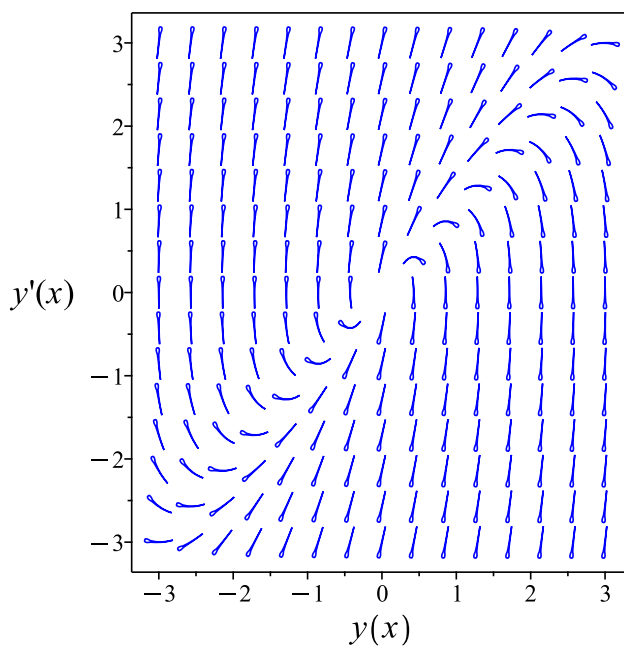


Figure 2.379: Slope field $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$

Solved as second order solved by an integrating factor

Time used: 0.041 (sec)

Solve

$$y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x))^2 + p'(x)}{2}y = f(x)$$

Where $p(x) = -4$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int -4 \, dx} \\ &= e^{-2x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$\begin{aligned} (M(x)y)'' &= \frac{e^{-2x}e^{2x}}{x^2} \\ (e^{-2x}y)'' &= \frac{e^{-2x}e^{2x}}{x^2} \end{aligned}$$

Integrating once gives

$$(e^{-2x}y)' = -\frac{1}{x} + c_1$$

Integrating again gives

$$(e^{-2x}y) = -\ln(x) + c_1x + c_2$$

Hence the solution is

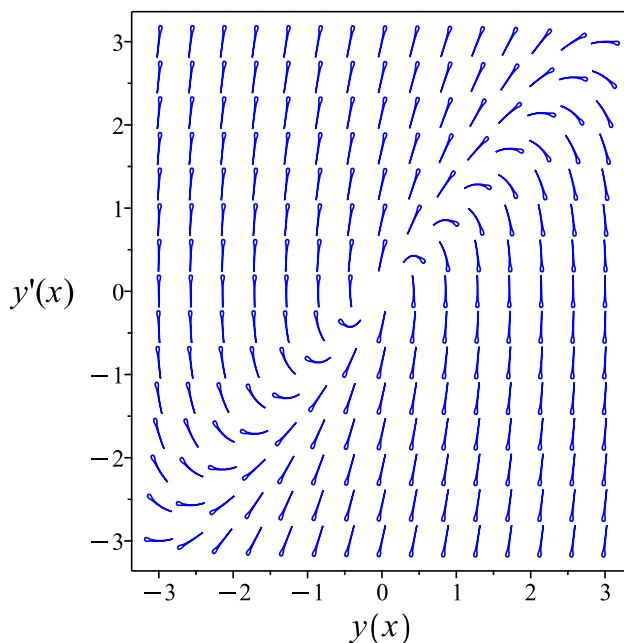
$$y = \frac{-\ln(x) + c_1x + c_2}{e^{-2x}}$$

Or

$$y = c_1x e^{2x} + e^{2x}c_2 - \ln(x) e^{2x}$$

Summary of solutions found

$$y = c_1x e^{2x} + e^{2x}c_2 - \ln(x) e^{2x}$$

Figure 2.380: Slope field $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$

Solved as second order ode using Kovacic algorithm

Time used: 0.215 (sec)

Solve

$$y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$$

Writing the ode as

$$y'' - 4y' + 4y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -4 \\ C &= 4 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 0 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.256: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4}{1} dx} \\ &= z_1 e^{2x} \\ &= z_1 (e^{2x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{2x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{2x}) + c_2 (e^{2x}(x)) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - 4y' + 4y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = e^{2x}c_1 + x e^{2x}c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{2x}$$

$$y_2 = x e^{2x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ \frac{d}{dx}(e^{2x}) & \frac{d}{dx}(x e^{2x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & e^{2x} + 2x e^{2x} \end{vmatrix}$$

Therefore

$$W = (e^{2x}) (e^{2x} + 2x e^{2x}) - (x e^{2x}) (2 e^{2x})$$

Which simplifies to

$$W = e^{4x}$$

Which simplifies to

$$W = e^{4x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^{4x}}{x}}{e^{4x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{1}{x} dx$$

Hence

$$u_1 = - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^{4x}}{x^2}}{e^{4x}} dx$$

Which simplifies to

$$u_2 = \int \frac{1}{x^2} dx$$

Hence

$$u_2 = -\frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\ln(x) e^{2x} - e^{2x}$$

Which simplifies to

$$y_p(x) = -e^{2x}(1 + \ln(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{2x}c_1 + x e^{2x}c_2) + (-e^{2x}(1 + \ln(x))) \end{aligned}$$

Summary of solutions found

$$y = -e^{2x}(1 + \ln(x)) + e^{2x}c_1 + x e^{2x}c_2$$

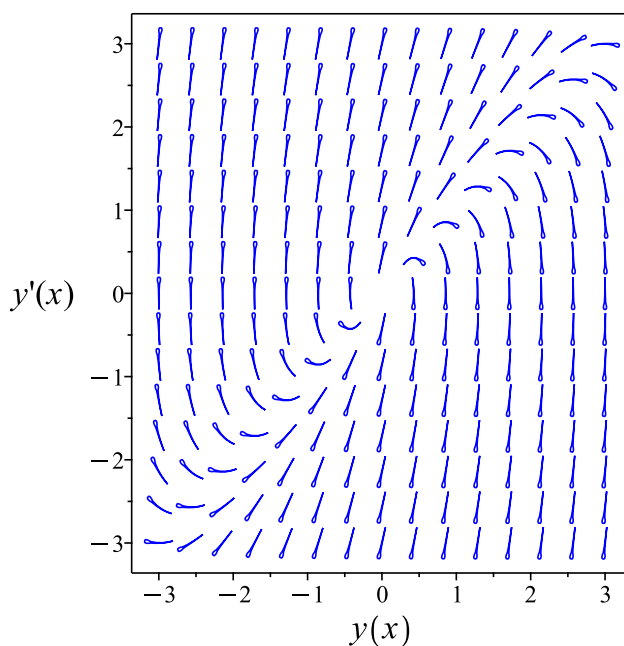


Figure 2.381: Slope field $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 19

```
ode:=diff(diff(y(x),x),x)-4*diff(y(x),x)+4*y(x) = exp(2*x)/x^2;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{2x}(c_1x - \ln(x) + c_2 - 1)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

```

trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - 4\frac{d}{dx}y(x) + 4y(x) = \frac{e^{2x}}{x^2}$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Characteristic polynomial of homogeneous ODE
 $r^2 - 4r + 4 = 0$
- Factor the characteristic polynomial
 $(r - 2)^2 = 0$
- Root of the characteristic polynomial
 $r = 2$
- 1st solution of the homogeneous ODE
 $y_1(x) = e^{2x}$
- Repeated root, multiply $y_1(x)$ by x to ensure linear independence
 $y_2(x) = x e^{2x}$
- General solution of the ODE
 $y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
 $y(x) = C_1 e^{2x} + C_2 x e^{2x} + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = \frac{e^{2x}}{x^2} \right]$$
 - Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & e^{2x} + 2x e^{2x} \end{bmatrix}$$
 - Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{4x}$$
 - Substitute functions into equation for $y_p(x)$

$$y_p(x) = e^{2x} \left(x \int \frac{1}{x^2} dx - \int \frac{1}{x} dx \right)$$
 - Compute integrals

$$y_p(x) = e^{2x} (-1 - \ln(x))$$
- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{2x} + C_2 x e^{2x} + e^{2x} (-1 - \ln(x))$$

✓ **Mathematica.** Time used: 0.015 (sec). Leaf size: 23

```
ode=D[y[x],{x,2}]-4*D[y[x],x]+4*y[x]==Exp[2*x]/x^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{2x}(-\log(x) + c_2x - 1 + c_1)$$

✓ **Sympy.** Time used: 0.167 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*y(x) - 4*Derivative(y(x), x) + Derivative(y(x), (x, 2)) - exp(2*x)/x
**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = (C_1 + C_2x - \log(x)) e^{2x}$$

2.11.13 Problem 39

Local contents

Solved as second order linear constant coeff ode	2003
Solved as second order ode using Kovacic algorithm	2006
✓ Maple	2010
✓ Mathematica	2012
✓ Sympy	2012

Internal problem ID [8023]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 39

Date solved : Thursday, November 27, 2025 at 07:32:25 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.095 (sec)

Solve

$$y'' - y = x e^{3x}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 0, C = -1, f(x) = x e^{3x}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1 \end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{(1)x} + c_2 e^{(-1)x}$$

Or

$$y = c_1 e^x + e^{-x} c_2$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 e^x + e^{-x} c_2$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x e^{3x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x e^{3x}, e^{3x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{e^x, e^{-x}\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 x e^{3x} + A_2 e^{3x}$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$6A_1 e^{3x} + 8A_1 x e^{3x} + 8A_2 e^{3x} = x e^{3x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{8}, A_2 = -\frac{3}{32} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

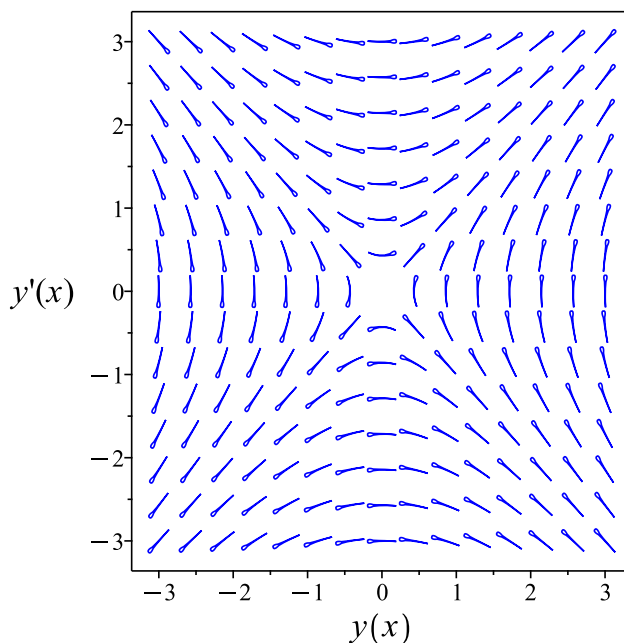
$$y_p = \frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^x + e^{-x} c_2) + \left(\frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32} + c_1 e^x + e^{-x} c_2$$

Figure 2.382: Slope field $y'' - y = x e^{3x}$

Solved as second order ode using Kovacic algorithm

Time used: 0.083 (sec)

Solve

$$y'' - y = x e^{3x}$$

Writing the ode as

$$y'' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= 0 \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = y e^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = r z(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.258: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 1$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$y_1 = z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx}$$

Since $B = 0$ then the above reduces to

$$\begin{aligned} y_1 &= z_1 \\ &= e^{-x} \end{aligned}$$

Which simplifies to

$$y_1 = e^{-x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Since $B = 0$ then the above becomes

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} dx \\ &= e^{-x} \int \frac{1}{e^{-2x}} dx \\ &= e^{-x} \left(\frac{e^{2x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-x}) + c_2 \left(e^{-x} \left(\frac{e^{2x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-x} + \frac{c_2 e^x}{2}$$

The particular solution is now found using the method of undetermined coefficients. Looking at the RHS of the ode, which is

$$x e^{3x}$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{x e^{3x}, e^{3x}\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\left\{ \frac{e^x}{2}, e^{-x} \right\}$$

Since there is no duplication between the basis function in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis in the UC_set.

$$y_p = A_1 x e^{3x} + A_2 e^{3x}$$

The unknowns $\{A_1, A_2\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$6A_1 e^{3x} + 8A_1 x e^{3x} + 8A_2 e^{3x} = x e^{3x}$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = \frac{1}{8}, A_2 = -\frac{3}{32} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{-x} + \frac{c_2 e^x}{2} \right) + \left(\frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x e^{3x}}{8} - \frac{3 e^{3x}}{32} + c_1 e^{-x} + \frac{c_2 e^x}{2}$$

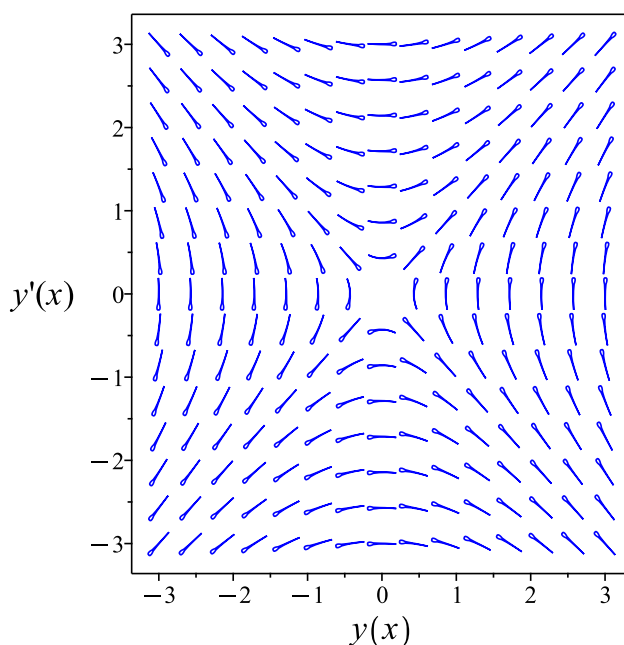


Figure 2.383: Slope field $y'' - y = x e^{3x}$

✓ Maple. Time used: 0.004 (sec). Leaf size: 26

```
ode:=diff(diff(y(x),x),x)-y(x) = x*exp(3*x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-x} c_2 + e^x c_1 + \frac{(4x - 3) e^{3x}}{32}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
```

```

trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - y(x) = x e^{3x}$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 - 1 = 0$$

- Factor the characteristic polynomial

$$(r - 1)(r + 1) = 0$$

- Roots of the characteristic polynomial

$$r = (-1, 1)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^x$$

- General solution of the ODE

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = x e^{3x} \right]$$

- Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-x} & e^x \\ -e^{-x} & e^x \end{bmatrix}$$

- Compute Wronskian

$$W(y_1(x), y_2(x)) = 2$$

- Substitute functions into equation for $y_p(x)$

$$y_p(x) = -\frac{e^{-x} \int x e^{4x} dx}{2} + \frac{e^x \int x e^{2x} dx}{2}$$

- Compute integrals

$$y_p(x) = \frac{(4x-3)e^{3x}}{32}$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 e^x + \frac{(4x-3)e^{3x}}{32}$$

✓ **Mathematica.** Time used: 0.011 (sec). Leaf size: 34

```
ode=D[y[x],{x,2}]-y[x]==x*Exp[3*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{32}e^{3x}(4x-3) + c_1e^x + c_2e^{-x}$$

✓ **Sympy.** Time used: 0.065 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*exp(3*x) - y(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 e^{-x} + C_2 e^x + \frac{(4x-3)e^{3x}}{32}$$

2.11.14 Problem 40

Local contents

Solved as second order linear constant coeff ode	2013
Solved as second order ode using Kovacic algorithm	2017
✓ Maple	2022
✓ Mathematica	2024
✓ Sympy	2024

Internal problem ID [8024]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 16. Linear equations with constant coefficients (Short methods). Supplementary problems. Page 107

Problem number : 40

Date solved : Thursday, November 27, 2025 at 07:32:25 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order linear constant coeff ode

Time used: 0.220 (sec)

Solve

$$y'' + 5y' + 6y = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = 1, B = 5, C = 6, f(x) = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 5y' + 6y = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = 0$$

Where in the above $A = 1, B = 5, C = 6$. Let the solution be $y = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 5\lambda e^{x\lambda} + 6e^{x\lambda} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 5\lambda + 6 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 5, C = 6$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{-5}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{5^2 - (4)(1)(6)} \\ &= -\frac{5}{2} \pm \frac{1}{2} \end{aligned}$$

Hence

$$\begin{aligned} \lambda_1 &= -\frac{5}{2} + \frac{1}{2} \\ \lambda_2 &= -\frac{5}{2} - \frac{1}{2} \end{aligned}$$

Which simplifies to

$$\begin{aligned} \lambda_1 &= -2 \\ \lambda_2 &= -3 \end{aligned}$$

Since roots are distinct, then the solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ y &= c_1 e^{(-2)x} + c_2 e^{(-3)x} \end{aligned}$$

Or

$$y = e^{-2x} c_1 + c_2 e^{-3x}$$

Therefore the homogeneous solution y_h is

$$y_h = e^{-2x} c_1 + c_2 e^{-3x}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-2x}$$

$$y_2 = e^{-3x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-2x} & e^{-3x} \\ \frac{d}{dx}(e^{-2x}) & \frac{d}{dx}(e^{-3x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-2x} & e^{-3x} \\ -2e^{-2x} & -3e^{-3x} \end{vmatrix}$$

Therefore

$$W = (e^{-2x})(-3e^{-3x}) - (e^{-3x})(-2e^{-2x})$$

Which simplifies to

$$W = -e^{-2x}e^{-3x}$$

Which simplifies to

$$W = -e^{-5x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{-3x}e^{-2x} \sec(x)^2 (1 + 2 \tan(x))}{-e^{-5x}} dx$$

Which simplifies to

$$u_1 = - \int (-1 - 2 \tan(x)) \sec(x)^2 dx$$

Hence

$$u_1 = \tan(x)^2 + \tan(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{-4x} \sec(x)^2 (1 + 2 \tan(x))}{-e^{-5x}} dx$$

Which simplifies to

$$u_2 = \int (-1 - 2 \tan(x)) \sec(x)^2 e^x dx$$

Hence

$$u_2 = -\frac{e^x}{\cos(x)^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = (\tan(x)^2 + \tan(x)) e^{-2x} - \frac{e^{-3x} e^x}{\cos(x)^2}$$

Which simplifies to

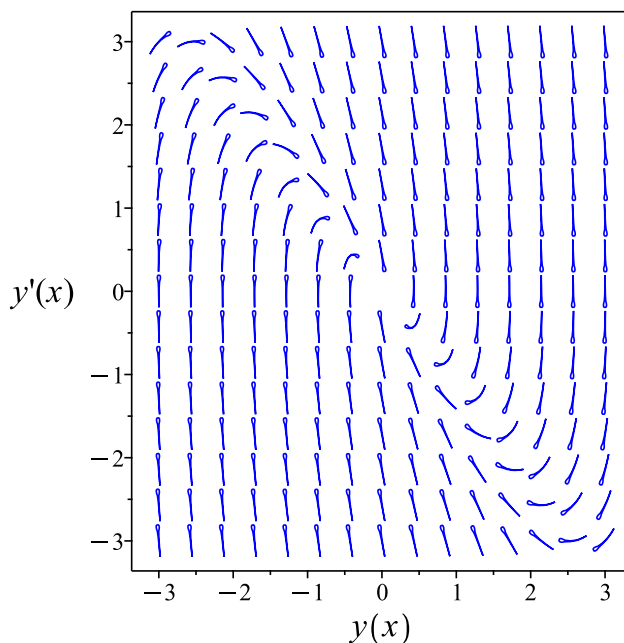
$$y_p(x) = e^{-2x}(-1 + \tan(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (e^{-2x} c_1 + c_2 e^{-3x}) + (e^{-2x}(-1 + \tan(x))) \end{aligned}$$

Summary of solutions found

$$y = e^{-2x}(-1 + \tan(x)) + e^{-2x} c_1 + c_2 e^{-3x}$$

Figure 2.384: Slope field $y'' + 5y' + 6y = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$

Solved as second order ode using Kovacic algorithm

Time used: 0.174 (sec)

Solve

$$y'' + 5y' + 6y = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$$

Writing the ode as

$$y'' + 5y' + 6y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$A = 1$$

$$B = 5 \quad (3)$$

$$C = 6$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{4} \quad (6)$$

Comparing the above to (5) shows that

$$s = 1$$

$$t = 4$$

Therefore eq. (4) becomes

$$z''(x) = \frac{z(x)}{4} \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.260: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = \frac{1}{4}$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-\frac{x}{2}}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{5}{1} dx} \\ &= z_1 e^{-\frac{5x}{2}} \\ &= z_1 \left(e^{-\frac{5x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{5}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-5x}}{(y_1)^2} dx \\ &= y_1 (e^{-5x} e^{6x}) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-3x}) + c_2 (e^{-3x} (e^{-5x} e^{6x})) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$y'' + 5y' + 6y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{-3x} + e^{-2x} c_2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{-3x}$$

$$y_2 = e^{-2x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{-3x} & e^{-2x} \\ \frac{d}{dx}(e^{-3x}) & \frac{d}{dx}(e^{-2x}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{-3x} & e^{-2x} \\ -3e^{-3x} & -2e^{-2x} \end{vmatrix}$$

Therefore

$$W = (e^{-3x})(-2e^{-2x}) - (e^{-2x})(-3e^{-3x})$$

Which simplifies to

$$W = e^{-2x}e^{-3x}$$

Which simplifies to

$$W = e^{-5x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{e^{-4x} \sec(x)^2 (1 + 2 \tan(x))}{e^{-5x}} dx$$

Which simplifies to

$$u_1 = - \int \sec(x)^2 e^x (1 + 2 \tan(x)) dx$$

Hence

$$u_1 = - \frac{e^x}{\cos(x)^2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{-3x} e^{-2x} \sec(x)^2 (1 + 2 \tan(x))}{e^{-5x}} dx$$

Which simplifies to

$$u_2 = \int \sec(x)^2 (1 + 2 \tan(x)) dx$$

Hence

$$u_2 = \tan(x)^2 + \tan(x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = (\tan(x)^2 + \tan(x)) e^{-2x} - \frac{e^{-3x} e^x}{\cos(x)^2}$$

Which simplifies to

$$y_p(x) = e^{-2x}(-1 + \tan(x))$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_1 e^{-3x} + e^{-2x} c_2) + (e^{-2x}(-1 + \tan(x))) \end{aligned}$$

Summary of solutions found

$$y = e^{-2x}(-1 + \tan(x)) + c_1 e^{-3x} + e^{-2x} c_2$$

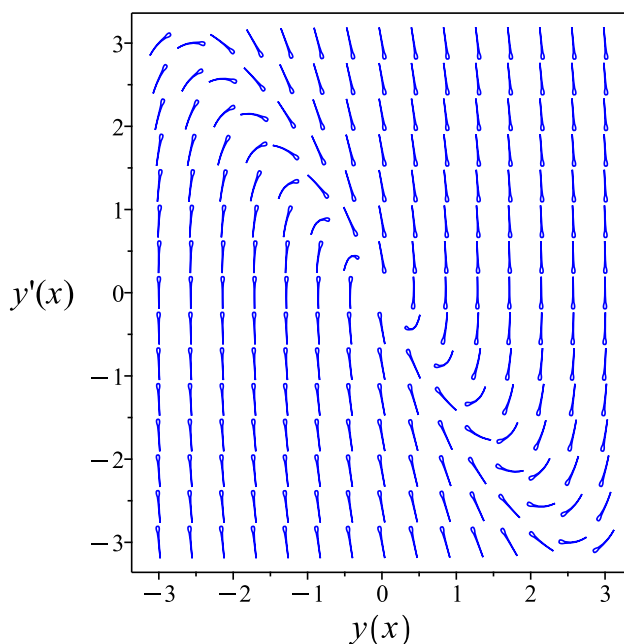


Figure 2.385: Slope field $y'' + 5y' + 6y = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 21

```
ode:=diff(diff(y(x),x),x)+5*diff(y(x),x)+6*y(x) = exp(-2*x)*sec(x)^2*(1+2*tan(x));
dsolve(ode,y(x), singsol=all);
```

$$y = (c_1 + \tan(x) - 1) e^{-2x} + e^{-3x} c_2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
```

```

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    <- constant coefficients successful
<- solving first the homogeneous part of the ODE successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 5\frac{d}{dx}y(x) + 6y(x) = e^{-2x} \sec(x)^2 (1 + 2 \tan(x))$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Characteristic polynomial of homogeneous ODE

$$r^2 + 5r + 6 = 0$$

- Factor the characteristic polynomial

$$(r + 3)(r + 2) = 0$$

- Roots of the characteristic polynomial

$$r = (-3, -2)$$

- 1st solution of the homogeneous ODE

$$y_1(x) = e^{-3x}$$

- 2nd solution of the homogeneous ODE

$$y_2(x) = e^{-2x}$$

- General solution of the ODE

$$y(x) = C1y_1(x) + C2y_2(x) + y_p(x)$$

- Substitute in solutions of the homogeneous ODE

$$y(x) = C1 e^{-3x} + C2 e^{-2x} + y_p(x)$$

- Find a particular solution $y_p(x)$ of the ODE

- o Use variation of parameters to find y_p here $f(x)$ is the forcing function

$$\left[y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1(x), y_2(x))} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1(x), y_2(x))} dx, f(x) = e^{-2x} \sec(x)^2 (1 + 2 \tan(x)) \right]$$

- o Wronskian of solutions of the homogeneous equation

$$W(y_1(x), y_2(x)) = \begin{bmatrix} e^{-3x} & e^{-2x} \\ -3e^{-3x} & -2e^{-2x} \end{bmatrix}$$

- o Compute Wronskian

$$W(y_1(x), y_2(x)) = e^{-5x}$$

- o Substitute functions into equation for $y_p(x)$

$$y_p(x) = \left(\int \sec(x)^2 (1 + 2 \tan(x)) dx \right) e^{-3x} - \left(\int \sec(x)^2 e^x (1 + 2 \tan(x)) dx \right) e^{-3x}$$

- o Compute integrals

$$y_p(x) = e^{-2x}(-1 + \tan(x))$$

- Substitute particular solution into general solution to ODE

$$y(x) = C1 e^{-3x} + C2 e^{-2x} + e^{-2x}(-1 + \tan(x))$$

✓ **Mathematica.** Time used: 0.078 (sec). Leaf size: 26

```
ode=D[y[x],{x,2}]+5*D[y[x],x]+6*y[x]==Exp[-2*x]*Sec[x]^2*(1+2*Tan[x]);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-3x}(e^x \tan(x) + c_2 e^x + c_1)$$

✓ **Sympy.** Time used: 2.104 (sec). Leaf size: 41

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-2*tan(x) - 1)*exp(-2*x)/cos(x)**2 + 6*y(x) + 5*Derivative(y(x), x) +
        Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = \left(C_1 + \left(C_2 - \int \frac{(2 \tan(x) + 1) e^x}{\cos^2(x)} dx \right) e^{-x} + \tan(x) + \frac{1}{\cos^2(x)} \right) e^{-2x}$$

2.12 Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre). Supplementary problems. Page 110

Local contents

2.12.1 Problem 6	2026
2.12.2 Problem 7	2052
2.12.3 Problem 8	2081
2.12.4 Problem 9	2087
2.12.5 Problem 10	2093
2.12.6 Problem 11	2122

2.12.1 Problem 6

Local contents

Solved as second order Euler type ode	2026
Solved as second order ode using change of variable on x method 2	2029
Solved as second order ode using change of variable on x method 1	2034
Solved as second order ode using change of variable on y method 2	2038
Solved as second order ode using Kovacic algorithm	2043
✓ Maple	2050
✓ Mathematica	2050
✓ Sympy	2051

Internal problem ID [8025]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 6

Date solved : Thursday, November 27, 2025 at 07:32:26 AM

CAS classification : [[_2nd_order, _linear, _nonhomogeneous]]

Solved as second order Euler type ode

Time used: 0.141 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -3x$, $C = 4$, $f(x) = x(\ln(x)x + 1)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2 y'' - 3y'x + 4y = 0$$

This is Euler second order ODE. Let the solution be $y = x^r$, then $y' = rx^{r-1}$ and $y'' = r(r-1)x^{r-2}$. Substituting these back into the given ODE gives

$$x^2(r(r-1))x^{r-2} - 3rx^{r-1} + 4x^r = 0$$

Simplifying gives

$$r(r-1)x^r - 3rx^r + 4x^r = 0$$

Since $x^r \neq 0$ then dividing throughout by x^r gives

$$r(r-1) - 3r + 4 = 0$$

Or

$$r^2 - 4r + 4 = 0 \quad (1)$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = 2$$

$$r_2 = 2$$

Since the roots are equal, then the general solution is

$$y = c_1 y_1 + c_2 y_2$$

Where $y_1 = x^r$ and $y_2 = x^r \ln(x)$. Hence

$$y = c_1 x^2 + c_2 \ln(x) x^2$$

Next, we find the particular solution to the ODE

$$x^2 y'' - 3y'x + 4y = x(\ln(x)x + 1)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x^2$$

$$y_2 = x^2 \ln(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ \frac{d}{dx}(x^2) & \frac{d}{dx}(x^2 \ln(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ 2x & 2 \ln(x) x + x \end{vmatrix}$$

Therefore

$$W = (x^2) (2 \ln(x) x + x) - (x^2 \ln(x)) (2x)$$

Which simplifies to

$$W = x^3$$

Which simplifies to

$$W = x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^3 \ln(x) (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x) (\ln(x) x + 1)}{x^2} dx$$

Hence

$$u_1 = -\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^3 (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x) x + 1}{x^2} dx$$

Hence

$$u_2 = \frac{\ln(x)^2}{2} - \frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x} \right) x^2 + x^2 \ln(x) \left(\frac{\ln(x)^2}{2} - \frac{1}{x} \right)$$

Which simplifies to

$$y_p(x) = \frac{x(\ln(x)^3 x + 6)}{6}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \frac{x^2 \ln(x)^3}{6} + x + c_1 x^2 + c_2 \ln(x) x^2 \end{aligned}$$

Summary of solutions found

$$y = \frac{x^2 \ln(x)^3}{6} + x + c_1 x^2 + c_2 \ln(x) x^2$$

Solved as second order ode using change of variable on x method 2

Time used: 0.372 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2 y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2 y'' - 3y'x + 4y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{4}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x)dx} dx \\ &= \int e^{-\int -\frac{3}{x}dx} dx \\ &= \int e^{3\ln(x)} dx \\ &= \int x^3 dx \\ &= \frac{x^4}{4} \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{\frac{4}{x^2}}{x^6} \\ &= \frac{4}{x^8} \end{aligned} \quad (7)$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned}\frac{d^2}{d\tau^2}y(\tau) + q_1y(\tau) &= 0 \\ \frac{d^2}{d\tau^2}y(\tau) + \frac{4y(\tau)}{x^8} &= 0\end{aligned}$$

But in terms of τ

$$\frac{4}{x^8} = \frac{1}{4\tau^2}$$

Hence the above ode becomes

$$\frac{d^2}{d\tau^2}y(\tau) + \frac{y(\tau)}{4\tau^2} = 0$$

The above ode is now solved for $y(\tau)$. This is Euler second order ODE. Let the solution be $y(\tau) = \tau^r$, then $y' = r\tau^{r-1}$ and $y'' = r(r-1)\tau^{r-2}$. Substituting these back into the given ODE gives

$$4\tau^2(r(r-1))\tau^{r-2} + 0r\tau^{r-1} + \tau^r = 0$$

Simplifying gives

$$4r(r-1)\tau^r + 0\tau^r + \tau^r = 0$$

Since $\tau^r \neq 0$ then dividing throughout by τ^r gives

$$4r(r-1) + 0 + 1 = 0$$

Or

$$4r^2 - 4r + 1 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$\begin{aligned}r_1 &= \frac{1}{2} \\ r_2 &= \frac{1}{2}\end{aligned}$$

Since the roots are equal, then the general solution is

$$y(\tau) = c_1y_1 + c_2y_2$$

Where $y_1 = \tau^r$ and $y_2 = \tau^r \ln(\tau)$. Hence

$$y(\tau) = c_1\sqrt{\tau} + c_2\sqrt{\tau} \ln(\tau)$$

The above solution is now transformed back to y using (6) which results in

$$y = \frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \sqrt{x^4}$$

$$y_2 = -\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \sqrt{x^4} & -\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4} \\ \frac{d}{dx}(\sqrt{x^4}) & \frac{d}{dx}\left(-\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \sqrt{x^4} & -\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4} \\ \frac{2x^3}{\sqrt{x^4}} & -\frac{\sqrt{4} \ln(2)x^3}{\sqrt{x^4}} + \frac{\sqrt{4} \ln(x^4)x^3}{2\sqrt{x^4}} + \frac{\sqrt{4} \sqrt{x^4}}{x} \end{vmatrix}$$

Therefore

$$W = \left(\sqrt{x^4}\right) \left(-\frac{\sqrt{4} \ln(2) x^3}{\sqrt{x^4}} + \frac{\sqrt{4} \ln(x^4) x^3}{2\sqrt{x^4}} + \frac{\sqrt{4} \sqrt{x^4}}{x}\right) \\ - \left(-\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4}\right) \left(\frac{2x^3}{\sqrt{x^4}}\right)$$

Which simplifies to

$$W = \sqrt{4} x^3$$

Which simplifies to

$$W = 2x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\left(-\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4}\right) x(\ln(x) x + 1)}{2x^5} dx$$

Which simplifies for $0 < x$ to

$$u_1 = - \int - \frac{\left(\ln(2) - \frac{\ln(x^4)}{2}\right) (\ln(x) x + 1)}{2x^2} dx$$

Hence

$$u_1 = \frac{\ln(2) \ln(x)^2}{4} - \frac{\ln(2)}{2x} + \frac{\ln(x^4)}{4x} + \frac{1}{x} - \frac{\ln(x)^2 \ln(x^4)}{8} + \frac{\ln(x)^3}{6}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\sqrt{x^4} x(\ln(x) x + 1)}{2x^5} dx$$

Which simplifies for $0 < x$ to

$$u_2 = \int \frac{\ln(x) x + 1}{2x^2} dx$$

Hence

$$u_2 = \frac{\ln(x)^2}{4} - \frac{1}{2x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(2) \ln(x)^2}{4} - \frac{\ln(2)}{2x} + \frac{\ln(x^4)}{4x} + \frac{1}{x} - \frac{\ln(x)^2 \ln(x^4)}{8} + \frac{\ln(x)^3}{6} \right) \sqrt{x^4} \\ + \left(-\frac{\sqrt{4} \sqrt{x^4} \ln(2)}{2} + \frac{\sqrt{4} \sqrt{x^4} \ln(x^4)}{4} \right) \left(\frac{\ln(x)^2}{4} - \frac{1}{2x} \right)$$

Which simplifies to

$$y_p(x) = \frac{x(\ln(x)^3 x + 6)}{6}$$

Therefore the general solution is

$$y = y_h + y_p \\ = \left(\frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4} \right) + \left(\frac{x(\ln(x)^3 x + 6)}{6} \right)$$

Summary of solutions found

$$y = \frac{c_1 \sqrt{4} \sqrt{x^4}}{4} + \frac{c_2 \sqrt{4} \sqrt{x^4} \ln\left(\frac{x^4}{4}\right)}{4} + \frac{x(\ln(x)^3 x + 6)}{6}$$

Solved as second order ode using change of variable on x method 1

Time used: 0.206 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -3x$, $C = 4$, $f(x) = x(\ln(x)x + 1)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2 y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2 y'' - 3y'x + 4y = 0 \quad (1)$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{4}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) results

$$\frac{d^2}{d\tau^2} y(\tau) + p_1 \left(\frac{d}{d\tau} y(\tau) \right) + q_1 y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $q_1 = c^2$ where c is some constant. Therefore from (5)

$$\begin{aligned} \tau' &= \frac{1}{c} \sqrt{q} \\ &= \frac{2\sqrt{\frac{1}{x^2}}}{c} \\ \tau'' &= -\frac{2}{c\sqrt{\frac{1}{x^2}} x^3} \end{aligned} \quad (6)$$

Substituting the above into (4) results in

$$\begin{aligned} p_1(\tau) &= \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \\ &= \frac{-\frac{2}{c\sqrt{\frac{1}{x^2}} x^3} - \frac{3}{x} \frac{2\sqrt{\frac{1}{x^2}}}{c}}{\left(\frac{2\sqrt{\frac{1}{x^2}}}{c}\right)^2} \\ &= -2c \end{aligned}$$

Therefore ode (3) now becomes

$$\begin{aligned} y(\tau)'' + p_1 y(\tau)' + q_1 y(\tau) &= 0 \\ \frac{d^2}{d\tau^2} y(\tau) - 2c \left(\frac{d}{d\tau} y(\tau) \right) + c^2 y(\tau) &= 0 \end{aligned} \quad (7)$$

The above ode is now solved for $y(\tau)$. Since the ode is now constant coefficients, it can be easily solved to give

$$y(\tau) = c_1 e^{c\tau} + c_2 \tau e^{c\tau}$$

Now from (6)

$$\begin{aligned} \tau &= \int \frac{1}{c} \sqrt{q} dx \\ &= \frac{\int 2\sqrt{\frac{1}{x^2}} dx}{c} \\ &= \frac{2 \ln(x)}{c} \end{aligned}$$

Substituting the above into the solution obtained gives

$$y = c_1 x^2 + \frac{2c_2 \ln(x) x^2}{c}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned} y_1 &= x^2 \\ y_2 &= \frac{2 \ln(x) x^2}{c} \end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x^2 & \frac{2 \ln(x) x^2}{c} \\ \frac{d}{dx}(x^2) & \frac{d}{dx} \left(\frac{2 \ln(x) x^2}{c} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x^2 & \frac{2 \ln(x) x^2}{c} \\ 2x & \frac{2x}{c} + \frac{4 \ln(x) x}{c} \end{vmatrix}$$

Therefore

$$W = (x^2) \left(\frac{2x}{c} + \frac{4 \ln(x) x}{c} \right) - \left(\frac{2 \ln(x) x^2}{c} \right) (2x)$$

Which simplifies to

$$W = \frac{2x^3}{c}$$

Which simplifies to

$$W = \frac{2x^3}{c}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{2 \ln(x) x^3 (\ln(x) x + 1)}{c}}{\frac{2x^5}{c}} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x) (\ln(x) x + 1)}{x^2} dx$$

Hence

$$u_1 = -\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^3 (\ln(x) x + 1)}{\frac{2x^5}{c}} dx$$

Which simplifies to

$$u_2 = \int \frac{(\ln(x) x + 1) c}{2x^2} dx$$

Hence

$$u_2 = \frac{c \ln(x)^2}{4} - \frac{c}{2x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x} \right) x^2 + \frac{2 \ln(x) x^2 \left(\frac{c \ln(x)^2}{4} - \frac{c}{2x} \right)}{c}$$

Which simplifies to

$$y_p(x) = \frac{x(\ln(x)^3 x + 6)}{6}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 x^2 + \frac{2c_2 \ln(x) x^2}{c} \right) + \left(\frac{x(\ln(x)^3 x + 6)}{6} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x(\ln(x)^3 x + 6)}{6} + c_1 x^2 + \frac{2c_2 \ln(x) x^2}{c}$$

Solved as second order ode using change of variable on y method 2

Time used: 0.328 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -3x$, $C = 4$, $f(x) = x(\ln(x)x + 1)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2y'' - 3y'x + 4y = 0$$

In normal form the ode

$$x^2y'' - 3y'x + 4y = 0 \quad (1)$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \quad (2)$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{4}{x^2}$$

Applying change of variables on the dependent variable $y = v(x)x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right)v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right)v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{3n}{x^2} + \frac{4}{x^2} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 2 \quad (6)$$

Substituting this value in (3) gives

$$v''(x) + \frac{v'(x)}{x} = 0$$

$$v''(x) + \frac{v'(x)}{x} = 0 \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{u(x)}{x} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{1}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{1}{x} \, dx} \\ &= x \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} (ux) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} ux &= \int 0 \, dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor x gives the final solution

$$u(x) = \frac{c_1}{x}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) \, dx + c_2 \\ &= c_1 \ln(x) + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= (c_1 \ln(x) + c_2) x^2 \\ &= (c_1 \ln(x) + c_2) x^2 \end{aligned}$$

Now the particular solution to this ODE is found

$$x^2 y'' - 3y'x + 4y = x(\ln(x)x + 1)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned} y_1 &= x^2 \\ y_2 &= x^2 \ln(x) \end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ \frac{d}{dx}(x^2) & \frac{d}{dx}(x^2 \ln(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ 2x & 2 \ln(x)x + x \end{vmatrix}$$

Therefore

$$W = (x^2) (2 \ln(x) x + x) - (x^2 \ln(x)) (2x)$$

Which simplifies to

$$W = x^3$$

Which simplifies to

$$W = x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^3 \ln(x) (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x) (\ln(x) x + 1)}{x^2} dx$$

Hence

$$u_1 = -\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^3 (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x) x + 1}{x^2} dx$$

Hence

$$u_2 = \frac{\ln(x)^2}{2} - \frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x} \right) x^2 + x^2 \ln(x) \left(\frac{\ln(x)^2}{2} - \frac{1}{x} \right)$$

Which simplifies to

$$y_p(x) = \frac{x(\ln(x)^3 x + 6)}{6}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= ((c_1 \ln(x) + c_2) x^2) + \left(\frac{x(\ln(x)^3 x + 6)}{6} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{x(\ln(x)^3 x + 6)}{6} + (c_1 \ln(x) + c_2) x^2$$

Solved as second order ode using Kovacic algorithm

Time used: 0.194 (sec)

Solve

$$x^2 y'' - 3y'x + 4y = x + x^2 \ln(x)$$

Writing the ode as

$$x^2 y'' - 3y'x + 4y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 \\ B &= -3x \\ C &= 4 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = y e^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = r z(x) \tag{4}$$

Where r is given by

$$r = \frac{s}{t} = \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{1}{4x^2}\right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.262: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -\frac{1}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = -\frac{1}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{1}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{1}{2}$	$\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{1}{2}$	$\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = \frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= \frac{1}{2} - \left(\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= \frac{1}{2x} + (-)(0) \\ &= \frac{1}{2x} \\ &= \frac{1}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(\frac{1}{2x}\right)(0) + \left(\left(-\frac{1}{2x^2}\right) + \left(\frac{1}{2x}\right)^2 - \left(-\frac{1}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \frac{1}{2x} dx} \\ &= \sqrt{x} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3x}{x^2} dx} \\ &= z_1 e^{\frac{3 \ln(x)}{2}} \\ &= z_1 (x^{3/2}) \end{aligned}$$

Which simplifies to

$$y_1 = x^2$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3 \ln(x)}}{(y_1)^2} dx \\ &= y_1 (\ln(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (x^2) + c_2 (x^2 (\ln(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2y'' - 3y'x + 4y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 x^2 + c_2 \ln(x) x^2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x^2$$

$$y_2 = x^2 \ln(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ \frac{d}{dx}(x^2) & \frac{d}{dx}(x^2 \ln(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x^2 & x^2 \ln(x) \\ 2x & 2 \ln(x) x + x \end{vmatrix}$$

Therefore

$$W = (x^2) (2 \ln(x) x + x) - (x^2 \ln(x)) (2x)$$

Which simplifies to

$$W = x^3$$

Which simplifies to

$$W = x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^3 \ln(x) (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x) (\ln(x) x + 1)}{x^2} dx$$

Hence

$$u_1 = -\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^3 (\ln(x) x + 1)}{x^5} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x) x + 1}{x^2} dx$$

Hence

$$u_2 = \frac{\ln(x)^2}{2} - \frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{\ln(x)^3}{3} + \frac{\ln(x)}{x} + \frac{1}{x} \right) x^2 + x^2 \ln(x) \left(\frac{\ln(x)^2}{2} - \frac{1}{x} \right)$$

Which simplifies to

$$y_p(x) = \frac{x(\ln(x)^3 x + 6)}{6}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= (c_1 x^2 + c_2 \ln(x) x^2) + \left(\frac{x(\ln(x)^3 x + 6)}{6} \right)$$

Summary of solutions found

$$y = \frac{x(\ln(x)^3 x + 6)}{6} + c_1 x^2 + c_2 \ln(x) x^2$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 27

```
ode:=x^2*diff(diff(y(x),x),x)-3*diff(y(x),x)*x+4*y(x) = x+x^2*ln(x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^2 \ln(x)^3}{6} + x^2 \ln(x) c_1 + c_2 x^2 + x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    <- LODE of Euler type successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.024 (sec). Leaf size: 30

```
ode=x^2*D[y[x],{x,2}]-3*x*D[y[x],x]+4*y[x]==x+x^2*Log[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{6}x(x \log^3(x) + 6c_1x + 12c_2x \log(x) + 6)$$

✓ Sympy. Time used: 0.202 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2*log(x) + x**2*Derivative(y(x), (x, 2)) - 3*x*Derivative(y(x), x) -
        x + 4*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = \frac{x(C_1x + C_2x \log(x) + x \log(x)^3 + 6)}{6}$$

2.12.2 Problem 7

Local contents

Solved as second order Euler type ode	2052
Solved as second order solved by an integrating factor	2055
Solved as second order ode using change of variable on x method 2 .	2057
Solved as second order ode using change of variable on y method 1 .	2062
Solved as second order ode using change of variable on y method 2 .	2066
Solved as second order ode using non constant coeff transformation on B method	2071
Solved as second order ode using Kovacic algorithm	2075
✓ Maple	2079
✓ Mathematica	2080
✓ Sympy	2080

Internal problem ID [8026]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 7

Date solved : Thursday, November 27, 2025 at 07:32:27 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order Euler type ode

Time used: 0.168 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -2x$, $C = 2$, $f(x) = \ln(x)^2 - \ln(x^2)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2 y'' - 2y'x + 2y = 0$$

This is Euler second order ODE. Let the solution be $y = x^r$, then $y' = rx^{r-1}$ and $y'' = r(r-1)x^{r-2}$. Substituting these back into the given ODE gives

$$x^2(r(r-1))x^{r-2} - 2xrx^{r-1} + 2x^r = 0$$

Simplifying gives

$$r(r-1)x^r - 2rx^r + 2x^r = 0$$

Since $x^r \neq 0$ then dividing throughout by x^r gives

$$r(r-1) - 2r + 2 = 0$$

Or

$$r^2 - 3r + 2 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = 1$$

$$r_2 = 2$$

Since the roots are real and distinct, then the general solution is

$$y = c_1y_1 + c_2y_2$$

Where $y_1 = x^{r_1}$ and $y_2 = x^{r_2}$. Hence

$$y = c_2x^2 + c_1x$$

Next, we find the particular solution to the ODE

$$x^2y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2)(1)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2(\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x)^2 - \ln(x^2)}{x^2} dx$$

Hence

$$u_1 = \frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(\ln(x))^2 - \ln(x^2)}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x)^2 - \ln(x^2)}{x^3} dx$$

Hence

$$u_2 = -\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x} \right) x + x^2 \left(-\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} + c_2 x^2 + c_1 x \end{aligned}$$

Summary of solutions found

$$y = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} + c_2 x^2 + c_1 x$$

Solved as second order solved by an integrating factor

Time used: 0.058 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

The ode satisfies this form

$$y'' + p(x)y' + \frac{(p(x)^2 + p'(x))y}{2} = f(x)$$

Where $p(x) = -\frac{2}{x}$. Therefore, there is an integrating factor given by

$$\begin{aligned} M(x) &= e^{\frac{1}{2} \int p \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x} \end{aligned}$$

Multiplying both sides of the ODE by the integrating factor $M(x)$ makes the left side of the ODE a complete differential

$$\begin{aligned} (M(x)y)'' &= \frac{\ln(x)^2 - \ln(x^2)}{x^3} \\ \left(\frac{y}{x}\right)'' &= \frac{\ln(x)^2 - \ln(x^2)}{x^3} \end{aligned}$$

Integrating once gives

$$\left(\frac{y}{x}\right)' = \frac{-2 \ln(x)^2 - 2 \ln(x) + 2 \ln(x^2) + 1}{4x^2} + c_1$$

Integrating again gives

$$\left(\frac{y}{x}\right) = \frac{4c_1 x^2 + 2 \ln(x)^2 + 6 \ln(x) - 2 \ln(x^2) + 1}{4x} + c_2$$

Hence the solution is

$$y = \frac{\frac{4c_1 x^2 + 2 \ln(x)^2 + 6 \ln(x) - 2 \ln(x^2) + 1}{4x} + c_2}{\frac{1}{x}}$$

Or

$$y = c_1 x^2 + c_2 x + \frac{\ln(x)^2}{2} + \frac{3 \ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Summary of solutions found

$$y = c_1 x^2 + c_2 x + \frac{\ln(x)^2}{2} + \frac{3 \ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Solved as second order ode using change of variable on x method 2

Time used: 0.447 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2 y'' - 2y'x + 2y = 0$$

In normal form the ode

$$x^2 y'' - 2y'x + 2y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \tag{2}$$

Where

$$p(x) = -\frac{2}{x}$$

$$q(x) = \frac{2}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2} y(\tau) + p_1 \left(\frac{d}{d\tau} y(\tau) \right) + q_1 y(\tau) = 0 \tag{3}$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \tag{4}$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \tag{5}$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned}
 \tau &= \int e^{-\int p(x)dx} dx \\
 &= \int e^{-\int -\frac{2}{x}dx} dx \\
 &= \int e^{2\ln(x)} dx \\
 &= \int x^2 dx \\
 &= \frac{x^3}{3}
 \end{aligned} \tag{6}$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned}
 q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\
 &= \frac{\frac{2}{x^2}}{x^4} \\
 &= \frac{2}{x^6}
 \end{aligned} \tag{7}$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned}
 \frac{d^2}{d\tau^2}y(\tau) + q_1y(\tau) &= 0 \\
 \frac{d^2}{d\tau^2}y(\tau) + \frac{2y(\tau)}{x^6} &= 0
 \end{aligned}$$

But in terms of τ

$$\frac{2}{x^6} = \frac{2}{9\tau^2}$$

Hence the above ode becomes

$$\frac{d^2}{d\tau^2}y(\tau) + \frac{2y(\tau)}{9\tau^2} = 0$$

The above ode is now solved for $y(\tau)$. This is Euler second order ODE. Let the solution be $y(\tau) = \tau^r$, then $y' = r\tau^{r-1}$ and $y'' = r(r-1)\tau^{r-2}$. Substituting these back into the given ODE gives

$$9\tau^2(r(r-1))\tau^{r-2} + 0r\tau^{r-1} + 2\tau^r = 0$$

Simplifying gives

$$9r(r-1)\tau^r + 0\tau^r + 2\tau^r = 0$$

Since $\tau^r \neq 0$ then dividing throughout by τ^r gives

$$9r(r-1) + 0 + 2 = 0$$

Or

$$9r^2 - 9r + 2 = 0 \quad (1)$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = \frac{1}{3}$$

$$r_2 = \frac{2}{3}$$

Since the roots are real and distinct, then the general solution is

$$y(\tau) = c_1 y_1 + c_2 y_2$$

Where $y_1 = \tau^{r_1}$ and $y_2 = \tau^{r_2}$. Hence

$$y(\tau) = c_1 \tau^{1/3} + c_2 \tau^{2/3}$$

The above solution is now transformed back to y using (6) which results in

$$y = \frac{c_1 3^{2/3} (x^3)^{1/3}}{3} + \frac{c_2 3^{1/3} (x^3)^{2/3}}{3}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_1 3^{2/3} (x^3)^{1/3}}{3} + \frac{c_2 3^{1/3} (x^3)^{2/3}}{3}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = (x^3)^{1/3}$$

$$y_2 = (x^3)^{2/3}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} (x^3)^{1/3} & (x^3)^{2/3} \\ \frac{d}{dx}((x^3)^{1/3}) & \frac{d}{dx}((x^3)^{2/3}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} (x^3)^{1/3} & (x^3)^{2/3} \\ \frac{x^2}{(x^3)^{2/3}} & \frac{2x^2}{(x^3)^{1/3}} \end{vmatrix}$$

Therefore

$$W = ((x^3)^{1/3}) \left(\frac{2x^2}{(x^3)^{1/3}} \right) - ((x^3)^{2/3}) \left(\frac{x^2}{(x^3)^{2/3}} \right)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(x^3)^{2/3} (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{(x^3)^{2/3} (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Hence

$$u_1 = - \int_0^x \frac{(\alpha^3)^{2/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha$$

And Eq. (3) becomes

$$u_2 = \int \frac{(x^3)^{1/3} (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{(x^3)^{1/3} (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Hence

$$u_2 = \int_0^x \frac{(\alpha^3)^{1/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = - \int_0^x \frac{(\alpha^3)^{2/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha (x^3)^{1/3} + (x^3)^{2/3} \int_0^x \frac{(\alpha^3)^{1/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(\frac{c_1 3^{2/3} (x^3)^{1/3}}{3} + \frac{c_2 3^{1/3} (x^3)^{2/3}}{3} \right) + \left(- \int_0^x \frac{(\alpha^3)^{2/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha (x^3)^{1/3} + (x^3)^{2/3} \int_0^x \frac{(\alpha^3)^{1/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha \right)$$

Summary of solutions found

$$y = \frac{c_1 3^{2/3} (x^3)^{1/3}}{3} + \frac{c_2 3^{1/3} (x^3)^{2/3}}{3} - \int_0^x \frac{(\alpha^3)^{2/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha (x^3)^{1/3} + (x^3)^{2/3} \int_0^x \frac{(\alpha^3)^{1/3} (\ln(\alpha)^2 - \ln(\alpha^2))}{\alpha^4} d\alpha$$

Solved as second order ode using change of variable on y method 1

Time used: 0.251 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2 y'' - 2y'x + 2y = 0$$

In normal form the given ode is written as

$$y'' + p(x)y' + q(x)y = 0 \quad (2)$$

Where

$$p(x) = -\frac{2}{x}$$

$$q(x) = \frac{2}{x^2}$$

Calculating the Liouville ode invariant Q given by

$$\begin{aligned} Q &= q - \frac{p'}{2} - \frac{p^2}{4} \\ &= \frac{2}{x^2} - \frac{\left(-\frac{2}{x}\right)'}{2} - \frac{\left(-\frac{2}{x}\right)^2}{4} \\ &= \frac{2}{x^2} - \frac{\left(\frac{2}{x^2}\right)}{2} - \frac{\left(\frac{4}{x^2}\right)}{4} \\ &= \frac{2}{x^2} - \left(\frac{1}{x^2}\right) - \frac{1}{x^2} \\ &= 0 \end{aligned}$$

Since the Liouville ode invariant does not depend on the independent variable x then the transformation

$$y = v(x)z(x) \quad (3)$$

is used to change the original ode to a constant coefficients ode in v . In (3) the term $z(x)$ is given by

$$\begin{aligned} z(x) &= e^{-\int \frac{p(x)}{2} dx} \\ &= e^{-\int \frac{-\frac{2}{x}}{2}} \\ &= x \end{aligned} \tag{5}$$

Hence (3) becomes

$$y = v(x) x \tag{4}$$

Applying this change of variable to the original ode results in

$$x^3 v''(x) = 0$$

Which is now solved for $v(x)$.

The above ode can be simplified to

$$v''(x) = 0$$

Integrating twice gives the solution

$$v(x) = c_3 x + c_4$$

Now that $v(x)$ is known, then

$$\begin{aligned} y &= v(x) z(x) \\ &= (c_3 x + c_4) (z(x)) \end{aligned} \tag{7}$$

But from (5)

$$z(x) = x$$

Hence (7) becomes

$$y = (c_3 x + c_4) x$$

Therefore the homogeneous solution y_h is

$$y_h = (c_3 x + c_4) x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2)(1)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2 (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x)^2 - \ln(x^2)}{x^2} dx$$

Hence

$$u_1 = \frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x)^2 - \ln(x^2)}{x^3} dx$$

Hence

$$u_2 = -\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x} \right) x + x^2 \left(-\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= ((c_3x + c_4)x) + \left(\frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = (c_3x + c_4)x + \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Solved as second order ode using change of variable on y method 2

Time used: 0.219 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -2x$, $C = 2$, $f(x) = \ln(x)^2 - \ln(x^2)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2 y'' - 2y'x + 2y = 0$$

In normal form the ode

$$x^2 y'' - 2y'x + 2y = 0 \tag{1}$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = -\frac{2}{x}$$

$$q(x) = \frac{2}{x^2}$$

Applying change of variables on the dependent variable $y = v(x)x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right)v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right)v(x) = 0 \tag{3}$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \tag{4}$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{2n}{x^2} + \frac{2}{x^2} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 2 \quad (6)$$

Substituting this value in (3) gives

$$\begin{aligned} v''(x) + \frac{2v'(x)}{x} &= 0 \\ v''(x) + \frac{2v'(x)}{x} &= 0 \end{aligned} \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{2u(x)}{x} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x} \, dx} \\ &= x^2 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} (u x^2) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} u x^2 &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor x^2 gives the final solution

$$u(x) = \frac{c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= -\frac{c_1}{x} + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(-\frac{c_1}{x} + c_2\right) x^2 \\ &= (c_2 x - c_1) x \end{aligned}$$

Now the particular solution to this ODE is found

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned} y_1 &= x \\ y_2 &= x^2 \end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2)(1)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2(\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x)^2 - \ln(x^2)}{x^2} dx$$

Hence

$$u_1 = \frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(\ln(x))^2 - \ln(x^2)}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x)^2 - \ln(x^2)}{x^3} dx$$

Hence

$$u_2 = -\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x} \right) x + x^2 \left(-\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(-\frac{c_1}{x} + c_2 \right) x^2 \right) + \left(\frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} + \left(-\frac{c_1}{x} + c_2 \right) x^2$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.254 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \tag{1}$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= x^2 \\ B &= -2x \\ C &= 2 \\ F &= \ln(x)^2 - \ln(x^2) \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x^2)(0) + (-2x)(-2) + (2)(-2x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-2x^3v'' + (0)v' = 0$$

Now by applying $v' = u$ the above becomes

$$-2x^3u'(x) = 0$$

Which is now solved for u . Solve Since the ode has the form $u'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int du &= \int 0 dx + c_1 \\ u(x) &= c_1 \end{aligned}$$

The ode for v now becomes

$$v'(x) = c_1$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dv &= \int c_1 dx \\ v(x) &= c_1x + c_2 \end{aligned}$$

Replacing $v(x)$ above by $-\frac{y}{2x}$, then the homogeneous solution is

$$\begin{aligned} y_h(x) &= Bv \\ &= -2(c_1x + c_2)x \end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2)(1)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2(\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x)^2 - \ln(x^2)}{x^2} dx$$

Hence

$$u_1 = \frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(\ln(x))^2 - \ln(x^2)}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x)^2 - \ln(x^2)}{x^3} dx$$

Hence

$$u_2 = -\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x} \right) x + x^2 \left(-\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= (-2(c_1x + c_2)x) + \left(\frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} \right) \\ &= -2c_1x^2 + \frac{\ln(x)^2}{2} - 2c_2x + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} \end{aligned}$$

Summary of solutions found

$$y = -2c_1x^2 + \frac{\ln(x)^2}{2} - 2c_2x + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.132 (sec)

Solve

$$x^2 y'' - 2y'x + 2y = \ln(x)^2 - \ln(x^2)$$

Writing the ode as

$$x^2 y'' - 2y'x + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$A = x^2$$

$$B = -2x \quad (3)$$

$$C = 2$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$r = \frac{s}{t} \quad (5)$$

$$= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{0}{1} \quad (6)$$

Comparing the above to (5) shows that

$$s = 0$$

$$t = 1$$

Therefore eq. (4) becomes

$$z''(x) = 0 \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.263: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - -\infty \\ &= \infty \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is *infinity* then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 0$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = 1$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2x}{x^2} dx} \\ &= z_1 e^{\ln(x)} \\ &= z_1(x) \end{aligned}$$

Which simplifies to

$$y_1 = x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{2\ln(x)}}{(y_1)^2} dx \\ &= y_1(x) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(x) + c_2(x(x)) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2 y'' - 2y'x + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_2 x^2 + c_1 x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2)(1)$$

Which simplifies to

$$W = x^2$$

Which simplifies to

$$W = x^2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2 (\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{\ln(x)^2 - \ln(x^2)}{x^2} dx$$

Hence

$$u_1 = \frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(\ln(x)^2 - \ln(x^2))}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(x)^2 - \ln(x^2)}{x^3} dx$$

Hence

$$u_2 = -\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{\ln(x)^2}{x} + \frac{2\ln(x)}{x} - \frac{\ln(x^2)}{x} \right) x + x^2 \left(-\frac{\ln(x)^2}{2x^2} - \frac{\ln(x)}{2x^2} + \frac{1}{4x^2} + \frac{\ln(x^2)}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (c_2 x^2 + c_1 x) + \left(\frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4} + c_2 x^2 + c_1 x$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 30

```
ode:=x^2*diff(diff(y(x),x),x)-2*diff(y(x),x)*x+2*y(x) = ln(x)^2-ln(x^2);
dsolve(ode,y(x), singsol=all);
```

$$y = c_2 x^2 + c_1 x + \frac{\ln(x)^2}{2} + \frac{3\ln(x)}{2} - \frac{\ln(x^2)}{2} + \frac{1}{4}$$

Maple trace

Methods for second order ODEs:

```
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    <- LODE of Euler type successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.021 (sec). Leaf size: 38

```
ode=x^2*D[y[x],{x,2}]-2*x*D[y[x],x]+2*y[x]==(Log[x])^2-Log[x^2];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{4}(-2\log(x^2) + 2\log^2(x) + 6\log(x) + 1) + c_2x^2 + c_1x$$

✓ **Sympy.** Time used: 0.161 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) - 2*x*Derivative(y(x), x) + 2*y(x) - log(x)
    )**2 + log(x**2),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1x + C_2x^2 + \frac{\log(x)^2}{2} + \frac{\log(x)}{2} + \frac{1}{4}$$

2.12.3 Problem 8

Local contents

Solved as higher order Euler type ode	2081
✓ Maple	2085
✓ Mathematica	2086
✓ Sympy	2086

Internal problem ID [8027]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 8

Date solved : Thursday, November 27, 2025 at 07:32:29 AM

CAS classification : `[_3rd_order, _missing_y]`

Solve

$$x^3 y''' + 2x^2 y'' = x + \sin(\ln(x))$$

Solved as higher order Euler type ode

Time used: 0.267 (sec)

This is Euler ODE of higher order. Let $y = x^\lambda$. Hence

$$\begin{aligned} y' &= \lambda x^{\lambda-1} \\ y'' &= \lambda(\lambda-1) x^{\lambda-2} \\ y''' &= \lambda(\lambda-1)(\lambda-2) x^{\lambda-3} \end{aligned}$$

Substituting these back into

$$x^3 y''' + 2x^2 y'' = x + \sin(\ln(x))$$

gives

$$2x^2 \lambda(\lambda-1) x^{\lambda-2} + x^3 \lambda(\lambda-1)(\lambda-2) x^{\lambda-3} = 0$$

Which simplifies to

$$2\lambda(\lambda-1) x^\lambda + \lambda(\lambda-1)(\lambda-2) x^\lambda = 0$$

And since $x^\lambda \neq 0$ then dividing through by x^λ , the above becomes

$$2\lambda(\lambda - 1) + \lambda(\lambda - 1)(\lambda - 2) = 0$$

Simplifying gives the characteristic equation as

$$\lambda^2(\lambda - 1) = 0$$

Solving the above gives the following roots

$$\lambda_1 = 1$$

$$\lambda_2 = 0$$

$$\lambda_3 = 0$$

This table summarises the result

root	multiplicity	type of root
0	2	real root
1	1	real root

The solution is generated by going over the above table. For each real root λ of multiplicity one generates a $c_1 x^\lambda$ basis solution. Each real root of multiplicity two, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ basis solutions. Each real root of multiplicity three, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ and $c_3 x^\lambda \ln(x)^2$ basis solutions, and so on. Each complex root $\alpha \pm i\beta$ of multiplicity one generates $x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity two generates $\ln(x) x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity three generates $\ln(x)^2 x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And so on. Using the above show that the solution is

$$y = c_1 + c_2 \ln(x) + c_3 x$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = \ln(x)$$

$$y_3 = x$$

This is higher order nonhomogeneous Euler type ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous Euler ODE And y_p is a particular solution to the nonhomogeneous Euler ODE. y_h is the solution to

$$x^3 y''' + 2x^2 y'' = 0$$

Now the particular solution to the given ODE is found

$$x^3 y''' + 2x^2 y'' = x + \sin(\ln(x))$$

Let the particular solution be

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Where y_i are the basis solutions found above for the homogeneous solution y_h and $U_i(x)$ are functions to be determined as follows

$$U_i = (-1)^{n-i} \int \frac{F(x)W_i(x)}{aW(x)} dx$$

Where $W(x)$ is the Wronskian and $W_i(x)$ is the Wronskian that results after deleting the last row and the i -th column of the determinant and n is the order of the ODE or equivalently, the number of basis solutions, and a is the coefficient of the leading derivative in the ODE, and $F(x)$ is the RHS of the ODE. Therefore, the first step is to find the Wronskian $W(x)$. This is given by

$$W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Substituting the fundamental set of solutions y_i found above in the Wronskian gives

$$W = \begin{bmatrix} 1 & \ln(x) & x \\ 0 & \frac{1}{x} & 1 \\ 0 & -\frac{1}{x^2} & 0 \end{bmatrix}$$

$$|W| = \frac{1}{x^2}$$

The determinant simplifies to

$$|W| = \frac{1}{x^2}$$

Now we determine W_i for each U_i .

$$W_1(x) = \det \begin{bmatrix} \ln(x) & x \\ \frac{1}{x} & 1 \end{bmatrix}$$

$$= \ln(x) - 1$$

$$W_2(x) = \det \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}$$

$$= 1$$

$$\begin{aligned}
 W_3(x) &= \det \begin{bmatrix} 1 & \ln(x) \\ 0 & \frac{1}{x} \end{bmatrix} \\
 &= \frac{1}{x}
 \end{aligned}$$

Now we are ready to evaluate each $U_i(x)$.

$$\begin{aligned}
 U_1 &= (-1)^{3-1} \int \frac{F(x)W_1(x)}{aW(x)} dx \\
 &= (-1)^2 \int \frac{(x + \sin(\ln(x))) (\ln(x) - 1)}{(x^3) \left(\frac{1}{x^2}\right)} dx \\
 &= \int \frac{(x + \sin(\ln(x))) (\ln(x) - 1)}{x} dx \\
 &= \int \left(\frac{(x + \sin(\ln(x))) (\ln(x) - 1)}{x} \right) dx \\
 &= -2x + \cos(\ln(x)) + \ln(x)x + \sin(\ln(x)) - \cos(\ln(x))\ln(x)
 \end{aligned}$$

$$\begin{aligned}
 U_2 &= (-1)^{3-2} \int \frac{F(x)W_2(x)}{aW(x)} dx \\
 &= (-1)^1 \int \frac{(x + \sin(\ln(x))) (1)}{(x^3) \left(\frac{1}{x^2}\right)} dx \\
 &= - \int \frac{x + \sin(\ln(x))}{x} dx \\
 &= - \int \left(\frac{x + \sin(\ln(x))}{x} \right) dx \\
 &= -x + \cos(\ln(x))
 \end{aligned}$$

$$\begin{aligned}
 U_3 &= (-1)^{3-3} \int \frac{F(x)W_3(x)}{aW(x)} dx \\
 &= (-1)^0 \int \frac{(x + \sin(\ln(x))) \left(\frac{1}{x}\right)}{(x^3) \left(\frac{1}{x^2}\right)} dx \\
 &= \int \frac{\frac{x + \sin(\ln(x))}{x}}{x} dx \\
 &= \int \left(\frac{x + \sin(\ln(x))}{x^2} \right) dx \\
 &= \ln(x) + \frac{-\frac{1}{2} + \frac{\tan\left(\frac{\ln(x)}{2}\right)^2}{2} - \tan\left(\frac{\ln(x)}{2}\right)}{\left(1 + \tan\left(\frac{\ln(x)}{2}\right)^2\right)x}
 \end{aligned}$$

Now that all the U_i functions have been determined, the particular solution is found from

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Hence

$$\begin{aligned}
 y_p &= (-2x + \cos(\ln(x)) + \ln(x)x + \sin(\ln(x)) - \cos(\ln(x))\ln(x)) \\
 &\quad + (-x + \cos(\ln(x)))(\ln(x)) \\
 &\quad + \left(\ln(x) + \frac{-\frac{1}{2} + \frac{\tan\left(\frac{\ln(x)}{2}\right)^2}{2} - \tan\left(\frac{\ln(x)}{2}\right)}{\left(1 + \tan\left(\frac{\ln(x)}{2}\right)^2\right)x} \right) (x)
 \end{aligned}$$

Therefore the particular solution is

$$y_p = \frac{\cos(\ln(x))}{2} - 2x + \ln(x)x + \frac{\sin(\ln(x))}{2}$$

Therefore the general solution is

$$\begin{aligned}
 y &= y_h + y_p \\
 &= (c_1 + c_2 \ln(x) + c_3 x) + \left(\frac{\cos(\ln(x))}{2} - 2x + \ln(x)x + \frac{\sin(\ln(x))}{2} \right)
 \end{aligned}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 40

```
ode:=x^3*diff(diff(diff(y(x),x),x),x)+2*x^2*diff(diff(y(x),x),x) = x+sin(ln(x));
dsolve(ode,y(x), singsol=all);
```

$$y = -c_1 \ln(x) + \ln(x)x + c_2 x + c_3 - x + \frac{\tan\left(\frac{\ln(x)}{2}\right) + 1}{1 + \tan\left(\frac{\ln(x)}{2}\right)^2}$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
-> Calling odsolve with the ODE, diff(_b(_a),_a) = -(2*_a^2*_b(_a)-sin(ln(_a)))-_a)/_a^3, _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- high order exact linear fully integrable successful
```

Maple step by step

Let's solve

$$x^3 \left(\frac{d^3}{dx^3} y(x) \right) + 2x^2 \left(\frac{d^2}{dx^2} y(x) \right) = x + \sin(\ln(x))$$

- Highest derivative means the order of the ODE is 3
 $\frac{d^3}{dx^3} y(x)$

✓ **Mathematica.** Time used: 32.715 (sec). Leaf size: 54

```
ode=x^3*D[y[x],{x,3}]+2*x^2*D[y[x],{x,2}]==x+Sin[Log[x]];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x \int_1^{K[3]} \frac{c_1 + \int_1^{K[2]} \left(\frac{\sin(\log(K[1]))}{K[1]} + 1 \right) dK[1]}{K[2]^2} dK[2] dK[3] + c_3 x + c_2$$

✓ **Sympy.** Time used: 0.193 (sec). Leaf size: 32

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**3*Derivative(y(x), (x, 3)) + 2*x**2*Derivative(y(x), (x, 2)) - x -
    sin(log(x)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 + C_2 x + C_3 \log(x) + x \log(x) + \frac{\sqrt{2} \sin\left(\log(x) + \frac{\pi}{4}\right)}{2}$$

2.12.4 Problem 9

Local contents

Solved as higher order Euler type ode	2087
✓ Maple	2091
✓ Mathematica	2092
✓ Sympy	2092

Internal problem ID [8028]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 9

Date solved : Thursday, November 27, 2025 at 07:32:30 AM

CAS classification : `[[_3rd_order, _with_linear_symmetries]]`

Solve

$$x^3 y''' + y'x - y = 3x^4$$

Solved as higher order Euler type ode

Time used: 0.214 (sec)

This is Euler ODE of higher order. Let $y = x^\lambda$. Hence

$$\begin{aligned} y' &= \lambda x^{\lambda-1} \\ y'' &= \lambda(\lambda-1) x^{\lambda-2} \\ y''' &= \lambda(\lambda-1)(\lambda-2) x^{\lambda-3} \end{aligned}$$

Substituting these back into

$$x^3 y''' + y'x - y = 3x^4$$

gives

$$x\lambda x^{\lambda-1} + x^3\lambda(\lambda-1)(\lambda-2)x^{\lambda-3} - x^\lambda = 0$$

Which simplifies to

$$\lambda x^\lambda + \lambda(\lambda-1)(\lambda-2)x^\lambda - x^\lambda = 0$$

And since $x^\lambda \neq 0$ then dividing through by x^λ , the above becomes

$$\lambda + \lambda(\lambda-1)(\lambda-2) - 1 = 0$$

Simplifying gives the characteristic equation as

$$(\lambda - 1)^3 = 0$$

Solving the above gives the following roots

$$\lambda_1 = 1$$

$$\lambda_2 = 1$$

$$\lambda_3 = 1$$

This table summarises the result

root	multiplicity	type of root
1	3	real root

The solution is generated by going over the above table. For each real root λ of multiplicity one generates a $c_1 x^\lambda$ basis solution. Each real root of multiplicity two, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ basis solutions. Each real root of multiplicity three, generates $c_1 x^\lambda$ and $c_2 x^\lambda \ln(x)$ and $c_3 x^\lambda \ln(x)^2$ basis solutions, and so on. Each complex root $\alpha \pm i\beta$ of multiplicity one generates $x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity two generates $\ln(x) x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And each complex root $\alpha \pm i\beta$ of multiplicity three generates $\ln(x)^2 x^\alpha (c_1 \cos(\beta \ln(x)) + c_2 \sin(\beta \ln(x)))$ basis solutions. And so on. Using the above show that the solution is

$$y = c_1 x + c_2 \ln(x) x + c_3 \ln(x)^2 x$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = x$$

$$y_2 = \ln(x) x$$

$$y_3 = \ln(x)^2 x$$

This is higher order nonhomogeneous Euler type ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous Euler ODE And y_p is a particular solution to the nonhomogeneous Euler ODE. y_h is the solution to

$$x^3 y''' + y' x - y = 0$$

Now the particular solution to the given ODE is found

$$x^3 y''' + y' x - y = 3x^4$$

Let the particular solution be

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Where y_i are the basis solutions found above for the homogeneous solution y_h and $U_i(x)$ are functions to be determined as follows

$$U_i = (-1)^{n-i} \int \frac{F(x)W_i(x)}{aW(x)} dx$$

Where $W(x)$ is the Wronskian and $W_i(x)$ is the Wronskian that results after deleting the last row and the i -th column of the determinant and n is the order of the ODE or equivalently, the number of basis solutions, and a is the coefficient of the leading derivative in the ODE, and $F(x)$ is the RHS of the ODE. Therefore, the first step is to find the Wronskian $W(x)$. This is given by

$$W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Substituting the fundamental set of solutions y_i found above in the Wronskian gives

$$W = \begin{bmatrix} x & \ln(x)x & \ln(x)^2 x \\ 1 & 1 + \ln(x) & \ln(x)(\ln(x) + 2) \\ 0 & \frac{1}{x} & \frac{2+2\ln(x)}{x} \end{bmatrix}$$

$$|W| = 2$$

The determinant simplifies to

$$|W| = 2$$

Now we determine W_i for each U_i .

$$W_1(x) = \det \begin{bmatrix} \ln(x)x & \ln(x)^2 x \\ 1 + \ln(x) & \ln(x)(\ln(x) + 2) \end{bmatrix}$$

$$= \ln(x)^2 x$$

$$W_2(x) = \det \begin{bmatrix} x & \ln(x)^2 x \\ 1 & \ln(x)(\ln(x) + 2) \end{bmatrix}$$

$$= 2 \ln(x)x$$

$$W_3(x) = \det \begin{bmatrix} x & \ln(x)x \\ 1 & 1 + \ln(x) \end{bmatrix}$$

$$= x$$

Now we are ready to evaluate each $U_i(x)$.

$$\begin{aligned}
 U_1 &= (-1)^{3-1} \int \frac{F(x)W_1(x)}{aW(x)} dx \\
 &= (-1)^2 \int \frac{(3x^4)(\ln(x)^2 x)}{(x^3)(2)} dx \\
 &= \int \frac{3x^5 \ln(x)^2}{2x^3} dx \\
 &= \int \left(\frac{3x^2 \ln(x)^2}{2} \right) dx \\
 &= \frac{x^3 \ln(x)^2}{2} - \frac{x^3 \ln(x)}{3} + \frac{x^3}{9}
 \end{aligned}$$

$$\begin{aligned}
 U_2 &= (-1)^{3-2} \int \frac{F(x)W_2(x)}{aW(x)} dx \\
 &= (-1)^1 \int \frac{(3x^4)(2 \ln(x) x)}{(x^3)(2)} dx \\
 &= - \int \frac{6x^5 \ln(x)}{2x^3} dx \\
 &= - \int (3x^2 \ln(x)) dx \\
 &= -x^3 \ln(x) + \frac{x^3}{3}
 \end{aligned}$$

$$\begin{aligned}
 U_3 &= (-1)^{3-3} \int \frac{F(x)W_3(x)}{aW(x)} dx \\
 &= (-1)^0 \int \frac{(3x^4)(x)}{(x^3)(2)} dx \\
 &= \int \frac{3x^5}{2x^3} dx \\
 &= \int \left(\frac{3x^2}{2} \right) dx \\
 &= \frac{x^3}{2}
 \end{aligned}$$

Now that all the U_i functions have been determined, the particular solution is found from

$$y_p = U_1 y_1 + U_2 y_2 + U_3 y_3$$

Hence

$$\begin{aligned}
 y_p &= \left(\frac{x^3 \ln(x)^2}{2} - \frac{x^3 \ln(x)}{3} + \frac{x^3}{9} \right) (x) \\
 &\quad + \left(-x^3 \ln(x) + \frac{x^3}{3} \right) (\ln(x) x) \\
 &\quad + \left(\frac{x^3}{2} \right) (\ln(x)^2 x)
 \end{aligned}$$

Therefore the particular solution is

$$y_p = \frac{x^4}{9}$$

Therefore the general solution is

$$\begin{aligned}
 y &= y_h + y_p \\
 &= (c_1 x + c_2 \ln(x) x + c_3 \ln(x)^2 x) + \left(\frac{x^4}{9} \right)
 \end{aligned}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 26

```
ode:=-y(x)+diff(y(x),x)*x+x^3*diff(diff(diff(y(x),x),x),x) = 3*x^4;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x(9c_3 \ln(x)^2 + x^3 + 9c_2 \ln(x) + 9c_1)}{9}$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 3; linear nonhomogeneous with symmetry [0,1]
trying high order linear exact nonhomogeneous
trying differential order: 3; missing the dependent variable
checking if the LODE is of Euler type
<- LODE of Euler type successful
Euler equation successful
```

Maple step by step

Let's solve

$$x^3 \left(\frac{d^3}{dx^3} y(x) \right) + x \left(\frac{d}{dx} y(x) \right) - y(x) = 3x^4$$

- Highest derivative means the order of the ODE is 3
 $\frac{d^3}{dx^3} y(x)$

✓ **Mathematica.** Time used: 0.004 (sec). Leaf size: 31

```
ode=x^3*D[y[x],{x,3}]+x*D[y[x],x]-y[x]==3*x^4;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^4}{9} + c_1 x + c_3 x \log^2(x) + c_2 x \log(x)$$

✓ **Sympy.** Time used: 0.165 (sec). Leaf size: 22

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-3*x**4 + x**3*Derivative(y(x), (x, 3)) + x*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x \left(C_1 + C_2 \log(x) + C_3 \log(x)^2 + \frac{x^3}{9} \right)$$

2.12.5 Problem 10

Local contents

Solved as second order Euler type ode	2093
Solved as second order linear exact ode	2097
Solved as second order integrable as is ode	2099
Solved as second order integrable as is ode (ABC method)	2100
Solved as second order ode using change of variable on x method 2	2103
Solved as second order ode using non constant coeff transformation on B method	2108
Solved as second order ode using Kovacic algorithm	2112
✓ Maple	2120
✓ Mathematica	2120
✗ Sympy	2121

Internal problem ID [8029]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 10

Date solved : Thursday, November 27, 2025 at 07:32:30 AM

CAS classification : [[_2nd_order, _exact, _linear, _nonhomogeneous]]

Solved as second order Euler type ode

Time used: 0.183 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

This is shifted Euler ode. We start by converting the ode to standard non shifted Euler ode using the substitution

$$X = 1 + x$$

Which results in the ode

$$X^2 \left(\frac{d^2}{dX^2} y(X) \right) + X \left(\frac{d}{dX} y(X) \right) - y(X) = \ln(X)^2 - 2 + X$$

Which is now solved for $y(X)$.

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(X) + By'(X) + Cy(X) = f(X)$$

Where $A = X^2, B = X, C = -1, f(X) = \ln(X)^2 - 2 + X$. Let the solution be

$$y(X) = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(X) + By'(X) + Cy(X) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(X) + By'(X) + Cy(X) = f(X)$. Solving for y_h from

$$X^2 \left(\frac{d^2}{dX^2} y(X) \right) + X \left(\frac{d}{dX} y(X) \right) - y(X) = 0$$

This is Euler second order ODE. Let the solution be $y(X) = X^r$, then $y' = rX^{r-1}$ and $y'' = r(r-1)X^{r-2}$. Substituting these back into the given ODE gives

$$X^2(r(r-1))X^{r-2} + XrX^{r-1} - X^r = 0$$

Simplifying gives

$$r(r-1)X^r + rX^r - X^r = 0$$

Since $X^r \neq 0$ then dividing throughout by X^r gives

$$r(r-1) + r - 1 = 0$$

Or

$$r^2 - 1 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$\begin{aligned} r_1 &= -1 \\ r_2 &= 1 \end{aligned}$$

Since the roots are real and distinct, then the general solution is

$$y(X) = c_1 y_1 + c_2 y_2$$

Where $y_1 = X^{r_1}$ and $y_2 = X^{r_2}$. Hence

$$y(X) = \frac{c_1}{X} + c_2 X$$

Next, we find the particular solution to the ODE

$$X^2 \left(\frac{d^2}{dX^2} y(X) \right) + X \left(\frac{d}{dX} y(X) \right) - y(X) = \ln(X)^2 - 2 + X$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on X as well. Let

$$y_p(X) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{1}{X}$$

$$y_2 = X$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(X)}{aW(X)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(X)}{aW(X)} \quad (3)$$

Where $W(X)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{X} & X \\ \frac{d}{dX}\left(\frac{1}{X}\right) & \frac{d}{dX}(X) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{1}{X} & X \\ -\frac{1}{X^2} & 1 \end{vmatrix}$$

Therefore

$$W = \left(\frac{1}{X}\right)(1) - (X)\left(-\frac{1}{X^2}\right)$$

Which simplifies to

$$W = \frac{2}{X}$$

Which simplifies to

$$W = \frac{2}{X}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{X(\ln(X))^2 - 2 + X}{2X} dX$$

Which simplifies to

$$u_1 = - \int \left(\frac{\ln(X)^2}{2} - 1 + \frac{X}{2} \right) dX$$

Hence

$$u_1 = -\frac{X^2}{4} - \frac{\ln(X)^2 X}{2} + \ln(X) X$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{\ln(X)^2 - 2 + X}{X}}{2X} dX$$

Which simplifies to

$$u_2 = \int \frac{\ln(X)^2 - 2 + X}{2X^2} dX$$

Hence

$$u_2 = -\frac{\ln(X)^2}{2X} - \frac{\ln(X)}{X} + \frac{\ln(X)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(X) = \frac{-\frac{X^2}{4} - \frac{\ln(X)^2 X}{2} + \ln(X) X}{X} + X \left(-\frac{\ln(X)^2}{2X} - \frac{\ln(X)}{X} + \frac{\ln(X)}{2} \right)$$

Which simplifies to

$$y_p(X) = -\ln(X)^2 + \frac{\ln(X) X}{2} - \frac{X}{4}$$

Therefore the general solution is

$$\begin{aligned} y(X) &= y_h + y_p \\ &= -\ln(X)^2 + \frac{\ln(X) X}{2} - \frac{X}{4} + \frac{c_1}{X} + c_2 X \end{aligned}$$

Changing back from X to x results in

$$y = -\ln(1+x)^2 + \frac{(1+x)\ln(1+x)}{2} - \frac{x}{4} - \frac{1}{4} + \frac{c_1}{1+x} + c_2(1+x)$$

Summary of solutions found

$$y = -\ln(1+x)^2 + \frac{(1+x)\ln(1+x)}{2} - \frac{x}{4} - \frac{1}{4} + \frac{c_1}{1+x} + c_2(1+x)$$

Solved as second order linear exact ode

Time used: 0.123 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

An ode of the form

$$p(x) y'' + q(x) y' + r(x) y = s(x)$$

is exact if

$$p''(x) - q'(x) + r(x) = 0 \tag{1}$$

For the given ode we have

$$\begin{aligned} p(x) &= (1+x)^2 \\ q(x) &= 1+x \\ r(x) &= -1 \\ s(x) &= \ln(1+x)^2 + x - 1 \end{aligned}$$

Hence

$$\begin{aligned} p''(x) &= 2 \\ q'(x) &= 1 \end{aligned}$$

Therefore (1) becomes

$$2 - (1) + (-1) = 0$$

Hence the ode is exact. Since we now know the ode is exact, it can be written as

$$(p(x) y' + (q(x) - p'(x)) y)' = s(x)$$

Integrating gives

$$p(x)y' + (q(x) - p'(x))y = \int s(x) dx$$

Substituting the above values for p, q, r, s gives

$$(1+x)^2 y' + (-x-1)y = \int \ln(1+x)^2 + x - 1 dx$$

We now have a first order ode to solve which is

$$(1+x)^2 y' + (-x-1)y = x + \frac{x^2}{2} + \ln(1+x)^2(1+x) - 2(1+x)\ln(1+x) + 2 + c_1$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1}{1+x}$$

$$p(x) = \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{1+x} dx} \\ &= \frac{1}{1+x}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{1+x} \right) &= \left(\frac{1}{1+x} \right) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2} \right) \\ d \left(\frac{y}{1+x} \right) &= \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{1+x} &= \int \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^3} dx \\ &= \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - 2c_1 - 3}{4(1+x)^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{1+x}$ gives the final solution

$$y = \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 + 4c_2x^2 + 8c_2x - 2c_1 + 4c_2 - 3}{4x+4}$$

Summary of solutions found

$$y = \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 + 4c_2x^2 + 8c_2x - 2c_1 + 4c_2 - 3}{4x+4}$$

Solved as second order integrable as is ode

Time used: 0.093 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

Integrating both sides of the ODE w.r.t x gives

$$\int ((1+x)^2 y'' + (1+x) y' - y) dx = \int (\ln(1+x)^2 + x - 1) dx$$

$$(-x-1)y + (x^2 + 2x + 1)y' = x + \frac{x^2}{2} + \ln(1+x)^2(1+x) - 2(1+x)\ln(1+x) + 2 + c_1$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -\frac{1}{1+x} \\ p(x) &= \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2}\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{1+x} dx} \\ &= \frac{1}{1+x}\end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu) \left(\frac{(2x+2) \ln(1+x)^2 + (-4x-4) \ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2} \right)$$

$$\frac{d}{dx} \left(\frac{y}{1+x} \right) = \left(\frac{1}{1+x} \right) \left(\frac{(2x+2) \ln(1+x)^2 + (-4x-4) \ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^2} \right)$$

$$d \left(\frac{y}{1+x} \right) = \left(\frac{(2x+2) \ln(1+x)^2 + (-4x-4) \ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^3} \right) dx$$

Integrating gives

$$\begin{aligned} \frac{y}{1+x} &= \int \frac{(2x+2) \ln(1+x)^2 + (-4x-4) \ln(1+x) + x^2 + 2x + 2c_1 + 4}{2(1+x)^3} dx \\ &= \frac{(-4x-4) \ln(1+x)^2 + 2 \ln(1+x) (1+x)^2 - 2c_1 - 3}{4(1+x)^2} + c_2 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{1+x}$ gives the final solution

$$y = \frac{(-4x-4) \ln(1+x)^2 + 2 \ln(1+x) (1+x)^2 + 4c_2 x^2 + 8c_2 x - 2c_1 + 4c_2 - 3}{4x+4}$$

Summary of solutions found

$$y = \frac{(-4x-4) \ln(1+x)^2 + 2 \ln(1+x) (1+x)^2 + 4c_2 x^2 + 8c_2 x - 2c_1 + 4c_2 - 3}{4x+4}$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.201 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

Writing the ode as

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

Integrating both sides of the ODE w.r.t x gives

$$\int ((1+x)^2 y'' + (1+x) y' - y) dx = \int (\ln(1+x))^2 + (-x-1)y + (x^2+2x+1)y' = \frac{(1+x)^2}{2} + \ln(1+x)^2(1+x) - 2(1+x)\ln(1+x) + c_1$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1}{1+x}$$

$$p(x) = \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{1+x} dx} \\ &= \frac{1}{1+x}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{1+x} \right) &= \left(\frac{1}{1+x} \right) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2} \right) \\ d \left(\frac{y}{1+x} \right) &= \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{1+x} &= \int \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^3} dx \\ &= -\frac{\ln(1+x)^2}{1+x} + \frac{\ln(1+x)}{2} - \frac{c_1}{2(1+x)^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{1+x}$ gives the final solution

$$y = (1+x) \left(-\frac{\ln(1+x)^2}{1+x} + \frac{\ln(1+x)}{2} - \frac{c_1}{2(1+x)^2} + c_2 \right)$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1}{1+x}$$

$$p(x) = \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{1}{1+x} \, dx} \\ &= \frac{1}{1+x} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{1+x} \right) &= \left(\frac{1}{1+x} \right) \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^2} \right) \\ d \left(\frac{y}{1+x} \right) &= \left(\frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^3} \right) dx \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{y}{1+x} &= \int \frac{(2x+2)\ln(1+x)^2 + (-4x-4)\ln(1+x) + x^2 + 2x + 2c_1 + 1}{2(1+x)^3} \, dx \\ &= -\frac{\ln(1+x)^2}{1+x} + \frac{\ln(1+x)}{2} - \frac{c_1}{2(1+x)^2} + c_2 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{1+x}$ gives the final solution

$$y = (1+x) \left(-\frac{\ln(1+x)^2}{1+x} + \frac{\ln(1+x)}{2} - \frac{c_1}{2(1+x)^2} + c_2 \right)$$

Solved as second order ode using change of variable on x method 2

Time used: 0.209 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(1+x)^2 y'' + (1+x) y' - y = 0$$

In normal form the ode

$$(1+x)^2 y'' + (1+x) y' - y = 0 \quad (1)$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = \frac{1}{1+x}$$

$$q(x) = -\frac{1}{(1+x)^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2} y(\tau) + p_1 \left(\frac{d}{d\tau} y(\tau) \right) + q_1 y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned}
 \tau &= \int e^{-\int p(x)dx} dx \\
 &= \int e^{-\int \frac{1}{1+x} dx} dx \\
 &= \int e^{-\ln(1+x)} dx \\
 &= \int \frac{1}{1+x} dx \\
 &= \ln(1+x)
 \end{aligned} \tag{6}$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned}
 q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\
 &= \frac{-\frac{1}{(1+x)^2}}{\frac{1}{(1+x)^2}} \\
 &= -1
 \end{aligned} \tag{7}$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned}
 \frac{d^2}{d\tau^2}y(\tau) + q_1y(\tau) &= 0 \\
 \frac{d^2}{d\tau^2}y(\tau) - y(\tau) &= 0
 \end{aligned}$$

The above ode is now solved for $y(\tau)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(\tau) + By'(\tau) + Cy(\tau) = 0$$

Where in the above $A = 1, B = 0, C = -1$. Let the solution be $y(\tau) = e^{\lambda\tau}$. Substituting this into the ODE gives

$$\lambda^2 e^{\lambda\tau} - e^{\lambda\tau} = 0 \tag{1}$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda\tau}$ gives

$$\lambda^2 - 1 = 0 \tag{2}$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -1$ into the above gives

$$\begin{aligned}\lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-1)} \\ &= \pm 1\end{aligned}$$

Hence

$$\lambda_1 = +1$$

$$\lambda_2 = -1$$

Which simplifies to

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Since roots are distinct, then the solution is

$$y(\tau) = c_1 e^{\lambda_1 \tau} + c_2 e^{\lambda_2 \tau}$$

$$y(\tau) = c_1 e^{(1)\tau} + c_2 e^{(-1)\tau}$$

Or

$$y(\tau) = c_1 e^{\tau} + c_2 e^{-\tau}$$

The above solution is now transformed back to y using (6) which results in

$$y = c_1(1+x) + \frac{c_2}{1+x}$$

Therefore the homogeneous solution y_h is

$$y_h = c_1(1+x) + \frac{c_2}{1+x}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{1}{1+x}$$

$$y_2 = 1+x$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{1+x} & 1+x \\ \frac{d}{dx} \left(\frac{1}{1+x} \right) & \frac{d}{dx} (1+x) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{1}{1+x} & 1+x \\ -\frac{1}{(1+x)^2} & 1 \end{vmatrix}$$

Therefore

$$W = \left(\frac{1}{1+x} \right) (1) - (1+x) \left(-\frac{1}{(1+x)^2} \right)$$

Which simplifies to

$$W = \frac{2}{1+x}$$

Which simplifies to

$$W = \frac{2}{1+x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(1+x) (\ln(1+x))^2 + x - 1}{2x + 2} dx$$

Which simplifies to

$$u_1 = - \int \left(\frac{\ln(1+x)^2}{2} + \frac{x}{2} - \frac{1}{2} \right) dx$$

Hence

$$u_1 = -\frac{x}{2} - \frac{x^2}{4} - \frac{\ln(1+x)^2 (1+x)}{2} + (1+x) \ln(1+x) - 1$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{\ln(1+x)^2 + x - 1}{1+x}}{2x + 2} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(1+x)^2 + x - 1}{2(1+x)^2} dx$$

Hence

$$u_2 = -\frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} + \frac{\ln(1+x)}{2}$$

Therefore the particular solution, from equation (1) is

$$\begin{aligned} y_p(x) = & \frac{-\frac{x}{2} - \frac{x^2}{4} - \frac{\ln(1+x)^2(1+x)}{2} + (1+x)\ln(1+x) - 1}{1+x} \\ & + (1+x) \left(-\frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} + \frac{\ln(1+x)}{2} \right) \end{aligned}$$

Which simplifies to

$$y_p(x) = \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - x^2 - 2x - 4}{4x+4}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1(1+x) + \frac{c_2}{1+x} \right) + \left(\frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - x^2 - 2x - 4}{4x+4} \right) \end{aligned}$$

Summary of solutions found

$$y = c_1(1+x) + \frac{c_2}{1+x} + \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - x^2 - 2x - 4}{4x+4}$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.309 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \quad (1)$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= (1+x)^2 \\ B &= 1+x \\ C &= -1 \\ F &= \ln(1+x)^2 + x - 1 \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= ((1+x)^2)(0) + (1+x)(1) + (-1)(1+x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$(1+x)^3 v'' + (3(1+x)^2) v' = 0$$

Now by applying $v' = u$ the above becomes

$$(1+x)^2 ((1+x) u'(x) + 3u(x)) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{3}{1+x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{3}{1+x} dx} \\ &= (1+x)^3 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} (u(1+x)^3) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} u(1+x)^3 &= \int 0 dx + c_3 \\ &= c_3 \end{aligned}$$

Dividing throughout by the integrating factor $(1+x)^3$ gives the final solution

$$u(x) = \frac{c_3}{(1+x)^3}$$

The ode for v now becomes

$$v'(x) = \frac{c_3}{(1+x)^3}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dv &= \int \frac{c_3}{(1+x)^3} dx \\ v(x) &= -\frac{c_3}{2(1+x)^2} + c_4\end{aligned}$$

Replacing $v(x)$ above by $\frac{y}{1+x}$, then the homogeneous solution is

$$\begin{aligned}y_h(x) &= Bv \\ &= -\frac{-2c_4x^2 - 4c_4x + c_3 - 2c_4}{2(1+x)}\end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned}y_1 &= \frac{1}{1+x} \\ y_2 &= \frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x}\end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = -\int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The

Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{1+x} & \frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x} \\ \frac{d}{dx}\left(\frac{1}{1+x}\right) & \frac{d}{dx}\left(\frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x}\right) \end{vmatrix}$$

Which gives

$$W = \left| -\frac{\frac{1}{1+x}}{(1+x)^2} - \frac{x^2}{(1+x)^2} + \frac{\frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x}}{(1+x)^2} + \frac{2}{1+x} - \frac{1}{(1+x)^2} \right|$$

Therefore

$$\begin{aligned} W &= \left(\frac{1}{1+x} \right) \left(-\frac{x^2}{(1+x)^2} + \frac{2x}{1+x} - \frac{2x}{(1+x)^2} + \frac{2}{1+x} - \frac{1}{(1+x)^2} \right) \\ &\quad - \left(\frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x} \right) \left(-\frac{1}{(1+x)^2} \right) \end{aligned}$$

Which simplifies to

$$W = \frac{2}{1+x}$$

Which simplifies to

$$W = \frac{2}{1+x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\left(\frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x} \right) (\ln(1+x)^2 + x - 1)}{2x + 2} dx$$

Which simplifies to

$$u_1 = - \int \left(\frac{\ln(1+x)^2}{2} + \frac{x}{2} - \frac{1}{2} \right) dx$$

Hence

$$u_1 = -\frac{x}{2} - \frac{x^2}{4} - \frac{\ln(1+x)^2(1+x)}{2} + (1+x)\ln(1+x) - 1$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{\ln(1+x)^2 + x - 1}{1+x}}{2x + 2} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(1+x)^2 + x - 1}{2(1+x)^2} dx$$

Hence

$$u_2 = -\frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} + \frac{\ln(1+x)}{2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{-\frac{x}{2} - \frac{x^2}{4} - \frac{\ln(1+x)^2(1+x)}{2} + (1+x)\ln(1+x) - 1}{1+x} + \left(\frac{x^2}{1+x} + \frac{2x}{1+x} + \frac{1}{1+x} \right) \left(-\frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} + \frac{\ln(1+x)}{2} \right)$$

Which simplifies to

$$y_p(x) = \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - x^2 - 2x - 4}{4x+4}$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= \left(-\frac{-2c_4x^2 - 4c_4x + c_3 - 2c_4}{2(1+x)} \right) + \left(\frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 - x^2 - 2x - 4}{4x+4} \right) \\ &= \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 + (4c_4-1)x^2 + (8c_4-2)x - 2c_3 + 4c_4 - 4}{4x+4} \end{aligned}$$

Summary of solutions found

$$y = \frac{(-4x-4)\ln(1+x)^2 + 2\ln(1+x)(1+x)^2 + (4c_4-1)x^2 + (8c_4-2)x - 2c_3 + 4c_4 - 4}{4x+4}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.322 (sec)

Solve

$$(1+x)^2 y'' + (1+x) y' - y = \ln(1+x)^2 + x - 1$$

Writing the ode as

$$(1+x)^2 y'' + (1+x) y' - y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= (1+x)^2 \\ B &= 1+x \\ C &= -1 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{3}{4(1+x)^2} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 3 \\ t &= 4(1+x)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{3}{4(1+x)^2} \right) z(x) \tag{7}$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.266: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 O(\infty) &= \deg(t) - \deg(s) \\
 &= 2 - 0 \\
 &= 2
 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4(1+x)^2$. There is a pole at $x = -1$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4(1+x)^2}$$

For the pole at $x = -1$ let b be the coefficient of $\frac{1}{(1+x)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned}
 [\sqrt{r}]_c &= 0 \\
 \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\
 \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2}
 \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{3}{4(1+x)^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{3}{4(1+x)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
-1	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x-c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned}
 \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_{\infty} \\
 &= -\frac{1}{2(1+x)} + (-)(0) \\
 &= -\frac{1}{2(1+x)} \\
 &= -\frac{1}{2(1+x)}
 \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned}
 (0) + 2\left(-\frac{1}{2(1+x)}\right)(0) + \left(\left(\frac{1}{2(1+x)^2}\right) + \left(-\frac{1}{2(1+x)}\right)^2 - \left(\frac{3}{4(1+x)^2}\right)\right) &= 0 \\
 0 &= 0
 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned}
 z_1(x) &= pe^{\int \omega dx} \\
 &= e^{\int -\frac{1}{2(1+x)} dx} \\
 &= \frac{1}{\sqrt{1+x}}
 \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{1+x}{(1+x)^2} dx} \\
 &= z_1 e^{-\frac{\ln(1+x)}{2}} \\
 &= z_1 \left(\frac{1}{\sqrt{1+x}} \right)
 \end{aligned}$$

Which simplifies to

$$y_1 = \frac{1}{1+x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{1+x}{(1+x)^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-\ln(1+x)}}{(y_1)^2} dx \\ &= y_1 \left(x + \frac{1}{2}x^2 \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{1}{1+x} \right) + c_2 \left(\frac{1}{1+x} \left(x + \frac{1}{2}x^2 \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(1+x)^2 y'' + (1+x)y' - y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1}{1+x} + \frac{c_2 x(2+x)}{2x+2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{1}{1+x}$$

$$y_2 = \frac{x(2+x)}{2x+2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{1+x} & \frac{x(2+x)}{2x+2} \\ \frac{d}{dx} \left(\frac{1}{1+x} \right) & \frac{d}{dx} \left(\frac{x(2+x)}{2x+2} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{1}{1+x} & \frac{x(2+x)}{2x+2} \\ -\frac{1}{(1+x)^2} & \frac{2+x}{2x+2} + \frac{x}{2x+2} - \frac{2x(2+x)}{(2x+2)^2} \end{vmatrix}$$

Therefore

$$W = \left(\frac{1}{1+x} \right) \left(\frac{2+x}{2x+2} + \frac{x}{2x+2} - \frac{2x(2+x)}{(2x+2)^2} \right) - \left(\frac{x(2+x)}{2x+2} \right) \left(-\frac{1}{(1+x)^2} \right)$$

Which simplifies to

$$W = \frac{1}{1+x}$$

Which simplifies to

$$W = \frac{1}{1+x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{x(2+x)(\ln(1+x)^2+x-1)}{2x+2}}{1+x} dx$$

Which simplifies to

$$u_1 = - \int \frac{x(2+x)(\ln(1+x)^2+x-1)}{2(1+x)^2} dx$$

Hence

$$u_1 = -\frac{\ln(1+x)^2(1+x)}{2} + (1+x)\ln(1+x) - \frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} - \frac{(1+x)^2}{4} + \frac{\ln(1+x)}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{\ln(1+x)^2+x-1}{1+x}}{1+x} dx$$

Which simplifies to

$$u_2 = \int \frac{\ln(1+x)^2+x-1}{(1+x)^2} dx$$

Hence

$$u_2 = -\frac{\ln(1+x)^2}{1+x} - \frac{2\ln(1+x)}{1+x} + \ln(1+x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{-\frac{\ln(1+x)^2(1+x)}{2} + (1+x)\ln(1+x) - \frac{\ln(1+x)^2}{2(1+x)} - \frac{\ln(1+x)}{1+x} - \frac{(1+x)^2}{4} + \frac{\ln(1+x)}{2}}{1+x} + \frac{x(2+x)\left(-\frac{\ln(1+x)^2}{1+x} - \frac{2\ln(1+x)}{1+x} + \ln(1+x)\right)}{2x+2}$$

Which simplifies to

$$y_p(x) = -\ln(1+x)^2 + \frac{(2x+2)\ln(1+x)}{4} - \frac{1}{4} - \frac{x}{4}$$

Therefore the general solution is

$$y = y_h + y_p$$

$$= \left(\frac{c_1}{1+x} + \frac{c_2 x(2+x)}{2x+2} \right) + \left(-\ln(1+x)^2 + \frac{(2x+2)\ln(1+x)}{4} - \frac{1}{4} - \frac{x}{4} \right)$$

Summary of solutions found

$$y = -\ln(1+x)^2 + \frac{(2x+2)\ln(1+x)}{4} - \frac{1}{4} - \frac{x}{4} + \frac{c_1}{1+x} + \frac{c_2 x(2+x)}{2x+2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 53

```
ode:=(1+x)^2*diff(diff(y(x),x),x)+(1+x)*diff(y(x),x)-y(x) = ln(1+x)^2+x-1;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(-4x-4)\ln(x+1)^2 + 2(x+1)^2\ln(x+1) + 4c_2x^2 + 8c_2x + 4c_1 + 4c_2 - 3}{4x+4}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
<- high order exact linear fully integrable successful
```

✓ **Mathematica.** Time used: 0.211 (sec). Leaf size: 72

```
ode=(x+1)^2*D[y[x],{x,2}]+(x+1)*D[y[x],x]-y[x]==(Log[x+1])^2+x-1;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{(-1 + 2c_1 + 2ic_2)x^2 - 4(x+1)\log^2(x+1) + 2(x+1)^2\log(x+1) + (-2 + 4c_1 + 4ic_2)x - 1 + 4c_1}{4(x+1)}$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x + (x + 1)**2*Derivative(y(x), (x, 2)) + (x + 1)*Derivative(y(x), x) -
        y(x) - log(x + 1)**2 + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (-x**2*Derivative(y(x),
(x, 2)) - 2*x*Derivative(y(x), (x, 2)) + x + y(x) + log(x + 1)**2 -
Derivative(y(x), (x, 2)) - 1)/(x + 1) cannot be solved by the factorable group
method
```

2.12.6 Problem 11

Local contents

Solved as second order Euler type ode	2122
Solved as second order linear exact ode	2126
Solved as second order integrable as is ode	2128
Solved as second order integrable as is ode (ABC method)	2129
Solved as second order ode using Kovacic algorithm	2131
✓ Maple	2139
✓ Mathematica	2139
✗ Sympy	2139

Internal problem ID [8030]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 17. Linear equations with variable coefficients (Cauchy and Legendre).

Supplementary problems. Page 110

Problem number : 11

Date solved : Thursday, November 27, 2025 at 07:32:33 AM

CAS classification : [[_2nd_order, _exact, _linear, _nonhomogeneous]]

Solved as second order Euler type ode

Time used: 0.180 (sec)

Solve

$$(2x + 1)^2 y'' - 2(2x + 1) y' - 12y = 6x$$

This is shifted Euler ode. We start by converting the ode to standard non shifted Euler ode using the substitution

$$X = x + \frac{1}{2}$$

Which results in the ode

$$4X^2 \left(\frac{d^2}{dX^2} y(X) \right) - 4X \left(\frac{d}{dX} y(X) \right) - 12y(X) = -3 + 6X$$

Which is now solved for $y(X)$.

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(X) + By'(X) + Cy(X) = f(X)$$

Where $A = 4X^2$, $B = -4X$, $C = -12$, $f(X) = -3 + 6X$. Let the solution be

$$y(X) = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(X) + By'(X) + Cy(X) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(X) + By'(X) + Cy(X) = f(X)$. Solving for y_h from

$$4X^2 \left(\frac{d^2}{dX^2} y(X) \right) - 4X \left(\frac{d}{dX} y(X) \right) - 12y(X) = 0$$

This is Euler second order ODE. Let the solution be $y(X) = X^r$, then $y' = rX^{r-1}$ and $y'' = r(r-1)X^{r-2}$. Substituting these back into the given ODE gives

$$4X^2(r(r-1))X^{r-2} - 4XrX^{r-1} - 12X^r = 0$$

Simplifying gives

$$4r(r-1)X^r - 4rX^r - 12X^r = 0$$

Since $X^r \neq 0$ then dividing throughout by X^r gives

$$4r(r-1) - 4r - 12 = 0$$

Or

$$4r^2 - 8r - 12 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = -1$$

$$r_2 = 3$$

Since the roots are real and distinct, then the general solution is

$$y(X) = c_1y_1 + c_2y_2$$

Where $y_1 = X^{r_1}$ and $y_2 = X^{r_2}$. Hence

$$y(X) = \frac{c_1}{X} + c_2X^3$$

Next, we find the particular solution to the ODE

$$4X^2 \left(\frac{d^2}{dX^2} y(X) \right) - 4X \left(\frac{d}{dX} y(X) \right) - 12y(X) = -3 + 6X$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on X as well. Let

$$y_p(X) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{1}{X}$$

$$y_2 = X^3$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(X)}{aW(X)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(X)}{aW(X)} \quad (3)$$

Where $W(X)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{X} & X^3 \\ \frac{d}{dX}\left(\frac{1}{X}\right) & \frac{d}{dX}(X^3) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{1}{X} & X^3 \\ -\frac{1}{X^2} & 3X^2 \end{vmatrix}$$

Therefore

$$W = \left(\frac{1}{X}\right)(3X^2) - (X^3)\left(-\frac{1}{X^2}\right)$$

Which simplifies to

$$W = 4X$$

Which simplifies to

$$W = 4X$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{X^3(-3 + 6X)}{16X^3} dX$$

Which simplifies to

$$u_1 = - \int \left(-\frac{3}{16} + \frac{3X}{8} \right) dX$$

Hence

$$u_1 = \frac{3}{16}X - \frac{3}{16}X^2$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{-3+6X}{X}}{16X^3} dX$$

Which simplifies to

$$u_2 = \int \frac{-3+6X}{16X^4} dX$$

Hence

$$u_2 = \frac{1}{16X^3} - \frac{3}{16X^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(X) = \frac{\frac{3}{16}X - \frac{3}{16}X^2}{X} + X^3 \left(\frac{1}{16X^3} - \frac{3}{16X^2} \right)$$

Which simplifies to

$$y_p(X) = \frac{1}{4} - \frac{3X}{8}$$

Therefore the general solution is

$$\begin{aligned} y(X) &= y_h + y_p \\ &= \frac{1}{4} - \frac{3X}{8} + \frac{c_1}{X} + c_2 X^3 \end{aligned}$$

Changing back from X to x results in

$$y = \frac{1}{16} - \frac{3x}{8} + \frac{c_1}{x + \frac{1}{2}} + c_2 \left(x + \frac{1}{2} \right)^3$$

Summary of solutions found

$$y = \frac{1}{16} - \frac{3x}{8} + \frac{c_1}{x + \frac{1}{2}} + c_2 \left(x + \frac{1}{2} \right)^3$$

Solved as second order linear exact ode

Time used: 0.105 (sec)

Solve

$$(2x + 1)^2 y'' - 2(2x + 1) y' - 12y = 6x$$

An ode of the form

$$p(x) y'' + q(x) y' + r(x) y = s(x)$$

is exact if

$$p''(x) - q'(x) + r(x) = 0 \tag{1}$$

For the given ode we have

$$p(x) = 4\left(x + \frac{1}{2}\right)^2$$

$$q(x) = -4x - 2$$

$$r(x) = -12$$

$$s(x) = 6x$$

Hence

$$p''(x) = 8$$

$$q'(x) = -4$$

Therefore (1) becomes

$$8 - (-4) + (-12) = 0$$

Hence the ode is exact. Since we now know the ode is exact, it can be written as

$$(p(x) y' + (q(x) - p'(x)) y)' = s(x)$$

Integrating gives

$$p(x) y' + (q(x) - p'(x)) y = \int s(x) dx$$

Substituting the above values for p, q, r, s gives

$$4\left(x + \frac{1}{2}\right)^2 y' + (-12x - 6) y = \int 6x dx$$

We now have a first order ode to solve which is

$$4\left(x + \frac{1}{2}\right)^2 y' + (-12x - 6)y = 3x^2 + c_1$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{6}{2x+1}$$

$$p(x) = \frac{3x^2 + c_1}{(2x+1)^2}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{6}{2x+1} dx} \\ &= \frac{1}{(2x+1)^3}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{3x^2 + c_1}{(2x+1)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{1}{(2x+1)^3} \right) \left(\frac{3x^2 + c_1}{(2x+1)^2} \right) \\ d \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{3x^2 + c_1}{(2x+1)^5} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{(2x+1)^3} &= \int \frac{3x^2 + c_1}{(2x+1)^5} dx \\ &= \frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{(2x+1)^3}$ gives the final solution

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Summary of solutions found

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Solved as second order integrable as is ode

Time used: 0.085 (sec)

Solve

$$(2x + 1)^2 y'' - 2(2x + 1) y' - 12y = 6x$$

Integrating both sides of the ODE w.r.t x gives

$$\int ((2x + 1)^2 y'' - 2(2x + 1) y' - 12y) dx = \int 6x dx$$

$$(-12x - 6) y + (4x^2 + 4x + 1) y' = 3x^2 + c_1$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{6}{2x + 1}$$

$$p(x) = \frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2}$$

The integrating factor μ is

$$\mu = e^{\int q dx}$$

$$= e^{\int -\frac{6}{2x+1} dx}$$

$$= \frac{1}{(2x + 1)^3}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right)$$

$$\frac{d}{dx} \left(\frac{y}{(2x + 1)^3} \right) = \left(\frac{1}{(2x + 1)^3} \right) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right)$$

$$d \left(\frac{y}{(2x + 1)^3} \right) = \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2 (2x + 1)^3} \right) dx$$

Integrating gives

$$\begin{aligned}\frac{y}{(2x+1)^3} &= \int \frac{3x^2 + c_1}{4\left(x + \frac{1}{2}\right)^2 (2x+1)^3} dx \\ &= \frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{(2x+1)^3}$ gives the final solution

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Summary of solutions found

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.177 (sec)

Solve

$$(2x+1)^2 y'' - 2(2x+1) y' - 12y = 6x$$

Writing the ode as

$$(2x+1)^2 y'' - 2(2x+1) y' - 12y = 6x$$

Integrating both sides of the ODE w.r.t x gives

$$\begin{aligned}\int ((2x+1)^2 y'' - 2(2x+1) y' - 12y) dx &= \int 6x dx \\ (-12x - 6)y + (4x^2 + 4x + 1) y' &= 3x^2 + c_1\end{aligned}$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -\frac{6}{2x+1} \\ p(x) &= \frac{3x^2 + c_1}{4\left(x + \frac{1}{2}\right)^2}\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{6}{2x+1} \, dx} \\ &= \frac{1}{(2x+1)^3}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{1}{(2x+1)^3} \right) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right) \\ d \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{(2x+1)^3} &= \int \frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)^3} \, dx \\ &= \frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{(2x+1)^3}$ gives the final solution

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -\frac{6}{2x+1} \\ p(x) &= \frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2}\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{6}{2x+1} dx} \\ &= \frac{1}{(2x+1)^3}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right) \\ \frac{d}{dx} \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{1}{(2x+1)^3} \right) \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2} \right) \\ d \left(\frac{y}{(2x+1)^3} \right) &= \left(\frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{(2x+1)^3} &= \int \frac{3x^2 + c_1}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)^3} dx \\ &= \frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{(2x+1)^3}$ gives the final solution

$$y = (2x+1)^3 \left(\frac{-24x^2 - 4c_1 - 8x - 1}{32(2x+1)^4} + c_2 \right)$$

Solved as second order ode using Kovacic algorithm

Time used: 0.217 (sec)

Solve

$$(2x+1)^2 y'' - 2(2x+1) y' - 12y = 6x$$

Writing the ode as

$$4 \left(x + \frac{1}{2} \right)^2 y'' + (-4x - 2) y' - 12y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 4\left(x + \frac{1}{2}\right)^2 \\ B &= -4x - 2 \\ C &= -12 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{15}{(2x+1)^2} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 15 \\ t &= (2x+1)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{15}{(2x+1)^2}\right) z(x) \tag{7}$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.267: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\
 &= 2 - 0 \\
 &= 2
 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = (2x + 1)^2$. There is a pole at $x = -\frac{1}{2}$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{15}{4 \left(x + \frac{1}{2}\right)^2}$$

For the pole at $x = -\frac{1}{2}$ let b be the coefficient of $\frac{1}{(x + \frac{1}{2})^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{15}{4}$. Hence

$$\begin{aligned}
 [\sqrt{r}]_c &= 0 \\
 \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{5}{2} \\
 \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{3}{2}
 \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{15}{(2x+1)^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{15}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{5}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{3}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{15}{(2x+1)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
$-\frac{1}{2}$	2	0	$\frac{5}{2}$	$-\frac{3}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{5}{2}$	$-\frac{3}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{3}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{3}{2} - \left(-\frac{3}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x-c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned}
 \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_{\infty} \\
 &= -\frac{3}{2\left(x + \frac{1}{2}\right)} + (-)(0) \\
 &= -\frac{3}{2\left(x + \frac{1}{2}\right)} \\
 &= -\frac{3}{2x + 1}
 \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned}
 (0) + 2\left(-\frac{3}{2\left(x + \frac{1}{2}\right)}\right)(0) + \left(\left(\frac{3}{2\left(x + \frac{1}{2}\right)^2}\right) + \left(-\frac{3}{2\left(x + \frac{1}{2}\right)}\right)^2 - \left(\frac{15}{(2x + 1)^2}\right)\right) &= 0 \\
 0 &= 0
 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned}
 z_1(x) &= pe^{\int \omega dx} \\
 &= e^{\int -\frac{3}{2\left(x + \frac{1}{2}\right)} dx} \\
 &= \frac{1}{(2x + 1)^{3/2}}
 \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{-4x-2}{4\left(x + \frac{1}{2}\right)^2} dx} \\
 &= z_1 e^{\frac{\ln(2x+1)}{2}} \\
 &= z_1 \left(\sqrt{2x + 1}\right)
 \end{aligned}$$

Which simplifies to

$$y_1 = \frac{1}{2x+1}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4x-2}{4(x+\frac{1}{2})^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(2x+1)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{(2x+1)^4}{8} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{1}{2x+1} \right) + c_2 \left(\frac{1}{2x+1} \left(\frac{(2x+1)^4}{8} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$4 \left(x + \frac{1}{2} \right)^2 y'' + (-4x - 2) y' - 12y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1}{2x+1} + \frac{c_2(2x+1)^3}{8}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{1}{2x+1}$$

$$y_2 = \frac{(2x+1)^3}{8}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{1}{2x+1} & \frac{(2x+1)^3}{8} \\ \frac{d}{dx} \left(\frac{1}{2x+1} \right) & \frac{d}{dx} \left(\frac{(2x+1)^3}{8} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{1}{2x+1} & \frac{(2x+1)^3}{8} \\ -\frac{2}{(2x+1)^2} & \frac{3(2x+1)^2}{4} \end{vmatrix}$$

Therefore

$$W = \left(\frac{1}{2x+1} \right) \left(\frac{3(2x+1)^2}{4} \right) - \left(\frac{(2x+1)^3}{8} \right) \left(-\frac{2}{(2x+1)^2} \right)$$

Which simplifies to

$$W = 2x + 1$$

Which simplifies to

$$W = 2x + 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{3(2x+1)^3 x}{4}}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)} dx$$

Which simplifies to

$$u_1 = - \int \frac{3x}{4} dx$$

Hence

$$u_1 = -\frac{3x^2}{8}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{6x}{2x+1}}{4 \left(x + \frac{1}{2}\right)^2 (2x+1)} dx$$

Which simplifies to

$$u_2 = \int \frac{6x}{(2x+1)^4} dx$$

Hence

$$u_2 = \frac{1}{2(2x+1)^3} - \frac{3}{4(2x+1)^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\frac{3x^2}{8(2x+1)} + \frac{(2x+1)^3 \left(\frac{1}{2(2x+1)^3} - \frac{3}{4(2x+1)^2} \right)}{8}$$

Which simplifies to

$$y_p(x) = \frac{-24x^2 - 8x - 1}{32 + 64x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1}{2x+1} + \frac{c_2(2x+1)^3}{8} \right) + \left(\frac{-24x^2 - 8x - 1}{32 + 64x} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{-24x^2 - 8x - 1}{32 + 64x} + \frac{c_1}{2x + 1} + \frac{c_2(2x + 1)^3}{8}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 41

```
ode:=-12*y(x)-2*(2*x+1)*diff(y(x),x)+(2*x+1)^2*diff(diff(y(x),x),x) = 6*x;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{c_1}{2x + 1} + (2x + 1)^3 c_2 + \frac{-24x^2 - 8x - 1}{64x + 32}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

<- high order exact linear fully integrable successful

✓ **Mathematica.** Time used: 0.035 (sec). Leaf size: 41

```
ode=(2*x+1)^2*D[y[x],{x,2}]-2*(2*x+1)*D[y[x],x]-12*y[x]==6*x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{-24x^2 - 8x + 32c_1(2x + 1)^4 - 1 + 32c_2}{32(2x + 1)}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-6*x + (2*x + 1)**2*Derivative(y(x), (x, 2)) - (4*x + 2)*Derivative(y(x),
    x) - 12*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (4*x**2*Derivative(y(x),
    (x, 2)) + 4*x*Derivative(y(x), (x, 2)) - 6*x - 12*y(x) + Derivative(y(x), (x,
    2)))/(2*(2*x + 1)) cannot be solved by the factorable group method
```

2.13 Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Local contents

2.13.1 Problem 21	2141
2.13.2 Problem 22	2150
2.13.3 Problem 23	2168
2.13.4 Problem 24	2186
2.13.5 Problem 25	2195
2.13.6 Problem 26	2202
2.13.7 Problem 27	2211
2.13.8 Problem 28	2219
2.13.9 Problem 29	2228
2.13.10 Problem 31	2241
2.13.11 Problem 32	2260
2.13.12 Problem 33	2267
2.13.13 Problem 35	2288
2.13.14 Problem 36	2303
2.13.15 Problem 37	2313
2.13.16 Problem 38	2328

2.13.1 Problem 21

Local contents

Solved as second order ode using Kovacic algorithm	2141
✓ Maple	2147
✓ Mathematica	2149
✗ Sympy	2149

Internal problem ID [8031]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 21

Date solved : Thursday, November 27, 2025 at 07:32:34 AM

CAS classification : [_Laguerre]

Solved as second order ode using Kovacic algorithm

Time used: 0.164 (sec)

Solve

$$xy'' - (2 + x)y' + 2y = 0$$

Writing the ode as

$$xy'' + (-2 - x)y' + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= -2 - x \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{x^2 - 4x + 8}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= x^2 - 4x + 8 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{x^2 - 4x + 8}{4x^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.268: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Therefore

$$L = [1, 2]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{1}{4} - \frac{1}{x} + \frac{2}{x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = 2$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = 2 \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -1 \end{aligned}$$

Since the order of r at ∞ is $O_r(\infty) = 0$ then

$$v = \frac{-O_r(\infty)}{2} = \frac{0}{2} = 0$$

$[\sqrt{r}]_\infty$ is the sum of terms involving x^i for $0 \leq i \leq v$ in the Laurent series for $\sqrt{r}$ at ∞ . Therefore

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^v a_i x^i \\ &= \sum_{i=0}^0 a_i x^i \end{aligned} \tag{8}$$

Let a be the coefficient of $x^v = x^0$ in the above sum. The Laurent series of $\sqrt{r}$ at ∞ is

$$\sqrt{r} \approx \frac{1}{2} - \frac{1}{x} + \frac{1}{x^2} + \frac{2}{x^3} + \frac{3}{x^4} + \frac{2}{x^5} - \frac{6}{x^6} - \frac{28}{x^7} + \dots \tag{9}$$

Comparing Eq. (9) with Eq. (8) shows that

$$a = \frac{1}{2}$$

From Eq. (9) the sum up to $v = 0$ gives

$$\begin{aligned} [\sqrt{r}]_{\infty} &= \sum_{i=0}^0 a_i x^i \\ &= \frac{1}{2} \end{aligned} \quad (10)$$

Now we need to find b , where b be the coefficient of $x^{v-1} = x^{-1} = \frac{1}{x}$ in r minus the coefficient of same term but in $([\sqrt{r}]_{\infty})^2$ where $[\sqrt{r}]_{\infty}$ was found above in Eq (10). Hence

$$([\sqrt{r}]_{\infty})^2 = \frac{1}{4}$$

This shows that the coefficient of $\frac{1}{x}$ in the above is 0. Now we need to find the coefficient of $\frac{1}{x}$ in r . How this is done depends on if $v = 0$ or not. Since $v = 0$ then starting from $r = \frac{s}{t}$ and doing long division in the form

$$r = Q + \frac{R}{t}$$

Where Q is the quotient and R is the remainder. Then the coefficient of $\frac{1}{x}$ in r will be the coefficient in R of the term in x of degree of t minus one, divided by the leading coefficient in t . Doing long division gives

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{x^2 - 4x + 8}{4x^2} \\ &= Q + \frac{R}{4x^2} \\ &= \left(\frac{1}{4}\right) + \left(\frac{-4x + 8}{4x^2}\right) \\ &= \frac{1}{4} + \frac{-4x + 8}{4x^2} \end{aligned}$$

Since the degree of t is 2, then we see that the coefficient of the term x in the remainder R is -4 . Dividing this by leading coefficient in t which is 4 gives -1 . Now b can be found.

$$\begin{aligned} b &= (-1) - (0) \\ &= -1 \end{aligned}$$

Hence

$$\begin{aligned} [\sqrt{r}]_{\infty} &= \frac{1}{2} \\ \alpha_{\infty}^{+} &= \frac{1}{2} \left(\frac{b}{a} - v \right) = \frac{1}{2} \left(\frac{-1}{\frac{1}{2}} - 0 \right) = -1 \\ \alpha_{\infty}^{-} &= \frac{1}{2} \left(-\frac{b}{a} - v \right) = \frac{1}{2} \left(-\frac{-1}{\frac{1}{2}} - 0 \right) = 1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{x^2 - 4x + 8}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	2	-1

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
0	$\frac{1}{2}$	-1	1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^+ = -1$ then

$$\begin{aligned} d &= \alpha_\infty^+ - (\alpha_{c_1}^-) \\ &= -1 - (-1) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c) [\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty) [\sqrt{r}]_\infty$$

Substituting the above values in the above results in

$$\begin{aligned} \omega &= \left((-) [\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (+) [\sqrt{r}]_\infty \\ &= -\frac{1}{x} + \left(\frac{1}{2} \right) \\ &= \frac{1}{2} - \frac{1}{x} \\ &= \frac{x - 2}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$(0) + 2\left(\frac{1}{2} - \frac{1}{x}\right)(0) + \left(\left(\frac{1}{x^2}\right) + \left(\frac{1}{2} - \frac{1}{x}\right)^2 - \left(\frac{x^2 - 4x + 8}{4x^2}\right)\right) = 0$$

$$0 = 0$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int (\frac{1}{2} - \frac{1}{x}) dx} \\ &= \frac{e^{\frac{x}{2}}}{x} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2-x}{x} dx} \\ &= z_1 e^{\frac{x}{2} + \ln(x)} \\ &= z_1 \left(x e^{\frac{x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2-x}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{x+2\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{(x^2 + 2x + 2) e^{x+2\ln(x)} e^{-2x}}{x^2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(e^x) + c_2 \left(e^x \left(-\frac{(x^2 + 2x + 2)e^{x+2\ln(x)}e^{-2x}}{x^2} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^x + c_2(-x^2 - 2x - 2)$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 19

```
ode:=x*diff(diff(y(x),x),x)-(x+2)*diff(y(x),x)+2*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 e^x + c_2(x^2 + 2x + 2)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Reducible group (found another exponential solution)
<- Kovacic's algorithm successful
```

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) - (x + 2) \left(\frac{d}{dx} y(x) \right) + 2y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2} y(x)$
- Isolate 2nd derivative

- $$\frac{d^2}{dx^2}y(x) = -\frac{2y(x)}{x} + \frac{(x+2)\left(\frac{d}{dx}y(x)\right)}{x}$$
 - Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear
- $$\frac{d^2}{dx^2}y(x) - \frac{(x+2)\left(\frac{d}{dx}y(x)\right)}{x} + \frac{2y(x)}{x} = 0$$
- Check to see if $x_0 = 0$ is a regular singular point
 - Define functions

$$\left[P_2(x) = -\frac{x+2}{x}, P_3(x) = \frac{2}{x} \right]$$
 - $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -2$$
 - $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$
 - $x = 0$ is a regular singular point
Check to see if $x_0 = 0$ is a regular singular point
 $x_0 = 0$
- Multiply by denominators

$$x\left(\frac{d^2}{dx^2}y(x)\right) + (-x-2)\left(\frac{d}{dx}y(x)\right) + 2y(x) = 0$$
- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$
- Rewrite ODE with series expansions
 - Convert $x^m \cdot \left(\frac{d}{dx}y(x)\right)$ to series expansion for $m = 0..1$

$$x^m \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1+m}$$
 - Shift index using $k- > k+1-m$

$$x^m \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) x^{k+r}$$
 - Convert $x \cdot \left(\frac{d^2}{dx^2}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-1}$$
 - Shift index using $k- > k+1$

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r)(k+r) x^{k+r}$$
- Rewrite ODE with series expansions

$$a_0 r(-3+r) x^{-1+r} + \left(\sum_{k=0}^{\infty} (a_{k+1} (k+1+r)(k+r-2) - a_k (k+r-2)) x^{k+r} \right) = 0$$
- a_0 cannot be 0 by assumption, giving the indicial equation

$$r(-3+r) = 0$$
- Values of r that satisfy the indicial equation

$$r \in \{0, 3\}$$
- Each term in the series must be 0, giving the recursion relation

- $(k+r-2)(a_{k+1}(k+1+r)-a_k)=0$
 Recursion relation that defines series solution to ODE
 $a_{k+1} = \frac{a_k}{k+1+r}$
- Recursion relation for $r=0$
 $a_{k+1} = \frac{a_k}{k+1}$
- Solution for $r=0$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+1} = \frac{a_k}{k+1} \right]$
- Recursion relation for $r=3$
 $a_{k+1} = \frac{a_k}{k+4}$
- Solution for $r=3$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+3}, a_{k+1} = \frac{a_k}{k+4} \right]$
- Combine solutions and rename parameters
 $\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^k \right) + \left(\sum_{k=0}^{\infty} b_k x^{k+3} \right), a_{k+1} = \frac{a_k}{k+1}, b_{k+1} = \frac{b_k}{k+4} \right]$

✓ **Mathematica.** Time used: 0.129 (sec). Leaf size: 35

```
ode=x*D[y[x],{x,2}]- (x+2)*D[y[x],x]+2*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{x+1} \left(c_2 \int_1^x e^{-K[1]} K[1]^2 dK[1] + c_1 \right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), (x, 2)) - (x + 2)*Derivative(y(x), x) + 2*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

False

2.13.2 Problem 22

Local contents

Solved as second order ode using change of variable on y method 2 .	2150
Solved as second order ode using non constant coeff transformation on	
B method	2155
Solved as second order ode using Kovacic algorithm	2159
✓ Maple	2166
✓ Mathematica	2167
✗ Sympy	2167

Internal problem ID [8032]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 22

Date solved : Thursday, November 27, 2025 at 07:32:34 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using change of variable on y method 2

Time used: 0.233 (sec)

Solve

$$(x^2 + 1) y'' - 2y'x + 2y = 2$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2 + 1$, $B = -2x$, $C = 2$, $f(x) = 2$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$(x^2 + 1) y'' - 2y'x + 2y = 0$$

In normal form the ode

$$(x^2 + 1) y'' - 2y'x + 2y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{2x}{x^2 + 1}$$

$$q(x) = \frac{2}{x^2 + 1}$$

Applying change of variables on the dependent variable $y = v(x) x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p \right) v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q \right) v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{2n}{x^2 + 1} + \frac{2}{x^2 + 1} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$v''(x) + \left(\frac{2}{x} - \frac{2x}{x^2 + 1} \right) v'(x) = 0$$

$$v''(x) + \frac{2v'(x)}{x(x^2 + 1)} = 0 \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{2u(x)}{x(x^2 + 1)} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x(x^2 + 1)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{2}{x(x^2+1)} dx} \\ &= \frac{x^2}{x^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 1} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x^2 + 1} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2+1}$ gives the final solution

$$u(x) = \frac{(x^2 + 1) c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= c_1 \left(x - \frac{1}{x} \right) + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(c_1 \left(x - \frac{1}{x} \right) + c_2 \right) x \\ &= c_1 x^2 + c_2 x - c_1 \end{aligned}$$

Now the particular solution to this ODE is found

$$(x^2 + 1) y'' - 2y'x + 2y = 2$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2 - 1$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 - 1 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2 - 1) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 - 1 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2 - 1)(1)$$

Which simplifies to

$$W = x^2 + 1$$

Which simplifies to

$$W = x^2 + 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{2x^2 - 2}{(x^2 + 1)^2} dx$$

Which simplifies to

$$u_1 = - \int \frac{2x^2 - 2}{(x^2 + 1)^2} dx$$

Hence

$$u_1 = \frac{2x}{x^2 + 1}$$

And Eq. (3) becomes

$$u_2 = \int \frac{2x}{(x^2 + 1)^2} dx$$

Which simplifies to

$$u_2 = \int \frac{2x}{(x^2 + 1)^2} dx$$

Hence

$$u_2 = -\frac{1}{x^2 + 1}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{2x^2}{x^2 + 1} - \frac{x^2 - 1}{x^2 + 1}$$

Which simplifies to

$$y_p(x) = 1$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(c_1 \left(x - \frac{1}{x} \right) + c_2 \right) x \right) + (1) \end{aligned}$$

Summary of solutions found

$$y = 1 + \left(c_1 \left(x - \frac{1}{x} \right) + c_2 \right) x$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.273 (sec)

Solve

$$(x^2 + 1) y'' - 2y'x + 2y = 2$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \quad (1)$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= x^2 + 1 \\ B &= -2x \\ C &= 2 \\ F &= 2 \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x^2 + 1)(0) + (-2x)(-2) + (2)(-2x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-2x(x^2 + 1)v'' + (-4)v' = 0$$

Now by applying $v' = u$ the above becomes

$$(-2x^3 - 2x)u'(x) - 4u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x(x^2 + 1)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{2}{x(x^2 + 1)} dx} \\ &= \frac{x^2}{x^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 1} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x^2 + 1} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2 + 1}$ gives the final solution

$$u(x) = \frac{(x^2 + 1)c_1}{x^2}$$

The ode for v now becomes

$$v'(x) = \frac{(x^2 + 1) c_1}{x^2}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\int dv = \int \frac{(x^2 + 1) c_1}{x^2} dx$$

$$v(x) = c_1 \left(x - \frac{1}{x} \right) + c_2$$

$$v(x) = \frac{c_1(x^2 - 1)}{x} + c_2$$

Replacing $v(x)$ above by $-\frac{y}{2x}$, then the homogeneous solution is

$$\begin{aligned} y_h(x) &= Bv \\ &= -2c_1 x^2 - 2c_2 x + 2c_1 \end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = -2x^2 + 2$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & -2x^2 + 2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(-2x^2 + 2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & -2x^2 + 2 \\ 1 & -4x \end{vmatrix}$$

Therefore

$$W = (x)(-4x) - (-2x^2 + 2)(1)$$

Which simplifies to

$$W = -2x^2 - 2$$

Which simplifies to

$$W = -2x^2 - 2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{-4x^2 + 4}{(x^2 + 1)(-2x^2 - 2)} dx$$

Which simplifies to

$$u_1 = - \int \frac{2x^2 - 2}{(x^2 + 1)^2} dx$$

Hence

$$u_1 = \frac{2x}{x^2 + 1}$$

And Eq. (3) becomes

$$u_2 = \int \frac{2x}{(x^2 + 1)(-2x^2 - 2)} dx$$

Which simplifies to

$$u_2 = \int -\frac{x}{(x^2 + 1)^2} dx$$

Hence

$$u_2 = \frac{1}{2x^2 + 2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{2x^2}{x^2 + 1} + \frac{-2x^2 + 2}{2x^2 + 2}$$

Which simplifies to

$$y_p(x) = 1$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= (-2c_1 x^2 - 2c_2 x + 2c_1) + (1) \\ &= -2c_1 x^2 - 2c_2 x + 2c_1 + 1 \end{aligned}$$

Summary of solutions found

$$y = -2c_1 x^2 - 2c_2 x + 2c_1 + 1$$

Solved as second order ode using Kovacic algorithm

Time used: 0.369 (sec)

Solve

$$(x^2 + 1) y'' - 2y'x + 2y = 2$$

Writing the ode as

$$(x^2 + 1) y'' - 2y'x + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 + 1 \\ B &= -2x \\ C &= 2 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-3}{(x^2 + 1)^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -3 \\ t &= (x^2 + 1)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{3}{(x^2 + 1)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.270: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 4 - 0 \\ &= 4 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = (x^2 + 1)^2$. There is a pole at $x = i$ of order 2. There is a pole at $x = -i$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 4 then the necessary

conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 4 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4(x-i)^2} + \frac{3}{4(x+i)^2} + \frac{3i}{4(x-i)} - \frac{3i}{4(x+i)}$$

For the pole at $x = i$ let b be the coefficient of $\frac{1}{(x-i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

For the pole at $x = -i$ let b be the coefficient of $\frac{1}{(x+i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $4 > 2$ then

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= 0 \\ \alpha_\infty^- &= 1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{3}{(x^2+1)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
i	2	0	$\frac{3}{2}$	$-\frac{1}{2}$
$-i$	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
4	0	0	1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = 1$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^- + \alpha_{c_2}^+) \\ &= 1 - (1) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + \left((+)[\sqrt{r}]_{c_2} + \frac{\alpha_{c_2}^+}{x - c_2} \right) + (-)[\sqrt{r}]_\infty \\ &= -\frac{1}{2(x-i)} + \frac{3}{2(x+i)} + (-)(0) \\ &= -\frac{1}{2(x-i)} + \frac{3}{2(x+i)} \\ &= \frac{x-2i}{x^2+1} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$(0) + 2 \left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)} \right) (0) + \left(\left(\frac{1}{2(x-i)^2} - \frac{3}{2(x+i)^2} \right) + \left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)} \right)^2 - \left(-\frac{1}{x^2} \right) \right) = 0$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)} \right) dx} \\ &= \frac{(x^2 + 1)^{3/2}}{(ix + 1)^2} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2x}{x^2+1} dx} \\ &= z_1 e^{\frac{\ln(x^2+1)}{2}} \\ &= z_1 \left(\sqrt{x^2 + 1} \right) \end{aligned}$$

Which simplifies to

$$y_1 = \frac{(x^2 + 1)^2}{(ix + 1)^2}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x}{x^2+1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x^2+1)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{x}{(x+i)^2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{(x^2 + 1)^2}{(ix + 1)^2} \right) + c_2 \left(\frac{(x^2 + 1)^2}{(ix + 1)^2} \left(-\frac{x}{(x+i)^2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(x^2 + 1)y'' - 2y'x + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1(x^2 + 1)^2}{(ix + 1)^2} + \frac{c_2(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{(x^2 + 1)^2}{(ix + 1)^2}$$

$$y_2 = \frac{(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{(x^2+1)^2}{(ix+1)^2} & \frac{(x^2+1)^2 x}{(x-i)^2 (x+i)^2} \\ \frac{d}{dx} \left(\frac{(x^2+1)^2}{(ix+1)^2} \right) & \frac{d}{dx} \left(\frac{(x^2+1)^2 x}{(x-i)^2 (x+i)^2} \right) \end{vmatrix}$$

Which gives

$$W = \left| \frac{\frac{(x^2+1)^2}{(ix+1)^2}}{\frac{4(x^2+1)x}{(ix+1)^2} - \frac{2i(x^2+1)^2}{(ix+1)^3}} - \frac{\frac{(x^2+1)^2 x}{(x-i)^2(x+i)^2}}{\frac{4(x^2+1)x^2}{(x-i)^2(x+i)^2} + \frac{(x^2+1)^2}{(x-i)^2(x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^3(x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^2(x+i)^3}} \right|$$

Therefore

$$W = \left(\frac{(x^2+1)^2}{(ix+1)^2} \right) \left(\frac{4(x^2+1)x^2}{(x-i)^2(x+i)^2} + \frac{(x^2+1)^2}{(x-i)^2(x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^3(x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^2(x+i)^3} \right) - \left(\frac{(x^2+1)^2 x}{(x-i)^2(x+i)^2} \right) \left(\frac{4(x^2+1)x}{(ix+1)^2} - \frac{2i(x^2+1)^2}{(ix+1)^3} \right)$$

Which simplifies to

$$W = \frac{(x^2+1)^3 (ix^5 + 3x^4 - 2ix^3 + 2x^2 - 3ix - 1)}{(ix+1)^3 (-x+i)^3 (x+i)^3}$$

Which simplifies to

$$W = \frac{(x^2+1)^4 (-x^3 + 3ix^2 + 3x - i)}{(-ix-1)^6 (x+i)^3}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{2(x^2+1)^2 x}{(x-i)^2(x+i)^2}}{\frac{(x^2+1)^5 (-x^3 + 3ix^2 + 3x - i)}{(-ix-1)^6 (x+i)^3}} dx$$

Which simplifies to

$$u_1 = - \int \frac{2x(x+i)(x-i)^4}{(x^3 - 3ix^2 - 3x + i)(x^2+1)^3} dx$$

Hence

$$u_1 = -\frac{i}{2(x-i)} + \frac{i}{2x+2i}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{2(x^2+1)^2}{(ix+1)^2}}{\frac{(x^2+1)^5 (-x^3 + 3ix^2 + 3x - i)}{(-ix-1)^6 (x+i)^3}} dx$$

Which simplifies to

$$u_2 = \int \frac{2(x+i)^3(x-i)^4}{(x^2+1)^3(-x^3+3ix^2+3x-i)} dx$$

Hence

$$u_2 = \frac{2}{x-i}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\left(-\frac{i}{2(x-i)} + \frac{i}{2x+2i}\right)(x^2+1)^2}{(ix+1)^2} + \frac{2(x^2+1)^2 x}{(x-i)^3(x+i)^2}$$

Which simplifies to

$$y_p(x) = \frac{(x^2+1)^2(-x^3+3ix^2+3x-i)}{(-x+i)^5(x+i)^2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1(x^2+1)^2}{(ix+1)^2} + \frac{c_2(x^2+1)^2 x}{(x-i)^2(x+i)^2} \right) + \left(\frac{(x^2+1)^2(-x^3+3ix^2+3x-i)}{(-x+i)^5(x+i)^2} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{(x^2+1)^2(-x^3+3ix^2+3x-i)}{(-x+i)^5(x+i)^2} + \frac{c_1(x^2+1)^2}{(ix+1)^2} + \frac{c_2(x^2+1)^2 x}{(x-i)^2(x+i)^2}$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 17

```
ode:=2*y(x)-2*diff(y(x),x)*x+(x^2+1)*diff(diff(y(x),x),x) = 2;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x^2 + c_2 x - c_1 + 1$$

Maple trace

Methods for second order ODEs:

```

--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    <- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful

```

✓ **Mathematica.** Time used: 0.185 (sec). Leaf size: 113

```

ode=(1+x^2)*D[y[x],{x,2}]-2*x*D[y[x],x]+2*y[x]==2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow c_1 \sqrt{x^2 + 1} \exp \left(\int_1^x \frac{K[1] + 2i}{K[1]^2 + 1} dK[1] \right) + c_2 \sqrt{x^2 + 1} \exp \left(\int_1^x \frac{K[1] + 2i}{K[1]^2 + 1} dK[1] \right) \int_1^x \exp \left(-2 \int_1^{K[2]} \frac{K}{K^2 + 1} dK \right) dK[2]$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*Derivative(y(x), x) + (x**2 + 1)*Derivative(y(x), (x, 2)) + 2*y(x) - 2, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)

```

```

NotImplementedError : The given ODE Derivative(y(x), x) - (x**2*Derivative(y(x), (x, 2)) + 2*y(x) + Derivative(y(x), (x, 2)) - 2)/(2*x) cannot be solved by the factorable group method

```

2.13.3 Problem 23

Local contents

Solved as second order ode using change of variable on y method 2 .	2168
Solved as second order ode using non constant coeff transformation on	
B method	2173
Solved as second order ode using Kovacic algorithm	2177
✓ Maple	2184
✓ Mathematica	2185
✗ Sympy	2185

Internal problem ID [8033]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 23

Date solved : Thursday, November 27, 2025 at 07:32:35 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using change of variable on y method 2

Time used: 0.235 (sec)

Solve

$$(x^2 + 4) y'' - 2y'x + 2y = 8$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2 + 4$, $B = -2x$, $C = 2$, $f(x) = 8$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$(x^2 + 4) y'' - 2y'x + 2y = 0$$

In normal form the ode

$$(x^2 + 4) y'' - 2y'x + 2y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{2x}{x^2 + 4}$$

$$q(x) = \frac{2}{x^2 + 4}$$

Applying change of variables on the dependent variable $y = v(x) x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right) v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right) v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{2n}{x^2 + 4} + \frac{2}{x^2 + 4} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$v''(x) + \left(\frac{2}{x} - \frac{2x}{x^2 + 4}\right) v'(x) = 0$$

$$v''(x) + \frac{8v'(x)}{x(x^2 + 4)} = 0 \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{8u(x)}{x(x^2 + 4)} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{8}{x(x^2 + 4)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{8}{x(x^2+4)} dx} \\ &= \frac{x^2}{x^2 + 4} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 4} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x^2 + 4} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2+4}$ gives the final solution

$$u(x) = \frac{(x^2 + 4) c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= c_1 \left(x - \frac{4}{x} \right) + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(c_1 \left(x - \frac{4}{x} \right) + c_2 \right) x \\ &= c_1 x^2 + c_2 x - 4c_1 \end{aligned}$$

Now the particular solution to this ODE is found

$$(x^2 + 4) y'' - 2y'x + 2y = 8$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2 - 4$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 - 4 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2 - 4) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 - 4 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2 - 4)(1)$$

Which simplifies to

$$W = x^2 + 4$$

Which simplifies to

$$W = x^2 + 4$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{8x^2 - 32}{(x^2 + 4)^2} dx$$

Which simplifies to

$$u_1 = - \int \frac{8x^2 - 32}{(x^2 + 4)^2} dx$$

Hence

$$u_1 = \frac{8x}{x^2 + 4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{8x}{(x^2 + 4)^2} dx$$

Which simplifies to

$$u_2 = \int \frac{8x}{(x^2 + 4)^2} dx$$

Hence

$$u_2 = -\frac{4}{x^2 + 4}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{8x^2}{x^2 + 4} - \frac{4(x^2 - 4)}{x^2 + 4}$$

Which simplifies to

$$y_p(x) = 4$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(c_1 \left(x - \frac{4}{x} \right) + c_2 \right) x \right) + (4) \end{aligned}$$

Summary of solutions found

$$y = 4 + \left(c_1 \left(x - \frac{4}{x} \right) + c_2 \right) x$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.276 (sec)

Solve

$$(x^2 + 4) y'' - 2y'x + 2y = 8$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \tag{1}$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= x^2 + 4 \\ B &= -2x \\ C &= 2 \\ F &= 8 \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x^2 + 4)(0) + (-2x)(-2) + (2)(-2x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-2x(x^2 + 4)v'' + (-16)v' = 0$$

Now by applying $v' = u$ the above becomes

$$(-2x^3 - 8x)u'(x) - 16u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{8}{x(x^2 + 4)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{8}{x(x^2 + 4)} dx} \\ &= \frac{x^2}{x^2 + 4} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 4} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x^2 + 4} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2 + 4}$ gives the final solution

$$u(x) = \frac{(x^2 + 4)c_1}{x^2}$$

The ode for v now becomes

$$v'(x) = \frac{(x^2 + 4) c_1}{x^2}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\int dv = \int \frac{(x^2 + 4) c_1}{x^2} dx$$

$$v(x) = c_1 \left(x - \frac{4}{x} \right) + c_2$$

$$v(x) = \frac{c_1(x^2 - 4)}{x} + c_2$$

Replacing $v(x)$ above by $-\frac{y}{2x}$, then the homogeneous solution is

$$\begin{aligned} y_h(x) &= Bv \\ &= -2c_1 x^2 - 2c_2 x + 8c_1 \end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = -2x^2 + 8$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & -2x^2 + 8 \\ \frac{d}{dx}(x) & \frac{d}{dx}(-2x^2 + 8) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & -2x^2 + 8 \\ 1 & -4x \end{vmatrix}$$

Therefore

$$W = (x)(-4x) - (-2x^2 + 8)(1)$$

Which simplifies to

$$W = -2x^2 - 8$$

Which simplifies to

$$W = -2x^2 - 8$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{-16x^2 + 64}{(x^2 + 4)(-2x^2 - 8)} dx$$

Which simplifies to

$$u_1 = - \int \frac{8x^2 - 32}{(x^2 + 4)^2} dx$$

Hence

$$u_1 = \frac{8x}{x^2 + 4}$$

And Eq. (3) becomes

$$u_2 = \int \frac{8x}{(x^2 + 4)(-2x^2 - 8)} dx$$

Which simplifies to

$$u_2 = \int -\frac{4x}{(x^2 + 4)^2} dx$$

Hence

$$u_2 = \frac{2}{x^2 + 4}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{8x^2}{x^2 + 4} + \frac{-4x^2 + 16}{x^2 + 4}$$

Which simplifies to

$$y_p(x) = 4$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= (-2c_1 x^2 - 2c_2 x + 8c_1) + (4) \\ &= -2c_1 x^2 - 2c_2 x + 8c_1 + 4 \end{aligned}$$

Summary of solutions found

$$y = -2c_1 x^2 - 2c_2 x + 8c_1 + 4$$

Solved as second order ode using Kovacic algorithm

Time used: 0.332 (sec)

Solve

$$(x^2 + 4) y'' - 2y'x + 2y = 8$$

Writing the ode as

$$(x^2 + 4) y'' - 2y'x + 2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 + 4 \\ B &= -2x \\ C &= 2 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-12}{(x^2 + 4)^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -12 \\ t &= (x^2 + 4)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{12}{(x^2 + 4)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.271: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 4 - 0 \\ &= 4 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = (x^2 + 4)^2$. There is a pole at $x = 2i$ of order 2. There is a pole at $x = -2i$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 4 then the necessary

conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 4 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4(x-2i)^2} + \frac{3}{4(x+2i)^2} + \frac{3i}{8(x-2i)} - \frac{3i}{8(x+2i)}$$

For the pole at $x = 2i$ let b be the coefficient of $\frac{1}{(x-2i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

For the pole at $x = -2i$ let b be the coefficient of $\frac{1}{(x+2i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $4 > 2$ then

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= 0 \\ \alpha_\infty^- &= 1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{12}{(x^2 + 4)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
$2i$	2	0	$\frac{3}{2}$	$-\frac{1}{2}$
$-2i$	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
4	0	0	1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = 1$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^- + \alpha_{c_2}^+) \\ &= 1 - (1) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + \left((+)[\sqrt{r}]_{c_2} + \frac{\alpha_{c_2}^+}{x - c_2} \right) + (-)[\sqrt{r}]_\infty \\ &= -\frac{1}{2(x - 2i)} + \frac{3}{2(x + 2i)} + (-)(0) \\ &= -\frac{1}{2(x - 2i)} + \frac{3}{2(x + 2i)} \\ &= \frac{x - 4i}{x^2 + 4} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$(0) + 2 \left(-\frac{1}{2(x - 2i)} + \frac{3}{2(x + 2i)} \right) (0) + \left(\left(\frac{1}{2(x - 2i)^2} - \frac{3}{2(x + 2i)^2} \right) + \left(-\frac{1}{2(x - 2i)} + \frac{3}{2(x + 2i)} \right)^2 - \right)$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2(x-2i)} + \frac{3}{2(x+2i)} \right) dx} \\ &= \frac{(x^2 + 4)^{3/2}}{(ix + 2)^2} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2x}{x^2+4} dx} \\ &= z_1 e^{\frac{\ln(x^2+4)}{2}} \\ &= z_1 \left(\sqrt{x^2 + 4} \right) \end{aligned}$$

Which simplifies to

$$y_1 = \frac{(x^2 + 4)^2}{(ix + 2)^2}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x}{x^2+4} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x^2+4)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{x}{(x + 2i)^2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{(x^2 + 4)^2}{(ix + 2)^2} \right) + c_2 \left(\frac{(x^2 + 4)^2}{(ix + 2)^2} \left(-\frac{x}{(x + 2i)^2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(x^2 + 4)y'' - 2y'x + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1(x^2 + 4)^2}{(ix + 2)^2} + c_2x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{(x^2 + 4)^2}{(ix + 2)^2}$$

$$y_2 = x$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{(x^2+4)^2}{(ix+2)^2} & x \\ \frac{d}{dx} \left(\frac{(x^2+4)^2}{(ix+2)^2} \right) & \frac{d}{dx}(x) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{(x^2+4)^2}{(ix+2)^2} & x \\ \frac{4(x^2+4)x}{(ix+2)^2} - \frac{2i(x^2+4)^2}{(ix+2)^3} & 1 \end{vmatrix}$$

Therefore

$$W = \left(\frac{(x^2+4)^2}{(ix+2)^2} \right) (1) - (x) \left(\frac{4(x^2+4)x}{(ix+2)^2} - \frac{2i(x^2+4)^2}{(ix+2)^3} \right)$$

Which simplifies to

$$W = -\frac{(x^2+4)(ix^3+6x^2-12ix-8)}{(ix+2)^3}$$

Which simplifies to

$$W = -\frac{(x^2+4)(ix^3+6x^2-12ix-8)}{(ix+2)^3}$$

Therefore Eq. (2) becomes

$$u_1 = -\int \frac{8x}{-\frac{(x^2+4)^2(ix^3+6x^2-12ix-8)}{(ix+2)^3}} dx$$

Which simplifies to

$$u_1 = -\int -\frac{8x(ix+2)^3}{(x^2+4)^2(ix^3+6x^2-12ix-8)} dx$$

Hence

$$u_1 = \frac{i}{x+2i} - \frac{i}{x-2i}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{8(x^2+4)^2}{(ix+2)^2}}{-\frac{(x^2+4)^2(ix^3+6x^2-12ix-8)}{(ix+2)^3}} dx$$

Which simplifies to

$$u_2 = \int \frac{8x-16i}{-x^3+6ix^2+12x-8i} dx$$

Hence

$$u_2 = \frac{8}{x - 2i}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\left(\frac{i}{x+2i} - \frac{i}{x-2i}\right)(x^2+4)^2}{(ix+2)^2} + \frac{8x}{x-2i}$$

Which simplifies to

$$y_p(x) = \frac{-4x^4 + 16ix^3 + 64ix + 64}{(ix+2)^2(x^2+4)}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1(x^2+4)^2}{(ix+2)^2} + c_2x \right) + \left(\frac{-4x^4 + 16ix^3 + 64ix + 64}{(ix+2)^2(x^2+4)} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{-4x^4 + 16ix^3 + 64ix + 64}{(ix+2)^2(x^2+4)} + \frac{c_1(x^2+4)^2}{(ix+2)^2} + c_2x$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 17

```
ode:=(x^2+4)*diff(diff(y(x),x),x)-2*diff(y(x),x)*x+2*y(x) = 8;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x^2 + c_2 x - 4c_1 + 4$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
```

```

checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful

```

✓ **Mathematica.** Time used: 0.192 (sec). Leaf size: 113

```

ode=(x^2+4)*D[y[x],{x,2}]-2*x*D[y[x],x]+2*y[x]==8;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow c_1 \sqrt{x^2 + 4} \exp \left(\int_1^x \frac{K[1] + 4i}{K[1]^2 + 4} dK[1] \right) + c_2 \sqrt{x^2 + 4} \exp \left(\int_1^x \frac{K[1] + 4i}{K[1]^2 + 4} dK[1] \right) \int_1^x \exp \left(-2 \int_1^{K[2]} \frac{K}{K^2 + 4} dK \right) dK[2]$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*Derivative(y(x), x) + (x**2 + 4)*Derivative(y(x), (x, 2)) + 2*y(x) - 8, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)

```

```

NotImplementedError : The given ODE Derivative(y(x), x) - (x**2*Derivative(y(x), (x, 2)))/2 + y(x) + 2*Derivative(y(x), (x, 2)) - 4)/x cannot be solved by the factorable group method

```

2.13.4 Problem 24

Local contents

Solved as second order ode using Kovacic algorithm	2186
✓ Maple	2193
✓ Mathematica	2194
✗ Sympy	2194

Internal problem ID [8034]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 24

Date solved : Thursday, November 27, 2025 at 07:32:36 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order ode using Kovacic algorithm

Time used: 0.217 (sec)

Solve

$$(1+x)y'' - (2x+3)y' + (2+x)y = (x^2 + 2x + 1)e^{2x}$$

Writing the ode as

$$(1+x)y'' + (-2x-3)y' + (2+x)y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1+x \\ B &= -2x-3 \\ C &= 2+x \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{3}{4(1+x)^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 3 \\ t &= 4(1+x)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{3}{4(1+x)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.272: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4(1+x)^2$. There is a pole at $x = -1$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4(1+x)^2}$$

For the pole at $x = -1$ let b be the coefficient of $\frac{1}{(1+x)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{3}{4(1+x)^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{3}{4(1+x)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
-1	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= -\frac{1}{2(1+x)} + (-)(0) \\ &= -\frac{1}{2(1+x)} \\ &= -\frac{1}{2(1+x)} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(-\frac{1}{2(1+x)}\right)(0) + \left(\left(\frac{1}{2(1+x)^2}\right) + \left(-\frac{1}{2(1+x)}\right)^2 - \left(\frac{3}{4(1+x)^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int -\frac{1}{2(1+x)} dx} \\ &= \frac{1}{\sqrt{1+x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2x-3}{1+x} dx} \\ &= z_1 e^{x + \frac{\ln(1+x)}{2}} \\ &= z_1 \left(\sqrt{1+x} e^x \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x-3}{1+x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{2x+\ln(1+x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{x(2+x) e^{2x+\ln(1+x)} e^{-2x}}{2x+2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^x) + c_2 \left(e^x \left(\frac{x(2+x) e^{2x+\ln(1+x)} e^{-2x}}{2x+2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(1+x)y'' + (-2x-3)y' + (2+x)y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^x + \frac{c_2 e^x x(2+x)}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^x$$

$$y_2 = \frac{e^x x(2+x)}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^x & \frac{e^x x(2+x)}{2} \\ \frac{d}{dx}(e^x) & \frac{d}{dx}\left(\frac{e^x x(2+x)}{2}\right) \end{vmatrix}$$

Which gives

$$W = \left| \begin{array}{c} e^x \\ e^x \frac{e^x x(2+x)}{2} + \frac{e^x x(2+x)}{2} + \frac{x e^x}{2} \end{array} \right|$$

Therefore

$$W = (e^x) \left(\frac{e^x x(2+x)}{2} + \frac{e^x(2+x)}{2} + \frac{x e^x}{2} \right) - \left(\frac{e^x x(2+x)}{2} \right) (e^x)$$

Which simplifies to

$$W = x e^{2x} + e^{2x}$$

Which simplifies to

$$W = (1+x) e^{2x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^x x(2+x)(1+x)^2 e^{2x}}{2}}{(1+x)^2 e^{2x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{e^x x(2+x)}{2} dx$$

Hence

$$u_1 = - \frac{e^x x^2}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^x (1+x)^2 e^{2x}}{(1+x)^2 e^{2x}} dx$$

Which simplifies to

$$u_2 = \int e^x dx$$

Hence

$$u_2 = e^x$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = - \frac{x^2 e^{2x}}{2} + \frac{e^{2x} x(2+x)}{2}$$

Which simplifies to

$$y_p(x) = x e^{2x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^x + \frac{c_2 e^x x(2+x)}{2} \right) + (x e^{2x}) \end{aligned}$$

Summary of solutions found

$$y = x e^{2x} + c_1 e^x + \frac{c_2 e^x x(2+x)}{2}$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 22

```
ode:=(1+x)*diff(diff(y(x),x),x)-(2*x+3)*diff(y(x),x)+(x+2)*y(x) = (x^2+2*x+1)*exp
(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^x (c_1 x^2 + x e^x + 2c_1 x + c_2)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.434 (sec). Leaf size: 161

```
ode=(x+1)*D[y[x],{x,2}]-(2*x+3)*D[y[x],x]+(x+2)*y[x]==(x^2+2*x+1)*Exp[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{\exp\left(-\frac{1}{2} \int_1^x \left(-2 - \frac{1}{K[1]+1}\right) dK[1]\right) \left((x+2)x \int_1^x \exp\left(\frac{1}{2} \left(4K[3] + \int_1^{K[3]} \left(-2 - \frac{1}{K[1]+1}\right) dK[1]\right)\right) \sqrt{K}\right)}{\sqrt{K}}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x + 1)*Derivative(y(x), (x, 2)) + (x + 2)*y(x) - (2*x + 3)*Derivative(y
(x), x) - (x**2 + 2*x + 1)*exp(2*x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (-x**2*exp(2*x) + x*y(x)
) - 2*x*exp(2*x) + x*Derivative(y(x), (x, 2)) + 2*y(x) - exp(2*x) + Derivative
(y(x), (x, 2)))/(2*x + 3) cannot be solved by the factorable group method
```

2.13.5 Problem 25

Local contents

Solved as second order ode using change of variable on y method 1	2195
Solved as second order ode using Kovacic algorithm	2197
✓ Maple	2200
✓ Mathematica	2201
✗ Sympy	2201

Internal problem ID [8035]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 25

Date solved : Thursday, November 27, 2025 at 07:32:37 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using change of variable on y method 1

Time used: 0.117 (sec)

Solve

$$y'' - 2 \tan(x) y' - 10y = 0$$

In normal form the given ode is written as

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -2 \tan(x)$$

$$q(x) = -10$$

Calculating the Liouville ode invariant Q given by

$$\begin{aligned}
 Q &= q - \frac{p'}{2} - \frac{p^2}{4} \\
 &= -10 - \frac{(-2 \tan(x))'}{2} - \frac{(-2 \tan(x))^2}{4} \\
 &= -10 - \frac{(-2 - 2 \tan(x)^2)}{2} - \frac{(4 \tan(x)^2)}{4} \\
 &= -10 - (-1 - \tan(x)^2) - \tan(x)^2 \\
 &= -9
 \end{aligned}$$

Since the Liouville ode invariant does not depend on the independent variable x then the transformation

$$y = v(x) z(x) \quad (3)$$

is used to change the original ode to a constant coefficients ode in v . In (3) the term $z(x)$ is given by

$$\begin{aligned} z(x) &= e^{-\int \frac{p(x)}{2} dx} \\ &= e^{-\int \frac{-2 \tan(x)}{2} dx} \\ &= \sec(x) \end{aligned} \quad (5)$$

Hence (3) becomes

$$y = v(x) \sec(x) \quad (4)$$

Applying this change of variable to the original ode results in

$$\sec(x) (v''(x) - 9v(x)) = 0$$

Which is now solved for $v(x)$.

The above ode can be simplified to

$$v''(x) - 9v(x) = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Av''(x) + Bv'(x) + Cv(x) = 0$$

Where in the above $A = 1, B = 0, C = -9$. Let the solution be $v(x) = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} - 9e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 - 9 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = -9$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(-9)} \\ &= \pm 3 \end{aligned}$$

Hence

$$\lambda_1 = +3$$

$$\lambda_2 = -3$$

Which simplifies to

$$\lambda_1 = 3$$

$$\lambda_2 = -3$$

Since roots are distinct, then the solution is

$$v(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$v(x) = c_1 e^{(3)x} + c_2 e^{(-3)x}$$

Or

$$v(x) = c_1 e^{3x} + c_2 e^{-3x}$$

Now that $v(x)$ is known, then

$$\begin{aligned} y &= v(x) z(x) \\ &= (c_1 e^{3x} + c_2 e^{-3x}) (z(x)) \end{aligned} \tag{7}$$

But from (5)

$$z(x) = \sec(x)$$

Hence (7) becomes

$$y = (c_1 e^{3x} + c_2 e^{-3x}) \sec(x)$$

Summary of solutions found

$$y = (c_1 e^{3x} + c_2 e^{-3x}) \sec(x)$$

Solved as second order ode using Kovacic algorithm

Time used: 0.081 (sec)

Solve

$$y'' - 2 \tan(x) y' - 10y = 0$$

Writing the ode as

$$y'' - 2 \tan(x) y' - 10y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1 \\ B &= -2 \tan(x) \\ C &= -10 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{9}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 9 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = 9z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.273: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\
 &= 0 - 0 \\
 &= 0
 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = 9$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = e^{-3x}$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{-2 \tan(x)}{1} dx} \\
 &= z_1 e^{-\ln(\cos(x))} \\
 &= z_1 (\sec(x))
 \end{aligned}$$

Which simplifies to

$$y_1 = \sec(x) e^{-3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2 \tan(x)}{1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-2 \ln(\cos(x))}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{6x}}{6} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (\sec(x) e^{-3x}) + c_2 \left(\sec(x) e^{-3x} \left(\frac{e^{6x}}{6} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 \sec(x) e^{-3x} + \frac{c_2 \sec(x) e^{3x}}{6}$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 20

```
ode:=diff(diff(y(x),x),x)-2*tan(x)*diff(y(x),x)-10*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \sec(x) (c_1 \sinh(3x) + c_2 \cosh(3x))$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Reducible group (found another exponential solution)
<- Kovacic's algorithm successful
```

✓ **Mathematica.** Time used: 0.039 (sec). Leaf size: 29

```
ode=D[y[x],{x,2}]-2*Tan[x]*D[y[x],x]-10*y[x]==0;  
ic={};  
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{6}e^{-3x}(c_2e^{6x} + 6c_1)\sec(x)$$

✗ **Sympy**

```
from sympy import *  
x = symbols("x")  
y = Function("y")  
ode = Eq(-10*y(x) - 2*tan(x)*Derivative(y(x), x) + Derivative(y(x), (x, 2)),0)  
ics = {}  
dsolve(ode,func=y(x),ics=ics)
```

False

2.13.6 Problem 26

Local contents

Solved as second order ode using Kovacic algorithm	2202
✓ Maple	2209
✓ Mathematica	2210
✗ Sympy	2210

Internal problem ID [8036]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 26

Date solved : Thursday, November 27, 2025 at 07:32:37 AM

CAS classification : [[_2nd_order, _linear, _nonhomogeneous]]

Solved as second order ode using Kovacic algorithm

Time used: 0.234 (sec)

Solve

$$x^2 y'' - x(2x + 3) y' + (x^2 + 3x + 3) y = (-x^2 + 6) e^x$$

Writing the ode as

$$x^2 y'' + (-2x^2 - 3x) y' + (x^2 + 3x + 3) y = 0 \quad (1)$$

$$A y'' + B y' + C y = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 \\ B &= -2x^2 - 3x \\ C &= x^2 + 3x + 3 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = y e^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = r z(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{3}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 3 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{3}{4x^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.274: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{3}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{3}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= -\frac{1}{2x} + (-)(0) \\ &= -\frac{1}{2x} \\ &= -\frac{1}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(-\frac{1}{2x}\right)(0) + \left(\left(\frac{1}{2x^2}\right) + \left(-\frac{1}{2x}\right)^2 - \left(\frac{3}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int -\frac{1}{2x} dx} \\ &= \frac{1}{\sqrt{x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-2x^2-3x}{x^2} dx} \\ &= z_1 e^{x + \frac{3 \ln(x)}{2}} \\ &= z_1 (x^{3/2} e^x) \end{aligned}$$

Which simplifies to

$$y_1 = x e^x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x^2-3x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{2x+3 \ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{e^{2x+3 \ln(x)} e^{-2x}}{2x} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (x e^x) + c_2 \left(x e^x \left(\frac{e^{2x+3 \ln(x)} e^{-2x}}{2x} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2y'' + (-2x^2 - 3x)y' + (x^2 + 3x + 3)y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1x e^x + \frac{c_2 e^x x^3}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x e^x$$

$$y_2 = \frac{e^x x^3}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x e^x & \frac{e^x x^3}{2} \\ \frac{d}{dx}(x e^x) & \frac{d}{dx}\left(\frac{e^x x^3}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x e^x & \frac{e^x x^3}{2} \\ e^x + x e^x & \frac{e^x x^3}{2} + \frac{3 e^x x^2}{2} \end{vmatrix}$$

Therefore

$$W = (x e^x) \left(\frac{e^x x^3}{2} + \frac{3 e^x x^2}{2} \right) - \left(\frac{e^x x^3}{2} \right) (e^x + x e^x)$$

Which simplifies to

$$W = x^3 e^{2x}$$

Which simplifies to

$$W = x^3 e^{2x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{e^{2x} x^3 (-x^2 + 6)}{2}}{x^5 e^{2x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{-x^2 + 6}{2x^2} dx$$

Hence

$$u_1 = \frac{x}{2} + \frac{3}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x e^{2x} (-x^2 + 6)}{x^5 e^{2x}} dx$$

Which simplifies to

$$u_2 = \int \frac{-x^2 + 6}{x^4} dx$$

Hence

$$u_2 = -\frac{2}{x^3} + \frac{1}{x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{x}{2} + \frac{3}{x} \right) x e^x + \frac{e^x x^3 \left(-\frac{2}{x^3} + \frac{1}{x} \right)}{2}$$

Which simplifies to

$$y_p(x) = e^x(x^2 + 2)$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 x e^x + \frac{c_2 e^x x^3}{2} \right) + (e^x(x^2 + 2)) \end{aligned}$$

Summary of solutions found

$$y = e^x(x^2 + 2) + c_1 x e^x + \frac{c_2 e^x x^3}{2}$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 20

```
ode:=x^2*diff(diff(y(x),x),x)-x*(2*x+3)*diff(y(x),x)+(x^2+3*x+3)*y(x) = (-x^2+6)*
exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^x(c_1 x^3 + c_2 x + x^2 + 2)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.032 (sec). Leaf size: 30

```
ode=x^2*D[y[x],{x,2}]-x*(2*x+3)*D[y[x],x]+(x^2+3*x+3)*y[x]==(6-x^2)*Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}e^x(c_2x^3 + 2x^2 + 2c_1x + 4)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) - x*(2*x + 3)*Derivative(y(x), x) - (6 - x
**2)*exp(x) + (x**2 + 3*x + 3)*y(x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (x**2*y(x) + x**2*exp(x)
) + x**2*Derivative(y(x), (x, 2)) + 3*x*y(x) + 3*y(x) - 6*exp(x))/(x*(2*x + 3)
) cannot be solved by the factorable group method
```

2.13.7 Problem 27

Local contents

Solved as second order ode using Kovacic algorithm	2211
✓ Maple	2216
✓ Mathematica	2218
✗ Sympy	2218

Internal problem ID [8037]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 27

Date solved : Thursday, November 27, 2025 at 07:32:38 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using Kovacic algorithm

Time used: 0.128 (sec)

Solve

$$4x^2y'' + 4y'x^3 + (x^2 + 1)^2y = 0$$

Writing the ode as

$$4x^2y'' + 4y'x^3 + (x^2 + 1)^2y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 4x^2 \\ B &= 4x^3 \\ C &= (x^2 + 1)^2 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-1}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -1 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{1}{4x^2}\right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.275: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -\frac{1}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = -\frac{1}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{1}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{1}{2}$	$\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{1}{2}$	$\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = \frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= \frac{1}{2} - \left(\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= \frac{1}{2x} + (-)(0) \\ &= \frac{1}{2x} \\ &= \frac{1}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(\frac{1}{2x}\right)(0) + \left(\left(-\frac{1}{2x^2}\right) + \left(\frac{1}{2x}\right)^2 - \left(-\frac{1}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \frac{1}{2x} dx} \\ &= \sqrt{x} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{4x^3}{4x^2} dx} \\ &= z_1 e^{-\frac{x^2}{4}} \\ &= z_1 \left(e^{-\frac{x^2}{4}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-\frac{x^2}{4}} \sqrt{x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{4x^3}{4x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-\frac{x^2}{2}}}{(y_1)^2} dx \\ &= y_1 (\ln(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(e^{-\frac{x^2}{4}} \sqrt{x} \right) + c_2 \left(e^{-\frac{x^2}{4}} \sqrt{x} (\ln(x)) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^{-\frac{x^2}{4}} \sqrt{x} + c_2 e^{-\frac{x^2}{4}} \sqrt{x} \ln(x)$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 20

```
ode:=4*x^2*diff(diff(y(x),x),x)+4*x^3*diff(y(x),x)+(x^2+1)^2*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-\frac{x^2}{4}} \sqrt{x} (\ln(x) c_2 + c_1)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Group is reducible, not completely reducible
<- Kovacic's algorithm successful
```

Maple step by step

Let's solve

$$4x^2 \left(\frac{d^2}{dx^2} y(x) \right) + 4x^3 \left(\frac{d}{dx} y(x) \right) + (x^2 + 1)^2 y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{(x^2+1)^2 y(x)}{4x^2} - x \left(\frac{d}{dx} y(x) \right)$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + x \left(\frac{d}{dx} y(x) \right) + \frac{(x^2+1)^2 y(x)}{4x^2} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- o Define functions

$$\left[P_2(x) = x, P_3(x) = \frac{(x^2+1)^2}{4x^2} \right]$$

- o $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = 0$$

- o $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = \frac{1}{4}$$

- $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$4x^2 \left(\frac{d^2}{dx^2} y(x) \right) + 4x^3 \left(\frac{d}{dx} y(x) \right) + (x^2 + 1)^2 y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $x^m \cdot y(x)$ to series expansion for $m = 0..4$

$$x^m \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+m}$$

- Shift index using $k \rightarrow k - m$

$$x^m \cdot y(x) = \sum_{k=m}^{\infty} a_{k-m} x^{k+r}$$

- Convert $x^3 \cdot \left(\frac{d}{dx} y(x) \right)$ to series expansion

$$x^3 \cdot \left(\frac{d}{dx} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r+2}$$

- Shift index using $k \rightarrow k - 2$

$$x^3 \cdot \left(\frac{d}{dx} y(x) \right) = \sum_{k=2}^{\infty} a_{k-2} (k-2+r) x^{k+r}$$

- Convert $x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion

$$x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0(-1+2r)^2 x^r + a_1(1+2r)^2 x^{1+r} + (a_2(3+2r)^2 + 2a_0(1+2r)) x^{2+r} + (a_3(5+2r)^2 + 2a_1(3+2r)) x^{3+r} + \dots = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$(-1+2r)^2 = 0$$

- Values of r that satisfy the indicial equation

$$r = \frac{1}{2}$$

- The coefficients of each power of x must be 0

$$[a_1(1+2r)^2 = 0, a_2(3+2r)^2 + 2a_0(1+2r) = 0, a_3(5+2r)^2 + 2a_1(3+2r) = 0]$$

- Solve for the dependent coefficient(s)

$$\left\{ a_1 = 0, a_2 = -\frac{2a_0(1+2r)}{4r^2+12r+9}, a_3 = 0 \right\}$$

- Each term in the series must be 0, giving the recursion relation

$$a_k(2k+2r-1)^2 + (4k+4r-6) a_{k-2} + a_{k-4} = 0$$

- Shift index using $k \rightarrow k+4$

$$a_{k+4}(2k+7+2r)^2 + (4k+10+4r) a_{k+2} + a_k = 0$$

- Recursion relation that defines series solution to ODE

$$a_{k+4} = -\frac{4ka_{k+2} + 4ra_{k+2} + a_k + 10a_{k+2}}{(2k+7+2r)^2}$$

- Recursion relation for $r = \frac{1}{2}$

$$a_{k+4} = -\frac{4ka_{k+2} + a_k + 12a_{k+2}}{(2k+8)^2}$$
- Solution for $r = \frac{1}{2}$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+\frac{1}{2}}, a_{k+4} = -\frac{4ka_{k+2} + a_k + 12a_{k+2}}{(2k+8)^2}, a_1 = 0, a_2 = -\frac{a_0}{4}, a_3 = 0 \right]$$

✓ **Mathematica.** Time used: 0.024 (sec). Leaf size: 28

```
ode=4*x^2*D[y[x],{x,2}]+4*x^3*D[y[x],x]+(x^2+1)^2*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{-\frac{x^2}{4}} \sqrt{x}(c_2 \log(x) + c_1)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x**3*Derivative(y(x), x) + 4*x**2*Derivative(y(x), (x, 2)) + (x**2 +
1)**2*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

False

2.13.8 Problem 28

Local contents

Solved as second order ode using Kovacic algorithm	2219
✓ Maple	2226
✓ Mathematica	2227
✗ Sympy	2227

Internal problem ID [8038]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 28

Date solved : Thursday, November 27, 2025 at 07:32:38 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order ode using Kovacic algorithm

Time used: 0.276 (sec)

Solve

$$x^2 y'' + (-4x^2 + x) y' + (4x^2 - 2x + 1) y = (x^2 - x + 1) e^x$$

Writing the ode as

$$x^2 y'' + (-4x^2 + x) y' + (4x^2 - 2x + 1) y = 0 \quad (1)$$

$$A y'' + B y' + C y = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 \\ B &= -4x^2 + x \\ C &= 4x^2 - 2x + 1 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = y e^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = r z(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-5}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -5 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{5}{4x^2}\right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.277: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -\frac{5}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{5}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} + i \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} - i \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = -\frac{5}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = -\frac{5}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{1}{2} + i \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = \frac{1}{2} - i \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{5}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{1}{2} + i$	$\frac{1}{2} - i$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{1}{2} + i$	$\frac{1}{2} - i$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = \frac{1}{2} - i$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= \frac{1}{2} - i - \left(\frac{1}{2} - i \right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= \frac{\frac{1}{2} - i}{x} + (-)(0) \\ &= \frac{\frac{1}{2} - i}{x} \\ &= \frac{\frac{1}{2} - i}{x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2 \left(\frac{\frac{1}{2} - i}{x} \right) (0) + \left(\left(\frac{-\frac{1}{2} + i}{x^2} \right) + \left(\frac{\frac{1}{2} - i}{x} \right)^2 - \left(-\frac{5}{4x^2} \right) \right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \frac{\frac{1}{2}-i}{x} dx} \\ &= x^{\frac{1}{2}-i} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{\frac{1}{2} - 4x^2 + x}{x^2} dx} \\ &= z_1 e^{2x - \frac{\ln(x)}{2}} \\ &= z_1 \left(\frac{e^{2x}}{\sqrt{x}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = x^{-i} e^{2x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-4x^2+x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4x-\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{ix e^{4x-\ln(x)} x^{2i} e^{-4x}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (x^{-i} e^{2x}) + c_2 \left(x^{-i} e^{2x} \left(-\frac{ix e^{4x-\ln(x)} x^{2i} e^{-4x}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^2 y'' + (-4x^2 + x) y' + (4x^2 - 2x + 1) y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 x^{-i} e^{2x} - \frac{ic_2 x^i e^{2x}}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x^{-i} e^{2x}$$

$$y_2 = -\frac{ix^i e^{2x}}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = -\int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x^{-i} e^{2x} & -\frac{ix^i e^{2x}}{2} \\ \frac{d}{dx}(x^{-i} e^{2x}) & \frac{d}{dx}\left(-\frac{ix^i e^{2x}}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x^{-i}e^{2x} & -\frac{ix^ie^{2x}}{2} \\ -\frac{ix^{-i}e^{2x}}{x} + 2x^{-i}e^{2x} & \frac{x^ie^{2x}}{2x} - ix^ie^{2x} \end{vmatrix}$$

Therefore

$$W = (x^{-i}e^{2x}) \left(\frac{x^ie^{2x}}{2x} - ix^ie^{2x} \right) - \left(-\frac{ix^ie^{2x}}{2} \right) \left(-\frac{ix^{-i}e^{2x}}{x} + 2x^{-i}e^{2x} \right)$$

Which simplifies to

$$W = \frac{x^{-i}e^{4x}x^i}{x}$$

Which simplifies to

$$W = \frac{e^{4x}}{x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{-\frac{ix^ie^{2x}(x^2-x+1)e^x}{2}}{e^{4x}x} dx$$

Which simplifies to

$$u_1 = - \int -\frac{ix^{-1+i}e^{-x}(x^2-x+1)}{2} dx$$

Hence

$$u_1 = -\frac{ix^i(x+i)e^{-x}}{2}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^{-i}e^{2x}(x^2-x+1)e^x}{e^{4x}x} dx$$

Which simplifies to

$$u_2 = \int x^{-1-i}e^{-x}(x^2-x+1) dx$$

Hence

$$u_2 = x^{-i}(-x+i)e^{-x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = -\frac{ix^i(x+i)e^{-x}x^{-i}e^{2x}}{2} - \frac{ix^ie^{2x}x^{-i}(-x+i)e^{-x}}{2}$$

Which simplifies to

$$y_p(x) = e^x$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 x^{-i} e^{2x} - \frac{ic_2 x^i e^{2x}}{2} \right) + (e^x) \end{aligned}$$

Summary of solutions found

$$y = e^x + c_1 x^{-i} e^{2x} - \frac{ic_2 x^i e^{2x}}{2}$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 27

```
ode:=x^2*diff(diff(y(x),x),x)+(-4*x^2+x)*diff(y(x),x)+(4*x^2-2*x+1)*y(x) = (x^2-x
+1)*exp(x);
dsolve(ode,y(x), singsol=all);
```

$$y = e^{2x} x^i c_2 + e^{2x} x^{-i} c_1 + e^x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
```

```

A Liouvillian solution exists
Group is reducible or imprimitive
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful

```

✓ **Mathematica.** Time used: 0.159 (sec). Leaf size: 104

```

ode=x^2*D[y[x],{x,2}]+(x-4*x^2)*D[y[x],x]+(1-2*x+4*x^2)*y[x]==(x^2-x+1)*Exp[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow \frac{1}{2}e^{2x}x^{-i}\left(ix^{2i}\Gamma(-i,x) - ix^{2i}\Gamma(1-i,x) + ix^{2i}\Gamma(2-i,x) - ic_2x^{2i} - i\Gamma(i,x) + i\Gamma(1+i,x) - i\Gamma(2+i,x)\right)$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) + (-4*x**2 + x)*Derivative(y(x), x) - (x
    **2 - x + 1)*exp(x) + (4*x**2 - 2*x + 1)*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)

```

```

NotImplementedError : The given ODE Derivative(y(x), x) - (4*x**2*y(x) - x**2*exp
(x) + x**2*Derivative(y(x), (x, 2)) - 2*x*y(x) + x*exp(x) + y(x) - exp(x))/(x
*(4*x - 1)) cannot be solved by the factorable group method

```

2.13.9 Problem 29

Local contents

Solved as second order ode using change of variable on x method 2 .	2228
Solved as second order Bessel ode	2231
Solved as second order ode using Kovacic algorithm	2232
✓ Maple	2238
✓ Mathematica	2240
✓ Sympy	2240

Internal problem ID [8039]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 29

Date solved : Thursday, November 27, 2025 at 07:32:39 AM

CAS classification :

[[_Emden, _Fowler], [_2nd_order, _linear, '_with_symmetry_[0,F(x)]']]

Solved as second order ode using change of variable on x method 2

Time used: 0.123 (sec)

Solve

$$xy'' - y' + 4x^3y = 0$$

In normal form the ode

$$xy'' - y' + 4x^3y = 0 \quad (1)$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \quad (2)$$

Where

$$p(x) = -\frac{1}{x}$$

$$q(x) = 4x^2$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x) \tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x) \tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x) dx} dx \\ &= \int e^{-\int -\frac{1}{x} dx} dx \\ &= \int e^{\ln(x)} dx \\ &= \int x dx \\ &= \frac{x^2}{2} \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{4x^2}{x^2} \\ &= 4 \end{aligned} \quad (7)$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2} y(\tau) + q_1 y(\tau) &= 0 \\ \frac{d^2}{d\tau^2} y(\tau) + 4y(\tau) &= 0 \end{aligned}$$

The above ode is now solved for $y(\tau)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(\tau) + By'(\tau) + Cy(\tau) = 0$$

Where in the above $A = 1, B = 0, C = 4$. Let the solution be $y(\tau) = e^{\lambda\tau}$. Substituting this into the ODE gives

$$\lambda^2 e^{\tau\lambda} + 4e^{\tau\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda\tau}$ gives

$$\lambda^2 + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(4)} \\ &= \pm 2i \end{aligned}$$

Hence

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

Which simplifies to

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 2$. Therefore the final solution, when using Euler relation, can be written as

$$y(\tau) = e^{\alpha\tau} (c_1 \cos(\beta\tau) + c_2 \sin(\beta\tau))$$

Which becomes

$$y(\tau) = e^0 (c_1 \cos(2\tau) + c_2 \sin(2\tau))$$

Or

$$y(\tau) = c_1 \cos(2\tau) + c_2 \sin(2\tau)$$

The above solution is now transformed back to y using (6) which results in

$$y = c_1 \cos(x^2) + c_2 \sin(x^2)$$

Summary of solutions found

$$y = c_1 \cos(x^2) + c_2 \sin(x^2)$$

Solved as second order Bessel ode

Time used: 0.060 (sec)

Solve

$$xy'' - y' + 4x^3y = 0$$

Writing the ode as

$$x^2y'' - y'x + 4yx^4 = 0 \quad (1)$$

Bessel ode has the form

$$x^2y'' + y'x + (-n^2 + x^2)y = 0 \quad (2)$$

The generalized form of Bessel ode is given by Bowman (1958) as the following

$$x^2y'' + (1 - 2\alpha)xy' + (\beta^2\gamma^2x^{2\gamma} - n^2\gamma^2 + \alpha^2)y = 0 \quad (3)$$

With the standard solution

$$y = x^\alpha(c_3 \text{BesselJ}(n, \beta x^\gamma) + c_4 \text{BesselY}(n, \beta x^\gamma)) \quad (4)$$

Comparing (3) to (1) and solving for α, β, n, γ gives

$$\alpha = 1$$

$$\beta = 1$$

$$n = \frac{1}{2}$$

$$\gamma = 2$$

Substituting all the above into (4) gives the solution as

$$y = \frac{c_3x\sqrt{2} \sin(x^2)}{\sqrt{\pi} \sqrt{x^2}} - \frac{c_4x\sqrt{2} \cos(x^2)}{\sqrt{\pi} \sqrt{x^2}}$$

Summary of solutions found

$$y = \frac{c_3x\sqrt{2} \sin(x^2)}{\sqrt{\pi} \sqrt{x^2}} - \frac{c_4x\sqrt{2} \cos(x^2)}{\sqrt{\pi} \sqrt{x^2}}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.199 (sec)

Solve

$$xy'' - y' + 4x^3y = 0$$

Writing the ode as

$$xy'' - y' + 4x^3y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= -1 \\ C &= 4x^3 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-16x^4 + 3}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -16x^4 + 3 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{-16x^4 + 3}{4x^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.278: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 4 \\ &= -2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is -2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Therefore

$$L = [1, 2]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -4x^2 + \frac{3}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r

given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $O_r(\infty) = -2$ then

$$v = \frac{-O_r(\infty)}{2} = \frac{2}{2} = 1$$

$[\sqrt{r}]_\infty$ is the sum of terms involving x^i for $0 \leq i \leq v$ in the Laurent series for $\sqrt{r}$ at ∞ . Therefore

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^v a_i x^i \\ &= \sum_{i=0}^1 a_i x^i \end{aligned} \quad (8)$$

Let a be the coefficient of $x^v = x^1$ in the above sum. The Laurent series of $\sqrt{r}$ at ∞ is

$$\begin{aligned} \sqrt{r} \approx 2ix &- \frac{3i}{16x^3} - \frac{9i}{1024x^7} - \frac{27i}{32768x^{11}} - \frac{405i}{4194304x^{15}} \\ &- \frac{1701i}{134217728x^{19}} - \frac{15309i}{8589934592x^{23}} - \frac{72171i}{274877906944x^{27}} + \dots \end{aligned} \quad (9)$$

Comparing Eq. (9) with Eq. (8) shows that

$$a = 2i$$

From Eq. (9) the sum up to $v = 1$ gives

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^1 a_i x^i \\ &= 2ix \end{aligned} \quad (10)$$

Now we need to find b , where b be the coefficient of $x^{v-1} = x^0 = 1$ in r minus the coefficient of same term but in $([\sqrt{r}]_\infty)^2$ where $[\sqrt{r}]_\infty$ was found above in Eq (10). Hence

$$([\sqrt{r}]_\infty)^2 = -4x^2$$

This shows that the coefficient of 1 in the above is 0. Now we need to find the coefficient of 1 in r . How this is done depends on if $v = 0$ or not. Since $v = 1$ which is not zero, then starting $r = \frac{s}{t}$, we do long division and write this in the form

$$r = Q + \frac{R}{t}$$

Where Q is the quotient and R is the remainder. Then the coefficient of 1 in r will be the coefficient this term in the quotient. Doing long division gives

$$\begin{aligned}
 r &= \frac{s}{t} \\
 &= \frac{-16x^4 + 3}{4x^2} \\
 &= Q + \frac{R}{4x^2} \\
 &= (-4x^2) + \left(\frac{3}{4x^2}\right) \\
 &= -4x^2 + \frac{3}{4x^2}
 \end{aligned}$$

We see that the coefficient of the term x in the quotient is 0. Now b can be found.

$$\begin{aligned}
 b &= (0) - (0) \\
 &= 0
 \end{aligned}$$

Hence

$$\begin{aligned}
 [\sqrt{r}]_{\infty} &= 2ix \\
 \alpha_{\infty}^{+} &= \frac{1}{2} \left(\frac{b}{a} - v \right) = \frac{1}{2} \left(\frac{0}{2i} - 1 \right) = -\frac{1}{2} \\
 \alpha_{\infty}^{-} &= \frac{1}{2} \left(-\frac{b}{a} - v \right) = \frac{1}{2} \left(-\frac{0}{2i} - 1 \right) = -\frac{1}{2}
 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{-16x^4 + 3}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^{+}	α_c^{-}
0	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_{\infty}$	α_{∞}^{+}	α_{∞}^{-}
-2	$2ix$	$-\frac{1}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^{\pm}$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_{\infty}$ and its associated $\alpha_{\infty}^{\pm}$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_{\infty}^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_{\infty}^{\pm}$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying

$\alpha_{\infty}^{-} = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_{\infty}^{-} - (\alpha_{c_1}^{-}) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_{\infty}$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^{-}}{x - c_1} \right) + (-)[\sqrt{r}]_{\infty} \\ &= -\frac{1}{2x} + (-)(2ix) \\ &= -\frac{1}{2x} - 2ix \\ &= -\frac{1}{2x} - 2ix \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(-\frac{1}{2x} - 2ix\right)(0) + \left(\left(\frac{1}{2x^2} - 2i\right) + \left(-\frac{1}{2x} - 2ix\right)^2 - \left(\frac{-16x^4 + 3}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2x} - 2ix\right) dx} \\ &= \frac{e^{-ix^2}}{\sqrt{x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-1}{x} dx} \\ &= z_1 e^{\frac{\ln(x)}{2}} \\ &= z_1 (\sqrt{x}) \end{aligned}$$

Which simplifies to

$$y_1 = e^{-ix^2}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-1}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{ie^{2ix^2}}{4} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{-ix^2}) + c_2 \left(e^{-ix^2} \left(-\frac{ie^{2ix^2}}{4} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 e^{-ix^2} - \frac{ic_2 e^{ix^2}}{4}$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 17

```
ode:=x*diff(diff(y(x),x),x)-diff(y(x),x)+4*x^3*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 \sin(x^2) + c_2 \cos(x^2)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

<- linear_1 successful

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) - \frac{d}{dx} y(x) + 4x^3 y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -4x^2 y(x) + \frac{\frac{d}{dx} y(x)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{\frac{d}{dx} y(x)}{x} + 4x^2 y(x) = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- o Define functions

$$[P_2(x) = -\frac{1}{x}, P_3(x) = 4x^2]$$

- o $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -1$$

- o $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$

- o $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$x \left(\frac{d^2}{dx^2} y(x) \right) - \frac{d}{dx} y(x) + 4x^3 y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $x^3 \cdot y(x)$ to series expansion

$$x^3 \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+3}$$

- Shift index using $k \rightarrow k-3$

$$x^3 \cdot y(x) = \sum_{k=3}^{\infty} a_{k-3} x^{k+r}$$

- Convert $\frac{d}{dx}y(x)$ to series expansion

$$\frac{d}{dx}y(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}$$

- Shift index using $k \rightarrow k+1$

$$\frac{d}{dx}y(x) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) x^{k+r}$$

- Convert $x \cdot \left(\frac{d^2}{dx^2}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-1}$$

- Shift index using $k \rightarrow k+1$

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r)(k+r) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0 r(-2+r) x^{-1+r} + a_1 (1+r)(-1+r) x^r + a_2 (2+r)r x^{1+r} + a_3 (3+r)(1+r) x^{2+r} + \left(\sum_{k=3}^{\infty} (a_{k+1} - 4a_{k-3})\right) x^{k+r} = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$r(-2+r) = 0$$

- Values of r that satisfy the indicial equation

$$r \in \{0, 2\}$$

- The coefficients of each power of x must be 0

$$[a_1(1+r)(-1+r) = 0, a_2(2+r)r = 0, a_3(3+r)(1+r) = 0]$$

- Solve for the dependent coefficient(s)

$$\{a_1 = 0, a_2 = 0, a_3 = 0\}$$

- Each term in the series must be 0, giving the recursion relation

$$a_{k+1}(k+1+r)(k+r-1) + 4a_{k-3} = 0$$

- Shift index using $k \rightarrow k+3$

$$a_{k+4}(k+4+r)(k+2+r) + 4a_k = 0$$

- Recursion relation that defines series solution to ODE

$$a_{k+4} = -\frac{4a_k}{(k+4+r)(k+2+r)}$$

- Recursion relation for $r = 0$

$$a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}$$

- Solution for $r = 0$

- $$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}, a_1 = 0, a_2 = 0, a_3 = 0 \right]$$
- Recursion relation for $r = 2$

$$a_{k+4} = -\frac{4a_k}{(k+6)(k+4)}$$
- Solution for $r = 2$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+2}, a_{k+4} = -\frac{4a_k}{(k+6)(k+4)}, a_1 = 0, a_2 = 0, a_3 = 0 \right]$$
- Combine solutions and rename parameters
$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^k \right) + \left(\sum_{k=0}^{\infty} b_k x^{k+2} \right), a_{k+4} = -\frac{4a_k}{(k+4)(k+2)}, a_1 = 0, a_2 = 0, a_3 = 0, b_{k+4} = -\frac{4b_k}{(k+6)(k+4)} \right]$$

✓ **Mathematica.** Time used: 0.012 (sec). Leaf size: 20

```
ode=x*D[y[x],{x,2}]-D[y[x],x]+4*x^3*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 \cos(x^2) + c_2 \sin(x^2)$$

✓ **Sympy.** Time used: 0.109 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x**3*y(x) + x*Derivative(y(x), (x, 2)) - Derivative(y(x), x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x \left(C_1 J_{\frac{1}{2}}(x^2) + C_2 Y_{\frac{1}{2}}(x^2) \right)$$

2.13.10 Problem 31

Local contents

Solved as second order ode using change of variable on x method 2	2241
Solved as second order Bessel ode	2246
Solved as second order ode using Kovacic algorithm	2250
✓ Maple	2258
✓ Mathematica	2258
✗ Sympy	2259

Internal problem ID [8040]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 31

Date solved : Thursday, November 27, 2025 at 07:32:39 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order ode using change of variable on x method 2

Time used: 0.253 (sec)

Solve

$$x^8 y'' + 4x^7 y' + y = \frac{1}{x^3}$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^8 y'' + 4x^7 y' + y = 0$$

In normal form the ode

$$x^8 y'' + 4x^7 y' + y = 0 \tag{1}$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \tag{2}$$

Where

$$p(x) = \frac{4}{x}$$

$$q(x) = \frac{1}{x^8}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x)\tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x)\tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x)dx} dx \\ &= \int e^{-\int \frac{4}{x} dx} dx \\ &= \int e^{-4\ln(x)} dx \\ &= \int \frac{1}{x^4} dx \\ &= -\frac{1}{3x^3} \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{\frac{1}{x^8}}{\frac{1}{x^8}} \\ &= 1 \end{aligned} \quad (7)$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2}y(\tau) + q_1y(\tau) &= 0 \\ \frac{d^2}{d\tau^2}y(\tau) + y(\tau) &= 0 \end{aligned}$$

The above ode is now solved for $y(\tau)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Ay''(\tau) + By'(\tau) + Cy(\tau) = 0$$

Where in the above $A = 1, B = 0, C = 1$. Let the solution be $y(\tau) = e^{\lambda\tau}$. Substituting this into the ODE gives

$$\lambda^2 e^{\lambda\tau} + e^{\lambda\tau} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda\tau}$ gives

$$\lambda^2 + 1 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 1$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(1)} \\ &= \pm i \end{aligned}$$

Hence

$$\lambda_1 = +i$$

$$\lambda_2 = -i$$

Which simplifies to

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 1$. Therefore the final solution, when using Euler relation, can be written as

$$y(\tau) = e^{\alpha\tau} (c_1 \cos(\beta\tau) + c_2 \sin(\beta\tau))$$

Which becomes

$$y(\tau) = e^0 (c_1 \cos(\tau) + c_2 \sin(\tau))$$

Or

$$y(\tau) = c_1 \cos(\tau) + c_2 \sin(\tau)$$

The above solution is now transformed back to y using (6) which results in

$$y = c_1 \cos\left(\frac{1}{3x^3}\right) - c_2 \sin\left(\frac{1}{3x^3}\right)$$

Therefore the homogeneous solution y_h is

$$y_h = c_1 \cos\left(\frac{1}{3x^3}\right) - c_2 \sin\left(\frac{1}{3x^3}\right)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \cos\left(\frac{1}{3x^3}\right)$$

$$y_2 = \sin\left(\frac{1}{3x^3}\right)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \tag{2}$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \tag{3}$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \cos\left(\frac{1}{3x^3}\right) & \sin\left(\frac{1}{3x^3}\right) \\ \frac{d}{dx}\left(\cos\left(\frac{1}{3x^3}\right)\right) & \frac{d}{dx}\left(\sin\left(\frac{1}{3x^3}\right)\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \cos\left(\frac{1}{3x^3}\right) & \sin\left(\frac{1}{3x^3}\right) \\ \frac{\sin\left(\frac{1}{3x^3}\right)}{x^4} & -\frac{\cos\left(\frac{1}{3x^3}\right)}{x^4} \end{vmatrix}$$

Therefore

$$W = \left(\cos\left(\frac{1}{3x^3}\right)\right) \left(-\frac{\cos\left(\frac{1}{3x^3}\right)}{x^4}\right) - \left(\sin\left(\frac{1}{3x^3}\right)\right) \left(\frac{\sin\left(\frac{1}{3x^3}\right)}{x^4}\right)$$

Which simplifies to

$$W = -\frac{\cos\left(\frac{1}{3x^3}\right)^2 + \sin\left(\frac{1}{3x^3}\right)^2}{x^4}$$

Which simplifies to

$$W = -\frac{1}{x^4}$$

Therefore Eq. (2) becomes

$$u_1 = -\int \frac{\sin\left(\frac{1}{3x^3}\right)}{-x^4} dx$$

Which simplifies to

$$u_1 = -\int -\frac{\sin\left(\frac{1}{3x^3}\right)}{x^7} dx$$

Hence

$$u_1 = -3 \sin\left(\frac{1}{3x^3}\right) + \frac{\cos\left(\frac{1}{3x^3}\right)}{x^3}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\cos\left(\frac{1}{3x^3}\right)}{-x^4} dx$$

Which simplifies to

$$u_2 = \int -\frac{\cos\left(\frac{1}{3x^3}\right)}{x^7} dx$$

Hence

$$u_2 = 3 \cos\left(\frac{1}{3x^3}\right) + \frac{\sin\left(\frac{1}{3x^3}\right)}{x^3}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-3 \sin \left(\frac{1}{3x^3} \right) + \frac{\cos \left(\frac{1}{3x^3} \right)}{x^3} \right) \cos \left(\frac{1}{3x^3} \right) + \sin \left(\frac{1}{3x^3} \right) \left(3 \cos \left(\frac{1}{3x^3} \right) + \frac{\sin \left(\frac{1}{3x^3} \right)}{x^3} \right)$$

Which simplifies to

$$y_p(x) = \frac{1}{x^3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 \cos \left(\frac{1}{3x^3} \right) - c_2 \sin \left(\frac{1}{3x^3} \right) \right) + \left(\frac{1}{x^3} \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 \cos \left(\frac{1}{3x^3} \right) - c_2 \sin \left(\frac{1}{3x^3} \right) + \frac{1}{x^3}$$

Solved as second order Bessel ode

Time used: 0.270 (sec)

Solve

$$x^8 y'' + 4x^7 y' + y = \frac{1}{x^3}$$

Writing the ode as

$$x^2 y'' + 4y'x + \frac{y}{x^6} = \frac{1}{x^9} \quad (1)$$

Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE and y_p is a particular solution to the non-homogeneous ODE. Bessel ode has the form

$$x^2 y'' + y'x + (-n^2 + x^2) y = 0 \quad (2)$$

The generalized form of Bessel ode is given by Bowman (1958) as the following

$$x^2 y'' + (1 - 2\alpha) xy' + (\beta^2 \gamma^2 x^{2\gamma} - n^2 \gamma^2 + \alpha^2) y = 0 \quad (3)$$

With the standard solution

$$y = x^\alpha (c_3 \text{BesselJ}(n, \beta x^\gamma) + c_4 \text{BesselY}(n, \beta x^\gamma)) \quad (4)$$

Comparing (3) to (1) and solving for α, β, n, γ gives

$$\begin{aligned} \alpha &= -\frac{3}{2} \\ \beta &= \frac{1}{3} \\ n &= \frac{1}{2} \\ \gamma &= -3 \end{aligned}$$

Substituting all the above into (4) gives the solution as

$$y = \frac{c_3 \sqrt{2} \sqrt{3} \sin\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} - \frac{c_4 \sqrt{2} \sqrt{3} \cos\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_3 \sqrt{2} \sqrt{3} \sin\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} - \frac{c_4 \sqrt{2} \sqrt{3} \cos\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned} y_1 &= \frac{\sqrt{2} \sqrt{3} \sin\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} \\ y_2 &= -\frac{\sqrt{2} \sqrt{3} \cos\left(\frac{1}{3x^3}\right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} \end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} & -\frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} \\ \frac{d}{dx}\left(\frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}\right) & \frac{d}{dx}\left(-\frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} & -\frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} \\ -\frac{3\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{2x^{5/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} + \frac{3\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{2x^{11/2}\sqrt{\pi}\left(\frac{1}{x^3}\right)^{3/2}} - \frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{11/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} & \frac{3\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{2x^{5/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} - \frac{3\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{2x^{11/2}\sqrt{\pi}\left(\frac{1}{x^3}\right)^{3/2}} - \frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{11/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} \end{vmatrix}$$

Therefore

$$W = \left(\frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}\right)\left(\frac{3\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{2x^{5/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} - \frac{3\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{2x^{11/2}\sqrt{\pi}\left(\frac{1}{x^3}\right)^{3/2}}\right) \\ - \frac{\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{x^{11/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} - \left(-\frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{3/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}\right)\left(-\frac{3\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{2x^{5/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}} + \frac{3\sqrt{2}\sqrt{3}\sin\left(\frac{1}{3x^3}\right)}{2x^{11/2}\sqrt{\pi}\left(\frac{1}{x^3}\right)^{3/2}} - \frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{11/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}\right)$$

Which simplifies to

$$W = -\frac{6\left(\cos\left(\frac{1}{3x^3}\right)^2 + \sin\left(\frac{1}{3x^3}\right)^2\right)}{x^4\pi}$$

Which simplifies to

$$W = -\frac{6}{x^4\pi}$$

Therefore Eq. (2) becomes

$$u_1 = -\int \frac{-\frac{\sqrt{2}\sqrt{3}\cos\left(\frac{1}{3x^3}\right)}{x^{21/2}\sqrt{\pi}\sqrt{\frac{1}{x^3}}}}{-\frac{6}{x^2\pi}} dx$$

Which simplifies to

$$u_1 = -\int \frac{\sqrt{\pi}\cos\left(\frac{1}{3x^3}\right)\sqrt{6}}{6x^{17/2}\sqrt{\frac{1}{x^3}}} dx$$

Hence

$$u_1 = \frac{\sqrt{6} \sqrt{\pi} \left(6 \cos \left(\frac{1}{3x^3} \right) x^3 + 2 \sin \left(\frac{1}{3x^3} \right) \right)}{12x^{9/2} \sqrt{\frac{1}{x^3}}}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{\sqrt{2} \sqrt{3} \sin \left(\frac{1}{3x^3} \right)}{x^{21/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}}}{-\frac{6}{x^2 \pi}} dx$$

Which simplifies to

$$u_2 = \int -\frac{\sqrt{\pi} \sin \left(\frac{1}{3x^3} \right) \sqrt{6}}{6x^{17/2} \sqrt{\frac{1}{x^3}}} dx$$

Hence

$$u_2 = \frac{i\sqrt{6} \sqrt{\pi} \left(2i \cos \left(\frac{1}{3x^3} \right) - 6ix^3 \sin \left(\frac{1}{3x^3} \right) \right)}{12x^{9/2} \sqrt{\frac{1}{x^3}}}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\sqrt{6} \left(6 \cos \left(\frac{1}{3x^3} \right) x^3 + 2 \sin \left(\frac{1}{3x^3} \right) \right) \sqrt{2} \sqrt{3} \sin \left(\frac{1}{3x^3} \right)}{12x^3} - \frac{i\sqrt{2} \sqrt{3} \cos \left(\frac{1}{3x^3} \right) \sqrt{6} \left(2i \cos \left(\frac{1}{3x^3} \right) - 6ix^3 \sin \left(\frac{1}{3x^3} \right) \right)}{12x^3}$$

Which simplifies to

$$y_p(x) = \frac{1}{x^3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_3 \sqrt{2} \sqrt{3} \sin \left(\frac{1}{3x^3} \right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} - \frac{c_4 \sqrt{2} \sqrt{3} \cos \left(\frac{1}{3x^3} \right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} \right) + \left(\frac{1}{x^3} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{c_3 \sqrt{2} \sqrt{3} \sin \left(\frac{1}{3x^3} \right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} - \frac{c_4 \sqrt{2} \sqrt{3} \cos \left(\frac{1}{3x^3} \right)}{x^{3/2} \sqrt{\pi} \sqrt{\frac{1}{x^3}}} + \frac{1}{x^3}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.401 (sec)

Solve

$$x^8 y'' + 4x^7 y' + y = \frac{1}{x^3}$$

Writing the ode as

$$x^8 y'' + 4x^7 y' + y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^8 \\ B &= 4x^7 \\ C &= 1 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{2x^6 - 1}{x^8} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 2x^6 - 1 \\ t &= x^8 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{2x^6 - 1}{x^8} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.280: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 8 - 6 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = x^8$. There is a pole at $x = 0$ of order 8. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Attempting to find a solution using case $n = 1$.

Looking at higher order poles of order $2v \geq 4$ (must be even order for case one). Then for each pole c , $[\sqrt{r}]_c$ is the sum of terms $\frac{1}{(x-c)^i}$ for $2 \leq i \leq v$ in the Laurent series expansion of $\sqrt{r}$ expanded around each pole c . Hence

$$[\sqrt{r}]_c = \sum_2^v \frac{a_i}{(x-c)^i} \quad (1B)$$

Let a be the coefficient of the term $\frac{1}{(x-c)^v}$ in the above where v is the pole order divided by 2. Let b be the coefficient of $\frac{1}{(x-c)^{v+1}}$ in r minus the coefficient of $\frac{1}{(x-c)^{v+1}}$ in $[\sqrt{r}]_c$. Then

$$\alpha_c^+ = \frac{1}{2} \left(\frac{b}{a} + v \right)$$

$$\alpha_c^- = \frac{1}{2} \left(-\frac{b}{a} + v \right)$$

The partial fraction decomposition of r is

$$r = -\frac{1}{x^8} + \frac{2}{x^2}$$

There is pole in r at $x = 0$ of order 8, hence $v = 4$. Expanding $\sqrt{r}$ as Laurent series about this pole $c = 0$ gives

$$[\sqrt{r}]_c \approx \frac{i}{x^4} + \dots \quad (2B)$$

Using eq. (1B), taking the sum up to $v = 4$ the above becomes

$$[\sqrt{r}]_c = \frac{i}{x^4} \quad (3V)$$

The above shows that the coefficient of $\frac{1}{(x-0)^4}$ is

$$a = i$$

Now we need to find b . let b be the coefficient of the term $\frac{1}{(x-c)^{v+1}}$ in r minus the coefficient of the same term but in the sum $[\sqrt{r}]_c$ found in eq. (3B). Here c is current pole which is $c = 0$. This term becomes $\frac{1}{x^5}$. The coefficient of this term in the sum $[\sqrt{r}]_c$ is seen to be 0 and the coefficient of this term r is found from the partial fraction decomposition from above to be 0. Therefore

$$b = (0) - (0)$$

$$= 0$$

Hence

$$[\sqrt{r}]_c = \frac{i}{x^4}$$

$$\alpha_c^+ = \frac{1}{2} \left(\frac{b}{a} + v \right) = \frac{1}{2} \left(\frac{0}{i} + 4 \right) = 2$$

$$\alpha_c^- = \frac{1}{2} \left(-\frac{b}{a} + v \right) = \frac{1}{2} \left(-\frac{0}{i} + 4 \right) = 2$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{2x^6 - 1}{x^8}$$

Since the $\gcd(s, t) = 1$. This gives $b = 2$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = 2 \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = -1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{2x^6 - 1}{x^8}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	8	$\frac{i}{x^4}$	2	2

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	2	-1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^+ = 2$ then

$$\begin{aligned} d &= \alpha_\infty^+ - (\alpha_{c_1}^-) \\ &= 2 - (2) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

Substituting the above values in the above results in

$$\begin{aligned}
 \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (+)[\sqrt{r}]_{\infty} \\
 &= -\frac{i}{x^4} + \frac{2}{x} + (0) \\
 &= -\frac{i}{x^4} + \frac{2}{x} \\
 &= \frac{2x^3 - i}{x^4}
 \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned}
 (0) + 2\left(-\frac{i}{x^4} + \frac{2}{x}\right)(0) + \left(\left(\frac{4i}{x^5} - \frac{2}{x^2}\right) + \left(-\frac{i}{x^4} + \frac{2}{x}\right)^2 - \left(\frac{2x^6 - 1}{x^8}\right)\right) &= 0 \\
 0 &= 0
 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned}
 z_1(x) &= pe^{\int \omega dx} \\
 &= e^{\int \left(-\frac{i}{x^4} + \frac{2}{x}\right) dx} \\
 &= x^2 e^{\frac{i}{3x^3}}
 \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{4x^7}{x^8} dx} \\
 &= z_1 e^{-2 \ln(x)} \\
 &= z_1 \left(\frac{1}{x^2}\right)
 \end{aligned}$$

Which simplifies to

$$y_1 = e^{\frac{i}{3x^3}}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{4x^7}{x^8} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-4 \ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{ie^{-\frac{2i}{3x^3}}}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(e^{\frac{i}{3x^3}} \right) + c_2 \left(e^{\frac{i}{3x^3}} \left(-\frac{ie^{-\frac{2i}{3x^3}}}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$x^8 y'' + 4x^7 y' + y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{\frac{i}{3x^3}} - \frac{ic_2 e^{-\frac{i}{3x^3}}}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{\frac{i}{3x^3}}$$

$$y_2 = -\frac{ie^{-\frac{i}{3x^3}}}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = -\int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{\frac{i}{3x^3}} & -\frac{ie^{-\frac{i}{3x^3}}}{2} \\ \frac{d}{dx}\left(e^{\frac{i}{3x^3}}\right) & \frac{d}{dx}\left(-\frac{ie^{-\frac{i}{3x^3}}}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{\frac{i}{3x^3}} & -\frac{ie^{-\frac{i}{3x^3}}}{2} \\ -\frac{ie^{\frac{i}{3x^3}}}{x^4} & \frac{e^{-\frac{i}{3x^3}}}{2x^4} \end{vmatrix}$$

Therefore

$$W = \left(e^{\frac{i}{3x^3}}\right) \left(\frac{e^{-\frac{i}{3x^3}}}{2x^4}\right) - \left(-\frac{ie^{-\frac{i}{3x^3}}}{2}\right) \left(-\frac{ie^{\frac{i}{3x^3}}}{x^4}\right)$$

Which simplifies to

$$W = \frac{e^{\frac{i}{3x^3}} e^{-\frac{i}{3x^3}}}{x^4}$$

Which simplifies to

$$W = \frac{1}{x^4}$$

Therefore Eq. (2) becomes

$$u_1 = -\int \frac{-\frac{ie^{-\frac{i}{3x^3}}}{2x^3}}{x^4} dx$$

Which simplifies to

$$u_1 = - \int - \frac{i e^{-\frac{i}{3x^3}}}{2x^7} dx$$

Hence

$$u_1 = - \frac{i(3x^3 + i) e^{-\frac{i}{3x^3}}}{2x^3}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{e^{\frac{i}{3x^3}}}{x^3}}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{e^{\frac{i}{3x^3}}}{x^7} dx$$

Hence

$$u_2 = \frac{(-3x^3 + i) e^{\frac{i}{3x^3}}}{x^3}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = - \frac{i(3x^3 + i) e^{-\frac{i}{3x^3}} e^{\frac{i}{3x^3}}}{2x^3} - \frac{i e^{-\frac{i}{3x^3}} (-3x^3 + i) e^{\frac{i}{3x^3}}}{2x^3}$$

Which simplifies to

$$y_p(x) = \frac{1}{x^3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{\frac{i}{3x^3}} - \frac{ic_2 e^{-\frac{i}{3x^3}}}{2} \right) + \left(\frac{1}{x^3} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{1}{x^3} + c_1 e^{\frac{i}{3x^3}} - \frac{ic_2 e^{-\frac{i}{3x^3}}}{2}$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 24

```
ode:=x^8*diff(diff(y(x),x),x)+4*x^7*diff(y(x),x)+y(x) = 1/x^3;
dsolve(ode,y(x), singsol=all);
```

$$y = \sin\left(\frac{1}{3x^3}\right) c_2 + \cos\left(\frac{1}{3x^3}\right) c_1 + \frac{1}{x^3}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

-> Try solving first the homogeneous part of the ODE

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

<- linear_1 successful

<- solving first the homogeneous part of the ODE successful

✓ **Mathematica.** Time used: 0.041 (sec). Leaf size: 90

```
ode=x^8*D[y[x],{x,2}]+4*x^7*D[y[x],x]+y[x]==1/x^3;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2} \left(-2 \sin\left(\frac{1}{3x^3}\right) \int_1^x \frac{\cos\left(\frac{1}{3K[1]^3}\right)}{K[1]^7} dK[1] + \frac{1}{x^3} - 3 \sin\left(\frac{2}{3x^3}\right) + \frac{\cos\left(\frac{2}{3x^3}\right)}{x^3} + 2c_1 \cos\left(\frac{1}{3x^3}\right) - 2c_2 \sin\left(\frac{1}{3x^3}\right) \right)$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**8*Derivative(y(x), (x, 2)) + 4*x**7*Derivative(y(x), x) + y(x) - 1/x
**3,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (-x**11*Derivative(y(x),
(x, 2)) - x**3*y(x) + 1)/(4*x**10) cannot be solved by the factorable group
method
```

2.13.11 Problem 32

Local contents

Solved as second order ode using change of variable on y method 2	2260
✓ Maple	2265
✓ Mathematica	2265
✗ Sympy	2266

Internal problem ID [8041]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 32

Date solved : Thursday, November 27, 2025 at 07:32:41 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using change of variable on y method 2

Time used: 1.033 (sec)

Solve

$$(x \sin(x) + \cos(x)) y'' - y' x \cos(x) + y \cos(x) = x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x \sin(x) + \cos(x)$, $B = -x \cos(x)$, $C = \cos(x)$, $f(x) = x$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$(x \sin(x) + \cos(x)) y'' - y' x \cos(x) + y \cos(x) = 0$$

In normal form the ode

$$(x \sin(x) + \cos(x)) y'' - y' x \cos(x) + y \cos(x) = 0 \quad (1)$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{x \cos(x)}{x \sin(x) + \cos(x)}$$

$$q(x) = \frac{\cos(x)}{x \sin(x) + \cos(x)}$$

Applying change of variables on the dependant variable $y = v(x) x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right) v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right) v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{n \cos(x)}{x \sin(x) + \cos(x)} + \frac{\cos(x)}{x \sin(x) + \cos(x)} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$v''(x) + \left(\frac{2}{x} - \frac{x \cos(x)}{x \sin(x) + \cos(x)}\right) v'(x) = 0$$

$$v''(x) + \left(\frac{2}{x} - \frac{x \cos(x)}{x \sin(x) + \cos(x)}\right) v'(x) = 0 \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \left(\frac{2}{x} - \frac{x \cos(x)}{x \sin(x) + \cos(x)}\right) u(x) = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{x^2 \cos(x) - 2x \sin(x) - 2 \cos(x)}{x(x \sin(x) + \cos(x))}$$

$$p(x) = 0$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{x^2 \cos(x) - 2x \sin(x) - 2 \cos(x)}{x(x \sin(x) + \cos(x))} dx} \\ &= \frac{x^2}{x \sin(x) + \cos(x)} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x \sin(x) + \cos(x)} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x \sin(x) + \cos(x)} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x \sin(x) + \cos(x)}$ gives the final solution

$$u(x) = \frac{(x \sin(x) + \cos(x)) c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= -\frac{\cos(x) c_1}{x} + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(-\frac{\cos(x) c_1}{x} + c_2 \right) x \\ &= -c_1 \cos(x) + c_2 x \end{aligned}$$

Now the particular solution to this ODE is found

$$(x \sin(x) + \cos(x)) y'' - y' x \cos(x) + y \cos(x) = x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = \cos(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & \cos(x) \\ \frac{d}{dx}(x) & \frac{d}{dx}(\cos(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & \cos(x) \\ 1 & -\sin(x) \end{vmatrix}$$

Therefore

$$W = (x)(-\sin(x)) - (\cos(x))(1)$$

Which simplifies to

$$W = -x \sin(x) - \cos(x)$$

Which simplifies to

$$W = -x \sin(x) - \cos(x)$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x \cos(x)}{(x \sin(x) + \cos(x))(-x \sin(x) - \cos(x))} dx$$

Which simplifies to

$$u_1 = - \int - \frac{x \cos(x)}{(x \sin(x) + \cos(x))^2} dx$$

Hence

$$u_1 = - \frac{1}{x \sin(x) + \cos(x)}$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^2}{(x \sin(x) + \cos(x))(-x \sin(x) - \cos(x))} dx$$

Which simplifies to

$$u_2 = \int - \frac{x^2}{(x \sin(x) + \cos(x))^2} dx$$

Hence

$$u_2 = \frac{x - x \tan\left(\frac{x}{2}\right)^2 - 2 \tan\left(\frac{x}{2}\right)}{2x \tan\left(\frac{x}{2}\right) - \tan\left(\frac{x}{2}\right)^2 + 1}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = - \frac{x}{x \sin(x) + \cos(x)} + \frac{\cos(x) \left(x - x \tan\left(\frac{x}{2}\right)^2 - 2 \tan\left(\frac{x}{2}\right) \right)}{2x \tan\left(\frac{x}{2}\right) - \tan\left(\frac{x}{2}\right)^2 + 1}$$

Which simplifies to

$$y_p(x) = - \sin(x)$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(- \frac{\cos(x) c_1}{x} + c_2 \right) x \right) + (- \sin(x)) \end{aligned}$$

Summary of solutions found

$$y = -\sin(x) + \left(-\frac{\cos(x)c_1}{x} + c_2 \right) x$$

✓ **Maple.** Time used: 0.032 (sec). Leaf size: 17

```
ode:=(x*sin(x)+cos(x))*diff(diff(y(x),x),x)-x*cos(x)*diff(y(x),x)+y(x)*cos(x) = x;
dsolve(ode,y(x), singsol=all);
```

$$y = -c_1 \cos(x) + c_2 x - \sin(x)$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

trying symmetries linear in x and y(x)

Try integration with the canonical coordinates of the symmetry [0, x]

-> Calling odsolve with the ODE, diff(_b(_a),_a) = (cos(_a)*_a^2*_b(_a)-2*sin(_a)*_a*_b(_a)-2*cos(_a)*_b(_a)+_a)/(_a*sin(_a)+cos(_a))/_a, _b(_a), explicit

*** Sublevel 2 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

<- differential order: 2; canonical coordinates successful

✓ **Mathematica.** Time used: 0.725 (sec). Leaf size: 108

```
ode=(x*Sin[x]+Cos[x])*D[y[x],{x,2}]-x*Cos[x]*D[y[x],x]+y[x]*Cos[x]==x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow x \left(\int_1^x -\frac{K[2]^2 \int_1^{K[2]} \frac{\cos(K[1])+K[1]\sin(K[1])}{K[1]^2} dK[1]}{(\cos(K[2])+K[2]\sin(K[2]))^2} dK[2] + \int_1^x \frac{\cos(K[1])+K[1]\sin(K[1])}{K[1]^2} dK[1] \left(\int_1^x \frac{1}{(c} \right. \right.$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x*cos(x)*Derivative(y(x), x) - x + (x*sin(x) + cos(x))*Derivative(y(x),
    (x, 2)) + y(x)*cos(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (x*tan(x)*Derivative(y(
    x), (x, 2)) - x/cos(x) + y(x) + Derivative(y(x), (x, 2)))/x cannot be solved
    by the factorable group method
```

2.13.12 Problem 33

Local contents

Solved as second order Euler type ode	2267
Solved as second order ode using change of variable on x method 2	2270
Solved as second order ode using change of variable on y method 2	2275
Solved as second order ode using Kovacic algorithm	2279
✓ Maple	2286
✓ Mathematica	2287
✓ Sympy	2287

Internal problem ID [8042]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 33

Date solved : Thursday, November 27, 2025 at 07:32:42 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order Euler type ode

Time used: 0.155 (sec)

Solve

$$xy'' - 3y' + \frac{3y}{x} = 2 + x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2$, $B = -3x$, $C = 3$, $f(x) = x(2 + x)$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$x^2y'' - 3y'x + 3y = 0$$

This is Euler second order ODE. Let the solution be $y = x^r$, then $y' = rx^{r-1}$ and $y'' = r(r-1)x^{r-2}$. Substituting these back into the given ODE gives

$$x^2(r(r-1))x^{r-2} - 3xr x^{r-1} + 3x^r = 0$$

Simplifying gives

$$r(r-1)x^r - 3rx^r + 3x^r = 0$$

Since $x^r \neq 0$ then dividing throughout by x^r gives

$$r(r-1) - 3r + 3 = 0$$

Or

$$r^2 - 4r + 3 = 0 \quad (1)$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = 1$$

$$r_2 = 3$$

Since the roots are real and distinct, then the general solution is

$$y = c_1y_1 + c_2y_2$$

Where $y_1 = x^{r_1}$ and $y_2 = x^{r_2}$. Hence

$$y = c_2x^3 + c_1x$$

Next, we find the particular solution to the ODE

$$x^2y'' - 3y'x + 3y = x(2+x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1y_1 + u_2y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^3$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^3 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^3) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix}$$

Therefore

$$W = (x)(3x^2) - (x^3)(1)$$

Which simplifies to

$$W = 2x^3$$

Which simplifies to

$$W = 2x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^4(2+x)}{2x^5} dx$$

Which simplifies to

$$u_1 = - \int \frac{2+x}{2x} dx$$

Hence

$$u_1 = -\frac{x}{2} - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{x^2(2+x)}{2x^5} dx$$

Which simplifies to

$$u_2 = \int \frac{2+x}{2x^3} dx$$

Hence

$$u_2 = -\frac{1}{2x} - \frac{1}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{x}{2} - \ln(x)\right)x + x^3\left(-\frac{1}{2x} - \frac{1}{2x^2}\right)$$

Which simplifies to

$$y_p(x) = -x^2 - \ln(x)x - \frac{x}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= -x^2 - \ln(x)x - \frac{x}{2} + c_2 x^3 + c_1 x \end{aligned}$$

Summary of solutions found

$$y = -x^2 - \ln(x)x - \frac{x}{2} + c_2 x^3 + c_1 x$$

Solved as second order ode using change of variable on x method 2

Time used: 0.356 (sec)

Solve

$$xy'' - 3y' + \frac{3y}{x} = 2 + x$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$xy'' - 3y' + \frac{3y}{x} = 0$$

In normal form the ode

$$xy'' - 3y' + \frac{3y}{x} = 0 \tag{1}$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{3}{x^2}$$

Applying change of variables $\tau = g(x)$ to (2) gives

$$\frac{d^2}{d\tau^2}y(\tau) + p_1\left(\frac{d}{d\tau}y(\tau)\right) + q_1y(\tau) = 0 \quad (3)$$

Where τ is the new independent variable, and

$$p_1(\tau) = \frac{\tau''(x) + p(x)\tau'(x)}{\tau'(x)^2} \quad (4)$$

$$q_1(\tau) = \frac{q(x)}{\tau'(x)^2} \quad (5)$$

Let $p_1 = 0$. Eq (4) simplifies to

$$\tau''(x) + p(x)\tau'(x) = 0$$

This ode is solved resulting in

$$\begin{aligned} \tau &= \int e^{-\int p(x)dx} dx \\ &= \int e^{-\int -\frac{3}{x}dx} dx \\ &= \int e^{3\ln(x)} dx \\ &= \int x^3 dx \\ &= \frac{x^4}{4} \end{aligned} \quad (6)$$

Using (6) to evaluate q_1 from (5) gives

$$\begin{aligned} q_1(\tau) &= \frac{q(x)}{\tau'(x)^2} \\ &= \frac{\frac{3}{x^2}}{x^6} \\ &= \frac{3}{x^8} \end{aligned} \quad (7)$$

Substituting the above in (3) and noting that now $p_1 = 0$ results in

$$\begin{aligned} \frac{d^2}{d\tau^2}y(\tau) + q_1y(\tau) &= 0 \\ \frac{d^2}{d\tau^2}y(\tau) + \frac{3y(\tau)}{x^8} &= 0 \end{aligned}$$

But in terms of τ

$$\frac{3}{x^8} = \frac{3}{16\tau^2}$$

Hence the above ode becomes

$$\frac{d^2}{d\tau^2}y(\tau) + \frac{3y(\tau)}{16\tau^2} = 0$$

The above ode is now solved for $y(\tau)$. This is Euler second order ODE. Let the solution be $y(\tau) = \tau^r$, then $y' = r\tau^{r-1}$ and $y'' = r(r-1)\tau^{r-2}$. Substituting these back into the given ODE gives

$$16\tau^2(r(r-1))\tau^{r-2} + 0r\tau^{r-1} + 3\tau^r = 0$$

Simplifying gives

$$16r(r-1)\tau^r + 0\tau^r + 3\tau^r = 0$$

Since $\tau^r \neq 0$ then dividing throughout by τ^r gives

$$16r(r-1) + 0 + 3 = 0$$

Or

$$16r^2 - 16r + 3 = 0 \tag{1}$$

Equation (1) is the characteristic equation. Its roots determine the form of the general solution. Using the quadratic equation the roots are

$$r_1 = \frac{1}{4}$$

$$r_2 = \frac{3}{4}$$

Since the roots are real and distinct, then the general solution is

$$y(\tau) = c_1y_1 + c_2y_2$$

Where $y_1 = \tau^{r_1}$ and $y_2 = \tau^{r_2}$. Hence

$$y(\tau) = c_1\tau^{1/4} + c_2\tau^{3/4}$$

The above solution is now transformed back to y using (6) which results in

$$y = \frac{c_14^{3/4}(x^4)^{1/4}}{4} + \frac{c_24^{1/4}(x^4)^{3/4}}{4}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_14^{3/4}(x^4)^{1/4}}{4} + \frac{c_24^{1/4}(x^4)^{3/4}}{4}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = (x^4)^{1/4}$$

$$y_2 = (x^4)^{3/4}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} (x^4)^{1/4} & (x^4)^{3/4} \\ \frac{d}{dx}((x^4)^{1/4}) & \frac{d}{dx}((x^4)^{3/4}) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} (x^4)^{1/4} & (x^4)^{3/4} \\ \frac{x^3}{(x^4)^{3/4}} & \frac{3x^3}{(x^4)^{1/4}} \end{vmatrix}$$

Therefore

$$W = ((x^4)^{1/4}) \left(\frac{3x^3}{(x^4)^{1/4}} \right) - ((x^4)^{3/4}) \left(\frac{x^3}{(x^4)^{3/4}} \right)$$

Which simplifies to

$$W = 2x^3$$

Which simplifies to

$$W = 2x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(x^4)^{3/4} (2+x)}{2x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{(x^4)^{3/4} (2+x)}{2x^4} dx$$

Hence

$$u_1 = -\frac{(x^4)^{3/4}}{2x^2} - \frac{(x^4)^{3/4} \ln(x)}{x^3}$$

And Eq. (3) becomes

$$u_2 = \int \frac{(x^4)^{1/4} (2+x)}{2x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{(x^4)^{1/4} (2+x)}{2x^4} dx$$

Hence

$$u_2 = -\frac{(1+x)(x^4)^{1/4}}{2x^3}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{(x^4)^{3/4}}{2x^2} - \frac{(x^4)^{3/4} \ln(x)}{x^3} \right) (x^4)^{1/4} - \frac{x(1+x)}{2}$$

Which simplifies to

$$y_p(x) = -x^2 - \ln(x)x - \frac{x}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1 4^{3/4} (x^4)^{1/4}}{4} + \frac{c_2 4^{1/4} (x^4)^{3/4}}{4} \right) + \left(-x^2 - \ln(x)x - \frac{x}{2} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{c_1 4^{3/4} (x^4)^{1/4}}{4} + \frac{c_2 4^{1/4} (x^4)^{3/4}}{4} - x^2 - \ln(x)x - \frac{x}{2}$$

Solved as second order ode using change of variable on y method 2

Time used: 0.329 (sec)

Solve

$$xy'' - 3y' + \frac{3y}{x} = 2 + x$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x$, $B = -3$, $C = \frac{3}{x}$, $f(x) = 2 + x$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$xy'' - 3y' + \frac{3y}{x} = 0$$

In normal form the ode

$$xy'' - 3y' + \frac{3y}{x} = 0 \tag{1}$$

Becomes

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = -\frac{3}{x}$$

$$q(x) = \frac{3}{x^2}$$

Applying change of variables on the dependent variable $y = v(x)x^n$ to (2) gives the following ode where the dependent variable is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right)v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right)v(x) = 0 \tag{3}$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \tag{4}$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{3n}{x^2} + \frac{3}{x^2} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 3 \quad (6)$$

Substituting this value in (3) gives

$$\begin{aligned} v''(x) + \frac{3v'(x)}{x} &= 0 \\ v''(x) + \frac{3v'(x)}{x} &= 0 \end{aligned} \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{3u(x)}{x} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{3}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{3}{x} \, dx} \\ &= x^3 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} (u x^3) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} u x^3 &= \int 0 dx + c_3 \\ &= c_3 \end{aligned}$$

Dividing throughout by the integrating factor x^3 gives the final solution

$$u(x) = \frac{c_3}{x^3}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_4 \\ &= -\frac{c_3}{2x^2} + c_4 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(-\frac{c_3}{2x^2} + c_4 \right) x^3 \\ &= c_4 x^3 - \frac{1}{2} c_3 x \end{aligned}$$

Now the particular solution to this ODE is found

$$xy'' - 3y' + \frac{3y}{x} = 2 + x$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^3$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^3 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^3) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix}$$

Therefore

$$W = (x)(3x^2) - (x^3)(1)$$

Which simplifies to

$$W = 2x^3$$

Which simplifies to

$$W = 2x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^3(2+x)}{2x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{2+x}{2x} dx$$

Hence

$$u_1 = -\frac{x}{2} - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(2+x)}{2x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{2+x}{2x^3} dx$$

Hence

$$u_2 = -\frac{1}{2x} - \frac{1}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{x}{2} - \ln(x)\right)x + x^3\left(-\frac{1}{2x} - \frac{1}{2x^2}\right)$$

Which simplifies to

$$y_p(x) = -x^2 - \ln(x)x - \frac{x}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(-\frac{c_3}{2x^2} + c_4\right)x^3\right) + \left(-x^2 - \ln(x)x - \frac{x}{2}\right) \end{aligned}$$

Summary of solutions found

$$y = -x^2 - \ln(x)x - \frac{x}{2} + \left(-\frac{c_3}{2x^2} + c_4\right)x^3$$

Solved as second order ode using Kovacic algorithm

Time used: 0.178 (sec)

Solve

$$xy'' - 3y' + \frac{3y}{x} = 2 + x$$

Writing the ode as

$$xy'' - 3y' + \frac{3y}{x} = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= -3 \\ C &= \frac{3}{x} \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{3}{4x^2} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 3 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{3}{4x^2} \right) z(x) \tag{7}$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.281: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\
 &= 2 - 0 \\
 &= 2
 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned}
 [\sqrt{r}]_c &= 0 \\
 \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\
 \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2}
 \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{3}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{3}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned}
 \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_{\infty} \\
 &= -\frac{1}{2x} + (-)(0) \\
 &= -\frac{1}{2x} \\
 &= -\frac{1}{2x}
 \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned}
 (0) + 2\left(-\frac{1}{2x}\right)(0) + \left(\left(\frac{1}{2x^2}\right) + \left(-\frac{1}{2x}\right)^2 - \left(\frac{3}{4x^2}\right)\right) &= 0 \\
 0 &= 0
 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned}
 z_1(x) &= pe^{\int \omega dx} \\
 &= e^{\int -\frac{1}{2x} dx} \\
 &= \frac{1}{\sqrt{x}}
 \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{-3}{x} dx} \\
 &= z_1 e^{\frac{3 \ln(x)}{2}} \\
 &= z_1 (x^{3/2})
 \end{aligned}$$

Which simplifies to

$$y_1 = x$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3 \ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{x^2}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(x) + c_2 \left(x \left(\frac{x^2}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$xy'' - 3y' + \frac{3y}{x} = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 x + \frac{1}{2} c_2 x^3$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = \frac{x^3}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & \frac{x^3}{2} \\ \frac{d}{dx}(x) & \frac{d}{dx}\left(\frac{x^3}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & \frac{x^3}{2} \\ 1 & \frac{3x^2}{2} \end{vmatrix}$$

Therefore

$$W = (x) \left(\frac{3x^2}{2} \right) - \left(\frac{x^3}{2} \right) \quad (1)$$

Which simplifies to

$$W = x^3$$

Which simplifies to

$$W = x^3$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{x^3(2+x)}{2}}{x^4} dx$$

Which simplifies to

$$u_1 = - \int \frac{2+x}{2x} dx$$

Hence

$$u_1 = -\frac{x}{2} - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(2+x)}{x^4} dx$$

Which simplifies to

$$u_2 = \int \frac{2+x}{x^3} dx$$

Hence

$$u_2 = -\frac{1}{x} - \frac{1}{x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(-\frac{x}{2} - \ln(x)\right)x + \frac{x^3\left(-\frac{1}{x} - \frac{1}{x^2}\right)}{2}$$

Which simplifies to

$$y_p(x) = -x^2 - \ln(x)x - \frac{x}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1x + \frac{1}{2}c_2x^3\right) + \left(-x^2 - \ln(x)x - \frac{x}{2}\right) \end{aligned}$$

Summary of solutions found

$$y = -x^2 - \ln(x)x - \frac{x}{2} + c_1x + \frac{c_2x^3}{2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 23

```
ode:=x*diff(diff(y(x),x),x)-3*diff(y(x),x)+3*y(x)/x = x+2;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(c_1x^2 - 2x - 2\ln(x) + 2c_2)x}{2}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 2; linear nonhomogeneous with symmetry [0,1]

trying a double symmetry of the form [xi=0, eta=F(x)]

<- double symmetry of the form [xi=0, eta=F(x)] successful

✓ **Mathematica.** Time used: 0.023 (sec). Leaf size: 30

```
ode=x*D[y[x],{x,2}]-3*D[y[x],x]+3*y[x]/x==x+2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{2}x(2c_2x^2 - 2x - 2\log(x) - 3 + 2c_1)$$

✓ **Sympy.** Time used: 0.153 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), (x, 2)) - x - 3*Derivative(y(x), x) - 2 + 3*y(x)/x,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x(C_1 + C_2x^2 - x - \log(x))$$

2.13.13 Problem 35

Local contents

Solved as second order ode using non constant coeff transformation on	
B method	2288
Solved as second order ode using Kovacic algorithm	2293
✓ Maple	2301
✓ Mathematica	2301
✗ Sympy	2302

Internal problem ID [8043]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 35

Date solved : Thursday, November 27, 2025 at 07:32:44 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.243 (sec)

Solve

$$(1+x)y'' - (3x+4)y' + 3y = (3x+2)e^{3x}$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \tag{1}$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= 1 + x \\ B &= -3x - 4 \\ C &= 3 \\ F &= (3x + 2)e^{3x} \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (1 + x)(0) + (-3x - 4)(-3) + (3)(-3x - 4) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-3x^2 - 7x - 4v'' + (9x^2 + 18x + 10)v' = 0$$

Now by applying $v' = u$ the above becomes

$$(-3x^2 - 7x - 4)u'(x) + 9\left(x^2 + 2x + \frac{10}{9}\right)u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{9x^2 + 18x + 10}{3x^2 + 7x + 4} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{9x^2 + 18x + 10}{3x^2 + 7x + 4} dx} \\ &= \frac{(3x + 4)^2 e^{-3x}}{1 + x} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu u &= 0 \\ \frac{d}{dx}\left(\frac{u(3x+4)^2 e^{-3x}}{1+x}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{u(3x+4)^2 e^{-3x}}{1+x} &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{(3x+4)^2 e^{-3x}}{1+x}$ gives the final solution

$$u(x) = \frac{(1+x) e^{3x} c_1}{(3x+4)^2}$$

The ode for v now becomes

$$v'(x) = \frac{(1+x) e^{3x} c_1}{(3x+4)^2}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dv &= \int \frac{(1+x) e^{3x} c_1}{(3x+4)^2} dx \\ v(x) &= \frac{c_1 e^{3x}}{27x+36} + c_2\end{aligned}$$

$$v(x) = \frac{c_1 e^{3x}}{27x+36} + c_2$$

Replacing $v(x)$ above by $\frac{y}{-3x-4}$, then the homogeneous solution is

$$\begin{aligned}y_h(x) &= Bv \\ &= -\frac{c_1 e^{3x}}{9} - 3c_2 x - 4c_2\end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{3x}$$

$$y_2 = -3x - 4$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{3x} & -3x - 4 \\ \frac{d}{dx}(e^{3x}) & \frac{d}{dx}(-3x - 4) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{3x} & -3x - 4 \\ 3e^{3x} & -3 \end{vmatrix}$$

Therefore

$$W = (e^{3x})(-3) - (-3x - 4)(3e^{3x})$$

Which simplifies to

$$W = 9xe^{3x} + 9e^{3x}$$

Which simplifies to

$$W = 9(1 + x)e^{3x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(-3x - 4)(3x + 2)e^{3x}}{9(1 + x)^2 e^{3x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{-9x^2 - 18x - 8}{9(1 + x)^2} dx$$

Hence

$$u_1 = x + \frac{1}{9x+9}$$

And Eq. (3) becomes

$$u_2 = \int \frac{(e^{3x})^2 (3x+2)}{9(1+x)^2 e^{3x}} dx$$

Which simplifies to

$$u_2 = \int \frac{e^{3x}(3x+2)}{9(1+x)^2} dx$$

Hence

$$u_2 = \frac{e^{3x}}{9x+9}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(x + \frac{1}{9x+9}\right) e^{3x} + \frac{(-3x-4)e^{3x}}{9x+9}$$

Which simplifies to

$$y_p(x) = \frac{(3x-1)e^{3x}}{3}$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= \left(-\frac{c_1 e^{3x}}{9} - 3c_2 x - 4c_2\right) + \left(\frac{(3x-1)e^{3x}}{3}\right) \\ &= \frac{(9x - c_1 - 3)e^{3x}}{9} + (-3x - 4)c_2 \end{aligned}$$

Summary of solutions found

$$y = \frac{(9x - c_1 - 3)e^{3x}}{9} + (-3x - 4)c_2$$

Solved as second order ode using Kovacic algorithm

Time used: 0.381 (sec)

Solve

$$(1+x)y'' - (3x+4)y' + 3y = (3x+2)e^{3x}$$

Writing the ode as

$$(1+x)y'' + (-3x-4)y' + 3y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= 1+x \\ B &= -3x-4 \\ C &= 3 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{9x^2 + 12x + 6}{4(1+x)^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 9x^2 + 12x + 6 \\ t &= 4(1+x)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{9x^2 + 12x + 6}{4(1+x)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.282: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4(1+x)^2$. There is a pole at $x = -1$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Therefore

$$L = [1, 2]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{9}{4} + \frac{3}{4(1+x)^2} - \frac{3}{2(1+x)}$$

For the pole at $x = -1$ let b be the coefficient of $\frac{1}{(1+x)^2}$ in the partial fractions decomposition

of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $O_r(\infty) = 0$ then

$$v = \frac{-O_r(\infty)}{2} = \frac{0}{2} = 0$$

$[\sqrt{r}]_\infty$ is the sum of terms involving x^i for $0 \leq i \leq v$ in the Laurent series for $\sqrt{r}$ at ∞ . Therefore

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^v a_i x^i \\ &= \sum_{i=0}^0 a_i x^i \end{aligned} \tag{8}$$

Let a be the coefficient of $x^v = x^0$ in the above sum. The Laurent series of $\sqrt{r}$ at ∞ is

$$\sqrt{r} \approx \frac{3}{2} - \frac{1}{2x} + \frac{2}{3x^2} - \frac{7}{9x^3} + \frac{91}{108x^4} - \frac{283}{324x^5} + \frac{143}{162x^6} - \frac{214}{243x^7} + \dots \tag{9}$$

Comparing Eq. (9) with Eq. (8) shows that

$$a = \frac{3}{2}$$

From Eq. (9) the sum up to $v = 0$ gives

$$\begin{aligned} [\sqrt{r}]_\infty &= \sum_{i=0}^0 a_i x^i \\ &= \frac{3}{2} \end{aligned} \tag{10}$$

Now we need to find b , where b be the coefficient of $x^{v-1} = x^{-1} = \frac{1}{x}$ in r minus the coefficient of same term but in $([\sqrt{r}]_\infty)^2$ where $[\sqrt{r}]_\infty$ was found above in Eq (10). Hence

$$([\sqrt{r}]_\infty)^2 = \frac{9}{4}$$

This shows that the coefficient of $\frac{1}{x}$ in the above is 0. Now we need to find the coefficient of $\frac{1}{x}$ in r . How this is done depends on if $v = 0$ or not. Since $v = 0$ then starting from $r = \frac{s}{t}$ and doing long division in the form

$$r = Q + \frac{R}{t}$$

Where Q is the quotient and R is the remainder. Then the coefficient of $\frac{1}{x}$ in r will be the coefficient in R of the term in x of degree of t minus one, divided by the leading coefficient in t . Doing long division gives

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{9x^2 + 12x + 6}{4x^2 + 8x + 4} \\ &= Q + \frac{R}{4x^2 + 8x + 4} \\ &= \left(\frac{9}{4}\right) + \left(\frac{-6x - 3}{4x^2 + 8x + 4}\right) \\ &= \frac{9}{4} + \frac{-6x - 3}{4x^2 + 8x + 4} \end{aligned}$$

Since the degree of t is 2, then we see that the coefficient of the term x in the remainder R is -6 . Dividing this by leading coefficient in t which is 4 gives $-\frac{3}{2}$. Now b can be found.

$$\begin{aligned} b &= \left(-\frac{3}{2}\right) - (0) \\ &= -\frac{3}{2} \end{aligned}$$

Hence

$$\begin{aligned} [\sqrt{r}]_{\infty} &= \frac{3}{2} \\ \alpha_{\infty}^{+} &= \frac{1}{2} \left(\frac{b}{a} - v \right) = \frac{1}{2} \left(\frac{-\frac{3}{2}}{\frac{3}{2}} - 0 \right) = -\frac{1}{2} \\ \alpha_{\infty}^{-} &= \frac{1}{2} \left(-\frac{b}{a} - v \right) = \frac{1}{2} \left(-\frac{-\frac{3}{2}}{\frac{3}{2}} - 0 \right) = \frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{9x^2 + 12x + 6}{4(1+x)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^{+}	α_c^{-}
-1	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_{\infty}$	α_{∞}^{+}	α_{∞}^{-}
0	$\frac{3}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^{\pm}$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_{\infty}$ and its associated $\alpha_{\infty}^{\pm}$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_{\infty}^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^+ = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^+ - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c) [\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty) [\sqrt{r}]_\infty$$

Substituting the above values in the above results in

$$\begin{aligned} \omega &= \left((-) [\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (+) [\sqrt{r}]_\infty \\ &= -\frac{1}{2(1+x)} + \left(\frac{3}{2} \right) \\ &= -\frac{1}{2(1+x)} + \frac{3}{2} \\ &= \frac{3x+2}{2x+2} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2 \left(-\frac{1}{2(1+x)} + \frac{3}{2} \right) (0) + \left(\left(\frac{1}{2(1+x)^2} \right) + \left(-\frac{1}{2(1+x)} + \frac{3}{2} \right)^2 - \left(\frac{9x^2 + 12x + 6}{4(1+x)^2} \right) \right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int \left(-\frac{1}{2(1+x)} + \frac{3}{2} \right) dx} \\ &= \frac{e^{\frac{3x}{2}}}{\sqrt{1+x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-3x-4}{1+x} dx} \\ &= z_1 e^{\frac{3x}{2} + \frac{\ln(1+x)}{2}} \\ &= z_1 \left(\sqrt{1+x} e^{\frac{3x}{2}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = e^{3x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-3x-4}{1+x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{3x+\ln(1+x)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{(3x+4) e^{3x+\ln(1+x)} e^{-6x}}{9(1+x)} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (e^{3x}) + c_2 \left(e^{3x} \left(-\frac{(3x+4) e^{3x+\ln(1+x)} e^{-6x}}{9(1+x)} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(1+x)y'' + (-3x-4)y' + 3y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 e^{3x} + c_2 \left(-\frac{x}{3} - \frac{4}{9} \right)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = e^{3x}$$

$$y_2 = -\frac{x}{3} - \frac{4}{9}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} e^{3x} & -\frac{x}{3} - \frac{4}{9} \\ \frac{d}{dx}(e^{3x}) & \frac{d}{dx}\left(-\frac{x}{3} - \frac{4}{9}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} e^{3x} & -\frac{x}{3} - \frac{4}{9} \\ 3e^{3x} & -\frac{1}{3} \end{vmatrix}$$

Therefore

$$W = (e^{3x}) \left(-\frac{1}{3} \right) - \left(-\frac{x}{3} - \frac{4}{9} \right) (3e^{3x})$$

Which simplifies to

$$W = x e^{3x} + e^{3x}$$

Which simplifies to

$$W = (1 + x) e^{3x}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\left(-\frac{x}{3} - \frac{4}{9}\right) (3x + 2) e^{3x}}{(1 + x)^2 e^{3x}} dx$$

Which simplifies to

$$u_1 = - \int \frac{-9x^2 - 18x - 8}{9(1 + x)^2} dx$$

Hence

$$u_1 = x + \frac{1}{9x + 9}$$

And Eq. (3) becomes

$$u_2 = \int \frac{e^{6x}(3x + 2)}{(1 + x)^2 e^{3x}} dx$$

Which simplifies to

$$u_2 = \int \frac{e^{3x}(3x + 2)}{(1 + x)^2} dx$$

Hence

$$u_2 = \frac{e^{3x}}{1 + x}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(x + \frac{1}{9x + 9}\right) e^{3x} + \frac{\left(-\frac{x}{3} - \frac{4}{9}\right) e^{3x}}{1 + x}$$

Which simplifies to

$$y_p(x) = \frac{(3x - 1) e^{3x}}{3}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 e^{3x} + c_2 \left(-\frac{x}{3} - \frac{4}{9}\right)\right) + \left(\frac{(3x - 1) e^{3x}}{3}\right) \end{aligned}$$

Summary of solutions found

$$y = \frac{(3x-1)e^{3x}}{3} + c_1 e^{3x} + c_2 \left(-\frac{x}{3} - \frac{4}{9} \right)$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 21

```
ode:=(1+x)*diff(diff(y(x),x),x)-(3*x+4)*diff(y(x),x)+3*y(x) = (3*x+2)*exp(3*x);
dsolve(ode,y(x), singsol=all);
```

$$y = (c_1 + x)e^{3x} + \frac{(3x+4)c_2}{3}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Reducible group (found another exponential solution)
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.223 (sec). Leaf size: 273

```
ode=(x+1)*D[y[x],{x,2}]- (3*x+4)*D[y[x],x]+3*y[x]==(3*x+2)*Exp[3*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \exp \left(\int_1^x \frac{3K[1]+2}{2K[1]+2} dK[1] - \frac{1}{2} \int_1^x \left(-3 - \frac{1}{K[2]+1} \right) dK[2] \right) \left(\int_1^x -\frac{\exp \left(3K[4] + \int_1^{K[4]} \frac{3K[1]+2}{2K[1]+2} dK[1] \right)}{\dots} dK[4] \right)$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x + 1)*Derivative(y(x), (x, 2)) - (3*x + 2)*exp(3*x) - (3*x + 4)*
    Derivative(y(x), x) + 3*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (-3*x*exp(3*x) + x*
    Derivative(y(x), (x, 2)) + 3*y(x) - 2*exp(3*x) + Derivative(y(x), (x, 2)))/(3*
    x + 4) cannot be solved by the factorable group method
```

2.13.14 Problem 36**Local contents**

Solved as second order ode using change of variable on y method 1 .	2303
Solved as second order Bessel ode	2306
Solved as second order ode using Kovacic algorithm	2307
✓ Maple	2309
✓ Mathematica	2312
✓ Sympy	2312

Internal problem ID [8044]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 36

Date solved : Thursday, November 27, 2025 at 07:32:45 AM

CAS classification : [[_2nd_order, _with_linear_symmetries]]

Solved as second order ode using change of variable on y method 1

Time used: 0.113 (sec)

Solve

$$x^2 y'' - 4y'x + (9x^2 + 6)y = 0$$

In normal form the given ode is written as

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = -\frac{4}{x}$$

$$q(x) = \frac{9x^2 + 6}{x^2}$$

Calculating the Liouville ode invariant Q given by

$$\begin{aligned}
 Q &= q - \frac{p'}{2} - \frac{p^2}{4} \\
 &= \frac{9x^2 + 6}{x^2} - \frac{\left(-\frac{4}{x}\right)'}{2} - \frac{\left(-\frac{4}{x}\right)^2}{4} \\
 &= \frac{9x^2 + 6}{x^2} - \frac{\left(\frac{4}{x^2}\right)}{2} - \frac{\left(\frac{16}{x^2}\right)}{4} \\
 &= \frac{9x^2 + 6}{x^2} - \left(\frac{2}{x^2}\right) - \frac{4}{x^2} \\
 &= 9
 \end{aligned}$$

Since the Liouville ode invariant does not depend on the independent variable x then the transformation

$$y = v(x) z(x) \quad (3)$$

is used to change the original ode to a constant coefficients ode in v . In (3) the term $z(x)$ is given by

$$\begin{aligned}
 z(x) &= e^{-\int \frac{p(x)}{2} dx} \\
 &= e^{-\int \frac{-4}{2} dx} \\
 &= x^2
 \end{aligned} \quad (5)$$

Hence (3) becomes

$$y = v(x) x^2 \quad (4)$$

Applying this change of variable to the original ode results in

$$x^4(v''(x) + 9v(x)) = 0$$

Which is now solved for $v(x)$.

The above ode can be simplified to

$$v''(x) + 9v(x) = 0$$

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Av''(x) + Bv'(x) + Cv(x) = 0$$

Where in the above $A = 1, B = 0, C = 9$. Let the solution be $v(x) = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 9e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 9 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 9$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(9)} \\ &= \pm 3i \end{aligned}$$

Hence

$$\lambda_1 = +3i$$

$$\lambda_2 = -3i$$

Which simplifies to

$$\lambda_1 = 3i$$

$$\lambda_2 = -3i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 3$. Therefore the final solution, when using Euler relation, can be written as

$$v(x) = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$v(x) = e^0 (c_1 \cos(3x) + c_2 \sin(3x))$$

Or

$$v(x) = c_1 \cos(3x) + c_2 \sin(3x)$$

Now that $v(x)$ is known, then

$$\begin{aligned} y &= v(x) z(x) \\ &= (c_1 \cos(3x) + c_2 \sin(3x)) (z(x)) \end{aligned} \quad (7)$$

But from (5)

$$z(x) = x^2$$

Hence (7) becomes

$$y = (c_1 \cos(3x) + c_2 \sin(3x)) x^2$$

Summary of solutions found

$$y = (c_1 \cos(3x) + c_2 \sin(3x)) x^2$$

Solved as second order Bessel ode

Time used: 0.050 (sec)

Solve

$$x^2 y'' - 4y'x + (9x^2 + 6)y = 0$$

Writing the ode as

$$x^2 y'' - 4y'x + (9x^2 + 6)y = 0 \quad (1)$$

Bessel ode has the form

$$x^2 y'' + y'x + (-n^2 + x^2)y = 0 \quad (2)$$

The generalized form of Bessel ode is given by Bowman (1958) as the following

$$x^2 y'' + (1 - 2\alpha)xy' + (\beta^2 \gamma^2 x^{2\gamma} - n^2 \gamma^2 + \alpha^2)y = 0 \quad (3)$$

With the standard solution

$$y = x^\alpha (c_1 \text{BesselJ}(n, \beta x^\gamma) + c_2 \text{BesselY}(n, \beta x^\gamma)) \quad (4)$$

Comparing (3) to (1) and solving for α, β, n, γ gives

$$\begin{aligned} \alpha &= \frac{5}{2} \\ \beta &= 3 \\ n &= -\frac{1}{2} \\ \gamma &= 1 \end{aligned}$$

Substituting all the above into (4) gives the solution as

$$y = \frac{c_1 x^2 \sqrt{2} \sqrt{3} \cos(3x)}{3\sqrt{\pi}} + \frac{c_2 x^2 \sqrt{2} \sqrt{3} \sin(3x)}{3\sqrt{\pi}}$$

Summary of solutions found

$$y = \frac{c_1 x^2 \sqrt{2} \sqrt{3} \cos(3x)}{3\sqrt{\pi}} + \frac{c_2 x^2 \sqrt{2} \sqrt{3} \sin(3x)}{3\sqrt{\pi}}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.086 (sec)

Solve

$$x^2 y'' - 4y'x + (9x^2 + 6)y = 0$$

Writing the ode as

$$x^2 y'' - 4y'x + (9x^2 + 6)y = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 \\ B &= -4x \\ C &= 9x^2 + 6 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-9}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -9 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -9z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.283: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -9$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(3x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-4x}{x^2} dx} \\ &= z_1 e^{2 \ln(x)} \\ &= z_1 (x^2) \end{aligned}$$

Which simplifies to

$$y_1 = x^2 \cos(3x)$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{4x}{x^2} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{4\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{\tan(3x)}{3} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (x^2 \cos(3x)) + c_2 \left(x^2 \cos(3x) \left(\frac{\tan(3x)}{3} \right) \right) \end{aligned}$$

Summary of solutions found

$$y = c_1 x^2 \cos(3x) + \frac{c_2 x^2 \sin(3x)}{3}$$

✓ Maple. Time used: 0.003 (sec). Leaf size: 21

```
ode:=x^2*diff(diff(y(x),x),x)-4*diff(y(x),x)*x+(9*x^2+6)*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = x^2(c_1 \sin(3x) + c_2 \cos(3x))$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
```

```

checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Group is reducible or imprimitive
<- Kovacic's algorithm successful

```

Maple step by step

Let's solve

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - 4x \left(\frac{d}{dx} y(x) \right) + (9x^2 + 6) y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{3(3x^2+2)y(x)}{x^2} + \frac{4\left(\frac{d}{dx} y(x)\right)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{4\left(\frac{d}{dx} y(x)\right)}{x} + \frac{3(3x^2+2)y(x)}{x^2} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- Define functions

$$\left[P_2(x) = -\frac{4}{x}, P_3(x) = \frac{3(3x^2+2)}{x^2} \right]$$

- $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -4$$

- $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 6$$

- $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - 4x \left(\frac{d}{dx} y(x) \right) + (9x^2 + 6) y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $x^m \cdot y(x)$ to series expansion for $m = 0..2$

$$x^m \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+m}$$

- Shift index using $k \rightarrow k - m$

$$x^m \cdot y(x) = \sum_{k=m}^{\infty} a_{k-m} x^{k+r}$$

- Convert $x \cdot \left(\frac{d}{dx}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=0}^{\infty} a_k(k+r) x^{k+r}$$

- Convert $x^2 \cdot \left(\frac{d^2}{dx^2}y(x)\right)$ to series expansion

$$x^2 \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=0}^{\infty} a_k(k+r)(k+r-1) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0(-2+r)(-3+r)x^r + a_1(-1+r)(-2+r)x^{1+r} + \left(\sum_{k=2}^{\infty} (a_k(k+r-2)(k+r-3) + 9a_{k-2})x^{k+r}\right)$$

- a_0 cannot be 0 by assumption, giving the indicial equation
 $(-2+r)(-3+r) = 0$
- Values of r that satisfy the indicial equation
 $r \in \{2, 3\}$
- Each term must be 0
 $a_1(-1+r)(-2+r) = 0$
- Solve for the dependent coefficient(s)
 $a_1 = 0$
- Each term in the series must be 0, giving the recursion relation
 $a_k(k+r-2)(k+r-3) + 9a_{k-2} = 0$
- Shift index using $k \rightarrow k+2$
 $a_{k+2}(k+r)(k+r-1) + 9a_k = 0$
- Recursion relation that defines series solution to ODE
 $a_{k+2} = -\frac{9a_k}{(k+r)(k+r-1)}$
- Recursion relation for $r = 2$
 $a_{k+2} = -\frac{9a_k}{(k+2)(k+1)}$
- Solution for $r = 2$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+2}, a_{k+2} = -\frac{9a_k}{(k+2)(k+1)}, a_1 = 0 \right]$
- Recursion relation for $r = 3$
 $a_{k+2} = -\frac{9a_k}{(k+3)(k+2)}$
- Solution for $r = 3$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+3}, a_{k+2} = -\frac{9a_k}{(k+3)(k+2)}, a_1 = 0 \right]$
- Combine solutions and rename parameters
 $\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^{k+2}\right) + \left(\sum_{k=0}^{\infty} b_k x^{k+3}\right), a_{k+2} = -\frac{9a_k}{(k+2)(k+1)}, a_1 = 0, b_{k+2} = -\frac{9b_k}{(k+3)(k+2)}, b_1 = 0 \right]$

✓ **Mathematica.** Time used: 0.035 (sec). Leaf size: 37

```
ode=x^2*D[y[x],{x,2}]-4*x*D[y[x],x]+(6+9*x^2)*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{1}{6}e^{-3ix}x^2(6c_1 - ic_2e^{6ix})$$

✓ **Sympy.** Time used: 0.139 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) - 4*x*Derivative(y(x), x) + (9*x**2 + 6)*y
      (x),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = x^{\frac{5}{2}} \left(C_1 J_{\frac{1}{2}}(3x) + C_2 Y_{\frac{1}{2}}(3x) \right)$$

2.13.15 Problem 37

Local contents

Solved as second order ode using change of variable on y method 1	2313
Solved as second order Bessel ode	2318
Solved as second order ode using Kovacic algorithm	2321
✓ Maple	2326
✓ Mathematica	2327
✗ Sympy	2327

Internal problem ID [8045]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 37

Date solved : Thursday, November 27, 2025 at 07:32:45 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

Solved as second order ode using change of variable on y method 1

Time used: 0.214 (sec)

Solve

$$xy'' + 2y' + 4yx = 4$$

This is second order non-homogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$xy'' + 2y' + 4yx = 0$$

In normal form the given ode is written as

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

Where

$$p(x) = \frac{2}{x}$$

$$q(x) = 4$$

Calculating the Liouville ode invariant Q given by

$$\begin{aligned}
 Q &= q - \frac{p'}{2} - \frac{p^2}{4} \\
 &= 4 - \frac{\left(\frac{2}{x}\right)'}{2} - \frac{\left(\frac{2}{x}\right)^2}{4} \\
 &= 4 - \frac{\left(-\frac{2}{x^2}\right)}{2} - \frac{\left(\frac{4}{x^2}\right)}{4} \\
 &= 4 - \left(-\frac{1}{x^2}\right) - \frac{1}{x^2} \\
 &= 4
 \end{aligned}$$

Since the Liouville ode invariant does not depend on the independent variable x then the transformation

$$y = v(x) z(x) \quad (3)$$

is used to change the original ode to a constant coefficients ode in v . In (3) the term $z(x)$ is given by

$$\begin{aligned}
 z(x) &= e^{-\int \frac{p(x)}{2} dx} \\
 &= e^{-\int \frac{2}{x} dx} \\
 &= \frac{1}{x}
 \end{aligned} \quad (5)$$

Hence (3) becomes

$$y = \frac{v(x)}{x} \quad (4)$$

Applying this change of variable to the original ode results in

$$v''(x) + 4v(x) = 0$$

Which is now solved for $v(x)$.

This is second order with constant coefficients homogeneous ODE. In standard form the ODE is

$$Av''(x) + Bv'(x) + Cv(x) = 0$$

Where in the above $A = 1, B = 0, C = 4$. Let the solution be $v(x) = e^{\lambda x}$. Substituting this into the ODE gives

$$\lambda^2 e^{x\lambda} + 4e^{x\lambda} = 0 \quad (1)$$

Since exponential function is never zero, then dividing Eq(2) throughout by $e^{\lambda x}$ gives

$$\lambda^2 + 4 = 0 \quad (2)$$

Equation (2) is the characteristic equation of the ODE. Its roots determine the general solution form. Using the quadratic formula

$$\lambda_{1,2} = \frac{-B}{2A} \pm \frac{1}{2A} \sqrt{B^2 - 4AC}$$

Substituting $A = 1, B = 0, C = 4$ into the above gives

$$\begin{aligned} \lambda_{1,2} &= \frac{0}{(2)(1)} \pm \frac{1}{(2)(1)} \sqrt{0^2 - (4)(1)(4)} \\ &= \pm 2i \end{aligned}$$

Hence

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

Which simplifies to

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

Since roots are complex conjugate of each others, then let the roots be

$$\lambda_{1,2} = \alpha \pm i\beta$$

Where $\alpha = 0$ and $\beta = 2$. Therefore the final solution, when using Euler relation, can be written as

$$v(x) = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Which becomes

$$v(x) = e^0 (c_1 \cos(2x) + c_2 \sin(2x))$$

Or

$$v(x) = c_1 \cos(2x) + c_2 \sin(2x)$$

Now that $v(x)$ is known, then

$$\begin{aligned} y &= v(x) z(x) \\ &= (c_1 \cos(2x) + c_2 \sin(2x)) (z(x)) \end{aligned} \tag{7}$$

But from (5)

$$z(x) = \frac{1}{x}$$

Hence (7) becomes

$$y = \frac{c_1 \cos(2x) + c_2 \sin(2x)}{x}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_1 \cos(2x) + c_2 \sin(2x)}{x}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{\sin(x) \cos(x)}{x}$$

$$y_2 = \frac{2 \cos(x)^2 - 1}{x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{\sin(x) \cos(x)}{x} & \frac{2 \cos(x)^2 - 1}{x} \\ \frac{d}{dx} \left(\frac{\sin(x) \cos(x)}{x} \right) & \frac{d}{dx} \left(\frac{2 \cos(x)^2 - 1}{x} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{\sin(x) \cos(x)}{x} & \frac{2 \cos(x)^2 - 1}{x} \\ \frac{\cos(x)^2}{x} - \frac{\sin(x)}{x^2} - \frac{\sin(x) \cos(x)}{x^2} & -\frac{4 \sin(x) \cos(x)}{x} - \frac{2 \cos(x)^2 - 1}{x^2} \end{vmatrix}$$

Therefore

$$W = \left(\frac{\sin(x) \cos(x)}{x} \right) \left(-\frac{4 \sin(x) \cos(x)}{x} - \frac{2 \cos(x)^2 - 1}{x^2} \right) \\ - \left(\frac{2 \cos(x)^2 - 1}{x} \right) \left(\frac{\cos(x)^2}{x} - \frac{\sin(x)^2}{x} - \frac{\sin(x) \cos(x)}{x^2} \right)$$

Which simplifies to

$$W = -\frac{2 \cos(x)^2 \sin(x)^2 + 2 \cos(x)^4 + \sin(x)^2 - \cos(x)^2}{x^2}$$

Which simplifies to

$$W = -\frac{1}{x^2}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{8 \cos(x)^2 - 4}{x}}{-\frac{1}{x}} dx$$

Which simplifies to

$$u_1 = - \int -4 \cos(2x) dx$$

Hence

$$u_1 = 2 \sin(2x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{4 \sin(x) \cos(x)}{x}}{-\frac{1}{x}} dx$$

Which simplifies to

$$u_2 = \int -2 \sin(2x) dx$$

Hence

$$u_2 = \cos(2x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{2 \sin(2x) \sin(x) \cos(x)}{x} + \frac{(2 \cos(x)^2 - 1) \cos(2x)}{x}$$

Which simplifies to

$$y_p(x) = \frac{1}{x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1 \cos(2x) + c_2 \sin(2x)}{x} \right) + \left(\frac{1}{x} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{c_1 \cos(2x) + c_2 \sin(2x)}{x} + \frac{1}{x}$$

Solved as second order Bessel ode

Time used: 0.147 (sec)

Solve

$$xy'' + 2y' + 4yx = 4$$

Writing the ode as

$$x^2y'' + 2y'x + 4yx^2 = 4x \quad (1)$$

Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE and y_p is a particular solution to the non-homogeneous ODE. Bessel ode has the form

$$x^2y'' + y'x + (-n^2 + x^2)y = 0 \quad (2)$$

The generalized form of Bessel ode is given by Bowman (1958) as the following

$$x^2y'' + (1 - 2\alpha)xy' + (\beta^2\gamma^2x^{2\gamma} - n^2\gamma^2 + \alpha^2)y = 0 \quad (3)$$

With the standard solution

$$y = x^\alpha (c_1 \text{BesselJ}(n, \beta x^\gamma) + c_2 \text{BesselY}(n, \beta x^\gamma)) \quad (4)$$

Comparing (3) to (1) and solving for α, β, n, γ gives

$$\begin{aligned}\alpha &= -\frac{1}{2} \\ \beta &= 2 \\ n &= \frac{1}{2} \\ \gamma &= 1\end{aligned}$$

Substituting all the above into (4) gives the solution as

$$y = \frac{c_1 \sin(2x)}{x\sqrt{\pi}} - \frac{c_2 \cos(2x)}{x\sqrt{\pi}}$$

Therefore the homogeneous solution y_h is

$$y_h = \frac{c_1 \sin(2x)}{x\sqrt{\pi}} - \frac{c_2 \cos(2x)}{x\sqrt{\pi}}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned}y_1 &= \frac{\sin(2x)}{x\sqrt{\pi}} \\ y_2 &= -\frac{\cos(2x)}{x\sqrt{\pi}}\end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{\sin(2x)}{x\sqrt{\pi}} & -\frac{\cos(2x)}{x\sqrt{\pi}} \\ \frac{d}{dx} \left(\frac{\sin(2x)}{x\sqrt{\pi}} \right) & \frac{d}{dx} \left(-\frac{\cos(2x)}{x\sqrt{\pi}} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{\sin(2x)}{x\sqrt{\pi}} & -\frac{\cos(2x)}{x\sqrt{\pi}} \\ -\frac{\sin(2x)}{x^2\sqrt{\pi}} + \frac{2\cos(2x)}{x\sqrt{\pi}} & \frac{\cos(2x)}{x^2\sqrt{\pi}} + \frac{2\sin(2x)}{x\sqrt{\pi}} \end{vmatrix}$$

Therefore

$$W = \left(\frac{\sin(2x)}{x\sqrt{\pi}} \right) \left(\frac{\cos(2x)}{x^2\sqrt{\pi}} + \frac{2\sin(2x)}{x\sqrt{\pi}} \right) - \left(-\frac{\cos(2x)}{x\sqrt{\pi}} \right) \left(-\frac{\sin(2x)}{x^2\sqrt{\pi}} + \frac{2\cos(2x)}{x\sqrt{\pi}} \right)$$

Which simplifies to

$$W = \frac{2\cos(2x)^2 + 2\sin(2x)^2}{x^2\pi}$$

Which simplifies to

$$W = \frac{2}{x^2\pi}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{-\frac{4\cos(2x)}{\sqrt{\pi}}}{\frac{2}{\pi}} dx$$

Which simplifies to

$$u_1 = - \int -2\sqrt{\pi} \cos(2x) dx$$

Hence

$$u_1 = \sqrt{\pi} \sin(2x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{4\sin(2x)}{\sqrt{\pi}}}{\frac{2}{\pi}} dx$$

Which simplifies to

$$u_2 = \int 2\sqrt{\pi} \sin(2x) dx$$

Hence

$$u_2 = -\sqrt{\pi} \cos(2x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\sin(2x)^2}{x} + \frac{\cos(2x)^2}{x}$$

Which simplifies to

$$y_p(x) = \frac{1}{x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1 \sin(2x)}{x\sqrt{\pi}} - \frac{c_2 \cos(2x)}{x\sqrt{\pi}} \right) + \left(\frac{1}{x} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{c_1 \sin(2x)}{x\sqrt{\pi}} - \frac{c_2 \cos(2x)}{x\sqrt{\pi}} + \frac{1}{x}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.257 (sec)

Solve

$$xy'' + 2y' + 4yx = 4$$

Writing the ode as

$$xy'' + 2y' + 4yx = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= 2 \\ C &= 4x \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-4}{1} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -4 \\ t &= 1 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = -4z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.285: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

There are no poles in r . Therefore the set of poles Γ is empty. Since there is no odd order pole larger than 2 and the order at ∞ is 0 then the necessary conditions for case one are met. Therefore

$$L = [1]$$

Since $r = -4$ is not a function of x , then there is no need run Kovacic algorithm to obtain a solution for transformed ode $z'' = rz$ as one solution is

$$z_1(x) = \cos(2x)$$

Using the above, the solution for the original ode can now be found. The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{2}{x} dx} \\ &= z_1 e^{-\ln(x)} \\ &= z_1 \left(\frac{1}{x} \right) \end{aligned}$$

Which simplifies to

$$y_1 = \frac{\cos(2x)}{x}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{2}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-2\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{\tan(2x)}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{\cos(2x)}{x} \right) + c_2 \left(\frac{\cos(2x)}{x} \left(\frac{\tan(2x)}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$xy'' + 2y' + 4yx = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1 \cos(2x)}{x} + \frac{c_2 \sin(2x)}{2x}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{\cos(2x)}{x}$$

$$y_2 = \frac{\sin(2x)}{2x}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{\cos(2x)}{x} & \frac{\sin(2x)}{2x} \\ \frac{d}{dx} \left(\frac{\cos(2x)}{x} \right) & \frac{d}{dx} \left(\frac{\sin(2x)}{2x} \right) \end{vmatrix}$$

Which gives

$$W = \left| \begin{array}{cc} \frac{\cos(2x)}{x} & \frac{\sin(2x)}{2x} \\ -\frac{\cos(2x)}{x^2} - \frac{2\sin(2x)}{x} & -\frac{\sin(2x)}{2x^2} + \frac{\cos(2x)}{x} \end{array} \right|$$

Therefore

$$W = \left(\frac{\cos(2x)}{x} \right) \left(-\frac{\sin(2x)}{2x^2} + \frac{\cos(2x)}{x} \right) - \left(\frac{\sin(2x)}{2x} \right) \left(-\frac{\cos(2x)}{x^2} - \frac{2\sin(2x)}{x} \right)$$

Which simplifies to

$$W = \frac{\cos(2x)^2 + \sin(2x)^2}{x^2}$$

Which simplifies to

$$W = \frac{1}{x^2}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{2\sin(2x)}{x}}{\frac{1}{x}} dx$$

Which simplifies to

$$u_1 = - \int 2 \sin(2x) dx$$

Hence

$$u_1 = \cos(2x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{4\cos(2x)}{x}}{\frac{1}{x}} dx$$

Which simplifies to

$$u_2 = \int 4 \cos(2x) dx$$

Hence

$$u_2 = 2 \sin(2x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\sin(2x)^2}{x} + \frac{\cos(2x)^2}{x}$$

Which simplifies to

$$y_p(x) = \frac{1}{x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\frac{c_1 \cos(2x)}{x} + \frac{c_2 \sin(2x)}{2x} \right) + \left(\frac{1}{x} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{1}{x} + \frac{c_1 \cos(2x)}{x} + \frac{c_2 \sin(2x)}{2x}$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 22

```
ode:=x*diff(diff(y(x),x),x)+2*diff(y(x),x)+4*y(x)*x = 4;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{1 + \sin(2x) c_2 + \cos(2x) c_1}{x}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
  checking if the LODE has constant coefficients
  checking if the LODE is of Euler type
  trying a symmetry of the form [xi=0, eta=F(x)]
  checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
  A Liouvillian solution exists
  Group is reducible or imprimitive
  <- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.032 (sec). Leaf size: 38

```
ode=x*D[y[x],{x,2}]+2*D[y[x],x]+4*x*y[x]==4;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{4c_1 e^{-2ix} - ic_2 e^{2ix} + 4}{4x}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x*y(x) + x*Derivative(y(x), (x, 2)) + 2*Derivative(y(x), x) - 4, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE 2*x*y(x) + x*Derivative(y(x), (x, 2))/2 +
Derivative(y(x), x) - 2 cannot be solved by the factorable group method
```

2.13.16 Problem 38**Local contents**

Solved as second order ode using change of variable on y method 2 .	2328
Solved as second order ode using non constant coeff transformation on	
B method	2333
Solved as second order ode using Kovacic algorithm	2337
✓ Maple	2345
✓ Mathematica	2346
✗ Sympy	2346

Internal problem ID [8046]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 18. Linear equations with variable coefficients (Equations of second order).

Supplementary problems. Page 120

Problem number : 38

Date solved : Thursday, November 27, 2025 at 07:32:46 AM

CAS classification : `[_2nd_order, _with_linear_symmetries]`

Solved as second order ode using change of variable on y method 2

Time used: 0.243 (sec)

Solve

$$(x^2 + 1)y'' - 2y'x + 2y = \frac{-x^2 + 1}{x}$$

This is second order non-homogeneous ODE. In standard form the ODE is

$$Ay''(x) + By'(x) + Cy(x) = f(x)$$

Where $A = x^2 + 1$, $B = -2x$, $C = 2$, $f(x) = -x + \frac{1}{x}$. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the non-homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. Solving for y_h from

$$(x^2 + 1)y'' - 2y'x + 2y = 0$$

In normal form the ode

$$(x^2 + 1) y'' - 2y'x + 2y = 0 \quad (1)$$

Becomes

$$y'' + p(x) y' + q(x) y = 0 \quad (2)$$

Where

$$p(x) = -\frac{2x}{x^2 + 1}$$

$$q(x) = \frac{2}{x^2 + 1}$$

Applying change of variables on the dependent variable $y = v(x) x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not y .

$$v''(x) + \left(\frac{2n}{x} + p\right) v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right) v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{2n}{x^2 + 1} + \frac{2}{x^2 + 1} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$v''(x) + \left(\frac{2}{x} - \frac{2x}{x^2 + 1}\right) v'(x) = 0$$

$$v''(x) + \frac{2v'(x)}{x(x^2 + 1)} = 0 \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{2u(x)}{x(x^2 + 1)} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x(x^2 + 1)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int \frac{2}{x(x^2+1)} dx} \\ &= \frac{x^2}{x^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 1} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u x^2}{x^2 + 1} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2+1}$ gives the final solution

$$u(x) = \frac{(x^2 + 1) c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned} v'(x) &= u(x) \\ v(x) &= \int u(x) dx + c_2 \\ &= c_1 \left(x - \frac{1}{x} \right) + c_2 \end{aligned}$$

Hence

$$\begin{aligned} y &= v(x) x^n \\ &= \left(c_1 \left(x - \frac{1}{x} \right) + c_2 \right) x \\ &= c_1 x^2 + c_2 x - c_1 \end{aligned}$$

Now the particular solution to this ODE is found

$$(x^2 + 1) y'' - 2y'x + 2y = -x + \frac{1}{x}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = x$$

$$y_2 = x^2 - 1$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & x^2 - 1 \\ \frac{d}{dx}(x) & \frac{d}{dx}(x^2 - 1) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & x^2 - 1 \\ 1 & 2x \end{vmatrix}$$

Therefore

$$W = (x)(2x) - (x^2 - 1)(1)$$

Which simplifies to

$$W = x^2 + 1$$

Which simplifies to

$$W = x^2 + 1$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(x^2 - 1) \left(-x + \frac{1}{x}\right)}{(x^2 + 1)^2} dx$$

Which simplifies to

$$u_1 = - \int - \frac{(x^2 - 1)^2}{x (x^2 + 1)^2} dx$$

Hence

$$u_1 = \frac{2}{x^2 + 1} + \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{x \left(-x + \frac{1}{x}\right)}{(x^2 + 1)^2} dx$$

Which simplifies to

$$u_2 = \int \frac{-x^2 + 1}{(x^2 + 1)^2} dx$$

Hence

$$u_2 = \frac{x}{x^2 + 1}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{2}{x^2 + 1} + \ln(x) \right) x + \frac{(x^2 - 1)x}{x^2 + 1}$$

Which simplifies to

$$y_p(x) = (1 + \ln(x)) x$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(\left(c_1 \left(x - \frac{1}{x} \right) + c_2 \right) x \right) + ((1 + \ln(x)) x) \end{aligned}$$

Summary of solutions found

$$y = (1 + \ln(x))x + \left(c_1 \left(x - \frac{1}{x}\right) + c_2\right)x$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.286 (sec)

Solve

$$(x^2 + 1)y'' - 2y'x + 2y = \frac{-x^2 + 1}{x}$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \tag{1}$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$A = x^2 + 1$$

$$B = -2x$$

$$C = 2$$

$$F = -x + \frac{1}{x}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x^2 + 1)(0) + (-2x)(-2) + (2)(-2x) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-2x(x^2 + 1)v'' + (-4)v' = 0$$

Now by applying $v' = u$ the above becomes

$$(-2x^3 - 2x)u'(x) - 4u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2}{x(x^2 + 1)} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2}{x(x^2 + 1)} \, dx} \\ &= \frac{x^2}{x^2 + 1} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u x^2}{x^2 + 1} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{u x^2}{x^2 + 1} &= \int 0 dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{x^2+1}$ gives the final solution

$$u(x) = \frac{(x^2 + 1) c_1}{x^2}$$

The ode for v now becomes

$$v'(x) = \frac{(x^2 + 1) c_1}{x^2}$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dv &= \int \frac{(x^2 + 1) c_1}{x^2} dx \\ v(x) &= c_1 \left(x - \frac{1}{x} \right) + c_2\end{aligned}$$

$$v(x) = \frac{c_1(x^2 - 1)}{x} + c_2$$

Replacing $v(x)$ above by $-\frac{y}{2x}$, then the homogeneous solution is

$$\begin{aligned}y_h(x) &= Bv \\ &= -2c_1 x^2 - 2c_2 x + 2c_1\end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \tag{1}$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned}y_1 &= x \\ y_2 &= -2x^2 + 2\end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} x & -2x^2 + 2 \\ \frac{d}{dx}(x) & \frac{d}{dx}(-2x^2 + 2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} x & -2x^2 + 2 \\ 1 & -4x \end{vmatrix}$$

Therefore

$$W = (x)(-4x) - (-2x^2 + 2)(1)$$

Which simplifies to

$$W = -2x^2 - 2$$

Which simplifies to

$$W = -2x^2 - 2$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{(-2x^2 + 2)(-x + \frac{1}{x})}{(x^2 + 1)(-2x^2 - 2)} dx$$

Which simplifies to

$$u_1 = - \int - \frac{(x^2 - 1)^2}{x(x^2 + 1)^2} dx$$

Hence

$$u_1 = \frac{2}{x^2 + 1} + \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{x(-x + \frac{1}{x})}{(x^2 + 1)(-2x^2 - 2)} dx$$

Which simplifies to

$$u_2 = \int \frac{x^2 - 1}{2(x^2 + 1)^2} dx$$

Hence

$$u_2 = -\frac{x}{2(x^2 + 1)}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \left(\frac{2}{x^2 + 1} + \ln(x) \right) x - \frac{(-2x^2 + 2)x}{2(x^2 + 1)}$$

Which simplifies to

$$y_p(x) = (1 + \ln(x))x$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= (-2c_1 x^2 - 2c_2 x + 2c_1) + ((1 + \ln(x))x) \\ &= -2c_1 x^2 + \ln(x)x - 2c_2 x + 2c_1 + x \end{aligned}$$

Summary of solutions found

$$y = -2c_1 x^2 + \ln(x)x - 2c_2 x + 2c_1 + x$$

Solved as second order ode using Kovacic algorithm

Time used: 0.335 (sec)

Solve

$$(x^2 + 1)y'' - 2y'x + 2y = \frac{-x^2 + 1}{x}$$

Writing the ode as

$$(x^2 + 1)y'' - 2y'x + 2y = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 + 1 \\ B &= -2x \\ C &= 2 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \tag{5}$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{-3}{(x^2 + 1)^2} \tag{6}$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= -3 \\ t &= (x^2 + 1)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(-\frac{3}{(x^2 + 1)^2} \right) z(x) \tag{7}$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.286: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned}
 \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\
 &= 4 - 0 \\
 &= 4
 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = (x^2 + 1)^2$. There is a pole at $x = i$ of order 2. There is a pole at $x = -i$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 4 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 4 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4(x-i)^2} + \frac{3}{4(x+i)^2} + \frac{3i}{4(x-i)} - \frac{3i}{4(x+i)}$$

For the pole at $x = i$ let b be the coefficient of $\frac{1}{(x-i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned}
 [\sqrt{r}]_c &= 0 \\
 \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\
 \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2}
 \end{aligned}$$

For the pole at $x = -i$ let b be the coefficient of $\frac{1}{(x+i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $4 > 2$ then

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= 0 \\ \alpha_\infty^- &= 1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = -\frac{3}{(x^2 + 1)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
i	2	0	$\frac{3}{2}$	$-\frac{1}{2}$
$-i$	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
4	0	0	1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = 1$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^- + \alpha_{c_2}^+) \\ &= 1 - (1) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x-c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned}
 \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + \left((+)[\sqrt{r}]_{c_2} + \frac{\alpha_{c_2}^+}{x - c_2} \right) + (-)[\sqrt{r}]_{\infty} \\
 &= -\frac{1}{2(x-i)} + \frac{3}{2(x+i)} + (-)(0) \\
 &= -\frac{1}{2(x-i)} + \frac{3}{2(x+i)} \\
 &= \frac{x-2i}{x^2+1}
 \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$(0) + 2\left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)}\right)(0) + \left(\left(\frac{1}{2(x-i)^2} - \frac{3}{2(x+i)^2}\right) + \left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)}\right)^2 - \left(-\frac{1}{x^2}\right)\right) = 0$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned}
 z_1(x) &= p e^{\int \omega dx} \\
 &= e^{\int \left(-\frac{1}{2(x-i)} + \frac{3}{2(x+i)}\right) dx} \\
 &= \frac{(x^2+1)^{3/2}}{(ix+1)^2}
 \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned}
 y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\
 &= z_1 e^{-\int \frac{1}{2} \frac{-2x}{x^2+1} dx} \\
 &= z_1 e^{\frac{\ln(x^2+1)}{2}} \\
 &= z_1 \left(\sqrt{x^2+1} \right)
 \end{aligned}$$

Which simplifies to

$$y_1 = \frac{(x^2 + 1)^2}{(ix + 1)^2}$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-2x}{x^2+1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x^2+1)}}{(y_1)^2} dx \\ &= y_1 \left(-\frac{x}{(x+i)^2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \left(\frac{(x^2 + 1)^2}{(ix + 1)^2} \right) + c_2 \left(\frac{(x^2 + 1)^2}{(ix + 1)^2} \left(-\frac{x}{(x+i)^2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(x^2 + 1) y'' - 2y'x + 2y = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = \frac{c_1(x^2 + 1)^2}{(ix + 1)^2} + \frac{c_2(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be

used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = \frac{(x^2 + 1)^2}{(ix + 1)^2}$$

$$y_2 = \frac{(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} \frac{(x^2+1)^2}{(ix+1)^2} & \frac{(x^2+1)^2 x}{(x-i)^2 (x+i)^2} \\ \frac{d}{dx} \left(\frac{(x^2+1)^2}{(ix+1)^2} \right) & \frac{d}{dx} \left(\frac{(x^2+1)^2 x}{(x-i)^2 (x+i)^2} \right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} \frac{(x^2+1)^2}{(ix+1)^2} & \frac{(x^2+1)^2 x}{(x-i)^2 (x+i)^2} \\ \frac{4(x^2+1)x}{(ix+1)^2} - \frac{2i(x^2+1)^2}{(ix+1)^3} & \frac{4(x^2+1)x^2}{(x-i)^2 (x+i)^2} + \frac{(x^2+1)^2}{(x-i)^2 (x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^3 (x+i)^2} - \frac{2(x^2+1)^2 x}{(x-i)^2 (x+i)^3} \end{vmatrix}$$

Therefore

$$W = \left(\frac{(x^2 + 1)^2}{(ix + 1)^2} \right) \left(\frac{4(x^2 + 1)x^2}{(x - i)^2 (x + i)^2} + \frac{(x^2 + 1)^2}{(x - i)^2 (x + i)^2} - \frac{2(x^2 + 1)^2 x}{(x - i)^3 (x + i)^2} - \frac{2(x^2 + 1)^2 x}{(x - i)^2 (x + i)^3} \right) - \left(\frac{(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2} \right) \left(\frac{4(x^2 + 1)x}{(ix + 1)^2} - \frac{2i(x^2 + 1)^2}{(ix + 1)^3} \right)$$

Which simplifies to

$$W = \frac{(x^2 + 1)^3 (ix^5 + 3x^4 - 2ix^3 + 2x^2 - 3ix - 1)}{(ix + 1)^3 (-x + i)^3 (x + i)^3}$$

Which simplifies to

$$W = \frac{(x^2 + 1)^4 (-x^3 + 3ix^2 + 3x - i)}{(-ix - 1)^6 (x + i)^3}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{(x^2+1)^2 x(-x+\frac{1}{x})}{(x-i)^2(x+i)^2}}{\frac{(x^2+1)^5(-x^3+3ix^2+3x-i)}{(-ix-1)^6(x+i)^3}} dx$$

Which simplifies to

$$u_1 = - \int \frac{(x^2 - 1)(x + i)(x - i)^4}{(x^2 + 1)^3 (-x^3 + 3ix^2 + 3x - i)} dx$$

Hence

$$u_1 = -\frac{1}{2(x-i)} - \frac{1}{2(x+i)}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{(x^2+1)^2(-x+\frac{1}{x})}{(ix+1)^2}}{\frac{(x^2+1)^5(-x^3+3ix^2+3x-i)}{(-ix-1)^6(x+i)^3}} dx$$

Which simplifies to

$$u_2 = \int -\frac{(x+i)^3(ix+1)^4(x^2-1)}{(-x^3+3ix^2+3x-i)x(x^2+1)^3} dx$$

Hence

$$u_2 = -\frac{2i}{x-i} + \ln(x)$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\left(-\frac{1}{2(x-i)} - \frac{1}{2(x+i)}\right)(x^2+1)^2}{(ix+1)^2} + \frac{(x^2+1)^2 x \left(-\frac{2i}{x-i} + \ln(x)\right)}{(x-i)^2(x+i)^2}$$

Which simplifies to

$$y_p(x) = \frac{(x^2 + 1)^2 (1 + \ln(x)) x(-x^3 + 3ix^2 + 3x - i)}{(-x + i)^5 (x + i)^2}$$

Therefore the general solution is

$$y = y_h + y_p \\ = \left(\frac{c_1(x^2 + 1)^2}{(ix + 1)^2} + \frac{c_2(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2} \right) + \left(\frac{(x^2 + 1)^2 (1 + \ln(x)) x(-x^3 + 3ix^2 + 3x - i)}{(-x + i)^5 (x + i)^2} \right)$$

Summary of solutions found

$$y = \frac{(x^2 + 1)^2 (1 + \ln(x)) x(-x^3 + 3ix^2 + 3x - i)}{(-x + i)^5 (x + i)^2} + \frac{c_1(x^2 + 1)^2}{(ix + 1)^2} + \frac{c_2(x^2 + 1)^2 x}{(x - i)^2 (x + i)^2}$$

✓ **Maple.** Time used: 0.005 (sec). Leaf size: 21

```
ode:=2*y(x)-2*diff(y(x),x)*x+(x^2+1)*diff(diff(y(x),x),x) = (-x^2+1)/x;
dsolve(ode,y(x), singsol=all);
```

$$y = c_1 x^2 + x \ln(x) + c_2 x - c_1 + x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
<- Kovacic's algorithm successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.329 (sec). Leaf size: 222

```
ode=(1+x^2)*D[y[x],{x,2}]-2*x*D[y[x],x]+2*y[x]==(1-x^2)/x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \sqrt{x^2 + 1} \exp \left(\int_1^x \frac{K[1] + 2i}{K[1]^2 + 1} dK[1] \right) \left(\int_1^x \frac{\exp \left(\int_1^{K[3]} \frac{K[1] + 2i}{K[1]^2 + 1} dK[1] \right) (K[3]^2 - 1) \int_1^{K[3]} \exp \left(-2 \int_1^{K[2]} \right)}{K[3] (K[3]^2 + 1)^{3/2}} \right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x*Derivative(y(x), x) + (x**2 + 1)*Derivative(y(x), (x, 2)) + 2*y(x) -
        (1 - x**2)/x, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (x**2*(x*Derivative(y(x),
(x, 2)) + 1) + x*(2*y(x) + Derivative(y(x), (x, 2))) - 1)/(2*x**2) cannot
be solved by the factorable group method
```

2.14 Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Local contents

2.14.1 Problem 22	2348
2.14.2 Problem 23	2356
2.14.3 Problem 24	2371
2.14.4 Problem 25	2393
2.14.5 Problem 26	2399
2.14.6 Problem 27	2406
2.14.7 Problem 28	2415
2.14.8 Problem 29	2422
2.14.9 Problem 30	2425
2.14.10 Problem 31	2432
2.14.11 Problem 32	2438
2.14.12 Problem 33	2448
2.14.13 Problem 34	2451
2.14.14 Problem 35	2454
2.14.15 Problem 36	2456

2.14.1 Problem 22

Local contents

Solved as second order missing x ode	2348
Solved as second order missing y ode	2351
✓ Maple	2354
✓ Mathematica	2355
✓ Sympy	2355

Internal problem ID [8047]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 22

Date solved : Thursday, November 27, 2025 at 07:32:47 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_xy]]

Solved as second order missing x ode

Time used: 2.954 (sec)

Solve

$$y'' + y'^2 + 1 = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$p(y) \left(\frac{d}{dy} p(y) \right) + p(y)^2 + 1 = 0$$

Which is now solved as first order ode for $p(y)$.

Solve Integrating gives

$$\int -\frac{p}{p^2 + 1} dp = dy$$

$$-\frac{\ln(p^2 + 1)}{2} = y + c_1$$

Singular solutions are found by solving

$$-\frac{p^2 + 1}{p} = 0$$

for p . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$p = -i$$

$$p = i$$

Solving for p gives

$$p = -i$$

$$p = i$$

$$p = \sqrt{-1 + e^{-2y-2c_1}}$$

$$p = -\sqrt{-1 + e^{-2y-2c_1}}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -i dx$$

$$y = -ix + c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int i dx$$

$$y = ix + c_3$$

For solution (3) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \sqrt{-1 + e^{-2y-2c_1}}$$

Solve Integrating gives

$$\int \frac{1}{\sqrt{-1 + e^{-2y-2c_1}}} dy = dx$$

$$-\arctan\left(\sqrt{-1 + e^{-2y-2c_1}}\right) = x + c_4$$

Singular solutions are found by solving

$$\sqrt{-1 + e^{-2y-2c_1}} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -c_1$$

Simplifying the above gives

$$-\arctan\left(\sqrt{-e^{-2y-2c_1}}(e^{2y+2c_1} - 1)\right) = x + c_4$$

$$y = -c_1$$

For solution (4) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\sqrt{-1 + e^{-2y-2c_1}}$$

Solve Integrating gives

$$\int -\frac{1}{\sqrt{-1 + e^{-2y-2c_1}}} dy = dx$$

$$\arctan\left(\sqrt{-1 + e^{-2y-2c_1}}\right) = x + c_5$$

Singular solutions are found by solving

$$-\sqrt{-1 + e^{-2y-2c_1}} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -c_1$$

Simplifying the above gives

$$\arctan \left(\sqrt{-e^{-2y-2c_1} (e^{2y+2c_1} - 1)} \right) = x + c_5$$

$$y = -c_1$$

Solving for y from the above solution(s) gives (after possible removing of solutions that do not verify)

$$y = -c_1$$

$$y = -c_1 - \frac{\ln (\tan (x+c_4)^2+1)}{2}$$

$$y = -c_1 - \frac{\ln (\tan (x+c_5)^2+1)}{2}$$

$$y = -ix + c_2$$

$$y = ix + c_3$$

The solution

$$y = -c_1$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = -c_1 - \frac{\ln (\tan (x+c_4)^2+1)}{2}$$

$$y = -c_1 - \frac{\ln (\tan (x+c_5)^2+1)}{2}$$

$$y = -ix + c_2$$

$$y = ix + c_3$$

Solved as second order missing y ode

Time used: 0.272 (sec)

Solve

$$y'' + y'^2 + 1 = 0$$

This is second order ode with missing dependent variable y . Let

$$u(x) = y'$$

Then

$$u'(x) = y''$$

Hence the ode becomes

$$u'(x) + u(x)^2 + 1 = 0$$

Which is now solved for $u(x)$ as first order ode.

Solve Integrating gives

$$\begin{aligned} \int \frac{1}{-u^2 - 1} du &= dx \\ -\arctan(u) &= x + c_1 \end{aligned}$$

Singular solutions are found by solving

$$-u^2 - 1 = 0$$

for $u(x)$. This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \end{aligned}$$

Solving for $u(x)$ gives

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \\ u(x) &= -\tan(x + c_1) \end{aligned}$$

In summary, these are the solution found for y

$$\begin{aligned} u(x) &= -i \\ u(x) &= i \\ u(x) &= -\tan(x + c_1) \end{aligned}$$

For solution $u(x) = -i$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = -i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dy &= \int -i dx \\ y &= -ix + c_2 \end{aligned}$$

In summary, these are the solution found for (y)

$$y = -ix + c_2$$

For solution $u(x) = i$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = i$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int i dx$$

$$y = ix + c_3$$

In summary, these are the solution found for (y)

$$y = ix + c_3$$

For solution $u(x) = -\tan(x + c_1)$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = -\tan(x + c_1)$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int -\tan(x + c_1) dx$$

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

In summary, these are the solution found for (y)

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

Summary of solutions found

$$y = -ix + c_2$$

$$y = ix + c_3$$

$$y = -\frac{\ln(1 + \tan(x + c_1)^2)}{2} + c_4$$

✓ **Maple.** Time used: 0.009 (sec). Leaf size: 15

```
ode:=diff(diff(y(x),x),x)+diff(y(x),x)^2+1 = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = \ln(-c_1 \sin(x) + c_2 \cos(x))$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying 2nd order Liouville

trying 2nd order WeierstrassP

trying 2nd order JacobiSN

differential order: 2; trying a linearization to 3rd order

trying 2nd order ODE linearizable_by_differentiation

trying 2nd order, 2 integrating factors of the form $\mu(x,y)$

trying a quadrature

checking if the LODE has constant coefficients

<- constant coefficients successful

<- 2nd order, 2 integrating factors of the form $\mu(x,y)$ successful

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + \left(\frac{d}{dx}y(x)\right)^2 + 1 = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Make substitution $u = \frac{d}{dx}y(x)$ to reduce order of ODE

$$\frac{d}{dx}u(x) + u(x)^2 + 1 = 0$$

- Solve for the highest derivative

$$\frac{d}{dx}u(x) = -u(x)^2 - 1$$

- Separate variables

$$\frac{\frac{d}{dx}u(x)}{-u(x)^2-1} = 1$$

- Integrate both sides with respect to x

$$\int \frac{\frac{d}{dx}u(x)}{-u(x)^2-1} dx = \int 1 dx + C1$$

- Evaluate integral

$$-\arctan(u(x)) = x + C1$$

- Solve for $u(x)$

$$u(x) = -\tan(x + C1)$$

- Solve 1st ODE for $u(x)$

- $u(x) = -\tan(x + C1)$
- Make substitution $u = \frac{d}{dx}y(x)$
 $\frac{d}{dx}y(x) = -\tan(x + C1)$
- Integrate both sides to solve for $y(x)$
 $\int \left(\frac{d}{dx}y(x)\right) dx = \int -\tan(x + C1) dx + C2$
- Compute integrals
 $y(x) = -\frac{\ln(1+\tan(x+C1)^2)}{2} + C2$

✓ **Mathematica.** Time used: 0.423 (sec). Leaf size: 40

```
ode=D[y[x],{x,2}]+D[y[x],x]^2+1==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x \text{InverseFunction} \left[\int_1^{\#1} \frac{1}{K[1]^2 + 1} dK[1] \& \right] [c_1 - K[2]] dK[2] + c_2$$

✓ **Sympy.** Time used: 0.720 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(Derivative(y(x), x)**2 + Derivative(y(x), (x, 2)) + 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$\left[y(x) = C_1 - \frac{\log(\tan^2(C_2 - x) + 1)}{2}, y(x) = C_1 - \frac{\log(\tan^2(C_2 - x) + 1)}{2} \right]$$

2.14.2 Problem 23

Local contents

Solved as second order linear exact ode	2356
Solved as second order missing y ode	2357
Solved as second order integrable as is ode	2359
Solved as second order integrable as is ode (ABC method)	2360
Solved as second order ode using Kovacic algorithm	2361
✓ Maple	2368
✓ Mathematica	2369
✓ Sympy	2370

Internal problem ID [8048]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 23

Date solved : Thursday, November 27, 2025 at 07:32:51 AM

CAS classification : [[_2nd_order, _missing_y]]

Solved as second order linear exact ode

Time used: 0.089 (sec)

Solve

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

An ode of the form

$$p(x) y'' + q(x) y' + r(x) y = s(x)$$

is exact if

$$p''(x) - q'(x) + r(x) = 0 \tag{1}$$

For the given ode we have

$$p(x) = x^2 + 1$$

$$q(x) = 2x$$

$$r(x) = 0$$

$$s(x) = \frac{2}{x^3}$$

Hence

$$\begin{aligned} p''(x) &= 2 \\ q'(x) &= 2 \end{aligned}$$

Therefore (1) becomes

$$2 - (2) + (0) = 0$$

Hence the ode is exact. Since we now know the ode is exact, it can be written as

$$(p(x) y' + (q(x) - p'(x)) y)' = s(x)$$

Integrating gives

$$p(x) y' + (q(x) - p'(x)) y = \int s(x) dx$$

Substituting the above values for p, q, r, s gives

$$(x^2 + 1) y' = \int \frac{2}{x^3} dx$$

We now have a first order ode to solve which is

$$(x^2 + 1) y' = -\frac{1}{x^2} + c_1$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dy &= \int \frac{c_1 x^2 - 1}{x^2 (x^2 + 1)} dx \\ y &= (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2 \end{aligned}$$

Summary of solutions found

$$y = \frac{1}{x} + (c_1 + 1) \arctan(x) + c_2$$

Solved as second order missing y ode

Time used: 0.115 (sec)

Solve

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

This is second order ode with missing dependent variable y . Let

$$u(x) = y'$$

Then

$$u'(x) = y''$$

Hence the ode becomes

$$(x^2 + 1) u'(x) + 2u(x) x - \frac{2}{x^3} = 0$$

Which is now solved for $u(x)$ as first order ode.

Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= \frac{2x}{x^2 + 1} \\ p(x) &= \frac{2}{(x^2 + 1)x^3} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int \frac{2x}{x^2+1} \, dx} \\ &= x^2 + 1 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx}(\mu u) &= \mu p \\ \frac{d}{dx}(\mu u) &= (\mu) \left(\frac{2}{(x^2 + 1)x^3} \right) \\ \frac{d}{dx}(u(x^2 + 1)) &= (x^2 + 1) \left(\frac{2}{(x^2 + 1)x^3} \right) \\ d(u(x^2 + 1)) &= \left(\frac{2}{x^3} \right) dx \end{aligned}$$

Integrating gives

$$\begin{aligned} u(x^2 + 1) &= \int \frac{2}{x^3} \, dx \\ &= -\frac{1}{x^2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $x^2 + 1$ gives the final solution

$$u(x) = \frac{-\frac{1}{x^2} + c_1}{x^2 + 1}$$

In summary, these are the solution found for y

$$u(x) = \frac{-\frac{1}{x^2} + c_1}{x^2 + 1}$$

For solution $u(x) = \frac{-\frac{1}{x^2} + c_1}{x^2 + 1}$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{-\frac{1}{x^2} + c_1}{x^2 + 1}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned} \int dy &= \int \frac{c_1 x^2 - 1}{x^2 (x^2 + 1)} dx \\ y &= (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2 \end{aligned}$$

In summary, these are the solution found for (y)

$$y = \frac{1}{x} + (c_1 + 1) \arctan(x) + c_2$$

Summary of solutions found

$$y = \frac{1}{x} + (c_1 + 1) \arctan(x) + c_2$$

Solved as second order integrable as is ode

Time used: 0.063 (sec)

Solve

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

Integrating both sides of the ODE w.r.t x gives

$$\begin{aligned} \int ((x^2 + 1) y'' + 2y'x) dx &= \int \frac{2}{x^3} dx \\ (x^2 + 1) y' &= -\frac{1}{x^2} + c_1 \end{aligned}$$

Which is now solved for y . Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \frac{c_1 x^2 - 1}{x^2 (x^2 + 1)} dx$$

$$y = (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2$$

Summary of solutions found

$$y = \frac{1}{x} + (c_1 + 1) \arctan(x) + c_2$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.056 (sec)

Solve

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

Writing the ode as

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

Integrating both sides of the ODE w.r.t x gives

$$\int ((x^2 + 1) y'' + 2y'x) dx = \int \frac{2}{x^3} dx$$

$$(x^2 + 1) y' = -\frac{1}{x^2} + c_1$$

Which is now solved for y . Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \frac{c_1 x^2 - 1}{x^2 (x^2 + 1)} dx$$

$$y = (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int \frac{c_1 x^2 - 1}{x^2 (x^2 + 1)} dx$$

$$y = (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2$$

Solved as second order ode using Kovacic algorithm

Time used: 0.377 (sec)

Solve

$$(x^2 + 1) y'' + 2y'x = \frac{2}{x^3}$$

Writing the ode as

$$(x^2 + 1) y'' + 2y'x = 0 \quad (1)$$

$$Ay'' + By' + Cy = 0 \quad (2)$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x^2 + 1 \\ B &= 2x \\ C &= 0 \end{aligned} \quad (3)$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \quad (4)$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{1}{(x^2 + 1)^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 1 \\ t &= (x^2 + 1)^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{1}{(x^2 + 1)^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.288: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} \mathcal{O}(\infty) &= \deg(t) - \deg(s) \\ &= 4 - 0 \\ &= 4 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = (x^2 + 1)^2$. There is a pole at $x = i$ of order 2. There is a pole at $x = -i$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 4 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 4 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = -\frac{1}{4(x-i)^2} - \frac{1}{4(x+i)^2} - \frac{i}{4(x-i)} + \frac{i}{4x+4i}$$

For the pole at $x = i$ let b be the coefficient of $\frac{1}{(x-i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{1}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = \frac{1}{2} \end{aligned}$$

For the pole at $x = -i$ let b be the coefficient of $\frac{1}{(x+i)^2}$ in the partial fractions decomposition of r given above. Therefore $b = -\frac{1}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1+4b} = \frac{1}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1+4b} = \frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is $4 > 2$ then

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= 0 \\ \alpha_\infty^- &= 1 \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{1}{(x^2 + 1)^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
i	2	0	$\frac{1}{2}$	$\frac{1}{2}$
$-i$	2	0	$\frac{1}{2}$	$\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
4	0	0	1

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = 1$

then

$$\begin{aligned} d &= \alpha_{\infty}^{-} - (\alpha_{c_1}^{+} + \alpha_{c_2}^{+}) \\ &= 1 - (1) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_{\infty}$$

The above gives

$$\begin{aligned} \omega &= \left((+)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^{+}}{x - c_1} \right) + \left((+)[\sqrt{r}]_{c_2} + \frac{\alpha_{c_2}^{+}}{x - c_2} \right) + (-)[\sqrt{r}]_{\infty} \\ &= \frac{1}{2x - 2i} + \frac{1}{2x + 2i} + (-)(0) \\ &= \frac{1}{2x - 2i} + \frac{1}{2x + 2i} \\ &= \frac{x}{x^2 + 1} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r)p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$(0) + 2 \left(\frac{1}{2x - 2i} + \frac{1}{2x + 2i} \right) (0) + \left(\left(-\frac{1}{2(x - i)^2} - \frac{1}{2(x + i)^2} \right) + \left(\frac{1}{2x - 2i} + \frac{1}{2x + 2i} \right)^2 - \left(\frac{1}{(x^2 + 1)^2} \right) \right) 1 = 0$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= p e^{\int \omega dx} \\ &= e^{\int \left(\frac{1}{2x - 2i} + \frac{1}{2x + 2i} \right) dx} \\ &= \sqrt{-x^2 - 1} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{2x}{x^2+1} dx} \\ &= z_1 e^{-\frac{\ln(x^2+1)}{2}} \\ &= z_1 \left(\frac{1}{\sqrt{x^2+1}} \right) \end{aligned}$$

Which simplifies to

$$y_1 = i$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{2x}{x^2+1} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{-\ln(x^2+1)}}{(y_1)^2} dx \\ &= y_1 (-\arctan(x)) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1(i) + c_2(i(-\arctan(x))) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$(x^2 + 1) y'' + 2y'x = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = ic_1 - ic_2 \arctan(x)$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = i$$

$$y_2 = -i \arctan(x)$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} i & -i \arctan(x) \\ \frac{d}{dx}(i) & \frac{d}{dx}(-i \arctan(x)) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} i & -i \arctan(x) \\ 0 & -\frac{i}{x^2+1} \end{vmatrix}$$

Therefore

$$W = (i) \left(-\frac{i}{x^2+1} \right) - (-i \arctan(x)) (0)$$

Which simplifies to

$$W = \frac{1}{x^2+1}$$

Which simplifies to

$$W = \frac{1}{x^2 + 1}$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{-\frac{2i \arctan(x)}{x^3}}{1} dx$$

Which simplifies to

$$u_1 = - \int -\frac{2i \arctan(x)}{x^3} dx$$

Hence

$$u_1 = -\frac{i \arctan(x)}{x^2} - i \arctan(x) - \frac{i}{x}$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{2i}{x^3}}{1} dx$$

Which simplifies to

$$u_2 = \int \frac{2i}{x^3} dx$$

Hence

$$u_2 = -\frac{i}{x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = i \left(-\frac{i \arctan(x)}{x^2} - i \arctan(x) - \frac{i}{x} \right) - \frac{\arctan(x)}{x^2}$$

Which simplifies to

$$y_p(x) = \frac{\arctan(x) x + 1}{x}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= (ic_1 - ic_2 \arctan(x)) + \left(\frac{\arctan(x) x + 1}{x} \right) \end{aligned}$$

Summary of solutions found

$$y = \frac{\arctan(x)x + 1}{x} + ic_1 - ic_2 \arctan(x)$$

✓ **Maple.** Time used: 0.000 (sec). Leaf size: 15

```
ode:=(x^2+1)*diff(diff(y(x),x),x)+2*diff(y(x),x)*x = 2/x^3;
dsolve(ode,y(x), singsol=all);
```

$$y = (c_1 + 1) \arctan(x) + \frac{1}{x} + c_2$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

-> Calling odsolve with the ODE, $\text{diff}(_b(_a), _a) = -2*(_a^4*_b(_a)-1)/_a^3/(_a^2+1), _b(_a)$

*** Sublevel 2 ***

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

<- high order exact linear fully integrable successful

Maple step by step

Let's solve

$$(x^2 + 1) \left(\frac{d^2}{dx^2} y(x) \right) + 2x \left(\frac{d}{dx} y(x) \right) = \frac{2}{x^3}$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Make substitution $u = \frac{d}{dx} y(x)$ to reduce order of ODE

$$(x^2 + 1) \left(\frac{d}{dx} u(x) \right) + 2xu(x) = \frac{2}{x^3}$$

- Isolate the derivative

$$\frac{d}{dx} u(x) = -\frac{2xu(x)}{x^2+1} + \frac{2}{x^3(x^2+1)}$$

- Group terms with $u(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx} u(x) + \frac{2xu(x)}{x^2+1} = \frac{2}{x^3(x^2+1)}$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx} u(x) + \frac{2xu(x)}{x^2+1} \right) = \frac{2\mu(x)}{x^3(x^2+1)}$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(u(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx} u(x) + \frac{2xu(x)}{x^2+1} \right) = \left(\frac{d}{dx} u(x) \right) \mu(x) + u(x) \left(\frac{d}{dx} \mu(x) \right)$$
- Isolate $\frac{d}{dx} \mu(x)$

$$\frac{d}{dx} \mu(x) = \frac{2\mu(x)x}{x^2+1}$$
- Solve to find the integrating factor

$$\mu(x) = x^2 + 1$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx} (u(x)\mu(x)) \right) dx = \int \frac{2\mu(x)}{x^3(x^2+1)} dx + C1$$
- Evaluate the integral on the lhs

$$u(x)\mu(x) = \int \frac{2\mu(x)}{x^3(x^2+1)} dx + C1$$
- Solve for $u(x)$

$$u(x) = \frac{\int \frac{2\mu(x)}{x^3(x^2+1)} dx + C1}{\mu(x)}$$
- Substitute $\mu(x) = x^2 + 1$

$$u(x) = \frac{\int \frac{2}{x^3} dx + C1}{x^2+1}$$
- Evaluate the integrals on the rhs

$$u(x) = \frac{-\frac{1}{x^2} + C1}{x^2+1}$$
- Solve 1st ODE for $u(x)$

$$u(x) = \frac{-\frac{1}{x^2} + C1}{x^2+1}$$
- Make substitution $u = \frac{d}{dx} y(x)$

$$\frac{d}{dx} y(x) = \frac{-\frac{1}{x^2} + C1}{x^2+1}$$
- Integrate both sides to solve for $y(x)$

$$\int \left(\frac{d}{dx} y(x) \right) dx = \int \frac{-\frac{1}{x^2} + C1}{x^2+1} dx + C2$$
- Compute integrals

$$y(x) = \frac{1}{x} + (C1 + 1) \arctan(x) + C2$$

✓ **Mathematica.** Time used: 60.026 (sec). Leaf size: 36

```
ode=(1+x^2)*D[y[x],{x,2}]+2*x*D[y[x],x]==2*x^(-3);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \int_1^x \frac{c_1 K[1]^2 - 1}{K[1]^4 + K[1]^2} dK[1] + c_2$$

✓ Sympy. Time used: 0.376 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*Derivative(y(x), x) + (x**2 + 1)*Derivative(y(x), (x, 2)) - 2/x
**3,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 - C_2 \log(x - i) + C_2 \log(x + i) + \frac{1}{x}$$

2.14.3 Problem 24

Local contents

Solved as second order linear exact ode	2371
Solved as second order missing y ode	2373
Solved as second order integrable as is ode	2375
Solved as second order integrable as is ode (ABC method)	2376
Solved as second order ode using non constant coeff transformation on B method	2379
Solved as second order ode using Kovacic algorithm	2383
✓ Maple	2390
✓ Mathematica	2392
✓ Sympy	2392

Internal problem ID [8049]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 24

Date solved : Thursday, November 27, 2025 at 07:32:52 AM

CAS classification : `[_2nd_order, _missing_y]`

Solved as second order linear exact ode

Time used: 0.108 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

An ode of the form

$$p(x)y'' + q(x)y' + r(x)y = s(x)$$

is exact if

$$p''(x) - q'(x) + r(x) = 0 \tag{1}$$

For the given ode we have

$$\begin{aligned} p(x) &= x \\ q(x) &= -1 \\ r(x) &= 0 \\ s(x) &= -\frac{2}{x} - \ln(x) \end{aligned}$$

Hence

$$\begin{aligned} p''(x) &= 0 \\ q'(x) &= 0 \end{aligned}$$

Therefore (1) becomes

$$0 - (0) + (0) = 0$$

Hence the ode is exact. Since we now know the ode is exact, it can be written as

$$(p(x) y' + (q(x) - p'(x)) y)' = s(x)$$

Integrating gives

$$p(x) y' + (q(x) - p'(x)) y = \int s(x) dx$$

Substituting the above values for p, q, r, s gives

$$y'x - 2y = \int -\frac{2}{x} - \ln(x) dx$$

We now have a first order ode to solve which is

$$y'x - 2y = -2 \ln(x) - \ln(x)x + x + c_1$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{2}{x} \\ p(x) &= \frac{(-2 - x) \ln(x) + x + c_1}{x} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{2}{x} dx} \\ &= \frac{1}{x^2} \end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(-2-x) \ln(x) + x + c_1}{x} \right) \\ \frac{d}{dx} \left(\frac{y}{x^2} \right) &= \left(\frac{1}{x^2} \right) \left(\frac{(-2-x) \ln(x) + x + c_1}{x} \right) \\ d \left(\frac{y}{x^2} \right) &= \left(\frac{(-2-x) \ln(x) + x + c_1}{x^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^2} &= \int \frac{(-2-x) \ln(x) + x + c_1}{x^3} dx \\ &= \frac{(2x+2) \ln(x) - c_1 + 1}{2x^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_2 x^2 + \ln(x) x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Summary of solutions found

$$y = c_2 x^2 + \ln(x) x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Solved as second order missing y ode

Time used: 0.143 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

This is second order ode with missing dependent variable y . Let

$$u(x) = y'$$

Then

$$u'(x) = y''$$

Hence the ode becomes

$$u'(x) x - u(x) + \frac{2}{x} + \ln(x) = 0$$

Which is now solved for $u(x)$ as first order ode.

Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{1}{x}$$

$$p(x) = \frac{-\ln(x)x - 2}{x^2}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{x} dx} \\ &= \frac{1}{x}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu u) &= \mu p \\ \frac{d}{dx}(\mu u) &= (\mu) \left(\frac{-\ln(x)x - 2}{x^2} \right) \\ \frac{d}{dx} \left(\frac{u}{x} \right) &= \left(\frac{1}{x} \right) \left(\frac{-\ln(x)x - 2}{x^2} \right) \\ d \left(\frac{u}{x} \right) &= \left(\frac{-\ln(x)x - 2}{x^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{u}{x} &= \int \frac{-\ln(x)x - 2}{x^3} dx \\ &= \frac{\ln(x)x + x + 1}{x^2} + c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x}$ gives the final solution

$$u(x) = \frac{c_1 x^2 + \ln(x)x + x + 1}{x}$$

In summary, these are the solution found for y

$$u(x) = \frac{c_1 x^2 + \ln(x)x + x + 1}{x}$$

For solution $u(x) = \frac{c_1 x^2 + \ln(x)x + x + 1}{x}$, since $u = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{c_1 x^2 + \ln(x)x + x + 1}{x}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dy &= \int \frac{c_1 x^2 + \ln(x)x + x + 1}{x} dx \\ y &= \frac{c_1 x^2}{2} + \ln(x)x + \ln(x) + c_2\end{aligned}$$

In summary, these are the solution found for (y)

$$y = \frac{c_1 x^2}{2} + \ln(x)x + \ln(x) + c_2$$

Summary of solutions found

$$y = \frac{c_1 x^2}{2} + \ln(x)x + \ln(x) + c_2$$

Solved as second order integrable as is ode

Time used: 0.079 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

Integrating both sides of the ODE w.r.t x gives

$$\begin{aligned}\int (xy'' - y') dx &= \int \left(-\frac{2}{x} - \ln(x)\right) dx \\ y'x - 2y &= -2\ln(x) - \ln(x)x + x + c_1\end{aligned}$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned}q(x) &= -\frac{2}{x} \\ p(x) &= \frac{(-2 - x)\ln(x) + x + c_1}{x}\end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(-2-x) \ln(x) + x + c_1}{x} \right) \\ \frac{d}{dx} \left(\frac{y}{x^2} \right) &= \left(\frac{1}{x^2} \right) \left(\frac{(-2-x) \ln(x) + x + c_1}{x} \right) \\ d \left(\frac{y}{x^2} \right) &= \left(\frac{(-2-x) \ln(x) + x + c_1}{x^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^2} &= \int \frac{(-2-x) \ln(x) + x + c_1}{x^3} \, dx \\ &= \frac{(2x+2) \ln(x) - c_1 + 1}{2x^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_2 x^2 + \ln(x) x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Summary of solutions found

$$y = c_2 x^2 + \ln(x) x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.075 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

Writing the ode as

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

Integrating both sides of the ODE w.r.t x gives

$$\int (xy'' - y') dx = \int \left(-\frac{2}{x} - \ln(x) \right) dx$$

$$y'x - 2y = -2\ln(x) - \ln(x)x + x + c_1$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{2}{x}$$

$$p(x) = \frac{(-2-x)\ln(x) + x + c_1}{x}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dx} \\ &= e^{\int -\frac{2}{x} dx} \\ &= \frac{1}{x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(-2-x)\ln(x) + x + c_1}{x} \right) \\ \frac{d}{dx} \left(\frac{y}{x^2} \right) &= \left(\frac{1}{x^2} \right) \left(\frac{(-2-x)\ln(x) + x + c_1}{x} \right) \\ d \left(\frac{y}{x^2} \right) &= \left(\frac{(-2-x)\ln(x) + x + c_1}{x^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^2} &= \int \frac{(-2-x)\ln(x) + x + c_1}{x^3} dx \\ &= \frac{(2x+2)\ln(x) - c_1 + 1}{2x^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_2 x^2 + \ln(x)x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Solve In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$q(x) = -\frac{2}{x}$$

$$p(x) = \frac{(-2-x)\ln(x) + x + c_1}{x}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} dx} \\ &= \frac{1}{x^2}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}(\mu y) &= \mu p \\ \frac{d}{dx}(\mu y) &= (\mu) \left(\frac{(-2-x)\ln(x) + x + c_1}{x} \right) \\ \frac{d}{dx} \left(\frac{y}{x^2} \right) &= \left(\frac{1}{x^2} \right) \left(\frac{(-2-x)\ln(x) + x + c_1}{x} \right) \\ d \left(\frac{y}{x^2} \right) &= \left(\frac{(-2-x)\ln(x) + x + c_1}{x^3} \right) dx\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{y}{x^2} &= \int \frac{(-2-x)\ln(x) + x + c_1}{x^3} dx \\ &= \frac{(2x+2)\ln(x) - c_1 + 1}{2x^2} + c_2\end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = c_2 x^2 + \ln(x)x + \ln(x) - \frac{c_1}{2} + \frac{1}{2}$$

Solved as second order ode using non constant coeff transformation on B method

Time used: 0.293 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

Given an ode of the form

$$Ay'' + By' + Cy = F(x)$$

This method reduces the order ode the ODE by one by applying the transformation

$$y = Bv$$

This results in

$$\begin{aligned} y' &= B'v + v'B \\ y'' &= B''v + B'v' + v''B + v'B' \\ &= v''B + 2v' + B' + B''v \end{aligned}$$

And now the original ode becomes

$$\begin{aligned} A(v''B + 2v'B' + B''v) + B(B'v + v'B) + CBv &= 0 \\ ABv'' + (2AB' + B^2)v' + (AB'' + BB' + CB)v &= 0 \end{aligned} \quad (1)$$

If the term $AB'' + BB' + CB$ is zero, then this method works and can be used to solve

$$ABv'' + (2AB' + B^2)v' = 0$$

By Using $u = v'$ which reduces the order of the above ode to one. The new ode is

$$ABu' + (2AB' + B^2)u = 0$$

The above ode is first order ode which is solved for u . Now a new ode $v' = u$ is solved for v as first order ode. Then the final solution is obtain from $y = Bv$.

This method works only if the term $AB'' + BB' + CB$ is zero. The given ODE shows that

$$\begin{aligned} A &= x \\ B &= -1 \\ C &= 0 \\ F &= -\frac{2}{x} - \ln(x) \end{aligned}$$

The above shows that for this ode

$$\begin{aligned} AB'' + BB' + CB &= (x)(0) + (-1)(0) + (0)(-1) \\ &= 0 \end{aligned}$$

Hence the ode in v given in (1) now simplifies to

$$-xv'' + (1)v' = 0$$

Now by applying $v' = u$ the above becomes

$$-xu'(x) + u(x) = 0$$

Which is now solved for u . Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{1}{x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q dx} \\ &= e^{\int -\frac{1}{x} dx} \\ &= \frac{1}{x} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dx} \mu u &= 0 \\ \frac{d}{dx} \left(\frac{u}{x} \right) &= 0 \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{u}{x} &= \int 0 dx + c_1 \\ &= c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x}$ gives the final solution

$$u(x) = c_1 x$$

The ode for v now becomes

$$v'(x) = c_1 x$$

Which is now solved for v . Solve Since the ode has the form $v'(x) = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dv &= \int c_1 x dx \\ v(x) &= \frac{c_1 x^2}{2} + c_2\end{aligned}$$

Replacing $v(x)$ above by $-y$, then the homogeneous solution is

$$\begin{aligned}y_h(x) &= Bv \\ &= -\frac{c_1 x^2}{2} - c_2\end{aligned}$$

And now the particular solution $y_p(x)$ will be found. The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$\begin{aligned}y_1 &= -1 \\ y_2 &= x^2\end{aligned}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} -1 & x^2 \\ \frac{d}{dx}(-1) & \frac{d}{dx}(x^2) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} -1 & x^2 \\ 0 & 2x \end{vmatrix}$$

Therefore

$$W = (-1)(2x) - (x^2)(0)$$

Which simplifies to

$$W = -2x$$

Which simplifies to

$$W = -2x$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{x^2(-\frac{2}{x} - \ln(x))}{-2x^2} dx$$

Which simplifies to

$$u_1 = - \int \left(\frac{1}{x} + \frac{\ln(x)}{2} \right) dx$$

Hence

$$u_1 = -\frac{\ln(x)x}{2} + \frac{x}{2} - \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{\frac{2}{x} + \ln(x)}{-2x^2} dx$$

Which simplifies to

$$u_2 = \int \frac{-\ln(x)x - 2}{2x^3} dx$$

Hence

$$u_2 = \frac{\ln(x)}{2x} + \frac{1}{2x} + \frac{1}{2x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\ln(x)x}{2} - \frac{x}{2} + \ln(x) + x^2 \left(\frac{\ln(x)}{2x} + \frac{1}{2x} + \frac{1}{2x^2} \right)$$

Which simplifies to

$$y_p(x) = \ln(x)x + \ln(x) + \frac{1}{2}$$

Hence the complete solution is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= \left(-\frac{c_1 x^2}{2} - c_2\right) + \left(\ln(x)x + \ln(x) + \frac{1}{2}\right) \\ &= -\frac{c_1 x^2}{2} - c_2 + \ln(x)x + \ln(x) + \frac{1}{2} \end{aligned}$$

Summary of solutions found

$$y = -\frac{c_1 x^2}{2} - c_2 + \ln(x)x + \ln(x) + \frac{1}{2}$$

Solved as second order ode using Kovacic algorithm

Time used: 0.182 (sec)

Solve

$$xy'' - y' = -\frac{2}{x} - \ln(x)$$

Writing the ode as

$$xy'' - y' = 0 \tag{1}$$

$$Ay'' + By' + Cy = 0 \tag{2}$$

Comparing (1) and (2) shows that

$$\begin{aligned} A &= x \\ B &= -1 \\ C &= 0 \end{aligned} \tag{3}$$

Applying the Liouville transformation on the dependent variable gives

$$z(x) = ye^{\int \frac{B}{2A} dx}$$

Then (2) becomes

$$z''(x) = rz(x) \tag{4}$$

Where r is given by

$$\begin{aligned} r &= \frac{s}{t} \\ &= \frac{2AB' - 2BA' + B^2 - 4AC}{4A^2} \end{aligned} \quad (5)$$

Substituting the values of A, B, C from (3) in the above and simplifying gives

$$r = \frac{3}{4x^2} \quad (6)$$

Comparing the above to (5) shows that

$$\begin{aligned} s &= 3 \\ t &= 4x^2 \end{aligned}$$

Therefore eq. (4) becomes

$$z''(x) = \left(\frac{3}{4x^2} \right) z(x) \quad (7)$$

Equation (7) is now solved. After finding $z(x)$ then y is found using the inverse transformation

$$y = z(x) e^{-\int \frac{B}{2A} dx}$$

The first step is to determine the case of Kovacic algorithm this ode belongs to. There are 3 cases depending on the order of poles of r and the order of r at ∞ . The following table summarizes these cases.

Case	Allowed pole order for r	Allowed value for $\mathcal{O}(\infty)$
1	$\{0, 1, 2, 4, 6, 8, \dots\}$	$\{\dots, -6, -4, -2, 0, 2, 3, 4, 5, 6, \dots\}$
2	Need to have at least one pole that is either order 2 or odd order greater than 2. Any other pole order is allowed as long as the above condition is satisfied. Hence the following set of pole orders are all allowed. $\{1, 2\}, \{1, 3\}, \{2\}, \{3\}, \{3, 4\}, \{1, 2, 5\}$.	no condition
3	$\{1, 2\}$	$\{2, 3, 4, 5, 6, 7, \dots\}$

Table 2.290: Necessary conditions for each Kovacic case

The order of r at ∞ is the degree of t minus the degree of s . Therefore

$$\begin{aligned} O(\infty) &= \deg(t) - \deg(s) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

The poles of r in eq. (7) and the order of each pole are determined by solving for the roots of $t = 4x^2$. There is a pole at $x = 0$ of order 2. Since there is no odd order pole larger than 2 and the order at ∞ is 2 then the necessary conditions for case one are met. Since there is a pole of order 2 then necessary conditions for case two are met. Since pole order is not larger than 2 and the order at ∞ is 2 then the necessary conditions for case three are met. Therefore

$$L = [1, 2, 4, 6, 12]$$

Attempting to find a solution using case $n = 1$.

Looking at poles of order 2. The partial fractions decomposition of r is

$$r = \frac{3}{4x^2}$$

For the pole at $x = 0$ let b be the coefficient of $\frac{1}{x^2}$ in the partial fractions decomposition of r given above. Therefore $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_c &= 0 \\ \alpha_c^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_c^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

Since the order of r at ∞ is 2 then $[\sqrt{r}]_\infty = 0$. Let b be the coefficient of $\frac{1}{x^2}$ in the Laurent series expansion of r at ∞ . which can be found by dividing the leading coefficient of s by the leading coefficient of t from

$$r = \frac{s}{t} = \frac{3}{4x^2}$$

Since the $\gcd(s, t) = 1$. This gives $b = \frac{3}{4}$. Hence

$$\begin{aligned} [\sqrt{r}]_\infty &= 0 \\ \alpha_\infty^+ &= \frac{1}{2} + \sqrt{1 + 4b} = \frac{3}{2} \\ \alpha_\infty^- &= \frac{1}{2} - \sqrt{1 + 4b} = -\frac{1}{2} \end{aligned}$$

The following table summarizes the findings so far for poles and for the order of r at ∞ where r is

$$r = \frac{3}{4x^2}$$

pole c location	pole order	$[\sqrt{r}]_c$	α_c^+	α_c^-
0	2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Order of r at ∞	$[\sqrt{r}]_\infty$	α_∞^+	α_∞^-
2	0	$\frac{3}{2}$	$-\frac{1}{2}$

Now that the all $[\sqrt{r}]_c$ and its associated $\alpha_c^\pm$ have been determined for all the poles in the set Γ and $[\sqrt{r}]_\infty$ and its associated $\alpha_\infty^\pm$ have also been found, the next step is to determine possible non negative integer d from these using

$$d = \alpha_\infty^{s(\infty)} - \sum_{c \in \Gamma} \alpha_c^{s(c)}$$

Where $s(c)$ is either $+$ or $-$ and $s(\infty)$ is the sign of $\alpha_\infty^\pm$. This is done by trial over all set of families $s = (s(c))_{c \in \Gamma \cup \infty}$ until such d is found to work in finding candidate ω . Trying $\alpha_\infty^- = -\frac{1}{2}$ then

$$\begin{aligned} d &= \alpha_\infty^- - (\alpha_{c_1}^-) \\ &= -\frac{1}{2} - \left(-\frac{1}{2}\right) \\ &= 0 \end{aligned}$$

Since d an integer and $d \geq 0$ then it can be used to find ω using

$$\omega = \sum_{c \in \Gamma} \left(s(c)[\sqrt{r}]_c + \frac{\alpha_c^{s(c)}}{x - c} \right) + s(\infty)[\sqrt{r}]_\infty$$

The above gives

$$\begin{aligned} \omega &= \left((-)[\sqrt{r}]_{c_1} + \frac{\alpha_{c_1}^-}{x - c_1} \right) + (-)[\sqrt{r}]_\infty \\ &= -\frac{1}{2x} + (-)(0) \\ &= -\frac{1}{2x} \\ &= -\frac{1}{2x} \end{aligned}$$

Now that ω is determined, the next step is find a corresponding minimal polynomial $p(x)$ of degree $d = 0$ to solve the ode. The polynomial $p(x)$ needs to satisfy the equation

$$p'' + 2\omega p' + (\omega' + \omega^2 - r) p = 0 \quad (1A)$$

Let

$$p(x) = 1 \quad (2A)$$

Substituting the above in eq. (1A) gives

$$\begin{aligned} (0) + 2\left(-\frac{1}{2x}\right)(0) + \left(\left(\frac{1}{2x^2}\right) + \left(-\frac{1}{2x}\right)^2 - \left(\frac{3}{4x^2}\right)\right) &= 0 \\ 0 &= 0 \end{aligned}$$

The equation is satisfied since both sides are zero. Therefore the first solution to the ode $z'' = rz$ is

$$\begin{aligned} z_1(x) &= pe^{\int \omega dx} \\ &= e^{\int -\frac{1}{2x} dx} \\ &= \frac{1}{\sqrt{x}} \end{aligned}$$

The first solution to the original ode in y is found from

$$\begin{aligned} y_1 &= z_1 e^{\int -\frac{1}{2} \frac{B}{A} dx} \\ &= z_1 e^{-\int \frac{1}{2} \frac{-1}{x} dx} \\ &= z_1 e^{\frac{\ln(x)}{2}} \\ &= z_1 (\sqrt{x}) \end{aligned}$$

Which simplifies to

$$y_1 = 1$$

The second solution y_2 to the original ode is found using reduction of order

$$y_2 = y_1 \int \frac{e^{\int -\frac{B}{A} dx}}{y_1^2} dx$$

Substituting gives

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{\int -\frac{-1}{x} dx}}{(y_1)^2} dx \\ &= y_1 \int \frac{e^{\ln(x)}}{(y_1)^2} dx \\ &= y_1 \left(\frac{x^2}{2} \right) \end{aligned}$$

Therefore the solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 (1) + c_2 \left(1 \left(\frac{x^2}{2} \right) \right) \end{aligned}$$

This is second order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE $Ay''(x) + By'(x) + Cy(x) = 0$, and y_p is a particular solution to the nonhomogeneous ODE $Ay''(x) + By'(x) + Cy(x) = f(x)$. y_h is the solution to

$$xy'' - y' = 0$$

The homogeneous solution is found using the Kovacic algorithm which results in

$$y_h = c_1 + \frac{c_2 x^2}{2}$$

The particular solution y_p can be found using either the method of undetermined coefficients, or the method of variation of parameters. The method of variation of parameters will be used as it is more general and can be used when the coefficients of the ODE depend on x as well. Let

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad (1)$$

Where u_1, u_2 to be determined, and y_1, y_2 are the two basis solutions (the two linearly independent solutions of the homogeneous ODE) found earlier when solving the homogeneous ODE as

$$y_1 = 1$$

$$y_2 = \frac{x^2}{2}$$

In the Variation of parameters u_1, u_2 are found using

$$u_1 = - \int \frac{y_2 f(x)}{aW(x)} \quad (2)$$

$$u_2 = \int \frac{y_1 f(x)}{aW(x)} \quad (3)$$

Where $W(x)$ is the Wronskian and a is the coefficient in front of y'' in the given ODE. The Wronskian is given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$. Hence

$$W = \begin{vmatrix} 1 & \frac{x^2}{2} \\ \frac{d}{dx}(1) & \frac{d}{dx}\left(\frac{x^2}{2}\right) \end{vmatrix}$$

Which gives

$$W = \begin{vmatrix} 1 & \frac{x^2}{2} \\ 0 & x \end{vmatrix}$$

Therefore

$$W = (1)(x) - \left(\frac{x^2}{2}\right)(0)$$

Which simplifies to

$$W = x$$

Which simplifies to

$$W = x$$

Therefore Eq. (2) becomes

$$u_1 = - \int \frac{\frac{x^2(-\frac{2}{x} - \ln(x))}{2}}{x^2} dx$$

Which simplifies to

$$u_1 = - \int \left(-\frac{1}{x} - \frac{\ln(x)}{2} \right) dx$$

Hence

$$u_1 = \frac{\ln(x)x}{2} - \frac{x}{2} + \ln(x)$$

And Eq. (3) becomes

$$u_2 = \int \frac{-\frac{2}{x} - \ln(x)}{x^2} dx$$

Which simplifies to

$$u_2 = \int \frac{-\ln(x)x - 2}{x^3} dx$$

Hence

$$u_2 = \frac{\ln(x)}{x} + \frac{1}{x} + \frac{1}{x^2}$$

Therefore the particular solution, from equation (1) is

$$y_p(x) = \frac{\ln(x)x}{2} - \frac{x}{2} + \ln(x) + \frac{x^2 \left(\frac{\ln(x)}{x} + \frac{1}{x} + \frac{1}{x^2} \right)}{2}$$

Which simplifies to

$$y_p(x) = \ln(x)x + \ln(x) + \frac{1}{2}$$

Therefore the general solution is

$$\begin{aligned} y &= y_h + y_p \\ &= \left(c_1 + \frac{c_2 x^2}{2} \right) + \left(\ln(x)x + \ln(x) + \frac{1}{2} \right) \end{aligned}$$

Summary of solutions found

$$y = \ln(x)x + \ln(x) + \frac{1}{2} + c_1 + \frac{c_2 x^2}{2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 18

```
ode:=-diff(y(x),x)+x*diff(diff(y(x),x),x) = -2/x-ln(x);
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{c_1 x^2}{2} + \ln(x)x + \ln(x) + c_2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
-> Calling odsolve with the ODE, diff(_b(_a),_a) = (_b(_a)*_a-ln(_a)*_a-2)/_a^2
, _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- high order exact linear fully integrable successful
```

Maple step by step

Let's solve

$$x\left(\frac{d^2}{dx^2}y(x)\right) - \frac{d}{dx}y(x) = -\frac{2}{x} - \ln(x)$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Make substitution $u = \frac{d}{dx}y(x)$ to reduce order of ODE

$$x\left(\frac{d}{dx}u(x)\right) - u(x) = -\frac{2}{x} - \ln(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}u(x) = \frac{u(x) - \frac{2}{x} - \ln(x)}{x}$$

- Collect w.r.t. $u(x)$ and simplify

$$\frac{d}{dx}u(x) = \frac{u(x)}{x} - \frac{\ln(x)x+2}{x^2}$$

- Group terms with $u(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}u(x) - \frac{u(x)}{x} = -\frac{\ln(x)x+2}{x^2}$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}u(x) - \frac{u(x)}{x} \right) = -\frac{\mu(x)(\ln(x)x+2)}{x^2}$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(u(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}u(x) - \frac{u(x)}{x} \right) = \left(\frac{d}{dx}u(x) \right) \mu(x) + u(x) \left(\frac{d}{dx}\mu(x) \right)$$

- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{\mu(x)}{x}$$

- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x}$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(u(x)\mu(x)) \right) dx = \int -\frac{\mu(x)(\ln(x)x+2)}{x^2} dx + C1$$

- Evaluate the integral on the lhs

$$u(x)\mu(x) = \int -\frac{\mu(x)(\ln(x)x+2)}{x^2} dx + C1$$

- Solve for $u(x)$

$$u(x) = \frac{\int -\frac{\mu(x)(\ln(x)x+2)}{x^2} dx + C1}{\mu(x)}$$

- Substitute $\mu(x) = \frac{1}{x}$

$$u(x) = x \left(\int -\frac{\ln(x)x+2}{x^3} dx + C1 \right)$$

- Evaluate the integrals on the rhs

$$u(x) = x \left(\frac{\ln(x)}{x} + \frac{1}{x} + \frac{1}{x^2} + C1 \right)$$

- Simplify

$$u(x) = \frac{C1 x^2 + \ln(x)x + x + 1}{x}$$

- Solve 1st ODE for $u(x)$

$$u(x) = \frac{C1 x^2 + \ln(x)x + x + 1}{x}$$

- Make substitution $u = \frac{d}{dx}y(x)$

$$\frac{d}{dx}y(x) = \frac{C1 x^2 + \ln(x)x + x + 1}{x}$$

- Integrate both sides to solve for $y(x)$

$$\int \left(\frac{d}{dx}y(x) \right) dx = \int \frac{C1 x^2 + \ln(x)x + x + 1}{x} dx + C2$$

- Compute integrals

$$y(x) = \frac{C_1 x^2}{2} + \ln(x) x + \ln(x) + C_2$$

✓ **Mathematica.** Time used: 0.024 (sec). Leaf size: 23

```
ode=x*D[y[x],{x,2}]-D[y[x],x]==-2/x-Log[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{c_1 x^2}{2} + (x+1) \log(x) + c_2$$

✓ **Sympy.** Time used: 0.130 (sec). Leaf size: 17

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), (x, 2)) + log(x) - Derivative(y(x), x) + 2/x, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2 x^2 + x \log(x) + \log(x)$$

2.14.4 Problem 25

Local contents

Solved as higher order constant coeff ode	2393
✓ Maple	2395
✓ Mathematica	2397
✓ Sympy	2398

Internal problem ID [8050]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 25

Date solved : Thursday, November 27, 2025 at 07:32:53 AM

CAS classification : `[[_3rd_order, _missing_y]]`

Solve

$$y''' + y'' = x^2$$

Solved as higher order constant coeff ode

Time used: 0.071 (sec)

The characteristic equation is

$$\lambda^3 + \lambda^2 = 0$$

The roots of the above equation are

$$\lambda_1 = -1$$

$$\lambda_2 = 0$$

$$\lambda_3 = 0$$

Therefore the homogeneous solution is

$$y_h(x) = c_2 x + c_1 + e^{-x} c_3$$

The fundamental set of solutions for the homogeneous solution are the following

$$y_1 = 1$$

$$y_2 = x$$

$$y_3 = e^{-x}$$

This is higher order nonhomogeneous ODE. Let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ODE And y_p is a particular solution to the nonhomogeneous ODE. y_h is the solution to

$$y''' + y'' = 0$$

Now the particular solution to the given ODE is found

$$y''' + y'' = x^2$$

The particular solution is now found using the method of undetermined coefficients.

Looking at the RHS of the ode, which is

$$x^2$$

Shows that the corresponding undetermined set of the basis functions (UC_set) for the trial solution is

$$[\{1, x, x^2\}]$$

While the set of the basis functions for the homogeneous solution found earlier is

$$\{1, x, e^{-x}\}$$

Since 1 is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x, x^2, x^3\}]$$

Since x is duplicated in the UC_set, then this basis is multiplied by extra x . The UC_set becomes

$$[\{x^2, x^3, x^4\}]$$

Since there was duplication between the basis functions in the UC_set and the basis functions of the homogeneous solution, the trial solution is a linear combination of all the basis function in the above updated UC_set.

$$y_p = A_3x^4 + A_2x^3 + A_1x^2$$

The unknowns $\{A_1, A_2, A_3\}$ are found by substituting the above trial solution y_p into the ODE and comparing coefficients. Substituting the trial solution into the ODE and simplifying gives

$$12x^2A_3 + 6xA_2 + 24xA_3 + 2A_1 + 6A_2 = x^2$$

Solving for the unknowns by comparing coefficients results in

$$\left[A_1 = 1, A_2 = -\frac{1}{3}, A_3 = \frac{1}{12} \right]$$

Substituting the above back in the above trial solution y_p , gives the particular solution

$$y_p = \frac{1}{12}x^4 - \frac{1}{3}x^3 + x^2$$

Therefore the general solution is

$$y = y_h + y_p \\ = (c_2x + c_1 + e^{-x}c_3) + \left(\frac{1}{12}x^4 - \frac{1}{3}x^3 + x^2 \right)$$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 28

```
ode:=diff(diff(diff(y(x),x),x),x)+diff(diff(y(x),x),x) = x^2;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{x^4}{12} + x^2 - \frac{x^3}{3} + e^{-x}c_1 + c_2x + c_3$$

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
-> Calling odsolve with the ODE, diff(_b(_a),_a) = _a^2-_b(_a), _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
<- high order exact linear fully integrable successful
```

Maple step by step

Let's solve

- $\frac{d^3}{dx^3}y(x) + \frac{d^2}{dx^2}y(x) = x^2$
- Highest derivative means the order of the ODE is 3
- $\frac{d^3}{dx^3}y(x)$
- Characteristic polynomial of homogeneous ODE
- $r^3 + r^2 = 0$
- Roots of the characteristic polynomial and corresponding multiplicities
- $r = [-1, 1], [0, 2]$
- Homogeneous solution from $r = -1$
- $y_1(x) = e^{-x}$
- 1st homogeneous solution from $r = 0$
- $y_2(x) = 1$
- 2nd homogeneous solution from $r = 0$
- $y_3(x) = x$
- General solution of the ODE
- $y(x) = C_1 y_1(x) + C_2 y_2(x) + C_3 y_3(x) + y_p(x)$
- Substitute in solutions of the homogeneous ODE
- $y(x) = C_1 e^{-x} + C_2 + C_3 x + y_p(x)$
- Find a particular solution $y_p(x)$ of the ODE
 - Define the forcing function of the ODE
 - $f(x) = x^2$
 - Form of the particular solution to the ODE where the $u_i(x)$ are to be found
 - $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$
 - Calculate the 1st derivative of $y_p(x)$
 - $\frac{d}{dx}y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) y_i(x) + u_i(x) \left(\frac{d}{dx}y_i(x) \right) \right)$
 - Choose equation to add to a system of equations in $\frac{d}{dx}u_i(x)$
 - $\sum_{i=1}^3 \left(\frac{d}{dx}u_i(x) \right) y_i(x) = 0$
 - Calculate the 2nd derivative of $y_p(x)$
 - $\frac{d^2}{dx^2}y_p(x) = \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) \left(\frac{d}{dx}y_i(x) \right) + u_i(x) \left(\frac{d^2}{dx^2}y_i(x) \right) \right)$
 - Choose equation to add to a system of equations in $\frac{d}{dx}u_i(x)$
 - $\sum_{i=1}^3 \left(\frac{d}{dx}u_i(x) \right) \left(\frac{d}{dx}y_i(x) \right) = 0$
 - The ODE is of the following form where the $P_i(x)$ in this situation are the coefficients of the derivative
 - $\frac{d^3}{dx^3}y(x) + \left(\sum_{i=0}^2 P_i(x) \left(\frac{d^i}{dx^i}y(x) \right) \right) = f(x)$
 - Substitute $y_p(x) = \sum_{i=1}^3 u_i(x) y_i(x)$ into the ODE
 - $\left(\sum_{j=0}^2 P_j(x) \left(\sum_{i=1}^3 u_i(x) \left(\frac{d^j}{dx^j}y_i(x) \right) \right) \right) + \sum_{i=1}^3 \left(\left(\frac{d}{dx}u_i(x) \right) \left(\frac{d^2}{dx^2}y_i(x) \right) + u_i(x) \left(\frac{d^3}{dx^3}y_i(x) \right) \right) = f(x)$

- Rearrange the ODE

$$\sum_{i=1}^3 \left(u_i(x) \cdot \left(\sum_{j=0}^2 P_j(x) \left(\frac{d^j}{dx^j} y_i(x) \right) \right) + \frac{d^3}{dx^3} y_i(x) \right) + \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- Notice that $y_i(x)$ are solutions to the homogeneous equation so the first term in the sum is 0

$$\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x)$$

- We have now made a system of 3 equations in 3 unknowns $\left(\frac{d}{dx} u_i(x) \right)$

$$\left[\sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) y_i(x) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d}{dx} y_i(x) \right) = 0, \sum_{i=1}^3 \left(\frac{d}{dx} u_i(x) \right) \left(\frac{d^2}{dx^2} y_i(x) \right) = f(x) \right]$$

- Convert the system to linear algebra format, notice that the matrix is the wronskian W

$$\begin{bmatrix} y_1(x) & y_2(x) & y_3(x) \\ \frac{d}{dx} y_1(x) & \frac{d}{dx} y_2(x) & \frac{d}{dx} y_3(x) \\ \frac{d^2}{dx^2} y_1(x) & \frac{d^2}{dx^2} y_2(x) & \frac{d^2}{dx^2} y_3(x) \end{bmatrix} \cdot \begin{bmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \\ \frac{d}{dx} u_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix}$$

- Solve for the varied parameters

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \int W^{-1} \cdot \begin{bmatrix} 0 \\ 0 \\ f(x) \end{bmatrix} dx$$

- Substitute in the homogeneous solutions and forcing function and solve

$$\begin{bmatrix} u_1(x) \\ u_2(x) \\ u_3(x) \end{bmatrix} = \begin{bmatrix} \frac{x^2}{e^{-x}} - \frac{2x}{e^{-x}} + \frac{2}{e^{-x}} \\ -\frac{1}{4}x^4 - \frac{1}{3}x^3 \\ \frac{x^3}{3} \end{bmatrix}$$

Find a particular solution $y_p(x)$ of the ODE

$$y_p(x) = x^2 - 2x + 2 + \frac{1}{12}x^4 - \frac{1}{3}x^3$$

- Substitute particular solution into general solution to ODE

$$y(x) = C_1 e^{-x} + C_2 + C_3 x + x^2 - 2x + 2 + \frac{x^4}{12} - \frac{x^3}{3}$$

✓ **Mathematica.** Time used: 0.055 (sec). Leaf size: 37

```
ode=D[y[x],{x,3}]+D[y[x],{x,2}]==x^2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x^4}{12} - \frac{x^3}{3} + x^2 + c_3 x + c_1 e^{-x} + c_2$$

✓ Sympy. Time used: 0.055 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + Derivative(y(x), (x, 2)) + Derivative(y(x), (x, 3)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

$$y(x) = C_1 + C_2x + C_3e^{-x} + \frac{x^4}{12} - \frac{x^3}{3} + x^2$$

2.14.5 Problem 26

Local contents

Solved as second order missing x ode	2399
Solved by factoring the differential equation	2400
✓ Maple	2403
✓ Mathematica	2405
✗ Sympy	2405

Internal problem ID [8051]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 26

Date solved : Thursday, November 27, 2025 at 07:32:53 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_x_y1], [_2nd_order, _reducible,

Solved as second order missing x ode

Time used: 0.306 (sec)

Solve

$$yy'' + y'^3 = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$yp(y) \left(\frac{d}{dy} p(y) \right) + p(y)^3 = 0$$

Which is now solved as first order ode for $p(y)$.

Solved by factoring the differential equation

Time used: 0.041 (sec)

Solve

$$ypp' + p^3 = 0$$

Writing the ode as

$$(p) (p'y + p^2) = 0$$

Therefore we need to solve the following equations

$$p = 0 \tag{1}$$

$$p'y + p^2 = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)Solving for p from

$$p = 0$$

Solving gives

$$p = 0$$

Solving equation (2)

Solve The ode

$$p' = -\frac{p^2}{y} \tag{2.121}$$

is separable as it can be written as

$$\begin{aligned} p' &= -\frac{p^2}{y} \\ &= f(y)g(p) \end{aligned}$$

Where

$$\begin{aligned} f(y) &= -\frac{1}{y} \\ g(p) &= p^2 \end{aligned}$$

Integrating gives

$$\int \frac{1}{g(p)} dp = \int f(y) dy$$

$$\int \frac{1}{p^2} dp = \int -\frac{1}{y} dy$$

$$-\frac{1}{p} = \ln \left(\frac{1}{y} \right) + c_1$$

We now need to find the singular solutions, these are found by finding for what values $g(p)$ is zero, since we had to divide by this above. Solving $g(p) = 0$ or

$$p^2 = 0$$

for p gives

$$p = 0$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-\frac{1}{p} = \ln \left(\frac{1}{y} \right) + c_1$$

$$p = 0$$

Solving for p gives

$$p = 0$$

$$p = -\frac{1}{\ln \left(\frac{1}{y} \right) + c_1}$$

Summary of solutions found

$$p = 0$$

$$p = -\frac{1}{\ln \left(\frac{1}{y} \right) + c_1}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = 0$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int 0 dx + c_2$$

$$y = c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\frac{1}{\ln\left(\frac{1}{y}\right) + c_1}$$

Solve Integrating gives

$$\int \left(-\ln\left(\frac{1}{y}\right) - c_1 \right) dy = dx$$

$$-yc_1 - \ln\left(\frac{1}{y}\right)y - y = x + c_3$$

Simplifying the above gives

$$-y\left(\ln\left(\frac{1}{y}\right) + c_1 + 1\right) = x + c_3$$

Solving for y from the above solution(s) gives (after possible removing of solutions that do not verify)

$$y = c_2$$

$$y = \frac{x + c_3}{\text{LambertW}((x + c_3)e^{-1-c_1})}$$

Summary of solutions found

$$y = c_2$$

$$y = \frac{x + c_3}{\text{LambertW}((x + c_3)e^{-1-c_1})}$$

✓ **Maple.** Time used: 0.013 (sec). Leaf size: 27

```
ode:=diff(y(x),x)^3+y(x)*diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = c_1$$

$$y = \frac{c_2 + x}{\text{LambertW}((c_2 + x)e^{c_1-1})}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying differential order: 2; missing variables
-> Calling odsolve with the ODE, diff(_b(_a),_a)*_b(_a)+_b(_a)^3/_a = 0, _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
<- differential order: 2; canonical coordinates successful
<- differential order 2; missing variables successful
```

Maple step by step

Let's solve

$$y(x) \left(\frac{d^2}{dx^2} y(x) \right) + \left(\frac{d}{dx} y(x) \right)^3 = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Define new dependent variable u

$$u(x) = \frac{d}{dx} y(x)$$

- Compute $\frac{d^2}{dx^2} y(x)$

$$\frac{d}{dx} u(x) = \frac{d^2}{dx^2} y(x)$$

- Use chain rule on the lhs

$$\left(\frac{d}{dx}y(x)\right)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Substitute in the definition of u

$$u(y)\left(\frac{d}{dy}u(y)\right) = \frac{d^2}{dx^2}y(x)$$
- Make substitutions $\frac{d}{dx}y(x) = u(y)$, $\frac{d^2}{dx^2}y(x) = u(y)\left(\frac{d}{dy}u(y)\right)$ to reduce order of ODE

$$yu(y)\left(\frac{d}{dy}u(y)\right) + u(y)^3 = 0$$
- Solve for the highest derivative

$$\frac{d}{dy}u(y) = -\frac{u(y)^2}{y}$$
- Separate variables

$$\frac{\frac{d}{dy}u(y)}{u(y)^2} = -\frac{1}{y}$$
- Integrate both sides with respect to y

$$\int \frac{\frac{d}{dy}u(y)}{u(y)^2} dy = \int -\frac{1}{y} dy + C1$$
- Evaluate integral

$$-\frac{1}{u(y)} = -\ln(y) + C1$$
- Solve for $u(y)$

$$u(y) = \frac{1}{\ln(y) - C1}$$
- Redefine the integration constant(s)

$$u(y) = \frac{1}{\ln(y) + C1}$$
- Solve 1st ODE for $u(y)$

$$u(y) = \frac{1}{\ln(y) + C1}$$
- Revert to original variables with substitution $u(y) = \frac{d}{dx}y(x)$, $y = y(x)$

$$\frac{d}{dx}y(x) = \frac{1}{\ln(y(x)) + C1}$$
- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{1}{\ln(y(x)) + C1}$$
- Separate variables

$$\left(\frac{d}{dx}y(x)\right)(\ln(y(x)) + C1) = 1$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}y(x)\right)(\ln(y(x)) + C1) dx = \int 1 dx + C2$$
- Evaluate integral

$$y(x) C1 + y(x) \ln(y(x)) - y(x) = x + C2$$
- Solve for $y(x)$

$$y(x) = e^{\text{Lambert } W((x+C2)e^{C1-1}) - C1 + 1}$$
- Simplify

$$y(x) = \frac{x+C2}{\text{Lambert } W((x+C2)e^{C1-1})}$$
- Redefine the integration constant(s)

$$y(x) = \frac{x+C2}{\text{Lambert } W((x+C2)C1)}$$

✓ **Mathematica.** Time used: 60.063 (sec). Leaf size: 26

```
ode=y[x]*D[y[x],{x,2}]+D[y[x],x]^3==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \frac{x + c_2}{W(e^{-1-c_1}(x + c_2))}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x)*Derivative(y(x), (x, 2)) + Derivative(y(x), x)**3,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE (-y(x)*Derivative(y(x), (x, 2))**(1/3)/2 -
sqrt(3)*I*(-y(x)*Derivative(y(x), (x, 2))**(1/3)/2 + Derivative(y(x), x)
cannot be solved by the factorable group method
```

2.14.6 Problem 27

Local contents

Solved as second order nonlinear exact ode	2406
Solved as second order missing x ode	2408
Solved by factoring the differential equation	2408
Solved as second order integrable as is ode	2410
Solved as second order integrable as is ode (ABC method)	2411
✓ Maple	2412
✓ Mathematica	2414
✗ Sympy	2414

Internal problem ID [8052]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 27

Date solved : Thursday, November 27, 2025 at 07:32:53 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _exact, _nonlinear], _Liouville, [_2nd_order, _r

Solved as second order nonlinear exact ode

Time used: 0.141 (sec)

Solve

$$yy'' + y'^2 = 0$$

An exact non-linear second order ode has the form

$$a_2(x, y, y') y'' + a_1(x, y, y') y' + a_0(x, y, y') = 0$$

Where the following conditions are satisfied

$$\begin{aligned} \frac{\partial a_2}{\partial y} &= \frac{\partial a_1}{\partial y'} \\ \frac{\partial a_2}{\partial x} &= \frac{\partial a_0}{\partial y'} \\ \frac{\partial a_1}{\partial x} &= \frac{\partial a_0}{\partial y} \end{aligned}$$

Looking at the the ode given we see that

$$a_2 = y$$

$$a_1 = y'$$

$$a_0 = 0$$

Applying the conditions to the above shows this is a nonlinear exact second order ode. Therefore it can be reduced to first order ode given by

$$\int a_2 dy' + \int a_1 dy + \int a_0 dx = c_1$$

$$\int y dy' + \int y' dy + \int 0 dx = c_1$$

Which results in

$$y'y = c_1$$

Which is now solved.

Solve Integrating gives

$$\int \frac{y}{c_1} dy = dx$$

$$\frac{y^2}{2c_1} = x + c_2$$

Solving for y gives

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

Summary of solutions found

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

Solved as second order missing x ode

Time used: 0.130 (sec)

Solve

$$yy'' + y'^2 = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$yp(y) \left(\frac{d}{dy} p(y) \right) + p(y)^2 = 0$$

Which is now solved as first order ode for $p(y)$.

Solved by factoring the differential equation

Time used: 0.025 (sec)

Solve

$$ypp' + p^2 = 0$$

Writing the ode as

$$(p) (p'y + p) = 0$$

Therefore we need to solve the following equations

$$p = 0 \tag{1}$$

$$p'y + p = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)

Solving for p from

$$p = 0$$

Solving gives

$$p = 0$$

Solving equation (2)

Solve In canonical form a linear first order is

$$p' + q(y)p = p(y)$$

Comparing the above to the given ode shows that

$$q(y) = \frac{1}{y}$$

$$p(y) = 0$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dy} \\ &= e^{\int \frac{1}{y} dy} \\ &= y\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dy}\mu p &= 0 \\ \frac{d}{dy}(py) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}py &= \int 0 dy + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor y gives the final solution

$$p = \frac{c_1}{y}$$

Summary of solutions found

$$p = 0$$

$$p = \frac{c_1}{y}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = 0$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dy &= \int 0 dx + c_2 \\ y &= c_2\end{aligned}$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{c_1}{y}$$

Solve Integrating gives

$$\begin{aligned}\int \frac{y}{c_1} dy &= dx \\ \frac{y^2}{2c_1} &= x + c_3\end{aligned}$$

Solving for y gives

$$\begin{aligned}y &= \sqrt{2c_3c_1 + 2c_1x} \\ y &= -\sqrt{2c_3c_1 + 2c_1x}\end{aligned}$$

Summary of solutions found

$$\begin{aligned}y &= c_2 \\ y &= \sqrt{2c_3c_1 + 2c_1x} \\ y &= -\sqrt{2c_3c_1 + 2c_1x}\end{aligned}$$

Solved as second order integrable as is ode

Time used: 0.049 (sec)

Solve

$$yy'' + y'^2 = 0$$

Integrating both sides of the ODE w.r.t x gives

$$\begin{aligned}\int (yy'' + y'^2) dx &= 0 \\ y'y &= c_1\end{aligned}$$

Which is now solved for y . Solve Integrating gives

$$\int \frac{y}{c_1} dy = dx$$

$$\frac{y^2}{2c_1} = x + c_2$$

Solving for y gives

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

Summary of solutions found

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.042 (sec)

Solve

$$yy'' + y'^2 = 0$$

Writing the ode as

$$yy'' + y'^2 = 0$$

Integrating both sides of the ODE w.r.t x gives

$$\int (yy'' + y'^2) dx = 0$$

$$y'y = c_1$$

Which is now solved for y . Solve Integrating gives

$$\int \frac{y}{c_1} dy = dx$$

$$\frac{y^2}{2c_1} = x + c_2$$

Solving for y gives

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

Solve Integrating gives

$$\int \frac{y}{c_1} dy = dx$$

$$\frac{y^2}{2c_1} = x + c_2$$

Solving for y gives

$$y = \sqrt{2c_1c_2 + 2c_1x}$$

$$y = -\sqrt{2c_1c_2 + 2c_1x}$$

✓ **Maple.** Time used: 0.012 (sec). Leaf size: 33

```
ode:=diff(y(x),x)^2+y(x)*diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = 0$$

$$y = \sqrt{2c_1x + 2c_2}$$

$$y = -\sqrt{2c_1x + 2c_2}$$

Maple trace

Methods for second order ODEs:

--- Trying classification methods ---

trying 2nd order Liouville

<- 2nd_order Liouville successful

Maple step by step

Let's solve

$$y(x) \left(\frac{d^2}{dx^2} y(x) \right) + \left(\frac{d}{dx} y(x) \right)^2 = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Define new dependent variable u

$$u(x) = \frac{d}{dx} y(x)$$

- Compute $\frac{d^2}{dx^2} y(x)$

$$\frac{d}{dx} u(x) = \frac{d^2}{dx^2} y(x)$$

- Use chain rule on the lhs

$$\left(\frac{d}{dx} y(x) \right) \left(\frac{d}{dy} u(y) \right) = \frac{d^2}{dx^2} y(x)$$

- Substitute in the definition of u

$$u(y) \left(\frac{d}{dy} u(y) \right) = \frac{d^2}{dx^2} y(x)$$

- Make substitutions $\frac{d}{dx}y(x) = u(y)$, $\frac{d^2}{dx^2}y(x) = u(y) \left(\frac{d}{dy}u(y) \right)$ to reduce order of ODE
 $yu(y) \left(\frac{d}{dy}u(y) \right) + u(y)^2 = 0$
- Solve for the highest derivative
 $\frac{d}{dy}u(y) = -\frac{u(y)}{y}$
- Separate variables
 $\frac{\frac{d}{dy}u(y)}{u(y)} = -\frac{1}{y}$
- Integrate both sides with respect to y
 $\int \frac{\frac{d}{dy}u(y)}{u(y)} dy = \int -\frac{1}{y} dy + C1$
- Evaluate integral
 $\ln(u(y)) = -\ln(y) + C1$
- Solve for $u(y)$
 $u(y) = \frac{e^{C1}}{y}$
- Redefine the integration constant(s)
 $u(y) = \frac{C1}{y}$
- Solve 1st ODE for $u(y)$
 $u(y) = \frac{C1}{y}$
- Revert to original variables with substitution $u(y) = \frac{d}{dx}y(x)$, $y = y(x)$
 $\frac{d}{dx}y(x) = \frac{C1}{y(x)}$
- Solve for the highest derivative
 $\frac{d}{dx}y(x) = \frac{C1}{y(x)}$
- Separate variables
 $\left(\frac{d}{dx}y(x) \right) y(x) = C1$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}y(x) \right) y(x) dx = \int C1 dx + C2$
- Evaluate integral
 $\frac{y(x)^2}{2} = C1x + C2$
- Solve for $y(x)$
 $\{y(x) = \sqrt{2C1x + 2C2}, y(x) = -\sqrt{2C1x + 2C2}\}$
- Redefine the integration constant(s)
 $\{y(x) = \sqrt{2C1x + C2}, y(x) = -\sqrt{2C1x + C2}\}$
- Redefine the integration constant(s)
 $\{y(x) = \sqrt{C1x + C2}, y(x) = C1\sqrt{x + C2}, y(x) = C2\sqrt{C1x + 1}\}$

✓ **Mathematica.** Time used: 0.109 (sec). Leaf size: 20

```
ode=y[x]*D[y[x],{x,2}]+D[y[x],x]^2==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_2 \sqrt{2x - c_1}$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x)*Derivative(y(x), (x, 2)) + Derivative(y(x), x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE -sqrt(-y(x)*Derivative(y(x), (x, 2))) +
Derivative(y(x), x) cannot be solved by the factorable group method
```

2.14.7 Problem 28

Local contents

Solved as second order missing x ode	2415
Solved by factoring the differential equation	2416
✓ Maple	2420
✓ Mathematica	2421
✗ Sympy	2421

Internal problem ID [8053]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 28

Date solved : Thursday, November 27, 2025 at 07:32:54 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_y_y1]]

Solved as second order missing x ode

Time used: 0.461 (sec)

Solve

$$yy'' = y'^2(1 - y' \cos(y) + yy' \sin(y))$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$yp(y) \left(\frac{d}{dy} p(y) \right) = p(y)^2 (1 - p(y) \cos(y) + yp(y) \sin(y))$$

Which is now solved as first order ode for $p(y)$.

Solved by factoring the differential equation

Time used: 0.131 (sec)

Solve

$$ypp' = p^2(1 - p \cos(y) + yp \sin(y))$$

Writing the ode as

$$(p) (\sin(y) p^2 y - \cos(y) p^2 - p'y + p) = 0$$

Therefore we need to solve the following equations

$$p = 0 \tag{1}$$

$$\sin(y) p^2 y - \cos(y) p^2 - p'y + p = 0 \tag{2}$$

Now each of the above equations is solved in turn.

Solving equation (1)Solving for p from

$$p = 0$$

Solving gives

$$p = 0$$

Solving equation (2)

Solve

In canonical form, the ODE is

$$\begin{aligned} p' &= F(y, p) \\ &= \frac{p(1 - p \cos(y) + yp \sin(y))}{y} \end{aligned}$$

This is a Bernoulli ODE.

$$p' = \left(\frac{1}{y}\right) p + \left(\frac{\sin(y) y - \cos(y)}{y}\right) p^2 \tag{1}$$

The standard Bernoulli ODE has the form

$$p' = f_0(y)p + f_1(y)p^n \tag{2}$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1}{y} \\ f_1 &= \frac{\sin(y) y - \cos(y)}{y} \end{aligned}$$

The first step is to divide the above equation by p^n which gives

$$\frac{p'}{p^n} = f_0(y)p^{1-n} + f_1(y) \quad (3)$$

The next step is use the substitution $v = p^{1-n}$ in equation (3) which generates a new ODE in $v(y)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $p(y)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(y) &= \frac{1}{y} \\ f_1(y) &= \frac{\sin(y)y - \cos(y)}{y} \\ n &= 2 \end{aligned}$$

Dividing both sides of ODE (1) by $p^n = p^2$ gives

$$p' \frac{1}{p^2} = \frac{1}{yp} + \frac{\sin(y)y - \cos(y)}{y} \quad (4)$$

Let

$$\begin{aligned} v &= p^{1-n} \\ &= \frac{1}{p} \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t y gives

$$v' = -\frac{1}{p^2}p' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} -v'(y) &= \frac{v(y)}{y} + \frac{\sin(y)y - \cos(y)}{y} \\ v' &= -\frac{v}{y} - \frac{\sin(y)y - \cos(y)}{y} \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(y)$ which is now solved.

In canonical form a linear first order is

$$v'(y) + q(y)v(y) = p(y)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(y) &= \frac{1}{y} \\ p(y) &= \frac{-\sin(y)y + \cos(y)}{y} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q dy} \\ &= e^{\int \frac{1}{y} dy} \\ &= y\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dy}(\mu v) &= \mu p \\ \frac{d}{dy}(\mu v) &= (\mu) \left(\frac{-\sin(y) y + \cos(y)}{y} \right) \\ \frac{d}{dy}(vy) &= (y) \left(\frac{-\sin(y) y + \cos(y)}{y} \right) \\ d(vy) &= (-\sin(y) y + \cos(y)) dy\end{aligned}$$

Integrating gives

$$\begin{aligned}vy &= \int -\sin(y) y + \cos(y) dy \\ &= \cos(y) y + c_1\end{aligned}$$

Dividing throughout by the integrating factor y gives the final solution

$$v(y) = \frac{\cos(y) y + c_1}{y}$$

The substitution $v = p^{1-n}$ is now used to convert the above solution back to p which results in

$$\frac{1}{p} = \frac{\cos(y) y + c_1}{y}$$

Solving for p gives

$$p = \frac{y}{\cos(y) y + c_1}$$

Summary of solutions found

$$p = 0$$

$$p = \frac{y}{\cos(y) y + c_1}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = 0$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\int dy = \int 0 dx + c_2$$

$$y = c_2$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{y}{\cos(y)y + c_1}$$

Solve Integrating gives

$$\int \frac{\cos(y)y + c_1}{y} dy = dx$$

$$\sin(y) + c_1 \ln(y) = x + c_3$$

Singular solutions are found by solving

$$\frac{y}{\cos(y)y + c_1} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = 0$$

Solving for y from the above solution(s) gives (after possible removing of solutions that do not verify)

$$y = 0$$

$$y = c_2$$

$$y = \text{RootOf}\left(-Z - e^{\frac{c_3 + x - \sin(-Z)}{c_1}}\right)$$

Summary of solutions found

$$y = 0$$

$$y = c_2$$

$$y = \text{RootOf}\left(-Z - e^{\frac{c_3 + x - \sin(-Z)}{c_1}}\right)$$

✓ **Maple.** Time used: 0.048 (sec). Leaf size: 22

```
ode:=y(x)*diff(diff(y(x),x),x) = diff(y(x),x)^2*(1-cos(y(x))*diff(y(x),x)+y(x)*
  diff(y(x),x)*sin(y(x)));
dsolve(ode,y(x), singsol=all);
```

$$y = c_1$$

$$\sin(y) + c_1 \ln(y) - x - c_2 = 0$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
--- trying a change of variables {x -> y(x), y(x) -> x}
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying differential order: 2; missing variables
-> Calling odsolve with the ODE, diff(_b(_a),_a)*_b(_a)-_b(_a)^2*( _b(_a)*sin(_a
)*_a-_b(_a)*cos(_a)+1)/_a = 0, _b(_a)
*** Sublevel 2 ***
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful
<- differential order: 2; canonical coordinates successful
<- differential order 2; missing variables successful
```


✓ **Mathematica.** Time used: 0.247 (sec). Leaf size: 147

```
ode=y[x]*D[y[x],{x,2}]==D[y[x],x]^2*(1-D[y[x],x]*Cos[y[x]]+y[x]*D[y[x],x]*Sin[y[x]]);
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow \text{InverseFunction} \left[\int_1^{\#1} \frac{c_1 - \int_1^{K[2]} (K[1] \sin(K[1]) - \cos(K[1])) dK[1]}{K[2]} dK[2] \& \right] [x + c_2]$$

$$y(x) \rightarrow \text{InverseFunction} \left[\int_1^{\#1} \frac{-c_1 - \int_1^{K[2]} (K[1] \sin(K[1]) - \cos(K[1])) dK[1]}{K[2]} dK[2] \& \right] [x + c_2]$$

$$y(x) \rightarrow \text{InverseFunction} \left[\int_1^{\#1} \frac{c_1 - \int_1^{K[2]} (K[1] \sin(K[1]) - \cos(K[1])) dK[1]}{K[2]} dK[2] \& \right] [x + c_2]$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((-y(x)*sin(y(x))*Derivative(y(x), x) + cos(y(x))*Derivative(y(x), x) -
1)*Derivative(y(x), x)**2 + y(x)*Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.14.8 Problem 29

Local contents

✓ Maple	2422
✓ Mathematica	2423
✗ Sympy	2424

Internal problem ID [8054]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 29

Date solved : Thursday, November 27, 2025 at 07:32:55 AM

CAS classification : [[_3rd_order, _with_linear_symmetries]]

$$(2x - 3)y''' - (6x - 7)y'' + 4y'x - 4y = 8$$

Solve

$$(2x - 3)y''' - (6x - 7)y'' + 4y'x - 4y = 8$$

Failed to solve.

✓ **Maple.** Time used: 0.007 (sec). Leaf size: 19

```
ode:=(2*x-3)*diff(diff(diff(y(x),x),x),x)-(6*x-7)*diff(diff(y(x),x),x)+4*diff(y(x),x)*x-4*y(x) = 8;
dsolve(ode,y(x), singsol=all);
```

$$y = -2 + c_1x + c_2e^x + c_3e^{2x}$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

trying high order exact linear fully integrable

trying differential order: 3; linear nonhomogeneous with symmetry [0,1]

trying high order linear exact nonhomogeneous

trying differential order: 3; missing the dependent variable

checking if the LODE is of Euler type

Equation is the LCLM of $-1/x*y(x)+diff(y(x),x)$, $-y(x)+diff(y(x),x)$, $-2*y(x)+dif\backslash$

```
f(y(x),x)
trying differential order: 1; missing the dependent variable
checking if the LODE is of Euler type
<- LODE of Euler type successful
Euler equation successful
trying differential order: 1; missing the dependent variable
checking if the LODE has constant coefficients
<- constant coefficients successful
trying differential order: 1; missing the dependent variable
checking if the LODE has constant coefficients
<- constant coefficients successful
<- solving the LCLM ode successful
```

Maple step by step

Let's solve

$$(2x - 3) \left(\frac{d^3}{dx^3} y(x) \right) - (6x - 7) \left(\frac{d^2}{dx^2} y(x) \right) + 4x \left(\frac{d}{dx} y(x) \right) - 4y(x) = 8$$

- Highest derivative means the order of the ODE is 3
 $\frac{d^3}{dx^3} y(x)$

✓ **Mathematica.** Time used: 0.124 (sec). Leaf size: 219

```
ode=(2*x-3)*D[y[x],{x,3}]- (6*x-7)*D[y[x],{x,2}]+4*x*D[y[x],x]-4*y[x]==8;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_2 x \int_1^x \exp \left(\int_1^{K[4]} \left(-\frac{1}{2K[1]} - \frac{1}{2} + \frac{1}{3-2K[1]} \right) dK[1] - \frac{1}{2} \int_1^{K[4]} \left(\frac{3}{K[2]} - 3 + \frac{2}{3-2K[2]} \right) dK[2] \right) dK[4]$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x*Derivative(y(x), x) + (2*x - 3)*Derivative(y(x), (x, 3)) - (6*x - 7)
        *Derivative(y(x), (x, 2)) - 4*y(x) - 8, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics)
```

```
NotImplementedError : The given ODE Derivative(y(x), x) - (2*x*(3*Derivative(y(x),
(x, 2)) - Derivative(y(x), (x, 3)))) + 4*y(x) - 7*Derivative(y(x), (x, 2)) +
3*Derivative(y(x), (x, 3)) + 8)/(4*x) cannot be solved by the factorable group
method
```

2.14.9 Problem 30

Local contents

Solved as higher order missing y ode	2425
✓ Maple	2428
✓ Mathematica	2430
✗ Sympy	2431

Internal problem ID [8055]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 30

Date solved : Thursday, November 27, 2025 at 07:32:55 AM

CAS classification : `[[_3rd_order, _missing_y]]`

Solve

$$(2x^3 - 1) y''' - 6x^2 y'' + 6y'x = 0$$

Solved as higher order missing y ode

Time used: 0.250 (sec)

Since y is missing from the ode then we can use the substitution $y' = u(x)$ to reduce the order by one. The ODE becomes

$$(2x^3 - 1) u''(x) - 6u'(x)x^2 + 6u(x)x = 0$$

Which is now solved. In normal form the ode

$$(2x^3 - 1) u''(x) - 6u'(x)x^2 + 6u(x)x = 0 \tag{1}$$

Becomes

$$u''(x) + p(x) u'(x) + q(x) u(x) = 0 \tag{2}$$

Where

$$p(x) = -\frac{6x^2}{2x^3 - 1}$$

$$q(x) = \frac{6x}{2x^3 - 1}$$

Applying change of variables on the dependent variable $u(x) = v(x)x^n$ to (2) gives the following ode where the dependent variables is $v(x)$ and not $u(x)$.

$$v''(x) + \left(\frac{2n}{x} + p\right)v'(x) + \left(\frac{n(n-1)}{x^2} + \frac{np}{x} + q\right)v(x) = 0 \quad (3)$$

Let the coefficient of $v(x)$ above be zero. Hence

$$\frac{n(n-1)}{x^2} + \frac{np}{x} + q = 0 \quad (4)$$

Substituting the earlier values found for $p(x)$ and $q(x)$ into (4) gives

$$\frac{n(n-1)}{x^2} - \frac{6nx}{2x^3-1} + \frac{6x}{2x^3-1} = 0 \quad (5)$$

Solving (5) for n gives

$$n = 1 \quad (6)$$

Substituting this value in (3) gives

$$\begin{aligned} v''(x) + \left(\frac{2}{x} - \frac{6x^2}{2x^3-1}\right)v'(x) &= 0 \\ v''(x) + \frac{(-2x^3-2)v'(x)}{2x^4-x} &= 0 \end{aligned} \quad (7)$$

Using the substitution

$$u(x) = v'(x)$$

Then (7) becomes

$$u'(x) + \frac{(-2x^3-2)u(x)}{2x^4-x} = 0 \quad (8)$$

The above is now solved for $u(x)$. Solve In canonical form a linear first order is

$$u'(x) + q(x)u(x) = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{2x^3+2}{2x^4-x} \\ p(x) &= 0 \end{aligned}$$

The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2x^3+2}{2x^4-x} \, dx} \\ &= \frac{x^2}{2x^3-1}\end{aligned}$$

The ode becomes

$$\begin{aligned}\frac{d}{dx}\mu u &= 0 \\ \frac{d}{dx}\left(\frac{u x^2}{2x^3-1}\right) &= 0\end{aligned}$$

Integrating gives

$$\begin{aligned}\frac{u x^2}{2x^3-1} &= \int 0 \, dx + c_1 \\ &= c_1\end{aligned}$$

Dividing throughout by the integrating factor $\frac{x^2}{2x^3-1}$ gives the final solution

$$u(x) = \frac{(2x^3-1)c_1}{x^2}$$

Now that $u(x)$ is known, then

$$\begin{aligned}v'(x) &= u(x) \\ v(x) &= \int u(x) \, dx + c_2 \\ &= c_1\left(x^2 + \frac{1}{x}\right) + c_2\end{aligned}$$

Hence

$$\begin{aligned}u(x) &= v(x) x^n \\ &= \left(c_1\left(x^2 + \frac{1}{x}\right) + c_2\right) x \\ &= c_1 x^3 + c_2 x + c_1\end{aligned}$$

Solve Since the ode has the form $y' = f(x)$, then we only need to integrate $f(x)$.

$$\begin{aligned}\int dy &= \int c_1 x^3 + c_2 x + c_1 \, dx \\ y &= \frac{1}{4}c_1 x^4 + \frac{1}{2}c_2 x^2 + c_1 x + c_3\end{aligned}$$

✓ **Maple.** Time used: 0.008 (sec). Leaf size: 20

```
ode:=(2*x^3-1)*diff(diff(diff(y(x),x),x),x)-6*x^2*diff(diff(y(x),x),x)+6*diff(y(x),x)*x = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = c_3 x^4 + c_2 x^2 + 4c_3 x + c_1$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying high order exact linear fully integrable

trying to convert to a linear ODE with constant coefficients

trying differential order: 3; missing the dependent variable

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

checking if the LODE is missing y

-> Trying a Liouvillian solution using Kovacic's algorithm

A Liouvillian solution exists

Reducible group (found an exponential solution)

<- Kovacic's algorithm successful

<- missing the dependent variable successful

Maple step by step

Let's solve

$$(2x^3 - 1) \left(\frac{d^3}{dx^3} y(x) \right) - 6x^2 \left(\frac{d^2}{dx^2} y(x) \right) + 6x \left(\frac{d}{dx} y(x) \right) = 0$$

- Highest derivative means the order of the ODE is 3

$$\frac{d^3}{dx^3} y(x)$$

- Check to see if x_0 is a regular singular point

- Define functions

$$\left[P_2(x) = -\frac{6x^2}{2x^3-1}, P_3(x) = \frac{6x}{2x^3-1}, P_4(x) = 0 \right]$$

- $\left(x + \frac{2^{2/3}}{4} + \frac{1\sqrt{3}2^{2/3}}{4} \right) \cdot P_2(x)$ is analytic at $x = -\frac{2^{2/3}}{4} - \frac{1\sqrt{3}2^{2/3}}{4}$

$$\left(\left(x + \frac{2^{2/3}}{4} + \frac{1\sqrt{3}2^{2/3}}{4} \right) \cdot P_2(x) \right) \Big|_{x = -\frac{2^{2/3}}{4} - \frac{1\sqrt{3}2^{2/3}}{4}} = 0$$

- $\left(x + \frac{2^{2/3}}{4} + \frac{\text{I}\sqrt{3}2^{2/3}}{4}\right)^2 \cdot P_3(x)$ is analytic at $x = -\frac{2^{2/3}}{4} - \frac{\text{I}\sqrt{3}2^{2/3}}{4}$
 $\left(\left(x + \frac{2^{2/3}}{4} + \frac{\text{I}\sqrt{3}2^{2/3}}{4}\right)^2 \cdot P_3(x)\right) \Big|_{x=-\frac{2^{2/3}}{4}-\frac{\text{I}\sqrt{3}2^{2/3}}{4}} = 0$
- $\left(x + \frac{2^{2/3}}{4} + \frac{\text{I}\sqrt{3}2^{2/3}}{4}\right)^3 \cdot P_4(x)$ is analytic at $x = -\frac{2^{2/3}}{4} - \frac{\text{I}\sqrt{3}2^{2/3}}{4}$
 $\left(\left(x + \frac{2^{2/3}}{4} + \frac{\text{I}\sqrt{3}2^{2/3}}{4}\right)^3 \cdot P_4(x)\right) \Big|_{x=-\frac{2^{2/3}}{4}-\frac{\text{I}\sqrt{3}2^{2/3}}{4}} = 0$
- $x = -\frac{2^{2/3}}{4} - \frac{\text{I}\sqrt{3}2^{2/3}}{4}$ is a regular singular point
 Check to see if x_0 is a regular singular point
 $x_0 = -\frac{2^{2/3}}{4} - \frac{\text{I}\sqrt{3}2^{2/3}}{4}$
- Change variables using $x = u - \frac{2^{2/3}}{4} - \frac{\text{I}\sqrt{3}2^{2/3}}{4}$ so that the regular singular point is at $u = 0$
 $\left(2u^3 - \frac{3u^2 2^{2/3}}{2} - \frac{3\text{I}u^2 \sqrt{3} 2^{2/3}}{2} - \frac{3u 2^{1/3}}{2} + \frac{3\text{I}u 2^{1/3} \sqrt{3}}{2}\right) \left(\frac{d^3}{du^3}y(u)\right) + \left(-6u^2 + 3u 2^{2/3} + 3\text{I}u \sqrt{3} 2^{2/3} + \frac{3 2^{1/3}}{2}\right)$
- Assume series solution for $y(u)$

$$y(u) = \sum_{k=0}^{\infty} a_k u^{k+r}$$
- Rewrite ODE with series expansions
 - Convert $u^m \cdot \left(\frac{d}{du}y(u)\right)$ to series expansion for $m = 0..1$

$$u^m \cdot \left(\frac{d}{du}y(u)\right) = \sum_{k=0}^{\infty} a_k (k+r) u^{k+r-1+m}$$
 - Shift index using $k- > k+1-m$

$$u^m \cdot \left(\frac{d}{du}y(u)\right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) u^{k+r}$$
 - Convert $u^m \cdot \left(\frac{d^2}{du^2}y(u)\right)$ to series expansion for $m = 0..2$

$$u^m \cdot \left(\frac{d^2}{du^2}y(u)\right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) u^{k+r-2+m}$$
 - Shift index using $k- > k+2-m$

$$u^m \cdot \left(\frac{d^2}{du^2}y(u)\right) = \sum_{k=-2+m}^{\infty} a_{k+2-m} (k+2-m+r)(k+1-m+r) u^{k+r}$$
 - Convert $u^m \cdot \left(\frac{d^3}{du^3}y(u)\right)$ to series expansion for $m = 1..3$

$$u^m \cdot \left(\frac{d^3}{du^3}y(u)\right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1)(k+r-2) u^{k+r-3+m}$$
 - Shift index using $k- > k+3-m$

$$u^m \cdot \left(\frac{d^3}{du^3}y(u)\right) = \sum_{k=-3+m}^{\infty} a_{k+3-m} (k+3-m+r)(k+2-m+r)(k+1-m+r) u^{k+r}$$
- Rewrite ODE with series expansions

$$\frac{3 2^{1/3} (\text{I}\sqrt{3}-1)(-3+r)(r-1) r a_0 u^{-2+r}}{2} + \left(\frac{3 2^{1/3} (\text{I}\sqrt{3}-1)(-2+r) r (1+r) a_1}{2} - \frac{3 2^{2/3} (1+\text{I}\sqrt{3}) r (r^2-5r+5) a_0}{2} \right) u^{r-1} + \left(\sum_{k=0}^{\infty} \right)$$
- a_0 cannot be 0 by assumption, giving the indicial equation

$$\frac{3 2^{1/3} (\text{I}\sqrt{3}-1)(-3+r)(r-1) r}{2} = 0$$
- Values of r that satisfy the indicial equation
 $r \in \{0, 1, 3\}$

- [illegible]

✓ **Mathematica.** Time used: 0.258 (sec). Leaf size: 70

```
ode=(2*x^3-1)*D[y[x],{x,3}]-6*x^2*D[y[x],{x,2}]+6*x*D[y[x],x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$\begin{aligned} y(x) &\rightarrow \int_1^x (c_1 K[1] - c_2 (K[1]^3 + 1)) dK[1] + c_3 \\ y(x) &\rightarrow \frac{1}{2} c_1 (x^2 - 1) + c_3 \\ y(x) &\rightarrow c_3 - \frac{1}{4} c_2 (x^4 + 4x - 5) \end{aligned}$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-6*x**2*Derivative(y(x), (x, 2)) + 6*x*Derivative(y(x), x) + (2*x**3 -
    1)*Derivative(y(x), (x, 3)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.14.10 Problem 31

Local contents

Solved as second order missing x ode	2432
✓ Maple	2436
✓ Mathematica	2436
✗ Sympy	2437

Internal problem ID [8056]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 31

Date solved : Thursday, November 27, 2025 at 07:32:56 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_xy]]

Solved as second order missing x ode

Time used: 0.868 (sec)

Solve

$$yy'' - y'^2 = y^2 \ln(y)$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$yp(y) \left(\frac{d}{dy} p(y) \right) - p(y)^2 = y^2 \ln(y)$$

Which is now solved as first order ode for $p(y)$.

Solve

In canonical form, the ODE is

$$\begin{aligned} p' &= F(y, p) \\ &= \frac{p^2 + y^2 \ln(y)}{yp} \end{aligned}$$

This is a Bernoulli ODE.

$$p' = \left(\frac{1}{y}\right)p + (y \ln(y)) \frac{1}{p} \quad (1)$$

The standard Bernoulli ODE has the form

$$p' = f_0(y)p + f_1(y)p^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= \frac{1}{y} \\ f_1 &= y \ln(y) \end{aligned}$$

The first step is to divide the above equation by p^n which gives

$$\frac{p'}{p^n} = f_0(y)p^{1-n} + f_1(y) \quad (3)$$

The next step is use the substitution $v = p^{1-n}$ in equation (3) which generates a new ODE in $v(y)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $p(y)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(y) &= \frac{1}{y} \\ f_1(y) &= y \ln(y) \\ n &= -1 \end{aligned}$$

Dividing both sides of ODE (1) by $p^n = \frac{1}{p}$ gives

$$p'p = \frac{p^2}{y} + y \ln(y) \quad (4)$$

Let

$$\begin{aligned} v &= p^{1-n} \\ &= p^2 \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t y gives

$$v' = 2pp' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(y)}{2} &= \frac{v(y)}{y} + y \ln(y) \\ v' &= \frac{2v}{y} + 2y \ln(y) \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(y)$ which is now solved.

In canonical form a linear first order is

$$v'(y) + q(y)v(y) = p(y)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(y) &= -\frac{2}{y} \\ p(y) &= 2y \ln(y) \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dy} \\ &= e^{\int -\frac{2}{y} \, dy} \\ &= \frac{1}{y^2} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dy}(\mu v) &= \mu p \\ \frac{d}{dy}(\mu v) &= (\mu) (2y \ln(y)) \\ \frac{d}{dy} \left(\frac{v}{y^2} \right) &= \left(\frac{1}{y^2} \right) (2y \ln(y)) \\ d \left(\frac{v}{y^2} \right) &= \left(\frac{2 \ln(y)}{y} \right) dy \end{aligned}$$

Integrating gives

$$\begin{aligned} \frac{v}{y^2} &= \int \frac{2 \ln(y)}{y} \, dy \\ &= \ln(y)^2 + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{y^2}$ gives the final solution

$$v(y) = y^2(\ln(y)^2 + c_1)$$

The substitution $v = p^{1-n}$ is now used to convert the above solution back to p which results in

$$p^2 = y^2(\ln(y)^2 + c_1)$$

Solving for p gives

$$p = \sqrt{\ln(y)^2 + c_1} y$$

$$p = -\sqrt{\ln(y)^2 + c_1} y$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \sqrt{\ln(y)^2 + c_1} y$$

Solve Integrating gives

$$\int \frac{1}{\sqrt{\ln(y)^2 + c_1} y} dy = dx$$

$$\ln\left(\ln(y) + \sqrt{\ln(y)^2 + c_1}\right) = x + c_2$$

Solving for y gives

$$y = e^{\frac{(e^{2x+2c_2}-c_1)e^{-x-c_2}}{2}}$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\sqrt{\ln(y)^2 + c_1} y$$

Solve Integrating gives

$$\int -\frac{1}{\sqrt{\ln(y)^2 + c_1} y} dy = dx$$

$$-\ln\left(\ln(y) + \sqrt{\ln(y)^2 + c_1}\right) = x + c_3$$

Solving for y gives

$$y = e^{\frac{(e^{-2c_3-2x}-c_1)e^{x+c_3}}{2}}$$

Summary of solutions found

$$y = e^{\frac{(e^{-2c_3-2x}-c_1)e^x+c_3}{2}}$$

$$y = e^{\frac{(e^{2x+2c_2}-c_1)e^{-x}-c_2}{2}}$$

✓ **Maple.** Time used: 0.007 (sec). Leaf size: 18

```
ode:=y(x)*diff(diff(y(x),x),x)-diff(y(x),x)^2 = ln(y(x))*y(x)^2;
dsolve(ode,y(x), singsol=all);
```

$$y = e^{-\frac{c_1 e^x}{2} + \frac{c_2 e^{-x}}{2}}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
<- 2nd order, 2 integrating factors of the form mu(x,y) successful
```

✓ **Mathematica.** Time used: 0.622 (sec). Leaf size: 63

```
ode=y[x]*D[y[x],{x,2}]-D[y[x],x]^2==y[x]^2*Log[y[x]];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow e^{\frac{1}{2}(e^{x+c_2}-c_1e^{-x-c_2})}$$

$$y(x) \rightarrow e^{\frac{1}{2}(e^{-x-c_2}-c_1e^{x+c_2})}$$

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-y(x)**2*log(y(x)) + y(x)*Derivative(y(x), (x, 2)) - Derivative(y(x), x)
        **2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
TypeError : cannot determine truth value of Relational: 2 < _X0**2
```

2.14.11 Problem 32

Local contents

Solved as second order integrable as is ode	2438
Solved as second order integrable as is ode (ABC method)	2441
✓ Maple	2446
✓ Mathematica	2447
✗ Sympy	2447

Internal problem ID [8057]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 32

Date solved : Thursday, November 27, 2025 at 07:32:57 AM

CAS classification :

[[_2nd_order, _exact, _nonlinear], [_2nd_order, _with_linear_symmetries], [_2nd_order, _

Solved as second order integrable as is ode

Time used: 0.221 (sec)

Solve

$$(x + 2y) y'' + 2y'^2 + 2y' = 2$$

Integrating both sides of the ODE w.r.t x gives

$$\int \left((x + 2y) y'' + 2y'^2 + 2y' \right) dx = \int 2dx$$

$$y + (x + 2y) y' = 2x + c_1$$

Which is now solved for y .

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N\end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned}(x + 2y) dy &= (-y + 2x + c_1) dx \\ (y - 2x - c_1) dx + (x + 2y) dy &= 0\end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= y - 2x - c_1 \\ N(x, y) &= x + 2y\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y - 2x - c_1) \\ &= 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x + 2y) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int y - 2x - c_1 dx \\ \phi &= -x(c_1 + x - y) + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x + 2y$. Therefore equation (4) becomes

$$x + 2y = x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 2y$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (2y) dy \\ f(y) &= y^2 + c_2 \end{aligned}$$

Where c_2 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -x(c_1 + x - y) + y^2 + c_2$$

But since ϕ itself is a constant function, then let $\phi = c_3$ where c_2 is new constant and combining c_2 and c_3 constants into the constant c_2 gives the solution as

$$c_2 = -x(c_1 + x - y) + y^2$$

Solving for y gives

$$y = -\frac{x}{2} - \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

$$y = -\frac{x}{2} + \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

Summary of solutions found

$$y = -\frac{x}{2} - \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

$$y = -\frac{x}{2} + \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

Solved as second order integrable as is ode (ABC method)

Time used: 0.125 (sec)

Solve

$$(x + 2y)y'' + 2y'^2 + 2y' = 2$$

Writing the ode as

$$(x + 2y)y'' + 2y'^2 + 2y' = 2$$

Integrating both sides of the ODE w.r.t x gives

$$\int \left((x + 2y)y'' + 2y'^2 + 2y' \right) dx = \int 2dx$$

$$y + (x + 2y)y' = 2x + c_1$$

Which is now solved for y .

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \tag{A}$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (1A)$$

Therefore

$$\begin{aligned} (x + 2y) dy &= (-y + 2x + c_1) dx \\ (y - 2x - c_1) dx + (x + 2y) dy &= 0 \end{aligned} \quad (2A)$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= y - 2x - c_1 \\ N(x, y) &= x + 2y \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y - 2x - c_1) \\ &= 1 \end{aligned}$$

And

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x + 2y) \\ &= 1 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \quad (1)$$

$$\frac{\partial \phi}{\partial y} = N \quad (2)$$

Integrating (1) w.r.t. x gives

$$\begin{aligned} \int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int y - 2x - c_1 dx \\ \phi &= -x(c_1 + x - y) + f(y) \end{aligned} \quad (3)$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x + f'(y) \quad (4)$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x + 2y$. Therefore equation (4) becomes

$$x + 2y = x + f'(y) \quad (5)$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 2y$$

Integrating the above w.r.t y gives

$$\begin{aligned} \int f'(y) dy &= \int (2y) dy \\ f(y) &= y^2 + c_2 \end{aligned}$$

Where c_2 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -x(c_1 + x - y) + y^2 + c_2$$

But since ϕ itself is a constant function, then let $\phi = c_3$ where c_2 is new constant and combining c_2 and c_3 constants into the constant c_2 gives the solution as

$$c_2 = -x(c_1 + x - y) + y^2$$

Solving for y gives

$$y = -\frac{x}{2} - \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

$$y = -\frac{x}{2} + \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx}\phi(x, y) = 0$$

Hence

$$\frac{\partial\phi}{\partial x} + \frac{\partial\phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\frac{\partial\phi}{\partial x} = M$$

$$\frac{\partial\phi}{\partial y} = N$$

But since $\frac{\partial^2\phi}{\partial x\partial y} = \frac{\partial^2\phi}{\partial y\partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2\phi}{\partial x\partial y} = \frac{\partial^2\phi}{\partial y\partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$(x + 2y) dy = (-y + 2x + c_1) dx$$

$$(y - 2x - c_1) dx + (x + 2y) dy = 0 \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned}M(x, y) &= y - 2x - c_1 \\N(x, y) &= x + 2y\end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y - 2x - c_1) \\&= 1\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x + 2y) \\&= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the ODE is exact. The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = M \tag{1}$$

$$\frac{\partial \phi}{\partial y} = N \tag{2}$$

Integrating (1) w.r.t. x gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial x} dx &= \int M dx \\ \int \frac{\partial \phi}{\partial x} dx &= \int y - 2x - c_1 dx \\ \phi &= -x(c_1 + x - y) + f(y)\end{aligned} \tag{3}$$

Where $f(y)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t y gives

$$\frac{\partial \phi}{\partial y} = x + f'(y) \tag{4}$$

But equation (2) says that $\frac{\partial \phi}{\partial y} = x + 2y$. Therefore equation (4) becomes

$$x + 2y = x + f'(y) \tag{5}$$

Solving equation (5) for $f'(y)$ gives

$$f'(y) = 2y$$

Integrating the above w.r.t y gives

$$\int f'(y) dy = \int (2y) dy$$

$$f(y) = y^2 + c_2$$

Where c_2 is constant of integration. Substituting result found above for $f(y)$ into equation (3) gives ϕ

$$\phi = -x(c_1 + x - y) + y^2 + c_2$$

But since ϕ itself is a constant function, then let $\phi = c_3$ where c_2 is new constant and combining c_2 and c_3 constants into the constant c_2 gives the solution as

$$c_2 = -x(c_1 + x - y) + y^2$$

Solving for y gives

$$y = -\frac{x}{2} - \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

$$y = -\frac{x}{2} + \frac{\sqrt{4c_1x + 5x^2 + 4c_2}}{2}$$

✓ **Maple.** Time used: 0.014 (sec). Leaf size: 49

```
ode:=(x+2*y(x))*diff(diff(y(x),x),x)+2*diff(y(x),x)^2+2*diff(y(x),x) = 2;
dsolve(ode,y(x), singsol=all);
```

$$y = -\frac{x}{2} - \frac{\sqrt{-4c_1x + 5x^2 + 4c_2}}{2}$$

$$y = -\frac{x}{2} + \frac{\sqrt{-4c_1x + 5x^2 + 4c_2}}{2}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
```

```

trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying a quadrature
<- quadrature successful
<- 2nd order, 2 integrating factors of the form mu(x,y) successful

```

✓ **Mathematica.** Time used: 0.44 (sec). Leaf size: 77

```

ode=(x+2*y[x])*D[y[x],[x,2]]+2*D[y[x],x]^2+2*D[y[x],x]==2;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

$$y(x) \rightarrow -\frac{1}{2}x \left(1 + \sqrt{\frac{1}{x^2} \sqrt{5x^2 + 4c_2x + 4c_1}} \right)$$

$$y(x) \rightarrow \frac{1}{2}x \left(-1 + \sqrt{\frac{1}{x^2} \sqrt{5x^2 + 4c_2x + 4c_1}} \right)$$

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((x + 2*y(x))*Derivative(y(x), (x, 2)) + 2*Derivative(y(x), x)**2 + 2*
    Derivative(y(x), x) - 2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)

```

```

NotImplementedError : The given ODE -sqrt(-2*x*Derivative(y(x), (x, 2)) - 4*y(x)*
    Derivative(y(x), (x, 2)) + 5)/2 + Derivative(y(x), x) + 1/2 cannot be solved
    by the factorable group method

```

2.14.12 Problem 33

Local contents

✓ Maple	2448
✗ Mathematica	2449
✗ Sympy	2450

Internal problem ID [8058]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 33

Date solved : Thursday, November 27, 2025 at 07:32:57 AM

CAS classification : [[_3rd_order, _exact, _nonlinear]]

$$(1 + 2y + 3y^2) y''' + 6y'(y'' + y'^2 + 3yy'') = x$$

Solve

$$(1 + 2y + 3y^2) y''' + 6y'(y'' + y'^2 + 3yy'') = x$$

Failed to solve.

✓ **Maple.** Time used: 0.023 (sec). Leaf size: 1110

```
ode:=(1+2*y(x)+3*y(x)^2)*diff(diff(diff(y(x),x),x),x)+6*diff(y(x),x)*(diff(diff(y(x),x),x)+diff(y(x),x)^2+3*y(x)*diff(diff(y(x),x),x)) = x;
dsolve(ode,y(x), singsol=all);
```

Solution too large to show

Maple trace

```
Methods for third order ODEs:
--- Trying classification methods ---
trying 3rd order ODE linearizable_by_differentiation
differential order: 3; trying a linearization to 4th order
trying differential order: 3; missing variables
trying differential order: 3; exact nonlinear
-> Calling odsolve with the ODE, diff(diff(_b(_a),_a),_a)*(1+2*_b(_a)+3*_b(_a)^
```

```

2)+6*diff(_b(_a),_a)^2*_b(_a)+2*diff(_b(_a),_a)^2-1/2*_a^2+_C1 = 0, _b(_a)
*** Sublevel 2 ***
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying a quadrature
<- quadrature successful
<- 2nd order, 2 integrating factors of the form mu(x,y) successful
<- differential order: 3; exact nonlinear successful

```

✗ Mathematica

```

ode=(1+2*y[x]+3*y[x]^2)*D[y[x],{x,3}]+6*D[y[x],x]*( D[y[x],{x,2}]+D[y[x],x]^2+3*y
[x]*D[y[x],{x,2}] )==x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

Not solved

X Sympy

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x + (18*y(x)*Derivative(y(x), (x, 2)) + 6*Derivative(y(x), x)**2 + 6*
    Derivative(y(x), (x, 2)))*Derivative(y(x), x) + (3*y(x)**2 + 2*y(x) + 1)*
    Derivative(y(x), (x, 3)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)

```

```

NotImplementedError : The given ODE  $(-9y(x) \cdot \text{Derivative}(y(x), (x, 2)) - 3 \cdot \text{Derivative}(y(x), (x, 2))) / (3 \cdot (-1/2 + \sqrt{3})I/2) \cdot (-9x/4 + \sqrt{-4 \cdot (-9y(x) \cdot \text{Derivative}(y(x), (x, 2)) - 3 \cdot \text{Derivative}(y(x), (x, 2)))^3 + (-9x/2 + 27y(x) \cdot \text{Derivative}(y(x), (x, 3))/2 + 9y(x) \cdot \text{Derivative}(y(x), (x, 3)) + 9 \cdot \text{Derivative}(y(x), (x, 3))/2)^2)/2 + 27y(x) \cdot \text{Derivative}(y(x), (x, 3))/4 + 9y(x) \cdot \text{Derivative}(y(x), (x, 3))/2 + 9 \cdot \text{Derivative}(y(x), (x, 3))/4) \cdot (1/3) + (-1/2 + \sqrt{3})I/2) \cdot (-9x/4 + \sqrt{-4 \cdot (-9y(x) \cdot \text{Derivative}(y(x), (x, 2)) - 3 \cdot \text{Derivative}(y(x), (x, 2)))^3 + (-9x/2 + 27y(x) \cdot \text{Derivative}(y(x), (x, 3))/2 + 9y(x) \cdot \text{Derivative}(y(x), (x, 3)) + 9 \cdot \text{Derivative}(y(x), (x, 3))/2)^2)/2 + 27y(x) \cdot \text{Derivative}(y(x), (x, 3))/4 + 9y(x) \cdot \text{Derivative}(y(x), (x, 3))/2 + 9 \cdot \text{Derivative}(y(x), (x, 3))/4} \cdot (1/3)/3 + \text{Derivative}(y(x), x)$  cannot be solved by the factorable group method

```

2.14.13 Problem 34

Local contents

✓ Maple	2451
✗ Mathematica	2452
✗ Sympy	2453

Internal problem ID [8059]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 34

Date solved : Thursday, November 27, 2025 at 07:32:58 AM

CAS classification :

[[_3rd_order, _exact, _nonlinear], [_3rd_order, _with_linear_symmetries]]

$$3x\left(y^2y''' + 6yy'y'' + 2y'^3\right) - 3y\left(yy'' + 2y'^2\right) = -\frac{2}{x}$$

Solve

$$3x\left(y^2y''' + 6yy'y'' + 2y'^3\right) - 3y\left(yy'' + 2y'^2\right) = -\frac{2}{x}$$

Failed to solve.

✓ **Maple.** Time used: 0.015 (sec). Leaf size: 104

```
ode:=3*x*(y(x)^2*diff(diff(diff(y(x),x),x),x)+6*y(x)*diff(y(x),x)*diff(diff(y(x),
x),x)+2*diff(y(x),x)^3)-3*y(x)*(y(x)*diff(diff(y(x),x),x)+2*diff(y(x),x)^2) =
-2/x;
dsolve(ode,y(x), singsol=all);
```

$$y = \frac{(8 \ln(x) x - 8c_3 x^3 + (12c_1 - 4) x + 8c_2)^{1/3}}{2}$$

$$y = -\frac{(8 \ln(x) x - 8c_3 x^3 + (12c_1 - 4) x + 8c_2)^{1/3} (1 + i\sqrt{3})}{4}$$

$$y = \frac{(8 \ln(x) x - 8c_3 x^3 + (12c_1 - 4) x + 8c_2)^{1/3} (-1 + i\sqrt{3})}{4}$$

Maple trace

Methods for third order ODEs:

```

--- Trying classification methods ---
trying 3rd order ODE linearizable_by_differentiation
differential order: 3; trying a linearization to 4th order
trying differential order: 3; missing variables
trying differential order: 3; exact nonlinear
-> Calling odsolve with the ODE,  $\_b(\_a)^2 \text{diff}(\text{diff}(\_b(\_a), \_a), \_a) \_a + 2 \_b(\_a) \text{diff}(\_b(\_a), \_a)^2 \_a - 2 \_b(\_a)^2 \text{diff}(\_b(\_a), \_a) + 2/3 \ln(\_a) + \_C1 = 0, \_b(\_a)$ 
*** Sublevel 2 ***
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form  $\mu(x,y)$ 
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
<- LODE of Euler type successful
<- 2nd order, 2 integrating factors of the form  $\mu(x,y)$  successful
<- differential order: 3; exact nonlinear successful

```

✗ Mathematica

```

ode=3*x*( y[x]^2* D[y[x],{x,3}]+6*y[x]*D[y[x],x]*D[y[x],{x,2}]+2*D[y[x],x]^3 )
-3*y[x]*( y[x]*D[y[x],{x,2}]+2* D[y[x],x]^2 )== -2/x;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

Not solved

Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(3*x*(y(x)**2*Derivative(y(x), (x, 3)) + 6*y(x)*Derivative(y(x), x)*
    Derivative(y(x), (x, 2)) + 2*Derivative(y(x), x)**3) - (3*y(x)*Derivative(y(x)
    , (x, 2)) + 6*Derivative(y(x), x)**2)*y(x) + 2/x,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

Timed Out

2.14.14 Problem 35

Local contents

✓ Maple	2454
✗ Mathematica	2455
✗ Sympy	2455

Internal problem ID [8060]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 35

Date solved : Thursday, November 27, 2025 at 07:32:58 AM

CAS classification :

[[_3rd_order, _exact, _nonlinear], [_3rd_order, _with_linear_symmetries]]

$$yy''' + 3y'y'' - 2yy'' - 2y'^2 + y'y = e^{2x}$$

Solve

$$yy''' + 3y'y'' - 2yy'' - 2y'^2 + y'y = e^{2x}$$

Failed to solve.

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 51

```
ode:=y(x)*diff(diff(diff(y(x),x),x),x)+3*diff(y(x),x)*diff(diff(y(x),x),x)-2*y(x)
  *diff(diff(y(x),x),x)-2*diff(y(x),x)^2+diff(y(x),x)*y(x) = exp(2*x);
dsolve(ode,y(x), singsol=all);
```

$$y = \sqrt{-2c_3 e^x x + e^{2x} + 2c_2 e^x - 2c_1}$$

$$y = -\sqrt{e^{2x} + (-2c_3 x + 2c_2) e^x - 2c_1}$$

Maple trace

Methods for third order ODEs:

--- Trying classification methods ---

trying 3rd order ODE linearizable_by_differentiation

differential order: 3; trying a linearization to 4th order

trying differential order: 3; missing variables

trying differential order: 3; exact nonlinear

```
-> Calling odsolve with the ODE, 1/2*_b(_a)^2+diff(_b(_a),_a)^2-2*_b(_a)*diff(
_b(_a),_a)+_b(_a)*diff(diff(_b(_a),_a),_a)-1/2*exp(2*_a)+_C1 = 0, _b(_a)
*** Sublevel 2 ***
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
<- 2nd order, 2 integrating factors of the form mu(x,y) successful
<- differential order: 3; exact nonlinear successful
```

✗ Mathematica

```
ode=y[x]*D[y[x],{x,3}]+3*D[y[x],x]*D[y[x],{x,2}]-2*y[x]*D[y[x],{x,2}]-2*D[y[x],x
]^2+y[x]*D[y[x],x]==Exp[2*x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

Not solved

✗ Sympy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(y(x)*Derivative(y(x), x) - 2*y(x)*Derivative(y(x), (x, 2)) + y(x)*
Derivative(y(x), (x, 3)) - exp(2*x) - 2*Derivative(y(x), x)**2 + 3*Derivative(
y(x), x)*Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

```
NotImplementedError : The given ODE -sqrt(y(x)**2 - 10*y(x)*Derivative(y(x), (x,
2)) + 8*y(x)*Derivative(y(x), (x, 3)) - 8*exp(2*x) + 9*Derivative(y(x), (x, 2)
)**2)/4 - y(x)/4 + Derivative(y(x), x) - 3*Derivative(y(x), (x, 2))/4 cannot
be solved by the factorable group method
```

2.14.15 Problem 36

Local contents

Solved as second order missing x ode	2456
✓ Maple	2461
✓ Mathematica	2462
✗ Sympy	2462

Internal problem ID [8061]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 19. Linear equations with variable coefficients (Misc. types). Supplementary problems. Page 132

Problem number : 36

Date solved : Thursday, November 27, 2025 at 07:32:58 AM

CAS classification :

[[_2nd_order, _missing_x], [_2nd_order, _reducible, _mu_xy]]

Solved as second order missing x ode

Time used: 4.345 (sec)

Solve

$$2(y+1)y'' + 2y'^2 + y^2 + 2y = 0$$

This is missing independent variable second order ode. Solved by reduction of order by using substitution which makes the dependent variable y an independent variable. Using

$$y' = p$$

Then

$$\begin{aligned} y'' &= \frac{dp}{dx} \\ &= \frac{dp}{dy} \frac{dy}{dx} \\ &= p \frac{dp}{dy} \end{aligned}$$

Hence the ode becomes

$$2(y+1)p(y) \left(\frac{d}{dy} p(y) \right) + 2p(y)^2 + y^2 + 2y = 0$$

Which is now solved as first order ode for $p(y)$.

Solve

In canonical form, the ODE is

$$\begin{aligned} p' &= F(y, p) \\ &= -\frac{2p^2 + y^2 + 2y}{2(y+1)p} \end{aligned}$$

This is a Bernoulli ODE.

$$p' = \left(-\frac{1}{y+1}\right)p + \left(-\frac{y^2 + 2y}{2(y+1)}\right)\frac{1}{p} \quad (1)$$

The standard Bernoulli ODE has the form

$$p' = f_0(y)p + f_1(y)p^n \quad (2)$$

Comparing this to (1) shows that

$$\begin{aligned} f_0 &= -\frac{1}{y+1} \\ f_1 &= -\frac{y^2 + 2y}{2(y+1)} \end{aligned}$$

The first step is to divide the above equation by p^n which gives

$$\frac{p'}{p^n} = f_0(y)p^{1-n} + f_1(y) \quad (3)$$

The next step is use the substitution $v = p^{1-n}$ in equation (3) which generates a new ODE in $v(y)$ which will be linear and can be easily solved using an integrating factor. Backsubstitution then gives the solution $p(y)$ which is what we want.

This method is now applied to the ODE at hand. Comparing the ODE (1) With (2) Shows that

$$\begin{aligned} f_0(y) &= -\frac{1}{y+1} \\ f_1(y) &= -\frac{y^2 + 2y}{2(y+1)} \\ n &= -1 \end{aligned}$$

Dividing both sides of ODE (1) by $p^n = \frac{1}{p}$ gives

$$p'p = -\frac{p^2}{y+1} - \frac{y^2 + 2y}{2(y+1)} \quad (4)$$

Let

$$\begin{aligned} v &= p^{1-n} \\ &= p^2 \end{aligned} \quad (5)$$

Taking derivative of equation (5) w.r.t y gives

$$v' = 2pp' \quad (6)$$

Substituting equations (5) and (6) into equation (4) gives

$$\begin{aligned} \frac{v'(y)}{2} &= -\frac{v(y)}{y+1} - \frac{y^2+2y}{2(y+1)} \\ v' &= -\frac{2v}{y+1} - \frac{y^2+2y}{y+1} \end{aligned} \quad (7)$$

The above now is a linear ODE in $v(y)$ which is now solved.

In canonical form a linear first order is

$$v'(y) + q(y)v(y) = p(y)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(y) &= \frac{2}{y+1} \\ p(y) &= -\frac{y(y+2)}{y+1} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dy} \\ &= e^{\int \frac{2}{y+1} \, dy} \\ &= (y+1)^2 \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dy}(\mu v) &= \mu p \\ \frac{d}{dy}(\mu v) &= (\mu) \left(-\frac{y(y+2)}{y+1} \right) \\ \frac{d}{dy}(v(y+1)^2) &= ((y+1)^2) \left(-\frac{y(y+2)}{y+1} \right) \\ d(v(y+1)^2) &= (-y(y+2)(y+1)) \, dy \end{aligned}$$

Integrating gives

$$\begin{aligned} v(y+1)^2 &= \int -y(y+2)(y+1) \, dy \\ &= -\frac{y^2(y+2)^2}{4} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $(y+1)^2$ gives the final solution

$$v(y) = \frac{-\frac{y^2(y+2)^2}{4} + c_1}{(y+1)^2}$$

The substitution $v = p^{1-n}$ is now used to convert the above solution back to p which results in

$$p^2 = \frac{-\frac{y^2(y+2)^2}{4} + c_1}{(y+1)^2}$$

Solving for p gives

$$p = -\frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2(y+1)}$$

$$p = \frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2y+2}$$

For solution (1) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = -\frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2(y+1)}$$

Solve Integrating gives

$$\int -\frac{2(y+1)}{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}} dy = dx$$

$$i \ln \left(iy^2 + 2iy + \sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1} \right) = x + c_2$$

Singular solutions are found by solving

$$-\frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2(y+1)} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -1 - \sqrt{1 - 2\sqrt{c_1}}$$

$$y = -1 + \sqrt{1 - 2\sqrt{c_1}}$$

Solving for y gives

$$y = \frac{i \left(2ie^{-i(x+c_2)} - \sqrt{2ie^{-3i(x+c_2)} - 4e^{-2i(x+c_2)} - 8ie^{-i(x+c_2)}c_1} \right) e^{i(x+c_2)}}{2}$$

$$y = \frac{i \left(2ie^{-i(x+c_2)} + \sqrt{2ie^{-3i(x+c_2)} - 4e^{-2i(x+c_2)} - 8ie^{-i(x+c_2)}c_1} \right) e^{i(x+c_2)}}{2}$$

$$y = -1 - \sqrt{1 - 2\sqrt{c_1}}$$

$$y = -1 + \sqrt{1 - 2\sqrt{c_1}}$$

For solution (2) found earlier, since $p = y'$ then we now have a new first order ode to solve which is

$$y' = \frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2y + 2}$$

Solve Integrating gives

$$\int \frac{2y + 2}{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}} dy = dx$$

$$-i \ln \left(iy^2 + 2iy + \sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1} \right) = x + c_3$$

Singular solutions are found by solving

$$\frac{\sqrt{-y^4 - 4y^3 - 4y^2 + 4c_1}}{2y + 2} = 0$$

for y . This is because we had to divide by this in the above step. This gives the following singular solution(s), which also have to satisfy the given ODE.

$$y = -1 - \sqrt{1 - 2\sqrt{c_1}}$$

$$y = -1 + \sqrt{1 - 2\sqrt{c_1}}$$

Solving for y gives

$$y = \frac{i \left(2ie^{i(x+c_3)} - \sqrt{2ie^{3i(x+c_3)} - 8ie^{i(x+c_3)}c_1 - 4e^{2i(x+c_3)}} \right) e^{-i(x+c_3)}}{2}$$

$$y = \frac{i \left(2ie^{i(x+c_3)} + \sqrt{2ie^{3i(x+c_3)} - 8ie^{i(x+c_3)}c_1 - 4e^{2i(x+c_3)}} \right) e^{-i(x+c_3)}}{2}$$

$$y = -1 - \sqrt{1 - 2\sqrt{c_1}}$$

$$y = -1 + \sqrt{1 - 2\sqrt{c_1}}$$

The solution

$$y = -1 - \sqrt{1 - 2\sqrt{c_1}}$$

was found not to satisfy the ode or the IC. Hence it is removed. The solution

$$y = -1 + \sqrt{1 - 2\sqrt{c_1}}$$

was found not to satisfy the ode or the IC. Hence it is removed.

Summary of solutions found

$$y = \frac{i \left(2ie^{-i(x+c_2)} - \sqrt{2ie^{-3i(x+c_2)} - 4e^{-2i(x+c_2)} - 8ie^{-i(x+c_2)}c_1} \right) e^{i(x+c_2)}}{2}$$

$$y = \frac{i \left(2ie^{-i(x+c_2)} + \sqrt{2ie^{-3i(x+c_2)} - 4e^{-2i(x+c_2)} - 8ie^{-i(x+c_2)}c_1} \right) e^{i(x+c_2)}}{2}$$

$$y = \frac{i \left(2ie^{i(x+c_3)} - \sqrt{2ie^{3i(x+c_3)} - 8ie^{i(x+c_3)}c_1 - 4e^{2i(x+c_3)}} \right) e^{-i(x+c_3)}}{2}$$

$$y = \frac{i \left(2ie^{i(x+c_3)} + \sqrt{2ie^{3i(x+c_3)} - 8ie^{i(x+c_3)}c_1 - 4e^{2i(x+c_3)}} \right) e^{-i(x+c_3)}}{2}$$

✓ Maple. Time used: 0.014 (sec). Leaf size: 41

```
ode:=2*(1+y(x))*diff(diff(y(x),x),x)+2*diff(y(x),x)^2+y(x)^2+2*y(x) = 0;
dsolve(ode,y(x), singsol=all);
```

$$y = -1 - \sqrt{1 + 2c_2 \cos(x) - 2c_1 \sin(x)}$$

$$y = -1 + \sqrt{1 + 2c_2 \cos(x) - 2c_1 \sin(x)}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying 2nd order Liouville
trying 2nd order WeierstrassP
trying 2nd order JacobiSN
differential order: 2; trying a linearization to 3rd order
trying 2nd order ODE linearizable_by_differentiation
trying 2nd order, 2 integrating factors of the form mu(x,y)
```

```

trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
<- 2nd order, 2 integrating factors of the form mu(x,y) successful

```

✓ **Mathematica.** Time used: 13.516 (sec). Leaf size: 5629

```

ode=2*(y[x]+1)*D[y[x],{x,2}]+2*D[y[x],x]^2+y[x]^2+2*y[x]==0;
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]

```

Too large to display

✗ **Sympy**

```

from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((2*y(x) + 2)*Derivative(y(x), (x, 2)) + y(x)**2 + 2*y(x) + 2*Derivative(
    y(x), x)**2,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)

```

```

NotImplementedError : The given ODE -sqrt(-2*y(x)**2 - 4*y(x)*Derivative(y(x), (x,
    2)) - 4*y(x) - 4*Derivative(y(x), (x, 2)))/2 + Derivative(y(x), x) cannot be
    solved by the factorable group method

```

2.15 Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Local contents

2.15.1 Problem 10	2464
2.15.2 Problem 11	2472
2.15.3 Problem 12	2479
2.15.4 Problem 13	2487
2.15.5 Problem 17	2490

2.15.1 Problem 10

Local contents

Solution using Matrix exponential method	2464
Solution using explicit Eigenvalue and Eigenvector method	2465
✓ Maple	2470
✓ Mathematica	2470
✓ Sympy	2471

Internal problem ID [8062]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Problem number : 10

Date solved : Thursday, November 27, 2025 at 07:33:03 AM

CAS classification : system_of_ODEs

$$x' - y' + y = -e^t$$

$$x + y' - y = e^{2t}$$

Solution using Matrix exponential method

In this method, we will assume we have found the matrix exponential e^{At} already. There are different methods to determine this but will not be shown here. This is a system of linear ODE's given as

$$\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e^{2t} - e^t \\ e^{2t} \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix. For the above matrix A , the matrix exponential can be found to be

$$e^{At} = \begin{bmatrix} e^{-t} & 0 \\ -\sinh(t) & e^t \end{bmatrix}$$

Therefore the homogeneous solution is

$$\begin{aligned}\vec{x}_h(t) &= e^{At} \vec{c} \\ &= \begin{bmatrix} e^{-t} & 0 \\ -\sinh(t) & e^t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} e^{-t} c_1 \\ -\sinh(t) c_1 + e^t c_2 \end{bmatrix}\end{aligned}$$

The particular solution given by

$$\vec{x}_p(t) = e^{At} \int e^{-At} \vec{G}(t) dt$$

But

$$\begin{aligned}e^{-At} &= (e^{At})^{-1} \\ &= \begin{bmatrix} e^t & 0 \\ \sinh(t) & e^{-t} \end{bmatrix}\end{aligned}$$

Hence

$$\begin{aligned}\vec{x}_p(t) &= \begin{bmatrix} e^{-t} & 0 \\ -\sinh(t) & e^t \end{bmatrix} \int \begin{bmatrix} e^t & 0 \\ \sinh(t) & e^{-t} \end{bmatrix} \begin{bmatrix} e^{2t} - e^t \\ e^{2t} \end{bmatrix} dt \\ &= \begin{bmatrix} e^{-t} & 0 \\ -\sinh(t) & e^t \end{bmatrix} \begin{bmatrix} \frac{e^{3t}}{6} + \frac{e^t}{2} - \frac{e^{2t}}{4} - \frac{1}{4} + \frac{t}{2} \\ \frac{e^{2t}}{3} - \frac{e^t}{2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{e^{2t}}{3} - \frac{e^t}{2} \\ \frac{(4e^t - 3 + 3t)e^t}{6} \end{bmatrix}\end{aligned}$$

Hence the complete solution is

$$\begin{aligned}\vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\ &= \begin{bmatrix} \frac{e^{2t}}{3} - \frac{e^t}{2} + e^{-t} c_1 \\ \frac{2e^{2t}}{3} + \frac{(3t+6c_2-3)e^t}{6} - \sinh(t) c_1 \end{bmatrix}\end{aligned}$$

Solution using explicit Eigenvalue and Eigenvector method

This is a system of linear ODE's given as

$$\vec{x}'(t) = A \vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e^{2t} - e^t \\ e^{2t} \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix.

The first step is find the homogeneous solution. We start by finding the eigenvalues of A . This is done by solving the following equation for the eigenvalues λ

$$\det(A - \lambda I) = 0$$

Expanding gives

$$\det\left(\begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

Therefore

$$\det\left(\begin{bmatrix} -1 - \lambda & 0 \\ -1 & 1 - \lambda \end{bmatrix}\right) = 0$$

Since the matrix A is triangular matrix, then the determinant is the product of the elements along the diagonal. Therefore the above becomes

$$(-1 - \lambda)(1 - \lambda) = 0$$

The roots of the above are the eigenvalues.

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

This table summarises the above result

eigenvalue	algebraic multiplicity	type of eigenvalue
-1	1	real eigenvalue
1	1	real eigenvalue

Now the eigenvector for each eigenvalue are found.

Considering the eigenvalue $\lambda_1 = -1$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\begin{aligned} \left(\begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} - (-1)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ -1 & 2 & 0 \end{array} \right]$$

Since the current pivot $A(1, 1)$ is zero, then the current pivot row is replaced with a row with a non-zero pivot. Swapping row 1 and row 2 gives

$$\left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\left[\begin{array}{cc} -1 & 2 \\ 0 & 0 \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = 2t\}$

Hence the solution is

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = \left[\begin{array}{c} 2t \\ t \end{array} \right]$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = t \left[\begin{array}{c} 2 \\ 1 \end{array} \right]$$

Let $t = 1$ the eigenvector becomes

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = \left[\begin{array}{c} 2 \\ 1 \end{array} \right]$$

Considering the eigenvalue $\lambda_2 = 1$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\left[\begin{array}{cc} -1 & 0 \\ -1 & 1 \end{array} \right] - (1) \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \right) \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

$$\left[\begin{array}{cc} -2 & 0 \\ -1 & 0 \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} -2 & 0 & 0 \\ -1 & 0 & 0 \end{array} \right]$$

$$R_2 = R_2 - \frac{R_1}{2} \implies \left[\begin{array}{cc|c} -2 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\left[\begin{array}{cc} -2 & 0 \\ 0 & 0 \end{array} \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = 0\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ t \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The following table gives a summary of this result. It shows for each eigenvalue the algebraic multiplicity m , and its geometric multiplicity k and the eigenvectors associated with the eigenvalue. If $m > k$ then the eigenvalue is defective which means the number of normal linearly independent eigenvectors associated with this eigenvalue (called the geometric multiplicity k) does not equal the algebraic multiplicity m , and we need to determine an additional $m - k$ generalized eigenvectors for this eigenvalue.

eigenvalue	multiplicity		defective?	eigenvectors
	algebraic m	geometric k		
1	1	1	No	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
-1	1	1	No	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Now that we found the eigenvalues and associated eigenvectors, we will go over each eigenvalue and generate the solution basis. The only problem we need to take care of is if the eigenvalue is defective. Since eigenvalue 1 is real and distinct then the corresponding eigenvector solution

is

$$\begin{aligned}\vec{x}_1(t) &= \vec{v}_1 e^t \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^t\end{aligned}$$

Since eigenvalue -1 is real and distinct then the corresponding eigenvector solution is

$$\begin{aligned}\vec{x}_2(t) &= \vec{v}_2 e^{-t} \\ &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-t}\end{aligned}$$

Therefore the homogeneous solution is

$$\vec{x}_h(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t)$$

Which is written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 \begin{bmatrix} 0 \\ e^t \end{bmatrix} + c_2 \begin{bmatrix} 2e^{-t} \\ e^{-t} \end{bmatrix}$$

Now that we found homogeneous solution above, we need to find a particular solution $\vec{x}_p(t)$. We will use Variation of parameters. The fundamental matrix is

$$\Phi = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots]$$

Where $\vec{x}_i$ are the solution basis found above. Therefore the fundamental matrix is

$$\Phi(t) = \begin{bmatrix} 0 & 2e^{-t} \\ e^t & e^{-t} \end{bmatrix}$$

The particular solution is then given by

$$\vec{x}_p(t) = \Phi \int \Phi^{-1} \vec{G}(t) dt$$

But

$$\Phi^{-1} = \begin{bmatrix} -\frac{e^{-t}}{2} & e^{-t} \\ \frac{e^t}{2} & 0 \end{bmatrix}$$

Hence

$$\begin{aligned}
 \vec{x}_p(t) &= \begin{bmatrix} 0 & 2e^{-t} \\ e^t & e^{-t} \end{bmatrix} \int \begin{bmatrix} -\frac{e^{-t}}{2} & e^{-t} \\ \frac{e^t}{2} & 0 \end{bmatrix} \begin{bmatrix} e^{2t} - e^t \\ e^{2t} \end{bmatrix} dt \\
 &= \begin{bmatrix} 0 & 2e^{-t} \\ e^t & e^{-t} \end{bmatrix} \int \begin{bmatrix} \frac{e^t}{2} + \frac{1}{2} \\ \frac{e^{3t}}{2} - \frac{e^{2t}}{2} \end{bmatrix} dt \\
 &= \begin{bmatrix} 0 & 2e^{-t} \\ e^t & e^{-t} \end{bmatrix} \begin{bmatrix} \frac{e^t}{2} + \frac{t}{2} \\ \frac{e^{3t}}{6} - \frac{e^{2t}}{4} \end{bmatrix} \\
 &= \begin{bmatrix} \frac{e^{2t}}{3} - \frac{e^t}{2} \\ \frac{e^t(8e^t + 6t - 3)}{12} \end{bmatrix}
 \end{aligned}$$

Now that we found particular solution, the final solution is

$$\begin{aligned}
 \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\
 \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ c_1 e^t \end{bmatrix} + \begin{bmatrix} 2c_2 e^{-t} \\ c_2 e^{-t} \end{bmatrix} + \begin{bmatrix} \frac{e^{2t}}{3} - \frac{e^t}{2} \\ \frac{e^t(8e^t + 6t - 3)}{12} \end{bmatrix}
 \end{aligned}$$

Which becomes

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2c_2 e^{-t} + \frac{e^{2t}}{3} - \frac{e^t}{2} \\ c_2 e^{-t} + \frac{(t + 2c_1 + \frac{4e^t}{3} - \frac{1}{2})e^t}{2} \end{bmatrix}$$

✓ **Maple.** Time used: 0.107 (sec). Leaf size: 47

```
ode:=[diff(x(t),t)-diff(y(t),t)+y(t) = -exp(t), x(t)+diff(y(t),t)-y(t) = exp(2*t)
];
dsolve(ode);
```

$$\begin{aligned}
 x(t) &= -\frac{e^t}{2} + \frac{e^{2t}}{3} + e^{-t}c_2 \\
 y(t) &= \frac{e^{-t}c_2}{2} + \frac{2e^{2t}}{3} + c_1 e^t + \frac{e^t t}{2}
 \end{aligned}$$

✓ **Mathematica.** Time used: 0.053 (sec). Leaf size: 72

```
ode={D[x[t],t]-D[y[t],t]+y[t]==-Exp[t],x[t]+D[y[t],t]-y[t]==Exp[2*t]};
ic={};
DSolve[{ode,ic},{x[t],y[t]},t,IncludeSingularSolutions->True]
```

$$\begin{aligned}
 x(t) &\rightarrow \frac{1}{6}e^t(2e^t - 3) + c_1 e^{-t} \\
 y(t) &\rightarrow \frac{2e^{2t}}{3} + \frac{c_1 e^{-t}}{2} + \frac{1}{4}e^t(2t - 1 - 2c_1 + 4c_2)
 \end{aligned}$$

✓ Sympy. Time used: 0.110 (sec). Leaf size: 51

```
from sympy import *
t = symbols("t")
x = Function("x")
y = Function("y")
ode=[Eq(y(t) + exp(t) + Derivative(x(t), t) - Derivative(y(t), t),0),Eq(x(t) - y(
    t) - exp(2*t) + Derivative(y(t), t),0)]
ics = {}
dsolve(ode,func=[x(t),y(t)],ics=ics)
```

$$\left[x(t) = 2C_1 e^{-t} + \frac{e^{2t}}{3} - \frac{e^t}{2}, \quad y(t) = C_1 e^{-t} + \frac{t e^t}{2} + \left(C_2 - \frac{1}{4} \right) e^t + \frac{2e^{2t}}{3} \right]$$

2.15.2 Problem 11

Local contents

Solution using Matrix exponential method	2472
Solution using explicit Eigenvalue and Eigenvector method	2473
✓ Maple	2478
✓ Mathematica	2478
✓ Sympy	2478

Internal problem ID [8063]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Problem number : 11

Date solved : Thursday, November 27, 2025 at 07:33:03 AM

CAS classification : system_of_ODEs

$$x' + 2x + y' + y = t$$

$$5x + y' + 3y = t^2$$

Solution using Matrix exponential method

In this method, we will assume we have found the matrix exponential e^{At} already. There are different methods to determine this but will not be shown here. This is a system of linear ODE's given as

$$\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -t^2 + t \\ t^2 \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix. For the above matrix A , the matrix exponential can be found to be

$$e^{At} = \begin{bmatrix} \cos(t) + 3\sin(t) & 2\sin(t) \\ -5\sin(t) & \cos(t) - 3\sin(t) \end{bmatrix}$$

Therefore the homogeneous solution is

$$\begin{aligned}
 \vec{x}_h(t) &= e^{At} \vec{c} \\
 &= \begin{bmatrix} \cos(t) + 3 \sin(t) & 2 \sin(t) \\ -5 \sin(t) & \cos(t) - 3 \sin(t) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\
 &= \begin{bmatrix} (\cos(t) + 3 \sin(t)) c_1 + 2 \sin(t) c_2 \\ -5 \sin(t) c_1 + (\cos(t) - 3 \sin(t)) c_2 \end{bmatrix} \\
 &= \begin{bmatrix} (3c_1 + 2c_2) \sin(t) + c_1 \cos(t) \\ (-5c_1 - 3c_2) \sin(t) + c_2 \cos(t) \end{bmatrix}
 \end{aligned}$$

The particular solution given by

$$\vec{x}_p(t) = e^{At} \int e^{-At} \vec{G}(t) dt$$

But

$$\begin{aligned}
 e^{-At} &= (e^{At})^{-1} \\
 &= \begin{bmatrix} \cos(t) - 3 \sin(t) & -2 \sin(t) \\ 5 \sin(t) & \cos(t) + 3 \sin(t) \end{bmatrix}
 \end{aligned}$$

Hence

$$\begin{aligned}
 \vec{x}_p(t) &= \begin{bmatrix} \cos(t) + 3 \sin(t) & 2 \sin(t) \\ -5 \sin(t) & \cos(t) - 3 \sin(t) \end{bmatrix} \int \begin{bmatrix} \cos(t) - 3 \sin(t) & -2 \sin(t) \\ 5 \sin(t) & \cos(t) + 3 \sin(t) \end{bmatrix} \begin{bmatrix} -t^2 + t \\ t^2 \end{bmatrix} dt \\
 &= \begin{bmatrix} \cos(t) + 3 \sin(t) & 2 \sin(t) \\ -5 \sin(t) & \cos(t) - 3 \sin(t) \end{bmatrix} \begin{bmatrix} (-t^2 + t + 3) \cos(t) - \sin(t) (t^2 - 3t + 1) \\ (2t^2 - 3t - 4) \cos(t) + \sin(t) (-1 + t) (t - 3) \end{bmatrix} \\
 &= \begin{bmatrix} -t^2 + t + 3 \\ 2t^2 - 3t - 4 \end{bmatrix}
 \end{aligned}$$

Hence the complete solution is

$$\begin{aligned}
 \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\
 &= \begin{bmatrix} (3c_1 + 2c_2) \sin(t) - t^2 + c_1 \cos(t) + t + 3 \\ (-5c_1 - 3c_2) \sin(t) + 2t^2 + c_2 \cos(t) - 3t - 4 \end{bmatrix}
 \end{aligned}$$

Solution using explicit Eigenvalue and Eigenvector method

This is a system of linear ODE's given as

$$\vec{x}'(t) = A \vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -t^2 + t \\ t^2 \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix.

The first step is find the homogeneous solution. We start by finding the eigenvalues of A . This is done by solving the following equation for the eigenvalues λ

$$\det(A - \lambda I) = 0$$

Expanding gives

$$\det\left(\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

Therefore

$$\det\left(\begin{bmatrix} 3 - \lambda & 2 \\ -5 & -3 - \lambda \end{bmatrix}\right) = 0$$

Which gives the characteristic equation

$$\lambda^2 + 1 = 0$$

The roots of the above are the eigenvalues.

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

This table summarises the above result

eigenvalue	algebraic multiplicity	type of eigenvalue
i	1	complex eigenvalue
$-i$	1	complex eigenvalue

Now the eigenvector for each eigenvalue are found.

Considering the eigenvalue $\lambda_1 = -i$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\begin{aligned} \left(\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} - (-i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 3 + i & 2 \\ -5 & -3 + i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} 3+i & 2 & 0 \\ -5 & -3+i & 0 \end{array} \right]$$

$$R_2 = R_2 + \left(\frac{3}{2} - \frac{i}{2} \right) R_1 \Rightarrow \left[\begin{array}{cc|c} 3+i & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\left[\begin{array}{cc} 3+i & 2 \\ 0 & 0 \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = (-\frac{3}{5} + \frac{i}{5})t\}$

Hence the solution is

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = \left[\begin{array}{c} (-\frac{3}{5} + \frac{i}{5})t \\ t \end{array} \right]$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = t \left[\begin{array}{c} -\frac{3}{5} + \frac{i}{5} \\ 1 \end{array} \right]$$

Let $t = 1$ the eigenvector becomes

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = \left[\begin{array}{c} -\frac{3}{5} + \frac{i}{5} \\ 1 \end{array} \right]$$

Which is normalized to

$$\left[\begin{array}{c} v_1 \\ t \end{array} \right] = \left[\begin{array}{c} -3+i \\ 5 \end{array} \right]$$

Considering the eigenvalue $\lambda_2 = i$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\left[\begin{array}{cc} 3 & 2 \\ -5 & -3 \end{array} \right] - (i) \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \right) \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

$$\left[\begin{array}{cc} 3-i & 2 \\ -5 & -3-i \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \end{array} \right]$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} 3-i & 2 & 0 \\ -5 & -3-i & 0 \end{array} \right]$$

$$R_2 = R_2 + \left(\frac{3}{2} + \frac{i}{2}\right) R_1 \implies \left[\begin{array}{cc|c} 3-i & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\left[\begin{array}{cc} 3-i & 2 \\ 0 & 0 \end{array} \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = (-\frac{3}{5} - \frac{i}{5})t\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} (-\frac{3}{5} - \frac{i}{5})t \\ t \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = t \begin{bmatrix} -\frac{3}{5} - \frac{i}{5} \\ 1 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} - \frac{i}{5} \\ 1 \end{bmatrix}$$

Which is normalized to

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -3-i \\ 5 \end{bmatrix}$$

The following table gives a summary of this result. It shows for each eigenvalue the algebraic multiplicity m , and its geometric multiplicity k and the eigenvectors associated with the eigenvalue. If $m > k$ then the eigenvalue is defective which means the number of normal linearly independent eigenvectors associated with this eigenvalue (called the geometric multiplicity k) does not equal the algebraic multiplicity m , and we need to determine an additional $m - k$ generalized eigenvectors for this eigenvalue.

eigenvalue	multiplicity		defective?	eigenvectors
	algebraic m	geometric k		
i	1	1	No	$\begin{bmatrix} -\frac{3}{5} - \frac{i}{5} \\ 1 \end{bmatrix}$
$-i$	1	1	No	$\begin{bmatrix} -\frac{3}{5} + \frac{i}{5} \\ 1 \end{bmatrix}$

Now that we found the eigenvalues and associated eigenvectors, we will go over each eigenvalue and generate the solution basis. The only problem we need to take care of is if the eigenvalue is defective. Therefore the homogeneous solution is

$$\vec{x}_h(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t)$$

Which is written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) e^{it} \\ e^{it} \end{bmatrix} + c_2 \begin{bmatrix} \left(-\frac{3}{5} + \frac{i}{5}\right) e^{-it} \\ e^{-it} \end{bmatrix}$$

Now that we found homogeneous solution above, we need to find a particular solution $\vec{x}_p(t)$. We will use Variation of parameters. The fundamental matrix is

$$\Phi = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots]$$

Where $\vec{x}_i$ are the solution basis found above. Therefore the fundamental matrix is

$$\Phi(t) = \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) e^{it} & \left(-\frac{3}{5} + \frac{i}{5}\right) e^{-it} \\ e^{it} & e^{-it} \end{bmatrix}$$

The particular solution is then given by

$$\vec{x}_p(t) = \Phi \int \Phi^{-1} \vec{G}(t) dt$$

But

$$\Phi^{-1} = \begin{bmatrix} \frac{5ie^{-it}}{2} & \left(\frac{1}{2} + \frac{3i}{2}\right) e^{-it} \\ -\frac{5ie^{it}}{2} & \left(\frac{1}{2} - \frac{3i}{2}\right) e^{it} \end{bmatrix}$$

Hence

$$\begin{aligned} \vec{x}_p(t) &= \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) e^{it} & \left(-\frac{3}{5} + \frac{i}{5}\right) e^{-it} \\ e^{it} & e^{-it} \end{bmatrix} \int \begin{bmatrix} \frac{5ie^{-it}}{2} & \left(\frac{1}{2} + \frac{3i}{2}\right) e^{-it} \\ -\frac{5ie^{it}}{2} & \left(\frac{1}{2} - \frac{3i}{2}\right) e^{it} \end{bmatrix} \begin{bmatrix} -t^2 + t \\ t^2 \end{bmatrix} dt \\ &= \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) e^{it} & \left(-\frac{3}{5} + \frac{i}{5}\right) e^{-it} \\ e^{it} & e^{-it} \end{bmatrix} \int \begin{bmatrix} -\left(-\frac{5i}{2} + \left(-\frac{1}{2} + i\right)t\right) t e^{-it} \\ \frac{e^{it}t(2it+t-5i)}{2} \end{bmatrix} dt \\ &= \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) e^{it} & \left(-\frac{3}{5} + \frac{i}{5}\right) e^{-it} \\ e^{it} & e^{-it} \end{bmatrix} \begin{bmatrix} -\frac{(-5+(2+i)t)e^{-it}(-t^2+(2+i)t+1-2i)}{2t-4+2i} \\ \frac{(5+(-2+i)t)(t^2+(-2+i)t-1-2i)e^{it}}{-2t+4+2i} \end{bmatrix} \\ &= \begin{bmatrix} \frac{t^4-5t^3+6t^2+7t-15}{(-t+2+i)(t-2+i)} \\ \frac{-2t^4+11t^3-18t^2-t+20}{(-t+2+i)(t-2+i)} \end{bmatrix} \end{aligned}$$

Now that we found particular solution, the final solution is

$$\begin{aligned} \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\ \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} \left(-\frac{3}{5} - \frac{i}{5}\right) c_1 e^{it} \\ c_1 e^{it} \end{bmatrix} + \begin{bmatrix} \left(-\frac{3}{5} + \frac{i}{5}\right) c_2 e^{-it} \\ c_2 e^{-it} \end{bmatrix} + \begin{bmatrix} \frac{t^4-5t^3+6t^2+7t-15}{(-t+2+i)(t-2+i)} \\ \frac{-2t^4+11t^3-18t^2-t+20}{(-t+2+i)(t-2+i)} \end{bmatrix} \end{aligned}$$

Which becomes

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{((3-i)c_2 e^{-it} + (3+i)c_1 e^{it} + 5t^2 - 5t - 15)(t^2 - 4t + 5)}{5(-t+2+i)(t-2+i)} \\ - \frac{(c_1 e^{it} + c_2 e^{-it} + 2t^2 - 3t - 4)(t^2 - 4t + 5)}{(-t+2+i)(t-2+i)} \end{bmatrix}$$

✓ **Maple.** Time used: 0.080 (sec). Leaf size: 53

```
ode:=[diff(x(t),t)+2*x(t)+diff(y(t),t)+y(t) = t, 5*x(t)+diff(y(t),t)+3*y(t) = t
^2];
dsolve(ode);
```

$$\begin{aligned} x(t) &= \sin(t) c_2 + \cos(t) c_1 - t^2 + t + 3 \\ y(t) &= 2t^2 - \frac{3 \sin(t) c_2}{2} - \frac{3 \cos(t) c_1}{2} - 3t - 4 + \frac{\cos(t) c_2}{2} - \frac{\sin(t) c_1}{2} \end{aligned}$$

✓ **Mathematica.** Time used: 0.074 (sec). Leaf size: 176

```
ode={D[x[t],t]+2*x[t]+D[y[t],t]+y[t]==t,5*x[t]+D[y[t],t]+3*y[t]==t^2};
ic={};
DSolve[{ode,ic},{x[t],y[t]},t,IncludeSingularSolutions->True]
```

$$\begin{aligned} x(t) &\rightarrow 2 \sin(t) \int_1^t K[2](\cos(K[2])K[2] + (5 - 2K[2]) \sin(K[2]))dK[2] + (3 \sin(t) + \cos(t)) \int_1^t K[1]((K[1] - \\ y(t) &\rightarrow (\cos(t) - 3 \sin(t)) \int_1^t K[2](\cos(K[2])K[2] + (5 - 2K[2]) \sin(K[2]))dK[2] - 5 \sin(t) \int_1^t K[1]((K[1] - \end{aligned}$$

✓ **Sympy.** Time used: 0.151 (sec). Leaf size: 133

```
from sympy import *
t = symbols("t")
x = Function("x")
y = Function("y")
ode=[Eq(-t + 2*x(t) + y(t) + Derivative(x(t), t) + Derivative(y(t), t),0),Eq(-t
**2 + 5*x(t) + 3*y(t) + Derivative(y(t), t),0)]
ics = {}
dsolve(ode,func=[x(t),y(t)],ics=ics)
```

$$\begin{aligned} \left[x(t) &= -t^2 \sin^2(t) - t^2 \cos^2(t) + t \sin^2(t) + t \cos^2(t) + \left(\frac{C_1}{5} + \frac{3C_2}{5} \right) \sin(t) \right. \\ &\quad \left. - \left(\frac{3C_1}{5} - \frac{C_2}{5} \right) \cos(t) + 3 \sin^2(t) + 3 \cos^2(t), \quad y(t) = C_1 \cos(t) - C_2 \sin(t) \right. \\ &\quad \left. + 2t^2 \sin^2(t) + 2t^2 \cos^2(t) - 3t \sin^2(t) - 3t \cos^2(t) - 4 \sin^2(t) - 4 \cos^2(t) \right] \end{aligned}$$

2.15.3 Problem 12

Local contents

Solution using Matrix exponential method	2479
Solution using explicit Eigenvalue and Eigenvector method	2481
✓ Maple	2485
✓ Mathematica	2485
✓ Sympy	2486

Internal problem ID [8064]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Problem number : 12

Date solved : Thursday, November 27, 2025 at 07:33:04 AM

CAS classification : system_of_ODEs

$$x' + x + 2y' + 7y = e^t + 2$$

$$-2x + y' + 3y = e^t - 1$$

Solution using Matrix exponential method

In this method, we will assume we have found the matrix exponential e^{At} already. There are different methods to determine this but will not be shown here. This is a system of linear ODE's given as

$$\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 4 - e^t \\ e^t - 1 \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix. For the above matrix A , the matrix exponential

can be found to be

$$\begin{aligned}
 e^{At} &= \begin{bmatrix} \frac{e^{-4t} \cos(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \sin(t)(-1+\cos(2t)+\sin(2t))}{2} + i \left(-\frac{e^{-4t} \sin(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \cos(t)(-1+\cos(2t)+\sin(2t))}{2} \right) & \\ & 2e^{-4t} \sin(t) \end{bmatrix} \\
 &= \begin{bmatrix} e^{-4t}(-\sin(t) + \cos(t)) & -e^{-4t} \sin(t) \\ 2e^{-4t} \sin(t) & e^{-4t}(\sin(t) + \cos(t)) \end{bmatrix}
 \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned}
 \vec{x}_h(t) &= e^{At} \vec{c} \\
 &= \begin{bmatrix} \frac{e^{-4t} \cos(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \sin(t)(-1+\cos(2t)+\sin(2t))}{2} + i \left(-\frac{e^{-4t} \sin(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \cos(t)(-1+\cos(2t)+\sin(2t))}{2} \right) & \\ & 2e^{-4t} \sin(t) \end{bmatrix} \\
 &= \begin{bmatrix} \left(\frac{e^{-4t} \cos(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \sin(t)(-1+\cos(2t)+\sin(2t))}{2} \right) + i \left(-\frac{e^{-4t} \sin(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \cos(t)(-1+\cos(2t)+\sin(2t))}{2} \right) & \\ 2e^{-4t} \sin(t) c_1 + \left(-\frac{e^{-4t} \cos(t)(-\cos(2t)-\sin(2t)-1)}{2} - \frac{e^{-4t} \sin(t)(\cos(2t)-\sin(2t)-1)}{2} + i \left(\frac{e^{-4t} \sin(t)(-\cos(2t)-\sin(2t))}{2} \right) \right) & \end{bmatrix} \\
 &= \begin{bmatrix} e^{-4t}((-c_2 - c_1) \sin(t) + \cos(t) c_1) \\ ((2c_1 + c_2) \sin(t) + \cos(t) c_2) e^{-4t} \end{bmatrix}
 \end{aligned}$$

The particular solution given by

$$\vec{x}_p(t) = e^{At} \int e^{-At} \vec{G}(t) dt$$

But

$$\begin{aligned}
 e^{-At} &= (e^{At})^{-1} \\
 &= \begin{bmatrix} e^{4t}(\sin(t) + \cos(t)) & e^{4t} \sin(t) \\ -2e^{4t} \sin(t) & e^{4t}(-\sin(t) + \cos(t)) \end{bmatrix}
 \end{aligned}$$

Hence

$$\begin{aligned}
 \vec{x}_p(t) &= \begin{bmatrix} \frac{e^{-4t} \cos(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \sin(t)(-1+\cos(2t)+\sin(2t))}{2} + i \left(-\frac{e^{-4t} \sin(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \cos(t)(-1+\cos(2t)+\sin(2t))}{2} \right) & \\ & 2e^{-4t} \sin(t) \end{bmatrix} \\
 &= \begin{bmatrix} \frac{e^{-4t} \cos(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \sin(t)(-1+\cos(2t)+\sin(2t))}{2} + i \left(-\frac{e^{-4t} \sin(t)(1+\cos(2t)-\sin(2t))}{2} + \frac{e^{-4t} \cos(t)(-1+\cos(2t)+\sin(2t))}{2} \right) & \\ & 2e^{-4t} \sin(t) \end{bmatrix} \\
 &= \begin{bmatrix} \frac{13}{17} - \frac{5e^t}{26} \\ \frac{3}{17} + \frac{2e^t}{13} \end{bmatrix}
 \end{aligned}$$

Hence the complete solution is

$$\begin{aligned}
 \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\
 &= \begin{bmatrix} \frac{13}{17} + (-(c_2 + c_1) \sin(t) + \cos(t) c_1) e^{-4t} - \frac{5e^t}{26} \\ \frac{3}{17} + ((2c_1 + c_2) \sin(t) + \cos(t) c_2) e^{-4t} + \frac{2e^t}{13} \end{bmatrix}
 \end{aligned}$$

Solution using explicit Eigenvalue and Eigenvector method

This is a system of linear ODE's given as

$$\vec{x}'(t) = A \vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 4 - e^t \\ e^t - 1 \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A \vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A \vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix.

The first step is find the homogeneous solution. We start by finding the eigenvalues of A . This is done by solving the following equation for the eigenvalues λ

$$\det(A - \lambda I) = 0$$

Expanding gives

$$\det \left(\begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

Therefore

$$\det \left(\begin{bmatrix} -5 - \lambda & -1 \\ 2 & -3 - \lambda \end{bmatrix} \right) = 0$$

Which gives the characteristic equation

$$\lambda^2 + 8\lambda + 17 = 0$$

The roots of the above are the eigenvalues.

$$\lambda_1 = -4 + i$$

$$\lambda_2 = -4 - i$$

This table summarises the above result

eigenvalue	algebraic multiplicity	type of eigenvalue
$-4 + i$	1	complex eigenvalue
$-4 - i$	1	complex eigenvalue

Now the eigenvector for each eigenvalue are found.

Considering the eigenvalue $\lambda_1 = -4 - i$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix} - (-4 - i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 + i & -1 \\ 2 & 1 + i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} -1 + i & -1 & 0 \\ 2 & 1 + i & 0 \end{array} \right]$$

$$R_2 = R_2 + (1 + i) R_1 \implies \left[\begin{array}{cc|c} -1 + i & -1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\begin{bmatrix} -1 + i & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = (-\frac{1}{2} - \frac{i}{2}) t\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} (-\frac{1}{2} - \frac{i}{2}) t \\ t \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = t \begin{bmatrix} -\frac{1}{2} - \frac{i}{2} \\ 1 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} - \frac{i}{2} \\ 1 \end{bmatrix}$$

Which is normalized to

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -1 - i \\ 2 \end{bmatrix}$$

Considering the eigenvalue $\lambda_2 = -4 + i$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix} - (-4+i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1-i & -1 \\ 2 & 1-i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{cc|c} -1-i & -1 & 0 \\ 2 & 1-i & 0 \end{array} \right]$$

$$R_2 = R_2 + (1-i)R_1 \implies \left[\begin{array}{cc|c} -1-i & -1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\begin{bmatrix} -1-i & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = (-\frac{1}{2} + \frac{i}{2})t\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} (-\frac{1}{2} + \frac{i}{2})t \\ t \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = t \begin{bmatrix} -\frac{1}{2} + \frac{i}{2} \\ 1 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} + \frac{i}{2} \\ 1 \end{bmatrix}$$

Which is normalized to

$$\begin{bmatrix} v_1 \\ t \end{bmatrix} = \begin{bmatrix} -1+i \\ 2 \end{bmatrix}$$

The following table gives a summary of this result. It shows for each eigenvalue the algebraic multiplicity m , and its geometric multiplicity k and the eigenvectors associated with the eigenvalue. If $m > k$ then the eigenvalue is defective which means the number of normal linearly independent eigenvectors associated with this eigenvalue (called the geometric multiplicity k) does not equal the algebraic multiplicity m , and we need to determine an additional $m - k$ generalized eigenvectors for this eigenvalue.

eigenvalue	multiplicity		defective?	eigenvectors
	algebraic m	geometric k		
$-4 + i$	1	1	No	$\begin{bmatrix} -\frac{1}{2} + \frac{i}{2} \\ 1 \end{bmatrix}$
$-4 - i$	1	1	No	$\begin{bmatrix} -\frac{1}{2} - \frac{i}{2} \\ 1 \end{bmatrix}$

Now that we found the eigenvalues and associated eigenvectors, we will go over each eigenvalue and generate the solution basis. The only problem we need to take care of is if the eigenvalue is defective. Therefore the homogeneous solution is

$$\vec{x}_h(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t)$$

Which is written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) e^{(-4+i)t} \\ e^{(-4+i)t} \end{bmatrix} + c_2 \begin{bmatrix} \left(-\frac{1}{2} - \frac{i}{2}\right) e^{(-4-i)t} \\ e^{(-4-i)t} \end{bmatrix}$$

Now that we found homogeneous solution above, we need to find a particular solution $\vec{x}_p(t)$. We will use Variation of parameters. The fundamental matrix is

$$\Phi = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots]$$

Where $\vec{x}_i$ are the solution basis found above. Therefore the fundamental matrix is

$$\Phi(t) = \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) e^{(-4+i)t} & \left(-\frac{1}{2} - \frac{i}{2}\right) e^{(-4-i)t} \\ e^{(-4+i)t} & e^{(-4-i)t} \end{bmatrix}$$

The particular solution is then given by

$$\vec{x}_p(t) = \Phi \int \Phi^{-1} \vec{G}(t) dt$$

But

$$\Phi^{-1} = \begin{bmatrix} -ie^{(4-i)t} & \left(\frac{1}{2} - \frac{i}{2}\right) e^{(4-i)t} \\ ie^{(4+i)t} & \left(\frac{1}{2} + \frac{i}{2}\right) e^{(4+i)t} \end{bmatrix}$$

Hence

$$\begin{aligned}
 \vec{x}_p(t) &= \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) e^{(-4+i)t} & \left(-\frac{1}{2} - \frac{i}{2}\right) e^{(-4-i)t} \\ e^{(-4+i)t} & e^{(-4-i)t} \end{bmatrix} \int \begin{bmatrix} -ie^{(4-i)t} & \left(\frac{1}{2} - \frac{i}{2}\right) e^{(4-i)t} \\ ie^{(4+i)t} & \left(\frac{1}{2} + \frac{i}{2}\right) e^{(4+i)t} \end{bmatrix} \begin{bmatrix} 4 - e^t \\ e^t - 1 \end{bmatrix} dt \\
 &= \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) e^{(-4+i)t} & \left(-\frac{1}{2} - \frac{i}{2}\right) e^{(-4-i)t} \\ e^{(-4+i)t} & e^{(-4-i)t} \end{bmatrix} \int \begin{bmatrix} \frac{e^{(4-i)t}((1+i)e^t - 1 - 7i)}{2} \\ -\frac{e^{(4+i)t}((-1+i)e^t + 1 - 7i)}{2} \end{bmatrix} dt \\
 &= \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) e^{(-4+i)t} & \left(-\frac{1}{2} - \frac{i}{2}\right) e^{(-4-i)t} \\ e^{(-4+i)t} & e^{(-4-i)t} \end{bmatrix} \begin{bmatrix} \frac{e^{(4-i)t}(39 - 377i + (34 + 51i)e^t)}{442} \\ \frac{e^{(4+i)t}(39 + 377i + (34 - 51i)e^t)}{442} \end{bmatrix} \\
 &= \begin{bmatrix} \frac{13}{17} - \frac{5e^t}{26} \\ \frac{3}{17} + \frac{2e^t}{13} \end{bmatrix}
 \end{aligned}$$

Now that we found particular solution, the final solution is

$$\begin{aligned}
 \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\
 \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) c_1 e^{(-4+i)t} \\ c_1 e^{(-4+i)t} \end{bmatrix} + \begin{bmatrix} \left(-\frac{1}{2} - \frac{i}{2}\right) c_2 e^{(-4-i)t} \\ c_2 e^{(-4-i)t} \end{bmatrix} + \begin{bmatrix} \frac{13}{17} - \frac{5e^t}{26} \\ \frac{3}{17} + \frac{2e^t}{13} \end{bmatrix}
 \end{aligned}$$

Which becomes

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \left(-\frac{1}{2} + \frac{i}{2}\right) c_1 e^{(-4+i)t} + \left(-\frac{1}{2} - \frac{i}{2}\right) c_2 e^{(-4-i)t} - \frac{5e^t}{26} + \frac{13}{17} \\ c_1 e^{(-4+i)t} + c_2 e^{(-4-i)t} + \frac{2e^t}{13} + \frac{3}{17} \end{bmatrix}$$

✓ **Maple.** Time used: 0.191 (sec). Leaf size: 70

```
ode:=[diff(x(t),t)+x(t)+2*diff(y(t),t)+7*y(t) = exp(t)+2, -2*x(t)+diff(y(t),t)+3*
y(t) = exp(t)-1];
dsolve(ode);
```

$$x(t) = e^{-4t} \sin(t) c_2 + e^{-4t} \cos(t) c_1 + \frac{13}{17} - \frac{5e^t}{26}$$

$$y(t) = -e^{-4t} \sin(t) c_2 - e^{-4t} \cos(t) c_2 - e^{-4t} \cos(t) c_1 + e^{-4t} \sin(t) c_1 + \frac{2e^t}{13} + \frac{3}{17}$$

✓ **Mathematica.** Time used: 0.102 (sec). Leaf size: 209

```
ode={D[x[t],t]+x[t]+2*D[y[t],t]+7*y[t]==Exp[t]+2,-2*x[t]+D[y[t],t]+3*y[t]==Exp[t]
]-1};
ic={};
DSolve[{ode,ic},{x[t],y[t]},t,IncludeSingularSolutions->True]
```

$$\begin{aligned}
 x(t) &\rightarrow e^{-4t} \left((\cos(t) - \sin(t)) \int_1^t e^{4K[1]} (3 \sin(K[1]) - (-4 + e^{K[1]}) \cos(K[1])) dK[1] - \sin(t) \int_1^t e^{4K[2]} ((-1 - \cos(K[2])) \sin(K[2]) + (-4 + e^{K[2]}) \cos(K[2])) dK[2] \right. \\
 y(t) &\rightarrow e^{-4t} \left(2 \sin(t) \int_1^t e^{4K[1]} (3 \sin(K[1]) - (-4 + e^{K[1]}) \cos(K[1])) dK[1] + (\sin(t) + \cos(t)) \int_1^t e^{4K[2]} ((-1 - \cos(K[2])) \sin(K[2]) + (-4 + e^{K[2]}) \cos(K[2])) dK[2] \right.
 \end{aligned}$$

✓ Sympy. Time used: 2.868 (sec). Leaf size: 136

```
from sympy import *
t = symbols("t")
x = Function("x")
y = Function("y")
ode=[Eq(x(t) + 7*y(t) - exp(t) + Derivative(x(t), t) + 2*Derivative(y(t), t) -
      2,0),Eq(-2*x(t) + 3*y(t) - exp(t) + Derivative(y(t), t) + 1,0)]
ics = {}
dsolve(ode,func=[x(t),y(t)],ics=ics)
```

$$\left[x(t) = -\left(\frac{C_1}{2} - \frac{C_2}{2}\right) e^{-4t} \sin(t) - \left(\frac{C_1}{2} + \frac{C_2}{2}\right) e^{-4t} \cos(t) - \frac{5e^t \sin^2(t)}{26} - \frac{5e^t \cos^2(t)}{26} + \frac{13 \sin^2(t)}{17} + \frac{13 \cos^2(t)}{17}, y(t) = C_1 e^{-4t} \cos(t) - C_2 e^{-4t} \sin(t) + \frac{2e^t \sin^2(t)}{13} + \frac{2e^t \cos^2(t)}{13} + \frac{3 \sin^2(t)}{17} + \frac{3 \cos^2(t)}{17} \right]$$

2.15.4 Problem 13

Local contents

✓ Maple	2489
✓ Mathematica	2489
✗ Sympy	2489

Internal problem ID [8065]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Problem number : 13

Date solved : Thursday, November 27, 2025 at 07:33:06 AM

CAS classification : system_of_ODEs

$$x' - x + y' + 3y = e^{-t} - 1$$

$$x' + 2x + y' + 3y = e^{2t} + 1$$

The system is

$$x' + y' = x - 3y + e^{-t} - 1 \quad (1)$$

$$x' + y' = -2x - 3y + e^{2t} + 1 \quad (2)$$

Since the left side is the same, this implies

$$\begin{aligned} x - 3y + e^{-t} - 1 &= -2x - 3y + e^{2t} + 1 \\ x &= \frac{(e^{3t} + 2e^t - 1)e^{-t}}{3} \end{aligned} \quad (3)$$

Taking derivative of the above w.r.t. t gives

$$x' = \frac{(3e^{3t} + 2e^t)e^{-t}}{3} - \frac{(e^{3t} + 2e^t - 1)e^{-t}}{3} \quad (4)$$

Substituting (3,4) in (1) to eliminate x, x' gives

$$\begin{aligned} \frac{(3e^{3t} + 2e^t)e^{-t}}{3} - \frac{(e^{3t} + 2e^t - 1)e^{-t}}{3} + y' &= \frac{(e^{3t} + 2e^t - 1)e^{-t}}{3} - 3y + e^{-t} - 1 \\ x' &= -\frac{(e^{3t} + 9ye^t + e^t - 1)e^{-t}}{3} \end{aligned} \quad (5)$$

Which is now solved for y . Solve In canonical form a linear first order is

$$y' + q(t)y = p(t)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(t) &= 3 \\ p(t) &= \frac{e^{-t}}{3} - \frac{e^{2t}}{3} - \frac{1}{3} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dt} \\ &= e^{\int 3 \, dt} \\ &= e^{3t} \end{aligned}$$

The ode becomes

$$\begin{aligned} \frac{d}{dt}(\mu y) &= \mu p \\ \frac{d}{dt}(\mu y) &= (\mu) \left(\frac{e^{-t}}{3} - \frac{e^{2t}}{3} - \frac{1}{3} \right) \\ \frac{d}{dt}(y e^{3t}) &= (e^{3t}) \left(\frac{e^{-t}}{3} - \frac{e^{2t}}{3} - \frac{1}{3} \right) \\ d(y e^{3t}) &= \left(\left(\frac{e^{-t}}{3} - \frac{e^{2t}}{3} - \frac{1}{3} \right) e^{3t} \right) dt \end{aligned}$$

Integrating gives

$$\begin{aligned} y e^{3t} &= \int \left(\frac{e^{-t}}{3} - \frac{e^{2t}}{3} - \frac{1}{3} \right) e^{3t} dt \\ &= \frac{e^{2t}}{6} - \frac{e^{3t}}{9} - \frac{e^{5t}}{15} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor e^{3t} gives the final solution

$$y = -\frac{e^{2t}}{15} + e^{-3t} c_1 - \frac{1}{9} + \frac{e^{-t}}{6}$$

Given now that we have the solution

$$y = -\frac{e^{2t}}{15} + e^{-3t} c_1 - \frac{1}{9} + \frac{e^{-t}}{6} \tag{6}$$

Then substituting (6) into (3) gives

$$x = \frac{(e^{3t} + 2e^t - 1)e^{-t}}{3} \tag{7}$$

✓ **Maple.** Time used: 0.098 (sec). Leaf size: 41

```
ode:=[diff(x(t),t)-x(t)+diff(y(t),t)+3*y(t) = exp(-t)-1, diff(x(t),t)+2*x(t)+diff
      (y(t),t)+3*y(t) = 1+exp(2*t)];
dsolve(ode);
```

$$x(t) = \frac{e^{2t}}{3} + \frac{2}{3} - \frac{e^{-t}}{3}$$

$$y(t) = -\frac{1}{9} - \frac{e^{2t}}{15} + \frac{e^{-t}}{6} + e^{-3t}c_1$$

✓ **Mathematica.** Time used: 0.047 (sec). Leaf size: 62

```
ode={D[x[t],t]-x[t]+D[y[t],t]+3*y[t]==Exp[-t]-1,D[x[t],t]+2*x[t]+D[y[t],t]+3*y[t]
    ]==Exp[2*t]+1};
ic={};
DSolve[{ode,ic},{x[t],y[t]},t,IncludeSingularSolutions->True]
```

$$x(t) \rightarrow \frac{1}{3}e^{-t}(2e^t + e^{3t} - 1)$$

$$y(t) \rightarrow \frac{e^{-t}}{6} - \frac{e^{2t}}{15} + \frac{1}{16}c_1e^{-3t} - \frac{1}{9}$$

✗ **Sympy**

```
from sympy import *
t = symbols("t")
x = Function("x")
y = Function("y")
ode=[Eq(-x(t) + 3*y(t) + Derivative(x(t), t) + Derivative(y(t), t) + 1 - exp(-t)
      ,0),Eq(2*x(t) + 3*y(t) - exp(2*t) + Derivative(x(t), t) + Derivative(y(t), t)
      - 1,0)]
ics = {}
dsolve(ode,func=[x(t),y(t)],ics=ics)
```

2.15.5 Problem 17

Local contents

Solution using Matrix exponential method	2490
Solution using explicit Eigenvalue and Eigenvector method	2492
✓ Maple	2498
✓ Mathematica	2499
✓ Sympy	2499

Internal problem ID [8066]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 21. System of simultaneous linear equations. Supplementary problems. Page 163

Problem number : 17

Date solved : Thursday, November 27, 2025 at 07:33:06 AM

CAS classification : system_of_ODEs

$$x' - x + y' + 2y = 1 + e^t$$

$$y' + 2y + z' + z = e^t + 2$$

$$x' - x + z' + z = 3 + e^t$$

Solution using Matrix exponential method

In this method, we will assume we have found the matrix exponential e^{At} already. There are different methods to determine this but will not be shown here. This is a system of linear ODE's given as

$$\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 1 + \frac{e^t}{2} \\ \frac{e^t}{2} \\ 2 + \frac{e^t}{2} \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters

method applied to the fundamental matrix. For the above matrix A , the matrix exponential can be found to be

$$e^{At} = \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{-2t} & 0 \\ 0 & 0 & e^{-t} \end{bmatrix}$$

Therefore the homogeneous solution is

$$\begin{aligned} \vec{x}_h(t) &= e^{At} \vec{c} \\ &= \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{-2t} & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \\ &= \begin{bmatrix} e^t c_1 \\ e^{-2t} c_2 \\ e^{-t} c_3 \end{bmatrix} \end{aligned}$$

The particular solution given by

$$\vec{x}_p(t) = e^{At} \int e^{-At} \vec{G}(t) dt$$

But

$$\begin{aligned} e^{-At} &= (e^{At})^{-1} \\ &= \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^t \end{bmatrix} \end{aligned}$$

Hence

$$\begin{aligned} \vec{x}_p(t) &= \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{-2t} & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \int \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^t \end{bmatrix} \begin{bmatrix} 1 + \frac{e^t}{2} \\ \frac{e^t}{2} \\ 2 + \frac{e^t}{2} \end{bmatrix} dt \\ &= \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{-2t} & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} \frac{t}{2} - e^{-t} \\ \frac{e^{3t}}{6} \\ 2e^t + \frac{e^{2t}}{4} \end{bmatrix} \\ &= \begin{bmatrix} \frac{e^t t}{2} - 1 \\ \frac{e^t}{6} \\ 2 + \frac{e^t}{4} \end{bmatrix} \end{aligned}$$

Hence the complete solution is

$$\begin{aligned} \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\ &= \begin{bmatrix} -1 + \frac{(t+2c_1)e^t}{2} \\ \frac{e^t}{6} + e^{-2t}c_2 \\ 2 + \frac{e^t}{4} + e^{-t}c_3 \end{bmatrix} \end{aligned}$$

Solution using explicit Eigenvalue and Eigenvector method

This is a system of linear ODE's given as

$$\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$$

Or

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 1 + \frac{e^t}{2} \\ \frac{e^t}{2} \\ 2 + \frac{e^t}{2} \end{bmatrix}$$

Since the system is nonhomogeneous, then the solution is given by

$$\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$$

Where $\vec{x}_h(t)$ is the homogeneous solution to $\vec{x}'(t) = A\vec{x}(t)$ and $\vec{x}_p(t)$ is a particular solution to $\vec{x}'(t) = A\vec{x}(t) + \vec{G}(t)$. The particular solution will be found using variation of parameters method applied to the fundamental matrix.

The first step is find the homogeneous solution. We start by finding the eigenvalues of A . This is done by solving the following equation for the eigenvalues λ

$$\det(A - \lambda I) = 0$$

Expanding gives

$$\det \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) = 0$$

Therefore

$$\det \left(\begin{bmatrix} 1 - \lambda & 0 & 0 \\ 0 & -2 - \lambda & 0 \\ 0 & 0 & -1 - \lambda \end{bmatrix} \right) = 0$$

Since the matrix A is triangular matrix, then the determinant is the product of the elements along the diagonal. Therefore the above becomes

$$(1 - \lambda)(-2 - \lambda)(-1 - \lambda) = 0$$

The roots of the above are the eigenvalues.

$$\lambda_1 = -1$$

$$\lambda_2 = -2$$

$$\lambda_3 = 1$$

This table summarises the above result

eigenvalue	algebraic multiplicity	type of eigenvalue
-1	1	real eigenvalue
1	1	real eigenvalue
-2	1	real eigenvalue

Now the eigenvector for each eigenvalue are found.

Considering the eigenvalue $\lambda_1 = -2$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - (-2) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Since the current pivot $A(2, 3)$ is zero, then the current pivot row is replaced with a row with a non-zero pivot. Swapping row 2 and row 3 gives

$$\left[\begin{array}{ccc|c} 3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_2\}$ and the leading variables are $\{v_1, v_3\}$. Let $v_2 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = 0, v_3 = 0\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ t \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ t \\ v_3 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ t \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Considering the eigenvalue $\lambda_2 = -1$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - (-1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_3\}$ and the leading variables are $\{v_1, v_2\}$. Let $v_3 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_1 = 0, v_2 = 0\}$

Hence the solution is

$$\begin{bmatrix} v_1 \\ v_2 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ t \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} v_1 \\ v_2 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} v_1 \\ v_2 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Considering the eigenvalue $\lambda_3 = 1$

We need to solve $A\vec{v} = \lambda\vec{v}$ or $(A - \lambda I)\vec{v} = \vec{0}$ which becomes

$$\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - (1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now forward elimination is applied to solve for the eigenvector $\vec{v}$. The augmented matrix is

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right]$$

Since the current pivot $A(1, 2)$ is zero, then the current pivot row is replaced with a row with a non-zero pivot. Swapping row 1 and row 2 gives

$$\left[\begin{array}{ccc|c} 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right]$$

Since the current pivot $A(2, 3)$ is zero, then the current pivot row is replaced with a row with a non-zero pivot. Swapping row 2 and row 3 gives

$$\left[\begin{array}{ccc|c} 0 & -3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore the system in Echelon form is

$$\begin{bmatrix} 0 & -3 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The free variables are $\{v_1\}$ and the leading variables are $\{v_2, v_3\}$. Let $v_1 = t$. Now we start back substitution. Solving the above equation for the leading variables in terms of free variables gives equation $\{v_2 = 0, v_3 = 0\}$

Hence the solution is

$$\begin{bmatrix} t \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix}$$

Since there is one free Variable, we have found one eigenvector associated with this eigenvalue. The above can be written as

$$\begin{bmatrix} t \\ v_2 \\ v_3 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Let $t = 1$ the eigenvector becomes

$$\begin{bmatrix} t \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The following table gives a summary of this result. It shows for each eigenvalue the algebraic multiplicity m , and its geometric multiplicity k and the eigenvectors associated with the eigenvalue. If $m > k$ then the eigenvalue is defective which means the number of normal linearly independent eigenvectors associated with this eigenvalue (called the geometric multiplicity k) does not equal the algebraic multiplicity m , and we need to determine an additional $m - k$ generalized eigenvectors for this eigenvalue.

eigenvalue	multiplicity		defective?	eigenvectors
	algebraic m	geometric k		
-1	1	1	No	$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
-2	1	1	No	$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
1	1	1	No	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Now that we found the eigenvalues and associated eigenvectors, we will go over each eigenvalue and generate the solution basis. The only problem we need to take care of is if the eigenvalue is defective. Since eigenvalue -1 is real and distinct then the corresponding eigenvector

solution is

$$\begin{aligned}\vec{x}_1(t) &= \vec{v}_1 e^{-t} \\ &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t}\end{aligned}$$

Since eigenvalue -2 is real and distinct then the corresponding eigenvector solution is

$$\begin{aligned}\vec{x}_2(t) &= \vec{v}_2 e^{-2t} \\ &= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-2t}\end{aligned}$$

Since eigenvalue 1 is real and distinct then the corresponding eigenvector solution is

$$\begin{aligned}\vec{x}_3(t) &= \vec{v}_3 e^t \\ &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t\end{aligned}$$

Therefore the homogeneous solution is

$$\vec{x}_h(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + c_3 \vec{x}_3(t)$$

Which is written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = c_1 \begin{bmatrix} 0 \\ 0 \\ e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ e^{-2t} \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} e^t \\ 0 \\ 0 \end{bmatrix}$$

Now that we found homogeneous solution above, we need to find a particular solution $\vec{x}_p(t)$. We will use Variation of parameters. The fundamental matrix is

$$\Phi = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots]$$

Where $\vec{x}_i$ are the solution basis found above. Therefore the fundamental matrix is

$$\Phi(t) = \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{-2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix}$$

The particular solution is then given by

$$\vec{x}_p(t) = \Phi \int \Phi^{-1} \vec{G}(t) dt$$

But

$$\Phi^{-1} = \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix}$$

Hence

$$\begin{aligned} \vec{x}_p(t) &= \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{-2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix} \int \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 + \frac{e^t}{2} \\ \frac{e^t}{2} \\ 2 + \frac{e^t}{2} \end{bmatrix} dt \\ &= \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{-2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix} \int \begin{bmatrix} \frac{e^{2t}}{2} + 2e^t \\ \frac{e^{3t}}{2} \\ \frac{1}{2} + e^{-t} \end{bmatrix} dt \\ &= \begin{bmatrix} 0 & 0 & e^t \\ 0 & e^{-2t} & 0 \\ e^{-t} & 0 & 0 \end{bmatrix} \begin{bmatrix} 2e^t + \frac{e^{2t}}{4} \\ \frac{e^{3t}}{6} \\ \frac{t}{2} - e^{-t} \end{bmatrix} \\ &= \begin{bmatrix} \frac{e^t t}{2} - 1 \\ \frac{e^t}{6} \\ 2 + \frac{e^t}{4} \end{bmatrix} \end{aligned}$$

Now that we found particular solution, the final solution is

$$\begin{aligned} \vec{x}(t) &= \vec{x}_h(t) + \vec{x}_p(t) \\ \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ e^{-t}c_1 \end{bmatrix} + \begin{bmatrix} 0 \\ c_2 e^{-2t} \\ 0 \end{bmatrix} + \begin{bmatrix} c_3 e^t \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{e^t t}{2} - 1 \\ \frac{e^t}{6} \\ 2 + \frac{e^t}{4} \end{bmatrix} \end{aligned}$$

Which becomes

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 + \frac{(t+2c_3)e^t}{2} \\ c_2 e^{-2t} + \frac{e^t}{6} \\ e^{-t}c_1 + 2 + \frac{e^t}{4} \end{bmatrix}$$

✓ **Maple.** Time used: 0.099 (sec). Leaf size: 44

```
ode:=[diff(x(t),t)-x(t)+diff(y(t),t)+2*y(t) = 1+exp(t), diff(y(t),t)+2*y(t)+diff(
  z(t),t)+z(t) = exp(t)+2, diff(x(t),t)-x(t)+diff(z(t),t)+z(t) = 3+exp(t)];
dsolve(ode);
```

$$\begin{aligned} x(t) &= -1 + \frac{e^t(2c_3 + t)}{2} \\ y(t) &= \frac{e^t}{6} + e^{-2t}c_2 \\ z(t) &= 2 + \frac{e^t}{4} + e^{-t}c_1 \end{aligned}$$

✓ **Mathematica.** Time used: 0.044 (sec). Leaf size: 60

```
ode={D[x[t],t]-x[t]+D[y[t],t]+2*y[t]==1+Exp[t],D[y[t],t]+2*y[t]+D[z[t],t]+z[t]
]==2+Exp[t],D[x[t],t]-x[t]+D[z[t],t]+z[t]==3+Exp[t]};
ic={};
DSolve[{ode,ic},{x[t],y[t],z[t]},t,IncludeSingularSolutions->True]
```

$$\begin{aligned}x(t) &\rightarrow -1 + e^t \left(\frac{t}{2} + c_1 \right) \\ y(t) &\rightarrow \frac{e^t}{6} + c_2 e^{-2t} \\ z(t) &\rightarrow \frac{e^t}{4} + (4 + c_3) e^{-t} + 2\end{aligned}$$

✓ **Sympy.** Time used: 0.119 (sec). Leaf size: 42

```
from sympy import *
t = symbols("t")
x = Function("x")
y = Function("y")
z = Function("z")
ode=[Eq(-x(t) + 2*y(t) - exp(t) + Derivative(x(t), t) + Derivative(y(t), t) -
1,0),Eq(2*y(t) + z(t) - exp(t) + Derivative(y(t), t) + Derivative(z(t), t) -
2,0),Eq(-x(t) + z(t) - exp(t) + Derivative(x(t), t) + Derivative(z(t), t) -
3,0)]
ics = {}
dsolve(ode,func=[x(t),y(t),z(t)],ics=ics)
```

$$\left[x(t) = C_1 e^t + \frac{t e^t}{2} - 1, \quad y(t) = C_2 e^{-2t} + \frac{e^t}{6}, \quad z(t) = C_3 e^{-t} + \frac{e^t}{4} + 2 \right]$$

2.16 Chapter 25. Integration in series. Supplementary problems. Page 205

Local contents

2.16.1 Problem 9	2501
2.16.2 Problem 10A	2510
2.16.3 Problem 10B	2519
2.16.4 Problem 11	2536
2.16.5 Problem 12	2545
2.16.6 Problem 13	2554
2.16.7 Problem 14	2563
2.16.8 Problem 15	2572
2.16.9 Problem 16	2581
2.16.10 Problem 17	2591

2.16.1 Problem 9

Local contents

✓ Maple	2507
✓ Mathematica	2508
✓ Sympy	2509

Internal problem ID [8067]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 9

Date solved : Thursday, November 27, 2025 at 07:33:07 AM

CAS classification : [_linear]

$$(1 - x) y' = x^2 - y$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving first order ode. Let

$$y' = f(x, y)$$

Where $f(x, y)$ is analytic at expansion point x_0 . We can always shift to $x_0 = 0$ if x_0 is not zero. So from now we assume $x_0 = 0$. Assume also that $y(x_0) = y_0$. Using Taylor series

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\
 &= y_0 + x f + \frac{x^2}{2} \left. \frac{df}{dx} \right|_{x_0, y_0} + \frac{x^3}{3!} \left. \frac{d^2 f}{dx^2} \right|_{x_0, y_0} + \cdots \\
 &= y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} \left. \frac{d^n f}{dx^n} \right|_{x_0, y_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \quad (1)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y)$ then the above can be written as

$$F_0 = f(x, y) \quad (4)$$

$$\begin{aligned} F_n &= \frac{d}{dx}(F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) F_0 \end{aligned} \quad (5)$$

For example, for $n = 1$ we see that

$$\begin{aligned} F_1 &= \frac{d}{dx}(F_0) \\ &= \frac{\partial}{\partial x} F_0 + \left(\frac{\partial F_0}{\partial y} \right) F_0 \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \end{aligned}$$

Which is (1). And when $n = 2$

$$\begin{aligned} F_2 &= \frac{d}{dx}(F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) F_0 \\ &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) f \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned}$$

Which is (2) and so on. Therefore (4,5) can be used from now on along with

$$y(x) = y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} F_n|_{x_0, y_0} \quad (6)$$

Hence

$$\begin{aligned}
 F_0 &= \frac{-x^2 + y}{-1 + x} \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} F_0 \\
 &= -\frac{2x}{-1 + x} \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y} F_1 \\
 &= \frac{2}{(-1 + x)^2} \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y} F_2 \\
 &= -\frac{4}{(-1 + x)^3} \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y} F_3 \\
 &= \frac{12}{(-1 + x)^4}
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x(0) = 0$ and $y(0) = y(0)$ gives

$$\begin{aligned}
 F_0 &= -y(0) \\
 F_1 &= 0 \\
 F_2 &= 2 \\
 F_3 &= 4 \\
 F_4 &= 12
 \end{aligned}$$

Substituting all the above in (6) and simplifying gives the solution as

$$y = (1 - x)y(0) + \frac{x^3}{3} + \frac{x^4}{6} + \frac{x^5}{10} + O(x^6)$$

Since $x = 0$ is also an ordinary point, then standard power series can also be used. Writing

the ODE as

$$y' + q(x)y = p(x)$$

$$y' - \frac{y}{-1+x} = -\frac{x^2}{-1+x}$$

Where

$$q(x) = -\frac{1}{-1+x}$$

$$p(x) = -\frac{x^2}{-1+x}$$

Next, the type of the expansion point $x = 0$ is determined. This point can be an ordinary point, a regular singular point (also called removable singularity), or irregular singular point (also called non-removable singularity or essential singularity). When $x = 0$ is an ordinary point, then the standard power series is used. If the point is a regular singular point, Frobenius series is used instead. Irregular singular point requires more advanced methods (asymptotic methods) and is not supported now. Hopefully this will be added in the future. $x = 0$ is called an ordinary point $q(x)$ has a Taylor series expansion around the point $x = 0$. $x = 0$ is called a regular singular point if $q(x)$ is not analytic at $x = 0$ but $xq(x)$ has Taylor series expansion. And finally, $x = 0$ is an irregular singular point if the point is not ordinary and not regular singular. This is the most complicated case. Now the expansion point $x = 0$ is checked to see if it is an ordinary point or not. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Substituting the above back into the ode gives

$$(1-x) \left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) = x^2 - \left(\sum_{n=0}^{\infty} a_n x^n \right) \quad (1)$$

Expanding $-x^2$ as Taylor series around $x = 0$ and keeping only the first 6 terms gives

$$-x^2 = -x^2 + \dots$$

$$= -x^2$$

Which simplifies to

$$\sum_{n=1}^{\infty} (-n a_n x^{n-1}) + \left(\sum_{n=1}^{\infty} n a_n x^n \right) + \sum_{n=0}^{\infty} (-a_n x^n) = -x^2 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=1}^{\infty} (-na_n x^{n-1}) = \sum_{n=0}^{\infty} (-(n+1) a_{n+1} x^n)$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\sum_{n=0}^{\infty} (-(n+1) a_{n+1} x^n) + \left(\sum_{n=1}^{\infty} na_n x^n \right) + \sum_{n=0}^{\infty} (-a_n x^n) = -x^2 \quad (3)$$

For $1 \leq n$, the recurrence equation is

$$(-(n+1) a_{n+1} + na_n - a_n) x^n = -x^2 \quad (4)$$

For $n = 1$ the recurrence equation gives

$$\begin{aligned} (-2a_2) x &= 0 \\ -2a_2 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_2 = 0$$

For $n = 2$ the recurrence equation gives

$$\begin{aligned} (-3a_3 + a_2) x^2 &= -x^2 \\ -3a_3 + a_2 &= -1 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_3 = \frac{1}{3}$$

For $n = 3$ the recurrence equation gives

$$\begin{aligned} (-4a_4 + 2a_3) x^3 &= 0 \\ -4a_4 + 2a_3 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_4 = \frac{1}{6}$$

For $n = 4$ the recurrence equation gives

$$\begin{aligned} (-5a_5 + 3a_4)x^4 &= 0 \\ -5a_5 + 3a_4 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_5 = \frac{1}{10}$$

For $n = 5$ the recurrence equation gives

$$\begin{aligned} (-6a_6 + 4a_5)x^5 &= 0 \\ -6a_6 + 4a_5 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_6 = \frac{1}{15}$$

And so on. Therefore the solution is

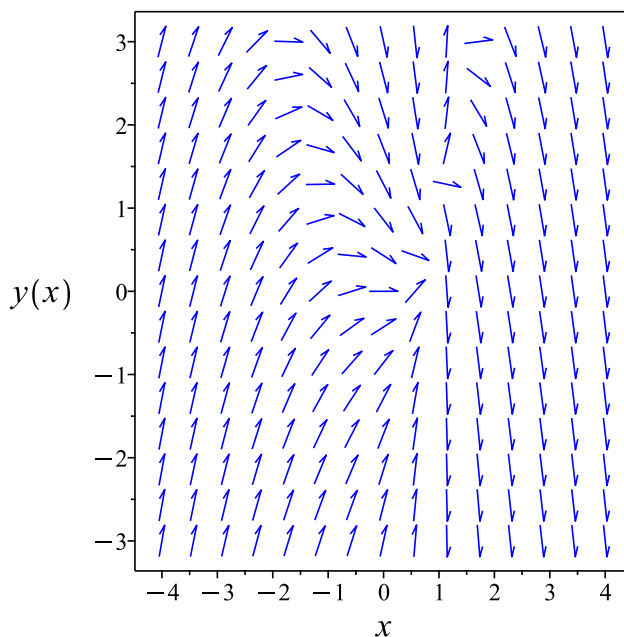
$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 - a_0 x + \frac{1}{3}x^3 + \frac{1}{6}x^4 + \frac{1}{10}x^5 + \dots$$

Collecting terms, the solution becomes

$$y = (1 - x)a_0 + \frac{x^3}{3} + \frac{x^4}{6} + \frac{x^5}{10} + O(x^6) \quad (3)$$

Figure 2.386: Slope field $(1 - x)y' = x^2 - y$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 32

```
Order:=6;
ode:=(1-x)*diff(y(x),x) = x^2-y(x);
dsolve(ode,y(x),type='series',x=0);
```

$$y = (1 - x)y(0) + \frac{x^3}{3} + \frac{x^4}{6} + \frac{x^5}{10} + O(x^6)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$(1 - x) \left(\frac{d}{dx} y(x) \right) = x^2 - y(x)$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx} y(x)$
- Solve for the highest derivative

- $\frac{d}{dx}y(x) = \frac{x^2 - y(x)}{1-x}$
- Collect w.r.t. $y(x)$ and simplify
 $\frac{d}{dx}y(x) = \frac{y(x)}{-1+x} - \frac{x^2}{-1+x}$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE
 $\frac{d}{dx}y(x) - \frac{y(x)}{-1+x} = -\frac{x^2}{-1+x}$
- The ODE is linear; multiply by an integrating factor $\mu(x)$
 $\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{-1+x} \right) = -\frac{\mu(x)x^2}{-1+x}$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$
 $\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{-1+x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$
- Isolate $\frac{d}{dx}\mu(x)$
 $\frac{d}{dx}\mu(x) = -\frac{\mu(x)}{-1+x}$
- Solve to find the integrating factor
 $\mu(x) = \frac{1}{-1+x}$
- Integrate both sides with respect to x
 $\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int -\frac{\mu(x)x^2}{-1+x} dx + C1$
- Evaluate the integral on the lhs
 $y(x)\mu(x) = \int -\frac{\mu(x)x^2}{-1+x} dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int -\frac{\mu(x)x^2}{-1+x} dx + C1}{\mu(x)}$
- Substitute $\mu(x) = \frac{1}{-1+x}$
 $y(x) = (-1+x) \left(\int -\frac{x^2}{(-1+x)^2} dx + C1 \right)$
- Evaluate the integrals on the rhs
 $y(x) = (-1+x) \left(-x + \frac{1}{-1+x} - 2 \ln(-1+x) + C1 \right)$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 33

```
ode=(1-x)*D[y[x],x]==x^2-y[x];
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow \frac{x^5}{10} + \frac{x^4}{6} + \frac{x^3}{3} + c_1(1-x)$$

✓ Sympy. Time used: 0.216 (sec). Leaf size: 26

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + (1 - x)*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="1st_power_series", x0=0, n=6)
```

$$y(x) = \frac{x^3}{3} + \frac{x^4}{6} + \frac{x^5}{10} + C_1 - C_1 x + O(x^6)$$

2.16.2 Problem 10A

Local contents

✓ Maple	2516
✓ Mathematica	2518
✗ Sympy	2518

Internal problem ID [8068]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 10A

Date solved : Thursday, November 27, 2025 at 07:33:07 AM

CAS classification : [_linear]

$$y'x = 1 - x + 2y$$

Series expansion around $x = 1$.

The ode does not have its expansion point at $x = 0$, therefore to simplify the computation of power series expansion, change of variable is made on the independent variable to shift the initial conditions and the expansion point back to zero. The new ode is then solved more easily since the expansion point is now at zero. The solution converted back to the original independent variable. Let

$$\alpha = -1 + x$$

The ode is converted to be in terms of the new independent variable α . This results in

$$\left(\frac{d}{d\alpha} y(\alpha) \right) (\alpha + 1) = -\alpha + 2y(\alpha)$$

With its expansion point and initial conditions now at $\alpha = 0$. Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving first order ode. Let

$$y' = f(x, y)$$

Where $f(x, y)$ is analytic at expansion point x_0 . We can always shift to $x_0 = 0$ if x_0 is not

zero. So from now we assume $x_0 = 0$. Assume also that $y(x_0) = y_0$. Using Taylor series

$$\begin{aligned} y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\ &= y_0 + x f + \frac{x^2}{2} \left. \frac{df}{dx} \right|_{x_0, y_0} + \frac{x^3}{3!} \left. \frac{d^2 f}{dx^2} \right|_{x_0, y_0} + \cdots \\ &= y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} \left. \frac{d^n f}{dx^n} \right|_{x_0, y_0} \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \quad (1)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

⋮

And so on. Hence if we name $F_0 = f(x, y)$ then the above can be written as

$$F_0 = f(x, y) \quad (4)$$

$$\begin{aligned} F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) F_0 \end{aligned} \quad (5)$$

For example, for $n = 1$ we see that

$$\begin{aligned} F_1 &= \frac{d}{dx} (F_0) \\ &= \frac{\partial}{\partial x} F_0 + \left(\frac{\partial F_0}{\partial y} \right) F_0 \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \end{aligned}$$

Which is (1). And when $n = 2$

$$\begin{aligned}
 F_2 &= \frac{d}{dx}(F_1) \\
 &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) F_0 \\
 &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) f \\
 &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f
 \end{aligned}$$

Which is (2) and so on. Therefore (4,5) can be used from now on along with

$$y(x) = y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} F_n|_{x_0, y_0} \quad (6)$$

Hence

$$\begin{aligned}
 F_0 &= \frac{-\alpha + 2y(\alpha)}{\alpha + 1} \\
 F_1 &= \frac{dF_0}{d\alpha} \\
 &= \frac{\partial F_0}{\partial \alpha} + \frac{\partial F_0}{\partial y} F_0 \\
 &= \frac{-1 - 2\alpha + 2y(\alpha)}{(\alpha + 1)^2} \\
 F_2 &= \frac{dF_1}{d\alpha} \\
 &= \frac{\partial F_1}{\partial \alpha} + \frac{\partial F_1}{\partial y} F_1 \\
 &= 0 \\
 F_3 &= \frac{dF_2}{d\alpha} \\
 &= \frac{\partial F_2}{\partial \alpha} + \frac{\partial F_2}{\partial y} F_2 \\
 &= 0 \\
 F_4 &= \frac{dF_3}{d\alpha} \\
 &= \frac{\partial F_3}{\partial \alpha} + \frac{\partial F_3}{\partial y} F_3 \\
 &= 0
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $\alpha(0) = 0$ and $y(0) = y(0)$ gives

$$\begin{aligned} F_0 &= 2y(0) \\ F_1 &= -1 + 2y(0) \\ F_2 &= 0 \\ F_3 &= 0 \\ F_4 &= 0 \end{aligned}$$

Substituting all the above in (6) and simplifying gives the solution as

$$y(\alpha) = (\alpha^2 + 2\alpha + 1) y(0) - \frac{\alpha^2}{2} + O(\alpha^6)$$

Since $\alpha = 0$ is also an ordinary point, then standard power series can also be used. Writing the ODE as

$$\begin{aligned} \frac{d}{d\alpha}y(\alpha) + q(\alpha)y(\alpha) &= p(\alpha) \\ \frac{d}{d\alpha}y(\alpha) - \frac{2y(\alpha)}{\alpha + 1} &= -\frac{\alpha}{\alpha + 1} \end{aligned}$$

Where

$$\begin{aligned} q(\alpha) &= -\frac{2}{\alpha + 1} \\ p(\alpha) &= -\frac{\alpha}{\alpha + 1} \end{aligned}$$

Next, the type of the expansion point $\alpha = 0$ is determined. This point can be an ordinary point, a regular singular point (also called removable singularity), or irregular singular point (also called non-removable singularity or essential singularity). When $\alpha = 0$ is an ordinary point, then the standard power series is used. If the point is a regular singular point, Frobenius series is used instead. Irregular singular point requires more advanced methods (asymptotic methods) and is not supported now. Hopefully this will be added in the future. $\alpha = 0$ is called an ordinary point $q(\alpha)$ has a Taylor series expansion around the point $\alpha = 0$. $\alpha = 0$ is called a regular singular point if $q(\alpha)$ is not not analytic at $\alpha = 0$ but $\alpha q(\alpha)$ has Taylor series expansion. And finally, $\alpha = 0$ is an irregular singular point if the point is not ordinary and not regular singular. This is the most complicated case. Now the expansion point $\alpha = 0$ is checked to see if it is an ordinary point or not. Let the solution be represented as power series of the form

$$y(\alpha) = \sum_{n=0}^{\infty} a_n \alpha^n$$

Then

$$\frac{d}{d\alpha}y(\alpha) = \sum_{n=1}^{\infty} n a_n \alpha^{n-1}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=1}^{\infty} n a_n \alpha^{n-1} \right) (\alpha + 1) = -\alpha + 2 \left(\sum_{n=0}^{\infty} a_n \alpha^n \right) \quad (1)$$

Expanding $-\alpha$ as Taylor series around $\alpha = 0$ and keeping only the first 6 terms gives

$$\begin{aligned} -\alpha &= -\alpha + \dots \\ &= -\alpha \end{aligned}$$

Which simplifies to

$$\left(\sum_{n=1}^{\infty} n a_n \alpha^n \right) + \left(\sum_{n=1}^{\infty} n a_n \alpha^{n-1} \right) + \sum_{n=0}^{\infty} (-2 a_n \alpha^n) = -\alpha \quad (2)$$

The next step is to make all powers of α be n in each summation term. Going over each summation term above with power of α in it which is not already α^n and adjusting the power and the corresponding index gives

$$\sum_{n=1}^{\infty} n a_n \alpha^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} \alpha^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of α are the same and equal to n .

$$\left(\sum_{n=1}^{\infty} n a_n \alpha^n \right) + \left(\sum_{n=0}^{\infty} (n+1) a_{n+1} \alpha^n \right) + \sum_{n=0}^{\infty} (-2 a_n \alpha^n) = -\alpha \quad (3)$$

For $1 \leq n$, the recurrence equation is

$$(n a_n + (n+1) a_{n+1} - 2 a_n) \alpha^n = -\alpha \quad (4)$$

For $n = 1$ the recurrence equation gives

$$\begin{aligned} (-a_1 + 2a_2) \alpha &= -\alpha \\ -a_1 + 2a_2 &= -1 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_2 = -\frac{1}{2} + a_0$$

For $n = 2$ the recurrence equation gives

$$\begin{aligned}(3a_3)\alpha^2 &= 0 \\ 3a_3 &= 0\end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_3 = 0$$

For $n = 3$ the recurrence equation gives

$$\begin{aligned}(a_3 + 4a_4)\alpha^3 &= 0 \\ a_3 + 4a_4 &= 0\end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_4 = 0$$

For $n = 4$ the recurrence equation gives

$$\begin{aligned}(2a_4 + 5a_5)\alpha^4 &= 0 \\ 2a_4 + 5a_5 &= 0\end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_5 = 0$$

For $n = 5$ the recurrence equation gives

$$\begin{aligned}(3a_5 + 6a_6)\alpha^5 &= 0 \\ 3a_5 + 6a_6 &= 0\end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_6 = 0$$

And so on. Therefore the solution is

$$\begin{aligned}y(\alpha) &= \sum_{n=0}^{\infty} a_n \alpha^n \\ &= a_3 \alpha^3 + a_2 \alpha^2 + a_1 \alpha + a_0 + \dots\end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y(\alpha) = a_0 + 2a_0\alpha + \left(-\frac{1}{2} + a_0\right)\alpha^2 + \dots$$

Collecting terms, the solution becomes

$$y(\alpha) = (\alpha^2 + 2\alpha + 1)a_0 - \frac{\alpha^2}{2} + O(\alpha^6) \quad (3)$$

Hence the solution is

$$y = ((-1 + x)^2 + 2x - 1)y(1) - \frac{(-1 + x)^2}{2} + O((-1 + x)^6)$$

Hence the solution is

$$y = ((-1 + x)^2 + 2x - 1)y(1) - \frac{(-1 + x)^2}{2} + O((-1 + x)^6)$$

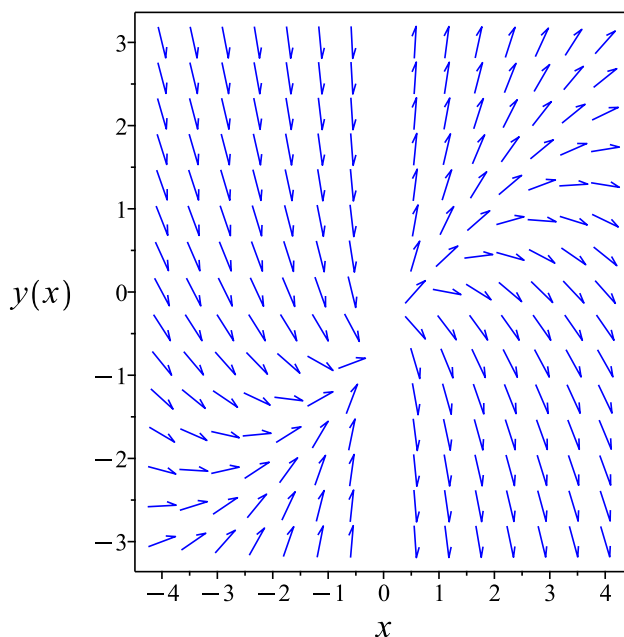


Figure 2.387: Slope field $y' = 1 - x + 2y$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 18

```
Order:=6;
ode:=diff(y(x),x)*x = 1-x+2*y(x);
dsolve(ode,y(x),type='series',x=1);
```

$$y = y(1)x^2 - \frac{(x-1)^2}{2}$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right) = 1 - x + 2y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{1-x+2y(x)}{x}$$

- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx}y(x) = \frac{2y(x)}{x} - \frac{-1+x}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - \frac{2y(x)}{x} = -\frac{-1+x}{x}$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{2y(x)}{x} \right) = -\frac{\mu(x)(-1+x)}{x}$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{2y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$

- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{2\mu(x)}{x}$$

- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x^2}$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int -\frac{\mu(x)(-1+x)}{x} dx + C1$$

- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int -\frac{\mu(x)(-1+x)}{x} dx + C1$$

- Solve for $y(x)$

$$y(x) = \frac{\int -\frac{\mu(x)(-1+x)}{x} dx + C1}{\mu(x)}$$

- Substitute $\mu(x) = \frac{1}{x^2}$

$$y(x) = x^2 \left(\int -\frac{-1+x}{x^3} dx + C1 \right)$$

- Evaluate the integrals on the rhs

$$y(x) = x^2 \left(\frac{1}{x} - \frac{1}{2x^2} + C1 \right)$$

- Simplify

$$y(x) = x - \frac{1}{2} + C1 x^2$$

✓ **Mathematica.** Time used: 0.013 (sec). Leaf size: 28

```
ode=x*D[y[x],x]==1-x+2*y[x];
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,1,5}]
```

$$y(x) \rightarrow -\frac{1}{2}(x-1)^2 + c_1((x-1)^2 + 2(x-1) + 1)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) + x - 2*y(x) - 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics,hint="1st_power_series",x0=1,n=6)
```

```
ValueError : ODE x*Derivative(y(x), x) + x - 2*y(x) - 1 does not match hint 1
st_power_series
```

2.16.3 Problem 10B

Local contents

Solved using first_order_ode_linear	2519
Solved using first_order_ode_exact	2521
Solved using first_order_ode_homog_type_maple_C	2525
Solved using first_order_ode_LIE	2529
✓ Maple	2534
✓ Mathematica	2535
✓ Sympy	2535

Internal problem ID [8069]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 10B

Date solved : Thursday, November 27, 2025 at 07:33:08 AM

CAS classification : [_linear]

Solved using first_order_ode_linear

Time used: 0.042 (sec)

Solve

$$y'x = 1 - x + 2y$$

In canonical form a linear first order is

$$y' + q(x)y = p(x)$$

Comparing the above to the given ode shows that

$$\begin{aligned} q(x) &= -\frac{2}{x} \\ p(x) &= \frac{1-x}{x} \end{aligned}$$

The integrating factor μ is

$$\begin{aligned} \mu &= e^{\int q \, dx} \\ &= e^{\int -\frac{2}{x} \, dx} \\ &= \frac{1}{x^2} \end{aligned}$$

The ode becomes

$$\frac{d}{dx}(\mu y) = \mu p$$

$$\frac{d}{dx}(\mu y) = (\mu) \left(\frac{1-x}{x} \right)$$

$$\frac{d}{dx} \left(\frac{y}{x^2} \right) = \left(\frac{1}{x^2} \right) \left(\frac{1-x}{x} \right)$$

$$d \left(\frac{y}{x^2} \right) = \left(\frac{1-x}{x^3} \right) dx$$

Integrating gives

$$\begin{aligned} \frac{y}{x^2} &= \int \frac{1-x}{x^3} dx \\ &= \frac{-1+2x}{2x^2} + c_1 \end{aligned}$$

Dividing throughout by the integrating factor $\frac{1}{x^2}$ gives the final solution

$$y = x - \frac{1}{2} + c_1 x^2$$

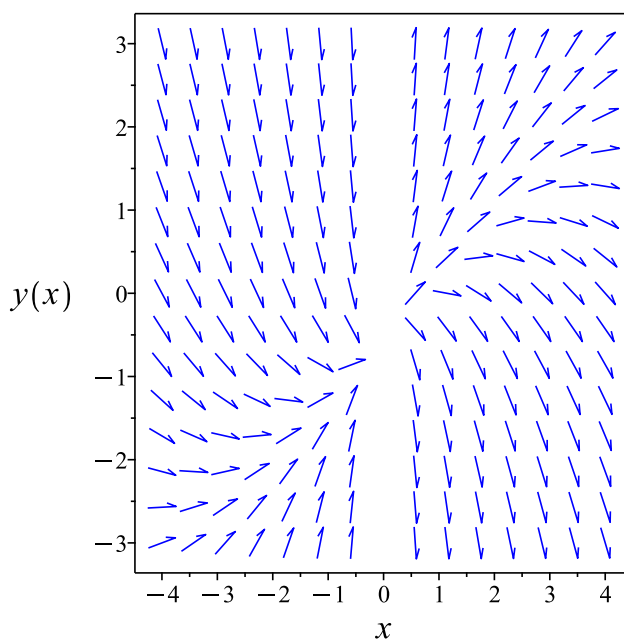


Figure 2.388: Slope field $y' = 1 - x + 2y$

Summary of solutions found

$$y = x - \frac{1}{2} + c_1 x^2$$

Solved using first_order_ode_exact

Time used: 0.105 (sec)

To solve an ode of the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{A})$$

We assume there exists a function $\phi(x, y) = c$ where c is constant, that satisfies the ode. Taking derivative of ϕ w.r.t. x gives

$$\frac{d}{dx} \phi(x, y) = 0$$

Hence

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} = 0 \quad (\text{B})$$

Comparing (A,B) shows that

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= M \\ \frac{\partial \phi}{\partial y} &= N \end{aligned}$$

But since $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ then for the above to be valid, we require that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If the above condition is satisfied, then the original ode is called exact. We still need to determine $\phi(x, y)$ but at least we know now that we can do that since the condition $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$ is satisfied. If this condition is not satisfied then this method will not work and we have to now look for an integrating factor to force this condition, which might or might not exist. The first step is to write the ODE in standard form to check for exactness, which is

$$M(x, y) dx + N(x, y) dy = 0 \quad (\text{1A})$$

Therefore

$$\begin{aligned} (x) dy &= (1 - x + 2y) dx \\ (-1 + x - 2y) dx + (x) dy &= 0 \end{aligned} \quad (\text{2A})$$

Comparing (1A) and (2A) shows that

$$\begin{aligned} M(x, y) &= -1 + x - 2y \\ N(x, y) &= x \end{aligned}$$

The next step is to determine if the ODE is exact or not. The ODE is exact when the following condition is satisfied

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Using result found above gives

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(-1 + x - 2y) \\ &= -2\end{aligned}$$

And

$$\begin{aligned}\frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(x) \\ &= 1\end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, then the ODE is not exact. Since the ODE is not exact, we will try to find an integrating factor to make it exact. Let

$$\begin{aligned}A &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \\ &= \frac{1}{x}((-2) - (1)) \\ &= -\frac{3}{x}\end{aligned}$$

Since A does not depend on y , then it can be used to find an integrating factor. The integrating factor μ is

$$\begin{aligned}\mu &= e^{\int A \, dx} \\ &= e^{\int -\frac{3}{x} \, dx}\end{aligned}$$

The result of integrating gives

$$\begin{aligned}\mu &= e^{-3 \ln(x)} \\ &= \frac{1}{x^3}\end{aligned}$$

M and N are multiplied by this integrating factor, giving new M and new N which are called $\overline{M}$ and $\overline{N}$ for now so not to confuse them with the original M and N .

$$\begin{aligned}\overline{M} &= \mu M \\ &= \frac{1}{x^3}(-1 + x - 2y) \\ &= \frac{-1 + x - 2y}{x^3}\end{aligned}$$

And

$$\begin{aligned}\bar{N} &= \mu N \\ &= \frac{1}{x^3}(x) \\ &= \frac{1}{x^2}\end{aligned}$$

Now a modified ODE is obtained from the original ODE, which is exact and can be solved. The modified ODE is

$$\begin{aligned}\bar{M} + \bar{N} \frac{dy}{dx} &= 0 \\ \left(\frac{-1+x-2y}{x^3} \right) + \left(\frac{1}{x^2} \right) \frac{dy}{dx} &= 0\end{aligned}$$

The following equations are now set up to solve for the function $\phi(x, y)$

$$\frac{\partial \phi}{\partial x} = \bar{M} \tag{1}$$

$$\frac{\partial \phi}{\partial y} = \bar{N} \tag{2}$$

Integrating (2) w.r.t. y gives

$$\begin{aligned}\int \frac{\partial \phi}{\partial y} dy &= \int \bar{N} dy \\ \int \frac{\partial \phi}{\partial y} dy &= \int \frac{1}{x^2} dy \\ \phi &= \frac{y}{x^2} + f(x)\end{aligned} \tag{3}$$

Where $f(x)$ is used for the constant of integration since ϕ is a function of both x and y . Taking derivative of equation (3) w.r.t x gives

$$\frac{\partial \phi}{\partial x} = -\frac{2y}{x^3} + f'(x) \tag{4}$$

But equation (1) says that $\frac{\partial \phi}{\partial x} = \frac{-1+x-2y}{x^3}$. Therefore equation (4) becomes

$$\frac{-1+x-2y}{x^3} = -\frac{2y}{x^3} + f'(x) \tag{5}$$

Solving equation (5) for $f'(x)$ gives

$$f'(x) = \frac{-1+x}{x^3}$$

Integrating the above w.r.t x gives

$$\int f'(x) dx = \int \left(\frac{-1+x}{x^3} \right) dx$$

$$f(x) = -\frac{1}{x} + \frac{1}{2x^2} + c_1$$

Where c_1 is constant of integration. Substituting result found above for $f(x)$ into equation (3) gives ϕ

$$\phi = \frac{y}{x^2} - \frac{1}{x} + \frac{1}{2x^2} + c_1$$

But since ϕ itself is a constant function, then let $\phi = c_2$ where c_2 is new constant and combining c_1 and c_2 constants into the constant c_1 gives the solution as

$$c_1 = \frac{y}{x^2} - \frac{1}{x} + \frac{1}{2x^2}$$

Simplifying the above gives

$$\frac{2y - 2x + 1}{2x^2} = c_1$$

Solving for y gives

$$y = x - \frac{1}{2} + c_1 x^2$$

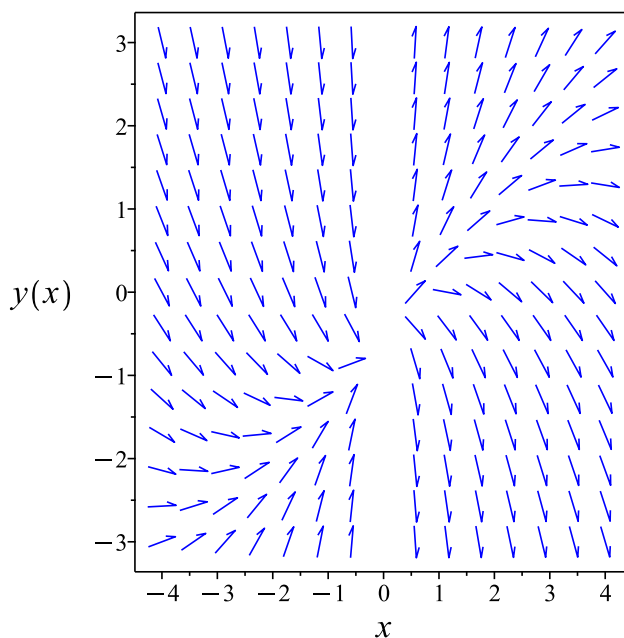


Figure 2.389: Slope field $y' = 1 - x + 2y$

Summary of solutions found

$$y = x - \frac{1}{2} + c_1 x^2$$

Solved using first_order_ode_homog_type_maple_C

Time used: 0.195 (sec)

Solve

$$y'x = 1 - x + 2y$$

Let $Y = y - y_0$ and $X = x - x_0$ then the above is transformed to new ode in $Y(X)$

$$\frac{d}{dX}Y(X) = \frac{1 - X - x_0 + 2Y(X) + 2y_0}{X + x_0}$$

Solving for possible values of x_0 and y_0 which makes the above ode a homogeneous ode results in

$$\begin{aligned} x_0 &= 0 \\ y_0 &= -\frac{1}{2} \end{aligned}$$

Using these values now it is possible to easily solve for $Y(X)$. The above ode now becomes

$$\frac{d}{dX}Y(X) = \frac{-X + 2Y(X)}{X}$$

In canonical form, the ODE is

$$\begin{aligned} Y' &= F(X, Y) \\ &= \frac{-X + 2Y}{X} \end{aligned} \tag{1}$$

An ode of the form $Y' = \frac{M(X,Y)}{N(X,Y)}$ is called homogeneous if the functions $M(X, Y)$ and $N(X, Y)$ are both homogeneous functions and of the same order. Recall that a function $f(X, Y)$ is homogeneous of order n if

$$f(t^n X, t^n Y) = t^n f(X, Y)$$

In this case, it can be seen that both $M = -X + 2Y$ and $N = X$ are both homogeneous and of the same order $n = 1$. Therefore this is a homogeneous ode. Since this ode is homogeneous, it is converted to separable ODE using the substitution $u = \frac{Y}{X}$, or $Y = uX$. Hence

$$\frac{dY}{dX} = \frac{du}{dX}X + u$$

Applying the transformation $Y = uX$ to the above ODE in (1) gives

$$\begin{aligned}\frac{du}{dX}X + u &= -1 + 2u \\ \frac{du}{dX} &= \frac{-1 + u(X)}{X}\end{aligned}$$

Or

$$\frac{d}{dX}u(X) - \frac{-1 + u(X)}{X} = 0$$

Or

$$\left(\frac{d}{dX}u(X)\right)X - u(X) + 1 = 0$$

Which is now solved as separable in $u(X)$.

The ode

$$\frac{d}{dX}u(X) = \frac{-1 + u(X)}{X} \quad (2.122)$$

is separable as it can be written as

$$\begin{aligned}\frac{d}{dX}u(X) &= \frac{-1 + u(X)}{X} \\ &= f(X)g(u)\end{aligned}$$

Where

$$\begin{aligned}f(X) &= \frac{1}{X} \\ g(u) &= -1 + u\end{aligned}$$

Integrating gives

$$\begin{aligned}\int \frac{1}{g(u)} du &= \int f(X) dX \\ \int \frac{1}{-1 + u} du &= \int \frac{1}{X} dX\end{aligned}$$

$$\ln(-1 + u(X)) = \ln(X) + c_1$$

Taking the exponential of both sides the solution becomes

$$-1 + u(X) = c_1 X$$

We now need to find the singular solutions, these are found by finding for what values $g(u)$ is zero, since we had to divide by this above. Solving $g(u) = 0$ or

$$-1 + u = 0$$

for $u(X)$ gives

$$u(X) = 1$$

Now we go over each such singular solution and check if it verifies the ode itself and any initial conditions given. If it does not then the singular solution will not be used.

Therefore the solutions found are

$$-1 + u(X) = c_1 X$$

$$u(X) = 1$$

Converting $-1 + u(X) = c_1 X$ back to $Y(X)$ gives

$$\frac{-X + Y(X)}{X} = c_1 X$$

Converting $u(X) = 1$ back to $Y(X)$ gives

$$Y(X) = X$$

Using the solution for $Y(X)$

$$\frac{-X + Y(X)}{X} = c_1 X \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - \frac{1}{2}$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$\frac{y - x + \frac{1}{2}}{x} = c_1 x$$

Using the solution for $Y(X)$

$$Y(X) = X \tag{A}$$

And replacing back terms in the above solution using

$$Y = y + y_0$$

$$X = x_0 + x$$

Or

$$Y = y - \frac{1}{2}$$

$$X = x$$

Then the solution in y becomes using EQ (A)

$$y + \frac{1}{2} = x$$

Solving for y gives

$$y = -\frac{1}{2} + x$$

$$y = x - \frac{1}{2} + c_1 x^2$$

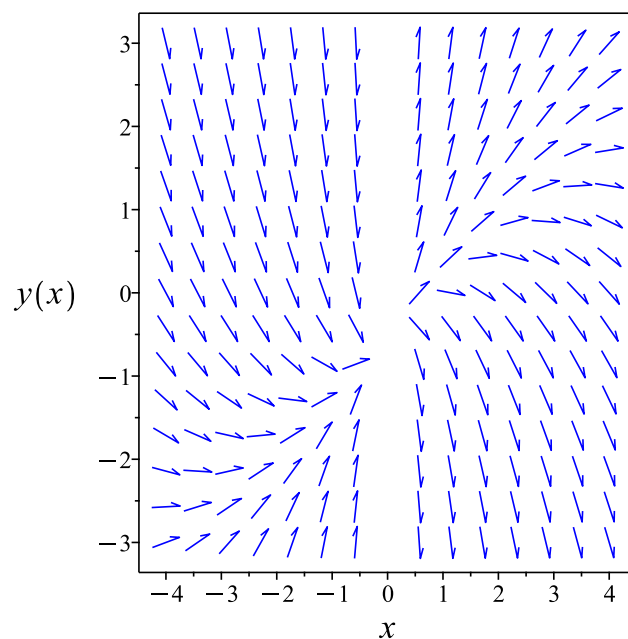


Figure 2.390: Slope field $y'x = 1 - x + 2y$

Summary of solutions found

$$y = -\frac{1}{2} + x$$

$$y = x - \frac{1}{2} + c_1 x^2$$

Solved using first_order_ode_LIE

Time used: 0.671 (sec)

Solve

$$y'x = 1 - x + 2y$$

Writing the ode as

$$y' = \frac{1 - x + 2y}{x}$$

$$y' = \omega(x, y)$$

The condition of Lie symmetry is the linearized PDE given by

$$\eta_x + \omega(\eta_y - \xi_x) - \omega^2 \xi_y - \omega_x \xi - \omega_y \eta = 0 \quad (\text{A})$$

To determine ξ, η then (A) is solved using ansatz. Making bivariate polynomials of degree 1 to use as anstaz gives

$$\xi = xa_2 + ya_3 + a_1 \quad (1\text{E})$$

$$\eta = xb_2 + yb_3 + b_1 \quad (2\text{E})$$

Where the unknown coefficients are

$$\{a_1, a_2, a_3, b_1, b_2, b_3\}$$

Substituting equations (1E,2E) and ω into (A) gives

$$b_2 + \frac{(1 - x + 2y)(b_3 - a_2)}{x} - \frac{(1 - x + 2y)^2 a_3}{x^2} - \left(-\frac{1}{x} - \frac{1 - x + 2y}{x^2} \right) (xa_2 + ya_3 + a_1) - \frac{2(xb_2 + yb_3 + b_1)}{x} = 0 \quad (5\text{E})$$

Putting the above in normal form gives

$$\frac{x^2 a_2 - x^2 a_3 - b_2 x^2 - x^2 b_3 + 4xy a_3 - 2y^2 a_3 + 2xa_3 - 2xb_1 + xb_3 + 2ya_1 - 3ya_3 + a_1 - a_3}{x^2} = 0$$

Setting the numerator to zero gives

$$x^2a_2 - x^2a_3 - b_2x^2 - x^2b_3 + 4xya_3 - 2y^2a_3 + 2xa_3 - 2xb_1 + xb_3 + 2ya_1 - 3ya_3 + a_1 - a_3 = 0 \quad (6E)$$

Looking at the above PDE shows the following are all the terms with $\{x, y\}$ in them.

$$\{x, y\}$$

The following substitution is now made to be able to collect on all terms with $\{x, y\}$ in them

$$\{x = v_1, y = v_2\}$$

The above PDE (6E) now becomes

$$\begin{aligned} a_2v_1^2 - a_3v_1^2 + 4a_3v_1v_2 - 2a_3v_2^2 - b_2v_1^2 - b_3v_1^2 + 2a_1v_2 \\ + 2a_3v_1 - 3a_3v_2 - 2b_1v_1 + b_3v_1 + a_1 - a_3 = 0 \end{aligned} \quad (7E)$$

Collecting the above on the terms v_i introduced, and these are

$$\{v_1, v_2\}$$

Equation (7E) now becomes

$$(a_2 - a_3 - b_2 - b_3)v_1^2 + 4a_3v_1v_2 + (2a_3 - 2b_1 + b_3)v_1 - 2a_3v_2^2 + (2a_1 - 3a_3)v_2 + a_1 - a_3 = 0 \quad (8E)$$

Setting each coefficients in (8E) to zero gives the following equations to solve

$$\begin{aligned} -2a_3 &= 0 \\ 4a_3 &= 0 \\ a_1 - a_3 &= 0 \\ 2a_1 - 3a_3 &= 0 \\ 2a_3 - 2b_1 + b_3 &= 0 \\ a_2 - a_3 - b_2 - b_3 &= 0 \end{aligned}$$

Solving the above equations for the unknowns gives

$$\begin{aligned} a_1 &= 0 \\ a_2 &= b_2 + 2b_1 \\ a_3 &= 0 \\ b_1 &= b_1 \\ b_2 &= b_2 \\ b_3 &= 2b_1 \end{aligned}$$

Substituting the above solution in the anstaz (1E,2E) (using 1 as arbitrary value for any unknown in the RHS) gives

$$\xi = x$$

$$\eta = x$$

Shifting is now applied to make $\xi = 0$ in order to simplify the rest of the computation

$$\begin{aligned}\eta &= \eta - \omega(x, y) \xi \\ &= x - \left(\frac{1 - x + 2y}{x} \right) (x) \\ &= 2x - 2y - 1 \\ \xi &= 0\end{aligned}$$

The next step is to determine the canonical coordinates R, S . The canonical coordinates map $(x, y) \rightarrow (R, S)$ where (R, S) are the canonical coordinates which make the original ode become a quadrature and hence solved by integration.

The characteristic pde which is used to find the canonical coordinates is

$$\frac{dx}{\xi} = \frac{dy}{\eta} = dS \quad (1)$$

The above comes from the requirements that $\left(\xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right) S(x, y) = 1$. Starting with the first pair of ode's in (1) gives an ode to solve for the independent variable R in the canonical coordinates, where $S(R)$. Since $\xi = 0$ then in this special case

$$R = x$$

S is found from

$$\begin{aligned}S &= \int \frac{1}{\eta} dy \\ &= \int \frac{1}{2x - 2y - 1} dy\end{aligned}$$

Which results in

$$S = -\frac{\ln(2x - 2y - 1)}{2}$$

Now that R, S are found, we need to setup the ode in these coordinates. This is done by evaluating

$$\frac{dS}{dR} = \frac{S_x + \omega(x, y)S_y}{R_x + \omega(x, y)R_y} \quad (2)$$

Where in the above R_x, R_y, S_x, S_y are all partial derivatives and $\omega(x, y)$ is the right hand side of the original ode given by

$$\omega(x, y) = \frac{1 - x + 2y}{x}$$

Evaluating all the partial derivatives gives

$$\begin{aligned} R_x &= 1 \\ R_y &= 0 \\ S_x &= -\frac{1}{2x - 2y - 1} \\ S_y &= \frac{1}{2x - 2y - 1} \end{aligned}$$

Substituting all the above in (2) and simplifying gives the ode in canonical coordinates.

$$\frac{dS}{dR} = -\frac{1}{x} \quad (2A)$$

We now need to express the RHS as function of R only. This is done by solving for x, y in terms of R, S from the result obtained earlier and simplifying. This gives

$$\frac{dS}{dR} = -\frac{1}{R}$$

The above is a quadrature ode. This is the whole point of Lie symmetry method. It converts an ode, no matter how complicated it is, to one that can be solved by integration when the ode is in the canonical coordinates R, S .

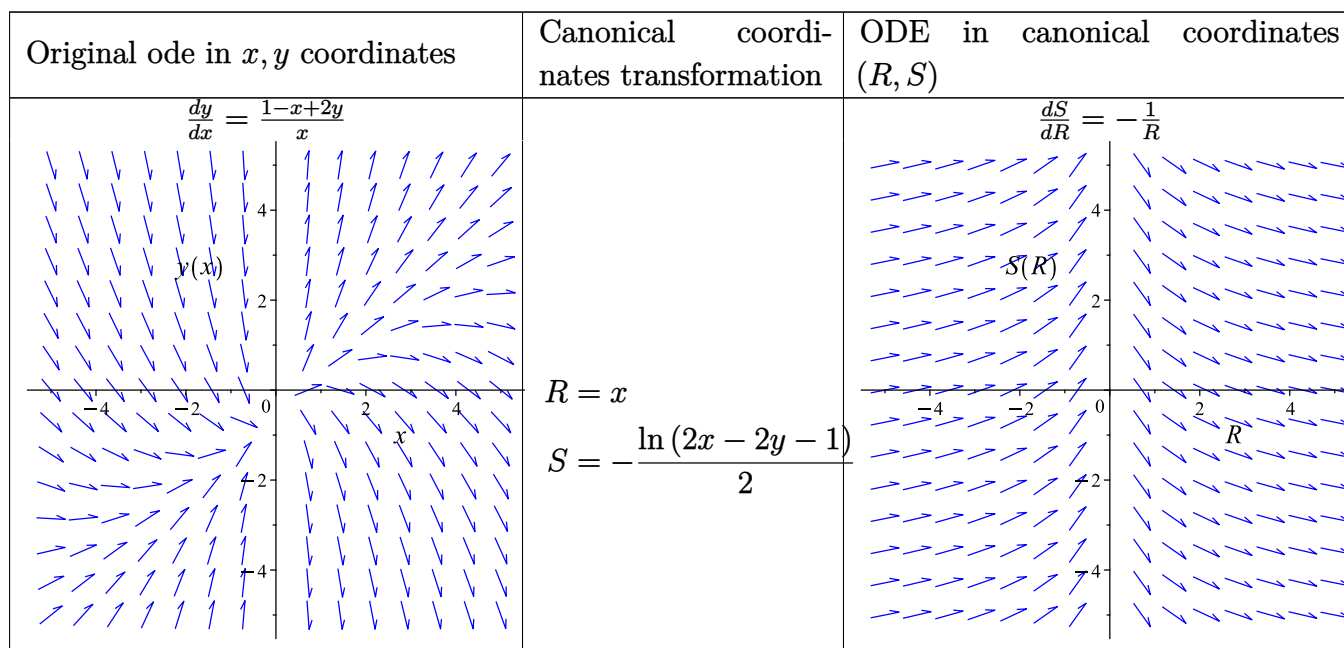
Since the ode has the form $\frac{d}{dR}S(R) = f(R)$, then we only need to integrate $f(R)$.

$$\begin{aligned} \int dS &= \int -\frac{1}{R} dR \\ S(R) &= -\ln(R) + c_2 \end{aligned}$$

To complete the solution, we just need to transform the above back to x, y coordinates. This results in

$$-\frac{\ln(2x - 2y - 1)}{2} = -\ln(x) + c_2$$

The following diagram shows solution curves of the original ode and how they transform in the canonical coordinates space using the mapping shown.



Solving for y gives

$$y = -\frac{e^{-2c_2}x^2}{2} + x - \frac{1}{2}$$

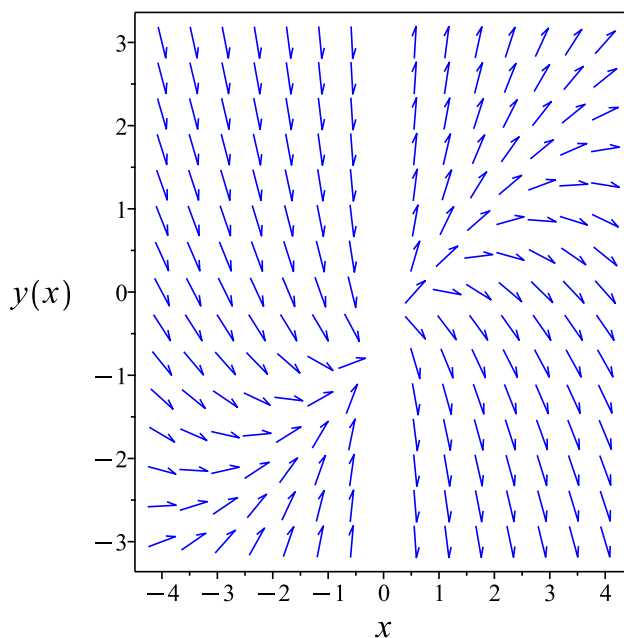


Figure 2.391: Slope field $y'x = 1 - x + 2y$

Summary of solutions found

$$y = -\frac{e^{-2c_2}x^2}{2} + x - \frac{1}{2}$$

✓ **Maple.** Time used: 0.001 (sec). Leaf size: 12

```
ode:=diff(y(x),x)*x = 1-x+2*y(x);
dsolve(ode,y(x), singsol=all);
```

$$y = x - \frac{1}{2} + c_1 x^2$$

Maple trace

Methods for first order ODEs:

--- Trying classification methods ---

trying a quadrature

trying 1st order linear

<- 1st order linear successful

Maple step by step

Let's solve

$$x\left(\frac{d}{dx}y(x)\right) = 1 - x + 2y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = \frac{1-x+2y(x)}{x}$$

- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx}y(x) = \frac{2y(x)}{x} - \frac{-1+x}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - \frac{2y(x)}{x} = -\frac{-1+x}{x}$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{2y(x)}{x} \right) = -\frac{\mu(x)(-1+x)}{x}$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{2y(x)}{x} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$

- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{2\mu(x)}{x}$$

- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x^2}$$

- Integrate both sides with respect to x

- $\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int -\frac{\mu(x)(-1+x)}{x} dx + C1$
- Evaluate the integral on the lhs
 $y(x)\mu(x) = \int -\frac{\mu(x)(-1+x)}{x} dx + C1$
- Solve for $y(x)$
 $y(x) = \frac{\int -\frac{\mu(x)(-1+x)}{x} dx + C1}{\mu(x)}$
- Substitute $\mu(x) = \frac{1}{x^2}$
 $y(x) = x^2 \left(\int -\frac{-1+x}{x^3} dx + C1 \right)$
- Evaluate the integrals on the rhs
 $y(x) = x^2 \left(-\frac{1}{2x^2} + \frac{1}{x} + C1 \right)$
- Simplify
 $y(x) = -\frac{1}{2} + x + C1 x^2$

✓ **Mathematica.** Time used: 0.02 (sec). Leaf size: 16

```
ode=x*D[y[x],x]==1-x+2*y[x];
ic={};
DSolve[{ode,ic},y[x],x,IncludeSingularSolutions->True]
```

$$y(x) \rightarrow c_1 x^2 + x - \frac{1}{2}$$

✓ **Sympy.** Time used: 0.149 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), x) + x - 2*y(x) - 1,0)
ics = {}
dsolve(ode,func=y(x),ics=ics)
```

$$y(x) = C_1 x^2 + x - \frac{1}{2}$$

2.16.4 Problem 11

Local contents

✓ Maple	2542
✓ Mathematica	2543
✓ Sympy	2544

Internal problem ID [8070]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 11

Date solved : Thursday, November 27, 2025 at 07:33:10 AM

CAS classification : `[_linear, 'class A']`

$$y' = 2x^2 + 3y$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving first order ode. Let

$$y' = f(x, y)$$

Where $f(x, y)$ is analytic at expansion point x_0 . We can always shift to $x_0 = 0$ if x_0 is not zero. So from now we assume $x_0 = 0$. Assume also that $y(x_0) = y_0$. Using Taylor series

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\
 &= y_0 + x f + \frac{x^2}{2} \frac{df}{dx} \Big|_{x_0, y_0} + \frac{x^3}{3!} \frac{d^2 f}{dx^2} \Big|_{x_0, y_0} + \cdots \\
 &= y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \quad (1)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y)$ then the above can be written as

$$F_0 = f(x, y) \quad (4)$$

$$\begin{aligned} F_n &= \frac{d}{dx}(F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) F_0 \end{aligned} \quad (5)$$

For example, for $n = 1$ we see that

$$\begin{aligned} F_1 &= \frac{d}{dx}(F_0) \\ &= \frac{\partial}{\partial x} F_0 + \left(\frac{\partial F_0}{\partial y} \right) F_0 \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \end{aligned}$$

Which is (1). And when $n = 2$

$$\begin{aligned} F_2 &= \frac{d}{dx}(F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) F_0 \\ &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) f \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned}$$

Which is (2) and so on. Therefore (4,5) can be used from now on along with

$$y(x) = y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} F_n|_{x_0, y_0} \quad (6)$$

Hence

$$\begin{aligned}
 F_0 &= 2x^2 + 3y \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} F_0 \\
 &= 4x + 6x^2 + 9y \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y} F_1 \\
 &= 4 + 12x + 18x^2 + 27y \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y} F_2 \\
 &= 12 + 36x + 54x^2 + 81y \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y} F_3 \\
 &= 36 + 108x + 162x^2 + 243y
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x(0) = 0$ and $y(0) = y(0)$ gives

$$\begin{aligned}
 F_0 &= 3y(0) \\
 F_1 &= 9y(0) \\
 F_2 &= 27y(0) + 4 \\
 F_3 &= 81y(0) + 12 \\
 F_4 &= 243y(0) + 36
 \end{aligned}$$

Substituting all the above in (6) and simplifying gives the solution as

$$y = \left(1 + 3x + \frac{9}{2}x^2 + \frac{9}{2}x^3 + \frac{27}{8}x^4 + \frac{81}{40}x^5 \right) y(0) + \frac{2x^3}{3} + \frac{x^4}{2} + \frac{3x^5}{10} + O(x^6)$$

Since $x = 0$ is also an ordinary point, then standard power series can also be used. Writing the ODE as

$$\begin{aligned}
 y' + q(x)y &= p(x) \\
 y' - 3y &= 2x^2
 \end{aligned}$$

Where

$$\begin{aligned} q(x) &= -3 \\ p(x) &= 2x^2 \end{aligned}$$

Next, the type of the expansion point $x = 0$ is determined. This point can be an ordinary point, a regular singular point (also called removable singularity), or irregular singular point (also called non-removable singularity or essential singularity). When $x = 0$ is an ordinary point, then the standard power series is used. If the point is a regular singular point, Frobenius series is used instead. Irregular singular point requires more advanced methods (asymptotic methods) and is not supported now. Hopefully this will be added in the future. $x = 0$ is called an ordinary point $q(x)$ has a Taylor series expansion around the point $x = 0$. $x = 0$ is called a regular singular point if $q(x)$ is not analytic at $x = 0$ but $xq(x)$ has Taylor series expansion. And finally, $x = 0$ is an irregular singular point if the point is not ordinary and not regular singular. This is the most complicated case. Now the expansion point $x = 0$ is checked to see if it is an ordinary point or not. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Substituting the above back into the ode gives

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = 2x^2 + 3 \left(\sum_{n=0}^{\infty} a_n x^n \right) \quad (1)$$

Expanding $2x^2$ as Taylor series around $x = 0$ and keeping only the first 6 terms gives

$$\begin{aligned} 2x^2 &= 2x^2 + \dots \\ &= 2x^2 \end{aligned}$$

Which simplifies to

$$\left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) + \sum_{n=0}^{\infty} (-3a_n x^n) = 2x^2 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n \right) + \sum_{n=0}^{\infty} (-3a_n x^n) = 2x^2 \quad (3)$$

For $0 \leq n$, the recurrence equation is

$$((n+1) a_{n+1} - 3a_n) x^n = 2x^2 \quad (4)$$

For $n = 0$ the recurrence equation gives

$$\begin{aligned} (a_1 - 3a_0) 1 &= 0 \\ a_1 - 3a_0 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_1 = 3a_0$$

For $n = 1$ the recurrence equation gives

$$\begin{aligned} (2a_2 - 3a_1) x &= 0 \\ 2a_2 - 3a_1 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_2 = \frac{9a_0}{2}$$

For $n = 2$ the recurrence equation gives

$$\begin{aligned} (3a_3 - 3a_2) x^2 &= 2x^2 \\ 3a_3 - 3a_2 &= 2 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_3 = \frac{2}{3} + \frac{9a_0}{2}$$

For $n = 3$ the recurrence equation gives

$$(4a_4 - 3a_3)x^3 = 0$$

$$4a_4 - 3a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = \frac{1}{2} + \frac{27a_0}{8}$$

For $n = 4$ the recurrence equation gives

$$(5a_5 - 3a_4)x^4 = 0$$

$$5a_5 - 3a_4 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = \frac{3}{10} + \frac{81a_0}{40}$$

For $n = 5$ the recurrence equation gives

$$(6a_6 - 3a_5)x^5 = 0$$

$$6a_6 - 3a_5 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = \frac{3}{20} + \frac{81a_0}{80}$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + 3a_0 x + \frac{9a_0 x^2}{2} + \left(\frac{2}{3} + \frac{9a_0}{2}\right)x^3 + \left(\frac{1}{2} + \frac{27a_0}{8}\right)x^4 + \left(\frac{3}{10} + \frac{81a_0}{40}\right)x^5 + \dots$$

Collecting terms, the solution becomes

$$y = \left(1 + 3x + \frac{9}{2}x^2 + \frac{9}{2}x^3 + \frac{27}{8}x^4 + \frac{81}{40}x^5\right) a_0 + \frac{2x^3}{3} + \frac{x^4}{2} + \frac{3x^5}{10} + O(x^6) \quad (3)$$

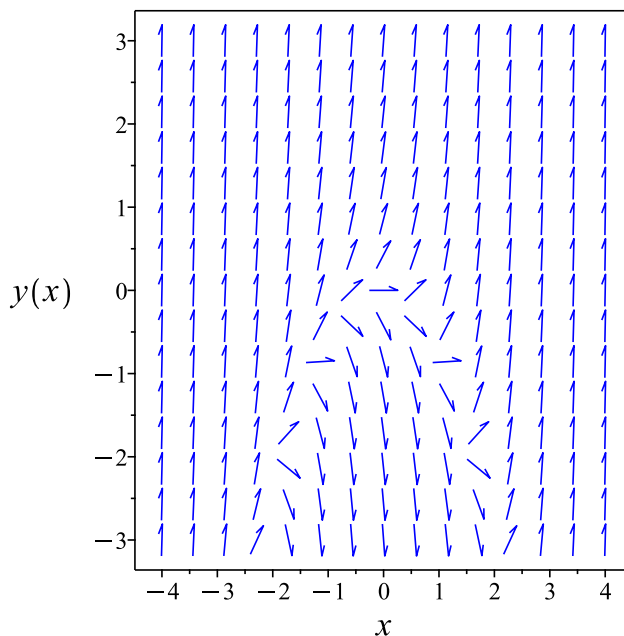


Figure 2.392: Slope field $y' = 2x^2 + 3y$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 52

```
Order:=6;
ode:=diff(y(x),x) = 2*x^2+3*y(x);
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 + 3x + \frac{9}{2}x^2 + \frac{9}{2}x^3 + \frac{27}{8}x^4 + \frac{81}{40}x^5\right) y(0) + \frac{2x^3}{3} + \frac{x^4}{2} + \frac{3x^5}{10} + O(x^6)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$\frac{d}{dx}y(x) = 2x^2 + 3y(x)$$

- Highest derivative means the order of the ODE is 1

$$\frac{d}{dx}y(x)$$

- Solve for the highest derivative

$$\frac{d}{dx}y(x) = 2x^2 + 3y(x)$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - 3y(x) = 2x^2$$

- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - 3y(x) \right) = 2\mu(x) x^2$$

- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - 3y(x) \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$

- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -3\mu(x)$$

- Solve to find the integrating factor

$$\mu(x) = e^{-3x}$$

- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int 2\mu(x) x^2 dx + C1$$

- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int 2\mu(x) x^2 dx + C1$$

- Solve for $y(x)$

$$y(x) = \frac{\int 2\mu(x) x^2 dx + C1}{\mu(x)}$$

- Substitute $\mu(x) = e^{-3x}$

$$y(x) = \frac{\int 2e^{-3x} x^2 dx + C1}{e^{-3x}}$$

- Evaluate the integrals on the rhs

$$y(x) = \frac{-\frac{2(9x^2+6x+2)e^{-3x}}{27} + C1}{e^{-3x}}$$

- Simplify

$$y(x) = -\frac{2x^2}{3} - \frac{4x}{9} - \frac{4}{27} + C1 e^{3x}$$

✓ **Mathematica.** Time used: 0.002 (sec). Leaf size: 61

```
ode=D[y[x],x]==2*x^2+3*y[x];
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow \frac{3x^5}{10} + \frac{x^4}{2} + \frac{2x^3}{3} + c_1 \left(\frac{81x^5}{40} + \frac{27x^4}{8} + \frac{9x^3}{2} + \frac{9x^2}{2} + 3x + 1 \right)$$

✓ Sympy. Time used: 0.203 (sec). Leaf size: 53

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-2*x**2 - 3*y(x) + Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="1st_power_series", x0=0, n=6)
```

$$y(x) = \frac{x^3(27C_1 + 4)}{6} + \frac{x^4(27C_1 + 4)}{8} + \frac{3x^5(27C_1 + 4)}{40} + C_1 + 3C_1x + \frac{9C_1x^2}{2} + O(x^6)$$

2.16.5 Problem 12

Local contents

✓ Maple	2551
✓ Mathematica	2552
✓ Sympy	2553

Internal problem ID [8071]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 12

Date solved : Thursday, November 27, 2025 at 07:33:10 AM

CAS classification : [_linear]

$$(1+x)y' = x^2 - 2x + y$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving first order ode. Let

$$y' = f(x, y)$$

Where $f(x, y)$ is analytic at expansion point x_0 . We can always shift to $x_0 = 0$ if x_0 is not zero. So from now we assume $x_0 = 0$. Assume also that $y(x_0) = y_0$. Using Taylor series

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0)y'(x_0) + \frac{(x - x_0)^2}{2}y''(x_0) + \frac{(x - x_0)^3}{3!}y'''(x_0) + \cdots \\
 &= y_0 + xf + \frac{x^2}{2} \frac{df}{dx} \Big|_{x_0, y_0} + \frac{x^3}{3!} \frac{d^2f}{dx^2} \Big|_{x_0, y_0} + \cdots \\
 &= y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \quad (1)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y)$ then the above can be written as

$$F_0 = f(x, y) \quad (4)$$

$$\begin{aligned} F_n &= \frac{d}{dx}(F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) F_0 \end{aligned} \quad (5)$$

For example, for $n = 1$ we see that

$$\begin{aligned} F_1 &= \frac{d}{dx}(F_0) \\ &= \frac{\partial}{\partial x} F_0 + \left(\frac{\partial F_0}{\partial y} \right) F_0 \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \end{aligned}$$

Which is (1). And when $n = 2$

$$\begin{aligned} F_2 &= \frac{d}{dx}(F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) F_0 \\ &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f \right) f \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) f \end{aligned}$$

Which is (2) and so on. Therefore (4,5) can be used from now on along with

$$y(x) = y_0 + \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!} F_n|_{x_0, y_0} \quad (6)$$

Hence

$$\begin{aligned}
 F_0 &= \frac{x^2 - 2x + y}{1 + x} \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} F_0 \\
 &= \frac{-2 + 2x}{1 + x} \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y} F_1 \\
 &= \frac{4}{(1 + x)^2} \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y} F_2 \\
 &= -\frac{8}{(1 + x)^3} \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y} F_3 \\
 &= \frac{24}{(1 + x)^4}
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x(0) = 0$ and $y(0) = y(0)$ gives

$$\begin{aligned}
 F_0 &= y(0) \\
 F_1 &= -2 \\
 F_2 &= 4 \\
 F_3 &= -8 \\
 F_4 &= 24
 \end{aligned}$$

Substituting all the above in (6) and simplifying gives the solution as

$$y = (1 + x)y(0) - x^2 + \frac{2x^3}{3} - \frac{x^4}{3} + \frac{x^5}{5} + O(x^6)$$

Since $x = 0$ is also an ordinary point, then standard power series can also be used. Writing

the ODE as

$$y' + q(x)y = p(x)$$

$$y' - \frac{y}{1+x} = \frac{x(x-2)}{1+x}$$

Where

$$q(x) = -\frac{1}{1+x}$$

$$p(x) = \frac{x(x-2)}{1+x}$$

Next, the type of the expansion point $x = 0$ is determined. This point can be an ordinary point, a regular singular point (also called removable singularity), or irregular singular point (also called non-removable singularity or essential singularity). When $x = 0$ is an ordinary point, then the standard power series is used. If the point is a regular singular point, Frobenius series is used instead. Irregular singular point requires more advanced methods (asymptotic methods) and is not supported now. Hopefully this will be added in the future. $x = 0$ is called an ordinary point $q(x)$ has a Taylor series expansion around the point $x = 0$. $x = 0$ is called a regular singular point if $q(x)$ is not analytic at $x = 0$ but $xq(x)$ has Taylor series expansion. And finally, $x = 0$ is an irregular singular point if the point is not ordinary and not regular singular. This is the most complicated case. Now the expansion point $x = 0$ is checked to see if it is an ordinary point or not. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Substituting the above back into the ode gives

$$(1+x) \left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) = x^2 - 2x + \left(\sum_{n=0}^{\infty} a_n x^n \right) \quad (1)$$

Expanding $x(x-2)$ as Taylor series around $x = 0$ and keeping only the first 6 terms gives

$$x(x-2) = x^2 - 2x + \dots$$

$$= x^2 - 2x$$

Which simplifies to

$$\left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) + \left(\sum_{n=1}^{\infty} n a_n x^n \right) + \sum_{n=0}^{\infty} (-a_n x^n) = x^2 - 2x \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n \right) + \left(\sum_{n=1}^{\infty} n a_n x^n \right) + \sum_{n=0}^{\infty} (-a_n x^n) = x^2 - 2x \quad (3)$$

For $1 \leq n$, the recurrence equation is

$$((n+1) a_{n+1} + n a_n - a_n) x^n = x^2 - 2x \quad (4)$$

For $n = 1$ the recurrence equation gives

$$\begin{aligned} (2a_2) x &= -2x \\ 2a_2 &= -2 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_2 = -1$$

For $n = 2$ the recurrence equation gives

$$\begin{aligned} (3a_3 + a_2) x^2 &= x^2 \\ 3a_3 + a_2 &= 1 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_3 = \frac{2}{3}$$

For $n = 3$ the recurrence equation gives

$$\begin{aligned} (4a_4 + 2a_3) x^3 &= 0 \\ 4a_4 + 2a_3 &= 0 \end{aligned}$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{1}{3}$$

For $n = 4$ the recurrence equation gives

$$(5a_5 + 3a_4)x^4 = 0$$

$$5a_5 + 3a_4 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = \frac{1}{5}$$

For $n = 5$ the recurrence equation gives

$$(6a_6 + 4a_5)x^5 = 0$$

$$6a_6 + 4a_5 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = -\frac{2}{15}$$

And so on. Therefore the solution is

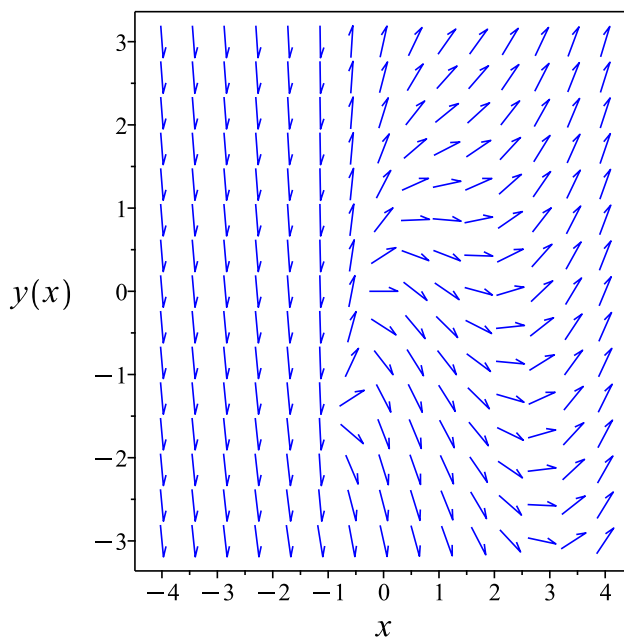
$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + a_0 x - x^2 + \frac{2}{3}x^3 - \frac{1}{3}x^4 + \frac{1}{5}x^5 + \dots$$

Collecting terms, the solution becomes

$$y = (1+x)a_0 - x^2 + \frac{2x^3}{3} - \frac{x^4}{3} + \frac{x^5}{5} + O(x^6) \quad (3)$$

Figure 2.393: Slope field $(1+x)y' = x^2 - 2x + y$

✓ **Maple.** Time used: 0.002 (sec). Leaf size: 35

```
Order:=6;
ode:=(1+x)*diff(y(x),x) = x^2-2*x+y(x);
dsolve(ode,y(x),type='series',x=0);
```

$$y = (x+1)y(0) - x^2 + \frac{2x^3}{3} - \frac{x^4}{3} + \frac{x^5}{5} + O(x^6)$$

Maple trace

```
Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
<- 1st order linear successful
```

Maple step by step

Let's solve

$$(x+1)\left(\frac{d}{dx}y(x)\right) = x^2 - 2x + y(x)$$

- Highest derivative means the order of the ODE is 1
- $\frac{d}{dx}y(x)$
- Solve for the highest derivative

- $\frac{d}{dx}y(x) = \frac{x^2-2x+y(x)}{x+1}$
- Collect w.r.t. $y(x)$ and simplify

$$\frac{d}{dx}y(x) = \frac{y(x)}{x+1} + \frac{x(x-2)}{x+1}$$
- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE

$$\frac{d}{dx}y(x) - \frac{y(x)}{x+1} = \frac{x(x-2)}{x+1}$$
- The ODE is linear; multiply by an integrating factor $\mu(x)$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x+1} \right) = \frac{\mu(x)x(x-2)}{x+1}$$
- Assume the lhs of the ODE is the total derivative $\frac{d}{dx}(y(x)\mu(x))$

$$\mu(x) \left(\frac{d}{dx}y(x) - \frac{y(x)}{x+1} \right) = \left(\frac{d}{dx}y(x) \right) \mu(x) + y(x) \left(\frac{d}{dx}\mu(x) \right)$$
- Isolate $\frac{d}{dx}\mu(x)$

$$\frac{d}{dx}\mu(x) = -\frac{\mu(x)}{x+1}$$
- Solve to find the integrating factor

$$\mu(x) = \frac{1}{x+1}$$
- Integrate both sides with respect to x

$$\int \left(\frac{d}{dx}(y(x)\mu(x)) \right) dx = \int \frac{\mu(x)x(x-2)}{x+1} dx + C1$$
- Evaluate the integral on the lhs

$$y(x)\mu(x) = \int \frac{\mu(x)x(x-2)}{x+1} dx + C1$$
- Solve for $y(x)$

$$y(x) = \frac{\int \frac{\mu(x)x(x-2)}{x+1} dx + C1}{\mu(x)}$$
- Substitute $\mu(x) = \frac{1}{x+1}$

$$y(x) = (x+1) \left(\int \frac{x(x-2)}{(x+1)^2} dx + C1 \right)$$
- Evaluate the integrals on the rhs

$$y(x) = (x+1) \left(x - \frac{3}{x+1} - 4 \ln(x+1) + C1 \right)$$
- Simplify

$$y(x) = (-4x - 4) \ln(x+1) + x^2 + (C1 + 1)x + C1 - 3$$

✓ **Mathematica.** Time used: 0.017 (sec). Leaf size: 36

```
ode=(x+1)*D[y[x],x]==x^2-2*x+y[x];
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow \frac{x^5}{5} - \frac{x^4}{3} + \frac{2x^3}{3} - x^2 + c_1(x+1)$$

✓ Sympy. Time used: 0.237 (sec). Leaf size: 31

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(-x**2 + 2*x + (x + 1)*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="1st_power_series", x0=0, n=6)
```

$$y(x) = -x^2 + \frac{2x^3}{3} - \frac{x^4}{3} + \frac{x^5}{5} + C_1 + C_1x + O(x^6)$$

2.16.6 Problem 13

Local contents

✓ Maple	2560
✓ Mathematica	2561
✓ Sympy	2562

Internal problem ID [8072]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 13

Date solved : Thursday, November 27, 2025 at 07:33:11 AM

CAS classification : [[_Emden, _Fowler]]

$$y'' + yx = 0$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\
 &= y_0 + xy'_0 + \frac{x^2}{2} f|_{x_0, y_0, y'_0} + \frac{x^3}{3!} f'|_{x_0, y_0, y'_0} + \cdots \\
 &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0, y'_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.123)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.124)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$\begin{aligned}
 F_0 &= -yx \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \\
 &= -y'x - y \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y} y' + \frac{\partial F_1}{\partial y'} F_1 \\
 &= yx^2 - 2y' \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y} y' + \frac{\partial F_2}{\partial y'} F_2 \\
 &= x(y'x + 4y) \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y} y' + \frac{\partial F_3}{\partial y'} F_3 \\
 &= -x^3y + 6y'x + 4y
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x = 0$ and $y(0) = y(0)$ and $y'(0) = y'(0)$ gives

$$\begin{aligned}
 F_0 &= 0 \\
 F_1 &= -y(0) \\
 F_2 &= -2y'(0) \\
 F_3 &= 0 \\
 F_4 &= 4y(0)
 \end{aligned}$$

Substituting all the above in (7) and simplifying gives the solution as

$$y = \left(1 - \frac{x^3}{6}\right) y(0) + \left(x - \frac{1}{12}x^4\right) y'(0) + O(x^6)$$

Since the expansion point $x = 0$ is an ordinary point, then this can also be solved using the standard power series method. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} a_n x^n \right) x = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} x^{1+n} a_n \right) = 0 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (1+n) x^n$$

$$\sum_{n=0}^{\infty} x^{1+n} a_n = \sum_{n=1}^{\infty} a_{n-1} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (1+n) x^n \right) + \left(\sum_{n=1}^{\infty} a_{n-1} x^n \right) = 0 \quad (3)$$

For $1 \leq n$, the recurrence equation is

$$(n+2) a_{n+2} (1+n) + a_{n-1} = 0 \quad (4)$$

Solving for a_{n+2} , gives

$$a_{n+2} = -\frac{a_{n-1}}{(n+2)(1+n)} \quad (5)$$

For $n = 1$ the recurrence equation gives

$$6a_3 + a_0 = 0$$

Which after substituting the earlier terms found becomes

$$a_3 = -\frac{a_0}{6}$$

For $n = 2$ the recurrence equation gives

$$12a_4 + a_1 = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{a_1}{12}$$

For $n = 3$ the recurrence equation gives

$$20a_5 + a_2 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = 0$$

For $n = 4$ the recurrence equation gives

$$30a_6 + a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = \frac{a_0}{180}$$

For $n = 5$ the recurrence equation gives

$$42a_7 + a_4 = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = \frac{a_1}{504}$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + a_1 x - \frac{1}{6} a_0 x^3 - \frac{1}{12} a_1 x^4 + \dots$$

Collecting terms, the solution becomes

$$y = \left(1 - \frac{x^3}{6}\right) a_0 + \left(x - \frac{1}{12} x^4\right) a_1 + O(x^6) \quad (3)$$

At $x = 0$ the solution above becomes

$$y = \left(1 - \frac{x^3}{6}\right) y(0) + \left(x - \frac{1}{12} x^4\right) y'(0) + O(x^6)$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 29

```
Order:=6;
ode:=diff(diff(y(x),x),x)+y(x)*x = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 - \frac{x^3}{6}\right) y(0) + \left(x - \frac{1}{12} x^4\right) y'(0) + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + xy(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

- Rewrite ODE with series expansions

- Convert $x \cdot y(x)$ to series expansion

$$x \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+1}$$

- Shift index using $k \rightarrow k-1$

$$x \cdot y(x) = \sum_{k=1}^{\infty} a_{k-1} x^k$$

- Convert $\frac{d^2}{dx^2}y(x)$ to series expansion

$$\frac{d^2}{dx^2}y(x) = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2}$$

- Shift index using $k \rightarrow k+2$

$$\frac{d^2}{dx^2}y(x) = \sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k$$

Rewrite ODE with series expansions

$$2a_2 + \left(\sum_{k=1}^{\infty} (a_{k+2}(k+2)(k+1) + a_{k-1}) x^k \right) = 0$$

- Each term must be 0

$$2a_2 = 0$$

- Each term in the series must be 0, giving the recursion relation

$$(k^2 + 3k + 2) a_{k+2} + a_{k-1} = 0$$

- Shift index using $k \rightarrow k+1$

$$((k+1)^2 + 3k + 5) a_{k+3} + a_k = 0$$

- Recursion relation that defines the series solution to the ODE

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+3} = -\frac{a_k}{k^2 + 5k + 6}, 2a_2 = 0 \right]$$

✓ **Mathematica.** Time used: 0.001 (sec). Leaf size: 28

```
ode=D[y[x],{x,2}]+x*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_2 \left(x - \frac{x^4}{12} \right) + c_1 \left(1 - \frac{x^3}{6} \right)$$

✓ Sympy. Time used: 0.161 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) + Derivative(y(x), (x, 2)),0)
ics = {}
dsolve(ode,func=y(x),ics=ics,hint="2nd_power_series_ordinary",x0=0,n=6)
```

$$y(x) = C_2 \left(1 - \frac{x^3}{6}\right) + C_1 x \left(1 - \frac{x^3}{12}\right) + O(x^6)$$

2.16.7 Problem 14

Local contents

✓ Maple	2569
✓ Mathematica	2570
✓ Sympy	2571

Internal problem ID [8073]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 14

Date solved : Thursday, November 27, 2025 at 07:33:11 AM

CAS classification : [[_Emden, _Fowler]]

$$y'' + 2yx^2 = 0$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\
 &= y_0 + xy'_0 + \frac{x^2}{2} f|_{x_0, y_0, y'_0} + \frac{x^3}{3!} f'|_{x_0, y_0, y'_0} + \cdots \\
 &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0, y'_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.125)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.126)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$\begin{aligned}
 F_0 &= -2yx^2 \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y}y' + \frac{\partial F_0}{\partial y'}F_0 \\
 &= -2x(y'x + 2y) \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y}y' + \frac{\partial F_1}{\partial y'}F_1 \\
 &= 4yx^4 - 8y'x - 4y \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y}y' + \frac{\partial F_2}{\partial y'}F_2 \\
 &= 4y'x^4 + 32x^3y - 12y' \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y}y' + \frac{\partial F_3}{\partial y'}F_3 \\
 &= 48y'x^3 - 8x^2y(x^4 - 15)
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x = 0$ and $y(0) = y(0)$ and $y'(0) = y'(0)$ gives

$$\begin{aligned}
 F_0 &= 0 \\
 F_1 &= 0 \\
 F_2 &= -4y(0) \\
 F_3 &= -12y'(0) \\
 F_4 &= 0
 \end{aligned}$$

Substituting all the above in (7) and simplifying gives the solution as

$$y = \left(1 - \frac{x^4}{6}\right) y(0) + \left(x - \frac{1}{10}x^5\right) y'(0) + O(x^6)$$

Since the expansion point $x = 0$ is an ordinary point, then this can also be solved using the standard power series method. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + 2 \left(\sum_{n=0}^{\infty} a_n x^n \right) x^2 = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} 2x^{n+2} a_n \right) = 0 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n$$

$$\sum_{n=0}^{\infty} 2x^{n+2} a_n = \sum_{n=2}^{\infty} 2a_{n-2} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n \right) + \left(\sum_{n=2}^{\infty} 2a_{n-2} x^n \right) = 0 \quad (3)$$

For $2 \leq n$, the recurrence equation is

$$(n+2) a_{n+2} (n+1) + 2a_{n-2} = 0 \quad (4)$$

Solving for a_{n+2} , gives

$$a_{n+2} = -\frac{2a_{n-2}}{(n+2)(n+1)} \quad (5)$$

For $n = 2$ the recurrence equation gives

$$12a_4 + 2a_0 = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{a_0}{6}$$

For $n = 3$ the recurrence equation gives

$$20a_5 + 2a_1 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = -\frac{a_1}{10}$$

For $n = 4$ the recurrence equation gives

$$30a_6 + 2a_2 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = 0$$

For $n = 5$ the recurrence equation gives

$$42a_7 + 2a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = 0$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + a_1 x - \frac{1}{6} a_0 x^4 - \frac{1}{10} a_1 x^5 + \dots$$

Collecting terms, the solution becomes

$$y = \left(1 - \frac{x^4}{6}\right) a_0 + \left(x - \frac{1}{10}x^5\right) a_1 + O(x^6) \quad (3)$$

At $x = 0$ the solution above becomes

$$y = \left(1 - \frac{x^4}{6}\right) y(0) + \left(x - \frac{1}{10}x^5\right) y'(0) + O(x^6)$$

✓ **Maple.** Time used: 0.004 (sec). Leaf size: 29

```
Order:=6;
ode:=diff(diff(y(x),x),x)+2*x^2*y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 - \frac{x^4}{6}\right) y(0) + \left(x - \frac{1}{10}x^5\right) y'(0) + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) + 2x^2y(x) = 0$$

- Highest derivative means the order of the ODE is 2
- $\frac{d^2}{dx^2}y(x)$
- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

□ Rewrite ODE with series expansions

- Convert $x^2 \cdot y(x)$ to series expansion

$$x^2 \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+2}$$

- Shift index using $k \rightarrow k-2$

$$x^2 \cdot y(x) = \sum_{k=2}^{\infty} a_{k-2} x^k$$

- Convert $\frac{d^2}{dx^2} y(x)$ to series expansion

$$\frac{d^2}{dx^2} y(x) = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2}$$

- Shift index using $k \rightarrow k+2$

$$\frac{d^2}{dx^2} y(x) = \sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k$$

Rewrite ODE with series expansions

$$6a_3x + 2a_2 + \left(\sum_{k=2}^{\infty} (a_{k+2}(k+2)(k+1) + 2a_{k-2}) x^k \right) = 0$$

- The coefficients of each power of x must be 0
 $[2a_2 = 0, 6a_3 = 0]$
- Solve for the dependent coefficient(s)
 $\{a_2 = 0, a_3 = 0\}$
- Each term in the series must be 0, giving the recursion relation
 $(k^2 + 3k + 2) a_{k+2} + 2a_{k-2} = 0$
- Shift index using $k \rightarrow k+2$
 $((k+2)^2 + 3k + 8) a_{k+4} + 2a_k = 0$
- Recursion relation that defines the series solution to the ODE

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+4} = -\frac{2a_k}{k^2 + 7k + 12}, a_2 = 0, a_3 = 0 \right]$$

✓ **Mathematica.** Time used: 0.001 (sec). Leaf size: 28

```
ode=D[y[x],{x,2}]+2*x^2*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_2 \left(x - \frac{x^5}{10} \right) + c_1 \left(1 - \frac{x^4}{6} \right)$$

✓ Sympy. Time used: 0.178 (sec). Leaf size: 24

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x**2*y(x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_ordinary", x0=0, n=6)
```

$$y(x) = C_2 \left(1 - \frac{x^4}{6}\right) + C_1 x \left(1 - \frac{x^4}{10}\right) + O(x^6)$$

2.16.8 Problem 15

Local contents

✓ Maple	2578
✓ Mathematica	2580
✓ Sympy	2580

Internal problem ID [8074]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 15

Date solved : Thursday, November 27, 2025 at 07:33:11 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$y'' - y'x + yx^2 = 0$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned}
 y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\
 &= y_0 + xy'_0 + \frac{x^2}{2} f|_{x_0, y_0, y'_0} + \frac{x^3}{3!} f'|_{x_0, y_0, y'_0} + \cdots \\
 &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \left. \frac{d^n f}{dx^n} \right|_{x_0, y_0, y'_0}
 \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.127)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.128)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$\begin{aligned}
 F_0 &= y'x - yx^2 \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \\
 &= y' + (-x^3 - 2x)y \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y} y' + \frac{\partial F_1}{\partial y'} F_1 \\
 &= -y'x^3 - 4yx^2 - y'x - 2y \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y} y' + \frac{\partial F_2}{\partial y'} F_2 \\
 &= (-x^4 - 8x^2 - 3)y' + yx(x^4 + x^2 - 8) \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y} y' + \frac{\partial F_3}{\partial y'} F_3 \\
 &= (-11x^3 - 27x)y' + y(x^6 + 13x^4 + 6x^2 - 8)
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x = 0$ and $y(0) = y(0)$ and $y'(0) = y'(0)$ gives

$$\begin{aligned}
 F_0 &= 0 \\
 F_1 &= y'(0) \\
 F_2 &= -2y(0) \\
 F_3 &= -3y'(0) \\
 F_4 &= -8y(0)
 \end{aligned}$$

Substituting all the above in (7) and simplifying gives the solution as

$$y = \left(1 - \frac{x^4}{12}\right) y(0) + \left(x + \frac{1}{6}x^3 - \frac{1}{40}x^5\right) y'(0) + O(x^6)$$

Since the expansion point $x = 0$ is an ordinary point, then this can also be solved using the standard power series method. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) - \left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) x + \left(\sum_{n=0}^{\infty} a_n x^n \right) x^2 = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \sum_{n=1}^{\infty} (-n x^n a_n) + \left(\sum_{n=0}^{\infty} x^{n+2} a_n \right) = 0 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n$$

$$\sum_{n=0}^{\infty} x^{n+2} a_n = \sum_{n=2}^{\infty} a_{n-2} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n \right) + \sum_{n=1}^{\infty} (-n x^n a_n) + \left(\sum_{n=2}^{\infty} a_{n-2} x^n \right) = 0 \quad (3)$$

$n = 1$ gives

$$6a_3 - a_1 = 0$$

Which after substituting earlier equations, simplifies to

$$a_3 = \frac{a_1}{6}$$

For $2 \leq n$, the recurrence equation is

$$(n+2)a_{n+2}(n+1) - na_n + a_{n-2} = 0 \quad (4)$$

Solving for a_{n+2} , gives

$$\begin{aligned} a_{n+2} &= \frac{na_n - a_{n-2}}{(n+2)(n+1)} \\ &= \frac{na_n}{(n+2)(n+1)} - \frac{a_{n-2}}{(n+2)(n+1)} \end{aligned} \quad (5)$$

For $n = 2$ the recurrence equation gives

$$12a_4 - 2a_2 + a_0 = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{a_0}{12}$$

For $n = 3$ the recurrence equation gives

$$20a_5 - 3a_3 + a_1 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = -\frac{a_1}{40}$$

For $n = 4$ the recurrence equation gives

$$30a_6 - 4a_4 + a_2 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = -\frac{a_0}{90}$$

For $n = 5$ the recurrence equation gives

$$42a_7 - 5a_5 + a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = -\frac{a_1}{144}$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + a_1 x + \frac{1}{6} a_1 x^3 - \frac{1}{12} a_0 x^4 - \frac{1}{40} a_1 x^5 + \dots$$

Collecting terms, the solution becomes

$$y = \left(1 - \frac{x^4}{12}\right) a_0 + \left(x + \frac{1}{6} x^3 - \frac{1}{40} x^5\right) a_1 + O(x^6) \quad (3)$$

At $x = 0$ the solution above becomes

$$y = \left(1 - \frac{x^4}{12}\right) y(0) + \left(x + \frac{1}{6} x^3 - \frac{1}{40} x^5\right) y'(0) + O(x^6)$$

✓ Maple. Time used: 0.003 (sec). Leaf size: 34

```
Order:=6;
ode:=diff(diff(y(x),x),x)-diff(y(x),x)*x+x^2*y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 - \frac{x^4}{12}\right) y(0) + \left(x + \frac{1}{6} x^3 - \frac{1}{40} x^5\right) y'(0) + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
```

```

-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exist
-> Trying a solution in terms of special functions:
  -> Bessel
  -> elliptic
  -> Legendre
  -> Kummer
      -> hyper3: Equivalence to 1F1 under a power @ Moebius
      <- hyper3 successful: received ODE is equivalent to the 1F1 ODE
      <- Kummer successful
<- special function solution successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dx^2}y(x) - x\left(\frac{d}{dx}y(x)\right) + x^2y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2}y(x)$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

- Rewrite ODE with series expansions

- Convert $x^2 \cdot y(x)$ to series expansion

$$x^2 \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+2}$$

- Shift index using $k \rightarrow k - 2$

$$x^2 \cdot y(x) = \sum_{k=2}^{\infty} a_{k-2} x^k$$

- Convert $x \cdot \left(\frac{d}{dx}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=0}^{\infty} a_k k x^k$$

- Convert $\frac{d^2}{dx^2}y(x)$ to series expansion

$$\frac{d^2}{dx^2}y(x) = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2}$$

- Shift index using $k \rightarrow k + 2$

$$\frac{d^2}{dx^2}y(x) = \sum_{k=0}^{\infty} a_{k+2}(k+2)(k+1) x^k$$

Rewrite ODE with series expansions

$$2a_2 + (6a_3 - a_1)x + \left(\sum_{k=2}^{\infty} (a_{k+2}(k+2)(k+1) - a_k k + a_{k-2}) x^k \right) = 0$$

- The coefficients of each power of x must be 0

$$[2a_2 = 0, 6a_3 - a_1 = 0]$$

- Solve for the dependent coefficient(s)

- $\{a_2 = 0, a_3 = \frac{a_1}{6}\}$
- Each term in the series must be 0, giving the recursion relation
 $(k^2 + 3k + 2) a_{k+2} - a_k k + a_{k-2} = 0$
- Shift index using $k \rightarrow k + 2$
 $((k + 2)^2 + 3k + 8) a_{k+4} - a_{k+2}(k + 2) + a_k = 0$
- Recursion relation that defines the series solution to the ODE

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+4} = \frac{k a_{k+2} - a_k + 2 a_{k-2}}{k^2 + 7k + 12}, a_2 = 0, a_3 = \frac{a_1}{6} \right]$$

✓ **Mathematica.** Time used: 0.001 (sec). Leaf size: 35

```
ode=D[y[x],{x,2}]-x*D[y[x],x]+x^2*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \left(1 - \frac{x^4}{12} \right) + c_2 \left(-\frac{x^5}{40} + \frac{x^3}{6} + x \right)$$

✓ **Sympy.** Time used: 0.226 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*y(x) - x*Derivative(y(x), x) + Derivative(y(x), (x, 2)), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_ordinary", x0=0, n=6)
```

$$y(x) = \frac{3x^5 r(3)}{20} + C_2 \left(-\frac{x^6}{90} - \frac{x^4}{12} + 1 \right) + C_1 x \left(1 - \frac{x^4}{20} \right) + O(x^6)$$

2.16.9 Problem 16

Local contents

✓ Maple	2588
✓ Mathematica	2590
✓ Sympy	2590

Internal problem ID [8075]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 16

Date solved : Thursday, November 27, 2025 at 07:33:12 AM

CAS classification : [_Gegenbauer]

$$(-x^2 + 1) y'' - 2y'x + p(p + 1) y = 0$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned} y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\ &= y_0 + xy'_0 + \frac{x^2}{2} f|_{x_0, y_0, y'_0} + \frac{x^3}{3!} f'|_{x_0, y_0, y'_0} + \cdots \\ &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0, y'_0} \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.130)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.131)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$F_0 = -\frac{-p^2y + 2y'x - py}{x^2 - 1}$$

$$\begin{aligned} F_1 &= \frac{dF_0}{dx} \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y}y' + \frac{\partial F_0}{\partial y'}F_0 \\ &= \frac{(p^2x^2 + px^2 - p^2 + 6x^2 - p + 2)y' - 4ypx(p+1)}{(x^2 - 1)^2} \end{aligned}$$

$$\begin{aligned} F_2 &= \frac{dF_1}{dx} \\ &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y}y' + \frac{\partial F_1}{\partial y'}F_1 \\ &= \frac{(1+x)(-1+x)(-8((p^2+p+3)x^2 - p^2 - p + 3)xy' + ((p^2+p+18)x^2 - p^2 - p + 6)yp(p+1))}{(x^2 - 1)^4} \end{aligned}$$

$$\begin{aligned} F_3 &= \frac{dF_2}{dx} \\ &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y}y' + \frac{\partial F_2}{\partial y'}F_2 \\ &= \frac{(((p^4 + 2p^3 + 59p^2 + 58p + 120)x^4 + (-2p^4 - 4p^3 - 46p^2 - 44p + 240)x^2 + p^4 + 2p^3 - 13p^2 - 14p + 20))y' + ((p^4 + 2p^3 - 13p^2 - 14p + 20)yp(p+1))}{(x^2 - 1)^5} \end{aligned}$$

$$\begin{aligned} F_4 &= \frac{dF_3}{dx} \\ &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y}y' + \frac{\partial F_3}{\partial y'}F_3 \\ &= \frac{(-18((p^4 + 2p^3 + \frac{77}{3}p^2 + \frac{74}{3}p + 40)x^4 - 2(p^2 + p - \frac{20}{3})(p^2 + p + 10)x^2 + p^4 + 2p^3 - 17p^2 - 18p + 40))y' + ((p^4 + 2p^3 - 17p^2 - 18p + 40)yp(p+1))}{(x^2 - 1)^6} \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x = 0$ and $y(0) = y(0)$ and $y'(0) = y'(0)$ gives

$$F_0 = -y(0)p(p+1)$$

$$F_1 = -y'(0)p^2 - y'(0)p + 2y'(0)$$

$$F_2 = y(0)p^4 + 2y(0)p^3 - 5y(0)p^2 - 6y(0)p$$

$$F_3 = y'(0)p^4 + 2y'(0)p^3 - 13y'(0)p^2 - 14y'(0)p + 24y'(0)$$

$$F_4 = -y(0)p^6 - 3y(0)p^5 + 23y(0)p^4 + 51y(0)p^3 - 94y(0)p^2 - 120y(0)p$$

Substituting all the above in (7) and simplifying gives the solution as

$$\begin{aligned} y &= \left(1 - \frac{1}{2}p^2x^2 - \frac{1}{2}p^2x^2 + \frac{1}{24}p^4x^4 + \frac{1}{12}p^3x^4 - \frac{5}{24}p^2x^4 - \frac{1}{4}p^2x^4\right)y(0) \\ &\quad + \left(x - \frac{1}{6}p^2x^3 - \frac{1}{6}p^2x^3 + \frac{1}{3}x^3 + \frac{1}{120}x^5p^4 + \frac{1}{60}x^5p^3 - \frac{13}{120}x^5p^2 - \frac{7}{60}x^5p + \frac{1}{5}x^5\right)y'(0) + O(x^6) \end{aligned}$$

Since the expansion point $x = 0$ is an ordinary point, then this can also be solved using the standard power series method. The ode is normalized to be

$$(-x^2 + 1)y'' - 2y'x + (p^2 + p)y = 0$$

Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting the above back into the ode gives

$$(-x^2 + 1) \left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) - 2 \left(\sum_{n=1}^{\infty} n a_n x^{n-1} \right) x + (p^2 + p) \left(\sum_{n=0}^{\infty} a_n x^n \right) = 0 \quad (1)$$

Which simplifies to

$$\sum_{n=2}^{\infty} (-x^n a_n n(n-1)) + \left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \sum_{n=1}^{\infty} (-2n a_n x^n) + \left(\sum_{n=0}^{\infty} a_n x^n (p^2 + p) \right) = 0 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\begin{aligned} & \sum_{n=2}^{\infty} (-x^n a_n n(n-1)) + \left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n \right) \\ & + \sum_{n=1}^{\infty} (-2n a_n x^n) + \left(\sum_{n=0}^{\infty} a_n x^n (p^2 + p) \right) = 0 \end{aligned} \quad (3)$$

$n = 0$ gives

$$2a_2 + a_0p(p+1) = 0$$

$$a_2 = -\frac{1}{2}a_0p^2 - \frac{1}{2}a_0p$$

$n = 1$ gives

$$6a_3 - 2a_1 + a_1p(p+1) = 0$$

Which after substituting earlier equations, simplifies to

$$a_3 = -\frac{1}{6}a_1p^2 - \frac{1}{6}a_1p + \frac{1}{3}a_1$$

For $2 \leq n$, the recurrence equation is

$$-na_n(n-1) + (n+2)a_{n+2}(n+1) - 2na_n + a_np(p+1) = 0 \quad (4)$$

Solving for a_{n+2} , gives

$$a_{n+2} = \frac{a_n(n^2 - p^2 + n - p)}{(n+2)(n+1)} \quad (5)$$

For $n = 2$ the recurrence equation gives

$$-6a_2 + 12a_4 + a_2p(p+1) = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{5}{24}a_0p^2 - \frac{1}{4}a_0p + \frac{1}{24}a_0p^4 + \frac{1}{12}a_0p^3$$

For $n = 3$ the recurrence equation gives

$$-12a_3 + 20a_5 + a_3p(p+1) = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = -\frac{13}{120}a_1p^2 - \frac{7}{60}a_1p + \frac{1}{5}a_1 + \frac{1}{120}a_1p^4 + \frac{1}{60}a_1p^3$$

For $n = 4$ the recurrence equation gives

$$-20a_4 + 30a_6 + a_4p(p+1) = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = -\frac{47}{360}a_0p^2 - \frac{1}{6}a_0p + \frac{23}{720}a_0p^4 + \frac{17}{240}a_0p^3 - \frac{1}{720}a_0p^6 - \frac{1}{240}a_0p^5$$

For $n = 5$ the recurrence equation gives

$$-30a_5 + 42a_7 + a_5p(p+1) = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = -\frac{5}{63}a_1p^2 - \frac{37}{420}a_1p + \frac{1}{7}a_1 + \frac{41}{5040}a_1p^4 + \frac{29}{1680}a_1p^3 - \frac{1}{5040}a_1p^6 - \frac{1}{1680}a_1p^5$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$\begin{aligned} y &= a_0 + a_1 x + \left(-\frac{1}{2}a_0p^2 - \frac{1}{2}a_0p\right)x^2 + \left(-\frac{1}{6}a_1p^2 - \frac{1}{6}a_1p + \frac{1}{3}a_1\right)x^3 \\ &\quad + \left(-\frac{5}{24}a_0p^2 - \frac{1}{4}a_0p + \frac{1}{24}a_0p^4 + \frac{1}{12}a_0p^3\right)x^4 \\ &\quad + \left(-\frac{13}{120}a_1p^2 - \frac{7}{60}a_1p + \frac{1}{5}a_1 + \frac{1}{120}a_1p^4 + \frac{1}{60}a_1p^3\right)x^5 + \dots \end{aligned}$$

Collecting terms, the solution becomes

$$\begin{aligned} y &= \left(1 + \left(-\frac{1}{2}p^2 - \frac{1}{2}p\right)x^2 + \left(-\frac{5}{24}p^2 - \frac{1}{4}p + \frac{1}{24}p^4 + \frac{1}{12}p^3\right)x^4\right)a_0 \\ &\quad + \left(x + \left(-\frac{1}{6}p^2 - \frac{1}{6}p + \frac{1}{3}\right)x^3 + \left(-\frac{13}{120}p^2 - \frac{7}{60}p + \frac{1}{5} + \frac{1}{120}p^4 + \frac{1}{60}p^3\right)x^5\right)a_1 + O(x^6) \end{aligned} \quad (3)$$

At $x = 0$ the solution above becomes

$$\begin{aligned} y &= \left(1 + \left(-\frac{1}{2}p^2 - \frac{1}{2}p\right)x^2 + \left(-\frac{5}{24}p^2 - \frac{1}{4}p + \frac{1}{24}p^4 + \frac{1}{12}p^3\right)x^4\right)y(0) \\ &\quad + \left(x + \left(-\frac{1}{6}p^2 - \frac{1}{6}p + \frac{1}{3}\right)x^3 + \left(-\frac{13}{120}p^2 - \frac{7}{60}p + \frac{1}{5} + \frac{1}{120}p^4 + \frac{1}{60}p^3\right)x^5\right)y'(0) + O(x^6) \end{aligned}$$

✓ **Maple.** Time used: 0.006 (sec). Leaf size: 81

```
Order:=6;
ode:=p*(1+p)*y(x)-2*diff(y(x),x)*x+(-x^2+1)*diff(diff(y(x),x),x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 - \frac{p(p+1)x^2}{2} + \frac{p(p^3 + 2p^2 - 5p - 6)x^4}{24}\right) y(0) + \left(x - \frac{(p^2 + p - 2)x^3}{6} + \frac{(p^4 + 2p^3 - 13p^2 - 14p + 24)x^5}{120}\right) y'(0) + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    -> elliptic
    -> Legendre
    <- Legendre successful
<- special function solution successful
```

Maple step by step

Let's solve

$$(-x^2 + 1) \left(\frac{d^2}{dx^2} y(x) \right) - 2x \left(\frac{d}{dx} y(x) \right) + p(p+1) y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = \frac{p(p+1)y(x)}{x^2-1} - \frac{2x \left(\frac{d}{dx} y(x) \right)}{x^2-1}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + \frac{2x \left(\frac{d}{dx} y(x) \right)}{x^2-1} - \frac{p(p+1)y(x)}{x^2-1} = 0$$

- Check to see if x_0 is a regular singular point

- Define functions

$$\left[P_2(x) = \frac{2x}{x^2-1}, P_3(x) = -\frac{p(p+1)}{x^2-1} \right]$$
- $(x+1) \cdot P_2(x)$ is analytic at $x = -1$

$$\left. ((x+1) \cdot P_2(x)) \right|_{x=-1} = 1$$
- $(x+1)^2 \cdot P_3(x)$ is analytic at $x = -1$

$$\left. ((x+1)^2 \cdot P_3(x)) \right|_{x=-1} = 0$$
- $x = -1$ is a regular singular point
Check to see if x_0 is a regular singular point
 $x_0 = -1$
- Multiply by denominators

$$(x^2 - 1) \left(\frac{d^2}{dx^2} y(x) \right) + 2x \left(\frac{d}{dx} y(x) \right) - p(p+1) y(x) = 0$$
- Change variables using $x = u - 1$ so that the regular singular point is at $u = 0$

$$(u^2 - 2u) \left(\frac{d^2}{du^2} y(u) \right) + (2u - 2) \left(\frac{d}{du} y(u) \right) + (-p^2 - p) y(u) = 0$$
- Assume series solution for $y(u)$

$$y(u) = \sum_{k=0}^{\infty} a_k u^{k+r}$$
- Rewrite ODE with series expansions
 - Convert $u^m \cdot \left(\frac{d}{du} y(u) \right)$ to series expansion for $m = 0..1$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) u^{k+r-1+m}$$
 - Shift index using $k- > k+1-m$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) u^{k+r}$$
 - Convert $u^m \cdot \left(\frac{d^2}{du^2} y(u) \right)$ to series expansion for $m = 1..2$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) u^{k+r-2+m}$$
 - Shift index using $k- > k+2-m$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=-2+m}^{\infty} a_{k+2-m} (k+2-m+r) (k+1-m+r) u^{k+r}$$
- Rewrite ODE with series expansions

$$-2a_0 r^2 u^{-1+r} + \left(\sum_{k=0}^{\infty} (-2a_{k+1} (k+1+r)^2 + a_k (r+1+p+k) (r-p+k)) u^{k+r} \right) = 0$$
- a_0 cannot be 0 by assumption, giving the indicial equation

$$-2r^2 = 0$$
- Values of r that satisfy the indicial equation

$$r = 0$$
- Each term in the series must be 0, giving the recursion relation

$$-2a_{k+1} (k+1)^2 + a_k (1+p+k) (-p+k) = 0$$
- Recursion relation that defines series solution to ODE

- $$a_{k+1} = \frac{a_k(1+p+k)(-p+k)}{2(k+1)^2}$$
- Recursion relation for $r = 0$

$$a_{k+1} = \frac{a_k(1+p+k)(-p+k)}{2(k+1)^2}$$
- Solution for $r = 0$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^k, a_{k+1} = \frac{a_k(1+p+k)(-p+k)}{2(k+1)^2} \right]$$
- Revert the change of variables $u = x + 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x+1)^k, a_{k+1} = \frac{a_k(1+p+k)(-p+k)}{2(k+1)^2} \right]$$

✓ **Mathematica.** Time used: 0.002 (sec). Leaf size: 120

```
ode=(1-x^2)*D[y[x],{x,2}]-2*x*D[y[x],x]+p*(p+1)*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_2 \left(\frac{1}{120} (p^2 + p)^2 x^5 + \frac{7}{60} (-p^2 - p) x^5 + \frac{1}{6} (-p^2 - p) x^3 + \frac{x^5}{5} + \frac{x^3}{3} + x \right) \\ + c_1 \left(\frac{1}{24} (p^2 + p)^2 x^4 + \frac{1}{4} (-p^2 - p) x^4 + \frac{1}{2} (-p^2 - p) x^2 + 1 \right)$$

✓ **Sympy.** Time used: 0.447 (sec). Leaf size: 83

```
from sympy import *
x = symbols("x")
p = symbols("p")
y = Function("y")
ode = Eq(p*(p + 1)*y(x) - 2*x*Derivative(y(x), x) + (1 - x**2)*Derivative(y(x), (
    x, 2)), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics,hint="2nd_power_series_ordinary",x0=0,n=6)
```

$$y(x) = C_2 \left(\frac{p^4 x^4}{24} + \frac{p^3 x^4}{12} - \frac{5p^2 x^4}{24} - \frac{p^2 x^2}{2} - \frac{px^4}{4} - \frac{px^2}{2} + 1 \right) \\ + C_1 x \left(-\frac{p^2 x^2}{6} - \frac{px^2}{6} + \frac{x^2}{3} + 1 \right) + O(x^6)$$

2.16.10 Problem 17

Local contents

✓ Maple	2598
✓ Mathematica	2598
✗ Sympy	2599

Internal problem ID [8076]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 25. Integration in series. Supplementary problems. Page 205

Problem number : 17

Date solved : Thursday, November 27, 2025 at 07:33:13 AM

CAS classification : `[[_2nd_order, _linear, _nonhomogeneous]]`

$$y'' + yx^2 = x^2 + x + 1$$

Series expansion around $x = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned} y(x) &= y(x_0) + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \cdots \\ &= y_0 + xy'_0 + \frac{x^2}{2} f|_{x_0, y_0, y'_0} + \frac{x^3}{3!} f'|_{x_0, y_0, y'_0} + \cdots \\ &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0, y'_0} \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.132)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.133)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$\begin{aligned}
 F_0 &= -yx^2 + x^2 + x + 1 \\
 F_1 &= \frac{dF_0}{dx} \\
 &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y}y' + \frac{\partial F_0}{\partial y'}F_0 \\
 &= -y'x^2 - 2yx + 2x + 1 \\
 F_2 &= \frac{dF_1}{dx} \\
 &= \frac{\partial F_1}{\partial x} + \frac{\partial F_1}{\partial y}y' + \frac{\partial F_1}{\partial y'}F_1 \\
 &= yx^4 - x^4 - x^3 - 4y'x - x^2 - 2y + 2 \\
 F_3 &= \frac{dF_2}{dx} \\
 &= \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial y}y' + \frac{\partial F_2}{\partial y'}F_2 \\
 &= (x^4 - 6)y' + 8x^3y - 8x^3 - 7x^2 - 6x \\
 F_4 &= \frac{dF_3}{dx} \\
 &= \frac{\partial F_3}{\partial x} + \frac{\partial F_3}{\partial y}y' + \frac{\partial F_3}{\partial y'}F_3 \\
 &= -yx^6 + x^6 + x^5 + 12y'x^3 + x^4 + 30yx^2 - 30x^2 - 20x - 12
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $x = 0$ and $y(0) = y(0)$ and $y'(0) = y'(0)$ gives

$$\begin{aligned}
 F_0 &= 1 \\
 F_1 &= 1 \\
 F_2 &= -2y(0) + 2 \\
 F_3 &= -6y'(0) \\
 F_4 &= -12
 \end{aligned}$$

Substituting all the above in (7) and simplifying gives the solution as

$$y = \left(1 - \frac{x^4}{12}\right)y(0) + \left(x - \frac{1}{20}x^5\right)y'(0) + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + O(x^6)$$

Since the expansion point $x = 0$ is an ordinary point, then this can also be solved using the standard power series method. Let the solution be represented as power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} a_n x^n \right) x^2 = x^2 + x + 1 \quad (1)$$

Expanding $x^2 + x + 1$ as Taylor series around $x = 0$ and keeping only the first 6 terms gives

$$x^2 + x + 1 = x^2 + x + 1 + \dots$$

$$= x^2 + x + 1$$

Hence the ODE in Eq (1) becomes

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} a_n x^n \right) x^2 = x^2 + x + 1$$

Which simplifies to

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \right) + \left(\sum_{n=0}^{\infty} x^{n+2} a_n \right) = x^2 + x + 1 \quad (2)$$

The next step is to make all powers of x be n in each summation term. Going over each summation term above with power of x in it which is not already x^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n$$

$$\sum_{n=0}^{\infty} x^{n+2} a_n = \sum_{n=2}^{\infty} a_{n-2} x^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of x are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) x^n \right) + \left(\sum_{n=2}^{\infty} a_{n-2} x^n \right) = x^2 + x + 1 \quad (3)$$

$n = 0$ gives

$$(2a_2) 1 = 1$$

$$2a_2 = 1$$

Or

$$a_2 = \frac{1}{2}$$

$n = 1$ gives

$$(6a_3) x = x$$

$$6a_3 = 1$$

Which after substituting earlier equations, simplifies to

$$6a_3 = 1$$

Or

$$a_3 = \frac{1}{6}$$

For $2 \leq n$, the recurrence equation is

$$((n+2)a_{n+2}(n+1) + a_{n-2})x^n = x^2 + x + 1 \quad (4)$$

For $n = 2$ the recurrence equation gives

$$(12a_4 + a_0)x^2 = x^2$$

$$12a_4 + a_0 = 1$$

Which after substituting the earlier terms found becomes

$$a_4 = \frac{1}{12} - \frac{a_0}{12}$$

For $n = 3$ the recurrence equation gives

$$(20a_5 + a_1)x^3 = 0$$

$$20a_5 + a_1 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = -\frac{a_1}{20}$$

For $n = 4$ the recurrence equation gives

$$(30a_6 + a_2)x^4 = 0$$

$$30a_6 + a_2 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = -\frac{1}{60}$$

For $n = 5$ the recurrence equation gives

$$(42a_7 + a_3)x^5 = 0$$

$$42a_7 + a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = -\frac{1}{252}$$

And so on. Therefore the solution is

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_3 x^3 + a_2 x^2 + a_1 x + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$y = a_0 + a_1 x + \frac{x^2}{2} + \frac{x^3}{6} + \left(\frac{1}{12} - \frac{a_0}{12}\right)x^4 - \frac{a_1 x^5}{20} + \dots$$

Collecting terms, the solution becomes

$$y = \left(1 - \frac{x^4}{12}\right)a_0 + \left(x - \frac{1}{20}x^5\right)a_1 + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + O(x^6) \quad (3)$$

At $x = 0$ the solution above becomes

$$y = \left(1 - \frac{x^4}{12}\right)y(0) + \left(x - \frac{1}{20}x^5\right)y'(0) + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + O(x^6)$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 44

```
Order:=6;
ode:=diff(diff(y(x),x),x)+x^2*y(x) = x^2+x+1;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left(1 - \frac{x^4}{12}\right) y(0) + \left(x - \frac{1}{20}x^5\right) y'(0) + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
    <- special function solution successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.01 (sec). Leaf size: 49

```
ode=D[y[x],{x,2}]+x^2*y[x]==1+x+x^2;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_2 \left(x - \frac{x^5}{20}\right) + \frac{x^4}{12} + c_1 \left(1 - \frac{x^4}{12}\right) + \frac{x^3}{6} + \frac{x^2}{2}$$

SymPy

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*y(x) - x**2 - x + Derivative(y(x), (x, 2)) - 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

```
ValueError : ODE x**2*y(x) - x**2 - x + Derivative(y(x), (x, 2)) - 1 does not
match hint 2nd_power_series_regular
```

2.17 Chapter 26. Integration in series (singular points). Supplemetary problems. Page 218

Local contents

2.17.1 Problem 11	2601
2.17.2 Problem 12	2613
2.17.3 Problem 13	2625
2.17.4 Problem 14	2636
2.17.5 Problem 15	2645
2.17.6 Problem 16	2655
2.17.7 Problem 17	2668
2.17.8 Problem 18	2679
2.17.9 Problem 19	2691
2.17.10 Problem 20	2704
2.17.11 Problem 21	2714

2.17.1 Problem 11

Local contents

✓ Maple	2609
✓ Mathematica	2611
✓ Sympy	2612

Internal problem ID [8077]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 11

Date solved : Thursday, November 27, 2025 at 07:33:13 AM

CAS classification : `[_2nd_order, _with_linear_symmetries]`

$$2(x^3 + x^2) y'' - (-3x^2 + x) y' + y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$(2x^3 + 2x^2) y'' + (3x^2 - x) y' + y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = \frac{3x - 1}{2x(1+x)}$$

$$q(x) = \frac{1}{2x^2(1+x)}$$

Table 2.307: Table $p(x), q(x)$ singularities.

$p(x) = \frac{3x-1}{2x(1+x)}$	
singularity	type
$x = -1$	“regular”
$x = 0$	“regular”

$q(x) = \frac{1}{2x^2(1+x)}$	
singularity	type
$x = -1$	“regular”
$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[-1, 0, \infty]$

Irregular singular points : $[\]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$2x^2(1+x)y'' + (3x^2 - x)y' + y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\begin{aligned} & 2x^2(1+x) \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) \\ & + (3x^2 - x) \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (1)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 2x^{1+n+r} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} 2x^{n+r} a_n (n+r)(n+r-1) \right) \\ & + \left(\sum_{n=0}^{\infty} 3x^{1+n+r} a_n (n+r) \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n (n+r)) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n+r$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r} and adjusting the

power and the corresponding index gives

$$\sum_{n=0}^{\infty} 2x^{1+n+r} a_n(n+r)(n+r-1) = \sum_{n=1}^{\infty} 2a_{n-1}(n+r-1)(n+r-2)x^{n+r}$$

$$\sum_{n=0}^{\infty} 3x^{1+n+r} a_n(n+r) = \sum_{n=1}^{\infty} 3a_{n-1}(n+r-1)x^{n+r}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n+r$.

$$\left(\sum_{n=1}^{\infty} 2a_{n-1}(n+r-1)(n+r-2)x^{n+r} \right) + \left(\sum_{n=0}^{\infty} 2x^{n+r} a_n(n+r)(n+r-1) \right) \quad (2B)$$

$$+ \left(\sum_{n=1}^{\infty} 3a_{n-1}(n+r-1)x^{n+r} \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n(n+r)) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0$$

The indicial equation is obtained from $n=0$. From Eq (2B) this gives

$$2x^{n+r} a_n(n+r)(n+r-1) - x^{n+r} a_n(n+r) + a_n x^{n+r} = 0$$

When $n=0$ the above becomes

$$2x^r a_0 r(-1+r) - x^r a_0 r + a_0 x^r = 0$$

Or

$$(2x^r r(-1+r) - x^r r + x^r) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(2r^2 - 3r + 1) x^r = 0$$

Since the above is true for all x then the indicial equation becomes

$$2r^2 - 3r + 1 = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 1$$

$$r_2 = \frac{1}{2}$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(2r^2 - 3r + 1) x^r = 0$$

Solving for r gives the roots of the indicial equation as $[1, \frac{1}{2}]$.

Since $r_1 - r_2 = \frac{1}{2}$ is not an integer, then we can construct two linearly independent solutions

$$y_1(x) = x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{1+n}$$

$$y_2(x) = \sum_{n=0}^{\infty} b_n x^{n+\frac{1}{2}}$$

We start by finding $y_1(x)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$2a_{n-1}(n+r-1)(n+r-2) + 2a_n(n+r)(n+r-1) + 3a_{n-1}(n+r-1) - a_n(n+r) + a_n = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -a_{n-1} \quad (4)$$

Which for the root $r = 1$ becomes

$$a_n = -a_{n-1} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 1$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = -1$$

Which for the root $r = 1$ becomes

$$a_1 = -1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	-1	-1

For $n = 2$, using the above recursive equation gives

$$a_2 = 1$$

Which for the root $r = 1$ becomes

$$a_2 = 1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	-1	-1
a_2	1	1

For $n = 3$, using the above recursive equation gives

$$a_3 = -1$$

Which for the root $r = 1$ becomes

$$a_3 = -1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	-1	-1
a_2	1	1
a_3	-1	-1

For $n = 4$, using the above recursive equation gives

$$a_4 = 1$$

Which for the root $r = 1$ becomes

$$a_4 = 1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	-1	-1
a_2	1	1
a_3	-1	-1
a_4	1	1

For $n = 5$, using the above recursive equation gives

$$a_5 = -1$$

Which for the root $r = 1$ becomes

$$a_5 = -1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	-1	-1
a_2	1	1
a_3	-1	-1
a_4	1	1
a_5	-1	-1

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= x(1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6)) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Eq (2B) derived above is now used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. For $1 \leq n$ the recursive equation is

$$2b_{n-1}(n+r-1)(n+r-2) + 2b_n(n+r)(n+r-1) + 3b_{n-1}(n+r-1) - b_n(n+r) + b_n = 0 \quad (3)$$

Solving for b_n from recursive equation (4) gives

$$b_n = -b_{n-1} \quad (4)$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_n = -b_{n-1} \quad (5)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = \frac{1}{2}$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1

For $n = 1$, using the above recursive equation gives

$$b_1 = -1$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_1 = -1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	-1	-1

For $n = 2$, using the above recursive equation gives

$$b_2 = 1$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_2 = 1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	-1	-1
b_2	1	1

For $n = 3$, using the above recursive equation gives

$$b_3 = -1$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_3 = -1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	-1	-1
b_2	1	1
b_3	-1	-1

For $n = 4$, using the above recursive equation gives

$$b_4 = 1$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_4 = 1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	-1	-1
b_2	1	1
b_3	-1	-1
b_4	1	1

For $n = 5$, using the above recursive equation gives

$$b_5 = -1$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_5 = -1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	-1	-1
b_2	1	1
b_3	-1	-1
b_4	1	1
b_5	-1	-1

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= x(b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots) \\ &= \sqrt{x} (1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6)) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1y_1(x) + c_2y_2(x) \\ &= c_1x(1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6)) + c_2\sqrt{x} (1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6)) \end{aligned}$$

Hence the final solution is

$$y = y_h \\ = c_1 x (1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6)) + c_2 \sqrt{x} (1 - x + x^2 - x^3 + x^4 - x^5 + O(x^6))$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 40

```
Order:=6;
ode:=2*(x^3+x^2)*diff(diff(y(x),x),x)-(-3*x^2+x)*diff(y(x),x)+y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = (-x^5 + x^4 - x^3 + x^2 - x + 1) (c_1 \sqrt{x} + c_2 x) + O(x^6)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Reducible group (found another exponential solution)
<- Kovacic's algorithm successful
```

Maple step by step

Let's solve

$$2(x^3 + x^2) \left(\frac{d^2}{dx^2} y(x) \right) - (-3x^2 + x) \left(\frac{d}{dx} y(x) \right) + y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{y(x)}{2x^2(x+1)} - \frac{(3x-1) \left(\frac{d}{dx} y(x) \right)}{2x(x+1)}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + \frac{(3x-1) \left(\frac{d}{dx} y(x) \right)}{2x(x+1)} + \frac{y(x)}{2x^2(x+1)} = 0$$

- Check to see if x_0 is a regular singular point

- o Define functions

- $\left[P_2(x) = \frac{3x-1}{2x(x+1)}, P_3(x) = \frac{1}{2x^2(x+1)} \right]$
 - $(x+1) \cdot P_2(x)$ is analytic at $x = -1$

$$\left. ((x+1) \cdot P_2(x)) \right|_{x=-1} = 2$$
 - $(x+1)^2 \cdot P_3(x)$ is analytic at $x = -1$

$$\left. ((x+1)^2 \cdot P_3(x)) \right|_{x=-1} = 0$$
 - $x = -1$ is a regular singular point
Check to see if x_0 is a regular singular point
 $x_0 = -1$
- Multiply by denominators
$$2x^2(x+1) \left(\frac{d^2}{dx^2} y(x) \right) + x(3x-1) \left(\frac{d}{dx} y(x) \right) + y(x) = 0$$
- Change variables using $x = u - 1$ so that the regular singular point is at $u = 0$

$$(2u^3 - 4u^2 + 2u) \left(\frac{d^2}{du^2} y(u) \right) + (3u^2 - 7u + 4) \left(\frac{d}{du} y(u) \right) + y(u) = 0$$
- Assume series solution for $y(u)$

$$y(u) = \sum_{k=0}^{\infty} a_k u^{k+r}$$

□ Rewrite ODE with series expansions

- Convert $u^m \cdot \left(\frac{d}{du} y(u) \right)$ to series expansion for $m = 0..2$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) u^{k+r-1+m}$$

- Shift index using $k- > k+1-m$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) u^{k+r}$$

- Convert $u^m \cdot \left(\frac{d^2}{du^2} y(u) \right)$ to series expansion for $m = 1..3$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) u^{k+r-2+m}$$

- Shift index using $k- > k+2-m$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=-2+m}^{\infty} a_{k+2-m} (k+2-m+r) (k+1-m+r) u^{k+r}$$

Rewrite ODE with series expansions

$$2a_0 r(1+r) u^{-1+r} + (2a_1(1+r)(2+r) - a_0(1+r)(-1+4r)) u^r + \left(\sum_{k=1}^{\infty} (2a_{k+1}(k+r+1)(k+2-m+r) - a_k(k+r)(k+r-1)) u^{k+r} \right)$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$2r(1+r) = 0$$

- Values of r that satisfy the indicial equation

$$r \in \{-1, 0\}$$

- Each term must be 0

$$2a_1(1+r)(2+r) - a_0(1+r)(-1+4r) = 0$$

- Each term in the series must be 0, giving the recursion relation

$$(-4a_k + 2a_{k-1} + 2a_{k+1}) k^2 + ((-8a_k + 4a_{k-1} + 4a_{k+1}) r - 3a_k - 3a_{k-1} + 6a_{k+1}) k + (-4a_k + 2a_{k-1} + 2a_{k+1}) = 0$$

- Shift index using $k \rightarrow k+1$
 $(-4a_{k+1} + 2a_k + 2a_{k+2})(k+1)^2 + ((-8a_{k+1} + 4a_k + 4a_{k+2})r - 3a_{k+1} - 3a_k + 6a_{k+2})(k+1) + (-$
- Recursion relation that defines series solution to ODE

$$a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} + 4kra_k - 8kra_{k+1} + 2r^2a_k - 4r^2a_{k+1} + ka_k - 11ka_{k+1} + ra_k - 11ra_{k+1} - 6a_{k+1}}{2(k^2 + 2kr + r^2 + 5k + 5r + 6)}$$
- Recursion relation for $r = -1$

$$a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} - 3ka_k - 3ka_{k+1} + a_k + a_{k+1}}{2(k^2 + 3k + 2)}$$
- Solution for $r = -1$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^{k-1}, a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} - 3ka_k - 3ka_{k+1} + a_k + a_{k+1}}{2(k^2 + 3k + 2)}, 0 = 0 \right]$$
- Revert the change of variables $u = x + 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x+1)^{k-1}, a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} - 3ka_k - 3ka_{k+1} + a_k + a_{k+1}}{2(k^2 + 3k + 2)}, 0 = 0 \right]$$
- Recursion relation for $r = 0$

$$a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} + ka_k - 11ka_{k+1} - 6a_{k+1}}{2(k^2 + 5k + 6)}$$
- Solution for $r = 0$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^k, a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} + ka_k - 11ka_{k+1} - 6a_{k+1}}{2(k^2 + 5k + 6)}, 4a_1 + a_0 = 0 \right]$$
- Revert the change of variables $u = x + 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x+1)^k, a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} + ka_k - 11ka_{k+1} - 6a_{k+1}}{2(k^2 + 5k + 6)}, 4a_1 + a_0 = 0 \right]$$
- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k (x+1)^{k-1} \right) + \left(\sum_{k=0}^{\infty} b_k (x+1)^k \right), a_{k+2} = -\frac{2k^2a_k - 4k^2a_{k+1} - 3ka_k - 3ka_{k+1} + a_k + a_{k+1}}{2(k^2 + 3k + 2)}, 0 = 0, \right.$$

✓ **Mathematica.** Time used: 0.006 (sec). Leaf size: 58

```
ode=2*(x^2+x^3)*D[y[x],{x,2}]-(x-3*x^2)*D[y[x],x]+y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 x(-x^5 + x^4 - x^3 + x^2 - x + 1) + c_2 \sqrt{x}(-x^5 + x^4 - x^3 + x^2 - x + 1)$$

✓ Sympy. Time used: 0.306 (sec). Leaf size: 15

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq((3*x**2 - x)*Derivative(y(x), x) + (2*x**3 + 2*x**2)*Derivative(y(x), (x,
    2)) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_2x + C_1\sqrt{x} + O(x^6)$$

2.17.2 Problem 12

Local contents

✓ Maple	2621
✓ Mathematica	2624
✓ Sympy	2624

Internal problem ID [8078]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 12

Date solved : Thursday, November 27, 2025 at 07:33:14 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$4xy'' + 2(1 - x)y' - y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$4xy'' + (-2x + 2)y' - y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = -\frac{-1+x}{2x}$$

$$q(x) = -\frac{1}{4x}$$

Table 2.309: Table $p(x), q(x)$ singularities.

$p(x) = -\frac{-1+x}{2x}$	
singularity	type
$x = 0$	“regular”

$q(x) = -\frac{1}{4x}$	
singularity	type
$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$4xy'' + (-2x + 2)y' - y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\begin{aligned} & 4 \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x \\ & + (-2x + 2) \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) - \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (1)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 4x^{n+r-1} a_n (n+r)(n+r-1) \right) + \sum_{n=0}^{\infty} (-2x^{n+r} a_n (n+r)) \\ & + \left(\sum_{n=0}^{\infty} 2(n+r) a_n x^{n+r-1} \right) + \sum_{n=0}^{\infty} (-a_n x^{n+r}) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n+r-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r-1} and adjusting the power and the corresponding index gives

$$\begin{aligned} \sum_{n=0}^{\infty} (-2x^{n+r} a_n (n+r)) &= \sum_{n=1}^{\infty} (-2a_{n-1} (n+r-1) x^{n+r-1}) \\ \sum_{n=0}^{\infty} (-a_n x^{n+r}) &= \sum_{n=1}^{\infty} (-a_{n-1} x^{n+r-1}) \end{aligned}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n + r - 1$.

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 4x^{n+r-1} a_n (n+r) (n+r-1) \right) + \sum_{n=1}^{\infty} (-2a_{n-1} (n+r-1) x^{n+r-1}) \\ & + \left(\sum_{n=0}^{\infty} 2(n+r) a_n x^{n+r-1} \right) + \sum_{n=1}^{\infty} (-a_{n-1} x^{n+r-1}) = 0 \end{aligned} \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$4x^{n+r-1} a_n (n+r) (n+r-1) + 2(n+r) a_n x^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$4x^{-1+r} a_0 r(-1+r) + 2r a_0 x^{-1+r} = 0$$

Or

$$(4x^{-1+r} r(-1+r) + 2r x^{-1+r}) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(4r^2 - 2r) x^{-1+r} = 0$$

Since the above is true for all x then the indicial equation becomes

$$4r^2 - 2r = 0$$

Solving for r gives the roots of the indicial equation as

$$\begin{aligned} r_1 &= \frac{1}{2} \\ r_2 &= 0 \end{aligned}$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(4r^2 - 2r) x^{-1+r} = 0$$

Solving for r gives the roots of the indicial equation as $[\frac{1}{2}, 0]$.

Since $r_1 - r_2 = \frac{1}{2}$ is not an integer, then we can construct two linearly independent solutions

$$\begin{aligned} y_1(x) &= x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right) \\ y_2(x) &= x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right) \end{aligned}$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+\frac{1}{2}}$$

$$y_2(x) = \sum_{n=0}^{\infty} b_n x^n$$

We start by finding $y_1(x)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$4a_n(n+r)(n+r-1) - 2a_{n-1}(n+r-1) + 2a_n(n+r) - a_{n-1} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = \frac{a_{n-1}}{2n+2r} \quad (4)$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_n = \frac{a_{n-1}}{2n+1} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = \frac{1}{2}$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = \frac{1}{2+2r}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_1 = \frac{1}{3}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2+2r}$	$\frac{1}{3}$

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{4(1+r)(2+r)}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_2 = \frac{1}{15}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2+2r}$	$\frac{1}{3}$
a_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{15}$

For $n = 3$, using the above recursive equation gives

$$a_3 = \frac{1}{8(1+r)(2+r)(3+r)}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_3 = \frac{1}{105}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2+2r}$	$\frac{1}{3}$
a_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{15}$
a_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{105}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{16(1+r)(2+r)(3+r)(4+r)}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_4 = \frac{1}{945}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2+2r}$	$\frac{1}{3}$
a_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{15}$
a_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{105}$
a_4	$\frac{1}{16(1+r)(2+r)(3+r)(4+r)}$	$\frac{1}{945}$

For $n = 5$, using the above recursive equation gives

$$a_5 = \frac{1}{32(1+r)(2+r)(3+r)(4+r)(5+r)}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_5 = \frac{1}{10395}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2+2r}$	$\frac{1}{3}$
a_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{15}$
a_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{105}$
a_4	$\frac{1}{16(1+r)(2+r)(3+r)(4+r)}$	$\frac{1}{945}$
a_5	$\frac{1}{32(1+r)(2+r)(3+r)(4+r)(5+r)}$	$\frac{1}{10395}$

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= \sqrt{x}(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{15} + \frac{x^3}{105} + \frac{x^4}{945} + \frac{x^5}{10395} + O(x^6) \right) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Eq (2B) derived above is now used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. For $1 \leq n$ the recursive equation is

$$4b_n(n+r)(n+r-1) - 2b_{n-1}(n+r-1) + 2(n+r)b_n - b_{n-1} = 0 \quad (3)$$

Solving for b_n from recursive equation (4) gives

$$b_n = \frac{b_{n-1}}{2n+2r} \quad (4)$$

Which for the root $r = 0$ becomes

$$b_n = \frac{b_{n-1}}{2n} \quad (5)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1

For $n = 1$, using the above recursive equation gives

$$b_1 = \frac{1}{2 + 2r}$$

Which for the root $r = 0$ becomes

$$b_1 = \frac{1}{2}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2+2r}$	$\frac{1}{2}$

For $n = 2$, using the above recursive equation gives

$$b_2 = \frac{1}{4(1+r)(2+r)}$$

Which for the root $r = 0$ becomes

$$b_2 = \frac{1}{8}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2+2r}$	$\frac{1}{2}$
b_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{8}$

For $n = 3$, using the above recursive equation gives

$$b_3 = \frac{1}{8(1+r)(2+r)(3+r)}$$

Which for the root $r = 0$ becomes

$$b_3 = \frac{1}{48}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2+2r}$	$\frac{1}{2}$
b_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{8}$
b_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{48}$

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{1}{16(1+r)(2+r)(3+r)(4+r)}$$

Which for the root $r = 0$ becomes

$$b_4 = \frac{1}{384}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2+2r}$	$\frac{1}{2}$
b_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{8}$
b_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{48}$
b_4	$\frac{1}{16(1+r)(2+r)(3+r)(4+r)}$	$\frac{1}{384}$

For $n = 5$, using the above recursive equation gives

$$b_5 = \frac{1}{32(1+r)(2+r)(3+r)(4+r)(5+r)}$$

Which for the root $r = 0$ becomes

$$b_5 = \frac{1}{3840}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2+2r}$	$\frac{1}{2}$
b_2	$\frac{1}{4(1+r)(2+r)}$	$\frac{1}{8}$
b_3	$\frac{1}{8(1+r)(2+r)(3+r)}$	$\frac{1}{48}$
b_4	$\frac{1}{16(1+r)(2+r)(3+r)(4+r)}$	$\frac{1}{384}$
b_5	$\frac{1}{32(1+r)(2+r)(3+r)(4+r)(5+r)}$	$\frac{1}{3840}$

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots \\ &= 1 + \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{48} + \frac{x^4}{384} + \frac{x^5}{3840} + O(x^6) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1 y_1(x) + c_2 y_2(x) \\ &= c_1 \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{15} + \frac{x^3}{105} + \frac{x^4}{945} + \frac{x^5}{10395} + O(x^6) \right) \\ &\quad + c_2 \left(1 + \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{48} + \frac{x^4}{384} + \frac{x^5}{3840} + O(x^6) \right) \end{aligned}$$

Hence the final solution is

$$\begin{aligned} y &= y_h \\ &= c_1 \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{15} + \frac{x^3}{105} + \frac{x^4}{945} + \frac{x^5}{10395} + O(x^6) \right) \\ &\quad + c_2 \left(1 + \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{48} + \frac{x^4}{384} + \frac{x^5}{3840} + O(x^6) \right) \end{aligned}$$

✓ **Maple.** Time used: 0.013 (sec). Leaf size: 44

```
Order:=6;
ode:=4*x*diff(diff(y(x),x),x)+2*(1-x)*diff(y(x),x)-y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$\begin{aligned} y &= c_1 \sqrt{x} \left(1 + \frac{1}{3}x + \frac{1}{15}x^2 + \frac{1}{105}x^3 + \frac{1}{945}x^4 + \frac{1}{10395}x^5 + O(x^6) \right) \\ &\quad + c_2 \left(1 + \frac{1}{2}x + \frac{1}{8}x^2 + \frac{1}{48}x^3 + \frac{1}{384}x^4 + \frac{1}{3840}x^5 + O(x^6) \right) \end{aligned}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
A Liouvillian solution exists
Reducible group (found an exponential solution)
Group is reducible, not completely reducible
Solution has integrals. Trying a special function solution free of integrals\
```

```

...
-> Trying a solution in terms of special functions:
  -> Bessel
  -> elliptic
  -> Legendre
  <- Kummer successful
<- special function solution successful
  -> Trying to convert hypergeometric functions to elementary form...
  <- elementary form is not straightforward to achieve - returning special \
function solution free of uncomputed integrals
  <- Kovacic's algorithm successful

```

Maple step by step

Let's solve

$$4x \left(\frac{d^2}{dx^2} y(x) \right) + 2(1-x) \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = \frac{y(x)}{4x} + \frac{(-1+x) \left(\frac{d}{dx} y(x) \right)}{2x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{(-1+x) \left(\frac{d}{dx} y(x) \right)}{2x} - \frac{y(x)}{4x} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- o Define functions

$$[P_2(x) = -\frac{-1+x}{2x}, P_3(x) = -\frac{1}{4x}]$$

- o $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = \frac{1}{2}$$

- o $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$

- o $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$4x \left(\frac{d^2}{dx^2} y(x) \right) + (-2x + 2) \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $x^m \cdot \left(\frac{d}{dx}y(x)\right)$ to series expansion for $m = 0.1$

$$x^m \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=0}^{\infty} a_k(k+r) x^{k+r-1+m}$$

- Shift index using $k \rightarrow k+1-m$

$$x^m \cdot \left(\frac{d}{dx}y(x)\right) = \sum_{k=-1+m}^{\infty} a_{k+1-m}(k+1-m+r) x^{k+r}$$

- Convert $x \cdot \left(\frac{d^2}{dx^2}y(x)\right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=0}^{\infty} a_k(k+r)(k+r-1) x^{k+r-1}$$

- Shift index using $k \rightarrow k+1$

$$x \cdot \left(\frac{d^2}{dx^2}y(x)\right) = \sum_{k=-1}^{\infty} a_{k+1}(k+1+r)(k+r) x^{k+r}$$

Rewrite ODE with series expansions

$$2a_0r(-1+2r)x^{-1+r} + \left(\sum_{k=0}^{\infty} (2a_{k+1}(k+1+r)(2k+2r+1) - a_k(2k+2r+1)) x^{k+r}\right) = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$2r(-1+2r) = 0$$

- Values of r that satisfy the indicial equation

$$r \in \left\{0, \frac{1}{2}\right\}$$

- Each term in the series must be 0, giving the recursion relation

$$4\left(a_{k+1}(k+1+r) - \frac{a_k}{2}\right)(k+r+\frac{1}{2}) = 0$$

- Recursion relation that defines series solution to ODE

$$a_{k+1} = \frac{a_k}{2(k+1+r)}$$

- Recursion relation for $r = 0$

$$a_{k+1} = \frac{a_k}{2(k+1)}$$

- Solution for $r = 0$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+1} = \frac{a_k}{2(k+1)}\right]$$

- Recursion relation for $r = \frac{1}{2}$

$$a_{k+1} = \frac{a_k}{2(k+\frac{3}{2})}$$

- Solution for $r = \frac{1}{2}$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+\frac{1}{2}}, a_{k+1} = \frac{a_k}{2(k+\frac{3}{2})}\right]$$

- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^k\right) + \left(\sum_{k=0}^{\infty} b_k x^{k+\frac{1}{2}}\right), a_{k+1} = \frac{a_k}{2(k+1)}, b_{k+1} = \frac{b_k}{2(k+\frac{3}{2})}\right]$$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 85

```
ode=4*x*D[y[x],{x,2}]+2*(1-x)*D[y[x],x]-y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \sqrt{x} \left(\frac{x^5}{10395} + \frac{x^4}{945} + \frac{x^3}{105} + \frac{x^2}{15} + \frac{x}{3} + 1 \right) + c_2 \left(\frac{x^5}{3840} + \frac{x^4}{384} + \frac{x^3}{48} + \frac{x^2}{8} + \frac{x}{2} + 1 \right)$$

✓ **Sympy.** Time used: 0.331 (sec). Leaf size: 60

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(4*x*Derivative(y(x), (x, 2)) + (2 - 2*x)*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_2 \left(\frac{x^5}{3840} + \frac{x^4}{384} + \frac{x^3}{48} + \frac{x^2}{8} + \frac{x}{2} + 1 \right) + C_1 \sqrt{x} \left(\frac{x^4}{945} + \frac{x^3}{105} + \frac{x^2}{15} + \frac{x}{3} + 1 \right) + O(x^6)$$

2.17.3 Problem 13

Local contents

✓ Maple	2632
✓ Mathematica	2634
✓ Sympy	2635

Internal problem ID [8079]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 13

Date solved : Thursday, November 27, 2025 at 07:33:15 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$2x^2y'' - y'x + (-x^2 + 1)y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$2x^2y'' - y'x + (-x^2 + 1)y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = -\frac{1}{2x}$$

$$q(x) = -\frac{x^2 - 1}{2x^2}$$

Table 2.311: Table $p(x), q(x)$ singularities.

$p(x) = -\frac{1}{2x}$		$q(x) = -\frac{x^2-1}{2x^2}$	
singularity	type	singularity	type
$x = 0$	“regular”	$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$2x^2y'' - y'x + (-x^2 + 1)y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\begin{aligned} & 2 \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x^2 \\ & - \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) x + (-x^2 + 1) \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (1)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 2x^{n+r} a_n (n+r)(n+r-1) \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n (n+r)) \\ & + \sum_{n=0}^{\infty} (-x^{n+r+2} a_n) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n+r$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} (-x^{n+r+2} a_n) = \sum_{n=2}^{\infty} (-a_{n-2} x^{n+r})$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n + r$.

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 2x^{n+r} a_n (n+r) (n+r-1) \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n (n+r)) \\ & + \sum_{n=2}^{\infty} (-a_{n-2} x^{n+r}) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$2x^{n+r} a_n (n+r) (n+r-1) - x^{n+r} a_n (n+r) + a_n x^{n+r} = 0$$

When $n = 0$ the above becomes

$$2x^r a_0 r(-1+r) - x^r a_0 r + a_0 x^r = 0$$

Or

$$(2x^r r(-1+r) - x^r r + x^r) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(2r^2 - 3r + 1) x^r = 0$$

Since the above is true for all x then the indicial equation becomes

$$2r^2 - 3r + 1 = 0$$

Solving for r gives the roots of the indicial equation as

$$\begin{aligned} r_1 &= 1 \\ r_2 &= \frac{1}{2} \end{aligned}$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(2r^2 - 3r + 1) x^r = 0$$

Solving for r gives the roots of the indicial equation as $[1, \frac{1}{2}]$.

Since $r_1 - r_2 = \frac{1}{2}$ is not an integer, then we can construct two linearly independent solutions

$$\begin{aligned} y_1(x) &= x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right) \\ y_2(x) &= x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right) \end{aligned}$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+1}$$

$$y_2(x) = \sum_{n=0}^{\infty} b_n x^{n+\frac{1}{2}}$$

We start by finding $y_1(x)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. Substituting $n = 1$ in Eq. (2B) gives

$$a_1 = 0$$

For $2 \leq n$ the recursive equation is

$$2a_n(n+r)(n+r-1) - a_n(n+r) - a_{n-2} + a_n = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = \frac{a_{n-2}}{2n^2 + 4nr + 2r^2 - 3n - 3r + 1} \quad (4)$$

Which for the root $r = 1$ becomes

$$a_n = \frac{a_{n-2}}{2n^2 + n} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 1$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{2r^2 + 5r + 3}$$

Which for the root $r = 1$ becomes

$$a_2 = \frac{1}{10}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{10}$

For $n = 3$, using the above recursive equation gives

$$a_3 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{10}$
a_3	0	0

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{4r^4 + 36r^3 + 113r^2 + 144r + 63}$$

Which for the root $r = 1$ becomes

$$a_4 = \frac{1}{360}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{10}$
a_3	0	0
a_4	$\frac{1}{4r^4+36r^3+113r^2+144r+63}$	$\frac{1}{360}$

For $n = 5$, using the above recursive equation gives

$$a_5 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{10}$
a_3	0	0
a_4	$\frac{1}{4r^4+36r^3+113r^2+144r+63}$	$\frac{1}{360}$
a_5	0	0

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= x\left(1 + \frac{x^2}{10} + \frac{x^4}{360} + O(x^6)\right) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Eq (2B) derived above is now used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. Substituting $n = 1$ in Eq. (2B) gives

$$b_1 = 0$$

For $2 \leq n$ the recursive equation is

$$2b_n(n+r)(n+r-1) - b_n(n+r) - b_{n-2} + b_n = 0 \quad (3)$$

Solving for b_n from recursive equation (4) gives

$$b_n = \frac{b_{n-2}}{2n^2 + 4nr + 2r^2 - 3n - 3r + 1} \quad (4)$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_n = \frac{b_{n-2}}{n(2n-1)} \quad (5)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = \frac{1}{2}$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0

For $n = 2$, using the above recursive equation gives

$$b_2 = \frac{1}{2r^2 + 5r + 3}$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_2 = \frac{1}{6}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{6}$

For $n = 3$, using the above recursive equation gives

$$b_3 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{6}$
b_3	0	0

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{1}{4r^4 + 36r^3 + 113r^2 + 144r + 63}$$

Which for the root $r = \frac{1}{2}$ becomes

$$b_4 = \frac{1}{168}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{6}$
b_3	0	0
b_4	$\frac{1}{4r^4+36r^3+113r^2+144r+63}$	$\frac{1}{168}$

For $n = 5$, using the above recursive equation gives

$$b_5 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$\frac{1}{2r^2+5r+3}$	$\frac{1}{6}$
b_3	0	0
b_4	$\frac{1}{4r^4+36r^3+113r^2+144r+63}$	$\frac{1}{168}$
b_5	0	0

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= x(b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots) \\ &= \sqrt{x} \left(1 + \frac{x^2}{6} + \frac{x^4}{168} + O(x^6) \right) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1 y_1(x) + c_2 y_2(x) \\ &= c_1 x \left(1 + \frac{x^2}{10} + \frac{x^4}{360} + O(x^6) \right) + c_2 \sqrt{x} \left(1 + \frac{x^2}{6} + \frac{x^4}{168} + O(x^6) \right) \end{aligned}$$

Hence the final solution is

$$\begin{aligned} y &= y_h \\ &= c_1 x \left(1 + \frac{x^2}{10} + \frac{x^4}{360} + O(x^6) \right) + c_2 \sqrt{x} \left(1 + \frac{x^2}{6} + \frac{x^4}{168} + O(x^6) \right) \end{aligned}$$

✓ **Maple.** Time used: 0.010 (sec). Leaf size: 33

```
Order:=6;
ode:=2*x^2*diff(diff(y(x),x),x)-diff(y(x),x)*x+(-x^2+1)*y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = c_1 \sqrt{x} \left(1 + \frac{1}{6}x^2 + \frac{1}{168}x^4 + O(x^6) \right) + c_2 x \left(1 + \frac{1}{10}x^2 + \frac{1}{360}x^4 + O(x^6) \right)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$2x^2 \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + (-x^2 + 1) y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = \frac{(x^2-1)y(x)}{2x^2} + \frac{\frac{d}{dx} y(x)}{2x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{\frac{d}{dx} y(x)}{2x} - \frac{(x^2-1)y(x)}{2x^2} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- o Define functions

$$\left[P_2(x) = -\frac{1}{2x}, P_3(x) = -\frac{x^2-1}{2x^2} \right]$$

- o $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -\frac{1}{2}$$

- o $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = \frac{1}{2}$$

- o $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$2x^2 \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + (-x^2 + 1) y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- o Convert $x^m \cdot y(x)$ to series expansion for $m = 0..2$

$$x^m \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+m}$$

- o Shift index using $k- > k - m$

$$x^m \cdot y(x) = \sum_{k=m}^{\infty} a_{k-m} x^{k+r}$$

- o Convert $x \cdot \left(\frac{d}{dx} y(x) \right)$ to series expansion

$$x \cdot \left(\frac{d}{dx} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r}$$

- o Convert $x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion

$$x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0(-1+2r)(-1+r)x^r + a_1(1+2r)r x^{1+r} + \left(\sum_{k=2}^{\infty} (a_k(2k+2r-1)(k+r-1) - a_{k-2}) x^{k+r} \right) = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation
 $(-1+2r)(-1+r) = 0$
- Values of r that satisfy the indicial equation
 $r \in \{1, \frac{1}{2}\}$
- Each term must be 0
 $a_1(1+2r)r = 0$
- Solve for the dependent coefficient(s)
 $a_1 = 0$
- Each term in the series must be 0, giving the recursion relation
 $2(k+r-\frac{1}{2})(k+r-1)a_k - a_{k-2} = 0$
- Shift index using $k \rightarrow k+2$
 $2(k+\frac{3}{2}+r)(k+1+r)a_{k+2} - a_k = 0$
- Recursion relation that defines series solution to ODE
 $a_{k+2} = \frac{a_k}{(2k+3+2r)(k+1+r)}$
- Recursion relation for $r = 1$
 $a_{k+2} = \frac{a_k}{(2k+5)(k+2)}$
- Solution for $r = 1$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+1}, a_{k+2} = \frac{a_k}{(2k+5)(k+2)}, a_1 = 0 \right]$
- Recursion relation for $r = \frac{1}{2}$
 $a_{k+2} = \frac{a_k}{(2k+4)(k+\frac{3}{2})}$
- Solution for $r = \frac{1}{2}$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+\frac{1}{2}}, a_{k+2} = \frac{a_k}{(2k+4)(k+\frac{3}{2})}, a_1 = 0 \right]$
- Combine solutions and rename parameters
 $\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^{k+1} \right) + \left(\sum_{k=0}^{\infty} b_k x^{k+\frac{1}{2}} \right), a_{k+2} = \frac{a_k}{(2k+5)(k+2)}, a_1 = 0, b_{k+2} = \frac{b_k}{(2k+4)(k+\frac{3}{2})}, b_1 = 0 \right]$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 48

```
ode=2*x^2*D[y[x],{x,2}]-x*D[y[x],x]+(1-x^2)*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 x \left(\frac{x^4}{360} + \frac{x^2}{10} + 1 \right) + c_2 \sqrt{x} \left(\frac{x^4}{168} + \frac{x^2}{6} + 1 \right)$$

✓ Sympy. Time used: 0.339 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x**2*Derivative(y(x), (x, 2)) - x*Derivative(y(x), x) + (1 - x**2)*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_2 x \left(\frac{x^4}{360} + \frac{x^2}{10} + 1 \right) + C_1 \sqrt{x} \left(\frac{x^4}{168} + \frac{x^2}{6} + 1 \right) + O(x^6)$$

2.17.4 Problem 14

Local contents

✓ Maple	2642
✓ Mathematica	2644
✓ Sympy	2644

Internal problem ID [8080]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 14

Date solved : Thursday, November 27, 2025 at 07:33:15 AM

CAS classification : [_Lienard]

$$xy'' + y' + yx = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$xy'' + y' + yx = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = \frac{1}{x}$$

$$q(x) = 1$$

Table 2.313: Table $p(x), q(x)$ singularities.

$p(x) = \frac{1}{x}$	
singularity	type
$x = 0$	“regular”

$q(x) = 1$	
singularity	type

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : [0]

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$xy'' + y' + yx = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) x = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} x^{1+n+r} a_n \right) = 0 \quad (2A)$$

The next step is to make all powers of x be $n+r-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r-1} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} x^{1+n+r} a_n = \sum_{n=2}^{\infty} a_{n-2} x^{n+r-1}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n+r-1$.

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=2}^{\infty} a_{n-2} x^{n+r-1} \right) = 0 \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$x^{n+r-1}a_n(n+r)(n+r-1) + (n+r)a_nx^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$x^{-1+r}a_0r(-1+r) + ra_0x^{-1+r} = 0$$

Or

$$(x^{-1+r}r(-1+r) + rx^{-1+r})a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$x^{-1+r}r^2 = 0$$

Since the above is true for all x then the indicial equation becomes

$$r^2 = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 0$$

$$r_2 = 0$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$x^{-1+r}r^2 = 0$$

Solving for r gives the roots of the indicial equation as $[0, 0]$.

Since the root of the indicial equation is repeated, then we can construct two linearly independent solutions. The first solution has the form

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+r} \quad (1A)$$

Now the second solution y_2 is found using

$$y_2(x) = y_1(x) \ln(x) + \left(\sum_{n=1}^{\infty} b_n x^{n+r} \right) \quad (1B)$$

Then the general solution will be

$$y = c_1 y_1(x) + c_2 y_2(x)$$

In Eq (1B) the sum starts from 1 and not zero. In Eq (1A), a_0 is never zero, and is arbitrary and is typically taken as $a_0 = 1$, and $\{c_1, c_2\}$ are two arbitrary constants of integration which can be found from initial conditions. We start by finding the first solution $y_1(x)$. Eq (2B)

derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. Substituting $n = 1$ in Eq. (2B) gives

$$a_1 = 0$$

For $2 \leq n$ the recursive equation is

$$a_n(n+r)(n+r-1) + a_n(n+r) + a_{n-2} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-2}}{n^2 + 2nr + r^2} \quad (4)$$

Which for the root $r = 0$ becomes

$$a_n = -\frac{a_{n-2}}{n^2} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0

For $n = 2$, using the above recursive equation gives

$$a_2 = -\frac{1}{(r+2)^2}$$

Which for the root $r = 0$ becomes

$$a_2 = -\frac{1}{4}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(r+2)^2}$	$-\frac{1}{4}$

For $n = 3$, using the above recursive equation gives

$$a_3 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(r+2)^2}$	$-\frac{1}{4}$
a_3	0	0

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{(r+2)^2(r+4)^2}$$

Which for the root $r = 0$ becomes

$$a_4 = \frac{1}{64}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(r+2)^2}$	$-\frac{1}{4}$
a_3	0	0
a_4	$\frac{1}{(r+2)^2(r+4)^2}$	$\frac{1}{64}$

For $n = 5$, using the above recursive equation gives

$$a_5 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(r+2)^2}$	$-\frac{1}{4}$
a_3	0	0
a_4	$\frac{1}{(r+2)^2(r+4)^2}$	$\frac{1}{64}$
a_5	0	0

Using the above table, then the first solution $y_1(x)$ becomes

$$\begin{aligned} y_1(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots \\ &= 1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \end{aligned}$$

Now the second solution is found. The second solution is given by

$$y_2(x) = y_1(x) \ln(x) + \left(\sum_{n=1}^{\infty} b_n x^{n+r} \right)$$

Where b_n is found using

$$b_n = \frac{d}{dr} a_{n,r}$$

And the above is then evaluated at $r = 0$. The above table for $a_{n,r}$ is used for this purpose. Computing the derivatives gives the following table

n	$b_{n,r}$	a_n	$b_{n,r} = \frac{d}{dr} a_{n,r}$	$b_n(r=0)$
b_0	1	1	N/A since b_n starts from 1	N/A
b_1	0	0	0	0
b_2	$-\frac{1}{(r+2)^2}$	$-\frac{1}{4}$	$\frac{2}{(r+2)^3}$	$\frac{1}{4}$
b_3	0	0	0	0
b_4	$\frac{1}{(r+2)^2(r+4)^2}$	$\frac{1}{64}$	$\frac{-4r-12}{(r+2)^3(r+4)^3}$	$-\frac{3}{128}$
b_5	0	0	0	0

The above table gives all values of b_n needed. Hence the second solution is

$$\begin{aligned} y_2(x) &= y_1(x) \ln(x) + b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots \\ &= \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + \frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1 y_1(x) + c_2 y_2(x) \\ &= c_1 \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) + c_2 \left(\left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + \frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \right) \end{aligned}$$

Hence the final solution is

$$\begin{aligned} y &= y_h \\ &= c_1 \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) + c_2 \left(\left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + \frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \right) \end{aligned}$$

✓ **Maple.** Time used: 0.015 (sec). Leaf size: 32

```
Order:=6;
ode:=x*diff(diff(y(x),x),x)+diff(y(x),x)+y(x)*x = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = (c_2 \ln(x) + c_1) \left(1 - \frac{1}{4}x^2 + \frac{1}{64}x^4 + O(x^6) \right) + \left(\frac{1}{4}x^2 - \frac{3}{128}x^4 + O(x^6) \right) c_2$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) + \frac{d}{dx} y(x) + xy(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -y(x) - \frac{\frac{d}{dx} y(x)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + \frac{\frac{d}{dx} y(x)}{x} + y(x) = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- Define functions

$$[P_2(x) = \frac{1}{x}, P_3(x) = 1]$$

- $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = 1$$

- $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$
- $x = 0$ is a regular singular point
Check to see if $x_0 = 0$ is a regular singular point
 $x_0 = 0$
- Multiply by denominators

$$x \left(\frac{d^2}{dx^2} y(x) \right) + \frac{d}{dx} y(x) + x y(x) = 0$$
- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$
- Rewrite ODE with series expansions
 - Convert $x \cdot y(x)$ to series expansion

$$x \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+1}$$
 - Shift index using $k- > k-1$

$$x \cdot y(x) = \sum_{k=1}^{\infty} a_{k-1} x^{k+r}$$
 - Convert $\frac{d}{dx} y(x)$ to series expansion

$$\frac{d}{dx} y(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}$$
 - Shift index using $k- > k+1$

$$\frac{d}{dx} y(x) = \sum_{k=-1}^{\infty} a_{k+1} (k+r+1) x^{k+r}$$
 - Convert $x \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) x^{k+r-1}$$
 - Shift index using $k- > k+1$

$$x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=-1}^{\infty} a_{k+1} (k+r+1) (k+r) x^{k+r}$$
- Rewrite ODE with series expansions

$$a_0 r^2 x^{-1+r} + a_1 (1+r)^2 x^r + \left(\sum_{k=1}^{\infty} (a_{k+1} (k+r+1)^2 + a_{k-1}) x^{k+r} \right) = 0$$
- a_0 cannot be 0 by assumption, giving the indicial equation

$$r^2 = 0$$
- Values of r that satisfy the indicial equation

$$r = 0$$
- Each term must be 0

$$a_1 (1+r)^2 = 0$$
- Each term in the series must be 0, giving the recursion relation

$$a_{k+1} (k+1)^2 + a_{k-1} = 0$$
- Shift index using $k- > k+1$

- $a_{k+2}(k+2)^2 + a_k = 0$
- Recursion relation that defines series solution to ODE

$$a_{k+2} = -\frac{a_k}{(k+2)^2}$$
- Recursion relation for $r = 0$

$$a_{k+2} = -\frac{a_k}{(k+2)^2}$$
- Solution for $r = 0$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+2} = -\frac{a_k}{(k+2)^2}, a_1 = 0 \right]$$

✓ **Mathematica.** Time used: 0.002 (sec). Leaf size: 60

```
ode=x*D[y[x],{x,2}]+D[y[x],x]+x*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) + c_2 \left(-\frac{3x^4}{128} + \frac{x^2}{4} + \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) \log(x) \right)$$

✓ **Sympy.** Time used: 0.219 (sec). Leaf size: 19

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) + x*Derivative(y(x), (x, 2)) + Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_1 \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) + O(x^6)$$

2.17.5 Problem 15

Local contents

✓ Maple	2651
✓ Mathematica	2653
✓ Sympy	2654

Internal problem ID [8081]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 15

Date solved : Thursday, November 27, 2025 at 07:33:16 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$x^2 y'' - y' x + (x^2 + 1) y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$x^2 y'' - y' x + (x^2 + 1) y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = -\frac{1}{x}$$

$$q(x) = \frac{x^2 + 1}{x^2}$$

Table 2.315: Table $p(x), q(x)$ singularities.

$p(x) = -\frac{1}{x}$		$q(x) = \frac{x^2+1}{x^2}$	
singularity	type	singularity	type
$x = 0$	“regular”	$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$x^2 y'' - y' x + (x^2 + 1) y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x^2 \\ & - \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) x + (x^2 + 1) \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (1)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} x^{n+r} a_n (n+r)(n+r-1) \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n (n+r)) \\ & + \left(\sum_{n=0}^{\infty} x^{n+r+2} a_n \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n+r$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} x^{n+r+2} a_n = \sum_{n=2}^{\infty} a_{n-2} x^{n+r}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n + r$.

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} x^{n+r} a_n (n+r) (n+r-1) \right) + \sum_{n=0}^{\infty} (-x^{n+r} a_n (n+r)) \\ & + \left(\sum_{n=2}^{\infty} a_{n-2} x^{n+r} \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$x^{n+r} a_n (n+r) (n+r-1) - x^{n+r} a_n (n+r) + a_n x^{n+r} = 0$$

When $n = 0$ the above becomes

$$x^r a_0 r(-1+r) - x^r a_0 r + a_0 x^r = 0$$

Or

$$(x^r r(-1+r) - x^r r + x^r) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(-1+r)^2 x^r = 0$$

Since the above is true for all x then the indicial equation becomes

$$(-1+r)^2 = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 1$$

$$r_2 = 1$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(-1+r)^2 x^r = 0$$

Solving for r gives the roots of the indicial equation as $[1, 1]$.

Since the root of the indicial equation is repeated, then we can construct two linearly independent solutions. The first solution has the form

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+r} \quad (1A)$$

Now the second solution y_2 is found using

$$y_2(x) = y_1(x) \ln(x) + \left(\sum_{n=1}^{\infty} b_n x^{n+r} \right) \quad (1B)$$

Then the general solution will be

$$y = c_1 y_1(x) + c_2 y_2(x)$$

In Eq (1B) the sum starts from 1 and not zero. In Eq (1A), a_0 is never zero, and is arbitrary and is typically taken as $a_0 = 1$, and $\{c_1, c_2\}$ are two arbitrary constants of integration which can be found from initial conditions. Using the value of the indicial root found earlier, $r = 1$, Eqs (1A,1B) become

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+1}$$

$$y_2(x) = y_1(x) \ln(x) + \left(\sum_{n=1}^{\infty} b_n x^{n+1} \right)$$

We start by finding the first solution $y_1(x)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. Substituting $n = 1$ in Eq. (2B) gives

$$a_1 = 0$$

For $2 \leq n$ the recursive equation is

$$a_n(n+r)(n+r-1) - a_n(n+r) + a_{n-2} + a_n = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-2}}{n^2 + 2nr + r^2 - 2n - 2r + 1} \quad (4)$$

Which for the root $r = 1$ becomes

$$a_n = -\frac{a_{n-2}}{n^2} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 1$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0

For $n = 2$, using the above recursive equation gives

$$a_2 = -\frac{1}{(1+r)^2}$$

Which for the root $r = 1$ becomes

$$a_2 = -\frac{1}{4}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(1+r)^2}$	$-\frac{1}{4}$

For $n = 3$, using the above recursive equation gives

$$a_3 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(1+r)^2}$	$-\frac{1}{4}$
a_3	0	0

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{(1+r)^2(r+3)^2}$$

Which for the root $r = 1$ becomes

$$a_4 = \frac{1}{64}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(1+r)^2}$	$-\frac{1}{4}$
a_3	0	0
a_4	$\frac{1}{(1+r)^2(r+3)^2}$	$\frac{1}{64}$

For $n = 5$, using the above recursive equation gives

$$a_5 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{(1+r)^2}$	$-\frac{1}{4}$
a_3	0	0
a_4	$\frac{1}{(1+r)^2(r+3)^2}$	$\frac{1}{64}$
a_5	0	0

Using the above table, then the first solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= x\left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6)\right) \end{aligned}$$

Now the second solution is found. The second solution is given by

$$y_2(x) = y_1(x) \ln(x) + \left(\sum_{n=1}^{\infty} b_n x^{n+r}\right)$$

Where b_n is found using

$$b_n = \frac{d}{dr} a_{n,r}$$

And the above is then evaluated at $r = 1$. The above table for $a_{n,r}$ is used for this purpose. Computing the derivatives gives the following table

n	$b_{n,r}$	a_n	$b_{n,r} = \frac{d}{dr} a_{n,r}$	$b_n(r = 1)$
b_0	1	1	N/A since b_n starts from 1	N/A
b_1	0	0	0	0
b_2	$-\frac{1}{(1+r)^2}$	$-\frac{1}{4}$	$\frac{2}{(1+r)^3}$	$\frac{1}{4}$
b_3	0	0	0	0
b_4	$\frac{1}{(1+r)^2(r+3)^2}$	$\frac{1}{64}$	$\frac{-4r-8}{(1+r)^3(r+3)^3}$	$-\frac{3}{128}$
b_5	0	0	0	0

The above table gives all values of b_n needed. Hence the second solution is

$$\begin{aligned} y_2(x) &= y_1(x) \ln(x) + b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots \\ &= x \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + x \left(\frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \right) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1 y_1(x) + c_2 y_2(x) \\ &= c_1 x \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) + c_2 \left(x \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + x \left(\frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \right) \right) \end{aligned}$$

Hence the final solution is

$$y = y_h = c_1 x \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) + c_2 \left(x \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6) \right) \ln(x) + x \left(\frac{x^2}{4} - \frac{3x^4}{128} + O(x^6) \right) \right)$$

✓ **Maple.** Time used: 0.012 (sec). Leaf size: 34

```
Order:=6;
ode:=x^2*diff(diff(y(x),x),x)-diff(y(x),x)*x+(x^2+1)*y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = \left((c_2 \ln(x) + c_1) \left(1 - \frac{1}{4}x^2 + \frac{1}{64}x^4 + O(x^6) \right) + \left(\frac{1}{4}x^2 - \frac{3}{128}x^4 + O(x^6) \right) c_2 \right) x$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + (x^2 + 1) y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{(x^2+1)y(x)}{x^2} + \frac{\frac{d}{dx} y(x)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{\frac{d}{dx} y(x)}{x} + \frac{(x^2+1)y(x)}{x^2} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- o Define functions

$$\left[P_2(x) = -\frac{1}{x}, P_3(x) = \frac{x^2+1}{x^2} \right]$$

- o $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -1$$

- o $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 1$$

- o $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$x^2 \left(\frac{d^2}{dx^2} y(x) \right) - x \left(\frac{d}{dx} y(x) \right) + (x^2 + 1) y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- o Convert $x^m \cdot y(x)$ to series expansion for $m = 0..2$

$$x^m \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+m}$$

- o Shift index using $k \rightarrow k - m$

$$x^m \cdot y(x) = \sum_{k=m}^{\infty} a_{k-m} x^{k+r}$$

- o Convert $x \cdot \left(\frac{d}{dx} y(x) \right)$ to series expansion

$$x \cdot \left(\frac{d}{dx} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r}$$

- o Convert $x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion

$$x^2 \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0(-1+r)^2 x^r + a_1 r^2 x^{1+r} + \left(\sum_{k=2}^{\infty} (a_k(k+r-1)^2 + a_{k-2}) x^{k+r} \right) = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation $(-1+r)^2 = 0$
- Values of r that satisfy the indicial equation
 $r = 1$
- Each term must be 0
 $a_1 r^2 = 0$
- Solve for the dependent coefficient(s)
 $a_1 = 0$
- Each term in the series must be 0, giving the recursion relation
 $a_k(k+r-1)^2 + a_{k-2} = 0$
- Shift index using $k \rightarrow k+2$
 $a_{k+2}(k+1+r)^2 + a_k = 0$
- Recursion relation that defines series solution to ODE
 $a_{k+2} = -\frac{a_k}{(k+1+r)^2}$
- Recursion relation for $r = 1$
 $a_{k+2} = -\frac{a_k}{(k+2)^2}$
- Solution for $r = 1$
 $\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+1}, a_{k+2} = -\frac{a_k}{(k+2)^2}, a_1 = 0 \right]$

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 65

```
ode=x^2*D[y[x],{x,2}]-x*D[y[x],x]+(x^2+1)*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 x \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) + c_2 \left(x \left(\frac{x^2}{4} - \frac{3x^4}{128} \right) + x \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) \log(x) \right)$$

✓ Sympy. Time used: 0.249 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*Derivative(y(x), (x, 2)) - x*Derivative(y(x), x) + (x**2 + 1)*y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_1 x \left(\frac{x^4}{64} - \frac{x^2}{4} + 1 \right) + O(x^6)$$

2.17.6 Problem 16

Local contents

✓ Maple	2665
✓ Mathematica	2667
✓ Sympy	2667

Internal problem ID [8082]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 16

Date solved : Thursday, November 27, 2025 at 07:33:17 AM

CAS classification : [[_Emden, _Fowler]]

$$xy'' - 2y' + y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$xy'' - 2y' + y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = -\frac{2}{x}$$

$$q(x) = \frac{1}{x}$$

Table 2.317: Table $p(x), q(x)$ singularities.

$p(x) = -\frac{2}{x}$	
singularity	type
$x = 0$	“regular”

$q(x) = \frac{1}{x}$	
singularity	type
$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$xy'' - 2y' + y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x - 2 \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \sum_{n=0}^{\infty} (-2(n+r) a_n x^{n+r-1}) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \quad (2A)$$

The next step is to make all powers of x be $n+r-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r-1} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} a_n x^{n+r} = \sum_{n=1}^{\infty} a_{n-1} x^{n+r-1}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n+r-1$.

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \sum_{n=0}^{\infty} (-2(n+r) a_n x^{n+r-1}) + \left(\sum_{n=1}^{\infty} a_{n-1} x^{n+r-1} \right) = 0 \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$x^{n+r-1}a_n(n+r)(n+r-1) - 2(n+r)a_nx^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$x^{-1+r}a_0r(-1+r) - 2ra_0x^{-1+r} = 0$$

Or

$$(x^{-1+r}r(-1+r) - 2rx^{-1+r})a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$rx^{-1+r}(-3+r) = 0$$

Since the above is true for all x then the indicial equation becomes

$$r(-3+r) = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 3$$

$$r_2 = 0$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$rx^{-1+r}(-3+r) = 0$$

Solving for r gives the roots of the indicial equation as $[3, 0]$.

Since $r_1 - r_2 = 3$ is an integer, then we can construct two linearly independent solutions

$$y_1(x) = x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = Cy_1(x) \ln(x) + x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Or

$$y_1(x) = x^3 \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = Cy_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+3}$$

$$y_2(x) = C y_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Where C above can be zero. We start by finding y_1 . Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$a_n(n+r)(n+r-1) - 2a_n(n+r) + a_{n-1} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-1}}{n^2 + 2nr + r^2 - 3n - 3r} \quad (4)$$

Which for the root $r = 3$ becomes

$$a_n = -\frac{a_{n-1}}{n(n+3)} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 3$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = -\frac{1}{r^2 - r - 2}$$

Which for the root $r = 3$ becomes

$$a_1 = -\frac{1}{4}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{r^2 - r - 2}$	$-\frac{1}{4}$

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{r^4 - 5r^2 + 4}$$

Which for the root $r = 3$ becomes

$$a_2 = \frac{1}{40}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{r^2-r-2}$	$-\frac{1}{4}$
a_2	$\frac{1}{r^4-5r^2+4}$	$\frac{1}{40}$

For $n = 3$, using the above recursive equation gives

$$a_3 = -\frac{1}{(r^4 - 5r^2 + 4)r(r+3)}$$

Which for the root $r = 3$ becomes

$$a_3 = -\frac{1}{720}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{r^2-r-2}$	$-\frac{1}{4}$
a_2	$\frac{1}{r^4-5r^2+4}$	$\frac{1}{40}$
a_3	$-\frac{1}{(r^4-5r^2+4)r(r+3)}$	$-\frac{1}{720}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{(r^4 - 5r^2 + 4)r(r+3)(r^2 + 5r + 4)}$$

Which for the root $r = 3$ becomes

$$a_4 = \frac{1}{20160}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{r^2-r-2}$	$-\frac{1}{4}$
a_2	$\frac{1}{r^4-5r^2+4}$	$\frac{1}{40}$
a_3	$-\frac{1}{(r^4-5r^2+4)r(r+3)}$	$-\frac{1}{720}$
a_4	$\frac{1}{(r^4-5r^2+4)r(r+3)(r^2+5r+4)}$	$\frac{1}{20160}$

For $n = 5$, using the above recursive equation gives

$$a_5 = -\frac{1}{(r+1)^2 (r-2) (r+2)^2 (-1+r) r (r+3) (r+4) (r+5)}$$

Which for the root $r = 3$ becomes

$$a_5 = -\frac{1}{806400}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{r^2-r-2}$	$-\frac{1}{4}$
a_2	$\frac{1}{r^4-5r^2+4}$	$\frac{1}{40}$
a_3	$-\frac{1}{(r^4-5r^2+4)r(r+3)}$	$-\frac{1}{720}$
a_4	$\frac{1}{(r^4-5r^2+4)r(r+3)(r^2+5r+4)}$	$\frac{1}{20160}$
a_5	$-\frac{1}{(r+1)^2(r-2)(r+2)^2(-1+r)r(r+3)(r+4)(r+5)}$	$-\frac{1}{806400}$

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= x^3(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= x^3\left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6)\right) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Let

$$r_1 - r_2 = N$$

Where N is positive integer which is the difference between the two roots. r_1 is taken as the larger root. Hence for this problem we have $N = 3$. Now we need to determine if C is zero or not. This is done by finding $\lim_{r \rightarrow r_2} a_3(r)$. If this limit exists, then $C = 0$, else we need to keep the log term and $C \neq 0$. The above table shows that

$$\begin{aligned} a_N &= a_3 \\ &= -\frac{1}{(r^4 - 5r^2 + 4)r(r+3)} \end{aligned}$$

Therefore

$$\begin{aligned} \lim_{r \rightarrow r_2} -\frac{1}{(r^4 - 5r^2 + 4)r(r+3)} &= \lim_{r \rightarrow 0} -\frac{1}{(r^4 - 5r^2 + 4)r(r+3)} \\ &= \text{undefined} \end{aligned}$$

Since the limit does not exist then the log term is needed. Therefore the second solution has the form

$$y_2(x) = Cy_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right)$$

Therefore

$$\begin{aligned}
 \frac{d}{dx}y_2(x) &= Cy_1'(x) \ln(x) + \frac{Cy_1(x)}{x} + \left(\sum_{n=0}^{\infty} \frac{b_n x^{n+r_2}(n+r_2)}{x} \right) \\
 &= Cy_1'(x) \ln(x) + \frac{Cy_1(x)}{x} + \left(\sum_{n=0}^{\infty} x^{-1+n+r_2} b_n (n+r_2) \right) \\
 \frac{d^2}{dx^2}y_2(x) &= Cy_1''(x) \ln(x) + \frac{2Cy_1'(x)}{x} - \frac{Cy_1(x)}{x^2} + \sum_{n=0}^{\infty} \left(\frac{b_n x^{n+r_2}(n+r_2)^2}{x^2} - \frac{b_n x^{n+r_2}(n+r_2)}{x^2} \right) \\
 &= Cy_1''(x) \ln(x) + \frac{2Cy_1'(x)}{x} - \frac{Cy_1(x)}{x^2} + \left(\sum_{n=0}^{\infty} x^{-2+n+r_2} b_n (n+r_2) (-1+n+r_2) \right)
 \end{aligned}$$

Substituting these back into the given ode $xy'' - 2y' + y = 0$ gives

$$\begin{aligned}
 &\left(Cy_1''(x) \ln(x) + \frac{2Cy_1'(x)}{x} - \frac{Cy_1(x)}{x^2} + \sum_{n=0}^{\infty} \left(\frac{b_n x^{n+r_2}(n+r_2)^2}{x^2} - \frac{b_n x^{n+r_2}(n+r_2)}{x^2} \right) \right) x \\
 &- 2Cy_1'(x) \ln(x) - \frac{2Cy_1(x)}{x} - 2 \left(\sum_{n=0}^{\infty} \frac{b_n x^{n+r_2}(n+r_2)}{x} \right) \\
 &+ Cy_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right) = 0
 \end{aligned}$$

Which can be written as

$$\begin{aligned}
 &\left((y_1''(x)x - 2y_1'(x) + y_1(x)) \ln(x) + \left(\frac{2y_1'(x)}{x} - \frac{y_1(x)}{x^2} \right) x - \frac{2y_1(x)}{x} \right) C \\
 &+ \left(\sum_{n=0}^{\infty} \left(\frac{b_n x^{n+r_2}(n+r_2)^2}{x^2} - \frac{b_n x^{n+r_2}(n+r_2)}{x^2} \right) \right) x \\
 &- 2 \left(\sum_{n=0}^{\infty} \frac{b_n x^{n+r_2}(n+r_2)}{x} \right) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right) = 0
 \end{aligned} \tag{7}$$

But since $y_1(x)$ is a solution to the ode, then

$$y_1''(x)x - 2y_1'(x) + y_1(x) = 0$$

Eq (7) simplifies to

$$\begin{aligned} & \left(\left(\frac{2y_1'(x)}{x} - \frac{y_1(x)}{x^2} \right) x - \frac{2y_1(x)}{x} \right) C \\ & + \left(\sum_{n=0}^{\infty} \left(\frac{b_n x^{n+r_2} (n+r_2)^2}{x^2} - \frac{b_n x^{n+r_2} (n+r_2)}{x^2} \right) \right) x \\ & - 2 \left(\sum_{n=0}^{\infty} \frac{b_n x^{n+r_2} (n+r_2)}{x} \right) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right) = 0 \end{aligned} \quad (8)$$

Substituting $y_1 = \sum_{n=0}^{\infty} a_n x^{n+r_1}$ into the above gives

$$\begin{aligned} & \frac{\left(2 \left(\sum_{n=0}^{\infty} x^{-1+n+r_1} a_n (n+r_1) \right) x - 3 \left(\sum_{n=0}^{\infty} a_n x^{n+r_1} \right) \right) C}{x} \\ & + \frac{\left(\sum_{n=0}^{\infty} x^{-2+n+r_2} b_n (n+r_2) (-1+n+r_2) \right) x^2 + \left(-2 \left(\sum_{n=0}^{\infty} x^{-1+n+r_2} b_n (n+r_2) \right) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right) \right) x}{x} \\ & = 0 \end{aligned} \quad (9)$$

Since $r_1 = 3$ and $r_2 = 0$ then the above becomes

$$\begin{aligned} & \frac{\left(2 \left(\sum_{n=0}^{\infty} x^{2+n} a_n (n+3) \right) x - 3 \left(\sum_{n=0}^{\infty} a_n x^{n+3} \right) \right) C}{x} \\ & + \frac{\left(\sum_{n=0}^{\infty} x^{-2+n} b_n n (n-1) \right) x^2 + \left(-2 \left(\sum_{n=0}^{\infty} x^{n-1} b_n n \right) + \left(\sum_{n=0}^{\infty} b_n x^n \right) \right) x}{x} = 0 \end{aligned} \quad (10)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} 2x^{2+n} a_n (n+3) C \right) + \sum_{n=0}^{\infty} (-3C x^{2+n} a_n) \\ & + \left(\sum_{n=0}^{\infty} n x^{n-1} b_n (n-1) \right) + \sum_{n=0}^{\infty} (-2x^{n-1} b_n n) + \left(\sum_{n=0}^{\infty} b_n x^n \right) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n-1} and adjusting the

power and the corresponding index gives

$$\begin{aligned}\sum_{n=0}^{\infty} 2x^{2+n}a_n(n+3)C &= \sum_{n=3}^{\infty} 2Ca_{n-3}nx^{n-1} \\ \sum_{n=0}^{\infty} (-3Cx^{2+n}a_n) &= \sum_{n=3}^{\infty} (-3Ca_{n-3}x^{n-1}) \\ \sum_{n=0}^{\infty} b_nx^n &= \sum_{n=1}^{\infty} b_{n-1}x^{n-1}\end{aligned}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n-1$.

$$\begin{aligned}\left(\sum_{n=3}^{\infty} 2Ca_{n-3}nx^{n-1}\right) + \sum_{n=3}^{\infty} (-3Ca_{n-3}x^{n-1}) + \left(\sum_{n=0}^{\infty} nx^{n-1}b_n(n-1)\right) \\ + \sum_{n=0}^{\infty} (-2x^{n-1}b_nn) + \left(\sum_{n=1}^{\infty} b_{n-1}x^{n-1}\right) = 0\end{aligned}\quad (2B)$$

For $n=0$ in Eq. (2B), we choose arbitray value for b_0 as $b_0=1$. For $n=1$, Eq (2B) gives

$$-2b_1 + b_0 = 0$$

Which when replacing the above values found already for b_n and the values found earlier for a_n and for C , gives

$$-2b_1 + 1 = 0$$

Solving the above for b_1 gives

$$b_1 = \frac{1}{2}$$

For $n=2$, Eq (2B) gives

$$-2b_2 + b_1 = 0$$

Which when replacing the above values found already for b_n and the values found earlier for a_n and for C , gives

$$-2b_2 + \frac{1}{2} = 0$$

Solving the above for b_2 gives

$$b_2 = \frac{1}{4}$$

For $n=N$, where $N=3$ which is the difference between the two roots, we are free to choose $b_3=0$. Hence for $n=3$, Eq (2B) gives

$$3C + \frac{1}{4} = 0$$

Which is solved for C . Solving for C gives

$$C = -\frac{1}{12}$$

For $n = 4$, Eq (2B) gives

$$5Ca_1 + b_3 + 4b_4 = 0$$

Which when replacing the above values found already for b_n and the values found earlier for a_n and for C , gives

$$4b_4 + \frac{5}{48} = 0$$

Solving the above for b_4 gives

$$b_4 = -\frac{5}{192}$$

For $n = 5$, Eq (2B) gives

$$7Ca_2 + b_4 + 10b_5 = 0$$

Which when replacing the above values found already for b_n and the values found earlier for a_n and for C , gives

$$10b_5 - \frac{13}{320} = 0$$

Solving the above for b_5 gives

$$b_5 = \frac{13}{3200}$$

Now that we found all b_n and C , we can calculate the second solution from

$$y_2(x) = Cy_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^{n+r_2} \right)$$

Using the above value found for $C = -\frac{1}{12}$ and all b_n , then the second solution becomes

$$\begin{aligned} y_2(x) = & -\frac{1}{12} \left(x^3 \left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6) \right) \right) \ln(x) \\ & + 1 + \frac{x}{2} + \frac{x^2}{4} - \frac{5x^4}{192} + \frac{13x^5}{3200} + O(x^6) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) = & c_1 y_1(x) + c_2 y_2(x) \\ = & c_1 x^3 \left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6) \right) \\ & + c_2 \left(-\frac{1}{12} \left(x^3 \left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6) \right) \right) \ln(x) + 1 + \frac{x}{2} \right. \\ & \left. + \frac{x^2}{4} - \frac{5x^4}{192} + \frac{13x^5}{3200} + O(x^6) \right) \end{aligned}$$

Hence the final solution is

$$y = y_h$$

$$= c_1 x^3 \left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6) \right) \\ + c_2 \left(-\frac{x^3 \left(1 - \frac{x}{4} + \frac{x^2}{40} - \frac{x^3}{720} + \frac{x^4}{20160} - \frac{x^5}{806400} + O(x^6) \right) \ln(x)}{12} + 1 + \frac{x}{2} + \frac{x^2}{4} - \frac{5x^4}{192} \right. \\ \left. + \frac{13x^5}{3200} + O(x^6) \right)$$

✓ **Maple.** Time used: 0.016 (sec). Leaf size: 56

```
Order:=6;
ode:=x*diff(diff(y(x),x),x)-2*diff(y(x),x)+y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = c_1 x^3 \left(1 - \frac{1}{4}x + \frac{1}{40}x^2 - \frac{1}{720}x^3 + \frac{1}{20160}x^4 - \frac{1}{806400}x^5 + O(x^6) \right) \\ + c_2 \left(\ln(x) \left(-x^3 + \frac{1}{4}x^4 - \frac{1}{40}x^5 + O(x^6) \right) + \left(12 + 6x + 3x^2 - \frac{5}{16}x^4 + \frac{39}{800}x^5 + O(x^6) \right) \right)$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
    -> Bessel
    <- Bessel successful
<- special function solution successful
```

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) - 2 \frac{d}{dx} y(x) + y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -\frac{y(x)}{x} + \frac{2 \left(\frac{d}{dx} y(x) \right)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) - \frac{2 \left(\frac{d}{dx} y(x) \right)}{x} + \frac{y(x)}{x} = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- Define functions

$$\left[P_2(x) = -\frac{2}{x}, P_3(x) = \frac{1}{x} \right]$$

- $x \cdot P_2(x)$ is analytic at $x = 0$

$$(x \cdot P_2(x)) \Big|_{x=0} = -2$$

- $x^2 \cdot P_3(x)$ is analytic at $x = 0$

$$(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$$

- $x = 0$ is a regular singular point

Check to see if $x_0 = 0$ is a regular singular point

$$x_0 = 0$$

- Multiply by denominators

$$x \left(\frac{d^2}{dx^2} y(x) \right) - 2 \frac{d}{dx} y(x) + y(x) = 0$$

- Assume series solution for $y(x)$

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$$

- Rewrite ODE with series expansions

- Convert $\frac{d}{dx} y(x)$ to series expansion

$$\frac{d}{dx} y(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}$$

- Shift index using $k- > k+1$

$$\frac{d}{dx} y(x) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) x^{k+r}$$

- Convert $x \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion

$$x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) x^{k+r-1}$$

- Shift index using $k- > k+1$

$$x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=-1}^{\infty} a_{k+1} (k+1+r) (k+r) x^{k+r}$$

Rewrite ODE with series expansions

$$a_0 r(-3+r) x^{-1+r} + \left(\sum_{k=0}^{\infty} (a_{k+1} (k+1+r) (k-2+r) + a_k) x^{k+r} \right) = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation
 $r(-3+r) = 0$
- Values of r that satisfy the indicial equation
 $r \in \{0, 3\}$
- Each term in the series must be 0, giving the recursion relation
 $a_{k+1}(k+1+r)(k-2+r) + a_k = 0$
- Recursion relation that defines series solution to ODE
 $a_{k+1} = -\frac{a_k}{(k+1+r)(k-2+r)}$
- Recursion relation for $r = 0$
 $a_{k+1} = -\frac{a_k}{(k+1)(k-2)}$
- Series not valid for $r = 0$, division by 0 in the recursion relation at $k = 2$
 $a_{k+1} = -\frac{a_k}{(k+1)(k-2)}$
- Recursion relation for $r = 3$
 $a_{k+1} = -\frac{a_k}{(k+4)(k+1)}$
- Solution for $r = 3$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k+3}, a_{k+1} = -\frac{a_k}{(k+4)(k+1)} \right]$$

✓ **Mathematica.** Time used: 0.015 (sec). Leaf size: 79

```
ode=x*D[y[x],{x,2}]-2*D[y[x],x]+y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \left(\frac{1}{48} (x-4)x^3 \log(x) + \frac{1}{576} (-19x^4 + 16x^3 + 144x^2 + 288x + 576) \right) \\ + c_2 \left(\frac{x^7}{20160} - \frac{x^6}{720} + \frac{x^5}{40} - \frac{x^4}{4} + x^3 \right)$$

✓ **Sympy.** Time used: 0.196 (sec). Leaf size: 20

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*Derivative(y(x), (x, 2)) + y(x) - 2*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_1 x^3 \left(\frac{x^2}{40} - \frac{x}{4} + 1 \right) + O(x^6)$$

2.17.7 Problem 17

Local contents

✓ Maple	2676
✓ Mathematica	2678
✓ Sympy	2678

Internal problem ID [8083]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 17

Date solved : Thursday, November 27, 2025 at 07:33:18 AM

CAS classification : [_Lienard]

$$xy'' + 2y' + yx = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$xy'' + 2y' + yx = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = \frac{2}{x}$$

$$q(x) = 1$$

Table 2.319: Table $p(x), q(x)$ singularities.

$p(x) = \frac{2}{x}$	
singularity	type
$x = 0$	“regular”

$q(x) = 1$	
singularity	type

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : [0]

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$xy'' + 2y' + yx = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x + 2 \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) x = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} 2(n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=0}^{\infty} x^{1+n+r} a_n \right) = 0 \quad (2A)$$

The next step is to make all powers of x be $n+r-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r-1} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} x^{1+n+r} a_n = \sum_{n=2}^{\infty} a_{n-2} x^{n+r-1}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n+r-1$.

$$\left(\sum_{n=0}^{\infty} x^{n+r-1} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} 2(n+r) a_n x^{n+r-1} \right) + \left(\sum_{n=2}^{\infty} a_{n-2} x^{n+r-1} \right) = 0 \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$x^{n+r-1} a_n (n+r) (n+r-1) + 2(n+r) a_n x^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$x^{-1+r} a_0 r (-1+r) + 2r a_0 x^{-1+r} = 0$$

Or

$$(x^{-1+r} r (-1+r) + 2r x^{-1+r}) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$r x^{-1+r} (1+r) = 0$$

Since the above is true for all x then the indicial equation becomes

$$r(1+r) = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 0$$

$$r_2 = -1$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$r x^{-1+r} (1+r) = 0$$

Solving for r gives the roots of the indicial equation as $[0, -1]$.

Since $r_1 - r_2 = 1$ is an integer, then we can construct two linearly independent solutions

$$y_1(x) = x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = C y_1(x) \ln(x) + x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$y_2(x) = C y_1(x) \ln(x) + \frac{\sum_{n=0}^{\infty} b_n x^n}{x}$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$y_2(x) = C y_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^{n-1} \right)$$

Where C above can be zero. We start by finding y_1 . Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. Substituting $n = 1$ in Eq. (2B) gives

$$a_1 = 0$$

For $2 \leq n$ the recursive equation is

$$a_n(n+r)(n+r-1) + 2a_n(n+r) + a_{n-2} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-2}}{n^2 + 2nr + r^2 + n + r} \quad (4)$$

Which for the root $r = 0$ becomes

$$a_n = -\frac{a_{n-2}}{n(1+n)} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0

For $n = 2$, using the above recursive equation gives

$$a_2 = -\frac{1}{r^2 + 5r + 6}$$

Which for the root $r = 0$ becomes

$$a_2 = -\frac{1}{6}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{6}$

For $n = 3$, using the above recursive equation gives

$$a_3 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{6}$
a_3	0	0

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{r^4 + 14r^3 + 71r^2 + 154r + 120}$$

Which for the root $r = 0$ becomes

$$a_4 = \frac{1}{120}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{6}$
a_3	0	0
a_4	$\frac{1}{r^4+14r^3+71r^2+154r+120}$	$\frac{1}{120}$

For $n = 5$, using the above recursive equation gives

$$a_5 = 0$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	0	0
a_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{6}$
a_3	0	0
a_4	$\frac{1}{r^4+14r^3+71r^2+154r+120}$	$\frac{1}{120}$
a_5	0	0

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots \\ &= 1 - \frac{x^2}{6} + \frac{x^4}{120} + O(x^6) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Let

$$r_1 - r_2 = N$$

Where N is positive integer which is the difference between the two roots. r_1 is taken as the larger root. Hence for this problem we have $N = 1$. Now we need to determine if C is zero or not. This is done by finding $\lim_{r \rightarrow r_2} a_1(r)$. If this limit exists, then $C = 0$, else we need to keep the log term and $C \neq 0$. The above table shows that

$$\begin{aligned} a_N &= a_1 \\ &= 0 \end{aligned}$$

Therefore

$$\begin{aligned} \lim_{r \rightarrow r_2} 0 &= \lim_{r \rightarrow -1} 0 \\ &= 0 \end{aligned}$$

The limit is 0. Since the limit exists then the log term is not needed and we can set $C = 0$. Therefore the second solution has the form

$$\begin{aligned} y_2(x) &= \sum_{n=0}^{\infty} b_n x^{n+r} \\ &= \sum_{n=0}^{\infty} b_n x^{n-1} \end{aligned}$$

Eq (3) derived above is used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. Substituting $n = 1$ in Eq(3) gives

$$b_1 = 0$$

For $2 \leq n$ the recursive equation is

$$b_n(n+r)(n+r-1) + 2(n+r)b_n + b_{n-2} = 0 \quad (4)$$

Which for the root $r = -1$ becomes

$$b_n(n-1)(n-2) + 2(n-1)b_n + b_{n-2} = 0 \quad (4A)$$

Solving for b_n from the recursive equation (4) gives

$$b_n = -\frac{b_{n-2}}{n^2 + 2nr + r^2 + n + r} \quad (5)$$

Which for the root $r = -1$ becomes

$$b_n = -\frac{b_{n-2}}{n^2 - n} \quad (6)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = -1$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0

For $n = 2$, using the above recursive equation gives

$$b_2 = -\frac{1}{r^2 + 5r + 6}$$

Which for the root $r = -1$ becomes

$$b_2 = -\frac{1}{2}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{2}$

For $n = 3$, using the above recursive equation gives

$$b_3 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{2}$
b_3	0	0

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{1}{(r^2 + 5r + 6)(r^2 + 9r + 20)}$$

Which for the root $r = -1$ becomes

$$b_4 = \frac{1}{24}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{2}$
b_3	0	0
b_4	$\frac{1}{r^4+14r^3+71r^2+154r+120}$	$\frac{1}{24}$

For $n = 5$, using the above recursive equation gives

$$b_5 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	0	0
b_2	$-\frac{1}{r^2+5r+6}$	$-\frac{1}{2}$
b_3	0	0
b_4	$\frac{1}{r^4+14r^3+71r^2+154r+120}$	$\frac{1}{24}$
b_5	0	0

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= 1(b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots) \\ &= \frac{1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)}{x} \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1y_1(x) + c_2y_2(x) \\ &= c_1\left(1 - \frac{x^2}{6} + \frac{x^4}{120} + O(x^6)\right) + \frac{c_2\left(1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)\right)}{x} \end{aligned}$$

Hence the final solution is

$$y = y_h$$

$$= c_1 \left(1 - \frac{x^2}{6} + \frac{x^4}{120} + O(x^6) \right) + \frac{c_2 \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6) \right)}{x}$$

✓ **Maple.** Time used: 0.016 (sec). Leaf size: 32

```
Order:=6;
ode:=x*diff(diff(y(x),x),x)+2*diff(y(x),x)+y(x)*x = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = c_1 \left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 + O(x^6) \right) + \frac{c_2 \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + O(x^6) \right)}{x}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Group is reducible or imprimitive
<- Kovacic's algorithm successful
```

Maple step by step

Let's solve

$$x \left(\frac{d^2}{dx^2} y(x) \right) + 2 \frac{d}{dx} y(x) + xy(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = -y(x) - \frac{2 \left(\frac{d}{dx} y(x) \right)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + \frac{2 \left(\frac{d}{dx} y(x) \right)}{x} + y(x) = 0$$

- Check to see if $x_0 = 0$ is a regular singular point

- Define functions
 $[P_2(x) = \frac{2}{x}, P_3(x) = 1]$
- $x \cdot P_2(x)$ is analytic at $x = 0$
 $(x \cdot P_2(x)) \Big|_{x=0} = 2$
- $x^2 \cdot P_3(x)$ is analytic at $x = 0$
 $(x^2 \cdot P_3(x)) \Big|_{x=0} = 0$
- $x = 0$ is a regular singular point
 Check to see if $x_0 = 0$ is a regular singular point
 $x_0 = 0$
- Multiply by denominators
 $x \left(\frac{d^2}{dx^2} y(x) \right) + 2 \frac{d}{dx} y(x) + xy(x) = 0$
- Assume series solution for $y(x)$
 $y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$
- Rewrite ODE with series expansions
- Convert $x \cdot y(x)$ to series expansion
 $x \cdot y(x) = \sum_{k=0}^{\infty} a_k x^{k+r+1}$
- Shift index using $k- > k-1$
 $x \cdot y(x) = \sum_{k=1}^{\infty} a_{k-1} x^{k+r}$
- Convert $\frac{d}{dx} y(x)$ to series expansion
 $\frac{d}{dx} y(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}$
- Shift index using $k- > k+1$
 $\frac{d}{dx} y(x) = \sum_{k=-1}^{\infty} a_{k+1} (k+r+1) x^{k+r}$
- Convert $x \cdot \left(\frac{d^2}{dx^2} y(x) \right)$ to series expansion
 $x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) x^{k+r-1}$
- Shift index using $k- > k+1$
 $x \cdot \left(\frac{d^2}{dx^2} y(x) \right) = \sum_{k=-1}^{\infty} a_{k+1} (k+r+1) (k+r) x^{k+r}$
- Rewrite ODE with series expansions
 $a_0 r(1+r) x^{-1+r} + a_1 (1+r) (2+r) x^r + \left(\sum_{k=1}^{\infty} (a_{k+1} (k+r+1) (k+2+r) + a_{k-1}) x^{k+r} \right) = 0$
- a_0 cannot be 0 by assumption, giving the indicial equation
 $r(1+r) = 0$
- Values of r that satisfy the indicial equation
 $r \in \{-1, 0\}$
- Each term must be 0

- $a_1(1+r)(2+r) = 0$
- Each term in the series must be 0, giving the recursion relation
 $a_{k+1}(k+r+1)(k+2+r) + a_{k-1} = 0$
- Shift index using $k \rightarrow k+1$
 $a_{k+2}(k+2+r)(k+3+r) + a_k = 0$
- Recursion relation that defines series solution to ODE
 $a_{k+2} = -\frac{a_k}{(k+2+r)(k+3+r)}$
- Recursion relation for $r = -1$
 $a_{k+2} = -\frac{a_k}{(k+1)(k+2)}$
- Solution for $r = -1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^{k-1}, a_{k+2} = -\frac{a_k}{(k+1)(k+2)}, 0 = 0 \right]$$
- Recursion relation for $r = 0$
 $a_{k+2} = -\frac{a_k}{(k+2)(k+3)}$
- Solution for $r = 0$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k x^k, a_{k+2} = -\frac{a_k}{(k+2)(k+3)}, 2a_1 = 0 \right]$$
- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k x^{k-1} \right) + \left(\sum_{k=0}^{\infty} b_k x^k \right), a_{k+2} = -\frac{a_k}{(k+1)(k+2)}, 0 = 0, b_{k+2} = -\frac{b_k}{(k+2)(k+3)}, 2b_1 = 0 \right]$$

✓ **Mathematica.** Time used: 0.007 (sec). Leaf size: 42

```
ode=x*D[y[x],{x,2}]+2*D[y[x],x]+x*y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \left(\frac{x^3}{24} - \frac{x}{2} + \frac{1}{x} \right) + c_2 \left(\frac{x^4}{120} - \frac{x^2}{6} + 1 \right)$$

✓ **Sympy.** Time used: 0.267 (sec). Leaf size: 39

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x*y(x) + x*Derivative(y(x), (x, 2)) + 2*Derivative(y(x), x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

$$y(x) = C_2 \left(\frac{x^4}{120} - \frac{x^2}{6} + 1 \right) + \frac{C_1 \left(-\frac{x^6}{720} + \frac{x^4}{24} - \frac{x^2}{2} + 1 \right)}{x} + O(x^6)$$

2.17.8 Problem 18

Local contents

✓ Maple	2688
✓ Mathematica	2690
✓ Sympy	2690

Internal problem ID [8084]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 18

Date solved : Thursday, November 27, 2025 at 07:33:19 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$x^2(1+x)y'' + x(1+x)y' - y = 0$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$(x^3 + x^2)y'' + (x^2 + x)y' - y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = \frac{1}{x}$$

$$q(x) = -\frac{1}{x^2(1+x)}$$

Table 2.321: Table $p(x), q(x)$ singularities.

$p(x) = \frac{1}{x}$	
singularity	type
$x = 0$	“regular”

$q(x) = -\frac{1}{x^2(1+x)}$	
singularity	type
$x = -1$	“regular”
$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0, -1, \infty]$

Irregular singular points : $[\]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$x^2(1+x)y'' + (x^2+x)y' - y = 0$$

Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$\begin{aligned} & x^2(1+x) \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) \\ & + (x^2+x) \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) - \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \end{aligned} \quad (1)$$

Which simplifies to

$$\begin{aligned} & \left(\sum_{n=0}^{\infty} x^{1+n+r} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} x^{n+r} a_n (n+r)(n+r-1) \right) \\ & + \left(\sum_{n=0}^{\infty} x^{1+n+r} a_n (n+r) \right) + \left(\sum_{n=0}^{\infty} x^{n+r} a_n (n+r) \right) + \sum_{n=0}^{\infty} (-a_n x^{n+r}) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of x be $n+r$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r} and adjusting the power and the corresponding index gives

$$\begin{aligned} \sum_{n=0}^{\infty} x^{1+n+r} a_n (n+r)(n+r-1) &= \sum_{n=1}^{\infty} a_{n-1} (n+r-1)(n+r-2) x^{n+r} \\ \sum_{n=0}^{\infty} x^{1+n+r} a_n (n+r) &= \sum_{n=1}^{\infty} a_{n-1} (n+r-1) x^{n+r} \end{aligned}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n + r$.

$$\begin{aligned} & \left(\sum_{n=1}^{\infty} a_{n-1}(n+r-1)(n+r-2)x^{n+r} \right) + \left(\sum_{n=0}^{\infty} x^{n+r} a_n(n+r)(n+r-1) \right) \\ & + \left(\sum_{n=1}^{\infty} a_{n-1}(n+r-1)x^{n+r} \right) + \left(\sum_{n=0}^{\infty} x^{n+r} a_n(n+r) \right) + \sum_{n=0}^{\infty} (-a_n x^{n+r}) = 0 \end{aligned} \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$x^{n+r} a_n(n+r)(n+r-1) + x^{n+r} a_n(n+r) - a_n x^{n+r} = 0$$

When $n = 0$ the above becomes

$$x^r a_0 r(-1+r) + x^r a_0 r - a_0 x^r = 0$$

Or

$$(x^r r(-1+r) + x^r r - x^r) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(r^2 - 1) x^r = 0$$

Since the above is true for all x then the indicial equation becomes

$$r^2 - 1 = 0$$

Solving for r gives the roots of the indicial equation as

$$\begin{aligned} r_1 &= 1 \\ r_2 &= -1 \end{aligned}$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(r^2 - 1) x^r = 0$$

Solving for r gives the roots of the indicial equation as $[1, -1]$.

Since $r_1 - r_2 = 2$ is an integer, then we can construct two linearly independent solutions

$$\begin{aligned} y_1(x) &= x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right) \\ y_2(x) &= C y_1(x) \ln(x) + x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right) \end{aligned}$$

Or

$$y_1(x) = x \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = C y_1(x) \ln(x) + \frac{\sum_{n=0}^{\infty} b_n x^n}{x}$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{1+n}$$

$$y_2(x) = C y_1(x) \ln(x) + \left(\sum_{n=0}^{\infty} b_n x^{n-1} \right)$$

Where C above can be zero. We start by finding y_1 . Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$a_{n-1}(n+r-1)(n+r-2) + a_n(n+r)(n+r-1) + a_{n-1}(n+r-1) + a_n(n+r) - a_n = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-1}(n+r-1)}{1+n+r} \quad (4)$$

Which for the root $r = 1$ becomes

$$a_n = -\frac{a_{n-1}n}{2+n} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 1$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = -\frac{r}{2+r}$$

Which for the root $r = 1$ becomes

$$a_1 = -\frac{1}{3}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{r}{2+r}$	$-\frac{1}{3}$

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{r(r+1)}{(2+r)(3+r)}$$

Which for the root $r = 1$ becomes

$$a_2 = \frac{1}{6}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{r}{2+r}$	$-\frac{1}{3}$
a_2	$\frac{r(r+1)}{(2+r)(3+r)}$	$\frac{1}{6}$

For $n = 3$, using the above recursive equation gives

$$a_3 = -\frac{r(r+1)}{(3+r)(4+r)}$$

Which for the root $r = 1$ becomes

$$a_3 = -\frac{1}{10}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{r}{2+r}$	$-\frac{1}{3}$
a_2	$\frac{r(r+1)}{(2+r)(3+r)}$	$\frac{1}{6}$
a_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	$-\frac{1}{10}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{r(r+1)}{(4+r)(5+r)}$$

Which for the root $r = 1$ becomes

$$a_4 = \frac{1}{15}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{r}{2+r}$	$-\frac{1}{3}$
a_2	$\frac{r(r+1)}{(2+r)(3+r)}$	$\frac{1}{6}$
a_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	$-\frac{1}{10}$
a_4	$\frac{r(r+1)}{(4+r)(5+r)}$	$\frac{1}{15}$

For $n = 5$, using the above recursive equation gives

$$a_5 = -\frac{r(r+1)}{(5+r)(6+r)}$$

Which for the root $r = 1$ becomes

$$a_5 = -\frac{1}{21}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{r}{2+r}$	$-\frac{1}{3}$
a_2	$\frac{r(r+1)}{(2+r)(3+r)}$	$\frac{1}{6}$
a_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	$-\frac{1}{10}$
a_4	$\frac{r(r+1)}{(4+r)(5+r)}$	$\frac{1}{15}$
a_5	$-\frac{r(r+1)}{(5+r)(6+r)}$	$-\frac{1}{21}$

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= x\left(1 - \frac{x}{3} + \frac{x^2}{6} - \frac{x^3}{10} + \frac{x^4}{15} - \frac{x^5}{21} + O(x^6)\right) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Let

$$r_1 - r_2 = N$$

Where N is positive integer which is the difference between the two roots. r_1 is taken as the larger root. Hence for this problem we have $N = 2$. Now we need to determine if C is zero or not. This is done by finding $\lim_{r \rightarrow r_2} a_2(r)$. If this limit exists, then $C = 0$, else we need to keep the log term and $C \neq 0$. The above table shows that

$$\begin{aligned} a_N &= a_2 \\ &= \frac{r(r+1)}{(2+r)(3+r)} \end{aligned}$$

Therefore

$$\begin{aligned}\lim_{r \rightarrow r_2} \frac{r(r+1)}{(2+r)(3+r)} &= \lim_{r \rightarrow -1} \frac{r(r+1)}{(2+r)(3+r)} \\ &= 0\end{aligned}$$

The limit is 0. Since the limit exists then the log term is not needed and we can set $C = 0$. Therefore the second solution has the form

$$\begin{aligned}y_2(x) &= \sum_{n=0}^{\infty} b_n x^{n+r} \\ &= \sum_{n=0}^{\infty} b_n x^{n-1}\end{aligned}$$

Eq (3) derived above is used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. For $1 \leq n$ the recursive equation is

$$b_{n-1}(n+r-1)(n+r-2) + b_n(n+r)(n+r-1) + b_{n-1}(n+r-1) + b_n(n+r) - b_n = 0 \quad (4)$$

Which for the root $r = -1$ becomes

$$b_{n-1}(n-2)(n-3) + b_n(n-1)(n-2) + b_{n-1}(n-2) + b_n(n-1) - b_n = 0 \quad (4A)$$

Solving for b_n from the recursive equation (4) gives

$$b_n = -\frac{b_{n-1}(n+r-1)}{1+n+r} \quad (5)$$

Which for the root $r = -1$ becomes

$$b_n = -\frac{b_{n-1}(n-2)}{n} \quad (6)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = -1$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1

For $n = 1$, using the above recursive equation gives

$$b_1 = -\frac{r}{2+r}$$

Which for the root $r = -1$ becomes

$$b_1 = 1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{r}{2+r}$	1

For $n = 2$, using the above recursive equation gives

$$b_2 = \frac{r(r+1)}{(2+r)(3+r)}$$

Which for the root $r = -1$ becomes

$$b_2 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{r}{2+r}$	1
b_2	$\frac{r(r+1)}{(2+r)(3+r)}$	0

For $n = 3$, using the above recursive equation gives

$$b_3 = -\frac{r(r+1)}{(3+r)(4+r)}$$

Which for the root $r = -1$ becomes

$$b_3 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{r}{2+r}$	1
b_2	$\frac{r(r+1)}{(2+r)(3+r)}$	0
b_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	0

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{r(r+1)}{(4+r)(5+r)}$$

Which for the root $r = -1$ becomes

$$b_4 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{r}{2+r}$	1
b_2	$\frac{r(r+1)}{(2+r)(3+r)}$	0
b_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	0
b_4	$\frac{r(r+1)}{(4+r)(5+r)}$	0

For $n = 5$, using the above recursive equation gives

$$b_5 = -\frac{r(r+1)}{(5+r)(6+r)}$$

Which for the root $r = -1$ becomes

$$b_5 = 0$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{r}{2+r}$	1
b_2	$\frac{r(r+1)}{(2+r)(3+r)}$	0
b_3	$-\frac{r(r+1)}{(3+r)(4+r)}$	0
b_4	$\frac{r(r+1)}{(4+r)(5+r)}$	0
b_5	$-\frac{r(r+1)}{(5+r)(6+r)}$	0

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= x(b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots) \\ &= \frac{1 + x + O(x^6)}{x} \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1y_1(x) + c_2y_2(x) \\ &= c_1x \left(1 - \frac{x}{3} + \frac{x^2}{6} - \frac{x^3}{10} + \frac{x^4}{15} - \frac{x^5}{21} + O(x^6) \right) + \frac{c_2(1 + x + O(x^6))}{x} \end{aligned}$$

Hence the final solution is

$$\begin{aligned} y &= y_h \\ &= c_1x \left(1 - \frac{x}{3} + \frac{x^2}{6} - \frac{x^3}{10} + \frac{x^4}{15} - \frac{x^5}{21} + O(x^6) \right) + \frac{c_2(1 + x + O(x^6))}{x} \end{aligned}$$

✓ **Maple.** Time used: 0.013 (sec). Leaf size: 37

```
Order:=6;
ode:=x^2*(1+x)*diff(diff(y(x),x),x)+x*(1+x)*diff(y(x),x)-y(x) = 0;
dsolve(ode,y(x),type='series',x=0);
```

$$y = c_1 x \left(1 - \frac{1}{3}x + \frac{1}{6}x^2 - \frac{1}{10}x^3 + \frac{1}{15}x^4 - \frac{1}{21}x^5 + O(x^6) \right) + \frac{c_2(-2 - 2x + O(x^6))}{x}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Group is reducible, not completely reducible
<- Kovacic's algorithm successful
```

Maple step by step

Let's solve

$$x^2(x+1) \left(\frac{d^2}{dx^2} y(x) \right) + x(x+1) \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dx^2} y(x)$$

- Isolate 2nd derivative

$$\frac{d^2}{dx^2} y(x) = \frac{y(x)}{x^2(x+1)} - \frac{\frac{d}{dx} y(x)}{x}$$

- Group terms with $y(x)$ on the lhs of the ODE and the rest on the rhs of the ODE; ODE is linear

$$\frac{d^2}{dx^2} y(x) + \frac{\frac{d}{dx} y(x)}{x} - \frac{y(x)}{x^2(x+1)} = 0$$

- Check to see if x_0 is a regular singular point

- Define functions

$$\left[P_2(x) = \frac{1}{x}, P_3(x) = -\frac{1}{x^2(x+1)} \right]$$

- $(x+1) \cdot P_2(x)$ is analytic at $x = -1$

$$\left. ((x+1) \cdot P_2(x)) \right|_{x=-1} = 0$$

- $(x+1)^2 \cdot P_3(x)$ is analytic at $x = -1$

$$\left. ((x+1)^2 \cdot P_3(x)) \right|_{x=-1} = 0$$

- $x = -1$ is a regular singular point

Check to see if x_0 is a regular singular point

$$x_0 = -1$$

- Multiply by denominators

$$x^2(x+1) \left(\frac{d^2}{dx^2} y(x) \right) + x(x+1) \left(\frac{d}{dx} y(x) \right) - y(x) = 0$$

- Change variables using $x = u - 1$ so that the regular singular point is at $u = 0$

$$(u^3 - 2u^2 + u) \left(\frac{d^2}{du^2} y(u) \right) + (u^2 - u) \left(\frac{d}{du} y(u) \right) - y(u) = 0$$

- Assume series solution for $y(u)$

$$y(u) = \sum_{k=0}^{\infty} a_k u^{k+r}$$

- Rewrite ODE with series expansions

- Convert $u^m \cdot \left(\frac{d}{du} y(u) \right)$ to series expansion for $m = 1..2$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) u^{k+r-1+m}$$

- Shift index using $k \rightarrow k+1-m$

$$u^m \cdot \left(\frac{d}{du} y(u) \right) = \sum_{k=-1+m}^{\infty} a_{k+1-m} (k+1-m+r) u^{k+r}$$

- Convert $u^m \cdot \left(\frac{d^2}{du^2} y(u) \right)$ to series expansion for $m = 1..3$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=0}^{\infty} a_k (k+r) (k+r-1) u^{k+r-2+m}$$

- Shift index using $k \rightarrow k+2-m$

$$u^m \cdot \left(\frac{d^2}{du^2} y(u) \right) = \sum_{k=-2+m}^{\infty} a_{k+2-m} (k+2-m+r) (k+1-m+r) u^{k+r}$$

Rewrite ODE with series expansions

$$a_0 r(-1+r) u^{-1+r} + (a_1(1+r)r - a_0(2r^2 - r + 1)) u^r + \left(\sum_{k=1}^{\infty} (a_{k+1}(k+1+r)(k+r) - a_k(2k^2 + 4kr + 2r^2 - k - r + 1)) u^{k+r} \right) = 0$$

- a_0 cannot be 0 by assumption, giving the indicial equation

$$r(-1+r) = 0$$

- Values of r that satisfy the indicial equation

$$r \in \{0, 1\}$$

- Each term must be 0

$$a_1(1+r)r - a_0(2r^2 - r + 1) = 0$$

- Each term in the series must be 0, giving the recursion relation

$$a_{k+1}(k+1+r)(k+r) - a_k(2k^2 + 4kr + 2r^2 - k - r + 1) + a_{k-1}(k+r-1)^2 = 0$$

- Shift index using $k \rightarrow k+1$

$$a_{k+2}(k+2+r)(k+1+r) - a_{k+1}(2(k+1)^2 + 4(k+1)r + 2r^2 - k - r) + a_k(k+r)^2 = 0$$

- Recursion relation that defines series solution to ODE

$$a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} + 2k r a_k - 4k r a_{k+1} + r^2 a_k - 2r^2 a_{k+1} - 3k a_{k+1} - 3r a_{k+1} - 2a_{k+1}}{(k+2+r)(k+1+r)}$$

- Recursion relation for $r = 0$

- $$a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} - 3k a_{k+1} - 2a_{k+1}}{(k+2)(k+1)}$$
- Solution for $r = 0$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^k, a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} - 3k a_{k+1} - 2a_{k+1}}{(k+2)(k+1)}, -a_0 = 0 \right]$$
- Revert the change of variables $u = x + 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x+1)^k, a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} - 3k a_{k+1} - 2a_{k+1}}{(k+2)(k+1)}, -a_0 = 0 \right]$$
- Recursion relation for $r = 1$

$$a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} + 2k a_k - 7k a_{k+1} + a_k - 7a_{k+1}}{(k+3)(k+2)}$$
- Solution for $r = 1$

$$\left[y(u) = \sum_{k=0}^{\infty} a_k u^{k+1}, a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} + 2k a_k - 7k a_{k+1} + a_k - 7a_{k+1}}{(k+3)(k+2)}, 2a_1 - 2a_0 = 0 \right]$$
- Revert the change of variables $u = x + 1$

$$\left[y(x) = \sum_{k=0}^{\infty} a_k (x+1)^{k+1}, a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} + 2k a_k - 7k a_{k+1} + a_k - 7a_{k+1}}{(k+3)(k+2)}, 2a_1 - 2a_0 = 0 \right]$$
- Combine solutions and rename parameters

$$\left[y(x) = \left(\sum_{k=0}^{\infty} a_k (x+1)^k \right) + \left(\sum_{k=0}^{\infty} b_k (x+1)^{k+1} \right), a_{k+2} = -\frac{k^2 a_k - 2k^2 a_{k+1} - 3k a_{k+1} - 2a_{k+1}}{(k+2)(k+1)}, -a_0 = 0, b_{k+2} \right]$$

✓ **Mathematica.** Time used: 0.016 (sec). Leaf size: 45

```
ode=x^2*(x+1)*D[y[x],{x,2}]+x*(x+1)*D[y[x],x]-y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_2 \left(\frac{x^5}{15} - \frac{x^4}{10} + \frac{x^3}{6} - \frac{x^2}{3} + x \right) + c_1 \left(\frac{1}{x} + 1 \right)$$

✓ **Sympy.** Time used: 0.313 (sec). Leaf size: 12

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**2*(x + 1)*Derivative(y(x), (x, 2)) + x*(x + 1)*Derivative(y(x), x) -
    y(x), 0)
ics = {}
dsolve(ode,func=y(x),ics=ics,hint="2nd_power_series_regular",x0=0,n=6)
```

$$y(x) = C_2 x + \frac{C_1}{x} + O(x^6)$$

2.17.9 Problem 19

Local contents

✓ Maple	2702
✓ Mathematica	2703
✗ Sympy	2703

Internal problem ID [8085]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 19

Date solved : Thursday, November 27, 2025 at 07:33:20 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$2xy'' + y' - y = 1 + x$$

Series expansion around $x = 0$.

The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$2xy'' + y' - y = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$y'' + p(x)y' + q(x)y = 0$$

Where

$$p(x) = \frac{1}{2x}$$

$$q(x) = -\frac{1}{2x}$$

Table 2.323: Table $p(x), q(x)$ singularities.

$p(x) = \frac{1}{2x}$	
singularity	type
$x = 0$	“regular”

$q(x) = -\frac{1}{2x}$	
singularity	type
$x = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $x = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$2xy'' + y' - y = 1 + x$$

Since this is an inhomogeneous, then let the solution be

$$y = y_h + y_p$$

Where y_h is the solution to the homogeneous ode $2xy'' + y' - y = 0$, and y_p is a particular solution to the inhomogeneous ode. which is found using the balance equation generated from indicial equation

First, we solve for y_h Let the solution be represented as Frobenius power series of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting the above back into the ode gives

$$2 \left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \right) x + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) - \left(\sum_{n=0}^{\infty} a_n x^{n+r} \right) = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=0}^{\infty} 2x^{n+r-1} a_n (n+r)(n+r-1) \right) + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \sum_{n=0}^{\infty} (-a_n x^{n+r}) = 0 \quad (2A)$$

The next step is to make all powers of x be $n+r-1$ in each summation term. Going over each summation term above with power of x in it which is not already x^{n+r-1} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} (-a_n x^{n+r}) = \sum_{n=1}^{\infty} (-a_{n-1} x^{n+r-1})$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of x are the same and equal to $n + r - 1$.

$$\left(\sum_{n=0}^{\infty} 2x^{n+r-1} a_n (n+r) (n+r-1) \right) + \left(\sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \right) + \sum_{n=1}^{\infty} (-a_{n-1} x^{n+r-1}) = 0 \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$2x^{n+r-1} a_n (n+r) (n+r-1) + (n+r) a_n x^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$2x^{-1+r} a_0 r (-1+r) + r a_0 x^{-1+r} = 0$$

Or

$$(2x^{-1+r} r (-1+r) + r x^{-1+r}) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$r x^{-1+r} (-1+2r) = 0$$

Since the above is true for all x then the indicial equation becomes

$$2r^2 - r = 0$$

Solving for r gives the roots of the indicial equation as

$$\begin{aligned} r_1 &= \frac{1}{2} \\ r_2 &= 0 \end{aligned}$$

The corresponding balance equation is found by replacing r by m and a by c to avoid confusing terms between particular solution and the homogeneous solution. Hence the balance equation is

$$(2x^{-1+m} m (-1+m) + m x^{-1+m}) c_0 = 1 + x$$

This equation will be used later to find the particular solution.

Since $a_0 \neq 0$ then the indicial equation becomes

$$r x^{-1+r} (-1+2r) = 0$$

Solving for r gives the roots of the indicial equation as $[\frac{1}{2}, 0]$.

Since $r_1 - r_2 = \frac{1}{2}$ is not an integer, then we can construct two linearly independent solutions

$$y_1(x) = x^{r_1} \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$y_2(x) = x^{r_2} \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

Or

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+\frac{1}{2}}$$

$$y_2(x) = \sum_{n=0}^{\infty} b_n x^n$$

We start by finding $y_1(x)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$2a_n(n+r)(n+r-1) + a_n(n+r) - a_{n-1} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = \frac{a_{n-1}}{2n^2 + 4nr + 2r^2 - n - r} \quad (4)$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_n = \frac{a_{n-1}}{2n^2 + n} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = \frac{1}{2}$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = \frac{1}{2r^2 + 3r + 1}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_1 = \frac{1}{3}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2r^2+3r+1}$	$\frac{1}{3}$

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{4r^4 + 20r^3 + 35r^2 + 25r + 6}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_2 = \frac{1}{30}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2r^2+3r+1}$	$\frac{1}{3}$
a_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{30}$

For $n = 3$, using the above recursive equation gives

$$a_3 = \frac{1}{8r^6 + 84r^5 + 350r^4 + 735r^3 + 812r^2 + 441r + 90}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_3 = \frac{1}{630}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2r^2+3r+1}$	$\frac{1}{3}$
a_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{30}$
a_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{630}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{16r^8 + 288r^7 + 2184r^6 + 9072r^5 + 22449r^4 + 33642r^3 + 29531r^2 + 13698r + 2520}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_4 = \frac{1}{22680}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2r^2+3r+1}$	$\frac{1}{3}$
a_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{30}$
a_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{630}$
a_4	$\frac{1}{16r^8+288r^7+2184r^6+9072r^5+22449r^4+33642r^3+29531r^2+13698r+2520}$	$\frac{1}{22680}$

For $n = 5$, using the above recursive equation gives

$$a_5 = \frac{1}{32r^{10} + 880r^9 + 10560r^8 + 72600r^7 + 315546r^6 + 902055r^5 + 1708465r^4 + 2102375r^3 + 1594197r^2 + 664290r + 113400}$$

Which for the root $r = \frac{1}{2}$ becomes

$$a_5 = \frac{1}{1247400}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{2r^2+3r+1}$	$\frac{1}{3}$
a_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{30}$
a_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{630}$
a_4	$\frac{1}{16r^8+288r^7+2184r^6+9072r^5+22449r^4+33642r^3+29531r^2+13698r+2520}$	$\frac{1}{22680}$
a_5	$\frac{1}{32r^{10}+880r^9+10560r^8+72600r^7+315546r^6+902055r^5+1708465r^4+2102375r^3+1594197r^2+664290r+113400}$	$\frac{1}{1247400}$

Using the above table, then the solution $y_1(x)$ is

$$\begin{aligned} y_1(x) &= \sqrt{x}(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \dots) \\ &= \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{30} + \frac{x^3}{630} + \frac{x^4}{22680} + \frac{x^5}{1247400} + O(x^6) \right) \end{aligned}$$

Now the second solution $y_2(x)$ is found. Eq (2B) derived above is now used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. For $1 \leq n$ the recursive equation is

$$2b_n(n+r)(n+r-1) + (n+r)b_n - b_{n-1} = 0 \quad (3)$$

Solving for b_n from recursive equation (4) gives

$$b_n = \frac{b_{n-1}}{2n^2 + 4nr + 2r^2 - n - r} \quad (4)$$

Which for the root $r = 0$ becomes

$$b_n = \frac{b_{n-1}}{n(2n-1)} \quad (5)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1

For $n = 1$, using the above recursive equation gives

$$b_1 = \frac{1}{2r^2 + 3r + 1}$$

Which for the root $r = 0$ becomes

$$b_1 = 1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2r^2+3r+1}$	1

For $n = 2$, using the above recursive equation gives

$$b_2 = \frac{1}{4r^4 + 20r^3 + 35r^2 + 25r + 6}$$

Which for the root $r = 0$ becomes

$$b_2 = \frac{1}{6}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2r^2+3r+1}$	1
b_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{6}$

For $n = 3$, using the above recursive equation gives

$$b_3 = \frac{1}{8r^6 + 84r^5 + 350r^4 + 735r^3 + 812r^2 + 441r + 90}$$

Which for the root $r = 0$ becomes

$$b_3 = \frac{1}{90}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2r^2+3r+1}$	1
b_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{6}$
b_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{90}$

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{1}{16r^8 + 288r^7 + 2184r^6 + 9072r^5 + 22449r^4 + 33642r^3 + 29531r^2 + 13698r + 2520}$$

Which for the root $r = 0$ becomes

$$b_4 = \frac{1}{2520}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2r^2+3r+1}$	1
b_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{6}$
b_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{90}$
b_4	$\frac{1}{16r^8+288r^7+2184r^6+9072r^5+22449r^4+33642r^3+29531r^2+13698r+2520}$	$\frac{1}{2520}$

For $n = 5$, using the above recursive equation gives

$$b_5 = \frac{1}{32r^{10} + 880r^9 + 10560r^8 + 72600r^7 + 315546r^6 + 902055r^5 + 1708465r^4 + 2102375r^3 + 1594197r^2 + 664290r + 113400}$$

Which for the root $r = 0$ becomes

$$b_5 = \frac{1}{113400}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$\frac{1}{2r^2+3r+1}$	1
b_2	$\frac{1}{4r^4+20r^3+35r^2+25r+6}$	$\frac{1}{6}$
b_3	$\frac{1}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$	$\frac{1}{90}$
b_4	$\frac{1}{16r^8+288r^7+2184r^6+9072r^5+22449r^4+33642r^3+29531r^2+13698r+2520}$	$\frac{1}{2520}$
b_5	$\frac{1}{32r^{10}+880r^9+10560r^8+72600r^7+315546r^6+902055r^5+1708465r^4+2102375r^3+1594197r^2+664290r+113400}$	$\frac{1}{113400}$

Using the above table, then the solution $y_2(x)$ is

$$\begin{aligned} y_2(x) &= b_0 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 \dots \\ &= 1 + x + \frac{x^2}{6} + \frac{x^3}{90} + \frac{x^4}{2520} + \frac{x^5}{113400} + O(x^6) \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(x) &= c_1 y_1(x) + c_2 y_2(x) \\ &= c_1 \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{30} + \frac{x^3}{630} + \frac{x^4}{22680} + \frac{x^5}{1247400} + O(x^6) \right) \\ &\quad + c_2 \left(1 + x + \frac{x^2}{6} + \frac{x^3}{90} + \frac{x^4}{2520} + \frac{x^5}{113400} + O(x^6) \right) \end{aligned}$$

The particular solution is found by solving for c, m the balance equation

$$(2x^{-1+m}m(-1+m) + mx^{-1+m})c_0 = F$$

Where $F(x)$ is the RHS of the ode. If $F(x)$ has more than one term, then this is done for each term one at a time and then all the particular solutions are added. The function $F(x)$ will be converted to series if needed. in order to solve for c_n, m for each term, the same recursive relation used to find $y_h(x)$ is used to find c_n, m which is used to find the particular solution $\sum_{n=0} c_n x^{n+m}$ by replacing a_n by c_n and r by m .

The following are the values of a_n found in terms of the indicial root r .

$a_1 = \frac{a_0}{2r^2+3r+1}$
$a_2 = \frac{a_0}{4r^4+20r^3+35r^2+25r+6}$
$a_3 = \frac{a_0}{8r^6+84r^5+350r^4+735r^3+812r^2+441r+90}$
$a_4 = \frac{a_0}{16r^8+288r^7+2184r^6+9072r^5+22449r^4+33642r^3+29531r^2+13698r+2520}$
$a_5 = \frac{a_0}{32r^{10}+880r^9+10560r^8+72600r^7+315546r^6+902055r^5+1708465r^4+2102375r^3+1594197r^2+664290r+113400}$

Since the $F = 1 + x$ has more than one term then we find a particular solution for each term and add the result to find the particular solution to the ode.

Now we determine the particular solution y_p associated with $F = 1$ by solving the balance equation

$$(2x^{-1+m}m(-1+m) + mx^{-1+m})c_0 = 1$$

For c_0 and x . This results in

$$c_0 = 1$$

$$m = 1$$

The particular solution is therefore

$$\begin{aligned} y_p &= \sum_{n=0}^{\infty} c_n x^{n+m} \\ &= \sum_{n=0}^{\infty} c_n x^{n+1} \end{aligned}$$

Where in the above $c_0 = 1$.

The remaining c_n values are found using the same recurrence relation given in the earlier table which was used to find the homogeneous solution but using c_0 in place of a_0 and using $m = 1$ in place of the root of the indicial equation used to find the homogeneous solution. By letting $a_0 = c_0$ or $a_0 = 1$ and $r = m$ or $r = 1$. The following table gives the resulting c_n values. These values will be used to find the particular solution. Values of c_n found not defined when doing the substitution will be discarded and not used

$c_0 = 1$
$c_1 = \frac{1}{6}$
$c_2 = \frac{1}{90}$
$c_3 = \frac{1}{2520}$
$c_4 = \frac{1}{113400}$
$c_5 = \frac{1}{7484400}$

The particular solution is now found using

$$\begin{aligned}
 y_p &= x^m \sum_{n=0}^{\infty} c_n x^n \\
 &= x \sum_{n=0}^{\infty} c_n x^n
 \end{aligned}$$

Using the values found above for c_n into the above sum gives

$$\begin{aligned}
 y_p &= x \left(1 + \frac{1}{6}x + \frac{1}{90}x^2 + \frac{1}{2520}x^3 + \frac{1}{113400}x^4 + \frac{1}{7484400}x^5 \right) \\
 &= x + \frac{1}{6}x^2 + \frac{1}{90}x^3 + \frac{1}{2520}x^4 + \frac{1}{113400}x^5 + \frac{1}{7484400}x^6
 \end{aligned}$$

Now we determine the particular solution y_p associated with $F = x$ by solving the balance equation

$$(2x^{-1+m}m(-1+m) + m x^{-1+m}) c_0 = x$$

For c_0 and x . This results in

$$\begin{aligned}
 c_0 &= \frac{1}{6} \\
 m &= 2
 \end{aligned}$$

The particular solution is therefore

$$\begin{aligned} y_p &= \sum_{n=0}^{\infty} c_n x^{n+m} \\ &= \sum_{n=0}^{\infty} c_n x^{n+2} \end{aligned}$$

Where in the above $c_0 = \frac{1}{6}$.

The remaining c_n values are found using the same recurrence relation given in the earlier table which was used to find the homogeneous solution but using c_0 in place of a_0 and using $m = 2$ in place of the root of the indicial equation used to find the homogeneous solution. By letting $a_0 = c_0$ or $a_0 = \frac{1}{6}$ and $r = m$ or $r = 2$. The following table gives the resulting c_n values. These values will be used to find the particular solution. Values of c_n found not defined when doing the substitution will be discarded and not used

$c_0 = \frac{1}{6}$
$c_1 = \frac{1}{90}$
$c_2 = \frac{1}{2520}$
$c_3 = \frac{1}{113400}$
$c_4 = \frac{1}{7484400}$
$c_5 = \frac{1}{681080400}$

The particular solution is now found using

$$\begin{aligned} y_p &= x^m \sum_{n=0}^{\infty} c_n x^n \\ &= x^2 \sum_{n=0}^{\infty} c_n x^n \end{aligned}$$

Using the values found above for c_n into the above sum gives

$$\begin{aligned} y_p &= x^2 \left(\frac{1}{6} + \frac{1}{90}x + \frac{1}{2520}x^2 + \frac{1}{113400}x^3 + \frac{1}{7484400}x^4 + \frac{1}{681080400}x^5 \right) \\ &= \frac{1}{6}x^2 + \frac{1}{90}x^3 + \frac{1}{2520}x^4 + \frac{1}{113400}x^5 + \frac{1}{7484400}x^6 + \frac{1}{681080400}x^7 \end{aligned}$$

Adding all the above particular solution(s) gives

$$y_p = x + \frac{x^2}{3} + \frac{x^3}{45} + \frac{x^4}{1260} + \frac{x^5}{56700} + \frac{x^6}{3742200} + \frac{x^7}{681080400} + O(x^6)$$

Truncating the particular solution to the order of series requested gives

$$y_p = x + \frac{x^2}{3} + \frac{x^3}{45} + \frac{x^4}{1260} + \frac{x^5}{56700} + O(x^6)$$

Hence the final solution is

$$\begin{aligned} y &= y_h + y_p \\ &= c_1 \sqrt{x} \left(1 + \frac{x}{3} + \frac{x^2}{30} + \frac{x^3}{630} + \frac{x^4}{22680} + \frac{x^5}{1247400} + O(x^6) \right) \\ &\quad + c_2 \left(1 + x + \frac{x^2}{6} + \frac{x^3}{90} + \frac{x^4}{2520} + \frac{x^5}{113400} + O(x^6) \right) + x + \frac{x^2}{3} + \frac{x^3}{45} + \frac{x^4}{1260} + \frac{x^5}{56700} \end{aligned}$$

✓ **Maple.** Time used: 0.013 (sec). Leaf size: 60

```
Order:=6;
ode:=2*x*diff(diff(y(x),x),x)+diff(y(x),x)-y(x) = 1+x;
dsolve(ode,y(x),type='series',x=0);
```

$$\begin{aligned} y &= c_1 \sqrt{x} \left(1 + \frac{1}{3}x + \frac{1}{30}x^2 + \frac{1}{630}x^3 + \frac{1}{22680}x^4 + \frac{1}{1247400}x^5 + O(x^6) \right) \\ &\quad + c_2 \left(1 + x + \frac{1}{6}x^2 + \frac{1}{90}x^3 + \frac{1}{2520}x^4 + \frac{1}{113400}x^5 + O(x^6) \right) \\ &\quad + x \left(1 + \frac{1}{3}x + \frac{1}{45}x^2 + \frac{1}{1260}x^3 + \frac{1}{56700}x^4 + O(x^5) \right) \end{aligned}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
trying high order exact linear fully integrable
trying differential order: 2; linear nonhomogeneous with symmetry [0,1]
trying a double symmetry of the form [xi=0, eta=F(x)]
-> Try solving first the homogeneous part of the ODE
    checking if the LODE has constant coefficients
    checking if the LODE is of Euler type
    trying a symmetry of the form [xi=0, eta=F(x)]
    <- linear_1 successful
<- solving first the homogeneous part of the ODE successful
```

✓ **Mathematica.** Time used: 0.021 (sec). Leaf size: 246

```
ode=2*x*D[y[x],{x,2}]+D[y[x],x]-y[x]==x+1;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,0,5}]
```

$$y(x) \rightarrow c_1 \left(\frac{x^5}{113400} + \frac{x^4}{2520} + \frac{x^3}{90} + \frac{x^2}{6} + x + 1 \right) + c_2 \sqrt{x} \left(\frac{x^5}{1247400} + \frac{x^4}{22680} + \frac{x^3}{630} + \frac{x^2}{30} + \frac{x}{3} + 1 \right) \\ + \sqrt{x} \left(\frac{x^5}{1247400} + \frac{x^4}{22680} + \frac{x^3}{630} + \frac{x^2}{30} + \frac{x}{3} + 1 \right) \left(\frac{23x^{11/2}}{311850} + \frac{29x^{9/2}}{11340} + \frac{16x^{7/2}}{315} + \frac{7x^{5/2}}{15} + \frac{4x^{3/2}}{3} \right. \\ \left. + 2\sqrt{x} \right) + \left(\frac{x^5}{113400} + \frac{x^4}{2520} + \frac{x^3}{90} + \frac{x^2}{6} + x + 1 \right) \left(-\frac{x^6}{133650} - \frac{37x^5}{113400} - \frac{11x^4}{1260} - \frac{11x^3}{90} - \frac{2x^2}{3} - x \right)$$

✗ **Sympy**

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x*Derivative(y(x), (x, 2)) - x - y(x) + Derivative(y(x), x) - 1, 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_regular", x0=0, n=6)
```

```
ValueError : ODE 2*x*Derivative(y(x), (x, 2)) - x - y(x) + Derivative(y(x), x) - 1 does not match hint 2nd_power_series_regular
```

2.17.10 Problem 20

Local contents

✓ Maple	2712
✓ Mathematica	2713
✓ Sympy	2713

Internal problem ID [8086]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 20

Date solved : Thursday, November 27, 2025 at 07:33:21 AM

CAS classification : [[_Emden, _Fowler]]

$$2x^3y'' + y'x^2 + y = 0$$

Series expansion around $x = \infty$.

Since expansion is around ∞ , then the independent variable x is replaced by $\frac{1}{\tau}$ and the expansion is made around $\tau = 0$ and after solving, the solution is changed back to x using $x = \frac{1}{\tau}$. Changing variables results in the new ode

$$-\frac{2\left(-\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau^2 - 2\left(\frac{d}{d\tau}y(\tau)\right)\tau\right)}{\tau} - \frac{d}{d\tau}y(\tau) + y(\tau) = 0$$

The transformed ODE is now solved. The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$2\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau + 3\frac{d}{d\tau}y(\tau) + y(\tau) = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$\frac{d^2}{d\tau^2}y(\tau) + p(\tau)\frac{d}{d\tau}y(\tau) + q(\tau)y(\tau) = 0$$

Where

$$p(\tau) = \frac{3}{2\tau}$$

$$q(\tau) = \frac{1}{2\tau}$$

Table 2.324: Table $p(\tau), q(\tau)$ singularities.

$p(\tau) = \frac{3}{2\tau}$	
singularity	type
$\tau = 0$	“regular”

$q(\tau) = \frac{1}{2\tau}$	
singularity	type
$\tau = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $\tau = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$2\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau + 3\frac{d}{d\tau}y(\tau) + y(\tau) = 0$$

Let the solution be represented as Frobenius power series of the form

$$y(\tau) = \sum_{n=0}^{\infty} a_n \tau^{n+r}$$

Then

$$\frac{d}{d\tau}y(\tau) = \sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1}$$

$$\frac{d^2}{d\tau^2}y(\tau) = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n \tau^{n+r-2}$$

Substituting the above back into the ode gives

$$2\left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n \tau^{n+r-2}\right)\tau + 3\left(\sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1}\right) + \left(\sum_{n=0}^{\infty} a_n \tau^{n+r}\right) = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=0}^{\infty} 2\tau^{n+r-1} a_n (n+r)(n+r-1)\right) + \left(\sum_{n=0}^{\infty} 3(n+r) a_n \tau^{n+r-1}\right) + \left(\sum_{n=0}^{\infty} a_n \tau^{n+r}\right) = 0 \quad (2A)$$

The next step is to make all powers of τ be $n+r-1$ in each summation term. Going over each summation term above with power of τ in it which is not already τ^{n+r-1} and adjusting the power and the corresponding index gives

$$\sum_{n=0}^{\infty} a_n \tau^{n+r} = \sum_{n=1}^{\infty} a_{n-1} \tau^{n+r-1}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of τ are the same and equal to $n + r - 1$.

$$\left(\sum_{n=0}^{\infty} 2\tau^{n+r-1} a_n (n+r) (n+r-1) \right) + \left(\sum_{n=0}^{\infty} 3(n+r) a_n \tau^{n+r-1} \right) + \left(\sum_{n=1}^{\infty} a_{n-1} \tau^{n+r-1} \right) = 0 \quad (2B)$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$2\tau^{n+r-1} a_n (n+r) (n+r-1) + 3(n+r) a_n \tau^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$2\tau^{-1+r} a_0 r(-1+r) + 3r a_0 \tau^{-1+r} = 0$$

Or

$$(2\tau^{-1+r} r(-1+r) + 3r \tau^{-1+r}) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$(2r^2 + r) \tau^{-1+r} = 0$$

Since the above is true for all τ then the indicial equation becomes

$$2r^2 + r = 0$$

Solving for r gives the roots of the indicial equation as

$$\begin{aligned} r_1 &= 0 \\ r_2 &= -\frac{1}{2} \end{aligned}$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$(2r^2 + r) \tau^{-1+r} = 0$$

Solving for r gives the roots of the indicial equation as $[0, -\frac{1}{2}]$.

Since $r_1 - r_2 = \frac{1}{2}$ is not an integer, then we can construct two linearly independent solutions

$$\begin{aligned} y_1(\tau) &= \tau^{r_1} \left(\sum_{n=0}^{\infty} a_n \tau^n \right) \\ y_2(\tau) &= \tau^{r_2} \left(\sum_{n=0}^{\infty} b_n \tau^n \right) \end{aligned}$$

Or

$$y_1(\tau) = \sum_{n=0}^{\infty} a_n \tau^n$$

$$y_2(\tau) = \sum_{n=0}^{\infty} b_n \tau^{n-\frac{1}{2}}$$

We start by finding $y_1(\tau)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$2a_n(n+r)(n+r-1) + 3a_n(n+r) + a_{n-1} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = -\frac{a_{n-1}}{2n^2 + 4nr + 2r^2 + n + r} \quad (4)$$

Which for the root $r = 0$ becomes

$$a_n = -\frac{a_{n-1}}{2n^2 + n} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = -\frac{1}{2r^2 + 5r + 3}$$

Which for the root $r = 0$ becomes

$$a_1 = -\frac{1}{3}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{2r^2+5r+3}$	$-\frac{1}{3}$

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{4r^4 + 28r^3 + 71r^2 + 77r + 30}$$

Which for the root $r = 0$ becomes

$$a_2 = \frac{1}{30}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{2r^2+5r+3}$	$-\frac{1}{3}$
a_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{30}$

For $n = 3$, using the above recursive equation gives

$$a_3 = -\frac{1}{8r^6 + 108r^5 + 590r^4 + 1665r^3 + 2552r^2 + 2007r + 630}$$

Which for the root $r = 0$ becomes

$$a_3 = -\frac{1}{630}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{2r^2+5r+3}$	$-\frac{1}{3}$
a_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{30}$
a_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{630}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{16r^8 + 352r^7 + 3304r^6 + 17248r^5 + 54649r^4 + 107338r^3 + 127251r^2 + 82962r + 22680}$$

Which for the root $r = 0$ becomes

$$a_4 = \frac{1}{22680}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{2r^2+5r+3}$	$-\frac{1}{3}$
a_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{30}$
a_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{630}$
a_4	$\frac{1}{16r^8+352r^7+3304r^6+17248r^5+54649r^4+107338r^3+127251r^2+82962r+22680}$	$\frac{1}{22680}$

For $n = 5$, using the above recursive equation gives

$$a_5 = -\frac{1}{32r^{10} + 1040r^9 + 14880r^8 + 123240r^7 + 653226r^6 + 2310945r^5 + 5514295r^4 + 8741785r^3 + 8786367r^2 + 4790016r + 1014000}$$

Which for the root $r = 0$ becomes

$$a_5 = -\frac{1}{1247400}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$-\frac{1}{2r^2+5r+3}$	$-\frac{1}{3}$
a_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{30}$
a_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{630}$
a_4	$\frac{1}{16r^8+352r^7+3304r^6+17248r^5+54649r^4+107338r^3+127251r^2+82962r+22680}$	$\frac{1}{22680}$
a_5	$-\frac{1}{32r^{10}+1040r^9+14880r^8+123240r^7+653226r^6+2310945r^5+5514295r^4+8741785r^3+8786367r^2+5039190r+1247400}$	$-\frac{1}{1247400}$

Using the above table, then the solution $y_1(\tau)$ is

$$\begin{aligned} y_1(\tau) &= a_0 + a_1\tau + a_2\tau^2 + a_3\tau^3 + a_4\tau^4 + a_5\tau^5 + a_6\tau^6 \dots \\ &= 1 - \frac{\tau}{3} + \frac{\tau^2}{30} - \frac{\tau^3}{630} + \frac{\tau^4}{22680} - \frac{\tau^5}{1247400} + O(\tau^6) \end{aligned}$$

Now the second solution $y_2(\tau)$ is found. Eq (2B) derived above is now used to find all b_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. b_0 is arbitrary and taken as $b_0 = 1$. For $1 \leq n$ the recursive equation is

$$2b_n(n+r)(n+r-1) + 3(n+r)b_n + b_{n-1} = 0 \quad (3)$$

Solving for b_n from recursive equation (4) gives

$$b_n = -\frac{b_{n-1}}{2n^2 + 4nr + 2r^2 + n + r} \quad (4)$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_n = -\frac{b_{n-1}}{n(2n-1)} \quad (5)$$

At this point, it is a good idea to keep track of b_n in a table both before substituting $r = -\frac{1}{2}$ and after as more terms are found using the above recursive equation.

n	$b_{n,r}$	b_n
b_0	1	1

For $n = 1$, using the above recursive equation gives

$$b_1 = -\frac{1}{2r^2 + 5r + 3}$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_1 = -1$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{1}{2r^2+5r+3}$	-1

For $n = 2$, using the above recursive equation gives

$$b_2 = \frac{1}{4r^4 + 28r^3 + 71r^2 + 77r + 30}$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_2 = \frac{1}{6}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{1}{2r^2+5r+3}$	-1
b_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{6}$

For $n = 3$, using the above recursive equation gives

$$b_3 = -\frac{1}{8r^6 + 108r^5 + 590r^4 + 1665r^3 + 2552r^2 + 2007r + 630}$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_3 = -\frac{1}{90}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{1}{2r^2+5r+3}$	-1
b_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{6}$
b_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{90}$

For $n = 4$, using the above recursive equation gives

$$b_4 = \frac{1}{16r^8 + 352r^7 + 3304r^6 + 17248r^5 + 54649r^4 + 107338r^3 + 127251r^2 + 82962r + 22680}$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_4 = \frac{1}{2520}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{1}{2r^2+5r+3}$	-1
b_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{6}$
b_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{90}$
b_4	$\frac{1}{16r^8+352r^7+3304r^6+17248r^5+54649r^4+107338r^3+127251r^2+82962r+22680}$	$\frac{1}{2520}$

For $n = 5$, using the above recursive equation gives

$$b_5 = -\frac{1}{32r^{10} + 1040r^9 + 14880r^8 + 123240r^7 + 653226r^6 + 2310945r^5 + 5514295r^4 + 8741785r^3 + 8786367r^2 + 5039190r + 1247400}$$

Which for the root $r = -\frac{1}{2}$ becomes

$$b_5 = -\frac{1}{113400}$$

And the table now becomes

n	$b_{n,r}$	b_n
b_0	1	1
b_1	$-\frac{1}{2r^2+5r+3}$	-1
b_2	$\frac{1}{4r^4+28r^3+71r^2+77r+30}$	$\frac{1}{6}$
b_3	$-\frac{1}{8r^6+108r^5+590r^4+1665r^3+2552r^2+2007r+630}$	$-\frac{1}{90}$
b_4	$\frac{1}{16r^8+352r^7+3304r^6+17248r^5+54649r^4+107338r^3+127251r^2+82962r+22680}$	$\frac{1}{2520}$
b_5	$-\frac{1}{32r^{10}+1040r^9+14880r^8+123240r^7+653226r^6+2310945r^5+5514295r^4+8741785r^3+8786367r^2+5039190r+1247400}$	$-\frac{1}{113400}$

Using the above table, then the solution $y_2(\tau)$ is

$$\begin{aligned} y_2(\tau) &= 1(b_0 + b_1\tau + b_2\tau^2 + b_3\tau^3 + b_4\tau^4 + b_5\tau^5 + b_6\tau^6 \dots) \\ &= \frac{1 - \tau + \frac{\tau^2}{6} - \frac{\tau^3}{90} + \frac{\tau^4}{2520} - \frac{\tau^5}{113400} + O(\tau^6)}{\sqrt{\tau}} \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned} y_h(\tau) &= c_1 y_1(\tau) + c_2 y_2(\tau) \\ &= c_1 \left(1 - \frac{\tau}{3} + \frac{\tau^2}{30} - \frac{\tau^3}{630} + \frac{\tau^4}{22680} - \frac{\tau^5}{1247400} + O(\tau^6) \right) \\ &\quad + \frac{c_2 \left(1 - \tau + \frac{\tau^2}{6} - \frac{\tau^3}{90} + \frac{\tau^4}{2520} - \frac{\tau^5}{113400} + O(\tau^6) \right)}{\sqrt{\tau}} \end{aligned}$$

Hence the final solution is

$$\begin{aligned}
 y(\tau) &= y_h \\
 &= c_1 \left(1 - \frac{\tau}{3} + \frac{\tau^2}{30} - \frac{\tau^3}{630} + \frac{\tau^4}{22680} - \frac{\tau^5}{1247400} + O(\tau^6) \right) \\
 &\quad + \frac{c_2 \left(1 - \tau + \frac{\tau^2}{6} - \frac{\tau^3}{90} + \frac{\tau^4}{2520} - \frac{\tau^5}{113400} + O(\tau^6) \right)}{\sqrt{\tau}}
 \end{aligned}$$

Replacing τ by $\frac{1}{x}$ gives

$$y = c_1 \left(1 - \frac{1}{3x} + \frac{1}{30x^2} - \frac{1}{630x^3} + \frac{1}{22680x^4} - \frac{1}{1247400x^5} + O\left(\frac{1}{x^6}\right) \right) + \frac{c_2 \left(1 - \frac{1}{x} + \frac{1}{6x^2} - \frac{1}{90x^3} + \frac{1}{2520x^4} - \frac{1}{113400x^5} + O\left(\frac{1}{x^6}\right) \right)}{\sqrt{\frac{1}{x}}}$$

✓ **Maple.** Time used: 0.014 (sec). Leaf size: 76

```
Order:=6;
ode:=2*x^3*diff(diff(y(x),x),x)+x^2*diff(y(x),x)+y(x) = 0;
dsolve(ode,y(x),type='series',x=infinity);
```

$$\begin{aligned}
 y &= \frac{c_1 \left(1 - \frac{1}{x} + \frac{1}{6x^2} - \frac{1}{90x^3} + \frac{1}{2520x^4} - \frac{1}{113400x^5} + O\left(\frac{1}{x^6}\right) \right)}{\sqrt{\frac{1}{x}}} \\
 &\quad + c_2 \left(1 - \frac{1}{3x} + \frac{1}{30x^2} - \frac{1}{630x^3} + \frac{1}{22680x^4} - \frac{1}{1247400x^5} + O\left(\frac{1}{x^6}\right) \right)
 \end{aligned}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Group is reducible or imprimitive
<- Kovacic's algorithm successful
```

✓ **Mathematica.** Time used: 0.003 (sec). Leaf size: 96

```
ode=2*x^3*D[y[x],{x,2}]+x^2*D[y[x],x]+y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,Infinity,5}]
```

$$y(x) \rightarrow c_2 \left(\frac{1}{6x^{3/2}} - \frac{1}{90x^{5/2}} + \frac{1}{2520x^{7/2}} - \frac{1}{113400x^{9/2}} + \sqrt{x} - \frac{1}{\sqrt{x}} \right) + c_1 \left(-\frac{1}{1247400x^5} + \frac{1}{22680x^4} - \frac{1}{630x^3} + \frac{1}{30x^2} - \frac{1}{3x} + 1 \right)$$

✓ **Sympy.** Time used: 0.472 (sec). Leaf size: 110

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(2*x**3*Derivative(y(x), (x, 2)) + x**2*Derivative(y(x), x) + y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_ordinary", x0=inf, n=6)
```

$$y(x) = C_3 \left(-C_2 + x - \frac{(C_2 - x)^2}{4C_2} - \frac{(C_2 - x)^3}{8C_2^2} - \frac{5(C_2 - x)^4}{64C_2^3} + \frac{(C_2 - x)^3}{12C_2^3} + \frac{7(C_2 - x)^4}{48C_2^4} \right) + C_1 \left(1 - \frac{(C_2 - x)^2}{4C_2^3} - \frac{7(C_2 - x)^3}{24C_2^4} - \frac{59(C_2 - x)^4}{192C_2^5} + \frac{(C_2 - x)^4}{96C_2^6} \right) + O(x^6)$$

2.17.11 Problem 21**Local contents**

✓ Maple	2721
✓ Mathematica	2721
✓ Sympy	2722

Internal problem ID [8087]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 26. Integration in series (singular points). Supplementary problems. Page 218

Problem number : 21

Date solved : Thursday, November 27, 2025 at 07:33:22 AM

CAS classification : `[_2nd_order, _with_linear_symmetries]`

$$x^3 y'' + (x^2 + x) y' - y = 0$$

Series expansion around $x = \infty$.

Since expansion is around ∞ , then the independent variable x is replaced by $\frac{1}{\tau}$ and the expansion is made around $\tau = 0$ and after solving, the solution is changed back to x using $x = \frac{1}{\tau}$. Changing variables results in the new ode

$$-\frac{\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau^2 - 2\left(\frac{d}{d\tau}y(\tau)\right)\tau}{\tau} - \left(\frac{1}{\tau^2} + \frac{1}{\tau}\right)\left(\frac{d}{d\tau}y(\tau)\right)\tau^2 - y(\tau) = 0$$

The transformed ODE is now solved. The type of the expansion point is first determined. This is done on the homogeneous part of the ODE.

$$\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau + (-\tau + 1)\left(\frac{d}{d\tau}y(\tau)\right) - y(\tau) = 0$$

The following is summary of singularities for the above ode. Writing the ode as

$$\frac{d^2}{d\tau^2}y(\tau) + p(\tau)\frac{d}{d\tau}y(\tau) + q(\tau)y(\tau) = 0$$

Where

$$p(\tau) = -\frac{\tau - 1}{\tau}$$

$$q(\tau) = -\frac{1}{\tau}$$

Table 2.325: Table $p(\tau), q(\tau)$ singularities.

$p(\tau) = -\frac{\tau-1}{\tau}$	
singularity	type
$\tau = 0$	“regular”

$q(\tau) = -\frac{1}{\tau}$	
singularity	type
$\tau = 0$	“regular”

Combining everything together gives the following summary of singularities for the ode as

Regular singular points : $[0]$

Irregular singular points : $[\infty]$

Since $\tau = 0$ is regular singular point, then Frobenius power series is used. The ode is normalized to be

$$\left(\frac{d^2}{d\tau^2}y(\tau)\right)\tau + (-\tau + 1)\left(\frac{d}{d\tau}y(\tau)\right) - y(\tau) = 0$$

Let the solution be represented as Frobenius power series of the form

$$y(\tau) = \sum_{n=0}^{\infty} a_n \tau^{n+r}$$

Then

$$\frac{d}{d\tau}y(\tau) = \sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1}$$

$$\frac{d^2}{d\tau^2}y(\tau) = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n \tau^{n+r-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n \tau^{n+r-2}\right)\tau + (-\tau + 1)\left(\sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1}\right) - \left(\sum_{n=0}^{\infty} a_n \tau^{n+r}\right) = 0 \quad (1)$$

Which simplifies to

$$\begin{aligned} &\left(\sum_{n=0}^{\infty} \tau^{n+r-1} a_n (n+r)(n+r-1)\right) + \sum_{n=0}^{\infty} (-a_n \tau^{n+r} (n+r)) \\ &+ \left(\sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1}\right) + \sum_{n=0}^{\infty} (-a_n \tau^{n+r}) = 0 \end{aligned} \quad (2A)$$

The next step is to make all powers of τ be $n + r - 1$ in each summation term. Going over each summation term above with power of τ in it which is not already τ^{n+r-1} and adjusting the power and the corresponding index gives

$$\begin{aligned}\sum_{n=0}^{\infty} (-a_n \tau^{n+r} (n+r)) &= \sum_{n=1}^{\infty} (-a_{n-1} (n+r-1) \tau^{n+r-1}) \\ \sum_{n=0}^{\infty} (-a_n \tau^{n+r}) &= \sum_{n=1}^{\infty} (-a_{n-1} \tau^{n+r-1})\end{aligned}$$

Substituting all the above in Eq (2A) gives the following equation where now all powers of τ are the same and equal to $n + r - 1$.

$$\begin{aligned}&\left(\sum_{n=0}^{\infty} \tau^{n+r-1} a_n (n+r) (n+r-1) \right) + \sum_{n=1}^{\infty} (-a_{n-1} (n+r-1) \tau^{n+r-1}) \\ &+ \left(\sum_{n=0}^{\infty} (n+r) a_n \tau^{n+r-1} \right) + \sum_{n=1}^{\infty} (-a_{n-1} \tau^{n+r-1}) = 0\end{aligned}\tag{2B}$$

The indicial equation is obtained from $n = 0$. From Eq (2B) this gives

$$\tau^{n+r-1} a_n (n+r) (n+r-1) + (n+r) a_n \tau^{n+r-1} = 0$$

When $n = 0$ the above becomes

$$\tau^{-1+r} a_0 r (-1+r) + r a_0 \tau^{-1+r} = 0$$

Or

$$(\tau^{-1+r} r (-1+r) + r \tau^{-1+r}) a_0 = 0$$

Since $a_0 \neq 0$ then the above simplifies to

$$r^2 \tau^{-1+r} = 0$$

Since the above is true for all τ then the indicial equation becomes

$$r^2 = 0$$

Solving for r gives the roots of the indicial equation as

$$r_1 = 0$$

$$r_2 = 0$$

Since $a_0 \neq 0$ then the indicial equation becomes

$$r^2\tau^{-1+r} = 0$$

Solving for r gives the roots of the indicial equation as $[0, 0]$.

Since the root of the indicial equation is repeated, then we can construct two linearly independent solutions. The first solution has the form

$$y_1(\tau) = \sum_{n=0}^{\infty} a_n \tau^{n+r} \quad (1A)$$

Now the second solution y_2 is found using

$$y_2(\tau) = y_1(\tau) \ln(\tau) + \left(\sum_{n=1}^{\infty} b_n \tau^{n+r} \right) \quad (1B)$$

Then the general solution will be

$$y(\tau) = c_1 y_1(\tau) + c_2 y_2(\tau)$$

In Eq (1B) the sum starts from 1 and not zero. In Eq (1A), a_0 is never zero, and is arbitrary and is typically taken as $a_0 = 1$, and $\{c_1, c_2\}$ are two arbitrary constants of integration which can be found from initial conditions. We start by finding the first solution $y_1(\tau)$. Eq (2B) derived above is now used to find all a_n coefficients. The case $n = 0$ is skipped since it was used to find the roots of the indicial equation. a_0 is arbitrary and taken as $a_0 = 1$. For $1 \leq n$ the recursive equation is

$$a_n(n+r)(n+r-1) - a_{n-1}(n+r-1) + a_n(n+r) - a_{n-1} = 0 \quad (3)$$

Solving for a_n from recursive equation (4) gives

$$a_n = \frac{a_{n-1}}{n+r} \quad (4)$$

Which for the root $r = 0$ becomes

$$a_n = \frac{a_{n-1}}{n} \quad (5)$$

At this point, it is a good idea to keep track of a_n in a table both before substituting $r = 0$ and after as more terms are found using the above recursive equation.

n	$a_{n,r}$	a_n
a_0	1	1

For $n = 1$, using the above recursive equation gives

$$a_1 = \frac{1}{1+r}$$

Which for the root $r = 0$ becomes

$$a_1 = 1$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{1+r}$	1

For $n = 2$, using the above recursive equation gives

$$a_2 = \frac{1}{(2+r)(1+r)}$$

Which for the root $r = 0$ becomes

$$a_2 = \frac{1}{2}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{1+r}$	1
a_2	$\frac{1}{(2+r)(1+r)}$	$\frac{1}{2}$

For $n = 3$, using the above recursive equation gives

$$a_3 = \frac{1}{(3+r)(2+r)(1+r)}$$

Which for the root $r = 0$ becomes

$$a_3 = \frac{1}{6}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{1+r}$	1
a_2	$\frac{1}{(2+r)(1+r)}$	$\frac{1}{2}$
a_3	$\frac{1}{(3+r)(2+r)(1+r)}$	$\frac{1}{6}$

For $n = 4$, using the above recursive equation gives

$$a_4 = \frac{1}{(3+r)(2+r)(1+r)(4+r)}$$

Which for the root $r = 0$ becomes

$$a_4 = \frac{1}{24}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{1+r}$	1
a_2	$\frac{1}{(2+r)(1+r)}$	$\frac{1}{2}$
a_3	$\frac{1}{(3+r)(2+r)(1+r)}$	$\frac{1}{6}$
a_4	$\frac{1}{(3+r)(2+r)(1+r)(4+r)}$	$\frac{1}{24}$

For $n = 5$, using the above recursive equation gives

$$a_5 = \frac{1}{(3+r)(2+r)(1+r)(4+r)(5+r)}$$

Which for the root $r = 0$ becomes

$$a_5 = \frac{1}{120}$$

And the table now becomes

n	$a_{n,r}$	a_n
a_0	1	1
a_1	$\frac{1}{1+r}$	1
a_2	$\frac{1}{(2+r)(1+r)}$	$\frac{1}{2}$
a_3	$\frac{1}{(3+r)(2+r)(1+r)}$	$\frac{1}{6}$
a_4	$\frac{1}{(3+r)(2+r)(1+r)(4+r)}$	$\frac{1}{24}$
a_5	$\frac{1}{(3+r)(2+r)(1+r)(4+r)(5+r)}$	$\frac{1}{120}$

Using the above table, then the first solution $y_1(\tau)$ becomes

$$\begin{aligned} y_1(\tau) &= a_0 + a_1\tau + a_2\tau^2 + a_3\tau^3 + a_4\tau^4 + a_5\tau^5 + a_6\tau^6 \dots \\ &= 1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6) \end{aligned}$$

Now the second solution is found. The second solution is given by

$$y_2(\tau) = y_1(\tau) \ln(\tau) + \left(\sum_{n=1}^{\infty} b_n \tau^{n+r} \right)$$

Where b_n is found using

$$b_n = \frac{d}{dr} a_{n,r}$$

And the above is then evaluated at $r = 0$. The above table for $a_{n,r}$ is used for this purpose. Computing the derivatives gives the following table

n	$b_{n,r}$	a_n	$b_{n,r} = \frac{d}{dr} a_{n,r}$	$b_n(r = 0)$
b_0	1	1	N/A since b_n starts from 1	N/A
b_1	$\frac{1}{1+r}$	1	$-\frac{1}{(1+r)^2}$	-1
b_2	$\frac{1}{(2+r)(1+r)}$	$\frac{1}{2}$	$\frac{-3-2r}{(2+r)^2(1+r)^2}$	$-\frac{3}{4}$
b_3	$\frac{1}{(3+r)(2+r)(1+r)}$	$\frac{1}{6}$	$\frac{-3r^2-12r-11}{(3+r)^2(2+r)^2(1+r)^2}$	$-\frac{11}{36}$
b_4	$\frac{1}{(3+r)(2+r)(1+r)(4+r)}$	$\frac{1}{24}$	$\frac{-4r^3-30r^2-70r-50}{(3+r)^2(2+r)^2(1+r)^2(4+r)^2}$	$-\frac{25}{288}$
b_5	$\frac{1}{(3+r)(2+r)(1+r)(4+r)(5+r)}$	$\frac{1}{120}$	$\frac{-5r^4-60r^3-255r^2-450r-274}{(3+r)^2(2+r)^2(1+r)^2(4+r)^2(5+r)^2}$	$-\frac{137}{7200}$

The above table gives all values of b_n needed. Hence the second solution is

$$\begin{aligned}
 y_2(\tau) &= y_1(\tau) \ln(\tau) + b_0 + b_1\tau + b_2\tau^2 + b_3\tau^3 + b_4\tau^4 + b_5\tau^5 + b_6\tau^6 \dots \\
 &= \left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6)\right) \ln(\tau) - \tau - \frac{3\tau^2}{4} - \frac{11\tau^3}{36} - \frac{25\tau^4}{288} - \frac{137\tau^5}{7200} + O(\tau^6)
 \end{aligned}$$

Therefore the homogeneous solution is

$$\begin{aligned}
 y_h(\tau) &= c_1 y_1(\tau) + c_2 y_2(\tau) \\
 &= c_1 \left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6)\right) \\
 &\quad + c_2 \left(\left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6)\right) \ln(\tau) - \tau - \frac{3\tau^2}{4} \right. \\
 &\quad \left. - \frac{11\tau^3}{36} - \frac{25\tau^4}{288} - \frac{137\tau^5}{7200} + O(\tau^6) \right)
 \end{aligned}$$

Hence the final solution is

$$\begin{aligned}
 y(\tau) &= y_h \\
 &= c_1 \left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6)\right) \\
 &\quad + c_2 \left(\left(1 + \tau + \frac{\tau^2}{2} + \frac{\tau^3}{6} + \frac{\tau^4}{24} + \frac{\tau^5}{120} + O(\tau^6)\right) \ln(\tau) - \tau - \frac{3\tau^2}{4} \right. \\
 &\quad \left. - \frac{11\tau^3}{36} - \frac{25\tau^4}{288} - \frac{137\tau^5}{7200} + O(\tau^6) \right)
 \end{aligned}$$

Replacing τ by $\frac{1}{x}$ gives

$$y = c_1 \left(1 + \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} + \frac{1}{24x^4} + \frac{1}{120x^5} + O\left(\frac{1}{x^6}\right) \right) + c_2 \left(\left(1 + \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} + \frac{1}{24x^4} + \frac{1}{120x^5} + O\left(\frac{1}{x^6}\right) \right) \right)$$

✓ **Maple.** Time used: 0.018 (sec). Leaf size: 112

```
Order:=6;
ode:=x^3*diff(diff(y(x),x),x)+(x^2+x)*diff(y(x),x)-y(x) = 0;
dsolve(ode,y(x),type='series',x=infinity);
```

$$y = \frac{7200 \left(O\left(\frac{1}{x^6}\right) x^5 + x^5 + x^4 + \frac{x^3}{2} + \frac{x^2}{6} + \frac{x}{24} + \frac{1}{120} \right) c_2 \ln\left(\frac{1}{x}\right) + 7200x^5(c_1 + c_2) O\left(\frac{1}{x^6}\right) + 7200c_1 x^5 + (7200c_1 + 7200c_2)}{7200x^5}$$

Maple trace

```
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
    A Liouvillian solution exists
    Reducible group (found an exponential solution)
    Group is reducible, not completely reducible
<- Kovacic's algorithm successful
```

✓ **Mathematica.** Time used: 0.004 (sec). Leaf size: 124

```
ode=x^3*D[y[x],{x,2}]+(x^2+x)*D[y[x],x]-y[x]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},y[x],{x,Infinity,5}]
```

$$y(x) \rightarrow c_1 \left(\frac{1}{120x^5} + \frac{1}{24x^4} + \frac{1}{6x^3} + \frac{1}{2x^2} + \frac{1}{x} + 1 \right) + c_2 \left(-\frac{137}{7200x^5} - \frac{\log(x)}{120x^5} - \frac{25}{288x^4} - \frac{\log(x)}{24x^4} - \frac{11}{36x^3} - \frac{\log(x)}{6x^3} - \frac{3}{4x^2} - \frac{\log(x)}{2x^2} - \frac{1}{x} - \frac{\log(x)}{x} - \log(x) \right)$$

✓ Sympy. Time used: 0.657 (sec). Leaf size: 175

```
from sympy import *
x = symbols("x")
y = Function("y")
ode = Eq(x**3*Derivative(y(x), (x, 2)) + (x**2 + x)*Derivative(y(x), x) - y(x), 0)
ics = {}
dsolve(ode, func=y(x), ics=ics, hint="2nd_power_series_ordinary", x0=inf, n=6)
```

$$y(x) = C_3 \left(-C_2 + x - \frac{(C_2 - x)^2}{2C_2} - \frac{(C_2 - x)^3}{3C_2^2} - \frac{(C_2 - x)^2}{2C_2^2} - \frac{(C_2 - x)^4}{4C_2^3} - \frac{5(C_2 - x)^3}{6C_2^3} \right. \\ \left. - \frac{13(C_2 - x)^4}{12C_2^4} - \frac{(C_2 - x)^3}{6C_2^4} - \frac{11(C_2 - x)^4}{24C_2^5} - \frac{(C_2 - x)^4}{24C_2^6} \right) + C_1 \left(1 + \frac{(C_2 - x)^2}{2C_2^3} \right. \\ \left. + \frac{2(C_2 - x)^3}{3C_2^4} + \frac{3(C_2 - x)^4}{4C_2^5} + \frac{(C_2 - x)^3}{6C_2^5} + \frac{5(C_2 - x)^4}{12C_2^6} + \frac{(C_2 - x)^4}{24C_2^7} \right) + O(x^6)$$

2.18 Chapter 27. The Legendre, Bessel and Gauss Equations. Supplemetary problems. Page 230

Local contents

2.18.1 Problem 20	2724
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2.18.1 Problem 20

Local contents

✓ Maple	2730
✓ Mathematica	2732
✓ Sympy	2732

Internal problem ID [8088]

Book : Schaums Outline. Theory and problems of Differential Equations, 1st edition. Frank Ayres. McGraw Hill 1952

Section : Chapter 27. The Legendre, Bessel and Gauss Equations. Supplementary problems. Page 230

Problem number : 20

Date solved : Thursday, November 27, 2025 at 07:33:22 AM

CAS classification : `[[_2nd_order, _with_linear_symmetries]]`

$$z'' + tz' + \left(t^2 - \frac{1}{9}\right)z = 0$$

Series expansion around $t = 0$.

Solving ode using Taylor series method. This gives review on how the Taylor series method works for solving second order ode.

Let

$$y'' = f(x, y, y')$$

Assuming expansion is at $x_0 = 0$ (we can always shift the actual expansion point to 0 by change of variables) and assuming $f(x, y, y')$ is analytic at x_0 which must be the case for an ordinary point. Let initial conditions be $y(x_0) = y_0$ and $y'(x_0) = y'_0$. Using Taylor series gives

$$\begin{aligned} y(x) &= y(x_0) + (x - x_0)y'(x_0) + \frac{(x - x_0)^2}{2}y''(x_0) + \frac{(x - x_0)^3}{3!}y'''(x_0) + \cdots \\ &= y_0 + xy'_0 + \frac{x^2}{2}f|_{x_0, y_0, y'_0} + \frac{x^3}{3!}f'|_{x_0, y_0, y'_0} + \cdots \\ &= y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} \frac{d^n f}{dx^n} \Big|_{x_0, y_0, y'_0} \end{aligned}$$

But

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y'} \frac{dy'}{dx} \quad (1)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \quad (2.134)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \quad (2.135)$$

$$\begin{aligned} \frac{d^2 f}{dx^2} &= \frac{d}{dx} \left(\frac{df}{dx} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{df}{dx} \right) + \frac{\partial}{\partial y} \left(\frac{df}{dx} \right) y' + \frac{\partial}{\partial y'} \left(\frac{df}{dx} \right) f \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d^3 f}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{d^2 f}{dx^2} \right) + \left(\frac{\partial}{\partial y} \frac{d^2 f}{dx^2} \right) y' + \frac{\partial}{\partial y'} \left(\frac{d^2 f}{dx^2} \right) f \end{aligned} \quad (3)$$

$\vdots$

And so on. Hence if we name $F_0 = f(x, y, y')$ then the above can be written as

$$F_0 = f(x, y, y') \quad (4)$$

$$\begin{aligned} F_1 &= \frac{df}{dx} \\ &= \frac{dF_0}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} f \\ &= \frac{\partial F_0}{\partial x} + \frac{\partial F_0}{\partial y} y' + \frac{\partial F_0}{\partial y'} F_0 \end{aligned} \quad (5)$$

$$\begin{aligned} F_2 &= \frac{d}{dx} \left(\frac{d}{dx} f \right) \\ &= \frac{d}{dx} (F_1) \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_1 + \left(\frac{\partial F_1}{\partial y} \right) y' + \left(\frac{\partial F_1}{\partial y'} \right) F_0 \\ &\vdots \\ F_n &= \frac{d}{dx} (F_{n-1}) \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) y'' \\ &= \frac{\partial}{\partial x} F_{n-1} + \left(\frac{\partial F_{n-1}}{\partial y} \right) y' + \left(\frac{\partial F_{n-1}}{\partial y'} \right) F_0 \end{aligned} \quad (6)$$

Therefore (6) can be used from now on along with

$$y(x) = y_0 + xy'_0 + \sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)!} F_n|_{x_0, y_0, y'_0} \quad (7)$$

To find $y(x)$ series solution around $x = 0$. Hence

$$\begin{aligned}
 F_0 &= -tz' - zt^2 + \frac{z}{9} \\
 F_1 &= \frac{dF_0}{dt} \\
 &= \frac{\partial F_0}{\partial t} + \frac{\partial F_0}{\partial z} z' + \frac{\partial F_0}{\partial z'} F_0 \\
 &= -\frac{8z'}{9} + \frac{(9t^3 - 19t)z}{9} \\
 F_2 &= \frac{dF_1}{dt} \\
 &= \frac{\partial F_1}{\partial t} + \frac{\partial F_1}{\partial z} z' + \frac{\partial F_1}{\partial z'} F_1 \\
 &= z't^3 + \frac{35zt^2}{9} - \frac{11tz'}{9} - \frac{179z}{81} \\
 F_3 &= \frac{dF_2}{dt} \\
 &= \frac{\partial F_2}{\partial t} + \frac{\partial F_2}{\partial z} z' + \frac{\partial F_2}{\partial z'} F_2 \\
 &= \frac{(-81t^4 + 657t^2 - 278)z'}{81} - t\left(t^4 - \frac{4}{3}t^2 - \frac{619}{81}\right)z \\
 F_4 &= \frac{dF_3}{dt} \\
 &= \frac{\partial F_3}{\partial t} + \frac{\partial F_3}{\partial z} z' + \frac{\partial F_3}{\partial z'} F_3 \\
 &= \frac{(-291t^3 + 737t)z'}{27} + \left(t^6 - \frac{119}{9}t^4 + \frac{25}{3}t^2 + \frac{5293}{729}\right)z
 \end{aligned}$$

And so on. Evaluating all the above at initial conditions $t = 0$ and $z(0) = z(0)$ and $z'(0) = z'(0)$ gives

$$\begin{aligned}
 F_0 &= \frac{z(0)}{9} \\
 F_1 &= -\frac{8z'(0)}{9} \\
 F_2 &= -\frac{179z(0)}{81} \\
 F_3 &= -\frac{278z'(0)}{81} \\
 F_4 &= \frac{5293z(0)}{729}
 \end{aligned}$$

Substituting all the above in (7) and simplifying gives the solution as

$$z = \left(1 + \frac{1}{18}t^2 - \frac{179}{1944}t^4\right)z(0) + \left(t - \frac{4}{27}t^3 - \frac{139}{4860}t^5\right)z'(0) + O(t^6)$$

Since the expansion point $t = 0$ is an ordinary point, then this can also be solved using the standard power series method. Let the solution be represented as power series of the form

$$z = \sum_{n=0}^{\infty} a_n t^n$$

Then

$$z' = \sum_{n=1}^{\infty} n a_n t^{n-1}$$

$$z'' = \sum_{n=2}^{\infty} n(n-1) a_n t^{n-2}$$

Substituting the above back into the ode gives

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n t^{n-2} \right) + t \left(\sum_{n=1}^{\infty} n a_n t^{n-1} \right) + \left(t^2 - \frac{1}{9} \right) \left(\sum_{n=0}^{\infty} a_n t^n \right) = 0 \quad (1)$$

Which simplifies to

$$\left(\sum_{n=2}^{\infty} n(n-1) a_n t^{n-2} \right) + \left(\sum_{n=1}^{\infty} n t^n a_n \right) + \left(\sum_{n=0}^{\infty} t^{n+2} a_n \right) + \sum_{n=0}^{\infty} \left(-\frac{a_n t^n}{9} \right) = 0 \quad (2)$$

The next step is to make all powers of t be n in each summation term. Going over each summation term above with power of t in it which is not already t^n and adjusting the power and the corresponding index gives

$$\sum_{n=2}^{\infty} n(n-1) a_n t^{n-2} = \sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) t^n$$

$$\sum_{n=0}^{\infty} t^{n+2} a_n = \sum_{n=2}^{\infty} a_{n-2} t^n$$

Substituting all the above in Eq (2) gives the following equation where now all powers of t are the same and equal to n .

$$\left(\sum_{n=0}^{\infty} (n+2) a_{n+2} (n+1) t^n \right) + \left(\sum_{n=1}^{\infty} n t^n a_n \right) + \left(\sum_{n=2}^{\infty} a_{n-2} t^n \right) + \sum_{n=0}^{\infty} \left(-\frac{a_n t^n}{9} \right) = 0 \quad (3)$$

$n = 0$ gives

$$2a_2 - \frac{a_0}{9} = 0$$

$$a_2 = \frac{a_0}{18}$$

$n = 1$ gives

$$6a_3 + \frac{8a_1}{9} = 0$$

Which after substituting earlier equations, simplifies to

$$a_3 = -\frac{4a_1}{27}$$

For $2 \leq n$, the recurrence equation is

$$(n+2)a_{n+2}(n+1) + na_n + a_{n-2} - \frac{a_n}{9} = 0 \quad (4)$$

Solving for a_{n+2} , gives

$$\begin{aligned} a_{n+2} &= -\frac{9na_n - a_n + 9a_{n-2}}{9(n+2)(n+1)} \\ &= -\frac{(9n-1)a_n}{9(n+2)(n+1)} - \frac{a_{n-2}}{(n+2)(n+1)} \end{aligned} \quad (5)$$

For $n = 2$ the recurrence equation gives

$$12a_4 + \frac{17a_2}{9} + a_0 = 0$$

Which after substituting the earlier terms found becomes

$$a_4 = -\frac{179a_0}{1944}$$

For $n = 3$ the recurrence equation gives

$$20a_5 + \frac{26a_3}{9} + a_1 = 0$$

Which after substituting the earlier terms found becomes

$$a_5 = -\frac{139a_1}{4860}$$

For $n = 4$ the recurrence equation gives

$$30a_6 + \frac{35a_4}{9} + a_2 = 0$$

Which after substituting the earlier terms found becomes

$$a_6 = \frac{5293a_0}{524880}$$

For $n = 5$ the recurrence equation gives

$$42a_7 + \frac{44a_5}{9} + a_3 = 0$$

Which after substituting the earlier terms found becomes

$$a_7 = \frac{3149a_1}{459270}$$

And so on. Therefore the solution is

$$\begin{aligned} z &= \sum_{n=0}^{\infty} a_n t^n \\ &= a_3 t^3 + a_2 t^2 + a_1 t + a_0 + \dots \end{aligned}$$

Substituting the values for a_n found above, the solution becomes

$$z = a_0 + a_1 t + \frac{1}{18} a_0 t^2 - \frac{4}{27} a_1 t^3 - \frac{179}{1944} a_0 t^4 - \frac{139}{4860} a_1 t^5 + \dots$$

Collecting terms, the solution becomes

$$z = \left(1 + \frac{1}{18} t^2 - \frac{179}{1944} t^4\right) a_0 + \left(t - \frac{4}{27} t^3 - \frac{139}{4860} t^5\right) a_1 + O(t^6) \quad (3)$$

At $t = 0$ the solution above becomes

$$z = \left(1 + \frac{1}{18} t^2 - \frac{179}{1944} t^4\right) z(0) + \left(t - \frac{4}{27} t^3 - \frac{139}{4860} t^5\right) z'(0) + O(t^6)$$

✓ **Maple.** Time used: 0.003 (sec). Leaf size: 39

```
Order:=6;
ode:=diff(diff(z(t),t),t)+t*diff(z(t),t)+(t^2-1/9)*z(t) = 0;
dsolve(ode,z(t),type='series',t=0);
```

$$z = \left(1 + \frac{1}{18} t^2 - \frac{179}{1944} t^4\right) z(0) + \left(t - \frac{4}{27} t^3 - \frac{139}{4860} t^5\right) z'(0) + O(t^6)$$

Maple trace

Methods for second order ODEs:

```

--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
checking if the LODE is of Euler type
trying a symmetry of the form [xi=0, eta=F(x)]
checking if the LODE is missing y
-> Trying a Liouvillian solution using Kovacic's algorithm
<- No Liouvillian solutions exists
-> Trying a solution in terms of special functions:
  -> Bessel
  -> elliptic
  -> Legendre
  -> Kummer
      -> hyper3: Equivalence to 1F1 under a power @ Moebius
      <- hyper3 successful: received ODE is equivalent to the 1F1 ODE
      <- Kummer successful
<- special function solution successful

```

Maple step by step

Let's solve

$$\frac{d^2}{dt^2}z(t) + t\left(\frac{d}{dt}z(t)\right) + \left(t^2 - \frac{1}{9}\right)z(t) = 0$$

- Highest derivative means the order of the ODE is 2

$$\frac{d^2}{dt^2}z(t)$$

- Multiply by denominators

$$9\frac{d^2}{dt^2}z(t) + 9t\left(\frac{d}{dt}z(t)\right) + (9t^2 - 1)z(t) = 0$$

- Assume series solution for $z(t)$

$$z(t) = \sum_{k=0}^{\infty} a_k t^k$$

- Rewrite ODE with series expansions

- o Convert $t^m \cdot z(t)$ to series expansion for $m = 0..2$

$$t^m \cdot z(t) = \sum_{k=\max(0, -m)}^{\infty} a_k t^{k+m}$$

- o Shift index using $k \rightarrow k - m$

$$t^m \cdot z(t) = \sum_{k=\max(0, -m)+m}^{\infty} a_{k-m} t^k$$

- o Convert $t \cdot \left(\frac{d}{dt}z(t)\right)$ to series expansion

$$t \cdot \left(\frac{d}{dt}z(t)\right) = \sum_{k=0}^{\infty} a_k k t^k$$

- Convert $\frac{d^2}{dt^2}z(t)$ to series expansion

$$\frac{d^2}{dt^2}z(t) = \sum_{k=2}^{\infty} a_k k(k-1) t^{k-2}$$

- Shift index using $k \rightarrow k+2$

$$\frac{d^2}{dt^2}z(t) = \sum_{k=0}^{\infty} a_{k+2}(k+2)(k+1) t^k$$

Rewrite ODE with series expansions

$$18a_2 - a_0 + (54a_3 + 8a_1)t + \left(\sum_{k=2}^{\infty} (9a_{k+2}(k+2)(k+1) + a_k(9k-1) + 9a_{k-2}) t^k \right) = 0$$

- The coefficients of each power of t must be 0
 $[18a_2 - a_0 = 0, 54a_3 + 8a_1 = 0]$
- Solve for the dependent coefficient(s)
 $\{a_2 = \frac{a_0}{18}, a_3 = -\frac{4a_1}{27}\}$
- Each term in the series must be 0, giving the recursion relation
 $9(k^2 + 3k + 2)a_{k+2} + 9a_k k - a_k + 9a_{k-2} = 0$
- Shift index using $k \rightarrow k+2$
 $9((k+2)^2 + 3k + 8)a_{k+4} + 9a_{k+2}(k+2) - a_{k+2} + 9a_k = 0$
- Recursion relation that defines the series solution to the ODE

$$\left[z(t) = \sum_{k=0}^{\infty} a_k t^k, a_{k+4} = -\frac{9ka_{k+2} + 9a_k + 17a_{k+2}}{9(k^2 + 7k + 12)}, a_2 = \frac{a_0}{18}, a_3 = -\frac{4a_1}{27} \right]$$

✓ **Mathematica.** Time used: 0.001 (sec). Leaf size: 42

```
ode=D[z[t],{t,2}]+t*D[z[t],t]+(t^2-1/9)*z[t]==0;
ic={};
AsymptoticDSolveValue[{ode,ic},z[t],{t,0,5}]
```

$$z(t) \rightarrow c_2 \left(-\frac{139t^5}{4860} - \frac{4t^3}{27} + t \right) + c_1 \left(-\frac{179t^4}{1944} + \frac{t^2}{18} + 1 \right)$$

✓ **Sympy.** Time used: 0.261 (sec). Leaf size: 41

```
from sympy import *
t = symbols("t")
z = Function("z")
ode = Eq(t*Derivative(z(t), t) + (t**2 - 1/9)*z(t) + Derivative(z(t), (t, 2)), 0)
ics = {}
dsolve(ode, func=z(t), ics=ics, hint="2nd_power_series_ordinary", x0=0, n=6)
```

$$z(t) = -\frac{13t^5 r(3)}{90} + C_2 \left(-\frac{179t^4}{1944} + \frac{t^2}{18} + 1 \right) + C_1 t \left(1 - \frac{t^4}{20} \right) + O(t^6)$$